

Curvature-Controlled Moiré Scaling and a Microscopic Magic–No-Magic Dichotomy in Hyperbolic Twisted Bilayers

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We establish a curvature-controlled crossover between Euclidean and hyperbolic moiré geometry while separating geometric effective counts, exact crystallographic sequences, and spectral flatness. The exact square five-state first-star Hamiltonian has no quadratic magic root. In contrast, a natural centered surface-group bilayer with geodesic intralayer hopping and full-distance exponential inter-layer tunneling has, in a declared character sector, an explicit positive simple dispersion-curvature root. Weighted Schur estimates continue this root to the full exponential kernel with a quantitative root interval, displacement, and derivative margin. Independently of character-sector completeness, the regular group-algebra operator has an isolated point island with a positive gap, active overlap, coherence, and stable Riesz projection. In an explicit centered regular-octagon symmetry class, the same coupling cancels every canonical Hodge tangent component and locks all principal Hessian eigenvalues at zero with a common nonzero slope; controlled symmetry breaking yields quantitative residual-anisotropy, root-displacement, and eigenvalue-spread bounds. Global finite-cover edges and gaps follow only on retained sectors with established no pollution, automatic tower independence only in the declared operator-tempered or operator-Ramanujan class, and completeness only through faithful reduced representations. Full-shell collapse requires a balanced growing cutoff and normalized physical tails, while unsmoothed and coherence-weighted densities require separate local-law and regularity hypotheses. The hyperbolic geometric-effective exponent is one, but the explicit centered exact sequence is $\mathfrak{U}_1(4)$, with arithmetic correction three, rather than a generic exponent-one exact class. These results separate the geometric, arithmetic, and spectral scales of the model.

I. GEOMETRY, COMMENSURABILITY, AND CURVATURE-CONTROLLED MOIRÉ SCALING

A. Unified Geometric Formulation of Twisted Bilayers

The central question of this work is whether moiré physics is controlled solely by the relative orientation of two lattices, or whether the geometry of the space supporting those lattices constitutes an independent control parameter. To formulate this question without implicitly assuming Euclidean geometry, we begin from a one-parameter family of simply connected two-dimensional spaces of constant Gaussian curvature, $\mathcal{M}_K$, $K \leq 0$. Throughout this work, we parameterize the curvature according to $K = -\kappa^2$, $\kappa \geq 0$. The Euclidean limit is obtained for

$$\kappa = 0, \quad K = 0, \quad \mathcal{M}_0 = \mathbb{R}^2, \quad (1)$$

whereas the negatively curved case is parameterized by a curvature radius according to

$$\kappa = \frac{1}{R}, \quad K = -\frac{1}{R^2} < 0, \quad \mathcal{M}_K = \mathbb{H}_R^2. \quad (2)$$

This formulation allows the Euclidean and hyperbolic problems to be treated as two geometric regimes of a

common construction rather than as unrelated lattice models. Choose an arbitrary reference point that subsequently serves as the center of the relative twist, $o \in \mathcal{M}_K$. In geodesic polar coordinates centered at this point, the metric of a simply connected two-dimensional manifold of constant nonpositive curvature can be written as

$$ds_K^2 = dr^2 + \text{sn}_K^2(r) d\phi^2, \quad (3)$$

where the generalized sine function is defined by

$$\text{sn}_K(r) = \begin{cases} r, & K = 0, \\ \frac{1}{\sqrt{-K}} \sinh(\sqrt{-K} r), & K < 0. \end{cases} \quad (4)$$

In the negatively curved regime this becomes

$$\text{sn}_K(r) = R \sinh\left(\frac{r}{R}\right). \quad (5)$$

The corresponding Riemannian area element follows directly from the determinant of the metric tensor,

$$dA_K = \sqrt{\det g} dr d\phi = \text{sn}_K(r) dr d\phi. \quad (6)$$

The area of a geodesic disk centered at the twist origin is therefore

$$B_K(o, r) = \{x \in \mathcal{M}_K : d_K(o, x) \leq r\}, \quad (7)$$

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with area

$$A_K(r) = 2\pi \int_0^r \text{sn}_K(s) ds. \quad (8)$$

For vanishing curvature, one obtains

$$A_0(r) = 2\pi \int_0^r s ds = \pi r^2. \quad (9)$$

For negative curvature, one instead finds

$$\begin{aligned} A_K(r) &= 2\pi R \int_0^r \sinh\left(\frac{s}{R}\right) ds \\ &= 2\pi R^2 \left[\cosh\left(\frac{r}{R}\right) - 1 \right]. \end{aligned} \quad (10)$$

The two results may therefore be summarized as

$$A_K(r) = \begin{cases} \pi r^2, & K = 0, \\ 2\pi R^2 [\cosh(r/R) - 1], & K = -R^{-2} < 0. \end{cases} \quad (11)$$

The distinction between quadratic Euclidean area growth and exponential hyperbolic area growth will play a central role in the curvature-controlled reconstruction of the moiré scaling law. A crystal on the underlying geometric manifold is defined through a discrete, properly discontinuous, and cocompact subgroup of the orientation-preserving isometry group, $\Gamma_K \subset \text{Isom}^+(\mathcal{M}_K)$, together with a finite microscopic basis, $\mathcal{B} = \{b_1, \dots, b_{n_b}\}$. The microscopic lattice is then defined as the corresponding group orbit,

$$\Lambda_K = \bigcup_{\gamma \in \Gamma_K} \bigcup_{\alpha=1}^{n_b} \gamma b_\alpha. \quad (12)$$

For a one-site Bravais lattice, the basis cardinality reduces to $n_b = 1$. The Euclidean square lattice appearing in the original problem is recovered by taking $K = 0$, $\Gamma_0 \simeq \mathbb{Z}^2$, where the group is generated by two independent translations. For negative curvature, the crystal group is chosen to be a cocompact Fuchsian group. The corresponding compact quotient is $\Sigma_K = \mathcal{M}_K/\Gamma_K$. Let a fundamental domain of the group be denoted by $F_K \subset \mathcal{M}_K$. Its covolume is defined by

$$A_0(K) = \text{Area}_K(F_K). \quad (13)$$

A natural quotient-space microscopic site density is consequently

$$\rho_0 = \frac{n_b}{A_0(K)}. \quad (14)$$

This quotient density should not, in hyperbolic geometry, be automatically identified with the asymptotic number of sites contained in an open geodesic disk divided by its area. The reason is that the boundary of a hyperbolic region remains extensive in the thermodynamic limit, as demonstrated explicitly below. We next construct the twisted bilayer. Let the relative twist be represented by an orientation-preserving isometry satisfying $g_\theta \in \text{Isom}^+(\mathcal{M}_K)$. The twist center is fixed, $g_\theta o = o$, and the differential at the fixed point acts as an ordinary tangent-space rotation, $Dg_\theta|_o = R_\theta$. The notation used in the original formulation of the problem expresses the relative twist as twice an auxiliary angle. Throughout the present work we use the convention $\theta = 2r$. For Euclidean geometry, the twist is represented by $g_\theta \mathbf{x} = R_\theta \mathbf{x}$. In the Poincaré-disk representation of the hyperbolic plane, after placing the twist center at the origin, the corresponding rotation can be written as $z \mapsto e^{i\theta} z$. The two microscopic layers are defined according to $\Lambda_1 = \Lambda_K$, and $\Lambda_2 = g_\theta \Lambda_K$. The complete bilayer configuration is therefore $\mathcal{B}_K(\theta) = \Lambda_1 \sqcup \Lambda_2$. The disjoint union is essential: geometrically coincident sites belonging to different layers remain distinct microscopic degrees of freedom because they carry different layer labels. Three mathematically distinct objects must be separated from the outset. The first is the local moiré length, ξ_M , which quantifies the distance from the twist center at which the relative displacement between the two layers reaches a prescribed microscopic registry scale. The second is an effective moiré domain, Ω_M^{eff} , with corresponding geometric area A_M^{eff} . The third is an exact crystallographic moiré supercell. Such an object exists only if the two infinite lattices possess a common discrete translation subgroup of finite index. The distinction can be summarized schematically as local moiré length $\neq$ effective moiré domain $\neq$ exact moiré supercell. These notions are not merely different conventions for measuring the same object. They are mathematically inequivalent constructions, and only the last one requires exact commensurability. Consider an arbitrary point whose geodesic distance from the twist center is

$$x \in \mathcal{M}_K, \quad r = d_K(o, x). \quad (15)$$

Because the relative twist is an isometry fixing the center, $d_K(o, g_\theta x) = r$. We define the twist-induced geodesic displacement as

$$\delta_\theta(r) = d_K(x, g_\theta x). \quad (16)$$

Rotational homogeneity implies that this quantity depends on the point only through its radial coordinate. In the Euclidean case, the twist center and the two corresponding points form an isosceles triangle. The Euclidean cosine law gives

$$\delta_\theta^2 = r^2 + r^2 - 2r^2 \cos \theta. \quad (17)$$

Using the trigonometric identity

$$1 - \cos \theta = 2 \sin^2 \left(\frac{\theta}{2} \right), \quad (18)$$

we obtain

$$\delta_\theta^2 = 4r^2 \sin^2 \left(\frac{\theta}{2} \right). \quad (19)$$

Since geodesic distance is nonnegative,

$$\delta_\theta^{(E)}(r) = 2r \left| \sin \left(\frac{\theta}{2} \right) \right|. \quad (20)$$

For negative curvature, the hyperbolic law of cosines gives

$$\cosh \left(\frac{\delta_\theta}{R} \right) = \cosh^2 \left(\frac{r}{R} \right) - \sinh^2 \left(\frac{r}{R} \right) \cos \theta. \quad (21)$$

Using $\cosh^2 x - \sinh^2 x = 1$, the right-hand side becomes

$$\begin{aligned} \cosh \left(\frac{\delta_\theta}{R} \right) &= 1 + \sinh^2 \left(\frac{r}{R} \right) - \sinh^2 \left(\frac{r}{R} \right) \cos \theta \\ &= 1 + \sinh^2 \left(\frac{r}{R} \right) (1 - \cos \theta) \\ &= 1 + 2 \sinh^2 \left(\frac{r}{R} \right) \sin^2 \left(\frac{\theta}{2} \right). \end{aligned} \quad (22)$$

The identity

$$\cosh u - 1 = 2 \sinh^2 \left(\frac{u}{2} \right) \quad (23)$$

then yields

$$2 \sinh^2 \left(\frac{\delta_\theta}{2R} \right) = 2 \sinh^2 \left(\frac{r}{R} \right) \sin^2 \left(\frac{\theta}{2} \right). \quad (24)$$

Nonnegativity of the distances implies

$$\sinh \left(\frac{\delta_\theta^{(H)}(r)}{2R} \right) = \sinh \left(\frac{r}{R} \right) \left| \sin \left(\frac{\theta}{2} \right) \right|. \quad (25)$$

Equivalently,

$$\delta_\theta^{(H)}(r) = 2R \operatorname{arsinh} \left[\sinh \left(\frac{r}{R} \right) \left| \sin \left(\frac{\theta}{2} \right) \right| \right]. \quad (26)$$

The Euclidean and hyperbolic displacement laws admit a unified constant-curvature representation,

$$\operatorname{sn}_K \left(\frac{\delta_\theta(r)}{2} \right) = \operatorname{sn}_K(r) \left| \sin \left(\frac{\theta}{2} \right) \right|. \quad (27)$$

For vanishing curvature, this reduces to

$$\frac{\delta_\theta}{2} = r \left| \sin \left(\frac{\theta}{2} \right) \right|, \quad (28)$$

whereas for negative curvature it gives

$$R \sinh \left(\frac{\delta_\theta}{2R} \right) = R \sinh \left(\frac{r}{R} \right) \left| \sin \left(\frac{\theta}{2} \right) \right|. \quad (29)$$

Equation (27) is exact for the full constant-curvature family considered here and constitutes the elementary geometric identity from which the local moiré scaling theory follows. To define a characteristic local moiré length unambiguously, we first establish monotonicity of the twist displacement. For Euclidean geometry,

$$\frac{d\delta_\theta^{(E)}}{dr} = 2 \left| \sin \left(\frac{\theta}{2} \right) \right| > 0 \quad (30)$$

for every nonzero nontrivial twist. For hyperbolic geometry, introduce the abbreviation

$$s = \left| \sin \left(\frac{\theta}{2} \right) \right|. \quad (31)$$

The displacement relation is

$$\sinh \left(\frac{\delta_\theta}{2R} \right) = s \sinh \left(\frac{r}{R} \right). \quad (32)$$

Differentiation with respect to the radial coordinate gives

$$\cosh \left(\frac{\delta_\theta}{2R} \right) \frac{1}{2R} \frac{d\delta_\theta}{dr} = \frac{s}{R} \cosh \left(\frac{r}{R} \right). \quad (33)$$

Solving for the derivative yields

$$\frac{d\delta_\theta}{dr} = \frac{2s \cosh(r/R)}{\cosh[\delta_\theta(r)/(2R)]} > 0. \quad (34)$$

Thus the displacement is strictly increasing with the distance from the twist center for every nonzero twist. Let the microscopic registry-resolution scale satisfy $\ell_* > 0$. Strict monotonicity implies that the equation $\delta_\theta(r) = \ell_*$ possesses a unique positive solution whenever the twist is nonzero. This establishes the mathematical well-posedness of the local moiré scale. We define the local moiré length by the condition $\delta_\theta(\xi_M) = \ell_*$. For a simple one-site lattice, a natural microscopic convention is

$\ell_* = a$, where the right-hand side denotes a microscopic lattice scale. Keeping the registry scale explicit, however, separates universal scaling from convention-dependent microscopic prefactors. Substituting the defining relation $\delta_\theta(\xi_M) = \ell_*$ into Eq. (27) yields

$$\text{sn}_K\left(\frac{\ell_*}{2}\right) = \text{sn}_K(\xi_M) \left| \sin\left(\frac{\theta}{2}\right) \right|. \quad (35)$$

Hence

$$\text{sn}_K(\xi_M) = \frac{\text{sn}_K(\ell_*/2)}{|\sin(\theta/2)|}. \quad (36)$$

Since the generalized sine function is strictly increasing on the nonnegative real axis, the exact moiré length is

$$\xi_M(\theta, K) = \text{sn}_K^{-1} \left[\frac{\text{sn}_K(\ell_*/2)}{|\sin(\theta/2)|} \right]. \quad (37)$$

This equation provides an exact constant-curvature interpolation between Euclidean and hyperbolic local moiré geometry. For vanishing curvature, $\text{sn}_0(r) = r$. Equation (37) becomes

$$\xi_M^{(E)} = \frac{\ell_*}{2 |\sin(\theta/2)|}. \quad (38)$$

For a small twist, the sine function has the expansion

$$\sin\left(\frac{\theta}{2}\right) = \frac{\theta}{2} - \frac{\theta^3}{48} + O(\theta^5). \quad (39)$$

Consequently,

$$2 \sin\left(\frac{\theta}{2}\right) = \theta \left[1 - \frac{\theta^2}{24} + O(\theta^4) \right]. \quad (40)$$

Inverting the series gives

$$\frac{1}{2 \sin(\theta/2)} = \frac{1}{\theta} \left[1 + \frac{\theta^2}{24} + O(\theta^4) \right]. \quad (41)$$

Therefore

$$\xi_M^{(E)} = \frac{\ell_*}{|\theta|} \left[1 + \frac{\theta^2}{24} + O(\theta^4) \right]. \quad (42)$$

In particular, $\xi_M^{(E)} \propto |\theta|^{-1}$. For negative curvature, the exact expression becomes

$$\xi_M^{(H)} = R \text{arsinh} \left[\frac{\sinh(\ell_*/2R)}{|\sin(\theta/2)|} \right]. \quad (43)$$

Introduce $b = \sinh\left(\frac{\ell_*}{2R}\right)$. Then

$$\frac{\xi_M^{(H)}}{R} = \text{arsinh} \left(\frac{b}{s} \right). \quad (44)$$

For fixed nonzero negative curvature and asymptotically small twist, $s \sim \frac{|\theta|}{2}$, and therefore

$$\frac{b}{s} \sim \frac{2b}{|\theta|} \rightarrow \infty. \quad (45)$$

Using the exact inverse-hyperbolic-sine identity,

$$\text{arsinh} x = \ln \left(x + \sqrt{1+x^2} \right), \quad (46)$$

and the large-argument expansion

$$\sqrt{1+x^2} = x \left[1 + \frac{1}{2x^2} + O(x^{-4}) \right], \quad (47)$$

we obtain

$$x + \sqrt{1+x^2} = 2x \left[1 + \frac{1}{4x^2} + O(x^{-4}) \right]. \quad (48)$$

Taking the logarithm gives

$$\text{arsinh} x = \ln(2x) + \frac{1}{4x^2} + O(x^{-4}). \quad (49)$$

Substitution into the exact moiré length yields

$$\xi_M^{(H)} = R \ln \left[\frac{4 \sinh(\ell_*/2R)}{|\theta|} \right] + O(R\theta^2). \quad (50)$$

Thus, for every fixed negative curvature, $\xi_M^{(H)} \sim R \ln |\theta|^{-1}$ in the small-angle limit. The Euclidean and hyperbolic asymptotic laws are therefore fundamentally different,

$$\xi_M^{(E)} \sim |\theta|^{-1}, \quad \xi_M^{(H)} \sim R \ln |\theta|^{-1}. \quad (51)$$

Negative curvature therefore changes the functional asymptotics of the local moiré length rather than merely renormalizing its prefactor. The crossover between locally Euclidean and curvature-dominated behavior is controlled by the dimensionless variable

$$\chi = \frac{\sinh(\ell_*/2R)}{|\sin(\theta/2)|}. \quad (52)$$

The exact hyperbolic moiré length can then be written as $\frac{\xi_M}{R} = \text{arsinh } \chi$. In the regime $\chi \ll 1$, the inverse hyperbolic sine admits the expansion

$$\text{arsinh } \chi = \chi - \frac{\chi^3}{6} + O(\chi^5). \quad (53)$$

Therefore $\xi_M = R\chi + O(R\chi^3)$. If the microscopic registry scale is additionally small compared with the curvature radius, $\ell_* \ll R$, then

$$\sinh\left(\frac{\ell_*}{2R}\right) = \frac{\ell_*}{2R} + O\left(\frac{\ell_*^3}{R^3}\right). \quad (54)$$

Consequently,

$$\xi_M \simeq \frac{\ell_*}{2|\sin(\theta/2)|}, \quad (55)$$

which recovers the Euclidean moiré scale. In the opposite regime, $\chi \gg 1$, one has

$$\text{arsinh } \chi = \ln(2\chi) + O(\chi^{-2}), \quad (56)$$

giving the curvature-dominated logarithmic regime. A natural crossover scale is determined parametrically by $\chi \simeq 1$. This implies

$$\left|\sin\left(\frac{\theta_c}{2}\right)\right| \simeq \sinh\left(\frac{\ell_*}{2R}\right). \quad (57)$$

For weak microscopic curvature, $\frac{\ell_*}{R} \ll 1$, the crossover angle satisfies $\theta_c \simeq \frac{\ell_*}{R}$. Negative curvature therefore introduces an independent geometric length scale, and the competition between twist and curvature is controlled by a dimensionless combination of the form $\frac{\ell_*}{R|\theta|}$. The local moiré length naturally defines an effective registry domain,

$$\Omega_M^{\text{eff}} = \{x \in \mathcal{M}_K : d_K(x, g_\theta x) \leq \ell_*\}. \quad (58)$$

Because the twist-induced displacement depends only on the radial coordinate and is strictly increasing, the defining condition is equivalent to $r \leq \xi_M$. Hence the effective moiré domain is exactly a geodesic disk, $\Omega_M^{\text{eff}} = B_K(o, \xi_M)$. Its area is consequently $A_M^{\text{eff}} = A_K(\xi_M)$. For Euclidean geometry,

$$A_M^{(E)} = \pi \left(\xi_M^{(E)}\right)^2. \quad (59)$$

Substitution of the exact Euclidean moiré length gives

$$A_M^{(E)} = \frac{\pi \ell_*^2}{4 \sin^2(\theta/2)}. \quad (60)$$

Using the small-angle expansion yields

$$A_M^{(E)} = \frac{\pi \ell_*^2}{\theta^2} \left[1 + \frac{\theta^2}{12} + O(\theta^4)\right]. \quad (61)$$

Therefore $A_M^{(E)} \propto |\theta|^{-2}$. For hyperbolic geometry,

$$A_M^{(H)} = 2\pi R^2 \left[\cosh\left(\frac{\xi_M}{R}\right) - 1\right]. \quad (62)$$

From the exact moiré-length relation,

$$\sinh\left(\frac{\xi_M}{R}\right) = \frac{b}{s}. \quad (63)$$

Using the hyperbolic identity gives

$$\cosh\left(\frac{\xi_M}{R}\right) = \sqrt{1 + \frac{b^2}{s^2}}. \quad (64)$$

Therefore

$$A_M^{(H)} = 2\pi R^2 \left[\sqrt{1 + \frac{\sinh^2(\ell_*/2R)}{\sin^2(\theta/2)}} - 1\right]. \quad (65)$$

Equation (65) is an exact crossover relation and does not rely on a small-angle approximation. In the curvature-dominated regime, $\chi \gg 1$. The square root admits the expansion

$$\sqrt{1 + \chi^2} = \chi + \frac{1}{2\chi} + O(\chi^{-3}). \quad (66)$$

The exact area therefore becomes

$$A_M^{(H)} = 2\pi R^2 \left[\chi - 1 + \frac{1}{2\chi} + O(\chi^{-3})\right]. \quad (67)$$

At small twist,

$$\chi = \frac{2 \sinh(\ell_*/2R)}{|\theta|} \left[1 + \frac{\theta^2}{24} + O(\theta^4)\right]. \quad (68)$$

Consequently,

$$A_M^{(H)} = \frac{4\pi R^2 \sinh(\ell_*/2R)}{|\theta|} - 2\pi R^2 + O(|\theta|). \quad (69)$$

The leading behavior is therefore

$$A_M^{(H)} \sim 4\pi R^2 \sinh\left(\frac{\ell_*}{2R}\right) |\theta|^{-1}. \quad (70)$$

Thus the effective moiré-domain area exhibits the asymptotic crossover

$$A_M^{(E)} \propto |\theta|^{-2} \longrightarrow A_M^{(H)} \propto |\theta|^{-1}. \quad (71)$$

This statement concerns the effective geometric registry domain. It does not by itself imply the same scaling for the index of an exact hyperbolic coincidence supercell. Define the dimensionless scaling variable

$$X = \frac{|\sin(\theta/2)|}{\sinh(\ell_*/2R)}. \quad (72)$$

The exact hyperbolic area relation becomes

$$\frac{A_M^{(H)}}{2\pi R^2} = \mathcal{F}(X), \quad (73)$$

where

$$\mathcal{F}(X) = \sqrt{1 + X^{-2}} - 1. \quad (74)$$

The entire geometric crossover is therefore controlled by a single dimensionless scaling function. To quantify the continuous evolution of the effective scaling exponent, define

$$\beta_M = -\frac{d \ln A_M}{d \ln s}. \quad (75)$$

Introduce $q = \sqrt{1 + X^{-2}}$. Since the effective area is proportional to $A_M \propto q - 1$, we calculate

$$\frac{dq}{dX} = -\frac{X^{-3}}{q}. \quad (76)$$

It follows that

$$\frac{d \ln A_M}{d \ln X} = \frac{X}{q-1} \frac{dq}{dX} = -\frac{X^{-2}}{q(q-1)}. \quad (77)$$

The identity

$$X^{-2} = q^2 - 1 = (q-1)(q+1) \quad (78)$$

therefore gives

$$\frac{d \ln A_M}{d \ln X} = -\frac{q+1}{q}. \quad (79)$$

Since the scaling variable is proportional to the sine of the half-twist angle, the running exponent becomes

$$\beta_M = 1 + \frac{1}{q} = 1 + \frac{1}{\sqrt{1 + X^{-2}}}. \quad (80)$$

In the locally Euclidean regime, $X \gg 1$, $\beta_M \rightarrow 2$, whereas in the curvature-dominated regime, $X \ll 1$, $\beta_M \rightarrow 1$. Thus the effective moiré-area exponent evolves continuously according to $2 \rightarrow 1$. This running exponent provides a direct observable for numerical tests of the predicted curvature-controlled crossover. Using the quotient-space primitive-cell covolume and the number of basis sites, define the area-normalized effective microscopic count per layer by

$$N_{M,1}^{\text{eff}} = n_b \frac{A_M^{\text{eff}}}{A_0(K)}. \quad (81)$$

For the complete bilayer,

$$N_M^{\text{eff}} = 2n_b \frac{A_M^{\text{eff}}}{A_0(K)}. \quad (82)$$

This quantity is not required to be an integer because it characterizes an effective geometric domain rather than an exact crystallographic supercell. The Euclidean scaling is $N_M^{\text{eff},E} \propto |\theta|^{-2}$, whereas the curvature-dominated hyperbolic regime gives $N_M^{\text{eff},H} \propto |\theta|^{-1}$. The distinction between quotient density and open-cluster counting is particularly important in hyperbolic geometry. For a Euclidean disk, the perimeter and area are $P_E(r) = 2\pi r$, and $A_E(r) = \pi r^2$. Their ratio is

$$\frac{P_E(r)}{A_E(r)} = \frac{2}{r}, \quad (83)$$

and hence

$$\lim_{r \rightarrow \infty} \frac{P_E(r)}{A_E(r)} = 0. \quad (84)$$

Euclidean boundary corrections therefore become subextensive. For a hyperbolic disk, the circumference is

$$P_H(r) = 2\pi R \sinh\left(\frac{r}{R}\right), \quad (85)$$

while its area is

$$A_H(r) = 2\pi R^2 \left[\cosh\left(\frac{r}{R}\right) - 1 \right]. \quad (86)$$

Consequently,

$$\frac{P_H(r)}{A_H(r)} = \frac{1}{R} \frac{\sinh(r/R)}{\cosh(r/R) - 1}. \quad (87)$$

Using the identities

$$\sinh x = 2 \sinh\left(\frac{x}{2}\right) \cosh\left(\frac{x}{2}\right) \quad (88)$$

and

$$\cosh x - 1 = 2 \sinh^2\left(\frac{x}{2}\right), \quad (89)$$

we obtain

$$\frac{P_H(r)}{A_H(r)} = \frac{1}{R} \coth\left(\frac{r}{2R}\right). \quad (90)$$

Therefore

$$\lim_{r \rightarrow \infty} \frac{P_H(r)}{A_H(r)} = \frac{1}{R} \neq 0. \quad (91)$$

The boundary of a large hyperbolic region remains extensive. Accordingly, the integer number of microscopic sites in an open effective moiré domain,

$$N_M^{\text{disk}} = \sum_{\ell=1}^2 \#(\Lambda_\ell \cap \Omega_M^{\text{eff}}), \quad (92)$$

cannot be replaced without additional proof by the quotient-density estimate $N_M^{\text{disk}} \simeq 2\rho_0 A_M^{\text{eff}}$ with an asymptotically vanishing relative boundary correction. This nonamenable boundary structure is one of the reasons why finite hyperbolic clusters, finite covers, and exact compact quotients must be treated separately. We now introduce the genuinely crystallographic moiré object. The translation group of the first layer is $\Gamma_1 = \Gamma_K$. The translation group of the rotated layer is its conjugate, $\Gamma_2 = g_\theta \Gamma_K g_\theta^{-1}$. An isometry is a translation symmetry of the complete bilayer precisely when it is simultaneously a translation symmetry of both microscopic layers. The maximal common translation subgroup is therefore

$$\Gamma_M = \Gamma_K \cap g_\theta \Gamma_K g_\theta^{-1}. \quad (93)$$

The common translation symmetry group of the twisted bilayer is precisely the subgroup defined in Eq. (93). Suppose first that a group element belongs to the common subgroup, $h \in \Gamma_M$. By definition, $h \in \Gamma_K$, and therefore $h\Lambda_1 = \Lambda_1$. At the same time, $h \in g_\theta \Gamma_K g_\theta^{-1}$, so there exists an element of the original microscopic translation group satisfying $\gamma \in \Gamma_K$ and $h = g_\theta \gamma g_\theta^{-1}$. Acting on the second layer gives

$$\begin{aligned} h\Lambda_2 &= g_\theta \gamma g_\theta^{-1} g_\theta \Lambda_1 = g_\theta \gamma \Lambda_1 \\ &= g_\theta \Lambda_1 = \Lambda_2. \end{aligned} \quad (94)$$

Hence every element of the common subgroup preserves both layers. Conversely, any translation belonging to the microscopic crystal symmetry class that preserves both layers must belong simultaneously to the two layer translation groups. Therefore it belongs to their intersection,

$$h \in \Gamma_1 \cap \Gamma_2 = \Gamma_M. \quad (95)$$

Thus the maximal common translation symmetry group is exactly

$$\Gamma_M = \Gamma_K \cap g_\theta \Gamma_K g_\theta^{-1}. \quad (96)$$

An exact finite moiré supercell exists if and only if the common translation subgroup contains only finitely many primitive cells of either microscopic layer. This condition is expressed by finite group index,

$$[\Gamma_1 : \Gamma_M] < \infty, \quad [\Gamma_2 : \Gamma_M] < \infty. \quad (97)$$

a. Theorem 1. Exact moiré commensurability. For a twisted bilayer generated by two conjugate cocompact crystal groups,

$$\Gamma_1 = \Gamma_K, \quad \Gamma_2 = g_\theta \Gamma_K g_\theta^{-1}, \quad (98)$$

an exact finite moiré supercell exists if and only if $\Gamma_K \cap g_\theta \Gamma_K g_\theta^{-1}$ has finite index in both microscopic translation groups. Equivalently, the relative twist must belong to the commensurator, $g_\theta \in \text{Comm}(\Gamma_K)$, where

$$\text{Comm}(\Gamma_K) = \left\{ g : \begin{aligned} &[\Gamma_K : \Gamma_K \cap g\Gamma_K g^{-1}] < \infty, \\ &[g\Gamma_K g^{-1} : \Gamma_K \cap g\Gamma_K g^{-1}] < \infty \end{aligned} \right\}. \quad (99)$$

Assume first that the common subgroup has finite index in the first microscopic translation group, $q_1 = [\Gamma_1 : \Gamma_M] < \infty$. Choose a complete set of coset representatives, $\gamma_1, \dots, \gamma_{q_1}$, such that

$$\Gamma_1 = \bigcup_{j=1}^{q_1} \Gamma_M \gamma_j. \quad (100)$$

Up to sets of measure zero, a fundamental region of the common subgroup can consequently be constructed as the union of finitely many translated primitive fundamental regions of the first layer. Therefore the common cell contains finitely many microscopic primitive cells. If likewise $q_2 = [\Gamma_2 : \Gamma_M] < \infty$, the same region contains finitely many primitive cells of the second layer.

Hence the common subgroup defines a finite exact bilayer supercell. Conversely, suppose that a finite common supercell exists, with translation group denoted by H . Because it is a symmetry of both microscopic layers, $H \subset \Gamma_1$, $H \subset \Gamma_2$. Because the common cell contains only finitely many primitive microscopic cells,

$$[\Gamma_1 : H] < \infty, \quad [\Gamma_2 : H] < \infty. \quad (101)$$

Furthermore,

$$H \subset \Gamma_1 \cap \Gamma_2 = \Gamma_M. \quad (102)$$

A subgroup containing a finite-index subgroup is itself of finite index. Hence

$$[\Gamma_1 : \Gamma_M] < \infty, \quad [\Gamma_2 : \Gamma_M] < \infty. \quad (103)$$

The condition is therefore both necessary and sufficient. Define the two finite indices as

$$q_1 = [\Gamma_1 : \Gamma_M], \quad q_2 = [\Gamma_2 : \Gamma_M]. \quad (104)$$

Since the two microscopic crystal groups are related by an isometric conjugation, their covolumes coincide,

$$\text{covol}(\Gamma_1) = \text{covol}(\Gamma_2) = A_0. \quad (105)$$

The common subgroup has covolume $\text{covol}(\Gamma_M) = q_1 A_0$, and simultaneously $\text{covol}(\Gamma_M) = q_2 A_0$. Consequently, $q_1 = q_2$. We denote the common value by q_M . The exact moiré-supercell area is therefore $A_M^{\text{sc}} = q_M A_0$. If every microscopic primitive cell contains the same number of basis sites per layer, the exact bilayer microscopic count is $N_M^{\text{sc}} = 2n_b q_M$. Unlike the effective geometric count, the exact supercell count is necessarily an integer. For a torsion-free cocompact hyperbolic translation group, the primitive quotient is a closed orientable surface, $\Sigma_0 = \mathbb{H}_R^2 / \Gamma_K$. Its genus satisfies $g_0 \geq 2$. The Gauss–Bonnet theorem gives

$$\int_{\Sigma_0} K dA = 2\pi\chi(\Sigma_0). \quad (106)$$

For a closed orientable surface, $\chi(\Sigma_0) = 2 - 2g_0$. Since the curvature is constant, $K = -\frac{1}{R^2}$, we obtain

$$-\frac{A_0}{R^2} = 2\pi(2 - 2g_0). \quad (107)$$

Hence $A_0 = 4\pi R^2(g_0 - 1)$. Suppose now that the common moiré subgroup has finite index, $[\Gamma_K : \Gamma_M] = q_M$. The quotient $\Sigma_M = \mathbb{H}_R^2 / \Gamma_M$ is then a finite covering of

the primitive quotient. Its Euler characteristic satisfies $\chi(\Sigma_M) = q_M \chi(\Sigma_0)$. Therefore

$$2 - 2g_M = q_M(2 - 2g_0), \quad (108)$$

which yields

$$g_M = 1 + q_M(g_0 - 1). \quad (109)$$

Applying Gauss–Bonnet to the enlarged quotient gives

$$A_M^{\text{sc}} = 4\pi R^2(g_M - 1). \quad (110)$$

Using Eq. (109), this becomes

$$A_M^{\text{sc}} = 4\pi R^2 q_M(g_0 - 1) = q_M A_0, \quad (111)$$

thereby reproducing $A_M^{\text{sc}} = q_M A_0$ independently. Thus hyperbolic supercell enlargement has a topological interpretation: enlarging the crystallographic fundamental domain corresponds simultaneously to increasing the covering degree and increasing the genus of the compact quotient. For crystal groups containing torsion, the corresponding statements must be formulated using orbifold Euler characteristic rather than the ordinary Euler characteristic of a smooth closed surface. The preceding construction now allows the distinct moiré quantities to be classified precisely. The local moiré length is a metric quantity, $\xi_M = \xi_M(\theta, K, \ell_*)$. It exists for every nonzero relative twist and does not require global lattice commensurability. The effective moiré area is likewise a local geometric quantity, $A_M^{\text{eff}} = A_K(\xi_M)$. The corresponding area-normalized effective count is

$$N_M^{\text{eff}} = 2n_b \frac{A_M^{\text{eff}}}{A_0(K)}. \quad (112)$$

By contrast, exact supercell quantities exist only under the commensurability condition $g_\theta \in \text{Comm}(\Gamma_K)$. When this condition holds, $A_M^{\text{sc}} = q_M A_0$, and $N_M^{\text{sc}} = 2n_b q_M$. The hierarchy can therefore be summarized as follows. The quantity ξ_M is a metric quantity characterizing the local moiré length. The quantity A_M^{eff} is a local geometric-domain quantity describing the effective moiré area, while N_M^{eff} denotes the corresponding area-normalized effective microscopic count. By contrast, A_M^{sc} and N_M^{sc} are exact group-theoretic supercell quantities defined only when an exact crystallographic moiré supercell exists. In general there is no identity of the form $A_M^{\text{eff}} = A_M^{\text{sc}}$, nor of the form $N_M^{\text{eff}} = N_M^{\text{sc}}$. Any asymptotic relation between these objects must be established rather than assumed. The distinction is already unavoidable in the simplest limiting cases. Consider first exactly zero twist, $\theta = 0$. Then $g_0 = \text{id}$, so that $\Gamma_M = \Gamma_K$. Consequently, $q_M = 1$, and therefore $A_M^{\text{sc}} = A_0$, while

$N_M^{\text{sc}} = 2n_b$. There is thus no enlarged exact crystallographic unit cell at zero twist. On the other hand, the local displacement vanishes identically, $\delta_0(r) = 0$ for every radial coordinate. Therefore no finite solution of the registry-threshold equation exists for a positive registry scale. Equivalently,

$$\lim_{\theta \rightarrow 0} \xi_M(\theta, K) = \infty. \quad (113)$$

Thus one simultaneously has

$$\lim_{\theta \rightarrow 0} \xi_M = \infty, \quad N_M^{\text{sc}}(\theta = 0) = 2n_b. \quad (114)$$

The divergence of the local interference scale near zero twist therefore cannot be interpreted as the literal size of the exact crystallographic unit cell at zero twist. As a second example, consider a generic incommensurate twist satisfying $g_\theta \notin \text{Comm}(\Gamma_K)$. No finite exact supercell exists in this case, and hence N_M^{sc} is undefined. Nevertheless, both local geometric quantities remain finite,

$$\xi_M(\theta, K) < \infty, \quad A_M^{\text{eff}}(\theta, K) < \infty. \quad (115)$$

Local moiré physics therefore does not require exact global crystallographic commensurability. The analysis above proves the Euclidean local scaling laws

$$\xi_M^{(E)} \sim \ell_* |\theta|^{-1}, \quad A_M^{(E)} \sim \pi \ell_*^2 |\theta|^{-2}. \quad (116)$$

For every fixed negative curvature, it also proves

$$\xi_M^{(H)} \sim R \ln |\theta|^{-1}, \quad (117)$$

and

$$A_M^{(H)} \sim 4\pi R^2 \sinh\left(\frac{\ell_*}{2R}\right) |\theta|^{-1}. \quad (118)$$

Consequently, the area-normalized effective microscopic counts satisfy

$$N_M^{\text{eff}, E} \propto |\theta|^{-2}, \quad N_M^{\text{eff}, H} \propto |\theta|^{-1}. \quad (119)$$

The local geometric argument alone does not determine an exact crystallographic law of the form $N_M^{\text{sc}, H} \propto |\theta|^{-1}$. The exact problem is instead controlled by the finite-index function

$$q_M(\theta) = [\Gamma_K : \Gamma_K \cap g_\theta \Gamma_K g_\theta^{-1}]. \quad (120)$$

A rigorous transfer of the geometric scaling law to the exact crystallographic supercell therefore requires information about commensurate sequences satisfying $\theta_j \rightarrow 0$

together with quantitative control of the corresponding indices, $q_M(\theta_j)$. A particularly strong geometric-arithmetic locking result would establish constants satisfying $0 < c_- \leq c_+ < \infty$ such that

$$c_- \frac{A_M^{\text{eff}}(\theta_j)}{A_0} \leq q_M(\theta_j) \leq c_+ \frac{A_M^{\text{eff}}(\theta_j)}{A_0}. \quad (121)$$

Such an estimate would imply

$$q_M(\theta_j) = \Theta(A_M^{\text{eff}}(\theta_j)), \quad (122)$$

and would transfer the geometric hyperbolic scaling law to the exact crystallographic index, $q_M(\theta_j) \propto |\theta_j|^{-1}$. Only after establishing such a result would it be mathematically justified to conclude $N_M^{\text{sc}, H} \propto |\theta|^{-1}$ along an appropriate family of exact commensurate twists. The geometric content of the preceding construction can be summarized in the following theorem.

b. Theorem 2. Curvature-controlled local moiré geometry. Let the underlying manifold be a simply connected two-dimensional space of constant nonpositive curvature, $\mathcal{M}_K$, $K \leq 0$. Let the relative twist be a rotation fixing a point of the manifold, $g_\theta o = o$, $\theta \neq 0$, and let the microscopic registry threshold satisfy $\ell_* > 0$. Then the twist-induced displacement satisfies the exact identity

$$\text{sn}_K\left(\frac{\delta_\theta(r)}{2}\right) = \text{sn}_K(r) \left| \sin\left(\frac{\theta}{2}\right) \right|. \quad (123)$$

The local moiré length exists uniquely and is given by

$$\xi_M = \text{sn}_K^{-1} \left[\frac{\text{sn}_K(\ell_*/2)}{|\sin(\theta/2)|} \right]. \quad (124)$$

The effective registry domain is exactly the geodesic disk $\Omega_M^{\text{eff}} = B_K(o, \xi_M)$, with area $A_M^{\text{eff}} = A_K(\xi_M)$. In Euclidean geometry, the small-angle asymptotics are

$$\xi_M \sim |\theta|^{-1}, \quad A_M^{\text{eff}} \sim |\theta|^{-2}. \quad (125)$$

For every fixed negative curvature, the asymptotic laws become

$$\xi_M \sim R \ln |\theta|^{-1}, \quad A_M^{\text{eff}} \sim |\theta|^{-1}. \quad (126)$$

By contrast, an exact crystallographic moiré supercell exists if and only if $g_\theta \in \text{Comm}(\Gamma_K)$. Whenever this condition holds, the exact supercell area and microscopic count are $A_M^{\text{sc}} = q_M A_0$, and $N_M^{\text{sc}} = 2n_b q_M$, where

$$q_M = [\Gamma_K : \Gamma_K \cap g_\theta \Gamma_K g_\theta^{-1}]. \quad (127)$$

Hence curvature-controlled local moiré scaling and exact crystallographic commensurability constitute mathematically distinct structures, and their quantitative relation must be established rather than assumed. The local geometric hierarchy established in this section is

$$K \longrightarrow \delta_\theta(r) \longrightarrow \xi_M \longrightarrow \Omega_M^{\text{eff}} \longrightarrow A_M^{\text{eff}} \longrightarrow N_M^{\text{eff}}. \quad (128)$$

The exact crystallographic hierarchy is instead

$$(\Gamma_K, g_\theta) \longrightarrow \Gamma_M = \Gamma_K \cap g_\theta \Gamma_K g_\theta^{-1} \longrightarrow q_M \longrightarrow A_M^{\text{sc}} \longrightarrow N_M^{\text{sc}}. \quad (129)$$

The central mathematical problem that follows from this formulation is therefore to determine whether, and under what assumptions, the local geometric and exact crystallographic chains become asymptotically locked. Establishing such a geometric–arithmetic locking theorem is the necessary next step before promoting the effective hyperbolic scaling exponent to an exact crystallographic universality law.

B. Exact Commensurability of Twisted Euclidean Square Bilayers

Having established a geometry-independent distinction between local moiré scales and exact crystallographic supercells, we now specialize to the Euclidean square bilayer. This case serves three purposes. First, it provides an exact solution of the commensurability problem appearing in the original square-lattice construction. Second, it supplies the Euclidean benchmark against which the hyperbolic problem will subsequently be compared. Third, it exposes an important arithmetic structure that is invisible in a purely geometric estimate of the moiré length: even in flat space, the size of an exact coincidence supercell is determined jointly by the twist geometry and by an integer commensurability condition. Throughout this subsection we retain a general microscopic lattice constant and set it to unity only when evaluating the examples appearing in the original problem. The primitive Euclidean square lattice is

$$\Lambda = a\mathbb{Z}^2 = \left\{ a \begin{pmatrix} j \\ k \end{pmatrix} : j, k \in \mathbb{Z} \right\}. \quad (130)$$

A convenient primitive basis is

$$\mathbf{a}_1 = a \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{a}_2 = a \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (131)$$

The primitive-cell area is therefore

$$A_0 = |\det(\mathbf{a}_1 \ \mathbf{a}_2)| = a^2. \quad (132)$$

We choose the first layer to be unrotated, $\Lambda_1 = \Lambda$, while the second layer is rotated through a relative angle, $\Lambda_2 = R_\theta \Lambda$. The rotation matrix is

$$R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}. \quad (133)$$

The exact coincidence-site lattice is defined by the intersection $\Lambda_M = \Lambda \cap R_\theta \Lambda$. An exact finite moiré supercell exists precisely when this common lattice has finite index in either microscopic lattice,

$$\Sigma(\theta) = [\Lambda : \Lambda_M] < \infty. \quad (134)$$

The integer quantity in Eq. (134) will be called the coincidence index. Because the microscopic square lattice has one site per primitive cell, an exact coincidence supercell contains $N_1^{\text{sc}} = \Sigma$ sites from each individual layer. For a bilayer, the two layer labels correspond to independent microscopic degrees of freedom even at geometrically coincident positions. Therefore the total number of atoms in the bilayer supercell is $N_M^{\text{sc}} = 2\Sigma$. The exact supercell area is similarly $A_M^{\text{sc}} = \Sigma a^2$. Thus determining the exact Euclidean moiré cell reduces to determining the coincidence index. We first establish a necessary and sufficient condition for the existence of a finite coincidence lattice. For a nontrivial rotation of the square lattice in the angular interval $0 < \theta < \frac{\pi}{2}$, the following statements are equivalent:

$$\begin{aligned} \text{(i)} \quad & [\Lambda : \Lambda \cap R_\theta \Lambda] < \infty, \quad \text{(ii)} \quad R_\theta \in SO(2, \mathbb{Q}), \\ \text{(iii)} \quad & \cos \theta \in \mathbb{Q}, \quad \sin \theta \in \mathbb{Q}, \quad \text{(iv)} \quad \tan\left(\frac{\theta}{2}\right) \in \mathbb{Q}. \end{aligned} \quad (135)$$

Assume that the coincidence lattice has finite index, $[\Lambda : \Lambda_M] < \infty$. A finite-index subgroup of a rank-two lattice is itself a rank-two lattice. Hence there exist two linearly independent coincidence vectors $\mathbf{u}_1, \mathbf{u}_2 \in \Lambda_M$. Since these vectors belong to the first microscopic lattice, there exists an integer matrix with nonzero determinant such that

$$(\mathbf{u}_1 \ \mathbf{u}_2) = aU, \quad U \in M_2(\mathbb{Z}), \quad \det U \neq 0. \quad (136)$$

The same vectors also belong to the rotated lattice. Consequently, there exist two integer vectors in the unrotated coordinate representation of the second layer, or equivalently an integer matrix satisfying

$$(\mathbf{u}_1 \ \mathbf{u}_2) = aR_\theta V, \quad V \in M_2(\mathbb{Z}), \quad \det V \neq 0. \quad (137)$$

Equating Eqs. (136) and (137) gives $U = R_\theta V$. Since the matrix on the right is invertible, we obtain $R_\theta = UV^{-1}$. The inverse of a nonsingular integer matrix is a rational matrix because

$$V^{-1} = \frac{\text{adj } V}{\det V}. \quad (138)$$

Hence $V^{-1} \in M_2(\mathbb{Q})$, and therefore $R_\theta \in M_2(\mathbb{Q})$. Because the matrix is a rotation, $R_\theta \in SO(2, \mathbb{Q})$. Inspection of the matrix entries then immediately implies

$$\cos \theta \in \mathbb{Q}, \quad \sin \theta \in \mathbb{Q}. \quad (139)$$

Finally, the half-angle identity gives

$$\tan\left(\frac{\theta}{2}\right) = \frac{\sin \theta}{1 + \cos \theta}. \quad (140)$$

In the angular interval considered here, the denominator is nonzero. Therefore

$$\tan\left(\frac{\theta}{2}\right) \in \mathbb{Q}. \quad (141)$$

This proves the necessity of the rational half-angle condition. Conversely, assume

$$\tan\left(\frac{\theta}{2}\right) \in \mathbb{Q}. \quad (142)$$

The standard half-angle identities are

$$\cos \theta = \frac{1 - \tan^2(\theta/2)}{1 + \tan^2(\theta/2)}, \quad (143)$$

and

$$\sin \theta = \frac{2 \tan(\theta/2)}{1 + \tan^2(\theta/2)}. \quad (144)$$

Thus both trigonometric functions are rational,

$$\cos \theta \in \mathbb{Q}, \quad \sin \theta \in \mathbb{Q}. \quad (145)$$

Hence there exists a positive integer denominator satisfying $qR_\theta \in M_2(\mathbb{Z})$. Because the inverse of a rotation is its transpose, $R_\theta^{-1} = R_\theta^T$, the same denominator may be chosen such that $qR_\theta^{-1} \in M_2(\mathbb{Z})$. For every integer vector, $\mathbf{z} \in \mathbb{Z}^2$, consider the lattice vector $\mathbf{x} = aq\mathbf{z}$. It clearly belongs to the first lattice, $\mathbf{x} \in \Lambda$. Moreover,

$$R_\theta^{-1}\mathbf{x} = a(qR_\theta^{-1})\mathbf{z} \in a\mathbb{Z}^2 = \Lambda. \quad (146)$$

Therefore $\mathbf{x} \in R_\theta\Lambda$. It follows that

$$aq\mathbb{Z}^2 \subset \Lambda \cap R_\theta\Lambda. \quad (147)$$

The sublattice on the left has finite index,

$$[\Lambda : aq\mathbb{Z}^2] = q^2. \quad (148)$$

Since the coincidence lattice contains a finite-index sublattice, it is itself finite index. Thus

$$[\Lambda : \Lambda \cap R_\theta\Lambda] < \infty. \quad (149)$$

The four conditions in Eq. (135) are therefore equivalent. This completes the proof. The preceding proposition implies that every nontrivial commensurate rotation can be parameterized by a rational half-angle. Write that rational number in lowest terms,

$$\tan\left(\frac{\theta}{2}\right) = \frac{n}{m}, \quad m > n > 0, \quad \gcd(m, n) = 1. \quad (150)$$

The notation of the original project is recovered by defining $r = \frac{\theta}{2}$. Hence $\tan r = \frac{n}{m}$. Substitution into the half-angle identities gives

$$\cos \theta = \frac{m^2 - n^2}{m^2 + n^2}. \quad (151)$$

Similarly,

$$\sin \theta = \frac{2mn}{m^2 + n^2}. \quad (152)$$

Thus the exact rotation matrix is

$$R_\theta = \frac{1}{m^2 + n^2} \begin{pmatrix} m^2 - n^2 & -2mn \\ 2mn & m^2 - n^2 \end{pmatrix}. \quad (153)$$

The corresponding twist angle is

$$\theta_{m,n} = 2 \arctan\left(\frac{n}{m}\right). \quad (154)$$

This establishes a complete integer parameterization of square-lattice commensurate rotations. The integer parameterization has an immediate geometric interpretation. Define two integer vectors symmetrically placed about the horizontal axis,

$$\mathbf{v}_- = a \begin{pmatrix} m \\ -n \end{pmatrix}, \quad \mathbf{v}_+ = a \begin{pmatrix} m \\ n \end{pmatrix}. \quad (155)$$

Their respective angles relative to the horizontal axis are $-r$ and $+r$, because $\tan r = \frac{n}{m}$. The angular separation between the two vectors is therefore $2r = \theta$. Using the exact rotation matrix, one obtains

$$R_\theta \mathbf{v}_- = \frac{a}{m^2 + n^2} \begin{pmatrix} (m^2 - n^2)m + 2mn^2 \\ 2m^2n - (m^2 - n^2)n \end{pmatrix}. \quad (156)$$

The first component simplifies according to

$$(m^2 - n^2)m + 2mn^2 = m(m^2 + n^2), \quad (157)$$

while the second satisfies

$$2m^2n - (m^2 - n^2)n = n(m^2 + n^2). \quad (158)$$

Therefore

$$R_\theta \begin{pmatrix} m \\ -n \end{pmatrix} = \begin{pmatrix} m \\ n \end{pmatrix}. \quad (159)$$

Hence

$$a \begin{pmatrix} m \\ n \end{pmatrix} \in \Lambda \cap R_\theta \Lambda. \quad (160)$$

An orthogonal companion is obtained from

$$\mathbf{w}_- = a \begin{pmatrix} n \\ m \end{pmatrix}. \quad (161)$$

A direct calculation gives

$$R_\theta \begin{pmatrix} n \\ m \end{pmatrix} = \begin{pmatrix} -n \\ m \end{pmatrix}. \quad (162)$$

Thus a natural pair of coincidence vectors is

$$\mathbf{T}_1 = a \begin{pmatrix} m \\ n \end{pmatrix}, \quad \mathbf{T}_2 = a \begin{pmatrix} -n \\ m \end{pmatrix}. \quad (163)$$

These vectors are mutually orthogonal,

$$\mathbf{T}_1 \cdot \mathbf{T}_2 = a^2 (-mn + mn) = 0, \quad (164)$$

and have equal norm,

$$|\mathbf{T}_1|^2 = |\mathbf{T}_2|^2 = a^2 (m^2 + n^2). \quad (165)$$

The corresponding cell area is

$$A_{\text{naive}} = a^2 (m^2 + n^2). \quad (166)$$

However, this cell is not always primitive. A parity-dependent reduction must still be performed. Define $S = m^2 + n^2$. Because the two integers are coprime, they

cannot both be even. There are therefore only two parity classes: opposite parity, both odd. The common divisor of the three integer quantities entering the rational rotation matrix is

$$d = \gcd(m^2 - n^2, 2mn, m^2 + n^2). \quad (167)$$

We now determine this divisor exactly. Suppose an odd prime divides both the first two nontrivial numerators,

$$p \mid m^2 - n^2, \quad p \mid 2mn, \quad p \text{ odd}. \quad (168)$$

Since the prime is odd, the second condition implies $p \mid m$ or $p \mid n$. If the first possibility holds, then $m \equiv 0 \pmod{p}$. The condition on the difference of squares then gives $-n^2 \equiv 0 \pmod{p}$, and therefore $p \mid n$. This contradicts coprimality. The same contradiction follows if the roles of the two integers are interchanged. Thus no odd prime can divide the three quantities simultaneously. The only possible common divisor is therefore a power of two. If the two coprime integers have opposite parity, then

$$m^2 + n^2 \equiv 1 \pmod{2}. \quad (169)$$

Hence $d = 1$. If both integers are odd, then

$$m^2 \equiv n^2 \equiv 1 \pmod{8}. \quad (170)$$

Consequently,

$$m^2 + n^2 \equiv 2 \pmod{4}. \quad (171)$$

Thus the common divisor contains precisely one factor of two and no higher power, $d = 2$. The complete parity factor is therefore

$$d = \begin{cases} 1, & m \not\equiv n \pmod{2}, \\ 2, & m \equiv n \equiv 1 \pmod{2}. \end{cases} \quad (172)$$

This factor determines whether the geometrically obvious coincidence cell constructed above is already primitive. The primitive reduction can be established most transparently and completely by identifying the square lattice with the Gaussian-integer ring. Introduce the correspondence

$$\mathbb{Z}^2 \longleftrightarrow \mathbb{Z}[i], \quad \begin{pmatrix} x \\ y \end{pmatrix} \longleftrightarrow x + iy. \quad (173)$$

Define the Gaussian integers

$$\alpha = m + in, \quad \bar{\alpha} = m - in. \quad (174)$$

The rotation phase may then be written exactly as

$$e^{i\theta} = \frac{m^2 - n^2 + 2imn}{m^2 + n^2}. \quad (175)$$

Since

$$(m + in)(m - in) = m^2 + n^2, \quad (176)$$

one obtains $e^{i\theta} = \frac{\alpha}{\bar{\alpha}}$. Thus the rotated square lattice is represented by

$$e^{i\theta}\mathbb{Z}[i] = \frac{\alpha}{\bar{\alpha}}\mathbb{Z}[i]. \quad (177)$$

The coincidence lattice is consequently

$$\mathbb{Z}[i] \cap \frac{\alpha}{\bar{\alpha}}\mathbb{Z}[i]. \quad (178)$$

The numerator and denominator in the rotation phase may possess a common Gaussian divisor. Let

$$g = \gcd_{\mathbb{Z}[i]}(\alpha, \bar{\alpha}), \quad (179)$$

and define the reduced Gaussian integers

$$\alpha_0 = \frac{\alpha}{g}, \quad \beta_0 = \frac{\bar{\alpha}}{g}. \quad (180)$$

They are coprime in the Gaussian-integer ring,

$$\gcd_{\mathbb{Z}[i]}(\alpha_0, \beta_0) = 1 \quad (181)$$

up to multiplication by a Gaussian unit. The rotation is now $e^{i\theta} = \frac{\alpha_0}{\beta_0}$. Consider a Gaussian integer belonging to the coincidence lattice,

$$z \in \mathbb{Z}[i] \cap \frac{\alpha_0}{\beta_0}\mathbb{Z}[i]. \quad (182)$$

By definition, there exists another Gaussian integer satisfying

$$w \in \mathbb{Z}[i], \quad z = \frac{\alpha_0}{\beta_0}w. \quad (183)$$

Multiplying by the denominator yields $\beta_0 z = \alpha_0 w$. Since the reduced numerator and denominator are coprime and the Gaussian integers form a unique-factorization domain, Euclid's lemma implies $\alpha_0 \mid z$. Hence $z \in \alpha_0 \mathbb{Z}[i]$. Conversely, if $z = \alpha_0 u$, $u \in \mathbb{Z}[i]$, then

$$z = \frac{\alpha_0}{\beta_0}(\beta_0 u), \quad (184)$$

and therefore the same element belongs to both lattices. Consequently, the exact primitive coincidence lattice is

$$\Lambda_M \cong \alpha_0 \mathbb{Z}[i]. \quad (185)$$

The coincidence index equals the Gaussian norm of the primitive generator, $\Sigma = N_{\mathbb{Z}[i]}(\alpha_0)$. The Gaussian norm is

$$N_{\mathbb{Z}[i]}(x + iy) = x^2 + y^2. \quad (186)$$

We now determine the Gaussian common divisor. If the two integer parameters have opposite parity, the Gaussian integer introduced above is not divisible by the unique Gaussian prime above two. The numerator and denominator are therefore coprime up to units, $N_{\mathbb{Z}[i]}(g) = 1$. If both parameters are odd, their real and imaginary parts have the same parity. Hence both Gaussian integers are divisible by $1 + i$. They cannot both be divisible by its square, because $(1+i)^2 = 2i$ would require both ordinary integer components to be even, contradicting coprimality. Thus, up to multiplication by a Gaussian unit, $g = 1 + i$ for the both-odd parity sector. Its norm is $N_{\mathbb{Z}[i]}(1 + i) = 2$. Therefore the primitive coincidence index is

$$\Sigma(m, n) = \frac{m^2 + n^2}{d}, \quad (187)$$

where the parity factor is precisely the one derived above,

$$d = \begin{cases} 1, & m, n \text{ of opposite parity,} \\ 2, & m, n \text{ both odd.} \end{cases} \quad (188)$$

This proves that the factor-of-two reduction is not a convention but an exact consequence of Gaussian-integer primitiveness. The Gaussian construction also gives the primitive supercell vectors explicitly. For opposite parity, no reduction occurs, $\alpha_0 = m + in$. A primitive coincidence basis is therefore

$$\mathbf{T}_1^M = a \begin{pmatrix} m \\ n \end{pmatrix}, \quad \mathbf{T}_2^M = a \begin{pmatrix} -n \\ m \end{pmatrix}. \quad (189)$$

The corresponding basis matrix is

$$C_M = a \begin{pmatrix} m & -n \\ n & m \end{pmatrix}. \quad (190)$$

Its determinant is

$$\det C_M = a^2 (m^2 + n^2). \quad (191)$$

Thus

$$A_M^{\text{sc}} = a^2 (m^2 + n^2) = a^2 \Sigma. \quad (192)$$

When both parameters are odd, division by the common Gaussian factor gives

$$\begin{aligned} \alpha_0 &= \frac{m+in}{1+i} = \frac{(m+in)(1-i)}{2} \\ &= \frac{m+n}{2} + i \frac{n-m}{2}. \end{aligned} \quad (193)$$

Thus one primitive vector is

$$\mathbf{T}_1^M = \frac{a}{2} \begin{pmatrix} m+n \\ n-m \end{pmatrix}. \quad (194)$$

Multiplication by the Gaussian unit corresponding to a ninety-degree rotation gives the second vector,

$$\mathbf{T}_2^M = \frac{a}{2} \begin{pmatrix} m-n \\ m+n \end{pmatrix}. \quad (195)$$

Therefore

$$C_M = \frac{a}{2} \begin{pmatrix} m+n & m-n \\ n-m & m+n \end{pmatrix}. \quad (196)$$

Its determinant is

$$\begin{aligned} \det C_M &= \frac{a^2}{4} [(m+n)^2 + (m-n)^2] \\ &= \frac{a^2}{4} [2m^2 + 2n^2] = \frac{a^2}{2} (m^2 + n^2). \end{aligned} \quad (197)$$

Hence

$$A_M^{\text{sc}} = a^2 \frac{m^2 + n^2}{2} = a^2 \Sigma. \quad (198)$$

The two parity sectors are therefore unified by

$$A_M^{\text{sc}} = a^2 \Sigma = a^2 \frac{m^2 + n^2}{d}. \quad (199)$$

Moreover, the primitive coincidence cell remains square. Its two primitive vectors are orthogonal, $\mathbf{T}_1^M \cdot \mathbf{T}_2^M = 0$, and have equal length,

$$L_M^{\text{sc}} = |\mathbf{T}_1^M| = |\mathbf{T}_2^M| = a\sqrt{\Sigma}. \quad (200)$$

Thus the exact coincidence cell of two commensurately twisted square lattices is itself a square lattice, with lattice constant enlarged by the square root of the coincidence index. Since each microscopic layer has one atom per primitive square cell, the number of sites contributed by either layer is

$$N_1^{\text{sc}} = \frac{A_M^{\text{sc}}}{a^2} = \Sigma. \quad (201)$$

The complete bilayer contains

$$N_M^{\text{sc}} = 2\Sigma = \frac{2(m^2 + n^2)}{d}. \quad (202)$$

Consequently, the correct primitive-cell counting formula is

$$N_M^{\text{sc}} = \begin{cases} 2(m^2 + n^2), & m, n \text{ of opposite parity,} \\ m^2 + n^2, & m, n \text{ both odd.} \end{cases} \quad (203)$$

A formula that always uses twice the sum of two squares therefore overcounts the primitive supercell by a factor of two in the both-odd sector. We now recover the structure displayed in the original problem. The visible common lattice vector has the integer components $(m, n) = (2, 1)$.

The half-angle therefore satisfies $\tan r = \frac{1}{2}$. Thus

$$r = \arctan\left(\frac{1}{2}\right) = 26.565051177 \dots^\circ. \quad (204)$$

The relative twist between the two layers is twice this value,

$$\theta = 2r = 2 \arctan\left(\frac{1}{2}\right) = 53.130102354 \dots^\circ. \quad (205)$$

The corresponding trigonometric values are

$$\cos \theta = \frac{3}{5}, \quad \sin \theta = \frac{4}{5}. \quad (206)$$

Because the two integer parameters have opposite parity, $d = 1$. Hence the coincidence index is $\Sigma = 2^2 + 1^2 = 5$. The primitive coincidence vectors are

$$\mathbf{T}_1^M = a \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad \mathbf{T}_2^M = a \begin{pmatrix} -1 \\ 2 \end{pmatrix}. \quad (207)$$

Their common length is $L_M^{\text{sc}} = \sqrt{5} a$, and the exact supercell area is $A_M^{\text{sc}} = 5a^2$. Each layer therefore contributes five microscopic sites, $N_1^{\text{sc}} = N_2^{\text{sc}} = 5$. The complete bilayer consequently contains $N_M^{\text{sc}} = 5 + 5 = 10$. This also explains the statement in the original figure that the cell

contains ten rather than nine atoms. At a coincidence point, one atom from the first layer and one atom from the second layer occupy the same in-plane coordinate, but they are not the same microscopic degree of freedom. The layer-resolved state space distinguishes them, so their contribution to the bilayer atom count is $1+1 = 2$ rather than one. Thus geometric coincidence does not reduce the Hilbert-space dimension of the bilayer. The square lattice possesses a fourfold point-group rotation. Therefore a relative rotation is physically equivalent, up to a crystallographic symmetry of either layer, under

$$\theta \mapsto \theta + k\frac{\pi}{2}, \quad k \in \mathbb{Z}. \quad (208)$$

Interchanging the two layers or applying a reflection additionally gives $\theta \mapsto -\theta$. Hence inequivalent square-lattice twists may be restricted to $0 \leq \hat{\theta} \leq \frac{\pi}{4}$. The angle in the original figure is therefore equivalent to $\hat{\theta} = \frac{\pi}{2} - \theta$. For the ten-atom structure, $\hat{\theta} = 36.869897646 \dots^\circ$. The corresponding half-angle rational parameter is $\tan\left(\frac{\hat{\theta}}{2}\right) = \frac{1}{3}$. Thus the same physical coincidence structure may be represented either by $(m, n) = (2, 1)$ at the angle displayed in the original figure, or by $(m, n) = (3, 1)$ inside the fundamental square-lattice angular domain. For the latter pair, both parameters are odd, and hence

$$\Sigma = \frac{3^2 + 1^2}{2} = 5. \quad (209)$$

The primitive coincidence index is therefore unchanged, as required. The preceding equivalence can be derived algebraically. Let

$$t = \tan\left(\frac{\theta}{2}\right) = \frac{n}{m}. \quad (210)$$

The complementary square-symmetry angle is $\theta' = \frac{\pi}{2} - \theta$. Its half-angle tangent is

$$\tan\left(\frac{\theta'}{2}\right) = \tan\left(\frac{\pi}{4} - \frac{\theta}{2}\right) = \frac{1-t}{1+t}. \quad (211)$$

Substitution of the rational representation gives

$$\tan\left(\frac{\theta'}{2}\right) = \frac{m-n}{m+n}. \quad (212)$$

Hence the transformation between symmetry-related integer data is generated by

$$(m, n) \mapsto (m+n, m-n) \quad (213)$$

followed by reduction to coprime integers. This transformation exchanges the two parity descriptions that frequently represent the same physical square-lattice coincidence structure. We now impose the condition requested in the original second question, $N_M^{\text{sc}} = 34$. Using the exact bilayer counting formula, $N_M^{\text{sc}} = 2\Sigma$, we obtain $\Sigma = 17$. There are two parity representations of this coincidence index within the full ninety-degree angular interval. For opposite parity, $m^2 + n^2 = 17$. The positive coprime solution with the larger integer listed first is $(m, n) = (4, 1)$, because $4^2 + 1^2 = 17$. Consequently,

$$r = \arctan\left(\frac{1}{4}\right) = 14.036243468 \dots^\circ, \quad (214)$$

and

$$\theta = 2 \arctan\left(\frac{1}{4}\right) = 28.072486936 \dots^\circ. \quad (215)$$

The exact rational trigonometric values are

$$\cos \theta = \frac{15}{17}, \quad \sin \theta = \frac{8}{17}. \quad (216)$$

The exact supercell vectors may be chosen as

$$\mathbf{T}_1^M = a \begin{pmatrix} 4 \\ 1 \end{pmatrix}, \quad \mathbf{T}_2^M = a \begin{pmatrix} -1 \\ 4 \end{pmatrix}. \quad (217)$$

The cell area is $A_M^{\text{sc}} = 17a^2$, and therefore $N_M^{\text{sc}} = 2 \times 17 = 34$. The square-symmetry partner lies at $\theta' = \frac{\pi}{2} - \theta$. Its numerical value is $\theta' = 61.927513064 \dots^\circ$. The corresponding rational parameter is

$$\tan\left(\frac{\theta'}{2}\right) = \frac{3}{5}. \quad (218)$$

Thus $(m, n) = (5, 3)$. Because both integers are odd, $d = 2$. The primitive index is therefore

$$\Sigma = \frac{5^2 + 3^2}{2} = \frac{34}{2} = 17. \quad (219)$$

Hence the two angles

$$28.072486936 \dots^\circ \quad \text{and} \quad 61.927513064 \dots^\circ \quad (220)$$

represent the same square-lattice coincidence class under fourfold crystallographic symmetry. Within the irreducible angular interval, the answer is therefore uniquely

$$\hat{\theta}_{34} = 28.072486936 \dots^\circ. \quad (221)$$

The Gaussian-integer description provides a much stronger result than the individual construction above: it allows the complete set of possible coincidence indices to be classified. Every nontrivial coincidence index of the square lattice has a prime factorization involving only rational primes congruent to one modulo four. Equivalently,

$$\Sigma = \prod_j p_j^{e_j}, \quad p_j \equiv 1 \pmod{4}, \quad e_j \in \mathbb{N}_0. \quad (222)$$

Conversely, every positive integer of the form Eq. (222) occurs as a square-lattice coincidence index. From the Gaussian-integer construction, $\Sigma = N_{\mathbb{Z}[i]}(\alpha_0)$, where the primitive Gaussian numerator and its conjugate are coprime,

$$\gcd_{\mathbb{Z}[i]}(\alpha_0, \bar{\alpha}_0) = 1. \quad (223)$$

The rational prime two is ramified in the Gaussian integers, $2 = -i(1+i)^2$. The primitive reduction removes the common factor above two. Consequently, $2 \nmid \Sigma$. Now consider a rational prime satisfying $p \equiv 3 \pmod{4}$. Such a prime remains prime in the Gaussian-integer ring. Suppose it divides the coincidence index, $p \mid \Sigma$. Since $\Sigma = \alpha_0 \bar{\alpha}_0$, Gaussian primality implies $p \mid \alpha_0$ or $p \mid \bar{\alpha}_0$. Because the prime is a real Gaussian integer, conjugation implies that either condition forces the other. Thus

$$p \mid \gcd_{\mathbb{Z}[i]}(\alpha_0, \bar{\alpha}_0), \quad (224)$$

contradicting primitiveness. Hence no prime congruent to three modulo four can divide the coincidence index,

$$p \equiv 3 \pmod{4} \implies p \nmid \Sigma. \quad (225)$$

The only nontrivial prime factors that remain are therefore primes satisfying $p \equiv 1 \pmod{4}$. Every rational prime congruent to one modulo four splits into two conjugate Gaussian primes,

$$p = \pi \bar{\pi}, \quad p \equiv 1 \pmod{4}. \quad (226)$$

For a target coincidence index of the form

$$\Sigma = \prod_j p_j^{e_j}, \quad (227)$$

choose one Gaussian prime from each conjugate pair and define

$$\alpha_0 = \prod_j \pi_j^{e_j}. \quad (228)$$

Its Gaussian norm is

$$N_{\mathbb{Z}[i]}(\alpha_0) = \prod_j N_{\mathbb{Z}[i]}(\pi_j)^{e_j} = \prod_j p_j^{e_j} = \Sigma. \quad (229)$$

Because only one member of each conjugate prime pair was selected, the constructed Gaussian integer and its conjugate possess no nonunit common factor,

$$\gcd_{\mathbb{Z}[i]}(\alpha_0, \bar{\alpha}_0) = 1. \quad (230)$$

It therefore defines a primitive coincidence rotation with index equal to the desired integer. This proves both necessity and sufficiency. The original problem requests every commensurate twist whose bilayer supercell contains fewer than one hundred atoms. The condition is $N_M^{\text{sc}} < 100$. Using the exact bilayer relation, $N_M^{\text{sc}} = 2\Sigma$, we obtain $\Sigma < 50$. The crystallographically trivial index is $\Sigma = 1$. It corresponds to rotations that map the square lattice onto itself and therefore does not generate an enlarged moiré cell. The nontrivial coincidence indices below fifty are determined completely by the arithmetic classification above. They are $\Sigma \in \{5, 13, 17, 25, 29, 37, 41\}$. The corresponding bilayer atom numbers are

$$N_M^{\text{sc}} \in \{10, 26, 34, 50, 58, 74, 82\}. \quad (231)$$

There are no other nontrivial possibilities below one hundred atoms. We now list the complete angular realizations in the interval $0 < \theta < \frac{\pi}{2}$. For coincidence index five,

$$\begin{aligned} \Sigma = 5, \quad N_M^{\text{sc}} = 10, \\ (m, n) = (3, 1), \quad \theta = 2 \arctan\left(\frac{1}{3}\right) = 36.869897646 \dots^\circ, \\ (m, n) = (2, 1), \quad \theta = 2 \arctan\left(\frac{1}{2}\right) = 53.130102354 \dots^\circ. \end{aligned} \quad (232)$$

For coincidence index thirteen,

$$\begin{aligned} \Sigma = 13, \quad N_M^{\text{sc}} = 26, \\ (m, n) = (5, 1), \quad \theta = 2 \arctan\left(\frac{1}{5}\right) = 22.619864948 \dots^\circ, \\ (m, n) = (3, 2), \quad \theta = 2 \arctan\left(\frac{2}{3}\right) = 67.380135052 \dots^\circ. \end{aligned} \quad (233)$$

For coincidence index seventeen,

$$\begin{aligned} \Sigma = 17, \quad N_M^{\text{sc}} = 34, \\ (m, n) = (4, 1), \quad \theta = 2 \arctan\left(\frac{1}{4}\right) = 28.072486936 \dots^\circ, \\ (m, n) = (5, 3), \quad \theta = 2 \arctan\left(\frac{3}{5}\right) = 61.927513064 \dots^\circ. \end{aligned} \quad (234)$$

For coincidence index twenty-five,

$$\begin{aligned}\Sigma &= 25, \quad N_M^{\text{sc}} = 50, \\ (m, n) &= (7, 1), \quad \theta = 2 \arctan\left(\frac{1}{7}\right) = 16.260204708 \dots^\circ, \\ (m, n) &= (4, 3), \quad \theta = 2 \arctan\left(\frac{3}{4}\right) = 73.739795292 \dots^\circ.\end{aligned}\quad (235)$$

For coincidence index twenty-nine,

$$\begin{aligned}\Sigma &= 29, \quad N_M^{\text{sc}} = 58, \\ (m, n) &= (5, 2), \quad \theta = 2 \arctan\left(\frac{2}{5}\right) = 43.602818973 \dots^\circ, \\ (m, n) &= (7, 3), \quad \theta = 2 \arctan\left(\frac{3}{7}\right) = 46.397181027 \dots^\circ.\end{aligned}\quad (236)$$

For coincidence index thirty-seven,

$$\begin{aligned}\Sigma &= 37, \quad N_M^{\text{sc}} = 74, \\ (m, n) &= (6, 1), \quad \theta = 2 \arctan\left(\frac{1}{6}\right) = 18.924644416 \dots^\circ, \\ (m, n) &= (7, 5), \quad \theta = 2 \arctan\left(\frac{5}{7}\right) = 71.075355584 \dots^\circ.\end{aligned}\quad (237)$$

Finally, for coincidence index forty-one,

$$\begin{aligned}\Sigma &= 41, \quad N_M^{\text{sc}} = 82, \\ (m, n) &= (9, 1), \quad \theta = 2 \arctan\left(\frac{1}{9}\right) = 12.680383492 \dots^\circ, \\ (m, n) &= (5, 4), \quad \theta = 2 \arctan\left(\frac{4}{5}\right) = 77.319616508 \dots^\circ.\end{aligned}\quad (238)$$

Thus the complete set of nontrivial twist angles in the ninety-degree interval that produce fewer than one hundred bilayer atoms is

$$\begin{aligned}\theta \in \{12.680383492 \dots^\circ, 16.260204708 \dots^\circ, 18.924644416 \dots^\circ, 22.619864948 \dots^\circ, \\ 28.072486936 \dots^\circ, 36.869897646 \dots^\circ, 43.602818973 \dots^\circ, 46.397181027 \dots^\circ, 53.130102354 \dots^\circ, \\ 61.927513064 \dots^\circ, 67.380135052 \dots^\circ, 71.075355584 \dots^\circ, 73.739795292 \dots^\circ, 77.319616508 \dots^\circ\}.\end{aligned}\quad (239)$$

Each angle above is paired with its complement, $\theta' = \frac{\pi}{2} - \theta$. Therefore only seven are crystallographically inequivalent. Inside the irreducible square-lattice angular interval, the complete list is

$$\begin{aligned}\hat{\theta} \in \{12.680383492 \dots^\circ, 16.260204708 \dots^\circ, 18.924644416 \dots^\circ, 22.619864948 \dots^\circ, \\ 28.072486936 \dots^\circ, 36.869897646 \dots^\circ, 43.602818973 \dots^\circ\}.\end{aligned}\quad (240)$$

Their associated atom numbers, ordered in the same sequence, are

$$N_M^{\text{sc}} \in \{82, 50, 74, 26, 34, 10, 58\}.\quad (241)$$

The nonmonotonic ordering of the atom numbers already demonstrates that exact commensurate supercell size is not a smooth function of twist angle. Its dominant geometric growth is modulated by a discrete arithmetic structure. We now make the preceding statement quantitative. From the rational half-angle parameterization,

$$\tan\left(\frac{\theta}{2}\right) = \frac{n}{m},\quad (242)$$

we obtain

$$m = \frac{n}{\tan(\theta/2)}.\quad (243)$$

The coincidence index is

$$\Sigma = \frac{m^2 + n^2}{d}.\quad (244)$$

Substituting the expression for the denominator parameter gives

$$\Sigma = \frac{n^2}{d} \left[\frac{1}{\tan^2(\theta/2)} + 1 \right].\quad (245)$$

The identity $1 + \cot^2 x = \csc^2 x$ therefore yields the exact relation

$$\Sigma(\theta; m, n) = \frac{n^2}{d} \csc^2\left(\frac{\theta}{2}\right).\quad (246)$$

Consequently, the exact bilayer atom number obeys

$$N_M^{\text{sc}} = \frac{2n^2}{d} \csc^2\left(\frac{\theta}{2}\right).\quad (247)$$

This equation displays an exact factorization into a continuous geometric factor and a discrete arithmetic factor. Define the arithmetic multiplier

$$\mathcal{A}_{m,n} = \frac{n^2}{d}.\quad (248)$$

$$\Sigma = \mathcal{A}_{m,n} \csc^2\left(\frac{\theta}{2}\right),\quad (249)$$

and

$$N_M^{\text{sc}} = 2\mathcal{A}_{m,n} \csc^2\left(\frac{\theta}{2}\right). \quad (250)$$

Thus the familiar inverse-square moiré scaling is the geometric part of the exact crystallographic result, while exact commensurability introduces the arithmetic multiplier. For small twist, the cosecant has the expansion

$$\csc^2 x = \frac{1}{x^2} + \frac{1}{3} + \frac{x^2}{15} + O(x^4). \quad (251)$$

Setting $x = \frac{\theta}{2}$ gives

$$\csc^2\left(\frac{\theta}{2}\right) = \frac{4}{\theta^2} + \frac{1}{3} + \frac{\theta^2}{60} + O(\theta^4). \quad (252)$$

The exact coincidence index therefore has the asymptotic expansion

$$\Sigma = \frac{4n^2}{d\theta^2} + \frac{n^2}{3d} + \frac{n^2\theta^2}{60d} + O(n^2\theta^4). \quad (253)$$

The corresponding bilayer atom number is

$$N_M^{\text{sc}} = \frac{8n^2}{d\theta^2} + \frac{2n^2}{3d} + \frac{n^2\theta^2}{30d} + O(n^2\theta^4). \quad (254)$$

For a commensurate sequence in which the reduced numerator remains bounded, the leading behavior is therefore $\Sigma = \Theta(\theta^{-2})$, and $N_M^{\text{sc}} = \Theta(\theta^{-2})$. The exact linear supercell dimension then behaves as $L_M^{\text{sc}} = a\sqrt{\Sigma}$. Using the exact factorization,

$$L_M^{\text{sc}} = \frac{an}{\sqrt{d}} \left| \csc\left(\frac{\theta}{2}\right) \right|. \quad (255)$$

Thus

$$L_M^{\text{sc}} \sim \frac{2an}{\sqrt{d}|\theta|}. \quad (256)$$

Accordingly, the familiar geometric scaling hierarchy is recovered,

$$L_M^{\text{sc}} \sim |\theta|^{-1}, \quad \mathcal{A}_M^{\text{sc}} \sim |\theta|^{-2}, \quad N_M^{\text{sc}} \sim |\theta|^{-2}, \quad (257)$$

along arithmetic branches for which the reduced numerator remains of order unity. The distinction between geometric and arithmetic scaling is essential at high precision. The reduced numerator satisfies $n \geq 1$, while the parity divisor satisfies $1 \leq d \leq 2$. Consequently,

$\frac{2n^2}{d} \geq 1$. Equation (247) therefore implies the universal lower bound

$$N_M^{\text{sc}} \geq \csc^2\left(\frac{\theta}{2}\right). \quad (258)$$

At small twist,

$$N_M^{\text{sc}} \geq \frac{4}{\theta^2} + \frac{1}{3} + O(\theta^2). \quad (259)$$

Thus every exact nontrivial commensurate supercell must grow at least as rapidly as the inverse square of the small twist angle. However, there is no corresponding universal upper bound depending only on the twist angle. The reason is that rational half-angle approximants may have arbitrarily large reduced numerators and denominators. For an arbitrary commensurate sequence,

$$\tan\left(\frac{\theta_j}{2}\right) = \frac{n_j}{m_j}, \quad \gcd(m_j, n_j) = 1, \quad (260)$$

the exact atom count is

$$N_{M,j}^{\text{sc}} = \frac{2n_j^2}{d_j} \csc^2\left(\frac{\theta_j}{2}\right). \quad (261)$$

If $n_j = O(1)$, then the conventional inverse-square law follows. If instead $n_j \rightarrow \infty$, the arithmetic multiplier itself diverges,

$$\mathcal{A}_{m_j, n_j} = \frac{n_j^2}{d_j} \rightarrow \infty, \quad (262)$$

and the exact supercell may grow parametrically faster than the geometric inverse-square scale. Therefore the strongest universally correct statement is not that every possible commensurate sequence has a unique asymptotic prefactor multiplying the inverse-square law. Rather, the exact Euclidean result is the exact supercell size = geometric twist factor $\times$ arithmetic commensurability factor. factorization Quantitatively,

$$N_M^{\text{sc}} = 2 \underbrace{\csc^2\left(\frac{\theta}{2}\right)}_{\text{geometric factor}} \underbrace{\frac{n^2}{d}}_{\text{arithmetic factor}}. \quad (263)$$

This arithmetic modulation explains why the finite list obtained above is not monotonically ordered in atom number as the angle decreases. The smallest possible arithmetic multiplier is obtained when $n = 1$ and both integer parameters are odd, so that $d = 2$. The arithmetic multiplier then takes its minimum value, $\mathcal{A}_{\min} = \frac{1}{2}$. Accordingly, the lower-envelope atom count is

$$N_{\text{env}} = \csc^2\left(\frac{\theta}{2}\right). \quad (264)$$

Its small-angle form is $N_{\text{env}} \sim \frac{4}{\theta^2}$. The corresponding coincidence index obeys

$$\Sigma_{\text{env}} = \frac{1}{2} \csc^2\left(\frac{\theta}{2}\right), \quad (265)$$

and therefore $\Sigma_{\text{env}} \sim \frac{2}{\theta^2}$. A second elementary branch occurs for $n = 1$, $d = 1$, for which

$$N_M^{\text{sc}} = 2 \csc^2\left(\frac{\theta}{2}\right) \quad (266)$$

and hence $N_M^{\text{sc}} \sim \frac{8}{\theta^2}$. These parity branches explain the factor-of-two variations that appear between different integer representations of square-symmetry-related commensurate rotations. The geometric contribution to the inverse-square scaling can also be derived directly from reciprocal space. The primitive reciprocal-lattice basis of the unrotated square lattice is

$$B = \frac{2\pi}{a} I_2. \quad (267)$$

The rotated reciprocal basis is $B_\theta = R_\theta B$. The elementary reciprocal mismatch matrix is

$$\Delta B = B_\theta - B = \frac{2\pi}{a} (R_\theta - I_2). \quad (268)$$

Its determinant is

$$\det(R_\theta - I_2) = 2 - 2 \cos \theta. \quad (269)$$

Using the half-angle identity gives

$$\det(R_\theta - I_2) = 4 \sin^2\left(\frac{\theta}{2}\right). \quad (270)$$

Therefore the area of the reciprocal beat cell is

$$A_{\text{beat}}^* = \left(\frac{2\pi}{a}\right)^2 4 \sin^2\left(\frac{\theta}{2}\right). \quad (271)$$

The corresponding real-space beat-cell area is

$$A_{\text{beat}} = \frac{(2\pi)^2}{A_{\text{beat}}^*}. \quad (272)$$

Hence

$$A_{\text{beat}} = \frac{a^2}{4 \sin^2(\theta/2)}. \quad (273)$$

At small twist, $A_{\text{beat}} \sim \frac{a^2}{\theta^2}$. The exact crystallographic supercell area is

$$A_M^{\text{sc}} = a^2 \frac{n^2}{d} \csc^2\left(\frac{\theta}{2}\right). \quad (274)$$

Comparison with the beat-cell area yields the exact relation

$$\frac{A_M^{\text{sc}}}{A_{\text{beat}}} = \frac{4n^2}{d}. \quad (275)$$

Thus even in Euclidean space the reciprocal-space moiré beat cell and the exact crystallographic coincidence cell are not generally identical. Their relation contains a purely arithmetic multiplier. This observation provides the Euclidean prototype for the more difficult geometric–arithmetic locking problem encountered later in negatively curved lattices. The results above may be summarized in a precise form. Let a sequence of primitive commensurate square-lattice twists satisfy $\theta_j \rightarrow 0$, with reduced rational half-angle representation

$$\tan\left(\frac{\theta_j}{2}\right) = \frac{n_j}{m_j}, \quad \gcd(m_j, n_j) = 1. \quad (276)$$

Then the exact coincidence index and bilayer atom number satisfy

$$\Sigma_j = \frac{n_j^2}{d_j} \csc^2\left(\frac{\theta_j}{2}\right), \quad (277)$$

and

$$N_{M,j}^{\text{sc}} = \frac{2n_j^2}{d_j} \csc^2\left(\frac{\theta_j}{2}\right). \quad (278)$$

The universal geometric lower bound is

$$N_{M,j}^{\text{sc}} \geq \csc^2\left(\frac{\theta_j}{2}\right) \sim \frac{4}{\theta_j^2}. \quad (279)$$

If the reduced numerators remain uniformly bounded, $\sup_j n_j < \infty$, then

$$N_{M,j}^{\text{sc}} = \Theta(\theta_j^{-2}), \quad (280)$$

while the linear supercell size satisfies

$$L_{M,j}^{\text{sc}} = \Theta(\theta_j^{-1}). \quad (281)$$

For unrestricted commensurate sequences, the exact cell may grow faster because the arithmetic multiplier need not remain bounded. This completes the derivation. The complete Euclidean square-bilayer commensurability problem can now be expressed through the following chain. A twist is exactly commensurate if and only if

$$\tan\left(\frac{\theta}{2}\right) = \frac{n}{m} \in \mathbb{Q}, \quad \gcd(m, n) = 1. \quad (282)$$

The rotation is then

$$R_\theta = \frac{1}{m^2 + n^2} \begin{pmatrix} m^2 - n^2 & -2mn \\ 2mn & m^2 - n^2 \end{pmatrix}, \quad (283)$$

the primitive coincidence index is

$$\Sigma = \frac{m^2 + n^2}{d}, \quad (284)$$

the exact supercell area is $A_M^{\text{sc}} = a^2 \Sigma$, the linear supercell scale is $L_M^{\text{sc}} = a\sqrt{\Sigma}$, and the exact bilayer atom number is $N_M^{\text{sc}} = 2\Sigma$. The parity divisor is

$$d = \begin{cases} 1, & m, n \text{ of opposite parity,} \\ 2, & m, n \text{ both odd.} \end{cases} \quad (285)$$

At the same time, the exact atom count admits the geometric–arithmetic decomposition

$$N_M^{\text{sc}} = 2 \csc^2\left(\frac{\theta}{2}\right) \frac{n^2}{d}. \quad (286)$$

The Euclidean hierarchy established in this subsection is therefore rational half-angle $\rightarrow$ integer pair $\rightarrow$ primitive Gaussian generator $\rightarrow$ coincidence index $\rightarrow$ exact supercell. More explicitly,

$$\frac{n}{m} \rightarrow R_\theta \rightarrow \alpha_0 \rightarrow \Sigma \rightarrow (A_M^{\text{sc}}, L_M^{\text{sc}}, N_M^{\text{sc}}). \quad (287)$$

This exact solution provides the flat-space reference point for the hyperbolic theory developed below. In Euclidean space, the geometric inverse-square moiré growth and the arithmetic coincidence condition can be separated exactly. The corresponding question in negative curvature is substantially more difficult: one must determine whether the local curvature-controlled geometric scale can be quantitatively locked to the finite-index structure of a non-Abelian Fuchsian group. It is this geometric–arithmetic correspondence, rather than the existence of a local interference length alone, that distinguishes exact hyperbolic moiré crystallography from its Euclidean counterpart.

C. Hyperbolic Twist Geometry and Local Moiré Length

We now turn to the first genuinely curvature-sensitive component of the moiré geometry. The Euclidean construction developed above already shows that a relative twist produces a displacement field whose magnitude grows linearly with the distance from the twist center. Negative curvature changes this geometric amplification mechanism at a fundamental level: angularly separated radial geodesics diverge exponentially rather than linearly. As a consequence, the inverse problem of determining the radial distance required to accumulate a fixed microscopic registry mismatch no longer produces the Euclidean inverse-angle law in the curvature-dominated regime. The purpose of the present subsection is to establish this statement exactly. We first derive the full finite-angle displacement law directly from intrinsic hyperbolic geometry. We then verify the same result independently in the hyperboloid and Poincaré-disk representations, identify its infinitesimal origin through the Jacobi equation, prove its monotonicity and convexity properties, and finally derive the exact local moiré length together with its Euclidean, curvature-dominated, and double-scaling limits. Throughout this subsection, the underlying space is the hyperbolic plane with constant Gaussian curvature

$$K = -\frac{1}{R^2}, \quad R > 0. \quad (288)$$

The positive parameter appearing in Eq. (288) is the curvature radius. For definiteness, we initially restrict the relative twist angle to $0 < \theta < \pi$. Within this interval, $\sin\left(\frac{\theta}{2}\right) > 0$. All results extend immediately to arbitrary signed twist by replacing the half-angle sine by its absolute value. Choose a point that acts as the twist center, $o \in \mathbb{H}_R^2$. Introduce geodesic polar coordinates centered at this point, (r, ϕ) , $r \geq 0$. The intrinsic hyperbolic metric is

$$ds^2 = dr^2 + R^2 \sinh^2\left(\frac{r}{R}\right) d\phi^2. \quad (289)$$

A centered relative twist is represented by the angular transformation

$$g_\theta : (r, \phi) \mapsto (r, \phi + \theta). \quad (290)$$

The radial coordinate is unchanged, $g_\theta^* dr = dr$, while the angular differential satisfies

$$g_\theta^* d\phi = d(\phi + \theta) = d\phi. \quad (291)$$

Consequently, the pullback of the metric is

$$\begin{aligned} g_\theta^* ds^2 &= dr^2 + R^2 \sinh^2\left(\frac{r}{R}\right) d(\phi + \theta)^2 \\ &= dr^2 + R^2 \sinh^2\left(\frac{r}{R}\right) d\phi^2 = ds^2. \end{aligned} \quad (292)$$

Thus the centered angular transformation is an exact orientation-preserving isometry of the hyperbolic plane. Let a point in the first layer be denoted by $x = (r, \phi)$, and let the corresponding point produced by the relative twist be

$$x_\theta = g_\theta x = (r, \phi + \theta). \quad (293)$$

Both points lie on the same geodesic circle,

$$d(o, x) = d(o, x_\theta) = r. \quad (294)$$

We define the exact twist-induced geodesic displacement by

$$d_\theta(r) = d_{\mathbb{H}}(x, x_\theta). \quad (295)$$

Rotational homogeneity implies that this displacement is independent of the initial polar angle and depends only on the radial coordinate, the curvature radius, and the relative twist. The three points consisting of the twist center, the original point, and the rotated point form an isosceles hyperbolic triangle. Its two equal radial sides have length r , the included angle at the twist center is θ , and the side opposite this angle has length $d_\theta(r)$. For a hyperbolic triangle with side lengths and opposite angle arranged in the present geometry, the hyperbolic cosine law gives

$$\cosh\left(\frac{d_\theta(r)}{R}\right) = \cosh\left(\frac{r}{R}\right) \cosh\left(\frac{r}{R}\right) - \sinh\left(\frac{r}{R}\right) \sinh\left(\frac{r}{R}\right) \cos \theta. \quad (296)$$

Therefore,

$$\cosh\left(\frac{d_\theta(r)}{R}\right) = \cosh^2\left(\frac{r}{R}\right) - \sinh^2\left(\frac{r}{R}\right) \cos \theta. \quad (297)$$

Using the fundamental hyperbolic identity $\cosh^2 x - \sinh^2 x = 1$, we rewrite the first term according to

$$\cosh^2\left(\frac{r}{R}\right) = 1 + \sinh^2\left(\frac{r}{R}\right). \quad (298)$$

Substitution into Eq. (297) yields

$$\begin{aligned} \cosh\left(\frac{d_\theta(r)}{R}\right) &= 1 + \sinh^2\left(\frac{r}{R}\right) - \sinh^2\left(\frac{r}{R}\right) \cos \theta \\ &= 1 + \sinh^2\left(\frac{r}{R}\right) (1 - \cos \theta). \end{aligned} \quad (299)$$

The half-angle identity gives

$$1 - \cos \theta = 2 \sin^2\left(\frac{\theta}{2}\right). \quad (300)$$

Therefore,

$$\cosh\left(\frac{d_\theta(r)}{R}\right) = 1 + 2 \sinh^2\left(\frac{r}{R}\right) \sin^2\left(\frac{\theta}{2}\right). \quad (301)$$

On the left-hand side, the identity

$$\cosh y = 1 + 2 \sinh^2\left(\frac{y}{2}\right) \quad (302)$$

implies

$$\cosh\left(\frac{d_\theta(r)}{R}\right) = 1 + 2 \sinh^2\left(\frac{d_\theta(r)}{2R}\right). \quad (303)$$

Equating Eqs. (301) and (303) gives

$$2 \sinh^2\left(\frac{d_\theta(r)}{2R}\right) = 2 \sinh^2\left(\frac{r}{R}\right) \sin^2\left(\frac{\theta}{2}\right). \quad (304)$$

All quantities are nonnegative in the angular interval under consideration. Taking the positive square root therefore gives the exact finite-angle relation

$$\sinh\left(\frac{d_\theta(r)}{2R}\right) = \sinh\left(\frac{r}{R}\right) \sin\left(\frac{\theta}{2}\right). \quad (305)$$

Solving explicitly for the displacement yields

$$d_\theta(r) = 2R \operatorname{arsinh} \left[\sinh\left(\frac{r}{R}\right) \sin\left(\frac{\theta}{2}\right) \right]. \quad (306)$$

Equation (305) is the central local geometric identity of the present subsection. It is exact at finite curvature, finite radial distance, and finite twist, and requires neither a continuum small-angle approximation nor a tight-binding description. Because the subsequent lattice theory may be implemented in different coordinate models of the hyperbolic plane, it is useful to verify that Eq. (305) is intrinsic rather than an artifact of geodesic polar coordinates. Consider the hyperboloid embedded in three-dimensional Minkowski space. Introduce the Minkowski bilinear form

$$\langle X, Y \rangle_L = -X_0 Y_0 + X_1 Y_1 + X_2 Y_2. \quad (307)$$

The hyperbolic plane of curvature radius equal to the chosen curvature radius is represented by

$$\mathbb{H}_R^2 = \{X : \langle X, X \rangle_L = -R^2, \quad X_0 > 0\}. \quad (308)$$

A point with intrinsic polar coordinates can be embedded as

$$X(r, \phi) = R \begin{pmatrix} \cosh(r/R) \\ \sinh(r/R) \cos \phi \\ \sinh(r/R) \sin \phi \end{pmatrix}. \quad (309)$$

The rotated point is

$$X_\theta = X(r, \phi + \theta). \quad (310)$$

Their Lorentzian inner product is

$$-\frac{\langle X, X_\theta \rangle_L}{R^2} = \cosh^2\left(\frac{r}{R}\right) - \sinh^2\left(\frac{r}{R}\right) [\cos \phi \cos(\phi + \theta) + \sin \phi \sin(\phi + \theta)]. \quad (311)$$

The trigonometric combination satisfies

$$\cos \phi \cos(\phi + \theta) + \sin \phi \sin(\phi + \theta) = \cos \theta. \quad (312)$$

Hence,

$$-\frac{\langle X, X_\theta \rangle_L}{R^2} = \cosh^2\left(\frac{r}{R}\right) - \sinh^2\left(\frac{r}{R}\right) \cos \theta. \quad (313)$$

Intrinsic geodesic distance on the hyperboloid satisfies

$$\cosh\left(\frac{d_{\mathbb{H}}(X, Y)}{R}\right) = -\frac{\langle X, Y \rangle_L}{R^2}. \quad (314)$$

Substitution gives

$$\cosh\left(\frac{d_\theta(r)}{R}\right) = \cosh^2\left(\frac{r}{R}\right) - \sinh^2\left(\frac{r}{R}\right) \cos \theta. \quad (315)$$

Repeating the half-angle reduction therefore reproduces

$$\sinh\left(\frac{d_\theta(r)}{2R}\right) = \sinh\left(\frac{r}{R}\right) \sin\left(\frac{\theta}{2}\right). \quad (316)$$

The intrinsic and embedding-space derivations are therefore exactly consistent. The Poincaré disk is particularly useful for numerical visualization of hyperbolic lattices. Introduce a complex disk coordinate satisfying $|z| < 1$. The metric of curvature given in Eq. (288) is

$$ds^2 = \frac{4R^2|dz|^2}{(1 - |z|^2)^2}. \quad (317)$$

A point at intrinsic geodesic radius r from the disk origin has Euclidean disk radius

$$\rho = \tanh\left(\frac{r}{2R}\right). \quad (318)$$

Thus it may be written as $z = \rho e^{i\phi}$. The centered relative twist acts according to

$$z_\theta = e^{i\theta} z = \rho e^{i(\phi + \theta)}. \quad (319)$$

For two points in the disk, the hyperbolic distance obeys

$$\sinh\left(\frac{d_{\mathbb{H}}(z, w)}{2R}\right) = \frac{|z - w|}{\sqrt{(1 - |z|^2)(1 - |w|^2)}}. \quad (320)$$

For the two rotated points under consideration, $|z| = |z_\theta| = \rho$. Their Euclidean chord separation is

Equation (320) therefore gives

$$\sinh\left(\frac{d_\theta(r)}{2R}\right) = \frac{2\rho}{1 - \rho^2} \sin\left(\frac{\theta}{2}\right). \quad (322)$$

Using

$$\rho = \tanh\left(\frac{r}{2R}\right) \quad (323)$$

and the identity

$$\frac{2 \tanh x}{1 - \tanh^2 x} = \sinh(2x), \quad (324)$$

we obtain

$$\frac{2\rho}{1 - \rho^2} = \sinh\left(\frac{r}{R}\right). \quad (325)$$

Consequently,

$$\sinh\left(\frac{d_\theta(r)}{2R}\right) = \sinh\left(\frac{r}{R}\right) \sin\left(\frac{\theta}{2}\right), \quad (326)$$

which independently reproduces the same intrinsic result. The agreement among the geodesic-triangle, hyperboloid, and Poincaré-disk derivations establishes that the displacement law is coordinate-model independent. The exact finite-angle formula reveals the geometry, but the physical mechanism by which negative curvature changes the moire scale is most transparent in the infinitesimal-twist limit. Consider a one-parameter family of radial geodesics emanating from the twist center. The norm of the transverse Jacobi field associated with an infinitesimal change of initial angle will be denoted by $\mathcal{J}(r)$. For a surface of constant curvature, the transverse Jacobi equation is

$$\frac{d^2 \mathcal{J}}{dr^2} + K \mathcal{J} = 0. \quad (327)$$

Using the negative curvature in Eq. (288), this becomes

$$\frac{d^2 \mathcal{J}}{dr^2} - \frac{1}{R^2} \mathcal{J} = 0. \quad (328)$$

Radial geodesics originate at the same point, so the first initial condition is $\mathcal{J}(0) = 0$. A unit infinitesimal angular variation fixes the second initial condition, $\mathcal{J}'(0) = 1$. The general solution of Eq. (328) is

$$\mathcal{J}(r) = C_+ e^{r/R} + C_- e^{-r/R}. \quad (329)$$

The first initial condition implies $C_+ + C_- = 0$. Hence, $C_- = -C_+$. The solution can therefore be written as

$$\mathcal{J}(r) = 2C_+ \sinh\left(\frac{r}{R}\right). \quad (330)$$

Differentiation gives

$$\mathcal{J}'(r) = \frac{2C_+}{R} \cosh\left(\frac{r}{R}\right). \quad (331)$$

The second initial condition therefore requires $\frac{2C_+}{R} = 1$. Thus, $C_+ = \frac{R}{2}$. The exact transverse Jacobi amplitude is consequently

$$\mathcal{J}(r) = R \sinh\left(\frac{r}{R}\right). \quad (332)$$

For comparison, in Euclidean space the Jacobi equation becomes

$$\frac{d^2 \mathcal{J}_E}{dr^2} = 0, \quad (333)$$

with the same initial conditions, and therefore $\mathcal{J}_E(r) = r$. Thus negative curvature replaces the linear radial divergence r by the hyperbolic divergence

$$R \sinh\left(\frac{r}{R}\right). \quad (334)$$

To recover the same result from the exact finite-angle displacement, take the small-angle limit of Eq. (305). Using

$$\sin\left(\frac{\theta}{2}\right) = \frac{\theta}{2} + O(\theta^3), \quad (335)$$

one finds

$$\sinh\left(\frac{d_\theta(r)}{2R}\right) = \frac{\theta}{2} \sinh\left(\frac{r}{R}\right) + O(\theta^3). \quad (336)$$

Because the displacement itself vanishes with the angle,

$$\sinh\left(\frac{d_\theta(r)}{2R}\right) = \frac{d_\theta(r)}{2R} + O(\theta^3). \quad (337)$$

Hence,

$$d_\theta(r) = R \sinh\left(\frac{r}{R}\right) \theta + O(\theta^3). \quad (338)$$

Therefore,

$$\lim_{\theta \rightarrow 0^+} \frac{d_\theta(r)}{\theta} = R \sinh\left(\frac{r}{R}\right) = \mathcal{J}(r). \quad (339)$$

The Jacobi field is therefore not merely an analogy: it is precisely the linear-response coefficient of the finite-angle twist displacement. This provides the geometric mechanism underlying the modified moire scaling. The deviation from Euclidean angular separation can be quantified by normalizing the Jacobi amplitude to its Euclidean value. Define the dimensionless radial coordinate $x = \frac{r}{R}$. The infinitesimal curvature-amplification factor is

$$\mathcal{C}(x) = \frac{R \sinh(r/R)}{r} = \frac{\sinh x}{x}. \quad (340)$$

The Taylor expansion is

$$\mathcal{C}(x) = 1 + \frac{x^2}{6} + \frac{x^4}{120} + O(x^6). \quad (341)$$

Thus the first correction to flat-space angular separation is positive and quadratic in the ratio of radial distance to curvature radius. At large radial distance,

$$\sinh x = \frac{e^x - e^{-x}}{2}, \quad (342)$$

so

$$\mathcal{C}(x) \sim \frac{e^x}{2x}. \quad (343)$$

Therefore infinitesimal angular mismatch grows exponentially faster than its Euclidean counterpart once the radial distance becomes comparable to or larger than the curvature radius. We now establish that the finite-angle displacement increases strictly with radial distance. For compact notation define

$$s = \sin\left(\frac{\theta}{2}\right), \quad c = \cos\left(\frac{\theta}{2}\right). \quad (344)$$

Within the chosen angular interval, $0 < s \leq 1$. Equation (306) becomes

$$d_\theta(r) = 2R \operatorname{arsinh} \left[s \sinh \left(\frac{r}{R} \right) \right]. \quad (345)$$

Differentiating gives

$$\frac{dd_\theta}{dr} = 2R \frac{1}{\sqrt{1 + s^2 \sinh^2(r/R)}} s \cosh \left(\frac{r}{R} \right) \frac{1}{R}. \quad (346)$$

Thus,

$$\frac{dd_\theta}{dr} = \frac{2s \cosh(r/R)}{\sqrt{1 + s^2 \sinh^2(r/R)}}. \quad (347)$$

Every factor is positive for positive radius and nonzero twist. Consequently,

$$\frac{dd_\theta}{dr} > 0. \quad (348)$$

The displacement field is therefore strictly increasing away from the twist center. At the origin, $d_\theta(0) = 0$. The initial slope is

$$\left. \frac{dd_\theta}{dr} \right|_{r=0} = 2s. \quad (349)$$

At asymptotically large radius,

$$\lim_{r \rightarrow \infty} \frac{\cosh(r/R)}{\sqrt{1 + s^2 \sinh^2(r/R)}} = \frac{1}{s}. \quad (350)$$

Therefore,

$$\lim_{r \rightarrow \infty} \frac{dd_\theta}{dr} = 2. \quad (351)$$

Thus the radial growth rate continuously increases from the Euclidean local slope determined by the twist angle toward the maximal triangle-inequality slope. The radial displacement is not merely monotone; for a generic twist it is strictly convex. Differentiate Eq. (347) once more. Let

$$D(r) = \sqrt{1 + s^2 \sinh^2(r/R)}. \quad (352)$$

Then

$$\frac{dd_\theta}{dr} = 2s \frac{\cosh(r/R)}{D(r)}. \quad (353)$$

The derivative of the denominator is

$$\frac{dD}{dr} = \frac{s^2 \sinh(r/R) \cosh(r/R)}{RD(r)}. \quad (354)$$

Therefore,

$$\begin{aligned} \frac{d^2 d_\theta}{dr^2} &= \frac{2s \sinh(r/R)}{R} \frac{1}{D(r)} \\ &\quad - 2s \cosh \left(\frac{r}{R} \right) \frac{1}{D^2(r)} \frac{s^2 \sinh(r/R) \cosh(r/R)}{RD(r)}. \end{aligned} \quad (355)$$

Combining the terms gives

$$\frac{d^2 d_\theta}{dr^2} = \frac{2s \sinh(r/R)}{RD^3(r)} \left[D^2(r) - s^2 \cosh^2 \left(\frac{r}{R} \right) \right]. \quad (356)$$

Using

$$D^2(r) = 1 + s^2 \sinh^2 \left(\frac{r}{R} \right), \quad (357)$$

and

$$\cosh^2 \left(\frac{r}{R} \right) = 1 + \sinh^2 \left(\frac{r}{R} \right), \quad (358)$$

the bracket simplifies to

$$\begin{aligned} D^2(r) - s^2 \cosh^2 \left(\frac{r}{R} \right) &= 1 + s^2 \sinh^2 \left(\frac{r}{R} \right) \\ &\quad - s^2 - s^2 \sinh^2 \left(\frac{r}{R} \right) \\ &= 1 - s^2 = c^2. \end{aligned} \quad (359)$$

Hence,

$$\frac{d^2 d_\theta}{dr^2} = \frac{2sc^2 \sinh(r/R)}{R [1 + s^2 \sinh^2(r/R)]^{3/2}}. \quad (360)$$

For a generic twist satisfying $0 < \theta < \pi$, and positive radial coordinate, $r > 0$, one has

$$\frac{d^2 d_\theta}{dr^2} > 0. \quad (361)$$

The hyperbolic twist mismatch is therefore strictly convex as a function of radial distance. This property has no Euclidean analogue, since the Euclidean displacement is exactly linear in radius. For the same radial coordinate and the same relative angle, the Euclidean displacement is $d_\theta^{(E)}(r) = 2sr$. We now prove that negative curvature always amplifies the finite-angle displacement. Define

$$F(x) = \operatorname{arsinh}(s \sinh x) - sx, \quad x \geq 0. \quad (362)$$

At the origin, $F(0) = 0$. Differentiation gives

$$F'(x) = s \left[\frac{\cosh x}{\sqrt{1 + s^2 \sinh^2 x}} - 1 \right]. \quad (363)$$

The difference between the square of the numerator and the square of the denominator appearing inside the bracket is

$$\begin{aligned} \cosh^2 x - [1 + s^2 \sinh^2 x] &= 1 + \sinh^2 x - 1 - s^2 \sinh^2 x \\ &= (1 - s^2) \sinh^2 x = c^2 \sinh^2 x. \end{aligned} \quad (364)$$

Therefore,

$$\cosh x \geq \sqrt{1 + s^2 \sinh^2 x}. \quad (365)$$

Hence, $F'(x) \geq 0$. Together with $F(0) = 0$, this proves $F(x) \geq 0$. Consequently,

$$d_\theta^{(H)}(r) \geq d_\theta^{(E)}(r) = 2sr. \quad (366)$$

The inequality is strict for positive radius and a generic angle strictly between zero and pi. An upper bound follows from $s \leq 1$. Monotonicity of the inverse hyperbolic sine gives

$$\operatorname{arsinh} \left[s \sinh \left(\frac{r}{R} \right) \right] \leq \operatorname{arsinh} \left[\sinh \left(\frac{r}{R} \right) \right]. \quad (367)$$

Because the hyperbolic sine is one-to-one on the real line,

$$\operatorname{arsinh} \left[\sinh \left(\frac{r}{R} \right) \right] = \frac{r}{R}. \quad (368)$$

Therefore, $d_\theta^{(H)}(r) \leq 2r$. Combining the two inequalities gives the exact global bounds

$$2r \sin \left(\frac{\theta}{2} \right) \leq d_\theta^{(H)}(r) \leq 2r. \quad (369)$$

The lower bound is precisely the Euclidean displacement, whereas the upper bound is the maximal distance allowed by the triangle inequality for two points lying at equal radial distance from the center. The exact curvature correction can be exposed by expanding around the Euclidean regime. Introduce $x = \frac{r}{R}$. The relevant expansions are

$$\sinh x = x + \frac{x^3}{6} + \frac{x^5}{120} + O(x^7), \quad (370)$$

and

$$\operatorname{arsinh} y = y - \frac{y^3}{6} + \frac{3y^5}{40} + O(y^7). \quad (371)$$

Substituting the first series into the second and collecting powers gives

$$\begin{aligned} \frac{d_\theta^{(H)}(r)}{R} &= 2sx + \frac{s(1-s^2)}{3} x^3 \\ &+ \frac{s(1-10s^2+9s^4)}{60} x^5 + O(x^7). \end{aligned} \quad (372)$$

Using $1 - s^2 = c^2$, and

$$1 - 10s^2 + 9s^4 = c^2 (1 - 9s^2), \quad (373)$$

the physical-distance expansion becomes

$$d_\theta^{(H)}(r) = 2sr + \frac{sc^2}{3} \frac{r^3}{R^2} + \frac{sc^2(1-9s^2)}{60} \frac{r^5}{R^4} + O\left(\frac{r^7}{R^6}\right). \quad (374)$$

The leading curvature correction is strictly positive for a generic twist,

$$\frac{sc^2}{3} \frac{r^3}{R^2} > 0. \quad (375)$$

Thus negative curvature increases the registry mismatch already at the first non-Euclidean order. A complementary asymptotic regime holds at fixed curvature and fixed radial distance while the twist angle becomes small. From the exact expression,

$$d_\theta(r) = 2R \operatorname{arsinh} \left[\sinh \left(\frac{r}{R} \right) \sin \left(\frac{\theta}{2} \right) \right], \quad (376)$$

define

$$A_r = \sinh \left(\frac{r}{R} \right). \quad (377)$$

The half-angle sine satisfies

$$\sin \left(\frac{\theta}{2} \right) = \frac{\theta}{2} - \frac{\theta^3}{48} + O(\theta^5). \quad (378)$$

Therefore,

$$A_r \sin \left(\frac{\theta}{2} \right) = \frac{A_r \theta}{2} - \frac{A_r \theta^3}{48} + O(\theta^5). \quad (379)$$

Using the inverse-hyperbolic-sine expansion gives

$$d_\theta(r) = RA_r\theta - \frac{RA_r(1+A_r^2)}{24}\theta^3 + O(\theta^5). \quad (380)$$

Since

$$1 + A_r^2 = \cosh^2\left(\frac{r}{R}\right), \quad (381)$$

this can be written as

$$d_\theta(r) = R \sinh\left(\frac{r}{R}\right)\theta - \frac{R \sinh(r/R) \cosh^2(r/R)}{24}\theta^3 + O(\theta^5). \quad (382)$$

The leading term is exactly the Jacobi-field result derived above. The finite-angle displacement has a qualitatively different asymptotic behavior at radial distances far beyond the curvature radius. For $\frac{r}{R} \gg 1$, one has

$$\sinh\left(\frac{r}{R}\right) = \frac{1}{2}e^{r/R} \left[1 - e^{-2r/R}\right]. \quad (383)$$

For a fixed nonzero twist,

$$s \sinh\left(\frac{r}{R}\right) \gg 1. \quad (384)$$

The large-argument expansion

$$\operatorname{arsinh} y = \ln(2y) + \frac{1}{4y^2} + O(y^{-4}) \quad (385)$$

therefore applies. Substitution yields

$$d_\theta(r) = 2r + 2R \ln s + 2R \frac{c^2}{s^2} e^{-2r/R} + O\left(R e^{-4r/R}\right). \quad (386)$$

Since $0 < s \leq 1$, the logarithmic offset satisfies $\ln s \leq 0$. Thus the displacement approaches the maximal radial slope from below,

$$d_\theta(r) = 2r + O(R), \quad (387)$$

while retaining a twist-dependent finite offset. We now introduce the local moire scale. Let the microscopic registry threshold be the geodesic length $a > 0$. For a nearest-neighbor lattice realization, this parameter may naturally be identified with the microscopic geodesic lattice spacing. More generally, it can be interpreted as the fixed displacement at which two locally corresponding layer registries cease to be regarded as approximately aligned. The local moire length is defined as the unique radial coordinate at which the twist-induced displacement reaches this threshold, $d_\theta(\xi_M) = a$. This definition is local and metric. It does not assume that the

bilayer possesses an exact global coincidence supercell. We now prove that this definition is well posed. At the twist center, $d_\theta(0) = 0$. At asymptotically large radius, the large-radius result implies

$$\lim_{r \rightarrow \infty} d_\theta(r) = \infty. \quad (388)$$

The displacement is continuous,

$$d_\theta \in C^\infty([0, \infty)), \quad (389)$$

and strictly increasing by Eq. (348). Therefore, for every fixed positive threshold, there exists exactly one positive radius satisfying the defining equation. The local moire length is consequently unique. Substituting $d_\theta(\xi_M) = a$ into the master displacement relation gives

$$\sinh\left(\frac{a}{2R}\right) = \sinh\left(\frac{\xi_M}{R}\right) \sin\left(\frac{\theta}{2}\right). \quad (390)$$

Solving for the hyperbolic sine of the local moire length gives

$$\sinh\left(\frac{\xi_M}{R}\right) = \frac{\sinh(a/2R)}{\sin(\theta/2)}. \quad (391)$$

Since the hyperbolic sine is strictly monotone, inversion is unique and yields

$$\xi_M(\theta, R) = R \operatorname{arsinh} \left[\frac{\sinh(a/2R)}{\sin(\theta/2)} \right]. \quad (392)$$

For an arbitrary signed twist, the corresponding fully symmetric formula is

$$\xi_M(\theta, R) = R \operatorname{arsinh} \left[\frac{\sinh(a/2R)}{|\sin(\theta/2)|} \right]. \quad (393)$$

Equation (392) is the exact local curvature-controlled moire-length law. No small-angle approximation has been made. The displacement bounds in Eq. (369) immediately imply bounds on the inverse problem. At the defining radius, $d_\theta(\xi_M) = a$. Therefore,

$$2s\xi_M \leq a \leq 2\xi_M. \quad (394)$$

Rearranging gives

$$\frac{a}{2} \leq \xi_M(\theta, R) \leq \frac{a}{2 \sin(\theta/2)}. \quad (395)$$

The upper bound is precisely the Euclidean local moire length defined using the same displacement threshold.

Thus, $\xi_M^{(H)} \leq \xi_M^{(E)}$. The physical meaning is direct: because negative curvature amplifies a given twist-induced displacement at fixed radius, a smaller radial distance is sufficient to accumulate the same microscopic registry mismatch. Define

$$b = \sinh\left(\frac{a}{2R}\right). \quad (396)$$

Then

$$\xi_M = R \operatorname{arsinh}\left(\frac{b}{s}\right). \quad (397)$$

Differentiating with respect to the twist angle gives

$$\frac{\partial \xi_M}{\partial \theta} = R \frac{1}{\sqrt{1 + b^2/s^2}} \left(-\frac{bc}{2s^2}\right). \quad (398)$$

Since

$$\sqrt{1 + \frac{b^2}{s^2}} = \frac{\sqrt{s^2 + b^2}}{s}, \quad (399)$$

the derivative simplifies to

$$\frac{\partial \xi_M}{\partial \theta} = -\frac{Rbc}{2s\sqrt{s^2 + b^2}}. \quad (400)$$

For a generic angle in the interval under consideration,

$$\frac{\partial \xi_M}{\partial \theta} < 0. \quad (401)$$

Thus increasing the twist angle always decreases the local moire length. The dependence on curvature is more subtle because the curvature radius appears both outside and inside the inverse hyperbolic sine. Introduce the dimensionless microscopic curvature parameter $A = \frac{a}{2R}$.

Then $R = \frac{a}{2A}$. Define

$$G_s(A) = \operatorname{arsinh}\left(\frac{\sinh A}{s}\right). \quad (402)$$

The local moire length becomes

$$\xi_M = \frac{a}{2} \frac{G_s(A)}{A}. \quad (403)$$

Differentiating the auxiliary function gives

$$G'_s(A) = \frac{\cosh A}{\sqrt{s^2 + \sinh^2 A}}. \quad (404)$$

Differentiating again gives

$$G''_s(A) = -\frac{c^2 \sinh A}{[s^2 + \sinh^2 A]^{3/2}}. \quad (405)$$

For a generic twist and positive microscopic curvature parameter, $G''_s(A) < 0$. Thus the auxiliary function is strictly concave. Moreover, $G_s(0) = 0$. Consider

$$J(A) = AG'_s(A) - G_s(A). \quad (406)$$

Its derivative is $J'(A) = AG''_s(A)$. Hence, $J'(A) < 0$ for positive argument, while $J(0) = 0$. Therefore, $J(A) < 0$. Now,

$$\frac{d}{dA} \left[\frac{G_s(A)}{A} \right] = \frac{AG'_s(A) - G_s(A)}{A^2} = \frac{J(A)}{A^2}. \quad (407)$$

Consequently,

$$\frac{d}{dA} \left[\frac{G_s(A)}{A} \right] < 0. \quad (408)$$

Because increasing the magnitude of negative curvature corresponds to decreasing the curvature radius and therefore increasing the parameter $A = a/(2R)$, the result implies

$$\frac{\partial \xi_M}{\partial R} > 0. \quad (409)$$

Equivalently,

$$\frac{\partial \xi_M}{\partial |K|} < 0. \quad (410)$$

Stronger negative curvature therefore strictly reduces the radial distance required to accumulate a fixed registry mismatch. A mandatory consistency requirement is that the hyperbolic result reduce continuously to the Euclidean moire length as the curvature vanishes. The Euclidean limit corresponds to $R \rightarrow \infty$ at fixed microscopic threshold and fixed twist. The numerator of the exact formula satisfies

$$\sinh\left(\frac{a}{2R}\right) = \frac{a}{2R} + O(R^{-3}). \quad (411)$$

Therefore the argument of the inverse hyperbolic sine tends to zero,

$$\frac{\sinh(a/2R)}{s} = \frac{a}{2Rs} + O(R^{-3}). \quad (412)$$

Using

$$\operatorname{arsinh} x = x + O(x^3), \quad (413)$$

one obtains

$$\xi_M = R \left[\frac{a}{2Rs} + O(R^{-3}) \right]. \quad (414)$$

Hence,

$$\lim_{R \rightarrow \infty} \xi_M(\theta, R) = \frac{a}{2 \sin(\theta/2)}. \quad (415)$$

This is precisely the Euclidean local moire length. The first several curvature corrections can be obtained systematically. The expansion of the exact inverse problem in powers of the dimensionless parameter $A = a/(2R)$ gives

$$\begin{aligned} \operatorname{arsinh} \left(\frac{\sinh A}{s} \right) &= \frac{A}{s} + A^3 \left[\frac{1}{6s} - \frac{1}{6s^3} \right] \\ &+ A^5 \left[\frac{1}{120s} - \frac{1}{12s^3} + \frac{3}{40s^5} \right] + O(A^7). \end{aligned} \quad (416)$$

Using $A = \frac{a}{2R}$, one obtains

$$\xi_M = \frac{a}{2s} - \frac{a^3 c^2}{48R^2 s^3} + \frac{a^5 c^2 (9 - s^2)}{3840R^4 s^5} + O(R^{-6}). \quad (417)$$

The leading curvature correction is negative,

$$-\frac{a^3 c^2}{48R^2 s^3} < 0. \quad (418)$$

This is the inverse-problem counterpart of the positive curvature correction to the displacement field: negative curvature increases the displacement at fixed radius and therefore decreases the radius at which a prescribed displacement threshold is reached. The most important departure from Euclidean moire geometry occurs when the twist becomes asymptotically small at fixed nonzero negative curvature. Retain the abbreviation

$$b = \sinh \left(\frac{a}{2R} \right), \quad (419)$$

and define the dimensionless crossover variable

$$\chi = \frac{b}{\sin(\theta/2)}. \quad (420)$$

The exact local moire law is then

$$\frac{\xi_M}{R} = \operatorname{arsinh} \chi. \quad (421)$$

At fixed curvature and fixed microscopic threshold, $\theta \rightarrow 0$ implies $\chi \rightarrow \infty$. The inverse hyperbolic sine has the large-argument expansion

$$\operatorname{arsinh} \chi = \ln(2\chi) + \frac{1}{4\chi^2} - \frac{3}{32\chi^4} + O(\chi^{-6}). \quad (422)$$

Therefore,

$$\frac{\xi_M}{R} = \ln \left[\frac{2b}{\sin(\theta/2)} \right] + \frac{\sin^2(\theta/2)}{4b^2} - \frac{3\sin^4(\theta/2)}{32b^4} + O(\theta^6). \quad (423)$$

The logarithm can itself be expanded using

$$\ln \left[\sin \left(\frac{\theta}{2} \right) \right] = \ln \left(\frac{\theta}{2} \right) - \frac{\theta^2}{24} - \frac{\theta^4}{2880} + O(\theta^6). \quad (424)$$

Furthermore,

$$\sin^2 \left(\frac{\theta}{2} \right) = \frac{\theta^2}{4} - \frac{\theta^4}{48} + O(\theta^6), \quad (425)$$

and

$$\sin^4 \left(\frac{\theta}{2} \right) = \frac{\theta^4}{16} + O(\theta^6). \quad (426)$$

Collecting the terms yields

$$\begin{aligned} \frac{\xi_M}{R} &= \ln \left(\frac{4b}{\theta} \right) + \theta^2 \left[\frac{1}{24} + \frac{1}{16b^2} \right] \\ &+ \theta^4 \left[\frac{1}{2880} - \frac{1}{192b^2} - \frac{3}{512b^4} \right] + O(\theta^6). \end{aligned} \quad (427)$$

The leading asymptotic behavior is therefore

$$\xi_M \sim R \ln \left[\frac{4 \sinh(a/2R)}{\theta} \right]. \quad (428)$$

In particular,

$$\xi_M = R \ln \theta^{-1} + O(R) \quad (429)$$

for fixed nonzero negative curvature. This is qualitatively different from the Euclidean law $\xi_M^{(E)} \sim \frac{a}{\theta}$. The functional dependence therefore changes from algebraic divergence to logarithmic divergence. When the microscopic spacing is small compared with the curvature radius, $\frac{a}{R} \ll 1$, the coefficient appearing inside the logarithm satisfies

$$\sinh \left(\frac{a}{2R} \right) = \frac{a}{2R} + O \left(\frac{a^3}{R^3} \right). \quad (430)$$

Consequently, within the curvature-dominated small-angle regime,

$$\xi_M \sim R \ln \left(\frac{2a}{R\theta} \right). \quad (431)$$

The validity condition is not merely that the twist be small; the twist must be sufficiently small relative to the curvature-controlled crossover scale derived below. The dimensionless variable in Eq. (420) fully controls the crossover. The locally Euclidean regime corresponds to $\chi \ll 1$. Using

$$\operatorname{arsinh} \chi = \chi - \frac{\chi^3}{6} + O(\chi^5), \quad (432)$$

one obtains

$$\xi_M = R\chi + O(R\chi^3). \quad (433)$$

For weak microscopic curvature,

$$R\chi = \frac{a}{2 \sin(\theta/2)} + O(R^{-2}), \quad (434)$$

which reproduces the Euclidean result. The curvature-dominated regime is $\chi \gg 1$, for which

$$\xi_M = R \ln(2\chi) + O(R\chi^{-2}). \quad (435)$$

A natural crossover point is defined by $\chi = 1$. If

$$\sinh\left(\frac{a}{2R}\right) \leq 1, \quad (436)$$

the corresponding exact crossover angle is

$$\theta_c = 2 \arcsin \left[\sinh\left(\frac{a}{2R}\right) \right]. \quad (437)$$

The condition in Eq. (436) is equivalent to

$$\frac{a}{R} \leq 2 \operatorname{arsinh}(1). \quad (438)$$

If this inequality is violated, the crossover variable is already larger than unity even at the maximal half-angle sine, and no parametrically Euclidean local regime exists within this criterion. For small microscopic curvature, expansion of the exact crossover angle gives

$$\theta_c = \frac{a}{R} + \frac{a^3}{12R^3} + \frac{a^5}{96R^5} + O(R^{-7}). \quad (439)$$

To leading order,

$$\theta_c \sim \frac{a}{R}. \quad (440)$$

Thus even when the local lattice spacing is much smaller than the curvature radius, negative curvature becomes asymptotically dominant once the twist angle is reduced below the ratio of those two microscopic geometric scales. The crossover becomes particularly transparent in a simultaneous weak-curvature and small-angle limit. Define the dimensionless scaling variable

$$u = \frac{R\theta}{a}. \quad (441)$$

Consider the double limit

$$\frac{a}{R} \rightarrow 0, \quad \theta \rightarrow 0, \quad u \text{ fixed}. \quad (442)$$

Introduce again $A = \frac{a}{2R}$. Then $\theta = 2Au$. The numerator of the crossover variable is

$$\sinh A = A + \frac{A^3}{6} + O(A^5), \quad (443)$$

while the denominator is

$$\sin(Au) = Au - \frac{A^3 u^3}{6} + O(A^5). \quad (444)$$

Therefore,

$$\chi = \frac{1}{u} \left[1 + \frac{A^2}{6} (1 + u^2) + O(A^4) \right]. \quad (445)$$

At leading order, $\chi \rightarrow \frac{1}{u}$. Consequently,

$$\frac{\xi_M}{R} \rightarrow \operatorname{arsinh} \left(\frac{1}{u} \right). \quad (446)$$

The first finite-curvature correction at fixed scaling variable is

$$\frac{\xi_M}{R} = \operatorname{arsinh} \left(\frac{1}{u} \right) + \frac{a^2}{24R^2} \sqrt{1 + u^2} + O\left(\frac{a^4}{R^4}\right). \quad (447)$$

The Euclidean small-angle moire scale is $\xi_E \sim \frac{a}{\theta}$. Using the definition of the scaling variable, $\frac{a}{\theta} = \frac{R}{u}$. Therefore the normalized hyperbolic moire length has the universal double-scaling form

$$\frac{\xi_M}{\xi_E} \rightarrow \mathcal{G}(u), \quad (448)$$

where

$$\mathcal{G}(u) = u \operatorname{arsinh} \left(\frac{1}{u} \right). \quad (449)$$

For $u \gg 1$, one has

$$\operatorname{arsinh} \left(\frac{1}{u} \right) = \frac{1}{u} - \frac{1}{6u^3} + O(u^{-5}), \quad (450)$$

and hence

$$\mathcal{G}(u) = 1 - \frac{1}{6u^2} + O(u^{-4}). \quad (451)$$

Thus the Euclidean regime is recovered. For $u \ll 1$, the inverse hyperbolic sine satisfies

$$\operatorname{arsinh} \left(\frac{1}{u} \right) = \ln \left(\frac{2}{u} \right) + \frac{u^2}{4} + O(u^4), \quad (452)$$

and therefore

$$\mathcal{G}(u) = u \ln \left(\frac{2}{u} \right) + \frac{u^3}{4} + O(u^5). \quad (453)$$

In particular,

$$\lim_{u \rightarrow 0} \mathcal{G}(u) = 0. \quad (454)$$

The hyperbolic local moire scale is therefore parametrically smaller than the Euclidean inverse-angle scale in the deeply curvature-dominated regime. The double-scaling structure clarifies an important issue concerning the order of asymptotic limits. At fixed nonzero twist, the flat-space limit gives $u \rightarrow \infty$, and hence $\frac{\xi_M}{\xi_E} \rightarrow 1$. At fixed finite curvature radius, the small-angle limit instead gives $u \rightarrow 0$, and therefore $\frac{\xi_M}{\xi_E} \rightarrow 0$. Thus the Euclidean expansion is nonuniform in the twist angle. The controlling quantity is not the twist angle or curvature independently, but the combined scaling variable defined in Eq. (441). The crossover from algebraic to logarithmic growth can be characterized by a scale-dependent effective exponent. Using the crossover variable in Eq. (420), define

$$\nu_\xi = -\frac{d \ln \xi_M}{d \ln s}. \quad (455)$$

Since

$$\xi_M = R \operatorname{arsinh} \chi, \quad (456)$$

and $\chi = \frac{b}{s}$, one has

$$\frac{d\chi}{d \ln s} = -\chi. \quad (457)$$

Therefore,

$$\frac{d \ln \xi_M}{d \ln s} = -\frac{\chi}{\sqrt{1 + \chi^2} \operatorname{arsinh} \chi}. \quad (458)$$

Hence,

$$\nu_\xi(\chi) = \frac{\chi}{\sqrt{1 + \chi^2} \operatorname{arsinh} \chi}. \quad (459)$$

For the locally Euclidean regime, $\chi \ll 1$, expansion gives

$$\nu_\xi = 1 - \frac{\chi^2}{3} + O(\chi^4). \quad (460)$$

Thus, $\nu_\xi \rightarrow 1$. This is the conventional inverse-angle exponent. In the curvature-dominated regime, $\chi \gg 1$, one obtains

$$\nu_\xi \sim \frac{1}{\ln(2\chi)}. \quad (461)$$

Therefore, ν_ξ as the twist approaches zero at fixed negative curvature. The local length exponent thus exhibits the continuous flow $1 \rightarrow 0$, reflecting the crossover from inverse-angle growth to logarithmic growth. The local displacement law does not depend on the arbitrary choice of origin in the homogeneous hyperbolic plane. Let a second twist center be related to the original center by an isometry, $o' = ho$. The rotation of the same angle about the new center is obtained by conjugation,

$$g_\theta^{(o')} = h g_\theta^{(o)} h^{-1}. \quad (462)$$

For a point transformed according to $x' = hx$, the new displacement is

$$\begin{aligned} d_{\mathbb{H}}(x', g_\theta^{(o')} x') &= d_{\mathbb{H}}(hx, h g_\theta^{(o)} h^{-1} hx) \\ &= d_{\mathbb{H}}(hx, h g_\theta^{(o)} x). \end{aligned} \quad (463)$$

Because the transformation is an isometry,

$$d_{\mathbb{H}}(hx, hy) = d_{\mathbb{H}}(x, y). \quad (464)$$

Consequently,

$$d_{\mathbb{H}}(x', g_\theta^{(o')} x') = d_{\mathbb{H}}(x, g_\theta^{(o)} x). \quad (465)$$

Thus the continuum displacement law is independent of the chosen twist center up to isometric conjugation. Once a discrete lattice is imposed, different placements of the center relative to the microscopic sites can lead to different discrete registry patterns, but the intrinsic radial metric law itself remains unchanged. For later numerical calculations it is useful to express the local moire length directly in terms of the combinatorial parameters of a regular hyperbolic tessellation. Consider a regular tessellation in which each face has a fixed number of edges and a fixed number of faces meet at every vertex. Denote these integers by p and q . The hyperbolic regime satisfies

$$\frac{1}{p} + \frac{1}{q} < \frac{1}{2}. \quad (466)$$

Let the nearest-neighbor geodesic edge length be a . Divide one regular polygon into right hyperbolic triangles by joining its center to a vertex and to the midpoint of an adjacent edge. The three triangle angles are $\frac{\pi}{p}$, $\frac{\pi}{q}$, $\frac{\pi}{2}$. The side opposite the first angle has length $\frac{a}{2}$. The hyperbolic cosine law for angles gives

$$\cos\left(\frac{\pi}{p}\right) = \sin\left(\frac{\pi}{q}\right) \cosh\left(\frac{a}{2R}\right). \quad (467)$$

Therefore,

$$\cosh\left(\frac{a}{2R}\right) = \frac{\cos(\pi/p)}{\sin(\pi/q)}. \quad (468)$$

It follows that

$$\begin{aligned} \sinh^2\left(\frac{a}{2R}\right) &= \cosh^2\left(\frac{a}{2R}\right) - 1 \\ &= \frac{\cos^2(\pi/p)}{\sin^2(\pi/q)} - 1. \end{aligned} \quad (469)$$

Hence,

$$\sinh\left(\frac{a}{2R}\right) = \frac{\sqrt{\cos^2(\pi/p) - \sin^2(\pi/q)}}{\sin(\pi/q)}. \quad (470)$$

Substitution into the exact local moire law gives

$$\xi_M^{\{p,q\}}(\theta) = R \operatorname{arsinh} \left[\frac{\sqrt{\cos^2(\pi/p) - \sin^2(\pi/q)}}{\sin(\pi/q) \sin(\theta/2)} \right]. \quad (471)$$

Thus, for a strictly regular tessellation, the dimensionless ratio of microscopic spacing to curvature radius is not an independent continuous parameter; it is fixed by the tessellation data through Eq. (468). This distinction

becomes important when comparing different numerical hyperbolic lattices. The results established in this subsection can be summarized in a single statement. Consider the hyperbolic plane of constant Gaussian curvature

$$K = -\frac{1}{R^2}, \quad (472)$$

and a centered relative rotation of nonzero angle. For two points at equal radial distance from the twist center, the exact twist-induced displacement satisfies

$$\sinh\left(\frac{d_\theta(r)}{2R}\right) = \sinh\left(\frac{r}{R}\right) \left| \sin\left(\frac{\theta}{2}\right) \right|. \quad (473)$$

For any fixed positive microscopic registry threshold, the defining equation $d_\theta(\xi_M) = a$ has a unique solution,

$$\xi_M(\theta, R) = R \operatorname{arsinh} \left[\frac{\sinh(a/2R)}{|\sin(\theta/2)|} \right]. \quad (474)$$

The local moire length obeys the global bounds

$$\frac{a}{2} \leq \xi_M(\theta, R) \leq \frac{a}{2|\sin(\theta/2)|}. \quad (475)$$

It decreases monotonically with increasing twist angle and with increasing magnitude of negative curvature. In the flat-space limit,

$$\xi_M \longrightarrow \frac{a}{2|\sin(\theta/2)|}. \quad (476)$$

At fixed nonzero negative curvature and asymptotically small twist,

$$\xi_M \sim R \ln \left[\frac{4 \sinh(a/2R)}{|\theta|} \right]. \quad (477)$$

Thus the asymptotic divergence changes from $\xi_M^{(E)} \propto |\theta|^{-1}$ to $\xi_M^{(H)} \propto \ln |\theta|^{-1}$. The crossover is controlled by the dimensionless ratio

$$\chi = \frac{\sinh(a/2R)}{|\sin(\theta/2)|}. \quad (478)$$

In the simultaneous weak-curvature and small-angle limit, the normalized moire length is governed by the universal scaling function

$$\mathcal{G}(u) = u \operatorname{arsinh} \left(\frac{1}{u} \right), \quad u = \frac{R|\theta|}{a}. \quad (479)$$

This completes the proof of the curvature-controlled local moire-length law. The complete local geometric mechanism established above can be written as

$$K < 0 \longrightarrow \mathcal{J}(r) = R \sinh(r/R) \longrightarrow d_\theta(r) \longrightarrow \xi_M(\theta, R). \quad (480)$$

The first arrow expresses geodesic deviation: negative curvature produces exponential separation of initially nearby radial geodesics. The second arrow upgrades this infinitesimal geometric instability to the exact finite-angle displacement law. The third arrow inverts the displacement field at a fixed microscopic registry threshold. As a result, the Euclidean relation $d_\theta^{(E)}(r) \sim r\theta$ implies $\xi_M^{(E)} \sim \theta^{-1}$, whereas the hyperbolic relation

$$d_\theta^{(H)}(r) \sim R e^{r/R} \theta \quad (481)$$

in the curvature-dominated regime implies

$$\xi_M^{(H)} \sim R \ln \theta^{-1}. \quad (482)$$

This establishes the geometric origin of the altered local moire scaling without invoking any electronic Hamiltonian, hopping parameter, or band structure. The result is therefore purely geometric. At this stage, however, no statement has yet been made concerning the area of an effective moire domain or the index of an exact hyperbolic crystallographic supercell. Those are logically distinct questions. The next step is to determine how the logarithmic local length law combines with exponential hyperbolic area growth and whether the resulting effective moire scaling can be locked to the finite-index structure of exact hyperbolic commensurability.

D. Curvature-Induced Crossover of Moiré Scaling

The preceding subsection established that negative curvature changes the local moiré length from the Euclidean inverse-angle law to a logarithmic small-angle law. The local length alone is not the quantity that controls the number of microscopic degrees of freedom participating in a moire domain. In two dimensions the relevant geometric measure is the area enclosed by the registry region, and the relation between radius and area is itself strongly curvature dependent. This distinction is essential. In Euclidean geometry, a disk of radius equal to the local moire length has an area that grows quadratically with that length. In hyperbolic geometry, by contrast, the area of a geodesic disk grows exponentially once its radius becomes comparable to or larger than the curvature radius. The logarithmic divergence of the hyperbolic local moire length must therefore be combined with exponential hyperbolic-area growth before the final twist-angle scaling can be determined. The central result of the present subsection is that these two geometric effects combine to produce an exact crossover of the effective moire-area exponent from its Euclidean value to a distinct curvature-dominated value. More precisely, the ef-

fective two-dimensional scaling changes continuously according to $2 \longrightarrow 1$. The result is derived below without assuming an exact crystallographic supercell. Accordingly, all quantities in the present subsection refer first to the effective geometric moire domain. The relation between this geometric scaling law and exact finite-index hyperbolic supercells is treated separately at the end of the subsection and in the following sections. The microscopic registry threshold introduced in the preceding subsection is denoted by $a > 0$. For a centered relative twist, define the effective registry domain by

$$\Omega_M^{\text{eff}} = \{x \in \mathbb{H}_R^2 : d_{\mathbb{H}}(x, g_\theta x) \leq a\}. \quad (483)$$

The twist-induced displacement depends only on the radial distance from the twist center,

$$d_{\mathbb{H}}(x, g_\theta x) = d_\theta(r), \quad r = d_{\mathbb{H}}(o, x). \quad (484)$$

The previous subsection established strict radial monotonicity,

$$\frac{dd_\theta(r)}{dr} > 0. \quad (485)$$

The local moire length is uniquely determined by $d_\theta(\xi_M) = a$. Therefore, $d_\theta(r) \leq a$ is equivalent to $r \leq \xi_M$. Hence the effective moire domain is exactly the geodesic disk

$$\Omega_M^{\text{eff}} = B_{\mathbb{H}}(o, \xi_M). \quad (486)$$

This statement follows from the radial symmetry and strict monotonicity of the displacement field and does not constitute an approximation. The effective moire area is therefore defined by

$$A_M^{\text{eff}} = \text{Area}[B_{\mathbb{H}}(o, \xi_M)]. \quad (487)$$

For a hyperbolic plane of constant Gaussian curvature $K = -\frac{1}{R^2}$, the geodesic-polar metric is

$$ds^2 = dr^2 + R^2 \sinh^2\left(\frac{r}{R}\right) d\phi^2. \quad (488)$$

The corresponding area element is

$$dA = R \sinh\left(\frac{r}{R}\right) dr d\phi. \quad (489)$$

The area of a geodesic disk of radius equal to an arbitrary positive value is consequently

$$A_{\mathbb{H}}(r) = \int_0^{2\pi} d\phi \int_0^r R \sinh\left(\frac{r'}{R}\right) dr'. \quad (490)$$

Performing the radial integral gives

$$\begin{aligned} A_{\mathbb{H}}(r) &= 2\pi R \int_0^r \sinh\left(\frac{r'}{R}\right) dr' \\ &= 2\pi R^2 \left[\cosh\left(\frac{r}{R}\right) - 1 \right]. \end{aligned} \quad (491)$$

Thus,

$$A_{\mathbb{H}}(r) = 2\pi R^2 \left[\cosh\left(\frac{r}{R}\right) - 1 \right]. \quad (492)$$

For radii much smaller than the curvature radius, $\frac{r}{R} \ll 1$, the hyperbolic cosine has the expansion

$$\cosh x = 1 + \frac{x^2}{2} + \frac{x^4}{24} + \frac{x^6}{720} + O(x^8). \quad (493)$$

Substitution gives

$$A_{\mathbb{H}}(r) = \pi r^2 + \frac{\pi r^4}{12R^2} + \frac{\pi r^6}{360R^4} + O\left(\frac{r^8}{R^6}\right). \quad (494)$$

Thus the Euclidean area law is recovered locally, $A_{\mathbb{H}}(r) \sim \pi r^2$. At large radial distance, $\frac{r}{R} \gg 1$, the hyperbolic cosine satisfies

$$\cosh\left(\frac{r}{R}\right) = \frac{1}{2}e^{r/R} + \frac{1}{2}e^{-r/R}. \quad (495)$$

Therefore,

$$A_{\mathbb{H}}(r) \sim \pi R^2 e^{r/R}. \quad (496)$$

The geometric area growth thus crosses over from quadratic to exponential as the radial scale becomes comparable to the curvature radius. The exact local moire length obtained previously is

$$\xi_M = R \operatorname{arsinh} \left[\frac{\sinh(a/2R)}{|\sin(\theta/2)|} \right]. \quad (497)$$

Introduce the abbreviations

$$b = \sinh\left(\frac{a}{2R}\right), \quad s = \left| \sin\left(\frac{\theta}{2}\right) \right|. \quad (498)$$

Then

$$\frac{\xi_M}{R} = \operatorname{arsinh} \left(\frac{b}{s} \right). \quad (499)$$

Consequently,

$$\sinh\left(\frac{\xi_M}{R}\right) = \frac{b}{s}. \quad (500)$$

Using $\cosh^2 x - \sinh^2 x = 1$, one obtains

$$\cosh\left(\frac{\xi_M}{R}\right) = \sqrt{1 + \frac{b^2}{s^2}}. \quad (501)$$

Substituting into the exact disk-area formula gives

$$A_M^{\text{eff}} = 2\pi R^2 \left[\sqrt{1 + \frac{b^2}{s^2}} - 1 \right]. \quad (502)$$

Restoring the original parameters yields

$$A_M^{\text{eff}}(\theta, R) = 2\pi R^2 \left[\sqrt{1 + \frac{\sinh^2(a/2R)}{\sin^2(\theta/2)}} - 1 \right]. \quad (503)$$

Equation (503) is exact. It contains the Euclidean, crossover, and curvature-dominated regimes in a single expression. Define the dimensionless geometric ratio

$$\chi = \frac{\sinh(a/2R)}{|\sin(\theta/2)|}. \quad (504)$$

The exact local moire length becomes

$$\frac{\xi_M}{R} = \operatorname{arsinh} \chi, \quad (505)$$

while the exact effective area becomes

$$\frac{A_M^{\text{eff}}}{2\pi R^2} = \sqrt{1 + \chi^2} - 1. \quad (506)$$

Define the dimensionless crossover function

$$\mathcal{F}_A(\chi) = \sqrt{1 + \chi^2} - 1. \quad (507)$$

Then

$$A_M^{\text{eff}} = 2\pi R^2 \mathcal{F}_A(\chi). \quad (508)$$

Thus the entire geometric crossover is governed by one dimensionless variable. The locally Euclidean regime is defined by $\chi \ll 1$. The square-root expansion gives

$$\sqrt{1 + \chi^2} = 1 + \frac{\chi^2}{2} - \frac{\chi^4}{8} + \frac{\chi^6}{16} - \frac{5\chi^8}{128} + O(\chi^{10}). \quad (509)$$

Therefore,

$$\mathcal{F}_A(\chi) = \frac{\chi^2}{2} - \frac{\chi^4}{8} + \frac{\chi^6}{16} - \frac{5\chi^8}{128} + O(\chi^{10}). \quad (510)$$

The effective area becomes

$$A_M^{\text{eff}} = \pi R^2 \chi^2 \left[1 - \frac{\chi^2}{4} + \frac{\chi^4}{8} - \frac{5\chi^6}{64} + O(\chi^8) \right]. \quad (511)$$

Using the definition of the crossover variable gives

$$A_M^{\text{eff}} = \pi R^2 \frac{\sinh^2(a/2R)}{\sin^2(\theta/2)} \left[1 - \frac{\sinh^2(a/2R)}{4\sin^2(\theta/2)} + O(\chi^4) \right]. \quad (512)$$

If the microscopic scale is simultaneously much smaller than the curvature radius, $\frac{a}{R} \ll 1$, then

$$\sinh\left(\frac{a}{2R}\right) = \frac{a}{2R} + O(R^{-3}). \quad (513)$$

The leading area therefore reduces to

$$A_M^{\text{eff}} \simeq \frac{\pi a^2}{4\sin^2(\theta/2)}. \quad (514)$$

For asymptotically small twist within the locally Euclidean regime,

$$\sin\left(\frac{\theta}{2}\right) \sim \frac{|\theta|}{2}, \quad (515)$$

and hence

$$A_M^{\text{eff}} \sim \frac{\pi a^2}{\theta^2}. \quad (516)$$

Thus the Euclidean effective-area exponent is $\beta_M^{(E)} = 2$. The exact formula permits the first curvature corrections to be obtained without introducing an independent continuum approximation. Define $\alpha = \frac{a}{2R}$. The exact Euclidean reference area associated with the same registry threshold is

$$A_M^{(E)} = \frac{\pi a^2}{4s^2}. \quad (517)$$

From the exact hyperbolic area formula one obtains

$$\frac{A_M^{\text{eff}}}{A_M^{(E)}} = \frac{2\sinh^2\alpha}{\alpha^2 \left[1 + \sqrt{1 + \sinh^2\alpha/s^2} \right]} \quad (518)$$

Expanding at fixed nonzero twist gives

$$\begin{aligned} \frac{A_M^{\text{eff}}}{A_M^{(E)}} &= 1 + \alpha^2 \left[\frac{1}{3} - \frac{1}{4s^2} \right] \\ &+ \alpha^4 \left[\frac{2}{45} - \frac{1}{6s^2} + \frac{1}{8s^4} \right] + O(\alpha^6). \end{aligned} \quad (519)$$

Thus curvature modifies not only the asymptotic scaling regime but also the finite-angle prefactor before the curvature-dominated limit is reached. The curvature-dominated regime is characterized by $\chi \gg 1$. The square root then satisfies

$$\sqrt{1 + \chi^2} = \chi + \frac{1}{2\chi} - \frac{1}{8\chi^3} + \frac{1}{16\chi^5} + O(\chi^{-7}). \quad (520)$$

Hence,

$$\mathcal{F}_A(\chi) = \chi - 1 + \frac{1}{2\chi} - \frac{1}{8\chi^3} + \frac{1}{16\chi^5} + O(\chi^{-7}). \quad (521)$$

The effective area therefore obeys

$$A_M^{\text{eff}} = 2\pi R^2 \left[\chi - 1 + \frac{1}{2\chi} - \frac{1}{8\chi^3} + O(\chi^{-5}) \right]. \quad (522)$$

Returning to the microscopic parameters gives

$$A_M^{\text{eff}} = 2\pi R^2 \left[\frac{b}{s} - 1 + \frac{s}{2b} - \frac{s^3}{8b^3} + O(s^5) \right]. \quad (523)$$

For small twist,

$$\frac{1}{s} = \frac{2}{|\theta|} + \frac{|\theta|}{12} + \frac{7|\theta|^3}{2880} + O(|\theta|^5). \quad (524)$$

Using also

$$s = \frac{|\theta|}{2} - \frac{|\theta|^3}{48} + O(|\theta|^5), \quad (525)$$

one obtains

$$\begin{aligned} A_M^{\text{eff}} &= \frac{4\pi R^2 b}{|\theta|} - 2\pi R^2 \\ &+ 2\pi R^2 |\theta| \left[\frac{b}{12} + \frac{1}{4b} \right] \\ &+ 2\pi R^2 |\theta|^3 \left[\frac{7b}{2880} - \frac{1}{96b} - \frac{1}{64b^3} \right] + O(|\theta|^5). \end{aligned} \quad (526)$$

The leading curvature-dominated result is therefore

$$A_M^{\text{eff}} \sim \frac{4\pi R^2 \sinh(a/2R)}{|\theta|}. \quad (527)$$

Thus, $A_M^{\text{eff}} \propto |\theta|^{-1}$ at fixed nonzero negative curvature. This is the first precise statement of the curvature-induced change of effective moiré scaling. The origin of the inverse-angle law can be displayed without using the closed-form square-root expression. In the curvature-dominated regime, the local moiré length obeys

$$\xi_M \sim R \ln \left[\frac{4 \sinh(a/2R)}{|\theta|} \right]. \quad (528)$$

At the same time, the area of a large hyperbolic disk obeys

$$A_{\mathbb{H}}(r) \sim \pi R^2 e^{r/R}. \quad (529)$$

Substitution gives

$$A_M^{\text{eff}} \sim \pi R^2 \exp \left[\ln \left(\frac{4 \sinh(a/2R)}{|\theta|} \right) \right]. \quad (530)$$

Therefore,

$$A_M^{\text{eff}} \sim \frac{4\pi R^2 \sinh(a/2R)}{|\theta|}. \quad (531)$$

The change of exponent is therefore generated by a compound geometric mechanism. Negative curvature produces an exponentially enhanced separation of initially nearby geodesics, so that the local moiré length no longer follows the Euclidean inverse-angle law but instead scales asymptotically as $\xi_M \sim \ln |\theta|^{-1}$. At the same time, the area of a hyperbolic disk grows exponentially with its geodesic radius according to $A_{\mathbb{H}} \sim e^{\xi_M/R}$. Combining these two effects gives the effective moiré-area scaling $A_M^{\text{eff}} \sim |\theta|^{-1}$. This compound relation constitutes the geometric origin of the curvature-induced change in the effective moiré scaling exponent. Let the primitive microscopic cell have hyperbolic covolume $A_0(K) > 0$, and contain a fixed number of microscopic basis sites per layer, $n_b \in \mathbb{N}$. The quotient-space microscopic density per layer is

$$\rho_0 = \frac{n_b}{A_0(K)}. \quad (532)$$

The area-normalized effective microscopic count of the bilayer is defined by

$$N_M^{\text{eff}} = 2n_b \frac{A_M^{\text{eff}}}{A_0(K)}. \quad (533)$$

Substitution of the exact area gives

$$N_M^{\text{eff}} = \frac{4\pi n_b R^2}{A_0(K)} \left[\sqrt{1 + \frac{\sinh^2(a/2R)}{\sin^2(\theta/2)}} - 1 \right]. \quad (534)$$

Define the curvature-dependent normalization

$$\mathcal{N}_0 = \frac{4\pi n_b R^2}{A_0(K)}. \quad (535)$$

Then

$$N_M^{\text{eff}} = \mathcal{N}_0 \left[\sqrt{1 + \chi^2} - 1 \right]. \quad (536)$$

In the locally Euclidean regime,

$$N_M^{\text{eff}} = \frac{\mathcal{N}_0}{2} \chi^2 + O(\chi^4), \quad (537)$$

whereas in the curvature-dominated regime,

$$N_M^{\text{eff}} = \mathcal{N}_0 \chi - \mathcal{N}_0 + O(\chi^{-1}). \quad (538)$$

At fixed negative curvature and asymptotically small twist,

$$N_M^{\text{eff}} \sim \frac{8\pi n_b R^2 \sinh(a/2R)}{A_0(K)} |\theta|^{-1}. \quad (539)$$

Thus, $N_M^{\text{eff},H} \propto |\theta|^{-1}$. By contrast, the Euclidean effective count obeys $N_M^{\text{eff},E} \propto |\theta|^{-2}$. The effective microscopic-count scaling therefore undergoes the same curvature-induced exponent crossover as the effective area. The crossover need not be characterized only by its two asymptotic endpoints. An exact running exponent can be defined throughout the entire parameter range. Define

$$\beta_M = -\frac{d \ln A_M^{\text{eff}}}{d \ln s}. \quad (540)$$

Since $\chi = \frac{b}{s}$, one has

$$\frac{d \ln \chi}{d \ln s} = -1. \quad (541)$$

Therefore,

$$\beta_M = \frac{d \ln A_M^{\text{eff}}}{d \ln \chi}. \quad (542)$$

Introduce $q = \sqrt{1 + \chi^2}$. Then $A_M^{\text{eff}} \propto q - 1$. Differentiation gives

$$\frac{dq}{d\chi} = \frac{\chi}{q}. \quad (543)$$

Hence,

$$\beta_M = \frac{\chi}{q-1} \frac{dq}{d\chi} = \frac{\chi^2}{q(q-1)}. \quad (544)$$

Since

$$\chi^2 = q^2 - 1 = (q-1)(q+1), \quad (545)$$

the exact running exponent reduces to

$$\beta_M(\chi) = 1 + \frac{1}{\sqrt{1+\chi^2}}. \quad (546)$$

This is an exact closed-form crossover law. For $\chi \ll 1$, one obtains

$$\beta_M = 2 - \frac{\chi^2}{2} + \frac{3\chi^4}{8} - \frac{5\chi^6}{16} + O(\chi^8). \quad (547)$$

Thus, $\beta_M \rightarrow 2$. For $\chi \gg 1$, one obtains

$$\beta_M = 1 + \frac{1}{\chi} - \frac{1}{2\chi^3} + \frac{3}{8\chi^5} + O(\chi^{-7}). \quad (548)$$

Therefore, $\beta_M \rightarrow 1$. The effective-area scaling exponent thus flows continuously according to $2 \rightarrow 1$ as the system crosses from the locally Euclidean regime to the curvature-dominated regime. The running exponent itself satisfies a closed differential equation. Differentiating Eq. (546) with respect to the logarithm of the crossover variable gives

$$\frac{d\beta_M}{d\ln \chi} = -\frac{\chi^2}{(1+\chi^2)^{3/2}}. \quad (549)$$

Define $y = \beta_M - 1$. Equation (546) implies

$$y = \frac{1}{\sqrt{1+\chi^2}}. \quad (550)$$

Therefore,

$$\chi^2 = \frac{1-y^2}{y^2}. \quad (551)$$

Substitution into Eq. (549) gives

$$\frac{d\beta_M}{d\ln \chi} = (\beta_M - 1)^3 - (\beta_M - 1). \quad (552)$$

The two asymptotic scaling regimes are therefore fixed points of the exact geometric flow,

$$\beta_M^* = 2, \quad \beta_M^* = 1. \quad (553)$$

As the crossover variable increases, the flow proceeds from the Euclidean fixed point toward the curvature-dominated fixed point. The exponent crossover admits an exact mechanistic decomposition. Define the running local-length exponent by

$$\nu_\xi = -\frac{d\ln \xi_M}{d\ln s}. \quad (554)$$

Since

$$\frac{\xi_M}{R} = \operatorname{arsinh} \chi, \quad (555)$$

one obtains

$$\nu_\xi = \frac{\chi}{\sqrt{1+\chi^2} \operatorname{arsinh} \chi}. \quad (556)$$

Next define the effective radial area-growth dimension,

$$D_A = \frac{d\ln A_{\mathbb{H}}(\xi_M)}{d\ln \xi_M}. \quad (557)$$

Introduce $y = \frac{\xi_M}{R}$. The area is

$$A_{\mathbb{H}} = 2\pi R^2 (\cosh y - 1). \quad (558)$$

Differentiating with respect to the local length gives

$$\frac{dA_{\mathbb{H}}}{d\xi_M} = 2\pi R \sinh y. \quad (559)$$

Therefore,

$$D_A = \frac{\xi_M}{A_{\mathbb{H}}} \frac{dA_{\mathbb{H}}}{d\xi_M} = \frac{y \sinh y}{\cosh y - 1}. \quad (560)$$

Using

$$\frac{\sinh y}{\cosh y - 1} = \coth\left(\frac{y}{2}\right), \quad (561)$$

one obtains

$$D_A = y \coth\left(\frac{y}{2}\right). \quad (562)$$

Since $y = \operatorname{arsinh} \chi$, and

$$\sinh y = \chi, \quad \cosh y = \sqrt{1+\chi^2}, \quad (563)$$

the same quantity can be written as

$$D_A = \operatorname{arsinh} \chi \frac{\chi}{\sqrt{1 + \chi^2 - 1}}. \quad (564)$$

The chain rule now gives

$$-\frac{d \ln A_M^{\text{eff}}}{d \ln s} = \frac{d \ln A_M^{\text{eff}}}{d \ln \xi_M} \left[-\frac{d \ln \xi_M}{d \ln s} \right]. \quad (565)$$

Therefore, $\beta_M = D_A \nu_\xi$. This is an exact identity. Substituting the explicit expressions gives

$$\begin{aligned} D_A \nu_\xi &= \left[\operatorname{arsinh} \chi \frac{\chi}{\sqrt{1 + \chi^2 - 1}} \right] \\ &\times \left[\frac{\chi}{\sqrt{1 + \chi^2} \operatorname{arsinh} \chi} \right] \\ &= \frac{\chi^2}{\sqrt{1 + \chi^2} [\sqrt{1 + \chi^2 - 1}]}. \end{aligned} \quad (566)$$

Using the same algebraic reduction as above yields

$$D_A \nu_\xi = 1 + \frac{1}{\sqrt{1 + \chi^2}} = \beta_M. \quad (567)$$

This factorization isolates the two geometric mechanisms responsible for the crossover. For small crossover variable, the local-length exponent is

$$\nu_\xi = 1 - \frac{\chi^2}{3} + \frac{11\chi^4}{45} - \frac{191\chi^6}{945} + O(\chi^8). \quad (568)$$

The effective radial area dimension is

$$D_A = 2 + \frac{\chi^2}{6} - \frac{7\chi^4}{120} + \frac{53\chi^6}{1680} + O(\chi^8). \quad (569)$$

Thus, $\nu_\xi \rightarrow 1$, and $D_A \rightarrow 2$. Their product therefore gives $\beta_M \rightarrow 2$. This reproduces the ordinary Euclidean mechanism,

$$\xi_M \sim |\theta|^{-1}, \quad A_M \sim \xi_M^2, \quad A_M \sim |\theta|^{-2}. \quad (570)$$

For large crossover variable,

$$\nu_\xi \sim \frac{1}{\ln(2\chi)}. \quad (571)$$

Thus the effective power-law exponent associated with the local length vanishes asymptotically, $\nu_\xi \rightarrow 0$. At the same time,

$$D_A = \operatorname{arsinh} \chi \frac{\chi}{\sqrt{1 + \chi^2 - 1}}. \quad (572)$$

The second factor has the large-variable expansion

$$\frac{\chi}{\sqrt{1 + \chi^2 - 1}} = 1 + \frac{1}{\chi} + \frac{1}{2\chi^2} + O(\chi^{-4}). \quad (573)$$

Therefore, $D_A \sim \ln(2\chi)$. Thus the effective radial area dimension diverges logarithmically, $D_A \rightarrow \infty$. The two apparently singular tendencies compensate exactly,

$$\nu_\xi \sim \frac{1}{\ln(2\chi)}, \quad D_A \sim \ln(2\chi), \quad (574)$$

so that $D_A \nu_\xi \rightarrow 1$. Hence, $\beta_M \rightarrow 1$. This cancellation provides a mechanism-level explanation for the curvature-dominated exponent. The result is not simply caused by the logarithmic local moire length, nor simply by exponential hyperbolic area growth. It is generated by their precise compensation. The crossover can be expressed in a universal form in the simultaneous weak-curvature and small-angle limit. Define $u = \frac{R|\theta|}{a}$. Consider the double-scaling limit

$$\frac{a}{R} \rightarrow 0, \quad |\theta| \rightarrow 0, \quad u \text{ fixed}. \quad (575)$$

Introduce $\alpha = \frac{a}{2R}$. Then $|\theta| = 2\alpha u$. The crossover variable becomes

$$\chi = \frac{\sinh \alpha}{\sin(\alpha u)}. \quad (576)$$

Expanding both numerator and denominator gives

$$\chi = \frac{1}{u} \left[1 + \frac{\alpha^2}{6} (1 + u^2) + O(\alpha^4) \right]. \quad (577)$$

Thus, $\chi \rightarrow \frac{1}{u}$. The effective area therefore obeys

$$\frac{A_M^{\text{eff}}}{2\pi R^2} \rightarrow \mathcal{H}(u), \quad (578)$$

where

$$\mathcal{H}(u) = \sqrt{1 + u^{-2}} - 1. \quad (579)$$

This is the universal crossover function for the effective area in the double-scaling limit. The leading Euclidean small-angle area is

$$A_M^{(E,0)} = \frac{\pi a^2}{\theta^2}. \quad (580)$$

Using the double-scaling variable gives

$$A_M^{(E,0)} = \frac{\pi R^2}{u^2}. \quad (581)$$

The ratio between the hyperbolic and Euclidean effective areas therefore approaches

$$\frac{A_M^{\text{eff}}}{A_M^{(E,0)}} \longrightarrow \mathcal{R}_A(u), \quad (582)$$

with

$$\mathcal{R}_A(u) = 2u^2 \left[\sqrt{1 + u^{-2}} - 1 \right]. \quad (583)$$

Rationalization yields the equivalent form

$$\mathcal{R}_A(u) = \frac{2}{1 + \sqrt{1 + u^{-2}}}. \quad (584)$$

For $u \gg 1$, one finds

$$\mathcal{R}_A(u) = 1 - \frac{1}{4u^2} + \frac{1}{8u^4} - \frac{5}{64u^6} + O(u^{-8}). \quad (585)$$

Thus the Euclidean area is recovered. For $u \ll 1$, one obtains

$$\mathcal{R}_A(u) = 2u - 2u^2 + u^3 - \frac{u^5}{4} + O(u^6). \quad (586)$$

Consequently, $\mathcal{R}_A(u) \longrightarrow 0$. The Euclidean inverse-square estimate therefore increasingly overestimates the effective hyperbolic registry area in the deeply curvature-dominated small-angle regime. Because the crossover is continuous, no single angle is uniquely privileged without specifying a convention. A natural order-one geometric crossover may be defined through $\chi = 1$. This gives

$$\left| \sin \left(\frac{\theta_c^{(\chi)}}{2} \right) \right| = \sinh \left(\frac{a}{2R} \right). \quad (587)$$

Whenever the right-hand side does not exceed unity,

$$\theta_c^{(\chi)} = 2 \arcsin \left[\sinh \left(\frac{a}{2R} \right) \right]. \quad (588)$$

In the weak-curvature limit,

$$\theta_c^{(\chi)} = \frac{a}{R} + \frac{a^3}{12R^3} + \frac{a^5}{96R^5} + O(R^{-7}). \quad (589)$$

An alternative convention is to define the crossover by the midpoint of the running exponent, $\beta_M = \frac{3}{2}$. Using the exact exponent law gives

$$\frac{1}{\sqrt{1 + \chi^2}} = \frac{1}{2}. \quad (590)$$

Therefore, $\chi = \sqrt{3}$. The corresponding angle satisfies

$$\left| \sin \left(\frac{\theta_c^{(\beta)}}{2} \right) \right| = \frac{1}{\sqrt{3}} \sinh \left(\frac{a}{2R} \right). \quad (591)$$

Thus,

$$\theta_c^{(\beta)} = 2 \arcsin \left[\frac{\sinh(a/2R)}{\sqrt{3}} \right]. \quad (592)$$

For weak curvature,

$$\theta_c^{(\beta)} \sim \frac{a}{\sqrt{3}R}. \quad (593)$$

The scaling variable is therefore more fundamental than any particular numerical definition of the crossover angle. The exponent change does not depend on identifying the registry threshold exactly with the nearest-neighbor lattice spacing. Let an arbitrary fixed positive threshold be denoted by $\ell_* > 0$. The exact area then becomes

$$A_M^{\text{eff}} = 2\pi R^2 \left[\sqrt{1 + \frac{\sinh^2(\ell_*/2R)}{\sin^2(\theta/2)}} - 1 \right]. \quad (594)$$

At fixed negative curvature and fixed threshold, $|\theta| \longrightarrow 0$ gives

$$A_M^{\text{eff}} \sim 4\pi R^2 \sinh \left(\frac{\ell_*}{2R} \right) |\theta|^{-1}. \quad (595)$$

Thus the threshold changes the prefactor and the crossover position but not the asymptotic exponent, $\beta_M^{(H)} = 1$. In Euclidean geometry, the same generalized threshold gives

$$A_M^{(E)} = \frac{\pi \ell_*^2}{4 \sin^2(\theta/2)}, \quad (596)$$

and hence $\beta_M^{(E)} = 2$. The endpoint exponents are therefore independent of the microscopic registry convention as long as the threshold remains finite, positive, and twist independent. The twist-angle exponent crossover should not be confused with a simple monotonic dependence of the effective area on the magnitude of curvature. The local moiré length decreases monotonically with increasing negative curvature, as proved previously. The effective area contains a competing effect because the area associated with a fixed geodesic radius itself increases rapidly

with negative curvature. To quantify this competition, retain $\alpha = \frac{a}{2R}$, and define

$$q = \sqrt{1 + \frac{\sinh^2 \alpha}{s^2}}. \quad (597)$$

The exact area may be written as

$$A_M^{\text{eff}} = \frac{\pi a^2}{2\alpha^2} (q - 1). \quad (598)$$

Differentiation with respect to the microscopic curvature parameter gives

$$\frac{d \ln A_M^{\text{eff}}}{d \ln \alpha} = \alpha \coth \alpha \beta_M - 2. \quad (599)$$

Since

$$\alpha = \frac{a}{2} \sqrt{|K|}, \quad (600)$$

one has

$$\frac{d \ln \alpha}{d \ln |K|} = \frac{1}{2}. \quad (601)$$

Therefore the logarithmic curvature response is

$$\frac{d \ln A_M^{\text{eff}}}{d \ln |K|} = \frac{1}{2} [\alpha \coth \alpha \beta_M - 2]. \quad (602)$$

The sign is not fixed universally. Thus the robust statement result concerns the twist-angle scaling at fixed curvature, not a claim that the effective area must be monotonic in curvature at every fixed twist. In the asymptotically small-angle regime, $\beta_M \rightarrow 1$, and the curvature response reduces to

$$\frac{d \ln A_M^{\text{eff}}}{d \ln |K|} \rightarrow \frac{1}{2} [\alpha \coth \alpha - 2]. \quad (603)$$

The competition between contraction of the local moire radius and exponential growth of hyperbolic area is therefore quantitatively separable. The effective microscopic count defined above is intentionally an area-normalized quotient-space quantity. It must not be identified without additional proof with the literal integer number of lattice sites inside an open geodesic disk. The circumference of a hyperbolic disk is

$$P_{\mathbb{H}}(r) = 2\pi R \sinh\left(\frac{r}{R}\right). \quad (604)$$

The ratio of perimeter to area is therefore

$$\frac{P_{\mathbb{H}}(r)}{A_{\mathbb{H}}(r)} = \frac{1}{R} \frac{\sinh(r/R)}{\cosh(r/R) - 1}. \quad (605)$$

Using the half-angle identities gives

$$\frac{P_{\mathbb{H}}(r)}{A_{\mathbb{H}}(r)} = \frac{1}{R} \coth\left(\frac{r}{2R}\right). \quad (606)$$

Hence,

$$\lim_{r \rightarrow \infty} \frac{P_{\mathbb{H}}(r)}{A_{\mathbb{H}}(r)} = \frac{1}{R} \neq 0. \quad (607)$$

The boundary remains extensive. Therefore a literal open-disk site count,

$$N_M^{\text{disk}} = \sum_{\ell=1}^2 \# [\Lambda_{\ell} \cap B_{\mathbb{H}}(o, \xi_M)], \quad (608)$$

need not satisfy

$$N_M^{\text{disk}} = N_M^{\text{eff}} + o(N_M^{\text{eff}}). \quad (609)$$

This distinction is a consequence of hyperbolic nonamenability and must be preserved in all finite-cluster numerics. The result proved above is $N_M^{\text{eff}, H} \propto |\theta|^{-1}$. This geometric relation alone does not imply $N_M^{\text{sc}, H} \propto |\theta|^{-1}$. The exact hyperbolic supercell is controlled by the finite-index common subgroup

$$\Gamma_M = \Gamma \cap g_{\theta} \Gamma g_{\theta}^{-1}. \quad (610)$$

Whenever this subgroup has finite index, define

$$q_M(\theta) = [\Gamma : \Gamma \cap g_{\theta} \Gamma g_{\theta}^{-1}]. \quad (611)$$

The exact bilayer microscopic count is then $N_M^{\text{sc}} = 2n_b q_M$. A transfer from effective geometric scaling to exact crystallographic scaling therefore requires a separate geometric-arithmetic locking statement. The precise logical requirement can be stated as follows. Consider a sequence of exact commensurate twists satisfying $\theta_j \rightarrow 0$. Suppose that there exist finite positive constants satisfying $0 < c_- \leq c_+ < \infty$ such that

$$c_- \frac{A_M^{\text{eff}}(\theta_j)}{A_0} \leq q_M(\theta_j) \leq c_+ \frac{A_M^{\text{eff}}(\theta_j)}{A_0}. \quad (612)$$

Then the exact coincidence index satisfies

$$q_M(\theta_j) = \Theta(|\theta_j|^{-1}). \quad (613)$$

Consequently,

$$N_M^{\text{sc}}(\theta_j) = \Theta(|\theta_j|^{-1}). \quad (614)$$

The effective-area theorem gives

$$A_M^{\text{eff}}(\theta_j) \sim C_H |\theta_j|^{-1}, \quad (615)$$

where

$$C_H = 4\pi R^2 \sinh\left(\frac{a}{2R}\right) > 0. \quad (616)$$

Therefore there exists a sufficiently large sequence index such that

$$C_1 |\theta_j|^{-1} \leq A_M^{\text{eff}}(\theta_j) \leq C_2 |\theta_j|^{-1}, \quad (617)$$

for finite positive constants satisfying $0 < C_1 \leq C_2 < \infty$. Substitution into the assumed locking bound gives

$$\frac{c_- C_1}{A_0} |\theta_j|^{-1} \leq q_M(\theta_j) \leq \frac{c_+ C_2}{A_0} |\theta_j|^{-1}. \quad (618)$$

Hence,

$$q_M(\theta_j) = \Theta(|\theta_j|^{-1}). \quad (619)$$

Using $N_M^{\text{sc}} = 2n_b q_M$, immediately gives

$$N_M^{\text{sc}}(\theta_j) = \Theta(|\theta_j|^{-1}). \quad (620)$$

This completes the conditional proof. The distinction between the effective and exact statements can now be made precise. The following local geometric results have been proved without any commensurability assumption: $\xi_M^{(E)} \sim |\theta|^{-1}$, $A_M^{\text{eff},E} \sim |\theta|^{-2}$, $\xi_M^{(H)} \sim R \ln |\theta|^{-1}$, and $A_M^{\text{eff},H} \sim |\theta|^{-1}$. Consequently,

$$N_M^{\text{eff},E} \sim |\theta|^{-2}, \quad N_M^{\text{eff},H} \sim |\theta|^{-1}. \quad (621)$$

The exact running exponent has also been obtained,

$$\beta_M(\chi) = 1 + \frac{1}{\sqrt{1 + \chi^2}}, \quad (622)$$

with endpoint values

$$\beta_M(0) = 2, \quad \beta_M(\infty) = 1. \quad (623)$$

The historical unconditional exact-crystallographic question was the statement $q_M(\theta) \stackrel{?}{\sim} |\theta|^{-1}$. Establishing or disproving such a law requires information about the

finite-index intersections of the hyperbolic crystal group and its rotated conjugate. The explicit centered sequence constructed below answers this question negatively in general: it lies in $\mathfrak{U}_1(4)$, not $\mathfrak{U}_1(1)$. Exponent one remains a conditional locking class, while other exact height rays may belong to different arithmetic classes. The results of the present subsection can be summarized in the following theorem.

a. Theorem 3. Curvature-induced crossover of effective moire scaling. Consider a two-dimensional hyperbolic bilayer of constant negative Gaussian curvature, $K = -\frac{1}{R^2}$, with a centered relative twist and a fixed positive microscopic registry threshold. The effective moire domain is the geodesic disk of radius equal to the local moire length,

$$\Omega_M^{\text{eff}} = B_{\mathbb{H}}(o, \xi_M). \quad (624)$$

Its exact area is

$$A_M^{\text{eff}} = 2\pi R^2 \left[\sqrt{1 + \frac{\sinh^2(a/2R)}{\sin^2(\theta/2)}} - 1 \right]. \quad (625)$$

Introducing

$$\chi = \frac{\sinh(a/2R)}{|\sin(\theta/2)|}, \quad (626)$$

the exact running twist-angle exponent is

$$\beta_M = 1 + \frac{1}{\sqrt{1 + \chi^2}}. \quad (627)$$

In the locally Euclidean regime, $\chi \ll 1$, one has

$$A_M^{\text{eff}} \propto |\theta|^{-2}, \quad \beta_M \longrightarrow 2. \quad (628)$$

In the curvature-dominated regime, $\chi \gg 1$, one has

$$A_M^{\text{eff}} \propto |\theta|^{-1}, \quad \beta_M \longrightarrow 1. \quad (629)$$

The corresponding area-normalized effective microscopic count satisfies

$$N_M^{\text{eff},E} \propto |\theta|^{-2}, \quad N_M^{\text{eff},H} \propto |\theta|^{-1}. \quad (630)$$

Moreover, the running area exponent factorizes exactly as $\beta_M = D_A \nu_\xi$, where

$$\nu_\xi = \frac{\chi}{\sqrt{1 + \chi^2} \operatorname{arsinh} \chi}, \quad (631)$$

and

E. Universal Crossover Function

$$D_A = \operatorname{arsinh} \chi \frac{\chi}{\sqrt{1 + \chi^2} - 1}. \quad (632)$$

The Euclidean endpoint satisfies

$$\nu_\xi \longrightarrow 1, \quad D_A \longrightarrow 2, \quad \beta_M \longrightarrow 2, \quad (633)$$

whereas the curvature-dominated endpoint satisfies

$$\nu_\xi \sim \frac{1}{\ln(2\chi)}, \quad D_A \sim \ln(2\chi), \quad \beta_M \longrightarrow 1. \quad (634)$$

Thus the change in moiré-area scaling is generated by an exact compensation between the logarithmically weakening twist response of the local length and the exponentially strengthening radial area growth of hyperbolic geometry. The complete hierarchy established in the present subsection is

$$K < 0 \longrightarrow R \sinh(r/R) \longrightarrow d_\theta(r) \longrightarrow \xi_M \longrightarrow A_{\mathbb{H}}(\xi_M) \longrightarrow A_M^{\text{eff}} \longrightarrow N_M^{\text{eff}} \quad (635)$$

In the Euclidean regime this chain reduces to

$$d_\theta \sim r\theta \longrightarrow \xi_M \sim \theta^{-1} \longrightarrow A_M \sim \xi_M^2 \longrightarrow N_M^{\text{eff}} \sim \theta^{-2}. \quad (636)$$

In the curvature-dominated regime it becomes

$$d_\theta \sim \theta e^{r/R} \longrightarrow \xi_M \sim R \ln \theta^{-1} \longrightarrow A_M \sim e^{\xi_M/R} \longrightarrow N_M^{\text{eff}} \sim \theta^{-1}. \quad (637)$$

The resulting exponent change, $2 \longrightarrow 1$, is therefore a direct consequence of negative curvature and does not require any electronic Hamiltonian, interaction, hopping profile, or band structure. At the same time, the result remains explicitly geometric rather than arithmetic. The next problem is to determine whether the finite-index common subgroup

$$\Gamma_M = \Gamma \cap g_\theta \Gamma g_\theta^{-1} \quad (638)$$

inherits the same scaling. That question requires a theory of hyperbolic commensurability and of the commensurator of the underlying Fuchsian crystal group. Only after that geometric–arithmetic locking problem is resolved can the effective crossover

$$N_M^{\text{eff}} : |\theta|^{-2} \longrightarrow |\theta|^{-1} \quad (639)$$

be promoted, where justified, to an exact crystallographic scaling law for hyperbolic moiré supercells.

The preceding analysis established two asymptotic regimes of the effective moiré geometry. In the locally Euclidean regime, the effective moiré area obeys an inverse-square twist law, whereas in the curvature-dominated regime the same quantity obeys an inverse-linear twist law. The existence of two asymptotic exponents, however, does not by itself constitute a complete scaling theory. A stronger result requires a single dimensionless variable that interpolates between the two regimes and a universal crossover function that determines the entire evolution between them. The purpose of the present subsection is to construct this crossover theory exactly. We show that the effective hyperbolic moiré area and the corresponding area-normalized microscopic count admit an exact factorization into a locally Euclidean reference scale and a dimensionless suppression function. In an appropriate continuum limit, this suppression function depends on a single variable measuring the ratio between the Euclidean moiré length and the curvature radius. Its two asymptotic branches generate the inverse-square and inverse-linear twist laws automatically. The same construction yields an exact flow equation, a parameter-free relation between the crossover amplitude and the running scaling exponent, and a direct numerical data-collapse criterion. We finally generalize the argument to arbitrary spatial dimension and show that negative curvature induces a universal reduction of the asymptotic moiré-volume exponent by one spatial dimension. Introduce the half-angle variable

$$s = \left| \sin\left(\frac{\theta}{2}\right) \right|. \quad (640)$$

The dimensionless microscopic curvature parameter is defined by $\alpha = \frac{a}{2R}$. The exact curvature variable appearing in the hyperbolic registry problem is

$$\chi = \frac{\sinh \alpha}{s}. \quad (641)$$

The exact local moiré length obtained previously is therefore

$$\frac{\xi_M}{R} = \operatorname{arsinh} \chi. \quad (642)$$

The exact effective hyperbolic moiré area is

$$A_M^{\text{eff}} = 2\pi R^2 \left[\sqrt{1 + \chi^2} - 1 \right]. \quad (643)$$

We now introduce the quadratic reference area

$$A_{\text{ref}} = \pi R^2 \chi^2. \quad (644)$$

This is the area obtained by extending the locally Euclidean quadratic area law to the exact curvature-renormalized mismatch variable. Using Eq. (643), the ratio between the exact effective area and this reference area is

$$\frac{A_M^{\text{eff}}}{A_{\text{ref}}} = \frac{2 \left[\sqrt{1 + \chi^2} - 1 \right]}{\chi^2}. \quad (645)$$

Multiplying numerator and denominator by the conjugate factor gives

$$\begin{aligned} \frac{2 \left[\sqrt{1 + \chi^2} - 1 \right]}{\chi^2} &= \frac{2 \left[\sqrt{1 + \chi^2} - 1 \right] \left[\sqrt{1 + \chi^2} + 1 \right]}{\chi^2 \left[\sqrt{1 + \chi^2} + 1 \right]} \\ &= \frac{2 \left[1 + \chi^2 - 1 \right]}{\chi^2 \left[1 + \sqrt{1 + \chi^2} \right]}. \end{aligned} \quad (646)$$

Therefore,

$$\frac{A_M^{\text{eff}}}{A_{\text{ref}}} = \mathcal{F}(\chi), \quad (647)$$

where

$$\mathcal{F}(\chi) = \frac{2}{1 + \sqrt{1 + \chi^2}}. \quad (648)$$

Equation (648) is the exact finite-curvature crossover function. No small-angle approximation and no weak-curvature approximation has been made in obtaining it. The effective moiré area can now be written in the factorized form

$$A_M^{\text{eff}} = A_{\text{ref}} \mathcal{F}(\chi). \quad (649)$$

Explicitly,

$$A_M^{\text{eff}} = \pi R^2 \chi^2 \mathcal{F}(\chi). \quad (650)$$

Similarly, the area-normalized effective microscopic count satisfies

$$N_M^{\text{eff}} = 2n_b \frac{A_M^{\text{eff}}}{A_0(K)}. \quad (651)$$

Define the corresponding reference microscopic count by

$$N_{\text{ref}} = 2n_b \frac{A_{\text{ref}}}{A_0(K)}. \quad (652)$$

Then

$$N_M^{\text{eff}} = N_{\text{ref}} \mathcal{F}(\chi). \quad (653)$$

Thus the entire twist-curvature dependence separates into a reference quadratic scale and a universal geometric suppression factor. The crossover function is normalized so that the locally Euclidean regime corresponds to unity. At vanishing crossover variable, $\sqrt{1 + \chi^2} \rightarrow 1$. Therefore, $\mathcal{F}(0) = 1$. At asymptotically large crossover variable,

$$\sqrt{1 + \chi^2} \rightarrow \infty, \quad (654)$$

and hence

$$\lim_{\chi \rightarrow \infty} \mathcal{F}(\chi) = 0. \quad (655)$$

Thus, $0 < \mathcal{F}(\chi) \leq 1$. The curvature-dominated geometry therefore suppresses the effective moiré area relative to the quadratic reference extrapolation. Differentiate Eq. (648) with respect to the crossover variable. One obtains

$$\frac{d\mathcal{F}}{d\chi} = -\frac{2\chi}{\sqrt{1 + \chi^2} \left[1 + \sqrt{1 + \chi^2} \right]^2}. \quad (656)$$

For every positive crossover variable,

$$\frac{d\mathcal{F}}{d\chi} < 0. \quad (657)$$

Hence the crossover function decreases strictly and continuously from the Euclidean fixed value to the curvature-dominated asymptotic branch. There is therefore no intermediate reversal, overshoot, or additional geometric scale within the centered constant-curvature construction. For small crossover variable, $\chi \ll 1$, the square root admits the expansion

$$\sqrt{1 + \chi^2} = 1 + \frac{\chi^2}{2} - \frac{\chi^4}{8} + \frac{\chi^6}{16} - \frac{5\chi^8}{128} + O(\chi^{10}). \quad (658)$$

Substitution into the exact crossover function yields

$$\mathcal{F}(\chi) = 1 - \frac{\chi^2}{4} + \frac{\chi^4}{8} - \frac{5\chi^6}{64} + \frac{7\chi^8}{128} + O(\chi^{10}). \quad (659)$$

The leading correction is therefore quadratic,

$$1 - \mathcal{F}(\chi) = \frac{\chi^2}{4} + O(\chi^4). \quad (660)$$

Thus the Euclidean scaling regime is stable against weak curvature, with the first departure controlled by the

square of the dimensionless curvature-to-moiré ratio. For large crossover variable, $\chi \gg 1$, the universal function has the expansion

$$\mathcal{F}(\chi) = \frac{2}{\chi} - \frac{2}{\chi^2} + \frac{1}{\chi^3} - \frac{1}{4\chi^5} + \frac{1}{8\chi^7} + O(\chi^{-8}). \quad (661)$$

The leading branch is therefore

$$\mathcal{F}(\chi) \sim \frac{2}{\chi}. \quad (662)$$

Since the quadratic reference scale grows as the square of the crossover variable, multiplication by the inverse-linear crossover function leaves only a single power,

$$A_M^{\text{eff}} \sim 2\pi R^2 \chi. \quad (663)$$

This recovers the curvature-dominated inverse-angle law. The crossover variable can be related directly to the conventional Euclidean moiré length. For the same microscopic registry threshold, define the exact Euclidean local moiré length by

$$\xi_M^{(E)} = \frac{a}{2s}. \quad (664)$$

The ratio between this Euclidean moiré length and the curvature radius is

$$x = \frac{\xi_M^{(E)}}{R} = \frac{a}{2Rs} = \frac{\alpha}{s}. \quad (665)$$

This quantity has a direct geometric interpretation. The regime $x \ll 1$ means that the moiré scale predicted from flat geometry is much smaller than the curvature radius. The regime $x \gg 1$ means that the flat-space moiré prediction extends far beyond the curvature radius and therefore necessarily probes non-Euclidean geometry. The exact crossover variable and this intuitive curvature variable are related by $\chi = \eta(\alpha)x$, where

$$\eta(\alpha) = \frac{\sinh \alpha}{\alpha}. \quad (666)$$

The factor in Eq. (666) measures the finite microscopic-curvature correction. Its expansion is

$$\eta(\alpha) = 1 + \frac{\alpha^2}{6} + \frac{\alpha^4}{120} + \frac{\alpha^6}{5040} + O(\alpha^8). \quad (667)$$

Thus,

$$\chi = x + O(\alpha^2 x). \quad (668)$$

The intuitive ratio between the Euclidean moiré length and the curvature radius therefore becomes the exact crossover variable in the continuum lattice limit. The strict Euclidean reference area is

$$A_E^{\text{ref}} = \pi \left[\xi_M^{(E)} \right]^2. \quad (669)$$

Using Eq. (665), this can be written as

$$A_E^{\text{ref}} = \pi R^2 x^2. \quad (670)$$

The exact hyperbolic area divided by this strict Euclidean reference is

$$\frac{A_M^{\text{eff}}}{A_E^{\text{ref}}} = \frac{2 \left[\sqrt{1 + \eta^2(\alpha)x^2} - 1 \right]}{x^2}. \quad (671)$$

Hence define the finite-microscopic-curvature scaling function

$$\mathcal{F}_x(x, \alpha) = \frac{2\eta^2(\alpha)}{1 + \sqrt{1 + \eta^2(\alpha)x^2}}. \quad (672)$$

The exact scaling relation is then

$$A_M^{\text{eff}} = A_E^{\text{ref}} \mathcal{F}_x(x, \alpha). \quad (673)$$

Likewise, defining

$$N_E^{\text{ref}} = 2n_b \frac{A_E^{\text{ref}}}{A_0(K)}, \quad (674)$$

one obtains

$$N_M^{\text{eff}} = N_E^{\text{ref}} \mathcal{F}_x(x, \alpha). \quad (675)$$

This is the exact finite-curvature version of the desired unified scaling relation. Consider the controlled microscopic continuum limit $\alpha = \frac{a}{2R} \rightarrow 0$ while the macroscopic crossover variable is held fixed. Then $\eta(\alpha) \rightarrow 1$. Consequently,

$$\mathcal{F}_x(x, \alpha) \rightarrow \mathcal{F}_\infty(x), \quad (676)$$

where the one-parameter universal function is

$$\mathcal{F}_\infty(x) = \frac{2}{1 + \sqrt{1 + x^2}}. \quad (677)$$

The effective microscopic count therefore assumes the universal scaling form

$$N_M^{\text{eff}} = N_E^{\text{ref}} \mathcal{F}_\infty \left(\frac{\xi_M^{(E)}}{R} \right) \quad (678)$$

up to controlled finite-microscopic-curvature corrections. This is the desired single-variable crossover law. The first correction away from the continuum scaling limit can be obtained analytically. Using the expansion of the finite-curvature renormalization factor gives

$$\mathcal{F}_x(x, \alpha) = \mathcal{F}_\infty(x) + \frac{\alpha^2}{3\sqrt{1+x^2}} + O(\alpha^4). \quad (679)$$

Restoring physical parameters yields

$$\mathcal{F}_x(x, \alpha) = \mathcal{F}_\infty(x) + \frac{a^2}{12R^2\sqrt{1+x^2}} + O\left(\frac{a^4}{R^4}\right). \quad (680)$$

The one-variable crossover is therefore asymptotically controlled with a leading correction quadratic in the microscopic curvature ratio. The Euclidean reference microscopic count has twist dependence $N_E^{\text{ref}} \propto s^{-2}$. At small angle, $s \sim \frac{|\theta|}{2}$, and hence $N_E^{\text{ref}} \propto |\theta|^{-2}$. In the Euclidean crossover limit, $x \ll 1$, the universal function obeys

$$\mathcal{F}_\infty(x) = 1 - \frac{x^2}{4} + O(x^4). \quad (681)$$

Therefore,

$$N_M^{\text{eff}} \sim N_E^{\text{ref}} \propto |\theta|^{-2}. \quad (682)$$

By contrast, in the curvature-dominated regime, $x \gg 1$, one has

$$\mathcal{F}_\infty(x) \sim \frac{2}{x}. \quad (683)$$

Since

$$x = \frac{a}{2Rs} \propto |\theta|^{-1}, \quad (684)$$

the crossover factor contributes one positive power of the twist angle,

$$\mathcal{F}_\infty(x) \propto |\theta|. \quad (685)$$

Multiplication by the Euclidean reference law therefore gives

$$N_M^{\text{eff}} \propto |\theta|^{-2}|\theta| = |\theta|^{-1}. \quad (686)$$

The exponent change is therefore encoded entirely in the large-variable tail of the universal crossover function. The universal scaling form provides an alternative derivation of the running exponent. At fixed curvature, define

$$\beta_M = -\frac{d \ln N_M^{\text{eff}}}{d \ln s}. \quad (687)$$

Using the universal factorization gives

$$\ln N_M^{\text{eff}} = \ln N_E^{\text{ref}} + \ln \mathcal{F}_\infty(x). \quad (688)$$

Since

$$\frac{d \ln N_E^{\text{ref}}}{d \ln s} = -2, \quad (689)$$

and

$$\frac{d \ln x}{d \ln s} = -1, \quad (690)$$

one obtains

$$\beta_M = 2 + \frac{d \ln \mathcal{F}_\infty}{d \ln x}. \quad (691)$$

Differentiating the explicit function yields

$$\frac{d \ln \mathcal{F}_\infty}{d \ln x} = -1 + \frac{1}{\sqrt{1+x^2}}. \quad (692)$$

Therefore,

$$\beta_M(x) = 1 + \frac{1}{\sqrt{1+x^2}}. \quad (693)$$

Thus, $\lim_{x \rightarrow 0} \beta_M = 2$, while $\lim_{x \rightarrow \infty} \beta_M = 1$. The same one-parameter function therefore controls both the amplitude crossover and the scaling-exponent crossover. The universal amplitude and running exponent are not independent observables. From

$$\mathcal{F}_\infty = \frac{2}{1 + \sqrt{1+x^2}}, \quad (694)$$

one obtains

$$\sqrt{1+x^2} = \frac{2}{\mathcal{F}_\infty} - 1. \quad (695)$$

Therefore,

$$\frac{1}{\sqrt{1+x^2}} = \frac{\mathcal{F}_\infty}{2 - \mathcal{F}_\infty}. \quad (696)$$

Substitution into the running-exponent formula gives

$$\beta_M = \frac{2}{2 - \mathcal{F}_\infty}. \quad (697)$$

Equivalently,

$$\mathcal{F}_\infty = 2 \left(1 - \frac{1}{\beta_M} \right). \quad (698)$$

This is a parameter-free relation between the normalized moiré-area suppression and the running twist exponent. It provides a direct internal consistency test for both analytical and numerical calculations. The universal crossover relation can also be inverted directly. Starting from

$$\frac{2}{\mathcal{F}_\infty} - 1 = \sqrt{1 + x^2}, \quad (699)$$

squaring gives

$$\left(\frac{2}{\mathcal{F}_\infty} - 1 \right)^2 = 1 + x^2. \quad (700)$$

Therefore,

$$x^2 = \frac{4(1 - \mathcal{F}_\infty)}{\mathcal{F}_\infty^2}. \quad (701)$$

Taking the positive square root gives

$$x = \frac{2\sqrt{1 - \mathcal{F}_\infty}}{\mathcal{F}_\infty}. \quad (702)$$

Thus the effective curvature variable can be reconstructed uniquely from the normalized geometric response. The crossover function satisfies a closed autonomous differential equation. From the explicit derivative,

$$\frac{d\mathcal{F}_\infty}{dx} = -\frac{2x}{\sqrt{1 + x^2} [1 + \sqrt{1 + x^2}]^2}, \quad (703)$$

multiplication by the crossover variable gives

$$\frac{d\mathcal{F}_\infty}{d \ln x} = x \frac{d\mathcal{F}_\infty}{dx}. \quad (704)$$

Using the exact inversion relation to eliminate the crossover variable, one finds

$$\frac{d\mathcal{F}_\infty}{d \ln x} = -\frac{2\mathcal{F}_\infty(1 - \mathcal{F}_\infty)}{2 - \mathcal{F}_\infty}. \quad (705)$$

The flow has two fixed points,

$$\mathcal{F}_\infty^* = 1, \quad \mathcal{F}_\infty^* = 0. \quad (706)$$

The first corresponds to the Euclidean regime, while the second corresponds to the curvature-dominated asymptotic regime. Near the Euclidean fixed point, define

$$\epsilon = 1 - \mathcal{F}_\infty, \quad \epsilon \ll 1. \quad (707)$$

Substitution into the exact flow equation gives

$$\frac{d\epsilon}{d \ln x} = 2\epsilon + O(\epsilon^2). \quad (708)$$

Therefore, $\epsilon \propto x^2$. This reproduces the leading small-variable correction

$$1 - \mathcal{F}_\infty \sim \frac{x^2}{4}. \quad (709)$$

The crossover perturbation away from Euclidean geometry therefore has quadratic onset. Near the curvature-dominated fixed point, $\mathcal{F}_\infty \ll 1$. The flow equation reduces to

$$\frac{d\mathcal{F}_\infty}{d \ln x} = -\mathcal{F}_\infty + O(\mathcal{F}_\infty^2). \quad (710)$$

Therefore, $\mathcal{F}_\infty \propto x^{-1}$. This immediately produces the inverse-linear hyperbolic moiré scaling when combined with the inverse-square Euclidean reference count. Thus the two endpoint power laws can be recovered directly from the linearized universal flow. The running exponent varies according to

$$\beta_M(x) = 1 + \frac{1}{\sqrt{1 + x^2}}. \quad (711)$$

Its logarithmic derivative is

$$\frac{d\beta_M}{d \ln x} = -\frac{x^2}{(1 + x^2)^{3/2}}. \quad (712)$$

The magnitude of this crossover rate is

$$\mathcal{S}(x) = \frac{x^2}{(1 + x^2)^{3/2}}. \quad (713)$$

Differentiation gives the extremum condition

$$\frac{d\mathcal{S}}{dx} = 0. \quad (714)$$

The nonzero solution is $x_{\max} = \sqrt{2}$. At this point,

$$\left| \frac{d\beta_M}{d \ln x} \right|_{\max} = \frac{2}{3\sqrt{3}}. \quad (715)$$

The corresponding running exponent is

$$\beta_M(\sqrt{2}) = 1 + \frac{1}{\sqrt{3}}. \quad (716)$$

This provides a parameter-free operational definition of the center of the most rapid crossover region. A second natural crossover convention is obtained by taking the arithmetic midpoint of the two endpoint exponents, $\beta_M = \frac{3}{2}$. Using the exact exponent law gives

$$\frac{1}{\sqrt{1+x^2}} = \frac{1}{2}. \quad (717)$$

Therefore, $x_\beta = \sqrt{3}$. At this point the universal suppression factor is

$$\mathcal{F}_\infty(\sqrt{3}) = \frac{2}{3}. \quad (718)$$

Thus the midpoint of the scaling-exponent flow does not coincide with a fifty-percent suppression of the geometric amplitude. The condition of fifty-percent suppression is $\mathcal{F}_\infty = \frac{1}{2}$. Using the inversion relation gives $x^2 = 8$. Hence, $x_{1/2} = 2\sqrt{2}$. The running exponent at this point is $\beta_M = \frac{4}{3}$. The distinctions among the maximal-slope point, exponent midpoint, and half-amplitude point demonstrate that the crossover is smooth and cannot be assigned a unique angle without a specified operational criterion. The universal function provides a direct numerical test of the geometric theory. For every set of microscopic parameters, define the normalized response

$$Y = \frac{A_M^{\text{eff}}}{A_{\text{ref}}}. \quad (719)$$

The theory predicts $Y = \mathcal{F}(\chi)$. Equivalently, using the effective microscopic count,

$$Y = \frac{N_M^{\text{eff}}}{N_{\text{ref}}}. \quad (720)$$

Thus all datasets obtained at different twist angles, curvature radii, registry thresholds, and microscopic densities must collapse onto the same curve when plotted against the exact variable in Eq. (641). No fitting exponent is required. No crossover amplitude is required. No phenomenological interpolation function is required. The collapse curve is fixed analytically by Eq. (648). The

exact crossover relation can be transformed into a particularly simple falsification test. From Eq. (695), one obtains

$$\left(\frac{2}{Y} - 1 \right)^2 = 1 + \chi^2. \quad (721)$$

Equivalently,

$$\left(\frac{2}{Y} - 1 \right)^2 - 1 = \chi^2. \quad (722)$$

Thus a transformed plot of the left-hand side against the squared crossover variable must lie on the parameter-free straight line $y = x$. Systematic violation of this identity would demonstrate that at least one assumption of the centered constant-curvature registry model has failed. Possible sources of such failure include nonconstant curvature, a position-dependent microscopic threshold, lattice anisotropy, noncentered deformation, or a breakdown of the radial registry approximation. Let the microscopic registry threshold be replaced by an arbitrary fixed positive length, $\ell_* > 0$. The exact crossover variable becomes

$$\chi_\ell = \frac{\sinh(\ell_*/2R)}{|\sin(\theta/2)|}. \quad (723)$$

The effective area is

$$A_M^{\text{eff}} = 2\pi R^2 \left[\sqrt{1 + \chi_\ell^2} - 1 \right]. \quad (724)$$

Define the corresponding reference area by

$$A_{\text{ref},\ell} = \pi R^2 \chi_\ell^2. \quad (725)$$

Then

$$\frac{A_M^{\text{eff}}}{A_{\text{ref},\ell}} = \frac{2}{1 + \sqrt{1 + \chi_\ell^2}}. \quad (726)$$

Therefore changing the microscopic registry convention changes only the map from physical parameters to the dimensionless crossover variable. It does not change the universal function itself. This establishes threshold universality of the centered geometric crossover. The two-dimensional exponent change can be embedded in a broader geometric statement. Consider a hyperbolic space of dimension $D \geq 2$ and constant curvature $K = -\frac{1}{R^2}$. The geodesic-polar metric is

$$ds^2 = dr^2 + R^2 \sinh^2\left(\frac{r}{R}\right) d\Omega_{D-1}^2. \quad (727)$$

Any two radial geodesics subtending an angle lie in a totally geodesic two-dimensional hyperbolic subspace. Therefore the local twist-displacement law remains

$$\sinh\left(\frac{d_\theta(r)}{2R}\right) = \sinh\left(\frac{r}{R}\right) s. \quad (728)$$

Consequently, the local moiré length remains

$$\frac{\xi_M}{R} = \operatorname{arsinh} \chi. \quad (729)$$

The volume element is

$$dV_D = R^{D-1} \sinh^{D-1}\left(\frac{r}{R}\right) dr d\Omega_{D-1}. \quad (730)$$

Let the surface area of the unit sphere in dimension one lower be denoted by

$$\Omega_{D-1} = \frac{2\pi^{D/2}}{\Gamma(D/2)}. \quad (731)$$

The volume of a hyperbolic ball is therefore

$$V_D(r) = \Omega_{D-1} R^D \int_0^{r/R} \sinh^{D-1} t dt. \quad (732)$$

For small radius, $\sinh t \sim t$. Hence,

$$V_D(r) \sim \frac{\Omega_{D-1}}{D} r^D. \quad (733)$$

Since the Euclidean local moiré length scales inversely with the twist, the corresponding Euclidean volume satisfies $V_M^{(E)} \propto |\theta|^{-D}$. At large radius,

$$\sinh^{D-1} t \sim 2^{-(D-1)} e^{(D-1)t}. \quad (734)$$

Therefore,

$$V_D(r) \sim \frac{\Omega_{D-1} R^D}{2^{D-1}(D-1)} e^{(D-1)r/R}. \quad (735)$$

Using the curvature-dominated local moiré length,

$$\xi_M \sim R \ln\left(\frac{C}{|\theta|}\right), \quad (736)$$

one obtains

$$\begin{aligned} V_M^{(H)} &\propto \exp\left[(D-1)\frac{\xi_M}{R}\right] \\ &\propto \exp\left[(D-1)\ln\left(\frac{C}{|\theta|}\right)\right]. \end{aligned} \quad (737)$$

Hence,

$$V_M^{(H)} \propto |\theta|^{-(D-1)}. \quad (738)$$

Negative curvature therefore produces the general exponent transformation $D \rightarrow D-1$. The two-dimensional result $2 \rightarrow 1$ is the first nontrivial member of this more general geometric law. The dimensional generalization can be sharpened beyond the two asymptotic limits. Define

$$y = \frac{\xi_M}{R} = \operatorname{arsinh} \chi. \quad (739)$$

The logarithmic response of the local length to twist is

$$\nu_\xi = \frac{\tanh y}{y}. \quad (740)$$

The radial volume-growth dimension is defined by

$$D_V(y) = \frac{d \ln V_D}{d \ln r}. \quad (741)$$

Using the exact ball volume gives

$$D_V(y) = \frac{y \sinh^{D-1} y}{\int_0^y \sinh^{D-1} t dt}. \quad (742)$$

The running twist exponent of the effective moiré volume is consequently $\beta_D = D_V \nu_\xi$. Substituting the exact expressions yields

$$\beta_D(y) = \frac{\sinh^D y}{\cosh y \int_0^y \sinh^{D-1} t dt}. \quad (743)$$

For small radial scale,

$$\int_0^y \sinh^{D-1} t dt \sim \frac{y^D}{D}, \quad (744)$$

and hence

$$\lim_{y \rightarrow 0} \beta_D(y) = D. \quad (745)$$

For large radial scale,

$$\int_0^y \sinh^{D-1} t dt \sim \frac{e^{(D-1)y}}{2^{D-1}(D-1)}, \quad (746)$$

while

$$\sinh^D y \sim \frac{e^{Dy}}{2^D}, \quad \cosh y \sim \frac{e^y}{2}. \quad (747)$$

Therefore,

$$\lim_{y \rightarrow \infty} \beta_D(y) = D - 1. \quad (748)$$

Thus the dimensional exponent reduction is not merely an asymptotic dimensional-analysis argument; it follows from an exact running-exponent formula. For the physical case considered in the present work, $D = 2$. The volume integral becomes

$$\int_0^y \sinh t \, dt = \cosh y - 1. \quad (749)$$

Equation (743) then gives

$$\beta_2 = \frac{\sinh^2 y}{\cosh y (\cosh y - 1)}. \quad (750)$$

Using

$$\sinh^2 y = (\cosh y - 1)(\cosh y + 1), \quad (751)$$

one obtains

$$\beta_2 = 1 + \frac{1}{\cosh y}. \quad (752)$$

Since

$$\cosh y = \sqrt{1 + \chi^2}, \quad (753)$$

the general dimensional expression reduces exactly to

$$\beta_2 = 1 + \frac{1}{\sqrt{1 + \chi^2}}, \quad (754)$$

recovering the two-dimensional result derived independently above. The universal function derived above governs the effective geometric moire domain. Exact crystallographic supercells introduce an additional arithmetic structure. For an exact commensurate hyperbolic twist, define the coincidence index by

$$q_M(\theta) = [\Gamma : \Gamma \cap g_\theta \Gamma g_\theta^{-1}]. \quad (755)$$

The exact supercell area is

$$A_M^{\text{sc}} = q_M A_0(K). \quad (756)$$

Define the geometric–arithmetic locking factor by

$$\mathcal{L}(\theta) = \frac{q_M(\theta) A_0(K)}{A_M^{\text{eff}}(\theta)}. \quad (757)$$

This definition is exact whenever an exact commensurate supercell exists. It follows immediately that

$$A_M^{\text{sc}} = \mathcal{L}(\theta) A_M^{\text{eff}}. \quad (758)$$

Likewise,

$$N_M^{\text{sc}} = \mathcal{L}(\theta) N_M^{\text{eff}}. \quad (759)$$

Combining this identity with the universal geometric factorization gives

$$N_M^{\text{sc}} = N_{\text{ref}} \mathcal{F}(\chi) \mathcal{L}(\theta). \quad (760)$$

Equation (760) is the hyperbolic counterpart of the geometric–arithmetic decomposition obtained for the Euclidean square lattice. The geometric crossover and the arithmetic coincidence structure are therefore separated multiplicatively. The exact crystallographic scaling exponent is defined by

$$\beta_{\text{sc}} = -\frac{d \ln N_M^{\text{sc}}}{d \ln s}. \quad (761)$$

Define the arithmetic contribution by

$$\delta_{\text{arith}} = -\frac{d \ln \mathcal{L}}{d \ln s}. \quad (762)$$

Taking the logarithmic derivative of Eq. (759) yields the exact decomposition

$$\beta_{\text{sc}} = \beta_M + \delta_{\text{arith}}. \quad (763)$$

This relation precisely identifies what must be proved before the effective geometric exponent can be promoted to an exact crystallographic exponent. If the locking factor approaches a finite nonzero constant,

$$\mathcal{L}(\theta) \longrightarrow \mathcal{L}_*, \quad 0 < \mathcal{L}_* < \infty, \quad (764)$$

then $\delta_{\text{arith}} \longrightarrow 0$. Consequently, $\beta_{\text{sc}} \longrightarrow \beta_M$. In particular, the exact hyperbolic supercell would inherit $\beta_{\text{sc}} \longrightarrow 1$. If, on the other hand, the locking factor contains an additional power-law dependence, $\mathcal{L}(\theta) \propto |\theta|^{-\delta}$, then $\delta_{\text{arith}} = \delta$, and the exact curvature-dominated exponent becomes $\beta_{\text{sc}} = 1 + \delta$. Thus arithmetic commensurability can preserve, modulate, or alter the geometric scaling law depending on the asymptotic behavior of the locking factor. The preceding decomposition suggests a precise

criterion for an exact crystallographic universality class. A sequence of exact commensurate twists exhibits geometric locking if

$$0 < \liminf_{\theta \rightarrow 0} \mathcal{L}(\theta) \leq \limsup_{\theta \rightarrow 0} \mathcal{L}(\theta) < \infty. \quad (765)$$

Under this condition,

$$q_M(\theta) = \Theta(A_M^{\text{eff}}(\theta)). \quad (766)$$

Since the effective area obeys $A_M^{\text{eff}} = \Theta(|\theta|^{-1})$ in the curvature-dominated regime, one obtains

$$q_M(\theta) = \Theta(|\theta|^{-1}). \quad (767)$$

Consequently,

$$N_M^{\text{sc}} = \Theta(|\theta|^{-1}). \quad (768)$$

A stronger universality statement is obtained if the locking factor converges, $\mathcal{L}(\theta) \rightarrow \mathcal{L}_*$. In that case,

$$\frac{N_M^{\text{sc}}}{N_{\text{ref}}} \sim \mathcal{L}_* \mathcal{F}(\chi). \quad (769)$$

The exact crystallographic theory would then inherit not only the endpoint exponent but the full universal crossover function up to a single arithmetic normalization. The principal results of the present subsection can be summarized as follows.

a. Theorem 4. Universal curvature-controlled moiré crossover. For a centered twisted bilayer on a two-dimensional space of constant negative curvature, define the exact dimensionless crossover variable by

$$\chi = \frac{\sinh(a/2R)}{|\sin(\theta/2)|}. \quad (770)$$

Then the effective moiré area admits the exact factorization

$$A_M^{\text{eff}} = \pi R^2 \chi^2 \mathcal{F}(\chi), \quad (771)$$

with the parameter-free universal crossover function

$$\mathcal{F}(\chi) = \frac{2}{1 + \sqrt{1 + \chi^2}}. \quad (772)$$

The same function controls the area-normalized microscopic count,

$$N_M^{\text{eff}} = N_{\text{ref}} \mathcal{F}(\chi). \quad (773)$$

The crossover function satisfies

$$\mathcal{F}(0) = 1, \quad \mathcal{F}(\infty) = 0, \quad \frac{d\mathcal{F}}{d\chi} < 0. \quad (774)$$

Its asymptotic branches are

$$\mathcal{F}(\chi) = 1 - \frac{\chi^2}{4} + O(\chi^4) \quad (775)$$

and

$$\mathcal{F}(\chi) = \frac{2}{\chi} + O(\chi^{-2}). \quad (776)$$

Therefore the effective moiré scaling crosses continuously from $N_M^{\text{eff}} \propto |\theta|^{-2}$ to $N_M^{\text{eff}} \propto |\theta|^{-1}$. In the microscopic continuum limit, the exact crossover variable reduces to $x = \frac{\xi_M^{(E)}}{R}$, and the universal scaling function becomes

$$\mathcal{F}_\infty(x) = \frac{2}{1 + \sqrt{1 + x^2}}. \quad (777)$$

The running area exponent satisfies

$$\beta_M(x) = 1 + \frac{1}{\sqrt{1 + x^2}} = \frac{2}{2 - \mathcal{F}_\infty(x)}. \quad (778)$$

The crossover function obeys the autonomous flow equation

$$\frac{d\mathcal{F}_\infty}{d \ln x} = -\frac{2\mathcal{F}_\infty(1 - \mathcal{F}_\infty)}{2 - \mathcal{F}_\infty}, \quad (779)$$

with endpoint fixed points

$$\mathcal{F}_\infty^* = 1, \quad \mathcal{F}_\infty^* = 0. \quad (780)$$

For hyperbolic space of arbitrary spatial dimension, the corresponding effective moiré-volume exponent undergoes the universal transformation $D \rightarrow D - 1$. Finally, for exact commensurate hyperbolic supercells, the crystallographic running exponent decomposes as

$$\beta_{\text{sc}} = \beta_M + \delta_{\text{arith}}, \quad (781)$$

where the second term measures the asymptotic variation of the geometric-arithmetic locking factor. Thus the universal geometric crossover and the arithmetic structure of exact hyperbolic commensurability are mathematically separable. The central scaling structure established in this subsection may be written as

$$\frac{\xi_M^{(E)}}{R} \rightarrow \mathcal{F} \rightarrow \beta_M : 2 \rightarrow 1. \quad (782)$$

The essential result is stronger than the existence of two asymptotic power laws. The complete crossover is analytically determined. The crossover variable is dimensionless. The scaling function contains no fitted exponent. The endpoint laws follow from the same function. The running exponent is algebraically locked to the normalized amplitude. The crossover possesses an exact autonomous flow. The same geometric mechanism generalizes to arbitrary dimension. This geometric universality does not fix the arithmetic class of an exact Fuchsian sequence. Bounded nonzero locking defines the conditional class $\mathfrak{U}_1(1)$, whereas the explicit centered sequence below lies in $\mathfrak{U}_1(4)$ and has arithmetic correction three. Thus exponent-one locking is not generic. The remaining arithmetic problem is to classify the other commensurator height rays and their attainable exponents.

F. Hyperbolic Commensurability and the Moiré Group

The preceding sections established an exact geometric crossover theory for the local and effective moiré scales. Those results hold for every relative twist and require no global periodicity. Exact crystallographic moiré structure is fundamentally more restrictive. In negative curvature, the analogue of a Euclidean coincidence lattice is controlled not by the intersection of two Abelian translation lattices but by the intersection of two conjugate non-Abelian Fuchsian groups. The purpose of the present subsection is to formulate this exact commensurability problem and solve its group-theoretic part completely. We define the hyperbolic moiré group as the maximal common translation group of the two layers, establish a necessary and sufficient commensurability criterion, determine the exact supercell index, construct the associated common finite cover, derive its area, genus, and microscopic degree-of-freedom count, and identify the correct equivalence classes of exact hyperbolic moiré structures. A second objective is to determine when the small-angle limit required by the scaling theory can exist at the exact crystallographic level. This introduces a qualitative distinction between arithmetic and non-arithmetic hyperbolic crystal groups. In particular, exact small-angle commensurability is not generically available for an arbitrary hyperbolic lattice. The existence of such a sequence is itself a nontrivial group-theoretic property. Let the underlying space be the hyperbolic plane of constant Gaussian curvature

$$K = -\frac{1}{R^2}, \quad R > 0. \quad (783)$$

Its orientation-preserving isometry group is denoted by

$$G = \text{Isom}^+(\mathbb{H}_R^2). \quad (784)$$

After the conventional identification of hyperbolic orientation-preserving isometries with fractional linear

transformations, one has $G \simeq \text{PSL}(2, \mathbb{R})$. The curvature radius rescales the metric but does not modify the abstract isometry-group structure. We take the hyperbolic Bravais translation group to be a discrete, torsion-free, cocompact subgroup, $\Gamma < G$. Cocompactness means that the quotient $\Sigma_0 = \Gamma \backslash \mathbb{H}_R^2$ is compact. Torsion-freeness ensures that the quotient is a smooth closed hyperbolic surface rather than an orbifold. Its genus satisfies $g_0 \geq 2$. The primitive hyperbolic unit-cell area is the covolume of the translation group,

$$A_0 = \text{Area}(\Gamma \backslash \mathbb{H}_R^2). \quad (785)$$

The Gauss–Bonnet theorem gives

$$\int_{\Sigma_0} K dA = 2\pi\chi(\Sigma_0). \quad (786)$$

Since $\chi(\Sigma_0) = 2 - 2g_0$, and the curvature is constant, one obtains

$$-\frac{A_0}{R^2} = 2\pi(2 - 2g_0). \quad (787)$$

Consequently,

$$A_0 = 4\pi R^2(g_0 - 1). \quad (788)$$

Let each primitive quotient contain a finite microscopic basis of cardinality $n_b \in \mathbb{N}$. The corresponding single-layer crystal may be represented as a finite union of group orbits. The first layer is generated by the original translation group, $\Gamma_1 = \Gamma$. Let the relative placement of the second layer be described by an orientation-preserving hyperbolic isometry $g \in G$. The translation group of the second layer is the conjugate group $\Gamma_2 = g\Gamma g^{-1}$. For a centered pure twist, the relative isometry is specialized to $g = g_\theta$, where the transformation fixes a selected point, $g_\theta o = o$, and rotates the tangent space through the relative twist angle. The use of a general element at the beginning of the construction is important. Exact commensurability is fundamentally a property of relative isometries. Pure rotations constitute a physically important one-parameter subfamily that can be imposed after the group-theoretic structure has been established. A hyperbolic translation is a symmetry of the complete bilayer only if it preserves both microscopic layers. The maximal common translation subgroup is therefore

$$\Gamma_M(g) = \Gamma \cap g\Gamma g^{-1}. \quad (789)$$

For a centered twist,

$$\Gamma_M(\theta) = \Gamma \cap g_\theta \Gamma g_\theta^{-1}. \quad (790)$$

We refer to this subgroup as the hyperbolic moiré group.

a. Lemma 1. Maximal common-translation property.

The group in Eq. (789) is the unique maximal subgroup of orientation-preserving hyperbolic translations that preserves both layers. Let an element belong to the intersection, $h \in \Gamma_M(g)$. Then $h \in \Gamma$, so it preserves the first layer. At the same time, $h \in g\Gamma g^{-1}$, so it preserves the second layer. Hence every element of the intersection is a common translation symmetry. Conversely, suppose that a translation preserves both layers. Preservation of the first layer requires $h \in \Gamma$, while preservation of the second requires $h \in g\Gamma g^{-1}$. Therefore, $h \in \Gamma \cap g\Gamma g^{-1}$. Any common translation subgroup is consequently contained in the intersection. Thus the intersection is the unique maximal common translation subgroup. Two subgroups of the same ambient group are said to be commensurable when their intersection has finite index in both. For two groups denoted by $H_1, H_2 < G$, we write $H_1 \sim H_2$ when

$$[H_1 : H_1 \cap H_2] < \infty \quad (791)$$

and

$$[H_2 : H_1 \cap H_2] < \infty. \quad (792)$$

This relation is reflexive, symmetric, and transitive and therefore defines an equivalence relation on lattices in the ambient isometry group. The commensurator of the hyperbolic crystal group is defined by

$$\text{Comm}_G(\Gamma) = \{g \in G : \Gamma \sim g\Gamma g^{-1}\}. \quad (793)$$

Equivalently,

$$\text{Comm}_G(\Gamma) = \left\{ g \in G : \begin{array}{l} [\Gamma : \Gamma \cap g\Gamma g^{-1}] < \infty, \\ [g\Gamma g^{-1} : \Gamma \cap g\Gamma g^{-1}] < \infty \end{array} \right\}. \quad (794)$$

For completeness, we verify that the commensurator is itself a subgroup of the ambient isometry group. The identity satisfies $e\Gamma e^{-1} = \Gamma$, and therefore $e \in \text{Comm}_G(\Gamma)$. Suppose that two elements belong to the commensurator, $g, h \in \text{Comm}_G(\Gamma)$. Then $\Gamma \sim g\Gamma g^{-1}$, and $\Gamma \sim h\Gamma h^{-1}$. Conjugation preserves finite group indices and hence preserves commensurability. Therefore,

$$g\Gamma g^{-1} \sim gh\Gamma h^{-1}g^{-1}. \quad (795)$$

By transitivity, $\Gamma \sim gh\Gamma h^{-1}g^{-1}$. Thus, $gh \in \text{Comm}_G(\Gamma)$. Similarly, symmetry of commensurability gives $g^{-1} \in \text{Comm}_G(\Gamma)$. Therefore, $\text{Comm}_G(\Gamma) < G$. We now state the principal crystallographic result.

b. Theorem 5. Exact hyperbolic moiré commensurability. Consider two identical hyperbolic crystals whose translation groups are related by

$$\Gamma_1 = \Gamma, \quad \Gamma_2 = g\Gamma g^{-1}, \quad (796)$$

with a torsion-free cocompact Fuchsian translation group. An exact finite hyperbolic moiré supercell exists if and only if $g \in \text{Comm}_G(\Gamma)$. For a centered twist, the condition becomes $g_\theta \in \text{Comm}_G(\Gamma)$. Assume $g \in \text{Comm}_G(\Gamma)$. By definition, the common subgroup

$$\Gamma_M = \Gamma \cap g\Gamma g^{-1} \quad (797)$$

has finite index in both layer translation groups. Define $q_1 = [\Gamma : \Gamma_M]$, and $q_2 = [g\Gamma g^{-1} : \Gamma_M]$. Both are finite. Choose coset representatives satisfying

$$\Gamma = \bigcup_{j=1}^{q_1} \Gamma_M \gamma_j. \quad (798)$$

A fundamental domain for the common subgroup can therefore be constructed, up to sets of measure zero, as the union of a finite number of primitive fundamental domains of the first layer. Likewise, since $q_2 < \infty$, the same common quotient contains only finitely many primitive cells of the second layer. Thus the subgroup in Eq. (797) defines a finite periodic cell common to both layers. Hence an exact finite moiré supercell exists. Conversely, suppose that an exact finite common supercell exists. Let its translation group be $H < G$. Because the supercell is periodic for the first layer, $H \subset \Gamma$. Because it is simultaneously periodic for the second layer, $H \subset g\Gamma g^{-1}$. Therefore,

$$H \subset \Gamma \cap g\Gamma g^{-1} = \Gamma_M. \quad (799)$$

The assumed supercell contains only finitely many primitive cells of each layer. Hence $[\Gamma : H] < \infty$, and $[g\Gamma g^{-1} : H] < \infty$. Since the moiré group contains this finite-index subgroup, $H \subset \Gamma_M \subset \Gamma$, the index multiplication formula gives

$$[\Gamma : H] = [\Gamma : \Gamma_M] [\Gamma_M : H]. \quad (800)$$

The left-hand side is finite, so $[\Gamma : \Gamma_M] < \infty$. Similarly,

$$[g\Gamma g^{-1} : \Gamma_M] < \infty. \quad (801)$$

Thus the original crystal group and its conjugate are commensurable. Therefore, $g \in \text{Comm}_G(\Gamma)$. The condition is both necessary and sufficient. Because the two layers are conjugate copies of the same lattice, the two finite-index conditions are not independent. The two microscopic translation groups have equal covolume,

$$\text{covol}(\Gamma) = \text{covol}(g\Gamma g^{-1}) = A_0. \quad (802)$$

Suppose that

$$q = [\Gamma : \Gamma_M] < \infty. \quad (803)$$

The covolume of the common subgroup is then

$$\text{covol}(\Gamma_M) = qA_0. \quad (804)$$

On the other hand,

$$\text{covol}(\Gamma_M) = [g\Gamma g^{-1} : \Gamma_M] A_0. \quad (805)$$

Comparing the two expressions gives

$$[\Gamma : \Gamma_M] = [g\Gamma g^{-1} : \Gamma_M]. \quad (806)$$

Thus, for two identical conjugate layers, $[\Gamma : \Gamma_M] < \infty$ already implies finite index in the second layer. We therefore define the exact hyperbolic coincidence index by

$$q_M(g) = [\Gamma : \Gamma \cap g\Gamma g^{-1}]. \quad (807)$$

For a pure centered twist,

$$q_M(\theta) = [\Gamma : \Gamma \cap g_\theta \Gamma g_\theta^{-1}]. \quad (808)$$

The common quotient is

$$\Sigma_M = \Gamma_M \backslash \mathbb{H}_R^2. \quad (809)$$

Its area is $A_M^{\text{sc}} = q_M A_0$. If the primitive cell contains the same number of microscopic basis sites in each layer, then one layer contributes $N_{M,1}^{\text{sc}} = q_M n_b$. The two layer labels remain independent even when positions geometrically coincide. Consequently, $N_M^{\text{sc}} = 2q_M n_b$. Thus the finite-index function is simultaneously the crystallographic supercell enlargement factor, the covering degree, and the multiplicative increase of microscopic basis dimension per layer. The exact hyperbolic supercell has a natural topological interpretation. The primitive quotient is $\Sigma_0 = \Gamma \backslash \mathbb{H}_R^2$. The common quotient is

$$\Sigma_M = \Gamma_M \backslash \mathbb{H}_R^2. \quad (810)$$

Because $\Gamma_M \subset \Gamma$, there is a natural covering map $p_1 : \Sigma_M \rightarrow \Sigma_0$. Its covering degree is $\deg p_1 = q_M$. A second map is defined by applying the inverse relative isometry before projecting to the primitive quotient, $p_2 : \Sigma_M \rightarrow \Sigma_0$. At the level of equivalence classes it is given by

$$p_2([x]_{\Gamma_M}) = [g^{-1}x]_{\Gamma}. \quad (811)$$

To verify that this map is well defined, suppose

$$x' = hx, \quad h \in \Gamma_M. \quad (812)$$

Then $g^{-1}hg \in \Gamma$, because $h \in g\Gamma g^{-1}$. Therefore,

$$g^{-1}x' = (g^{-1}hg)g^{-1}x \quad (813)$$

represents the same point of the primitive quotient. Hence the second map is well defined. Its degree is likewise $\deg p_2 = q_M$. The exact moiré supercell therefore defines a common finite cover, $\Sigma_0 \leftarrow \Sigma_M \rightarrow \Sigma_0$, with equal covering degree on both sides. This correspondence provides the natural non-Abelian hyperbolic analogue of a Euclidean common superlattice. Euler characteristic is multiplicative under finite-sheeted covers. Therefore,

$$\chi(\Sigma_M) = q_M \chi(\Sigma_0). \quad (814)$$

For closed orientable surfaces,

$$\chi(\Sigma_M) = 2 - 2g_M, \quad (815)$$

and

$$\chi(\Sigma_0) = 2 - 2g_0. \quad (816)$$

Thus,

$$2 - 2g_M = q_M (2 - 2g_0). \quad (817)$$

Solving for the supercell genus gives

$$g_M = 1 + q_M (g_0 - 1). \quad (818)$$

The exact moiré enlargement therefore has three equivalent descriptions,

$$q_M = \frac{A_M^{\text{sc}}}{A_0} = \frac{g_M - 1}{g_0 - 1} = \frac{N_M^{\text{sc}}}{2n_b}. \quad (819)$$

Thus exact hyperbolic moiré enlargement is simultaneously a crystallographic, geometric, topological, and microscopic counting quantity. The torsion-free assumption is convenient but not fundamental. If the microscopic translation or space group contains elliptic elements, the quotient is a hyperbolic orbifold rather than a smooth surface. The appropriate topological invariant is then the orbifold Euler characteristic, χ_{orb} . For a finite-index subgroup, multiplicativity becomes

$$\chi_{\text{orb}}(\Sigma_M) = q_M \chi_{\text{orb}}(\Sigma_0). \quad (820)$$

The covolume relation remains $A_M^{\text{sc}} = q_M A_0$. Thus all index and area statements survive unchanged; only the

genus-based interpretation must be replaced by orbifold topology. The normalizer of the crystal group is

$$N_G(\Gamma) = \{g \in G : g\Gamma g^{-1} = \Gamma\}. \quad (821)$$

Every normalizer element belongs to the commensurator,

$$N_G(\Gamma) \subset \text{Comm}_G(\Gamma). \quad (822)$$

For a normalizer element, $\Gamma_M(g) = \Gamma$, and hence $q_M(g) = 1$. Conversely, suppose $q_M(g) = 1$. Then

$$\Gamma \cap g\Gamma g^{-1} = \Gamma. \quad (823)$$

Therefore, $\Gamma \subset g\Gamma g^{-1}$. The two groups have equal covolume. A proper finite-index inclusion would strictly change covolume. Thus the inclusion must be equality, $g\Gamma g^{-1} = \Gamma$. Hence, $g \in N_G(\Gamma)$. We have therefore proved

$$q_M(g) = 1 \iff g \in N_G(\Gamma). \quad (824)$$

A genuinely enlarged exact moiré cell consequently requires

$$g \in \text{Comm}_G(\Gamma) \setminus N_G(\Gamma). \quad (825)$$

The exact coincidence index is invariant under exchange of the two layers. For the inverse relative motion,

$$q_M(g^{-1}) = [\Gamma : \Gamma \cap g^{-1}\Gamma g]. \quad (826)$$

Conjugating the intersection by the original relative isometry gives

$$g(\Gamma \cap g^{-1}\Gamma g)g^{-1} = g\Gamma g^{-1} \cap \Gamma. \quad (827)$$

Therefore,

$$q_M(g^{-1}) = [g\Gamma g^{-1} : g\Gamma g^{-1} \cap \Gamma]. \quad (828)$$

The two layer indices are equal by Eq. (806). Hence, $q_M(g^{-1}) = q_M(g)$. Thus exchanging the two layers leaves the exact moiré cell size unchanged. Let two microscopic lattice translations be chosen, $\gamma_L, \gamma_R \in \Gamma$. Consider the modified relative isometry $g' = \gamma_L g \gamma_R$. Since conjugation by an element of the crystal group preserves the group, $\gamma_R \Gamma \gamma_R^{-1} = \Gamma$. Therefore,

$$g' \Gamma g'^{-1} = \gamma_L g \Gamma g^{-1} \gamma_L^{-1}. \quad (829)$$

The common subgroup becomes

$$\begin{aligned} \Gamma \cap g' \Gamma g'^{-1} &= \Gamma \cap \gamma_L g \Gamma g^{-1} \gamma_L^{-1} \\ &= \gamma_L [\Gamma \cap g \Gamma g^{-1}] \gamma_L^{-1}. \end{aligned} \quad (830)$$

Conjugation preserves subgroup index. Hence,

$$q_M(\gamma_L g \gamma_R) = q_M(g). \quad (831)$$

Thus the coincidence index is naturally a function on the double-coset space $\Gamma \setminus \text{Comm}_G(\Gamma) / \Gamma$. Exact moiré structures differing only by microscopic translations of either layer therefore belong to the same crystallographic commensurability class. The coincidence index also possesses a useful composition inequality. Let $g, h \in \text{Comm}_G(\Gamma)$. Define $A = \Gamma$, $B = g\Gamma g^{-1}$, and $C = gh\Gamma h^{-1}g^{-1}$. Consider the triple intersection $D = A \cap B \cap C$. The subgroup-index multiplication formula gives

$$[A : D] = [A : A \cap B] [A \cap B : D]. \quad (832)$$

Because

$$D = (A \cap B) \cap (B \cap C), \quad (833)$$

one has the general subgroup-index inequality

$$[A \cap B : D] \leq [B : B \cap C]. \quad (834)$$

The first factor equals

$$[A : A \cap B] = q_M(g), \quad (835)$$

while conjugation by the first relative isometry gives

$$[B : B \cap C] = q_M(h). \quad (836)$$

Consequently,

$$[A : D] \leq q_M(g)q_M(h). \quad (837)$$

Since

$$D \subset A \cap C, \quad (838)$$

one obtains

$$[A : A \cap C] \leq [A : D]. \quad (839)$$

Therefore,

$$q_M(gh) \leq q_M(g)q_M(h). \quad (840)$$

The logarithm of the coincidence index consequently satisfies

$$\ln q_M(gh) \leq \ln q_M(g) + \ln q_M(h). \quad (841)$$

The exact supercell complexity is therefore submultiplicative under composition of commensurate relative isometries. We now specialize the exact commensurability theory to genuine centered twists. For a chosen point in the hyperbolic plane, define its stabilizer in the orientation-preserving isometry group by $K_o = \text{Stab}_G(o)$. The stabilizer is the compact one-parameter rotation group, $K_o \simeq \text{SO}(2)$. The centered rotation of relative angle is an element satisfying $g_\theta \in K_o$. Define the fixed-center twist commensurator by

$$\mathcal{C}_o(\Gamma) = \text{Comm}_G(\Gamma) \cap K_o. \quad (842)$$

The exact centered-twist condition is therefore $g_\theta \in \mathcal{C}_o(\Gamma)$. Thus the question of exact twist commensurability at a fixed center reduces to the intersection of the full commensurator with a compact rotation subgroup. The fixed-center twist commensurator is a subgroup of the circle group. This immediately produces a sharp dichotomy.

c. Theorem 6. Finite-or-dense centered-twist dichotomy. For a fixed twist center, the exact centered-twist subgroup satisfies precisely one of the following alternatives: $\mathcal{C}_o(\Gamma)$ is finite, or $\mathcal{C}_o(\Gamma)$ is dense in K_o . Take the topological closure, $\overline{\mathcal{C}_o(\Gamma)} \subset K_o$. The closure of a subgroup is again a subgroup. Hence, $\overline{\mathcal{C}_o(\Gamma)}$ is a closed subgroup of the circle group. Every proper closed subgroup of the circle group is finite cyclic. Therefore, if the closure is proper, one has $|\overline{\mathcal{C}_o(\Gamma)}| < \infty$. This implies that the original subgroup is finite. If the subgroup is infinite, its closure cannot be a proper closed subgroup. Consequently, $\overline{\mathcal{C}_o(\Gamma)} = K_o$. Hence the subgroup is dense. The previous theorem gives a necessary and sufficient criterion for the existence of a sequence of exact nonzero twists approaching alignment. Such a sequence has the form

$$\theta_j \neq 0, \quad \theta_j \rightarrow 0, \quad (843)$$

with $g_{\theta_j} \in \text{Comm}_G(\Gamma)$. If the fixed-center twist commensurator is finite, the identity is isolated inside this finite subgroup, and no such nontrivial sequence exists. If the subgroup is infinite, it is dense by Theorem 6, so every neighborhood of the identity contains a nontrivial exact centered twist. Therefore,

$$\begin{aligned} \exists \theta_j \neq 0, \quad \theta_j \rightarrow 0, \quad g_{\theta_j} \in \text{Comm}_G(\Gamma) \\ \iff |\mathcal{C}_o(\Gamma)| = \infty. \end{aligned} \quad (844)$$

This criterion is stronger than the statement that the full commensurator is large. The relevant object for a

fixed-center twist is specifically its intersection with the rotation stabilizer. The size of the full commensurator is controlled by arithmeticity. For a finite-covolume Fuchsian lattice, the commensurator theorem gives the fundamental dichotomy

$$\Gamma \text{ arithmetic} \iff \overline{\text{Comm}_G(\Gamma)} = G. \quad (845)$$

Thus an arithmetic hyperbolic crystal has a dense commensurator. For a non-arithmetic lattice, the commensurator is discrete,

$$\Gamma \text{ non-arithmetic} \implies \text{Comm}_G(\Gamma) \text{ is discrete.} \quad (846)$$

Because the original lattice is contained in its commensurator, $\Gamma \subset \text{Comm}_G(\Gamma)$, and both are discrete finite-covolume groups in the non-arithmetic case, the original group has finite index in its commensurator,

$$[\text{Comm}_G(\Gamma) : \Gamma] < \infty. \quad (847)$$

The arithmeticity distinction has an immediate physical consequence.

d. Theorem 7. Non-arithmetic small-angle commensurability obstruction. Let the hyperbolic crystal translation group be a non-arithmetic cocompact Fuchsian lattice. Then, for every fixed twist center, $|\mathcal{C}_o(\Gamma)| < \infty$. Consequently, there exists no nontrivial exact centered commensurate sequence satisfying $\theta_j \rightarrow 0$. For a non-arithmetic lattice, the full commensurator is discrete. The rotation stabilizer is compact, $K_o \simeq \text{SO}(2)$. Suppose that the intersection were infinite. Then an infinite subset of the compact group would possess an accumulation point. This would contradict discreteness of the full commensurator. Therefore, $\text{Comm}_G(\Gamma) \cap K_o$ must be finite. The finite-or-dense theorem then excludes a nontrivial sequence approaching the identity. If the fixed-center exact-twist set contains nontrivial rotations, its finiteness implies the existence of a strictly positive angular gap. Define

$$\theta_{\min}^{\text{exact}}(o) = \min \{|\theta| : g_\theta \in \mathcal{C}_o(\Gamma), \quad g_\theta \neq e\}. \quad (848)$$

Whenever the set is nonempty, $\theta_{\min}^{\text{exact}}(o) > 0$. Thus a non-arithmetic hyperbolic crystal can possess exact nontrivial commensurate twists, but those twists cannot accumulate continuously at the aligned configuration. This is qualitatively different from the Euclidean square lattice, where commensurate rotations form sequences approaching zero angle. For an arithmetic lattice, the full commensurator is dense, $\overline{\text{Comm}_G(\Gamma)} = G$. Consequently, for every neighborhood of the identity, $U \subset G$, $e \in U$, there exists a nontrivial commensurating relative isometry satisfying $g \in U \cap \text{Comm}_G(\Gamma)$. Thus arithmeticity removes the discreteness obstruction to arbitrarily small

exact relative isometries. One must nevertheless distinguish this statement from fixed-center rotational commensurability. Density of the full commensurator does not logically imply

$$|\text{Comm}_G(\Gamma) \cap K_o| = \infty \quad (849)$$

for an arbitrarily prescribed center. The fixed-center question remains controlled by the subgroup $\mathcal{C}_o(\Gamma)$. If this subgroup is infinite, Theorem 6 immediately upgrades it to a dense set of exact centered twist angles. It is therefore useful to introduce a center-dependent property. We call a twist center commensurator-active when $|\mathcal{C}_o(\Gamma)| = \infty$. For such a center, $\mathcal{C}_o(\Gamma) = K_o$, and there exist arbitrarily small nonzero exact centered twists,

$$\exists \theta_j \neq 0, \quad \theta_j \rightarrow 0, \quad g_{\theta_j} \in \text{Comm}_G(\Gamma). \quad (850)$$

For a twist-inactive center, $|\mathcal{C}_o(\Gamma)| < \infty$, no such sequence exists. This center dependence must be kept distinct from arithmeticity of the full crystal group. Although a fixed-center sequence requires the additional condition above, density of an arithmetic commensurator does guarantee arbitrarily small elliptic commensurating transformations when their centers are allowed to vary. Choose a sequence of nontrivial rotations in the ambient isometry group satisfying $k_j \rightarrow e$. Each nontrivial sufficiently small rotation lies inside the open elliptic region of the isometry group. For every such rotation, choose an open neighborhood satisfying $V_j \subset G$ such that every element of the neighborhood is elliptic and $V_j \rightarrow e$. Density of the arithmetic commensurator guarantees

$$V_j \cap \text{Comm}_G(\Gamma) \neq \emptyset. \quad (851)$$

Choose

$$g_j \in V_j \cap \text{Comm}_G(\Gamma). \quad (852)$$

Then $g_j \rightarrow e$, and every element is elliptic. Every nontrivial elliptic orientation-preserving hyperbolic isometry has a unique fixed point in the hyperbolic plane. Hence there exist centers $o_j \in \mathbb{H}_R^2$ and angles satisfying $\theta_j \rightarrow 0$ such that g_j is a rotation through the corresponding angle about the corresponding center. Thus arithmetic lattices possess arbitrarily small exact rotational commensurations when the rotation center is not required to remain fixed. The existence of a small exact commensurate sequence does not by itself determine the asymptotic exponent of the supercell index. It does, however, force the exact supercell size to diverge.

e. Theorem 8. Divergence of the coincidence index near the identity. Let a sequence of distinct exact commensurating isometries satisfy $g_j \in \text{Comm}_G(\Gamma)$, $g_j \neq e$, and $g_j \rightarrow e$. Then $q_M(g_j) \rightarrow \infty$. Assume the contrary. Then there exists a finite positive integer satisfying $Q < \infty$ and an infinite subsequence for which $q_M(g_j) \leq Q$. Define

$$H_j = \Gamma \cap g_j \Gamma g_j^{-1}. \quad (853)$$

Then $[\Gamma : H_j] \leq Q$. The lattice group is finitely generated. A finitely generated group has only finitely many subgroups of index bounded by a fixed integer. Therefore, after passing to a further subsequence, there exists a single finite-index subgroup satisfying $H_j = H$ for all terms of that subsequence. Choose a finite generating set, $H = \langle h_1, \dots, h_m \rangle$. Since $H \subset g_j \Gamma g_j^{-1}$, one has

$$g_j^{-1} h_\mu g_j \in \Gamma \quad (854)$$

for every generator. Because $g_j \rightarrow e$, continuity of conjugation gives

$$g_j^{-1} h_\mu g_j \rightarrow h_\mu. \quad (855)$$

The original group is discrete. Hence, for each generator, sufficiently large sequence index implies

$$g_j^{-1} h_\mu g_j = h_\mu. \quad (856)$$

Because there are only finitely many generators, a single sufficiently large index works for all of them. Therefore, $g_j h = h g_j$ for every element of the finite-index subgroup. Thus $g_j \in C_G(H)$, where the centralizer is defined by

$$C_G(H) = \{g \in G : gh = hg \text{ for every } h \in H\}. \quad (857)$$

A finite-index subgroup of a cocompact Fuchsian surface group is non-elementary. It contains two hyperbolic elements with distinct invariant axes. Any orientation-preserving hyperbolic isometry commuting with both must preserve the endpoints of both axes. This fixes at least three distinct points on the ideal boundary. An orientation-preserving fractional linear transformation fixing three distinct boundary points is the identity. Therefore, $C_G(H) = \{e\}$. Hence $g_j = e$ for all sufficiently large sequence index, contradicting the assumption that the sequence consists of distinct nontrivial elements. The bounded-index assumption is therefore impossible. Consequently, $q_M(g_j) \rightarrow \infty$. For an exact centered sequence, $g_{\theta_j} \rightarrow e$ is equivalent to $\theta_j \rightarrow 0$. The previous theorem therefore gives

$$\theta_j \rightarrow 0 \implies q_M(\theta_j) \rightarrow \infty \quad (858)$$

for every nontrivial exact commensurate sequence that exists. The exact atom number consequently satisfies

$$N_M^{\text{sc}}(\theta_j) = 2n_b q_M(\theta_j) \rightarrow \infty. \quad (859)$$

Thus exact crystallography agrees qualitatively with the geometric expectation that alignment is accompanied by

a diverging moiré scale, but no universal rate of divergence follows from group theory alone. The preceding results reveal that the asymptotic question must be separated into two logically distinct problems. The first is an existence problem: Does there exist an exact commensurate sequence approaching $\theta = 0$? The second is a growth problem: If such a sequence exists, how rapidly does $q_M(\theta)$ diverge? For a non-arithmetic lattice at fixed twist center, the first question has a negative answer. For a commensurator-active center, the first question has a positive answer and the divergence theorem guarantees $q_M(\theta_j) \rightarrow \infty$. The exponent, however, remains an arithmetic problem. This distinction is essential for interpreting the geometric inverse-linear law derived previously. For every exact centered commensurate twist, define the locking factor by

$$\mathcal{L}(\theta) = \frac{A_M^{\text{sc}}(\theta)}{A_M^{\text{eff}}(\theta)}. \quad (860)$$

Using the exact supercell area gives

$$\mathcal{L}(\theta) = \frac{q_M(\theta)A_0}{A_M^{\text{eff}}(\theta)}. \quad (861)$$

The exact crystallographic count is therefore

$$N_M^{\text{sc}}(\theta) = \mathcal{L}(\theta)N_M^{\text{eff}}(\theta). \quad (862)$$

The arithmetic contribution to the running exponent is

$$\delta_{\text{arith}} = -\frac{d \ln \mathcal{L}}{d \ln |\sin(\theta/2)|}. \quad (863)$$

The exact crystallographic exponent then decomposes as

$$\beta_{\text{sc}} = \beta_M + \delta_{\text{arith}}. \quad (864)$$

The geometric theory has already determined $\beta_M \rightarrow 1$ in the curvature-dominated regime. Consequently, the exact crystallographic exponent is

$$\beta_{\text{sc}} = 1 + \delta_{\text{arith}} \quad (865)$$

whenever the small-angle exact sequence exists. Suppose there exist finite positive constants satisfying $0 < c_- \leq c_+ < \infty$ such that along an exact commensurate sequence, $c_- \leq \mathcal{L}(\theta_j) \leq c_+$. Then

$$q_M(\theta_j) = \Theta \left(\frac{A_M^{\text{eff}}(\theta_j)}{A_0} \right). \quad (866)$$

Since the geometric theory gives

$$A_M^{\text{eff}}(\theta_j) = \Theta(|\theta_j|^{-1}), \quad (867)$$

one obtains

$$q_M(\theta_j) = \Theta(|\theta_j|^{-1}). \quad (868)$$

Thus,

$$N_M^{\text{sc}}(\theta_j) = \Theta(|\theta_j|^{-1}). \quad (869)$$

This is the precise condition under which the effective geometric exponent is inherited by exact hyperbolic crystallography. An even stronger form of universality occurs if

$$\mathcal{L}(\theta_j) \rightarrow \mathcal{L}_*, \quad 0 < \mathcal{L}_* < \infty. \quad (870)$$

Then

$$q_M(\theta_j) \sim \frac{\mathcal{L}_*}{A_0} A_M^{\text{eff}}(\theta_j). \quad (871)$$

Substitution of the hyperbolic small-angle area law gives

$$q_M(\theta_j) \sim \frac{4\pi\mathcal{L}_*R^2 \sinh(a/2R)}{A_0} |\theta_j|^{-1}. \quad (872)$$

The exact bilayer count then satisfies

$$N_M^{\text{sc}}(\theta_j) \sim \frac{8\pi n_b \mathcal{L}_* R^2 \sinh(a/2R)}{A_0} |\theta_j|^{-1}. \quad (873)$$

Within the conditional class $\mathfrak{U}_1(1)$, a convergent locking factor determines both the exact exponent and its crystallographic prefactor. The explicit centered class $\mathfrak{U}_1(4)$ shows that this hypothesis is not automatic. Suppose instead that

$$\mathcal{L}(\theta_j) \sim C_{\text{arith}} |\theta_j|^{-\delta} \quad (874)$$

for some finite positive arithmetic amplitude and real exponent. Then $\delta_{\text{arith}} \rightarrow \delta$. Consequently,

$$q_M(\theta_j) \sim |\theta_j|^{-(1+\delta)}, \quad (875)$$

and

$$N_M^{\text{sc}}(\theta_j) \sim |\theta_j|^{-(1+\delta)}. \quad (876)$$

The arithmetic contribution therefore acts as an anomalous correction to the purely geometric scaling exponent. This is the exact hyperbolic analogue of the arithmetic multiplier found for Euclidean coincidence lattices,

but here the relevant arithmetic object is the finite-index structure of a non-Abelian Fuchsian commensurator. Many microscopic hyperbolic lattice models are constructed from regular tessellations. The orientation-preserving symmetry group of such a tessellation is commensurable with an orientation-preserving hyperbolic triangle group. A standard presentation is

$$\Delta^+(2, p, q) = \langle a, b, c \mid a^2 = b^p = c^q = abc = e \rangle. \quad (877)$$

The hyperbolic condition is

$$\frac{1}{2} + \frac{1}{p} + \frac{1}{q} < 1. \quad (878)$$

A torsion-free Bravais translation group may be chosen as a finite-index subgroup,

$$\Gamma < \Delta^+(2, p, q). \quad (879)$$

Arithmeticity is invariant under passage to finite-index subgroups. Therefore,

$$\Gamma \text{ is arithmetic} \iff \Delta^+(2, p, q) \text{ is arithmetic.} \quad (880)$$

Thus the choice of regular hyperbolic tessellation is not neutral with respect to exact twist physics. If the parent commensurability class is non-arithmetic, exact small-angle centered commensurate sequences are obstructed. If the parent class is arithmetic, the full commensurator is dense and small exact relative commensurations become possible, subject to the additional fixed-center condition derived above. This provides a concrete criterion for selecting hyperbolic lattices in the subsequent numerical study. The theory now naturally separates hyperbolic bilayers into three classes. The first class consists of generic incommensurate twists, $g_\theta \notin \text{Comm}_G(\Gamma)$. For these twists, N_M^{sc} is undefined because no finite exact supercell exists, while the local and effective quantities remain well defined. The second class consists of isolated exact commensurate twists, $g_\theta \in \text{Comm}_G(\Gamma)$ without an exact sequence accumulating at zero twist. This is necessarily the situation at fixed center for a non-arithmetic crystal. The third class consists of asymptotically commensurate twist families for which $\theta_j \rightarrow 0$ and $g_{\theta_j} \in \text{Comm}_G(\Gamma)$. Only the third class admits a meaningful comparison between the exact small-angle crystallographic exponent and the geometric crossover derived in the preceding sections. The exact commensurability condition is highly restrictive. It requires the relative isometry to belong to the commensurator, $g \in \text{Comm}_G(\Gamma)$. By contrast, an arbitrary relative isometry belongs only to the full continuous group, $g \in G$. Thus exact crystallographic moiré periodicity is a special subset of the full relative-configuration space. This fact does not weaken the geometric theory. The local mismatch field, $d_\theta(r)$,

the local moiré length, ξ_M , and the effective moiré area, A_M^{eff} , remain well defined for incommensurate configurations. Exact commensurability adds an additional discrete arithmetic layer rather than replacing the local geometric description. The principal group-theoretic results can now be summarized.

f. Theorem 9. Hyperbolic moiré-group structure. Let the microscopic hyperbolic Bravais translation group be a torsion-free cocompact Fuchsian lattice, $\Gamma < \text{PSL}(2, \mathbb{R})$, and let the second layer be related to the first by an orientation-preserving hyperbolic isometry. The maximal common translation group is

$$\Gamma_M(g) = \Gamma \cap g\Gamma g^{-1}. \quad (881)$$

A finite exact hyperbolic moiré supercell exists if and only if $g \in \text{Comm}_G(\Gamma)$. Whenever this condition holds, the two layer indices coincide and define

$$q_M(g) = [\Gamma : \Gamma_M(g)]. \quad (882)$$

The exact supercell area, genus, and microscopic count satisfy $A_M^{\text{sc}} = q_M A_0$, $g_M = 1 + q_M(g_0 - 1)$, and $N_M^{\text{sc}} = 2n_b q_M$. The coincidence index obeys

$$q_M(g^{-1}) = q_M(g), \quad q_M(\gamma_L g \gamma_R) = q_M(g), \quad (883)$$

and

$$q_M(gh) \leq q_M(g)q_M(h). \quad (884)$$

Moreover,

$$q_M(g) = 1 \iff g \in N_G(\Gamma). \quad (885)$$

For a fixed twist center, exact centered twists are classified by

$$\mathcal{C}_o(\Gamma) = \text{Comm}_G(\Gamma) \cap \text{Stab}_G(o). \quad (886)$$

This subgroup is either finite or dense in the rotation group. For a non-arithmetic lattice it is necessarily finite, and therefore no nontrivial exact centered commensurate sequence can approach zero twist. Whenever an exact nontrivial sequence approaches the identity, its coincidence index necessarily diverges,

$$g_j \rightarrow e \implies q_M(g_j) \rightarrow \infty. \quad (887)$$

Finally, the relation between geometric and exact crystallographic scaling is controlled by the locking factor,

$$\mathcal{L}(\theta) = \frac{q_M(\theta)A_0}{A_M^{\text{eff}}(\theta)}. \quad (888)$$

The exact running exponent decomposes into

$$\beta_{\text{sc}} = \beta_M + \delta_{\text{arith}}. \quad (889)$$

Thus negative-curvature geometry determines the effective moiré scaling, whereas the Fuchsian commensurator determines whether that scaling can be realized as an exact crystallographic law. The exact hyperbolic commensurability problem is therefore governed by the sequence

$$\Gamma \longrightarrow \text{Comm}_G(\Gamma) \longrightarrow g \longrightarrow \Gamma_M(g) \longrightarrow q_M(g) \longrightarrow \Sigma_M \longrightarrow A_M^{\text{sc}} \longrightarrow N_M^{\text{sc}}. \quad q_M = [\Gamma : \Gamma \cap g\theta\Gamma g\theta^{-1}]. \quad (890)$$

For centered twists an additional restriction appears,

$$\text{Comm}_G(\Gamma) \longrightarrow \mathcal{C}_o(\Gamma) = \text{Comm}_G(\Gamma) \cap K_o. \quad (891)$$

The complete geometry–arithmetic hierarchy of the theory is consequently

$$\begin{aligned} \text{negative curvature} &\longrightarrow d_\theta(r) \longrightarrow \xi_M \longrightarrow A_M^{\text{eff}} \\ &\longrightarrow \mathcal{L}(\theta) \longrightarrow q_M(\theta) \longrightarrow A_M^{\text{sc}} \longrightarrow N_M^{\text{sc}}. \end{aligned} \quad (892)$$

The first part of this hierarchy is universal and geometric. The second is arithmetic and sequence dependent. Along a declared small-angle sequence in the conditional class $\mathfrak{U}_1(1)$, finite nonzero locking gives

$$q_M(\theta) = \Theta(|\theta|^{-1}) \quad (893)$$

and hence

$$N_M^{\text{sc}}(\theta) = \Theta(|\theta|^{-1}). \quad (894)$$

The analysis above also reveals a more fundamental prerequisite: such an asymptotic exact sequence must exist in the first place. For non-arithmetic hyperbolic crystals at a fixed center, it does not. For arithmetic or otherwise commensurator-active constructions, the existence and scaling of such sequences become the central arithmetic problem. This arithmeticity-controlled distinction is therefore not a technical detail of the construction. It determines whether the curvature-induced effective moiré crossover can be promoted to an exact small-angle crystallographic universality law.

G. Exact Supercells versus Effective Moiré Domains

The preceding sections established two mathematically distinct structures. The first is the effective moiré domain generated by a local registry criterion. It exists for every nonzero relative twist and is determined entirely by hyperbolic geometry. The second is the exact

crystallographic supercell generated by the finite-index intersection of two conjugate Fuchsian crystal groups. It exists only at exact commensurate relative isometries and is therefore controlled by arithmetic and group-theoretic data. The purpose of the present subsection is to determine precisely how these two structures can be compared. This comparison is indispensable. The geometric theory has already proved the curvature-dominated scaling law $N_M^{\text{eff}} \propto |\theta|^{-1}$. However, the exact crystallographic count is $N_M^{\text{sc}} = 2n_b q_M$, where the coincidence index is

No identity between these two quantities follows from their definitions. A rigorous theory must therefore establish one of three logically distinct possibilities. The exact and effective scales may become asymptotically proportional; they may remain comparable only up to finite multiplicative bounds; or the arithmetic coincidence index may develop an additional asymptotic scaling that changes the exact crystallographic exponent. The present subsection develops a framework capable of distinguishing all three possibilities. To make the distinction explicit, we introduce the notation $N_M^{\text{geo}} \equiv N_M^{\text{eff}}$ for the area-normalized geometric count, and $N_M^{\text{exact}} \equiv N_M^{\text{sc}}$ for the exact crystallographic count. The geometric quantity is

$$N_M^{\text{geo}} = 2n_b \frac{A_M^{\text{eff}}}{A_0}. \quad (896)$$

The exact quantity is $N_M^{\text{exact}} = 2n_b q_M$. The corresponding exact supercell area is $A_M^{\text{sc}} = q_M A_0$. Consequently, the ratio of the exact and geometric microscopic counts is identical to the ratio of the corresponding areas,

$$\frac{N_M^{\text{exact}}}{N_M^{\text{geo}}} = \frac{A_M^{\text{sc}}}{A_M^{\text{eff}}}. \quad (897)$$

This ratio is the natural quantity controlling geometric–arithmetic locking. Define the geometric–arithmetic locking factor by

$$\mathcal{L}_M(\theta) = \frac{N_M^{\text{exact}}(\theta)}{N_M^{\text{geo}}(\theta)}. \quad (898)$$

Using Eq. (897), one equivalently has

$$\mathcal{L}_M(\theta) = \frac{A_M^{\text{sc}}(\theta)}{A_M^{\text{eff}}(\theta)} = \frac{q_M(\theta) A_0}{A_M^{\text{eff}}(\theta)}. \quad (899)$$

The exact microscopic count therefore factorizes as $N_M^{\text{exact}} = \mathcal{L}_M N_M^{\text{geo}}$. The locking factor contains every difference between the effective geometric theory and the exact crystallographic theory. If $\mathcal{L}_M = 1$, the two areas coincide. If $0 < \mathcal{L}_M < 1$, the exact crystallographic area is

smaller than the effective registry domain. If $\mathcal{L}_M > 1$, the exact crystallographic cell is larger. None of these three possibilities is excluded by geometry alone. Directly comparing an exact fundamental polygon with a geodesic disk can be misleading because the exact cell need not be radially symmetric. We therefore introduce a scalar radius that contains only the exact supercell area and makes no assumption concerning its shape. Define the area-equivalent exact-supercell radius by $A_{\mathbb{H}}(r_M^{\text{sc}}) = A_M^{\text{sc}}$. The hyperbolic disk-area relation is

$$A_{\mathbb{H}}(r) = 2\pi R^2 \left[\cosh\left(\frac{r}{R}\right) - 1 \right]. \quad (900)$$

Using the identity

$$\cosh x - 1 = 2 \sinh^2\left(\frac{x}{2}\right), \quad (901)$$

this becomes

$$A_{\mathbb{H}}(r) = 4\pi R^2 \sinh^2\left(\frac{r}{2R}\right). \quad (902)$$

Therefore the area-equivalent exact-supercell radius satisfies

$$q_M A_0 = 4\pi R^2 \sinh^2\left(\frac{r_M^{\text{sc}}}{2R}\right). \quad (903)$$

Solving gives

$$r_M^{\text{sc}} = 2R \operatorname{arsinh} \left[\sqrt{\frac{q_M A_0}{4\pi R^2}} \right]. \quad (904)$$

Equivalently,

$$r_M^{\text{sc}} = R \operatorname{arcosh} \left[1 + \frac{q_M A_0}{2\pi R^2} \right]. \quad (905)$$

The radius in Eq. (904) does not assert that the exact supercell is a disk. It is simply the unique radius of a hyperbolic disk having the same area as the exact crystallographic quotient. The effective geometric domain is already exactly a geodesic disk. Its radius is the local moiré length, $r_M^{\text{geo}} = \xi_M$. The exact formula is

$$\xi_M = R \operatorname{arsinh} \left[\frac{\sinh(a/2R)}{|\sin(\theta/2)|} \right]. \quad (906)$$

The effective area is therefore

$$A_M^{\text{eff}} = 4\pi R^2 \sinh^2\left(\frac{\xi_M}{2R}\right). \quad (907)$$

The comparison between exact and effective areas can consequently be reduced to a comparison of two scalar radial variables. Define the signed radial locking defect by

$$\Delta_M = r_M^{\text{sc}} - \xi_M. \quad (908)$$

A positive radial defect means that the area-equivalent exact supercell is larger than the effective registry disk. A negative radial defect means that it is smaller. The exact and effective radial scales are related by

$$r_M^{\text{sc}} = \xi_M + \Delta_M. \quad (909)$$

This variable provides a geometrically transparent representation of the arithmetic correction. Using Eqs. (902), (907), and (909), the locking factor becomes

$$\mathcal{L}_M = \left[\frac{\sinh[(\xi_M + \Delta_M)/(2R)]}{\sinh(\xi_M/2R)} \right]^2. \quad (910)$$

Introduce the dimensionless variables $y = \frac{\xi_M}{2R}$, and $\delta = \frac{\Delta_M}{2R}$. Then

$$\mathcal{L}_M = \left[\frac{\sinh(y + \delta)}{\sinh y} \right]^2. \quad (911)$$

Using the addition law for the hyperbolic sine gives

$$\sinh(y + \delta) = \sinh y \cosh \delta + \cosh y \sinh \delta. \quad (912)$$

Therefore,

$$\mathcal{L}_M = [\cosh \delta + \coth y \sinh \delta]^2. \quad (913)$$

An alternative exact representation is obtained from

$$\sinh\left(\frac{r}{2R}\right) = \frac{1}{2} e^{r/(2R)} \left[1 - e^{-r/R} \right]. \quad (914)$$

Substitution yields

$$\mathcal{L}_M = e^{\Delta_M/R} \left[\frac{1 - e^{-(\xi_M + \Delta_M)/R}}{1 - e^{-\xi_M/R}} \right]^2. \quad (915)$$

Equation (915) gives an exact connection between multiplicative area mismatch and additive radial mismatch. Along a small-angle sequence in the curvature-dominated regime, $\xi_M \rightarrow \infty$. Suppose first that the radial defect remains bounded, $|\Delta_M| \leq C$ for a finite constant independent of the sequence index. Then $e^{-\xi_M/R} \rightarrow 0$, and $e^{-(\xi_M + \Delta_M)/R} \rightarrow 0$. Consequently,

$$\mathcal{L}_M = e^{\Delta_M/R} \left[1 + O\left(e^{-\xi_M/R}\right) \right]. \quad (916)$$

Thus an additive shift of order one in hyperbolic radius becomes a finite multiplicative renormalization of the supercell area. This property is a direct consequence of exponential hyperbolic area growth. If the radial defect converges, $\Delta_M \rightarrow \Delta_*$, where $-\infty < \Delta_* < \infty$, then Eq. (916) gives

$$\mathcal{L}_M \rightarrow \exp\left(\frac{\Delta_*}{R}\right). \quad (917)$$

Therefore,

$$N_M^{\text{exact}} \sim \exp\left(\frac{\Delta_*}{R}\right) N_M^{\text{geo}}. \quad (918)$$

The desired relation $N_M^{\text{exact}} \sim C N_M^{\text{geo}}$ is therefore equivalent to convergence of the area-equivalent radial defect to a finite constant, with

$$C = \exp\left(\frac{\Delta_*}{R}\right). \quad (919)$$

This gives a geometric interpretation of the asymptotic proportionality constant. The preceding relation may also be inverted without approximation. From the definition of the locking factor,

$$A_{\mathbb{H}}(r_M^{\text{sc}}) = \mathcal{L}_M A_{\mathbb{H}}(\xi_M). \quad (920)$$

Using the hyperbolic disk-area formula gives

$$\cosh\left(\frac{r_M^{\text{sc}}}{R}\right) - 1 = \mathcal{L}_M \left[\cosh\left(\frac{\xi_M}{R}\right) - 1 \right]. \quad (921)$$

Thus,

$$\frac{r_M^{\text{sc}}}{R} = \text{arcosh} \left\{ 1 + \mathcal{L}_M \left[\cosh\left(\frac{\xi_M}{R}\right) - 1 \right] \right\}. \quad (922)$$

The exact radial defect is therefore

$$\frac{\Delta_M}{R} = \text{arcosh} \left\{ 1 + \mathcal{L}_M \left[\cosh\left(\frac{\xi_M}{R}\right) - 1 \right] \right\} - \frac{\xi_M}{R}. \quad (923)$$

This equation permits the radial arithmetic mismatch to be reconstructed directly from exact crystallographic data. The preceding expressions yield an asymptotic equivalence. Consider an exact commensurate sequence

satisfying $\theta_j \rightarrow 0$ and $q_M(\theta_j) \rightarrow \infty$. Then the following two statements are equivalent. First, there exist finite positive constants satisfying $0 < c_- \leq c_+ < \infty$ such that

$$c_- \leq \mathcal{L}_M(\theta_j) \leq c_+ \quad (924)$$

for all sufficiently large sequence index. Second, there exists a finite constant satisfying $C_\Delta < \infty$ such that

$$|\Delta_M(\theta_j)| \leq C_\Delta \quad (925)$$

for all sufficiently large sequence index. Suppose first that the radial defect is bounded. Equation (915) gives

$$\mathcal{L}_M = e^{\Delta_M/R} \left[\frac{1 - e^{-r_M^{\text{sc}}/R}}{1 - e^{-\xi_M/R}} \right]^2. \quad (926)$$

Both radial scales diverge along the sequence. Therefore the factor in square brackets converges to unity. The boundedness of the radial defect then implies that the exponential factor is bounded above and below by finite positive constants. Thus the locking factor satisfies the strong multiplicative bounds. Conversely, suppose that the strong multiplicative bounds hold. Because $A_M^{\text{eff}} \rightarrow \infty$, the positive lower bound implies $A_M^{\text{sc}} \rightarrow \infty$. Hence $r_M^{\text{sc}} \rightarrow \infty$. Using Eq. (915), one therefore has

$$\ln \mathcal{L}_M = \frac{\Delta_M}{R} + o(1). \quad (927)$$

The assumed upper and lower bounds imply that the logarithm of the locking factor remains bounded. Therefore the radial defect remains bounded. This proves the equivalence. The previous proposition immediately implies the desired comparison between the exact and geometric counts. If the radial defect is bounded, there exist finite positive constants satisfying $0 < c_- \leq c_+ < \infty$ such that

$$c_- N_M^{\text{geo}} \leq N_M^{\text{exact}} \leq c_+ N_M^{\text{geo}}. \quad (928)$$

Equivalently,

$$c_- \frac{A_M^{\text{eff}}}{A_0} \leq q_M \leq c_+ \frac{A_M^{\text{eff}}}{A_0}. \quad (929)$$

Since $A_M^{\text{eff}} = \Theta(|\theta|^{-1})$ in the curvature-dominated regime, the bounds imply $q_M = \Theta(|\theta|^{-1})$. Consequently,

$$N_M^{\text{exact}} = \Theta(|\theta|^{-1}). \quad (930)$$

This is the strong geometric–arithmetic locking result sought in the preceding sections. The bounded-ratio condition is stronger than what is required to preserve the

exponent. To formulate the sharp condition, consider an exact commensurate small-angle sequence, $\theta_j \rightarrow 0$. Define

$$s_j = \left| \sin\left(\frac{\theta_j}{2}\right) \right|. \quad (931)$$

The effective geometric count satisfies

$$N_M^{\text{geo}}(\theta_j) = \Theta(s_j^{-1}). \quad (932)$$

Thus,

$$\lim_{j \rightarrow \infty} \frac{\ln N_M^{\text{geo}}(\theta_j)}{\ln s_j^{-1}} = 1. \quad (933)$$

The exact crystallographic exponent should be defined on the sequence itself rather than by differentiating a function that need not be continuous in the twist angle. Define the lower and upper exact exponents by

$$\beta_{\text{exact}} = \liminf_{j \rightarrow \infty} \frac{\ln N_M^{\text{exact}}(\theta_j)}{\ln s_j^{-1}}, \quad (934)$$

and

$$\bar{\beta}_{\text{exact}} = \limsup_{j \rightarrow \infty} \frac{\ln N_M^{\text{exact}}(\theta_j)}{\ln s_j^{-1}}. \quad (935)$$

Similarly, define the lower arithmetic exponent by

$$\delta_{\text{arith}} = \liminf_{j \rightarrow \infty} \frac{\ln \mathcal{L}_M(\theta_j)}{\ln s_j^{-1}}, \quad (936)$$

and the upper arithmetic exponent by

$$\bar{\delta}_{\text{arith}} = \limsup_{j \rightarrow \infty} \frac{\ln \mathcal{L}_M(\theta_j)}{\ln s_j^{-1}}. \quad (937)$$

Taking logarithms of $N_M^{\text{exact}} = N_M^{\text{geo}} \mathcal{L}_M$ gives

$$\ln N_M^{\text{exact}} = \ln N_M^{\text{geo}} + \ln \mathcal{L}_M. \quad (938)$$

Since the geometric contribution has a true limit equal to unity, one obtains the exact relations

$$\beta_{\text{exact}} = 1 + \delta_{\text{arith}}, \quad (939)$$

and

$$\bar{\beta}_{\text{exact}} = 1 + \bar{\delta}_{\text{arith}}. \quad (940)$$

If the two arithmetic limits coincide,

$$\delta_{\text{arith}} = \bar{\delta}_{\text{arith}} = \delta_{\text{arith}}, \quad (941)$$

then the exact exponent exists and satisfies

$$\beta_{\text{exact}} = 1 + \delta_{\text{arith}}. \quad (942)$$

The exact crystallographic exponent equals the geometric value whenever

$$\frac{\ln \mathcal{L}_M(\theta_j)}{\ln s_j^{-1}} \rightarrow 0. \quad (943)$$

Equivalently, $\mathcal{L}_M(\theta_j) = s_j^{-o(1)}$. Under this weaker condition, $\beta_{\text{exact}} = 1$. Thus bounded multiplicative locking is sufficient but not necessary. For example, a locking factor that diverges only logarithmically, $\mathcal{L}_M \sim \ln(s^{-1})$ does not alter the exact power-law exponent. Likewise, any iterated logarithmic or other sub-power correction leaves the exponent unchanged. The sharp exponent criterion is therefore sub-power arithmetic growth, rather than convergence of the locking factor to a constant. The area-equivalent exact-supercell radius provides an even more geometric form of the same result. For large radius, the hyperbolic disk area obeys

$$A_{\mathbb{H}}(r) = \pi R^2 e^{r/R} \left[1 - e^{-r/R} \right]^2. \quad (944)$$

The exact microscopic count may therefore be written as

$$N_M^{\text{exact}} = \frac{2n_b \pi R^2}{A_0} e^{r_M^{\text{sc}}/R} \left[1 - e^{-r_M^{\text{sc}}/R} \right]^2. \quad (945)$$

Taking logarithms gives

$$\begin{aligned} \ln N_M^{\text{exact}} &= \frac{r_M^{\text{sc}}}{R} + \ln \left(\frac{2n_b \pi R^2}{A_0} \right) \\ &+ 2 \ln \left[1 - e^{-r_M^{\text{sc}}/R} \right]. \end{aligned} \quad (946)$$

Along a nontrivial exact sequence approaching alignment, $r_M^{\text{sc}} \rightarrow \infty$. Hence,

$$\ln N_M^{\text{exact}} = \frac{r_M^{\text{sc}}}{R} + O(1). \quad (947)$$

For the geometric moiré length,

$$\frac{\xi_M}{R} = \text{arsinh} \left(\frac{b}{s} \right), \quad (948)$$

where

$$b = \sinh\left(\frac{a}{2R}\right). \quad (949)$$

At small twist,

$$\frac{\xi_M}{R} = \ln s^{-1} + \ln(2b) + o(1). \quad (950)$$

Therefore,

$$\ln s^{-1} = \frac{\xi_M}{R} + O(1). \quad (951)$$

Combining Eqs. (947) and (951) gives the exact asymptotic identification

$$\beta_{\text{exact}} = \liminf_{j \rightarrow \infty} \frac{r_M^{\text{sc}}(\theta_j)}{\xi_M(\theta_j)}, \quad (952)$$

and

$$\bar{\beta}_{\text{exact}} = \limsup_{j \rightarrow \infty} \frac{r_M^{\text{sc}}(\theta_j)}{\xi_M(\theta_j)}. \quad (953)$$

If the radial ratio converges, then

$$\beta_{\text{exact}} = \lim_{j \rightarrow \infty} \frac{r_M^{\text{sc}}(\theta_j)}{\xi_M(\theta_j)}. \quad (954)$$

This provides a direct geometric interpretation of the exact crystallographic exponent. Using $r_M^{\text{sc}} = \xi_M + \Delta_M$, the radial exponent formula becomes

$$\frac{r_M^{\text{sc}}}{\xi_M} = 1 + \frac{\Delta_M}{\xi_M}. \quad (955)$$

Consequently,

$$\delta_{\text{arith}} = \liminf_{j \rightarrow \infty} \frac{\Delta_M(\theta_j)}{\xi_M(\theta_j)}, \quad (956)$$

and

$$\bar{\delta}_{\text{arith}} = \limsup_{j \rightarrow \infty} \frac{\Delta_M(\theta_j)}{\xi_M(\theta_j)}. \quad (957)$$

If the corresponding limit exists,

$$\delta_{\text{arith}} = \lim_{j \rightarrow \infty} \frac{\Delta_M(\theta_j)}{\xi_M(\theta_j)}. \quad (958)$$

Thus the arithmetic anomalous exponent has a precise radial interpretation. A bounded radial offset gives

$\frac{\Delta_M}{\xi_M} \rightarrow 0$, and therefore preserves the geometric exponent. More generally, even an unbounded radial defect preserves the exponent if $\Delta_M = o(\xi_M)$. This is the sharp radial criterion. Under this condition, $\beta_{\text{exact}} = 1$. Suppose the exact crystallographic exponent differs from the geometric value by a finite amount, $\beta_{\text{exact}} = 1 + \delta$. Then the radial formulation requires $\Delta_M \sim \delta \xi_M$. Since $\xi_M \sim R \ln |\theta|^{-1}$, one obtains

$$\Delta_M \sim \delta R \ln |\theta|^{-1}. \quad (959)$$

Substitution into the exponential relation between area and radius gives $\mathcal{L}_M \sim e^{\Delta_M/R}$. Hence, $\mathcal{L}_M \sim |\theta|^{-\delta}$. Thus a genuine arithmetic shift of the crystallographic power-law exponent is equivalent to a radial mismatch that grows linearly with the geometric moiré length. This is substantially stronger than an order-one mismatch. The preceding results define a hierarchy of increasingly strong forms of geometric–arithmetic locking. The weakest condition that preserves the exponent is $\Delta_M = o(\xi_M)$. Equivalently,

$$\ln \mathcal{L}_M = o(\ln |\theta|^{-1}). \quad (960)$$

This implies

$$N_M^{\text{exact}} = |\theta|^{-1+o(1)}. \quad (961)$$

A stronger condition is bounded radial locking, $\Delta_M = O(1)$. This is equivalent to two-sided multiplicative comparison, $N_M^{\text{exact}} = \Theta(N_M^{\text{geo}})$. An even stronger condition is convergent radial locking, $\Delta_M \rightarrow \Delta_*$. This implies an asymptotic proportionality law,

$$N_M^{\text{exact}} \sim e^{\Delta_*/R} N_M^{\text{geo}}. \quad (962)$$

The strongest possible case is asymptotically perfect locking, $\Delta_M \rightarrow 0$, for which

$$\frac{N_M^{\text{exact}}}{N_M^{\text{geo}}} \rightarrow 1. \quad (963)$$

These conditions should not be conflated. In particular, exact exponent inheritance does not require perfect or even bounded multiplicative locking. The area-equivalent radius captures the exact area but not the shape of the crystallographic supercell. To obtain a stronger geometric comparison, consider the Dirichlet fundamental domain of the exact moiré group centered at a chosen lift of the twist center,

$$\mathcal{D}_M(o) = \{x \in \mathbb{H}_R^2 : d(x, o) \leq d(x, \gamma o) \text{ for every } \gamma \in \Gamma_M\}. \quad (964)$$

Because the exact moiré group is cocompact whenever the coincidence index is finite, the Dirichlet domain is compact. Its area is exactly

$$\text{Area}[\mathcal{D}_M(o)] = q_M A_0. \quad (965)$$

Define the Dirichlet inradius by

$$r_{\text{in}} = \sup \{r : B_{\mathbb{H}}(o, r) \subset \mathcal{D}_M(o)\}. \quad (966)$$

For a torsion-free action and a generic base point, the first Dirichlet boundary is generated by the nearest nontrivial orbit point. Therefore,

$$r_{\text{in}} = \frac{1}{2} \min_{\gamma \in \Gamma_M \setminus \{e\}} d(o, \gamma o). \quad (967)$$

The inradius is the injectivity radius of the compact quotient at the chosen base point. Define the circumradius by

$$r_{\text{out}} = \inf \{r : \mathcal{D}_M(o) \subset B_{\mathbb{H}}(o, r)\}. \quad (968)$$

Compactness implies

$$r_{\text{out}} = \max_{x \in \mathcal{D}_M(o)} d(o, x). \quad (969)$$

The circumradius measures the covering radius of the chosen base point in the exact moiré quotient. By construction,

$$B_{\mathbb{H}}(o, r_{\text{in}}) \subset \mathcal{D}_M(o) \subset B_{\mathbb{H}}(o, r_{\text{out}}). \quad (970)$$

Taking areas gives the exact inequalities

$$A_{\mathbb{H}}(r_{\text{in}}) \leq q_M A_0 \leq A_{\mathbb{H}}(r_{\text{out}}). \quad (971)$$

These bounds connect the shape of the exact Dirichlet supercell directly to its index. We now formulate a sufficient geometric criterion for the exact supercell to track the effective registry domain. Suppose there exist finite nonnegative constants satisfying $C_- < \infty$, $C_+ < \infty$ such that, along an exact small-angle sequence,

$$r_{\text{in}} \geq \xi_M - C_-, \quad r_{\text{out}} \leq \xi_M + C_+. \quad (972)$$

We refer to this condition as registry-adapted shape locking. For sufficiently small twist, $\xi_M > C_-$, and monotonicity of the hyperbolic disk area gives

$$A_{\mathbb{H}}(\xi_M - C_-) \leq q_M A_0 \leq A_{\mathbb{H}}(\xi_M + C_+). \quad (973)$$

Dividing by the effective moiré area gives

$$\frac{A_{\mathbb{H}}(\xi_M - C_-)}{A_{\mathbb{H}}(\xi_M)} \leq \mathcal{L}_M \leq \frac{A_{\mathbb{H}}(\xi_M + C_+)}{A_{\mathbb{H}}(\xi_M)}. \quad (974)$$

Using the exact exponential representation of hyperbolic area gives

$$\begin{aligned} \mathcal{L}_M &\geq e^{-C_-/R} \left[\frac{1 - e^{-(\xi_M - C_-)/R}}{1 - e^{-\xi_M/R}} \right]^2, \\ \mathcal{L}_M &\leq e^{C_+/R} \left[\frac{1 - e^{-(\xi_M + C_+)/R}}{1 - e^{-\xi_M/R}} \right]^2. \end{aligned} \quad (975)$$

Taking the small-angle limit gives

$$e^{-C_-/R} \leq \liminf \mathcal{L}_M \leq \limsup \mathcal{L}_M \leq e^{C_+/R}. \quad (976)$$

Registry-adapted shape locking therefore implies strong multiplicative area locking. The preceding argument may be summarized as a formal result.

a. Theorem 10. Registry-adapted supercell locking.

Consider an exact commensurate small-angle sequence for which the Dirichlet fundamental domains satisfy $r_{\text{in}} \geq \xi_M - C_-$ and $r_{\text{out}} \leq \xi_M + C_+$, with finite constants independent of the sequence index. Then the exact coincidence index obeys

$$c_- \frac{A_M^{\text{eff}}}{A_0} \leq q_M \leq c_+ \frac{A_M^{\text{eff}}}{A_0} \quad (977)$$

for finite positive constants. The limiting constants can be chosen arbitrarily close to $e^{-C_-/R}$ and $e^{C_+/R}$ for sufficiently small twist. Consequently, $q_M = \Theta(|\theta|^{-1})$, and $N_M^{\text{exact}} = \Theta(|\theta|^{-1})$. Thus bounded radial distortion of the exact Dirichlet cell relative to the effective moiré disk is sufficient to transfer the geometric exponent to the exact crystallographic supercell. The converse of Theorem 10 does not follow from area information alone. The condition

$$c_- A_M^{\text{eff}} \leq A_M^{\text{sc}} \leq c_+ A_M^{\text{eff}} \quad (978)$$

controls the total covolume of the exact subgroup. It does not independently control the geometry of the fundamental polygon. In particular, an area constraint alone does not determine r_{in} , r_{out} , or the difference $r_{\text{out}} - r_{\text{in}}$. Therefore the theory distinguishes area locking from the stronger condition shape locking. Area locking is sufficient for the exact scaling exponent. Shape locking is additionally relevant when local registry structure, wavefunction support, real-space hopping networks, or finite-cover spectral convergence depend on the geometry of the exact fundamental domain rather than only on its area. A useful scalar diagnostic of exact-cell shape is the radial eccentricity

$$\mathcal{E}_M = r_{\text{out}} - r_{\text{in}}. \quad (979)$$

Another useful quantity is the radial midpoint,

$$\bar{r}_M = \frac{r_{\text{out}} + r_{\text{in}}}{2}. \quad (980)$$

The shape-locking condition can be rewritten as simultaneous control of $\mathcal{E}_M = O(1)$ and $\bar{r}_M - \xi_M = O(1)$. Together, these conditions imply

$$r_{\text{in}} = \xi_M + O(1), \quad (981)$$

and

$$r_{\text{out}} = \xi_M + O(1). \quad (982)$$

Hence the exact Dirichlet cell is radially trapped within an order-one hyperbolic shell around the effective moiré domain. The bounded-shape condition can itself be weakened if only the exact scaling exponent is sought. Suppose

$$r_{\text{in}} = \xi_M + o(\xi_M), \quad (983)$$

and

$$r_{\text{out}} = \xi_M + o(\xi_M). \quad (984)$$

Then

$$r_M^{\text{sc}} = \xi_M + o(\xi_M), \quad (985)$$

because the area-equivalent radius lies between the inradius and circumradius,

$$r_{\text{in}} \leq r_M^{\text{sc}} \leq r_{\text{out}}. \quad (986)$$

Consequently, $\beta_{\text{exact}} = 1$. Thus even sublinear radial distortions preserve the crystallographic exponent. The exact coincidence index exhibits an important singular behavior at perfect alignment. At zero relative twist, $g_0 = e$. Therefore, $\Gamma_M(0) = \Gamma$, and $q_M(0) = 1$. Hence, $N_M^{\text{exact}}(0) = 2n_b$. By contrast, the local geometric moiré length diverges in the punctured small-angle limit,

$$\lim_{\theta \rightarrow 0, \theta \neq 0} \xi_M(\theta) = \infty. \quad (987)$$

The effective geometric count likewise diverges,

$$\lim_{\theta \rightarrow 0, \theta \neq 0} N_M^{\text{geo}}(\theta) = \infty. \quad (988)$$

If a nontrivial exact commensurate sequence approaching the identity exists, the divergence theorem established previously gives $q_M(\theta_j) \rightarrow \infty$. Thus one simultaneously has

$$q_M(0) = 1, \quad q_M(\theta_j) \rightarrow \infty \quad (989)$$

along every nontrivial exact sequence approaching zero twist. The exact coincidence index is therefore highly noncontinuous at the aligned configuration. The small-angle crystallographic scaling law must consequently be understood as a punctured asymptotic statement along nonzero exact commensurate twists. For a non-arithmetic cocompact Fuchsian crystal at a fixed twist center, the centered exact commensurator is finite. Therefore there is no nontrivial exact sequence satisfying $\theta_j \rightarrow 0$. In this case the comparison developed above cannot be promoted into a small-angle exact crystallographic exponent because the required sequence does not exist. The geometric law $N_M^{\text{geo}} \propto |\theta|^{-1}$ remains valid. However, the exact crystallographic law $N_M^{\text{exact}} \propto |\theta|^{-1}$ has no asymptotic meaning at fixed center. This distinction is not a failure of the geometric theory. It is a genuine arithmetic obstruction to exact periodic realization. For a twist center whose exact centered commensurator is infinite, there exist nonzero exact twists satisfying $\theta_j \rightarrow 0$. Along such a sequence, $q_M(\theta_j) \rightarrow \infty$. The theory developed in this subsection then provides a complete hierarchy of possibilities. If $\Delta_M = O(1)$, one has strong multiplicative locking. If $\Delta_M = o(\xi_M)$, but the radial defect is unbounded, the exact and effective counts need not be asymptotically proportional, yet their scaling exponents remain equal. If $\Delta_M \sim \delta \xi_M$, the arithmetic structure generates an anomalous exponent. Thus the exact asymptotic problem reduces to determining the growth class of the radial locking defect. Arithmetic fluctuations may prevent a unique exact exponent from existing. The radial formulation handles this situation naturally. Define the lower radial locking ratio by

$$\lambda_- = \liminf_{j \rightarrow \infty} \frac{r_M^{\text{sc}}(\theta_j)}{\xi_M(\theta_j)}, \quad (990)$$

and the upper radial locking ratio by

$$\lambda_+ = \limsup_{j \rightarrow \infty} \frac{r_M^{\text{sc}}(\theta_j)}{\xi_M(\theta_j)}. \quad (991)$$

Then $\beta_{\text{exact}} = \lambda_-$, and $\bar{\beta}_{\text{exact}} = \lambda_+$. The width

$$\Delta\beta_{\text{arith}} = \lambda_+ - \lambda_- \quad (992)$$

quantifies asymptotic arithmetic intermittency. If $\Delta\beta_{\text{arith}} = 0$, a unique exact exponent exists. If $\Delta\beta_{\text{arith}} > 0$, different subsequences of exact commensurate twists

realize different asymptotic radial growth rates. This possibility must be tested rather than excluded a priori. Along every nontrivial exact commensurate sequence approaching alignment, $q_M \rightarrow \infty$. Therefore, $r_M^{\text{sc}} \rightarrow \infty$. Since $\xi_M \rightarrow \infty$, the lower radial locking ratio satisfies $\lambda_- \geq 0$. Consequently, $\delta_{\text{arith}} \geq -1$. There is no corresponding universal finite upper bound from group theory alone. Thus exact commensurability can in principle produce an arbitrarily strong enhancement of supercell growth unless additional arithmetic information is available. The exact exponential identity implies

$$\ln \mathcal{L}_M = \frac{\Delta_M}{R} + 2 \ln \left[\frac{1 - e^{-r_M^{\text{sc}}/R}}{1 - e^{-\xi_M/R}} \right]. \quad (993)$$

Along an exact small-angle sequence for which both radial scales diverge,

$$\ln \mathcal{L}_M = \frac{\Delta_M}{R} + o(1). \quad (994)$$

Since

$$\frac{\xi_M}{R} = \ln |\theta|^{-1} + O(1), \quad (995)$$

one obtains

$$\frac{\ln \mathcal{L}_M}{\ln |\theta|^{-1}} = \frac{\Delta_M}{\xi_M} + o(1). \quad (996)$$

This equation makes the equivalence between arithmetic scaling and radial mismatch explicit. The preceding analysis yields the following theorem.

b. Theorem 11. Exact exponent-locking criterion. Consider a nontrivial exact commensurate sequence satisfying $\theta_j \rightarrow 0$. Assume the exact crystallographic exponent exists. Then the following statements are equivalent: $\beta_{\text{exact}} = 1$, and

$$\begin{aligned} \frac{\ln \mathcal{L}_M}{\ln |\theta|^{-1}} &\rightarrow 0, \quad \mathcal{L}_M = |\theta|^{-o(1)}, \\ \frac{\Delta_M}{\xi_M} &\rightarrow 0, \quad \frac{r_M^{\text{sc}}}{\xi_M} \rightarrow 1. \end{aligned} \quad (997)$$

The exact count factorization gives

$$\frac{\ln N_M^{\text{exact}}}{\ln |\theta|^{-1}} = \frac{\ln N_M^{\text{geo}}}{\ln |\theta|^{-1}} + \frac{\ln \mathcal{L}_M}{\ln |\theta|^{-1}}. \quad (998)$$

The first term converges to unity. Therefore the exact exponent equals unity if and only if the second term vanishes. This proves equivalence of the first two statements. The second and third statements are equivalent by the definition of sub-power growth. Equation (996) proves equivalence with the fourth statement. Finally,

$$\frac{r_M^{\text{sc}}}{\xi_M} = 1 + \frac{\Delta_M}{\xi_M}, \quad (999)$$

which proves equivalence with the fifth statement. A direct corollary is obtained by imposing the stronger condition $\Delta_M = O(1)$. Then

$$N_M^{\text{exact}} = \Theta(N_M^{\text{geo}}), \quad (1000)$$

and therefore

$$N_M^{\text{exact}} = \Theta(|\theta|^{-1}). \quad (1001)$$

If the radial defect converges to a finite constant, the stronger proportionality follows,

$$N_M^{\text{exact}} \sim C N_M^{\text{geo}}, \quad (1002)$$

with

$$C = \exp\left(\frac{\Delta_*}{R}\right). \quad (1003)$$

The theory suggests a direct numerical program for exact commensurate hyperbolic lattices. For every exact commensurate twist, first compute the coincidence index, $q_M(\theta)$. Next compute the effective geometric area, $A_M^{\text{eff}}(\theta)$. The locking factor is then

$$\mathcal{L}_M(\theta) = \frac{q_M A_0}{A_M^{\text{eff}}}. \quad (1004)$$

The area-equivalent exact radius follows from

$$r_M^{\text{sc}} = 2R \operatorname{arsinh} \left[\sqrt{\frac{q_M A_0}{4\pi R^2}} \right]. \quad (1005)$$

The radial defect is

$$\Delta_M = r_M^{\text{sc}} - \xi_M. \quad (1006)$$

The strongest numerical test of exponent locking is therefore not merely a log-log fit of the exact index. One should directly examine $\frac{r_M^{\text{sc}}}{\xi_M}$ and test whether it approaches unity. A complementary test is

$$\frac{\ln \mathcal{L}_M}{\ln |\theta|^{-1}} \quad (1007)$$

and test whether it approaches zero. Strong multiplicative locking requires the more restrictive diagnostic $\Delta_M = O(1)$. Shape locking additionally requires direct computation of r_{in} , r_{out} , and verification that both

track the local moiré length with bounded or sublinear radial error. A numerical claim of exact crystallographic scaling should therefore be supported in stages. The weakest evidence is a direct regression of q_M against the twist angle. A stronger test is collapse of the normalized ratio

$$\frac{q_M A_0}{A_M^{\text{eff}}}. \quad (1008)$$

A still stronger test is convergence of the radial ratio $\frac{r_M^{\text{sc}}}{\xi_M}$. The strongest real-space test additionally verifies the Dirichlet-domain bounds,

$$r_{\text{in}} = \xi_M + o(\xi_M), \quad (1009)$$

and

$$r_{\text{out}} = \xi_M + o(\xi_M). \quad (1010)$$

Only the latter establishes that the exact crystallographic cell is geometrically adapted to the local moiré domain rather than merely having the same asymptotic area. The framework identifies several mathematically distinct failure modes. The first is failure of exact commensurability, $g_\theta \notin \text{Comm}_G(\Gamma)$. Then no finite exact supercell exists. The second is failure of an exact small-angle sequence. This occurs at a fixed center for non-arithmetic lattices. The third is arithmetic super-growth,

$$\frac{\Delta_M}{\xi_M} \longrightarrow \delta > 0. \quad (1011)$$

Then

$$\beta_{\text{exact}} = 1 + \delta > 1. \quad (1012)$$

The fourth is arithmetic under-growth,

$$-1 < \frac{\Delta_M}{\xi_M} \longrightarrow \delta < 0. \quad (1013)$$

Then

$$0 < \beta_{\text{exact}} = 1 + \delta < 1. \quad (1014)$$

A limiting value

$$\frac{\Delta_M}{\xi_M} \longrightarrow -1 \quad (1015)$$

corresponds to exact supercell growth that is slower than every positive power of the inverse twist. Finally, persistent oscillation of the radial ratio produces distinct lower and upper exact exponents rather than a unique scaling law. Thus failure of exact exponent locking need not have a single origin. The complete relation between exact and effective moiré scales may now be summarized.

c. Theorem 12. Exact supercells versus effective moiré domains. Consider a centered hyperbolic bilayer and an exact nontrivial commensurate sequence satisfying $\theta_j \rightarrow 0$. Define the effective geometric count by

$$N_M^{\text{geo}} = 2n_b \frac{A_M^{\text{eff}}}{A_0}, \quad (1016)$$

and the exact crystallographic count by $N_M^{\text{exact}} = 2n_b q_M$. Their ratio is the exact locking factor,

$$\mathcal{L}_M = \frac{N_M^{\text{exact}}}{N_M^{\text{geo}}} = \frac{q_M A_0}{A_M^{\text{eff}}}. \quad (1017)$$

Let the area-equivalent exact-supercell radius satisfy

$$r_M^{\text{sc}} = 2R \operatorname{arsinh} \left[\sqrt{\frac{q_M A_0}{4\pi R^2}} \right], \quad (1018)$$

and define the radial defect by

$$\Delta_M = r_M^{\text{sc}} - \xi_M. \quad (1019)$$

Then

$$\mathcal{L}_M = e^{\Delta_M/R} \left[\frac{1 - e^{-r_M^{\text{sc}}/R}}{1 - e^{-\xi_M/R}} \right]^2. \quad (1020)$$

Consequently,

$$\ln \mathcal{L}_M = \frac{\Delta_M}{R} + o(1) \quad (1021)$$

in the small-angle exact limit. The exact lower and upper crystallographic exponents satisfy

$$\underline{\beta}_{\text{exact}} = \liminf \frac{r_M^{\text{sc}}}{\xi_M}, \quad (1022)$$

and

$$\bar{\beta}_{\text{exact}} = \limsup \frac{r_M^{\text{sc}}}{\xi_M}. \quad (1023)$$

If the limit exists, the exact exponent is

$$\beta_{\text{exact}} = 1 + \lim \frac{\Delta_M}{\xi_M}. \quad (1024)$$

In particular, $\beta_{\text{exact}} = 1$ if and only if $\Delta_M = o(\xi_M)$, or equivalently, $\mathcal{L}_M = |\theta|^{-o(1)}$. The stronger condition $\Delta_M = O(1)$ is equivalent to two-sided multiplicative locking,

$$c_- N_M^{\text{geo}} \leq N_M^{\text{exact}} \leq c_+ N_M^{\text{geo}}, \quad (1025)$$

with finite positive constants. If the radial defect converges, $\Delta_M \rightarrow \Delta_*$, then

$$N_M^{\text{exact}} \sim e^{\Delta_*/R} N_M^{\text{geo}}. \quad (1026)$$

Thus exact hyperbolic crystallographic scaling is controlled by the asymptotic radial mismatch between the area-equivalent exact supercell and the effective geometric moiré domain. The geometric theory determines

$$\theta \rightarrow \xi_M \rightarrow A_M^{\text{eff}} \rightarrow N_M^{\text{geo}}. \quad (1027)$$

The arithmetic theory independently determines

$$\theta \rightarrow g_\theta \rightarrow \Gamma_M \rightarrow q_M \rightarrow N_M^{\text{exact}}. \quad (1028)$$

The two chains meet through

$$\mathcal{L}_M = \frac{N_M^{\text{exact}}}{N_M^{\text{geo}}}. \quad (1029)$$

Equivalently, they meet through the radial defect

$$\Delta_M = r_M^{\text{sc}} - \xi_M. \quad (1030)$$

The exact exponent is inherited from the geometric theory precisely when

$$\frac{\Delta_M}{\xi_M} \rightarrow 0. \quad (1031)$$

The stronger proportionality relation $N_M^{\text{exact}} \sim C N_M^{\text{geo}}$ requires the stronger radial condition $\Delta_M \rightarrow R \ln C$. Thus the original problem of comparing exact and effective moiré-cell sizes is reduced to a sharply defined geometric-arithmetic question: whether the area-equivalent radius generated by the finite-index Fuchsian intersection tracks the local curvature-controlled moiré radius with a sub-linear, bounded, or convergent radial defect. This distinction closes the logical gap between local hyperbolic interference geometry and exact crystallographic supercell scaling without assuming that the two objects are identical.

H. Exact locking reduction in the arithmetic (2, 3, 8) class

We now isolate the arithmetic input required for exact locking in a concrete cocompact arithmetic commensurability class. Let

$$\Delta_{238}^+ = \langle x, y, z \mid x^2 = y^3 = z^8 = xyz = e \rangle < \text{PSL}(2, \mathbb{R}). \quad (1032)$$

The group is cocompact because $1/2 + 1/3 + 1/8 < 1$, and it belongs to the compact arithmetic list in Takeuchi's classification.[1] Fix a torsion-free subgroup $\Gamma < \Delta_{238}^+$ of finite index

$$d = [\Delta_{238}^+ : \Gamma]. \quad (1033)$$

We do not require Γ to be normal or congruence. The orbifold Euler characteristic of the (2, 3, 8) quotient is

$$\chi_{\text{orb}} = 2 - \left[\left(1 - \frac{1}{2}\right) + \left(1 - \frac{1}{3}\right) + \left(1 - \frac{1}{8}\right) \right] = -\frac{1}{24}. \quad (1034)$$

Consequently, Gauss–Bonnet fixes the primitive area of Γ to be

$$A_0 = d(-2\pi R^2 \chi_{\text{orb}}) = \frac{d\pi R^2}{12}. \quad (1035)$$

Fix a twist center o and write $K_o = \text{Stab}_G(o)$. The existence of a fixed-center exact small-angle sequence is logically separate from arithmeticity of the full commensurator. We therefore state the precise center assumption

$$|\text{Comm}_G(\Gamma) \cap K_o| = \infty. \quad (1036)$$

Under Eq. (1036), choose any nonzero exact centered sequence

$$g_{\theta_j} \in \text{Comm}_G(\Gamma) \cap K_o, \quad \theta_j \rightarrow 0. \quad (1037)$$

The lattice, center, curvature, and registry threshold remain fixed along the sequence. We use the first-layer index convention

$$\Gamma_{M,j} = \Gamma \cap g_{\theta_j} \Gamma g_{\theta_j}^{-1}, \quad q_j = [\Gamma : \Gamma_{M,j}]. \quad (1038)$$

Set

$$s_j = \left| \sin\left(\frac{\theta_j}{2}\right) \right|, \quad b_* = \sinh\left(\frac{\ell_*}{2R}\right) > 0. \quad (1039)$$

Theorem 1 (Exact index–angle reduction). *For the family and exact sequence defined above, the effective moiré area, the exact area-equivalent radius, and the signed radial locking defect are given by*

$$A_{M,j}^{\text{eff}} = 2\pi R^2 \left(\sqrt{1 + \frac{b_*^2}{s_j^2}} - 1 \right), \quad (1040)$$

$$r_{M,j}^{\text{sc}} = R \text{arcosh} \left(1 + \frac{dq_j}{24} \right), \quad (1041)$$

$$\frac{\Delta_j}{R} = \text{arcosh} \left(1 + \frac{dq_j}{24} \right) - \text{arsinh} \left(\frac{b_*}{s_j} \right). \quad (1042)$$

Moreover,

$$\frac{\Delta_j}{R} = \log \left(\frac{dq_j s_j}{24 b_*} \right) + \rho_j, \quad (1043)$$

where the remainder satisfies the explicit bounds

$$-\frac{s_j^2}{4b_*^2} \leq \rho_j \leq \frac{24}{dq_j}, \quad |\rho_j| \leq \frac{24}{dq_j} + \frac{s_j^2}{4b_*^2}. \quad (1044)$$

In particular, $\rho_j \rightarrow 0$.

Proof. The local geometric radius is

$$\xi_j = R \operatorname{arsinh} \left(\frac{b_*}{s_j} \right). \quad (1045)$$

Substituting Eq. (1045) into the disk-area identity $A_{\mathbb{H}}(r) = 2\pi R^2 [\cosh(r/R) - 1]$ gives Eq. (1040). On the arithmetic side, $A_{M,j}^{\text{sc}} = q_j A_0$. Equations (1035) and $A_{\mathbb{H}}(r_{M,j}^{\text{sc}}) = A_{M,j}^{\text{sc}}$ give Eq. (1041); subtracting Eq. (1045) proves Eq. (1042). For $x > 0$, write

$$\operatorname{arsinh} x = \log(2x) + \epsilon_s(x), \quad \epsilon_s(x) = \log \left(\frac{1 + \sqrt{1 + x^{-2}}}{2} \right). \quad (1046)$$

The elementary inequality $1 \leq \sqrt{1+u} \leq 1 + u/2$ for $u \geq 0$ yields

$$0 \leq \epsilon_s(x) \leq \frac{1}{4x^2}. \quad (1047)$$

Similarly, for $y > 0$,

$$\operatorname{arcosh}(1+y) = \log(2y) + \epsilon_c(y), \quad (1048)$$

where

$$\epsilon_c(y) = \log \left(\frac{1 + y^{-1} + \sqrt{1 + 2y^{-1}}}{2} \right), \quad 0 \leq \epsilon_c(y) \leq \frac{1}{y}. \quad (1049)$$

We apply these identities with $x = b_*/s_j$ and $y = dq_j/24$. The logarithmic main terms obey

$$\log \left(\frac{dq_j}{12} \right) - \log \left(\frac{2b_*}{s_j} \right) = \log \left(\frac{dq_j s_j}{24b_*} \right). \quad (1050)$$

Thus Eq. (1043) holds with

$$\rho_j = \epsilon_c \left(\frac{dq_j}{24} \right) - \epsilon_s \left(\frac{b_*}{s_j} \right), \quad (1051)$$

and Eqs. (1047) and (1049) give Eq. (1044). Every nontrivial exact sequence approaching the identity has $q_j \rightarrow \infty$, while $s_j \rightarrow 0$ by construction. Hence the explicit remainder tends to zero. $\square$

The corresponding multiplicative factor is also available without an asymptotic approximation. Direct division of $A_{M,j}^{\text{sc}} = dq_j \pi R^2 / 12$ by Eq. (1040) gives

$$\begin{aligned} \mathcal{L}_j &= \frac{dq_j}{24 \left(\sqrt{1 + b_*^2/s_j^2} - 1 \right)} \\ &= \frac{dq_j s_j}{24b_*} \left(\sqrt{1 + \frac{s_j^2}{b_*^2}} + \frac{s_j}{b_*} \right) \\ &= \frac{dq_j s_j}{24b_*} \exp \left[\operatorname{arsinh} \left(\frac{s_j}{b_*} \right) \right]. \end{aligned} \quad (1052)$$

Consequently,

$$\frac{dq_j s_j}{24b_*} \leq \mathcal{L}_j \leq \frac{dq_j s_j}{24b_*} \exp \left(\frac{s_j}{b_*} \right). \quad (1053)$$

Proposition 1 (Sharp locking criteria). *Along the exact sequence in Eq. (1037), the following statements hold.*

(i) *For every finite Δ_* ,*

$$\Delta_j \rightarrow \Delta_* \iff q_j s_j \rightarrow \frac{24b_*}{d} \exp \left(\frac{\Delta_*}{R} \right). \quad (1054)$$

(ii) *Bounded radial locking is equivalent to*

$$\Delta_j = O(1) \iff 0 < \liminf_{j \rightarrow \infty} q_j s_j \leq \limsup_{j \rightarrow \infty} q_j s_j < \infty. \quad (1055)$$

(iii) *The sharp exponent-locking condition is*

$$\Delta_j = o(\xi_j) \iff \frac{\log q_j}{\log s_j^{-1}} \rightarrow 1 \iff q_j = s_j^{-1+o(1)}. \quad (1056)$$

Under these equivalent conditions the exact exponent exists and equals $\beta_{\text{exact}} = 1$.

Proof. Define $\kappa_j = dq_j s_j / (24b_*)$. Theorem 1 gives $\Delta_j/R = \log \kappa_j + o(1)$, which proves parts (i) and (ii). Furthermore,

$$\frac{\xi_j}{R} = \log \left(\frac{2b_*}{s_j} \right) + \epsilon_s \left(\frac{b_*}{s_j} \right), \quad 0 \leq \epsilon_s \left(\frac{b_*}{s_j} \right) \leq \frac{s_j^2}{4b_*^2}. \quad (1057)$$

Hence $\xi_j/R = \log s_j^{-1} + O(1)$ and

$$\frac{\Delta_j}{\xi_j} = \frac{\log(q_j s_j) + O(1) + o(1)}{\log s_j^{-1} + O(1)}. \quad (1058)$$

The ratio in Eq. (1058) tends to zero if and only if $\log q_j / \log s_j^{-1} \rightarrow 1$, proving part (iii). Since $N_{M,j}^{\text{exact}} = 2n_b q_j$, the same condition gives

$$\frac{\log N_{M,j}^{\text{exact}}}{\log s_j^{-1}} \rightarrow 1, \quad (1059)$$

so the exact exponent exists and equals unity. $\square$

The constants in Proposition 1 also determine the exact prefactor whenever the strongest limit exists. Because $s_j = |\theta_j|/2 [1 + O(\theta_j^2)]$, Eq. (1054) gives

$$q_j \sim \frac{48b_*}{d} \exp \left(\frac{\Delta_*}{R} \right) |\theta_j|^{-1}, \quad (1060)$$

$$A_{M,j}^{\text{sc}} \sim 4\pi R^2 b_* \exp \left(\frac{\Delta_*}{R} \right) |\theta_j|^{-1}, \quad (1061)$$

$$N_{M,j}^{\text{exact}} \sim \frac{96n_b b_*}{d} \exp \left(\frac{\Delta_*}{R} \right) |\theta_j|^{-1}. \quad (1062)$$

If instead $|\Delta_j| \leq C$ eventually, then for every $\eta > 0$ and all sufficiently large j ,

$$\frac{24b_*}{d} e^{-C/R-\eta} s_j^{-1} \leq q_j \leq \frac{24b_*}{d} e^{C/R+\eta} s_j^{-1}. \quad (1063)$$

a. Sharp arithmetic obstruction. For the arithmetic $(2, 3, 8)$ class, density of the full commensurator does not by itself establish Eq. (1036) for an arbitrarily prescribed center. After fixed-center activity is assumed, the available group-theoretic conclusions are the existence of a sequence in Eq. (1037) and divergence $q_j \rightarrow \infty$. Neither conclusion controls $q_j s_j$. Proposition 1 shows that such control is exactly, rather than merely technically, what the three locking targets require. The minimal additional arithmetic inputs are respectively

$$\log q_j = \log s_j^{-1} + o(\log s_j^{-1}), \quad \text{for } \Delta_j = o(\xi_j), \quad (1064)$$

$$0 < c_- \leq q_j s_j \leq c_+ < \infty, \quad \text{for } \Delta_j = O(1), \quad (1065)$$

$$q_j s_j \rightarrow \kappa \in (0, \infty), \quad \text{for } \Delta_j \rightarrow R \log \left(\frac{d\kappa}{24b_*} \right). \quad \text{and} \quad (1066)$$

We have not proved any of Eqs. (1064)–(1066) for this lattice family. Accordingly, we retain the conditional classification of the exact crystallographic universality law. This conclusion leaves all unconditional effective-geometric results unchanged and identifies the precise arithmetic estimate needed for a later unconditional upgrade.

I. A constructive commensurator-active center in the $(2, 3, 8)$ class

We now remove the fixed-center existence assumption for the arithmetic commensurability class used above. Work in the upper-half-plane model

$$\mathbb{H}_R^2 = \{z \in \mathbb{C} : \text{Im } z > 0\}, \quad ds_R^2 = R^2 \frac{|dz|^2}{(\text{Im } z)^2}, \quad (1067)$$

and put $G = \text{PSL}(2, \mathbb{R})$. We choose the upper-half-plane realization of the abstract triangle group in the conjugacy class specified by the split quaternion embedding below.

Let $t_r = 2 \cos(\pi/e_r)$. Takeuchi's formula for the invariant quaternion algebra of an arithmetic triangle group gives

$$\begin{aligned} k &= \mathbb{Q}(t_1^2, t_2^2, t_3^2, t_1 t_2 t_3), \\ a &= t_2^2(t_2^2 - 4), \\ b &= t_2^2 t_3^2(t_1^2 + t_2^2 + t_3^2 + t_1 t_2 t_3 - 4). \end{aligned} \quad (1068)$$

For $(e_1, e_2, e_3) = (2, 3, 8)$, one has $t_1 = 0$, $t_2 = 1$, and $t_3^2 = 2 + \sqrt{2}$. Hence

$$k = \mathbb{Q}(\sqrt{2}), \quad B = \begin{pmatrix} -3, \sqrt{2} \\ k \end{pmatrix}. \quad (1069)$$

This specialization agrees with class III and discriminant $\mathfrak{p}_2$ in Takeuchi's table.[2]

Write

$$B = k + k\iota + k\mathbf{j} + k\iota\mathbf{j}, \quad \iota^2 = -3, \quad \mathbf{j}^2 = \sqrt{2}, \quad \iota\mathbf{j} = -\mathbf{j}\iota. \quad (1070)$$

At the distinguished real place, where $\sqrt{2} > 0$, a complete splitting is

$$\rho(\iota) = \begin{pmatrix} 0 & -\sqrt{3} \\ \sqrt{3} & 0 \end{pmatrix}, \quad \rho(\mathbf{j}) = \begin{pmatrix} 2^{1/4} & 0 \\ 0 & -2^{1/4} \end{pmatrix}. \quad (1071)$$

Indeed, the two displayed matrices have the required squares and anticommute. If $q = a + b\iota + c\mathbf{j} + d\iota\mathbf{j}$, then

$$\rho(q) = \begin{pmatrix} a + 2^{1/4}c & -\sqrt{3}b + 2^{1/4}\sqrt{3}d \\ \sqrt{3}b + 2^{1/4}\sqrt{3}d & a - 2^{1/4}c \end{pmatrix} \quad (1072)$$

$$\text{nrd}(q) = \det \rho(q) = a^2 + 3b^2 - \sqrt{2}c^2 - 3\sqrt{2}d^2. \quad (1073)$$

At the other real place both entries of the Hilbert symbol in Eq. (1069) are negative, so B is ramified there.

This model also supplies a concrete lattice. With $\mathcal{O}_k = \mathbb{Z}[\sqrt{2}]$, define the order

$$\mathcal{O}_0 = \mathcal{O}_k[1, \iota, \mathbf{j}, \iota\mathbf{j}], \quad \Lambda_0 = P\rho(\mathcal{O}_0^1), \quad (1074)$$

where $\mathcal{O}_0^1 = \{u \in \mathcal{O}_0 : \text{nrd}(u) = 1\}$ and P denotes projectivization. The defining relations show directly that $\mathcal{O}_0$ is an order. The usual order construction makes Λ_0 a cocompact arithmetic Fuchsian lattice, and Takeuchi's invariant-algebra classification places it in the same commensurability class as Δ_{238}^+ . We use the conjugate realization in which the two groups are commensurable. Any torsion-free finite-index subgroup $\Gamma < \Delta_{238}^+$ is then commensurable with Λ_0 .

Lemma 1 (Quaternion elements commensurate the order lattice). *Let $\mathcal{O}$ be an order in B , and let $x \in B^\times$ have positive reduced norm at the distinguished real place. Then*

$$P\rho(x) \in \text{Comm}_G(P\rho(\mathcal{O}^1)). \quad (1075)$$

Consequently, $P\rho(x) \in \text{Comm}_G(\Gamma)$ for every finite-index subgroup $\Gamma < \Delta_{238}^+$ in the realization fixed above.

Proof. Set $\mathcal{O}_x = \mathcal{O} \cap x\mathcal{O}x^{-1}$. The intersection of the two full $\mathcal{O}_k$ -lattices is an order of finite additive index in each. Choose a nonzero integer m such that $m\mathcal{O} \subset \mathcal{O}_x$. Reduction modulo $m\mathcal{O}$ gives a homomorphism

$$\mathcal{O}^1 \rightarrow (\mathcal{O}/m\mathcal{O})^\times \quad (1076)$$

with finite target. Its kernel is contained in $1 + m\mathcal{O} \subset \mathcal{O}_x$, and hence in $\mathcal{O}_x^1$. Thus $\mathcal{O}_x^1$ has finite index in $\mathcal{O}^1$. The same argument applied to $x\mathcal{O}x^{-1}$ shows that it also has finite index in $(x\mathcal{O}x^{-1})^1$. Therefore

$$P\rho(\mathcal{O}_x^1) \subset P\rho(\mathcal{O}^1) \cap P\rho(x)P\rho(\mathcal{O}^1)P\rho(x)^{-1} \quad (1077)$$

has finite index on both sides, proving Eq. (1075).

Finally, commensurable subgroups have identical commensurators. Indeed, if $H_1 \sim H_2$ and g commensurates H_1 , then $H_2 \sim H_1 \sim gH_1g^{-1} \sim gH_2g^{-1}$. Applying this observation to $\Gamma \sim \Delta_{238}^+ \sim \Lambda_0$ proves the last assertion. $\square$

Theorem 2 (Explicit commensurator-active center). *For every torsion-free finite-index subgroup $\Gamma < \Delta_{238}^+$, the point*

$$o = i \in \mathbb{H}_R^2 \quad (1078)$$

is commensurator-active in the realization above. More precisely, for $j \geq 1$ let

$$x_j = j - i \in B^\times, \quad g_j = P\rho(x_j). \quad (1079)$$

Then

$$g_j \in \text{Comm}_G(\Gamma) \cap K_o, \quad g_j \neq e, \quad g_j \longrightarrow e. \quad (1080)$$

The oriented rotation angle is exactly

$$\theta_j = 2 \arctan\left(\frac{\sqrt{3}}{j}\right) = 2 \arccos\left(\frac{j}{\sqrt{j^2+3}}\right), \quad (1081)$$

and, for $j \geq 2$,

$$\left| \theta_j - \left(\frac{2\sqrt{3}}{j} - \frac{2\sqrt{3}}{j^3} \right) \right| \leq \frac{18\sqrt{3}}{5j^5}. \quad (1082)$$

Proof. Equation (1073) gives $\text{nrd}(x_j) = j^2 + 3 > 0$. Lemma 1 therefore proves commensurator membership. A determinant-one representative and its Möbius action are

$$G_j = \frac{1}{\sqrt{j^2+3}} \begin{pmatrix} j & \sqrt{3} \\ -\sqrt{3} & j \end{pmatrix}, \quad g_j(z) = \frac{jz + \sqrt{3}}{-\sqrt{3}z + j}. \quad (1083)$$

The fixed-point equation for the second expression is $z^2 + 1 = 0$. Its unique solution in $\mathbb{H}_R^2$ is $z = i$, so every g_j belongs to the same stabilizer K_o .

Put $\phi_j = \arctan(\sqrt{3}/j)$. Then

$$G_j = \begin{pmatrix} \cos \phi_j & \sin \phi_j \\ -\sin \phi_j & \cos \phi_j \end{pmatrix}, \quad \text{spec}(G_j) = \{e^{i\phi_j}, e^{-i\phi_j}\}, \quad (1084)$$

and

$$\text{tr}(G_j) = \frac{2j}{\sqrt{j^2+3}} = 2 \cos \phi_j < 2. \quad (1085)$$

Thus g_j is a nontrivial elliptic element. Moreover, the derivative at the fixed point is

$$g'_j(i) = \left(\frac{j - i\sqrt{3}}{\sqrt{j^2+3}} \right)^{-2} = e^{2i\phi_j}. \quad (1086)$$

The conformal tangent action is therefore rotation through $\theta_j = 2\phi_j$, proving Eq. (1081) from either the

trace or the eigenvalues. This angle is positive and tends to zero, while Eq. (1083) converges entrywise to the identity.

For $u \geq 0$,

$$0 \leq u - \arctan u = \int_0^u \frac{t^2}{1+t^2} dt \leq \frac{u^3}{3}. \quad (1087)$$

With $u = \sqrt{3}/j$, this gives

$$0 \leq \frac{2\sqrt{3}}{j} - \theta_j \leq \frac{2\sqrt{3}}{j^3}. \quad (1088)$$

For $j \geq 2$, one has $u < 1$, and the alternating Taylor remainder after $u - u^3/3$ is bounded by $u^5/5$. Multiplication by two yields Eq. (1082). $\square$

The theorem proves, rather than assumes, fixed-center activity and an exact small-angle sequence for the chosen arithmetic class. It implies only that each coincidence index $[\Gamma : \Gamma \cap g_j \Gamma g_j^{-1}]$ is finite. No estimate of that index, no height-index relation, and no locking or exponent claim is made here; those are separate arithmetic questions.

J. Arithmetic height and coincidence growth of the explicit sequence

We now determine the coincidence-index exponent of the centered sequence constructed above. Retain $k = \mathbb{Q}(\sqrt{2})$, $B = (-3, \sqrt{2})_k$, and $x_j = j - i$, and put

$$n_j = \text{nrd}(x_j) = j^2 + 3, \quad s_j = \sin\left(\frac{\theta_j}{2}\right) > 0. \quad (1089)$$

The exact angle formula in Eq. (1081) gives

$$s_j^2 = \frac{3}{j^2+3}, \quad n_j = 3s_j^{-2}. \quad (1090)$$

Choose a maximal order $\mathcal{M} \supset \mathcal{O}_0$ and write

$$\Lambda = P\rho(\mathcal{M}^1). \quad (1091)$$

At a split finite prime $\mathfrak{p}$, choose an identification $\mathcal{M}_{\mathfrak{p}} \simeq M_2(\mathcal{O}_{k,\mathfrak{p}})$. If $a_{\mathfrak{p}}(x)$ is the minimum valuation of the four entries of the resulting matrix, define

$$\delta_{\mathfrak{p}}(x) = v_{\mathfrak{p}}(\text{nrd } x) - 2a_{\mathfrak{p}}(x). \quad (1092)$$

At a ramified finite prime set $\delta_{\mathfrak{p}}(x) = 0$, and define the projective denominator ideal and height by

$$\mathfrak{D}_{\mathcal{M}}(x) = \prod_{\mathfrak{p} \nmid \text{disc } B} \mathfrak{p}^{\delta_{\mathfrak{p}}(x)}, \quad H_{\mathcal{M}}(x) = N_{k/\mathbb{Q}} \mathfrak{D}_{\mathcal{M}}(x). \quad (1093)$$

Changing the local matrix identification multiplies on the left and right by integral invertible matrices and hence does not change $a_{\mathfrak{p}}$. Moreover, $a_{\mathfrak{p}}(\lambda x) = a_{\mathfrak{p}}(x) + v_{\mathfrak{p}}(\lambda)$, whereas $v_{\mathfrak{p}}(\text{nrd}(\lambda x)) = v_{\mathfrak{p}}(\text{nrd } x) + 2v_{\mathfrak{p}}(\lambda)$. Thus Eqs. (1092) and (1093) depend only on $P\rho(x)$.

Lemma 2 (Local degree and global product). *For $x \in B^\times$, let*

$$Q_{\mathcal{M}}(x) = [\Lambda : \Lambda \cap P\rho(x)\Lambda P\rho(x)^{-1}]. \quad (1094)$$

Then

$$Q_{\mathcal{M}}(x) = \prod_{\substack{\mathfrak{p} \nmid \text{disc } B \\ \delta_{\mathfrak{p}}(x) > 0}} (N\mathfrak{p} + 1)(N\mathfrak{p})^{\delta_{\mathfrak{p}}(x)-1}. \quad (1095)$$

Proof. At a split prime, the two maximal orders $\mathcal{M}_{\mathfrak{p}}$ and $x\mathcal{M}_{\mathfrak{p}}x^{-1}$ are vertices of the Bruhat–Tits tree. Elementary divisors show that their distance is $\delta_{\mathfrak{p}}(x)$. The local norm-one maximal compact fixes the first vertex and is transitive on the sphere of radius e . For $e \geq 1$ that sphere is parametrized by $\mathbb{P}^1(\mathcal{O}_{k,\mathfrak{p}}/\mathfrak{p}^e)$ and has cardinality

$$\#\mathbb{P}^1(\mathcal{O}_{k,\mathfrak{p}}/\mathfrak{p}^e) = (N\mathfrak{p} + 1)(N\mathfrak{p})^{e-1}. \quad (1096)$$

The stabilizer of the second vertex is the conjugate maximal compact, so the local index is Eq. (1096). At a ramified finite prime the quaternion division algebra has a unique maximal order; conjugation preserves it and the local index is one.

Finally, $\text{SL}_1(B)$ satisfies strong approximation away from the distinguished real place because B splits there. Its global norm-one group is therefore dense in the product of the finite maximal compacts. The quotient by their open intersection with the conjugate compacts is finite, so density is surjectivity onto that quotient. Its kernel is precisely the global coincidence subgroup, proving the product formula.[3] $\square$

Let $\mathfrak{p}_2 = (\sqrt{2})$ be the unique prime ramified in B . Primitive cancellation does not occur at any split prime for the present sequence. Indeed, if $x_j \in \mathfrak{p}\mathcal{M}_{\mathfrak{p}}$, then

$$\mathfrak{p} \mid \text{trd}(x_j) = 2j, \quad \mathfrak{p}^2 \mid \text{nrd}(x_j) = j^2 + 3. \quad (1097)$$

For $\mathfrak{p} \neq \mathfrak{p}_2$, the first divisibility implies $\mathfrak{p} \mid j$. The second would then imply $\mathfrak{p}^2 \mid 3$. The only candidate is the prime $\mathfrak{p}_3$ above 3, but 3 is inert in k and $v_{\mathfrak{p}_3}(3) = 1$. This contradiction proves

$$\delta_{\mathfrak{p}}(x_j) = v_{\mathfrak{p}}(n_j) \quad (\mathfrak{p} \neq \mathfrak{p}_2). \quad (1098)$$

For rational integers m , $v_{\mathfrak{p}_2}(m) = 2v_2(m)$. If j is even, n_j is odd; if j is odd, then $j^2 + 3 \equiv 4 \pmod{8}$ and $v_2(n_j) = 2$. Define

$$\varepsilon_j = \begin{cases} 1, & j \text{ even,} \\ 2^{-4}, & j \text{ odd.} \end{cases} \quad (1099)$$

Removing the ramified prime from (n_j) now gives the exact height

$$H_{\mathcal{M}}(x_j) = \varepsilon_j n_j^2. \quad (1100)$$

Introduce the explicit squarefree Euler factor

$$\mathcal{F}_j = \prod_{\substack{\mathfrak{p} \mid (n_j) \\ \mathfrak{p} \neq \mathfrak{p}_2}} \left(1 + \frac{1}{N\mathfrak{p}}\right). \quad (1101)$$

Lemma 2 and Eq. (1098) give

$$Q_{\mathcal{M}}(x_j) = H_{\mathcal{M}}(x_j)\mathcal{F}_j = \varepsilon_j n_j^2 \mathcal{F}_j. \quad (1102)$$

In terms of rational primes, this factor is

$$\mathcal{F}_j = \prod_{\substack{p \mid n_j, \, p \text{ odd} \\ (2/p)=1}} \left(1 + \frac{1}{p}\right)^2 \prod_{\substack{p \mid n_j \\ (2/p)=-1}} \left(1 + \frac{1}{p^2}\right), \quad (1103)$$

where $(2/p)$ is the Legendre symbol; the second product includes $p = 3$ when $3 \mid j$. In particular,

$$1 \leq \mathcal{F}_j \leq \prod_{p \mid n_j} \left(1 + \frac{1}{p}\right)^2 \leq \left(\frac{n_j}{\varphi(n_j)}\right)^2, \quad (1104)$$

$$\left(\frac{n_j}{\varphi(n_j)}\right)^2 \leq \exp\left[O\left(\frac{\log n_j}{\log \log n_j}\right)\right] = n_j^{o(1)}.$$

For completeness, the sub-power bound follows without an analytic prime estimate: if $r = \omega(n)$, then $n \geq \prod_{m=1}^r p_m \geq r!$, whence $r = O(\log n / \log \log n)$, while the product in Eq. (1104) is at most 4^r .

We next pass to the prescribed torsion-free group. If H is a subgroup of finite index a in a group A , direct coset comparison gives, for every commensurator element g ,

$$a^{-1}q_A(g) \leq q_H(g) \leq aq_A(g), \quad (1105)$$

$$q_A(g) = [A : A \cap gAg^{-1}].$$

Indeed, the lower inequality follows by viewing $H \cap gHg^{-1}$ as a subgroup of A , and the upper inequality follows by inserting $H \cap gAg^{-1}$ and using $[gAg^{-1} : gHg^{-1}] = a$. Apply this twice with

$$H_{\Gamma} = \Gamma \cap \Lambda, \quad C_{\Gamma} = [\Gamma : H_{\Gamma}][\Lambda : H_{\Gamma}]. \quad (1106)$$

No normality or congruence assumption is needed. We obtain

$$C_{\Gamma}^{-1}Q_{\mathcal{M}}(x_j) \leq q_j \leq C_{\Gamma}Q_{\mathcal{M}}(x_j). \quad (1107)$$

Theorem 3 (Exact leading coincidence exponent). *For the centered sequence $g_j = P\rho(j - \mathfrak{t})$ and every fixed torsion-free finite-index $\Gamma < \Delta_{238}^+$ in the realization of the centered commensurator construction,*

$$\frac{9\varepsilon_j \mathcal{F}_j}{C_{\Gamma}} s_j^{-4} \leq q_j \leq 9C_{\Gamma} \varepsilon_j \mathcal{F}_j s_j^{-4}. \quad (1108)$$

Consequently,

$$\lim_{j \rightarrow \infty} \frac{\log q_j}{\log s_j^{-1}} = 4, \quad q_j = s_j^{-4+o(1)} = |\theta_j|^{-4+o(1)}. \quad (1109)$$

More explicitly,

$$\begin{aligned} C_\Gamma^{-1} \left(\frac{\theta_j}{2s_j} \right)^4 &\leq \frac{q_j}{144\epsilon_j \mathcal{F}_j |\theta_j|^{-4}} \\ &\leq C_\Gamma \left(\frac{\theta_j}{2s_j} \right)^4, \\ \left(\frac{\theta_j}{2s_j} \right)^4 &= 1 + \frac{\theta_j^2}{6} + O(\theta_j^4). \end{aligned} \quad (1110)$$

Proof. Substitute Eqs. (1090) and (1102) into Eq. (1107). The parity factor is bounded between $1/16$ and 1 , while Eq. (1104) shows that $\log \mathcal{F}_j = o(\log n_j)$. This proves Eqs. (1108) and (1109). Finally, $s_j = (\theta_j/2)[1 - \theta_j^2/24 + O(\theta_j^4)]$ gives Eq. (1110). $\square$

The role of the original order is now transparent. In the basis $1, \iota, \mathbf{j}, \iota\mathbf{j}$, the reduced-trace Gram determinant is -288 , so

$$\text{discrd}(\mathcal{O}_0) = (12\sqrt{2}) = \mathfrak{p}_2^5 \mathfrak{p}_3. \quad (1111)$$

Thus $\mathcal{O}_0$ is nonmaximal only at the primes over 2 and 3 . Its norm-one group has finite index in $\mathcal{M}^1$, and its co-incidence indices satisfy bounds of the form Eq. (1107). Exact indices for that nonmaximal group require the local embeddings of $(\mathcal{O}_0)_{\mathfrak{p}_2}$ and $(\mathcal{O}_0)_{\mathfrak{p}_3}$ in a specified maximal overorder. Exact values for the arbitrary group Γ require, in addition, its actual finite coset action relative to Λ ; the single number $[\Delta_{238}^+ : \Gamma]$ does not determine them. The two-sided bounds above are therefore the strongest uniform conclusion from the specified data. Finally, this explicit sequence does not satisfy the locking criterion of Proposition 1. Equation (1109) gives a limit of four rather than one, and Eq. (1108) gives $q_j s_j \rightarrow \infty$. Combining the exact logarithmic reduction Eq. (1043) with $\xi_j/R = \log s_j^{-1} + O(1)$ yields

$$\frac{\Delta_j}{\xi_j} \rightarrow 3, \quad \frac{r_{M,j}^{\text{sc}}}{\xi_j} \rightarrow 4. \quad (1112)$$

Hence the exact crystallographic exponent along this sequence exists and equals four, while bounded, convergent, and exponent locking to the geometric inverse-linear law all fail.

K. Exact subsequential sampling of the magic surface

We now combine the normalized magic equation with the explicit arithmetic sequence. Fix $K = -R^{-2} < 0$, $\zeta = a/(2R) > 0$, a positive finite root α_ν^* , and the normalization $\gamma_1 = \gamma_0 > 0$. The existence of α_ν^* is an assumption here. Define

$$\mathfrak{c}_\nu = \frac{2\pi\zeta}{n_c} \sqrt{\frac{d\alpha_\nu^*}{\gamma_0}} = \frac{\pi a}{n_c R} \sqrt{\frac{d\alpha_\nu^*}{\gamma_0}} > 0. \quad (1113)$$

Thus Eq. (5348) reads

$$s_\nu^*(\omega) := \sin\left(\frac{\theta_\nu^*(K, \omega)}{2}\right) = \frac{\sinh \zeta}{\sinh(\mathfrak{c}_\nu/\sqrt{\omega})}. \quad (1114)$$

For $j \geq 1$, set

$$\lambda_j^{\text{ar}} = \text{arsinh} \left[\sinh \zeta \sqrt{\frac{j^2 + 3}{3}} \right], \quad (1115)$$

$$\omega_j = \frac{\mathfrak{c}_\nu^2}{(\lambda_j^{\text{ar}})^2}. \quad (1116)$$

Theorem 4 (Exact arithmetic sampling of the magic surface). *For the centered commensurator elements $g_j = P\rho(j - \iota)$, the couplings in Eq. (1116) form a strictly decreasing positive sequence with $\omega_j \rightarrow 0$, and*

$$\theta_\nu^*(K, \omega_j) = \theta_j = 2 \arctan\left(\frac{\sqrt{3}}{j}\right) \quad (1117)$$

for every $j \geq 1$. Moreover,

$$(\lambda_j^{\text{ar}}) = \log j + \log\left(\frac{2 \sinh \zeta}{\sqrt{3}}\right) + O(j^{-2}), \quad (1118)$$

$$\omega_j^{-1/2} = \frac{1}{\mathfrak{c}_\nu} \left[\log j + \log\left(\frac{2 \sinh \zeta}{\sqrt{3}}\right) + O(j^{-2}) \right], \quad (1119)$$

$$\frac{\omega_{j+1}}{\omega_j} = 1 - \frac{2 + o(1)}{j \log j} \rightarrow 1, \quad (1120)$$

$$\omega_j - \omega_{j+1} = \frac{2\mathfrak{c}_\nu^2 + o(1)}{j(\log j)^3}. \quad (1121)$$

For sufficiently small $\Omega > 0$, the exact counting function is

$$\begin{aligned} N_\nu(\Omega) &:= \#\{j \geq 1 : \omega_j \geq \Omega\} \\ &= \left\lfloor \sqrt{\frac{3 \sinh^2(\mathfrak{c}_\nu/\sqrt{\Omega})}{\sinh^2 \zeta}} - 3 \right\rfloor \\ &\sim \frac{\sqrt{3}}{2 \sinh \zeta} \exp\left(\frac{\mathfrak{c}_\nu}{\sqrt{\Omega}}\right). \end{aligned} \quad (1122)$$

Proof. Equations (1115) and (1116) give

$$\frac{\mathfrak{c}_\nu}{\sqrt{\omega_j}} = \lambda_j^{\text{ar}}. \quad (1123)$$

Using Eq. (1090) in Eq. (1114) therefore yields

$$s_\nu^*(\omega_j) = \frac{\sinh \zeta}{\sinh \lambda_j^{\text{ar}}} = \sqrt{\frac{3}{j^2 + 3}} = s_j. \quad (1124)$$

Since the relevant angles lie in $(0, \pi)$, $2 \arcsin s_j = 2 \arctan(\sqrt{3}/j)$, proving Eq. (1117). The sequence λ_j^{ar} is strictly increasing, so ω_j is strictly decreasing. For fixed $\zeta > 0$, $\text{arsinh } x = \log(2x) + O(x^{-2})$ gives Eq. (1118), and hence Eq. (1119) and $\omega_j \rightarrow 0$. Furthermore,

$$\lambda_{j+1}^{\text{ar}} - \lambda_j^{\text{ar}} = \frac{1}{j} + O(j^{-2}). \quad (1125)$$

Substitution into $\omega_{j+1}/\omega_j = (\lambda_j^{\text{ar}}/\lambda_{j+1}^{\text{ar}})^2$ proves Eqs. (1120) and (1121). Finally, $\omega_j \geq \Omega$ is equivalent to

$$j^2 + 3 \leq \frac{3 \sinh^2(\mathbf{c}_\nu/\sqrt{\Omega})}{\sinh^2 \zeta}. \quad (1126)$$

Counting the positive integers that satisfy Eq. (1126) proves the exact formula in Eq. (1122); its asymptotic follows from $\sinh x \sim e^x/2$. $\square$

We next retain the full arithmetic information from the arithmetic-height result. Denote the maximal-order coincidence degree at the sampled magic point by $Q_{\mathcal{M},\nu,j}^* = Q_{\mathcal{M}}(x_j)$ and the degree for the prescribed torsion-free group by $q_{M,\nu,j}^* = q_j$.

Theorem 5 (Exact exponent-four magic-supercell complexity). *Along the exact sampled sequence of Theorem 4, the maximal-order degree is*

$$\begin{aligned} Q_{\mathcal{M},\nu,j}^* &= \varepsilon_j(j^2 + 3)^2 \mathcal{F}_j \\ &= 9\varepsilon_j \mathcal{F}_j \left[\frac{\sinh(\mathbf{c}_\nu/\sqrt{\omega_j})}{\sinh \zeta} \right]^4. \end{aligned} \quad (1127)$$

For the prescribed group Γ , the fixed comparison constant C_Γ gives the sharp retained-factor bounds

$$\begin{aligned} \frac{9\varepsilon_j \mathcal{F}_j}{C_\Gamma} \left[\frac{\sinh(\mathbf{c}_\nu/\sqrt{\omega_j})}{\sinh \zeta} \right]^4 &\leq q_{M,\nu,j}^* \\ &\leq 9C_\Gamma \varepsilon_j \mathcal{F}_j \left[\frac{\sinh(\mathbf{c}_\nu/\sqrt{\omega_j})}{\sinh \zeta} \right]^4. \end{aligned} \quad (1128)$$

Equivalently, define the exact exponential factor

$$\begin{aligned} \mathcal{E}_{\nu,j} &= \exp\left(\frac{4\mathbf{c}_\nu}{\sqrt{\omega_j}}\right) \\ &\times \left[1 - \exp\left(-\frac{2\mathbf{c}_\nu}{\sqrt{\omega_j}}\right) \right]^4. \end{aligned} \quad (1129)$$

Then

$$\frac{9\varepsilon_j \mathcal{F}_j}{16C_\Gamma \sinh^4 \zeta} \mathcal{E}_{\nu,j} \leq q_{M,\nu,j}^* \leq \frac{9C_\Gamma \varepsilon_j \mathcal{F}_j}{16 \sinh^4 \zeta} \mathcal{E}_{\nu,j}. \quad (1130)$$

In particular, both the maximal-order and prescribed-group degrees have the sharp logarithmic asymptotic

$$\lim_{j \rightarrow \infty} \sqrt{\omega_j} \log Q_{\mathcal{M},\nu,j}^* = \lim_{j \rightarrow \infty} \sqrt{\omega_j} \log q_{M,\nu,j}^* = 4\mathbf{c}_\nu. \quad (1131)$$

By contrast, the continuous geometric magic area is

$$A_{M,\nu}^{\text{eff},*}(\omega_j) = 2\pi R^2 \left[\cosh\left(\frac{\mathbf{c}_\nu}{\sqrt{\omega_j}}\right) - 1 \right], \quad (1132)$$

and has logarithmic coefficient $\mathbf{c}_\nu$ rather than $4\mathbf{c}_\nu$.

Proof. Equation (1102) gives $Q_{\mathcal{M}}(x_j) = \varepsilon_j(j^2 + 3)^2 \mathcal{F}_j$. Equations (1090) and (1124) then give Eq. (1127). The fixed-index transfer in Eq. (1107) proves Eq. (1128). The identity $\sinh x = e^x(1 - e^{-2x})/2$ gives Eq. (1130) without absorbing ε_j , $\mathcal{F}_j$, or C_Γ into asymptotic constants. The parity factor ε_j is bounded above and below by positive constants, C_Γ is fixed, and $\log \mathcal{F}_j = o(\log(j^2 + 3))$ by Eq. (1104). Also, $\log(j^2 + 3) = 2\mathbf{c}_\nu/\sqrt{\omega_j} + O(1)$ by Eqs. (1090) and (1123). Taking logarithms in Eq. (1128) proves Eq. (1131). At a magic point, $\text{arsinh}(\sinh \zeta/s_\nu^*) = \mathbf{c}_\nu/\sqrt{\omega}$. Substitution into the hyperbolic disk-area formula proves Eq. (1132), and $\cosh x \sim e^x/2$ gives its coefficient $\mathbf{c}_\nu$. $\square$

The exact sampling statement cannot be extended to all weak couplings. Indeed, Eq. (1114) is strictly increasing in ω , while the arithmetic half-angles form a discrete set. Thus $\theta_\nu^*(K, \omega) = \theta_j$ for some j if and only if $\omega = \omega_j$.

Proposition 2 (Off-subsequence detuning). *If $\Omega \in [\omega_{j+1}, \omega_j]$, then*

$$\theta_{j+1} \leq \theta_\nu^*(K, \Omega) \leq \theta_j, \quad (1133)$$

and hence

$$\min_{k \in \{j, j+1\}} |\theta_\nu^*(K, \Omega) - \theta_k| \leq \frac{\theta_j - \theta_{j+1}}{2} = \frac{\sqrt{3} + o(1)}{j^2}. \quad (1134)$$

Possibly choosing the other endpoint, write $\alpha_{\square,k}^{\text{geo}}(\Omega) = \alpha_{\square}^{\text{geo}}(K, \theta_k, \Omega)$ and define

$$\mathcal{D}_{\nu,j}(\Omega) := \min_{k \in \{j, j+1\}} \left| \log \frac{\alpha_{\square,k}^{\text{geo}}(\Omega)}{\alpha_\nu^*} \right|. \quad (1135)$$

Then

$$\begin{aligned} \mathcal{D}_{\nu,j}(\Omega) &\leq \frac{1}{2} \log \frac{\omega_j}{\omega_{j+1}} \\ &= \frac{1 + o(1)}{j \log j} \\ &= O\left(\sqrt{\Omega} e^{-\mathbf{c}_\nu/\sqrt{\Omega}}\right). \end{aligned} \quad (1136)$$

Proof. Monotonicity of the continuous magic angle and the endpoint identities in Eq. (1117) give Eq. (1133). The nearest endpoint is at most half the interval length away. Equation (1082) then gives Eq. (1134). At fixed arithmetic angle θ_k , the normalized coupling in Eq. (5342) is linear in its last argument. Since it equals α_ν^* at ω_k , one has the exact identity

$$\frac{\alpha_{\square}^{\text{geo}}(K, \theta_k, \Omega)}{\alpha_\nu^*} = \frac{\Omega}{\omega_k}. \quad (1137)$$

Choosing the nearer endpoint in logarithmic coupling and using Eq. (1120) proves the first two estimates in Eq. (1136). Equations (1119) and (1122) give the last estimate. $\square$

a. Subsequent repair of Theorem 130. For this explicit $(2, 3, 8)$ sequence, the exact crystallographic statement is the subsequential law

$$q_{M,\nu,j}^* = \exp\left[\frac{4\mathbf{c}_\nu + o(1)}{\sqrt{\omega_j}}\right], \quad j \rightarrow \infty, \quad (1138)$$

with the retained-factor bounds (1128). Exponent-one locking would give the geometric coefficient $\mathbf{c}_\nu$; the proved arithmetic exponent four gives $4\mathbf{c}_\nu$. The general exponent-one form of Theorem 130 therefore remains conditional for other exact sequences and does not apply to $g_j = P\rho(j - \iota)$. Away from the tuned set $\{\omega_j\}$, Eqs. (1134) and (1136) quantify near-sampling but do not assert exact magic.

L. Exact spectral analysis of the minimal square first star

We now test the finite microscopic Hamiltonian underlying the perturbative square-lattice benchmark. Momentum is measured in units of G_M , energy is measured in units of AG_M^2 , and $\mathbf{p} = (p_x, p_y) = \mathbf{k}/G_M$ and $\alpha = w_1/(AG_M^2)$. In the ordered basis $(|c\rangle, |x+\rangle, |x-\rangle, |y+\rangle, |y-\rangle)$, define

$$\begin{aligned} d_0 &= p_x^2 + p_y^2, \\ d_{x\pm} &= p_x^2 + p_y^2 + 1 \pm 2p_x, \\ d_{y\pm} &= p_x^2 + p_y^2 + 1 \pm 2p_y. \end{aligned} \quad (1139)$$

The exact equal-amplitude five-state Hamiltonian is

$$\hat{H}_\square(\mathbf{p}, \alpha) = \begin{pmatrix} d_0 & \alpha & \alpha & \alpha & \alpha \\ \alpha & d_{x+} & 0 & 0 & 0 \\ \alpha & 0 & d_{x-} & 0 & 0 \\ \alpha & 0 & 0 & d_{y+} & 0 \\ \alpha & 0 & 0 & 0 & d_{y-} \end{pmatrix}. \quad (1140)$$

Thus no expansion in α and no unlisted satellite–satellite matrix element enters the following calculation.

At $\mathbf{p} = 0$, introduce the symmetric satellite state

$$|s\rangle = \frac{1}{2}(|x+\rangle + |x-\rangle + |y+\rangle + |y-\rangle). \quad (1141)$$

Its three-dimensional satellite-space orthogonal complement is dark and has eigenvalue one. On $\text{span}\{|c\rangle, |s\rangle\}$, the remaining block is

$$\begin{pmatrix} 0 & 2\alpha \\ 2\alpha & 1 \end{pmatrix}. \quad (1142)$$

With

$$r(\alpha) = \sqrt{1 + 16\alpha^2}, \quad (1143)$$

the complete point spectrum is

$$\lambda_-(\alpha) = \frac{1-r}{2}, \quad 1, 1, 1, \quad \lambda_+(\alpha) = \frac{1+r}{2}. \quad (1144)$$

The target branch is λ_- , since it continues the central state at $\alpha = 0$. A normalized target vector, continuously extended to $\alpha = 0$, is

$$|\psi_-(\alpha)\rangle = \sqrt{\frac{r+1}{2r}}|c\rangle - \sqrt{\frac{r-1}{2r}}|s\rangle. \quad (1145)$$

Theorem 6 (Sharp no-go theorem for the equal-amplitude square first star). *For every $\alpha \geq 0$, the branch of Eq. (1140) continued from the central state is simple at $\mathbf{p} = 0$ and has the expansion*

$$E_-(\mathbf{p}, \alpha) = \lambda_-(\alpha) + c_\square(\alpha)|\mathbf{p}|^2 + O(|\mathbf{p}|^4), \quad (1146)$$

where

$$c_\square(\alpha) = \frac{r^2 - r + 2}{r(r+1)}. \quad (1147)$$

Write $\hat{H}_\square^0(\alpha) := \hat{H}_\square(0, \alpha)$. Its exact isolation gap at $\mathbf{p} = 0$ is

$$\begin{aligned} \delta_0(\alpha) &= \text{dist}\left(\lambda_-(\alpha), \text{spec } \hat{H}_\square^0(\alpha) \setminus \{\lambda_-(\alpha)\}\right) \\ &= \frac{1+r}{2} \geq 1. \end{aligned} \quad (1148)$$

Nevertheless,

$$c_\square(\alpha) \geq 4\sqrt{2} - 5 > 0, \quad \alpha \geq 0, \quad (1149)$$

and equality holds when $\alpha^2 = (1 + \sqrt{2})/8$. Hence this finite Hamiltonian has no exact quadratic-flattening root and admits no interval whose endpoint curvature coefficients have opposite signs.

Proof. The diagonalization in Eqs. (1142)–(1144) proves simplicity of λ_- . Its distance to a dark level is $1 - \lambda_- = (1+r)/2$, whereas its distance to λ_+ is r . Since $r \geq (1+r)/2$ for $r \geq 1$, this proves Eq. (1148). For the dispersion, write $E_-(\mathbf{p}, \alpha) = |\mathbf{p}|^2 + z(\mathbf{p}, \alpha)$. Eliminating the four satellites from the exact eigenvalue equation gives

$$\begin{aligned} 0 &= F(z, \mathbf{p}; \alpha) \\ &:= z - 2\alpha^2(z-1) \left[\frac{1}{(z-1)^2 - 4p_x^2} + \frac{1}{(z-1)^2 - 4p_y^2} \right]. \end{aligned} \quad (1150)$$

At $\mathbf{p} = 0$, this becomes $z - 4\alpha^2/(z-1) = 0$, whose lower root is λ_- . Moreover,

$$\partial_z F(\lambda_-, 0; \alpha) = 1 + \frac{4\alpha^2}{(\lambda_- - 1)^2} > 0. \quad (1151)$$

The target eigenvalue is therefore analytic near $\mathbf{p} = 0$. Expansion of the exact Schur equation in momentum, but not in α , yields

$$F(z, \mathbf{p}; \alpha) = z - \frac{4\alpha^2}{z-1} - \frac{8\alpha^2}{(z-1)^3}|\mathbf{p}|^2 + O(|\mathbf{p}|^4). \quad (1152)$$

Substituting $z = \lambda_- + \beta|\mathbf{p}|^2 + O(|\mathbf{p}|^4)$ and using $4\alpha^2 = \lambda_-(\lambda_- - 1)$ gives

$$\beta = \frac{8\alpha^2/(\lambda_- - 1)^3}{1 + 4\alpha^2/(\lambda_- - 1)^2} = -\frac{2(r-1)}{r(r+1)}. \quad (1153)$$

Since $c_\square = 1 + \beta$, Eq. (1147) follows. In physical units it is exactly the ratio A_{eff}/A . This algebra also has an independent eigenvector check. For momentum along x , let $D_x = \text{diag}(0, 1, -1, 0, 0)$, so that $\hat{H}_\square((p_x, 0), \alpha) = \hat{H}_\square(0, \alpha) + 2p_x D_x + p_x^2 I$. With $|d_x\rangle = (|x+\rangle - |x-\rangle)/\sqrt{2}$, one has $D_x|s\rangle = |d_x\rangle/\sqrt{2}$; no other eigenstate couples to $|\psi_-\rangle$ through D_x . The target satellite weight is $(r-1)/(2r)$, and exact second-order perturbation in momentum gives

$$1 + \frac{|\langle d_x | 2D_x | \psi_- \rangle|^2}{\lambda_- - 1} = 1 - \frac{2(r-1)}{r(r+1)}, \quad (1154)$$

confirming Eq. (1147). The y calculation is identical, and the mixed quadratic term vanishes. Finally, the numerator of Eq. (1147) satisfies $r^2 - r + 2 = (r-1/2)^2 + 7/4 > 0$. The sharp lower bound follows from

$$\frac{d}{dr} \left(\frac{r^2 - r + 2}{r^2 + r} \right) = \frac{2(r^2 - 2r - 1)}{(r^2 + r)^2}. \quad (1155)$$

On $r \geq 1$, the unique critical point is $r = 1 + \sqrt{2}$; it is the global minimum and its value is $4\sqrt{2} - 5$. The relation $r^2 = 1 + 16\alpha^2$ gives the stated equality point. This proves the strict sign obstruction and the absence of a root. $\square$

The open point gap extends to a genuine momentum neighborhood. Indeed,

$$\left\| \hat{H}_\square(\mathbf{p}, \alpha) - \hat{H}_\square(0, \alpha) \right\| \leq |\mathbf{p}|^2 + 2|\mathbf{p}|. \quad (1156)$$

Weyl's inequality therefore gives the target-to-complement bound

$$\delta_{\text{loc}}(\mathbf{p}, \alpha) \geq \delta_0(\alpha) - 2(|\mathbf{p}|^2 + 2|\mathbf{p}|). \quad (1157)$$

For a fixed α , any radius ρ satisfying $2(\rho^2 + 2\rho) < \delta_0(\alpha)$ preserves a positive gap on $|\mathbf{p}| \leq \rho$. Since $\delta_0(\alpha) \geq 1$, every

$$0 < \rho < \sqrt{\frac{3}{2}} - 1 \quad (1158)$$

works uniformly for all $\alpha \geq 0$. Thus the obstruction is not caused by a hidden local gap closure. The weak-coupling expansion of the exact result is

$$c_\square(\alpha) = 1 - 8\alpha^2 + 128\alpha^4 + O(\alpha^6). \quad (1159)$$

The second-order term is therefore correct, but its formal zero $\alpha_{\text{pc}} = 1/\sqrt{8}$ is outside the regime in which cancellation of the retained terms controls the omitted terms. At that value the exact answer is

$$c_\square\left(\frac{1}{\sqrt{8}}\right) = 3 - \frac{4\sqrt{3}}{3} > 0. \quad (1160)$$

The scale $1/\sqrt{8}$ records the initial second-order mass suppression, but it is not an exact spectral root. For clarity, the obstruction above does not preclude a root in a genuinely richer active-shell Hamiltonian. It does, however, identify the exact data that such a model must supply.

Theorem 7 (Finite-dimensional sign-change-and-gap certificate). *Let $H_R(\mathbf{p}, \alpha)$ be a finite-dimensional Hermitian Hamiltonian on $I = [\alpha_-, \alpha_+]$. Assume that it is continuous in α , that its momentum derivatives through second order are jointly continuous near $\mathbf{p} = 0$, and that C_4 symmetry makes the tracked target dispersion isotropic. Suppose the target eigenvalue $E_b(0, \alpha)$ is simple and obeys the uniform gap condition*

$$g := \inf_{\alpha \in I} \text{dist}(E_b(0, \alpha), \text{spec } H_R(0, \alpha) \setminus \{E_b(0, \alpha)\}) > 0. \quad (1161)$$

Write

$$E_b(\mathbf{p}, \alpha) = E_b(0, \alpha) + c_R(\alpha)|\mathbf{p}|^2 + o(|\mathbf{p}|^2). \quad (1162)$$

If

$$c_R(\alpha_-)c_R(\alpha_+) < 0, \quad (1163)$$

then at least one $\alpha^* \in (\alpha_-, \alpha_+)$ satisfies $c_R(\alpha^*) = 0$. There is also a radius $\rho > 0$ for which the target branch remains separated from the other eigenvalues by at least $g/2$ for every $\alpha \in I$ and $|\mathbf{p}| \leq \rho$.

Proof. The uniformly isolated finite-dimensional spectral projector can be written as a Riesz contour integral. Equation (1161) and the assumed regularity make this projector, the tracked eigenvalue, and its quadratic coefficient continuous in α . Equation (1163) and the intermediate value theorem then give α^* . Uniform continuity on the compact interval permits $\rho > 0$ such that

$$\sup_{\substack{\alpha \in I \\ |\mathbf{p}| \leq \rho}} \|H_R(\mathbf{p}, \alpha) - H_R(0, \alpha)\| < \frac{g}{4}. \quad (1164)$$

Weyl's inequality shifts each of the two spectral edges defining the gap by less than $g/4$, leaving a separation of at least $g/2$. $\square$

The certificate in Theorem 7 is a requirement, not an instantiated result. Additional shells, satellite-satellite tunneling, unequal channels, or internal orbital structure could change the exact curvature, but a richer microscopic model must actually exhibit the endpoint sign change in Eq. (1163) while maintaining Eq. (1161); no such data are supplied by the five-state truncation.

a. Subsequent classification of the sampled magic surface. The positive finite root assumed in the magic-surface sampling result is not discharged by the minimal square first-star Hamiltonian. Under the exact quadratic-flattening criterion, that model supplies no value that can be substituted for α_ν^* , so the magic-surface sampling result sampling and complexity statements remain conditional on a richer Hamiltonian satisfying the separate certificate above. the root-stability

theorem may address uniqueness and stability only after such a root has first been established; neither property is asserted here.

M. Quantitative isolation and structural stability of the magic condition

The exact five-state calculation above has an empty curvature-root set, so there is no root of that model to continue. Its positive margin instead gives a sharp no-root stability theorem. We then state the quantitative continuation result that applies if a richer Hamiltonian independently supplies a gapped, simple crossing.

Define

$$m_{\square} := 4\sqrt{2} - 5. \quad (1165)$$

Theorem 8 (Sharp C^0 stability of the five-state no-go result). *Let $\tilde{c}$ be any bounded real curvature profile on $[0, \infty)$ such that*

$$\|\tilde{c} - c_{\square}\|_{\infty, [0, \infty)} < m_{\square}. \quad (1166)$$

Then $\tilde{c}(\alpha) > 0$ for every $\alpha \geq 0$. Moreover, if

$$\mathcal{L} := \{c \in C_b([0, \infty)) : c(\alpha) = 0 \text{ for some } \alpha \geq 0\}, \quad (1167)$$

then

$$\text{dist}_{\infty}(c_{\square}, \mathcal{L}) = m_{\square}. \quad (1168)$$

Thus $m_{\square}$ is the sharp functional C^0 stability radius of the five-state no-root result.

Proof. Theorem 6 gives $c_{\square}(\alpha) \geq m_{\square}$ for all $\alpha \geq 0$. Hence

$$\tilde{c}(\alpha) \geq m_{\square} - \|\tilde{c} - c_{\square}\|_{\infty, [0, \infty)} > 0. \quad (1169)$$

If $c \in \mathcal{L}$ vanishes at α_* , then

$$\|c - c_{\square}\|_{\infty} \geq c_{\square}(\alpha_*) \geq m_{\square}. \quad (1170)$$

The minimum is attained at $\alpha_m^2 = (1 + \sqrt{2})/8$. The profile $c_{\square} - m_{\square}$ vanishes at α_m and is exactly a uniform distance $m_{\square}$ from $c_{\square}$, proving sharpness. $\square$

In particular, any microscopic enrichment that creates a curvature zero must change the exact curvature by at least $m_{\square}$ somewhere in coupling space. This statement is a distance in curvature-function space, not an operator-norm radius. An operator-norm version would require a separate operator-to-curvature derivative estimate.

Theorem 9 (Quantitative local continuation of a simple gapped root). *Let $I = [\alpha_0 - h, \alpha_0 + h]$, where $h > 0$ and I lies in the physical coupling domain. Suppose $f_0 \in C^1(I)$, $f_0(\alpha_0) = 0$, and, for some $\mu > 0$ and $\sigma \in \{-1, 1\}$,*

$$\sigma f'_0(\alpha) \geq \mu, \quad \alpha \in I. \quad (1171)$$

Let $f \in C^1(I)$ obey

$$\|f - f_0\|_{\infty, I} \leq \eta < \mu h, \quad \|f' - f'_0\|_{\infty, I} \leq \frac{\mu}{2}. \quad (1172)$$

Then f has exactly one root $\alpha_ \in I$. It is interior and simple, and*

$$|f'(\alpha_*)| \geq \frac{\mu}{2}, \quad |\alpha_* - \alpha_0| \leq \frac{\eta}{\mu}. \quad (1173)$$

Suppose in addition that $H_0(\alpha)$ is a continuous finite-dimensional Hermitian family whose tracked target eigenvalue is simple on I , with uniform gap

$$g_0 := \inf_{\alpha \in I} \text{dist}(\lambda_0(\alpha), \text{spec } H_0(\alpha) \setminus \{\lambda_0(\alpha)\}) > 0. \quad (1174)$$

If a perturbed Hermitian family H satisfies

$$\varepsilon_H := \sup_{\alpha \in I} \|H(\alpha) - H_0(\alpha)\| < \frac{g_0}{2}, \quad (1175)$$

then the continued target eigenvalue remains simple throughout I and obeys

$$g_H(\alpha) \geq g_0 - 2\varepsilon_H > 0. \quad (1176)$$

In particular, this gap bound holds at α_ .*

Proof. Integrating Eq. (1171) from α_0 to the endpoints gives

$$\sigma f_0(\alpha_0 - h) \leq -\mu h, \quad \sigma f_0(\alpha_0 + h) \geq \mu h. \quad (1177)$$

Equation (1172) leaves opposite endpoint signs with residual margin at least $\mu h - \eta > 0$, so continuity gives an interior root. The same bounds imply

$$\sigma f'(\alpha) \geq \mu - \frac{\mu}{2} = \frac{\mu}{2}, \quad \alpha \in I. \quad (1178)$$

Thus σf is strictly increasing, the root is unique in I , and it is simple. The reference derivative bound also gives $|f_0(\alpha)| \geq \mu|\alpha - \alpha_0|$ on I . At the perturbed root, $|f_0(\alpha_*)| = |f_0(\alpha_*) - f(\alpha_*)| \leq \eta$, which proves the displacement estimate.

For the spectral statement, Weyl's inequality moves the target eigenvalue and each adjacent spectral edge by at most ε_H . Their separation can therefore decrease by at most $2\varepsilon_H$, proving Eq. (1176). $\square$

For a momentum-dependent family, if

$$\sup_{\substack{\alpha \in I \\ |\mathbf{p}| \leq \rho}} \|H(\mathbf{p}, \alpha) - H(0, \alpha)\| \leq q(\rho), \quad (1179)$$

then the same argument yields

$$g_H(\mathbf{p}, \alpha) \geq g_0 - 2\varepsilon_H - 2q(\rho). \quad (1180)$$

Hence any ρ satisfying $\varepsilon_H + q(\rho) < g_0/2$ preserves the gap on the momentum ball. The operator threshold and

the scalar C^1 thresholds are independent; no operator-to-curvature implication is used.

The theorem applies to a richer quadratic-flattening model by taking $f_0 = c_R$, provided that the richer model first establishes $c_R(\alpha_0) = 0$, Eq. (1171), and the gap in Eq. (1174). The sign-change certificate in Theorem 7 supplies existence under its assumptions but does not by itself supply the derivative margin.

Corollary 1 (Explicit profile-space continuation radius). *Let X be a Banach space, let $s_0 \in X$, and let $F(\alpha, s_0) = f_0(\alpha)$, with f_0 satisfying the hypotheses of Theorem 9. Suppose that, whenever $\|s - s_0\|_X < r$,*

$$\begin{aligned} \|F(\cdot, s) - f_0\|_{\infty, I} &\leq L_0 \|s - s_0\|_X, \\ \|\partial_\alpha F(\cdot, s) - f'_0\|_{\infty, I} &\leq L_1 \|s - s_0\|_X, \\ \sup_{\alpha \in I} \|H(\alpha, s) - H_0(\alpha)\| &\leq L_H \|s - s_0\|_X. \end{aligned} \quad (1181)$$

Assume also that a coherence diagnostic has reference margin $C_0(\alpha) \geq q_0 > 0$ and satisfies

$$\|C(\cdot, s) - C_0\|_{\infty, I} \leq L_C \|s - s_0\|_X. \quad (1182)$$

With a quotient by zero interpreted as $+\infty$, set

$$r_{\text{adm}} := \min \left\{ r, \frac{\mu h}{L_0}, \frac{\mu}{2L_1}, \frac{g_0}{2L_H}, \frac{q_0}{L_C} \right\}. \quad (1183)$$

For $\|s - s_0\|_X < r_{\text{adm}}$, there is exactly one root in I and

$$\begin{aligned} |\alpha_*(s) - \alpha_0| &\leq \frac{L_0}{\mu} \|s - s_0\|_X, \\ g(\alpha, s) &\geq g_0 - 2L_H \|s - s_0\|_X > 0, \\ C(\alpha, s) &\geq q_0 - L_C \|s - s_0\|_X > 0. \end{aligned} \quad (1184)$$

If F is continuously Fréchet differentiable in s and $\|D_s F\| \leq L_s$, then

$$D_s \alpha_*(s) = -\frac{D_s F(\alpha_*(s), s)}{\partial_\alpha F(\alpha_*(s), s)}, \quad \|D_s \alpha_*(s)\| \leq \frac{2L_s}{\mu}. \quad (1185)$$

Proof. Equations (1181) and (1183) imply the two scalar perturbation bounds and the operator perturbation bound in Theorem 9. The theorem gives the root, its uniqueness, and its displacement. Weyl's inequality gives the stated gap, while Eq. (1182) gives the coherence bound directly. The final formula is the implicit-function identity, and Eq. (1178) bounds its denominator from below. $\square$

Corollary 2 (Stability of a strict operational maximum). *Let $M_0 \in C^2(I)$ satisfy*

$$M'_0(\alpha_0) = 0, \quad M''_0(\alpha) \leq -\kappa < 0, \quad \alpha \in I. \quad (1186)$$

If $M \in C^2(I)$ obeys the explicit C^1 stationary-equation and C^2 curvature controls

$$\|M' - M'_0\|_{\infty, I} \leq \eta < \kappa h, \quad \|M'' - M''_0\|_{\infty, I} \leq \frac{\kappa}{2}, \quad (1187)$$

then M has exactly one stationary point $\alpha_ \in I$, and*

$$|\alpha_* - \alpha_0| \leq \frac{\eta}{\kappa}, \quad M''(\alpha) \leq -\frac{\kappa}{2} < 0 \quad (\alpha \in I). \quad (1188)$$

Thus α_ is the unique strict maximum of M on I .*

Proof. Apply Theorem 9 to $f_0 = -M'_0$ and $f = -M'$, with $\mu = \kappa$ and $\sigma = 1$. This gives the unique stationary point and its displacement. The second bound in Eq. (1187) gives $M'' \leq -\kappa/2$ throughout I , so M is strictly concave there and the stationary point is its strict maximum. $\square$

For a profile-dependent operational score, Corollary 1 applies with $F = -\partial_\alpha M$ only after bounds on the profile dependence of both $\partial_\alpha M$ and $\partial_\alpha^2 M$ have been supplied. No profile radius is implied in the absence of those constants.

a. Local scope and subsequent classification. The uniqueness statements above hold only on the declared interval I and for the tracked admissible branch. They do not exclude roots at other couplings or on other branches; global uniqueness would require global derivative and admissibility bounds. The positive derivative margin keeps the continuation neighborhood away from folds and cusps, the gap estimate keeps it away from gap closure, and Eq. (1184) keeps it away from a coherence boundary. No continuation is asserted across any of those loci or across a physical parameter boundary.

The five-state model remains root-free by Theorem 8. The continuation and operational-score results therefore specify conditional tests for a richer Hamiltonian; they do not fabricate the positive finite microscopic root needed by subsequent magic-surface statements. The explicit calculation of active-scattering prefactors is a separate problem and is not used here.

N. Analytic active tunneling factors for fixed radial sectors

We now separate the exactly identifiable radial scattering power from the additional band-projector data required by a physical active sector. Write

$$s = |\sin(\theta/2)|, \quad \zeta = \frac{a}{2R}, \quad \chi = \frac{\sinh \zeta}{s}, \quad L = \text{arsinh } \chi. \quad (1189)$$

Theorem 10 (Exact radial probability reduction). *Under the change of variable*

$$t = \frac{\sinh(uL)}{\sinh L}, \quad 0 \leq t \leq 1, \quad (1190)$$

the normalized radial measure is

$$p_L(u) du = \rho_\chi(t) dt, \quad \rho_\chi(t) = \frac{(\sqrt{1+\chi^2}+1)t}{\sqrt{1+\chi^2 t^2}}, \quad (1191)$$

and its cumulative distribution is

$$F_\chi(t) = \frac{\sqrt{1 + \chi^2 t^2} - 1}{\sqrt{1 + \chi^2} - 1}. \quad (1192)$$

The continuous endpoint conventions are $\rho_0(t) = 2t$, $F_0(t) = t^2$ and $\rho_\infty(t) = 1$, $F_\infty(t) = t$. If $x = F_\chi(t)$, then x is uniform on $[0, 1]$ and

$$t_\chi(x)^2 = x^2 + \varepsilon_\chi x(1 - x), \quad \varepsilon_\chi = \frac{2}{\sqrt{1 + \chi^2} + 1}. \quad (1193)$$

The exact normalized mismatch is

$$q_\zeta(t) := \frac{d_M}{a} = \frac{\text{arsinh}(\sinh \zeta t)}{\zeta}, \quad q_0(t) := t. \quad (1194)$$

Thus exact finite-curvature tunneling depends on ζ in addition to χ .

Proof. Differentiating Eq. (1190) and using $\sinh L = \chi$ gives

$$\frac{dt}{du} = \frac{L\sqrt{1 + \chi^2 t^2}}{\chi}. \quad (1195)$$

Substitution in the exact measure p_L proves Eq. (1191); integration gives Eq. (1192). Solving $x = F_\chi(t)$ and using $\chi^2 = (\sqrt{1 + \chi^2} - 1)(\sqrt{1 + \chi^2} + 1)$ proves Eq. (1193). Finally, substitution of $t = \sinh(uL)/\sinh L$ in the exact mismatch gives Eq. (1194).

For $0 < t < 1$, put $g(\zeta) = \text{arsinh}(t \sinh \zeta)$. Direct differentiation gives

$$g''(\zeta) = \frac{t(1 - t^2) \sinh \zeta}{(1 + t^2 \sinh^2 \zeta)^{3/2}} > 0. \quad (1196)$$

Since $g(0) = 0$, $q_\zeta(t) = g(\zeta)/\zeta$ is strictly increasing in ζ . At fixed χ the measure is unchanged while the mismatch and, for either kernel below, the first tunneling moment change strictly. Therefore ζ cannot be removed from the exact data. The replacement $q_\zeta(t) = t + O(\zeta^2)$ is only a weak-microscopic-curvature reduction. $\square$

Let

$$H = \frac{h}{a}, \quad \Lambda = \frac{\lambda_\perp}{a}. \quad (1197)$$

The normalized exponential and lateral-Gaussian profiles are

$$\begin{aligned} \Phi_X(q; H, \Lambda) &= \exp \left[-\frac{\sqrt{H^2 + q^2} - H}{\Lambda} \right], \\ \Phi_G(q; \Lambda) &= \exp \left[-\frac{q^2}{2\Lambda^2} \right]. \end{aligned} \quad (1198)$$

The Gaussian corresponds to $T_G(D) = w \exp[-(D^2 - h^2)/(2\lambda_\perp^2)]$ and hence satisfies $T_G(h) = w$ exactly.

Proposition 3 (Non-identifiability from total Parseval power). *The total nontrivial scattering power does not determine w_ν^{eff} when P_ν and Q_ν^{act} are unspecified. In particular, no strictly positive band-specific lower bound follows from a positive total Parseval variance alone.*

Proof. On a three-dimensional space let $A = S|e_1\rangle\langle e_2|$. It has fixed total power $\|A\|_F^2 = S^2$. With $P_1 = |e_1\rangle\langle e_1|$ and $Q_1 = |e_2\rangle\langle e_2|$, one has $\|P_1 A Q_1\|_F = S$. With $P_0 = |e_3\rangle\langle e_3|$ and the same Q_1 , one has $P_0 A Q_1 = 0$. The example embeds into larger prescribed ranks. Projection is contractive in Frobenius norm, so, after matching the Parseval normalization,

$$0 \leq w_\nu^{\text{eff}} \leq \frac{\|\delta\tau\|_F}{\sqrt{r_\nu z_\nu}}, \quad (1199)$$

but there is no corresponding positive lower bound without projector or overlap data. $\square$

Theorem 11 (Canonical scalar active factor). *Let $\mathcal{H} = L^2([0, 1], dx)$, let P_0 project onto the constant function 1, and set $Q_0 = I - P_0$. Define τ_j to be multiplication by $w\Phi_j(q_\zeta(t_\chi(x)))$, for $j \in \{X, G\}$, and remove its scalar mean as the trivial component. With $r = z = 1$, the resulting active factor is*

$$\gamma_j^{\text{can}}(\chi, \zeta; H, \Lambda) = [\mathcal{M}_{j,2} - \mathcal{M}_{j,1}^2]^{1/2}, \quad (1200)$$

where the exact analytic one-dimensional moments are

$$\begin{aligned} \mathcal{M}_{j,n}(\chi, \zeta; H, \Lambda) &= \int_0^1 \rho_\chi(t) \Phi_j(q_\zeta(t))^n dt \\ &= \int_0^1 \Phi_j(q_\zeta(t_\chi(x)))^n dx, \quad n = 1, 2. \end{aligned} \quad (1201)$$

For finite positive Λ , $\gamma_j^{\text{can}} > 0$; the Gaussian value is independent of H .

Proof. Let $m_j = \mathcal{M}_{j,1}$ and let V_j be $P_0(\tau_j - w m_j I)Q_0$. Since the multiplication function $w(\Phi_j - m_j)$ has zero mean,

$$\text{Tr}(V_j V_j^\dagger) = w^2 \int_0^1 (\Phi_j - m_j)^2 dx = w^2 (\mathcal{M}_{j,2} - m_j^2). \quad (1202)$$

This proves Eqs. (1200) and (1201). Both q_ζ and the support of the measure span a nontrivial interval, and both kernels are strictly decreasing there. Their variances are therefore strictly positive. $\square$

This scalar construction captures all nontrivial radial Parseval power by a fixed mean-zero projector rule. It is an analytic benchmark, not an identification of an unspecified microscopic band projector.

Proposition 4 (Explicit nondegenerate bounds). *Fix*

$$\begin{aligned} 0 &\leq \zeta \leq Z < \infty, \\ 0 &< \Lambda_- \leq \Lambda \leq \Lambda_+ < \infty, \\ 0 &\leq H \leq H_+ < \infty. \end{aligned} \quad (1203)$$

and allow every physical $\chi \in [0, \infty]$. Put

$$dz = \frac{1}{2\sqrt{1 + (9/16)\sinh^2 Z}}. \quad (1204)$$

Then

$$\frac{d_Z e^{-1/\Lambda_-}}{32\Lambda_+ \sqrt{H_+^2 + 1}} \leq \gamma_X^{\text{can}} \leq \frac{1 - e^{-[\sqrt{H^2 + 1} - H]/\Lambda}}{2}, \quad (1205)$$

$$\frac{d_Z e^{-1/(2\Lambda_-^2)}}{32\Lambda_+^2} \leq \gamma_G^{\text{can}} \leq \frac{1 - e^{-1/(2\Lambda_-^2)}}{2}. \quad (1206)$$

In particular, both factors are bounded strictly away from zero and are smaller than 1/2 on this parameter set.

Proof. Equation (1193) implies $t^2 \leq F_\chi(t) \leq t$. Hence the intervals $A = [0, 1/4]$ and $B = [3/4, 1]$ have probabilities at least 1/16 and 1/4, respectively. For independent copies Y, Y' of $\Phi_j(q_\zeta(t))$,

$$\text{Var } Y = \frac{1}{2} \mathbb{E}(Y - Y')^2 \geq \frac{\Delta_j^2}{64}, \quad (1207)$$

where $\Delta_j = \Phi_j(q_\zeta(1/4)) - \Phi_j(q_\zeta(3/4))$. Moreover,

$$q_\zeta(3/4) - q_\zeta(1/4) \geq dz. \quad (1208)$$

Indeed, $q'_\zeta(t) = \sinh \zeta / [\zeta \sqrt{1 + \sinh^2 \zeta t^2}]$, and its minimum on the indicated rectangle gives Eq. (1208).

On $q \in [1/4, 1]$, the magnitudes of the kernel derivatives are bounded below by

$$\frac{e^{-1/\Lambda_-}}{4\Lambda_+ \sqrt{H_+^2 + 1}} \quad \text{and} \quad \frac{e^{-1/(2\Lambda_-^2)}}{4\Lambda_+^2}, \quad (1209)$$

respectively. The lower bounds follow from the mean-value theorem and Eq. (1207). Since each kernel takes values in $[\Phi_j(1), 1]$, Popoviciu's inequality yields the stated upper bounds. $\square$

No uniform positive lower bound survives a constant-kernel limit $\Lambda \rightarrow \infty$, the concentration limit $\Lambda \rightarrow 0$, the limit $H \rightarrow \infty$ for the exponential kernel, unbounded microscopic curvature, or vanishing physical active overlap.

Theorem 12 (Ordered curvature and twist asymptotics). *For fixed $K < 0$, hence fixed $\zeta > 0$, define*

$$\mathcal{B}_{j,n}(\zeta) = \int_0^1 \Phi_j(q_\zeta(t))^n dt, \quad \gamma_{j,\infty} = \sqrt{\mathcal{B}_{j,2} - \mathcal{B}_{j,1}^2}. \quad (1210)$$

If $D_{j,n} = \|\partial_t[\Phi_j(q_\zeta(t))^n]\|_\infty$, then

$$|\mathcal{M}_{j,n} - \mathcal{B}_{j,n}| \leq \frac{\varepsilon_\chi}{2} D_{j,n},$$

$$|\gamma_j^{\text{can}} - \gamma_{j,\infty}| \leq \frac{\varepsilon_\chi}{\gamma_{j,\infty}} \left(D_{j,1} + \frac{D_{j,2}}{2} \right), \quad (1211)$$

where

$$\varepsilon_\chi = \frac{2s}{\sqrt{s^2 + \sinh^2 \zeta} + s} = \frac{2s}{\sinh \zeta} + O(s^2). \quad (1212)$$

Consequently,

$$\gamma_j^{\text{can}}(K, \theta) = \gamma_{j,\infty}(\zeta) + O(|\theta|) \quad (K < 0 \text{ fixed}). \quad (1213)$$

For fixed $s > 0$ and $\zeta \rightarrow 0$, put

$$\mathcal{M}_{j,n}^E = \int_0^1 2t \Phi_j(t)^n dt \quad (1214)$$

and

$$\mathcal{A}_{j,n}(s) = \int_0^1 dt \left[\frac{t(1/2 - t^2)}{s^2} \Phi_j(t)^n + \frac{n}{3} t(t - t^3) \Phi_j(t)^{n-1} \Phi'_j(t) \right]. \quad (1215)$$

Then

$$\mathcal{M}_{j,n} = \mathcal{M}_{j,n}^E + \zeta^2 \mathcal{A}_{j,n}(s) + O(\zeta^4). \quad (1216)$$

Writing $(\gamma_j^E)^2 = \mathcal{M}_{j,2}^E - (\mathcal{M}_{j,1}^E)^2$, one has

$$\gamma_j^{\text{can}} = \gamma_j^E + \frac{\zeta^2}{2\gamma_j^E} [\mathcal{A}_{j,2} - 2\mathcal{M}_{j,1}^E \mathcal{A}_{j,1}] + O(\zeta^4). \quad (1217)$$

The remainders are uniform for $s \geq s_- > 0$, Λ in a compact subset of $(0, \infty)$, and either H in a compact subset of $(0, \infty)$ or fixed $H = 0$. Since $\zeta^2 = a^2|K|/4$, the last remainder is $O(|K|^2)$.

Proof. Equation (1193) implies $0 \leq t_\chi(x) - x \leq \varepsilon_\chi/2$. The mean-value theorem gives the first line of Eq. (1211). Since all moments lie in $[0, 1]$, the difference of the two variances is at most $\varepsilon_\chi(D_{j,1} + D_{j,2}/2)$. Division by the sum of the two positive square roots proves the second line. At fixed $s > 0$, direct Taylor expansion gives, uniformly on $[0, 1]$,

$$q_\zeta(t) = t + \frac{\zeta^2}{6}(t - t^3) + O(\zeta^4),$$

$$\rho_\chi(t) = 2t + \frac{\zeta^2}{s^2} t \left(\frac{1}{2} - t^2 \right) + O(\zeta^4). \quad (1218)$$

Taylor expansion of the kernel under the integral proves Eq. (1216). Expanding the positive square root of the variance proves Eq. (1217). $\square$

Corollary 3 (Closed moments in the Gaussian reductions). *For the fixed-curvature boundary limit in Eq. (1210), set*

$$A_n = \frac{\sqrt{n}}{\sqrt{2\Lambda}}, \quad B_n = \frac{\Lambda\zeta}{\sqrt{2n}}. \quad (1219)$$

Then the exact finite- ζ Gaussian moment is

$$\mathcal{B}_{G,n}(\zeta) = \frac{\sqrt{\pi}\Lambda\zeta}{2\sqrt{2n}\sinh\zeta} e^{\Lambda^2\zeta^2/(2n)}$$

$$\times [\operatorname{erf}(A_n - B_n) + \operatorname{erf}(A_n + B_n)]. \quad (1220)$$

In the distinct weak-microscopic-curvature reduction $\zeta \rightarrow 0$ with χ fixed, put

$$C_n = \frac{n}{2\Lambda^2\chi^2}, \quad D = \sqrt{1 + \chi^2}. \quad (1221)$$

The reduced moment is

$$\begin{aligned} \mathcal{M}_{G,n}^{(0)}(\chi) &= \frac{e^{C_n}\sqrt{\pi}}{2(D-1)\sqrt{C_n}} \\ &\times \left[\operatorname{erf}(\sqrt{C_n}D) - \operatorname{erf}(\sqrt{C_n}) \right]. \end{aligned} \quad (1222)$$

It extends continuously at $\chi = 0$ and differs from the exact moment by $O(\zeta^2)$ on compact nondegenerate parameter sets.

Proof. In the first integral, use $x = \operatorname{arsinh}(\sinh \zeta t)$ to obtain

$$\mathcal{B}_{G,n} = \frac{1}{\sinh \zeta} \int_0^\zeta \cosh x e^{-nx^2/(2\Lambda^2\zeta^2)} dx. \quad (1223)$$

Completing the square in the two exponential terms of $\cosh x$ gives Eq. (1220). For the second formula, use $x = \sqrt{1 + \chi^2}t$, for which $\rho_\chi(t)dt = dx/(D-1)$, and integrate the Gaussian. The $O(\zeta^2)$ estimate follows from the uniform expansion of q_ζ . $\square$

For reference, the exact Euclidean moments are

$$\begin{aligned} \mathcal{M}_{X,n}^E &= 2 \left[\frac{H(1 - e^{-a_n\delta_H})}{a_n} \right. \\ &\quad \left. + \frac{1 - (1 + a_n\delta_H)e^{-a_n\delta_H}}{a_n^2} \right], \end{aligned} \quad (1224)$$

$$\mathcal{M}_{G,n}^E = \frac{2\Lambda^2}{n} \left(1 - e^{-n/(2\Lambda^2)} \right), \quad (1225)$$

where $a_n = n/\Lambda$ and $\delta_H = \sqrt{H^2 + 1} - H$.

a. Order of limits. Taking $\theta \rightarrow 0$ first at fixed $K < 0$ produces the measure dt and the finite- ζ mismatch $q_\zeta(t)$. Sending $K \rightarrow 0$ afterward leaves dt and gives $q = t$. Reversing the order gives the Euclidean disk measure $2t dt$ and $q = t$; a later small-twist limit does not alter that rescaled law. For the Gaussian, the respective limiting moments are

$$\frac{\sqrt{\pi}\Lambda}{\sqrt{2n}} \operatorname{erf}\left(\frac{\sqrt{n}}{\sqrt{2\Lambda}}\right) \quad \text{and} \quad \frac{2\Lambda^2}{n} \left(1 - e^{-n/(2\Lambda^2)} \right), \quad (1226)$$

which are generally unequal. The two limits therefore cannot be interchanged.

b. Spectral scope. Equations (1200)–(1217) compute a tunneling factor for the specified scalar radial sector. They do not create a quadratic root in the equal-amplitude five-state Hamiltonian, whose root set remains empty by Theorem 6. Nor do they discharge the simple-root, derivative, gap, and coherence hypotheses

required for the conditional richer-model continuation in Theorem 9. A physical band-specific factor still requires fixed microscopic projectors or a quantified active-overlap margin; total Parseval power alone is insufficient by Proposition 3.

O. Reference-point construction of the active scattering sector

An active subspace selected after inspecting the final narrow band is not an independent microscopic input. We therefore register the target and active spaces at a noninteracting reference point and keep that prescription fixed throughout the scan.

Let $\mathcal{H}_0$ be the Hilbert space of one fixed finite cover or one fixed representation fiber. Before the interacting scan, declare an anchor $\bar{p}_0 = (K_0, \theta_0, \dots)$ and set

$$H_{\text{ref}} = H(\bar{p}_0, w = 0). \quad (1227)$$

Let a finite exact symmetry group G act through a fixed unitary representation U_0 , with $[H_{\text{ref}}, U_0(g)] = 0$. Choose a contour Γ_ν that encloses the declared target cluster and no other reference eigenvalue, and define

$$P_\nu^0 = \frac{1}{2\pi i} \oint_{\Gamma_\nu} (z - H_{\text{ref}})^{-1} dz, \quad r_\nu = \operatorname{rank} P_\nu^0. \quad (1228)$$

Its specified center is

$$E_\nu^0 = \frac{1}{r_\nu} \operatorname{Tr}(P_\nu^0 H_{\text{ref}}). \quad (1229)$$

Fix $E_{\text{ref}} > 0$ in the noninteracting reference convention and fixed numbers $\Lambda_- < \Lambda_+$. Neither may be recomputed from a later optimized branch. Set

$$\begin{aligned} a_\nu &= E_\nu^0 + \Lambda_- E_{\text{ref}}, \\ b_\nu &= E_\nu^0 + \Lambda_+ E_{\text{ref}}, \\ W_\nu^0 &= \mathbf{1}_{[a_\nu, b_\nu]}(H_{\text{ref}}). \end{aligned} \quad (1230)$$

The intrinsic normalization of E_{ref} is not changed here; that separate normalization repair is deferred to the intrinsic-normalization result.

For each irreducible representation σ of G , the exact isotypic projector is

$$\Pi_\sigma = \frac{d_\sigma}{|G|} \sum_{g \in G} \overline{\chi_\sigma(g)} U_0(g). \quad (1231)$$

Let R_τ be the representation carried by the predeclared scattering operator components, and define

$$\begin{aligned} S_\nu &= \{\rho : \Pi_\rho P_\nu^0 \neq 0\}, \\ \mathcal{A}_\nu &= \{\sigma : \text{there is } \rho \in S_\nu \text{ such that} \\ &\quad \operatorname{Hom}_G(V_\sigma, R_\tau \otimes V_\rho) \neq \{0\}\}, \\ \Pi_{\mathcal{A}_\nu} &= \sum_{\sigma \in \mathcal{A}_\nu} \Pi_\sigma. \end{aligned} \quad (1232)$$

For invariant scalar scattering, R_τ is trivial and Schur's lemma reduces this rule to equality of source and target irreducible types.

Theorem 13 (Pre-specified active projector and no look-ahead). *Define*

$$Q_\nu^{\text{pre},0} = (I - P_\nu^0)\Pi_{\mathcal{A}_\nu}W_\nu^0. \quad (1233)$$

Then $Q_\nu^{\text{pre},0}$ is an orthogonal projector determined entirely by the specified reference energy and exact symmetry data. Let $J_p : \mathcal{H}_0 \rightarrow \mathcal{H}_p$ be unitaries declared before the scan, with $J_{p_0} = I$ and $J_p U_0(g) = U_p(g)J_p$. Set

$$\begin{aligned} P_\nu^{\text{pre}}(p) &= J_p P_\nu^0 J_p^\dagger, \\ Q_\nu^{\text{pre}}(p) &= J_p Q_\nu^{\text{pre},0} J_p^\dagger. \end{aligned} \quad (1234)$$

Their ranks are constant throughout the scan, and their construction is independent of the final bandwidth, operational score, interacting eigenvectors, observed magic point, and any post-selected energy window.

Proof. The Riesz projector, the window projector, and every isotypic projector commute because they are spectral or symmetry projectors of the same G -invariant reference Hamiltonian. A product of commuting orthogonal projectors is an orthogonal projector, proving the first statement. Unitary conjugation preserves orthogonality and rank. Every datum entering Eqs. (1233) and (1234) is fixed before the interacting scan, which proves no look-ahead by construction. $\square$

Write

$$q_\nu = \text{rank } Q_\nu^{\text{pre},0}, \quad z_\nu^{\text{pre}} = \frac{q_\nu}{r_\nu}. \quad (1235)$$

For complete symmetry-equivalent multiplets, z_ν^{pre} is the usual integer channel count. Otherwise the exact ratio is retained. If $q_\nu = 0$, the specified rule has selected no active channel and no positive active factor is asserted.

If no such family J_p exists, for example because scan points use inequivalent covers, unequal-dimensional fibers, or unmatched boundary conditions, Eq. (1234) is undefined. The construction must then be restricted to one fixed cover or fiber, where $J_p = I$, rather than claiming a common active sector.

Theorem 14 (Quantitative reference-projector stability). *Let Γ_W enclose exactly the reference spectrum in the fixed window, and, for $i \in \{\nu, W\}$, define*

$$\delta_i = \text{dist}(\Gamma_i, \text{spec } H_{\text{ref}}) > 0, \quad \ell_i = \text{length}(\Gamma_i). \quad (1236)$$

Let $\tilde{H}_{\text{ref}} = H_{\text{ref}} + E$, where $[E, U_0(g)] = 0$ and

$$\epsilon := \|E\| < \delta_* := \min\{\delta_\nu, \delta_W\}. \quad (1237)$$

Using the same contours, boundaries, and allowed isotypic set, define the perturbed target, window, and active projectors. Then

$$\|\tilde{P}_\nu^0 - P_\nu^0\| \leq C_\nu(\epsilon),$$

$$\begin{aligned} \|\tilde{W}_\nu^0 - W_\nu^0\| &\leq C_W(\epsilon), \\ \|\tilde{Q}_\nu^{\text{pre},0} - Q_\nu^{\text{pre},0}\| &\leq C_\nu(\epsilon) + C_W(\epsilon), \end{aligned} \quad (1238)$$

where

$$C_i(\epsilon) = \frac{\ell_i}{2\pi} \frac{\epsilon}{\delta_i(\delta_i - \epsilon)}. \quad (1239)$$

If $C_\nu(\epsilon) + C_W(\epsilon) < 1$, the target and active ranks are unchanged. In particular, with

$$S_\Gamma = \frac{\ell_\nu}{\delta_\nu^2} + \frac{\ell_W}{\delta_W^2}, \quad (1240)$$

the explicit sufficient condition

$$\epsilon < r_{\text{ref}} := \min\left\{\frac{\delta_*}{2}, \frac{\pi}{S_\Gamma}\right\} \quad (1241)$$

preserves both ranks.

Proof. For $z \in \Gamma_i$, Weyl's spectral bound and Eq. (1237) give

$$\|(z - H_{\text{ref}})^{-1}\| \leq \delta_i^{-1}, \quad \|(z - \tilde{H}_{\text{ref}})^{-1}\| \leq (\delta_i - \epsilon)^{-1}. \quad (1242)$$

The resolvent identity and contour integration give the first two bounds in Eq. (1238). Since the symmetry projector is fixed, adding and subtracting one mixed product gives

$$\begin{aligned} \tilde{Q} - Q &= (I - \tilde{P})\Pi_{\mathcal{A}_\nu}(\tilde{W} - W) \\ &\quad + (P - \tilde{P})\Pi_{\mathcal{A}_\nu}W, \end{aligned} \quad (1243)$$

and proves the third bound. Orthogonal projections at operator-norm distance less than one have equal finite rank. Finally, if $\epsilon < \delta_*/2$, then $C_i(\epsilon) \leq \ell_i \epsilon / (\pi \delta_i^2)$. The second part of Eq. (1241) makes their sum smaller than one. $\square$

Proposition 5 (Sharp zero-buffer rank obstruction). *Suppose a fixed window boundary is an eigenvalue of H_{ref} and has a nonzero G -invariant eigenspace*

$$S \subset \text{Ran } \Pi_{\mathcal{A}_\nu} \cap \ker P_\nu^0. \quad (1244)$$

Then no positive operator-norm radius can guarantee constancy of the active rank under symmetry-preserving reference perturbations.

Proof. For every $\epsilon > 0$, let $E_\pm = \pm \epsilon \Pi_S$. These perturbations preserve G , have norm ϵ , and move the boundary eigenspace to opposite sides of the fixed window boundary. The two active ranks differ by $\dim S$. Since ϵ is arbitrary, the zero-buffer instability has no positive norm radius. An unisolated target cluster has the analogous obstruction. $\square$

Corollary 4 (Preselected RMS factor and overlap margin). *Assume $q_\nu > 0$. With $\delta\tau(p) = \tau(p) - \tau_{\text{triv}}(p)$, define*

$$V_\nu^{\text{pre}}(p) = P_\nu^{\text{pre}}(p) \delta\tau(p) Q_\nu^{\text{pre}}(p),$$

$$[w_\nu^{\text{pre}}(p)]^2 = \frac{\text{Tr}[V_\nu^{\text{pre}}(p)V_\nu^{\text{pre}}(p)^\dagger]}{r_\nu z_\nu^{\text{pre}}}. \quad (1245)$$

Let

$$\begin{aligned} [\mathcal{W}_\nu^{\text{pre}}(p)]^2 &= \frac{\|\delta\tau(p)\|_F^2}{r_\nu z_\nu^{\text{pre}}}, \\ \eta_\nu^{\text{pre}}(p) &= \frac{\|P_\nu^{\text{pre}}(p)\delta\tau(p)Q_\nu^{\text{pre}}(p)\|_F^2}{\|\delta\tau(p)\|_F^2}. \end{aligned} \quad (1246)$$

Whenever $\delta\tau(p) \neq 0$,

$$0 \leq \eta_\nu^{\text{pre}}(p) \leq 1, \quad [w_\nu^{\text{pre}}]^2 = \eta_\nu^{\text{pre}}[\mathcal{W}_\nu^{\text{pre}}]^2. \quad (1247)$$

If a scan region $\mathcal{D}$ with $w(p) \neq 0$ satisfies the explicit active-overlap margin

$$\|P_\nu^{\text{pre}}(p)\delta\tau(p)Q_\nu^{\text{pre}}(p)\|_F \geq c_\nu \sqrt{r_\nu z_\nu^{\text{pre}}} |w(p)|, \quad p \in \mathcal{D}, \quad (1248)$$

for some $c_\nu > 0$, then

$$w_\nu^{\text{pre}}(p) \geq c_\nu |w(p)|, \quad \gamma_\nu^{\text{pre}}(p) := \frac{w_\nu^{\text{pre}}(p)}{|w(p)|} \geq c_\nu. \quad (1249)$$

Proof. Projection is contractive in Frobenius norm, which proves $0 \leq \eta_\nu^{\text{pre}} \leq 1$. The remaining statements follow by substitution in Eqs. (1245) and (1248). Without the overlap margin, positivity does not follow from total scattering power. $\square$

The quantity $\mathcal{W}_\nu^{\text{pre}}$ matches a scalar W_{sc} only after the Hilbert-space and channel normalizations are identified. This preserves the band-active non-identifiability in Proposition 3 rather than hiding it in a fitted active fraction.

a. Scope and subsequent classification. The construction is reference dependent but contains no information from the completed scan. It does not prove that the specified sector is the unique physically dominant one. The five-state benchmark remains root-free by Theorem 6, and the richer-model root, derivative, gap, and coherence requirements in Theorem 9 remain conditional. The fixed scale E_{ref} removes look-ahead here but does not settle the intrinsic normalization problem reserved for the intrinsic-normalization result.

P. Intrinsic normalization and convention-invariant magic coordinate

The inverse local registry length is the intrinsic spectral scale. Exact Euclidean branch arithmetic is retained as a secondary conversion rather than built into the primary hyperbolic normalization. With

$$L = \text{arsinh} \left[\frac{\sinh \zeta}{|\sin(\theta/2)|} \right], \quad \zeta = \frac{a}{2R}, \quad (1250)$$

define

$$\bar{Q}_M := \frac{2\pi}{\xi_M} = \frac{2\pi}{RL} = \frac{4\pi\zeta}{aL}. \quad (1251)$$

Theorem 15 (Intrinsic quadratic normalization). *For the quadratic benchmark, the primary kinetic scale and effective coupling are*

$$\bar{E}_M := ta^2 \bar{Q}_M^2 = \frac{16\pi^2 t \zeta^2}{L^2}, \quad (1252)$$

$$\bar{\alpha}_\nu := \frac{w_\nu^{\text{pre}}}{\bar{E}_M} = \frac{\gamma_\nu^{\text{pre}}}{16\pi^2} \omega \left(\frac{L}{\zeta} \right)^2, \quad \omega = \frac{w}{t}, \quad (1253)$$

where $w_\nu^{\text{pre}} = \gamma_\nu^{\text{pre}} w$ may be the preselected the active-sector construction numerator. For the selected exact Euclidean branch, put

$$c_{\text{ar}} := \frac{\sqrt{d}}{2n_c}. \quad (1254)$$

Then

$$\begin{aligned} Q_{M,\text{ar}} &= c_{\text{ar}} \bar{Q}_M = \frac{\pi\sqrt{d}}{n_c R L}, \\ E_{M,\text{ar}} &= c_{\text{ar}}^2 \bar{E}_M = \frac{4\pi^2 dt \zeta^2}{n_c^2 L^2}, \\ \alpha_{\nu,\text{ar}} &= c_{\text{ar}}^{-2} \bar{\alpha}_\nu = \frac{\gamma_\nu^{\text{pre}} n_c^2}{4\pi^2 d} \omega \left(\frac{L}{\zeta} \right)^2. \end{aligned} \quad (1255)$$

Thus the earlier coefficient is recovered exactly as a secondary branch conversion.

Proof. Substituting $R = a/(2\zeta)$ in Eq. (1251) gives the first line of Eq. (1252). Division of $w_\nu^{\text{pre}} = \gamma_\nu^{\text{pre}} w$ by that energy gives Eq. (1253). Multiplication by c_{ar} and its second power gives Eq. (1255). $\square$

The factor γ_ν^{pre} retains its exact kernel and projector dependence, including the finite-curvature shape variable ζ . Normalization does not remove that physical dependence.

Theorem 16 (Exact normalization-gauge covariance). *Let $c > 0$ be a convention factor applied to the same physical quadratic family, and define*

$$Q_c = c \bar{Q}_M, \quad E_c = c^2 \bar{E}_M, \quad \alpha_{\nu,c} = c^{-2} \bar{\alpha}_\nu. \quad (1256)$$

If the physical family has a specified root or strict operational maximum whose intrinsic coordinate is $\bar{\alpha}_\nu^$, then its c -coordinate is necessarily*

$$\alpha_{\nu,c}^* = c^{-2} \bar{\alpha}_\nu^*. \quad (1257)$$

Consequently,

$$X_{\nu,c} := \frac{\alpha_{\nu,c}}{\alpha_{\nu,c}^*} = \frac{\bar{\alpha}_\nu}{\bar{\alpha}_\nu^*} =: \bar{X}_\nu \quad (1258)$$

exactly.

Proof. The energy identity follows from $E_c = ta^2 (c \bar{Q}_M)^2$. The numerator is a fixed physical matrix element and is not changed by a normalization convention, so division by

E_c gives the coupling transformation. Equation (1257) is the same coordinate transformation applied at the same physical root. Its substitution proves Eq. (1258). Keeping the numerical root fixed while changing c would instead compare different physical points. $\square$

The theorem is conditional on a specified root; it does not assert that one exists. The invariance concerns the horizontal master coordinate. A convention-normalized ordinate such as W/E_c transforms as c^{-2} in the quadratic case (or c^{-p} at order p) unless its plotted normalization is converted as well. Master curves must therefore use the intrinsic ordinate or display their convention explicitly.

Corollary 5 (Homogeneous order- p covariance). *Let $p > 0$ and let the declared scalar kinetic scale be homogeneous,*

$$\bar{E}_{M,p} = \kappa_p \bar{Q}_M^p = \kappa_p \left(\frac{4\pi\zeta}{aL} \right)^p, \quad \kappa_p > 0. \quad (1259)$$

Then

$$\begin{aligned} E_{M,p,c} &= c^p \bar{E}_{M,p}, \\ \alpha_{\nu,c}^{(p)} &= c^{-p} \bar{\alpha}_{\nu}^{(p)}, \\ \alpha_{\nu,c}^{*,(p)} &= c^{-p} \bar{\alpha}_{\nu}^{*,(p)}, \end{aligned} \quad (1260)$$

and

$$X_{\nu,c}^{(p)} = \frac{\alpha_{\nu,c}^{(p)}}{\alpha_{\nu,c}^{*,(p)}} = \frac{\bar{\alpha}_{\nu}^{(p)}}{\bar{\alpha}_{\nu}^{*,(p)}}. \quad (1261)$$

Proof. Positive homogeneity gives $\kappa_p (c\bar{Q}_M)^p = c^p \bar{E}_{M,p}$. Division of the same physical numerator and application of the same coordinate change at the root prove the remaining statements. $\square$

This homogeneous covariance does not choose the scalar invariant of an anisotropic Hessian or matrix principal symbol. That tensor-kinetic problem is deferred to the invariant-Hessian result.

Proposition 6 (Derivative law and branch conventions). *For a positive convention $c = c(p)$,*

$$\begin{aligned} d \ln \alpha_{\nu,c}^{(p)} &= d \ln w_{\nu}^{\text{pre}} - p d \ln \bar{Q}_M - p d \ln c \\ &= d \ln \bar{\alpha}_{\nu}^{(p)} - p d \ln c. \end{aligned} \quad (1262)$$

For the quadratic formula,

$$d \ln \bar{\alpha}_{\nu} = d \ln \gamma_{\nu}^{\text{pre}} + d \ln \omega + 2 d \ln L - 2 d \ln \zeta. \quad (1263)$$

On a fixed branch, c is constant and its derivative term vanishes. A smoothly parameter-dependent convention contributes the displayed term. A branch change $c_1 \rightarrow c_2$ produces the coordinate jump

$$\alpha_{\nu,c_2}^{(p)} = \left(\frac{c_2}{c_1} \right)^{-p} \alpha_{\nu,c_1}^{(p)} \quad (1264)$$

at fixed intrinsic data. The root has the identical jump, so X remains unchanged.

Proof. Take the logarithmic differential of $\alpha_{\nu,c}^{(p)} = c^{-p} \bar{\alpha}_{\nu}^{(p)}$. Equation (1263) follows directly from Eq. (1253). Applying the same identity on the two sides of a discrete branch change proves Eq. (1264). Because

$$d \ln \alpha_{\nu,c}^{*,(p)} = d \ln \bar{\alpha}_{\nu}^{*,(p)} - p d \ln c, \quad (1265)$$

the convention term cancels from $d \ln X$. $\square$

The factor c_{ar} remains genuine in the exact reciprocal vector of a selected Euclidean commensurate branch. Likewise, exact coincidence indices, supercell areas, parity factors, and arithmetic fluctuations retain their branch data. Equation (1258) removes arithmetic from the primary spectral coordinate; it does not erase arithmetic from exact crystallography or alter the arithmetic conclusions of the arithmetic-height result.

Proposition 7 (Intrinsic relabeling of a specified window). *At the the active-sector construction anchor p_0 , let*

$$\bar{E}_{\text{ref}} = t a^2 \bar{Q}_M(p_0)^2 \quad (1266)$$

for the quadratic benchmark, with the order- p analogue defined by Eq. (1259). If a window has already been specified through physical offsets $\Delta_{\pm} = \Lambda_{\pm} E_{\text{ref}}$, define

$$\bar{\Lambda}_{\pm} := \frac{\Delta_{\pm}}{\bar{E}_{\text{ref}}}. \quad (1267)$$

Then

$$E_{\nu}^0 + \Lambda_{\pm} E_{\text{ref}} = E_{\nu}^0 + \bar{\Lambda}_{\pm} \bar{E}_{\text{ref}}, \quad (1268)$$

so the window and every specified projector are unchanged. If $E_{\text{ref},c} = c^p \bar{E}_{\text{ref}}$, then $\bar{\Lambda}_{\pm} = c^p \Lambda_{\pm}$.

Proof. Both descriptions use the same physical offsets $\Delta_{\pm}$, so their spectral boundaries coincide exactly. Borel functional calculus then gives the same window projector. The target, symmetry, and transported active projectors are consequently unchanged. $\square$

Changing only the energy unit or only the dimensionless window endpoints would move the physical boundaries and reselect the active subspace; that is not a normalization gauge transformation.

a. *Subsequent replacements and classification.* Primary spectral formulas should use

$$Q_M^{\text{eff}} \mapsto \bar{Q}_M, \quad E_M^{\text{eff}} \mapsto \bar{E}_M, \quad \alpha_{\nu}^{\text{eff}} \mapsto \bar{\alpha}_{\nu}, \quad X_{\nu} = \frac{\bar{\alpha}_{\nu}}{\bar{\alpha}_{\nu}^*}. \quad (1269)$$

Use c_{ar} only when an exact Euclidean branch conversion is required, and transform the root in the same convention. The five-state root set remains empty by Theorem 6. For a richer model, Theorem 9 remains conditional on an independently established root and its derivative, gap, and coherence margins. the active-sector construction's specified projectors are not changed by the intrinsic relabeling above.

Q. Coordinate-invariant Hessian kinetic scales

Let ρ_0 be a stationary point of the bare band on the physical parameter manifold, let $V = T_{\rho_0}\mathcal{B}_{\text{phys}}$ have dimension D , and fix a positive Hodge or ordinary momentum metric G on V . Define the symmetric covariant Hessian

$$H_0(X, Y) := \nabla^2 E_{\text{bare}}(\rho_0)(X, Y). \quad (1270)$$

The metric self-adjoint endomorphism A and its symmetric frame representative are

$$A = G^{-1}H_0, \quad \hat{A} = G^{-1/2}H_0G^{-1/2}. \quad (1271)$$

The metric is held fixed here; choosing or renormalizing it is a separate problem.

Theorem 17 (Invariant directional kinetic spectrum). *Let the generalized eigenvalues be ordered by*

$$H_0 c_a = \lambda_a G c_a, \quad \lambda_1 \leq \dots \leq \lambda_D. \quad (1272)$$

They are invariant under tangent-coordinate changes. For a G -unit vector u and the intrinsic-normalization result's intrinsic scale $\bar{Q}_M = 2\pi/\xi_M$, define

$$\bar{E}_M(u) := \frac{\bar{Q}_M^2}{2} H_0(u, u). \quad (1273)$$

The principal scales and exact quadratic directional interval are

$$\bar{E}_{M,a} = \frac{\bar{Q}_M^2}{2} \lambda_a, \quad \bar{E}_M(u) \in \frac{\bar{Q}_M^2}{2} [\lambda_1, \lambda_D]. \quad (1274)$$

The full signed spectrum is lossless quadratic data; a single scalar is not unless an additional role or isotropy assumption is declared.

Proof. Under a basis change $X = SX'$, the bilinear-form matrices obey

$$H'_0 = S^T H_0 S, \quad G' = S^T G S. \quad (1275)$$

Consequently,

$$(G')^{-1} H'_0 = S^{-1} (G^{-1} H_0) S, \quad (1276)$$

so the generalized spectrum is invariant. The Rayleigh quotient is also unchanged. At the stationary point, Taylor's theorem gives

$$E_{\text{bare}}(\exp_{\rho_0} \delta\rho) - E_{\text{bare}}(\rho_0) = \frac{1}{2} H_0(\delta\rho, \delta\rho) + o(\|\delta\rho\|_G^2). \quad (1277)$$

Setting $\delta\rho = \bar{Q}_M u$ proves the directional formula. The min-max theorem for the self-adjoint endomorphism A proves the principal interval. $\square$

Corollary 6 (Exact square-lattice reduction). *For*

$$E(\mathbf{k}) - E_0 = ta^2 |\mathbf{k}|^2 + O(a^4 |\mathbf{k}|^4), \quad (1278)$$

with the Euclidean metric,

$$H_0 = 2ta^2 G, \quad \lambda_a = 2ta^2, \quad (1279)$$

and every directional and principal scale equals

$$\bar{E}_M(u) = \bar{E}_{M,a} = ta^2 \bar{Q}_M^2. \quad (1280)$$

Proof. Compare Eq. (1278) with the quadratic term $H_0(\mathbf{k}, \mathbf{k})/2$ in Eq. (1277). The factor $1/2$ gives $H_0 = 2ta^2 G$, and substitution in Eq. (1274) proves the result. $\square$

Proposition 8 (Role-labeled scalarizations and obstructions). *The following invariant scalars have distinct roles:*

$$\bar{E}_M^{\text{mean}} = \frac{\bar{Q}_M^2}{2D} \text{tr } A, \quad (1281)$$

$$\bar{E}_M^{\text{op}} = \frac{\bar{Q}_M^2}{2} \|A\|_{G, \text{op}}, \quad (1282)$$

$$\bar{E}_M^{\text{coer}} = \frac{\bar{Q}_M^2}{2} \lambda_1, \quad \lambda_1 > 0. \quad (1283)$$

The mean is the signed isotropic directional average. The operator scale is the uniform magnitude bound $|\bar{E}_M(u)| \leq \bar{E}_M^{\text{op}}$. The coercive scale alone gives a uniform positive lower bound. The optional RMS magnitude

$$\bar{E}_M^{\text{rms}} = \frac{\bar{Q}_M^2}{2} \left(\frac{1}{D} \sum_a \lambda_a^2 \right)^{1/2} \quad (1284)$$

does not replace the signed spectrum.

If $\lambda_1 > 0$, the Hessian is positive definite. If $\lambda_D < 0$, it is negative definite and becomes a positive denominator only after an explicit sign-reversed convention. A semidefinite Hessian has a zero quadratic direction and no positive coercive scale. If $\lambda_1 < 0 < \lambda_D$, the point is a saddle; neither a positive operator norm nor trace or RMS scalarization turns it into a positive kinetic denominator.

Proof. The G -spherical average of $H_0(u, u)$ is $\text{tr } A/D$. Self-adjointness gives $|H_0(u, u)| \leq \max_a |\lambda_a|$ and, in the positive-definite case, $H_0(u, u) \geq \lambda_1$. These prove the stated roles. The sign classification follows directly from the generalized eigenvalues. $\square$

Theorem 18 (Weyl stability and positivity radius). *Keep G fixed and let ΔH_0 be symmetric. Set*

$$\Delta \hat{A} = G^{-1/2} \Delta H_0 G^{-1/2}, \quad \varepsilon_A = \|\Delta \hat{A}\|_{\text{op}}. \quad (1285)$$

For the ordered perturbed eigenvalues and principal scales,

$$|\tilde{\lambda}_a - \lambda_a| \leq \varepsilon_A,$$

$$|\tilde{E}_{M,a} - \bar{E}_{M,a}| \leq \frac{\bar{Q}_M^2}{2} \varepsilon_A. \quad (1286)$$

If $\lambda_1 > 0$ and

$$\varepsilon_A < \lambda_1, \quad (1287)$$

then the perturbed Hessian remains positive definite and

$$\tilde{E}_M^{\text{coer}} \geq \frac{\bar{Q}_M^2}{2} (\lambda_1 - \varepsilon_A) > 0. \quad (1288)$$

The radius in Eq. (1287) is sharp in the metric-normalized operator norm.

Proof. Weyl's inequality for the symmetric representatives $\hat{A}$ and $\hat{A} + \Delta\hat{A}$ proves Eq. (1286). It also gives $\tilde{\lambda}_1 \geq \lambda_1 - \varepsilon_A$, proving positivity and Eq. (1288). At equality, subtracting λ_1 times the orthogonal projector onto a minimum eigendirection creates a zero mode, so no larger uniform radius is possible. $\square$

Proposition 9 (the intrinsic-normalization result convention covariance). *Under $Q_c = c\bar{Q}_M$, with $c > 0$, the tensor A , its signed spectrum, eigenspaces, and eigenvalue ratios are unchanged. Every quadratic scale defined above obeys*

$$E_{M,c}^{\text{role}} = c^2 \bar{E}_M^{\text{role}}. \quad (1289)$$

For the fixed the active-sector construction numerator and any nonzero admissible denominator,

$$\alpha_{\nu,c}^{\text{role}} := \frac{w_\nu^{\text{pre}}}{E_{M,c}^{\text{role}}} = c^{-2} \bar{\alpha}_\nu^{\text{role}}. \quad (1290)$$

A conditional root coordinate transforms by the same factor; hence the the intrinsic-normalization result ratio X is invariant within each consistently chosen role.

Proof. Changing $\bar{Q}_M$ does not alter G or H_0 , so it does not alter A . Every quadratic scale contains exactly the common prefactor $\bar{Q}_M^2/2$, proving Eq. (1289). Division of the fixed numerator proves Eq. (1290); the root and X statements then follow from Theorem 16. $\square$

a. Subsequent replacements and scope. For an anisotropic quadratic band, replace a single denominator first by

$$\{\bar{E}_M(u), \bar{E}_{M,a}\}, \quad \bar{E}_{M,a} = \frac{\bar{Q}_M^2}{2} \lambda_a. \quad (1291)$$

If $H_0 > 0$ and a uniform positive denominator is required, use

$$\bar{\alpha}_\nu^{\text{coer}} = \frac{2w_\nu^{\text{pre}}}{\bar{Q}_M^2 \lambda_1}. \quad (1292)$$

The signed principal couplings

$$\bar{\alpha}_{\nu,a} = \frac{2w_\nu^{\text{pre}}}{\bar{Q}_M^2 \lambda_a} \quad (1293)$$

exist only for $\lambda_a \neq 0$. Mean and operator scales must retain their role labels and may not be substituted for coercivity.

the tunneling-factor analysis's kernel variables remain in the numerator, the active-sector construction's pre-selected projectors are unchanged, and the intrinsic-normalization result's intrinsic momentum scale remains primary. The metric G is fixed; its canonical normalization is deferred to the representation-velocity result. No self-energy cancellation, magic root, or higher-order principal-symbol statement is proved here. The five-state no-root result and richer-model conditional classification remain unchanged.

R. Controlled operator collapse on the fixed first star

We now replace the assumed normalized-operator limit by an explicit estimate for the nearest-neighbor square benchmark. The active sector is selected at the reference point and transported, by the declared unitaries of the reference construction, to the common scalar fiber $\mathcal{F} = \mathbb{C}^5$. Its dimensionless reciprocal offsets are

$$\begin{aligned} g_0 &= 0, & g_1 &= (1, 0), & g_2 &= (-1, 0), \\ g_3 &= (0, 1), & g_4 &= (0, -1). \end{aligned} \quad (1294)$$

Let K_P be a fixed bounded momentum domain such that $|p + g_a| \leq P$ for all $p \in K_P$ and $0 \leq a \leq 4$. We use the intrinsic scales

$$\begin{aligned} \bar{Q}_{M,j} &= \frac{2\pi}{\xi_{M,j}}, \\ \bar{E}_{M,j} &= ta^2 \bar{Q}_{M,j}^2, \\ y_{\text{lat},j} &= a^2 \bar{Q}_{M,j}^2. \end{aligned} \quad (1295)$$

The fixed energy-origin shift $-4t$ is removed before normalization.

For $q_a = p + g_a$ and $y_{\text{lat},j} > 0$, define the diagonal matrices

$$\begin{aligned} [D_{Q,j}(p)]_{aa} &= \frac{2[1 - \cos(\sqrt{y_{\text{lat},j}} q_{a,x})]}{y_{\text{lat},j}} \\ &\quad + \frac{2[1 - \cos(\sqrt{y_{\text{lat},j}} q_{a,y})]}{y_{\text{lat},j}}, \end{aligned} \quad (1296)$$

$$[D_2(p)]_{aa} = |q_a|^2. \quad (1297)$$

At $y_{\text{lat},j} = 0$, the first definition is understood by continuity.

Lemma 3 (Sharp uniform cosine remainder). *For every real x ,*

$$|2(1 - \cos x) - x^2| \leq \frac{x^4}{12}. \quad (1298)$$

Consequently,

$$\sup_{p \in K_P} \|D_{Q,j}(p) - D_2(p)\|_{\text{op}} \leq \frac{P^4}{12} y_{\text{lat},j}. \quad (1299)$$

The uniform square-lattice coefficient is therefore $C_{\text{lat}} = P^4/12$.

Proof. Taylor's theorem through degree three gives

$$2(1 - \cos x) - x^2 = -\frac{x^4}{12} \cos \vartheta_x \quad (1300)$$

for some ϑ_x between 0 and x . This proves Eq. (1298); the constant is sharp because the quotient by x^4 tends to $1/12$ in absolute value as $x \rightarrow 0$. Applying the bound in both coordinate directions gives

$$\begin{aligned} |[D_{Q,j} - D_2]_{aa}| &\leq \frac{y_{\text{lat},j}}{12} (q_{a,x}^4 + q_{a,y}^4) \\ &\leq \frac{y_{\text{lat},j}}{12} |q_a|^4 \leq \frac{P^4}{12} y_{\text{lat},j}. \end{aligned} \quad (1301)$$

The operator norm of a diagonal Hermitian matrix is the largest absolute diagonal entry. $\square$

Let $B_j(p)$ be the normalized Hermitian active hopping block of the untruncated microscopic operator after transport to $\mathcal{F}$, and let $B_\star(p)$ be a fixed Hermitian class profile. Set

$$\begin{aligned} y_{T,j} &:= \sup_{p \in K_P} \|B_j(p) - B_\star(p)\|_{\text{op}}, \\ \|B_\star\|_\infty &:= \sup_{p \in K_P} \|B_\star(p)\|_{\text{op}}. \end{aligned} \quad (1302)$$

Choose an arbitrary $\alpha_{\text{ref}} > 0$ and write

$$\alpha_j = \alpha_{\text{ref}} X_j, \quad 0 \leq X_j \leq X_{\text{max}} < \infty. \quad (1303)$$

Thus the explicit profile coefficient is $C_T = \alpha_{\text{ref}} X_{\text{max}}$.

The cutoff error is kept separate from the profile error. If $P_j(p)$ is the specified active projector and $J_{j,p}$ its declared unitary transport, define exactly once

$$R_{\text{tail},j}^{(L_c)}(p) := \frac{J_{j,p} P_j(p) [H_j^{(L_c)} - H_j] P_j(p) J_{j,p}^\dagger}{\bar{E}_{M,j}}. \quad (1304)$$

Unitary transport and orthogonal compression do not increase the norm. Accordingly, the physical Schur estimate gives

$$\sup_{p \in K_P} \|R_{\text{tail},j}^{(L_c)}(p)\|_{\text{op}} \leq y_{\text{tail},j}, \quad (1305)$$

where the dimensionless physical tail field is

$$\begin{aligned} y_{\text{tail},j} &:= \frac{2C_{\parallel}^{\text{tr}} e^{-\delta_{\parallel,j} L_{c,j}}}{\bar{E}_{M,j}} \\ &\quad + \frac{C_{\perp}^{\text{tr}} e^{-\delta_{\perp,j} L_{c,j}}}{\bar{E}_{M,j}}, \end{aligned} \quad (1306)$$

$$\delta_{\parallel,j} := \mu_{\parallel} - R_j^{-1}, \quad \delta_{\perp,j} := \mu_{\perp} - R_j^{-1}. \quad (1307)$$

Proposition 10 (Normalized logarithmic cutoff). *Assume*

$$\delta_{\parallel,j} \geq \delta_0, \quad \delta_{\perp,j} \geq \delta_0, \quad \delta_0 > 0, \quad (1308)$$

and put $C_{\text{tail}} = 2C_{\parallel}^{\text{tr}} + C_{\perp}^{\text{tr}} > 0$. For any $\eta_j > 0$, the cutoff condition

$$L_{c,j} \geq \frac{1}{\delta_0} \left[\log \left(\frac{C_{\text{tail}}}{\eta_j \bar{E}_{M,j}} \right) \right]_+, \quad [s]_+ := \max\{s, 0\}, \quad (1309)$$

implies $y_{\text{tail},j} \leq \eta_j$. In particular, for fixed $E_0 > 0$, $\sigma > 0$, and $\bar{E}_{M,j} \leq E_0$, the explicit choice

$$\begin{aligned} \eta_j &= \left(\frac{\bar{E}_{M,j}}{E_0} \right)^\sigma, \\ L_{c,j} &\geq \frac{1}{\delta_0} \left[\log \left(\frac{C_{\text{tail}} E_0^\sigma}{\bar{E}_{M,j}^{1+\sigma}} \right) \right]_+ \end{aligned} \quad (1310)$$

gives $y_{\text{tail},j} \leq (\bar{E}_{M,j}/E_0)^\sigma \rightarrow 0$ whenever $\bar{E}_{M,j} \rightarrow 0$.

Proof. Equation (1308) yields

$$y_{\text{tail},j} \leq \frac{C_{\text{tail}}}{\bar{E}_{M,j}} e^{-\delta_0 L_{c,j}}. \quad (1311)$$

Substitution of Eq. (1309) proves the first assertion, and the displayed specialization proves the second. The division by $\bar{E}_{M,j}$ is indispensable. Hence a fixed physical cutoff does not, in general, establish normalized convergence as $\bar{E}_{M,j} \rightarrow 0$. If either $\mu_{\parallel} \leq R_j^{-1}$ or $\mu_{\perp} \leq R_j^{-1}$, the corresponding shell-counting integral is not summable; the present estimate then supplies no cutoff convergence theorem. $\square$

Define the cutoff active-shell operator and its master family by

$$\hat{H}_j^{(L_c)}(p) := D_{Q,j}(p) + \alpha_{\text{ref}} X_j B_j(p) + R_{\text{tail},j}^{(L_c)}(p), \quad (1312)$$

$$\hat{H}_\star(p; X) := D_2(p) + \alpha_{\text{ref}} X B_\star(p). \quad (1313)$$

Theorem 19 (Uniform fixed-shell operator bound). *Under the definitions above, the comparison at the matched horizontal coordinate satisfies*

$$\begin{aligned} \sup_{p \in K_P} \|\hat{H}_j^{(L_c)}(p) - \hat{H}_\star(p; X_j)\|_{\text{op}} \\ \leq \frac{P^4}{12} y_{\text{lat},j} + \alpha_{\text{ref}} X_{\text{max}} y_{T,j} + y_{\text{tail},j} =: \varepsilon_j^{\text{mat}}. \end{aligned} \quad (1314)$$

For every fixed X it also satisfies

$$\begin{aligned} \sup_{p \in K_P} \|\hat{H}_j^{(L_c)}(p) - \hat{H}_\star(p; X)\|_{\text{op}} \\ \leq \varepsilon_j^{\text{mat}} + \alpha_{\text{ref}} \|B_\star\|_\infty |X_j - X| =: \varepsilon_j(X). \end{aligned} \quad (1315)$$

Consequently, $y_{\text{lat},j} \rightarrow 0$, $y_{T,j} \rightarrow 0$, $y_{\text{tail},j} \rightarrow 0$, and $X_j \rightarrow X$ imply uniform operator convergence to $\hat{H}_\star(\cdot; X)$.

Proof. Subtracting Eq. (1313) at X_j from Eq. (1312) produces exactly three terms: $D_{Q,j} - D_2$, $\alpha_{\text{ref}} X_j (B_j - B_\star)$, and $R_{\text{tail},j}^{(L_c)}$. The triangle inequality, Lemma 3, Eq. (1302), and Eq. (1305) prove Eq. (1314). Finally,

$$\widehat{H}_\star(p; X_j) - \widehat{H}_\star(p; X) = \alpha_{\text{ref}}(X_j - X)B_\star(p), \quad (1316)$$

which proves Eq. (1315). $\square$

Corollary 7 (Direct-integral interpretation). *On the common Hilbert space*

$$\mathcal{H}_P = \int_{K_P}^\oplus \mathbb{C}^5 dp = L^2(K_P; \mathbb{C}^5), \quad (1317)$$

let $\mathcal{H}_j^{(L_c)}$ and $\mathcal{H}_\star(X)$ be multiplication by the corresponding measurable matrix fibers. Then

$$\begin{aligned} \Delta_j(p; X) &:= \widehat{H}_j^{(L_c)}(p) \\ &\quad - \widehat{H}_\star(p; X), \\ \|\mathcal{H}_j^{(L_c)} - \mathcal{H}_\star(X)\|_{\text{op}} &= \text{ess sup}_{p \in K_P} \|\Delta_j(p; X)\|_{\text{op}} \\ &\leq \varepsilon_j(X). \end{aligned} \quad (1318)$$

Thus the fiber estimate is genuine operator-norm convergence, rather than only pointwise convergence of matrices.

Proof. The norm of a bounded decomposable multiplication operator is the essential supremum of its fiber norms. The uniform bound in Eq. (1315) dominates that essential supremum. $\square$

Corollary 8 (Weyl-stable edges, bandwidths, and gaps). *Let $\lambda_{r,j}(p)$ and $\lambda_{r,\star}(p; X)$ be the increasingly ordered eigenvalues, $1 \leq r \leq 5$. Then*

$$\sup_{p \in K_P} |\lambda_{r,j}(p) - \lambda_{r,\star}(p; X)| \leq \varepsilon_j(X). \quad (1319)$$

For the r th bandwidth $W_r = \sup_p \lambda_r(p) - \inf_p \lambda_r(p)$ and the adjacent gap $\Delta_r = \inf_p \lambda_{r+1}(p) - \sup_p \lambda_r(p)$,

$$|W_{r,j} - W_{r,\star}(X)| \leq 2\varepsilon_j(X), \quad (1320)$$

$$|\Delta_{r,j} - \Delta_{r,\star}(X)| \leq 2\varepsilon_j(X). \quad (1321)$$

In particular, if $\Delta_{r,\star}(X) > 2\varepsilon_j(X)$, then

$$\Delta_{r,j} \geq \Delta_{r,\star}(X) - 2\varepsilon_j(X) > 0. \quad (1322)$$

Proof. Equation (1319) is Weyl's inequality applied fiberwise. A supremum or infimum of a uniformly perturbed function moves by at most the uniform perturbation. Each bandwidth and gap is a difference of two such edges, proving the factor two and the positivity assertion. $\square$

Operator C^0 convergence is insufficient for a velocity statement. The additional hypothesis can be stated quantitatively as follows.

Proposition 11 (Derivative convergence required for velocity). *Fix a unit tangent direction v in K_P , and suppose all displayed fibers are differentiable in that direction. Define*

$$\begin{aligned} z_{\text{lat},j}^{(v)} &:= \sup_p \|\partial_v(D_{Q,j} - D_2)(p)\|_{\text{op}}, \\ z_{T,j}^{(v)} &:= \sup_p \|\partial_v(B_j - B_\star)(p)\|_{\text{op}}, \\ z_{\text{tail},j}^{(v)} &:= \sup_p \|\partial_v R_{\text{tail},j}^{(L_c)}(p)\|_{\text{op}}. \end{aligned} \quad (1323)$$

Then

$$\begin{aligned} \sup_p \|\partial_v \widehat{H}_j^{(L_c)}(p) - \partial_v \widehat{H}_\star(p; X)\|_{\text{op}} \\ \leq z_{\text{lat},j}^{(v)} + \alpha_{\text{ref}} X_{\text{max}} z_{T,j}^{(v)} + z_{\text{tail},j}^{(v)} \\ + \alpha_{\text{ref}} \|\partial_v B_\star\|_\infty |X_j - X| =: z_j^{(v)}. \end{aligned} \quad (1324)$$

Assume a master eigenvalue $\lambda_{r,\star}(p; X)$ is simple and separated from all other master eigenvalues by a uniform gap $g > 0$. Put $M_v = \sup_p \|\partial_v \widehat{H}_\star(p; X)\|_{\text{op}}$. If $\varepsilon_j(X) < g/2$, then the corresponding normalized band slopes obey

$$\sup_p |\partial_v \lambda_{r,j}(p) - \partial_v \lambda_{r,\star}(p; X)| \leq z_j^{(v)} + \frac{4M_v \varepsilon_j(X)}{g - 2\varepsilon_j(X)}. \quad (1325)$$

Thus velocity convergence requires $z_j^{(v)} \rightarrow 0$ in addition to the C^0 conditions of Theorem 19.

Proof. Differentiating Eqs. (1312) and (1313) and applying the triangle inequality proves Eq. (1324). On the circle of radius $g/2$ about the isolated master eigenvalue, the resolvent identity gives the rank-one spectral-projector estimate

$$\|P_{r,j}(p) - P_{r,\star}(p)\|_{\text{op}} \leq \frac{2\varepsilon_j(X)}{g - 2\varepsilon_j(X)}. \quad (1326)$$

The Hellmann–Feynman formula, together with $\|P_{r,j} - P_{r,\star}\|_1 = 2\|P_{r,j} - P_{r,\star}\|_{\text{op}}$ for rank-one projectors, proves Eq. (1325). The scalar sequence $h_j(p) = j^{-1} \sin(j^2 p)$ converges uniformly to zero whereas its derivative does not; hence no derivative conclusion follows from C^0 convergence alone. $\square$

Corollary 9 (Explicit repair of the earlier convergence premise). *For a sequence on the fixed transported first-star fiber, assume*

$$\begin{aligned} y_{\text{lat},j} &\rightarrow 0, y_{T,j} \rightarrow 0, \\ X_j &\rightarrow X, 0 \leq X_j \leq X_{\text{max}}. \end{aligned} \quad (1327)$$

the uniform margin in Eq. (1308), and the cutoff law in Eq. (1309) for some $\eta_j \rightarrow 0$. Then the operator-norm premise used in the earlier Theorems 160 and 169 holds, with the explicit error $\varepsilon_j(X) \rightarrow 0$. Their eigenvalue, edge, bandwidth, and gap consequences are therefore controlled by Corollary 8. A velocity or generalized-dispersion conclusion additionally requires Proposition 11.

For the equal-amplitude profile, Eq. (1313) is the five-state family in the raw coordinate $\alpha = \alpha_{\text{ref}}X$. Its operator family exhibits one-parameter collapse, but its quadratic curvature is strictly positive for every $\alpha > 0$; it has no magic root. Consequently, α_{ref} is merely a coordinate choice in that model, and $X = 1$ is not a magic level. If a richer model independently supplies an isolated simple, gapped root α_* , one may instead set $\alpha_{\text{ref}} = \alpha_*$; only in that conditional setting does $X = 1$ denote the magic coordinate.

The result is confined to the fixed finite active shell. A growing shell or full infinite-volume limit requires a separate embedding and tightness argument. The componentwise conditions above are sufficient, not necessary, because operator errors can cancel. Whether the small fields vanish, and at which rates, is a separate correction-to-scaling problem. Explicit nonperturbative master functions are also separate. Finally, the tensor kinetic theory applies to anisotropic bands, whereas the sharp coefficient $P^4/12$ above is the concrete nearest-neighbor square-lattice benchmark.

S. Class labels and path-dependent irrelevant fields

The horizontal coupling is a relevant coordinate, whereas a shape variable is irrelevant only after its convergence to a fixed class value has been proved. Put

$$s = |\sin(\theta/2)|, \quad \zeta = \frac{a}{2R}, \quad \chi = \frac{\sinh \zeta}{s}, \quad L = \text{arsinh } \chi. \quad (1328)$$

A convenient class-label vector is

$$\ell = (\mathbf{g}_*, H_*, \Lambda_*, r_2^*, r_3^*, \eta_*^{\text{act}}, \mathcal{K}_*, \mathcal{R}_*). \quad (1329)$$

Here $H = h/a$, $\Lambda = \lambda_{\perp}/a$, $r_m = T_m/T_1$, $\mathcal{K}_*$ is the kinetic-tensor class, and $\mathcal{R}_*$ is the specified active-fiber representation together with its fixed transport. At fixed negative curvature, $\mathbf{g}_* = (\zeta, \chi = \infty)$ with $\zeta > 0$. On a weak-curvature crossover path, $\mathbf{g}_* = (\zeta = 0, \chi_*)$ with $0 \leq \chi_* < \infty$.

When the indicated limits exist, define the deviations

$$\begin{aligned} \delta r_{m,j} &= r_{m,j} - r_m^*, \delta \eta_j = \eta_j^{\text{act}} - \eta_*^{\text{act}}, \\ \delta H_j &= H_j - H_*, \delta \Lambda_j = \Lambda_j - \Lambda_*. \end{aligned} \quad (1330)$$

The limiting values in Eq. (1329) are class labels, not irrelevant fields. The deviations in Eq. (1330) become irrelevant fields only when an independent microscopic estimate proves that they vanish. The coupling α , or X when a root has been independently established, is the relevant horizontal coordinate.

Theorem 20 (Exact geometry fields and pathwise rates). *Let*

$$\epsilon_{\chi} := \frac{2}{\sqrt{1 + \chi^2 + 1}}. \quad (1331)$$

In the common uniform coordinate $x \in [0, 1]$, the exact radial quantile satisfies

$$t_{\chi}(x)^2 = x^2 + \epsilon_{\chi}x(1-x), \quad 0 \leq t_{\chi}(x) - x \leq \frac{\epsilon_{\chi}}{2}. \quad (1332)$$

If $\zeta > 0$ is fixed and $\theta \rightarrow 0$, then

$$\epsilon_{\chi} = \frac{|\theta|}{\sinh \zeta} + O(\theta^2). \quad (1333)$$

For every Lipschitz function f_{ζ} on $[0, 1]$, multiplication by $f_{\zeta}(t_{\chi}(x))$ converges in operator norm to multiplication by $f_{\zeta}(x)$, with

$$\|M_{f_{\zeta} \circ t_{\chi}} - M_{f_{\zeta}}\|_{\text{op}} \leq \frac{\text{Lip}(f_{\zeta})}{2} \epsilon_{\chi}. \quad (1334)$$

Its bounded moment obeys the distinct estimate

$$\left| \int_0^1 f_{\zeta}(t_{\chi}(x)) dx - \int_0^1 f_{\zeta}(x) dx \right| \leq \frac{\text{Lip}(f_{\zeta})}{2} \epsilon_{\chi}. \quad (1335)$$

Thus both established rates are $O(|\theta|)$, although one is an operator sup-norm statement and the other is an integral statement. The intrinsic lattice field on the same path is

$$\begin{aligned} y_{\text{lat}} &= \frac{16\pi^2 \zeta^2}{L^2} \\ &= \frac{16\pi^2 \zeta^2}{\log^2(2 \sinh \zeta/s)} \left[1 + O\left(\frac{s^2}{\log(2 \sinh \zeta/s)}\right) \right]. \end{aligned} \quad (1336)$$

In particular, $y_{\text{lat}} = O(\log^{-2}(1/|\theta|))$. If instead $0 < \chi < \infty$ is fixed and $\zeta \rightarrow 0$, then

$$|\theta| = \frac{2\zeta}{\chi} + O(\zeta^3), \quad \zeta = \frac{\chi}{2}|\theta| + O(|\theta|^3), \quad (1337)$$

$$q_{\zeta}(t) = t + \frac{\zeta^2}{6}(t - t^3) + O(\zeta^4) \quad (1338)$$

uniformly for $0 \leq t \leq 1$. Hence a Lipschitz multiplication profile, each of its bounded moments, and every nondegenerate smooth function of finitely many such moments converge at $O(\zeta^2) = O(\theta^2)$. Moreover,

$$y_{\text{lat}} = \frac{4\pi^2 \chi^2}{\text{arsinh}^2 \chi} \theta^2 + O(\theta^4). \quad (1339)$$

At the Euclidean endpoint $\chi = 0$, this is replaced by its continuous limit

$$y_{\text{lat}} = 16\pi^2 s^2 = 4\pi^2 \theta^2 + O(\theta^4). \quad (1340)$$

All profile estimates are uniform when H and Λ range over compact sets with $0 < \Lambda_- \leq \Lambda \leq \Lambda_+ < \infty$ and $0 \leq H \leq H_+ < \infty$.

Proof. The quantile identity gives $t_{\chi}(x) \geq x$. For $x > 0$,

$$t_{\chi}(x) - x = \frac{\epsilon_{\chi}x(1-x)}{t_{\chi}(x) + x} \leq \frac{\epsilon_{\chi}(1-x)}{2} \leq \frac{\epsilon_{\chi}}{2}; \quad (1341)$$

at $x = 0$ both sides vanish. Multiplication-operator norm is essential supremum, so the Lipschitz estimate proves Eq. (1334). Integration proves the separate moment estimate. Notice that moment convergence alone would not imply the operator estimate; the latter uses the common uniform-coordinate identification and Lipschitz regularity.

For fixed ζ , rationalization of Eq. (1331) gives

$$\epsilon_\chi = \frac{2s}{\sqrt{s^2 + \sinh^2 \zeta} + s} = \frac{2s}{\sinh \zeta} + O(s^2), \quad (1342)$$

which proves Eq. (1333). The large-argument expansion of arsinh gives

$$L = \log \frac{2 \sinh \zeta}{s} + O(s^2). \quad (1343)$$

Substitution in $y_{\text{lat}} = 16\pi^2 \zeta^2 / L^2$ proves Eq. (1336).

At fixed positive χ , expand $s = \sinh \zeta / \chi$ and $s = \sin(|\theta|/2)$, and invert the resulting odd series. This proves Eq. (1337). Uniform Taylor expansion of $\operatorname{arsinh}(t \sinh \zeta) / \zeta$ proves Eq. (1338). Composition with a uniformly Lipschitz profile proves the operator and moment rates. A smooth function of the moments has the same rate on a compact nondegenerate set; for a square-root variance this requires a positive limiting variance. Substitution of the parameter rate in $16\pi^2 \zeta^2 / \operatorname{arsinh}^2 \chi$ proves Eq. (1339). Finally, $\zeta / L \rightarrow s$ at the Euclidean endpoint, proving Eq. (1340). $\square$

The exact equality of the operator and moment exponents in the fixed- ζ case follows from Eq. (1341); the weaker $O(\sqrt{\epsilon_\chi})$ estimate is unnecessary. The topologies remain distinct despite this sharper common rate.

Let the normalized transported profile be locally Lipschitz on a compact label set, in the explicit form

$$y_{T,j} \leq C_{\text{geo}} r_{\text{geo},j} + L_2 |\delta r_{2,j}| + L_3 |\delta r_{3,j}| + L_\eta |\delta \eta_j| + L_H |\delta H_j| + L_\Lambda |\delta \Lambda_j|. \quad (1344)$$

Here

$$r_{\text{geo},j} = \begin{cases} \epsilon_{\chi,j}, & \zeta > 0 \text{ fixed}, \\ \zeta_j^2, & 0 \leq \chi < \infty \text{ fixed}, \end{cases} \quad (1345)$$

The metric-normalized kinetic-tensor deviation $y_{K,j}$ enters the Hamiltonian estimate separately, with local Lipschitz constant L_K . The reference kinetic tensor is assumed positive definite, and $y_{K,j}$ must remain below its coercivity radius. The active projectors, their ranks, and their unitary transports are fixed in advance; a positive active-overlap margin is retained throughout the class.

Theorem 21 (Classwise irrelevance criterion). *Consider the fixed central-plus-first-star fiber on a compact interval $0 \leq X \leq X_{\text{max}}$. Assume Eq. (1344), and define the matched- X error*

$$\delta_j^{\text{mat}} := \frac{P^4}{12} y_{\text{lat},j} + \alpha_{\text{ref}} X_{\text{max}} y_{T,j} + L_K y_{K,j} + y_{\text{tail},j}. \quad (1346)$$

Then

$$\sup_p \|\hat{H}_j(p; X_j, \ell) - \hat{H}_*(p; X_j, \ell)\|_{\text{op}} \leq \delta_j^{\text{mat}}. \quad (1347)$$

If $X_j \rightarrow X$ and $B_{*,\infty} = \sup_p \|B_*(p; \ell)\|_{\text{op}}$, then

$$\sup_p \|\hat{H}_j(p; X_j, \ell) - \hat{H}_*(p; X, \ell)\|_{\text{op}} \leq \delta_j, \quad (1348)$$

where

$$\delta_j := \delta_j^{\text{mat}} + \alpha_{\text{ref}} B_{*,\infty} |X_j - X|. \quad (1349)$$

Thus one-parameter operator collapse holds at fixed ℓ if every term on the right tends to zero. It does not identify different values of ℓ .

Proof. The exact cosine remainder contributes $(P^4/12)y_{\text{lat},j}$. The profile estimate contributes at most $\alpha_{\text{ref}} X_{\text{max}} y_{T,j}$, the kinetic-tensor deviation contributes $L_K y_{K,j}$, and the physical cutoff remainder, after division by $\bar{E}_{M,j}$, contributes $y_{\text{tail},j}$. These are the four terms in Eq. (1346). The triangle inequality proves Eq. (1347). The master family is $\alpha_{\text{ref}} B_{*,\infty}$ -Lipschitz in X , proving the actual-limit bound. $\square$

Assume the microscopic family supplies rates

$$\begin{aligned} |\delta r_{2,j}| &\leq C_2 \rho_{2,j}, |\delta r_{3,j}| \leq C_3 \rho_{3,j}, \\ |\delta \eta_j| &\leq C_\eta \rho_{\eta,j}, |\delta H_j| \leq C_H \rho_{H,j}, \\ |\delta \Lambda_j| &\leq C_\Lambda \rho_{\Lambda,j}, y_{K,j} \leq C_K \rho_{K,j}, \end{aligned} \quad (1350)$$

where every displayed $\rho_{a,j} \rightarrow 0$. Define

$$\rho_{\text{mic},j} := \rho_{2,j} + \rho_{3,j} + \rho_{\eta,j} + \rho_{H,j} + \rho_{\Lambda,j} + \rho_{K,j}. \quad (1351)$$

Corollary 10 (Explicit total rates and tail targets). *Assume Eq. (1350), that the square-lattice hopping t is fixed and positive, and the uniform summability margins $\delta_{\parallel,j}, \delta_{\perp,j} \geq \delta_0 > 0$, and choose*

$$\eta_{\text{tail},j} = y_{\text{lat},j}^\sigma, \quad \sigma > 0. \quad (1352)$$

Since $\bar{E}_{M,j} = t y_{\text{lat},j}$, the sufficient cutoff

$$L_{c,j} \geq \frac{1}{\delta_0} \left[\log \left(\frac{C_{\text{tail}}}{t y_{\text{lat},j}^{1+\sigma}} \right) \right]_+ \quad (1353)$$

gives $y_{\text{tail},j} \leq y_{\text{lat},j}^\sigma$. At matched X , the resulting operator error satisfies

$$\begin{aligned} \delta_j^{\text{mat}} &= O\left(\log^{-2} \frac{1}{|\theta_j|}\right) + O(|\theta_j|) + O(\rho_{\text{mic},j}) \\ &\quad + O\left(\log^{-2\sigma} \frac{1}{|\theta_j|}\right) \end{aligned} \quad (1354)$$

at fixed $\zeta > 0$, and

$$\delta_j^{\text{mat}} = O(|\theta_j|^2) + O(\rho_{\text{mic},j}) + O(|\theta_j|^{2\sigma}) \quad (1355)$$

at fixed finite χ . Equivalently, the geometry-plus-regulator parts are respectively

$$O\left(\log^{-2\min\{1,\sigma\}} \frac{1}{|\theta_j|}\right), \quad O(|\theta_j|^{2\min\{1,\sigma\}}). \quad (1356)$$

A proved rate for $|X_j - X|$ is added through Eq. (1349). A sufficient cutoff chosen at the lower bound in Eq. (1353) has the asymptotics

$$L_{c,j}^{\text{sup}} = \begin{cases} \frac{2(1+\sigma)}{\delta_0} \log \log(1/|\theta_j|) + O(1), & \zeta > 0 \text{ fixed,} \\ \frac{2(1+\sigma)}{\delta_0} \log(1/|\theta_j|) + O(1), & \chi < \infty \text{ fixed.} \end{cases} \quad (1357)$$

Proof. The cutoff bound follows by inserting $\eta_{\text{tail},j} = y_{\text{lat},j}^\sigma$ and $\bar{E}_{M,j} = ty_{\text{lat},j}$ in the normalized Schur-tail estimate. Theorem 20 supplies the geometry and lattice rates. Equations (1344) and (1350) supply the inherited profile rates. Their substitution in Eq. (1346) proves Eqs. (1354) and (1355). At fixed ζ , twist decays faster than every inverse power of $\log(1/|\theta|)$, giving the first simplified rate. Taking logarithms of the two lattice asymptotics proves Eq. (1357). $\square$

If any estimate in Eq. (1350) is absent, its exponent is absent as well. A convergent variable still labels the limiting class; a nonconvergent variable remains an uncontrolled field and prevents the conclusion of Theorem 21.

Proposition 12 (Non-collapse obstruction). *Let ℓ and ℓ' be two fixed label values, and let $\mathcal{U}(\ell, \ell')$ be a nonempty fixed set of symmetry-compatible unitary intertwiners declared before the scan. Write $A_\ell(p) = \hat{H}_*(p; X_0, \ell)$. At some X_0 , suppose*

$$\begin{aligned} d_*(X_0; \ell, \ell') &:= \inf_{U \in \mathcal{U}(\ell, \ell')} D_U > 0, \\ D_U &:= \sup_{p \in K_P} d_U(p), \\ d_U(p) &:= \|A_\ell(p) - UA_{\ell'}(p)U^\dagger\|_{\text{op}}. \end{aligned} \quad (1358)$$

Then the two classes cannot converge to one common master operator under the allowed identifications. If $\mathcal{U}(\ell, \ell')$ is empty, the cross-class operator comparison is undefined. Distinct labels may share a master only when the limiting Hamiltonian map itself identifies them.

Proof. If both classes converged uniformly to the same master after allowed fixed intertwiners, the triangle inequality would force the left-hand side of Eq. (1358) to vanish. This contradicts its positive value. Without an allowed intertwiner, the fibers have no declared identification on which an operator norm could be based. $\square$

Theorem 22 (Weyl and simple-root rate transfer). *Suppose Theorem 21 gives total operator error δ_j on a fixed class. Each ordered normalized eigenvalue and each spectral edge then has error at most δ_j ; a bandwidth or gap*

formed from two edges has error at most $2\delta_j$. A limiting gap $\Delta_ > 2\delta_j$ remains open with size at least $\Delta_* - 2\delta_j$.*

For root transfer, let $I = [X_ - h, X_* + h]$, and let a richer reference model have a scalar magic functional $F_* \in C^1(I)$ satisfying*

$$F_*(X_*) = 0, \quad \varsigma F'_*(X) \geq \mu > 0 \quad (X \in I), \quad (1359)$$

for one $\varsigma \in \{-1, 1\}$. Assume its target band is simple on I with gap $g_0 > 0$, and an admissibility diagnostic obeys $C_(X) \geq q_0 > 0$. Suppose, for constants $L_0, L_1, L_C > 0$,*

$$\begin{aligned} \|F_j - F_*\|_{\infty, I} &\leq L_0 \delta_j, \\ \|F'_j - F'_*\|_{\infty, I} &\leq L_1 \delta_j, \\ \|\mathcal{C}_j - C_*\|_{\infty, I} &\leq L_C \delta_j, \end{aligned} \quad (1360)$$

and the Hamiltonian error on I is at most δ_j . If

$$\delta_j < \min \left\{ \frac{\mu h}{L_0}, \frac{\mu}{2L_1}, \frac{g_0}{2}, \frac{q_0}{L_C} \right\}, \quad (1361)$$

then F_j has exactly one root $X_j^ \in I$, and*

$$|X_j^* - X_*| \leq \frac{L_0}{\mu} \delta_j, \quad (1362)$$

$$g_j(X) \geq g_0 - 2\delta_j > 0, \quad \mathcal{C}_j(X) \geq q_0 - L_C \delta_j > 0. \quad (1363)$$

Proof. Weyl's inequality gives the eigenvalue bound. Each edge is a supremum or infimum of one uniformly controlled eigenvalue, and a width or gap is a difference of two edges, proving the factor two.

For the root, Eq. (1359) gives opposite endpoint values with magnitude at least μh . The first threshold in Eq. (1361) preserves their signs. The second threshold gives $\varsigma F'_j \geq \mu/2$, hence existence, local uniqueness, and simplicity. At the perturbed root, $|F_*(X_j^*)| \leq L_0 \delta_j$, while the reference derivative margin gives $|F_*(X)| \geq \mu|X - X_*|$; this proves Eq. (1362). Weyl moves the two gap edges by at most δ_j , and the last hypothesis controls the admissibility margin, proving Eq. (1363). $\square$

The scalar estimates in Eq. (1360) are additional differentiability or response hypotheses; operator C^0 convergence alone does not control a momentum curvature or stationary equation. The theorem does not cross a fold, cusp, switching point, closed gap, coherence boundary, lost active-overlap margin, or physical boundary. The positivity and fixed-metric assumptions on the kinetic class also exclude semidefinite and saddle limits.

Corollary 11 (Repairs of Theorems 151 and 163). *The one-parameter statement of Theorem 151 is valid only classwise: fix ℓ , verify the specified fiber, overlap, kinetic, gap, and regularity assumptions, and prove that the irrelevant fields vanish. If a richer model independently supplies the simple root in Eq. (1359), its class-dependent root inherits the rate in Eq. (1362). The equal-amplitude five-state model remains root-free even when every irrelevant field vanishes, so it has no magic level $X = 1$.*

The correction claim of Theorem 163 is replaced by the rigorous bounds (1354) and (1355), together with the conditional inherited rates in Eq. (1350). A Lipschitz profile map gives these first-order bounds but does not by itself give a Taylor coefficient or a quadratic remainder. Such an expansion requires the corresponding Fréchet differentiability. No explicit master function is asserted here.

All statements above concern a fixed transported finite-rank active fiber. Rank jumps, unidentified fibers, or loss of the specified overlap margin fall outside the theorem. A fixed physical cutoff is also insufficient: the normalized tail must be controlled relative to the vanishing intrinsic kinetic scale as in Eq. (1353).

T. Exact master curves for a fixed bright–dark class

We first fix the active fiber, momentum geometry, and measure. On the specified transported fiber

$$\mathcal{F} = \text{span}\{|c\rangle, |s\rangle, |d_1\rangle, |d_2\rangle, |d_3\rangle\} \cong \mathbb{C}^5, \quad (1364)$$

take $q \in [0, 1]$, metric dq^2 , normalized measure dq , and diameter $D_G = 1$. Define, for the raw normalized coupling $X \geq 0$,

$$H_{\text{bd}}(q, X) = \begin{pmatrix} q & 2X \\ 2X & 2 \end{pmatrix} \oplus 2I_3. \quad (1365)$$

In the original central-plus-four-satellite basis, all four satellite energies are 2 and all central-satellite matrix elements are X . Thus the symmetric satellite couples with amplitude $2X$, while three orthogonal combinations are dark. This reduction holds for every q .

Theorem 23 (Exact bright–dark spectral resolution). *Set*

$$\begin{aligned} D(z; q, X) &= (z - q)(z - 2) - 4X^2, \\ R(q, X) &= \sqrt{(2 - q)^2 + 16X^2}. \end{aligned} \quad (1366)$$

The characteristic polynomial and resolvent are

$$\det(z - H_{\text{bd}}) = (z - 2)^3 D(z; q, X), \quad (1367)$$

and

$$(z - H_{\text{bd}})^{-1} = \frac{1}{D} \begin{pmatrix} z - 2 & 2X \\ 2X & z - q \end{pmatrix} \oplus \frac{1}{z - 2} I_3. \quad (1368)$$

The bright branches and dark levels are

$$e_{\pm}(q, X) = \frac{q + 2 \pm R(q, X)}{2}, \quad e_{da} = 2. \quad (1369)$$

The target e_- is simple and strictly increasing on $[0, 1]$, with

$$\partial_q e_- = \frac{1}{2} \left(1 + \frac{2 - q}{R} \right) > 0. \quad (1370)$$

Its maximum differential occurs at $q = 0$. Its true pointwise distance to the spectral complement is

$$g(q, X) = 2 - e_-(q, X) = \frac{2 - q + R(q, X)}{2}, \quad (1371)$$

and the uniform isolation gap occurs at $q = 1$.

Proof. Taking the determinant and inverse of the bright 2×2 block proves Eqs. (1367) and (1368); solving its quadratic polynomial proves Eq. (1369). Differentiation gives Eq. (1370) and

$$\partial_q^2 e_- = -\frac{8X^2}{R^3} \leq 0. \quad (1372)$$

For $X = 0$, $e_-(q, 0) = q$; for $X > 0$, the last inequality is strict. Thus the target is monotone and its derivative is maximal at $q = 0$.

The distances to a dark level and to the upper bright level are

$$g_d = \frac{2 - q + R}{2}, \quad g_b = R. \quad (1373)$$

Because $R \geq 2 - q$, one has $g_d \leq g_b$, including equality at $X = 0$. Hence the dark triplet is the closest complement. Finally, $\partial_q g_d = -[1 + (2 - q)/R]/2 < 0$, so its minimum is at $q = 1$. $\square$

Put

$$r_0(X) = \sqrt{4 + 16X^2}, \quad r_1(X) = \sqrt{1 + 16X^2}, \quad (1374)$$

and define the exact target support

$$I_X = \left[1 - \frac{r_0(X)}{2}, \frac{3 - r_1(X)}{2} \right]. \quad (1375)$$

Corollary 12 (Exact nonperturbative master curves). *For the fixed class, metric, and measure above, the bandwidth, isolation gap, and maximum fixed-metric differential are*

$$\Phi_W(X) = \frac{1 + r_0(X) - r_1(X)}{2}, \quad (1376)$$

$$\Phi_{\Delta}(X) = \frac{1 + r_1(X)}{2}, \quad (1377)$$

$$\begin{aligned} \Phi_V(X) &= \frac{1}{2} \left(1 + \frac{2}{r_0(X)} \right) \\ &= \frac{1}{2} \left(1 + \frac{1}{\sqrt{1 + 4X^2}} \right). \end{aligned} \quad (1378)$$

The full normalized target-band DOS is, almost everywhere,

$$\Phi_{\rho}(\epsilon, X) = \left[1 + \frac{4X^2}{(2 - \epsilon)^2} \right] \mathbf{1}_{I_X}(\epsilon), \quad (1379)$$

and its peak is

$$\Phi_{\rho}^{\max}(X) = \frac{2r_1(X)}{r_1(X) + 1}. \quad (1380)$$

These formulas are exact for every $X \geq 0$ and contain no fitted coefficient.

Proof. Monotonicity gives $\Phi_W = e_-(1, X) - e_-(0, X)$, which is Eq. (1376). Theorem 23 gives $\Phi_\Delta = g(1, X)$ and $\Phi_V = D_G \partial_q e_-(0, X)$; since $D_G = 1$, these are Eqs. (1377) and (1378).

The secular equation on the lower branch can be inverted exactly:

$$q_X(\epsilon) = \epsilon + \frac{4X^2}{2 - \epsilon}, \quad \epsilon \in I_X. \quad (1381)$$

The pushforward of dq therefore has density $dq_X/d\epsilon$, proving Eq. (1379). Its normalization is

$$\int_{\mathbb{R}} \Phi_\rho(\epsilon, X) d\epsilon = q_X(e_-(1, X)) - q_X(e_-(0, X)) = 1. \quad (1382)$$

For $X > 0$, the density increases with ϵ ; for $X = 0$, it is constant on $[0, 1]$. At the upper endpoint,

$$\begin{aligned} \Phi_\rho^{\max} &= \frac{1}{\partial_q e_-(1, X)} \\ &= 1 + \frac{4X^2}{[2 - e_-(1, X)]^2} = \frac{2r_1}{r_1 + 1}, \end{aligned} \quad (1383)$$

which proves the peak formula. $\square$

Corollary 13 (Endpoint checks and plot-ready classification). *At zero coupling,*

$$\begin{aligned} (\Phi_W, \Phi_\Delta, \Phi_V, \Phi_\rho^{\max})(0) &= (1, 1, 1, 1), \\ \Phi_\rho(\epsilon, 0) &= \mathbf{1}_{[0,1]}(\epsilon). \end{aligned} \quad (1384)$$

At strong coupling,

$$\Phi_W \rightarrow \frac{1}{2}, \quad \frac{\Phi_\Delta}{2X} \rightarrow 1, \quad \Phi_V \rightarrow \frac{1}{2}, \quad \Phi_\rho^{\max} \rightarrow 2. \quad (1385)$$

Moreover, $1/2 < \Phi_W(X) \leq 1$ for finite $X \geq 0$. Thus Eqs. (1375)–(1380) are plot-ready on $X \in [0, \infty)$ and $\epsilon \in \mathbb{R}$.

Proof. Substitution of $r_0(0) = 2$ and $r_1(0) = 1$ proves Eq. (1384). The asymptotics $r_0 = 4X + O(X^{-1})$ and $r_1 = 4X + O(X^{-1})$ prove Eq. (1385). Finally,

$$\Phi'_W(X) = 8X \left(\frac{1}{r_0} - \frac{1}{r_1} \right) < 0 \quad (X > 0), \quad (1386)$$

and its limiting values are 1 and 1/2. $\square$

The symbol X in this benchmark is a raw normalized coupling coordinate. There is no α_* with which to form $X = \alpha/\alpha_*$, and $X = 1$ is not a magic point.

Proposition 13 (Exact implicit data for the full square first star). *Let $d_0(p)$ be the central diagonal, let $d_a(p)$, $1 \leq a \leq 4$, be the satellite diagonals of the momentum-dependent equal-amplitude square first star, and let every central-satellite coupling be α . Its characteristic polynomial is*

$$P_5(z, p; \alpha) = (z - d_0) \prod_{a=1}^4 (z - d_a)$$

$$- \alpha^2 \sum_{a=1}^4 \prod_{\substack{b=1 \\ b \neq a}}^4 (z - d_b). \quad (1387)$$

Define

$$S(z, p; \alpha) = z - d_0 - \alpha^2 \sum_{a=1}^4 \frac{1}{z - d_a}. \quad (1388)$$

Where the Schur representation is regular, the exact resolvent entries are

$$G_{cc} = S^{-1} = \frac{\prod_a (z - d_a)}{P_5(z, p; \alpha)}, \quad (1389)$$

$$G_{ca} = \frac{\alpha S^{-1}}{z - d_a}, \quad (1390)$$

$$G_{ab} = \frac{\delta_{ab}}{z - d_a} + \frac{\alpha^2 S^{-1}}{(z - d_a)(z - d_b)}. \quad (1391)$$

The polynomial identity distinguishes removable Schur poles from genuine dark-state poles: when satellite diagonals coincide, a surviving pole at that common value is physical. At generic momentum the spectral data are an implicit quintic solution; they supply neither a radical formula nor unproved global-extremum formulas.

Proof. Write $z - H$ in central/satellite block form. The determinant identity $\det(z - H) = \det(z - D_s)S$, with $D_s = \text{diag}(d_1, \dots, d_4)$, gives Eq. (1387). The block-inverse identity gives Eqs. (1389)–(1391). Multiplication by the satellite product gives polynomial identities that separate removable factors from genuine spectral factors. $\square$

The exact square first star remains root-free: its target quadratic curvature is positive for every $\alpha \geq 0$. The implicit formulas above do not create a magic root.

1. established envelopes for a richer fixed-fiber model

Let $H_f(q, X)$ be a Hermitian family on the same fixed transported fiber and define

$$\epsilon_0(X) := \sup_{q \in [0,1]} \|H_f(q, X) - H_{\text{bd}}(q, X)\|_{\text{op}}. \quad (1392)$$

Theorem 24 (C^0 spectral and C^1 velocity envelopes). *Track the lowest simple branch of H_f against e_- . Its bandwidth and uniform closest-complement gap obey*

$$\max\{0, \Phi_W - 2\epsilon_0\} \leq \Phi_W^f \leq \Phi_W + 2\epsilon_0, \quad (1393)$$

$$\max\{0, \Phi_\Delta - 2\epsilon_0\} \leq \Phi_\Delta^f \leq \Phi_\Delta + 2\epsilon_0. \quad (1394)$$

In particular, $2\epsilon_0 < \Phi_\Delta$ preserves uniform isolation.

Assume additionally that both families are C^1 in q and set

$$\epsilon_1(X) := \sup_q \|\partial_q H_f(q, X) - \partial_q H_{\text{bd}}(q, X)\|_{\text{op}}. \quad (1395)$$

If $2\epsilon_0 < \Phi_\Delta$, then

$$|\Phi_V^f(X) - \Phi_V(X)| \leq \epsilon_1(X) + \frac{4\epsilon_0(X)}{\Phi_\Delta(X) - 2\epsilon_0(X)}. \quad (1396)$$

No velocity conclusion follows from ϵ_0 alone.

Proof. Weyl's inequality gives uniform error at most ϵ_0 for every ordered eigenvalue. Each band edge moves by at most ϵ_0 , so a bandwidth moves by at most $2\epsilon_0$. At fixed q , each of the two eigenvalues defining the isolation gap moves by at most ϵ_0 ; taking the infimum in q proves the gap envelope.

For the derivative statement, a circle of radius $\Phi_\Delta/2$ about $e_-(q, X)$ and the resolvent identity give the uniform rank-one projector bound

$$\|P_f(q, X) - P_{bd}(q, X)\|_{\text{op}} \leq \frac{2\epsilon_0}{\Phi_\Delta - 2\epsilon_0}. \quad (1397)$$

Since $\partial_q H_{bd} = |c\rangle\langle c|$ has norm one, the Hellmann–Feynman formula and $\|P_f - P_{bd}\|_1 = 2\|P_f - P_{bd}\|_{\text{op}}$ yield the pointwise slope error on the right of Eq. (1396). Taking suprema, and using $|\sup |u| - \sup |v|| \leq \sup |u - v|$, proves the result because $D_G = 1$. $\square$

For $\eta > 0$, let

$$L_\eta(x) = \frac{\eta}{\pi(x^2 + \eta^2)}, \quad \Phi_{\rho, \eta}(\epsilon) = \int_0^1 L_\eta(\epsilon - e(q)) dq. \quad (1398)$$

Proposition 14 (Lorentzian DOS envelope). *For the matched target branches in Theorem 24,*

$$\|\Phi_{\rho, \eta}^f - \Phi_{\rho, \eta}^{bd}\|_\infty \leq \frac{3\sqrt{3}}{8\pi\eta^2} \epsilon_0. \quad (1399)$$

The broadened peak heights obey the same bound, and each broadened density integrates to one on $\mathbb{R}$. No uniform envelope for the unbroadened DOS follows from C^0 operator control alone.

Proof. Direct differentiation gives

$$|L'_\eta(x)| = \frac{2\eta|x|}{\pi(x^2 + \eta^2)^2}. \quad (1400)$$

Its maximum occurs at $|x| = \eta/\sqrt{3}$ and equals

$$\|L'_\eta\|_\infty = \frac{3\sqrt{3}}{8\pi\eta^2}. \quad (1401)$$

Weyl gives $|e_f(q) - e_-(q)| \leq \epsilon_0$. The mean-value theorem, followed by integration over the unit-mass measure dq , proves Eq. (1399). The supremum of two functions changes by at most their uniform distance. Fubini's theorem and $\int_{\mathbb{R}} L_\eta = 1$ prove normalization.

For the obstruction, use the zero-coupling target $e(q) = q$ and

$$e_n(q) = q + \frac{1}{\pi n} \sin(2\pi nq). \quad (1402)$$

Then $\|e_n - e\|_\infty \rightarrow 0$, but $e'_n(q) = 1 + 2\cos(2\pi nq)$ has nondegenerate zeros. The pushforward density therefore has divergent one-dimensional van Hove peaks, while the reference density is bounded. This excludes an unbroadened uniform-DOS bound based only on C^0 spectral error. $\square$

Corollary 14 (Repair of Theorem 162). *The perturbative expression based on $|1 - X_{\text{pc}}^2|$ is not the exact five-state master curve, and its formal zero is not an exact magic point. For the fixed bright–dark class it is replaced by the exact root-free curves in Corollary 12. For a richer model, Eqs. (1393), (1394), (1396), and (1399) give established statements instead of invented closed forms. Equality of master curves across microscopic models still requires actual unitary equivalence, or a proved vanishing residual under one fixed identification; equality of low-order coefficients is insufficient.*

The exact benchmark has $\Phi_W > 0$ and hence belongs to a finite-scaled-bandwidth class. Classification of the full physical model at a separately established magic root is deferred. A root-normalized coordinate is available only for a richer model satisfying the independent simple-root, gap, fixed-fiber, overlap, and classwise convergence hypotheses.

U. Exact flatness obstruction and quartic normal form

The equal-amplitude square first star has no quadratic-flattening root. Accordingly, a root-normalized coordinate $X = \alpha/\alpha_\star$ and the phrase “at the magic root $X = 1$ ” are unavailable for this target. We can, however, decide the manuscript's formal perturbative point exactly.

Write

$$E_-(p, \alpha) = |p|^2 + z(p, \alpha), \quad r = \sqrt{1 + 16\alpha^2}, \quad (1403)$$

$$\lambda_- = \frac{1 - r}{2}.$$

Eliminating the four satellites gives the exact Schur equation

$$0 = F(z, p; \alpha) := z - 2\alpha^2(z - 1) \times \left[\frac{1}{(z - 1)^2 - 4p_x^2} + \frac{1}{(z - 1)^2 - 4p_y^2} \right]. \quad (1404)$$

The target root $z(0, \alpha) = \lambda_-$ is simple for every $\alpha \geq 0$.

Theorem 25 (Exact quartic normal form). *Without expanding in α , the analytic target branch has the complete C_4 -allowed expansion*

$$E_-(p, \alpha) = \lambda_- + c(r)(p_x^2 + p_y^2) + A_4(r)(p_x^4 + p_y^4) + B_4(r)p_x^2 p_y^2 + O(|p|^6), \quad (1405)$$

where

$$c(r) = \frac{r^2 - r + 2}{r(r+1)}, \quad (1406)$$

$$A_4(r) = -\frac{4(r-1)(3r+1)}{r^3(r+1)^2}, \quad (1407)$$

$$B_4(r) = \frac{8(r-1)^2(5r+1)}{r^3(r+1)^3}. \quad (1408)$$

In the tensor convention

$$E_- = \lambda_- + \frac{1}{2}H_{ij}p_i p_j + \frac{1}{24}T_{ijkl}p_i p_j p_k p_l + O(|p|^6), \quad (1409)$$

the nonzero independent components are

$$H_{xx} = H_{yy} = 2c, \quad H_{xy} = 0, \quad (1410)$$

$$T_{xxxx} = T_{yyyy} = 24A_4, \quad T_{xxyy} = 4B_4. \quad (1411)$$

Every permutation of T_{xxyy} has the displayed value. Components with an odd multiplicity of either coordinate vanish, as do all odd-order derivatives at the origin.

Proof. Put

$$S_2 = p_x^2 + p_y^2, \quad S_4 = p_x^4 + p_y^4, \quad P_{22} = p_x^2 p_y^2, \quad (1412)$$

and $a = \alpha^2$. Expanding only the momentum denominators in Eq. (1404) gives

$$F = f(z) + g(z)S_2 + h(z)S_4 + O(|p|^6), \quad (1413)$$

with

$$f = z - \frac{4a}{z-1}, \quad g = -\frac{8a}{(z-1)^3}, \quad h = -\frac{32a}{(z-1)^5}. \quad (1414)$$

At $z = \lambda_-$, use $4a = \lambda_- (\lambda_- - 1)$ to obtain

$$\begin{aligned} f' &= \frac{2r}{r+1}, f'' = g = \frac{4(r-1)}{(r+1)^2}, \\ g' &= \frac{24(r-1)}{(r+1)^3}, h = \frac{64(r-1)}{(r+1)^4}. \end{aligned} \quad (1415)$$

Set

$$z = \lambda_- + \beta S_2 + A_4 S_4 + B_4 P_{22} + O(|p|^6). \quad (1416)$$

The coefficients of S_2 , S_4 , and P_{22} respectively give

$$f'\beta + g = 0, \quad (1417)$$

$$f'A_4 + \frac{1}{2}f''\beta^2 + g'\beta + h = 0, \quad (1418)$$

$$f'B_4 + f''\beta^2 + 2g'\beta = 0. \quad (1419)$$

Solving first yields

$$\beta = -\frac{2(r-1)}{r(r+1)}, \quad c = 1 + \beta, \quad (1420)$$

and substitution in the last two equations gives Eqs. (1407) and (1408). Inversion removes odd powers; C_4 symmetry leaves precisely S_2 , S_4 , and P_{22} through fourth order. Differentiating the normal form proves the tensor components. $\square$

The manuscript's perturbative square-star coefficient is 8. Thus

$$\alpha_{\text{pc}} = \frac{1}{\sqrt{8}}, \quad X_{\text{pc}} = \frac{\alpha}{\alpha_{\text{pc}}}, \quad X_{\text{pc}} = 1 \implies r = \sqrt{3}. \quad (1421)$$

Corollary 15 (Obstruction at the formal perturbative point). *At $X_{\text{pc}} = 1$,*

$$c_{\text{pc}} = 3 - \frac{4\sqrt{3}}{3} > 0, \quad (1422)$$

$$A_{4,\text{pc}} = 8 - \frac{44\sqrt{3}}{9} \neq 0, \quad (1423)$$

$$B_{4,\text{pc}} = -112 + \frac{584\sqrt{3}}{9} \neq 0. \quad (1424)$$

Hence

$$\text{Hess}_p E_-(0, \alpha_{\text{pc}}) = 2c_{\text{pc}} I_2 > 0. \quad (1425)$$

The quartic tensor is nonzero but subleading. On every momentum domain that contains the origin as an interior point, the exact target bandwidth is strictly positive. Therefore the fixed binary question at the formal coordinate has the answer

$$\Phi_W^\square(X_{\text{pc}} = 1) > 0, \quad (1426)$$

not zero.

Proof. Substitution of $r = \sqrt{3}$ in Eqs. (1406)–(1408) proves the displayed values. The irrational parts show directly that both quartic coefficients are nonzero. Equation (1425) and analyticity imply a radius $\rho > 0$ such that

$$E_-(p, \alpha_{\text{pc}}) - E_-(0, \alpha_{\text{pc}}) \geq \frac{c_{\text{pc}}}{2}|p|^2 > 0 \quad (1427)$$

whenever $0 < |p| \leq \rho$. A domain containing the origin as an interior point contains such a momentum, so its maximum and minimum target energies cannot coincide. $\square$

Theorem 26 (Sharp all-coupling Hessian floor). *For every $\alpha \geq 0$,*

$$c(\alpha) \geq m_\square := 4\sqrt{2} - 5 > 0, \quad (1428)$$

and equality holds exactly when

$$\alpha_m^2 = \frac{1 + \sqrt{2}}{8}, \quad r_m = 1 + \sqrt{2}. \quad (1429)$$

Consequently, in quadratic-form order,

$$\text{Hess}_p E_-(0, \alpha) \succeq 2(4\sqrt{2} - 5)I_2, \quad (1430)$$

and this constant is sharp. At the optimizer,

$$A_4 = 28 - 20\sqrt{2}, \quad B_4 = 600 - 424\sqrt{2}, \quad (1431)$$

but the quartic tensor never becomes the leading local dispersion.

Proof. Differentiating Eq. (1406) with respect to r gives

$$\frac{dc}{dr} = \frac{2(r^2 - 2r - 1)}{(r^2 + r)^2}. \quad (1432)$$

On $r \geq 1$, its unique zero is $r_m = 1 + \sqrt{2}$, where the derivative changes from negative to positive. Substitution gives $c(r_m) = 4\sqrt{2} - 5$ and $r_m^2 = 1 + 16\alpha_m^2$, proving the minimum and its coupling. Equation (1410) proves the sharp matrix bound. Substitution of r_m in the quartic formulas gives Eq. (1431); the strictly positive Hessian floor keeps them subleading. $\square$

Corollary 16 (Exact target classification). *On every compact momentum domain containing the origin as an interior point, the target branch has finite, strictly positive normalized bandwidth for every $\alpha \geq 0$. The exact equal-amplitude square first star is therefore a root-free residual-quadratic, finite-scaled-bandwidth class. It is neither continuum-exact-flat nor a established magic band, and no global closed bandwidth formula is inferred from its generic quintic characteristic polynomial.*

Proof. Equation (1430) makes the origin a strict local minimum of the analytic branch at every coupling. The argument in Eq. (1427) proves positive bandwidth. Continuity on a compact momentum domain proves finiteness. Since the curvature never vanishes, there is no quadratic magic root and no quartic-leading local regime. $\square$

Proposition 15 (Separate bright–dark raw-coordinate check). *For the exactly solvable bright–dark benchmark,*

$$\Phi_W^{\text{bd}}(1) = \frac{1 + 2\sqrt{5} - \sqrt{17}}{2} > \frac{1}{2}. \quad (1433)$$

This is a raw-coupling evaluation in a different fixed class, not a magic-root classification of the momentum-dependent square first star.

Proof. Insert $X = 1$ in $\Phi_W^{\text{bd}} = [1 + \sqrt{4 + 16X^2} - \sqrt{1 + 16X^2}]/2$. The strict inequality follows from $2\sqrt{5} > \sqrt{17}$. $\square$

Theorem 27 (Richer-model residual and identity decisions). *Suppose a richer model independently supplies a simple, gapped root, a fixed transported fiber, and a declared identification with the bright–dark benchmark at root-normalized $X = 1$. Define*

$$\epsilon_0(1) := \sup_p \|H_{\text{full}}(p, 1) - H_{\text{bd}}(p, 1)\|_{\text{op}}. \quad (1434)$$

If

$$\epsilon_0(1) < \frac{1}{2}\Phi_W^{\text{bd}}(1), \quad (1435)$$

then

$$\Phi_W^{\text{full}}(1) \geq \Phi_W^{\text{bd}}(1) - 2\epsilon_0(1) > 0. \quad (1436)$$

Equation (1435) is a sufficient test for positive bandwidth.

For the full square-star characteristic polynomial P_5 , exact flatness of a simple, uniformly isolated target branch at energy E_ is equivalent to the polynomial identity*

$$P_5(E_*, p; \alpha_*) \equiv 0 \quad (1437)$$

on a momentum domain with nonempty interior, together with continuous target selection. Equivalently, every coefficient in p_x, p_y vanishes, or $(z - E_)$ divides $P_5(z, p; \alpha_*)$ as a polynomial. Finite-order curvature cancellation and a small nonzero residual do not prove this identity.*

Proof. The bandwidth envelope for two uniformly ϵ_0 -close Hermitian families gives $|\Phi_W^{\text{full}} - \Phi_W^{\text{bd}}| \leq 2\epsilon_0$, proving the first claim. If the target is exactly flat, its constant eigenvalue makes the characteristic polynomial vanish for every momentum, proving necessity of Eq. (1437). Conversely, the identity makes E_* an eigenvalue at every momentum; continuous target selection, simplicity, and the uniform gap identify it with one constant target branch. A polynomial that vanishes on an open set is identically zero, which is equivalent to coefficientwise vanishing and to the stated factorization. $\square$

Corollary 17 (Scope repairs for Theorems 162, 164, and 173). *The zero in Theorem 162's truncated $|1 - X_{\text{pc}}^2|$ curve is a perturbative artifact; the exact result is Eq. (1425). The exact-flat/quartic premise of Theorem 164 is false for the stated target, so its quartic residual-bandwidth law does not apply. Neither magic subclass in Theorem 173 applies literally, because this target has no magic root. Along a nonmagic path of fixed raw coupling, the surviving quadratic scale may yield inverse-square length scaling, but it is not a magic length–bandwidth law.*

For an independently enriched model, the finite-bandwidth versus exact-flat dichotomy remains conditional on the simple-root, derivative, gap, fixed-fiber, overlap, residual, and polynomial-identity hypotheses. Complete hyperbolic tangent-space cancellation and thermodynamic gap isolation are separate problems.

V. Multidirectional self-energy cancellation analysis

Let T_{phys} be the gauge-quotiented physical tangent space equipped with its Hodge metric G . Three objects must be kept distinct: the exact energy covector $dE \in T_{\text{phys}}^*$, the exact covariant Hessian $\nabla^2 E$ at a stationary representation, and the perturbative normalized Hessian response $I - \alpha^2 C$. A path derivative tests only one component of the first object, while the third is only a leading coupling expansion of the second.

On the fixed transported active fiber, write

$$H = \begin{pmatrix} H_P & \alpha V \\ \alpha V^* & H_Q \end{pmatrix}, \quad R(E) = (E - H_Q)^{-1}, \quad (1438)$$

where $E \notin \text{spec } H_Q$. Let D_a denote differentiation along a Hodge-orthonormal physical tangent basis X_a , after the fixed fiber transport.

Proposition 16 (Exact Feshbach tangent derivative). *At fixed E and fixed scalar α , the self-energy*

$$\Sigma(E) = \alpha^2 V R(E) V^* \quad (1439)$$

has the exact derivative

$$D_a \Sigma = \alpha^2 [(D_a V) R V^* + V R (D_a H_Q) R V^* + V R (D_a V)^*]. \quad (1440)$$

For a normalized P -sector eigenvector of the effective problem,

$$dE[X_a] = Z_{\text{eff}} \langle \psi_P, (D_a H_P + D_a \Sigma) \psi_P \rangle, \quad (1441)$$

where

$$Z_{\text{eff}} = \langle \psi_P, (I - \partial_E \Sigma) \psi_P \rangle^{-1} > 0, \quad (1442)$$

$$\partial_E \Sigma = -\alpha^2 V R(E)^2 V^*. \quad (1443)$$

Thus Z_{eff} handles energy dependence and does not change the zero set of dE . Neither the profile derivatives $D_a V$ nor the complementary derivative $D_a H_Q$ may be omitted.

Proof. For $A = E - H_Q$, differentiation of A^{-1} at fixed E gives

$$D_a R = -R(D_a A)R = R(D_a H_Q)R. \quad (1444)$$

The product rule proves Eq. (1440). Differentiating the nonlinear effective eigenproblem $(H_P + \Sigma(E))\psi_P = E\psi_P$ and applying the Hellmann–Feynman identity gives Eq. (1441). Since $\partial_E R = -R^2$, Eq. (1443) follows. Off resonance, $R = R^*$ and $I - \partial_E \Sigma = I + \alpha^2 V R^2 V^* > 0$, proving positivity of Z_{eff} . $\square$

Write the leading fixed-energy expansion as $\Sigma = \alpha^2 \Sigma^{(2)} + O(\alpha^4)$ and define the Hodge-dual covectors

$$b_a = \langle \psi_P, D_a H_P \psi_P \rangle, \quad s_a = \langle \psi_P, D_a \Sigma^{(2)} \psi_P \rangle. \quad (1445)$$

Theorem 28 (Leading-order one-scalar Hodge-covector cancellation). *Let $t = \alpha^2 \geq 0$. If $b \neq 0$, a positive scalar solves*

$$b + ts = 0 \quad \text{in } T_{\text{phys}}^* \quad (1446)$$

if and only if

$$s = -cb \quad \text{for one } c > 0; \quad t = \frac{1}{c}. \quad (1447)$$

In components, $b_a = 0$ requires $s_a = 0$, and all nonzero ratios $-b_a/s_a$ must be the same positive number. If $b = s = 0$, every t solves Eq. (1446); if $b = 0 \neq s$, only $t = 0$ solves it; if $b \neq 0 = s$, there is no solution.

For $s \neq 0$, the best nonnegative least-squares scalar is

$$t_{\text{ls}} = \max \left\{ 0, -\frac{\langle b, s \rangle_{G^{-1}}}{\|s\|_{G^{-1}}^2} \right\}. \quad (1448)$$

If $\langle b, s \rangle_{G^{-1}} < 0$, its squared residual is

$$\min_{t \geq 0} \|b + ts\|_{G^{-1}}^2 = \|b\|_{G^{-1}}^2 - \frac{\langle b, s \rangle_{G^{-1}}^2}{\|s\|_{G^{-1}}^2}. \quad (1449)$$

If that inner product is nonnegative, the minimum is $\|b\|_{G^{-1}}^2$ at $t = 0$. In every case,

$$\|b + ts\|_{G^{-1}} = \sup_{\|X\|_G=1} |b[X] + ts[X]|. \quad (1450)$$

Proof. For $b \neq 0$ and $t > 0$, Eq. (1446) is equivalent to $s = -b/t$, proving the collinearity, sign, component, and ratio conditions. The zero cases follow by direct substitution. For $s \neq 0$, expand

$$\|b + ts\|_{G^{-1}}^2 = \|b\|_{G^{-1}}^2 + 2t\langle b, s \rangle_{G^{-1}} + t^2\|s\|_{G^{-1}}^2. \quad (1451)$$

Projection of its unconstrained minimizer onto $[0, \infty)$ proves Eqs. (1448) and (1449). The final identity is the defining duality between the Hodge norm and its dual norm. $\square$

At an exact stationary representation $dE = 0$, define the exact Hessian $\mathcal{M} = \nabla^2 E$. After a declared positive normalization, its leading coupling response may take the form

$$K(t) = I - tC + O(t^2), \quad t = \alpha^2, \quad (1452)$$

where C is Hodge-self-adjoint and positive semidefinite. The following statement concerns the leading operator $I - tC$, not the exact Hessian.

Theorem 29 (Sharp response-operator anisotropy bound). *Complete leading quadratic cancellation occurs at one $t \geq 0$ if and only if*

$$C = cI, \quad c > 0, \quad t = \frac{1}{c}. \quad (1453)$$

If the active eigenvalues satisfy $0 < c_{\min} \leq c_a \leq c_{\max}$, then

$$\inf_{t \geq 0} \|I - tC\|_{\text{op}} = \frac{c_{\max} - c_{\min}}{c_{\max} + c_{\min}}, \quad (1454)$$

attained at

$$t_{\text{opt}} = \frac{2}{c_{\min} + c_{\max}}. \quad (1455)$$

If C has a zero eigenvalue, then

$$\inf_{t \geq 0} \|I - tC\|_{\text{op}} = 1; \quad (1456)$$

the zero-eigenvalue direction is uncancellable.

Proof. The identity $I - tC = 0$ requires $t > 0$ and $C = t^{-1}I$, proving the first claim. By the spectral theorem,

$$\|I - tC\|_{\text{op}} = \max_a |1 - tc_a| = \max\{|1 - tc_{\min}|, |1 - tc_{\max}|\}. \quad (1457)$$

The minimum occurs when the two extreme residuals have equal magnitude and opposite sign, which gives Eqs. (1454) and (1455). A zero eigenvalue leaves an eigenvalue one in $I - tC$ for every t , while $t = 0$ attains norm one. $\square$

Corollary 18 (Exact square-star analysis). *For the four unit-amplitude channels $g_a \in \{+e_x, -e_x, +e_y, -e_y\}$,*

$$\begin{aligned} C_{mn} &= 4 \sum_a \frac{g_{am} g_{an}}{|g_a|^6} \\ &= 8\delta_{mn}. \end{aligned} \quad (1458)$$

Thus C_4 symmetry locks the two perturbative directional roots at $\alpha = 1/\sqrt{8}$. Nonperturbatively, however,

$$\begin{aligned} \text{Hess}_p E_-(0, \alpha) &= 2c(\alpha)I_2, \\ c(\alpha) &\geq 4\sqrt{2} - 5 > 0 \quad (\alpha \geq 0). \end{aligned} \quad (1459)$$

Moreover,

$$c(1/\sqrt{8}) = 3 - \frac{4\sqrt{3}}{3} > 0. \quad (1460)$$

Hence the exact target square star has no directional root and no complete root.

Proof. The two x channels contribute $2e_x e_x^T$, the two y channels contribute $2e_y e_y^T$, and the prefactor four gives $C = 8I_2$. The leading response $I - \alpha^2 C$ therefore vanishes at $\alpha = 1/\sqrt{8}$. The exact Hessian bound and its value at that coupling prove that this perturbative zero does not survive nonlinear reconstruction. $\square$

Theorem 30 (Symmetry and isotypic locking). *Let a finite or compact stabilizer K act orthogonally on the active Hodge tangent space, and suppose the Hamiltonian and response construction are K -equivariant. Then C lies in the commutant of the K -action. On an isotypic decomposition*

$$T_{\text{act}} \cong \bigoplus_{\lambda} W_{\lambda} \otimes M_{\lambda}, \quad (1461)$$

it has the form

$$C = \bigoplus_{\lambda} I_{W_{\lambda}} \otimes A_{\lambda}, \quad (1462)$$

where A_{λ} is self-adjoint on the multiplicity space over the appropriate real Schur algebra. Symmetry makes a multiplicity-one irreducible response scalar within that block. It neither equates scalars on inequivalent blocks nor removes a nontrivial multiplicity-space matrix. Global locking requires $A_{\lambda} = cI$ with the same c for every active λ .

Proof. Equivariance and differentiation imply $CU(k) = U(k)C$ for every $k \in K$. The real isotypic decomposition and Schur's lemma identify the commutant as Eq. (1462). Self-adjointness restricts each multiplicity operator to its self-adjoint part. No intertwiner connects inequivalent isotypic components, so symmetry supplies no equality between their response eigenvalues. $\square$

Proposition 17 (Trace-normalization insufficiency). *Let the active tangent dimension be $D \geq 2$ and fix $S > 0$. The matrices*

$$C_{\text{iso}} = \frac{S}{D}I, \quad (1463)$$

$$C_{\text{aniso}} = \text{diag}(S - (D-1)\varepsilon, \varepsilon, \dots, \varepsilon), \quad (1464)$$

with $0 < \varepsilon < S/D$, are positive definite and have the same trace S , but only the first is scalar. Therefore a trace, total Parseval power, or mean-square tunneling normalization cannot certify eigenvalue locking or complete cancellation.

Proof. Both traces equal S . All eigenvalues of C_{aniso} are positive, and its first eigenvalue differs from the remaining ones because $\varepsilon < S/D$. Theorem 29 then distinguishes its nonzero worst-direction residual from exact cancellation for C_{iso} . $\square$

The affine equation $b + ts = 0$ is strictly a leading-order, fixed-profile test. In the full Feshbach problem, $E(t)$, $\psi_P(t)$, and the profile derivatives entering $D_a V(t)$ and $D_a H_Q(t)$ generally depend on t . Exact cancellation therefore requires a common zero of the full covector $dE(t)$, not merely a zero of its affine truncation.

Proposition 18 (Nonlinear block-root obstruction). *Let*

$$f_{\lambda}(t) = 1 - c_{\lambda}t + R_{\lambda}(t), \quad c_{\lambda} > 0, \quad |R_{\lambda}(t)| \leq \eta < 1. \quad (1465)$$

Every zero of f_{λ} lies in

$$I_{\lambda} = \left[\frac{1 - \eta}{c_{\lambda}}, \frac{1 + \eta}{c_{\lambda}} \right]. \quad (1466)$$

A common nonlinear root requires $\bigcap_{\lambda} I_{\lambda} \neq \emptyset$. Empty intersection proves that the bounded nonlinear corrections cannot align the block roots. In particular, for $c_{\lambda} < c_{\mu}$, the explicit condition

$$\frac{c_{\mu}}{c_{\lambda}} > \frac{1 + \eta}{1 - \eta} \quad (1467)$$

makes I_{λ} and I_{μ} disjoint. Nonempty intersection is necessary but not sufficient for a common root.

Proof. At a zero, $c_{\lambda}t = 1 + R_{\lambda}(t)$ lies in $[1 - \eta, 1 + \eta]$, proving the interval. A common zero must belong to every interval. Under Eq. (1467), $(1 - \eta)/c_{\lambda} > (1 + \eta)/c_{\mu}$, so the two intervals are disjoint. The remainder functions need not vanish at the same point even when their intervals overlap, proving the final qualification. $\square$

Corollary 19 (Classification and repairs of Theorems 152 and 153). *The exact square target is perturbatively locked but nonperturbatively root-free; neither directional nor complete magic exists. For the full hyperbolic target presently specified, complete cancellation is not established: the matrices $D_a H_P$, $D_a H_Q$, $D_a V$, or equivalent channel covectors are not supplied over all of $H^1(\Gamma_M, \text{Ad } \rho)$. Consequently, no numerical full hyperbolic C_{ab} , all-block locking, or nonlinear common root follows. Only directional candidates may be reported after those data are computed.*

For a richer fixed-fiber model, complete magic remains conditional on an exact common zero of the full Hodge covector and the required exact Hessian/operator response, together with independent simple-root, transport, overlap, gap, and C^1 hypotheses. Theorem 152 is therefore a conditional implication. Pointwise $dE = 0$ gives pointwise stationarity; if $\sup_{\rho \in \mathcal{D}} \|dE(\rho)\|_{G^{-1}} \leq \varepsilon$ on a connected scan domain of intrinsic Hodge diameter $L_{\mathcal{D}}$, then $\text{BW}_{\mathcal{D}}(E) \leq L_{\mathcal{D}}\varepsilon$. This is not a model-specific proof of cancellation. Theorem 153 is a directional leading-order criterion and becomes complete only after global locking or a proved nonlinear common root. No thermodynamic isolation-gap statement is made here.

W. Quantitative thermodynamic isolation-gap verification

The exact equal-amplitude square first star has no quadratic-flattening root. Hence a gap at its magic root has no evaluation point. The formal perturbative coordinate $\alpha_{\text{pc}} = 1/\sqrt{8}$ remains useful for an exact local calculation, but it is not a magic coupling.

Proposition 19 (Formal-point local gap, without a magic root). *At $\alpha_{\text{pc}} = 1/\sqrt{8}$, the five-state point gap is*

$$\delta_{\text{pc}} = \frac{1 + \sqrt{3}}{2}. \quad (1468)$$

For $|p| \leq \rho$, its local target-to-complement gap obeys

$$g_{\square}(p, \alpha_{\text{pc}}) \geq \delta_{\text{pc}} - 2(\rho^2 + 2\rho). \quad (1469)$$

It is therefore positive whenever

$$0 \leq \rho < \rho_{\text{pc}}, \quad \rho_{\text{pc}} := \frac{\sqrt{5 + \sqrt{3}}}{2} - 1. \quad (1470)$$

This is a local finite five-state estimate, not a thermodynamic full-domain gap theorem.

Proof. The exact point gap $[1 + \sqrt{1 + 16\alpha^2}]/2$ gives Eq. (1468). The momentum perturbation satisfies

$$\|H_{\square}(p, \alpha) - H_{\square}(0, \alpha)\|_{\text{op}} \leq |p|^2 + 2|p|. \quad (1471)$$

Weyl's inequality can move both the target edge and the closest complementary edge, proving Eq. (1469).

Its right-hand side is positive exactly when $(\rho + 1)^2 < (5 + \sqrt{3})/4$, which proves Eq. (1470). The already established strictly positive curvature excludes a magic-root interpretation. $\square$

The thermodynamic comparison uses a distinct fixed-class reference.

Proposition 20 (Uniform bright-dark gap at raw $X = 1$). *For $q \in [0, 1]$, the exact bright-dark target gap at raw coupling $X = 1$ is*

$$g_{\text{bd}}(q, 1) = \frac{2 - q + \sqrt{(2 - q)^2 + 16}}{2}. \quad (1472)$$

It is decreasing in q and has the uniform value

$$g_{\text{bd}}(1) := \inf_{q \in [0, 1]} g_{\text{bd}}(q, 1) = \frac{1 + \sqrt{17}}{2} > 0. \quad (1473)$$

This statement does not identify raw $X = 1$ with a magic point.

Proof. The closest complement of the lower bright branch is the dark level at energy two. Differentiating Eq. (1472) gives

$$\partial_q g_{\text{bd}} = -\frac{1}{2} \left[1 + \frac{2 - q}{\sqrt{(2 - q)^2 + 16}} \right] < 0. \quad (1474)$$

The minimum is therefore attained at $q = 1$, which proves the result. $\square$

For the N th symmetry-resolved finite cover, let $E_{\text{kin}, N} > 0$ and define

$$\hat{H}_N(\rho) := \frac{H_N(\rho)}{E_{\text{kin}, N}}, \quad \rho \in \mathcal{D}_N. \quad (1475)$$

After the predeclared unitary transport, take the dimensionless reference

$$\hat{H}_{\text{ref}, N}(\rho) = H_{\text{bd}}(q_N(\rho), 1) \oplus D_N(\rho), \quad q_N(\rho) \in [0, 1]. \quad (1476)$$

The reference target is the simple lower bright branch. Require the global added-sector separation

$$d_{\perp} := \inf_N \inf_{\rho, \rho' \in \mathcal{D}_N} \text{dist}(e_{-}(q_N(\rho), 1), \text{spec } D_N(\rho')) > 0, \quad (1477)$$

$$g_0 := \min \left\{ \frac{1 + \sqrt{17}}{2}, d_{\perp} \right\} > 0. \quad (1478)$$

The two scan variables in Eq. (1477) ensure an indirect, full-energy-set separation rather than only a pointwise one.

Assume the normalized discrepancy decomposes as

$$\hat{H}_N - \hat{H}_{\text{ref}, N} = R_{\text{act}, N} + R_{\text{off}, N} + R_{\text{tail}, N} + R_{\text{cov}, N}, \quad (1479)$$

where the terms denote active mismatch, off-sector coupling, normalized physical tail/cutoff, and cover/embedding error. Define

$$\begin{aligned} \varepsilon_N := & \varepsilon_{\text{act},N} + \varepsilon_{\text{off},N} \\ & + \varepsilon_{\text{tail},N} + \varepsilon_{\text{cov},N}, \end{aligned} \quad (1480)$$

with each summand bounding the uniform operator norm of its corresponding term. In particular,

$$\sup_{\rho \in \mathcal{D}_N} \|\hat{H}_N(\rho) - \hat{H}_{\text{ref},N}(\rho)\|_{\text{op}} \leq \varepsilon_N. \quad (1481)$$

The physical tail is divided by $E_{\text{kin},N}$ and occurs exactly once in this budget.

Theorem 31 (Finite-cover min-max gap and rank stability). *Let Δ_N be the physical isolation gap of the target continued from the reference lower bright branch, with value zero if the target meets its complement. Under Eqs. (1476)–(1481),*

$$\frac{\Delta_N}{E_{\text{kin},N}} \geq \max\{0, g_0 - 2\varepsilon_N\}. \quad (1482)$$

If $\varepsilon_N < g_0/2$, the target remains isolated on the entire cover and its transported Riesz projector has the same rank as the pre-specified reference target.

Proof. Finite-dimensional min-max, equivalently Weyl's inequality, moves every ordered eigenvalue by at most ε_N . Hence the target edge can move toward a complementary edge by ε_N , while that complementary edge can move toward the target by another ε_N . Taking the infimum over all cover fibers and both complementary sides proves Eq. (1482).

For rank stability, set

$$\hat{H}_N^{(s)} = \hat{H}_{\text{ref},N} + s(\hat{H}_N - \hat{H}_{\text{ref},N}), \quad 0 \leq s \leq 1. \quad (1483)$$

Its target separation is at least $g_0 - 2s\varepsilon_N > 0$. The target Riesz projector is therefore continuous throughout the homotopy, and its integer-valued rank is constant. The fixed transport and symmetry resolution identify this projector with the pre-specified target rather than a post-selected band. $\square$

Proposition 21 (Sharpness of the factor two). *The coefficient two in Eq. (1482) is optimal for unrestricted Hermitian operator perturbations. Indeed, for*

$$H_{\text{ref}} = \text{diag}(-g_0/2, g_0/2), \quad R = \text{diag}(\varepsilon, -\varepsilon), \quad (1484)$$

one has $\|R\|_{\text{op}} = \varepsilon$ and

$$\text{gap}(H_{\text{ref}} + R) = g_0 - 2\varepsilon \quad (0 \leq \varepsilon < g_0/2). \quad (1485)$$

Thus a factor-one estimate requires an independently proved one-sided order, monotonicity, or fixed-edge structure.

Proof. The perturbed eigenvalues are $-g_0/2 + \varepsilon$ and $g_0/2 - \varepsilon$; their difference is Eq. (1485). $\square$

Corollary 20 (Positive normalized thermodynamic gap). *Put*

$$\bar{\varepsilon} := \limsup_{N \rightarrow \infty} \varepsilon_N. \quad (1486)$$

Then

$$\liminf_{N \rightarrow \infty} \frac{\Delta_N}{E_{\text{kin},N}} \geq g_0 - 2\bar{\varepsilon}. \quad (1487)$$

In particular,

$$\bar{\varepsilon} < \frac{g_0}{2} \quad (1488)$$

implies a strictly positive normalized thermodynamic isolation gap and eventual stability of the target rank.

Proof. For every $\eta > 0$, eventually $\varepsilon_N \leq \bar{\varepsilon} + \eta$. Apply Eq. (1482), take the lower limit, and then let $\eta \downarrow 0$. Under Eq. (1488), the right-hand side is positive and the strict finite-cover threshold holds for all sufficiently large N . $\square$

Proposition 22 (Conditional off-resonant self-energy budget). *Consider, in one consistent energy convention,*

$$H = \begin{pmatrix} H_P & V \\ V^* & H_Q \end{pmatrix}, \quad \Sigma_P(E) = V(E - H_Q)^{-1}V^*. \quad (1489)$$

Let a fixed real target window J satisfy

$$\text{dist}(J, \text{spec } H_Q) \geq d_Q > 0, \quad \|V\|_{\text{op}} \leq v. \quad (1490)$$

Then, uniformly for $E \in J$,

$$\|\Sigma_P(E)\|_{\text{op}} \leq \frac{v^2}{d_Q}. \quad (1491)$$

In physical units this contributes at most $v^2/(d_Q E_{\text{kin},N})$ to a normalized effective residual.

Proof. The spectral theorem gives $\|(E - H_Q)^{-1}\|_{\text{op}} \leq d_Q^{-1}$. Submultiplicativity proves Eq. (1491). $\square$

This estimate may replace an off-sector term in Eq. (1480) only after one proves that the same fixed window contains the matched target throughout the comparison, the target rank is fixed, and the excluded sector cannot enter the window. It must not be counted in addition to a residual that already contains the full off-diagonal block. No conclusion across $d_Q = 0$ follows; resonant-shell Riesz/Feshbach analysis is deferred.

Proposition 23 (Finite positive gaps do not imply a thermodynamic gap). *Let*

$$\hat{H}_N = \text{diag}(0, N^{-1}) \quad (1492)$$

on $\mathbb{C}^2$, with the first coordinate as target. Every finite N has a simple rank-one target, a constant target projector, and positive gap N^{-1} , but

$$\liminf_{N \rightarrow \infty} \text{gap}(\hat{H}_N) = 0. \quad (1493)$$

Proof. The eigenvalues are 0 and N^{-1} , so both assertions are immediate. $\square$

Corollary 21 (Conditional richer-root verification). *Suppose a richer fixed-class model independently supplies a simple root normalized to $X = 1$, a fixed transport, a stable pre-specified target rank and positive active-overlap margin, and the complement detuning $d_\perp > 0$. If its complete normalized residual budget satisfies*

$$\limsup_{N \rightarrow \infty} \varepsilon_N < \frac{1}{2} \min \left\{ \frac{1 + \sqrt{17}}{2}, d_\perp \right\}, \quad (1494)$$

then

$$\liminf_{N \rightarrow \infty} \frac{\Delta_N}{E_{\text{kin},N}} \geq \min \left\{ \frac{1 + \sqrt{17}}{2}, d_\perp \right\} - 2 \limsup_{N \rightarrow \infty} \varepsilon_N > 0. \quad (1495)$$

Proof. The independent root makes $X = 1$ a root-normalized coordinate for this richer model. The transport, rank, and overlap assumptions identify its target with the fixed reference cluster. The conclusion is Corollary 20 with the displayed g_0 . $\square$

Corollary 22 (Repairs of Theorems 79 and 101). *The positive lower thermodynamic gap in Theorem 79 remains an admissibility premise unless Eqs. (1477) and (1488), together with fixed target identification, are verified. The same is true of the isolation premise in Theorem 101. The exact square target cannot instantiate either theorem at a magic root because it has no such root. Raw $X = 1$ in the bright-dark benchmark supplies a reference gap, not a magic point.*

For an independently rooted richer model, Corollary 21 supplies the required gap premise quantitatively, but it does not prove the other operational residuals. In particular, coherence and delocalization remain separate, and no resonant-shell conclusion is included here.

X. Translation coherence from an isolated complete branch

Flat eigenvalues and a positive spectral gap do not determine real-space participation. The missing implication requires translation covariance of the isolated target projector and completeness across the declared representation sectors.

Proposition 24 (Spectral-data obstruction). *On one orbital per cell, put*

$$u_N = N_c^{-1/2}(1, \dots, 1), \quad e_1 = (1, 0, \dots, 0), \quad (1496)$$

and define

$$\begin{aligned} H_{\text{ext},N} &= I - |u_N\rangle\langle u_N|, \\ H_{\text{loc},N} &= I - |e_1\rangle\langle e_1|. \end{aligned} \quad (1497)$$

These Hamiltonians are unitarily equivalent. Each has target eigenvalue zero, complement eigenvalue one, zero target bandwidth, and unit gap, but

$$p_{\text{cell}}[|u_N\rangle\langle u_N|] = 1, \quad p_{\text{cell}}[|e_1\rangle\langle e_1|] = N_c^{-1}. \quad (1498)$$

Thus spectral data alone cannot imply a positive thermodynamic participation bound.

Proof. Choose a unitary W_N with $W_N u_N = e_1$. Then $H_{\text{loc},N} = W_N H_{\text{ext},N} W_N^*$. The two spectra and gaps are therefore identical. The extended projector has cell weights N_c^{-1} , whereas the localized projector has weights $(1, 0, \dots, 0)$. Substitution in $p_{\text{cell}} = [N_c \sum_x Q_x^2]^{-1}$ proves Eq. (1498). $\square$

Let a normal finite cover have

$$\mathcal{H}_N = \bigoplus_{x \in X_N} \mathbb{C}^\nu, \quad |X_N| = N_c, \quad (1499)$$

with rank- ν cell projectors Π_x . Let a finite translation group G_N act transitively on X_N and satisfy

$$U_g \Pi_x U_g^* = \Pi_{gx}. \quad (1500)$$

Theorem 32 (Translation-covariant Riesz participation). *Suppose $[H_N, U_g] = 0$ for every $g \in G_N$, and let a complete pre-specified isolated target cluster have Riesz projector*

$$P_N = \frac{1}{2\pi i} \oint_\Gamma (z - H_N)^{-1} dz, \quad r_N = \text{rank } P_N. \quad (1501)$$

Then $[P_N, U_g] = 0$ for all g . Its normalized cell weights

$$Q_x(P_N) := \frac{\text{Tr}(\Pi_x P_N)}{r_N} \quad (1502)$$

satisfy

$$\begin{aligned} Q_x(P_N) &= \frac{1}{N_c}, \\ \text{IPR}_{\text{cell}}[P_N] &= \frac{1}{N_c}, \\ p_{\text{cell}}[P_N] &= 1. \end{aligned} \quad (1503)$$

If the target is the complete branch over every required finite-quotient irrep, then

$$C_{\text{rep},N} = 1, \quad \epsilon_{\text{cov},N}^{\text{all}} = 0, \quad C_{\text{coh},N} = 1. \quad (1504)$$

All conclusions hold on every cover and are independent of the basis chosen inside the cluster.

Proof. Every resolvent in Eq. (1501) commutes with U_g , so functional calculus gives $[P_N, U_g] = 0$. Cyclicity of the trace and Eq. (1500) yield

$$Q_{gx}(P_N) = \frac{\text{Tr}(U_g \Pi_x U_g^* P_N)}{r_N} = Q_x(P_N). \quad (1505)$$

Transitivity makes every weight equal. Since $\sum_x \Pi_x = I$ and $\text{Tr } P_N = r_N$, their sum is one; hence $Q_x = N_c^{-1}$. The definitions of cell IPR and normalized participation give Eq. (1503).

For a complete branch, every indicator in the representation-coverage sum is one. The regular-representation weights sum to one, so $\mathcal{C}_{\text{rep},N} = 1$. Projector covariance makes the full-group defect zero. Substitution in the degenerate projector coherence factor proves Eq. (1504). The Riesz projector itself is unchanged by a unitary basis rotation within its range. $\square$

A local energy or curvature root does not establish $\mathcal{C}_{\text{rep},N} = 1$. Completeness across the declared quotient irreps is an independent branch hypothesis in Theorem 32.

Corollary 23 (Simple-character caveat). *Let an eigenspace be one dimensional and invariant under every translation. For a normalized spanning vector ψ ,*

$$U_g \psi = \chi(g) \psi, \quad |\chi(g)| = 1, \quad (1506)$$

where χ is a one-dimensional character. Then

$$\|\Pi_x \psi\|^2 = \frac{1}{N_c}, \quad p_{\text{cell}}[\psi] = 1. \quad (1507)$$

No analogous statement applies to an arbitrary basis vector in a degenerate eigenspace. In particular, a non-Abelian higher-dimensional irrep must be treated through its full spectral projector.

Proof. Invariance of a one-dimensional space makes U_g act by a scalar of unit modulus. The representation law makes those scalars a character. Equations (1500) and (1506) show that

$$\|\Pi_{gx} \psi\| = \|U_g \Pi_x U_g^* \psi\| = \|\Pi_x \psi\|. \quad (1508)$$

Transitivity and normalization prove Eq. (1507). In a degenerate space, translations can mix basis vectors, so only the full invariant projector has the preceding covariance. $\square$

Lemma 4 (Exact participation-defect identity). *For any rank- r orthogonal projector P , define*

$$Q_x = \frac{\text{Tr}(\Pi_x P)}{r}, \quad d_{\text{cell}}(P)^2 = N_c \sum_x \left(Q_x - \frac{1}{N_c} \right)^2. \quad (1509)$$

Then

$$p_{\text{cell}}[P] = \frac{1}{1 + d_{\text{cell}}(P)^2}. \quad (1510)$$

Proof. Since $\sum_x Q_x = 1$,

$$d_{\text{cell}}(P)^2 = N_c \sum_x Q_x^2 - 1. \quad (1511)$$

The definition $p_{\text{cell}}[P] = [N_c \sum_x Q_x^2]^{-1}$ proves the identity. $\square$

For a finite transitive group G , introduce the group average and full-group covariance defect

$$P^G := \frac{1}{|G|} \sum_{g \in G} U_g P U_g^*, \quad (1512)$$

$$\epsilon_{\text{cov}}^{\text{all}} := \max_{g \in G} \frac{\|[P, U_g]\|_F}{2\sqrt{r}}. \quad (1513)$$

Theorem 33 (Quantitative full-group participation and coherence). *For the cell decomposition in Eq. (1499),*

$$d_{\text{cell}}(P)^2 \leq \frac{N_c \nu}{r^2} \|P - P^G\|_F^2 \leq \frac{4N_c \nu}{r} (\epsilon_{\text{cov}}^{\text{all}})^2. \quad (1514)$$

Consequently,

$$p_{\text{cell}}[P] \geq \left[1 + \frac{4N_c \nu}{r} (\epsilon_{\text{cov}}^{\text{all}})^2 \right]^{-1}. \quad (1515)$$

If $\mathcal{C}_{\text{rep}} \geq c_{\text{rep}} > 0$, the degenerate projector coherence factor satisfies

$$\mathcal{C}_{\text{coh}} \geq \frac{c_{\text{rep}}}{(1 + \epsilon_{\text{cov}}^{\text{all}}) \left[1 + \frac{4N_c \nu}{r} (\epsilon_{\text{cov}}^{\text{all}})^2 \right]}. \quad (1516)$$

For a cover sequence, if

$$\frac{N_c \nu}{r} (\epsilon_{\text{cov}}^{\text{all}})^2 \leq M < \infty, \quad (1517)$$

then

$$\mathcal{C}_{\text{coh}} \geq \frac{c_{\text{rep}}}{2(1 + 4M)} > 0. \quad (1518)$$

If the left side of Eq. (1517) tends to zero and $\mathcal{C}_{\text{rep}} \rightarrow 1$, then $\mathcal{C}_{\text{coh}} \rightarrow 1$.

Proof. The average P^G commutes with G , has trace r , and therefore has uniform normalized cell density. Put $A = P - P^G$. Then

$$Q_x - \frac{1}{N_c} = \frac{\text{Tr}(\Pi_x A \Pi_x)}{r}. \quad (1519)$$

Hilbert–Schmidt Cauchy–Schwarz on the rank- ν cell block gives

$$|\text{Tr}(\Pi_x A \Pi_x)|^2 \leq \nu \|\Pi_x A \Pi_x\|_F^2. \quad (1520)$$

The diagonal blocks are Hilbert–Schmidt orthogonal, so their squared norms sum to at most $\|A\|_F^2$. Multiplying the summed inequality by N_c/r^2 proves the first bound in Eq. (1514).

Moreover,

$$\begin{aligned} \|P - P^G\|_F &\leq \frac{1}{|G|} \sum_g \|P - U_g P U_g^*\|_F \\ &= \frac{1}{|G|} \sum_g \|[P, U_g]\|_F \leq 2\sqrt{r} \epsilon_{\text{cov}}^{\text{all}}. \end{aligned} \quad (1521)$$

This proves the second defect bound, and Lemma 4 proves Eq. (1515). Substitution in

$$\mathcal{C}_{\text{coh}} = \frac{p_{\text{cell}}[P]\mathcal{C}_{\text{rep}}}{1 + \epsilon_{\text{cov}}^{\text{all}}} \quad (1522)$$

gives Eq. (1516).

The triangle inequality gives $0 \leq \epsilon_{\text{cov}}^{\text{all}} \leq 1$, proving Eq. (1518). Finally, $r \leq N_c \nu$ implies $(\epsilon_{\text{cov}}^{\text{all}})^2$ is bounded above by the scaled defect. Its vanishing, together with coverage convergence, makes every factor in Eq. (1522) tend to one. $\square$

The manuscript's test-set defect can replace Eq. (1513) only if the test set is the full group, or if a cover-uniform word-length, mixing, or Poincaré estimate controls the full group. For generators S , a telescoping commutator estimate gives

$$\epsilon_{\text{cov}}^{\text{all}} \leq L_N \epsilon_{\text{cov}}^S \quad (1523)$$

when every group element has S -word length at most L_N . An uncontrolled L_N on growing covers gives no uniform implication.

Theorem 34 (Hamiltonian-to-projector covariance transfer). *Let H be finite-dimensional and Hermitian, and let Γ be a Riesz contour of length ℓ with*

$$\begin{aligned} \delta &:= \text{dist}(\Gamma, \text{spec } H) > 0, \\ \eta_{\text{sym}} &:= \max_{g \in G} \|U_g H U_g^* - H\|_{\text{op}}. \end{aligned} \quad (1524)$$

If P is the rank- r target projector enclosed by Γ , then

$$\|U_g P U_g^* - P\|_{\text{op}} \leq \frac{\ell \eta_{\text{sym}}}{2\pi \delta^2}, \quad (1525)$$

$$\|U_g P U_g^* - P\|_F \leq \sqrt{2r} \frac{\ell \eta_{\text{sym}}}{2\pi \delta^2}, \quad (1526)$$

$$\epsilon_{\text{cov}}^{\text{all}} \leq a_H := \frac{\ell \eta_{\text{sym}}}{2\sqrt{2} \pi \delta^2}. \quad (1527)$$

Hence

$$p_{\text{cell}}[P] \geq \left(1 + \frac{4N_c \nu}{r} a_H^2\right)^{-1}. \quad (1528)$$

If $\mathcal{C}_{\text{rep}} \geq c_{\text{rep}} > 0$, then also

$$\mathcal{C}_{\text{coh}} \geq \frac{c_{\text{rep}}}{(1 + a_H)(1 + 4N_c \nu a_H^2 / r)}. \quad (1529)$$

Exact Hamiltonian covariance gives $\epsilon_{\text{cov}}^{\text{all}} = 0$ exactly.

Proof. Set $H_g = U_g H U_g^*$. It is unitarily equivalent to H , so both resolvents on Γ have norm at most δ^{-1} . The resolvent identity gives

$$\|(z - H_g)^{-1} - (z - H)^{-1}\|_{\text{op}} \leq \frac{\eta_{\text{sym}}}{\delta^2}. \quad (1530)$$

The Riesz projector of H_g is $U_g P U_g^*$. Contour integration proves Eq. (1525). The difference of two rank- r orthogonal projectors has rank at most $2r$, so its Frobenius norm is at most $\sqrt{2r}$ times its operator norm. This proves the remaining transfer bounds. Substitution in Eq. (1515) proves Eq. (1528); substitution in Eq. (1516) proves Eq. (1529). When $\eta_{\text{sym}} = 0$, functional calculus gives exact commutation directly. $\square$

Corollary 24 (Combined richer-branch flatness, gap, and coherence). *Consider one independently rooted richer branch, normalized so that its root is $X = 1$. Assume that the same pre-specified transported target:*

1. *is complete across all required finite-quotient irreps and has fixed rank and positive active-overlap margin;*
2. *is isolated and exactly covariant under a translation group acting transitively on every finite cover;*
3. *independently satisfies*

$$\frac{W_N}{E_{\text{kin},N}} \rightarrow 0, \quad \Omega_N^{\text{max}} \rightarrow 0; \quad (1531)$$

4. *has the complement detuning $g_0 > 0$ and complete residual budget*

$$\varepsilon_N < \frac{g_0}{2} \quad \text{for every } N, \quad \bar{\varepsilon} := \limsup_N \varepsilon_N < \frac{g_0}{2}. \quad (1532)$$

Then

$$\liminf_{N \rightarrow \infty} \frac{\Delta_N}{E_{\text{kin},N}} \geq g_0 - 2\bar{\varepsilon} > 0, \quad (1533)$$

and, on every cover,

$$\begin{aligned} p_{\text{cell}}[P_N] &= 1, \mathcal{C}_{\text{rep},N} = 1, \\ \epsilon_{\text{cov},N}^{\text{all}} &= 0, \mathcal{C}_{\text{coh},N} = 1. \end{aligned} \quad (1534)$$

Thus flatness, complete tangent suppression, positive thermodynamic isolation, and projector coherence hold for one and the same branch.

Proof. The strict residual inequality preserves the isolated target rank on every cover, and finite-cover min-max gives $\Delta_N / E_{\text{kin},N} \geq g_0 - 2\varepsilon_N$. Taking the lower limit proves Eq. (1533). Exact translation covariance, isolation, transitivity, and completeness permit Theorem 32 to be applied to the same projector, proving Eq. (1534). The flatness and tangent conclusions are the independent hypotheses in Eq. (1531); no scalar root is used to infer them. $\square$

Corollary 25 (Repairs of Theorems 95, 100, 101, and 103). *Theorem 95 is repaired by requiring exact Hamiltonian translation covariance, an isolated pre-specified complete cluster, and transitive cell action; projector covariance then follows from the Riesz formula rather than being an unsupported premise.*

For Theorem 100, projector participation, covariance, and representation coverage become conclusions under Theorem 32; outside those hypotheses they remain independent independent tests. In Theorem 101, the coherence residual is similarly derived only for the exact theorem or under the quantitative bounds (1516) and (1527). Its bandwidth, velocity, DOS, and gap conditions remain separate.

Theorem 103 remains a valid veto for nondegenerate localized sequences and incomplete clusters. It cannot use one compact-localized basis vector to veto a complete degenerate translation-covariant projector. The exact square target cannot instantiate the combined root theorem because it has no magic root, and complete hyperbolic tangent cancellation remains unproved from the presently available channel data. No resonant-shell conclusion is made here.

Proof. The first statement is the functional-calculus implication in Theorem 32. The second follows from its exact participation and coverage conclusions and from the quantitative near-covariant theorem. Proposition 24 shows why none follows from spectral flatness and a gap alone. The basis-invariant Riesz projector distinguishes a complete degenerate band from an arbitrary localized basis choice, proving the stated scope of the localization veto. The final root and resonance restrictions preserve the independently established obstructions. $\square$

Y. Exact Feshbach–Riesz treatment of a resonant shell

Near resonance, eliminating a bare complementary state is singular even though the full Hamiltonian is regular. Every such state must first be moved into an enlarged active block.

Proposition 25 (Resonant two-level obstruction). *For real δ, t , let*

$$H(\delta, t) = \begin{pmatrix} 0 & t \\ t & \delta \end{pmatrix}. \quad (1535)$$

Eliminating the second coordinate at $E \neq \delta$ gives

$$\Sigma(E) = \frac{t^2}{E - \delta}, \quad (1536)$$

whereas the exact eigenvalues are

$$E_{\pm} = \frac{\delta \pm \sqrt{\delta^2 + 4t^2}}{2}. \quad (1537)$$

At $\delta = 0$, $E_{\pm} = \pm|t|$. Thus resonant splitting is linear in $|t|$, and no uniform $O(t^2)$ virtual-correction expansion covers the resonance.

Proof. For an eigenvector $(u, v)^T$, the second row gives $v = tu/(E - \delta)$ when $E \neq \delta$. Substitution in the first row

gives $E - \Sigma(E) = 0$. Direct evaluation of $\det(E - H) = E(E - \delta) - t^2$ gives Eq. (1537). At $\delta = 0$ the exact eigenvalues are finite while Eq. (1536) has a pole at the bare resonant energy. $\square$

Let a finite-dimensional Hermitian Hamiltonian on $\mathcal{P}_0 \oplus \mathcal{Q}_0$ be

$$H = \begin{pmatrix} H_P & V \\ V^* & H_Q \end{pmatrix}. \quad (1538)$$

Fix a real target window J .

Theorem 35 (Buffered resonant shell and exact enlarged blocks). *Let $\mathcal{R}$ be a buffered spectral window containing every H_Q -eigenvalue declared resonant with J , and assume its boundary is spectrally separated. Define*

$$S = \mathbf{1}_{\mathcal{R}}(H_Q), \quad R = I_{\mathcal{Q}_0} - S. \quad (1539)$$

On

$$\widetilde{\mathcal{P}} = \mathcal{P}_0 \oplus S\mathcal{Q}_0, \quad \widetilde{\mathcal{Q}} = R\mathcal{Q}_0, \quad (1540)$$

the exact blocks are

$$A = \begin{pmatrix} H_P & VS \\ SV^* & SH_QS \end{pmatrix}, \quad (1541)$$

$$B = \begin{pmatrix} VR \\ 0 \end{pmatrix}, \quad D = RH_QR, \quad H = \begin{pmatrix} A & B \\ B^* & D \end{pmatrix}. \quad (1542)$$

Assume the genuine tail detuning

$$\text{dist}(J, \text{spec } D) \geq d_{\text{tail}} > 0. \quad (1543)$$

All resonant couplings VS are then retained in A and are diagonalized nonperturbatively.

Proof. Spectral functional calculus gives $[S, H_Q] = 0$ and hence

$$SH_QR = RH_QS = 0. \quad (1544)$$

Reordering the original decomposition as $(\mathcal{P}_0 \oplus S\mathcal{Q}_0) \oplus R\mathcal{Q}_0$ and inserting $S + R = I_{\mathcal{Q}_0}$ gives Eqs. (1541) and (1542). Equation (1543) is an independent uniform statement about the excluded spectrum. $\square$

The buffer is essential. A spectral gap between $\partial\mathcal{R}$ and $\text{spec } H_Q$ makes S a stable Riesz projector under perturbation. If an eigenvalue reaches the boundary, a fixed-rank continuation is not established: the shell must be enlarged or overlapping Riesz charts must be patched.

For $z \in \rho(D)$, define

$$\begin{aligned} R_D(z) &= (z - D)^{-1}, \\ \Sigma_{\text{tail}}(z) &= BR_D(z)B^*, \\ F(z) &= z - A - \Sigma_{\text{tail}}(z), \\ H_{\text{eff}}(z) &= A + \Sigma_{\text{tail}}(z). \end{aligned} \quad (1545)$$

Theorem 36 (Exact kernel and determinant equivalence). *For every $z \in \rho(D)$, the maps*

$$u \mapsto \begin{pmatrix} u \\ R_D(z)B^*u \end{pmatrix}, \quad \begin{pmatrix} u \\ v \end{pmatrix} \mapsto u \quad (1546)$$

are inverse linear isomorphisms between $\ker F(z)$ and $\ker(z - H)$. Moreover,

$$\det(z - H) = \det(z - D) \det F(z). \quad (1547)$$

These identities remain exact at resonances internal to A and contain no expansion in VS .

Proof. The lower block equation in $(z - H)(u, v)^T = 0$ is

$$(z - D)v = B^*u, \quad (1548)$$

so $v = R_D B^*u$. The upper equation then becomes $F(z)u = 0$. Conversely, this reconstruction satisfies both block equations. If $u = 0$, Eq. (1548) and $z \in \rho(D)$ force $v = 0$, so the maps are injective as well as surjective. Gaussian block factorization of $z - H$ with the invertible block $z - D$ gives Eq. (1547). $\square$

Theorem 37 (Full resolvent and full-space Riesz projector). *If $z \in \rho(D)$ and $F(z)$ is invertible, put $G(z) = F(z)^{-1}$. Then*

$$(z - H)^{-1} = \begin{pmatrix} G & GBR_D \\ R_D B^* G & R_D + R_D B^* GBR_D \end{pmatrix}. \quad (1549)$$

Let $\Gamma \subset \rho(D)$ be a positively oriented contour on which $F(z)$ is invertible. The exact full-space target projector is

$$\mathcal{P}_\Gamma(H) = \frac{1}{2\pi i} \oint_\Gamma \mathcal{G}(z) dz, \quad (1550)$$

where $\mathcal{G}(z)$ is the block matrix on the right-hand side of Eq. (1549).

Proof. Multiply the proposed inverse in Eq. (1549) by

$$z - H = \begin{pmatrix} z - A & -B \\ -B^* & z - D \end{pmatrix}. \quad (1551)$$

Using $F = z - A - BR_D B^*$ and $(z - D)R_D = I$ reduces both diagonal products to the identity and both off-diagonal products to zero. This proves the resolvent formula. The stated contour lies in $\rho(H)$ by the determinant identity. Integrating the full resolvent is the Riesz functional calculus, which proves Eq. (1550). $\square$

Theorem 38 (Quantitative enlarged-cluster rank certificate). *For a positively oriented contour $\Gamma \subset \rho(D)$ whose interior contains no point of $\text{spec } D$, define*

$$m_\Gamma := \inf_{z \in \Gamma} s_{\min}(z - A), \quad (1552)$$

$$s_\Gamma := \sup_{z \in \Gamma} \|\Sigma_{\text{tail}}(z)\|_{\text{op}}. \quad (1553)$$

If

$$s_\Gamma < m_\Gamma, \quad (1554)$$

then

$$F_\tau(z) := z - A - \tau \Sigma_{\text{tail}}(z) \quad (1555)$$

is invertible on Γ for every $0 \leq \tau \leq 1$. Consequently, the full target cluster inside Γ has the same algebraic rank as the enlarged-active A -cluster inside Γ . With

$$d_\Gamma := \text{dist}(\Gamma, \text{spec } D), \quad (1556)$$

one has the explicit bound

$$s_\Gamma \leq \frac{\|B\|_{\text{op}}^2}{d_\Gamma}. \quad (1557)$$

Proof. For every $z \in \Gamma$,

$$\begin{aligned} s_{\min}(F_\tau(z)) &\geq s_{\min}(z - A) - \tau \|\Sigma_{\text{tail}}(z)\|_{\text{op}} \\ &\geq m_\Gamma - \tau s_\Gamma > 0. \end{aligned} \quad (1558)$$

Thus F_τ is invertible on the contour. Introduce the Hermitian homotopy

$$H_\tau = \begin{pmatrix} A & \sqrt{\tau}B \\ \sqrt{\tau}B^* & D \end{pmatrix}. \quad (1559)$$

Its Schur complement is F_τ . Since $\Gamma \subset \rho(D)$, the determinant identity shows that $\Gamma \subset \rho(H_\tau)$ for all τ . The Riesz-projector rank, equivalently the determinant winding number, is constant along this homotopy. At $\tau = 0$, the contour encloses the selected A -cluster and no D -spectrum, proving the rank statement. Finally, $\|R_D(z)\|_{\text{op}} \leq d_\Gamma^{-1}$ and submultiplicativity prove Eq. (1557). $\square$

The Feshbach pencil, determinant identity, and full resolvent are exact whenever their stated inverses exist, even if $s_\Gamma \geq m_\Gamma$. Equation (1554) is needed only to identify the full cluster with the chosen A -cluster through this homotopy.

Theorem 39 (Exact energy metric, norm, and tangent derivative). *For real $E \in J$, put $R_E = (E - D)^{-1}$. Then*

$$\partial_E \Sigma_{\text{tail}}(E) = -BR_E^2 B^*, \quad (1560)$$

and

$$\mathcal{M}_E := I - \partial_E \Sigma_{\text{tail}}(E) = I + BR_E^2 B^* > 0. \quad (1561)$$

If $F(E)u = 0$, its reconstructed full eigenvector satisfies

$$\left\| \begin{pmatrix} u \\ R_E B^* u \end{pmatrix} \right\|^2 = \langle u, \mathcal{M}_E u \rangle. \quad (1562)$$

Let s be a differentiable tangent or model parameter. At fixed energy,

$$D_s \Sigma_{\text{tail}}(E) = (D_s B) R_E B^* + B R_E (D_s D) R_E B^*$$

$$+ BR_E(D_s B)^*. \quad (1563)$$

For a simple eigenvalue branch,

$$D_s E = \frac{\langle u, [D_s A + D_s \Sigma_{\text{tail}}(E)] u \rangle}{\langle u, \mathcal{M}_E u \rangle}. \quad (1564)$$

Proof. Differentiating $R_E = (E - D)^{-1}$ gives $\partial_E R_E = -R_E^2$ and, at fixed E , $D_s R_E = R_E(D_s D)R_E$. The product rule proves Eqs. (1560) and (1563). Since E is real and $E \in \rho(D)$, $R_E = R_E^*$ and $BR_E^2 B^*$ is positive semidefinite, proving Eq. (1561). Direct expansion of the full-vector norm proves Eq. (1562).

Differentiate $[A(s) + \Sigma_{\text{tail}}(E(s), s)]u(s) = E(s)u(s)$ and take the inner product with u . The eigenvector-derivative terms cancel by self-adjointness. Moving $\langle u, \partial_E \Sigma_{\text{tail}} u \rangle D_s E$ to the right gives Eq. (1564). $\square$

The positivity in Eq. (1561) belongs only to the detuned tail. Every near-resonant channel has already been included in A .

Theorem 40 (Uniform finite-cover resonant-shell control). *Let H_N be a finite-cover sequence with the enlarged decomposition above. Assume uniformly in N :*

1. *the buffered boundary has a positive spectral margin and the selected shell and A_N -cluster have their specified ranks;*
2. $\text{dist}(J, \text{spec } D_N) \geq d_{\text{tail}} > 0$;
3. $\text{length } \Gamma_N \leq \ell$, $d_{\Gamma, N} \geq d_\Gamma > 0$, *and the interior of Γ_N contains no point of $\text{spec } D_N$;*
4. $m_{\Gamma, N} \geq m > 0$, $\|B_N\|_{\text{op}} \leq b$, *and*

$$\frac{b^2}{d_\Gamma} \leq s < m; \quad (1565)$$

5. *for the chosen parameter s ,*

$$\|D_s A_N\| \leq a_s, \quad \|D_s B_N\| \leq b_s, \quad \|D_s D_N\| \leq d_s. \quad (1566)$$

Then the exact Feshbach pencil and full-space Riesz projector exist on every Γ_N , and the full target cluster has the specified A_N -cluster rank. Uniformly for real $E \in J$,

$$\|\Sigma_{\text{tail}, N}(E)\|_{\text{op}} \leq \frac{b^2}{d_{\text{tail}}}, \quad (1567)$$

$$I \preceq \mathcal{M}_{E, N} \preceq \left(1 + \frac{b^2}{d_{\text{tail}}^2}\right) I, \quad (1568)$$

$$\|D_s \Sigma_{\text{tail}, N}(E)\|_{\text{op}} \leq \frac{2bb_s}{d_{\text{tail}}} + \frac{b^2 d_s}{d_{\text{tail}}^2}. \quad (1569)$$

Every simple branch in J therefore obeys

$$|D_s E_N| \leq a_s + \frac{2bb_s}{d_{\text{tail}}} + \frac{b^2 d_s}{d_{\text{tail}}^2}. \quad (1570)$$

Proof. The contour-distance and coupling bounds give $s_{\Gamma, N} \leq b^2/d_\Gamma \leq s < m \leq m_{\Gamma, N}$. Theorem 38 therefore supplies the exact projector and rank certificate uniformly. On J , $\|R_{E, N}\| \leq d_{\text{tail}}^{-1}$, which proves Eq. (1567). Positivity and $\|B_N R_{E, N}^2 B_N^*\| \leq b^2/d_{\text{tail}}^2$ prove Eq. (1568). The three terms in Eq. (1563), together with Eq. (1566), prove Eq. (1569). Finally, the denominator in Eq. (1564) is at least $\|u\|^2$; the Rayleigh quotient and the triangle inequality prove Eq. (1570). $\square$

This theorem gives uniform exact construction and cluster-rank control. A positive thermodynamic isolation gap still requires the separate strict half-gap residual hypotheses, and projector coherence still requires the separate symmetry-completeness hypotheses.

Corollary 26 (Repairs of the Schur formula and Theorems 150, 157, 174). *In every representation block, the Schur formula must first retain the spectral projector of every state resonant with the target window. Only the detuned complement may appear in a resolvent denominator.*

Theorem 150's quadratic inverse-mass expansion is valid only when its eliminated denominators are uniformly off resonance. A resonant shell is part of A and must be diagonalized before its Hessian or any root is assessed. Theorem 157's $\Sigma/E_M = O(\alpha_{\text{eff}}^2)$ chain applies only to a uniformly detuned tail with $B = O(\alpha_{\text{eff}})$; internal resonant hybridization can instead split at $O(|\alpha_{\text{eff}}|)$. Theorem 174's resolvent bound and relative quartic amplification likewise refer only to the detuned excluded tail.

The equal-amplitude square target already retains its four first-star satellites in the exact five-state matrix and remains root-free. Shell enlargement does not restore its spurious perturbative root. For a richer model, a root remains conditional on the actual enlarged matrices A, B, D . No curvature-versus-tessellation realizability conclusion is made here.

Proof. The representation-block statement is Theorem 35 applied inside each conserved irrep. Proposition 25 excludes a uniform quadratic self-energy law at resonance. The exact separation between nonperturbative A and the detuned tail proves the restrictions on Theorems 150, 157, and 174. The exact five-state diagonalization supplies the square-model obstruction; the Feshbach identities alone impose no root condition on an unspecified richer A . $\square$

Continuous curvature families and discrete regular realizations

Curvature differentiation requires a continuous microscopic model. A collection of regular tessellations, indexed by integers, is a different object and in general supports only divided differences.

Proposition 26 (Fixed-regular similarity obstruction). *For a hyperbolic regular tessellation $\{p, q\}$, define*

$$\chi_{p,q} := 2 \operatorname{arccosh} \left[\frac{\cos(\pi/p)}{\sin(\pi/q)} \right]. \quad (1571)$$

Then

$$\frac{a}{R} = \chi_{p,q}, \quad \kappa_{\text{lat}} := a^2 K = -\chi_{p,q}^2, \quad K = -R^{-2}. \quad (1572)$$

Consequently, at fixed integers (p, q) , both $\chi_{p,q}$ and κ_{lat} are fixed. The ray

$$(R, a) \mapsto (cR, ca), \quad c > 0, \quad (1573)$$

is a similarity rescaling, not a sweep through dimensionless regular-lattice geometries.

Proof. The exact regular-tessellation identity is

$$\cosh\left(\frac{a}{2R}\right) = \frac{\cos(\pi/p)}{\sin(\pi/q)}. \quad (1574)$$

The hyperbolic inequality $p^{-1} + q^{-1} < 1/2$ makes the right-hand side larger than one. Applying 2 arcosh proves $a/R = \chi_{p,q}$, and multiplication by $a^2 K = -a^2/R^2$ proves Eq. (1572). Equation (1573) preserves a/R , proving the claim. $\square$

Physical curvature K and dimensionless lattice curvature κ_{lat} must therefore not be interchanged. A spectral change along Eq. (1573) must involve a fixed dimensional length, energy, coupling law, or normalization that was not co-rescaled; it cannot be attributed solely to a change of regular-lattice dimensionless geometry.

Theorem 41 (Transported operator-differentiable curvature family). *Let I be a curvature interval and let*

$$J_K : \mathcal{H}_K \longrightarrow \mathcal{H}_0 \quad (1575)$$

be declared unitary transports. Set

$$\hat{H}(K) = J_K H(K) J_K^*. \quad (1576)$$

Equivalently, one may use a fixed marked graph or group and take $J_K = I$. For a countable marking X , let $h_K(x, y)$ be the finite-orbital kernel of $\hat{H}(K)$, and define

$$\|h\|_{\text{Sch}} := \max \left\{ \sup_x \sum_y \|h(x, y)\|, \sup_y \sum_x \|h(x, y)\| \right\}. \quad (1577)$$

For some $r \in \{1, 2, 3\}$, assume that the entrywise derivatives $\partial_K^j h_K$ exist for $0 \leq j \leq r$ and that

$$K \mapsto \partial_K^j h_K \quad \text{is continuous in } \|\cdot\|_{\text{Sch}}. \quad (1578)$$

Then $K \mapsto \hat{H}(K)$ is C^r in operator norm and

$$\|\partial_K^j \hat{H}(K)\|_{\text{op}} \leq \|\partial_K^j h_K\|_{\text{Sch}}. \quad (1579)$$

Suppose, in addition, that one fixed contour Γ obeys

$$\inf_{K \in I} \operatorname{dist}(\Gamma, \operatorname{spec} \hat{H}(K)) \geq \delta > 0. \quad (1580)$$

Then the resolvent and the Riesz projector

$$P(K) = \frac{1}{2\pi i} \oint_{\Gamma} (z - \hat{H}(K))^{-1} dz \quad (1581)$$

are C^r . At first order,

$$\partial_K R(z, K) = R(z, K) \hat{H}'(K) R(z, K), \quad (1582)$$

$$P'(K) = \frac{1}{2\pi i} \oint_{\Gamma} R \hat{H}' R dz, \quad (1583)$$

and

$$\|P'(K)\|_{\text{op}} \leq \frac{\operatorname{length} \Gamma}{2\pi \delta^2} \|\hat{H}'(K)\|_{\text{op}}. \quad (1584)$$

Thus Hellmann–Feynman, resolvent, Riesz-projector, susceptibility, and implicit-function formulas are legitimate under the regularity order each formula requires.

Proof. The Schur test gives Eq. (1579). The fundamental theorem of calculus in the Schur-kernel Banach space, followed by the continuous embedding into bounded operators, proves operator-norm C^r regularity. In finite dimension the same conclusion follows from entrywise C^r regularity.

Equation (1580) gives $\|R(z, K)\| \leq \delta^{-1}$. Differentiating $(z - \hat{H})R = I$ proves Eq. (1582); iteration gives all derivatives through order r . Contour integration proves Eq. (1583), and its norm together with the contour length proves Eq. (1584). $\square$

A continuous microscopic family may keep one marked graph fixed while its embedding, distances, hopping matrices, twist registry, and couplings vary smoothly with K . The intermediate embeddings need not be regular tessellations. A moving resonant selection must retain a buffered contour or use compatible overlapping Riesz charts before Theorem 41 is applied.

Proposition 27 (Transport gauge and covariant response). *Replace Eq. (1575) by*

$$\tilde{J}_K = U_K J_K, \quad (1585)$$

where U_K is a C^1 unitary on $\mathcal{H}_0$. Put $A_K = U_K^ \partial_K U_K$. Then $A_K^* = -A_K$ and*

$$U_K^* (\partial_K \tilde{H}) U_K = \partial_K \hat{H} + [A_K, \hat{H}]. \quad (1586)$$

Hence a bare represented operator derivative is transport dependent. Simple-eigenvalue derivatives and correctly covariant spectral traces are transport invariant.

Proof. Differentiate $\tilde{H} = U_K \hat{H} U_K^*$, conjugate by U_K^* and U_K , and use $\partial_K U_K^* U_K = -U_K^* \partial_K U_K$. This proves Eq. (1586). If $\hat{H}u = Eu$, then

$$\langle u, [A_K, \hat{H}]u \rangle = 0. \quad (1587)$$

If P is a finite-rank spectral projector, cyclicity and $[P, \hat{H}] = 0$ similarly give $\text{Tr } P[A_K, \hat{H}] = 0$. These identities prove the invariance claims. An operator-valued response must instead report the transport or use the covariant derivative of the declared connection. $\square$

Theorem 42 (Discrete regular response and susceptibility criterion). *Fix a physical edge convention $a_0 > 0$. A regular sample $\{p, q\}$ then has*

$$K_{p,q} = -\frac{\chi_{p,q}^2}{a_0^2}. \quad (1588)$$

For two samples with $K_m \neq K_n$, the scalar response is

$$[X]_{mn;K} := \frac{X_m - X_n}{K_m - K_n}. \quad (1589)$$

An operator response is defined only after declared common-fiber transports:

$$\hat{H}_n = J_n H_n J_n^*, \quad [H]_{mn;K} := \frac{\hat{H}_m - \hat{H}_n}{K_m - K_n}. \quad (1590)$$

Here the J_n are unitaries, or isometries with a fixed common range and an explicit common extension convention.

A regular-lattice susceptibility at K_0 exists only if, for every admissible refining sequence,

$$K_{m_j}, K_{n_j} \rightarrow K_0, \quad K_{m_j} \neq K_{n_j}, \quad (1591)$$

the corresponding divided differences converge to the same limit, independently of the sequence and allowed transports, modulo the chosen covariant gauge convention. A finite collection of regular samples defines only discrete responses.

Proof. Equation (1588) follows from Eq. (1572) after setting $a = a_0$. The integer labels make the available values discrete. Subtraction of operators is meaningful only after both act on the same fiber, which proves the necessity of Eq. (1590). Without a common range or extension, an isometric embedding can turn a dimension or rank change into an artificial operator response. Finally, uniqueness of a derivative requires the same limit along every admissible approach; Eqs. (1591) and the stated independence are precisely that criterion. $\square$

Keeping q fixed while varying p controls vertex coordination, but it does not turn Eq. (1589) into a derivative. When Eq. (1590) is unavailable, one may compare scalar local observables or matched local resolvents, not matrices belonging to unmatched graphs or groups.

Theorem 43 (Continuum branch, fold, and cusp restrictions). *Let $F(\theta, K)$ be C^3 and arise from a microscopic family covered by Theorem 41. At a regular zero,*

$$F(\theta_0, K_0) = 0, \quad F_\theta(\theta_0, K_0) \neq 0, \quad (1592)$$

there is a unique local branch and

$$\frac{d\theta^*}{dK} = -\frac{F_K}{F_\theta}. \quad (1593)$$

At a nondegenerate fold (θ_c, K_c) , require

$$F = F_\theta = 0, \quad a := F_K \neq 0, \quad b := F_{\theta\theta} \neq 0. \quad (1594)$$

On the side where $-2a(K - K_c)/b > 0$, the two branches satisfy

$$\theta_\pm(K) - \theta_c = \pm \sqrt{-\frac{2a}{b}(K - K_c)} + O(|K - K_c|), \quad (1595)$$

$$\theta_+(K) - \theta_-(K) = 2\sqrt{-\frac{2a}{b}(K - K_c)} + O(|K - K_c|). \quad (1596)$$

If $F = \partial_\theta \mathcal{L}$, precisely one of these stationary branches is a strict local maximum near the fold.

For a generic cusp, let $F(\theta, K, \omega)$ be C^4 and require two independent continuous controls together with

$$F = F_\theta = F_{\theta\theta} = 0, \quad F_{\theta\theta\theta} \neq 0, \quad (1597)$$

and

$$\mathfrak{d}_c := \det \begin{pmatrix} F_K & F_\omega \\ F_{\theta K} & F_{\theta\omega} \end{pmatrix}_c \neq 0. \quad (1598)$$

After an invertible local change of the two controls and nonzero rescalings, the leading stationary equation is

$$x^3 - Ux - V = 0, \quad x = \theta - \theta_c. \quad (1599)$$

Thus a generic cusp is codimension two. Integer changes of (p, q) certify neither Eq. (1594) nor Eqs. (1597)–(1598).

Proof. Equation (1593) is the implicit-function theorem. At a fold, $F_K \neq 0$ permits solving $F = 0$ for $K = K(\theta)$. Implicit differentiation gives

$$K'(\theta_c) = 0, \quad K''(\theta_c) = -\frac{b}{a}. \quad (1600)$$

Taylor expansion and inversion of the two sides of this quadratic graph prove Eqs. (1595) and (1596). Moreover, $F_\theta = bx + O(x^2 + |K - K_c|)$, so its signs on the two branches are opposite. When $F = \partial_\theta \mathcal{L}$, only the negative sign is a strict maximum.

At a cusp, with $\lambda = (K - K_c, \omega - \omega_c)$, Taylor expansion gives

$$F = \frac{F_{\theta\theta\theta,c}}{6}x^3 + (F_K, F_\omega)_c\lambda + x(F_{\theta K}, F_{\theta\omega})_c\lambda + O(|x|^4 + |x|^2|\lambda| + |\lambda|^2). \quad (1601)$$

Equation (1598) says that the constant and linear coefficients are independent control coordinates. Their invertible linear redefinition, followed by nonzero scalings of x and F , gives Eq. (1599). Two independent controls are therefore necessary for the generic unfolding. $\square$

Gap closure, coherence loss, target-rank change, a resonant-shell boundary, or exit from the physical angle interval is not a fold. Such events remain separate admissibility boundaries. A change between two regular tessellations is called a discrete transition unless a smooth interpolating family or the following consistency theorem is supplied.

Theorem 44 (Conditional continuum–discrete consistency). *Let $H \in C^1(I; \mathcal{B}(\mathcal{H}_0))$, and let transported approximants at regular or near-regular sample curvatures satisfy*

$$\|\hat{H}_j - H(K_j)\|_{\text{op}} \leq \varepsilon_j. \quad (1602)$$

For $K_m \neq K_n$, put

$$h_{mn} := |K_m - K_n|, \quad r_{mn} := \max_{j=m,n} |K_j - K_0|. \quad (1603)$$

Then

$$\|[H]_{mn;K} - H'(K_0)\|_{\text{op}} \leq \omega_{H'}(r_{mn}) + \frac{\varepsilon_m + \varepsilon_n}{h_{mn}}, \quad (1604)$$

where

$$\omega_{H'}(r) := \sup_{|K - K_0| \leq r} \|H'(K) - H'(K_0)\|_{\text{op}}. \quad (1605)$$

Consequently, the divided differences converge to $H'(K_0)$ if

$$r_{mn} \longrightarrow 0, \quad \frac{\varepsilon_m + \varepsilon_n}{h_{mn}} \longrightarrow 0. \quad (1606)$$

The same estimate holds for scalar observables.

If a common contour has distance at least $\delta > 0$ from $\text{spec } H(K_j)$, has length ℓ_Γ , and $\varepsilon_j < \delta$, then the corresponding Riesz projectors obey

$$\|P_j - P(K_j)\|_{\text{op}} \leq \frac{\ell_\Gamma}{2\pi} \frac{\varepsilon_j}{\delta(\delta - \varepsilon_j)}. \quad (1607)$$

Thus projector divided differences require the right-hand side to be $o(h_{mn})$.

Finally, suppose smooth near-regular functionals F_n converge locally to F in C^2 at a fold satisfying Eq. (1594), or in C^3 at a cusp satisfying Eqs. (1597)–(1598). Then the corresponding nondegenerate fold or cusp persists uniquely nearby for all sufficiently large n . If only discrete regular samples are available, the conclusion is limited to convergent divided-difference jets and sampled branch counts; it is not a finite-sample proof of a continuum singularity.

Proof. Add and subtract $H(K_m)$ and $H(K_n)$. The continuum part satisfies

$$\frac{H(K_m) - H(K_n)}{K_m - K_n} = \int_0^1 H'(K_n + t(K_m - K_n)) dt. \quad (1608)$$

Taking norms after subtracting $H'(K_0)$ gives the first term in Eq. (1604); the two approximation errors give its second term. This proves Eqs. (1604) and (1606).

For the projectors, the contour resolvents satisfy

$$R_j - R = R_j(\hat{H}_j - H(K_j))R. \quad (1609)$$

Their norms are at most $(\delta - \varepsilon_j)^{-1}$ and δ^{-1} , respectively. Contour integration proves Eq. (1607).

At a fold, the Jacobian of (F, F_θ) with respect to (θ, K) has determinant

$$-F_K F_{\theta\theta} \neq 0. \quad (1610)$$

At a cusp, the Jacobian of $(F, F_\theta, F_{\theta\theta})$ with respect to (θ, K, ω) has determinant, up to sign,

$$F_{\theta\theta\theta} \mathfrak{d}_c \neq 0. \quad (1611)$$

The implicit-function theorem applied to these defining systems proves persistence under C^2 and C^3 convergence, respectively. Neither argument exists for an uninterpolated finite integer sample, proving the last statement. $\square$

The condition in Eq. (1606) is essential: operator convergence $\varepsilon_j \rightarrow 0$ alone need not imply derivative convergence when the curvature mesh closes faster than the approximation error. Uniform target isolation, the finite-cover half-gap budget, translation coherence, and buffered resonant-shell selection remain independent hypotheses.

Corollary 27 (Scope repairs for curvature-response theorems). *The block formula for $\partial_K H$ and Theorems 16–17 are continuum-family identities under Theorem 41 and a reported transport gauge. On regular $\{p, q\}$ data they become matched divided-difference statements.*

Theorems 70 and 72, extremum-motion formulas, every curvature susceptibility, and every curvature derivative of a Riesz projector require the same operator differentiability and uniform contour isolation. Theorems 122–123 and all K -derivatives of the operational magic functional are likewise continuum-family results and additionally require the stated simple-root or strict-maximum hypotheses.

Theorems 138–143 diagnose folds, cusps, hierarchy changes, and curvature-born branches only on a smooth continuous control family. Regular-tessellation samples alone use Eqs. (1589) and (1590) and are described as discrete branch-count, hierarchy, gap, coherence, rank, or representation transitions unless Theorem 44 is verified.

The exact equal-amplitude square five-state model remains root-free. A richer-model root remains conditional on its actual matrices and isolation hypotheses. No fixed- $\{p, q\}$ similarity ray is asserted to yield a square crystallographic lattice.

Proof. The first two paragraphs are direct applications of the fixed-fiber, operator-regularity, gauge, and Riesz

hypotheses in Theorem 41 and Proposition 27. Theorem 42 supplies the only intrinsic regular-sample replacement. Theorem 43 gives the missing continuous-control and nondegeneracy hypotheses, while Theorem 44 states the additional bridge needed to transfer them to approximants. None of these constructions changes an independently established spectral root obstruction or proves a crystallographic limit. $\square$

Metric flatness and a square-crystal endpoint

Three notions of convergence must be separated. Local metric convergence means convergence of the constant-curvature metric and its fixed-ball geometric observables in normal coordinates. Local marked-lattice convergence additionally requires a vertex identification, fixed adjacency, and convergence of microscopic coupling entries on every finite neighborhood. Global operator or crystallographic convergence further requires uniform control of the infinite Hamiltonian and an actual endpoint translation group. None of these levels implies the next without added hypotheses.

Proposition 28 (Positive regular-tessellation gap). *For integers $p, q \geq 3$ satisfying*

$$\frac{1}{p} + \frac{1}{q} < \frac{1}{2}, \quad (1612)$$

put

$$A_{p,q} := \frac{\cos(\pi/p)}{\sin(\pi/q)}, \quad \chi_{p,q} := 2 \operatorname{arccosh} A_{p,q}. \quad (1613)$$

Then

$$\chi_{p,q} \geq \chi_{7,3} = 2 \operatorname{arccosh} \left[\frac{2 \cos(\pi/7)}{\sqrt{3}} \right] > 0. \quad (1614)$$

Consequently, no sequence of strictly regular hyperbolic tessellations has $a/R = \chi_{p,q} \rightarrow 0$. With a common physical edge a_0 ,

$$|K_{p,q}|a_0^2 = \chi_{p,q}^2 \geq \chi_{7,3}^2 > 0. \quad (1615)$$

The Euclidean datum $\{4, 4\}$ has $\chi_{4,4} = 0$, but it cannot be approached through distinct integer regular hyperbolic pairs.

Proof. On integers at least three, $\cos(\pi/p)$ increases with p , while $\sin(\pi/q)$ decreases with q . Hence $A_{p,q}$ increases in either coordinate. The coordinatewise-minimal points in Eq. (1612) are

$$(3, 7), \quad (4, 5), \quad (5, 4), \quad (7, 3). \quad (1616)$$

Every admissible pair dominates one of these. Their ratios are

$$A_{3,7} = \frac{1}{2 \sin(\pi/7)}, \quad A_{4,5} = \frac{1}{\sqrt{2} \sin(\pi/5)},$$

$$A_{5,4} = \sqrt{2} \cos(\pi/5), \quad A_{7,3} = \frac{2 \cos(\pi/7)}{\sqrt{3}}. \quad (1617)$$

First, $A_{7,3} < A_{3,7}$ is equivalent to $2 \sin(2\pi/7) < \sqrt{3}$, which follows from $2\pi/7 < \pi/3$. Next,

$$8 \sin^2(\pi/5) = 5 - \sqrt{5} < 3 \quad (1618)$$

implies $2/\sqrt{3} < A_{4,5}$, while $A_{7,3} < 2/\sqrt{3}$. Finally, $\pi/7 > \pi/8$ gives

$$A_{7,3}^2 < \frac{2 + \sqrt{2}}{3} < \frac{3 + \sqrt{5}}{4} = A_{5,4}^2. \quad (1619)$$

The middle inequality follows, for example, from $4\sqrt{2} < 6$ and $3\sqrt{5} > 6$. Thus $(7, 3)$ is the strict minimizer. Since $\pi/7 < \pi/6$, $2 \cos(\pi/7)/\sqrt{3} > 1$, proving positivity in Eq. (1614). The regular identity $a/R = \chi_{p,q}$ proves the no-route and fixed-edge claims. At $(4, 4)$, numerator and denominator in Eq. (1613) are equal, so $\chi_{4,4} = 0$; the positive gap prevents an integer hyperbolic approach. $\square$

The fixed- $\{p, q\}$ similarity ray is already excluded as a dimensionless curvature sweep. Proposition 28 strengthens that obstruction by ruling out every sequence of distinct strictly regular hyperbolic integer data.

Theorem 45 (Explicit nonregular marked square bridge). *Fix $a_0 > 0$, the marked nearest-neighbor graph $\mathbb{Z}^2$, an origin o , and an orthonormal tangent frame (e_1, e_2) . For $K \leq 0$, set*

$$v_n := a_0(n_1 e_1 + n_2 e_2), \quad \Phi_K(n) := \operatorname{Exp}_o^K(v_n). \quad (1620)$$

Let $g_{\theta,K}$ be the centered orientation-preserving isometry whose differential at o is R_θ . The two layer embeddings are

$$\begin{aligned} \Phi_K^{(1)}(n) &= \Phi_K(n), \\ \Phi_K^{(2)}(n) &= g_{\theta,K} \Phi_K(n) \\ &= \operatorname{Exp}_o^K(R_\theta v_n). \end{aligned} \quad (1621)$$

Keep square nearest-neighbor adjacency on the fixed Hilbert space

$$\mathcal{H} = \ell^2(\mathbb{Z}^2 \times \{1, 2\}). \quad (1622)$$

Let the scalar hopping kernels be finite range and Lipschitz in the relevant in-plane geodesic separation. Uniformly on a normal-coordinate ball of radius L ,

$$\|g_K - g_0\|_{g_0} \leq C_g |K| L^2, \quad (1623)$$

$$|d_K(x, y) - d_0(x, y)| \leq C_d |K| L^3. \quad (1624)$$

If P_L is the two-layer compression to $|v_n| \leq L$, each retained row has at most N_L nonzero hopping entries, and L_{hop} is a common Lipschitz constant, then

$$\|P_L(H_K - H_0)P_L\|_{\text{op}} \leq N_L L_{\text{hop}} C_d |K| L^3. \quad (1625)$$

Thus every fixed finite compression converges in norm. At $K = 0$ the family is the centered twisted square bilayer. At $K < 0$ it is generally nonregular and is not asserted to be a cocompact Fuchsian crystal.

Proof. For $K = -\kappa^2$, geodesic polar coordinates give

$$g_K = dr^2 + \text{sn}_K^2(r) d\phi^2, \quad \text{sn}_K(r) = \frac{\sinh(\kappa r)}{\kappa}. \quad (1626)$$

On $r \leq L$, Taylor's theorem gives

$$\frac{\text{sn}_K^2(r)}{r^2} = 1 + \frac{|K|r^2}{3} + O(|K|^2 L^4), \quad (1627)$$

uniformly when $|K|L^2$ is bounded. In the Euclidean orthonormal polar frame this proves Eq. (1623). Uniform metric equivalence compares the length of every path with its Euclidean length. Taking infima over paths in both directions, and using that relevant path lengths are $O(L)$, proves Eq. (1624).

Centered isotropy intertwines the exponential map and rotation, proving the last equality in Eq. (1621); hence the same distance bound covers interlayer pairs. Each hopping-entry defect is at most $L_{\text{hop}} C_d |K| L^3$. The maximum row sum of the compressed Hermitian defect is therefore at most the right-hand side of Eq. (1625), and the Schur test proves the bound.

At zero curvature the exponential map is the affine identification of the tangent plane, so the square endpoint is exact. For $K < 0$, adjacent marked edges away from the origin generally have position-dependent lengths. More decisively, $d_K(o, \Phi_K(n)) = |v_n|$, so the marked-point count in a radius- L geodesic ball is $O(L^2)$. A cocompact Fuchsian orbit has point count comparable to hyperbolic area and hence grows exponentially in L . Therefore this bridge is not such an orbit. $\square$

The scalar interlayer kernel may include already-declared fixed layer parameters, but no alternative ambient product metric or orbital-matrix tunneling is introduced here.

Theorem 46 (Strong operator and strong-resolvent bridge). *Let $h_K(x, y)$ be the Hermitian kernel of the bridge Hamiltonian on the fixed marking $\mathcal{X} = \mathbb{Z}^2 \times \{1, 2\}$. Assume either finite range, or entrywise local convergence together with uniform Schur tail decay. Assume also*

$$\begin{aligned} \sup_{K \in [-K_*, 0]} \sup_x \sum_y |h_K(x, y)| &\leq M, \\ \sup_{K \in [-K_*, 0]} \sup_y \sum_x |h_K(x, y)| &\leq M < \infty. \end{aligned} \quad (1628)$$

Then

$$H_K \xrightarrow[K \rightarrow 0^-]{s} H_0 \quad (1629)$$

and, for every $z \notin \mathbb{R}$,

$$(H_K - z)^{-1} \xrightarrow[K \rightarrow 0^-]{s} (H_0 - z)^{-1}. \quad (1630)$$

Consequently, for $f \in C_0(\mathbb{R})$ and $\psi \in \mathcal{H}$,

$$\langle \psi, f(H_K) \psi \rangle \longrightarrow \langle \psi, f(H_0) \psi \rangle. \quad (1631)$$

Strong-resolvent convergence alone does not imply Hausdorff convergence of full spectra, convergence of band edges or gaps, fixed Riesz rank, Chern data, or convergence of global Bloch fibers.

Proof. For nonpositive curvature, exponential-map distance comparison gives

$$d_K(\text{Exp}_o^K v, \text{Exp}_o^K w) \geq |v - w|. \quad (1632)$$

Thus a finite physical hopping range couples the support of a finitely supported vector to a curvature-independent finite set of markings. Theorem 45 gives convergence of all entries on that set, so $H_K \psi \rightarrow H_0 \psi$ for finitely supported ψ . In the long-range alternative, uniform Schur-tail decay first removes the common tail and the same finite-set argument handles the remainder.

Equation (1628) and the Schur test give $\sup_K \|H_K\| \leq M$. Approximation by finitely supported vectors then extends convergence to all of $\mathcal{H}$, proving Eq. (1629). The resolvent identity is

$$R_K(z) - R_0(z) = -R_K(z)(H_K - H_0)R_0(z). \quad (1633)$$

Since $\|R_K(z)\| \leq |\text{Im } z|^{-1}$, strong operator convergence proves Eq. (1630). Strong resolvent functional calculus proves Eq. (1631). Spectral pollution and rank changes are possible in this topology, which proves the stated limitations. $\square$

Uniformly isolated spectral data can be added to the strong statement, but their persistence is a separate hypothesis. The following stronger criterion supplies direct quantitative control.

Theorem 47 (Global Schur-norm bridge and spectral stability). *Put $\Delta h_K = h_K - h_0$ and define*

$$\begin{aligned} r_K &:= \sup_x \sum_y |\Delta h_K(x, y)|, \\ c_K &:= \sup_y \sum_x |\Delta h_K(x, y)|. \end{aligned} \quad (1634)$$

If $r_K \rightarrow 0$ and $c_K \rightarrow 0$, then

$$\varepsilon_K := \|H_K - H_0\|_{\text{op}} \leq \sqrt{r_K c_K} \longrightarrow 0. \quad (1635)$$

For self-adjoint H_K ,

$$d_{\text{H}}(\text{spec } H_K, \text{spec } H_0) \leq \varepsilon_K. \quad (1636)$$

Let a positively oriented contour Γ have length ℓ_Γ and distance

$$\delta := \text{dist}(\Gamma, \text{spec } H_0) > 0. \quad (1637)$$

If $\varepsilon_K < \delta$, its Riesz projectors satisfy

$$\|P_K - P_0\|_{\text{op}} \leq \frac{\ell_\Gamma}{2\pi} \frac{\varepsilon_K}{\delta(\delta - \varepsilon_K)}. \quad (1638)$$

The rank of a finite isolated cluster is constant for sufficiently small $|K|$. If its endpoint separation from the complementary spectrum is $g_0 > 0$, then the perturbed separation is at least

$$g_K \geq g_0 - 2\varepsilon_K. \quad (1639)$$

Proof. The Schur test gives Eq. (1635); for Hermitian scalar kernels the row and column defects coincide. The standard resolvent characterization of self-adjoint spectra, or the min-max principle on spectral edges, gives Eq. (1636).

For $z \in \Gamma$, spectral stability gives

$$\|R_0(z)\| \leq \delta^{-1}, \quad \|R_K(z)\| \leq (\delta - \varepsilon_K)^{-1}. \quad (1640)$$

The resolvent identity and contour integration give Eq. (1638). Along $H_0 + t(H_K - H_0)$, the contour stays in the resolvent set; continuity of the finite-rank Riesz projector preserves its integer rank. Each of the two spectral edges defining an isolation gap can move by at most ε_K , proving Eq. (1639) and the sharp half-gap threshold $\varepsilon_K < g_0/2$. $\square$

Proposition 29 (Group and crystallography separation). *At $K = 0$, Eqs. (1620)–(1622) have the actual square translation group $\mathbb{Z}^2$ whenever the endpoint couplings are translation covariant. Ordinary square Bloch characters and diffraction formulas then follow algebraically; a commensurate bilayer uses its common finite-index translation subgroup.*

Neither local metric convergence nor strong or norm convergence of Hamiltonians contracts an unspecified cocompact Fuchsian group, its irreducible dual, subgroup indices, representation dimensions, or exact reciprocal data to those of $\mathbb{Z}^2$. The bridge at $K < 0$ is a nonregular marked model and carries no claim of exact Fuchsian crystallography.

Proof. At $K = 0$, $\text{Exp}_o^0(v) = o + v$, and marked shifts $n \mapsto n + m$ act by Euclidean translations. Translation-covariant couplings commute with these shifts, so Fourier decomposition over the characters of $\mathbb{Z}^2$ is exact. The common subgroup statement is the same algebra applied to the bilayer symmetry group.

Away from the endpoint, the bridge has polynomial rather than cocompact hyperbolic orbit growth, as proved in Theorem 45. More generally, convergence of metric tensors or bounded operators contains no homomorphisms identifying different discrete groups or their unitary duals. Such group data must be supplied separately, proving the negative statement. $\square$

Corollary 28 (Valid policies and manuscript repairs). *There are two valid policies.*

1. Under the bridge policy, use the explicit nonregular family above, or another family satisfying the same certificates, and attach only the local, strong-resolvent, or norm consequences supported by the topology actually proved.

2. Under the endpoint policy, treat exact regular hyperbolic crystals and the Euclidean square crystal as separate universality endpoints. Formula-level metric crossovers may be compared, but no direct microscopic crystallographic path is claimed.

Accordingly, Theorems 14, 53, and 144 and the local moire and reciprocal formulas are metric limits. The local finite-matrix convergence surrounding the bilayer Hamiltonian limit requires a common marking and coupling-entry convergence. Theorem 57 is an algebraic specialization to an actual Abelian endpoint, not an automatic Fuchsian-group limit. Theorems 113 and 115 remain independent square-endpoint benchmarks.

Theorems 122–123 require a smooth transported microscopic family and a verified square endpoint. Theorem 147 is a Euclidean-normalized formula-level limit unless an operator bridge is proved. Theorem 189’s phrase concerning all finite-twist observables is restricted to declared metric or local observables and observables continuous in the verified operator topology. The final square and hyperbolic scaling laws are universality-endpoint laws absent a bridge certificate.

The exact equal-amplitude square five-state target remains root-free. A bridge does not create a root. Transfer of root or no-root information requires the corresponding uniform root-functional, derivative, and spectral-isolation hypotheses in addition to the operator topology. No claim about alternative ambient product metrics or orbital-matrix tunneling is made here.

Proof. The bridge policy is exactly the hierarchy of conclusions in Theorems 45–47. The endpoint policy follows from the positive regular gap in Proposition 28 and the group obstruction in Proposition 29. Applying those distinctions to the hypotheses of the listed manuscript theorems gives each scope repair. Operator convergence controls the Hamiltonian only in its stated topology; it neither supplies a missing zero of a root functional nor identifies unprovided geometric or orbital structures. $\square$

Scalar separation-law universality

The in-plane mismatch and the physical interlayer separation play different roles. The former is intrinsic to the curved layer; the latter may be an effective constitutive law. This distinction permits a product-independent local theorem without assigning an ambient metric to every admissible law. Throughout this subsection the tunneling is scalar, isotropic, and single-orbital.

Definition 1 (Two radial regularity classes). *Fix $h > 0$. A minimal radial separation law is a positive function $\mathcal{D}(h, d)$, $d \geq 0$, which is C^2 from the right near $d = 0$ and satisfies*

$$\begin{aligned} \mathcal{D}(h, 0) &= h, & \mathcal{D}_d(h, 0) &= 0, \\ c_2(h) &:= \mathcal{D}_{dd}(h, 0) > 0. \end{aligned} \quad (1641)$$

It is squared-distance smooth if, near $d = 0$,

$$\mathcal{D}(h, d) = \mathcal{E}(h, d^2), \quad \mathcal{E}(h, \cdot) \in C^2. \quad (1642)$$

These are effective separation laws. An ambient geodesic realization is not assumed.

The exact mismatch $d_\theta(r) = d_K(x, g_\theta x)$, its Jacobi amplitude $J_K(r)$, the intrinsic moire length and area, and all intrinsic registry data are defined entirely in the layer metric. Choosing $\mathcal{D}$ affects only the physical separation and the scalar tunneling profile

$$T_{ij} = \mathcal{T}(\mathcal{D}[h, d_K(x_i, g_\theta x_j)]). \quad (1643)$$

Theorem 48 (Local weak-twist universality). *Suppose the intrinsic same-label mismatch obeys, at fixed K and r ,*

$$d_\theta(r) = J_K(r)|\theta| + O(|\theta|^3). \quad (1644)$$

For a minimal radial law,

$$D_{\text{loc}}(r, \theta) = h + \frac{c_2(h)}{2} J_K^2(r) \theta^2 + o(\theta^2). \quad (1645)$$

If $\mathcal{T}$ is differentiable at h , then

$$T_{\text{loc}}(r, \theta) = \mathcal{T}(h) + \frac{\mathcal{T}'(h)c_2(h)}{2} J_K^2(r) \theta^2 + o(\theta^2). \quad (1646)$$

If instead Eq. (1642) holds and $\mathcal{T} \in C^2$ near h , the two remainders in Eqs. (1645) and (1646) improve to $O(\theta^4)$.

Proof. Taylor expansion at the radial origin gives

$$\mathcal{D}(h, d) = h + \frac{c_2(h)}{2} d^2 + o(d^2). \quad (1647)$$

Equation (1644) implies

$$d_\theta^2(r) = J_K^2(r) \theta^2 + O(\theta^4). \quad (1648)$$

Substitution proves Eq. (1645). Differentiability of the scalar kernel gives

$$\mathcal{T}(h + \delta) = \mathcal{T}(h) + \mathcal{T}'(h)\delta + o(\delta).$$

Here $\delta = O(\theta^2)$, which proves Eq. (1646).

In the stronger class, Taylor expansion in $s = d^2$ gives

$$\mathcal{D}(h, d) = h + \mathcal{E}_s(h, 0)d^2 + O(d^4), \quad c_2(h) = 2\mathcal{E}_s(h, 0). \quad (1649)$$

Equations (1648) and (1649) give an $O(\theta^4)$ distance remainder. Since $\mathcal{T} \in C^2$, its Taylor remainder is $O(\delta^2) = O(\theta^4)$, proving the stronger claim. $\square$

Proposition 30 (Zero radial slope, cusps, and degeneracies). *Let $F_h(u) = \mathcal{D}(h, |u|)$ be isotropic in a signed in-plane displacement u . If F_h is differentiable at $u = 0$, then*

$$\mathcal{D}_d(h, 0) = 0. \quad (1650)$$

By contrast, if the one-sided slope $b(h) = \mathcal{D}_d(h, 0)$ is nonzero, then

$$\mathcal{D}(h, d_\theta) = h + b(h)J_K(r)|\theta| + o(|\theta|). \quad (1651)$$

Whenever $\mathcal{T}'(h)b(h)J_K(r) \neq 0$, the composed tunneling has a genuine $|\theta|$ cusp.

If $c_2(h) = 0$, the quadratic distance coefficient vanishes; in a smooth even law the first nonzero even derivative fixes the higher leading power. If $\mathcal{T}'(h) = 0$, the distance response can remain quadratic while the displayed quadratic tunneling coefficient vanishes.

Proof. Differentiability of F_h supplies a linear covector L at the origin. Isotropy makes L rotation invariant, hence $L = 0$. Along any unit vector e ,

$$\frac{F_h(te) - F_h(0)}{t} = \begin{cases} b(h) + o(1), & t \downarrow 0, \\ -b(h) + o(1), & t \uparrow 0. \end{cases}$$

The two limits agree precisely when $b(h) = 0$, proving Eq. (1650). If $b(h) \neq 0$, one-sided Taylor expansion followed by Eq. (1644) proves Eq. (1651). Composition with a differentiable $\mathcal{T}$ proves the cusp criterion.

For an even smooth law, let the first nonzero Taylor term in d be $a_{2m}d^{2m}$. Equation (1644) then makes its leading twist order $2m$. The statement about $\mathcal{T}'(h) = 0$ follows directly from the kernel Taylor expansion. $\square$

The product law and two useful alternatives are

$$\begin{aligned} \mathcal{D}_{\text{prod}}(h, d) &= \sqrt{h^2 + d^2}, & c_2(h) &= \frac{1}{h}, \\ \mathcal{D}_\lambda(h, d) &= \sqrt{h^2 + \lambda(h)d^2}, & c_2(h) &= \frac{\lambda(h)}{h}, \end{aligned} \quad (1652)$$

where $\lambda(h) > 0$. A genuinely non-product example is

$$\mathcal{D}_{\text{sat}}(h, d) = h + \ell(h) \left[1 - e^{-d^2/[2\ell^2(h)]} \right], \quad \ell(h) > 0, \quad (1653)$$

for which $c_2(h) = 1/\ell(h)$. It is locally admissible but not coercive, illustrating the independence of local and global assumptions.

Corollary 29 (Geometric response ratio). *Compare negative curvature $K = -R^{-2}$ with $K = 0$ at common h , $\mathcal{D}$, and $\mathcal{T}$. Suppose*

$$A(h) := \frac{\mathcal{T}'(h)c_2(h)}{2} \neq 0. \quad (1654)$$

For $r > 0$, with $\Delta T_K = T_{\text{loc},K} - \mathcal{T}(h)$,

$$\lim_{\theta \rightarrow 0} \frac{\Delta T_K(r, \theta)}{\Delta T_0(r, \theta)} = \left[\frac{R \sinh(r/R)}{r} \right]^2. \quad (1655)$$

The ratio is not asserted when $A(h) = 0$, or when the kernel or separation law changes explicitly with curvature.

Proof. Theorem 48 gives $\Delta T_K = A(h)J_K^2(r)\theta^2 + o(\theta^2)$. The same common coefficient multiplies $J_0^2(r) = r^2$ at $K = 0$. Division and $J_K(r) = R \sinh(r/R)$ prove Eq. (1655). The qualifications are exactly the cases in which this common nonzero coefficient is unavailable. $\square$

Proposition 31 (Complete curvature chain rule and local vertex). *Let $K \mapsto h(K)$, $K \mapsto d_K$, and $K \mapsto \mathcal{D}_K$ be differentiable on a declared common fiber or under a declared unitary transport. Then*

$$\begin{aligned} \frac{d}{dK} \mathcal{D}_K(h(K), d_K) &= (\partial_K \mathcal{D}_K)_{\text{exp}} \\ &+ \mathcal{D}_h h'(K) + \mathcal{D}_d d'_K(K). \end{aligned} \quad (1656)$$

All partial derivatives on the right are evaluated at $(h(K), d_K)$. If $\mathcal{T}_K$ also depends explicitly on K ,

$$\frac{d}{dK} \mathcal{T}_K(S_K) = (\partial_K \mathcal{T}_K)_{\text{exp}} + \mathcal{T}'_K(S_K) S'_K, \quad (1657)$$

where $S_K = \mathcal{D}_K(h(K), d_K)$.

At fixed h , for K -independent laws, the geometric factor is $\mathcal{D}_d(h, d)d'_K$. In particular,

$$(\mathcal{D}_{\text{prod}})_d(h, d) = \frac{d}{\sqrt{h^2 + d^2}}, \quad (1658)$$

so the familiar d/D factor is product-specific.

More generally, suppose $\mathcal{D}(h, \cdot)$ and $\mathcal{T}$ are C^2 near d_E and $\mathcal{D}(h, d_E)$, respectively. If a small curvature parameter ε gives

$$d_\varepsilon = d_E + \varepsilon \dot{d}_E + O(\varepsilon^2), \quad (1659)$$

then

$$\begin{aligned} \mathcal{T}(\mathcal{D}(h, d_\varepsilon)) &= \mathcal{T}(\mathcal{D}(h, d_E)) \\ &+ \varepsilon \mathcal{T}'(\mathcal{D}(h, d_E)) \mathcal{D}_d(h, d_E) \dot{d}_E \\ &+ O(\varepsilon^2). \end{aligned} \quad (1660)$$

For the constant-curvature pair expansion and $d_E > 0$,

$$\dot{d}_E = \frac{r_i^2 r_j^2 \sin^2 \varphi_{ij}}{6d_E}. \quad (1661)$$

Proof. Equations (1656) and (1657) are the multivariable chain rule. Equation (1658) follows by differentiation. Taylor expansion of $\mathcal{D}(h, d_\varepsilon)$, followed by Taylor expansion of $\mathcal{T}$, proves Eq. (1660). Substitution of the constant-curvature first variation gives Eq. (1661). For the product law, Eq. (1658) cancels the d_E denominator and recovers the product expression. At $d_E = 0$, the vertex must instead be evaluated from the exact pair expansion or from its continuous limit. $\square$

The differentiability in Proposition 31 is an operator-family hypothesis, not a consequence of sampling discrete regular tessellations. Without a continuous transported family, curvature responses are matched divided differences.

Theorem 49 (Global scalar summability certificates). *Let X_1, X_2 be countable marked layers and*

$$q_{ij} = \mathcal{T}(\mathcal{D}(h, d_{ij})) \quad (1662)$$

the scalar interlayer kernel. If

$$M_r := \sup_i \sum_j |q_{ij}| < \infty, \quad M_c := \sup_j \sum_i |q_{ij}| < \infty, \quad (1663)$$

then the interlayer operator satisfies

$$\|\tau\|_{\text{op}} \leq \sqrt{M_r M_c}. \quad (1664)$$

A sufficient geometric certificate is the following. For some $\rho > 0$, suppose both layer orientations obey the uniform shell bound

$$\#\{j : n\rho \leq d_{ij} < (n+1)\rho\} \leq C_X e^{\sigma n\rho}. \quad (1665)$$

Assume, for $a, \mu > 0$ and $b \geq 0$,

$$\begin{aligned} \mathcal{D}(h, d) &\geq ad - b, \\ |\mathcal{T}(s)| &\leq C_T e^{-\mu s}. \end{aligned} \quad (1666)$$

If $\mu a > \sigma$, then

$$M_r, M_c \leq \frac{C_T C_X e^{\mu b}}{1 - e^{-(\mu a - \sigma)\rho}}. \quad (1667)$$

For a uniformly bounded-density layer at $K = -R^{-2}$, one may take $\sigma = R^{-1}$ after adjusting C_X . The product law has $a = 1$, $b = 0$, and recovers the sufficient threshold $\mu > R^{-1}$.

If $\mathcal{T}(s) = 0$ for $s > D_c$, finite in-plane range follows when the sublevel $\{d : \mathcal{D}(h, d) \leq D_c\}$ is bounded. Properness of $d \mapsto \mathcal{D}(h, d)$ guarantees this for every cutoff. Under the coercive part of Eq. (1666),

$$\mathcal{D}(h, d) \leq D_c \implies d \leq \frac{D_c + b}{a}. \quad (1668)$$

Proof. The Schur test applied to the rectangular kernel gives Eq. (1664). For a point in shell n , Eq. (1666) gives

$$|q_{ij}| \leq C_T e^{\mu b} e^{-\mu a n \rho}.$$

Multiplication by Eq. (1665) and summation of the geometric series proves Eq. (1667). The same argument in the opposite orientation bounds columns. Hyperbolic shell growth is $O(e^{n\rho/R})$, and $\sqrt{h^2 + d^2} \geq d$, proving the product specialization.

Properness makes every compact sublevel bounded. Thus it guarantees that a physical cutoff has bounded in-plane support. For one fixed cutoff, the corresponding sublevel condition is the sharp requirement. The affine bound yields Eq. (1668) directly. $\square$

The local hypotheses in Definition 1 do not imply Theorem 49. In particular, Eq. (1653) is smooth and has positive c_2 , but it is not proper. The product inequality $\sqrt{h^2 + d^2} \geq d$ therefore cannot be assigned to the entire local universality class.

Corollary 30 (Product-independent scope repairs). *The scalar tunneling sector is repaired as follows.*

1. *The exact mismatch, J_K , intrinsic moire scales, and intrinsic area remain independent of the effective separation law.*
2. *Every exact scalar radial profile uses $\mathcal{T}(\mathcal{D}[h, d_K(x, g_\theta y)])$.*
3. *Theorem 16's product coefficient $\mathcal{T}'(h)/(2h)$ becomes $\mathcal{T}'(h)c_2(h)/2$. Minimal C^2 regularity gives $o(\theta^2)$; the $O(\theta^4)$ claim requires the squared-distance-smooth tier.*
4. *The product curvature multiplier d/D becomes $\mathcal{D}_d$, together with the spacing and explicit-curvature terms in Eq. (1656).*
5. *The flat-curvature tunneling vertex is Eq. (1660).*
6. *Infinite-volume boundedness uses Eq. (1663), or a separately verified coercivity-and-decay certificate. Local weak-twist universality alone does not bound the Hamiltonian.*
7. *In every boundary-layer radial formula, $\sqrt{h^2 + d_M^2(u)}$ is replaced by $\mathcal{D}(h, d_M(u))$. Product closed forms and their numerical prefactors remain examples.*

The marked Euclidean bridge does not select a unique $\mathcal{D}$; operator convergence must be proved for the chosen composite kernel. No matrix, orbital, or angular-harmonic tunneling statement is made. None of these local or global replacements supplies a cancellation root; the exact square five-state model remains root-free.

Proof. Items 1 and 2 are the intrinsic-effective separation preceding Eq. (1643). Item 3 is Theorem 48; items 4 and 5 are Proposition 31. Item 6 is Theorem 49. Item 7 is direct substitution in the exact radial profile. The remaining limitations follow because neither a marked metric bridge nor a local scalar composition specifies additional orbital data, global operator convergence, or a zero of an independent root functional. $\square$

Orbital-covariant tunneling profiles

Finite orbital structure requires more than replacing scalar entries by matrices. The source and target fibers, frame transformations, adjoint direction, symmetry intertwiners, and active-rank convention must all be declared.

Definition 2 (Directed orbital kernel and frame covariance). *At marked sites i and j , let*

$$E_{1i} \simeq \mathbb{C}^{n_1}, \quad E_{2j} \simeq \mathbb{C}^{n_2}. \quad (1669)$$

The directed interlayer kernel is the rectangular map

$$\mathbf{T}_{ij} = \mathbf{T}(D_{ij}, \Omega_{ij}) : E_{2j} \longrightarrow E_{1i}. \quad (1670)$$

Here Ω_{ij} contains the bond direction in the chosen orbital frames, the relative frames, and every other declared internal orientation datum. Under local frame changes,

$$\mathbf{T}_{ij} \longmapsto U_{1i} \mathbf{T}_{ij} U_{2j}^\dagger. \quad (1671)$$

Thus entries are frame dependent, while singular values and the operator and Frobenius norms are invariant.

On

$$\mathcal{H}_\ell = \bigoplus_i E_{\ell i}, \quad \tau : \mathcal{H}_2 \longrightarrow \mathcal{H}_1, \quad (1672)$$

the block of τ from E_{2j} to E_{1i} is $\mathbf{T}_{ij}$.

Proposition 32 (Hermiticity and conditional symmetries). *If $h_\ell = h_\ell^\dagger$, then*

$$H_{\text{BL}}(\theta) = \begin{pmatrix} h_1(\theta) & \tau_\theta \\ \tau_\theta^\dagger & h_2(\theta) \end{pmatrix} \quad (1673)$$

is Hermitian for rectangular τ_θ . A reverse directed kernel is

$$\mathbf{T}_{ji}^{21} = \mathbf{T}_{ij}^\dagger; \quad (1674)$$

the directed matrix $\mathbf{T}_{ij}$ need not itself be Hermitian.

Point-group covariance is the separate intertwining requirement

$$\mathbf{T}_{gi,gj} = R_{1,g;i} \mathbf{T}_{ij} R_{2,g;j}^\dagger, \quad (1675)$$

with transformed (D, Ω) understood. Time reversal requires declared antiunitaries Θ_ℓ , including

$$\Theta_1 \tau_\theta \Theta_2^{-1} = \tau_\theta, \quad (1676)$$

with momentum and all time-reversal-odd internal data transformed, together with the analogous intralayer laws. Layer exchange requires its own unitary identification of the layer fibers. An even signed-twist spectrum follows only if layer exchange, a mirror, or another declared spatial unitary L_θ satisfies

$$L_\theta H_{\text{BL}}(\theta) L_\theta^{-1} = H_{\text{BL}}(-\theta). \quad (1677)$$

Proof. Taking the adjoint of Eq. (1673) exchanges the two off-diagonal blocks and leaves the diagonal blocks fixed. This proves Hermiticity and Eq. (1674) without comparing a rectangular matrix with its own adjoint.

Equation (1675) is precisely the condition that the directed kernel intertwine the two orbital representations. Equation (1676) is the antiunitary time-reversal condition at fixed structural twist. Equation (1677) is a separate spatial equivalence and proves spectral evenness. Without these additional identities, Hermiticity alone implies none of the symmetry conclusions. Scalar reality and transpose formulas are special choices of fibers and frames. $\square$

Theorem 50 (Full local distance–orientation expansion). *Choose a parallel orbital-frame trivialization near $\theta = 0$. Suppose*

$$D_\theta = h + A_2\theta^2 + o(\theta^2), \quad A_2 = \frac{c_2(h)}{2} J_K^2(r), \quad (1678)$$

$$\Omega_\theta = \Omega_0 + \theta\dot{\Omega} + \frac{\theta^2}{2}\ddot{\Omega} + o(\theta^2). \quad (1679)$$

If $\mathbf{T}(D, \Omega)$ is C^2 , then

$$\begin{aligned} \mathbf{T}_\theta &= \mathbf{T}_0 + \theta \nabla_\Omega \mathbf{T}_0[\dot{\Omega}] \\ &+ \theta^2 \left\{ A_2 \partial_D \mathbf{T}_0 + \frac{1}{2} \nabla_\Omega \mathbf{T}_0[\ddot{\Omega}] \right. \\ &\quad \left. + \frac{1}{2} \nabla_\Omega^2 \mathbf{T}_0[\dot{\Omega}, \dot{\Omega}] \right\} + o(\theta^2). \end{aligned} \quad (1680)$$

All derivatives are evaluated at (h, Ω_0) . Thus orbital orientation can produce an $O(\theta)$ response. The quadratic distance law controls only the term $A_2 \partial_D \mathbf{T}_0$.

If $\Omega_\theta = \Omega_0$, the separation law is squared-distance smooth, and the kernel is C^2 in D , then the distance-only remainder improves to $O(\theta^4)$.

Proof. Apply the covariant two-variable Taylor formula. The first orientation increment is $O(\theta)$, whereas $D_\theta - h = O(\theta^2)$. Consequently, the mixed D - Ω term is $O(\theta^3)$, and the second distance derivative first contributes at $O(\theta^4)$. The remaining terms through order two are exactly those in Eq. (1680). In the fixed-orientation case, the squared-distance separation theorem gives $D_\theta - h = A_2\theta^2 + O(\theta^4)$; ordinary Taylor expansion in D then gives the final claim. $\square$

Proposition 33 (Inverse-free matrix logarithmic slope). *Let P, Q be fixed or covariantly transported finite-rank projectors and*

$$B_\mu(D, \Omega) = P \mathbf{T}_\mu(D, \Omega) Q. \quad (1681)$$

A basis-covariant common logarithmic slope is the aligned derivative

$$\nabla_D B_\mu(h, \Omega) = -\frac{\zeta}{h} B_\mu(h, \Omega), \quad (1682)$$

with the same real ζ for every profile μ . No inverse of B_μ is used.

At fixed orientation, Eq. (1682) gives

$$B_{\mu, \theta} = \left[1 - \frac{\zeta c_2(h)}{2h} J_K^2(r) \theta^2 \right] B_{\mu, 0} + o(\theta^2). \quad (1683)$$

For the product law $c_2(h) = 1/h$, the coefficient is $-\zeta J_K^2(r)/(2h^2)$. If the vertical active matrices are identified across profiles by symmetry-compatible left and right unitaries, Eq. (1683) is one common matrix response.

Without that identification, only its relative multiplier is common. Every singular value satisfies

$$\begin{aligned} \sigma_a(B_{\mu, \theta}) &= \left[1 - \frac{\zeta c_2(h)}{2h} J_K^2(r) \theta^2 \right] \\ &\times \sigma_a(B_{\mu, 0}) + o(\theta^2). \end{aligned} \quad (1684)$$

Proof. The covariant derivative in Eq. (1682) transforms by the same left and right unitaries as B_μ , proving basis covariance even when the rectangular matrix is singular. Substitution of Eq. (1678) proves Eq. (1683). If its scalar bracket is $a(\theta)$, then

$$\|B_{\mu, \theta} - a(\theta) B_{\mu, 0}\|_{\text{op}} = o(\theta^2).$$

The singular-value perturbation inequality and $a(\theta) > 0$ for small θ prove Eq. (1684). $\square$

Common orientation jets may be added to all profiles through Theorem 50; they do not convert the distance sector into a global spectral theorem. Local slope alignment alone does not control active-projector motion or the normalized full operator.

Theorem 51 (Invariant active coupling scales). *On one preselected fixed or transported active fiber, define*

$$V_\nu = P_\nu \delta \tau Q_\nu^{\text{act}}. \quad (1685)$$

Let

$$N_\nu = r_\nu z_\nu > 0, \quad k_\nu = \text{rank } V_\nu, \quad (1686)$$

and order the nonzero singular values as $\sigma_1 \geq \dots \geq \sigma_{k_\nu} > 0$. Define

$$w_{\nu, \text{op}}^{\text{eff}} := \|V_\nu\|_{\text{op}} = \sigma_1(V_\nu), \quad (1687)$$

$$w_{\nu, \text{RMS}}^{\text{eff}} := \frac{\|V_\nu\|_F}{\sqrt{N_\nu}}, \quad (1688)$$

$$w_{\nu, \min}^{(k)} := \sigma_k(V_\nu), \quad 1 \leq k \leq k_\nu. \quad (1689)$$

Then

$$\frac{w_{\nu, \text{op}}^{\text{eff}}}{\sqrt{N_\nu}} \leq w_{\nu, \text{RMS}}^{\text{eff}} \leq \sqrt{\frac{k_\nu}{N_\nu}} w_{\nu, \text{op}}^{\text{eff}}, \quad (1690)$$

and

$$\sqrt{\frac{k}{N_\nu}} w_{\nu, \min}^{(k)} \leq w_{\nu, \text{RMS}}^{\text{eff}}. \quad (1691)$$

The operator scale controls the strongest channel, the RMS scale controls total active power, and the minimum retained scale controls simultaneous coupling of all retained directions. Under covariant changes of orbital bases and complete-cluster bases,

$$V_\nu \mapsto U_1 V_\nu U_2^\dagger, \quad (1692)$$

so all these scales are invariant.

Proof. The singular-value decomposition gives

$$\|V_\nu\|_F^2 = \sum_{a=1}^{k_\nu} \sigma_a^2(V_\nu). \quad (1693)$$

Bounding this sum below by σ_1^2 , above by $k_\nu \sigma_1^2$, and below by $k \sigma_k^2$ proves Eqs. (1690) and (1691). Covariant transformation of P_ν , Q_ν^{act} , and $\delta\tau$ gives Eq. (1692); left and right unitary multiplication preserves singular values. $\square$

Fixed Frobenius power does not fix the singular distribution. For example, $\text{diag}(c, 0)$ and $\text{diag}(c/\sqrt{2}, c/\sqrt{2})$ have the same Frobenius norm, but different operator norms, minimum singular values, and ranks.

Proposition 34 (Frobenius Parseval and operator-norm obstruction). *For a square-integrable rectangular matrix field with mean removed, Euclidean Fourier analysis gives*

$$\frac{1}{A} \int_A \|\delta \mathbf{T}(x)\|_F^2 dx = \sum_{G \neq 0} \|\widehat{\delta \mathbf{T}}(G)\|_F^2. \quad (1694)$$

For a finite group G , define

$$\widehat{F}(\rho) = \frac{\sqrt{d_\rho}}{|G|} \sum_{g \in G} F(g) \otimes \rho(g)^*. \quad (1695)$$

Then

$$\frac{1}{|G|} \sum_{g \in G} \|F(g)\|_F^2 = \sum_{\rho} \|\widehat{F}(\rho)\|_F^2. \quad (1696)$$

For a mean-removed field the trivial term vanishes. There is no corresponding operator-norm Parseval equality.

Proof. Apply scalar Parseval to every matrix entry and sum the resulting identities; the sum of entry moduli squared is the Frobenius norm squared. For Eq. (1696), apply finite-group Schur orthogonality to each entry with the normalization in Eq. (1695). For the obstruction, take

$$F(t) = \text{diag}(e^{it}, e^{2it}).$$

Its pointwise operator norm is one, so the averaged squared norm is one. Its two nonzero coefficient matrices each have operator norm one, so the sum of their squared operator norms is two. $\square$

Theorem 52 (Thresholded-rank stability and exact-rank obstruction). *For equal-size rectangular matrices V, W and every singular-value index,*

$$|\sigma_a(V) - \sigma_a(W)| \leq \|V - W\|_{\text{op}}. \quad (1697)$$

Fix a threshold $\tau > 0$, let $1 \leq k \leq m = \min(r, q)$, and use the convention $\sigma_{m+1} = 0$. If, for some $g > 0$,

$$\sigma_k(V) \geq \tau + g, \quad \sigma_{k+1}(V) \leq \tau - g, \quad (1698)$$

then $\|V - W\|_{\text{op}} < g$ preserves the number k of singular values above τ .

If V is rank deficient, no positive perturbation radius preserves its exact algebraic rank against arbitrary perturbations. Exact rank k is instead established, for example, by

$$V(s) = L(s)C(s)R(s)^\dagger, \quad (1699)$$

where L, R are k -column isometries, $C(s) \in \text{GL}(k, \mathbb{C})$, and $\inf_s \sigma_k(C(s)) > 0$.

Proof. The min-max characterization of singular values, applied to $\|Vx\| \leq \|Wx\| + \|(V - W)x\|$ and then with V, W exchanged, proves Eq. (1697). It moves every singular value by less than g , proving the threshold statement.

For rank-deficient V , choose right and left null directions v, u compatible with an additional singular channel. Then $W = V + \varepsilon uv^\dagger$ has $\|W - V\|_{\text{op}} = \varepsilon$ and larger rank; block-adapted singular bases make this explicit. In Eq. (1699), the invertible middle factor and the two isometries give rank exactly k , while its lower singular bound prevents collapse. $\square$

Theorem 53 (Quantitative fixed-finite-channel endpoint exponents). *Let the preselected target and source projectors have fixed finite ranks r and q , with fixed channel normalization $N > 0$. Put $s = |\sin(\theta/2)|$, and let*

$$V(s) : \mathbb{C}^q \longrightarrow \mathbb{C}^r$$

be C^1 for $0 < s \leq s_0$, with fixed bare amplitude $w_0 > 0$. If an exact retained rank is used, assume a fixed-rank certificate such as Eq. (1699). Suppose

$$\delta(s) := \left\| \frac{V(s)}{w_0} - V_* \right\|_{\text{op}} \longrightarrow 0, \quad V_* \neq 0, \quad (1700)$$

$$\varepsilon(s) := \frac{\|s \partial_s V(s)\|_{\text{op}}}{w_0} \longrightarrow 0. \quad (1701)$$

Put $m = \min(r, q)$ and

$$\gamma_{\text{op}}(s) := \frac{\|V(s)\|_{\text{op}}}{w_0}, \quad (1702)$$

$$\gamma_{\text{RMS}}(s) := \frac{\|V(s)\|_F}{\sqrt{N} w_0}. \quad (1703)$$

Then

$$|\gamma_{\text{op}} - \|V_*\|_{\text{op}}| \leq \delta(s), \quad (1704)$$

$$\left| \gamma_{\text{RMS}} - \frac{\|V_*\|_F}{\sqrt{N}} \right| \leq \sqrt{\frac{m}{N}} \delta(s). \quad (1705)$$

Both scales therefore have finite positive limits.

At every point where the scalar norm is differentiable,

$$|s \partial_s \ln \gamma_{\text{op}}| \leq \frac{\varepsilon(s)}{\gamma_{\text{op}}(s)}, \quad (1706)$$

$$|s \partial_s \ln \gamma_{\text{RMS}}| \leq \frac{\sqrt{m} \varepsilon(s)}{\|V(s)\|_F / w_0}. \quad (1707)$$

The same bounds hold for the absolute upper Dini derivatives at singular-value crossings. Thus the logarithmic profile exponent tends to zero for both roles, classically almost everywhere and in the Dini sense.

If $\sigma_k(V_*) > 0$, then

$$\frac{\sigma_k(V(s))}{w_0} \longrightarrow \sigma_k(V_*) > 0, \quad (1708)$$

so all k retained directions remain coupled. The condition $V_* \neq 0$ alone guarantees only positive operator and RMS limits.

For dispersion order $p > 0$, let

$$\alpha_{\text{role}} = \frac{w_0 \gamma_{\text{role}}}{E_{M,p}}, \quad E_{M,p} \propto \xi_M^{-p}. \quad (1709)$$

With

$$\nu_\xi = -\partial_{\ln s} \ln \xi_M, \quad \eta_\gamma = -\partial_{\ln s} \ln \gamma_{\text{role}}, \quad (1710)$$

the exact effective-coupling exponent is

$$\beta_X := -\partial_{\ln s} \ln \alpha_{\text{role}} = p\nu_\xi + \eta_\gamma. \quad (1711)$$

Consequently,

$$\beta_{X,E} \longrightarrow p, \quad \beta_{X,H} \longrightarrow 0. \quad (1712)$$

Equivalently, the Euclidean effective coupling grows as s^{-p} , while at fixed negative curvature it grows as $\ln^p(1/s)$, up to finite positive role-dependent amplitudes.

Proof. The reverse triangle inequality proves Eq. (1704). For a matrix of size $r \times q$,

$$\|A\|_F \leq \sqrt{m} \|A\|_{\text{op}}.$$

Apply this to $V(s)/w_0 - V_*$ and use the reverse triangle inequality in Frobenius norm to prove Eq. (1705). Since $V_* \neq 0$, both limiting norms are positive.

Matrix norms are locally Lipschitz. At their almost-everywhere differentiability points,

$$\begin{aligned} |\partial_s \|V\|_{\text{op}}| &\leq \|\partial_s V\|_{\text{op}}, \\ |\partial_s \|V\|_F| &\leq \|\partial_s V\|_F \leq \sqrt{m} \|\partial_s V\|_{\text{op}}. \end{aligned}$$

Division by the positive scales proves Eqs. (1706) and (1707). The norm Lipschitz inequalities also bound the one-sided difference quotients, proving the Dini statement. Their right-hand sides vanish by Eqs. (1700) and (1701).

Equation (1697) gives Eq. (1708). Finally, $\alpha_{\text{role}} \propto \gamma_{\text{role}} \xi_M^p$, so logarithmic differentiation proves Eq. (1711). The Euclidean endpoint has $\nu_\xi \rightarrow 1$, while the fixed-negative-curvature logarithmic endpoint has $\nu_\xi \rightarrow 0$. Together with $\eta_\gamma \rightarrow 0$, this proves Eq. (1712) and the two stated growth laws. $\square$

Corollary 31 (Matrix-profile repairs and obstructions). *The orbital generalization has the following consequences.*

1. The early scalar-to-matrix promotion requires Definition 2, the adjoint direction, and separate symmetry intertwiners. Scalar time-reversal, transpose, layer exchange, and even-twist identities are conditional special cases.
2. Theorem 146 is a Frobenius Parseval statement as in Proposition 34; it has no operator-norm counterpart.
3. The active γ_ν^{eff} must be role labeled by Eqs. (1687)–(1703), with retained singular values reported when simultaneous channel coupling matters.
4. Fixed total mean-square coupling does not fix active singular redistribution, directional flattening, or a root.
5. Theorem 154 becomes the local matrix alignment condition Eq. (1682), together with fixed or common orientation jets. Local agreement does not imply global spectral universality.
6. Theorem 168 continues to require convergence of normalized active operators up to symmetry-compatible left and right unitary identifications, not merely agreement of slopes, norms, or singular values.
7. Theorems 148 and 191 retain their endpoint exponents under Theorem 53. Growing dimensions, vanishing active overlap, singular-value collapse, threshold crossing, rank change, or explicit angular/orbital scaling can instead yield $\eta_\gamma^* \neq 0$.

The exact scalar square five-state model remains root-free. A matrix orbital model is a different richer Hamiltonian and requires its own explicit root, derivative, active-overlap, gap, and coherence analysis. The global block kernel that must later be controlled is $\mathbf{T}(D_{ij}, \Omega_{ij})$; no matrix summability or tail-rate claim is made here.

Proof. Items 1–3 follow from Proposition 32, Proposition 34, and Theorem 51. The equal-Frobenius counterexample proves item 4. Proposition 33 and the independent orientation terms in Theorem 50 prove item 5. Equality of scalar summaries cannot imply unitary equivalence of operators, proving item 6. Theorem 53 proves item 7 under its hypotheses; failure of any listed hypothesis removes that conclusion. None of these statements supplies a zero of an independent root functional or a global block-kernel estimate. $\square$

Infinite-volume hopping admissibility and tail propagation

All infinite-volume spectral statements require an operator-theoretic admissibility condition. Local smoothness of a hopping profile does not provide that condition. We formulate one condition that applies to

scalar kernels and to finite-orbital matrix kernels, and then propagate its quantitative error through the spectral constructions used below.

Block-kernel admissibility

Let Λ_ℓ , $1 \leq \ell \leq m$, be finitely many uniformly discrete layers. At $x \in \Lambda_\ell$, let the orbital fiber $E_{\ell x}$ have uniformly bounded finite dimension, and set

$$\mathcal{H} = \bigoplus_{\ell=1}^m \bigoplus_{x \in \Lambda_\ell} E_{\ell x}. \quad (1713)$$

For every ordered layer pair, assume that the relevant in-plane distance has shells

$$S_n(x) = \{y : n\rho \leq d(x, y) < (n+1)\rho\} \quad (1714)$$

with $\rho > 0$, and that both row and column shell counts satisfy

$$\#S_n(x) \leq C_X e^{h_\Lambda n\rho}, \quad n \geq 0. \quad (1715)$$

The constants are uniform in the base point and layer pair. For a bounded-density layer in curvature $K = -R^{-2}$, one may use $h_\Lambda = R^{-1}$, after increasing C_X .

Definition 3 (Unified finite-range or exponential condition). *A block kernel*

$$K_{\ell x, \ell' y} : E_{\ell' y} \longrightarrow E_{\ell x}$$

is Hermitian if

$$K_{\ell' y, \ell x} = K_{\ell x, \ell' y}^\dagger. \quad (1716)$$

It is admissible if, for each layer pair, one of the following holds.

1. It is finite range: for constants $A_{\ell\ell'}$ and $L_{\ell\ell'} < \infty$,

$$\|K_{\ell x, \ell' y}\|_{\text{op}} \leq A_{\ell\ell'}, \quad K_{\ell x, \ell' y} = 0 \text{ if } d(x, y) > L_{\ell\ell'}. \quad (1717)$$

2. It has a direct composite exponential envelope

$$\|K_{\ell x, \ell' y}\|_{\text{op}} \leq A_{\ell\ell'} e^{-\mu_{\ell\ell'} d(x, y)}, \quad \delta_{\ell\ell'} := \mu_{\ell\ell'} - h_\Lambda > 0. \quad (1718)$$

The constants may depend on the layer pair but are uniform along every parameter path to which an infinite-volume theorem is applied.

Using block operator norms absorbs the finite orbital multiplicity into $A_{\ell\ell'}$; it does not change the decay exponent.

Theorem 54 (Explicit block Schur and tail bounds). *Under Definition 3, an exponential layer-pair block obeys*

$$M_{\ell\ell'} \leq \frac{A_{\ell\ell'} C_X}{1 - e^{-\delta_{\ell\ell'} \rho}} \quad (1719)$$

for both its row and column Schur sums. For an arbitrary cutoff $L \geq 0$, put

$$N_L = \lfloor L/\rho \rfloor, \quad q_{\ell\ell'} = e^{-\delta_{\ell\ell'} \rho}. \quad (1720)$$

The omitted blocks with $d(x, y) > L$ satisfy

$$M_{\ell\ell'}^{>L} \leq \frac{A_{\ell\ell'} C_X q_{\ell\ell'}^{N_L}}{1 - q_{\ell\ell'}}, \quad (1721)$$

$$M_{\ell\ell'}^{>L} \leq C_{\ell\ell'}^{\text{tr}} e^{-\delta_{\ell\ell'} L}, \quad (1722)$$

$$C_{\ell\ell'}^{\text{tr}} := \frac{A_{\ell\ell'} C_X e^{\delta_{\ell\ell'} \rho}}{1 - e^{-\delta_{\ell\ell'} \rho}}. \quad (1723)$$

For a finite-range pair, with $N_0 = \lfloor L_{\ell\ell'}/\rho \rfloor$,

$$M_{\ell\ell'} \leq A_{\ell\ell'} C_X \sum_{n=0}^{N_0} e^{h_\Lambda n\rho}, \quad (1724)$$

and its tail is zero for $L \geq L_{\ell\ell'}$.

Let $R(L)$ and $C(L)$ be the maximum full block-row and block-column sums of these pairwise tails. Then

$$\|K - K^{(L)}\|_{\text{op}} \leq \sqrt{R(L)C(L)}. \quad (1725)$$

For a Hermitian kernel with symmetric pairwise bounds, this becomes

$$\|K - K^{(L)}\|_{\text{op}} \leq \max_{\ell} \sum_{\ell'} M_{\ell\ell'}^{>L}. \quad (1726)$$

Proof. For $y \in S_n(x)$, the product of Eqs. (1715) and (1718) is bounded by

$$A_{\ell\ell'} C_X e^{-\delta_{\ell\ell'} n\rho}.$$

Summing the geometric series proves Eq. (1719). If $d(x, y) > L$, the shell $n = \lfloor L/\rho \rfloor$ can still contribute, so the tail must begin at N_L . This proves Eq. (1721). Since

$$\rho \lfloor L/\rho \rfloor \geq L - \rho,$$

the floor factor gives Eqs. (1722) and (1723). Summing the finitely many allowed shells proves Eq. (1724).

Apply the operator-valued Schur test to the scalar kernel $\|K_{\ell x, \ell' y}\|_{\text{op}}$. It gives Eq. (1725). Hermiticity identifies the row and column estimates and proves Eq. (1726). $\square$

Corollary 32 (Sharp bilayer block-row propagation). *For*

$$H = \begin{pmatrix} h_1 & \tau \\ \tau^\dagger & h_2 \end{pmatrix},$$

let the two intralayer tails be bounded by $\varepsilon_1(L)$ and $\varepsilon_2(L)$, and let $\|\tau - \tau^{(L)}\| \leq \varepsilon_\perp(L)$. Then

$$\|H - H^{(L)}\|_{\text{op}} \leq \max\{\varepsilon_1(L), \varepsilon_2(L)\} + \varepsilon_\perp(L). \quad (1727)$$

Each term is zero in its finite-range branch, or has its own explicit constant and margin from Eqs. (1722) and (1723).

Proof. The diagonal block has norm $\max\{\varepsilon_1, \varepsilon_2\}$. The off-diagonal Hermitian block $\begin{pmatrix} 0 & B \\ B^\dagger & 0 \end{pmatrix}$ has norm $\|B\|$. The triangle inequality proves Eq. (1727). $\square$

Proposition 35 (Bounded and domain-controlled operator tiers). *The Hermitian kernel in Theorem 54, initially defined on finitely supported vectors, extends uniquely to an everywhere-defined bounded self-adjoint operator on $\mathcal{H}$. If D is instead an unbounded self-adjoint reference operator and V is an admissible bounded Hermitian hopping operator, then*

$$H = D + V, \quad \text{Dom } H = \text{Dom } D, \quad (1728)$$

is self-adjoint. More singular or unbounded hopping is outside this tier unless a separate closed-form or domain certificate is supplied.

Proof. The Schur estimate gives a bounded extension from the dense finitely supported subspace. Kernel Hermiticity makes the extension symmetric. An everywhere-defined bounded symmetric operator is self-adjoint. The second assertion is the bounded self-adjoint perturbation theorem applied to D . $\square$

Separation laws and differentiated kernels

Theorem 55 (Propagation through a general separation law). *Let the scalar or matrix profile obey, uniformly in all orientation data,*

$$\|\mathbf{T}(D, \Omega)\|_{\text{op}} \leq C_T e^{-\mu D}. \quad (1729)$$

If the separation law satisfies

$$\mathcal{D}(h, d) \geq ad - b, \quad a > 0, \quad b \geq 0, \quad (1730)$$

then the composite block kernel satisfies

$$\|\mathbf{T}(\mathcal{D}(h, d), \Omega)\|_{\text{op}} \leq C_T e^{\mu b} e^{-\mu ad}. \quad (1731)$$

It is therefore admissible whenever

$$\mu a > h_\Lambda. \quad (1732)$$

The constants in Theorem 54 are obtained by replacing A by $C_T e^{\mu b}$, μ by μa , and δ by $\mu a - h_\Lambda$.

Without Eq. (1730), admissibility requires a direct composite block envelope such as Eq. (1718). A physical cutoff $\mathbf{T}(D, \Omega) = 0$ for $D > D_c$ gives finite in-plane range only when the sublevel $\{d : \mathcal{D}(h, d) \leq D_c\}$ is bounded. Under Eq. (1730), its range is at most $(D_c + b)/a$.

Proof. Equations (1729) and (1730) give

$$e^{-\mu \mathcal{D}(h, d)} \leq e^{\mu b} e^{-\mu ad},$$

which proves Eq. (1731). Theorem 54 applies with effective decay rate μa , proving the summability and tail claims. The cutoff statement is precisely the definition of the displayed sublevel; the affine range bound follows by rearranging Eq. (1730). $\square$

For a parameter family, the positive margin in Eq. (1732) must be uniform along the path. Matrix covariance alone does not imply Eq. (1729); its operator-norm constant must also be uniform in Ω .

Theorem 56 (Weighted Schur moments for derivatives). *Let $K(p)$ be represented on a fixed fiber, or on fibers identified by a declared unitary transport. For every multi-index α used in an observable, suppose*

$$\|\partial_p^\alpha K_{xy}(p)\|_{\text{op}} \leq A_\alpha (1 + d(x, y))^{s_\alpha} e^{-\mu_\alpha d(x, y)} \quad (1733)$$

uniformly in p , where s_α is a nonnegative integer and $\delta_\alpha := \mu_\alpha - h_\Lambda > 0$. With $N_L = \lfloor L/\rho \rfloor$, the derivative tail has the explicit Schur bound

$$\mathcal{E}_\alpha(L) := A_\alpha C_X \sum_{n=N_L}^{\infty} [1 + (n+1)\rho]^{s_\alpha} e^{-\delta_\alpha n\rho}. \quad (1734)$$

It also obeys the closed estimate

$$\mathcal{E}_\alpha(L) \leq C_\alpha e^{-\delta_\alpha L/2}, \quad (1735)$$

$$C_\alpha := \frac{A_\alpha C_X e^{\delta_\alpha \rho/2} (1 + \rho)^{s_\alpha} (1 + 2s_\alpha/\delta_\alpha)^{s_\alpha}}{1 - e^{-\delta_\alpha \rho/2}}. \quad (1736)$$

Consequently, the corresponding operator derivative exists in norm and its cutoff error is at most $\mathcal{E}_\alpha(L)$.

The case $|\alpha| = 0$ controls energies and spectra. Generalized velocities and first projector derivatives require the C^1 envelopes. Curvatures, Hessians, and second projector derivatives require the corresponding C^2 envelopes. Berry and Chern statements additionally require a uniform spectral gap. None of these differentiated conclusions follows from the base envelope alone.

Proof. On shell n , the polynomial is at most $[1 + (n+1)\rho]^{s_\alpha}$, while shell growth cancels h_Λ from the exponential. Summation proves Eq. (1734). For $s = s_\alpha$ and $\delta = \delta_\alpha$,

$$1 + (n+1)\rho \leq (1 + \rho)(1 + n\rho)$$

and

$$(1 + x)^s e^{-\delta x/2} \leq (1 + 2s/\delta)^s, \quad x \geq 0.$$

Extracting a remaining factor $e^{-\delta n\rho/2}$, summing the geometric series, and using $\rho N_L \geq L - \rho$ proves Eqs. (1735) and (1736). Operator-valued Schur gives norm convergence. Differentiating a phase or transported kernel produces the weighted derivatives in Eq. (1733); without their summability, termwise differentiation and differentiated-projector control are not justified. $\square$

Spectral propagation of the tail

Theorem 57 (Norm-error propagation). *Let H and H_L be bounded self-adjoint operators on an identified Hilbert space and put*

$$\varepsilon_L = \|H - H_L\|_{\text{op}}. \quad (1737)$$

Then

$$d_H(\text{spec } H, \text{spec } H_L) \leq \varepsilon_L, \quad (1738)$$

and each lower or upper spectral edge moves by at most ε_L . For finite fibers, or for an isolated finite-rank cluster tracked through the comparison, eigenvalue errors are at most ε_L , while bandwidth and isolation-gap errors are at most

$$|W - W_L| \leq 2\varepsilon_L, \quad |g - g_L| \leq 2\varepsilon_L. \quad (1739)$$

In particular,

$$g_L \geq g - 2\varepsilon_L. \quad (1740)$$

Let Γ have length ℓ_Γ and satisfy

$$\delta := \text{dist}(\Gamma, \text{spec } H) > 0, \quad \varepsilon_L < \delta. \quad (1741)$$

For $z \in \Gamma$,

$$\|(z - H_L)^{-1} - (z - H)^{-1}\| \leq \frac{\varepsilon_L}{\delta(\delta - \varepsilon_L)}. \quad (1742)$$

The associated Riesz projectors obey

$$\|P_L - P\|_{\text{op}} \leq \frac{\ell_\Gamma \varepsilon_L}{2\pi\delta(\delta - \varepsilon_L)}. \quad (1743)$$

If the right-hand side is below one and P has finite rank, the cluster rank is preserved on the identified fibers.

For fixed finite-rank contractions P_a, Q_a , the active blocks $V = P_a A Q_a$ and $V_L = P_a A_L Q_a$ satisfy

$$|\sigma_j(V_L) - \sigma_j(V)| \leq \|A_L - A\|_{\text{op}}. \quad (1744)$$

If the projectors also move, then

$$\|P_L A_L Q_L - P A Q\| \leq \|A_L - A\| + \|P_L - P\| \|A_L\| + \|Q_L - Q\| \|A\|. \quad (1745)$$

Derivative observables require the derivative-tail bounds of Theorem 56.

Proof. The resolvent characterization of the spectrum gives Eq. (1738); the min-max principle gives the finite-rank eigenvalue statement. Each bandwidth or gap is a difference of two edges, proving Eqs. (1739) and (1740). The resolvent identity

$$(z - H_L)^{-1} - (z - H)^{-1} = (z - H_L)^{-1} (H_L - H) (z - H)^{-1}$$

together with

$$\|(z - H)^{-1}\| \leq \delta^{-1}, \quad \|(z - H_L)^{-1}\| \leq (\delta - \varepsilon_L)^{-1}$$

proves Eq. (1742). Contour integration gives Eq. (1743). Two finite-rank orthogonal projectors at distance below one have isomorphic ranges, proving rank stability. No finite multiplicity conclusion is asserted for an uncontrolled infinite-rank cluster.

Compression is contractive, and singular values are one-Lipschitz in operator norm, which proves Eq. (1744). Insert and subtract $P A_L Q_L$ and $P A Q_L$ to obtain Eq. (1745). $\square$

If the Hamiltonian is normalized by an energy scale $E_M > 0$, the error entering a normalized theorem is ε_L/E_M , not ε_L . A cutoff adequate at fixed physical energy can therefore be inadequate when $E_M \rightarrow 0$.

Finite covers and the local-global boundary

Theorem 58 (Finite-cover compression and moment limit). *Let H satisfy the admissibility condition and let a boundaryless cover have injectivity radius $r_{\text{inj},N}$. Fix an observation radius L_{obs} .*

For a finite-range Hamiltonian of range L_0 , all local matrix elements are exact whenever

$$r_{\text{inj},N} > L_{\text{obs}} + L_0. \quad (1746)$$

For the exponential branch, choose

$$L_{c,N} < r_{\text{inj},N} - L_{\text{obs}}. \quad (1747)$$

After the declared local identification J_N , let $a_N(L_{\text{obs}}, L_{c,N})$ be the separate operator error caused by an angular approximant on the truncated patch. Then

$$\begin{aligned} & \|P_{\text{obs}}[J_N^\dagger H_N^{(L_{c,N})} J_N - H]P_{\text{obs}}\| \\ & \leq \varepsilon_{L_{c,N}} + a_N(L_{\text{obs}}, L_{c,N}), \end{aligned} \quad (1748)$$

where $\varepsilon_{L_{c,N}}$ is the explicit physical tail from Theorem 54.

For a fixed moment order r , quotient aliasing of the truncated Hamiltonian is absent when

$$2r_{\text{inj},N} > rL_{c,N}. \quad (1749)$$

If $\|H\|, \|H^{(L_{c,N})}\| \leq M$, then

$$\|H^r - (H^{(L_{c,N})})^r\| \leq rM^{r-1}\varepsilon_{L_{c,N}}. \quad (1750)$$

For full long-range operators on both sides of the cover comparison, the two truncations contribute at most $2rM^{r-1}\varepsilon_{L_{c,N}}$; for a simulated truncated cover, one of these two terms is absent.

A safe joint limit is

$$L_{c,N} \rightarrow \infty, \quad \frac{L_{c,N}}{r_{\text{inj},N}} \rightarrow 0, \quad a_N \rightarrow 0. \quad (1751)$$

It yields local compression convergence and, under the normalized-trace moment hypotheses, weak convergence of the density-of-states measures. Under a fixed marking, uniform Schur bounds and convergence on finitely supported vectors also yield strong operator, hence strong-resolvent, convergence.

None of these local, strong-resolvent, or weak-DOS conclusions alone gives global band-edge, isolation-gap, Riesz-rank, or magic-band convergence. Those conclusions require a global operator-norm or norm-resolvent comparison on identified spaces, or independent uniform two-edge bounds.

Proof. The injectivity buffers ensure that every retained path lies in an isometrically embedded patch, proving exactness in the finite-range branch and exactness of the truncated local matrix in the exponential branch. The triangle inequality then gives Eq. (1748). A closed path contributing to the r th moment has length at most $rL_{c,N}$, so Eq. (1749) excludes every nontrivial deck identification.

The telescoping identity

$$H^r - G^r = \sum_{j=0}^{r-1} H^{r-1-j} (H - G) G^j$$

proves Eq. (1750). Uniform compact spectral support and convergence of all fixed moments imply weak convergence of normalized spectral measures. Uniform boundedness plus convergence on the dense finitely supported subspace gives strong operator convergence. The resolvent identity then gives strong-resolvent convergence. None of these steps is a norm estimate for the full cover operator, so the final global limitation follows. $\square$

For an angular mismatch $\delta\theta_N$, the geometric displacement at radius L_{geo} is bounded by

$$b_N = 2R \operatorname{arsinh}[\sinh(L_{\text{geo}}/R) |\sin(\delta\theta_N/2)|]. \quad (1752)$$

Converting b_N to a_N requires the kernel's local Lipschitz and Schur bounds. This angular error is not part of the hopping tail.

Proposition 36 (Weak-DOS and local-convergence obstruction). *Let*

$$A_N = \operatorname{diag}(0, \dots, 0, 1)$$

on $\mathbb{C}^N$. Its normalized empirical measure converges weakly to δ_0 , and every fixed coordinate patch eventually sees the zero operator, but

$$\sup \operatorname{spec} A_N = 1 \quad (1753)$$

for every N . Thus local convergence and weak-DOS convergence do not control a global spectral edge or gap.

Proof. The empirical measure is $(1 - N^{-1})\delta_0 + N^{-1}\delta_1$, which converges weakly to δ_0 . The exceptional coordinate can be placed outside every fixed marked patch, whereas Eq. (1753) remains exact. $\square$

Subsequent assumption ledger

Corollary 33 (Infinite-volume repair ledger). *Every subsequent infinite-volume use of a hopping Hamiltonian is subject to Definition 3, with a uniform positive margin along the relevant parameter path. The named statements are repaired as follows.*

1. Theorem 13 is a local finite-cover statement. Its finite-range exactness uses Eq. (1746); the exponential extension uses Eqs. (1747) and (1748), with angular error separate.
2. Theorem 15 defines an infinite bilayer operator only after the finite-range or exponential condition is imposed on every intra- and interlayer block. Hermiticity and Proposition 35 then give the stated self-adjoint operator.
3. Theorem 16's local Taylor formulas are finite-matrix or local statements. Its global truncation claim uses separate intra- and interlayer margins and the sharper bilayer bound (1727). A general separation law uses $\mu a > h_\Lambda$, not merely $\mu > h_\Lambda$.
4. Theorem 31 requires absolute group-kernel summability

$$\sum_{\gamma \in \Gamma_M} \|h(\gamma)\|_{\text{op}} < \infty. \quad (1754)$$

The metric shell condition implies this only when the group displacement length and shell count entering the sum satisfy the declared bounds. Differentiation over representation space requires the corresponding weighted differentiated group sum.

5. Theorem 36 is exact for a bounded finite-range Hamiltonian and fixed moment order under its no-aliasing buffer. It is finite-cover and normalized-trace exactness, not global spectral-edge convergence.
6. Theorem 37 uses a uniform positive exponential margin, the joint limit (1751), and the explicit tail. Its conclusion is local and weak-DOS convergence unless a stronger global comparison is supplied.
7. Theorem 66 is finite-dimensional as written. An infinite-volume extension requires a common transported Hilbert space and norm continuity of a uniformly admissible self-adjoint family. Differentiable spectral claims require the matching C^1 or C^2 Schur tier.
8. Theorem 76 includes the hopping tail, cover, angular, symmetry, and solver contributions in its total operator error. Energy, bandwidth, and gap errors then follow from Theorem 57; derivative errors do not.
9. Theorem 79 remains a conditional thermodynamic criterion. A weak DOS or local-cover limit does not identify its global bandwidth and gap. Those quantities require stable target transport and a global norm, norm-resolvent, or direct two-edge certificate.

10. Theorems 105 and 108 require an admissible thermodynamic sequence, positive uniform isolation, stable cluster identification, and all separate cutoff, cover, angular, symmetry, solver, and coherence errors. Their velocity factors also require derivative-tail convergence.
11. Theorems 160 and 169 require uniform operator-norm convergence of the normalized admissible Hamiltonians on fixed transported fibers. If the normalization scale tends to zero, the required tail is $\varepsilon_L/E_M \rightarrow 0$. Gap and bandwidth collapse follow at the C^0 tier; generalized-velocity collapse requires C^1 convergence, and unsmoothed DOS does not follow from operator norm alone.
12. the fixed-star collapse theorem retains its fixed active shell, transport, and normalized tail counted exactly once. the isolation-gap result retains its complete residual budget and strict half-gap condition. the resonant-shell analysis retains its buffered resonant shell, detuned tail, and derivative budgets.
13. the metric-flatness result retains the distinction between strong local convergence and its separate global Schur-norm hypothesis. the separation-law theorem supplies Eq. (1730); the orbital-covariant result supplies covariant matrix blocks, but must add the orientation-uniform operator-norm and derivative envelopes used here.

Proof. Items 1, 5, and 6 are Theorem 58. Items 2, 3, 12, and 13 follow from Theorem 54, Corollary 32, Proposition 35, and Theorem 55. Absolute convergence proves item 4 by the Weierstrass test; differentiated representation sums require the same test after differentiation. Items 7–11 are Theorems 56 and 57, together with Proposition 36. These implications use exactly the topology stated in each item and no stronger one. $\square$

Corollary 34 (Scope repairs and obstructions). *The following limitations are sharp for this addendum.*

1. At zero or negative exponential margin, the displayed shell series does not yield a bounded operator or a vanishing tail. Another direct summability certificate would be needed.
2. Local separation-law regularity, matrix covariance, or finite orbital dimension does not imply global summability.
3. Base operator convergence controls energy and spectral sets but not velocity, curvature, Berry data, Chern data, or differentiated projectors.
4. Local finite-cover and weak-DOS convergence do not imply convergence of global edges, gaps, target ranks, or operational magic scores.

5. Summability and self-adjointness do not create a cancellation root, an isolation gap, or coherence. The exact scalar square five-state model remains root-free. A richer scalar or matrix model requires an independently established root, derivative, active overlap, gap, and coherence analysis.

Thus every magic or master-curve conclusion remains conditional on uniform admissibility, the topology required by the observable, and the independent spectral hypotheses. No stronger global thermodynamic claim is inferred from local cover data.

Proof. Items 1 and 2 follow from the geometric series and Theorem 55. Item 3 is Theorem 56. Item 4 is witnessed by Proposition 36. The admissibility conditions contain no equation imposing a zero of a root functional or a positive spectral separation, proving item 5 and the final conditional statement. $\square$

Intrinsic global spectral islands

For a generic incommensurate hyperbolic bilayer, a finite-cover bandwidth is not itself an infinite-volume observable. The global spectral object must first be defined for the bounded self-adjoint infinite Hamiltonian provided by the hopping-admissibility theorem.

Configuration and hull spectra

Fix a twist θ , and let H_θ be the admissible bounded self-adjoint operator on the infinite one-particle Hilbert space. Write

$$\Sigma_\theta = \text{spec } H_\theta. \quad (1755)$$

Because H_θ is bounded and self-adjoint, $\Sigma_\theta \subset \mathbb{R}$ is nonempty and compact.

Definition 4 (Intrinsic target spectral island). *A target spectral island is a proper nonempty compact subset $\mathcal{I}_C \subset \Sigma_\theta$ such that*

$$\Delta_C := \text{dist}(\mathcal{I}_C, \Sigma_\theta \setminus \mathcal{I}_C) > 0. \quad (1756)$$

Its global bandwidth and spectral projection are

$$W_C := \text{diam } \mathcal{I}_C, \quad (1757)$$

$$P_C := \mathbf{1}_{\mathcal{I}_C}(H_\theta). \quad (1758)$$

No finite-rank assumption is included.

Proposition 37 (Clopen and Riesz characterizations). *The positive-separation condition in Definition 4 is equivalent to $\mathcal{I}_C$ being a proper nonempty clopen subset of the relative spectrum Σ_θ . It is also equivalent to the existence of a bounded open set $\mathcal{U} \subset \mathbb{C}$ whose boundary Γ lies in the resolvent set and satisfies*

$$\mathcal{I}_C = \Sigma_\theta \cap \mathcal{U}. \quad (1759)$$

For a positively oriented such contour,

$$P_C = \frac{1}{2\pi i} \oint_{\Gamma} (z - H_{\theta})^{-1} dz. \quad (1760)$$

The range of P_C may be infinite dimensional.

Proof. Positive separation gives disjoint relative neighborhoods of the island and its complement, so the island is clopen. Conversely, a clopen subset and its complement are disjoint compact subsets of $\mathbb{R}$, hence have positive distance.

With $0 < r < \Delta_C/2$, compactness supplies finitely many intervals whose union contains the island and lies in its open r -neighborhood. Thin rectangles around their union give a finite rectifiable contour in the resolvent set and prove Eq. (1759). The continuous and holomorphic functional calculi agree on this separated spectral subset, proving Eq. (1760). Functional calculus imposes no rank restriction. $\square$

Origin-independent bulk language requires an explicit configuration space. Let $\mathbb{X}_{\theta}$ be the compact local-pattern hull of a declared incommensurate configuration, and let a group G act continuously on it. Suppose

$$\xi \mapsto H_{\theta,\xi}$$

is a uniformly bounded strongly continuous family of self-adjoint operators on a fixed identified Hilbert space, and suppose

$$H_{\theta,g\xi} = U_g H_{\theta,\xi} U_g^{\dagger} \quad (1761)$$

for a unitary representation U_g .

Theorem 59 (Common spectrum on a minimal covariant hull). *If the action of G on $\mathbb{X}_{\theta}$ is minimal, then*

$$\text{spec } H_{\theta,\xi} = \text{spec } H_{\theta,\eta} \quad (1762)$$

for all $\xi, \eta \in \mathbb{X}_{\theta}$. Thus W_C and Δ_C are registry independent when the same common spectral island is selected.

If minimality, strong continuity, or covariance is unavailable, the origin-independent robust spectrum is instead

$$\Sigma_{\theta}^{\text{hull}} := \overline{\bigcup_{\xi \in \mathbb{X}_{\theta}} \text{spec } H_{\theta,\xi}}. \quad (1763)$$

An island used in bulk language must then be isolated relative to $\Sigma_{\theta}^{\text{hull}}$. The union can close a gap that is present for an individual configuration.

Proof. Fix $\lambda \in \text{spec } H_{\theta,\xi}$ and $\varepsilon > 0$. Since a bounded self-adjoint operator has spectrum equal to its approximate point spectrum, choose a unit vector ψ with

$$\|(H_{\theta,\xi} - \lambda)\psi\| < \varepsilon.$$

Minimality gives $g \in G$ for which $g\eta$ is sufficiently close to ξ . Strong continuity, evaluated on the fixed vector ψ , gives

$$\|(H_{\theta,g\eta} - H_{\theta,\xi})\psi\| < \varepsilon.$$

Hence ψ has residual below 2ε for $H_{\theta,g\eta}$. By covariance, $U_g^{\dagger}\psi$ has the same residual for $H_{\theta,\eta}$, so $\lambda \in \text{spec } H_{\theta,\eta}$. Interchanging ξ and η proves Eq. (1762).

Without the hypotheses used in this argument, no equality follows. The closed union in Eq. (1763) contains every configuration spectrum and is compact by uniform boundedness. Adding spectral points can reduce or close a configuration-specific gap. $\square$

Quantitative transport from global approximants

Let $S_N \subset \mathbb{R}$ be the complete spectrum of an approximant. For compact nonempty subsets of $\mathbb{R}$, write d_H for Hausdorff distance.

Theorem 60 (Quantitative transport of a spectral island). *Assume the genuine global certificate*

$$d_H(S_N, \Sigma_{\theta}) \leq \eta_N, \quad 0 \leq \eta_N < \frac{\Delta_C}{3}. \quad (1764)$$

Put $r = \Delta_C/3$, and use the fixed neighborhood

$$\mathcal{I}_{C,N} := \{s \in S_N : \text{dist}(s, \mathcal{I}_C) \leq r\}, \quad (1765)$$

$$\mathcal{B}_{C,N} := S_N \setminus \mathcal{I}_{C,N}. \quad (1766)$$

Both sets are nonempty compact sets, and

$$d_H(\mathcal{I}_{C,N}, \mathcal{I}_C) \leq \eta_N, \quad (1767)$$

$$d_H(\mathcal{B}_{C,N}, \Sigma_{\theta} \setminus \mathcal{I}_C) \leq \eta_N. \quad (1768)$$

Consequently, with

$$W_{C,N} := \text{diam } \mathcal{I}_{C,N}, \quad (1769)$$

$$\Delta_{C,N} := \text{dist}(\mathcal{I}_{C,N}, \mathcal{B}_{C,N}), \quad (1770)$$

one has

$$|W_{C,N} - W_C| \leq 2\eta_N, \quad (1771)$$

$$|\Delta_{C,N} - \Delta_C| \leq 2\eta_N, \quad (1772)$$

$$\Delta_{C,N} \geq \Delta_C - 2\eta_N > 0. \quad (1773)$$

Proof. For $x \in \mathcal{I}_C$, the global certificate gives $s \in S_N$ with $|s - x| \leq \eta_N < r$, so $s \in \mathcal{I}_{C,N}$. Conversely, take $s \in \mathcal{I}_{C,N}$, and choose $x \in \Sigma_{\theta}$ with $|s - x| \leq \eta_N$. If $x \notin \mathcal{I}_C$, then

$$\Delta_C \leq \text{dist}(x, \mathcal{I}_C) \leq \eta_N + r < \Delta_C,$$

a contradiction. Thus $x \in \mathcal{I}_C$, proving Eq. (1767).

For $x \in \Sigma_{\theta} \setminus \mathcal{I}_C$, choose $s \in S_N$ within η_N . Then

$$\text{dist}(s, \mathcal{I}_C) \geq \Delta_C - \eta_N > r,$$

so $s \in \mathcal{B}_{C,N}$. Conversely, if $s \in \mathcal{B}_{C,N}$, a spectral point within η_N cannot belong to the island, because that would put s inside its r -neighborhood. This proves Eq. (1768), as well as compactness and nonemptiness of both selected sets.

The diameter of compact sets changes by at most twice their Hausdorff distance. More generally, if $d_H(A, A') \leq \eta$ and $d_H(B, B') \leq \eta$, then

$$|\text{dist}(A, B) - \text{dist}(A', B')| \leq 2\eta.$$

Apply these two facts to Eqs. (1767) and (1768). This proves Eqs. (1771)–(1773). $\square$

The fixed neighborhood is chosen from the intrinsic gap, rather than from an approximant. It therefore prevents post-selection of whichever finite-cover cluster looks narrowest.

Global certificates and a sharp topology obstruction

Proposition 38 (Sufficient global spectral certificates). *Each of the following independently supplies the hypothesis of Theorem 60.*

1. On a common Hilbert space,

$$\|H_N - H_\theta\|_{\text{op}} \leq \eta_N \quad (1774)$$

implies

$$d_H(\text{spec } H_N, \Sigma_\theta) \leq \eta_N.$$

2. On changing Hilbert spaces, suppose there are contractions $J_N : \mathcal{H} \rightarrow \mathcal{H}_N$ and $K_N : \mathcal{H}_N \rightarrow \mathcal{H}$, and numbers $0 \leq a_N < 1$, $b_N \geq 0$, such that

$$\begin{aligned} \|J_N u\| &\geq (1 - a_N)\|u\|, \\ \|K_N v\| &\geq (1 - a_N)\|v\|, \end{aligned} \quad (1775)$$

$$\begin{aligned} \|H_N J_N - J_N H_\theta\| &\leq b_N, \\ \|H_\theta K_N - K_N H_N\| &\leq b_N. \end{aligned} \quad (1776)$$

Then

$$d_H(\text{spec } H_N, \Sigma_\theta) \leq \frac{b_N}{1 - a_N}. \quad (1777)$$

The reverse map and second intertwining estimate are the explicit no-pollution half of this certificate.

3. A direct independently verified Hausdorff bound (1764) may be used without an operator comparison.

A generalized norm-resolvent or quasiunitary scheme on changing spaces is sufficient only when its stated hypotheses prove both spectral inclusion directions, equivalently no loss and no pollution, on the full energy range used in the island comparison.

Proof. Item 1 is the standard self-adjoint spectral stability bound. For $\lambda \in \Sigma_\theta$, take unit approximate eigenvectors u_j . Equations (1775) and (1776) give

$$\text{dist}(\lambda, \text{spec } H_N) \leq \frac{b_N}{1 - a_N}.$$

The same argument with an approximate eigenvector of H_N and the map K_N gives the reverse inclusion. Taking the two suprema proves Eq. (1777). Item 3 is exactly the hypothesis needed by Theorem 60. $\square$

Proposition 39 (Escaping rank-one spectral pollution). *On $\ell^2(\mathbb{N})$, let e_N be the standard basis and*

$$H_N = |e_N\rangle\langle e_N|, \quad H = 0. \quad (1778)$$

Then $H_N \rightarrow H$ strongly, every fixed finite-coordinate compression is eventually equal to that of H , and $H_N \rightarrow H$ in strong resolvent sense. Nevertheless,

$$\text{spec } H_N = \{0, 1\}, \quad \text{spec } H = \{0\}, \quad (1779)$$

so the spectral point 1 persists and the Hausdorff error is one.

Proof. For every $\psi \in \ell^2(\mathbb{N})$,

$$\|H_N \psi\| = |\langle e_N, \psi \rangle| \rightarrow 0.$$

If Q_m projects onto the first m coordinates, then $Q_m H_N Q_m = 0$ for $N > m$. For $z \in \mathbb{C} \setminus \mathbb{R}$,

$$(H_N - z)^{-1} = -z^{-1}(I - H_N) + (1 - z)^{-1}H_N, \quad (1780)$$

$$\begin{aligned} (H_N - z)^{-1} - (H - z)^{-1} \\ = \frac{1}{z(1 - z)} H_N. \end{aligned} \quad (1781)$$

The last expression converges strongly to zero. Direct diagonalization gives Eq. (1779). Thus strong-resolvent and local convergence do not exclude a persistent global spectral point. $\square$

verification policy and repairs

Theorem 61 (Global-island verification dichotomy). *For a generic incommensurate system, global bandwidth and isolation mean the intrinsic quantities*

$$W_{\mathcal{C}} = \text{diam } \mathcal{I}_{\mathcal{C}}, \quad \Delta_{\mathcal{C}} = \text{dist}(\mathcal{I}_{\mathcal{C}}, \Sigma_\theta \setminus \mathcal{I}_{\mathcal{C}})$$

of the infinite configuration spectrum, or of the common or closed-union hull spectrum when bulk origin independence is invoked.

A finite-cover value approximates these quantities only after a global certificate such as Proposition 38. Without it, the justified conclusions are restricted to fixed-radius local observables and, under the separate trace hypotheses, weak density-of-states limits. Strong-resolvent convergence is not a global certificate.

For an exact commensurate system, global claims may instead use a complete exact spectral realization. A selected or incomplete family of finite-dimensional automorphic sectors is not, by itself, the complete spectrum.

Proof. The first statement is Definition 4. Theorem 60 proves the established approximant statement. Proposition 39 disproves the upgrade from local or strong-resolvent convergence to global spectral convergence. In an exact commensurate realization, the complete decomposition is an exact representation of the operator spectrum; the same conclusion cannot follow from an undeclared subset of sectors. $\square$

Corollary 35 (Repairs of global magic and master-curve claims). *The global statements are repaired as follows.*

1. In Theorem 79, $\limsup_N W_{C,N}$ and $\liminf_N \Delta_{C,N}$ are conservative diagnostics of the chosen sequence. They equal the intrinsic island values only after $\eta_N \rightarrow 0$ in Eq. (1764), in which case Eqs. (1771) and (1772) give the rates.
2. Theorems 105, 108, and 109 use the intrinsic island and its full functional-calculus projector. Their finite-cover bandwidth, gap, coherence, and operational score describe a generic incommensurate bulk object only after global island transport and the independent coherence and derivative conditions have been verified.
3. Every subsequent magic or master-curve theorem using global W_C or Δ_C inherits the same certificate uniformly along its parameter path. Generalized velocities and projector derivatives additionally retain the C^1 or C^2 weighted Schur tiers of the hopping-admissibility theorem.
4. the isolation-gap result's positive gap bound remains valid when its residual is a genuine global two-edge or operator-norm comparison for the same target island. A local residual does not become global through this definition.
5. the translation-coherence theorem's complete-cluster and covariance hypotheses remain independent. The intrinsic Riesz projector can have infinite rank, so finite-rank participation formulas require their own finite-cover or finite-rank hypotheses.
6. the metric-flatness result's strong-local versus global-norm distinction and the hopping-admissibility theorem's finite-range or positive-margin admissibility, tail, domain, and derivative tiers remain unchanged.

A general theorem deriving edge and gap convergence from finite-range cover hypotheses remains to be supplied separately. Likewise, a theorem showing that arbitrary admissible cover towers give the same limit remains separate. Neither conclusion is proved here.

The spectral-island definition creates no root, gap, coherence, or universal curve. The exact scalar square five-state model remains root-free. A richer model still requires its own root, isolation, transport, and coherence certificates.

Proof. Item 1 is Theorem 60. Items 2 and 3 combine the intrinsic definition with the verification dichotomy; spectral transport does not control derivatives or coherence. Items 4–6 preserve the distinct hypotheses of the cited prior results. The two remaining requirements concern, respectively, a general finite-cover edge theorem and tower independence, neither of which follows from the present conditional transport theorem. Finally, none of the definitions or Hausdorff estimates imposes a zero of an independent root functional. $\square$

Corollary 36 (Obstruction list). *The following implications are invalid without additional hypotheses:*

$$\begin{aligned} \text{local cover} &\not\Rightarrow \text{global spectrum,} \\ \text{strong resolvent} &\not\Rightarrow \text{spectral Hausdorff,} \\ \text{weak DOS} &\not\Rightarrow \text{edges/gaps,} \\ \text{one tower} &\not\Rightarrow \text{all towers.} \end{aligned} \quad (1782)$$

Configuration-specific gaps are not origin-independent unless the common minimal-hull theorem applies or they survive in the closed-union hull spectrum. These obstructions apply equally to global flatness, magic scores, and master-curve claims.

Proof. The first two failures are explicit in Proposition 39. Weak convergence of probability measures does not control support extrema or gaps. The transport theorem is conditional on its Hausdorff hypothesis for each sequence and therefore contains no tower-independence conclusion. The hull qualification is Theorem 59. $\square$

DOS, spectral loss, and spectral pollution on finite covers

Finite hyperbolic covers can reproduce every fixed local pattern while retaining global spectral modes of vanishing normalized weight. Weak density-of-states convergence, approximation of the infinite spectrum, and exclusion of finite-cover outliers are therefore three distinct statements.

Finite propagation and weak DOS

Let X be an infinite regular cover with a cocompact deck action and a finite orbital fiber per fundamental cell. Let H be a periodic bounded self-adjoint operator on $\ell^2(X; \mathbb{C}^\nu)$ with propagation radius $L_0 < \infty$. Let X_N be finite boundaryless covers with injectivity radii $r_{\text{inj},N} \rightarrow \infty$, and let H_N be the induced periodic operators. Assume

$$\|H\|, \sup_N \|H_N\| \leq M < \infty. \quad (1783)$$

Write τ for the canonical normalized trace per fundamental cell and

$$\tau_N(A) = \frac{\text{Tr } A}{\dim \ell^2(X_N; \mathbb{C}^\nu)}. \quad (1784)$$

The spectral measures ν and ν_N are defined by

$$\tau[f(H)] = \int f d\nu, \quad \tau_N[f(H_N)] = \int f d\nu_N. \quad (1785)$$

Theorem 62 (Fixed moments and weak normalized DOS). *For every fixed integer $k \geq 0$,*

$$2r_{\text{inj},N} > kL_0 \quad (1786)$$

implies

$$\tau_N(H_N^k) = \tau(H^k). \quad (1787)$$

Consequently,

$$\nu_N \implies \nu, \quad (1788)$$

meaning

$$\lim_{N \rightarrow \infty} \int f d\nu_N = \int f d\nu \quad (1789)$$

for every $f \in C([-M, M])$.

Proof. Every diagonal matrix element of H^k is a sum over closed k -step hopping paths of physical length at most kL_0 . Under Eq. (1786), such a path has no nontrivial deck identification and lifts uniquely between the infinite and finite covers. Periodicity and normalized averaging therefore give Eq. (1787).

Equation (1783) places all measures on the common compact interval $[-M, M]$. Moment stabilization gives convergence for polynomials. Uniform polynomial approximation on $[-M, M]$ and the fact that all measures have total mass one prove Eq. (1789). $\square$

No spectral loss

Put

$$\Sigma = \text{spec } H, \quad S_N = \text{spec } H_N. \quad (1790)$$

Theorem 63 (Lifted Weyl vectors and no loss). *Let $\lambda \in \Sigma$. For every $\varepsilon > 0$, there is a finitely supported unit vector ψ such that*

$$\|(H - \lambda)\psi\| < \varepsilon. \quad (1791)$$

Once the support of ψ and its L_0 -neighborhood embed isometrically in X_N , its lift ψ_N satisfies

$$\|(H_N - \lambda)\psi_N\| = \|(H - \lambda)\psi\|, \quad (1792)$$

and hence

$$\text{dist}(\lambda, S_N) \leq \|(H_N - \lambda)\psi_N\| < \varepsilon. \quad (1793)$$

In fact the one-sided Hausdorff error

$$\ell_N := \sup_{\lambda \in \Sigma} \text{dist}(\lambda, S_N) \quad (1794)$$

obeys $\ell_N \rightarrow 0$.

Proof. For a bounded self-adjoint operator, every spectral point belongs to the approximate point spectrum. Approximate its unit Weyl vector by a finitely supported vector. Boundedness of $H - \lambda$, followed by normalization, gives Eq. (1791).

Finite propagation makes every matrix element used by $(H - \lambda)\psi$ lie in the L_0 -neighborhood of the support. Local cover exactness proves Eq. (1792). For a self-adjoint operator B and a unit vector u , the spectral theorem gives

$$\text{dist}(\lambda, \text{spec } B) \leq \|(B - \lambda)u\|,$$

which proves Eq. (1793).

To make the conclusion uniform, choose a finite $\varepsilon/3$ -net in the compact set Σ , lift one finite Weyl vector for each net point, and take the largest required cover index. The distance to S_N is one-Lipschitz in λ , so $\ell_N < \varepsilon$ eventually. $\square$

This theorem is only the inclusion $\Sigma \subset (S_N)_{\ell_N}$. It does not bound points of S_N that remain far from Σ .

The nonamenable outlier

Proposition 40 (Regular-tree spectrum and persistent quotient mode). *Let $d \geq 3$, $q = d - 1$, and let $A_{\mathbb{T}_d}$ be the adjacency operator of the infinite d -regular tree. Then*

$$\text{spec } A_{\mathbb{T}_d} = [-2\sqrt{q}, 2\sqrt{q}]. \quad (1795)$$

If G_N is a finite d -regular graph of girth tending to infinity, then its universal cover is $\mathbb{T}_d$, its normalized DOS converges weakly to the tree DOS, and

$$d \in \text{spec } A_{G_N} \quad (1796)$$

for every N . Therefore

$$\begin{aligned} \sup \text{spec } A_{G_N} - \sup \text{spec } A_{\mathbb{T}_d} &= d - 2\sqrt{d-1} \\ &= (\sqrt{d-1} - 1)^2 > 0. \end{aligned} \quad (1797)$$

Proof. Choose an end of the tree and orient every edge away from it. Every vertex has one incoming edge and q outgoing edges. For finitely supported f , apply

$$2|ab| \leq q^{-1/2}|a|^2 + q^{1/2}|b|^2$$

on every oriented edge. Each vertex receives total coefficient $\sqrt{q} + \sqrt{q}$, so

$$|\langle f, A_{\mathbb{T}_d} f \rangle| \leq 2\sqrt{q} \|f\|^2.$$

Thus $\|A_{\mathbb{T}_d}\| \leq 2\sqrt{q}$.

For $0 < t < \pi$, put $\lambda_t = 2\sqrt{q} \cos t$, choose a root, and define

$$u_L(x) = \begin{cases} q^{-r(x)/2} \sin[(r(x) + 1)t], & 0 \leq r(x) \leq L, \\ 0, & r(x) > L. \end{cases} \quad (1798)$$

The recurrence

$$u(r-1) + qu(r+1) = \lambda_t u(r)$$

holds on the interior shells. The squared norm of u_L grows linearly in L , whereas the root defect and the two boundary-shell defects have uniformly bounded squared norm. Hence

$$\frac{\|(A_{T_d} - \lambda_t)u_L\|}{\|u_L\|} \rightarrow 0.$$

Every point of the open interval in Eq. (1795) is therefore spectral; the endpoints follow by closedness. The norm bound proves equality.

For each fixed moment order, sufficiently large girth makes every relevant closed walk identical to its tree lift. Theorem 62 gives weak DOS convergence. The constant vector on G_N is an eigenvector with eigenvalue d , and the degree bound gives $\|A_{G_N}\| \leq d$. Thus d is the upper edge, proving Eqs. (1796) and (1797). Its normalized DOS weight is $|V(G_N)|^{-1}$, which vanishes. $\square$

Thus neither local exactness nor weak normalized DOS detects every global finite-cover mode.

Faithful DOS support and no pollution

Proposition 41 (What a faithful limiting trace proves). *Suppose the trace τ defining ν is faithful on the C^* -algebra generated by H . Then*

$$\text{supp } \nu = \Sigma. \quad (1799)$$

Together with weak convergence $\nu_N \Rightarrow \nu$, this gives the no-loss conclusion of Theorem 63. It does not give no pollution: spectral outliers whose normalized multiplicity tends to zero are invisible to the weak limit.

Proof. If an open interval U meets Σ , choose a nonzero continuous function $f \geq 0$ supported in U with $f(H) \neq 0$. Faithfulness gives $\int f d\nu = \tau[f(H)] > 0$, proving Eq. (1799). Weak convergence then gives $\int f d\nu_N > 0$ eventually, so $S_N \cap U \neq \emptyset$. Compactness upgrades this pointwise statement to uniform no loss. Proposition 40 proves the final limitation. $\square$

Define the pollution error

$$p_N := \sup_{s \in S_N} \text{dist}(s, \Sigma). \quad (1800)$$

Then

$$d_H(S_N, \Sigma) = \max\{\ell_N, p_N\}. \quad (1801)$$

Theorem 64 (No-pollution certificates). *Let $S_N, \Sigma \subset [-M, M]$. Each of the following is a sufficient no-pollution certificate.*

1. A direct upper inclusion

$$p_N \leq \varepsilon_N, \quad \varepsilon_N \rightarrow 0. \quad (1802)$$

2. The uniform residual lower bound

$$\inf_{\|v\|=1} \|(H_N - \lambda)v\| \geq \text{dist}(\lambda, \Sigma) - \varepsilon_N \quad (1803)$$

for every real $\lambda \in [-M, M]$. Outside the ε_N -neighborhood of Σ , this is equivalently the resolvent estimate

$$\|(H_N - \lambda)^{-1}\| \leq \frac{1}{\text{dist}(\lambda, \Sigma) - \varepsilon_N}. \quad (1804)$$

3. On a declared invariant reduced or tempered sector $\mathcal{K}_N$, put $\tilde{H}_N = H_N|_{\mathcal{K}_N}$. If

$$\limsup_{N \rightarrow \infty} \|p(\tilde{H}_N)\| \leq \|p(H)\| \quad (1805)$$

for every real polynomial p , then the retained spectra have no pollution.

In item 3, no loss must also be proved inside the same retained sector. Removing the trivial representation alone is not a universal certificate: other non-tempered sectors can remain.

Proof. Item 1 is the definition of no pollution. For self-adjoint H_N ,

$$\inf_{\|v\|=1} \|(H_N - \lambda)v\| = \text{dist}(\lambda, S_N).$$

Setting $\lambda = s \in S_N$ in Eq. (1803) gives $\text{dist}(s, \Sigma) \leq \varepsilon_N$, proving item 2. Conversely, the triangle inequality for distance functions shows that Eq. (1802) implies the residual lower bound. The resolvent formula follows from the spectral theorem.

For item 3, suppose pollution persists. Passing to a subsequence gives $s_N \in \text{spec } \tilde{H}_N$ and $s_N \rightarrow s \notin \Sigma$. On $[-M, M]$, choose a continuous function equal to one at s and zero on Σ , and approximate it uniformly by a real polynomial p . The approximation can be chosen so that

$$|p(s_N)| > \frac{1}{2}, \quad \sup_{\lambda \in \Sigma} |p(\lambda)| < \frac{1}{4}$$

eventually. Then

$$\|p(\tilde{H}_N)\| \geq |p(s_N)| > \frac{1}{2}, \quad \|p(H)\| < \frac{1}{4},$$

contradicting Eq. (1805). The last statement follows because deleting one representation does not establish polynomial norm domination on the entire retained sector. $\square$

Corollary 37 (Global Hausdorff, edges, and islands). *If $\ell_N \rightarrow 0$ and $p_N \rightarrow 0$, then*

$$\eta_N := d_H(S_N, \Sigma) \rightarrow 0. \quad (1806)$$

The full spectral edges satisfy

$$|\inf S_N - \inf \Sigma| \leq \eta_N, \quad (1807)$$

$$|\sup S_N - \sup \Sigma| \leq \eta_N. \quad (1808)$$

For an intrinsic spectral island of gap Δ_C , if $\eta_N < \Delta_C/3$, the fixed-neighborhood selection obeys

$$|W_{C,N} - W_C| \leq 2\eta_N, \quad (1809)$$

$$|\Delta_{C,N} - \Delta_C| \leq 2\eta_N. \quad (1810)$$

A local gap certificate requires no pollution in a neighborhood containing the island and the nearest complementary spectrum on both its lower and upper sides. No loss alone cannot protect either gap edge.

Proof. Equation (1801) proves Eq. (1806). Extrema of compact real sets are one-Lipschitz in Hausdorff distance, proving Eqs. (1807) and (1808). Quantitative intrinsic-island transport with the fixed $\Delta_C/3$ neighborhood gives Eqs. (1809) and (1810). That proof transports both the island and its complement, which is exactly the two-sided local no-pollution requirement in the final statement. $\square$

Exponential hopping and subsequent repairs

Theorem 65 (Cutoff extension with both physical tails). *Let $H^{(L)}$ and $H_N^{(L)}$ be finite-propagation truncations of the infinite and cover operators, respectively. Suppose*

$$\begin{aligned} \|H - H^{(L)}\| &\leq \varepsilon_\infty^{\text{tr}}(L), \\ \|H_N - H_N^{(L)}\| &\leq \varepsilon_N^{\text{tr}}(L), \end{aligned} \quad (1811)$$

where the explicit hopping bounds have positive decay margins. If

$$d_H(\text{spec } H_N^{(L)}, \text{spec } H^{(L)}) \leq \eta_N^{\text{fr}}(L), \quad (1812)$$

then

$$\begin{aligned} d_H(S_N, \Sigma) &\leq \varepsilon_N^{\text{tr}}(L) + \eta_N^{\text{fr}}(L) \\ &\quad + \varepsilon_\infty^{\text{tr}}(L). \end{aligned} \quad (1813)$$

A valid diagonal choice has

$$L = L_{c,N} \rightarrow \infty, \quad \frac{L_{c,N}}{r_{\text{inj},N}} \rightarrow 0, \quad (1814)$$

together with the finite-range no-loss and no-pollution certificate at that cutoff. For a positive intrinsic scale $\bar{E}_{M,N}$, a normalized claim uses

$$\frac{\varepsilon_N^{\text{tr}} + \eta_N^{\text{fr}} + \varepsilon_\infty^{\text{tr}}}{\bar{E}_{M,N}} \rightarrow 0. \quad (1815)$$

Velocity, curvature, and differentiated-projector conclusions additionally require their separately differentiated tails and C^1 or C^2 comparison errors.

Proof. Self-adjoint norm perturbation bounds the Hausdorff distance between each full operator and its truncation by the corresponding term in Eq. (1811). The triangle inequality for Hausdorff distance proves Eq. (1813). The cutoff condition supplies the injectivity buffer for every fixed moment and finitely supported Weyl vector while sending both physical tails to zero. Dividing the complete operator error by the normalization scale proves Eq. (1815). Undifferentiated norm convergence contains no derivative estimate. $\square$

Corollary 38 (Repair ledger). *The subsequent statements separate as follows.*

1. *Theorem 36 proves exact fixed-moment stabilization and weak normalized DOS for finite propagation. The lifted-Weyl-vector argument additionally proves no loss, but neither conclusion proves no pollution.*
2. *Theorem 37 inherits the two physical tails and injectivity limit in Theorem 65. Its native local and weak-DOS conclusions do not include global edges or gaps.*
3. *Theorem 76 may use its factor-two bandwidth and gap errors only after the total uncertainty contains a global no-pollution certificate in addition to tail, cover, angular, symmetry, and solver errors.*
4. *Theorem 79's limsup bandwidth and liminf gap remain conservative sequence diagnostics until Eq. (1806) identifies them with the intrinsic spectral island.*
5. *Theorem 105's thermodynamic bandwidth and positive gap, and every operational score built from them, require the same intrinsic-island transport. Weak DOS cannot replace it.*
6. *Theorems 160 and 169 require global normalized operator or Hausdorff convergence for bandwidth and gap master curves. Generalized-dispersion curves additionally require the differentiated comparison tier.*

No assertion is made that an arbitrary residual or Farber tower satisfies the no-pollution hypothesis; the admissible tower classes remain to be identified separately. Completeness of finite-dimensional automorphic or representation sectors is also separate.

The hopping admissibility and intrinsic-island definitions remain unchanged. Spectral convergence creates no cancellation root or coherence. The exact scalar square five-state model remains root-free.

Proof. Items 1 and 2 are Theorems 62, 63, and 65. Items 3–6 follow from Corollary 37 and the separation between undifferentiated and differentiated errors. The tree example shows why none can infer no pollution from weak

DOS. The remaining tower, representation, root, and coherence statements require hypotheses absent from the present convergence theorems. $\square$

Corollary 39 (Convergence obstructions). *The implications*

$$\begin{aligned} \nu_N \Rightarrow \nu &\not\Rightarrow p_N \rightarrow 0, \\ \ell_N \rightarrow 0 &\not\Rightarrow p_N \rightarrow 0, \\ \text{local or strong-resolvent convergence} &\not\Rightarrow p_N \rightarrow 0 \end{aligned} \quad (1816)$$

are all false in general. Removing one known outlier sector does not repair them without a no-pollution proof on the entire retained spectrum.

Proof. Proposition 40 proves the first two failures. For the standard basis of $\ell^2(\mathbb{N})$, the operators $H_N = |e_N\rangle\langle e_N|$ converge strongly, hence in strong resolvent sense, to zero, while $\text{spec } H_N = \{0, 1\}$. This gives the third failure. Polynomial norm domination in Theorem 64 shows the additional condition required after any sector removal. $\square$

Spectrally admissible towers and intrinsic master functions

Finite-cover independence is a theorem only after the cover class and the retained bulk sector have been fixed. Local convergence controls traces and compactly supported states; exclusion of global quotient modes is a separate spectral requirement.

Farber sequences and residual towers

Let Γ be a finitely generated group with a finite symmetric generating set and word length $|\cdot|$. For a finite-index subgroup $\Gamma_N < \Gamma$, let

$$X_N = \Gamma_N \backslash \Gamma \quad (1817)$$

carry the right action of Γ .

Definition 5 (Farber sequence and normal residual tower). *The sequence (Γ_N) is Farber if, for every $\gamma \neq e$,*

$$\frac{\#\{x \in X_N : x\gamma = x\}}{|X_N|} \rightarrow 0. \quad (1818)$$

A normal residual tower is a nested sequence

$$\Gamma = \Gamma_0 \triangleright \Gamma_1 \triangleright \Gamma_2 \triangleright \cdots \quad (1819)$$

of finite-index normal subgroups such that

$$\bigcap_{N \geq 1} \Gamma_N = \{e\}. \quad (1820)$$

For a fixed radius R , let $b_N(R)$ be the fraction of cosets whose rooted labelled R -ball is not isomorphic to the corresponding ball in the Cayley graph.

Proposition 42 (Typical-root versus global injectivity). *A Farber sequence satisfies*

$$b_N(R) \rightarrow 0 \quad (1821)$$

for every fixed R . Thus it gives typical-root local convergence, but it need not give an injective R -ball at every coset.

For a normal residual tower, put

$$r_{\text{inj},N} = \frac{1}{2} \min\{|\gamma| : \gamma \in \Gamma_N \setminus \{e\}\}. \quad (1822)$$

Then $r_{\text{inj},N} \rightarrow \infty$, and every rooted ball of any fixed radius is eventually exact. Every normal residual tower is Farber.

Proof. Failure of a rooted R -ball to embed produces a nonidentity element of the finite set $B_{2R}(e)$ that fixes the root. The union bound and Eq. (1818) prove Eq. (1821).

For a normal tower, residuality excludes every nonidentity element of a fixed finite word ball from all sufficiently deep subgroups. This proves Eq. (1822) tends to infinity. Normality also makes the fixed-point fraction of γ equal to zero or one, according as $\gamma \notin \Gamma_N$ or $\gamma \in \Gamma_N$. Residuality therefore proves the Farber property. $\square$

Proposition 43 (Farber convergence is not global spectral convergence). *Neither Farber convergence nor a normal residual tower implies convergence of complete finite-cover spectra to the reduced infinite spectrum on a nonamenable group.*

Proof. Take a free group of rank r , put $d = 2r \geq 4$, and choose a residual normal sequence with growing injectivity radius. The infinite adjacency spectrum is

$$[-2\sqrt{d-1}, 2\sqrt{d-1}], \quad (1823)$$

whereas every finite regular quotient has the constant eigenvalue d . The persistent distance is

$$d - 2\sqrt{d-1} = (\sqrt{d-1} - 1)^2 > 0. \quad (1824)$$

The outlier has vanishing normalized weight, so moment and weak-DOS convergence remain valid. Hence even global injectivity at every fixed radius does not exclude a non-tempered quotient sector. $\square$

A non-circular admissible tower class

Fix a compact interval J . Let $F = F^{-1} \subset \Gamma$ be finite, and consider the normalized finite-propagation family

$$\hat{H}(x) = \sum_{\gamma \in F} A_\gamma(x) \otimes \rho(\gamma), \quad x \in J, \quad (1825)$$

on $\ell^2(\Gamma) \otimes \mathbb{C}^q$. Here ρ is the right regular representation and

$$A_{\gamma^{-1}}(x) = A_{\gamma}(x)^{\dagger}. \quad (1826)$$

Assume the common coefficient modulus

$$\omega(\delta) := \sup_{\substack{x, y \in J \\ |x-y| \leq \delta}} \sum_{\gamma \in F} \|A_{\gamma}(x) - A_{\gamma}(y)\| \longrightarrow 0. \quad (1827)$$

Let π_N be the quasi-regular representation on $\ell^2(X_N)$, and define $\hat{H}_N(x)$ by replacing ρ with π_N in Eq. (1825). Then

$$\|\hat{H}_N(x) - \hat{H}_N(y)\| \leq \omega(|x - y|), \quad (1828)$$

uniformly in N .

Let $\mathcal{K}_N$ be a declared invariant subspace for every $\hat{H}_N(x)$, let P_N be its orthogonal projector, and put

$$\hat{H}_N^b(x) = \hat{H}_N(x)|_{\mathcal{K}_N}. \quad (1829)$$

Definition 6 (Spectrally admissible tower on J). *A tower with retained sectors $\mathcal{K}_N$ is spectrally admissible for the family (1825) if the following hold.*

1. *The subgroup sequence is Farber.*
2. *It is locally exhaustive in the retained sector. For every finite set $E \subset \Gamma$, put $E^+ = E \cup EF$. One can choose rooted patch isometries*

$$J_{N,E^+} : \ell^2(E^+) \otimes \mathbb{C}^q \longrightarrow \ell^2(X_N) \otimes \mathbb{C}^q \quad (1830)$$

that intertwine the right translations from E into E^+ and satisfy

$$\|(I - P_N)J_{N,E^+}\| \longrightarrow 0. \quad (1831)$$

3. *For every real polynomial p ,*

$$\limsup_{N \rightarrow \infty} \sup_{x \in J} \left[\|p(\hat{H}_N^b(x))\| - \|p(\hat{H}(x))\| \right] \leq 0. \quad (1832)$$

An independently proved uniform residual lower bound, or another explicit upper spectral inclusion on the same retained sector, may replace item 3. The definition does not assume Hausdorff convergence as an axiom.

For the full finite space, item 2 is automatic. It also holds for the mean-zero sector when the removed constant space has asymptotically negligible overlap with every fixed patch. Neither observation proves item 3. In particular, deleting only the trivial sector is not a universal no-pollution theorem.

Theorem 66 (Uniform two-sided spectral convergence). *Let a tower be spectrally admissible on J . With*

$$\begin{aligned} \Sigma(x) &= \text{spec } \hat{H}(x), \\ S_N^b(x) &= \text{spec } \hat{H}_N^b(x), \end{aligned} \quad (1833)$$

one has

$$\eta_N := \sup_{x \in J} d_H(S_N^b(x), \Sigma(x)) \longrightarrow 0. \quad (1834)$$

Proof. Both infinite and finite families obey the modulus (1827). Hence

$$d_H(\Sigma(x), \Sigma(y)) \leq \omega(|x - y|). \quad (1835)$$

The spectral graph $\{(x, s) : x \in J, s \in \Sigma(x)\}$ is therefore compact.

Fix $\varepsilon > 0$. Choose a finite net (x_j, s_j) in that graph and finitely supported unit vectors u_j such that

$$\|(\hat{H}(x_j) - s_j)u_j\| < \varepsilon. \quad (1836)$$

Farber convergence supplies good rooted patches for the finite union of their supports and propagation neighborhoods. By Eq. (1831), the normalized vectors

$$v_{j,N} = \frac{P_N J_{N,E^+} u_j}{\|P_N J_{N,E^+} u_j\|} \quad (1837)$$

are defined for large N , and their residuals at (x_j, s_j) are less than 2ε . Equations (1827) and (1828) transfer this estimate to every nearby (x, s) . A sufficiently fine fixed net gives

$$\sup_{x \in J} \sup_{s \in \Sigma(x)} \text{dist}(s, S_N^b(x)) \longrightarrow 0. \quad (1838)$$

Suppose the reverse inclusion is not uniform. Then, along a subsequence, there are $x_N \rightarrow x_*$ and $s_N \in S_N^b(x_N)$, with

$$\text{dist}(s_N, \Sigma(x_N)) \geq \varepsilon_0 > 0. \quad (1839)$$

Uniform boundedness gives a further subsequence $s_N \rightarrow s_*$. Equation (1835) implies $s_* \notin \Sigma(x_*)$. Choose a real polynomial p such that

$$|p(s_*)| > \sup_{s \in \Sigma(x_*)} |p(s)| + 3c \quad (1840)$$

for some $c > 0$. Coefficient continuity and $s_N \rightarrow s_*$ give, eventually,

$$\begin{aligned} \|p(\hat{H}_N^b(x_N))\| &\geq |p(s_N)|, \\ |p(s_N)| &> \|p(\hat{H}(x_N))\| + c. \end{aligned} \quad (1841)$$

This contradicts Eq. (1832). Thus the reverse one-sided error also tends to zero, and Eq. (1834) follows. $\square$

The conclusion concerns S_N^b . It concerns the exact complete finite-cover spectrum only when $\mathcal{K}_N$ is the full finite space and the full-space no-pollution hypothesis has actually been verified.

Intrinsic spectral master functions

Assume that $\Sigma(x)$ has a continuously tracked proper spectral island $\mathcal{I}_C(x)$ with

$$\Delta_0 := \inf_{x \in J} \text{dist}(\mathcal{I}_C(x), \Sigma(x) \setminus \mathcal{I}_C(x)) > 0. \quad (1842)$$

Define the intrinsic normalized functions

$$\Phi_W(x) := \text{diam } \mathcal{I}_C(x), \quad (1843)$$

$$\Phi_\Delta(x) := \text{dist}(\mathcal{I}_C(x), \Sigma(x) \setminus \mathcal{I}_C(x)). \quad (1844)$$

For a tower T , let $S_N^{b,T}(x)$ denote its retained spectrum and use the fixed radius $r = \Delta_0/3$ to define

$$\mathcal{I}_{C,N}^T(x) := \{s \in S_N^{b,T}(x) : \text{dist}(s, \mathcal{I}_C(x)) \leq r\}, \quad (1845)$$

$$\mathcal{B}_{C,N}^T(x) := S_N^{b,T}(x) \setminus \mathcal{I}_{C,N}^T(x). \quad (1846)$$

Set

$$\begin{aligned} \Phi_{W,N}^T(x) &:= \text{diam } \mathcal{I}_{C,N}^T(x), \\ \Phi_{\Delta,N}^T(x) &:= \text{dist}(\mathcal{I}_{C,N}^T(x), \mathcal{B}_{C,N}^T(x)). \end{aligned} \quad (1847)$$

Let η_N^T denote the uniform Hausdorff error (1834) for tower T . For two towers, define

$$\mathcal{R}_{\text{cover},N,M}^W := \sup_{x \in J} |\Phi_{W,N}^A(x) - \Phi_{W,M}^B(x)|, \quad (1848)$$

$$\mathcal{R}_{\text{cover},N,M}^\Delta := \sup_{x \in J} |\Phi_{\Delta,N}^A(x) - \Phi_{\Delta,M}^B(x)|. \quad (1849)$$

Theorem 67 (Uniform tower independence). *For every spectrally admissible tower T , once $\eta_N^T < \Delta_0/3$,*

$$\sup_{x \in J} |\Phi_{W,N}^T(x) - \Phi_W(x)| \leq 2\eta_N^T, \quad (1850)$$

$$\sup_{x \in J} |\Phi_{\Delta,N}^T(x) - \Phi_\Delta(x)| \leq 2\eta_N^T. \quad (1851)$$

For two admissible towers A and B , at depths N, M satisfying $\eta_N^A, \eta_M^B < \Delta_0/3$,

$$\mathcal{R}_{\text{cover},N,M}^W \leq 2\eta_N^A + 2\eta_M^B, \quad (1852)$$

$$\mathcal{R}_{\text{cover},N,M}^\Delta \leq 2\eta_N^A + 2\eta_M^B. \quad (1853)$$

Consequently, the corresponding uniform cross-cover residuals vanish.

Proof. At each x , the fixed-neighborhood island and its complement are both within Hausdorff distance η_N^T of their intrinsic counterparts. Diameter and the distance between two compact sets change by at most twice that error. This proves Eqs. (1850) and (1851), uniformly in x . Apply the triangle inequality through Φ_W or Φ_Δ to obtain Eqs. (1852) and (1853). $\square$

More generally, a spectral-set functional that is Lipschitz in Hausdorff distance is tower independent with its Lipschitz constant. Homogeneity in energy alone does not imply that continuity.

DOS, derivatives, and optimizer stability

Let $\nu_{N,x}$ be the full-cover normalized empirical measure and let ν_x be the canonical regular-trace DOS of $\hat{H}(x)$.

Theorem 68 (Canonical Farber DOS and fixed broadening). *For the finite-propagation family (1825), every Farber sequence satisfies*

$$\sup_{x \in J} \left| \int f d\nu_{N,x} - \int f d\nu_x \right| \longrightarrow 0 \quad (1854)$$

for every continuous f on a common compact spectral interval.

Let $K_\eta(E - s)$ be bounded and jointly continuous for fixed $\eta > 0$. Then the smoothed DOS

$$\rho_\eta(x, E) = \int K_\eta(E - s) d\nu_x(s) \quad (1855)$$

is tower independent, and its finite-cover versions converge uniformly on compact (x, E) -sets. This includes Lorentzian broadening at fixed positive η .

Proof. For $\gamma \in \Gamma$, the normalized trace of $\pi_N(\gamma)$ is the fixed-coset fraction in Eq. (1818). Expanding a fixed power of $\hat{H}_N(x)$ produces finitely many group words. Farber convergence removes every nonidentity word, while the identity terms give the regular trace. The coefficient modulus and finiteness of the word set make this moment convergence uniform in x . A common operator bound and uniform polynomial approximation prove Eq. (1854).

For fixed η , the kernels $K_\eta(E - \cdot)$ form a compact subset of the continuous functions as E ranges over a compact interval. A finite kernel net and Eq. (1854) prove uniform smoothed convergence. $\square$

The theorem concerns the canonical full normalized trace. A retained sector has the same DOS limit when its discarded dimension is $o(|X_N|)$; otherwise its trace requires a separate convergence theorem. Weak convergence does not control an unsmoothed density, its maximum, or a singular $\eta \downarrow 0$ limit. The safe order is first the tower limit at fixed $\eta > 0$, then $\eta \downarrow 0$, with additional uniform regularity before interchanging limits or taking peaks. A coherence-weighted DOS additionally requires convergence of its projector and coherence weight.

Proposition 44 (Observable and optimizer tiers). *Tower independence of a generalized-velocity function requires a fixed representation-space metric and an independently established comparison*

$$\sup_{x \in J} \left(\|\hat{H}_N(x) - \hat{H}(x)\| + \|D\hat{H}_N(x) - D\hat{H}(x)\| \right) \longrightarrow 0 \quad (1856)$$

on declared transported fibers. A uniform isolation gap then transfers the Riesz projectors and their first derivatives. Spectral Hausdorff convergence alone does not imply Eq. (1856). Participation and coherence require their independent projector-rank, covariance, and overlap hypotheses.

Let continuous functions f_N converge uniformly to f on J . If f has a unique maximizer x_* , then every exact

maximizer x_N of f_N converges to x_* . Quantitatively, for any open neighborhood U of x_* , put

$$m_U = f(x_*) - \sup_{x \in J \setminus U} f(x) > 0. \quad (1857)$$

If $\|f_N - f\|_\infty < m_U/3$, then every maximizer of f_N lies in U . Without such isolation, only maximum values and subsequential optimizer-set statements are justified.

Proof. The differentiated resolvent identity transfers first Riesz derivatives under Eq. (1856) and a fixed contour gap. The metric must be fixed because changing it changes the velocity norm. No spectral-set estimate controls the required eigenvector covariance or active overlap.

For the optimizer claim, if $y \notin U$, then

$$\begin{aligned} f_N(x_*) - f_N(y) &\geq m_U - 2\|f_N - f\|_\infty \\ &> m_U/3. \end{aligned} \quad (1858)$$

Thus no maximizer lies outside U . Without uniqueness, uniform convergence still gives

$$|\max_J f_N - \max_J f| \leq \|f_N - f\|_\infty, \quad (1859)$$

and every cluster point of maximizers belongs to $\arg\max_J f$, but a unique location need not exist. $\square$

Exponential extension and repairs

Theorem 69 (Uniform diagonal cutoff). *Let the hopping family have uniform positive summability margins and let $\hat{H}^{(L)}(x)$ and $\hat{H}_N^{(L)}(x)$ be its normalized finite-range truncations, with the same declared retained sector invariant before and after truncation. Suppose a diagonal choice $L_N \rightarrow \infty$ obeys*

$$\begin{aligned} \hat{\varepsilon}_\infty^{\text{tr}}(L_N) &:= \sup_{x \in J} \|\hat{H} - \hat{H}^{(L_N)}\| \rightarrow 0, \\ \hat{\varepsilon}_N^{\text{tr}}(L_N) &:= \sup_{x \in J} \|\hat{H}_N - \hat{H}_N^{(L_N)}\| \rightarrow 0, \end{aligned} \quad (1860)$$

and that the truncated retained spectra satisfy

$$\eta_N^{\text{fr}}(L_N) := \sup_{x \in J} d_H \left(\text{spec } \hat{H}_N^{(L_N),b}, \text{spec } \hat{H}^{(L_N)} \right) \rightarrow 0. \quad (1861)$$

If $a_N \rightarrow 0$ is a separate normalized parameter-matching error, then

$$\begin{aligned} \eta_N^{\text{exp}} &\leq \hat{\varepsilon}_N^{\text{tr}}(L_N) + \eta_N^{\text{fr}}(L_N) \\ &\quad + \hat{\varepsilon}_\infty^{\text{tr}}(L_N) + a_N. \end{aligned} \quad (1862)$$

For a normal residual tower, one may impose

$$L_N/r_{\text{inj},N} \rightarrow 0. \quad (1863)$$

For a general Farber sequence, the corresponding diagonal condition is $b_N(cL_N) \rightarrow 0$ for the finite propagation multiplier c , together with retained-sector local exhaustivity. The finite-range no-pollution certificate at L_N remains separate in both cases.

Proof. Self-adjoint norm perturbations move spectra by at most their operator norm. Apply this once on the finite cover, once on the infinite system, and use the triangle inequality with Eq. (1861). This proves Eq. (1862). The two stated diagonal conditions supply, respectively, global or typical-root local buffers. Neither condition proves the upper spectral inclusion at the growing cut-off. If the estimates originate in physical units, every term in Eq. (1862) must be divided by the same positive intrinsic energy scale before taking the supremum over J . Differentiated master functions require the corresponding uniform weighted derivative tails. $\square$

Corollary 40 (Repairs of tower-dependent claims). *The finite-cover and master-function statements are repaired as follows.*

1. Theorem 105 uses the intrinsic infinite spectral island. Its finite-cover bandwidth and gap are tower independent only within the spectrally admissible class. Velocity, coherent DOS, participation, and coherence retain their separate derivative and projector hypotheses.
2. Theorem 108's scalarization equivalence remains algebraic. A cross-tower magic score or maximizing location additionally requires uniform convergence of every constituent and the optimizer margin (1857).
3. Theorem 160 defines Φ_W and Φ_Δ intrinsically by Eqs. (1843) and (1844). Other degree-one observables require the continuity topology appropriate to that observable. Homogeneity alone is insufficient.
4. Theorem 169 and simultaneous multi-observable collapse inherit the same spectral, C^1 , DOS, coherence, cutoff, and optimizer tiers. The cross-cover residual is replaced by the quantitative bounds (1852) and (1853).
5. Exact compact-quotient spectra and retained reduced bulk spectra are distinct objects. They coincide in this theorem only when the complete finite space itself satisfies the no-pollution certificate.

No claim is made that every Farber or residual tower is spectrally admissible. No completeness claim is made for finite-dimensional automorphic sectors; the automorphic-sector result owns that question. Spectral convergence does not create a root. The exact scalar square five-state model remains root-free, and $X = 1$ is a magic locus only for a richer model with an independently established root.

Proof. Items 1 and 3 are Theorem 67. Item 2 is Proposition 44. Item 4 combines Theorems 68 and 69 with the independent derivative and coherence requirements. Item 5 is the sector restriction in Definition 6. None of these hypotheses asserts completeness of all finite representations or a zero of a cancellation functional. $\square$

Corollary 41 (Sharp tower-independence boundary). *The following implications are invalid without extra hypotheses:*

$$\begin{aligned} \text{Farber/residual} &\not\Rightarrow \text{global Hausdorff}, \\ \text{weak DOS} &\not\Rightarrow \text{edge/gap}, \\ \text{remove trivial sector} &\not\Rightarrow \text{no pollution}, \\ C^0 \text{ spectrum} &\not\Rightarrow \text{velocity/coherence}. \end{aligned} \quad (1864)$$

Arbitrary Farber admissibility is not proved, and the automorphic-sector result retains representation-completeness scope.

Proof. Proposition 43 proves the first two failures. Polynomial domination is imposed on the entire retained sector, not only on one removed representation. The last failure follows from the differentiated and projector tiers in Proposition 44. $\square$

Finite automorphic sectors and the reduced regular bulk

Finite-group Fourier theory is exactly complete on each finite quotient. That algebraic statement must be separated from automorphic boundary conditions for the infinite group and from the reduced regular bulk.

Four distinct representation layers

Let Γ be the infinite moiré translation group. Fix a finite internal dimension ν and a self-adjoint matrix group-algebra element

$$h = \sum_{\gamma \in F} A_\gamma \otimes \gamma, \quad A_{\gamma^{-1}} = A_\gamma^\dagger, \quad (1865)$$

where $F \subset \Gamma$ is finite. The same definitions extend to an absolutely summable kernel by norm convergence.

Definition 7 (The four representation layers). *The following objects are distinct.*

1. For a finite normal quotient

$$q_N : \Gamma \longrightarrow Q_N = \Gamma/\Gamma_N, \quad (1866)$$

the finite quotient dual $\widehat{Q}_N$ is the finite set of equivalence classes of irreducible unitary representations of Q_N .

2. A finite-image sector of Γ is a finite-dimensional unitary representation ρ for which $\rho(\Gamma)$ is finite. The tower samples only

$$\widehat{\Gamma}_{\text{fin}}^\tau := \bigcup_N q_N^*(\widehat{Q}_N), \quad q_N^* \pi = \pi \circ q_N. \quad (1867)$$

This may be smaller than the set of all finite-image sectors.

3. The finite-dimensional irreducible automorphic hierarchy is

$$\mathfrak{B}_{\text{fd}} = \bigsqcup_{d \geq 1} \text{Hom}^{\text{irr}}(\Gamma, U(d))/U(d). \quad (1868)$$

It contains finite-image sectors but can also contain representations with infinite image. Reducible boundary conditions are finite direct sums of these sectors.

4. The reduced regular bulk is defined by the left regular representation λ_Γ on $\ell^2(\Gamma)$,

$$\begin{aligned} C_r^*(\Gamma) &:= \overline{\lambda_\Gamma(\mathbb{C}\Gamma)}^{\|\cdot\|}, \\ L(\Gamma) &:= \lambda_\Gamma(\Gamma)'', \end{aligned} \quad (1869)$$

$$\tau(T) := \langle \delta_e, T \delta_e \rangle. \quad (1870)$$

The bulk Hamiltonian is

$$H_\lambda = (\text{id} \otimes \lambda_\Gamma)(h) \in M_\nu(C_r^*(\Gamma)). \quad (1871)$$

Its spectrum and regular-trace DOS are the intrinsic reduced-bulk objects.

There is no inclusion symbol identifying $\mathfrak{B}_{\text{fd}}$ with a decomposition of λ_Γ . Likewise, the tower union (1867) is not asserted to be dense in $\mathfrak{B}_{\text{fd}}$ or in the full unitary dual.

Exact finite-quotient completeness

Push h to $M_\nu(\mathbb{C}[Q_N])$ and call the result h_N . The strict primary finite-cover operator is

$$H_N^{\text{reg}} = (\text{id} \otimes \lambda_{Q_N})(h_N) \quad (1872)$$

on $\mathbb{C}^\nu \otimes \ell^2(Q_N)$. It is defined before any irreducible blocks are selected.

For $\pi \in \widehat{Q}_N$, let $d_\pi = \dim V_\pi$ and define

$$H_N[\pi] = (\text{id} \otimes \pi)(h_N) \quad \text{on} \quad \mathbb{C}^\nu \otimes V_\pi. \quad (1873)$$

Theorem 70 (Finite Peter–Weyl completeness and state weights). *For every finite quotient Q_N ,*

$$\ell^2(Q_N) \simeq \bigoplus_{\pi \in \widehat{Q}_N} V_\pi \otimes V_\pi^*, \quad (1874)$$

$$|Q_N| = \sum_{\pi \in \widehat{Q}_N} d_\pi^2, \quad (1875)$$

$$H_N^{\text{reg}} \simeq \bigoplus_{\pi \in \widehat{Q}_N} H_N[\pi] \otimes I_{d_\pi}. \quad (1876)$$

Thus $H_N[\pi]$ has dimension νd_π and occurs with multiplicity d_π . The aggregate normalized state weight is

$$w_{\pi,N} = \frac{d_\pi^2}{|Q_N|}, \quad \sum_{\pi \in \widehat{Q}_N} w_{\pi,N} = 1. \quad (1877)$$

For every continuous f on the finite spectral interval,

$$\frac{1}{\nu|Q_N|} \operatorname{Tr} f(H_N^{\text{reg}}) = \sum_{\pi \in \hat{Q}_N} \frac{d_\pi^2}{|Q_N|} \times \frac{1}{\nu d_\pi} \operatorname{Tr} f(H_N[\pi]). \quad (1878)$$

Proof. Maschke's theorem decomposes the regular representation as $\lambda_{Q_N} \simeq \bigoplus_{\pi \in \hat{Q}_N} m_\pi \pi$. Its character equals $|Q_N|$ at the identity and vanishes elsewhere, so character orthogonality gives $m_\pi = d_\pi$. This proves Eqs. (1874) and (1875).

Schur orthogonality gives, for irreducible π, σ ,

$$\sum_{q \in Q_N} \pi(q)_{ij} \overline{\sigma(q)_{kl}} = \frac{|Q_N|}{d_\pi} \delta_{\pi\sigma} \delta_{ik} \delta_{jl}. \quad (1879)$$

Together with the dimension identity, this shows that the normalized matrix coefficients form an orthonormal basis of $\ell^2(Q_N)$. Under the associated Fourier map, the regular representation is the sum of d_π copies of π . Applying the matrix-valued group-algebra element h_N proves Eq. (1876). Taking dimensions and then traces proves Eqs. (1877) and (1878). $\square$

For a scalar field $f : Q_N \rightarrow \mathbb{C}$, the transform

$$\hat{f}(\pi) = \sqrt{\frac{d_\pi}{|Q_N|}} \sum_{q \in Q_N} f(q) \pi(q) \quad (1880)$$

obeys the exact Parseval identity

$$\sum_{q \in Q_N} |f(q)|^2 = \sum_{\pi \in \hat{Q}_N} \|\hat{f}(\pi)\|_F^2. \quad (1881)$$

These formulas are finite-group identities; their weights are not an infinite-group Plancherel measure.

Corollary 42 (Primary numerical definition). *An exact finite-cover calculation means either direct diagonalization of H_N^{reg} , or evaluation of every block in $\hat{Q}_N$ with the multiplicities and weights in Eqs. (1876) and (1878). A selected subset of blocks is a declared sector calculation, not the complete finite-cover spectrum or DOS.*

Proof. Theorem 70 is a unitary equivalence of the full finite Hilbert space. Omitting a block removes its spectral states and its exact trace weight. $\square$

Automorphic blocks and the reduced-spectrum condition

For any finite-dimensional unitary representation $\rho : \Gamma \rightarrow U(d_\rho)$, define

$$H[\rho] = (\text{id} \otimes \rho)(h) = \sum_{\gamma \in F} A_\gamma \otimes \rho(\gamma). \quad (1882)$$

Theorem 71 (Exact automorphic block and weak-containment condition). *The matrix $H[\rho]$ is the exact Hamiltonian on the flat bundle*

$$E_\rho = (\tilde{X} \times \mathbb{C}^{d_\rho})/\Gamma \quad (1883)$$

with holonomy ρ . It is Hermitian, and conjugate representations give unitarily equivalent matrices. This statement alone does not make $H[\rho]$ a subblock of H_λ .

The following are equivalent:

1. ρ is weakly contained in λ_Γ , written $\rho \prec \lambda_\Gamma$.
2. For every $a \in \mathbb{C}\Gamma$,

$$\|\rho(a)\| \leq \|\lambda_\Gamma(a)\|. \quad (1884)$$

3. The integrated representation of $C^*(\Gamma)$ factors through a unital $*$ -representation

$$\tilde{\rho} : C_r^*(\Gamma) \longrightarrow B(V_\rho). \quad (1885)$$

Only under these equivalent conditions does one have, for every self-adjoint $h \in M_\nu(\mathbb{C}\Gamma)$,

$$\operatorname{spec} H[\rho] \subset \operatorname{spec} H_\lambda. \quad (1886)$$

Proof. A ρ -automorphic vector satisfies $\Psi(\gamma x) = \rho(\gamma)\Psi(x)$. Substitution into the group convolution factors out $\rho(\gamma)$ and leaves exactly Eq. (1882). The coefficient relation in Eq. (1865) and unitarity of ρ prove Hermiticity. Replacing ρ by $U\rho U^\dagger$ conjugates $H[\rho]$ by $I_\nu \otimes U$.

The reduced group algebra is the completion of $\mathbb{C}\Gamma$ in the regular norm. Therefore Eq. (1884) is exactly the condition for the integrated map to extend continuously to Eq. (1885); this is the standard weak-containment criterion. Matrix amplification gives

$$H[\rho] = (\text{id} \otimes \tilde{\rho})(H_\lambda). \quad (1887)$$

The spectrum of the image of an element under a unital $*$ -homomorphism is contained in its original spectrum, proving Eq. (1886). $\square$

Without the condition, $H[\rho]$ is an exact automorphic probe of the full group algebra. Even the union over all finite-dimensional ρ need not exhaust the full group C^* -algebra spectrum, and it is not the reduced spectrum.

Proposition 45 (Nonamenable trivial-sector obstruction). *If Γ is nonamenable, its trivial representation $\mathbf{1}_\Gamma$ is not weakly contained in λ_Γ . Hence a quotient or automorphic eigenvalue in the trivial sector need not belong to the reduced bulk spectrum.*

For the free group whose Cayley graph is d -regular, let

$$a = \sum_{s \in S} s, \quad |S| = d. \quad (1888)$$

Then

$$\begin{aligned} \operatorname{spec} \mathbf{1}_\Gamma(a) &= \{d\}, \\ \operatorname{spec} \lambda_\Gamma(a) &= [-2\sqrt{d-1}, 2\sqrt{d-1}], \end{aligned} \quad (1889)$$

so d lies outside the reduced spectrum.

Proof. Weak containment of $\mathbf{1}_\Gamma$ in the regular representation is equivalent to the existence of almost invariant regular vectors. Their squared moduli give Reiter probability vectors and hence an invariant mean. Thus weak containment would imply amenability, proving the first claim by contraposition. In the free-group example, the trivial representation sends every generator to one. The regular operator is adjacency on the d -regular tree, whose norm and spectrum are $2\sqrt{d-1}$ and the interval in Eq. (1889). This proves the explicit spectral separation. $\square$

Regular-trace Plancherel theory

Theorem 72 (Noncommutative regular Plancherel identity). *Let*

$$A = \sum_{\gamma \in F} a_\gamma \lambda_\Gamma(\gamma). \quad (1890)$$

Define $B = \sum_{\gamma \in F} b_\gamma \lambda_\Gamma(\gamma)$ on the same finite support, enlarging F if necessary. Then

$$\tau(A^*A) = \sum_{\gamma \in F} |a_\gamma|^2, \quad (1891)$$

$$\tau(A^*B) = \sum_{\gamma \in F} \overline{a_\gamma} b_\gamma. \quad (1892)$$

For matrix coefficients

$$H = \sum_{\gamma \in F} A_\gamma \otimes \lambda_\Gamma(\gamma), \quad (1893)$$

and the normalized internal trace $\text{tr}_\nu = \nu^{-1} \text{Tr}$,

$$(\text{tr}_\nu \otimes \tau)(H^*H) = \sum_{\gamma \in F} \text{tr}_\nu(A_\gamma^\dagger A_\gamma). \quad (1894)$$

The reduced-bulk DOS of a bounded self-adjoint H is characterized by

$$\int f(E) d\nu_H(E) = (\text{tr}_\nu \otimes \tau)[f(H)]. \quad (1895)$$

Proof. The coefficient of the identity in A^*B is $\sum_\gamma \overline{a_\gamma} b_\gamma$, and the regular trace extracts that coefficient. This proves Eqs. (1891) and (1892). Applying the same calculation entry by entry and then the normalized matrix trace proves Eq. (1894). Functional calculus and the faithful normal trace give the probability measure in Eq. (1895). $\square$

By completion, the scalar and matrix identities extend to the corresponding noncommutative L^2 spaces, without asserting that every square-summable coefficient field defines a bounded operator.

This is the complete regular-bulk L^2 identity. It needs no parameterization by finite-dimensional irreducibles and does not use the finite weights $d_\pi^2/|Q_N|$.

Proposition 46 (No ordinary irreducible Brillouin zone). *An ordinary irreducible direct-integral formula*

$$\lambda_\Gamma \simeq \int_{\widehat{\Gamma}_u}^\oplus \pi \otimes I_{m(\pi)} d\mu_{\text{Pl}}(\pi) \quad (1896)$$

uses the full unitary dual $\widehat{\Gamma}_u$ and a multiplicity field m . Treating Eq. (1896) as a canonical ordinary Brillouin-zone formula requires a declared standard measurable dual, measurable field, and Plancherel measure. The familiar essentially unique irreducible formulation is controlled in the type-I setting. No such ordinary irreducible Plancherel data are assumed or constructed here.

A countable discrete group is type I only if it is virtually Abelian. A non-elementary cocompact Fuchsian surface group is not virtually Abelian, so its finite-dimensional automorphic hierarchy is not an ordinary generalized Brillouin zone carrying a canonical irreducible Plancherel measure. The safe complete bulk object is $(C_r^(\Gamma), L(\Gamma), \tau)$. A central or factor decomposition, when used, must not be relabelled as Eq. (1868).*

Proof. The type-I hypothesis is what makes the unitary dual a standard measurable parameter space with essentially unique irreducible disintegration. For countable discrete groups, the type-I criterion is virtual Abelianity. A non-elementary cocompact Fuchsian group has a finite-index surface subgroup and is not virtually Abelian. The claimed ordinary irreducible parameterization therefore has no basis in this setting, whereas the regular algebra and trace are defined without it. $\square$

Finite-level completeness versus tower convergence

Theorem 73 (Tower completeness boundary). *At each level N , Theorem 70 exhausts the finite Hilbert space exactly. Even if the tower is residual and its finite-image representations separate all elements of Γ , the following do not follow:*

1. *density or completeness of finite-image sectors in $\mathfrak{B}_{\text{fd}}$ or $\widehat{\Gamma}_u$;*
2. *an infinite-group Plancherel formula obtained by taking a limit of the weights $d_\pi^2/|Q_N|$;*
3. *global Hausdorff convergence of complete quotient spectra to $\text{spec } H_\lambda$.*

For a reduced-bulk spectral limit one must retain a declared invariant sector $\mathcal{X}_N$ and verify the same-sector no-loss and no-pollution conditions. In particular, if

$$\eta_N^b := d_H(\text{spec}(H_N^{\text{reg}}|_{\mathcal{X}_N}), \text{spec } H_\lambda) \longrightarrow 0, \quad (1897)$$

then reduced spectral conclusions follow; finite-level completeness alone does not prove Eq. (1897).

Proof. Residuality separates group elements, which is an algebraic statement. It does not identify the reduced C^* -norm with a supremum or limit over the tower representations. The trivial representation occurs in every finite regular quotient, while Proposition 45 places its free group adjacency eigenvalue outside the reduced spectrum. Thus item 3 is false even though every finite level is exactly decomposed. Items 1 and 2 would require additional topology and measure theorems absent from finite-level Wedderburn theory. The no-loss/no-pollution hypotheses in Eq. (1897) are precisely the separate tower admissibility condition. $\square$

The full finite-cover operator remains the primary numerical definition. The retained sector is declared before extracting a reduced-bulk limit; it is not assembled after inspecting which quotient blocks resemble the desired spectrum.

Vocabulary and subsequent repairs

Corollary 43 (Completeness vocabulary). *The word “complete” has only the following scoped meanings here.*

1. “Complete finite-cover spectrum” means $\text{spec } H_N^{\text{reg}}$, equivalently all $\pi \in \widehat{Q}_N$ with their exact multiplicities.
2. “Complete finite quotient representation coverage” means coverage of all $\widehat{Q}_N$, weighted by $d_\pi^2/|Q_N|$. It does not mean coverage of the infinite unitary dual.
3. “Automorphic dispersion over $\mathcal{A}$ ” means extrema or differentials over an explicitly declared finite-dimensional test family $\mathcal{A} \subset \mathfrak{B}_{\text{fd}}$.
4. “Reduced bulk spectrum or DOS” means the objects defined from H_λ and the regular trace. It is not the union of unrestricted automorphic spectra.

The phrase “complete generalized Brillouin zone” is not used for $\mathfrak{B}_{\text{fd}}$ without the type-I data in Proposition 46.

Proof. Items 1 and 2 are Theorem 70. Item 3 is a declared family of exact boundary-condition blocks from Theorem 71. Item 4 is Definition 7 and Theorem 72. The non-type-I obstruction prevents these meanings from being merged. $\square$

Corollary 44 (Repairs of representation claims). *The named statements are repaired as follows.*

1. Theorem 31 is an exact finite-dimensional flat-bundle reduction for each declared ρ . It is a reduced-bulk block only under Eq. (1884).
2. Theorem 35 is complete for the finite-dimensional induced representation $\text{Ind}_{\Gamma_{\text{sc}}}^\Gamma \chi$ and its irreducible summands. It is not a decomposition of λ_Γ .

3. Theorem 48’s complete reciprocal content means the finite permutation representation $\text{Ind}_{\Gamma_M}^\Gamma \mathbf{1}$. In the normal case it is the regular representation of the finite quotient only.

4. Theorem 66’s complete finite spectrum means a fixed H_N^{reg} . Continuity of a fixed automorphic block remains valid, but does not establish infinite-bulk representation completeness.

5. Theorem 80’s complete generalized dispersion is relative to a declared smooth automorphic test domain, or is replaced by an intrinsic reduced-bulk observable. No representation-space metric is chosen here.

6. In Theorem 99, $C_{\text{rep}}^{(N)} = 1$ means exact coverage of $\widehat{Q}_N$ with the finite weights (1877). It does not certify infinite-dual coverage, reduced spectral convergence, or coherence by itself.

Every other use of “complete generalized dispersion” or “complete representation coverage” inherits Corollary 43. Tower limits retain the admissibility condition (1897). The representation-space metric normalization is deferred. None of these representation statements creates a cancellation root: the exact scalar square five-state model remains root-free.

Proof. Items 1–3 follow from the automorphic and finite-group theorems above. Item 4 is finite-matrix continuity with the finite-level scope preserved. Items 5 and 6 are vocabulary restrictions necessitated by Proposition 46 and Theorem 73. The final root statement is independent of representation completeness. $\square$

Corollary 45 (Representation-completeness obstructions). *Without additional hypotheses, all of the following implications fail:*

- $\text{finite-level Peter-Weyl} \not\Rightarrow \text{infinite Plancherel},$
 $\text{residual finite images} \not\Rightarrow \text{unitary-dual completeness},$
 $\text{automorphic block} \not\Rightarrow \text{reduced-bulk block},$
 $\text{finite coverage one} \not\Rightarrow \text{bulk spectral convergence}.$
(1898)

Proof. The finite and infinite Plancherel statements are Theorems 70 and 72, respectively. The weak-containment condition and its explicit failure are Theorem 71 and Proposition 45. The tower failure is Theorem 73. $\square$

Canonical normalization of representation-space velocity

The statement $dE = 0$ on the physical representation tangent space is metric independent. A numerical size for a nonzero dE , by contrast, requires both a base metric and a fiber pairing. We now fix these data and separate coordinate invariance from normalization covariance.

Fix a compact oriented hyperbolic quotient

$$\Sigma_M = \Gamma_M \backslash \mathbb{H}_R^2, \quad A_M = \text{Area}(\Sigma_M), \quad (1899)$$

a marking of Γ_M , a rank d , and a smooth irreducible unitary representation $\rho: \Gamma_M \rightarrow U(d)$. Let $\text{ad } E_\rho$ be the flat $\mathfrak{u}(d)$ -bundle with unitary flat connection ∇_ρ .

Definition 8 (Normalized fundamental pairing). *For $A, B \in \mathfrak{u}(d)$, declare*

$$\langle A, B \rangle_d = \frac{1}{d} \text{Re Tr}(A^\dagger B). \quad (1900)$$

This is the fiber convention used below.

The hyperbolic metric determines the volume form and Hodge star, and the unitary connection determines the twisted harmonic problem. Neither datum alone determines Eq. (1900).

Proposition 47 (Invariant-pairing freedom). *For $d \geq 2$, every positive real $\text{Ad } U(d)$ -invariant inner product on*

$$\mathfrak{u}(d) = \mathfrak{su}(d) \oplus i\mathbb{R}I \quad (1901)$$

has one positive weight on $\mathfrak{su}(d)$ and an independent positive weight on the center. The cross term vanishes. On the semisimple summand the invariant pairing is unique only up to scale. If $A = A_0 + iaI$ and $B = B_0 + ibI$, Definition 8 fixes the two weights by

$$\langle A, B \rangle_d = \frac{1}{d} \text{Re Tr}(A_0^\dagger B_0) + ab. \quad (1902)$$

For $d = 1$, only the central term remains.

Proof. The real adjoint module $\mathfrak{su}(d)$ is simple, so its invariant positive forms are positive multiples of one another. Pairing a central vector with $\mathfrak{su}(d)$ would define an invariant linear functional on $\mathfrak{su}(d)$, which vanishes because $[\mathfrak{su}(d), \mathfrak{su}(d)] = \mathfrak{su}(d)$. The center is a trivial one-dimensional module and therefore has its own positive weight. Expanding the normalized trace gives Eq. (1902). $\square$

Put

$$\mathcal{V}_\rho = H^1(\Sigma_M; \text{ad } E_\rho). \quad (1903)$$

Twisted Hodge theory assigns each $X \in \mathcal{V}_\rho$ a unique harmonic representative α_X .

Definition 9 (Canonical intensive metric). *The declared physical metric is*

$$\begin{aligned} G_\rho^{\text{can}}(X, Y) &= \frac{1}{A_M} \int_{\Sigma_M} \langle \alpha_X, \alpha_Y \rangle_d d \text{vol}_g \\ &= \frac{1}{d A_M} \text{Re} \int_{\Sigma_M} \text{Tr}(\alpha_X^\dagger \wedge * \alpha_Y). \end{aligned} \quad (1904)$$

The factors d^{-1} and A_M^{-1} fix, respectively, the rank and intensive-area normalizations.

Theorem 74 (Positivity, gauge invariance, and units). *The form G_ρ^{can} is a positive-definite real inner product on $\mathcal{V}_\rho$. It is invariant under unitary bundle gauge changes and depends only on cohomology classes. For dimensionless holonomy tangents,*

$$[G_\rho^{\text{can}}] = L^{-2}, \quad \left[\|dE\|_{(G_\rho^{\text{can}})^{-1}} \right] = EL. \quad (1905)$$

Consequently

$$\Omega_E^{\text{can}}(\rho) = \frac{1}{\hbar} \|dE(\rho)\|_{(G_\rho^{\text{can}})^{-1}} \quad (1906)$$

has units L/T : it is a generalized velocity, not a frequency. A frequency requires division by a separately declared physical length.

Proof. The pointwise pairing in Eq. (1900) is positive. Hence Eq. (1904) is nonnegative, and it vanishes only when the harmonic representative is zero, equivalently when the cohomology class is zero. Replacing a cocycle representative does not change its harmonic representative. Under a unitary gauge intertwiner, the twisted Laplacian and its harmonic kernel are conjugated, while the trace pairing and base Hodge star are preserved. This proves gauge invariance.

A harmonic one-form representing a dimensionless holonomy variation has units L^{-1} . Its squared norm times the volume element is dimensionless; division by A_M gives L^{-2} . Dualization and division by $\hbar$ give Eq. (1905) and velocity units for Eq. (1906). $\square$

Thus “canonical” means canonical after the base geometry, flat unitary connection, normalized fundamental pairing, and intensive-area convention have all been declared. It is not uniqueness from hyperbolic geometry alone.

Coordinate covariance and the variational norm

Let q^a and $\tilde{q}^\mu$ be two charts on the same smooth stratum, and write

$$J^a{}_\mu = \frac{\partial q^a}{\partial \tilde{q}^\mu}, \quad K^\mu{}_a = \frac{\partial \tilde{q}^\mu}{\partial q^a}. \quad (1907)$$

Thus $K = J^{-1}$.

Theorem 75 (Full coordinate transformation law). *For tangent bases $X_a = \partial/\partial q^a$, energy components $E_a = dE(X_a)$, and metric components $G_{ab} = G^{\text{can}}(X_a, X_b)$,*

$$\tilde{X}_\mu = J^a{}_\mu X_a, \quad \tilde{E}_\mu = J^a{}_\mu E_a, \quad (1908)$$

$$\tilde{G}_{\mu\nu} = J^a{}_\mu J^b{}_\nu G_{ab}, \quad \tilde{G}^{\mu\nu} = K^\mu{}_a K^\nu{}_b G^{ab}. \quad (1909)$$

In particular,

$$\tilde{G}^{\mu\nu} \tilde{E}_\mu \tilde{E}_\nu = G^{ab} E_a E_b. \quad (1910)$$

Moreover,

$$\|e\|_{G^{-1}} = \sup_{X \neq 0} \frac{|e(X)|}{\sqrt{G(X, X)}}. \quad (1911)$$

Both sides are invariant under chart changes, tangent-basis changes, and regular reparameterizations of a path.

Proof. The first three laws are the tensor transformation rules. Inverting $\tilde{G} = J^T G J$ gives the fourth. Substitution and $KJ = I$ prove Eq. (1910). Cauchy–Schwarz gives the upper bound in Eq. (1911); equality is attained at the metric dual vector $G^{-1}e$. A path-speed change multiplies both $|e(\dot{\gamma})|$ and $\sqrt{G(\dot{\gamma}, \dot{\gamma})}$ by the same positive factor. It therefore cannot change the quotient. $\square$

Let $\mathfrak{D}$ be a connected declared automorphic domain inside this fixed stratum. Let $\mathcal{C}$ consist of continuously tracked C^1 energy branches on $\mathfrak{D}$; at a degeneracy this hypothesis must be replaced by a separately established differentiable cluster observable. Define

$$\Omega_{\mathcal{C}}^{\max} = \sup_{\rho \in \mathfrak{D}, a \in \mathcal{C}} \frac{1}{\hbar} \|dE_a(\rho)\|_{(G_{\rho}^{\text{can}})^{-1}}. \quad (1912)$$

The domain is the same geometric automorphic test domain before and after a coordinate change. Theorem 75 therefore proves coordinate invariance of Eq. (1912). It does not enlarge that domain to an undeclared unitary dual or identify it with the reduced regular bulk.

Intrinsic diameter and energy oscillation

For piecewise C^1 curves contained in $\mathfrak{D}$, define

$$d_G(\rho_0, \rho_1) = \inf_{\gamma: \rho_0 \rightarrow \rho_1} \int_0^1 \sqrt{G_{\gamma(s)}^{\text{can}}(\dot{\gamma}, \dot{\gamma})} ds, \quad (1913)$$

$$D_G = \text{diam}_{G^{\text{can}}}(\mathfrak{D}). \quad (1914)$$

Here $[d_G] = [D_G] = L^{-1}$.

Theorem 76 (Velocity–oscillation bound). *Assume $D_G < \infty$ and $\Omega_{\mathcal{C}}^{\max} < \infty$. Then every tracked branch satisfies*

$$\text{osc}_{\mathfrak{D}} E_a \leq \hbar D_G \Omega_{\mathcal{C}}^{\max}. \quad (1915)$$

For a multi-branch cluster and a reference $\rho_* \in \mathfrak{D}$, put

$$w_*(\mathcal{C}) = \max_{a, b \in \mathcal{C}} |E_a(\rho_*) - E_b(\rho_*)|. \quad (1916)$$

Then its total energy diameter obeys

$$\text{diam}\{E_a(\rho) : \rho \in \mathfrak{D}, a \in \mathcal{C}\} \leq w_*(\mathcal{C}) + 2\hbar D_G \Omega_{\mathcal{C}}^{\max}. \quad (1917)$$

Thus Eq. (1915) is a full bandwidth bound for a single connected branch. Derivatives alone do not remove internal cluster splitting.

Proof. For any admissible path γ ,

$$\begin{aligned} |E_a(\rho_1) - E_a(\rho_0)| &\leq \int_0^1 |dE_a(\dot{\gamma})| ds \\ &\leq \hbar \Omega_{\mathcal{C}}^{\max} \text{Length}_G(\gamma). \end{aligned} \quad (1918)$$

Taking the infimum over paths and then the supremum over endpoints proves Eq. (1915); no minimizing geodesic is needed. For two branches and endpoints, insert and subtract their two values at ρ_* , apply the branch bound twice, and use the triangle inequality. This proves Eq. (1917). $\square$

Normalization covariance and dimensionless diagnostics

Theorem 77 (Constant metric-rescaling law). *Let $G' = cG$ for one constant $c > 0$ on the entire declared domain. Then*

$$\|dE\|_{(G')^{-1}} = c^{-1/2} \|dE\|_{G^{-1}}, \quad \Omega' = c^{-1/2} \Omega, \quad (1919)$$

$$d_{G'} = c^{1/2} d_G, \quad D_{G'} = c^{1/2} D_G. \quad (1920)$$

Consequently

$$D_{G'}(\Omega')_{\mathcal{C}}^{\max} = D_G \Omega_{\mathcal{C}}^{\max}. \quad (1921)$$

The same product is coordinate invariant by Theorems 75 and 76.

Proof. Since $(cG)^{-1} = c^{-1}G^{-1}$, the first line follows by taking square roots. Every curve length acquires the factor $c^{1/2}$, proving the second line and Eq. (1921). $\square$

This theorem is normalization covariance, not a coordinate change. If $c = c(\rho)$ varies over the domain, the pointwise velocity law still holds, but the diameter has no single compensating factor. A magic functional that uses $\Omega^{\max}$ alone must retain the canonical convention in Definitions 8 and 9. A globally scale-independent choice is

$$\mathcal{V}_{\mathcal{C}}^{\text{can}} = \frac{\hbar D_G \Omega_{\mathcal{C}}^{\max}}{E_{\text{ref}}}, \quad E_{\text{ref}} > 0, \quad (1922)$$

where the energy reference and spectral domain are declared independently. The exact condition $dE = 0$ remains independent of every positive metric.

Hessians and comparison across curvature

At a stationary point let M be the covariant Hessian of an energy branch and let $r = \dim \mathcal{V}_{\rho}$.

Proposition 48 (Hessian convention law). *The generalized pencil*

$$Mv = \Lambda G^{\text{can}} v \quad (1923)$$

has coordinate-invariant eigenvalues. Under $G' = cG$,

$$\Lambda'_j = c^{-1}\Lambda_j, \quad \frac{\det M}{\det G'} = c^{-r} \frac{\det M}{\det G}. \quad (1924)$$

In contrast, $D_G^2\Lambda_j$ is invariant under this constant rescaling. Determinant and eigenvalue comparisons require fixed tangent rank and the same declared metric convention.

Proof. A chart change sends the pencil to $(J^\top MJ, J^\top GJ)$, which has the same generalized eigenvalues. Replacing G by cG proves the first formula in Eq. (1924); taking determinants proves the second. Equation (1920) then proves invariance of $D_G^2\Lambda_j$. $\square$

For a curvature parameter κ , a derivative comparison is defined only when surface topology, marking, rank, and smooth irreducible stratum are fixed and a smooth transport

$$T_\kappa : \mathcal{V}_{\rho_0} \longrightarrow \mathcal{V}_{\rho_\kappa} \quad (1925)$$

has been declared. Pull all data back to the reference fiber:

$$\bar{e}_\kappa = T_\kappa^* dE_\kappa, \quad \bar{G}_\kappa = T_\kappa^* G_{\rho_\kappa}^{\text{can}}. \quad (1926)$$

Theorem 78 (Running and fixed metric comparison). *With $S_\kappa = \bar{G}_\kappa^{-1}$, define*

$$\Omega_{\text{run}}(\kappa) = \hbar^{-1} \|\bar{e}_\kappa\|_{S_\kappa}, \quad (1927)$$

$$\Omega_{\text{fr}}(\kappa) = \hbar^{-1} \|\bar{e}_\kappa\|_{S_0}. \quad (1928)$$

Where the data are differentiable,

$$\frac{d}{d\kappa} \Omega_{\text{run}}^2 = \frac{1}{\hbar^2} \left[\dot{S}_\kappa(\bar{e}_\kappa, \bar{e}_\kappa) + 2S_\kappa(\bar{e}_\kappa, \dot{\bar{e}}_\kappa) \right]. \quad (1929)$$

The fixed diagnostic contains only the transported covector derivative:

$$\frac{d}{d\kappa} \Omega_{\text{fr}}^2 = \frac{2}{\hbar^2} S_0(\bar{e}_\kappa, \dot{\bar{e}}_\kappa). \quad (1930)$$

Thus running-metric variation mixes spectral reconstruction with metric reconstruction, whereas the fixed diagnostic isolates the former relative to the chosen transport.

Proof. Both formulas follow by differentiating the displayed quadratic forms on the single fixed vector space $\mathcal{V}_{\rho_0}^*$. Without Eq. (1925), the derivatives would compare vectors in different spaces and would not be defined. $\square$

A change of genus, quotient topology, rank, marking without transport, or tangent dimension is outside Theorem 78. Such data cannot be compared by inserting unrelated Gram matrices into one formula.

Invariant directional locking

Let $e = dE^{(0)}$ and $\xi = \Xi^{(1)}$ be nonzero covectors in the same transported cotangent fiber. Define their dual-metric locking coefficient by

$$\mathcal{L}^{\text{dir}} = -\frac{\langle e, \xi \rangle_{G^{-1}}}{\|e\|_{G^{-1}} \|\xi\|_{G^{-1}}}. \quad (1931)$$

Theorem 79 (Coordinate-free linear locking bound). *One has $-1 \leq \mathcal{L}^{\text{dir}} \leq 1$, and*

$$\begin{aligned} \|e + \kappa \xi\|_{G^{-1}}^2 &= \|e\|_{G^{-1}}^2 - 2\kappa \mathcal{L}^{\text{dir}} \|e\|_{G^{-1}} \|\xi\|_{G^{-1}} \\ &\quad + \kappa^2 \|\xi\|_{G^{-1}}^2. \end{aligned} \quad (1932)$$

For unrestricted κ , the minimizer and residual are

$$\kappa_{\text{opt}} = \mathcal{L}^{\text{dir}} \frac{\|e\|_{G^{-1}}}{\|\xi\|_{G^{-1}}}, \quad (1933)$$

$$\min_{\kappa} \|e + \kappa \xi\|_{G^{-1}} = \|e\|_{G^{-1}} \sqrt{1 - (\mathcal{L}^{\text{dir}})^2}. \quad (1934)$$

For the physical restriction $\kappa \geq 0$, the interior formula applies only when $\mathcal{L}^{\text{dir}} > 0$; otherwise the minimum is at $\kappa = 0$. Exact positive- κ cancellation in this linear model requires $\mathcal{L}^{\text{dir}} = 1$ and $\kappa = \|e\|_{G^{-1}} / \|\xi\|_{G^{-1}}$.

Proof. Cauchy–Schwarz proves the range of $\mathcal{L}^{\text{dir}}$. Expanding the squared norm gives Eq. (1932); completing the square gives Eqs. (1933) and (1934). The constrained statement follows by projecting the unconstrained minimizer onto $[0, \infty)$. Vanishing of the residual with positive κ is precisely antiparallel equality in Cauchy–Schwarz. No orthonormal chart is required. $\square$

If $e = 0$, the branch is already first-order stationary. If $\xi = 0$, there is no leading curvature response. In either case the angle in Eq. (1931) is intentionally left undefined.

Geodesic finite differences and convergence

Proposition 49 (Metric-consistent finite differences). *Let $\gamma(s)$ be a unit-speed G^{can} -geodesic through ρ , parametrized by metric arclength. If $E \circ \gamma$ is C^3 and $|(E \circ \gamma)^{(3)}| \leq M_3$ for $|s| \leq h$, then*

$$\left| \frac{E(\gamma(h)) - E(\gamma(-h))}{2h} - dE_\rho(\dot{\gamma}(0)) \right| \leq \frac{M_3 h^2}{6}. \quad (1935)$$

For a chart displacement δq ,

$$d_G(\rho, \rho + \delta q)^2 = G_{ab}(\rho) \delta q^a \delta q^b + O(|\delta q|^3). \quad (1936)$$

If $q = q(\tilde{q})$, the leading physical step satisfies

$$\begin{aligned} \sigma_{\min}(G^{1/2}J)|\delta \tilde{q}| &\leq \sqrt{\delta \tilde{q}^\top J^\top G J \delta \tilde{q}} \\ &\leq \sigma_{\max}(G^{1/2}J)|\delta \tilde{q}|. \end{aligned} \quad (1937)$$

Thus raw coordinate increments do not define physical velocities, and an ill-conditioned chart magnifies finite-difference error.

Proof. Taylor's theorem with symmetric remainder proves Eq. (1935). The squared-distance expansion in normal coordinates and tensor covariance give Eq. (1936). Applying the extremal singular value bounds to $G^{1/2}J\delta\tilde{q}$ proves Eq. (1937). $\square$

Proposition 50 (Derivative-convergence condition). *Uniform C^0 spectral convergence does not imply velocity convergence. Indeed, on the one-dimensional Hilbert space and $q \in [-1, 1]$,*

$$H_N(q) = N^{-1} \sin(Nq) \quad (1938)$$

converges uniformly to zero, whereas $DH_N(0) = 1$.

Conversely, let H_N and H be C^1 bounded self-adjoint families on fixed transported fibers over a compact parameter domain. Suppose

$$\sup_q (\|H_N(q) - H(q)\| + \|DH_N(q) - DH(q)\|_{\text{op,dual}}) \longrightarrow 0, \quad (1939)$$

and suppose a tracked simple eigenvalue has a uniform isolation gap. If the transported canonical metrics converge uniformly with uniformly positive lower bounds, then its energy covectors and canonical velocities converge uniformly. For a higher-rank cluster the same hypotheses transfer its Riesz projector and projector derivative, but differentiable individual eigenbranches require an additional simplicity or splitting hypothesis.

Proof. The example proves the obstruction. Under Eq. (1939), the uniform gap and the resolvent identity give uniform convergence of the simple-branch Riesz projectors $P_N \rightarrow P$. The Feynman–Hellmann formula reads

$$dE_N = \text{Tr}(P_N DH_N). \quad (1940)$$

Projector convergence, boundedness of DH_N , and derivative convergence therefore give $dE_N \rightarrow dE$ uniformly. Uniform positivity and convergence of the metrics imply uniform convergence of their inverses and hence of the dual norms. The resolvent derivative gives the stated projector conclusion for a cluster; it does not choose differentiable eigenvalue labels inside a degenerate block. $\square$

Thus finite-cover or tower convergence of energies alone does not imply convergence of Eq. (1912). Metric and Hessian claims need the corresponding differentiated data. Local or strong-resolvent convergence alone is insufficient.

Repairs and retained obstructions

Corollary 46 (Scope repairs for generalized dispersion). *The following replacements apply.*

1. Theorem 83 uses G^{can} from Definition 9. Its Ω^{gen} is a velocity with units L/T , while $dE = 0$ remains metric independent.

2. Theorem 84's generalized Hessian eigenvalues and determinant ratio are coordinate invariant only within the fixed metric convention. They obey Eq. (1924); cross-rank determinant comparisons are not defined. The associated DOS comparison must use the same fixed-rank volume convention.
3. Theorems 88 and 89 replace component alignment by Eq. (1931) and retain the constrained-curvature qualification in Theorem 79.
4. Theorem 99's "complete generalized dispersion" is relative to the declared smooth automorphic test domain. Its velocity requires the canonical metric and the independent C^1 convergence condition stated above; it does not assert reduced-bulk or infinite-dual completeness.
5. Every subsequent occurrence of Ω^{max} alone retains the canonical normalization. Every occurrence of $\hbar D_G \Omega^{\text{max}}$ uses one common domain and metric. The preferred dimensionless operational factor is Eq. (1922).
6. Curvature comparisons use Eqs. (1925)–(1930); finite differences use Eqs. (1935)–(1937).

These repairs leave the finite-quotient, automorphic, and reduced-regular representation sectors distinct.

Corollary 47 (Sharp scope obstructions). *Hyperbolic geometry alone does not choose a fiber trace pairing, and $u(d)$ has independent central and semisimple invariant weights. A coordinate change cannot be identified with a rescaling of the physical metric. Tangent spaces of different rank, topology, marking, or dimension cannot be compared without a declared transport or embedding. A derivative bound does not eliminate internal cluster splitting, and base spectral convergence does not supply derivative convergence. No assertion here constructs a metric on the full non-type-I unitary dual, creates a fold or cusp continuation, or changes the exact square five-state no-root result.*

Proof. The pairing obstruction is Proposition 47; the coordinate and normalization distinction follows from Theorems 75 and 77. The transport restriction is necessary for Eqs. (1926) and (1929). The cluster and convergence obstructions are respectively Eq. (1917) and Proposition 50. The remaining exclusions are scope conditions, not conclusions inferred from the metric. $\square$

Microscopic verification of folds, cusps, and curvature-born branches

Generic catastrophe normal forms describe what follows after a fold or cusp has been established. They do not show that a microscopic bilayer enters the required

jet. We now give the complete operator-to-score derivative chain, quantitative certificates, and an explicit Hermitian active-sector family in which the certificates hold on a nonempty open coefficient set.

Transported operator chart and differentiated Riesz calculus

Let $p = (\theta, K, \omega)$ range in an open set $\mathcal{U}$. All Hilbert spaces are identified with one reference space by declared unitary transports.

Definition 10 (Admissible microscopic bifurcation chart). *A microscopic bifurcation chart consists of a bounded self-adjoint family*

$$p \mapsto H(p) \in \mathcal{B}(\mathcal{H}) \quad (1941)$$

that is jointly C^4 in operator norm, together with a fixed contour Γ satisfying

$$\inf_{p \in \mathcal{U}} \text{dist}(\Gamma, \text{spec } H(p)) \geq \delta_\Gamma > 0. \quad (1942)$$

The Riesz projector

$$P(p) = \frac{1}{2\pi i} \oint_\Gamma (z - H(p))^{-1} dz \quad (1943)$$

has fixed rank. The active sector, resonant shell, representation layer, and canonical representation metric are preselected and transported. The target gap, active overlap, coherence, and physical-angle distances have strictly positive margins on $\mathcal{U}$.

For an infinite hopping kernel, C^4 means weighted differentiated Schur/tail summability for every multi-index used below. Base norm or strong-resolvent convergence is not a substitute. On this smooth chart the operational score below is C^4 , so its force $F = \partial_\theta \ell$ is C^3 ; no fifth-order microscopic regularity is assumed.

The co-rotating microscopic bilayer has the form

$$H = \begin{pmatrix} h_0 & \tau \\ \tau^\dagger & h_0 \end{pmatrix}, \quad (1944)$$

with

$$H_\theta = \begin{pmatrix} 0 & \tau_\theta \\ \tau_\theta^\dagger & 0 \end{pmatrix}, \quad (1945)$$

$$H_K = \begin{pmatrix} (h_0)_K & \tau_K \\ \tau_K^\dagger & (h_0)_K \end{pmatrix}. \quad (1946)$$

Mixed and higher jets are obtained by differentiating the same blocks. Thus every jet is Hermitian on the fixed transported fiber; twist enters only through the off-diagonal tunneling channel in this frame.

Put $R(z, p) = (z - H(p))^{-1}$, and use subscripts for covariant transported partial derivatives.

Theorem 80 (Complete resolvent and Riesz jets). *For control indices $i, j, k \in \{\theta, K, \omega\}$,*

$$R_i = RH_i R, \quad (1947)$$

$$R_{ij} = RH_{ij} R + RH_i RH_j R + RH_j RH_i R, \quad (1948)$$

and

$$\begin{aligned} R_{ijk} &= RH_{ijk} R \\ &+ \sum_{\text{cyc}(i,j,k)} (RH_i RH_{jk} R + RH_{jk} RH_i R) \\ &+ \sum_{\sigma \in S_3} RH_{\sigma(i)} RH_{\sigma(j)} RH_{\sigma(k)} R. \end{aligned} \quad (1949)$$

In particular, with primes denoting repeated θ -derivatives,

$$\begin{aligned} R''' &= RH''' R + 3(RH'' RH' R + RH' RH'' R) \\ &+ 6RH' RH' RH' R. \end{aligned} \quad (1950)$$

The fourth repeated derivative required by a stationary cusp is

$$\begin{aligned} R'''' &= RH'''' R + 4(RH''' RH' R + RH' RH''' R) \\ &+ 6RH'' RH'' R \\ &+ 12(RH'' RH' RH' R + RH' RH'' RH' R \\ &\quad + RH' RH' RH'' R) \\ &+ 24RH' RH' RH' RH' R. \end{aligned} \quad (1951)$$

Every projector jet is obtained by contour integration:

$$P_I = \frac{1}{2\pi i} \oint_\Gamma R_I(z, p) dz, \quad 1 \leq |I| \leq 4. \quad (1952)$$

If $\|H_I\| \leq M_{|I|}$ and $\|R\| \leq \delta_\Gamma^{-1}$, then

$$\|R_{ij}\| \leq \delta_\Gamma^{-2} M_2 + 2\delta_\Gamma^{-3} M_1^2, \quad (1953)$$

$$\|R_{ijk}\| \leq \delta_\Gamma^{-2} M_3 + 6\delta_\Gamma^{-3} M_1 M_2 + 6\delta_\Gamma^{-4} M_1^3, \quad (1954)$$

while the right side for R'''' is

$$\begin{aligned} \mathcal{R}_4 &= \delta_\Gamma^{-2} M_4 + \delta_\Gamma^{-3} (8M_1 M_3 + 6M_2^2) \\ &+ 36\delta_\Gamma^{-4} M_1^2 M_2 + 24\delta_\Gamma^{-5} M_1^4. \end{aligned} \quad (1955)$$

Multiplying these resolvent bounds by $\text{length}(\Gamma)/(2\pi)$ bounds the corresponding projector jets.

Proof. Differentiating $(z - H)R = I$ gives Eq. (1947). A second differentiation gives Eq. (1948). Differentiating each of its three terms produces one third Hamiltonian jet, all six ordered first/second-jet products, and all six ordered triple products, proving Eq. (1949). Repeated indices give the displayed combinatorial coefficients. At fourth order the ordered compositions of four have coefficients 1, 4, 4, 6, 12, 12, 12, and 24, proving Eq. (1951). Contour differentiation is justified by Eq. (1942). Counting ordered terms and using submultiplicativity proves Eqs. (1953)–(1955). $\square$

Corollary 48 (Hellmann–Feynman and reduced-resolvent chain). *Let E be a simple eigenvalue inside Γ , with rank-one projector P . Then*

$$E_i = \text{Tr}(PH_i), \quad (1956)$$

$$E_{ij} = \text{Tr}(PH_{ij} + P_j H_i), \quad (1957)$$

$$E_{ijk} = \text{Tr}(PH_{ijk} + P_j H_{ik} + P_k H_{ij} + P_j H_i). \quad (1958)$$

For repeated θ -derivatives,

$$\begin{aligned} E'' &= \text{Tr}(PH'' + P'H'), \\ E''' &= \text{Tr}(PH''' + 2P'H'' + P''H'), \end{aligned} \quad (1959)$$

$$E'''' = \text{Tr}(PH'''' + 3P'H''' + 3P''H'' + P'''H'). \quad (1960)$$

If $Q = I - P$ and

$$S = Q(E - H)^{-1}Q, \quad (1961)$$

then

$$P_i = SH_i P + PH_i S, \quad (1962)$$

$$E_{ij} = \text{Tr}(PH_{ij}) + \text{Tr}[P(H_i SH_j + H_j SH_i)]. \quad (1963)$$

Proof. The first formula is Hellmann–Feynman. Differentiating it once, twice, and three times gives Eqs. (1957)–(1960); the apparent choice of the first index is immaterial because E is scalar. Differentiating the eigenvalue equation, projecting with Q , and choosing the parallel gauge gives Eq. (1962). Substitution into Eq. (1957) proves Eq. (1963). $\square$

From microscopic jets to the operational force

Work in a chart in which the active target edges, nearest complementary edge, simple active singular values, velocity maximizer, and coherent-density maximizer remain fixed. Put

$$y = (y_1, \dots, y_5) = (\Delta, W, V, X, C), \quad (1964)$$

where all five entries and all logarithm arguments below are positive. The logarithmic operational score is

$$\begin{aligned} \ell(y) &= \frac{1}{4} [2 \log \Delta - \log(\Delta + W) - \log(\Delta + V) \\ &\quad + \log X - \log(1 + X) + \log C]. \end{aligned} \quad (1965)$$

Define $F = \partial_\theta \ell$.

For $m = 1, \dots, 4$, all score derivatives are explicitly

$$\begin{aligned} \ell_{a_1 \dots a_m} &= \frac{(-1)^{m-1} (m-1)!}{4} \left[\frac{2e_{a_1}^\Delta \dots e_{a_m}^\Delta}{\Delta^m} \right. \\ &\quad - \frac{r_{a_1}^W \dots r_{a_m}^W}{(\Delta + W)^m} - \frac{r_{a_1}^V \dots r_{a_m}^V}{(\Delta + V)^m} \\ &\quad \left. + \frac{e_{a_1}^X \dots e_{a_m}^X}{X^m} - \frac{e_{a_1}^X \dots e_{a_m}^X}{(1 + X)^m} \right] \end{aligned}$$

$$+ \frac{e_{a_1}^C \dots e_{a_m}^C}{C^m} \Big], \quad (1966)$$

where e^A is the coordinate covector of ingredient A , $r^W = e^\Delta + e^W$, and $r^V = e^\Delta + e^V$.

Let $z = (z_\beta)$ be a finite list of simple eigenvalues, Riesz traces, simple singular values, metric components, and fixed-extremizer data from which $y_a = \Phi_a(z)$ is reconstructed. For a multi-index $I = (i_1, \dots, i_m)$, $m \leq 4$,

$$D_I y_a = \sum_{\pi \in \Pi(I)} D^{|\pi|} \Phi_a(z) [D_{B_1} z, \dots, D_{B_{|\pi|}} z]. \quad (1967)$$

Here $\Pi(I)$ is the set of set partitions of the labeled derivative positions, and B_j are its blocks. Equations (1952) and (1956)–(1960) determine every $D_B z$.

For example, fixed active edge labels give

$$D_I W = D_I E_+ - D_I E_-, \quad D_I \Delta = D_I E_c - D_I E_+, \quad (1968)$$

with the analogous lower-gap formula when that side is active. A simple singular value is a simple eigenvalue of the Hermitian dilation

$$\mathcal{D}(A) = \begin{pmatrix} 0 & A \\ A^\dagger & 0 \end{pmatrix}, \quad (1969)$$

so the same Riesz chain applies. The factor V uses the fixed canonical metric and a fixed active maximizer; C is a smooth trace functional of the transported projector.

Theorem 81 (Explicit microscopic force derivatives). *With repeated ingredient indices summed,*

$$F = \ell_a y_{a\theta}, \quad (1970)$$

$$F_K = \ell_{ab} y_{a\theta} y_{bK} + \ell_a y_{a\theta K}, \quad (1971)$$

$$F_{\theta\theta} = \ell_{abc} y_{a\theta} y_{b\theta} y_{c\theta} + 3\ell_{ab} y_{a\theta\theta} y_{b\theta} + \ell_a y_{a\theta\theta\theta}, \quad (1972)$$

$$\begin{aligned} F_{\theta\theta\theta} &= \ell_{abcd} y_{a\theta} y_{b\theta} y_{c\theta} y_{d\theta} \\ &\quad + 6\ell_{abc} y_{a\theta\theta} y_{b\theta} y_{c\theta} \\ &\quad + 3\ell_{ab} y_{a\theta\theta} y_{b\theta\theta} + 4\ell_{ab} y_{a\theta\theta\theta} y_{b\theta} \\ &\quad + \ell_a y_{a\theta\theta\theta\theta}. \end{aligned} \quad (1973)$$

For $u \in \{K, \omega\}$, the cusp unfolding also uses

$$\begin{aligned} F_{\theta u} &= \ell_{abc} y_{a\theta} y_{b\theta} y_{cu} \\ &\quad + \ell_{ab} (y_{a\theta\theta} y_{bu} + 2y_{a\theta} y_{b\theta u}) + \ell_a y_{a\theta\theta u}. \end{aligned} \quad (1974)$$

Thus F_K and $F_{\theta\theta}$ require the microscopic chain through orders two and three, while $F_{\theta\theta\theta}$ requires order four.

Proof. Equation (1970) is the ordinary chain rule applied to Eq. (1965). Differentiating it by K , twice by θ , three times by θ , and by (θ, u) gives the remaining formulas. The coefficients 3, 6, 3, 4 count the partitions of the repeated derivative positions. Equation (1967) and the Riesz/Hellmann–Feynman formulas close the chain at the microscopic Hamiltonian jets. $\square$

Proposition 51 (Nonsmooth-extremizer obstruction). *If a band-edge label, nearest-gap side, singular-value label, velocity maximizer, or coherent-density maximizer switches, the corresponding scalar ingredient need not be differentiable. Dini or one-sided derivatives may still exist, but Eqs. (1971)–(1974) do not certify a smooth fold or cusp there. Likewise, rank, resonant-shell, representation, gap, coherence, physical-angle, and exact-crystallographic boundaries are separate events.*

Proof. A maximum or minimum of smooth branches is generally only Lipschitz at an active-label switch. A repeated singular value has the same obstruction. The classical derivatives used by the fold and cusp Jacobians are then undefined. The other listed events leave the fixed admissible chart of Definition 10. $\square$

Quantitative fold and cusp certificates

Theorem 82 (Quantitative microscopic fold). *Suppose $F \in C^2$ on an admissible microscopic chart and, at $p_c = (\theta_c, K_c, \omega_c)$,*

$$\begin{aligned} F_c &= (F_\theta)_c = 0, \\ |F_K|_c &\geq a_* > 0, \\ |F_{\theta\theta}|_c &\geq b_* > 0. \end{aligned} \quad (1975)$$

Then p_c is a fold on the fixed- ω_c slice. The defining map (F, F_θ) has Jacobian determinant

$$\det D_{(\theta, K)}(F, F_\theta)_c = -(F_K F_{\theta\theta})_c, \quad |\det| \geq a_* b_*. \quad (1976)$$

Hence the fold persists uniquely under sufficiently small microscopic C^3 perturbations, equivalently C^2 perturbations of F . After shrinking that open neighborhood, the continued fold satisfies

$$|F_K| \geq a_*/2, \quad |F_{\theta\theta}| \geq b_*/2. \quad (1977)$$

Writing $x = \theta - \theta_c$, $\delta K = K - K_c$, $a = (F_K)_c$, and $b = (F_{\theta\theta})_c$, the two branches on the side $-2a\delta K/b > 0$ obey

$$x_\pm = \pm \sqrt{-\frac{2a}{b}\delta K [1 + o(1)]}. \quad (1978)$$

More quantitatively, solve the fold graph as $K = k(\theta)$, put

$$A_f = -\frac{b}{2a}, \quad (1979)$$

and choose $r_f > 0$ so that

$$|k''(\theta_c + x) - 2A_f| \leq |A_f|, \quad |x| \leq r_f. \quad (1980)$$

Then, for $|x| \leq r_f$,

$$\frac{|A_f|}{2}x^2 \leq |\delta K| \leq \frac{3|A_f|}{2}x^2. \quad (1981)$$

Exactly one of the two stationary branches has $F_\theta < 0$ and is a strict local score maximum.

Proof. Equation (1976) follows by evaluating the Jacobian at $F_\theta = 0$. The implicit-function theorem applied to the two defining equations proves persistence; continuity gives the half-margin bounds. Since $F_K \neq 0$, solve $F = 0$ as $K = k(\theta)$. Implicit differentiation gives $k'(\theta_c) = 0$ and $k''(\theta_c) = -b/a = 2A_f$. Continuity gives Eq. (1980), while the integral identity

$$\delta K = \int_0^x (x-t)k''(\theta_c+t)dt$$

proves Eq. (1981). Since $k(\theta_c + x) = K_c + A_f x^2 + o(x^2)$, inversion gives Eq. (1978). Finally, continuity and $F_{\theta\theta,c} = b \neq 0$ give opposite signs of F_θ on the two branches. $\square$

Theorem 83 (Quantitative microscopic cusp). *Suppose $F \in C^3$ and*

$$F_c = (F_\theta)_c = (F_{\theta\theta})_c = 0, \quad |F_{\theta\theta\theta}|_c \geq c_* > 0. \quad (1982)$$

Let

$$\mathfrak{d}_c = \det \begin{pmatrix} F_K & F_\omega \\ F_{\theta K} & F_{\theta\omega} \end{pmatrix}_c, \quad |\mathfrak{d}_c| \geq d_* > 0. \quad (1983)$$

Then the Jacobian of $(F, F_\theta, F_{\theta\theta})$ with respect to (θ, K, ω) has absolute determinant at least $c_ d_*$. The cusp persists uniquely under sufficiently small microscopic C^4 perturbations, equivalently C^3 perturbations of F , and the continued point has margins $c_*/2$ and $d_*/2$.*

With $x = \theta - \theta_c$ and $\lambda = (K - K_c, \omega - \omega_c)$,

$$\begin{aligned} F &= \frac{F_{\theta\theta\theta,c}}{6}x^3 + (F_K, F_\omega)_c \lambda \\ &\quad + x(F_{\theta K}, F_{\theta\omega})_c \lambda + \mathcal{R}(x, \lambda), \end{aligned} \quad (1984)$$

where there are $M_c < \infty$ and a modulus $\epsilon_c(r) \rightarrow 0$ as $r \downarrow 0$ such that, for $r = |x| + |\lambda|$ sufficiently small,

$$|\mathcal{R}(x, \lambda)| \leq \epsilon_c(r)|x|^3 + M_c(|x|^2|\lambda| + |\lambda|^2). \quad (1985)$$

An invertible control change and nonzero rescalings put the leading equation in the form

$$x^3 - Ux - V = 0. \quad (1986)$$

Its three-root region is $4U^3 > 27V^2$, with two outer strict maxima and one intervening minimum after orienting the score sign.

Proof. At the cusp, expansion of the three-by-three Jacobian along its first column gives $F_{\theta\theta\theta,c}\mathfrak{d}_c$, up to sign. The implicit-function theorem proves persistence and continuity gives the stated half margins. Apply one-variable third-order Taylor expansion to $F(x, 0)$, whose remainder is $o(|x|^3)$. Next expand $F(x, \lambda) - F(x, 0)$ first in λ and twice in x ; bounded third derivatives give the two M_c -terms in Eq. (1985). This proves Eqs. (1984) and (1985). The two row covectors in Eq. (1983) are independent, so they are valid local control coordinates for the constant and linear terms. Rescaling gives Eq. (1986); its discriminant is $4U^3 - 27V^2$. The signs of F_θ distinguish the two maxima from the middle minimum. $\square$

The cusp in Theorem 83 is a codimension-two stationary singularity of a smooth F . It is not the one-sided radial $|\theta|$ distance cusp that can occur in a nonsmooth separation law.

A nonempty microscopic active-sector region

Let $q \in [-1, 1]$ be a coordinate in one fixed automorphic chart, with normalized measure $dq/2$ and constant canonical metric $G_{qq} = g > 0$. Set $x = \theta - \theta_c$ and

$$U = -a(K - K_0), \quad Y = \beta(\omega - \omega_0), \quad (1987)$$

$$\Psi = -\frac{x^4}{4} + \frac{Ux^2}{2} + Yx, \quad W = S_0 e^{-4\Psi} - \Delta_0. \quad (1988)$$

Put $V_0 = 2s$, $\widetilde{W} = W - V_0$, and $d_0 = \Delta_0 + V_0$.

Definition 11 (Hermitian bilayer active block). *On $\mathbb{C}^2 \oplus \mathbb{C}^2$, define*

$$H_{\text{act}}(q; p) = \begin{pmatrix} sqI_2 & T(p) \\ T(p) & sqI_2 \end{pmatrix}, \quad (1989)$$

$$T(p) = \text{diag} \left(\frac{\widetilde{W}}{2}, \frac{\widetilde{W}}{2} + d_0 \right).$$

The first orbital channel is the specified target; the second is its remote complement. Within the target orbital, the two one-dimensional layer subspaces are the preselected tunneling-active channels. Their off-diagonal matrix element is $\widetilde{W}/2$; these layer channels are distinct from the spectral target and complement.

Proposition 52 (Exact operational data of the active block). *Whenever $\widetilde{W} > 0$, the target eigenvalues are*

$$E_{\pm}(q) = sq \pm \frac{\widetilde{W}}{2}, \quad (1990)$$

and the complement eigenvalues are

$$E_{c,\pm}(q) = sq \pm \left(\frac{\widetilde{W}}{2} + d_0 \right). \quad (1991)$$

The target rank is two, its global bandwidth is exactly W , and its global target-to-complement gap is exactly Δ_0 . Its projector is constant, hence $C = 1$. Moreover,

$$\hbar D_G \Omega^{\text{max}} = \hbar(2\sqrt{g}) \frac{s}{\hbar\sqrt{g}} = 2s = V_0. \quad (1992)$$

If $\widetilde{W} > 2s$, the two target energy intervals are disjoint and the normalized cluster DOS has maximum $1/(4s)$. Choosing $E_{\text{kin}} = 4s$ gives $X = 1$.

Proof. Each orbital channel of Eq. (1989) is a two-layer matrix $sqI_2 + t\sigma_x$, proving Eqs. (1990) and (1991). The target extrema are $-s - \widetilde{W}/2$ and $s + \widetilde{W}/2$, whose difference is $2s + \widetilde{W} = W$. The nearest complement edge is

separated by $d_0 - 2s = \Delta_0$ on both sides. The rank-two target projector selects the first orbital channel and is independent of q, p .

Both target slopes have magnitude s . The metric diameter of $[-1, 1]$ is $2\sqrt{g}$, proving Eq. (1992). With measure $dq/2$ and equal weight $1/2$ on each target band, each disjoint interval has density $1/(4s)$. $\square$

The block is of the co-rotating form Eq. (1944): $h_0 = sqI_2$ and $\tau = T$. Thus H_θ is purely off diagonal, all microscopic jets are Hermitian, and curvature reconstruction acts through an allowed tunneling channel.

Definition 12 (Open microscopic coefficient box). *Let $\mathfrak{R}$ be the open set*

$$\begin{aligned} \frac{1}{2} < a < \frac{3}{2}, \quad \frac{1}{2} < \beta < \frac{3}{2}, \quad \frac{1}{20} < x_f < \frac{3}{20}, \\ \frac{\pi}{4} < \theta_c < \frac{\pi}{2}, \quad \frac{1}{2} < s < \frac{3}{2}, \quad 9 < \Delta_0 < 11, \\ 19 < S_0 < 21, \quad 1 < \omega_0 < 2. \end{aligned} \quad (1993)$$

The physical controls are restricted to

$$0 < \theta < \pi, \quad -1 < K \leq 0, \quad \frac{1}{2} < \omega < \frac{5}{2}. \quad (1994)$$

For each coefficient point set

$$K_0 = -\frac{4x_f^2}{a}, \quad K_f = -\frac{7x_f^2}{a}, \quad \omega_f = \omega_0 - \frac{2x_f^3}{\beta}. \quad (1995)$$

Take sufficiently small neighborhoods of the cusp and fold contained in

$$|\Psi| < \eta, \quad \eta = \frac{1}{1000}. \quad (1996)$$

Theorem 84 (Nonempty analytic fold-cusp region). *For every coefficient point in $\mathfrak{R}$, the family Eq. (1989) has positive uniform bandwidth, remote-isolation, active-overlap, coherence, and physical-angle margins on the scan chart. Explicitly,*

$$W > \frac{1981}{250}, \quad \widetilde{W} > \frac{1231}{250}, \quad \widetilde{W} - 2s > \frac{481}{250}, \quad (1997)$$

$$\Delta = \Delta_0 > 9, \quad C = 1. \quad (1998)$$

The active overlap between the preselected target-layer channels is $\widetilde{W}/2 > 1231/500$. The variation of W in the chart is less than $1/5$, so one fixed local Riesz contour can be chosen with spectral distance greater than 2.

At

$$(x, K, \omega) = (0, K_0, \omega_0), \quad (1999)$$

the operational force is an exact cusp with

$$F = F_\theta = F_{\theta\theta} = 0, \quad |F_{\theta\theta\theta}| = 6, \quad |\mathfrak{d}_c| = a\beta > \frac{1}{4}. \quad (2000)$$

At

$$(x, K, \omega) = (x_f, K_f, \omega_f), \quad (2001)$$

it is an exact fold with

$$F = F_\theta = 0, \quad |F_K| = ax_f > \frac{1}{40}, \quad |F_{\theta\theta}| = 6x_f > \frac{3}{10}. \quad (2002)$$

The set $\mathfrak{R}$ is nonempty and all these inequalities persist on open coefficient neighborhoods.

Proof. On Eq. (1996), $e^{-4\Psi} > e^{-1/250} > 249/250$. The worst coefficients in Eq. (1993) give

$$W > 19 \frac{249}{250} - 11 = \frac{1981}{250}. \quad (2003)$$

Subtracting $2s < 3$ once and twice proves Eq. (1997). Proposition 52 proves the gap, rank, bandwidth, and coherence statements. Also, for $t = 1/250$, $e^t < (1-t)^{-1}$ and $e^{-t} > 1-t$, so

$$\text{osc } W < 21 \left(\frac{250}{249} - \frac{249}{250} \right) < \frac{1}{5}. \quad (2004)$$

The active-overlap statement uses the fixed target-layer channels in Definition 11; it is not a matrix element between the spectral target and its remote complement. The union target-to-complement separation is greater than $9 - 1/10$, allowing a fixed contour with distance greater than 2.

The fold interval

$$\theta_c + \frac{3x_f}{4} < \theta < \theta_c + \frac{5x_f}{4} \quad (2005)$$

has distance at least $\min\{\pi/4, \pi/2 - 3/16\} > 0$ from the physical-angle boundary. Equation (1995) lies in the physical curvature and ω ranges because

$$0 < |K_f| < \frac{63}{200} < 1, \quad |\omega_f - \omega_0| < \frac{27}{2000}. \quad (2006)$$

By Proposition 52, the only nonconstant score factor is W . Since $\Delta_0 + W = S_0 e^{-4\Psi}$,

$$\ell = \ell_0 + \Psi, \quad F = \Psi_\theta = -x^3 + Ux + Y. \quad (2007)$$

At Eq. (1999), $U = Y = 0$. Moreover

$$\begin{aligned} F_K &= -ax, F_\omega = \beta, \\ F_{\theta K} &= -a, F_{\theta\omega} = 0, \\ F_{\theta\theta\theta} &= -6. \end{aligned} \quad (2008)$$

which proves Eq. (2000).

At Eq. (2001), $U = 3x_f^2$ and $Y = -2x_f^3$. Direct substitution into Eq. (2007) gives $F = F_\theta = 0$, $F_K = -ax_f$, and $F_{\theta\theta} = -6x_f$, proving Eq. (2002). At the cusp $\Psi = 0$,

while at the fold $\Psi = -3x_f^4/4$; both satisfy Eq. (1996). The choices

$$\begin{aligned} a &= \beta = s = 1, x_f = \frac{1}{10}, \\ \theta_c &= \frac{\pi}{3}, \Delta_0 = 10, \\ S_0 &= 20, \omega_0 = \frac{3}{2}. \end{aligned} \quad (2009)$$

lie strictly inside $\mathfrak{R}$. Strict inequalities and Theorems 82 and 83 prove open persistence. $\square$

Model-specific curvature-born branch

Theorem 85 (established microscopic curvature birth). *Consider a fixed projector family on a physical angle interval I . Assume:*

1. at $K = 0$, F has no zero in I , with a uniform sign margin;
2. at $K_f < 0$, an interior fold satisfies Eq. (1975);
3. on the born side, the stable branch remains inside I , while the gap, rank, resonant-shell, active-overlap, and coherence margins stay positive;
4. the transported microscopic family and hopping tails converge in the C^3 tier required for the fold, and in C^4 when a cusp is claimed;
5. global target islands have no-loss and no-pollution verification on a spectrally admissible tower, the representation sector is declared, and any velocity term uses the fixed canonical metric and its independent C^1 condition.

Then the stable local maximum created at the fold is curvature born in this projector family. The local stable-branch count increases by one, and its angle satisfies Eq. (1978). It is not a shifted Euclidean branch.

The active family of Definition 11 satisfies these conditions on the fixed slice $\omega = \omega_f$. Indeed, throughout Eq. (2005), at $K = 0$,

$$F = -x^3 - 4x_f^2 x - 2x_f^3 \leq -\frac{347}{64} x_f^3 < 0. \quad (2010)$$

For $K = K_f - \varepsilon$, $\varepsilon > 0$, its two local roots are

$$x_\pm = x_f \pm \sqrt{\frac{a\varepsilon}{3}} + O(\varepsilon), \quad (2011)$$

and the $+$ branch has $F_\theta < 0$. All admissibility margins are the uniform bounds in Theorem 84.

Proof. The fold theorem gives one local maximum and one local minimum on exactly one side of K_f , changing the stable count by one. The Euclidean sign margin excludes a connected stationary branch in the same

interval and projector family. The remaining hypotheses exclude gap, coherence, shell, rank, representation, angle-boundary, and numerical events.

For the explicit family, $K = 0$ gives $U = -4x_f^2$ and the fixed slice gives $Y = -2x_f^3$. The left side of Eq. (2010) decreases for $x > 0$, and at $x = 3x_f/4$ it equals $-347x_f^3/64$. For $K = K_f - \varepsilon$, $U = 3x_f^2 + a\varepsilon$. Expanding Eq. (2007) at x_f gives

$$F = a\varepsilon x_f + a\varepsilon y - 3x_f y^2 - y^3, \quad y = x - x_f, \quad (2012)$$

which proves Eq. (2011). Since $F_\theta = -6x_f y + O(\varepsilon)$, the positive- y branch is the strict maximum. The gap and projectors are exact, so no admissibility boundary is crossed. $\square$

This is a model-specific sufficient theorem for an analytic active sector. It is not evidence that an undeclared material hopping table occupies $\mathfrak{R}$. Material verification requires the microscopic jets and all convergence margins in Theorem 85. Regular-tessellation integer samples alone provide divided differences, not curvature derivatives.

Repairs and sharp obstructions

Corollary 49 (Repairs of Theorems 138–143). *The following replacements apply.*

1. *Theorem 138 concerns a score-order exchange between two independently established stable branches. It neither proves a fold nor supplies microscopic existence.*
2. *Theorem 139 is valid only after the C^4 microscopic condition and the two quantitative margins in Eqs. (1982) and (1983) are verified.*
3. *Theorem 140's branch number is locally constant only inside one fixed admissible projector chart. Gap, coherence, rank, shell, representation, angle, and crystallographic boundaries remain separate.*
4. *Theorem 141 remains a necessary reduced-scaling condition. It is not a sufficient microscopic fold certificate.*
5. *Theorem 142 remains an event taxonomy. A fold or cusp label requires the operator-to-force chain of Theorem 81 and the corresponding quantitative certificate.*
6. *Theorem 143 is replaced by Theorem 85. A bulk or material claim also inherits differentiated-tail, global-island, tower, representation, and metric conditions.*

The generic square-root and cusp normal forms remain conditional consequences unless these microscopic hypotheses are met.

Corollary 50 (Obstructions retained). *Pure fixed-root hyperbolic registry remapping cannot create an interior fold. A radial $|\theta|$ distance cusp is not a stationary cusp of F . Gap closure, coherence loss, target-rank or resonant-shell change, representation change, physical-angle exit, crystallographic termination, and failure of numerical convergence are not folds. Norm, local, or strong-resolvent convergence does not imply the differentiated convergence used here. No result in this subsection changes the exact square five-state no-root conclusion or supplies an empirical fold/cusp location for an undeclared material.*

Proof. The remapping obstruction follows from its strict monotonicity when the root and normalized tunneling harmonic are fixed. The distance cusp is nonsmooth, whereas Theorem 83 assumes $F \in C^3$. The remaining events violate Definition 10 or its differentiated convergence hypotheses. The explicit existence family is the four-state active block in Eq. (1989), not the root-free square truncation. $\square$

Weight-stable bifurcations and Pareto-critical carriers

Exact vanishing of every residual is independent of any strictly positive scalarization. At finite parameters, however, stationary points of a weighted score can move, appear, or disappear as the weights change. We separate quantitative continuation near one scalarization from exact branch existence on a weight-independent Pareto carrier.

Admissible weights and exact weight jets

Let $p = (\theta, u)$, with $u = (K, \omega)$, range in one admissible microscopic chart. All target labels, maximizers, projectors, representation sectors, and metric conventions are fixed on this chart.

Definition 13 (Compact admissible weight simplex). *Let $m \geq 2$, choose $0 < \underline{\alpha} < 1/m$, and define*

$$\mathcal{A}_{\underline{\alpha}} = \left\{ \alpha \in \mathbb{R}^m : \sum_{a=1}^m \alpha_a = 1, \quad \alpha_a \geq \underline{\alpha} \right\}. \quad (2013)$$

Let $B_0, \dots, B_m \in C^4(\mathcal{U}; \mathbb{R})$. The weighted log score, positive score, and angular force are

$$\begin{aligned} \ell_\alpha(p) &= B_0(p) + \sum_{a=1}^m \alpha_a B_a(p), \\ \mathcal{M}_\alpha(p) &= e^{\ell_\alpha(p)}, \quad F_\alpha(p) = \partial_\theta \ell_\alpha(p). \end{aligned} \quad (2014)$$

For the multiplicative residual score one takes

$$B_0 = \log \mathcal{G}_\Delta, \quad B_a = -\log(1 + r_a), \quad (2015)$$

with $m = 4$. Thus the unweighted gap condition is retained explicitly.

The microscopic C^4 tier makes ℓ_α jointly C^4 and F_α jointly C^3 , uniformly on the compact weight simplex.

Proposition 53 (Exact weight derivatives). *Let $h \in \mathbb{R}^m$ satisfy $\sum_a h_a = 0$. For every control multi-index β and every $j \geq 0$ with $j + 1 + |\beta| \leq 4$,*

$$D_\alpha \left(\partial_\theta^j \partial_u^\beta F_\alpha \right) [h] = \sum_{a=1}^m h_a \partial_\theta^{j+1} \partial_u^\beta B_a, \quad (2016)$$

$$D_\alpha^2 \left(\partial_\theta^j \partial_u^\beta F_\alpha \right) = 0. \quad (2017)$$

Proof. Differentiate Eq. (2014). Its dependence on α is affine, giving both identities. The tangent condition merely restricts the ambient derivative to the affine hyperplane containing the simplex. $\square$

A quantitative continuation device

We use Euclidean norms in control space and the ℓ^1 norm in weight space.

Lemma 5 (Fixed-Jacobian zero continuation). *Let $\Phi : \overline{B}_r(z_0) \times \mathcal{N}_\rho \rightarrow \mathbb{R}^n$ be continuously differentiable in z , where*

$$\mathcal{N}_\rho = \{ \alpha \in \mathcal{A}_\alpha : \|\alpha - \alpha_0\|_1 \leq \rho \}. \quad (2018)$$

Suppose

$$\begin{aligned} \Phi(z_0, \alpha_0) &= 0, \quad A_0 = D_z \Phi(z_0, \alpha_0), \\ \sigma &= \|A_0^{-1}\|^{-1} > 0, \end{aligned} \quad (2019)$$

and

$$\sup_{\overline{B}_r \times \mathcal{N}_\rho} \|D_z \Phi - A_0\| \leq \frac{\sigma}{2}, \quad (2020)$$

$$\|\Phi(z_0, \alpha) - \Phi(z_0, \alpha_0)\| \leq B \|\alpha - \alpha_0\|_1, \quad (2021)$$

$$\frac{2B\rho}{\sigma} \leq r. \quad (2022)$$

Then every $\alpha \in \mathcal{N}_\rho$ has exactly one zero $z(\alpha) \in \overline{B}_r(z_0)$, and

$$\|z(\alpha) - z_0\| \leq \frac{2B}{\sigma} \|\alpha - \alpha_0\|_1. \quad (2023)$$

Where Φ is differentiable in α ,

$$D_\alpha z(\alpha)[h] = -[D_z \Phi(z(\alpha), \alpha)]^{-1} D_\alpha \Phi(z(\alpha), \alpha)[h]. \quad (2024)$$

Proof. For fixed α , define

$$T_\alpha(z) = z - A_0^{-1} \Phi(z, \alpha).$$

Equation (2020) makes T_α a contraction with constant at most $1/2$. Equations (2021) and (2022) give

$$\|T_\alpha(z) - z_0\| \leq \frac{1}{2} \|z - z_0\| + \frac{B\rho}{\sigma} \leq r,$$

so T_α maps the closed ball into itself. Banach's theorem gives existence and uniqueness. At its fixed point,

$$\|z(\alpha) - z_0\| \leq \frac{1}{2} \|z(\alpha) - z_0\| + \frac{B}{\sigma} \|\alpha - \alpha_0\|_1,$$

which proves Eq. (2023). Differentiating the zero equation gives Eq. (2024). $\square$

Folds and cusps throughout a weight neighborhood

Theorem 86 (Quantitative fold continuation in weight space). *Fix $\omega = \omega_0$, write $z = (\theta, K)$, and suppose that at (z_0, α_0)*

$$\begin{aligned} F_{\alpha_0} &= F_{\alpha_0, \theta} = 0, \\ |F_{\alpha_0, K}| &\geq a_* > 0, \quad |F_{\alpha_0, \theta\theta}| \geq b_* > 0. \end{aligned} \quad (2025)$$

Put

$$\begin{aligned} \Phi_f(z, \alpha) &= \begin{pmatrix} F_\alpha \\ F_{\alpha, \theta} \end{pmatrix}, \\ A_f &= \begin{pmatrix} 0 & F_{\alpha_0, K} \\ F_{\alpha_0, \theta\theta} & F_{\alpha_0, \theta K} \end{pmatrix}_{z_0}, \quad \sigma_f = \|A_f^{-1}\|^{-1}. \end{aligned} \quad (2026)$$

On $\overline{B}_{r_f}(z_0)$, define the explicit weight-forcing bound

$$B_f = \sup_z \max_{1 \leq a \leq m} (|B_{a, \theta}|^2 + |B_{a, \theta\theta}|^2)^{1/2}. \quad (2027)$$

Assume, on $\overline{B}_{r_f}(z_0) \times \mathcal{N}_{\rho_f}$,

$$\sup \|D_z \Phi_f - A_f\| \leq \frac{\sigma_f}{2}, \quad \frac{2B_f \rho_f}{\sigma_f} \leq r_f. \quad (2028)$$

For $J \in \{K, \theta\theta\}$, let

$$\varpi_{f, J}(s) = \sup \{ |F_{\alpha, J}(z) - F_{\alpha_0, J}(z_0)| : \|z - z_0\| + \|\alpha - \alpha_0\|_1 \leq s \}. \quad (2029)$$

where the supremum is restricted to the preceding cylinder. If

$$\begin{aligned} s_f &= \left(1 + \frac{2B_f}{\sigma_f} \right) \rho_f, \\ \varpi_{f, K}(s_f) &\leq \frac{a_*}{2}, \quad \varpi_{f, \theta\theta}(s_f) \leq \frac{b_*}{2}, \end{aligned} \quad (2030)$$

then every $\alpha \in \mathcal{N}_{\rho_f}$ has a unique fold $z_f(\alpha)$ in $\overline{B}_{r_f}(z_0)$, with

$$\|z_f(\alpha) - z_0\| \leq \frac{2B_f}{\sigma_f} \|\alpha - \alpha_0\|_1. \quad (2031)$$

The continued fold obeys

$$|F_{\alpha,K}| \geq \frac{a_*}{2}, \quad |F_{\alpha,\theta\theta}| \geq \frac{b_*}{2}, \quad |\det D_z \Phi_f| \geq \frac{a_* b_*}{4}. \quad (2032)$$

Its weight derivative is Eq. (2024) with

$$D_\alpha \Phi_f[h] = \sum_a h_a \begin{pmatrix} B_{a,\theta} \\ B_{a,\theta\theta} \end{pmatrix}. \quad (2033)$$

Proof. At the reference fold, $\det A_f = -F_{\alpha_0,K} F_{\alpha_0,\theta\theta}$, so A_f is invertible. Proposition 53 gives

$$\|\Phi_f(z_0, \alpha) - \Phi_f(z_0, \alpha_0)\| \leq B_f \|\alpha - \alpha_0\|_1.$$

Lemma 5 proves unique continuation and Eq. (2031). The joint displacement in the moduli is at most s_f , so Eq. (2030) gives the first two inequalities in Eq. (2032). At any continued fold,

$$\det D_z \Phi_f = -F_{\alpha,K} F_{\alpha,\theta\theta},$$

which gives the determinant bound. The final formula follows from Eqs. (2016) and (2024). $\square$

The theorem certifies the entire declared weight neighborhood, but it allows the fold coordinates to move with α .

Theorem 87 (Quantitative cusp continuation in weight space). *Let $z = (\theta, K, \omega)$. Suppose that at (z_0, α_0)*

$$F_{\alpha_0} = F_{\alpha_0,\theta} = F_{\alpha_0,\theta\theta} = 0, \quad |F_{\alpha_0,\theta\theta\theta}| \geq c_* > 0, \quad (2034)$$

and

$$|\mathfrak{d}_{\alpha_0}(z_0)| \geq d_* > 0, \quad \mathfrak{d}_\alpha = \det \begin{pmatrix} F_{\alpha,K} & F_{\alpha,\omega} \\ F_{\alpha,\theta K} & F_{\alpha,\theta\omega} \end{pmatrix}. \quad (2035)$$

Define

$$\Phi_c(z, \alpha) = \begin{pmatrix} F_\alpha \\ F_{\alpha,\theta} \\ F_{\alpha,\theta\theta} \end{pmatrix}, \quad A_c = D_z \Phi_c(z_0, \alpha_0), \quad \sigma_c = \|A_c^{-1}\|^{-1}, \quad (2036)$$

and

$$B_c = \sup_z \max_{1 \leq a \leq m} (|B_{a,\theta}|^2 + |B_{a,\theta\theta}|^2 + |B_{a,\theta\theta\theta}|^2)^{1/2}. \quad (2037)$$

Assume on $\overline{B}_{r_c}(z_0) \times \mathcal{N}_{\rho_c}$ that

$$\sup \|D_z \Phi_c - A_c\| \leq \frac{\sigma_c}{2}, \quad \frac{2B_c \rho_c}{\sigma_c} \leq r_c. \quad (2038)$$

Let $\varpi_{c,3}$ and $\varpi_{c,d}$ be the joint continuity moduli, defined as in Eq. (2029), of $F_{\alpha,\theta\theta\theta}$ and $\mathfrak{d}_\alpha$, respectively. If

$$s_c = \left(1 + \frac{2B_c}{\sigma_c}\right) \rho_c,$$

$$\varpi_{c,3}(s_c) \leq \frac{c_*}{2}, \quad \varpi_{c,d}(s_c) \leq \frac{d_*}{2}, \quad (2039)$$

then every $\alpha \in \mathcal{N}_{\rho_c}$ has a unique cusp $z_c(\alpha)$ in $\overline{B}_{r_c}(z_0)$, and

$$\|z_c(\alpha) - z_0\| \leq \frac{2B_c}{\sigma_c} \|\alpha - \alpha_0\|_1. \quad (2040)$$

At the continued cusp,

$$|F_{\alpha,\theta\theta\theta}| \geq \frac{c_*}{2}, \quad |\mathfrak{d}_\alpha| \geq \frac{d_*}{2}, \quad (2041)$$

$$|\det D_z \Phi_c| \geq \frac{c_* d_*}{4}. \quad (2042)$$

Moreover,

$$D_\alpha \Phi_c[h] = \sum_a h_a \begin{pmatrix} B_{a,\theta} \\ B_{a,\theta\theta} \\ B_{a,\theta\theta\theta} \end{pmatrix}. \quad (2043)$$

Proof. At a cusp the first column of A_c is $(0, 0, F_{\alpha_0,\theta\theta\theta})^\top$. Expansion along that column gives

$$\det A_c = F_{\alpha_0,\theta\theta\theta} \mathfrak{d}_{\alpha_0}(z_0),$$

up to the immaterial orientation sign. Thus A_c is invertible. The exact weight derivative in Eq. (2043) bounds the parameter forcing by $B_c \|\alpha - \alpha_0\|_1$. Lemma 5 gives the unique cusp and Eq. (2040). The moduli yield Eq. (2041); expansion of the continued Jacobian along its first column gives Eq. (2042).

Only continuity of the third force jet is used in the margin modulus. Thus the theorem is consistent with microscopic C^4 , which gives $F_\alpha \in C^3$, and does not assume a fifth microscopic derivative. $\square$

A weight-independent Pareto-critical carrier

We regard each B_a as a favorable objective: larger is better. This is equivalent to minimizing the corresponding positive residual because the logarithms in Eq. (2015) are monotone.

Definition 14 (Positive rank-one Pareto carrier). *An admissible chart has a positive rank-one Pareto carrier if there are $G \in C^3(\mathcal{U})$ and functions $q_0, \dots, q_m \in C^2(\mathcal{U}; \mathbb{R}_{\geq 0})$ such that*

$$\partial_\theta B_0 = q_0 G, \quad \partial_\theta B_a = q_a G \quad (1 \leq a \leq m), \quad (2044)$$

and some active index a_* satisfies

$$q_{a_*}(p) \geq q_* > 0 \quad (p \in \mathcal{U}). \quad (2045)$$

Theorem 88 (Weight-independent Pareto carrier). *Under Definition 14, put*

$$q_\alpha = q_0 + \sum_a \alpha_a q_a. \quad (2046)$$

Then, throughout $\mathcal{A}_\alpha$,

$$q_\alpha \geq \underline{\alpha} q_* > 0, \quad F_\alpha = q_\alpha G. \quad (2047)$$

Consequently all admissible weights have exactly the same stationary set, regular stability types, fold and cusp locations, and local branch counts.

At a regular carrier zero,

$$F_{\alpha,\theta} = q_\alpha G_\theta. \quad (2048)$$

At a fold of G with control K ,

$$\det D_{(\theta,K)} \begin{pmatrix} F_\alpha \\ F_{\alpha,\theta} \end{pmatrix} = q_\alpha^2 \det D_{(\theta,K)} \begin{pmatrix} G \\ G_\theta \end{pmatrix}. \quad (2049)$$

At a cusp of G ,

$$\mathfrak{d}_\alpha = q_\alpha^2 \det \begin{pmatrix} G_K & G_\omega \\ G_{\theta K} & G_{\theta\omega} \end{pmatrix}, \quad (2050)$$

$$\det D_{(\theta,K,\omega)} \Phi_c = q_\alpha^3 \det D_{(\theta,K,\omega)} \begin{pmatrix} G \\ G_\theta \\ G_{\theta\theta} \end{pmatrix}. \quad (2051)$$

If $G(p) \neq 0$, then p is locally dominated in the angular direction. If $G(p_*) = 0$ and $G_\theta(p_*) < 0$, then p_* strictly dominates every sufficiently near point with the same u , in at least the active objective. Hence stable carrier zeros are strict local Pareto branches.

Proof. Equations (2044) and (2014) give Eq. (2047); the active-weight floor gives its strict lower bound. Thus $F_\alpha = 0$ if and only if $G = 0$. At a carrier zero, differentiation gives Eq. (2048).

At a carrier fold, $G = G_\theta = 0$, and hence

$$F_{\alpha,K} = q_\alpha G_K, \quad F_{\alpha,\theta\theta} = q_\alpha G_{\theta\theta}.$$

This proves Eq. (2049). At a carrier cusp, $G = G_\theta = G_{\theta\theta} = 0$, so

$$F_{\alpha,\theta\theta\theta} = q_\alpha G_{\theta\theta\theta}.$$

The C^2 assumption on q_α is sufficient here: the three vanishing carrier jets imply $G(\theta_c + x) = O(|x|^3)$, so the third difference quotient of $q_\alpha G$ retains only $q_\alpha(\theta_c) G_{\theta\theta\theta}(\theta_c)$. Furthermore, with $u = (K, \omega)$,

$$\nabla_u F_\alpha = q_\alpha \nabla_u G, \quad \nabla_u F_{\alpha,\theta} = q_{\alpha,\theta} \nabla_u G + q_\alpha \nabla_u G_\theta.$$

The term proportional to the first row does not change the two-by-two determinant. This proves Eq. (2050), and expansion of the full Jacobian along its first column proves Eq. (2051).

If $G(p) \neq 0$, its sign is fixed on a short angular segment. Move along that segment in the sign of G . Every B_a is nondecreasing by Eq. (2044), and B_{a_*} increases strictly. Thus the original point is dominated. If $G_\theta(p_*) < 0$, then $G > 0$ immediately to the left and $G < 0$ immediately to the right. Integrating Eq. (2044) toward p_* shows that all objectives improve weakly and the active one improves strictly. This proves the Pareto statement. $\square$

The factorization is a neighborhood property. Membership in a Pareto front at one point does not imply Eq. (2044).

Corollary 51 (Existence versus ranking). *Let $p_j(u)$ and $p_k(u)$ be two carrier branches. Their locations and existence are weight independent, but their score difference is*

$$S_j(\alpha) - S_k(\alpha) = B_0(p_j) - B_0(p_k) + \sum_a \alpha_a [B_a(p_j) - B_a(p_k)]. \quad (2052)$$

It may change sign on a weight hyperplane. For a stationary branch that merely continues with the weights, the envelope identity is

$$D_\alpha S_j(\alpha)[h] = \sum_a h_a B_a(p_j(\alpha)). \quad (2053)$$

Thus scalarization ranks coexisting branches; a ranking wall is not a branch creation event.

Proof. Equation (2052) is obtained by evaluating Eq. (2014) on the two weight-independent branches. In the moving case, differentiating the optimized score produces an additional term $F_\alpha D_\alpha \theta_j$, which vanishes by stationarity. This gives Eq. (2053). $\square$

Exact specialization to the analytic four-state family

Theorem 89 (Exact weight independence in the active family). *In the analytic four-state family, set*

$$K_0 = -\frac{4x_f^2}{a} \quad (2054)$$

and

$$\begin{aligned} x &= \theta - \theta_c, & U &= -a(K - K_0), \\ Y &= \beta(\omega - \omega_0), \\ \Psi &= -\frac{x^4}{4} + \frac{Ux^2}{2} + Yx, \\ W &= S_0 e^{-4\Psi} - \Delta_0. \end{aligned} \quad (2055)$$

The remaining operational factors are

$$\Delta = \Delta_0, \quad V = 2s, \quad X = 1, \quad C = 1. \quad (2056)$$

For the bandwidth objective

$$B_W = -\log \left(1 + \frac{W}{\Delta_0} \right) = 4\Psi + \log \Delta_0 - \log S_0. \quad (2057)$$

Hence, for every $\alpha \in \mathcal{A}_\alpha$,

$$F_\alpha = 4\alpha_W G, \quad G = -x^3 + Ux + Y. \quad (2058)$$

The cusp

$$(x, K, \omega) = (0, K_0, \omega_0) \quad (2059)$$

and the fold

$$(x, K, \omega) = \left(x_f, -\frac{7x_f^2}{a}, \omega_0 - \frac{2x_f^3}{\beta} \right) \quad (2060)$$

therefore have exactly weight-independent locations and branch counts. Using

$$\frac{1}{2} < a, \beta < \frac{3}{2}, \quad \frac{1}{20} < x_f < \frac{3}{20},$$

their weighted margins satisfy

$$|F_{\alpha, \theta\theta\theta}| \geq 24\alpha, \quad |\mathfrak{D}_\alpha| > 4\alpha^2, \quad (2061)$$

$$|F_{\alpha, K}| > \frac{\alpha}{10}, \quad |F_{\alpha, \theta\theta}| > \frac{6\alpha}{5}. \quad (2062)$$

The full cusp Jacobian is larger than $96\alpha^3$ in absolute value.

On the Euclidean fold interval,

$$F_\alpha \leq -\frac{347}{16}\alpha x_f^3 < 0. \quad (2063)$$

For $K = K_f - \varepsilon$, $\varepsilon > 0$, the born roots remain

$$x_\pm = x_f \pm \sqrt{\frac{a\varepsilon}{3}} + O(\varepsilon), \quad (2064)$$

and the $+$ branch is a strict local maximum for every admissible weight.

Proof. Equation (2057) follows from $\Delta_0 + W = S_0 e^{-4\Psi}$. Every other operational factor and the unweighted gap condition are constant on the scan chart. Differentiating the weighted score proves Eq. (2058). Thus Definition 14 holds with

$$q_W = 4, \quad q_a = 0 \ (a \neq W), \quad q_0 = 0.$$

Theorem 88 proves exact invariance of the stationary set and its stability types.

At the cusp,

$$|G_{\theta\theta\theta}| = 6, \quad \left| \det \begin{pmatrix} G_K & G_\omega \\ G_{\theta K} & G_{\theta\omega} \end{pmatrix} \right| = a\beta > \frac{1}{4}.$$

Since $q_\alpha = 4\alpha_W$, the determinant scaling formulas give Eq. (2061) and the full-Jacobian bound. At the fold,

$$|G_K| = ax_f > \frac{1}{40}, \quad |G_{\theta\theta}| = 6x_f > \frac{3}{10},$$

which gives Eq. (2062).

The unweighted Euclidean force obeys $G \leq -347x_f^3/64$ on the fold interval. Multiplication by $4\alpha_W \geq 4\alpha$ proves Eq. (2063). Positive multiplication leaves the root expansion and the sign of G_θ unchanged, proving the final claim. In this particular one-varying-factor family, positive rescaling also preserves the ranking of the branches. $\square$

If the boundary value $\alpha_W = 0$ is admitted, all varying objectives in this family receive zero weight and $F_\alpha \equiv 0$. The scalar force then loses its diagnostic meaning, which is the sharp reason for the active weight floor.

Sharp counterexample and repairs

Proposition 54 (An interior weight can create a stationary pair). *On $-1 < \theta < 1$, let*

$$B_1(\theta) = \frac{\theta^3}{3} + \theta, \quad B_2(\theta) = \frac{\theta^3}{3} - \theta, \quad (2065)$$

and use weights $(\alpha, 1 - \alpha)$ with $1/4 \leq \alpha \leq 3/4$. Then

$$\ell_\alpha(\theta) = \frac{\theta^3}{3} + (2\alpha - 1)\theta, \quad F_\alpha(\theta) = \theta^2 + 2\alpha - 1. \quad (2066)$$

At $(\theta, \alpha) = (0, 1/2)$,

$$F_\alpha = F_{\alpha, \theta} = 0, \quad \partial_\alpha F_\alpha = 2, \quad F_{\alpha, \theta\theta} = 2. \quad (2067)$$

For $\alpha < 1/2$ there are two stationary points; for $\alpha > 1/2$ there are none. Thus strictly positive scalarization weights can create or remove a stationary pair when the carrier or uniform continuation hypotheses fail.

Proof. Differentiation gives Eqs. (2066) and (2067). For $\alpha < 1/2$, the roots are

$$\theta_\pm = \pm\sqrt{1 - 2\alpha}.$$

Since $F_{\alpha, \theta} = 2\theta$, the negative root is a strict score maximum and the positive root is a minimum. No root exists for $\alpha > 1/2$. At $\theta = 0$, $B'_1 = 1$ and $B'_2 = -1$, so their derivatives cannot share a nonnegative common factor with a positive active coefficient. $\square$

Corollary 52 (Repairs of scalarized magic statements). *The affected statements are repaired as follows.*

1. Theorem 107 is the exact stationarity balance for one declared weight. It neither reduces to bandwidth minimization nor proves a weight-independent Pareto branch. Coincident extrema remain sufficient but nongeneric.
2. Theorem 108's exact-limit equivalence remains algebraically valid for strictly positive weights. It does not imply equality of finite-parameter stationary sets or convergence of finite-cover optimizers.
3. Theorem 122 is a fixed-weight simple-root continuation result. Across weights, its branch generally moves according to Eq. (2024); exact location independence requires Theorem 88.
4. Theorem 138 concerns the ranking exchange in Eq. (2052), not branch creation.
5. Theorem 139 requires both the microscopic cusp certificate and either Theorem 87 or the common carrier.
6. Theorem 140 has a weight-independent local branch count only under Theorem 88. Otherwise its count is local to one scalarization and one admissible chart.

7. Theorem 141 remains a necessary reduced-scaling relation and is not an existence or scalarization-independence certificate.
8. Theorem 142 remains an event taxonomy. Weight folds and cusps are smooth events distinct from label, gap, rank, shell, representation, coherence, angle, and crystallographic boundaries.
9. Theorem 143 retains the microscopic curvature-born hypotheses. Exact weight independence is established by Theorem 89 for the analytic active family, not for an arbitrary richer model.

Proof. Items 1–3 separate fixed-weight stationarity, algebraic residual limits, and simple-root continuation from the weight continuation theorems above. Equation (2052) proves item 4. The quantitative cusp and carrier theorems prove items 5 and 6. The remaining items preserve the distinction between necessary scaling, event classification, and model-specific microscopic existence. $\square$

Corollary 53 (Inherited conditions and obstructions). *Every bulk branch under every scalarization retains the intrinsic global spectral-island definition, no-loss and no-pollution, a spectrally admissible retained tower sector, declared representation scope, the canonical metric, and the required differentiated hopping and projector convergence tiers. Active-label or maximizer switches and gap, coherence, rank, resonant-shell, representation, physical-angle, or crystallographic boundaries are not smooth weight folds or cusps.*

Positive weights alone do not imply equal stationary sets. Pareto-front membership alone does not imply the rank-one factorization. Boundary weights may erase the only active objective. Weight-neighborhood persistence permits coordinate motion, while a ranking wall does not create a branch. The exact square five-state truncation remains root-free, and the analytic four-state carrier is not an empirical certificate for an undeclared material.

Proof. The counterexample proves the first failure, and Definition 14 shows the additional structure needed beyond Pareto membership. The boundary failure follows from Eq. (2058). The continuation and ranking claims are respectively Theorems 86–87 and Corollary 51. The remaining conditions are independent hypotheses of the microscopic and infinite-volume constructions; changing scalarization does not supply them. $\square$

Three-layer universality and separation of implications

The word “universal” is used below only after its layer has been declared. This is essential because a local geometric identity, an operator identity, and an exact commensurate-cell statement have different hypotheses and do not imply one another.

Layer tags and admissible bridges

Definition 15 (Universality layers and identity tag). *We use the following tags.*

1. **G** denotes local geometric universality: metric displacement, local moire length, registry area or count, normalized geometric crossover, running geometric exponents, and separation-law response.
2. **S** denotes spectral universality: normalized operators, spectral islands, bandwidths, gaps, density of states, velocities, coherence, active-sector roots, and spectral fold or cusp response.
3. **C** denotes exact-crystallographic universality: exact common subgroups, exact commensurate sequences, exact cell or index counts, and their arithmetic locking to a geometric scale.
4. **I_L**, with $L \in \{G, S, C\}$, denotes an exact identity internal to layer L . It is not, by itself, a universality claim.
5. **N** denotes a definition, obstruction, counterexample, or convergence condition that makes no universality claim.

A local spectral fold or cusp class is tagged S_{NF} . This tag records a normal form under transversality hypotheses; it does not assert a global operator master curve.

Definition 16 (Hypothesis packages). *The geometric package $\mathcal{H}_G$ consists of a declared local metric, centered registry construction, fixed density and marking, and an admissible separation law on the parameter set under study.*

The spectral package $\mathcal{H}_S$ consists, as required by the observable, of the following data:

1. *a fixed transported Hilbert fiber and a bounded self-adjoint, or domain-controlled self-adjoint, Hamiltonian;*
2. *finite-range or exponentially admissible kernels, with the hyperbolic decay margin, and weighted C^1 or C^2 envelopes for differentiated observables;*
3. *an reference-point active Riesz or symmetry projector of constant rank, a positive active-overlap margin, and fixed energy and metric conventions;*
4. *an intrinsic isolated spectral island with a positive gap, together with no-loss and no-pollution verification whenever global finite-cover spectra are used;*
5. *the reduced-regular representation condition whenever “bulk” spectrum is meant, rather than an arbitrary automorphic or finite-image block;*
6. *operator-level one-parameter reduction for master functions, differentiated reduction for velocity or curvature, and separate hypotheses for participation and coherence;*

7. an actual simple root, or quantitative fold or cusp margins, whenever such a spectral feature is claimed.

The crystallographic package $\mathcal{H}_C$ consists of an actual exact commensurate sequence, a finite-index common subgroup or an equivalent exact cell construction, an exact index/count identity, and an arithmetic-locking estimate on the scale of the claimed asymptotic exponent.

For a spectral one-parameter claim, the required bridge has the form

$$\sup_{p \in A} \|E_p^{-1} U_p H_p U_p^* - \mathcal{H}(X(p))\| \leq \varepsilon_A, \quad \varepsilon_A \longrightarrow 0, \quad (2068)$$

on a fixed transported fiber. For exact-cell exponent transfer, write

$$\delta_{\text{ar}}(p) = \log \frac{N_{\text{ex}}(p)}{N_G(p)}. \quad (2069)$$

Theorem 90 (No cross-layer leakage). *The following are the only layer implications supported by the present analysis.*

1. The package $\mathcal{H}_G$ proves $\mathbf{G}$ statements only. It proves an $\mathbf{S}$ statement only after adding $\mathcal{H}_S$ and the operator or profile bridge appropriate to that statement.
2. Suppose $N_G(\theta) = |\theta|^{-\nu+o(1)}$ along an exact sequence satisfying $\mathcal{H}_C$. If

$$\delta_{\text{ar}}(\theta) = o(|\log |\theta||), \quad (2070)$$

then $N_{\text{ex}}(\theta) = |\theta|^{-\nu+o(1)}$. If $|\delta_{\text{ar}}|$ is uniformly bounded and $N_G = \Theta(|\theta|^{-\nu})$, then $N_{\text{ex}} = \Theta(|\theta|^{-\nu})$.

3. There is no implication

$$\mathbf{G} \implies \mathbf{S}, \quad \mathbf{G} \implies \mathbf{C}, \quad \mathbf{S} \implies \mathbf{C}, \quad \mathbf{C} \implies \mathbf{S} \quad (2071)$$

without the corresponding additional package.

4. A combined spectral-crystallographic conclusion requires both $\mathcal{H}_S$ and $\mathcal{H}_C$. Neither package substitutes for the other.

Proof. The first assertion is a dependence statement: geometric data do not specify the hopping profile, active projector, or spectral representation. On the same geometry one may choose zero tunneling or an admissible nonzero tunneling profile; their spectra differ, so geometry alone cannot imply a spectral master function.

For the second assertion, equations (2069) and (2070) give

$$\log N_{\text{ex}} = \nu |\log |\theta|| + o(|\log |\theta||). \quad (2072)$$

Exponentiation proves the first conclusion. If the two logarithmic errors are bounded, their exponentials are

bounded above and below by positive constants, which proves the Θ conclusion.

For the remaining nonimplications, a generic incommensurate angle can obey all local geometric formulas without possessing an exact common subgroup. A covariant incommensurate Hamiltonian may possess an isolated spectral island without any exact finite cell. Conversely, on a nonamenable group the trivial finite-dimensional representation can contribute a spectral point outside the reduced-regular bulk, so an exact finite quotient or automorphic block does not determine that bulk. These examples establish all four failures in (2071). A joint conclusion must therefore verify both packages separately. $\square$

Corollary 54 (The word “exact” has no automatic layer transfer). *An exact local distance formula is $\mathbf{l}_G$, an exact Schur–Feshbach, Riesz, resolvent, symmetry, or matrix identity is $\mathbf{l}_S$, and an exact finite-index formula is $\mathbf{l}_C$. None is a $\mathbf{C}$ universality claim unless an exact sequence and arithmetic locking are also present.*

Proof. Each identity follows from hypotheses internal to its own object. Theorem 90 shows that the missing bridge data cannot be inferred from algebraic exactness. $\square$

Scope of the principal result families

Theorem 91 (Scope of the principal result families). *The following replacements are mandatory.*

1. Theorem 4 is $\mathbf{G}$: its normalized area crossover is an exact local geometric identity. Spectral and exact-cell consequences require the separate bridges of Theorem 90.
2. Theorems 16 and 65 are local $\mathbf{l}_S$ kernel-response identities. Geometry enters them through a declared separation map, so their use as a $\mathbf{G}$ -to- $\mathbf{S}$ bridge is conditional. Smooth distance dependence, a nonzero logarithmic slope, and shell summability do not prove a global spectral master curve.
3. Theorem 59 is a local $\mathbf{G}$ registry-mode diffraction slope. It is neither a global spectral theorem nor an exact-cell theorem.
4. Theorem 114 is conditional $\mathbf{S}$: its scaling law applies only to a finite, independently established root in the declared active model.
5. Theorems 134 and 136 are $\mathbf{S}_{\text{NF}}$. They require fixed labels, differentiability, transversality, gap/coherence margins, and the microscopic derivative chain; they are not cross-model master curves.
6. Theorem 154 is a leading local profile equivalence under matching amplitude and logarithmic slope. Higher profile derivatives remain independent correction fields.

7. Theorem 159 is a conditional geometric-to-spectral coordinate bridge, not an operator equivalence.
8. Theorems 168 and 169 become strong $\mathbf{S}$ results only under normalized operator reduction, an intrinsic isolated spectral island, no-loss/no-pollution control, admissible reduced-bulk representation and tower, and the differentiated convergence required by the observable.
9. Theorem 183 is the $\mathbf{C}$ bridge only along an exact sequence with arithmetic locking. Its representation-dimension formulas are $\mathbf{l}_C$, not a completeness theorem for reduced bulk spectrum.
10. Theorems 185, 186, 187, 188, 189, and 190 are $\mathbf{G}$ fixed-point, basin, boundary, and registry-threshold results.
11. Theorems 191, 192, 193, 194, and 195 are conditional $\mathbf{S}$ inheritance results. They require the complete spectral bridge.
12. Theorem 196 is restricted to its effective-volume geometry. It does not supply a higher-dimensional rotation theorem.
13. Theorem 197 is $\mathbf{C}$ and requires existence of an exact sequence and the stated locking rate. Theorem 198 is conditional $\mathbf{S}$ and requires an actual spectral root and operator bridge.
14. Theorem 199 has three separate clauses: local $\mathbf{G}$, conditional $\mathbf{S}$, and conditional $\mathbf{C}$. It is not a single unconditional universality theorem.

Proof. Items 1–3 and 10 follow from the fact that the relevant quantities are functions of the local metric and registry map before any Hamiltonian or exact subgroup is chosen. Items 4–8, 11, and the spectral part of item 13 involve roots, bands, gaps, or operator limits and hence require $\mathcal{H}_S$. Items 9 and the crystallographic part of item 13 involve exact cells and hence require $\mathcal{H}_C$. Theorem 90 excludes every stronger reading. Item 12 is a scope boundary: its proof uses only the stated effective-volume geometry and contains no higher-dimensional rotation analysis. Splitting item 14 is therefore forced by the dependence of its three conclusions on different hypothesis packages. $\square$

Model separation and spectral conditions

Theorem 92 (Root, bulk, convergence, and metric separation theorem). *The layer tags do not relax any earlier obstruction.*

1. The exact square five-state star has no magic root. Its positive structural margin is

$$m_{\square} = 4\sqrt{2} - 5 > 0. \quad (2073)$$

A universality tag, a change of normalization, a finite cover, a representation block, or an exact sequence cannot create a root absent from that model. Richer-model continuation is conditional on an independently verified simple root or fold/cusp certificate.

2. Local or strong-resolvent convergence alone does not certify global band edges, gaps, or spectral islands. Such claims retain the norm/no-pollution conditions and the admissible-tower hypotheses.
3. Completeness of the irreducible blocks of each finite quotient does not imply completeness for the infinite reduced-regular bulk. The weak-containment or reduced- $\mathbf{C}^*$ -factorization condition remains mandatory.
4. Velocity requires uniform C^1 operator control in a fixed metric; curvature requires the corresponding C^2 control; coherence and participation require their separate projector estimates.
5. Metric-dependent spectral comparisons use a fixed rank, topology, marking, transport, and normalization convention. With the canonical metric, $\Omega_{\max}$ has units of length per time, and a metric-only magic score must use a fixed dimensionless combination such as

$$\frac{\hbar D_G \Omega_{\max}}{E_{\text{ref}}}. \quad (2074)$$

6. Exact-crystallographic scaling is asserted only on an existing exact sequence satisfying (2070). Its failure does not alter a valid local $\mathbf{G}$ result.

Proof. Item 1 is the square-star no-root theorem together with its sharp positive margin; the richer-model statements are continuation theorems and therefore cannot supply the missing premise. Items 2 and 3 follow from spectral pollution and nonamenable representation counterexamples. Differentiation is not continuous in the C^0 operator topology, which proves item 4. The coordinate and normalization covariance of the Hodge metric proves item 5, including the units in (2074). Item 6 is precisely the locking calculation in Theorem 90. $\square$

Corollary 55 (Repaired synthesis rule). *Every subsequent statement must state its layer. A local metric/registry claim may be stated as $\mathbf{G}$ under $\mathcal{H}_G$. A root, band, gap, density-of-states, velocity, coherence, fold/cusp, or master-function claim is $\mathbf{S}$ and must list the relevant parts of $\mathcal{H}_S$. An exact cell/index exponent is $\mathbf{C}$ and must list $\mathcal{H}_C$. A theorem using more than one layer must state and prove each bridge separately.*

Proof. This is the direct combination of Theorems 90, 91, and 92. It repairs the three-layer synthesis without changing any later result. $\square$

Symmetric arithmetic assumptions at the Euclidean and hyperbolic endpoints

Exact crystallographic counts are defined only on exact sequences. They must not be inferred from a local registry domain or from a spectral root. We now put the Euclidean and hyperbolic arithmetic corrections in the same notation.

Exact-sequence domains and paired factors

Definition 17 (Paired exact-sequence domains). *Write*

$$s_{X,j} := \left| \sin \frac{\theta_{X,j}}{2} \right|, \quad X \in \{E, H\}. \quad (2075)$$

An exact Euclidean square sequence consists of primitive positive pairs $(m_j, n_j) \in \mathbb{N}^2$, with $\mathbb{N} = \{1, 2, \dots\}$, such that

$$\tan \frac{\theta_{E,j}}{2} = \frac{n_j}{m_j}, \quad \gcd(m_j, n_j) = 1, \quad 0 < \theta_{E,j} \longrightarrow 0, \quad (2076)$$

with parity divisor $d_j \in \{1, 2\}$. Its primitive arithmetic multiplier is

$$\mathcal{A}_{E,j} := \frac{n_j^2}{d_j}. \quad (2077)$$

An exact centered hyperbolic sequence consists of

$$\begin{aligned} g_j &\in \text{Comm}_G(\Gamma) \cap K_o, \quad g_j \neq e, \quad g_j \longrightarrow e, \\ \theta_{H,j} &\longrightarrow 0, \\ q_j &:= [\Gamma : \Gamma \cap g_j \Gamma g_j^{-1}] < \infty. \end{aligned} \quad (2078)$$

Here $\theta_{H,j}$ is the centered rotation angle of g_j . The two domains are independent. In particular, neither the existence nor the arithmetic behavior of one sequence is an assumption about the other.

For the one-site square benchmark, let $A_{0,E}$ be the primitive-cell area and retain the fixed registry threshold $\ell_* > 0$. Put

$$C_E := \frac{\pi \ell_*^2}{2A_{0,E}}, \quad \kappa_E := \frac{2}{C_E} = \frac{4A_{0,E}}{\pi \ell_*^2}. \quad (2079)$$

For the hyperbolic endpoint, put

$$b_* := \sinh \frac{\ell_*}{2R}, \quad C_H := \frac{4\pi n_b R^2 b_*}{A_{0,H}}. \quad (2080)$$

Definition 18 (Effective counts and exact locking factors). *The density-normalized effective Euclidean count and the exact square count are*

$$N_{E,j}^{\text{eff}} = C_E s_{E,j}^{-2}, \quad (2081)$$

$$N_{E,j}^{\text{sc}} = 2\mathcal{A}_{E,j} s_{E,j}^{-2}. \quad (2082)$$

Thus the factor relative to the fixed effective-count convention is

$$\mathcal{L}_{E,j} := \frac{N_{E,j}^{\text{sc}}}{N_{E,j}^{\text{eff}}} = \kappa_E \mathcal{A}_{E,j}. \quad (2083)$$

The fixed constant κ_E affects amplitudes but not exponents.

At the hyperbolic endpoint, define

$$A_{H,j}^{\text{eff}} := 2\pi R^2 \left(\sqrt{1 + b_*^2 / s_{H,j}^2} - 1 \right). \quad (2084)$$

Then

$$N_{H,j}^{\text{eff}} = \frac{4\pi n_b R^2}{A_{0,H}} \left(\sqrt{1 + \frac{b_*^2}{s_{H,j}^2}} - 1 \right), \quad (2085)$$

$$N_{H,j}^{\text{sc}} = 2n_b q_j, \quad (2086)$$

$$\mathcal{L}_{H,j} := \frac{N_{H,j}^{\text{sc}}}{N_{H,j}^{\text{eff}}} = \frac{q_j A_{0,H}}{A_{H,j}^{\text{eff}}}. \quad (2087)$$

In both cases the exact identity is

$$N_{X,j}^{\text{sc}} = \mathcal{L}_{X,j} N_{X,j}^{\text{eff}}, \quad X \in \{E, H\}. \quad (2088)$$

Lower and upper exact exponents

Theorem 93 (Paired factorization and exponent envelopes). *For either exact sequence in Definition 17, define*

$$\begin{aligned} \lambda_{X,j} &:= \log s_{X,j}^{-1}, \\ \eta_{X,j}^{\text{ar}} &:= \frac{\log \mathcal{L}_{X,j}}{\lambda_{X,j}}, \\ b_{X,j}^{\text{sc}} &:= \frac{\log N_{X,j}^{\text{sc}}}{\lambda_{X,j}}. \end{aligned} \quad (2089)$$

Let $p_E = 2$ and $p_H = 1$. Then

$$b_{X,j}^{\text{sc}} = p_X + \eta_{X,j}^{\text{ar}} + o(1). \quad (2090)$$

Consequently, with limits in the extended real line,

$$\underline{\beta}_X^{\text{sc}} := \liminf_j b_{X,j}^{\text{sc}} = p_X + \underline{\eta}_X^{\text{ar}}, \quad (2091)$$

$$\overline{\beta}_X^{\text{sc}} := \limsup_j b_{X,j}^{\text{sc}} = p_X + \overline{\eta}_X^{\text{ar}}. \quad (2092)$$

Here $\underline{\eta}_X^{\text{ar}}$ and $\overline{\eta}_X^{\text{ar}}$ are the corresponding lower and upper limits of $\eta_{X,j}^{\text{ar}}$.

If both arithmetic exponent limits exist and are finite, then the exact endpoint shift is

$$\beta_E^{\text{sc}} - \beta_H^{\text{sc}} = 1 + \eta_E^{\text{ar}} - \eta_H^{\text{ar}}. \quad (2093)$$

It equals the geometric value one only when the two arithmetic corrections agree; in particular, it equals one when both corrections vanish.

Proof. Equation (2081) gives

$$\frac{\log N_{E,j}^{\text{eff}}}{\lambda_{E,j}} = 2 + \frac{\log C_E}{\lambda_{E,j}} \longrightarrow 2. \quad (2094)$$

From Eq. (2085),

$$N_{H,j}^{\text{eff}} = C_H s_{H,j}^{-1} [1 + o(1)], \quad (2095)$$

so its logarithmic exponent tends to one. Taking logarithms in Eq. (2088) proves Eq. (2090). Adding a convergent sequence does not change a lower or upper limit, which proves Eqs. (2091) and (2092). Subtracting the two finite limiting identities gives Eq. (2093). $\square$

Corollary 56 (Three symmetric locking tiers). *For either $X = E$ or $X = H$, the following statements are exact.*

1. *The exact exponent inherits p_X if and only if*

$$\log \mathcal{L}_{X,j} = o(\lambda_{X,j}). \quad (2096)$$

2. *The two-sided comparison*

$$N_{X,j}^{\text{sc}} = \Theta(N_{X,j}^{\text{eff}}) \quad (2097)$$

holds if and only if there are fixed $0 < c_- \leq c_+ < \infty$ such that

$$c_- \leq \mathcal{L}_{X,j} \leq c_+ \quad (2098)$$

eventually.

3. *If $s_{X,j}^{p_X} N_{X,j}^{\text{eff}} \rightarrow C_X \in (0, \infty)$, then a positive finite exact amplitude exists if and only if*

$$\mathcal{L}_{X,j} \longrightarrow \mathcal{L}_{X,*} \in (0, \infty). \quad (2099)$$

In that case

$$s_{X,j}^{p_X} N_{X,j}^{\text{sc}} \longrightarrow \mathcal{L}_{X,*} C_X. \quad (2100)$$

The amplitude tier implies the bounded tier, which implies the exponent tier. No reverse implication holds without further arithmetic information.

Proof. The first equivalence follows from Eq. (2090). Dividing Eq. (2097) by the positive effective count proves the second equivalence. For the last assertion, divide the normalized exact count by the normalized effective count. Their ratio is exactly $\mathcal{L}_{X,j}$, which proves both directions and Eq. (2100). The hierarchy is immediate. Subpower unbounded factors and bounded nonconvergent factors show that the reverse implications fail. $\square$

Sharp endpoint obstructions

Theorem 94 (Unrestricted Euclidean exact sequences). *Every primitive Euclidean exact sequence satisfies*

$$\mathcal{A}_{E,j} \geq \frac{1}{2}, \quad N_{E,j}^{\text{sc}} \geq \frac{\kappa_E}{2} N_{E,j}^{\text{eff}} = s_{E,j}^{-2}. \quad (2101)$$

Moreover,

$$n_j = O(1) \iff N_{E,j}^{\text{sc}} = \Theta(N_{E,j}^{\text{eff}}), \quad (2102)$$

and either condition gives $N_{E,j}^{\text{sc}} = \Theta(|\theta_{E,j}|^{-2})$. The weaker exponent-two condition is

$$\log n_j = o(\lambda_{E,j}). \quad (2103)$$

There is no sequence-uniform upper comparison by a constant multiple of the inverse-square geometric count. Indeed, the primitive sequence

$$n_j = 2^j, \quad m_j = j2^j + 1 \quad (2104)$$

has $d_j = 1$, $\theta_{E,j} \sim 2/j$, and

$$\frac{N_{E,j}^{\text{sc}}}{N_{E,j}^{\text{eff}}} = \kappa_E 4^j \longrightarrow \infty. \quad (2105)$$

In fact its upper exact exponent is infinite. Thus unrestricted exact Euclidean sequences need not have exponent two.

Proof. Because $n_j \geq 1$ and $d_j \leq 2$, $\mathcal{A}_{E,j} \geq 1/2$. Equations (2081)–(2083) then prove Eq. (2101). Since $d_j \in \{1, 2\}$, boundedness of n_j is equivalent to a two-sided bound on $\mathcal{L}_{E,j}$. Corollary 56 proves Eq. (2102). Also

$$\log \mathcal{L}_{E,j} = 2 \log n_j - \log d_j + \log \kappa_E, \quad (2106)$$

which proves Eq. (2103).

For Eq. (2104), $\gcd(2^j, j2^j + 1) = 1$, and the two integers have opposite parity, so $d_j = 1$. Furthermore,

$$\tan \frac{\theta_{E,j}}{2} = \frac{1}{j + 2^{-j}}, \quad (2107)$$

which gives $\theta_{E,j} \sim 2/j$. Equation (2105) follows from the exact factorization. Finally, $\lambda_{E,j} \sim \log j$, whereas $\log \mathcal{L}_{E,j} = j \log 4 + O(1)$; hence the exact upper exponent is infinite. $\square$

Theorem 95 (Hyperbolic existence and exponent-four obstruction). *Arithmeticity or density of a commensurator does not by itself provide a fixed-center exact sequence. Even after such a sequence is constructed, existence does not imply locking.*

On the explicit centered arithmetic sequence already constructed in the (2, 3, 8) class,

$$q_j = s_{H,j}^{-4+o(1)}, \quad N_{H,j}^{\text{sc}} = s_{H,j}^{-4+o(1)},$$

$$\mathcal{L}_{H,j} = s_{H,j}^{-3+o(1)}. \quad (2108)$$

Thus

$$\eta_H^{\text{ar}} = 3, \quad \beta_H^{\text{sc}} = 4, \quad (2109)$$

and none of the three locking tiers holds. Along the associated conditionally tuned exact magic subsequence, whose construction assumes a finite positive richer-model root, the exact-cell logarithmic coefficient is four times the geometric effective-area coefficient.

Proof. Fixed-center activity is an extra intersection condition with K_o , while locking requires an index-angle estimate; neither follows from density. For the stated explicit sequence, the proved local-to-global index formula gives $q_j = s_{H,j}^{-4+o(1)}$. Multiplication by the fixed factor $2n_b$ does not change the exponent. Equation (2095) has exponent one, so exact division gives the third line of Eq. (2108). The two limits in Eq. (2109) follow from Theorem 93. Substitution of the tuned magic relation into the same index estimate gives coefficients four and one for the exact and effective quantities, respectively. $\square$

Intrinsic spectral normalization and magic cells

Theorem 96 (Arithmetic is a secondary normalization coordinate). *The primary spectral scales remain*

$$\bar{Q}_M = \frac{2\pi}{\xi_M}, \quad \bar{E}_M = ta^2\bar{Q}_M^2. \quad (2110)$$

On a Euclidean exact branch,

$$c_{\text{ar},j} := \frac{\sqrt{d_j}}{2n_j}, \quad \mathcal{A}_{E,j} = \frac{1}{4c_{\text{ar},j}^2}. \quad (2111)$$

This branch factor is a secondary convention. Under any fixed positive conversion c ,

$$Q_c = c\bar{Q}_M, \quad E_c = c^2\bar{E}_M, \quad \alpha_c = c^{-2}\bar{\alpha}, \quad \alpha_c^* = c^{-2}\bar{\alpha}^*, \quad (2112)$$

and therefore

$$\frac{\alpha_c}{\alpha_c^*} = \frac{\bar{\alpha}}{\bar{\alpha}^*}. \quad (2113)$$

Neither $\mathcal{L}_E$ nor $\mathcal{L}_H$ is to be inserted into the primary spectral coordinate. Equations (2112) and (2113) are conditional on a specified root and do not assert its existence.

Proof. Equation (2111) is a rearrangement of Eq. (2077). Quadratic homogeneity gives the first two transformations in Eq. (2112); division of a fixed physical numerator gives the third. The same coordinate change at the same physical root gives the fourth, and their ratio proves Eq. (2113). Arithmetic factors count exact cells; they do not alter the local length from which $\bar{Q}_M$ is defined. $\square$

Proposition 55 (Effective and exact magic-cell statements). *At any operational or model-established magic point, the geometric count is $N_X^{\text{eff},*}$, whether or not the angle is commensurate. The exact square five-state target has no magic root, and no arithmetic sequence creates one; every magic-point statement below is conditional on a richer independently established root. If and only if that point lies on a declared exact sequence may one define*

$$N_{X,j}^{\text{sc},*} = \mathcal{L}_{X,j}^* N_{X,j}^{\text{eff},*}. \quad (2114)$$

Its lower and upper exact magic-cell exponents are obtained from Eqs. (2091) and (2092) with the arithmetic field restricted to that sampled magic subsequence. In particular, a geometric endpoint comparison gives exponents two and one, whereas an exact endpoint comparison gives

$$\beta_{-E}^{\text{sc},*} = 2 + \eta_E^{\text{ar},*}, \quad \beta_{-H}^{\text{sc},*} = 1 + \eta_H^{\text{ar},*}, \quad (2115)$$

with the analogous upper formulas. No exact count is assigned off the exact sampling set.

Proof. The effective count is defined by the local registry disk and hence exists at every twist. An exact count requires the finite-index data in Definition 17. Once those data exist, Eq. (2114) is the exact factorization, and Theorem 93 applied to the sampled subsequence gives all stated exponent formulas. $\square$

Subsequent theorem-family repairs

Theorem 97 (Symmetric repair ledger). *The following readings replace every asymmetric Euclidean-hyperbolic cell comparison in the checked theorem families.*

1. Theorems 1–13: Theorem 1 retains the exact Euclidean factor $\mathcal{A}_E$; Theorems 2–4 use N^{eff} ; Theorems 5–9 retain exact-sequence and existence conditions without locking; Theorems 10–11 use the three tiers of Corollary 56; Theorems 12–13 imply no equality between effective and exact counts.
2. Theorem 62: exact diffraction-space growth carries the applicable $\mathcal{L}_X$. A geometric endpoint shift alone is not an exact exponent statement.
3. Theorems 111–129: spectral magic statements use the intrinsic scale (2110). Fixed n_j, d_j are secondary exact sampling data and do not prove a crystallographic exponent.
4. Theorem 130: both endpoint exact counts are $\mathcal{L}_X N_X^{\text{eff}}$. The explicit hyperbolic sampled sequence has exact exponent four, not one.
5. Theorems 145–160: local geometry and effective coupling use the local length, N^{eff} , and intrinsic normalization. Any exact-cell conclusion adds $\mathcal{L}_X$ on a declared exact sequence.

6. Theorem 170 keeps $\overline{Q}_M$ primary and c_{ar} secondary. Theorems 171–176 use N^{eff} for common-length and spectral reconstruction statements and imply no arithmetic locking.
7. In Theorem 177, “exact” refers to the analytic effective count–twist identity unless an exact factor is displayed. Theorems 178–182 use N^{eff} for geometric magic-cell laws; a literal exact-cell law is Eq. (2114).
8. Theorem 183 applies the same exponent, bounded, or amplitude tier on both sides. Theorem 184 uses N^{eff} for geometry-to-spectrum correspondence; crystallographic promotion is a separate conditional step.
9. Theorem 196’s shift $2 \rightarrow 1$ is geometric-effective. For finite arithmetic exponent limits, the exact shift is Eq. (2093). Theorem 197 retains hyperbolic sequence existence and $\mathcal{L}_H$, and any exact endpoint comparison also displays $\mathcal{L}_E$.
10. Theorem 198 is spectral and root-conditional, not arithmetic. Theorem 199 must state the paired effective formulas and the paired exact lower/upper corrections; neither exact endpoint exponent is unconditional.

Proof. Items 1, 2, 4, and 7–10 follow directly from the exact factorization (2088) and the exponent theorem. Items 3, 5, and 6 follow from Theorem 96: a spectral coordinate is built from local length, not from an exact-cell multiplier. The explicit exception in item 4 is Theorem 95. Thus every geometric-effective comparison uses N^{eff} , and every literal exact comparison carries the appropriate arithmetic factor on both sides. $\square$

Corollary 57 (No-asymmetric-assumption separation theorem). *The repaired endpoint formulas are*

$$N_E^{\text{eff}} = s_E^{-2+o(1)}, \quad N_H^{\text{eff}} = s_H^{-1+o(1)}, \quad (2116)$$

$$\begin{aligned} \underline{\beta}_E^{\text{sc}} &= 2 + \underline{\eta}_E^{\text{ar}}, & \underline{\beta}_H^{\text{sc}} &= 1 + \underline{\eta}_H^{\text{ar}}, \\ \overline{\beta}_E^{\text{sc}} &= 2 + \overline{\eta}_E^{\text{ar}}, & \overline{\beta}_H^{\text{sc}} &= 1 + \overline{\eta}_H^{\text{ar}}. \end{aligned} \quad (2117)$$

A comparison may claim exponent inheritance, a Θ law, or amplitude locking only if the corresponding tier of Corollary 56 is imposed on each exact side separately.

Accordingly, geometric statements use N^{eff} , spectral statements retain their root/operator and intrinsic-normalization hypotheses, and exact crystallographic statements require an exact sequence and its measured $\mathcal{L}_X$. Bounded Euclidean numerators may not be silently paired with unrestricted hyperbolic arithmetic, and hyperbolic locking may not be paired with an unrestricted Euclidean exact sequence.

Proof. The first line is the local geometric calculation. The remaining lines are Theorem 93. The tier restriction follows from Corollary 56, while the separation of geometric, spectral, and crystallographic claims is enforced by Theorem 96 and Proposition 55. $\square$

Rotation-class conditions for the higher-dimensional geometric exponent shift

The higher-dimensional registry set is generally not radial. The correct object is an angular-profile integral. We state the exact geometry first and then isolate the rotation classes for which finite-volume asymptotics hold.

Exact angular-profile geometry

Fix an integer $D \geq 2$, a curvature radius $R > 0$, and a displacement threshold $\ell_* > 0$. Let $d\Omega$ be standard surface measure on $\mathbb{S}^{D-1}$, and put

$$\Omega_{D-1} := \frac{2\pi^{D/2}}{\Gamma(D/2)}, \quad b_* := \sinh \frac{\ell_*}{2R}. \quad (2118)$$

The parameter θ and every generator A below are dimensionless. Volumes have units L^D ; a fixed density ν_X has units L^{-D} . For $y \geq 0$, define

$$F_D(y) := \int_0^y \sinh^{D-1} t \, dt. \quad (2119)$$

Definition 19 (Angular displacement profile). *For $Q \in SO(D)$ and $n \in \mathbb{S}^{D-1}$, set*

$$\begin{aligned} \vartheta_Q(n) &:= \arccos(n^T Q n), \\ s_Q(n) &:= \sin \frac{\vartheta_Q(n)}{2} = \sqrt{\frac{1 - n^T Q n}{2}}. \end{aligned} \quad (2120)$$

The value $s_Q(n) = 0$ means that the direction n is fixed by Q . All integrals below are extended nonnegative integrals at such directions.

Theorem 98 (Exact direction-dependent registry geometry). *Let two points have common radius r and directions n and Qn . Their distances in $\mathbb{H}_R^D$ and $\mathbb{R}^D$ obey*

$$\sinh \frac{d_H(r, n)}{2R} = \sinh \frac{r}{R} s_Q(n), \quad (2121)$$

$$d_E(r, n) = 2r s_Q(n). \quad (2122)$$

Consequently, the radial threshold functions are

$$\rho_{H,Q}(n) = R \operatorname{arsinh} \frac{b_*}{s_Q(n)}, \quad (2123)$$

$$\rho_{E,Q}(n) = \frac{\ell_*}{2s_Q(n)}, \quad (2124)$$

where either expression is $+\infty$ when $s_Q(n) = 0$. The exact registry volumes are

$$V_H(Q) = R^D \int_{\mathbb{S}^{D-1}} F_D \left(\operatorname{arsinh} \frac{b_*}{s_Q(n)} \right) d\Omega(n), \quad (2125)$$

$$V_E(Q) = \frac{1}{D} \left(\frac{\ell_*}{2} \right)^D \int_{\mathbb{S}^{D-1}} s_Q(n)^{-D} d\Omega(n). \quad (2126)$$

Thus these sets are radial balls only when s_Q is direction independent.

Proof. The hyperbolic cosine law in the totally geodesic two-plane containing the two radial geodesics gives

$$\begin{aligned} \cosh \frac{d_H}{R} &= \cosh^2 \frac{r}{R} - \sinh^2 \frac{r}{R} n^\top Q n \\ &= 1 + 2 \sinh^2 \frac{r}{R} s_Q(n)^2. \end{aligned} \quad (2127)$$

Using $\cosh u = 1 + 2 \sinh^2(u/2)$ proves Eq. (2121). Euclidean subtraction gives

$$|rn - rQn|^2 = 2r^2(1 - n^\top Qn) = 4r^2 s_Q(n)^2, \quad (2128)$$

which proves Eq. (2122). Solving the two threshold inequalities proves Eqs. (2123) and (2124). Finally, integrate $R^{D-1} \sinh^{D-1}(r/R) dr d\Omega$ in the hyperbolic case and $r^{D-1} dr d\Omega$ in the Euclidean case. $\square$

Proposition 56 (Sharp angular finiteness tests). *For fixed $Q \in SO(D)$,*

$$V_E(Q) < \infty \iff s_Q^{-D} \in L^1(\mathbb{S}^{D-1}), \quad (2129)$$

$$V_H(Q) < \infty \iff s_Q^{-(D-1)} \in L^1(\mathbb{S}^{D-1}). \quad (2130)$$

In particular, a uniform positive lower bound on s_Q is sufficient, but zero directions may not be hidden inside an amplitude constant.

Proof. The Euclidean equivalence is Eq. (2126). Put $\Phi_D(x) = F_D(\operatorname{arsinh} x)$. As $x \rightarrow \infty$,

$$\Phi_D(x) \sim \frac{x^{D-1}}{D-1}. \quad (2131)$$

Hence continuity on every interval away from zero gives constants $0 < c < C < \infty$, depending only on D and b_* , such that

$$c(1 + s^{-(D-1)}) \leq \Phi_D(b_*/s) \leq C(1 + s^{-(D-1)}) \quad (2132)$$

for $0 < s \leq 1$. The sphere has finite measure, so Eq. (2132) proves the hyperbolic equivalence. $\square$

Isoclinic rotations and the exact radial crossover

Proposition 57 (Generator characterization of the isoclinic class). *Let $A \neq 0$ be a real skew-symmetric matrix and $Q_\theta = \exp(\theta A)$. The following are equivalent.*

1. *There is a $a > 0$ such that*

$$A^\top A = a^2 I. \quad (2133)$$

2. *The function $s_{Q_\theta}(n)$ is independent of n for every sufficiently small θ .*

Under these conditions,

$$Q_\theta = \cos(a\theta)I + \frac{\sin(a\theta)}{a}A, \quad s_{Q_\theta}(n) = \left| \sin \frac{a\theta}{2} \right|. \quad (2134)$$

Proof. Since $A^\top = -A$, Eq. (2133) is equivalent to $A^2 = -a^2 I$. Separating the even and odd terms in the exponential proves the first identity in Eq. (2134). The quadratic form of A vanishes, so $n^\top Q_\theta n = \cos(a\theta)$, which proves direction independence and the second identity.

Conversely, expand Definition 19 at zero:

$$s_{Q_\theta}(n)^2 = \frac{\theta^2}{4} |An|^2 + O(\theta^4) \quad (2135)$$

uniformly on the sphere. Direction independence for an interval of parameters makes $|An|^2$ constant on the unit sphere. The quadratic form of $A^\top A$ is therefore $a^2 |n|^2$; polarization proves Eq. (2133). $\square$

Theorem 99 (Exact isoclinic ball and running exponent). *Assume Eq. (2133), set*

$$s_\theta := \left| \sin \frac{a\theta}{2} \right|, \quad y_\theta := \operatorname{arsinh} \frac{b_*}{s_\theta}, \quad (2136)$$

and suppose $s_\theta > 0$. The registry domains are balls, with

$$V_H(\theta) = \Omega_{D-1} R^D F_D(y_\theta), \quad (2137)$$

$$V_E(\theta) = \frac{\Omega_{D-1}}{D} \left(\frac{\ell_*}{2s_\theta} \right)^D. \quad (2138)$$

The exact running exponent with respect to the common half-angle profile is

$$\beta_D(y) := \frac{d \log V_H}{d \log s_\theta^{-1}} = \frac{\sinh^D y}{\cosh y \int_0^y \sinh^{D-1} t dt}. \quad (2139)$$

It satisfies

$$\lim_{y \rightarrow 0} \beta_D(y) = D, \quad \lim_{y \rightarrow \infty} \beta_D(y) = D - 1. \quad (2140)$$

For $0 < a|\theta| < \pi$, the exponent with respect to the physical parameter is instead

$$\beta_D^{(\theta)}(\theta) = \beta_D(y_\theta) \frac{a|\theta|}{2} \cot \frac{a|\theta|}{2}. \quad (2141)$$

Thus Eq. (2139) is exact in s_θ ; the physical-angle Jacobian may be omitted only asymptotically as $\theta \rightarrow 0$.

Proof. Proposition 57 makes both cutoffs in Theorem 98 direction independent, proving the ball formulas. Since $s_\theta = b_*/\sinh y$,

$$\frac{d \log V_H}{dy} = \frac{\sinh^{D-1} y}{F_D(y)}, \quad \frac{d \log s_\theta^{-1}}{dy} = \coth y. \quad (2142)$$

Their quotient proves Eq. (2139). The expansions $F_D(y) = y^D/D + O(y^{D+2})$ at zero and

$$F_D(y) = \frac{e^{(D-1)y}}{2^{D-1}(D-1)} [1 + o(1)] \quad (2143)$$

at infinity prove Eq. (2140). Finally,

$$\frac{d \log s_\theta}{d \log |\theta|} = \frac{a|\theta|}{2} \cot \frac{a|\theta|}{2}, \quad (2144)$$

and the chain rule gives Eq. (2141). $\square$

Proposition 58 (Quantitative radial asymptotic). *For $0 < u < 1$, define*

$$\varepsilon_D(u) := \begin{cases} u, & D = 2, \\ u^2(1 + |\log u|), & D = 3, \\ u^2, & D \geq 4. \end{cases} \quad (2145)$$

Then, as $x \rightarrow \infty$,

$$F_D(\operatorname{arsinh} x) = \frac{x^{D-1}}{D-1} [1 + O_D(\varepsilon_D(x^{-1}))]. \quad (2146)$$

Proof. For $D = 2$,

$$F_2(\operatorname{arsinh} x) = \sqrt{1+x^2} - 1 = x[1 + O(x^{-1})]. \quad (2147)$$

For $D = 3$, the identity

$$F_3(y) = \frac{1}{4} \sinh(2y) - \frac{y}{2} \quad (2148)$$

gives a relative error $O(x^{-2} \log x)$ after $y = \operatorname{arsinh} x$. For $D \geq 4$, on $t \geq 1$,

$$\sinh^{D-1} t = 2^{-(D-1)} e^{(D-1)t} [1 + O_D(e^{-2t})]. \quad (2149)$$

Integration gives a relative $O_D(e^{-2y})$ error; the bounded initial interval is smaller. Finally, $e^y = x + \sqrt{1+x^2} = 2x[1 + O(x^{-2})]$, proving the claim. $\square$

Theorem 100 (Full-rank anisotropic endpoint theorem). *Let A be an invertible real skew-symmetric D -by- D matrix and $Q_\theta = e^{\theta A}$. Then $D = 2m$ is even. There are rotation rates*

$$0 < a_- \leq a_+ < \infty \quad (2150)$$

such that, in orthogonal two-dimensional blocks,

$$s_{Q_\theta}(n)^2 = \sum_{k=1}^m |z_k|^2 \sin^2 \frac{a_k \theta}{2}, \quad |An|^2 = \sum_{k=1}^m a_k^2 |z_k|^2. \quad (2151)$$

Here $a_- \leq a_k \leq a_+$ and $\sum_k |z_k|^2 = 1$. If $|\theta|a_+ \leq 2$, then uniformly in n ,

$$1 - \frac{a_+^2 \theta^2}{24} \leq \frac{2s_{Q_\theta}(n)}{|\theta||An|} \leq 1. \quad (2152)$$

For $p > 0$, put

$$I_p(A) := \int_{\mathbb{S}^{D-1}} |An|^{-p} d\Omega(n). \quad (2153)$$

Then

$$\Omega_{D-1} a_+^{-p} \leq I_p(A) \leq \Omega_{D-1} a_-^{-p}. \quad (2154)$$

As $\theta \rightarrow 0$ through nonzero values,

$$V_E(\theta) = \mathcal{A}_E(A) |\theta|^{-D} [1 + O_A(\theta^2)], \quad (2155)$$

$$\mathcal{A}_E(A) := \frac{\ell_*^D}{D} I_D(A), \quad (2156)$$

$$V_H(\theta) = \mathcal{A}_H(A) |\theta|^{-(D-1)} [1 + O_{A,b_*,D}(\varepsilon_D(|\theta|))], \quad (2157)$$

$$\mathcal{A}_H(A) := \frac{2^{D-1} R^D b_*^{D-1}}{D-1} I_{D-1}(A). \quad (2158)$$

The amplitudes are explicitly positive and finite. More precisely,

$$\begin{aligned} \frac{\ell_*^D \Omega_{D-1}}{D a_+^D} &\leq \mathcal{A}_E(A) \leq \frac{\ell_*^D \Omega_{D-1}}{D a_-^D}, \\ \frac{2^{D-1} R^D b_*^{D-1} \Omega_{D-1}}{(D-1) a_+^{D-1}} &\leq \mathcal{A}_H(A) \leq \frac{2^{D-1} R^D b_*^{D-1} \Omega_{D-1}}{(D-1) a_-^{D-1}}. \end{aligned} \quad (2159)$$

Hence the Euclidean and hyperbolic effective-volume exponents are D and $D-1$, respectively. Isoclinicity is sufficient for the exact radial crossover, but it is not necessary for this endpoint reduction.

Proof. The orthogonal normal form of a real skew matrix consists of blocks $a_k J$, where

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \quad (2161)$$

Invertibility removes zero blocks and forces even dimension. Taking the quadratic form of each planar rotation proves Eq. (2151). For $|x| \leq 1$,

$$1 - \frac{x^2}{6} \leq \frac{\sin x}{x} \leq 1. \quad (2162)$$

The square of the middle ratio in Eq. (2152) is a convex combination of $\{\sin(a_k \theta/2)/(a_k \theta/2)\}^2$, with weights proportional to $a_k^2 |z_k|^2$. The displayed sine bound proves the uniform estimate. The singular-rate inequalities imply Eq. (2154).

Write

$$q_\theta(n) := \frac{2s_{Q_\theta}(n)}{|\theta||An|}. \quad (2163)$$

Equation (2152) gives $q_\theta = 1 + O_A(\theta^2)$ uniformly. Substitution in Eq. (2126) yields Eqs. (2155) and (2156).

Moreover, $s_{Q_\theta}(n)$ is bounded above and below by positive constant multiples of $|\theta|$, uniformly in n . Thus $b_*/s_{Q_\theta}(n) \rightarrow \infty$ uniformly. Proposition 58 can therefore be inserted under the angular integral in Eq. (2125). It gives

$$\begin{aligned} F_D\left(\operatorname{arsinh} \frac{b_*}{s_{Q_\theta}(n)}\right) &= \frac{b_*^{D-1}}{D-1} \left(\frac{2}{|\theta||An|}\right)^{D-1} \\ &\quad \times [1 + O_{A,b_*,D}(\varepsilon_D(|\theta|))], \end{aligned} \quad (2164)$$

uniformly in n . Integration proves Eqs. (2157) and (2158). The amplitude bounds follow from Eq. (2154). Taking logarithms proves the two endpoint exponents. $\square$

Theorem 101 (Fixed-axis divergence). *Let A be real and skew-symmetric with $\ker A \neq \{0\}$. For every $\theta \in \mathbb{R}$, the uncut displacement-threshold registry domains of $Q_\theta = e^{\theta A}$ have*

$$V_E(Q_\theta) = V_H(Q_\theta) = \infty. \quad (2165)$$

In particular, every real skew generator in odd spatial dimension has a fixed axis and lies outside the finite-volume $D \rightarrow D-1$ theorem.

Proof. If $A = 0$, both distances vanish identically and the conclusion is immediate. Otherwise write

$$\mathbb{R}^D = \ker A \oplus (\ker A)^\perp, \quad r_A := \text{rank } A \leq D-1. \quad (2166)$$

Every unit $v \in \ker A$ is fixed by Q_θ , so the entire radial ray through v lies in each registry domain. Near the fixed sphere, use coordinates

$$n = (\sqrt{1 - |u|^2} v, u), \quad v \in \mathbb{S}(\ker A), \quad u \in (\ker A)^\perp. \quad (2167)$$

The spherical measure is comparable to $du d\Omega(v)$ for small u , and the fixed component is unchanged by Q_θ . Hence

$$s_{Q_\theta}(n) \leq |u|. \quad (2168)$$

The Euclidean angular integral is bounded below by a positive multiple of

$$\int_0^\epsilon \rho^{r_A-1-D} d\rho = \infty. \quad (2169)$$

Equation (2132) gives a hyperbolic lower bound by

$$\int_0^\epsilon \rho^{r_A-D} d\rho = \infty, \quad (2170)$$

because $r_A \leq D-1$. This proves Eq. (2165). Finally,

$$\det A = \det A^\top = \det(-A) = (-1)^D \det A, \quad (2171)$$

so $\det A = 0$ when D is odd. $\square$

A physical radial cutoff, a compact quotient, an excluded angular cone, or a different registry definition can make a fixed-axis model finite. Its exponent then depends on that added cutoff or model and is not an unconditional consequence of constant negative curvature.

Effective counts, arithmetic, and subsequent repairs

Corollary 58 (Effective counts and symmetric arithmetic fields). *Let $\nu_E, \nu_H > 0$ be fixed number densities and define $N_X^{\text{eff}} = \nu_X V_X$. Under Theorem 100, the effective endpoint exponents are*

$$\beta_E^{\text{eff}} = D, \quad \beta_H^{\text{eff}} = D-1. \quad (2172)$$

On independently declared exact sequences with $0 < |\theta_{X,j}| \rightarrow 0$, retain the factorization

$$N_{X,j}^{\text{sc}} = \mathcal{L}_{X,j} N_{X,j}^{\text{eff}}, \quad X \in \{E, H\}. \quad (2173)$$

With $\lambda_{X,j} = \log |\theta_{X,j}|^{-1}$ and

$$\eta_{X,j}^{\text{ar}} := \frac{\log \mathcal{L}_{X,j}}{\lambda_{X,j}}, \quad (2174)$$

Define

$$\begin{aligned} \underline{\eta}_X^{\text{ar}} &:= \liminf_j \eta_{X,j}^{\text{ar}}, & \bar{\eta}_X^{\text{ar}} &:= \limsup_j \eta_{X,j}^{\text{ar}}, \\ \underline{\beta}_X^{\text{sc}} &:= \liminf_j \frac{\log N_{X,j}^{\text{sc}}}{\lambda_{X,j}}, & \bar{\beta}_X^{\text{sc}} &:= \limsup_j \frac{\log N_{X,j}^{\text{sc}}}{\lambda_{X,j}}. \end{aligned} \quad (2175)$$

The lower and upper exact exponents are

$$\underline{\beta}_E^{\text{sc}} = D + \underline{\eta}_E^{\text{ar}}, \quad \bar{\beta}_E^{\text{sc}} = D + \bar{\eta}_E^{\text{ar}}, \quad (2176)$$

$$\underline{\beta}_H^{\text{sc}} = D-1 + \underline{\eta}_H^{\text{ar}}, \quad \bar{\beta}_H^{\text{sc}} = D-1 + \bar{\eta}_H^{\text{ar}}. \quad (2177)$$

If both arithmetic limits exist and are finite, then

$$\beta_E^{\text{sc}} - \beta_H^{\text{sc}} = 1 + \eta_E^{\text{ar}} - \eta_H^{\text{ar}}. \quad (2178)$$

Exponent inheritance, two-sided Θ locking, and amplitude locking require, separately on each side, the corresponding subpower, bounded, and convergent positive-factor hypotheses. Geometry supplies none of them.

Proof. Multiplication by a fixed density changes amplitudes but not exponents. Equations (2155) and (2157) prove Eq. (2172). Taking logarithms in Eq. (2173) proves the lower and upper envelopes. Subtraction gives Eq. (2178). The three locking tiers follow by dividing the exact count by its effective counterpart. $\square$

Theorem 102 (Higher-dimensional repair ledger). *The checked higher-dimensional theorem families are repaired as follows.*

1. *The early arbitrary-dimensional statement states an exact radial crossover only for the isoclinic class. The endpoint $D \rightarrow D-1$ statement uses the broader full-rank generator class and its finite-volume hypothesis.*
2. *In the main higher-dimensional derivation, the scalar displacement, radial cutoff, and ball volume are replaced by Eqs. (2121)–(2126). The original running formula is Eq. (2139) only in the isoclinic class; physical-angle differentiation also uses Eq. (2141).*
3. *The mid-manuscript arithmetic repeat uses Eq. (2173). It does not infer an exact sequence or locking factor from the effective-volume law.*

4. The late higher-dimensional magic-cell comparison may use $|\theta^*|^{-D}$ and $|\theta^*|^{-(D-1)}$ only on a full-rank rotation family and only after an independently established richer-model root defines the magic branch. Frequency and dispersion-order relations retain their independent hypotheses.
5. Original Theorem 196 is a geometric effective-volume theorem. It combines the isoclinic exact crossover, the full-rank even-dimensional endpoint result, and the fixed-axis obstruction. It is not a spectral-root or exact-cell theorem.

Proof. Item 1 follows from Proposition 57, Theorem 100, and Theorem 101. Item 2 is the exact angular-profile calculation and its chain rule. Item 3 is Corollary 58. The geometric calculation contains no Hamiltonian, root functional, or dispersion symbol, which forces the independent hypotheses in item 4 and the layer restriction in item 5. $\square$

Corollary 59 (Geometric, spectral, and crystallographic separation theorem). *The exact distance and volume identities in Theorem 98 are $\mathbb{I}_G$ statements. The isoclinic crossover and full-rank endpoint rotation-class laws in Theorems 99 and 100 are $\mathbb{G}$ statements. They do not imply a magic root, spectral island, band flattening, gap, coherence, operator limit, or representation completeness. Such conclusions retain their fixed-fiber, operator, differentiability, isolation, and convergence hypotheses. Nor do the geometric laws identify discrete regular data with a smooth continuum family, promote local geometry to global spectral convergence, identify Euclidean and hyperbolic groups or crystallography, or alter the fixed metric-normalization convention.*

An exact crystallographic count additionally requires an exact sequence and the arithmetic factor in Eq. (2173); that factor is not absorbed into the intrinsic spectral normalization. The exact square five-state model remains root-free, and no rotation-class exponent creates a root. No conclusion here concerns anisotropic dispersion symbols, a joint physical feasibility region, or a reverse universality theorem.

Proof. The geometric statements depend only on the ambient metric, centered rotation, threshold, and fixed density. Spectral and crystallographic data do not occur in their proofs. Corollary 58 supplies the only supported exact-cell bridge, while the absence of a Hamiltonian or root equation excludes every spectral implication. $\square$

Coordinate-invariant anisotropic homogeneous kinetic symbols

The dispersion order determines a radial power but does not select a scalar kinetic coefficient. An anisotropic or matrix principal symbol requires a declared metric, fiber, and scalarization role. We keep those data explicit.

Scalar and Hermitian principal symbols

Fix $p > 0$, a d -dimensional real vector space V , and a positive inner product G . Put

$$\|q\|_G := \sqrt{G(q, q)}, \quad S_G := \{n \in V : \|n\|_G = 1\}. \quad (2179)$$

Let $d\Omega_G$ be the induced sphere measure and

$$\omega_{d-1} := \int_{S_G} d\Omega_G = \frac{2\pi^{d/2}}{\Gamma(d/2)}. \quad (2180)$$

The spectral tangent dimension d is not the spatial registry dimension of the higher-dimensional rotation theorem unless a model identifies them.

Definition 20 (Homogeneous scalar and matrix symbols). *A scalar order- p principal symbol is a continuous map*

$$P_p : V \longrightarrow \mathbb{R}, \quad P_p(\lambda q) = \lambda^p P_p(q) \quad (\lambda \geq 0). \quad (2181)$$

For a finite-dimensional complex Hilbert space $\mathcal{F}$, a Hermitian order- p principal symbol is a continuous map

$$\mathcal{P}_p : V \longrightarrow \text{Herm}(\mathcal{F}), \quad \mathcal{P}_p(\lambda q) = \lambda^p \mathcal{P}_p(q). \quad (2182)$$

Continuity and $p > 0$ imply $P_p(0) = 0$ and $\mathcal{P}_p(0) = 0$.

Proposition 59 (Coordinate and fiber covariance). *Let $S : V' \rightarrow V$ be invertible and $U : \mathcal{F} \rightarrow \mathcal{F}'$ unitary. In the new tangent and fiber frames, set*

$$\begin{aligned} G' &= S^* G S, \\ P'_p(q') &= P_p(Sq'), \\ \mathcal{P}'_p(q') &= U \mathcal{P}_p(Sq') U^*. \end{aligned} \quad (2183)$$

Then homogeneity, scalar values, matrix spectra, singular values, and all metric-sphere integrals below are unchanged. Directional data transform by $n' = S^{-1}n$.

Proof. The equality $G'(n', n') = G(Sn', Sn')$ maps $S_{G'}$ isometrically onto S_G , including its induced measure. Scalar composition preserves values, and unitary conjugation preserves Hermitian eigenvalues and singular values. Changing variables in each angular integral proves invariance. Applying Eq. (2183) to $\lambda q'$ proves that the degree is unchanged. $\square$

The metric and fiber transport are held fixed in every comparison. A coordinate change is not a rescaling or reconstruction of the physical metric.

Role ledger and positive denominators

Definition 21 (Scalar role ledger). *For $P_p^S = P_p|_{S_G}$, define*

$$\kappa_p^{\text{dir}}(n) := P_p(n),$$

$$\begin{aligned} m_p &:= \min_{S_G} P_p, \\ M_p &:= \max_{S_G} P_p, \end{aligned} \quad (2184)$$

$$\kappa_p^{\text{op}} := \max_{S_G} |P_p|,$$

$$\kappa_p^{\text{mean}} := \frac{1}{\omega_{d-1}} \int_{S_G} P_p(n) d\Omega_G(n), \quad (2185)$$

$$\kappa_p^{\text{rms}} := \left[\frac{1}{\omega_{d-1}} \int_{S_G} |P_p(n)|^2 d\Omega_G(n) \right]^{1/2}. \quad (2186)$$

The directional and mean roles are signed. The operator and RMS roles are magnitude summaries. The coercive role is

$$\kappa_p^{\text{coer}} := m_p, \quad m_p > 0. \quad (2187)$$

Let $\lambda_1(n) \leq \dots \leq \lambda_r(n)$ be the eigenvalues of $\mathcal{P}_p(n)$, where $r = \dim \mathcal{F}$.

Definition 22 (Hermitian role ledger). *The matrix magnitude and ellipticity roles are*

$$\kappa_p^{\text{mat,op}} := \max_{n \in S_G} \|\mathcal{P}_p(n)\|_{\text{op}}, \quad (2188)$$

$$\kappa_p^{\text{sing}} := \min_{n \in S_G} \sigma_{\min}(\mathcal{P}_p(n)). \quad (2189)$$

The full fiber is uniformly positive only if

$$\kappa_p^{\text{mat,coer}} := \min_{n \in S_G} \lambda_1(n) > 0. \quad (2190)$$

An isolated positive eigencluster $\mathcal{C}$ consists of a continuous spectral projector $\Pi_{\mathcal{C}}(n)$ of fixed rank $r_{\mathcal{C}}$, a uniform separation

$$g_{\mathcal{C}} := \inf_{n \in S_G} \text{dist}(\sigma_{\mathcal{C}}(n), \sigma(\mathcal{P}_p(n)) \setminus \sigma_{\mathcal{C}}(n)) > 0, \quad (2191)$$

and positive cluster bounds

$$0 < \mu_{\mathcal{C},-} := \min_{\substack{n \in S_G \\ j \in \mathcal{C}}} \lambda_j(n) \leq \mu_{\mathcal{C},+} := \max_{\substack{n \in S_G \\ j \in \mathcal{C}}} \lambda_j(n) < \infty. \quad (2192)$$

Its coercive, mean, and RMS roles are

$$\kappa_{p,\mathcal{C}}^{\text{coer}} := \mu_{\mathcal{C},-},$$

$$\kappa_{p,\mathcal{C}}^{\text{mean}} := \frac{1}{r_{\mathcal{C}} \omega_{d-1}} \int_{S_G} \text{Tr}[\Pi_{\mathcal{C}}(n) \mathcal{P}_p(n)] d\Omega_G(n), \quad (2193)$$

$$\kappa_{p,\mathcal{C}}^{\text{rms}} := \left[\frac{1}{r_{\mathcal{C}} \omega_{d-1}} \int_{S_G} \text{Tr}[\Pi_{\mathcal{C}}(n) \mathcal{P}_p(n)^2] d\Omega_G(n) \right]^{1/2}. \quad (2194)$$

Proposition 60 (Role meanings and positivity obstruction). *The scalar coercive role in Eq. (2187), the matrix role in Eq. (2190), and the positive cluster coercive role provide uniform positive kinetic denominators. No other role does so without an additional sign or magnitude convention.*

In particular, $\kappa_p^{\text{sing}} > 0$ proves invertibility of the matrix symbol on S_G , but permits positive and negative

eigenvalues. A positive operator norm or RMS does not turn a scalar saddle or a sign-indefinite matrix symbol into a positive band edge. Cluster means and RMS values summarize a positive cluster but do not replace its lower bound $\mu_{\mathcal{C},-} > 0$.

Proof. Homogeneity gives

$$P_p(q) \geq m_p \|q\|_G^p \quad (2195)$$

when $m_p > 0$. The min-max principle gives the analogous full-fiber and cluster lower bounds. Singular values and absolute norms discard sign, while an average or RMS can stay positive in the presence of a zero or negative direction. Therefore none supplies the omitted lower spectral bound. $\square$

A fixed role functional Φ below is coordinate invariant, unitary invariant when matrix valued, and positively degree-one in the angular symbol:

$$\Phi(c\mathcal{P}) = c\Phi(\mathcal{P}), \quad c > 0. \quad (2196)$$

Its role label is retained even when its numerical value is positive.

Exact kinetic scaling and normalization covariance

Theorem 103 (Role-labeled Q^p scale). *Let $\bar{Q}_M = 2\pi/\xi_M > 0$, and let Φ be a fixed role for which*

$$\kappa_{p,\Phi} := \Phi(\mathcal{P}_p|_{S_G}) \quad (2197)$$

is defined. Here $\mathcal{P}_p$ denotes either P_p or $\mathcal{P}_p$ with its declared cluster. The exact role scale is

$$\bar{E}_{M,p,\Phi} := \bar{Q}_M^p \kappa_{p,\Phi}. \quad (2198)$$

Equivalently,

$$\Phi(\mathcal{P}_p(\bar{Q}_M \cdot)|_{S_G}) = \bar{Q}_M^p \kappa_{p,\Phi}. \quad (2199)$$

This identity chooses no preferred role. It is a positive kinetic denominator only under Proposition 60. When q and $\bar{Q}_M$ have momentum units and the symbol has energy units, $\kappa_{p,\Phi}$ has units energy times momentum $^{-p}$. Thus $\bar{E}_{M,p,\Phi}$ and w_{Ψ} have energy units, and $\bar{\alpha}_{\Psi|\Phi}$ is dimensionless.

Proof. Equations (2181) and (2182) multiply the angular symbol by $\bar{Q}_M^p$. Equation (2196) pulls out that factor. Coordinate and fiber invariance follow from Proposition 59. $\square$

Theorem 104 (Order- p convention covariance). *For a fixed admissible role and a fixed numerator $w_{\Psi} > 0$, define*

$$\bar{\alpha}_{\Psi|\Phi} := \frac{w_{\Psi}}{\bar{E}_{M,p,\Phi}}. \quad (2200)$$

Under $Q_c = c\bar{Q}_M$, with one fixed $c > 0$,

$$\begin{aligned} E_{M,p,\Phi,c} &= c^p \bar{E}_{M,p,\Phi}, \\ \alpha_{\Psi|\Phi,c} &= c^{-p} \bar{\alpha}_{\Psi|\Phi}. \end{aligned} \quad (2201)$$

If an actual root is specified in the same role pair, then

$$\alpha_{\Psi|\Phi,c}^* = c^{-p} \bar{\alpha}_{\Psi|\Phi}^*, \quad X_{\Psi|\Phi,c} := \frac{\alpha_{\Psi|\Phi,c}}{\alpha_{\Psi|\Phi,c}^*} = \bar{X}_{\Psi|\Phi}. \quad (2202)$$

A parameter-dependent c contributes its own logarithmic derivative and is not a fixed convention change.

Proof. Theorem 103 gives the energy law. Division of the same physical numerator gives the coupling law. Applying the same convention change at the same physical root gives the root law and cancels c^{-p} from the ratio. Nothing in this argument asserts that the root exists. $\square$

Corollary 60 (Recovery of the quadratic Hessian theory). At $p = 2$, let

$$P_2(q) = \frac{1}{2} H_0(q, q), \quad A_G = G^{-1} H_0, \quad (2203)$$

with generalized eigenvalues $\lambda_1 \leq \dots \leq \lambda_d$. Then

$$\begin{aligned} \kappa_2^{\text{dir}}(n) &= \frac{1}{2} H_0(n, n), & \kappa_2^{\text{coer}} &= \frac{\lambda_1}{2}, \\ \kappa_2^{\text{op}} &= \frac{1}{2} \max_a |\lambda_a|, & \kappa_2^{\text{mean}} &= \frac{\text{tr } A_G}{2d}. \end{aligned} \quad (2204)$$

The eigenvalue-RMS role used in the quadratic tensor theory is

$$\kappa_2^{\text{T10,rms}} := \frac{1}{2} \left(\frac{1}{d} \sum_{a=1}^d \lambda_a^2 \right)^{1/2}. \quad (2205)$$

It is a distinct magnitude role from the angular RMS in Eq. (2186). For $H_0 = 2ta^2 G$,

$$\bar{E}_{M,2} = ta^2 \bar{Q}_M^2 \quad (2206)$$

for every directional or principal role.

Proof. The Rayleigh quotient of A_G gives the directional extrema. The metric sphere average of $H_0(n, n)$ is $\text{tr } A_G/d$, which proves Eq. (2204). Equation (2205) is the quadratic principal-spectrum scalarization. Substituting $\lambda_a = 2ta^2$ proves Eq. (2206), including the Taylor factor $1/2$. $\square$

Anisotropic scalar and matrix density of states

Fix a phase-space normalization $\mathfrak{c}_{\text{ph}} > 0$. For standard momentum normalization one may take $(2\pi)^{-d}$.

Theorem 105 (Sharp scalar homogeneous DOS law). Assume $P_p \geq 0$, $P_p(n) > 0$ for almost every $n \in S_G$, and

$$I_{p,d}[P] := \int_{S_G} P_p(n)^{-d/p} d\Omega_G(n) < \infty. \quad (2207)$$

For $\epsilon > 0$, define the local integrated count

$$\mathcal{N}_P(\epsilon) := \mathfrak{c}_{\text{ph}} \int_V \mathbf{1}_{\{0 < P_p(q) \leq \epsilon\}} dq_G. \quad (2208)$$

Then

$$\mathcal{N}_P(\epsilon) = \frac{\mathfrak{c}_{\text{ph}}}{d} I_{p,d}[P] \epsilon^{d/p}, \quad (2209)$$

$$\rho_P(\epsilon) := \frac{d\mathcal{N}_P}{d\epsilon} = \frac{\mathfrak{c}_{\text{ph}}}{p} I_{p,d}[P] \epsilon^{d/p-1}. \quad (2210)$$

The moment condition is sharp. Under coercivity $0 < m_p \leq P_p(n) \leq M_p$,

$$\omega_{d-1} M_p^{-d/p} \leq I_{p,d}[P] \leq \omega_{d-1} m_p^{-d/p}. \quad (2211)$$

A zero direction destroys coercivity. It yields a finite DOS coefficient only when the negative moment in Eq. (2207) remains integrable; a positive-measure zero set makes it infinite.

Proof. Metric polar coordinates give $dq_G = r^{d-1} dr d\Omega_G$. For a direction with $P_p(n) > 0$, the indicator in Eq. (2208) restricts

$$0 < r \leq \left(\frac{\epsilon}{P_p(n)} \right)^{1/p}. \quad (2212)$$

Integrating r^{d-1} proves Eq. (2209); differentiation proves Eq. (2210). The integral is finite exactly under Eq. (2207). The pointwise bounds on $P_p^{-d/p}$ prove Eq. (2211). $\square$

A sign-changing scalar symbol describes a saddle rather than a positive edge. Its operator norm or RMS does not put it under Theorem 105.

Theorem 106 (Finite-fiber eigenchannel DOS). Let $\mathcal{C}$ be a declared positive eigencluster, not necessarily simple internally, and suppose

$$I_{p,d}[\mathcal{C}] := \int_{S_G} \sum_{j \in \mathcal{C}} \lambda_j(n)^{-d/p} d\Omega_G(n) < \infty. \quad (2213)$$

Then its multiplicity-counted local DOS is

$$\rho_{\mathcal{C}}(\epsilon) = \frac{\mathfrak{c}_{\text{ph}}}{p} I_{p,d}[\mathcal{C}] \epsilon^{d/p-1}, \quad \epsilon > 0. \quad (2214)$$

If Eq. (2192) holds, then

$$r_{\mathcal{C}} \omega_{d-1} \mu_{\mathcal{C},+}^{-d/p} \leq I_{p,d}[\mathcal{C}] \leq r_{\mathcal{C}} \omega_{d-1} \mu_{\mathcal{C},-}^{-d/p}. \quad (2215)$$

Without a fixed cluster gap, the unlabeled total positive trace

$$\sum_{\lambda_j(n)>0} \lambda_j(n)^{-d/p} \quad (2216)$$

may still be integrated when finite; it is symmetric through crossings among positive eigenvalues. A named channel requires a fixed isolated projector or a separate simple-eigenvalue gap. A zero crossing changes positive multiplicity and is outside the fixed positive-cluster state-ment.

Proof. Diagonalize the finite Hermitian matrix at each n . Homogeneity scales every eigenvalue by r^p . Applying the scalar polar calculation to each positive eigenvalue and summing proves Eq. (2214). The cluster bounds give Eq. (2215). Symmetric sums are invariant under permutations at crossings, whereas a labeled eigenvector or projector is not continued through a collision with the complement without a gap. $\square$

Effective coupling and running fields

Theorem 107 (Quantitative Q^{-p} coupling bounds). *Let $w_\Psi > 0$ be a fixed, role-labeled numerator, and let $\kappa_{p,\Phi} > 0$ be an admissible denominator. If on a parameter set*

$$0 < w_- \leq w_\Psi \leq w_+, \quad 0 < \kappa_- \leq \kappa_{p,\Phi} \leq \kappa_+, \quad (2217)$$

then

$$\frac{w_-}{\kappa_+ \bar{Q}_M^p} \leq \alpha_{\Psi|\Phi} \leq \frac{w_+}{\kappa_- \bar{Q}_M^p}. \quad (2218)$$

For a scalar directional denominator one may take $(\kappa_-, \kappa_+) = (m_p, M_p)$. For a positive matrix cluster one may take $(\mu_{C,-}, \mu_{C,+})$. If only singular-value ellipticity is known, Eq. (2218) defines a magnitude coupling, not an undeclared positive band-energy coupling.

Proof. Insert Eq. (2217) into $\alpha_{\Psi|\Phi} = w_\Psi / (\bar{Q}_M^p \kappa_{p,\Phi})$. The role interpretations follow from Proposition 60. $\square$

Theorem 108 (Exact numerator-length-shape composition). *Let $s > 0$ parametrize a differentiable interval on which the metric, fiber transport, numerator role Ψ , kinetic role Φ , and cluster rank are fixed. Define*

$$\begin{aligned} \nu_\xi &:= -\partial_{\ln s} \ln \xi_M, \\ \eta_w &:= -\partial_{\ln s} \ln w_\Psi, \\ \eta_{\text{sh}} &:= \partial_{\ln s} \ln \kappa_{p,\Phi}. \end{aligned} \quad (2219)$$

Where these derivatives exist,

$$\beta_\alpha := -\partial_{\ln s} \ln \alpha_{\Psi|\Phi} = p\nu_\xi + \eta_w + \eta_{\text{sh}}. \quad (2220)$$

For $w_\Psi = w_0 \gamma_\Psi$ with fixed w_0 , the numerator term is $-\partial_{\ln s} \ln \gamma_\Psi$. At norm or singular-value crossings, one may state one-sided Dini bounds, but not an undeclared classical derivative. A role change, gap closure, rank change, or metric change ends the interval of validity.

Proof. From $\bar{Q}_M = 2\pi/\xi_M$, $\partial_{\ln s} \ln \bar{Q}_M = \nu_\xi$. Differentiate

$$\ln \alpha_{\Psi|\Phi} = \ln w_\Psi - p \ln \bar{Q}_M - \ln \kappa_{p,\Phi}. \quad (2221)$$

Multiplication by -1 gives Eq. (2220). The crossing restrictions are the differentiability and fixed-role hypotheses used in that calculation. $\square$

Analytic checks and obstructions

Proposition 61 (Coercive anisotropic power polynomial). *Choose a G -orthonormal frame and let*

$$P_p(q) = \sum_{a=1}^d c_a |q^a|^p, \quad c_a > 0. \quad (2222)$$

This is continuous, homogeneous, and coercive for every $p > 0$; it is a polynomial when p is an even integer. Its angular moment is

$$\int_{S_G} P_p(n)^{-d/p} d\Omega_G(n) = \frac{d 2^d \Gamma(1 + 1/p)^d}{\Gamma(1 + d/p)} \prod_{a=1}^d c_a^{-1/p}. \quad (2223)$$

Consequently its DOS coefficient is finite and explicitly anisotropic. At $p = 2$, Eq. (2223) is the ellipsoidal quadratic coefficient.

Proof. The positive continuous function $P_p|_{S_G}$ has a positive minimum. Metric polar coordinates give

$$\text{Vol}\{q : P_p(q) \leq 1\} = \frac{1}{d} \int_{S_G} P_p(n)^{-d/p} d\Omega_G(n). \quad (2224)$$

The change $x_a = c_a^{1/p} q^a$ reduces the star body to the unit ℓ^p ball, whose volume is

$$\frac{2^d \Gamma(1 + 1/p)^d}{\Gamma(1 + d/p)}. \quad (2225)$$

Substitution in Eq. (2224) proves the result. $\square$

Proposition 62 (Anisotropic Dirac check). *Let $B : V \rightarrow \mathbb{R}^d$ be invertible and let Hermitian matrices Γ_a obey*

$$\Gamma_a \Gamma_b + \Gamma_b \Gamma_a = 2\delta_{ab} I. \quad (2226)$$

Assume $d \geq 2$, and put $r_+ := \frac{1}{2} \dim \mathcal{F}$. The Dirac symbol

$$\mathcal{D}_1(q) := \sum_{a=1}^d (Bq)_a \Gamma_a \quad (2227)$$

satisfies

$$\mathcal{D}_1(q)^2 = |Bq|^2 I. \quad (2228)$$

It is singular-value elliptic, but sign indefinite. Its positive projector and sphere gap are

$$\Pi_+(n) = \frac{1}{2} \left(I + \frac{\mathcal{D}_1(n)}{|Bn|} \right), \quad g_+ \geq 2 \min_{S_G} |Bn| > 0. \quad (2229)$$

Writing $|\det_G B|$ for the determinant in a G -orthonormal frame,

$$\int_{S_G} |Bn|^{-d} d\Omega_G(n) = \frac{\omega_{d-1}}{|\det_G B|}. \quad (2230)$$

Hence the positive-branch DOS has exponent $d-1$ and coefficient $\mathfrak{c}_{\text{ph}} r_+ \omega_{d-1} / |\det_G B|$. The operator and singular roles are magnitude scales; neither is an undeclared positive energy for the full sign-indefinite fiber.

Proof. The Clifford relation cancels all cross terms in $\mathcal{D}_1(q)^2$. For $y = Bq \neq 0$, choose a unit vector $z \perp y$. The Hermitian unitary $\Gamma(z) := \sum_a z_a \Gamma_a$ anticommutes with $\mathcal{D}_1(q)$, so it pairs the eigenvalues $+|Bq|$ and $-|Bq|$. Thus both have multiplicity $r_+ = \frac{1}{2} \dim \mathcal{F}$. Invertibility of B gives the uniform gap and Eq. (2229). The star body $\{q : |Bq| \leq 1\}$ has volume $\omega_{d-1} / (d |\det_G B|)$. Comparing with Eq. (2224) proves Eq. (2230); Theorem 106 gives the DOS. $\square$

The mixed expression $q_1^2 + q_2^4$ is not homogeneous under ordinary scalar dilation. It lies outside Definition 20 unless a weighted dilation, its homogeneous dimension, and its measure Jacobian are declared separately.

Conditional magic scaling and cancellation

Theorem 109 (Conditional anisotropic order- p magic laws). *Let $\omega \downarrow 0$. Fix one positive numerator-kinetic role pair and suppose*

$$\begin{aligned} w_\Psi(\omega) &= w_0 \omega^\sigma [1 + o(1)], \quad w_0 > 0, \quad \sigma > 0, \\ \kappa_{p,\Phi}(\omega) &\longrightarrow \kappa_0 \in (0, \infty), \\ \alpha_{\Psi|\Phi}^*(\omega) &\longrightarrow \alpha_0^* \in (0, \infty). \end{aligned} \quad (2231)$$

The last line is an independently established richer-model root. On the corresponding branch,

$$\overline{Q}_M^* = \left(\frac{w_\Psi}{\alpha_{\Psi|\Phi}^* \kappa_{p,\Phi}} \right)^{1/p}, \quad (2232)$$

$$\xi_M^* = 2\pi \left(\frac{\alpha_{\Psi|\Phi}^* \kappa_{p,\Phi}}{w_\Psi} \right)^{1/p} = C_\xi \omega^{-\sigma/p} [1 + o(1)], \quad (2233)$$

$$C_\xi := 2\pi \left(\frac{\alpha_0^* \kappa_0}{w_0} \right)^{1/p}. \quad (2234)$$

If the Euclidean momentum law is

$$\overline{Q}_{M,E}(\theta) = q_E |\theta| [1 + o(1)], \quad q_E > 0, \quad (2235)$$

then

$$|\theta_E^*| = \frac{1}{q_E} \left(\frac{w_0}{\alpha_0^* \kappa_0} \right)^{1/p} \omega^{\sigma/p} [1 + o(1)]. \quad (2236)$$

If at fixed negative curvature

$$\log |\theta_H|^{-1} = \frac{\xi_M}{R} + O(1), \quad (2237)$$

then

$$\log |\theta_H^*|^{-1} = \frac{C_\xi}{R} \omega^{-\sigma/p} [1 + o(1)], \quad (2238)$$

$$|\theta_H^*| = \exp \left[-\frac{C_\xi}{R} \omega^{-\sigma/p} [1 + o(1)] \right]. \quad (2239)$$

The exact curvature crossover coordinate is

$$g^* := |K| (\xi_M^*)^2 = 4\pi^2 |K| \left(\frac{\alpha_{\Psi|\Phi}^* \kappa_{p,\Phi}}{w_\Psi} \right)^{2/p}. \quad (2240)$$

It reduces to a fixed multiple of $a^2 |K| \omega^{-2/p}$ only when $\sigma = 1$ and the normalized numerator, symbol role, root, and microscopic length convention have fixed limits.

Proof. The branch condition $\alpha_{\Psi|\Phi} = \alpha_{\Psi|\Phi}^*$ and Eq. (2200) give Eq. (2232). Inversion of $\overline{Q}_M = 2\pi/\xi_M$ and Eq. (2231) prove Eqs. (2233) and (2234). Substitution into the two declared geometric momentum laws proves the Euclidean and hyperbolic angle formulas. Squaring the exact length gives Eq. (2240). $\square$

Theorem 110 (Common-length p -cancellation and failure modes). *Let u parametrize a differentiable magic branch on which one fixed role, metric, and transport are used. Suppose its effective count and half-angle are functions of the same intrinsic length:*

$$N_{\text{eff}}^*(u) = \mathcal{N}_G(\xi_M^*(u)), \quad s^*(u) = \mathcal{S}_G(\xi_M^*(u)). \quad (2241)$$

If all displayed logarithmic derivatives exist and

$$\frac{d \ln \xi_M^*}{d \ln u} \neq 0, \quad \frac{d \ln \mathcal{S}_G}{d \ln \xi} \neq 0, \quad (2242)$$

then

$$-\frac{d \ln N_{\text{eff}}^*}{d \ln s^*} = -\frac{d \ln \mathcal{N}_G / d \ln \xi}{d \ln \mathcal{S}_G / d \ln \xi}. \quad (2243)$$

Thus p , numerator scaling, angular-symbol drift, and root drift cancel from this direct effective-count-versus-angle derivative because they share one branch factor.

They do not cancel from the separate laws versus u . Equation (2243) remains a chain-rule identity whenever Eqs. (2241) and (2242) hold. If the kinetic and registry lengths differ or either required derivative vanishes, the displayed quotient is unavailable. A role or metric change, a cluster gap closure, or a rank change does not by itself falsify a separately re-established geometric chain

rule; it terminates the fixed-role spectral derivation and may destroy branch differentiability.

For an exact crystallographic sequence, interpret every logarithmic derivative below as a limit of matched two-point logarithmic difference quotients along that sequence; alternatively, assume a differentiable exact family. When those derivatives exist,

$$N_{\text{sc}}^* = \mathcal{L}_X^* N_{\text{eff}}^*, \quad -\frac{d \ln N_{\text{sc}}^*}{d \ln s^*} = -\frac{d \ln N_{\text{eff}}^*}{d \ln s^*} - \frac{d \ln \mathcal{L}_X^*}{d \ln s^*}. \quad (2244)$$

The arithmetic term is independent of dispersion order.

Proof. Apply the chain rule to both expressions in Eq. (2241). Division cancels $d \ln \xi_M^*/d \ln u$, proving Eq. (2243). The nonzero hypotheses condition the division. The stated role, metric, gap, and rank changes affect its fixed-branch spectral interpretation; once a new common differentiable length branch is established, the chain rule may be applied again. Taking logarithms of $N_{\text{sc}}^* = \mathcal{L}_X^* N_{\text{eff}}^*$ proves Eq. (2244). $\square$

For the admissible full-rank rotation class in spatial dimension D_g , the effective geometric endpoint exponents are D_g and $D_g - 1$. They are independent of the DOS spectral dimension d and of the symbol order p .

Subsequent repairs and firewalls

Theorem 111 (Order- p theorem-family repair ledger). *The checked statements are replaced as follows.*

1. Theorems 86–87 replace the radial model Ar^p by Theorems 105 and 106. The exponent $d/p - 1$ survives fixed anisotropy, while its coefficient and finiteness are the angular negative moment. Named matrix channels retain their gap condition.
2. Theorem 114 uses Eq. (2236) with $\sigma = 1$, a fixed positive role and shape, the Euclidean momentum law, and a finite independently established root.
3. The quadratic mass-cancellation statements distinguish a perturbative coefficient zero from an operational root. Anisotropic symbol shape is additional dimensionless data; the square five-state cancellation scale is not an exact root.
4. Theorem 155 replaces every scalar order- p denominator by $\bar{Q}_{M\kappa_{p,\Phi}}^p$. Its quadratic numerical denominator is only the square specialization in Eq. (2206).
5. Theorems 181 and 195 use Theorem 110. Their direct effective-count versus angle relation is geometric only under the common-length and fixed-role hypotheses. Literal exact-cell claims add Eq. (2244).
6. Theorems 192–193 use Theorem 109. The crossover field is Eq. (2240); numerator, symbol-shape, and root fields may not be suppressed.

7. Theorem 199 and related effective-coupling statements replace $\beta_X = p\nu_\xi$ by Eq. (2220). Numerator and denominator roles remain separately labeled.

Proof. Items 1 and 4 are the exact homogeneity and polar-integration results above. Items 2 and 6 are conditional consequences of a specified root and the two declared length-angle laws. Item 3 is the positivity and root distinction in Proposition 60. Item 5 is the common-factor chain rule plus the exact arithmetic factorization. Item 7 is Theorem 108. $\square$

Corollary 61 (Role, root, bulk, arithmetic, and metric separation theorem). *The exact homogeneity and DOS calculations are local $\mathbb{I}_S$ statements. They do not certify a global band, a DOS for an infinite reduced bulk, an isolated spectral island, a gap, coherence, finite-cover convergence, or representation completeness. Those conclusions retain their operator, summability, no-loss, no-pollution, representation, and projector hypotheses.*

Numerator operator, RMS, and singular roles remain distinct from kinetic coercive, operator, RMS, singular, and cluster roles. Coordinate invariance holds with the fixed metric and fiber transport; changing their normalization is a separate convention change. Exact crystallographic counts retain their Euclidean and hyperbolic arithmetic factors. The spatial rotation class and dimension of the registry theorem remain independent of d and p .

The exact square five-state target has no magic root. Homogeneous scaling, anisotropy, a role choice, or a normalization change cannot create one. Every magic law above is conditional on an independently established richer-model root and all of its gap, rank, coherence, angle, and operator admissibility conditions. No result here constructs a joint physical parameter region or a reverse universality theorem.

Proof. Definitions 20–22 contain only local symbol and role data. The global spectral and crystallographic objects listed in the first two paragraphs are absent from their hypotheses and therefore cannot follow. The root statement is an independent obstruction, preserved explicitly in Eq. (2231). $\square$

A nonempty joint physical admissibility window

Separate decay, root, gap, and coherence hypotheses do not prove that one Hamiltonian satisfies them at one point. We first give a reusable common budget and then construct one analytic bilayer for which a rational four-dimensional box lies inside the joint admissible set.

Definition 23 (Dimensionless physical variables). *Fix a lattice length $a > 0$, an intralayer energy $t > 0$, and put*

$$p = (\omega, r, H, \Lambda) := \left(\frac{w}{t}, \frac{a}{R}, \frac{h}{a}, \frac{\lambda_\perp}{a} \right) \in (0, \infty)^4. \quad (2245)$$

The full-distance exponential interlayer kernel is

$$\begin{aligned} T_X(d; p) &= t\omega \exp \left[-\frac{\sqrt{h^2 + d^2} - h}{\lambda_\perp} \right] \\ &= t\omega \Phi_X(d/a; H, \Lambda), \end{aligned} \quad (2246)$$

$$\Phi_X(u; H, \Lambda) = \exp \left[-\frac{\sqrt{H^2 + u^2} - H}{\Lambda} \right]. \quad (2247)$$

Intralayer hopping in the explicit witness below is finite range.

Proposition 63 (Exact summability equivalence and tail). *Define the dimensionless Schur and rate margins*

$$\delta_S(p) := 1 - \Lambda r, \quad \delta_X(p) := \Lambda^{-1} - r = \frac{\delta_S(p)}{\Lambda}. \quad (2248)$$

Then

$$\lambda_\perp < R \iff \mu_\perp := \lambda_\perp^{-1} > R^{-1} \iff \Lambda r < 1. \quad (2249)$$

Moreover,

$$|T_X(d; p)| \leq t\omega e^{H/\Lambda} e^{-d/(a\Lambda)}. \quad (2250)$$

If dimensionless shells have width one and count constant C_X , then, for every $L \geq 0$ and $\delta_X > 0$, the omitted interlayer operator obeys

$$\varepsilon_\perp(L; p) \leq \frac{t\omega C_X e^{H/\Lambda + \delta_X}}{1 - e^{-\delta_X}} e^{-\delta_X L/a}. \quad (2251)$$

Proof. The equivalences follow by substituting $\lambda_\perp = a\Lambda$ and $R = a/r$. Since $\sqrt{h^2 + d^2} \geq d$,

$$-\frac{\sqrt{h^2 + d^2} - h}{\lambda_\perp} \leq \frac{H}{\Lambda} - \frac{d}{a\Lambda},$$

which proves Eq. (2250). Hyperbolic shell growth in units of a contributes e^{rn} , whereas the kernel contributes $e^{-n/\Lambda}$. Their ratio is $e^{-\delta_X}$. The arbitrary-cutoff geometric-series estimate, including its floor correction e^{δ_X} , proves Eq. (2251). $\square$

One quantitative simultaneous-continuation budget

Definition 24 (Specified reference package). *Let $\mathcal{D} \subset (0, \infty)^4$ be open and let $p_0 \in \mathcal{D}$. Fix a transported Hilbert fiber, one specified target projector, of fixed finite rank on each finite fiber whenever fiber-rank language is used,*

one metric and kinetic-role convention, and one positive operational score $\mathcal{M}(\theta, p)$. Assume $H(\theta, p)$ is bounded self-adjoint on that fiber, or has the form $D + V(\theta, p)$, self-adjoint on one common Dom D , with e_H controlling the bounded perturbation $V(\theta, p) - V(\theta, p_0)$. Put

$$F(\theta, p) := \partial_\theta \log \mathcal{M}(\theta, p). \quad (2252)$$

Choose $0 < \theta_0 < \theta_{\max}$ and

$$\begin{aligned} I &= [\theta_0 - h_\theta, \theta_0 + h_\theta], \\ 0 < h_\theta &< d_{\theta,0} := \min\{\theta_0, \theta_{\max} - \theta_0\}. \end{aligned} \quad (2253)$$

Assume

$$F(\theta_0, p_0) = 0, \quad \partial_\theta F(\theta, p_0) \leq -m_F < 0 \quad (\theta \in I). \quad (2254)$$

On I , require for the same target a global isolation gap $\Delta_0 > 0$, a common Riesz contour Γ of length ℓ_Γ and spectral distance $\delta_\Gamma > 0$, a specified active-overlap margin $A_0 > 0$, coherence $C_0 > 0$, and a positive role coefficient $\kappa_0 > 0$. Finally, require

$$\delta_{S,0} := 1 - \Lambda_0 r_0 > 0. \quad (2255)$$

For $j \in \{\omega, r, H, \Lambda\}$, let the following nonnegative constants be displayed on the chosen chart:

$$L_{F,j}, \quad L_{F_1,j}, \quad L_{H,j}, \quad L_{A,j}, \quad L_{C,j}, \quad L_{\kappa,j}.$$

Let $\tau_F, \tau_{F_1}, \tau_H, \tau_A, \tau_C, \tau_\kappa$ be complete tail, cover, angular, transport, and numerical errors in the topology indicated by their subscripts. Define

$$\begin{aligned} e_F(p) &:= \sum_j L_{F,j} |p_j - p_{0,j}| + \tau_F, \\ e_{F_1}(p) &:= \sum_j L_{F_1,j} |p_j - p_{0,j}| + \tau_{F_1}, \\ e_H(p) &:= \sum_j L_{H,j} |p_j - p_{0,j}| + \tau_H. \end{aligned} \quad (2256)$$

Whenever $e_H < \delta_\Gamma$, put

$$\mathcal{R}_\Gamma(e_H) := \frac{\ell_\Gamma}{2\pi} \frac{e_H}{\delta_\Gamma(\delta_\Gamma - e_H)}. \quad (2257)$$

For $e_H \geq \delta_\Gamma$, set $\mathcal{R}_\Gamma(e_H) := +\infty$. For fixed diagnostic Lipschitz constants $c_A, c_C \geq 0$, set

$$\begin{aligned} e_A(p) &:= \sum_j L_{A,j} |p_j - p_{0,j}| + \tau_A + c_A \mathcal{R}_\Gamma(e_H), \\ e_C(p) &:= \sum_j L_{C,j} |p_j - p_{0,j}| + \tau_C + c_C \mathcal{R}_\Gamma(e_H), \\ e_\kappa(p) &:= \sum_j L_{\kappa,j} |p_j - p_{0,j}| + \tau_\kappa. \end{aligned} \quad (2258)$$

After the fixed transport, assume these quantities dominate the actual variations on I :

$$\sup_{\theta \in I} |F(\theta, p) - F(\theta, p_0)| \leq e_F,$$

$$\begin{aligned} \sup_{\theta \in I} |\partial_\theta F(\theta, p) - \partial_\theta F(\theta, p_0)| &\leq e_{F_1}, \\ \sup_{\theta \in I} \|H(\theta, p) - H(\theta, p_0)\| &\leq e_H. \end{aligned} \quad (2259)$$

The reference diagnostics obey $A(\theta, p_0) \geq A_0$, $C(\theta, p_0) \geq C_0$, and $\kappa(\theta, p_0) \geq \kappa_0$ on I , and

$$\begin{aligned} \sup_{\theta \in I} |A(\theta, p) - A(\theta, p_0)| &\leq e_A, \\ \sup_{\theta \in I} |C(\theta, p) - C(\theta, p_0)| &\leq e_C, \\ \sup_{\theta \in I} |\kappa(\theta, p) - \kappa(\theta, p_0)| &\leq e_\kappa. \end{aligned} \quad (2260)$$

Definition 25 (Combined admissibility budget). *The single dimensionless budget is*

$$\mathfrak{B}(p) := \max \left\{ \frac{e_F}{m_F h_\theta}, \frac{2e_{F_1}}{m_F}, \frac{4e_H}{\Delta_0}, \frac{e_H}{\delta_\Gamma}, \mathcal{R}_\Gamma(e_H), \frac{2e_A}{A_0}, \frac{2e_C}{C_0}, \frac{2e_\kappa}{\kappa_0}, \frac{2|\Lambda r - \Lambda_0 r_0|}{\delta_{S,0}} \right\}. \quad (2261)$$

The auxiliary condition $\mathfrak{G}_{\text{aux}}$ means:

1. every infinite-volume kernel is finite range or has a uniform positive exponential margin;
2. τ_H uses the base operator tail, while τ_F, τ_{F_1} use the differentiated tail tier required by the fixed score;
3. a finite-cover claim has an injectivity buffer and an independently established global norm or no-pollution comparison;
4. target transport, representation sector, scalarization, metric, and kinetic role are fixed.

Define the common-budget set

$$\mathcal{P}_{\text{cb}} := \{p \in \mathcal{D} : \Lambda r < 1, \quad \mathfrak{B}(p) < 1, \quad \mathfrak{G}_{\text{aux}} \text{ holds}\}. \quad (2262)$$

Theorem 112 (Simultaneous continuation on one physical budget). *Under Definitions 24 and 25, every $p \in \mathcal{P}_{\text{cb}}$ has exactly one operational root $\theta_\star(p) \in I$. It is simple and is the strict maximum of the fixed score on I . Quantitatively,*

$$|\theta_\star - \theta_0| \leq \frac{e_F}{m_F}, \quad (2263)$$

$$\partial_\theta F(\theta, p) \leq -\frac{m_F}{2} \quad (\theta \in I), \quad (2264)$$

$$\min\{\theta_\star, \theta_{\text{max}} - \theta_\star\} \geq d_{\theta,0} - \frac{e_F}{m_F} > 0. \quad (2265)$$

For the same continued target,

$$\Delta(\theta, p) \geq \Delta_0 - 2e_H > \frac{\Delta_0}{2}, \quad (2266)$$

$$\|P(\theta, p) - P(\theta, p_0)\| \leq \mathcal{R}_\Gamma(e_H) < 1, \quad (2267)$$

$$\begin{aligned} A(\theta, p) &> A_0/2, \\ C(\theta, p) &> C_0/2, \end{aligned} \quad (2268)$$

$$\begin{aligned} \kappa(\theta, p) &> \kappa_0/2, \\ 1 - \Lambda r &> \delta_{S,0}/2. \end{aligned} \quad (2269)$$

In particular, the root, global gap, Riesz rank, active overlap, coherence, role, and Schur conclusions hold simultaneously at $\theta_\star(p)$ for one Hamiltonian and one specified target.

Proof. Integrating Eq. (2254) from θ_0 to the endpoints of I gives

$$F(\theta_0 - h_\theta, p_0) \geq m_F h_\theta, \quad F(\theta_0 + h_\theta, p_0) \leq -m_F h_\theta.$$

The first entry of Eq. (2261) leaves strict opposite signs, while the second gives

$$\partial_\theta F(\theta, p) \leq -m_F + e_{F_1} < -\frac{m_F}{2}.$$

Thus $F(\cdot, p)$ has exactly one zero in I , and $\log \mathcal{M}$ increases before it and decreases after it. This proves existence, uniqueness, simplicity, and strict maximality.

The reference derivative bound also gives

$$|F(\theta, p_0)| \geq m_F |\theta - \theta_0|.$$

At the continued zero, its left side is at most e_F , proving Eq. (2263). The triangle inequality gives Eq. (2265).

Weyl's inequality moves both edges of a global gap by at most e_H , which proves Eq. (2266). The resolvent identity on Γ and contour integration give Eq. (2267). Projections at norm distance below one are unitarily equivalent; in particular, any finite fiber rank is preserved. Definitions (2258) and (2261) prove the active, coherence, and role bounds. Finally,

$$1 - \Lambda r \geq \delta_{S,0} - |\Lambda r - \Lambda_0 r_0| > \frac{\delta_{S,0}}{2}.$$

Every estimate uses the same p , $\theta_\star(p)$, and transported target. $\square$

Corollary 62 (Open common-budget region). *If all chart maps and quantitative auxiliary margins are continuous, then $\mathcal{P}_{\text{cb}}$ is open. A nonempty rectangular subset of it is therefore a robust joint physical window, not a collection of separately inhabited hypotheses.*

Proof. Every defining condition in Eq. (2262) is a strict inequality between continuous quantities. Their finite intersection is open. $\square$

An exact analytic witness

At the specified lateral mismatch $d = a$, define the actual full-distance active factor

$$\begin{aligned} \phi_X(H, \Lambda) &:= \Phi_X(1; H, \Lambda) \\ &= \exp \left[-\frac{\sqrt{H^2 + 1} - H}{\Lambda} \right], \\ A(p) &:= \omega \phi_X(H, \Lambda). \end{aligned} \quad (2270)$$

The preselected channel is fixed before the scan, so this is an explicit matrix element rather than an inference from total Parseval power.

Definition 26 (Rational physical witness box). *Let*

$$\mathcal{B} := \left(\frac{1}{4}, \frac{1}{3}\right) \times \left(\frac{1}{10}, \frac{1}{8}\right) \times \left(1, \frac{5}{4}\right) \times \left(\frac{1}{2}, \frac{3}{4}\right), \quad (2271)$$

in the order (ω, r, H, Λ) , *and let*

$$p_0 = \left(\frac{7}{24}, \frac{9}{80}, \frac{9}{8}, \frac{5}{8}\right) \in \mathcal{B}. \quad (2272)$$

Set

$$\Theta(p) := \frac{1}{8} + \frac{1}{4} \left[\omega - \frac{7}{24} + r - \frac{9}{80} + H - \frac{9}{8} + \Lambda - \frac{5}{8} \right], \quad (2273)$$

$$x := \theta - \Theta(p), \quad v(\theta, p) := A(p)(1 + x^2). \quad (2274)$$

Proposition 64 (Worst-corner kernel and Schur bounds). *Every* $p \in \mathcal{B}$ *obeys*

$$\Lambda r < \frac{3}{32}, \quad 1 - \Lambda r > \frac{29}{32}, \quad (2275)$$

$$\delta_X = \Lambda^{-1} - r > \frac{29}{24}, \quad (2276)$$

$$\frac{1}{3} < \phi_X(H, \Lambda) < 1, \quad \frac{1}{12} < A(p) < \frac{1}{3}. \quad (2277)$$

Thus H *and* Λ *enter the same active matrix element through the physical kernel, and* r *enters the same Hamiltonian through* $\Theta(p)$; *none is a spectator coordinate.*

Proof. The first two bounds use the worst corner $(\Lambda, r) = (3/4, 1/8)$. Also

$$\Lambda^{-1} - r > \frac{4}{3} - \frac{1}{8} = \frac{29}{24}.$$

For $H > 1$,

$$0 < \sqrt{H^2 + 1} - H = \frac{1}{\sqrt{H^2 + 1} + H} < \frac{1}{2}. \quad (2278)$$

Since $\Lambda > 1/2$, the exponent in Eq. (2270) lies strictly between -1 and zero. The elementary inequality $e < 3$ yields $e^{-1} > 1/3$. Multiplication by $1/4 < \omega < 1/3$ proves Eq. (2277). $\square$

Definition 27 (Same-target Hermitian bilayer witness). *Let* $q \in [-1, 1]$, *fix the metric* $G = g dq^2$ *with* $g > 0$, *set* $s = 1/24$, $d_0 = 25/12$, *and define on* $\mathbb{C}_{\text{layer}}^2 \otimes \mathbb{C}_{\text{orb}}^2$

$$H_{\text{wit}}(q, \theta, p) = t \begin{pmatrix} sqI_2 & T(\theta, p) \\ T(\theta, p) & sqI_2 \end{pmatrix}, \quad (2279)$$

$$T(\theta, p) = \text{diag}(v(\theta, p), v(\theta, p) + d_0). \quad (2280)$$

Before the scan, specify

$$P_{\text{tar}} = I_{\text{layer}} \otimes |1\rangle_{\text{orb}}\langle 1|, \quad (2281)$$

and use identity transport. The two layer states in $\text{Ran } P_{\text{tar}}$ *are the preselected active channels.*

Theorem 113 (Exact same-target spectrum, gap, and coherence). *For every* $p \in \mathcal{B}$, *every* $0 < \theta < 1/4$, *and every* $q \in [-1, 1]$, *the target eigenvalues are*

$$E_{\pm} = t(sq \pm v), \quad (2282)$$

and the complement eigenvalues are

$$E_{c,\pm} = t[sq \pm (v + d_0)]. \quad (2283)$$

The target rank is two, its global bandwidth and isolation gap are

$$W = t(2s + 2v), \quad \Delta = t(d_0 - 2s) = 2t, \quad (2284)$$

and its active overlap is $tv > t/12 = 2st$. *Hence the target bands are disjoint throughout the box. Their invariant velocity factor and normalized maximum density of states are*

$$V_{\text{vel}} = \hbar D_G \Omega^{\text{max}} = 2st = \frac{t}{12}, \quad \rho_{\text{tar}}^{\text{max}} = \frac{1}{4st} = \frac{6}{t}. \quad (2285)$$

Indeed, $D_G([-1, 1]) = 2\sqrt{g}$ *and* $\Omega^{\text{max}} = st/(\hbar\sqrt{g})$. *With the fixed reference* $E_{\text{kin}} = t/6$,

$$X = E_{\text{kin}} \rho_{\text{tar}}^{\text{max}} = 1. \quad (2286)$$

The projector is constant, so its exact covariance and coherence factors equal one.

The circle

$$\Gamma_{\text{wit}} := \{z \in \mathbb{C} : |z| = 3t/2\} \quad (2287)$$

is one common Riesz contour for the target on the whole box and angle domain, with

$$\text{dist}(\Gamma_{\text{wit}}, \text{spec } H_{\text{wit}}) > \frac{t}{2}. \quad (2288)$$

Proof. Each orbital block is $t(sqI_2 + \tau\sigma_x)$, giving Eqs. (2282) and (2283). Across all q , the target extrema are $t(-s - v)$ and $t(s + v)$, proving the bandwidth. The nearest upper and lower complement edges are each separated by $t(d_0 - 2s) = 2t$.

The half-widths of the four factors in $\mathcal{B}$ give

$$|\Theta(p) - 1/8| < \frac{1}{4} \left(\frac{1}{24} + \frac{1}{80} + \frac{1}{8} + \frac{1}{8} \right) = \frac{73}{960}. \quad (2289)$$

Thus $0 < \theta < 1/4$ implies $|x| < 193/960 < 1/4$. Consequently,

$$0 < v < \frac{1}{3} \left(1 + \frac{1}{16} \right) = \frac{17}{48}. \quad (2290)$$

Every target energy therefore has absolute value less than $19t/48$, while every complement energy has absolute value greater than $t(d_0 - s) = 49t/24$. The circle in Eq. (2287) has distance greater than $t/2$ from both sets. The active matrix element is $tv \geq tA > t/12 = 2st$, so the two target intervals are disjoint. With measure $dq/2$ and

equal band weights, each linear band contributes density $1/(4st)$, proving Eq. (2285); the metric calculation in the theorem gives its invariant velocity identity. The chosen kinetic reference then proves Eq. (2286). Equation (2281) is independent of every scan coordinate and commutes with the declared covariance action, so its rank and coherence are exact. $\square$

Theorem 114 (Explicit interior operational maximum). *Take the fixed physical angle domain*

$$0 < \theta < \theta_{\max} := \frac{1}{4}. \quad (2291)$$

Put $\bar{W} = W/t$, $\bar{\Delta} = \Delta/t = 2$, and $\bar{V} = V_{\text{vel}}/t = 1/12$. Using the derived value $X = 1$ and exact coherence $\mathcal{C} = 1$, take the fixed score

$$\mathcal{M}_{\text{wit}} = \left[\frac{2}{2 + \bar{W}} \frac{2}{2 + 1/12} \frac{1}{2} \right]^{1/4}. \quad (2292)$$

It has the unique root and global strict maximum

$$\theta_{\star}(p) = \Theta(p). \quad (2293)$$

Uniformly on $\mathcal{B}$,

$$\frac{47}{960} < \theta_{\star} < \frac{193}{960} < \frac{1}{4}, \quad (2294)$$

$$\min\{\theta_{\star}, 1/4 - \theta_{\star}\} > \frac{47}{960}, \quad (2295)$$

$$\partial_{\theta}^2 \log \mathcal{M}_{\text{wit}}(\theta_{\star}, p) < -\frac{1}{27}. \quad (2296)$$

Proof. Only the first factor of Eq. (2292) varies with θ . Since

$$\bar{W} = \frac{1}{12} + 2A(1 + x^2), \quad \partial_{\theta} \bar{W} = 4Ax,$$

the logarithmic force is

$$F_{\text{wit}} = \partial_{\theta} \log \mathcal{M}_{\text{wit}} = -\frac{Ax}{2 + \bar{W}}. \quad (2297)$$

Because $A > 0$, its only zero is $x = 0$. The bandwidth is a strictly increasing function of x^2 , so the zero is the unique global maximum of the score on the angle domain.

Equation (2289) proves the root interval and the end-point margin. At the root,

$$\partial_{\theta} F_{\text{wit}} = -\frac{A}{25/12 + 2A}.$$

The last quotient is strictly increasing in $A > 0$, so $A > 1/12$ gives

$$\frac{A}{25/12 + 2A} > \frac{1/12}{27/12} = \frac{1}{27},$$

which proves Eq. (2296). $\square$

Proposition 65 (Explicit four-coordinate sensitivities). *On $\mathcal{B} \times (0, 1/4)$,*

$$|\partial_{\omega} A| \leq 1, \quad \partial_r A = 0, \quad |\partial_H A|, |\partial_{\Lambda} A| \leq \frac{2}{3}, \quad (2298)$$

$$|\partial_{\omega} v| < \frac{12}{5}, \quad |\partial_r v| < \frac{1}{5}, \quad |\partial_H v|, |\partial_{\Lambda} v| < \frac{5}{3}. \quad (2299)$$

Hence

$$\|\partial_{p_j} H_{\text{wit}}\| = t|\partial_{p_j} v| \quad (2300)$$

with the four constants in Eq. (2299). The force obeys

$$\begin{aligned} |\partial_{\omega} F_{\text{wit}}| &< \frac{1}{2}, & |\partial_r F_{\text{wit}}| &< \frac{1}{16}, \\ |\partial_H F_{\text{wit}}|, |\partial_{\Lambda} F_{\text{wit}}| &< \frac{3}{8}, \end{aligned} \quad (2301)$$

$$\begin{aligned} |\partial_{\omega} \partial_{\theta} F_{\text{wit}}| &< \frac{13}{8}, & |\partial_r \partial_{\theta} F_{\text{wit}}| &< \frac{1}{8}, \\ |\partial_H \partial_{\theta} F_{\text{wit}}|, |\partial_{\Lambda} \partial_{\theta} F_{\text{wit}}| &< \frac{9}{8}. \end{aligned} \quad (2302)$$

Furthermore,

$$\partial_{p_j} \theta_{\star} = \frac{1}{4} \quad \text{for every } j, \quad (2303)$$

the exact gap and coherence derivatives vanish, and the root active-overlap derivatives are bounded by

$$t \left(1, 0, \frac{2}{3}, \frac{2}{3} \right) \quad (2304)$$

in the order (ω, r, H, Λ) .

Proof. Let $d_H = \sqrt{H^2 + 1} - H$. Direct differentiation gives

$$\begin{aligned} \partial_H \phi_X &= \frac{\phi_X}{\Lambda} \left(1 - \frac{H}{\sqrt{H^2 + 1}} \right), \\ \partial_{\Lambda} \phi_X &= \frac{\phi_X d_H}{\Lambda^2}. \end{aligned} \quad (2305)$$

Equations (2278) and (2277) make both derivatives positive and at most two. Multiplication by $\omega < 1/3$ proves Eq. (2298).

Every coordinate derivative of Θ is $1/4$, and

$$\partial_{p_j} v = (\partial_{p_j} A)(1 + x^2) - 2Ax \partial_{p_j} \Theta. \quad (2306)$$

Use $1 + x^2 < 17/16$, $A < 1/3$, and $|x| < 1/4$. The four resulting bounds are

$$\frac{53}{48} < \frac{12}{5}, \quad \frac{1}{24} < \frac{1}{5}, \quad \frac{3}{4} < \frac{5}{3},$$

which proves Eq. (2299). The off-diagonal Hermitian block has norm $|\partial v|$, proving Eq. (2300).

Put $D = 2 + \bar{W} = 25/12 + 2A(1 + x^2)$. From $F_{\text{wit}} = -Ax/D$, set $c = 25/12$ and $y = |x|$. Then

$$|\partial_A F_{\text{wit}}| = \frac{cy}{[c + 2A(1 + y^2)]^2} < \frac{cy}{(c + y/3)^2} < \frac{2}{5}. \quad (2307)$$

Here $2A(1 + y^2) > y/3$, and the middle expression is increasing for $0 \leq y < 3c$, so its value at $y = 1/4$ gives the last inequality. Since $|x| < 1$,

$$\partial_x F_{\text{wit}} = -\frac{A[c + 2A(1 - x^2)]}{D^2} < 0.$$

Moreover,

$$|\partial_\Theta F_{\text{wit}}| = |\partial_x F_{\text{wit}}| \leq \frac{A}{c + 2A} < \frac{4}{33} < \frac{1}{4}. \quad (2308)$$

The chain rule and Eq. (2298) prove Eq. (2301). Differentiating

$$\partial_\theta F_{\text{wit}} = -\frac{A}{D} + \frac{4A^2 x^2}{D^2}$$

gives the coarse uniform bounds

$$|\partial_A \partial_\theta F_{\text{wit}}| < \frac{22}{15}, \quad |\partial_x \partial_\theta F_{\text{wit}}| < \frac{1}{2}. \quad (2309)$$

Another chain-rule application proves Eq. (2302). The remaining statements follow directly from Eqs. (2293), (2284), and (2298). $\square$

Proposition 66 (Uniform tail and optional cover certificate). *On $\mathcal{B}$,*

$$\varepsilon_\perp(L; p) < \frac{182}{3} t C_X \exp\left[-\frac{29L}{24a}\right]. \quad (2310)$$

Consequently,

$$L_c > \frac{24a}{29} \log\left(\frac{5824C_X}{3}\right) \quad (2311)$$

makes $\varepsilon_\perp(L_c) < t/32$.

If a finite-cover realization also satisfies

$$L_c < r_{\text{inj}} - L_{\text{obs}} \quad (2312)$$

and its independently established global cover, angular, and numerical operator errors total less than $t/32$, then the complete operator error is less than $t/16$. The continued global gap is greater than $15t/8$, and the Riesz projectors on Γ_{wit} obey

$$\|P_{\text{cov}} - P_{\text{tar}}\| < \frac{3}{7} < 1. \quad (2313)$$

Active overlap and coherence on the cover retain the separate entries of the common budget; they do not follow from local injectivity alone.

Proof. On the box,

$$\omega e^{H/\Lambda} < \frac{e^{5/2}}{3} < \frac{13}{3}, \quad \frac{29}{24} < \delta_X < \frac{19}{10}.$$

Also $e^{19/10} < 7$ and $1 - e^{-29/24} > 1 - e^{-1} > 1/2$. Substitution in Eq. (2251) proves Eq. (2310). The cutoff in Eq. (2311) makes its right-hand side smaller than

$$\frac{182}{3} t C_X \frac{3}{5824C_X} = \frac{t}{32}.$$

The cover buffer makes the truncated local matrix exact, but the stated global comparison remains a separate hypothesis. Adding its error to the tail gives $e_H < t/16$. Weyl's inequality and Eq. (2284) give

$$\Delta_{\text{cov}} > 2t - \frac{2t}{16} = \frac{15t}{8}.$$

The contour has length $3\pi t$ and distance greater than $t/2$. Therefore

$$\frac{3t}{2} \frac{t/16}{(t/2)(t/2 - t/16)} = \frac{3}{7},$$

which proves Eq. (2313). $\square$

Theorem 115 (Nonempty open joint admissible region). *Let $\mathcal{P}_{\text{adm}}^{\text{wit}}$ be the set of physical points for which the Hamiltonian in Definition 27, the fixed target in Eq. (2281), and the fixed score in Eq. (2292) satisfy simultaneously*

$$\begin{aligned} \omega, r, H, \Lambda > 0, \quad 1 - \Lambda r > 0, \\ 0 < \theta_* < 1/4, \quad -\partial_\theta F_{\text{wit}}(\theta_*, p) > 0, \\ \Delta > 0, \quad A_{\text{act}} > 0, \quad C_{\text{coh}} > 0, \end{aligned} \quad (2314)$$

together with the complete declared q -domain, finite-range intralayer hopping, the exponential interlayer Schur and tail condition, and the fixed representation, scalarization, metric, transport, energy, and role conventions. No finite-cover conclusion is included in this definition. Then

$$\mathcal{B} \subset \mathcal{P}_{\text{adm}}^{\text{wit}}. \quad (2315)$$

In particular, $\mathcal{P}_{\text{adm}}^{\text{wit}}$ is nonempty and contains the rational witness p_0 . It is open.

Proof. Proposition 64 proves the positive physical coordinates and the worst-corner Schur margin. Theorem 114 proves the unique simple interior root, strict maximum, and endpoint margin. Theorem 113 proves the positive global gap, common Riesz contour, constant target rank, active overlap, and coherence for that same root and target. Proposition 66 proves the uniform physical tail; no cover approximation is needed for the complete exact witness. The score, metric, transport, and energy normalization are fixed by construction. Thus every point of the one rectangle satisfies the full exact-model intersection. Analyticity and strict inequalities prove openness. $\square$

Corollary 63 (Conditional finite-cover joint membership). *Consider a finite-cover realization of the witness with the cutoff and injectivity buffer in Proposition 66. If an independent global norm or no-pollution comparison holds and the complete operator, differentiated-score, active-overlap, coherence, transport, and role errors make $\mathfrak{B}(p) < 1$, then Theorem 112 gives the same joint conclusions on the cover. Under the explicit operator-error split in that proposition,*

$$\Delta_{\text{cov}} > \frac{15t}{8}, \quad \|P_{\text{cov}} - P_{\text{tar}}\| < \frac{3}{7} < 1. \quad (2316)$$

Proof. The buffer identifies the truncated local matrix, while the independent global comparison controls the full target and complement. The common budget supplies the differentiated and same-target diagnostic bounds. The stated numerical conclusions are Eqs. (2313) and the Weyl estimate in the proof of Proposition 66. $\square$

Subsequent repairs and scope firewalls

Corollary 64 (Joint-region repair ledger). *The affected manuscript families are repaired as follows.*

1. *The infinite-volume hopping and full-distance-kernel statements use Eq. (2249), with a uniform positive margin and the explicit tail in Eq. (2251). Finite-range intralayer hopping does not supply the interlayer margin.*
2. *Theorems 79 and 100–109 use one specified target, one global spectral island, and one fixed score. Their root, gap, overlap, and coherence claims must lie in one set such as $\mathcal{P}_{\text{cb}}$, not in separately established parameter sets.*
3. *Theorems 120–122 and every Euclidean benchmark root remain conditional for an arbitrary microscopic model. Theorem 115 supplies existence only for its explicit richer analytic witness.*
4. *Theorems 136–143 retain gap closure, coherence loss, rank change, resonant-shell change, and physical-angle endpoints as distinct boundary faces. None is identified with failure of the Schur margin.*
5. *The active-coupling and scaling families around Theorems 148 and 155 retain the actual full-distance factor, the specified overlap, the fixed positive kinetic role, and the independently established operational root.*
6. *The global-collapse families including Theorems 160 and 169 retain the operator topology, tail, no-loss, no-pollution, tower, representation, and differentiated-observable conditions. The local analytic witness does not remove them.*

Proof. Item 1 is Proposition 63. Items 2–4 are the single-budget dependence structure in Theorem 112. Item 5 follows from Eq. (2270) and the fixed target. Item 6 is the local–global distinction retained explicitly in Proposition 66. $\square$

Corollary 65 (Root, bulk, arithmetic, role, and material separation theorem). *The nonempty window is a forward spectral existence theorem for one analytic richer model. It is not a named-material claim. Its direct-integral gap is global for the complete declared q -domain, but it does not imply global finite-cover convergence without the separate norm or no-pollution condition. Local injectivity and weak density-of-states convergence remain insufficient.*

The fixed specified target is not replaced after inspecting the root. The operational scalarization, metric, energy normalization, and kinetic role are fixed throughout the box. Arbitrary positive reweighting need not preserve the root. Exact crystallographic counts retain their Euclidean or hyperbolic arithmetic multipliers and are not consequences of the spectral window.

The exact square five-state target has no magic root. The witness uses a different, richer four-state active Hamiltonian and cannot create a root in the square five-state truncation. No reverse one-parameter-universality obstruction is asserted here.

Proof. The first paragraph lists operator and cover hypotheses absent from the finite-fiber algebra alone. The second lists conventions fixed in Definitions 24 and 27. The final paragraph is a model distinction: existence in one richer Hamiltonian has no implication for a different root-free Hamiltonian or for a reverse universality theorem. $\square$

Reverse certificates for one-parameter spectral collapse

A forward scaling ansatz does not identify its falsification topology. We first formulate an operator-level reverse theorem. Root and cancellation tests enter only after the stronger differentiated hypotheses that make them stable have been imposed.

Fixed comparison package and master classes

Definition 28 (Transported thermodynamic data). *Fix a compact interval $K_X \subset \mathbb{R}$, a Hilbert space $\mathcal{H}$, a specified target and representation sector, a resonant shell, a canonical metric, an operational scalarization, an energy-origin functional $\mathfrak{o}$, and a positive kinetic-role convention. Only declared symmetry and gauge transports into $\mathcal{H}$ are allowed.*

Let $\mathfrak{A}$ be a directed thermodynamic index. It includes the hopping cutoff and, when used, an admissible cover or tower index. At $\alpha \in \mathfrak{A}$, let $\mathcal{D}_\alpha$ be a finite label set. The exact normalized data are

$$\hat{H}_{\alpha z} \in \mathcal{B}_{\text{sa}}(\mathcal{H}), \quad X_{\alpha z} \in K_X, \quad z \in \mathcal{D}_\alpha. \quad (2317)$$

Equivalently, one may use $D + V_{\alpha z}$, self-adjoint on one common $\text{Dom } D$, when every comparison below is a bounded perturbation comparison. Observed data satisfy

$$\|\tilde{H}_{\alpha z} - \hat{H}_{\alpha z}\| \leq u_{\alpha z}, \quad |\tilde{X}_{\alpha z} - X_{\alpha z}| \leq \xi_{\alpha z}. \quad (2318)$$

These uncertainties include all numerical, hopping-tail, angular, transport, and established cover errors relevant to the stated topology.

For the differentiated tier, each supported index i denotes an entire transported local scan

$$x \mapsto \hat{H}_i(x), \quad x \in I_X, \quad (2319)$$

not merely one operator value. Its specified root and covector data are evaluated from that same scan.

For $\lambda \in \mathfrak{A}$, write

$$\mathcal{Z}_{\succeq \lambda} := \{(\alpha, z) : \alpha \succeq \lambda, z \in \mathcal{D}_\alpha\}. \quad (2320)$$

Let $\mathcal{B}$ be the supported local-scan indices, with thermodynamic level $\alpha(i)$. Define

$$\mathcal{B}_{\succeq \lambda} := \{i \in \mathcal{B} : \alpha(i) \succeq \lambda\}. \quad (2321)$$

Definition 29 (The C^0 and required- C^1 master classes). Fix $L_\star < \infty$. The class $\mathfrak{M}_0(L_\star)$ consists of maps

$$H_\star : K_X \longrightarrow \mathcal{B}_{\text{sa}}(\mathcal{H}) \quad (2322)$$

that use the fixed comparison package and obey

$$\|H_\star(x) - H_\star(y)\| \leq L_\star |x - y|, \quad \mathfrak{o}(H_\star(x)) = 0. \quad (2323)$$

For $H_\star \in \mathfrak{M}_0(L_\star)$, put

$$e_{0,\lambda}(H_\star) := \sup_{i \in \mathcal{Z}_{\succeq \lambda}} \|\hat{H}_i - H_\star(X_i)\|, \quad (2324)$$

and

$$\mathcal{E}_0 := \inf_{H_\star \in \mathfrak{M}_0(L_\star)} \lim_{\lambda} e_{0,\lambda}(H_\star). \quad (2325)$$

The explaining class is

$$\mathfrak{M}_0^{\text{coll}} := \left\{ H_\star \in \mathfrak{M}_0(L_\star) : \lim_{\lambda} e_{0,\lambda}(H_\star) = 0 \right\}. \quad (2326)$$

On the fixed chart below, let D_{req} contain ∂_x and every transported tangent derivative entering the root or cancellation diagnostics. Define

$$\begin{aligned} \|A\|_{C_{\text{req}}^1} &:= \sup_q \{ \|A(q)\| \\ &\quad + \ell_{\text{req}} \|D_{\text{req}} A(q)\|_{\text{op,dual}} \}. \end{aligned} \quad (2327)$$

Here $\ell_{\text{req}} > 0$. For a root-only test, D_{req} may be restricted to ∂_x . Let $\mathfrak{M}_1 \subset \mathfrak{M}_0(L_\star)$ be the C_{req}^1 masters satisfying the root and cancellation conditions below. For the supported scan tail $\mathcal{B}_{\succeq \lambda}$, put

$$e_{1,\lambda}(H_\star) := \sup_{i \in \mathcal{B}_{\succeq \lambda}} \|\hat{H}_i - H_\star \circ X_i\|_{C_{\text{req}}^1}, \quad (2328)$$

$$\mathcal{E}_1 := \inf_{H_\star \in \mathfrak{M}_1} \lim_{\lambda} e_{1,\lambda}(H_\star). \quad (2329)$$

Here the master is pulled back along each scan coordinate map X_i . Let $\mathfrak{M}_1^{\text{coll}}$ contain the masters with zero limiting C_{req}^1 error. Whenever a C^0 component is included in the C^1 combined certificate, each of its points is an evaluation of a scan in $\mathcal{B}_{\succeq \lambda}$. Hence

$$\mathcal{E}_1 \geq \mathcal{E}_0 \quad (2330)$$

on the supported comparison domain.

The sharp same-coordinate obstruction

Definition 30 (Matched-coordinate operator diameter). For $\delta > 0$, define

$$\mathcal{D}_\lambda(\delta) := \sup \left\{ \|\hat{H}_i - \hat{H}_j\| : i, j \in \mathcal{Z}_{\succeq \lambda}, |X_i - X_j| \leq \delta \right\}, \quad (2331)$$

with the supremum of an empty set equal to zero. Its thermodynamic matched-coordinate diameter is

$$\mathcal{D}_X := \lim_{\delta \downarrow 0} \lim_{\lambda} \mathcal{D}_\lambda(\delta). \quad (2332)$$

The limits are decreasing tail infima, with values allowed in $[0, \infty]$.

Theorem 116 (Sharp half-diameter no-go bound). For every $H_\star \in \mathfrak{M}_0(L_\star)$,

$$e_{0,\lambda}(H_\star) \geq \frac{1}{2} [\mathcal{D}_\lambda(\delta) - L_\star \delta]_+. \quad (2333)$$

For an exact same- X cluster $\mathcal{S}_x$, its Chebyshev radius obeys

$$\text{rad}(\mathcal{S}_x) := \inf_{A \in \mathcal{B}_{\text{sa}}(\mathcal{H})} \sup_{H \in \mathcal{S}_x} \|H - A\| \geq \frac{1}{2} \text{diam}(\mathcal{S}_x). \quad (2334)$$

Consequently,

$$\boxed{\mathcal{E}_0 \geq \frac{1}{2} \mathcal{D}_X}. \quad (2335)$$

The factor $1/2$ is the largest universal constant.

Proof. For every admissible pair,

$$\begin{aligned} \|\hat{H}_i - \hat{H}_j\| &\leq \|\hat{H}_i - H_\star(X_i)\| + \|\hat{H}_j - H_\star(X_j)\| \\ &\quad + \|H_\star(X_i) - H_\star(X_j)\| \\ &\leq 2e_{0,\lambda}(H_\star) + L_\star \delta. \end{aligned} \quad (2336)$$

Taking the supremum proves Eq. (2333). The same triangle inequality with an arbitrary center A proves Eq. (2334). Taking the two limits and then the infimum over masters proves Eq. (2335). For two self-adjoint operators A, B , their midpoint is at distance $\|A - B\|/2$ from both. This proves sharpness. $\square$

Proposition 67 (Uncertainty-subtracted finite certificate). For two observed points i, j , define

$$\begin{aligned} c_{ij}^{\text{op}} &:= \frac{1}{2} [\|\tilde{H}_i - \tilde{H}_j\| - u_i - u_j \\ &\quad - L_\star (|\tilde{X}_i - \tilde{X}_j| + \xi_i + \xi_j)]_+. \end{aligned} \quad (2337)$$

Every $H_\star \in \mathfrak{M}_0(L_\star)$ satisfies

$$\max_{k \in \{i,j\}} \|\hat{H}_k - H_\star(X_k)\| \geq c_{ij}^{\text{op}}. \quad (2338)$$

For declared pair sets $\mathcal{P}_\lambda$ in the thermodynamic tails, put

$$C_{\mathcal{P},\lambda}^{\text{op}} := \sup_{(i,j) \in \mathcal{P}_\lambda} c_{ij}^{\text{op}}. \quad (2339)$$

Then

$$\mathcal{E}_0 \geq \liminf_{\lambda} C_{\mathcal{P},\lambda}^{\text{op}}. \quad (2340)$$

Proof. The observation and coordinate bounds give

$$\begin{aligned} \|\tilde{H}_i - \tilde{H}_j\| &\leq u_i + u_j + \|\hat{H}_i - H_\star(X_i)\| \\ &\quad + \|\hat{H}_j - H_\star(X_j)\| \\ &\quad + L_\star(|\tilde{X}_i - \tilde{X}_j| + \xi_i + \xi_j). \end{aligned} \quad (2341)$$

Rearrangement proves the finite statement. Every tail error dominates the supremum of its pair certificates. The tail liminf and infimum over masters prove Eq. (2340). $\square$

Shape and established cross-cover residuals

Definition 31 (Supported C^0 residual families). *Let $\mathcal{P}_\lambda^{\text{sh}}$ contain matched- X points with different declared normalized-shape or microscopic-path labels. Define*

$$R_{\text{sh},\lambda}^{\text{op}} := \sup_{(i,j) \in \mathcal{P}_\lambda^{\text{sh}}} \|\tilde{H}_i - \tilde{H}_j\|, \quad (2342)$$

$$c_{\text{sh},\lambda} := C_{\mathcal{P}_\lambda^{\text{sh}},\lambda}^{\text{op}}. \quad (2343)$$

Let $\mathcal{P}_\lambda^{\text{cov}}$ contain points from inequivalent cover towers only after both towers have no-loss and no-pollution certificates for the same transported reduced-bulk target. Define

$$R_{\text{cover},\lambda}^{\text{op}} := \sup_{(i,j) \in \mathcal{P}_\lambda^{\text{cov}}} \|\tilde{H}_i - \tilde{H}_j\|, \quad (2344)$$

$$c_{\text{cov},\lambda}^{\text{op}} := C_{\mathcal{P}_\lambda^{\text{cov}},\lambda}^{\text{op}}. \quad (2345)$$

An uncertified cover pair does not belong to $\mathcal{P}_\lambda^{\text{cov}}$.

Proposition 68 (Observable-to-operator conversion). *Let $\mathcal{O}$ be a diagnostic of the specified global island with*

$$|\mathcal{O}(H) - \mathcal{O}(K)| \leq L_{\mathcal{O}} \|H - K\|, \quad L_{\mathcal{O}} > 0. \quad (2346)$$

For supported cover pairs, define the raw residual

$$R_{\text{cover},\lambda}^{\mathcal{O}} := \sup_{(i,j) \in \mathcal{P}_\lambda^{\text{cov}}} |\tilde{\mathcal{O}}_i - \tilde{\mathcal{O}}_j|. \quad (2347)$$

If its observation errors are $v_i^{\mathcal{O}}$, put

$$\begin{aligned} c_{ij}^{\mathcal{O}} &:= \frac{1}{2L_{\mathcal{O}}} [|\tilde{\mathcal{O}}_i - \tilde{\mathcal{O}}_j| - v_i^{\mathcal{O}} - v_j^{\mathcal{O}} \\ &\quad - L_{\mathcal{O}} L_\star(|\tilde{X}_i - \tilde{X}_j| + \xi_i + \xi_j)]_+. \end{aligned} \quad (2348)$$

This is a lower bound for the best operator error on the pair. For the normalized bandwidth and isolation gap,

$$L_W = L_\Delta = 2. \quad (2349)$$

Thus a established excess scalar discrepancy $d > 0$ gives the operator-error lower bound $d/4$.

Proof. Apply Eq. (2346) between each datum and its master value, and then between the two master values. The observation and coordinate bounds give Eq. (2348). Both edges defining a bandwidth move by at most the operator error. The two sets defining an isolation gap also move by at most that error on each side. This proves Eq. (2349). $\square$

For supported cover diagnostics, set

$$c_{\text{cov},\lambda} := \max \left\{ c_{\text{cov},\lambda}^{\text{op}}, \sup_{(i,j) \in \mathcal{O}} c_{ij}^{\mathcal{O}} \right\}. \quad (2350)$$

Corollary 66 (The C^0 shape and cover no-go). *The best master error obeys*

$$\mathcal{E}_0 \geq \max \left\{ \liminf_{\lambda} c_{\text{sh},\lambda}, \liminf_{\lambda} c_{\text{cov},\lambda} \right\}. \quad (2351)$$

A positive right-hand side makes $\mathfrak{M}_0^{\text{coll}}$ empty. Weak density-of-states convergence, an unsmoothed density peak, a local compression, an uncertified cover outlier, or strong-resolvent convergence alone supplies neither component.

Proof. Apply Proposition 67 to shape and direct cover pairs, and Proposition 68 to every supported cover diagnostic. Taking the maximum preserves each lower bound. The excluded data do not identify one global operator target in norm. $\square$

The differentiated root certificate

Definition 32 (Specified simple-root chart). *Fix*

$$I_X = [1 - h_X, 1 + h_X] \subseteq K_X, \quad h_X > 0. \quad (2352)$$

Every supported scan has a C^1 coordinate map $\theta \mapsto x = X(\theta)$ of fixed orientation with

$$0 < \chi_X \leq |\partial_\theta X| \leq \Xi_X < \infty. \quad (2353)$$

Thus the chart contains no fold or cusp. The data force f_i , obtained from the same target and fixed score, has one specified root with

$$\begin{aligned} f_i(x_i^\star) &= 0, \\ -\partial_x f_i(x) &\geq m_d > 0, \\ \text{dist}(x_i^\star, \partial I_X) &\geq d_d > 0. \end{aligned} \quad (2354)$$

Let $\tilde{f}_i$ be the numerical force and $\tilde{x}_i^*$ its specified zero. Require

$$\begin{aligned} \varepsilon_i^{\text{root}} &:= \|\tilde{f}_i - f_i\|_{C^0(I_X)} < m_d d_d, \\ |\tilde{x}_i^* - x_i^*| &\leq \xi_i^{\text{root}} := \frac{\varepsilon_i^{\text{root}}}{m_d}. \end{aligned} \quad (2355)$$

The endpoint margin prevents boundary escape in this root estimate. The master force satisfies

$$f_\star(1) = 0, \quad -\partial_x f_\star(x) \geq m_\star > 0 \quad (x \in I_X). \quad (2356)$$

The chart has a target gap $g_0 > 0$, a Riesz contour of length ℓ_Γ and distance $\delta_\Gamma > 0$, and a perturbation radius $0 < r_\Gamma < \delta_\Gamma$. Its kinetic-role scale is at least $e_0 > 0$. Every argument of the fixed logarithmic score has a positive lower margin. For the five-factor score these are

$$\begin{aligned} \Delta &\geq d_\Delta, \quad \Delta + W \geq d_W, \quad \Delta + V \geq d_V, \\ Z_\rho &\geq d_\rho, \quad 1 + Z_\rho \geq d_{1+\rho}, \quad C \geq d_C, \end{aligned} \quad (2357)$$

with all lower bounds positive. Active labels, edge labels, velocity maximizers, and density maximizers remain specified.

The contour constant is

$$C_\Gamma := \frac{\ell_\Gamma}{2\pi} \frac{1}{\delta_\Gamma(\delta_\Gamma - r_\Gamma)}. \quad (2358)$$

On this uniformly bounded chart, define the established force constant

$$0 < C_F := \sup_{H \text{ in the chart}} \|D\mathcal{F}(H)\|_{\mathcal{L}(C_{\text{req}}^1, C^0)} < \infty, \quad (2359)$$

where $\mathcal{F}(H) = f_H$. Its verification uses Eq. (2358), χ_X^{-1} , uniform numerator bounds, and exactly the margins in Eq. (2357). Hence

$$\|f_i - f_\star\|_{C^0(I_X)} \leq C_F \|\hat{H}_i - H_\star \circ X_i\|_{C_{\text{req}}^1} + \eta_i^F. \quad (2360)$$

Here η_i^F is the complete differentiated numerical, tail, transport, and cover uncertainty. Base C^0 decay does not control it.

Theorem 117 (Root-residual reverse bound). *At tail λ , choose a finite family $\mathcal{B}_\lambda \subset \mathcal{B}_{\geq \lambda}$ and weights $w_i \geq 0$ with $\sum_i w_i = 1$. For a candidate master, write*

$$e^1 := e_{1,\lambda}(H_\star), \quad e_i^1 := \|\hat{H}_i - H_\star \circ X_i\|_{C_{\text{req}}^1} \leq e^1. \quad (2361)$$

Define

$$\begin{aligned} R_X &:= \left[\sum_i w_i |\tilde{x}_i^* - 1|^2 \right]^{1/2}, \\ u_X &:= \left[\sum_i w_i (\xi_i^{\text{root}})^2 \right]^{1/2}, \end{aligned}$$

$$u_F := \left[\sum_i w_i (\eta_i^F)^2 \right]^{1/2}. \quad (2362)$$

Equation (2355) makes u_X a established root-coordinate uncertainty. Every admissible C_{req}^1 master has tail error at least

$$c_X := \frac{[m_\star(R_X - u_X) - u_F]_+}{C_F}. \quad (2363)$$

Consequently,

$$\mathcal{E}_1 \geq \liminf_\lambda c_{X,\lambda}. \quad (2364)$$

Proof. The fixed sign in Eq. (2356) gives

$$\begin{aligned} m_\star |x_i^* - 1| &\leq |f_\star(x_i^*) - f_\star(1)| \\ &= |f_\star(x_i^*) - f_i(x_i^*)| \leq C_F e_i^1 + \eta_i^F. \end{aligned} \quad (2365)$$

Minkowski's inequality and the observation error give

$$R_X \leq u_X + \frac{C_F e^1 + u_F}{m_\star}. \quad (2366)$$

Rearrangement proves Eq. (2363). The tail liminf proves Eq. (2364). The data derivative and endpoint margins give the numerical root-error conversion by the same mean-value argument. $\square$

The differentiated cancellation certificate

Definition 33 (Metric cancellation package). *Along the same specified local scan, let*

$$\begin{aligned} a_i(x) &:= dE_{\text{bare},i}(x), \\ b_i(x) &:= dE_{\text{hyb},i}(x), \\ S_i(x) &:= a_i(x) + b_i(x). \end{aligned} \quad (2367)$$

At x_i^* , abbreviate $a_i = a_i(x_i^*)$ and $b_i = b_i(x_i^*)$. These covectors lie in the fixed transported cotangent fiber, and their dual-metric norms have separately established lower margins

$$\begin{aligned} \|a_i\|_{G^{-1}} &\geq \beta_b > 0, \\ \|b_i\|_{G^{-1}} &\geq \beta_h > 0, \\ \beta &:= \beta_b + \beta_h. \end{aligned} \quad (2368)$$

The master covectors obey exact first-order cancellation,

$$S_\star(1) = a_\star(1) + b_\star(1) = 0. \quad (2369)$$

Require the uniform class bounds

$$\|S_i(x_i^*) - S_\star(x_i^*)\|_{G^{-1}} \leq C_{ab} e_i^1 + \eta_i^{ab}, \quad (2370)$$

$$L_S := \sup_{H_\star \in \mathfrak{M}_1} \sup_{x \in I_X} \quad (2371)$$

$$\|\partial_x S_\star(x)\|_{G^{-1}} < \infty. \quad (2372)$$

The explicit L_S condition is the additional regularity needed to compare cancellation at x_i^* with cancellation at one. The positive constant C_{ab} is uniform over the declared class and is established by the fixed-gap, Riesz, Hellmann–Feynman, metric, and required- C^1 tail package.

Theorem 118 (Cancellation-residual reverse bound). Define the exact and observed point residuals by

$$\begin{aligned} r_i &:= \frac{\|a_i + b_i\|_{G^{-1}}}{\|a_i\|_{G^{-1}} + \|b_i\|_{G^{-1}}}, \\ |\tilde{r}_i - r_i| &\leq \eta_i^R. \end{aligned} \quad (2373)$$

With the weights from Theorem 117, put

$$\begin{aligned} R_{\text{cancel}} &:= \left[\sum_i w_i \tilde{r}_i^2 \right]^{1/2}, \\ u_R &:= \left[\sum_i w_i (\eta_i^R)^2 \right]^{1/2}, \\ u_{ab} &:= \left[\sum_i w_i (\eta_i^{ab})^2 \right]^{1/2}. \end{aligned} \quad (2374)$$

Define

$$C_{\text{cancel}} := C_{ab} + \frac{L_S C_F}{m_\star} > 0. \quad (2375)$$

The constants $m_\star, C_F, C_{ab}, L_S$ are fixed uniformly over $\mathfrak{M}_1$. If a diagnostic transfer constant vanishes identically, treat that rigid diagnostic separately instead of dividing by zero. Every admissible C_{req}^1 master has tail error at least

$$c_{\text{cancel}} := \frac{[\beta(R_{\text{cancel}} - u_R) - u_{ab} - L_S u_F / m_\star]_+}{C_{\text{cancel}}}. \quad (2376)$$

Therefore

$$\mathcal{E}_1 \geq \liminf_\lambda c_{\text{cancel}, \lambda}. \quad (2377)$$

Proof. At the data root, Eqs. (2370) and (2369) give

$$\begin{aligned} \|a_i + b_i\|_{G^{-1}} &\leq C_{ab} e_i^1 + \eta_i^{ab} + L_S |x_i^* - 1| \\ &\leq \left(C_{ab} + \frac{L_S C_F}{m_\star} \right) e_i^1 + \eta_i^{ab} + \frac{L_S \eta_i^F}{m_\star}, \end{aligned} \quad (2378)$$

where Eq. (2365) gives the second line. The denominator margins imply

$$\beta r_i \leq \|a_i + b_i\|_{G^{-1}}. \quad (2379)$$

Apply the weighted RMS triangle inequality and the observed-residual uncertainty. Rearrangement proves Eq. (2376); the tail $\liminf$ proves the last claim. $\square$

One combined reverse certificate

Definition 34 (Combined tail certificate). At each thermodynamic tail, all components are measured in units of operator approximation error. Shape and cover pair sets lie in $\mathcal{X}_{\succeq \lambda}$. Root and cancellation components use $\mathcal{B}_\lambda \subset \mathcal{B}_{\succeq \lambda}$. For $A \in \{\text{sh}, \text{cov}\}$, let $\mathcal{Q}_{A, \lambda}^{(1)}$ be the set of supported operator-error-unit pair certificates entering $c_{A, \lambda}$ whose two data points are evaluations of scans in $\mathcal{B}_{\succeq \lambda}$. Define

$$c_{A, \lambda}^{(1)} := \sup(\mathcal{Q}_{A, \lambda}^{(1)} \cup \{0\}). \quad (2380)$$

Thus a C^0 pair not supplied by the same scan tail is absent from the C_{req}^1 certificate. Define

$$\mathfrak{C}_\lambda^{(0)} := \max\{c_{\text{sh}, \lambda}, c_{\text{cov}, \lambda}\}, \quad (2381)$$

$$\mathfrak{C}_\lambda^{(1)} := \max\{c_{\text{sh}, \lambda}^{(1)}, c_{\text{cov}, \lambda}^{(1)}, c_{X, \lambda}, c_{\text{cancel}, \lambda}\}. \quad (2382)$$

An unlicensed component is omitted. If no component is supported, the corresponding algebraic maximum is zero and carries no empirical content.

Theorem 119 (If-any-residual reverse no-go theorem). For every thermodynamic tail,

$$\inf_{H_\star \in \mathfrak{M}_0(L_\star)} e_{0, \lambda}(H_\star) \geq \mathfrak{C}_\lambda^{(0)}, \quad (2383)$$

$$\inf_{H_\star \in \mathfrak{M}_1} e_{1, \lambda}(H_\star) \geq \mathfrak{C}_\lambda^{(1)}. \quad (2384)$$

In the thermodynamic limit,

$$\mathcal{E}_0 \geq \liminf_\lambda \mathfrak{C}_\lambda^{(0)}, \quad (2385)$$

$$\mathcal{E}_1 \geq \liminf_\lambda \mathfrak{C}_\lambda^{(1)}. \quad (2386)$$

Hence a positive uncertainty-subtracted thermodynamic lower bound for either the shape or established cover component excludes every asymptotically exact C^0 master in the declared class. A positive lower bound for either the specified root or metric cancellation component excludes every asymptotically exact master in the declared C_{req}^1 class. A root or cancellation obstruction is not promoted to a C^0 obstruction.

Proof. Corollary 66 gives the two C^0 entries. Theorems 117 and 118 give the additional C_{req}^1 entries. Each component uses points or scans inside the same tail controlled by the corresponding error. Every finite best error is therefore at least the indicated maximum. For each fixed master, take the tail limit on the left and $\liminf$ on the right, then take the infimum over masters. A positive best-error lower bound excludes a zero-error master. $\square$

Corollary 67 (Explicit excess-to-error conversions). After every declared uncertainty and X -matching term has been subtracted, let $d_{\text{op}}, d_W, d_\Delta, d_X, d_{\text{can}}$ denote the thermodynamic $\liminf$ excesses of, respectively, the operator-pair numerator, the bandwidth and gap numerators, the

root numerator $m_*(R_X - u_X) - u_F$, and the cancellation numerator $\beta(R_{\text{cancel}} - u_R) - u_{ab} - L_S u_F / m_*$. Then

$$\begin{aligned} d_{\text{op}} > 0 &\implies \mathcal{E}_0 \geq d_{\text{op}}/2, \\ d_W > 0 \text{ or } d_\Delta > 0 &\implies \mathcal{E}_0 \geq d_W/4 \text{ or } d_\Delta/4, \\ d_X > 0 &\implies \mathcal{E}_1 \geq d_X/C_F, \\ d_{\text{can}} > 0 &\implies \mathcal{E}_1 \geq d_{\text{can}}/C_{\text{cancel}}. \end{aligned} \quad (2387)$$

The same implications hold for one finite tail with $\mathcal{E}_r$ replaced by the corresponding finite best error.

Proof. These are Eqs. (2337), (2349), (2363), and (2376), respectively. $\square$

Repairs and final firewalls

Corollary 68 (Repair ledger for the collapse statements). *The reverse theorem imposes the following replacements.*

1. Theorem 160 and every master-function statement first fixes the intrinsic global island, transported bulk representation, normalization, and master class. A shape or cover residual is a C^0 rejection only through Theorem 116 or an explicitly Lipschitz diagnostic.
2. Theorem 167 subtracts numerical, tail, transport, cover, and coordinate-matching uncertainty. For bandwidth and gap data, its established operator-error lower bound is one quarter of the remaining pair discrepancy.
3. Theorem 169 separates bandwidth and gap collapse in C^0 from velocity, specified-root, and cancellation tests in C^1_{req} . The latter retain the chart, gap, simple-root, score-denominator, metric, and covector margins in Definitions 32 and 33.
4. Pairwise path-collapse and cross-cover prose uses Eq. (2337). Close rather than equal coordinates retains the explicit L_* mismatch subtraction.
5. Finite covers define a bulk obstruction only after no loss, no pollution, target registration, and representation compatibility. Local matrix convergence or weak density-of-states convergence is insufficient.

Proof. Items 1 and 4 follow from Proposition 67 and Theorem 116. Item 2 is Proposition 68. Item 3 is the separation between Corollary 66 and the differentiated reverse theorems. Item 5 is built into Definition 31. $\square$

Corollary 69 (Necessity-only and layer separation theorem). *After uncertainty and matching envelopes vanish, vanishing of all supported residuals is necessary for an asymptotically exact master in the tested class, but it is not sufficient. It does not construct a master, prove continuity or uniqueness, control an untested observable,*

identify different branches, or certify a material realization.

All data and masters retain the same energy origin, kinetic role, metric, scalarization, active target, resonant shell, symmetry transport, and representation sector. A convention change transports both sides together. Shape, anisotropy, higher-dimensional rotation class, and role changes remain explicit fields. Thermodynamic cover tests retain finite-range or exponential summability, differentiated tails when used, admissible towers, no loss, and no pollution. Exact Euclidean and hyperbolic arithmetic multipliers remain outside the spectral master. Spectral rejection is separate from geometric and exact-crystallographic universality.

The exact square five-state target has no operational root, so R_X and R_{cancel} are inapplicable to it. A richer model must first certify its root and every margin in Definition 32. The nonempty forward box of the richer analytic witness proves existence only for that model and does not imply one-parameter collapse or a named-material statement.

Proof. The reverse bounds test necessary consequences of a declared master and do not supply a compactness, extension, or reconstruction theorem. The second paragraph lists the fixed data needed before any norm or derivative residual is meaningful. The last paragraph preserves the root-free target and the forward-versus-reverse logical direction. $\square$

A natural microscopic hyperbolic bilayer with a positive root

The result below uses a genuine surface-group hopping operator. Its finite-shell calculation is not the five-state square model and is not an abstract carrier construction. The complete operator and a declared character sector are kept separate throughout.

Geometric crystal and complete operator

Definition 35 (Centered surface-group bilayer). *Let*

$$\Gamma = \left\langle s_1, \dots, s_{2g} \left| \prod_{a=1}^g [s_{2a-1}, s_{2a}] = e \right. \right\rangle, \quad g \geq 2, \quad (2388)$$

be the side-pairing group of a regular hyperbolic $4g$ -gon. Put

$$m = 2g, \quad S = \{s_a, s_a^{-1} : 1 \leq a \leq m\}. \quad (2389)$$

The orbit of the polygon center is a uniformly discrete hyperbolic lattice. All elements of S move the center through the same geodesic length $d_1 > 0$. Thus this equal-length first shell is realized by an explicit hyperbolic crystal rather than imposed on an arbitrary group presentation.

Fix two congruent centered layers at normal separation $h > 0$, and define $\ell_1 = (h^2 + d_1^2)^{1/2} - h$. For $\gamma \in \Gamma$, let

$$D_\gamma = (h^2 + d_{\mathbb{H}}(x_1, \gamma x_2)^2)^{1/2}, \quad r_\gamma = D_\gamma - h. \quad (2390)$$

The relative placement is chosen at the exact centered crystallographic configuration, so that

$$r_e = 0, \quad r_s = \ell_1 \quad (s \in S), \quad r_{\gamma^{-1}} = r_\gamma. \quad (2391)$$

For $\mu = \lambda_\perp^{-1} > 0$, define the physical exponential full-distance coefficients

$$q_\gamma = e^{-\mu r_\gamma}, \quad q_1 = e^{-\mu \ell_1}. \quad (2392)$$

Let λ_Γ be the left regular representation on $\ell^2(\Gamma)$, and set

$$A_S = \sum_{a=1}^m [\lambda_\Gamma(s_a) + \lambda_\Gamma(s_a)^*], \quad \|A_S\| \leq 2m. \quad (2393)$$

For $t > 0$, $w > 0$, put

$$h_0 = -tA_S, \quad Q_1 = I + q_1A_S, \quad (2394)$$

$$Q_\infty = \sum_{\gamma \in \Gamma} q_\gamma \lambda_\Gamma(\gamma), \quad H_Q(w) = \begin{pmatrix} h_0 & wQ \\ wQ & h_0 \end{pmatrix}. \quad (2395)$$

Here Q is either Q_1 or Q_∞ . The target is specified as the layer-even sector of the exact layer-exchange symmetry.

The intralayer terms in Eq. (2394) are geodesic nearest-neighbor hopping on the regular orbit. The interlayer coefficients in Eq. (2392) are values of one radial exponential law at the actual full distances in Eq. (2390). Hence both parts of H_Q arise from one microscopic hyperbolic bilayer geometry.

Theorem 120 (Schur convergence and differentiated physical tails). Assume that, for some $\rho > 0$, $C_\Gamma \geq 1$, and $h_\Gamma > 0$, the later shells satisfy

$$\#\{\gamma : n\rho \leq r_\gamma < (n+1)\rho\} \leq C_\Gamma e^{h_\Gamma n\rho}. \quad (2396)$$

Assume also that the first omitted shell starts at $N_2\rho > \ell_1$. If $\mu > h_\Gamma$, then Q_∞ is bounded and self-adjoint and

$$\|Q_\infty - Q_1\| \leq E_0(\mu), \quad E_0(\mu) = C_\Gamma \sum_{n=N_2}^{\infty} e^{-(\mu-h_\Gamma)n\rho}. \quad (2397)$$

Let $\mathbf{n} : \Gamma \rightarrow \mathbb{Z}^m$ be the abelianization map associated with the marked generators. Let G be the positive-definite canonical Hodge metric on $H^1(\Gamma; \mathbb{R})$, and assume

$$\|\mathbf{n}(\gamma)\|_{G^{-1}} \leq C_A[1 + (n+1)\rho] \quad (2398)$$

on the n -th shell. The character tail and its first two derivatives obey

$$\|D_k^j R(k)\|_G \leq E_j(\mu), \quad 0 \leq j \leq 2, \quad (2399)$$

where

$$E_j(\mu) = C_\Gamma C_A^j \sum_{n=N_2}^{\infty} [1 + (n+1)\rho]^j e^{-(\mu-h_\Gamma)n\rho}. \quad (2400)$$

For a distance cutoff $L \geq N_2\rho$, the same conclusions hold with the sums beginning at $\lfloor L/\rho \rfloor$. In particular, each tail is a polynomial in L times $e^{-(\mu-h_\Gamma)L}$, up to an explicit constant depending only on the displayed shell data.

Proof. The shell-row sum of the omitted kernel is bounded by

$$\sum_{n=N_2}^{\infty} C_\Gamma e^{h_\Gamma n\rho} e^{-\mu n\rho} = E_0(\mu). \quad (2401)$$

The column sum has the same bound because $r_{\gamma^{-1}} = r_\gamma$. The Schur test gives Eq. (2397); coefficient symmetry gives self-adjointness. Every character derivative inserts at most one factor of $\mathbf{n}(\gamma)$. Equation (2398), followed by the same shell sum, proves Eqs. (2399) and (2400). Replacing N_2 by $\lfloor L/\rho \rfloor$ proves the cutoff statement. The ratio of consecutive exponential factors is $e^{-(\mu-h_\Gamma)\rho} < 1$; differentiating the resulting geometric series in that ratio gives the asserted polynomial-times-exponential form. $\square$

Exact character block and first-shell root

Definition 36 (Declared character sector and Hodge scalar). For $k \in \mathbb{T}^m$, let $\chi_k(s_a) = e^{ik_a}$, and define

$$C(k) = \sum_{a=1}^m \cos k_a, \quad C_S = \sum_{a=1}^m dk_a \otimes dk_a, \quad (2402)$$

$$c_S = \frac{1}{m} \text{tr}_G C_S > 0.$$

For a twice differentiable energy branch $E(k; w)$, define its normalized Hodge curvature at the trivial character by

$$\mathcal{F}[E](w) = \frac{1}{2tmc_S} \text{tr}_G \text{Hess}_k E(0; w). \quad (2403)$$

Theorem 121 (Exact natural first-shell root and complete isolation). Assume

$$b_0 = 1 - 2mq_1 > 0, \quad \text{equivalently} \quad \mu\ell_1 > \log(2m). \quad (2404)$$

The exact character block of $H_{Q_1}(w)$ is

$$H_1(k; w) = \begin{pmatrix} -2tC(k) & w[1 + 2q_1C(k)] \\ w[1 + 2q_1C(k)] & -2tC(k) \end{pmatrix}. \quad (2405)$$

Its layer-even and layer-odd branches are

$$E_+(k; w) = w + 2(wq_1 - t)C(k), \quad (2406)$$

$$E_-(k; w) = -w - 2(wq_1 + t)C(k). \quad (2407)$$

The layer-even operational scalar and score are

$$F_1(w) = 1 - \frac{wq_1}{t}, \quad M_1(w) = \frac{1}{1 + F_1(w)^2}. \quad (2408)$$

They have the positive simple root and strict operational maximum

$$\begin{aligned} w_* &= \frac{t}{q_1} > 0, \quad F'_1(w_*) = -\frac{q_1}{t}, \\ M''_1(w_*) &= -2 \left(\frac{q_1}{t} \right)^2 < 0. \end{aligned} \quad (2409)$$

Moreover,

$$W_+(w) = 4m|wq_1 - t|, \quad W_+(w_*) = 0. \quad (2410)$$

This sector calculation has an independent complete-operator certificate. At w_* , layer parity diagonalizes the regular operator as

$$H_{1,+}(w_*) = \frac{t}{q_1}I, \quad H_{1,-}(w_*) = -\frac{t}{q_1}I - 2tA_S. \quad (2411)$$

Thus the specified target is the point island $\{E_* = t/q_1\}$, with complete-operator gap

$$g_* = 2t(q_1^{-1} - \|A_S\|) \geq 2t(q_1^{-1} - 2m) > 0. \quad (2412)$$

The circle $\Gamma_* = \{z : |z - E_*| = g_*/3\}$ is a valid Riesz contour, and its projector is exactly the layer-even projector P_+ .

Proof. Evaluation of the group-algebra coefficients in the character χ_k gives Eq. (2405). Exact layer parity then gives Eqs. (2406) and (2407). Since

$$\text{Hess}_k E_+(0; w) = 2(t - wq_1)C_S, \quad (2413)$$

Definition 36 gives F_1 , its root, and its derivative. Direct differentiation of $M_1 = (1 + F_1^2)^{-1}$ gives the strict-maximum formula. Since $C(k) \in [-m, m]$, the range of E_+ has diameter $4m|wq_1 - t|$.

On the full regular fiber, layer parity changes H_{Q_1} into $h_0 + wQ_1$ and $h_0 - wQ_1$. Substitution of $w_* = t/q_1$ proves Eq. (2411). For every $\lambda \in \text{spec}(A_S)$, the distance between E_* and $-t/q_1 - 2t\lambda$ is at least g_* . Equation (2404) and $\|A_S\| \leq 2m$ make this quantity positive. The contour and projector statements follow from the spectral theorem and exact layer parity. $\square$

Proposition 69 (Microscopic active overlap). *The first-shell active coefficient at the root is*

$$w_*q_1 = t > 0. \quad (2414)$$

The target is also perfectly layer coherent because it is the exact layer-even isotypic sector. These are quantitative facts, not inferences from symmetry allowance alone.

Full exponential kernel and continued root

Let the character series of the full kernel be

$$\tau_\infty(k) = 1 + 2q_1C(k) + R(k). \quad (2415)$$

Theorem 120 makes R twice continuously differentiable.

Theorem 122 (Quantitative root for the full physical kernel). *Define*

$$\beta_2 = \frac{1}{2mc_S} \text{tr}_G D_k^2 R(0), \quad b_2 = \frac{E_2(\mu)}{2c_S}, \quad q_{\text{eff}} = q_1 - \beta_2. \quad (2416)$$

Then $|\beta_2| \leq b_2$. If $b_2 < q_1$, the full layer-even Hodge scalar is

$$F_\infty(w) = 1 - \frac{wq_{\text{eff}}}{t}, \quad (2417)$$

and it has exactly one positive root w_∞ . Quantitatively,

$$\frac{t}{q_1 + b_2} \leq w_\infty \leq \frac{t}{q_1 - b_2}, \quad (2418)$$

$$|w_\infty - w_*| \leq d_w := \frac{tb_2}{q_1(q_1 - b_2)}, \quad (2419)$$

$$|F'_\infty(w)| \geq \frac{q_1 - b_2}{t}. \quad (2420)$$

If $b_2 \leq q_1/8$, set

$$h_w = \frac{w_*}{10} + 2d_w, \quad I_w = [w_* - h_w, w_* + h_w]. \quad (2421)$$

Then $I_w \subset (0, \infty)$, $w_\infty \in \text{int } I_w$, and

$$\text{dist}(w_\infty, \partial I_w) \geq \frac{w_*}{10} + d_w. \quad (2422)$$

Finally,

$$M_\infty(w) = \frac{1}{1 + F_\infty(w)^2} \quad (2423)$$

has its unique strict maximum on I_w at w_∞ , with

$$M''_\infty(w_\infty) = -2 \left(\frac{q_{\text{eff}}}{t} \right)^2 \leq -2 \left(\frac{q_1 - b_2}{t} \right)^2 < 0. \quad (2424)$$

Proof. The tensor trace is bounded by m times its G -operator norm, so Theorem 120 gives $|\beta_2| \leq b_2$. The full layer-even character branch is

$$E_{\infty,+}(k; w) = -2tC(k) + w\tau_\infty(k). \quad (2425)$$

Taking its Hodge-traced Hessian gives Eq. (2417). The inequality $b_2 < q_1$ makes $q_{\text{eff}} > 0$; hence $w_\infty = t/q_{\text{eff}}$ is the only positive zero. The bounds $q_1 - b_2 \leq q_{\text{eff}} \leq q_1 + b_2$ prove Eqs. (2418)–(2420).

If $b_2 \leq q_1/8$, then

$$\frac{d_w}{w_*} = \frac{b_2}{q_1 - b_2} \leq \frac{1}{7}, \quad \frac{h_w}{w_*} \leq \frac{1}{10} + \frac{2}{7} < 1. \quad (2426)$$

Thus I_w is positive. The displacement bound gives Eq. (2422). The score statements follow by direct differentiation exactly as in Theorem 121. $\square$

The scalar equation in Theorem 122 cancels the declared Hodge trace. It does not assert directional tensor cancellation after the later shells are restored.

Theorem 123 (Complete-operator persistence, gap, and coherence). *Assume the hypotheses of Theorem 122, and put*

$$Q_{\max} = 1 + 2mq_1, \quad w_+ = \frac{t}{q_1 - b_2}, \quad \varepsilon_H = w_+ E_0(\mu) + Q_{\max} d_w \quad (2427)$$

If

$$E_0(\mu) < b_0, \quad \varepsilon_H < \frac{g_*}{6}, \quad (2428)$$

then $H_{Q_\infty}(w_\infty)$ is bounded self-adjoint and

$$\|H_{Q_\infty}(w_\infty) - H_{Q_1}(w_*)\| \leq \varepsilon_H. \quad (2429)$$

Its complete regular spectrum has a layer-even target island satisfying

$$W_{\infty,+} \leq 2\varepsilon_H, \quad \Delta_{\infty,+} \geq g_* - 2\varepsilon_H > \frac{2g_*}{3}. \quad (2430)$$

The contour Γ_* remains valid, and the integrated Riesz projector is exactly P_+ . The full character kernel is uniformly positive:

$$\tau_\infty(k) \geq b_0 - E_0(\mu) > 0. \quad (2431)$$

The retained active overlap obeys

$$w_\infty q_1 \geq \frac{tq_1}{q_1 + b_2} > 0. \quad (2432)$$

Set

$$r_\Gamma = \frac{g_*}{3}, \quad \delta_\Gamma = r_\Gamma - \varepsilon_H > 0. \quad (2433)$$

For any further self-adjoint perturbation $\tilde{H} = H_{Q_\infty}(w_\infty) + V$ with $e = \|V\| < \delta_\Gamma$, the same contour defines $\tilde{P}$, and

$$\|\tilde{P} - P_+\| \leq \frac{r_\Gamma e}{\delta_\Gamma(\delta_\Gamma - e)}. \quad (2434)$$

In particular, if the right-hand side is less than one, the target rank is constant, including when that rank is infinite.

Proof. Theorem 120 gives

$$\|H_{Q_\infty}(w_\infty) - H_{Q_1}(w_*)\| \leq w_\infty E_0(\mu) + |w_\infty - w_*| \|Q_1\|. \quad (2435)$$

Use $w_\infty \leq w_+$, $\|Q_1\| \leq Q_{\max}$, and Eq. (2419) to obtain Eq. (2429). The spectral perturbation bound gives the target bandwidth and the factor-two gap loss in Eq. (2430).

Both full operators commute with exact layer exchange. The entire even spectrum stays inside the contour and the entire odd spectrum stays outside. Consequently the contour projector is the exact layer-even

projector P_+ , without a finite-rank assumption. The first-shell character kernel is at least b_0 , while the omitted scalar tail has modulus at most E_0 . This proves Eq. (2431). The root interval proves Eq. (2432).

On Γ_* , the full resolvent norm is at most δ_Γ^{-1} , and the perturbed resolvent norm is at most $(\delta_\Gamma - e)^{-1}$. The resolvent identity and the contour length $2\pi r_\Gamma$ give Eq. (2434). Projectors at operator-norm distance less than one have isomorphic ranges. $\square$

Corollary 70 (A nonempty open microscopic parameter class). *Fix the regular 4g-gon crystal, $t > 0$, $h > 0$, $\ell_1 > 0$, and shell constants in Theorem 120, with $N_2\rho > \ell_1$. There exists $\mu_0 < \infty$ such that every $\mu > \mu_0$ satisfies all strict hypotheses of Theorems 121–123. Equivalently, the admissible set contains a nonempty open interval of positive hopping lengths $0 < \lambda_\perp < \mu_0^{-1}$.*

Proof. The first-shell weight is $q_1 = e^{-\mu\ell_1}$. Because $N_2\rho > \ell_1$, Eq. (2400) gives

$$\frac{E_j(\mu)}{q_1} \rightarrow 0, \quad 0 \leq j \leq 2, \quad \mu \rightarrow \infty. \quad (2436)$$

Hence $b_0 > 0$, $b_2 \leq q_1/8$, and $E_0 < b_0$ eventually. Moreover, $d_w = o(w_*)$, while g_* and w_* are both asymptotic to $2t/q_1$ and t/q_1 , respectively, and $w_+ E_0 = o(t/q_1)$. Therefore $\varepsilon_H/g_* \rightarrow 0$, proving the remaining gap condition. All inequalities are strict and depend continuously on μ , so the resulting class is open. $\square$

Representation, normalization, and subsequent firewalls

Proposition 70 (Scope of the positive-root result). *The conclusions above have the following precise scope.*

1. The character calculation is an exact one-dimensional automorphic sector. For a nonamenable surface group it is not, by itself, a complete description of the reduced regular spectrum and need not be weakly contained in the regular representation.
2. The complete-operator gap and Riesz statements do not use character completeness. They follow independently from the regular group-algebra identity in Eq. (2411) and the Schur estimate in Eq. (2429).
3. A finite quotient is spectrally complete only after all of its irreducible blocks and their exact multiplicity weights are included. A single character block cannot certify such completeness.
4. A geometric normalization change rescales the displayed coupling coordinate but does not create or remove the operator identity. Any comparison across normalizations must transform the root and energy scale covariantly.

5. The square five-state truncation remains root-free. The present positive root belongs to a different, explicitly declared surface-group model with a complete first shell and a controlled physical exponential tail.
6. No named material is identified, and no cross-cover bulk claim follows without the separate admissibility and no-pollution hypotheses required for global spectral convergence.

Proof. The first statement is the distinction between a quotient character and the left regular representation. The second is exactly Theorems 121 and 123. The third follows from finite-group block decomposition and its multiplicity weights. The remaining statements record the hypotheses actually used in the proofs and prevent changing the model, normalization, or convergence topology after the root has been found. $\square$

The subsequent replacement is therefore conditional but constructive: a formula requiring an existing simple operational root may use w_∞ together with Eqs. (2418), (2420), and (2430). A formula requiring directional tensor cancellation, representation completeness, or cover-independent bulk edges must retain its separate hypotheses.

Full Hodge-space cancellation in a natural microscopic class

This subsection upgrades a scalar Hodge-trace zero to cancellation of the entire tangent Hessian. It uses the natural regular-polygon bilayer of Definition 35, its differentiated Schur estimate in Theorem 120, and its specified layer-even target. The invariant Hessian convention is the one fixed by Theorem 17 and Proposition 48.

The complete tangent response

Definition 37 (Full Hodge response tensor). *Let*

$$V = H^1(\Gamma; \mathbb{R}) \quad (2437)$$

carry the positive canonical Hodge metric G of Definition 9. Let $\mathbf{n} : \Gamma \rightarrow V^$ be the abelianization covector, and let $q_\gamma = q_{\gamma^{-1}} > 0$ be the full-distance exponential coefficient of the centered microscopic bilayer. Define*

$$C_S = \sum_{a=1}^m dk_a \otimes dk_a \quad (2438)$$

and

$$B_\infty = \sum_{\gamma \in \Gamma} q_\gamma \mathbf{n}(\gamma) \otimes \mathbf{n}(\gamma). \quad (2439)$$

The sum is interpreted in the operator norm induced by G .

For the layer-even character branch

$$E_+(k; w) = \epsilon(k) + w\tau_\infty(k), \quad (2440)$$

its full Hodge response is the bilinear form

$$K(w) = \text{Hess}_k E_+(0; w). \quad (2441)$$

Proposition 71 (Every tangent component). *The derivative-weighted Schur hypotheses of Theorems 120 and 49 make Eq. (2439) absolutely convergent. For arbitrary $X_A, X_B \in V$,*

$$K_{AB}(w) = 2t \sum_{a=1}^m dk_a(X_A) dk_a(X_B) - w \sum_{\gamma \in \Gamma} q_\gamma \mathbf{n}(\gamma)(X_A) \mathbf{n}(\gamma)(X_B). \quad (2442)$$

Equivalently,

$$K(w) = 2tC_S - wB_\infty. \quad (2443)$$

Every parameter derivative is therefore explicit:

$$\partial_w K_{AB}(w) = -(B_\infty)_{AB}, \quad \partial_w^2 K_{AB}(w) = 0. \quad (2444)$$

Moreover,

$$B_\infty \succeq 2q_1 C_S \succ 0. \quad (2445)$$

Proof. Inversion symmetry pairs γ and γ^{-1} , so

$$D_k \tau_\infty(0) = 0 \quad (2446)$$

and

$$D_k^2 \tau_\infty(0) = -B_\infty. \quad (2447)$$

The intralayer symbol is $\epsilon(k) = -2t \sum_a \cos k_a$, hence

$$D_k^2 \epsilon(0) = 2tC_S. \quad (2448)$$

The C^2 Schur sum justifies termwise differentiation and proves Eqs. (2442)–(2444). The two first-shell elements s_a and s_a^{-1} contribute $2q_1 dk_a \otimes dk_a$ for every a . Every later contribution is positive semidefinite, which proves Eq. (2445). $\square$

Equation (2442) is covariant under every coordinate Jacobian. Consequently the normalized endomorphism

$$\mathcal{K}(w) = G^{-1/2} K(w) G^{-1/2} \quad (2449)$$

has the coordinate-invariant principal spectrum specified by Theorem 75.

A finite-symmetry certificate

Definition 38 (Microscopic Hodge-isotropy certificate). *Let $\mathcal{P}$ be a finite group of actual centered-bilayer isometries. It is an admissible Hodge-isotropy group when all of the following hold.*

1. *Its action U_p on V is G -orthogonal.*
2. *It signed-permutes the marked first-shell covectors.*
3. *It preserves every full distance, and therefore $q_{p\gamma p^{-1}} = q_\gamma$.*
4. *Its G -self-adjoint commutant is scalar:*

$$\{A = A^{*G} : AU_p = U_p A \text{ for every } p \in \mathcal{P}\} = \mathbb{R}I. \quad (2450)$$

Theorem 124 (Point-group forcing of the scalar carrier). *If Definition 38 holds, then there are strictly positive scalars c_S, d_∞, b_∞ such that*

$$C_S = c_S G, \quad B_\infty = d_\infty G, \quad B_\infty = b_\infty C_S, \quad b_\infty = \frac{d_\infty}{c_S}. \quad (2451)$$

Proof. Signed permutation of the marked shell preserves C_S . Full-distance invariance and the change of variable $\gamma \mapsto p\gamma p^{-1}$ preserve B_∞ . Raising one index with G turns both forms into G -self-adjoint endomorphisms commuting with every U_p . Equation (2450) makes each endomorphism scalar. Positivity of G , C_S , and Eq. (2445) makes all three scalars positive. $\square$

Proposition 72 (Concrete nonempty octagon realization). *The centered genus-two surface obtained from a regular hyperbolic octagon with opposite-side pairing satisfies Definition 38. In a G -orthonormal marked basis, a quaternion subgroup of its centered isometry group acts by*

$$I_0 = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad (2452)$$

and

$$J_0 = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}. \quad (2453)$$

These matrices obey

$$I_0^2 = J_0^2 = -I, \quad I_0 J_0 = -J_0 I_0. \quad (2454)$$

The realization remains admissible for the nonempty open small- $\lambda_\perp$ region of Corollary 70.

Proof. The centered quotient isometries that induce I_0 and J_0 send each paired oriented side to another paired oriented side. They therefore descend to centered isometries of the quotient and signed-permute the four marked first-shell covectors. Direct multiplication proves Eq. (2454). For every nonzero $v \in V$,

$$\text{span}_{\mathbb{R}}\{v, I_0 v, J_0 v, I_0 J_0 v\} = V. \quad (2455)$$

Thus the quaternion subgroup acts real irreducibly. A G -self-adjoint commuting endomorphism has invariant eigenspaces, so irreducibility already forces it to be scalar.

Equivalently, identify the real Hodge space with the quaternions. The matrices I_0 and J_0 are left multiplication by the quaternion units. An endomorphism commuting with both is right multiplication by a quaternion r . Its G -adjoint is right multiplication by $\bar{r}$. It is self-adjoint exactly when $r = \bar{r}$, hence when $r \in \mathbb{R}$. This proves Eq. (2450). Centered isometries preserve $d_{\mathbb{H}}$, the fixed layer separation, and every full distance. Consequently they preserve the entire exponential kernel, not only its first shell. The strict Schur and gap inequalities remain true on the open region already constructed in Corollary 70. $\square$

Exact simultaneous locking

Theorem 125 (Necessary and sufficient full-cancellation criterion). *There exists $w > 0$ for which $K(w) = 0$ on all of $V \times V$ if and only if*

$$B_\infty = b_\infty C_S \quad (2456)$$

for one $b_\infty > 0$. In that event the root is unique and equals

$$w_H = \frac{2t}{b_\infty}. \quad (2457)$$

At w_H ,

$$K(w_H)(X, Y) = 0 \quad (2458)$$

for every $X, Y \in V$, and every normalized covector response $K(w_H)(X, \cdot)$ vanishes.

Under the symmetry certificate, all principal eigenvalues of $\mathcal{K}(w)$ are locked to

$$\lambda_j(w) = c_S(2t - w b_\infty), \quad 1 \leq j \leq m. \quad (2459)$$

They vanish simultaneously at w_H , with the common nonzero slope

$$\lambda'_j(w_H) = -c_S b_\infty < 0. \quad (2460)$$

The scalar root of Theorem 122 agrees with w_H through the exact relation

$$q_{\text{eff}} = \frac{b_\infty}{2}. \quad (2461)$$

Proof. If $K(w) = 0$, Eq. (2443) gives $B_\infty = (2t/w)C_S$. Conversely, Eq. (2456) and Eq. (2457) make the complete bilinear form zero. Positivity of b_∞ gives uniqueness. Under the symmetry certificate, substitution of Eq. (2451) into Eq. (2449) gives

$$\mathcal{K}(w) = c_S(2t - wb_\infty)I. \quad (2462)$$

This proves simultaneous locking and its slope. Taking the normalized Hodge trace gives $1 - wb_\infty/(2t)$, while Theorem 122 gives $1 - wq_{\text{eff}}/t$. Equality proves Eq. (2461). $\square$

Exact and controlled self-energy response

Proposition 73 (All Feshbach tangent terms). *Work in a fixed transported fiber and let P be a fixed target projector, $Q = I - P$,*

$$\begin{aligned} V &= PHQ, \quad H_Q = QH_Q, \\ R &= (E - H_Q)^{-1}, \quad \Sigma = VRV^*. \end{aligned} \quad (2463)$$

For tangent directions A, B , put

$$L_A = D_A H_Q - (D_A E)I \quad (2464)$$

and

$$L_{AB} = D_{AB} H_Q - (D_{AB} E)I. \quad (2465)$$

Then $D_A R = RL_A R$, and

$$D_A \Sigma = V_A R V^* + V R V_A^* + V R L_A R V^*. \quad (2466)$$

The complete second response is

$$\begin{aligned} D_{AB} \Sigma &= V_{AB} R V^* + V R V_{AB}^* + V_A R V_B^* + V_B R V_A^* \\ &+ V_A R L_B R V^* + V R L_B R V_A^* + V_B R L_A R V^* + V R L_A R V_B^* \\ &+ V R L_B R L_A R V^* + V R L_{AB} R V^* + V R L_A R L_B R V^*. \end{aligned} \quad (2467)$$

If

$$\|R\| \leq \delta^{-1}, \quad \|V\| \leq v_0, \quad \|DV\| \leq v_1, \quad \|D^2 V\| \leq v_2 \quad (2468)$$

and

$$\|L_A\| \leq \ell_1, \quad \|L_{AB}\| \leq \ell_2, \quad (2469)$$

then

$$\begin{aligned} \|D^2 \Sigma\| &\leq \frac{2v_0 v_2 + 2v_1^2}{\delta} + \frac{4v_0 v_1 \ell_1 + v_0^2 \ell_2}{\delta^2} \\ &\quad + \frac{2v_0^2 \ell_1^2}{\delta^3}. \end{aligned} \quad (2470)$$

For the the microscopic-root theorem layer-even target, layer parity is a reducing symmetry at every k and w . Hence

$$P_+ H P_- = 0, \quad D^j(P_+ H P_-) = 0 \quad (1 \leq j \leq 2), \quad (2471)$$

so Σ , $D\Sigma$, and $D^2\Sigma$ vanish exactly.

Proof. Differentiation of $R(E - H_Q) = I$ gives $D_A R = RL_A R$. The product rule gives Eq. (2466). Differentiating it once more in direction B gives exactly the eleven terms in Eq. (2467). Applying the displayed norm bounds to those terms gives Eq. (2470). Exact layer exchange commutes with the complete microscopic Hamiltonian and its tangent derivatives. Thus its two parity subspaces reduce every operator in the derivative chain, which proves Eq. (2471). $\square$

The estimate in Eq. (2470) is the controlled off-resonant alternative to the exact parity reduction and is consistent with Proposition 22. For a nonreducing target, exact full cancellation additionally requires $D^2\Sigma$ to share the same scalar carrier. Smallness alone yields approximate, not exact, cancellation.

The anisotropic obstruction

Theorem 126 (Sharp obstruction at a trace root). *Without assuming Eq. (2456), define*

$$A = G^{-1/2} C_S G^{-1/2}, \quad D = G^{-1/2} B_\infty G^{-1/2} \quad (2472)$$

and the Hodge-trace root

$$w_{\text{tr}} = 2t \frac{\text{tr } A}{\text{tr } D}. \quad (2473)$$

Then

$$\mathcal{K}(w_{\text{tr}}) = 2t \left[A - \frac{\text{tr } A}{\text{tr } D} D \right] \quad (2474)$$

is traceless. Put

$$a_\infty = \left\| A - \frac{\text{tr } A}{\text{tr } D} D \right\|_{\text{op}}. \quad (2475)$$

Full cancellation at w_{tr} occurs if and only if $a_\infty = 0$, equivalently if and only if $B_\infty = b_\infty C_S$. If proportionality fails, then

$$\|\mathcal{K}(w_{\text{tr}})\|_{\text{op}} = 2ta_\infty > 0. \quad (2476)$$

In that case the principal responses include both a positive and a negative value, so no scalar adjustment of w cancels every direction.

Proof. Taking the trace of Eq. (2443) proves that Eq. (2473) is the unique trace zero and gives Eq. (2474). The residual vanishes precisely when A and D are proportional, which is equivalent to proportionality of C_S and B_∞ . This proves the exact norm identity. A nonzero traceless self-adjoint endomorphism in dimension $m = 2g \geq 4$ cannot be semidefinite, and therefore has eigenvalues of both signs. $\square$

Theorem 127 (Quantitative C^2 suppression and gap persistence). *Let $H_0(w)$ be a symmetric microscopic realization satisfying Theorem 124, and write*

$$s_H = c_S b_\infty > 0. \quad (2477)$$

Let $\tilde{H}(w)$ be a C^2 -admissible perturbation on $I = [w_H - h, w_H + h]$. Suppose its normalized tangent Hessian satisfies

$$\sup_{w \in I} \|\tilde{\mathcal{K}}(w) - \mathcal{K}(w)\| \leq \eta_2 \quad (2478)$$

and

$$\sup_{w \in I} \|\partial_w \tilde{\mathcal{K}}(w) - \partial_w \mathcal{K}(w)\| \leq \eta_{2,1} < s_H. \quad (2479)$$

Then every ordered principal eigenvalue obeys

$$|\tilde{\lambda}_j(w) - c_S(2t - wb_\infty)| \leq \eta_2, \quad (2480)$$

and

$$\max_{j,l} |\tilde{\lambda}_j(w) - \tilde{\lambda}_l(w)| \leq 2\eta_2. \quad (2481)$$

If

$$h > \frac{\eta_2}{s_H - \eta_{2,1}}, \quad (2482)$$

the normalized trace has a unique zero $\tilde{w}_{\text{tr}} \in I$, with

$$|\tilde{w}_{\text{tr}} - w_H| \leq \frac{\eta_2}{s_H - \eta_{2,1}}. \quad (2483)$$

At that point

$$|\tilde{\lambda}_j(\tilde{w}_{\text{tr}})| \leq 2\eta_2. \quad (2484)$$

If the perturbation retains the certificate in Definition 38, the eigenvalues remain exactly locked and their common zero is exact.

Let the the microscopic-root theorem complete-operator constants be $g_ > 0$, $\varepsilon_H < g_*/6$,*

$$r_\Gamma = \frac{g_*}{3}, \quad \delta_\Gamma = r_\Gamma - \varepsilon_H. \quad (2485)$$

Assume the total common-space operator error at the selected perturbed root, including the scan displacement, is

$$\eta_0 = \|\tilde{H}(\tilde{w}_{\text{tr}}) - H_0(w_H)\| < \delta_\Gamma. \quad (2486)$$

Then the target island has

$$\tilde{W} \leq 2(\varepsilon_H + \eta_0) \quad (2487)$$

and

$$\tilde{\Delta} \geq g_* - 2(\varepsilon_H + \eta_0). \quad (2488)$$

Its Riesz projector satisfies

$$\|\tilde{P} - P_+\| \leq \frac{r_\Gamma \eta_0}{\delta_\Gamma(\delta_\Gamma - \eta_0)}. \quad (2489)$$

Thus rank and quantitative layer coherence persist whenever the right-hand side is less than one.

Proof. Equation (2480) is Weyl stability for the invariant Hessian spectrum, as in Theorem 18. Subtracting two copies gives Eq. (2481). The normalized mean eigenvalue has derivative at most $-(s_H - \eta_{2,1}) < 0$. Its value at w_H has modulus at most η_2 . The mean-value theorem, the endpoint condition, and strict monotonicity prove existence, uniqueness, and Eq. (2483). At the trace root the ideal scalar value has modulus at most η_2 ; another application of Eq. (2480) proves Eq. (2484). If the commutant certificate remains valid, Theorem 124 makes the entire perturbed pencil scalar, so the common zero is exact.

The complete operator in Theorem 123 is within ε_H of the point-island reference. Adding the total perturbation η_0 , the Hausdorff spectral bound and the sharp factor-two gap estimate give Eqs. (2487) and (2488). On the the microscopic-root theorem contour, the unperturbed full resolvent is bounded by δ_Γ^{-1} , while the perturbed resolvent is bounded by $(\delta_\Gamma - \eta_0)^{-1}$. The resolvent identity and contour length $2\pi r_\Gamma$ give Eq. (2489). Projectors at norm distance below one have isomorphic ranges, and their overlap changes by at most the same norm. $\square$

The C^2 hypothesis in Theorem 127 is not a consequence of base Schur convergence. It requires the derivative-weighted full-distance curvature chain of Proposition 31 and the local Hermitian expansion of Theorem 50. Active operator, RMS, and singular scales remain distinct as in Theorem 51.

Scope and subsequent replacement

Proposition 74 (Full-cancellation separation theorem). *The symmetric octagon class proves a nonempty natural microscopic realization with one positive root that cancels the complete canonical Hodge Hessian, while retaining the the microscopic-root theorem gap, active overlap, and layer coherence. The following limitations remain mandatory.*

1. *Outside the proportionality class, a Hodge-trace zero leaves the exact residual in Eq. (2476).*
2. *A controlled self-energy response is not an exact cancellation unless it has the same scalar carrier; exact parity reduction avoids this issue in the declared target.*
3. *The character tensor is an exact automorphic-sector calculation. It does not imply finite-cover completeness or reduced-regular completeness, in accord with Proposition 70.*
4. *The square five-state model remains root-free. The positive root here belongs to the regular-octagon surface-group bilayer with its full physical kernel.*
5. *Coordinate invariance does not remove normalization dependence. The canonical metric and fixed Hessian convention must be retained when comparing roots or slopes.*

6. *No named material, arbitrary-cover universality, or claim established elsewhere is made.*

Proof. The positive realization and its common slope are Proposition 72 and Theorem 125. Complete spectral persistence is Theorem 127. The obstruction and self-energy qualifications are Theorem 126 and Proposition 73. The remaining items state the representation, normalization, and model boundaries used in those proofs. $\square$

Subsequent statements that require cancellation of every Hodge direction may therefore use $w_H = 2t/b_\infty$ only after verifying Eq. (2450) or directly verifying Eq. (2456). A trace-root certificate alone does not condition that replacement.

established finite-cover approximation of global bulk spectra

The statement below combines the Schur and cover controls of Theorems 54 and 58 with the two directed spectral conditions isolated in Theorems 63 and 64. The no-pollution hypothesis is placed on the same retained sector as the no-loss construction.

Retained sectors and directed certificates

Definition 39 (Uniform retained-sector family). *Let J be a compact parameter interval and let*

$$H(p) = \sum_{\gamma \in \Gamma} h_\gamma(p) \lambda_\Gamma(\gamma) \quad (2490)$$

be a uniformly bounded self-adjoint matrix-valued group kernel on $\ell^2(\Gamma) \otimes \mathbb{C}^d$. For a finite quotient representation π_N , let $\mathcal{B}_N$ be a declared invariant retained sector and define

$$H_N^b(p) = \sum_{\gamma \in \Gamma} h_\gamma(p) \pi_N(\gamma) \Big|_{\mathcal{B}_N}. \quad (2491)$$

Write

$$\Sigma(p) = \text{spec } H(p), \quad S_N(p) = \text{spec } H_N^b(p). \quad (2492)$$

All spectra are contained in one fixed compact interval $K = [-M, M]$.

Definition 40 (Directed errors). *For nonempty compact subsets of $\mathbb{R}$, define*

$$\varepsilon_N^{\text{loss}}(p) = \sup_{\lambda \in \Sigma(p)} \text{dist}(\lambda, S_N(p)) \quad (2493)$$

and

$$\varepsilon_N^{\text{poll}}(p) = \sup_{s \in S_N(p)} \text{dist}(s, \Sigma(p)). \quad (2494)$$

Their maximum is exactly the Hausdorff distance:

$$d_H(S_N(p), \Sigma(p)) = \max\{\varepsilon_N^{\text{loss}}(p), \varepsilon_N^{\text{poll}}(p)\}. \quad (2495)$$

Suppose first that $H(p)$ has propagation at most L , uniformly on J . For $R > 0$, define the finite-support Weyl modulus

$$\omega(R) = \sup_{p \in J} \sup_{\lambda \in \Sigma(p)} \inf\{\|(H(p) - \lambda)u\| : \|u\| = 1, \text{supp } u \subset B_R(e)\}. \quad (2496)$$

Definition 41 (Quantitative retained lift). *Choose $R_N > 0$ with*

$$2(R_N + L) < r_{\text{inj}, N}. \quad (2497)$$

A retained lift has defect $\rho_N \geq 0$ if, for every $p \in J$, $\lambda \in \Sigma(p)$, and unit u supported in $B_{R_N}(e)$, there is a unit lift $v \in \mathcal{B}_N$ satisfying

$$\|(H_N^b(p) - \lambda)v\| \leq \|(H(p) - \lambda)u\| + \rho_N. \quad (2498)$$

The resulting computable no-loss certificate is

$$a_N = \omega(R_N) + \rho_N. \quad (2499)$$

For a norm-continuous finite-propagation family on compact J , finite support density and a finite parameter net give

$$\omega(R) \longrightarrow 0 \quad \text{as } R \longrightarrow \infty. \quad (2500)$$

The separate defect ρ_N records whether the lifted local vector is actually represented in the declared retained sector.

Definition 42 (Residual no-pollution certificate). *A retained operator has no-pollution error $b_N \geq 0$ when, uniformly for $p \in J$, $x \in K$, and $v \in \mathcal{B}_N$,*

$$\|(H_N^b(p) - x)v\| \geq [\text{dist}(x, \Sigma(p)) - b_N]_+ \|v\|. \quad (2501)$$

Here $[y]_+ = \max\{y, 0\}$. The inequality must hold on the entire retained sector, not merely after deleting one selected representation.

Proposition 75 (Checkable sufficient no-pollution mechanisms). *Each of the following is sufficient for Definition 42.*

1. *A direct verification of Eq. (2501).*
2. *For every $x \in K$, a transported finite-propagation parametrix $R_{p,x,N}$ with two-sided defect $\beta_{p,x,N} < 1$ and*

$$\frac{1 - \beta_{p,x,N}}{\|R_{p,x,N}\|} \geq [\text{dist}(x, \Sigma(p)) - b_N]_+. \quad (2502)$$
3. *Exact contractive reduced-norm domination on the retained representation, equivalently weak containment in the regular representation. In this case $b_N = 0$ for every finite group-algebra kernel.*
4. *A quantitative two-sided quasiunitary resolvent comparison that yields Eq. (2501) with the stated b_N .*

Approximate reduced polynomial-norm domination is also sufficient when its verified separation error is inserted into b_N .

Proof. The first item is the definition. For the second item, the Neumann lemma gives

$$\|(H_N^b(p) - x)^{-1}\| \leq \frac{\|R_{p,x,N}\|}{1 - \beta_{p,x,N}}. \quad (2503)$$

Equations (2502) and (2503), together with the spectral theorem, give Eq. (2501).

In the third item, weak containment means that the retained representation factors through the reduced group C^* -algebra. The spectrum of the image of a self-adjoint element is then contained in its reduced spectrum. The fourth item gives a uniform inverse whenever the infinite resolvent exists outside the declared error neighborhood. Polynomial separation is the same argument applied to a verified polynomial approximate inverse. $\square$

The weak-containment statement is precisely a condition, not an automatic property of automorphic blocks, as emphasized by Theorem 71.

The finite-propagation theorem

Theorem 128 (established two-sided finite-cover approximation). *Assume Definitions 39, 41, and 42. Then, uniformly for $p \in J$,*

$$\varepsilon_N^{\text{loss}}(p) \leq a_N, \quad \varepsilon_N^{\text{poll}}(p) \leq b_N \quad (2504)$$

and therefore

$$d_H(S_N(p), \Sigma(p)) \leq \eta_N^{\text{fp}} := \max\{a_N, b_N\}. \quad (2505)$$

If $R_N \rightarrow \infty$, $\rho_N \rightarrow 0$, and $b_N \rightarrow 0$, then the convergence is uniform on J .

Proof. Fix $p \in J$ and $\lambda \in \Sigma(p)$. Choose a finite-support unit vector in Eq. (2496) with residual arbitrarily close to $\omega(R_N)$. The injectivity buffer makes its patch and the action of $H(p)$ lift exactly. Equation (2498) gives a unit retained vector with residual at most a_N . The residual-to-spectrum inequality gives

$$\text{dist}(\lambda, S_N(p)) \leq a_N. \quad (2506)$$

Taking the supremum proves the no-loss bound.

For $s \in S_N(p)$, insert a normalized eigenvector with eigenvalue s in Eq. (2501). Its left side is zero, so

$$\text{dist}(s, \Sigma(p)) \leq b_N. \quad (2507)$$

Taking the supremum proves the no-pollution bound. The Hausdorff identity (2495) proves the sharp maximum estimate. Uniform decay follows from Eq. (2500) and the two assumed defect limits. $\square$

Normal residual injectivity supplies exact local patches. A Farber sequence only controls the fraction of bad rooted patches, as separated in Definition 5 and Proposition 42. Such typical-root control may contribute to a no-loss or weak-DOS argument, but it does not prove the uniform residual bound in Eq. (2501).

Physical exponential tails

Theorem 129 (Diagonal-cutoff exponential extension). *Assume the block kernel has shell growth*

$$\#S_n \leq C_X e^{h_\Lambda n \rho} \quad (2508)$$

and uniform operator envelope

$$\|h_\gamma(p)\| \leq C e^{-\mu d(e, \gamma)}, \quad \mu > h_\Lambda. \quad (2509)$$

Put $\delta = \mu - h_\Lambda > 0$. For the diagonal cutoff at distance L , the infinite and finite retained tails obey

$$\tau_\infty(L) = \sup_{p \in J} \|H(p) - H^{(L)}(p)\| \leq C_\infty e^{-\delta L} \quad (2510)$$

and

$$\tau_N(L) = \sup_{p \in J} \|H_N^b(p) - H_N^{b,(L)}(p)\| \leq C_N e^{-\delta L}. \quad (2511)$$

One admissible explicit infinite-volume constant is

$$C_\infty = \frac{CC_X e^{\delta \rho}}{1 - e^{-\delta \rho}}, \quad (2512)$$

and C_N is the verified finite block-row analogue.

Choose $L_N \rightarrow \infty$ and $R_N \rightarrow \infty$ so that

$$2(R_N + L_N) < r_{\text{inj}, N}, \quad \frac{L_N}{r_{\text{inj}, N}} \rightarrow 0. \quad (2513)$$

Suppose the common truncated pair has directed certificates $a_N^{(L_N)}$ and $b_N^{(L_N)}$. Then

$$d_H(S_N(p), \Sigma(p)) \leq \eta_N \quad (2514)$$

uniformly on J , where

$$\eta_N = \tau_\infty(L_N) + \tau_N(L_N) + \max\{a_N^{(L_N)}, b_N^{(L_N)}\}. \quad (2515)$$

Consequently $\eta_N \rightarrow 0$ whenever both truncated certificates vanish.

Proof. The shell Schur sum is

$$CC_X \sum_{n=\lfloor L/\rho \rfloor}^{\infty} e^{-\delta n \rho}. \quad (2516)$$

Summing the geometric series and using $\lfloor L/\rho \rfloor \rho > L - \rho$ proves Eqs. (2510) and (2512). The verified finite block-row sum proves Eq. (2511). The cutoff buffer makes the truncated local matrices lift exactly.

Theorem 57 moves each truncated spectrum by at most its operator tail. For either directed error, pass from the full infinite spectrum to its truncation at cost τ_∞ , apply the corresponding truncated certificate, and pass from the finite truncation to the full finite operator at cost τ_N . Thus

$$\varepsilon_N^{\text{loss}} \leq \tau_\infty + \tau_N + a_N^{(L_N)} \quad (2517)$$

and

$$\varepsilon_N^{\text{poll}} \leq \tau_\infty + \tau_N + b_N^{(L_N)}. \quad (2518)$$

Taking their maximum proves Eq. (2515). The exponential tails vanish because $L_N \rightarrow \infty$. $\square$

This is the physical-tail extension of Theorem 65. The entire error, including both tails, must be divided by one fixed positive intrinsic energy scale in every normalized statement.

Edges, islands, and gaps

Theorem 130 (Global spectral consequences). *Let S_N and Σ be nonempty compact subsets of $\mathbb{R}$ with*

$$d_H(S_N, \Sigma) \leq \eta_N. \quad (2519)$$

Then the global edges satisfy

$$|\inf S_N - \inf \Sigma| \leq \eta_N \quad (2520)$$

and

$$|\sup S_N - \sup \Sigma| \leq \eta_N. \quad (2521)$$

Let $I \subset \Sigma$ be a nonempty proper compact island with

$$\Delta = \text{dist}(I, \Sigma \setminus I) > 0. \quad (2522)$$

If $\eta_N < \Delta/3$, define the finite island by the fixed neighborhood

$$I_N = S_N \cap \{x : \text{dist}(x, I) < \Delta/2\}. \quad (2523)$$

Then

$$d_H(I_N, I) \leq \eta_N \quad (2524)$$

and

$$d_H(S_N \setminus I_N, \Sigma \setminus I) \leq \eta_N. \quad (2525)$$

Writing $W_N = \text{diam } I_N$, $W = \text{diam } I$, and

$$\Delta_N = \text{dist}(I_N, S_N \setminus I_N), \quad (2526)$$

one has

$$|\inf I_N - \inf I| \leq \eta_N, \quad |\sup I_N - \sup I| \leq \eta_N, \quad (2527)$$

$$|W_N - W| \leq 2\eta_N, \quad (2528)$$

and

$$|\Delta_N - \Delta| \leq 2\eta_N, \quad \Delta_N \geq \Delta - 2\eta_N > 0. \quad (2529)$$

All statements are uniform in $p \in J$ when $I(p)$, its partition, and a positive lower gap margin are continuous and uniform.

Proof. The extreme points of either compact set lie within η_N of the other, which proves Eqs. (2520) and (2521).

For $x \in I$, a point of S_N within η_N belongs to I_N . For $s \in I_N$, a point of Σ within η_N cannot belong to $\Sigma \setminus I$, because that complement is at distance at least Δ from I . This proves Eq. (2524). The same argument with the two parts interchanged proves Eq. (2525).

The edge estimates follow from island Hausdorff distance. The diameter map is 2-Lipschitz in Hausdorff distance. The distance between two compact sets changes by at most the sum of their two Hausdorff errors. These observations prove Eqs. (2528) and (2529). The same proof with suprema over p gives the uniform statement. $\square$

This theorem recovers the quantitative island transport of Theorem 60 and the global conclusions of Corollary 37 once the two directed certificates have been verified.

Riesz comparison and its topology boundary

Theorem 131 (Projector comparison requires an intertwiner). *Suppose first that self-adjoint H and $\hat{H}_N$ act on one Hilbert space and*

$$\|\hat{H}_N - H\| \leq e_N < \delta, \quad (2530)$$

where a rectifiable contour Γ_0 of length ℓ_{Γ_0} has distance at least δ from $\text{spec } H$. Then

$$\|(z - \hat{H}_N)^{-1} - (z - H)^{-1}\| \leq \frac{e_N}{\delta(\delta - e_N)} \quad (2531)$$

on Γ_0 , and the enclosed Riesz projectors satisfy

$$\|\hat{P}_N - P\| \leq \frac{\ell_{\Gamma_0} e_N}{2\pi\delta(\delta - e_N)}. \quad (2532)$$

For operators on different spaces, let $J_N : \mathcal{B}_N \rightarrow \mathcal{H}$ be an identified comparison map. If

$$\|J_N(z - H_N^b)^{-1} - (z - H)^{-1}J_N\| \leq r_N(z) \quad (2533)$$

on Γ_0 , then

$$\|J_N P_N - P J_N\| \leq \frac{\ell_{\Gamma_0}}{2\pi} \sup_{z \in \Gamma_0} r_N(z). \quad (2534)$$

Rank, bundle, or projector-topology convergence follows only when a reverse map and quantitative two-sided near-unitarity are also supplied on the relevant spectral subspaces. Hausdorff convergence alone gives none of these projector conclusions.

Proof. The spectral theorem gives $\|(z - H)^{-1}\| \leq \delta^{-1}$. Norm perturbation gives $\|(z - \widehat{H}_N)^{-1}\| \leq (\delta - e_N)^{-1}$. The resolvent identity proves Eq. (2531), and contour integration proves Eq. (2532). Integrating the assumed intertwined resolvent estimate proves Eq. (2534). Without maps between the Hilbert spaces, the expression comparing projectors is not defined. One-sided maps also cannot identify kernels or cokernels, which proves the final boundary statement. $\square$

The spectral-set alternatives in Proposition 75 need not provide the intertwiner required here. This is the distinction made in Theorem 61.

Sharp obstruction and application separation theorem

Proposition 76 (Injectivity without no pollution). *Let A_{T_d} be adjacency on the infinite d -regular tree, $d \geq 3$, and let A_N be adjacency on finite d -regular quotients whose girth tends to infinity. Then*

$$\text{spec } A_{T_d} = [-2\sqrt{d-1}, 2\sqrt{d-1}], \quad (2535)$$

while

$$d \in \text{spec } A_N \quad (2536)$$

for every N . Hence

$$\varepsilon_N^{\text{poll}} \geq d - 2\sqrt{d-1} > 0 \quad (2537)$$

despite diverging injectivity radius and local convergence.

Proof. Orient the tree toward one end. Each vertex has one incoming edge and $d-1$ outgoing edges. Applying the weighted inequality

$$2|ab| \leq (d-1)^{-1/2}|a|^2 + (d-1)^{1/2}|b|^2 \quad (2538)$$

on every oriented edge gives $\|A_{T_d}\| \leq 2\sqrt{d-1}$. Radial functions proportional to $(d-1)^{-n/2} \sin((n+1)\phi)$, cut off on increasingly long annuli, are Weyl vectors for $2\sqrt{d-1} \cos \phi$. This proves Eq. (2535). The constant vector on a finite quotient has eigenvalue d , proving Eqs. (2536) and (2537). $\square$

The same obstruction appears for a nonamenable Cayley graph: the trivial finite sector retains the row-sum eigenvalue while the reduced regular norm is strictly smaller. Removing only that sector is still insufficient because other non-tempered quotient sectors can survive. This strengthens the escaping-pollution warning in Proposition 39 and the Farber obstruction in Proposition 43.

Proposition 77 (Application and subsequent separation theorem). *Theorems 128 and 129 apply to the natural the microscopic-root theorem and the Hodge-cancellation theorem microscopic family on a compact parameter interval only if one declared finite retained sector satisfies both directed certificates uniformly. When that condition holds, the island and gap of Theorems 123 and 127 obey Theorem 130. The following boundaries remain.*

1. *No arbitrary full finite-quotient spectrum of a non-amenable surface group is asserted to converge to the reduced regular spectrum.*
2. *Residual or Farber geometry alone is not an admissible-tower theorem. The uniform theorem in Theorem 66 remains conditional on its retained-sector certificate.*
3. *Finite-quotient regular, automorphic, character, and reduced regular spectra remain distinct representation objects.*
4. *The theorem transports an existing spectral island. It creates no root, Hodge cancellation, active overlap, or coherence. The square five-state model remains root-free.*
5. *This is an undifferentiated C^0 spectral theorem. Velocities, Hessians, and projector derivatives require separate weighted derivative tails and a fixed metric.*
6. *Weak normalized DOS convergence does not imply no pollution, an unsmoothed DOS peak, or global edge convergence.*
7. *No automatic tower class, representation-completeness conclusion, or named-material statement is made.*

Proof. The natural kernel satisfies the physical tail hypothesis by Theorem 120. Once the two retained-sector certificates are added, Theorem 129 supplies the required Hausdorff error and Theorem 130 transports the already existing island. The obstruction above and Theorem 71 prove the representation and tower limitations. The derivative and DOS boundaries follow because none of their stronger topologies occurs in the hypotheses. $\square$

Verifiable operator-tempered tower classes

The definition below replaces the circular instruction to choose a spectrally convergent tower by finite coefficient, geometry, retained-lift, and polynomial tests. Its output is the pair of directed certificates needed by Theorem 128.

Let J be compact and let

$$H(p) = \sum_{\gamma \in F} A_\gamma(p) \lambda_\gamma(\gamma) \quad (2539)$$

be a self-adjoint matrix-valued family with one finite support F , continuous coefficients, and

$$\text{spec } H(p) \subset [-M, M]. \quad (2540)$$

Use the same verified box for every retained operator:

$$\sup_{N,p} \|H_N^b(p)\| \leq M. \quad (2541)$$

Assume a verified modulus

$$\|H(p) - H(p')\| \leq \nu(d_J(p, p')). \quad (2542)$$

The same coefficient estimate is required for every retained finite operator.

Definition 43 (Finite polynomial separation system). Choose $\delta_m \downarrow 0$. A level- m separation system consists of a finite set $\mathcal{P}_m \subset \mathbb{R}[y]$ and a computable $\sigma_m > 0$ such that, for every $p \in J$ and $x \in [-M, M]$ satisfying

$$\text{dist}(x, \text{spec } H(p)) \geq \delta_m, \quad (2543)$$

there is $q \in \mathcal{P}_m$ with

$$|q(x)| \geq 1 \quad (2544)$$

and

$$\|q(H(p))\| \leq 1 - \sigma_m. \quad (2545)$$

For $q(y) = \sum_{r=0}^{d_q} c_r y^r$, define

$$L_q(M) = \sum_{r=1}^{d_q} r |c_r| M^{r-1}, \quad L_m = \max_{q \in \mathcal{P}_m} L_q(M). \quad (2546)$$

Lemma 6 (Existence and finite robustness). Every family in Eq. (2539) admits finite separation systems. Their inequalities can be established from rational coefficient intervals and rigorous upper bounds on the finitely many reduced norms in Eq. (2545).

Proof. Fix m . The set

$$\mathcal{D}_m = \{(p, x) \in J \times [-M, M] : \text{dist}(x, \text{spec } H(p)) \geq \delta_m\} \quad (2547)$$

is compact because norm continuity implies Hausdorff continuity of the spectrum. For each $(p, x) \in \mathcal{D}_m$, polynomial approximation on the compact set $\text{spec } H(p) \cup \{x\}$ gives a real polynomial that is small on the spectrum and larger than one at x . Both inequalities remain strict on a neighborhood of (p, x) . A finite subcover of $\mathcal{D}_m$ gives $\mathcal{P}_m$, and the least strict margin gives $\sigma_m > 0$.

For self-adjoint A, B with norms at most M , the telescoping identity for each monomial gives

$$\|q(A) - q(B)\| \leq L_q(M) \|A - B\|. \quad (2548)$$

Thus outward-rounded coefficient intervals and established reduced-norm upper bounds verify every inequality on a finite parameter net, with the explicit interpolation budget below. $\square$

Let J_m be a finite parameter net of mesh κ_m . For a retained operator $H_N^b(p)$, the finite polynomial tests are

$$\|q(H_N^b(p_k))\| \leq \|q(H(p_k))\| + \epsilon_{N,m} \quad (2549)$$

for every $p_k \in J_m$ and $q \in \mathcal{P}_m$. Define the interpolated error

$$\zeta_{N,m} = \epsilon_{N,m} + 2L_m \nu(\kappa_m). \quad (2550)$$

Proposition 78 (Finite tests imply upper inclusion). If

$$\zeta_{N,m} < \sigma_m, \quad (2551)$$

then, uniformly for $p \in J$,

$$\sup_{s \in \text{spec } H_N^b(p)} \text{dist}(s, \text{spec } H(p)) < \delta_m. \quad (2552)$$

Proof. Equation (2548), applied once to the finite operator and once to the infinite operator, extends Eq. (2549) from the net to all $p \in J$:

$$\|q(H_N^b(p))\| \leq \|q(H(p))\| + \zeta_{N,m}. \quad (2553)$$

If a finite eigenvalue s were at distance at least δ_m , choose the separating polynomial from Definition 43. Spectral mapping gives $1 \leq |q(s)| \leq \|q(H_N^b(p))\|$, whereas Eqs. (2545), (2553), and (2551) make the same norm strictly less than one. This contradiction proves Eq. (2552). $\square$

An alternative finite test uses polynomial parametrices. On a finite parameter-energy net outside the δ_m -neighborhood of the infinite spectrum, verify both defects

$$\|R_{p,x}(H_N^b(p) - x) - I\| < 1 \quad (2554)$$

and

$$\|(H_N^b(p) - x)R_{p,x} - I\| < 1. \quad (2555)$$

Coefficient Lipschitz bounds and a verified positive defect margin extend the tests between net points. The Neumann lemma then proves Eq. (2552). These are the parametrix certificates of Proposition 75, expressed as finite tests. They also give the explicit upper-inclusion tier of Theorem 64.

Definition 44 (Uniformly kernel-tempered tower). *For the declared pair (H, J) , a tower with retained sectors $\mathcal{B}_N$ is uniformly kernel-tempered when it supplies the following verified data.*

1. Radii $R_N \rightarrow \infty$ satisfying the finite-propagation injectivity buffer, or a quantitative Farber good-patch construction that realizes every member of the finite parameter-spectral Weyl net.
2. Retained local exhaustivity in the sense of Definition 41, with defect $\rho_N \rightarrow 0$.
3. Levels $m_N \rightarrow \infty$, parameter meshes $\kappa_{m_N} \rightarrow 0$, and either the finite polynomial tests with

$$\zeta_{N,m_N} < \sigma_{m_N} \quad (2556)$$

or finite parametrix tests with a verified positive Neumann margin.

4. A no-loss interpolation budget $u_N \rightarrow 0$ for passing from the finite parameter-spectral Weyl net to all $p \in J$.

Define

$$a_N = \omega(R_N) + \rho_N + u_N \quad (2557)$$

and

$$b_N = \delta_{m_N}. \quad (2558)$$

Membership is determined by the displayed finite data and inequalities, not by a spectral-convergence conclusion. The theorem below proves that this finite-data class satisfies Definition 6; that admissibility is an output rather than a clause of the definition.

Definition 45 (Operator-Ramanujan tower). *A kernel-tempered tower is operator-Ramanujan for (H, J) when the retained representation satisfies exact reduced-norm domination for the declared finite-range family and all its real polynomial tests. Equivalently for this family,*

$$\|q(H_N^b(p))\| \leq \|q(H(p))\| \quad (2559)$$

for every real polynomial q and $p \in J$. Its no-pollution rate is $b_N = 0$.

The exact domination in Definition 45 is implied by weak containment of the retained representation, but weak containment is not assumed for arbitrary finite-dimensional automorphic sectors. The boundary is Theorem 71.

Theorem 132 (Automatic uniform bulk verification). *Every uniformly kernel-tempered tower satisfies, uniformly for $p \in J$,*

$$\varepsilon_N^{\text{loss}}(p) \leq a_N \quad (2560)$$

and

$$\varepsilon_N^{\text{poll}}(p) \leq b_N. \quad (2561)$$

Consequently,

$$\begin{aligned} \sup_{p \in J} d_H(\text{spec } H_N^b(p), \text{spec } H(p)) \\ \leq \eta_N := \max\{a_N, b_N\} \longrightarrow 0. \end{aligned} \quad (2562)$$

Every global edge has error at most η_N . Every continuously selected intrinsic island with a uniform gap $\Delta_0 > 0$ has edge error at most η_N , width error at most $2\eta_N$, and isolation-gap error at most $2\eta_N$, once $\eta_N < \Delta_0/3$.

Proof. The geometry and retained local-exhaustivity clauses give the retained lift used in the proof of Theorem 128. The finite-support Weyl modulus, retention defect, and interpolation budget give Eq. (2560). Proposition 78, or the finite parametrix alternative, gives Eq. (2561). The sharp maximum in Eq. (2562) is then Theorem 128. The edge, island, width, and gap conclusions follow from Theorem 130. $\square$

Normal residual injectivity can verify the first geometric clause directly. For a general Farber sequence, Definition 5 and Proposition 42 provide only a typical-root rate. Definition 44 additionally requires the finite Weyl list to be realized in the retained sector. This prevents the averaged bad-root fraction from being mistaken for no pollution.

Cross-tower independence

Corollary 71 (Same intrinsic reduced-bulk limit). *Let towers A and B be kernel-tempered for the same family (H, J) , with errors η_N^A and η_N^B . Then*

$$\sup_{p \in J} d_H(S_N^A(p), S_M^B(p)) \leq \eta_N^A + \eta_M^B. \quad (2563)$$

If Φ is a Hausdorff-Lipschitz spectral functional with constant L_Φ , then

$$\sup_{p \in J} |\Phi(S_N^A(p)) - \Phi(S_M^B(p))| \leq L_\Phi(\eta_N^A + \eta_M^B). \quad (2564)$$

For a fixed isolated island, lower and upper edges have $L_\Phi = 1$, while width and isolation gap have $L_\Phi = 2$ after the common island partition is valid.

Proof. The triangle inequality through the common intrinsic spectrum gives Eq. (2563). The definition of a Hausdorff-Lipschitz functional gives Eq. (2564). The constants for edges, widths, and gaps are those proved in Theorem 130. $\square$

This conclusion supplies a sufficient-condition version of Theorem 67. It does not apply to unsmoothed DOS peaks, velocities, projectors on unidentified spaces, coherence, or optimizer locations.

Exponential physical tails

Theorem 133 (Operator-tempered exponential extension). *Let the physical kernel satisfy the uniform exponential Schur hypotheses with decay margin $\mu - h_\Lambda > 0$. Choose $L_N \rightarrow \infty$ and $R_N \rightarrow \infty$ such that*

$$2(R_N + L_N) < r_{\text{inj},N}, \quad \frac{L_N}{r_{\text{inj},N}} \longrightarrow 0. \quad (2565)$$

Suppose the L_N -truncated family is kernel-tempered at a diagonal test level m_N , with directed errors $a_N^{(L_N)}$ and $b_N^{(L_N)}$. Let $\tau_{\infty,N}$ and $\tau_{f,N}$ be the verified infinite and finite physical tails. Then

$$\begin{aligned} & \sup_{p \in J} d_H(\text{spec } H_N^b(p), \text{spec } H(p)) \\ & \leq \eta_N^{\text{exp}} := \tau_{\infty,N} + \tau_{f,N} + \max\{a_N^{(L_N)}, b_N^{(L_N)}\}. \end{aligned} \quad (2566)$$

If the truncated errors vanish, then $\eta_N^{\text{exp}} \rightarrow 0$. The edge, island, gap, and cross-tower conclusions hold with η_N^{exp} . Every normalized version divides the complete right-hand side by the same fixed positive intrinsic energy scale.

Proof. Apply Theorem 132 to the common diagonal cutoff. The infinite full-to-cutoff comparison costs $\tau_{\infty,N}$, and the finite cutoff-to-full comparison costs $\tau_{f,N}$. The two directed triangle inequalities in Theorem 129 give exactly Eq. (2566); neither tail is duplicated. The Schur estimate makes both tails decay exponentially. The remaining conclusions follow from Theorem 130 and Corollary 71. $\square$

This theorem is consistent with Theorems 58, 57, 65, and 66. It supplies sufficient testable hypotheses; it does not declare every tower admissible.

Finite verification ledger

Proposition 79 (Depth-(N) verification algorithm). *Membership through depth N is established by the following finite ledger.*

1. *Verify coefficient Hermiticity, support, interval bounds, the common spectral box, and the parameter modulus.*
2. *Compute an injectivity buffer, or certify every required good patch and its retained local-exhaustivity defect.*
3. *Verify retained-sector invariance and compute ρ_N .*
4. *Choose m_N , list the finite parameter net and every polynomial or parametrix test, and compute all retained finite norms or defects.*
5. *Supply rigorous reduced-norm upper bounds for the finite polynomial list, or verify the two transported parametrix defects.*

6. *Check the strict budget $\zeta_{N,m_N} < \sigma_{m_N}$, or the positive Neumann margin, and output a_N , b_N , and η_N .*
7. *For an exponential kernel, verify the diagonal cutoff and both physical Schur tails before outputting η_N^{exp} .*

No item uses the desired Hausdorff conclusion as an input.

Proof. Items one through three verify the retained no-loss hypotheses. Items four through six verify Proposition 78 or its parametrix alternative. Theorem 132 then produces the finite-range error. Item seven supplies exactly the two additional terms in Theorem 133. $\square$

A concrete scalar graph-cover class

Proposition 80 (High-girth scalar operator-Ramanujan towers). *Let X_N be a finite d -regular graph-cover tower, $d \geq 2$, whose girth tends to infinity. Let A_N be adjacency. For $d \geq 3$, retain the orthogonal complement of the constant vector; when X_N is bipartite, also remove the alternating constant vector. For $d = 2$, retain the full space. Assume the finite, directly checkable bound*

$$\text{spec}(A_N|_{\mathcal{B}_N}) \subset [-2\sqrt{d-1}, 2\sqrt{d-1}]. \quad (2567)$$

Then the tower is operator-Ramanujan for scalar adjacency and is kernel-tempered. It remains kernel-tempered for every compact family

$$H(p) = P_p(A) \quad (2568)$$

whose polynomials have one fixed maximal degree and whose real coefficients vary continuously on J .

Proof. The infinite d -regular tree has adjacency spectrum $[-2\sqrt{d-1}, 2\sqrt{d-1}]$. Equation (2567) and spectral mapping give exact no pollution for adjacency and for every $P_p(A)$.

High girth gives exact lifts of balls whose radius is smaller than the injectivity radius. Let a lifted unit Weyl vector have support size s_N , let $n_N = |V(X_N)|$, and let $r_{\text{exc}} \leq 2$ be the removed exceptional rank. Its exceptional projection has norm at most

$$\xi_N = \sqrt{\frac{r_{\text{exc}} s_N}{n_N}}. \quad (2569)$$

After normalization of the retained component, the adjacency residual increases by at most $4d\xi_N$ when $\xi_N \leq 1/2$. Choose $R_N \rightarrow \infty$ much smaller than the girth. The Moore bound, with its linear $d = 2$ specialization, makes $s_N/n_N \rightarrow 0$, hence $\xi_N \rightarrow 0$. The tree Weyl modulus also vanishes, so retained no loss follows. Coefficient continuity and spectral mapping make these estimates uniform for the compact polynomial family. $\square$

The cycle tower $X_N = C_{2N}$ is an explicit nonempty $d = 2$ member: its girth diverges and its full adjacency

spectrum lies in $[-2, 2]$. For $d \geq 3$, any independently supplied high-girth Ramanujan graph-cover tower is established by the finite adjacency test (2567). The proposition does not assert existence of such a tower for an arbitrary surface group and does not transfer the scalar bound to Fuchsian, matrix-valued, or multi-orbital families.

Representation and physical firewalls

Proposition 81 (Scope of the automatic class). *The following restrictions are part of every the tempered-tower construction application.*

1. *Residual or Farber geometry is not automatically operator-tempered. The obstruction in Proposition 43 remains.*
2. *Removing the trivial sector alone is insufficient. In a bipartite scalar graph even the alternating exceptional sector must also be removed, and general non-tempered sectors require separate control.*
3. *No nonzero finite-dimensional representation of a nonamenable group is declared weakly contained in the regular representation. Finite-level block completeness does not prove infinite reduced completeness, as stated in Theorem 73.*
4. *The scalar graph class does not certify the natural the microscopic-root theorem or the Hodge-cancellation theorem matrix family. That application remains conditional on passing the finite tests of Proposition 79.*
5. *Spectral convergence transports already established edges and islands; it does not create a root, Hodge cancellation, active overlap, or coherence. The square five-state model remains root-free.*
6. *Continuous Hausdorff-Lipschitz spectral functionals are covered. Unsmoothed DOS, velocity, projector topology, optimizer location, arithmetic universality, and material identification retain their separate hypotheses.*
7. *No representation-completeness conclusion beyond this theorem is asserted.*

Proof. The first three items are the local-versus-global and representation obstructions retained from the tower-admissibility theorem and the automorphic-sector result. The fourth follows because the scalar interval test contains no matrix-kernel estimate. The remaining items list observables and conclusions whose required topology is absent from Definition 44. $\square$

Faithful reduced completeness without a fictitious Bloch dual

Finite Peter–Weyl completeness, one automorphic block, and the reduced regular bulk are different representation-theoretic objects. This subsection gives the exact infinite-volume replacement for ordinary Bloch language and then identifies what a the tempered-tower construction tower approximates.

Faithful measurable sectorizations

Definition 46 (Essential spectral closure). *Let (Ω, μ) be a standard measure space and let $S_\omega \subset \mathbb{R}$ be a measurable field of nonempty compact sets. Define*

$$\text{ecl}_\mu\{S_\omega\} = \{x \in \mathbb{R} : \mu(\{\omega : \text{dist}(x, S_\omega) < \varepsilon\}) > 0 \text{ for every } \varepsilon > 0\}. \quad (2570)$$

Fibers supported on a null set do not enter this closure.

Theorem 134 (Faithful reduced direct-integral spectrum). *Let Γ be countable, put*

$$\mathcal{A} = M_d(C_r^*(\Gamma)), \quad (2571)$$

and let $a = a^ \in \mathcal{A}$. Let*

$$\rho = \int_\Omega^\oplus \rho_\omega d\mu(\omega) \quad (2572)$$

be a measurable direct integral of representations of $\mathcal{A}$. If ρ is faithful, then

$$\text{spec } \lambda_d(a) = \text{spec } \rho(a) = \text{ecl}_\mu\{\text{spec } \rho_\omega(a)\}. \quad (2573)$$

Every fiber in Eq. (2572) factors through the reduced algebra and is therefore weakly contained in the regular representation.

Proof. The amplified regular representation λ_d is faithful by the definition of $C_r^*(\Gamma)$. An injective unital C^* -homomorphism is isometric and inverse-closed, so

$$\text{spec } \lambda_d(a) = \text{spec}_\mathcal{A}(a) = \text{spec } \rho(a). \quad (2574)$$

For real x , the decomposable operator $\rho(a) - x$ has a bounded inverse exactly when its fiber inverses exist almost everywhere and

$$\text{ess sup}_\omega \|(\rho_\omega(a) - x)^{-1}\| < \infty. \quad (2575)$$

By self-adjointness, the fiber inverse norm is the reciprocal of $\text{dist}(x, \text{spec } \rho_\omega(a))$. Thus Eq. (2575) holds exactly when x lies outside the essential closure in Definition 46. This proves Eq. (2573). Since each ρ_ω is a representation of $C_r^*(\Gamma)$, its underlying group representation is weakly contained in the regular representation. $\square$

Proposition 82 (Faithfulness and weak containment are separate conditions). *Let ρ be a representation of $\mathcal{A}$.*

1. *If ρ factors through $C_r^*(\Gamma)$ but is not faithful, then*

$$\text{spec } \rho(a) \subseteq \text{spec } \lambda_d(a), \quad (2576)$$

and the inclusion can be strict.

2. *If a representation of the full group algebra does not factor through the reduced algebra, it is not a reduced sector and can add spectrum outside $\text{spec } \lambda_d(a)$.*

Consequently, weak containment prevents reduced spectral pollution, while faithfulness prevents reduced spectral loss.

Proof. In the first case, $\rho(a)$ is the image of a in the quotient by $\ker \rho$. Spectra decrease under a unital quotient map, and a nonzero kernel can remove spectral values. In the second case there is no $C_r^*(\Gamma)$ -representation to which reduced spectral inclusion applies. The nonamenable example below realizes the extra-spectrum alternative. $\square$

This is the exact operator-algebraic version of the weak-containment condition in Theorem 71.

The non-type-I surface-group boundary

Theorem 135 (Factor and GNS completeness). *For a separable reduced group C^* -algebra, any faithful standard representation or faithful family of GNS representations may be decomposed measurably into factor representations. Theorem 134 applies to that factor decomposition.*

If Γ is a torsion-free cocompact surface group, then Γ is ICC and its group von Neumann algebra in the canonical trace representation is a type-II₁ factor. Its central decomposition is therefore a single factor. Neither this fact nor Theorem 134 supplies a classical irreducible Bloch-momentum space.

Proof. Countability of Γ makes $C_r^*(\Gamma)$ separable. Standard decomposition theory gives a measurable factor decomposition, and faithfulness of the total representation permits direct application of Theorem 134.

For a nontrivial element of a torsion-free cocompact surface group, the centralizer is cyclic, while the group is not virtually cyclic. Every nontrivial conjugacy class is therefore infinite, so Γ is ICC. The canonical group trace is finite and faithful, and the center of the group von Neumann algebra is trivial. Hence the algebra is a type-II₁ factor. The regular representation is therefore not type I, so Γ is not a type-I group. A one-point central factor decomposition is not a parameterization by irreducible momenta. $\square$

For a non-type-I group there is no supported ordinary direct integral over all irreducible representations with a standard unitary dual and classical Plancherel measure.

Faithful representation, factor, or GNS direct integrals are the correct measurable objects. The variable ω in Eq. (2572) is not called Bloch momentum unless an independently declared commuting or type-I structure proves that interpretation.

Finite quotient and operator-wise completeness

Proposition 83 (Exact finite Wedderburn completeness). *Let G_N be a finite quotient of Γ . Its left regular representation has the exact decomposition*

$$\ell^2(G_N) \cong \bigoplus_{\pi \in \widehat{G}_N} \mathbb{C}^{d_\pi} \otimes \mathbb{C}^{d_\pi}, \quad (2577)$$

where the left action is $\pi \otimes I_{d_\pi}$. Hence a finite group-kernel operator has

$$\text{spec } H_N = \bigcup_{\pi \in \widehat{G}_N} \text{spec } H_N[\pi] \quad (2578)$$

with exact multiplicity d_π for the block $H_N[\pi]$. This is complete finite-quotient spectrum only when every block and multiplicity is included.

Proof. Finite Peter–Weyl theory gives Eq. (2577); applying the group-kernel matrix to that direct sum gives Eq. (2578) and its multiplicities. This is Theorem 70 specialized to the present operator. $\square$

Definition 47 (Operator-wise reduced spectral completeness). *For one declared compact family $a(p) = a(p)^* \in M_d(C_r^*(\Gamma))$, a retained finite sequence is operator-wise reduced spectrally complete when*

$$\sup_{p \in J} d_H(\text{spec } H_N^b(p), \text{spec } \lambda_d(a(p))) \longrightarrow 0. \quad (2579)$$

Theorem 136 (the tempered-tower construction towers are operator-wise complete). *If a retained tower passes the finite tests in Theorem 132, then it is operator-wise reduced spectrally complete for that declared family, with*

$$\sup_{p \in J} d_H(\text{spec } H_N^b(p), \text{spec } \lambda_d(a(p))) \leq \eta_N. \quad (2580)$$

For two established towers A, B ,

$$\sup_{p \in J} d_H(S_N^A(p), S_M^B(p)) \leq \eta_N^A + \eta_M^B. \quad (2581)$$

These statements do not imply density of finite-image representations, Fell density in the full dual, or completeness for an untested operator.

Proof. Equation (2580) is the conclusion of Theorem 132, whose no-loss and no-pollution certificates refer to the reduced regular spectrum. This supplies the finite-data sufficient-condition version of Theorem 66. The triangle inequality through that common spectrum gives Eq. (2581), equivalently Corollary 71. Both proofs are operator-wise and contain no assertion about convergence of representations as objects of a unitary dual. $\square$

The finite Riesz participation theorem 32 and the finite-cover block theorem 40 remain exact within their declared finite fibers. They do not add omitted Wedderburn blocks or prove an infinite representation theorem.

The amenable benchmark

Theorem 137 (Amenable residual full-quotient convergence). *Let Γ be finitely generated, amenable, and residually finite. Let (Γ_N) be a normal residual tower with injectivity radii tending to infinity. For any norm-continuous compact family of finite-range self-adjoint matrix group kernels, the full finite-quotient operators satisfy*

$$\text{spec } H_N(p) \subseteq \text{spec } \lambda_d(a(p)) \quad (2582)$$

and

$$\sup_{p \in J} d_H(\text{spec } H_N(p), \text{spec } \lambda_d(a(p))) \longrightarrow 0. \quad (2583)$$

For a uniformly exponentially summable family, choose diagonal cutoffs $L_N \rightarrow \infty$ below the injectivity scale. Then Eq. (2583) remains true, with total error

$$\eta_N^{\text{exp}} = a_N^{(L_N)} + \tau_{\infty, N} + \tau_{f, N}, \quad (2584)$$

where the two physical tails occur exactly once.

Proof. Amenability gives $C^*(\Gamma) = C_r^*(\Gamma)$. Every finite quotient representation is a representation of the full algebra and therefore factors contractively through the reduced algebra. Spectra decrease under a unital C^* -homomorphism, which proves Eq. (2582). Residual injectivity lifts every fixed finite-support Weyl vector exactly, as in Theorem 58. Compact parameter nets make the finite-support no-loss modulus uniform, so Theorem 128 proves Eq. (2583).

For the exponential family, apply the finite-range argument to the common cutoff and use Theorem 129. Exact finite-range no pollution removes the truncated pollution term and leaves precisely Eq. (2584). $\square$

This benchmark uses the group identity $C^*(\Gamma) = C_r^*(\Gamma)$; it is not an exact-crystallographic statement. The separation between group and crystallographic claims remains Proposition 29.

The nonamenable obstruction

Theorem 138 (Full-quotient pollution for a nonamenable group). *Let $S = S^{-1}$ be a finite generating set of a nonamenable group, put $d_S = |S|$, and define*

$$A_S = \sum_{s \in S} \lambda_\Gamma(s). \quad (2585)$$

Every finite quotient adjacency has the constant-vector eigenvalue d_S , whereas

$$\|A_S\| < d_S. \quad (2586)$$

Consequently every full finite quotient has pollution at least

$$d_S - \|A_S\| > 0, \quad (2587)$$

even when the injectivity radius diverges. Deleting the trivial sector removes this eigenvalue but does not certify all other retained sectors.

Proof. Each row of finite quotient adjacency sums to d_S , so the constant vector has that eigenvalue. The strict norm inequality in Eq. (2586) is the spectral-radius characterization of nonamenability for the symmetric generating random walk, after multiplication by d_S . The directed pollution bound follows. Injectivity affects local lifts, not this global vector. Other non-tempered quotient representations are not controlled by removal of the trivial block, so a retained-sector certificate remains necessary. $\square$

This is the representation-theoretic obstruction behind Theorem 64 and the completeness boundary in Theorem 73.

Vocabulary and application repair

Proposition 84 (Supported completeness vocabulary). *Every occurrence of complete generalized Bloch spectrum is replaced by exactly one of the following phrases.*

1. Complete finite-quotient spectrum: *all finite Wedderburn blocks and their exact multiplicities are included.*
2. Declared automorphic-sector spectrum: *one specified representation block is being studied, without a completeness claim.*
3. Complete reduced-regular bulk spectrum: *the spectrum is $\text{spec } \lambda_d(a)$, or a faithful reduced direct integral as in Theorem 134.*
4. established retained-sector approximation: *a retained finite sequence has passed the no-loss and no-pollution tests for the declared operator family.*

No other completeness wording is supported without a new representation theorem.

Proof. Proposition 83 proves the first meaning. Proposition 82 proves that a single sector need not be faithful. Theorem 134 proves the third meaning, and Theorem 136 proves the fourth. These are the four cases distinguished in Corollary 43. $\square$

Proposition 85 (the microscopic-root theorem and the Hodge-cancellation theorem scope). *For the natural microscopic family, the following statements remain logically distinct.*

1. The character formula and positive root in Theorem 121 are declared automorphic-sector results.
2. The layer-parity point-island identity and its persisted gap in Theorem 123 are complete reduced-regular bulk results, proved independently of character completeness.
3. The full Hodge cancellation criterion in Theorem 125 is an exact tangent statement for its declared natural symmetry class. It does not manufacture missing representations.
4. A finite retained approximation to the regular island is supported only after the tempered-tower construction verification for that operator family.

Proof. The first statement evaluates one character representation. The second uses the faithful regular group-algebra operator and its exact layer-parity identity. The third is a tensor cancellation theorem in the declared model. The fourth is precisely Definition 47. None implies another representation-completeness statement. $\square$

Proposition 86 (Final separation theorem). *The faithful spectrum theorem does not provide a classical Plancherel formula over all finite-dimensional or irreducible representations of a non-type-I surface group. Finite-image representations are not declared dense or complete, and arbitrary finite quotients are not asserted to recover the reduced bulk. Spectral completeness creates no root, Hodge cancellation, coherence, unsmoothed DOS conclusion, arithmetic identity, exact crystallographic statement, or material identification. The square five-state model remains root-free. the balanced full-shell theorem growing/full-shell collapse and the density-limit theorem unsmoothed/coherent DOS remain outside this subsection.*

Proof. The non-type-I restriction is Theorem 135. The quotient obstruction is Theorem 138. All remaining conclusions require structures or topologies absent from the hypotheses of Theorem 134. $\square$

Balanced full-shell normalized operator collapse

The fixed-shell theorem of the fixed-star collapse theorem does not by itself control a growing microscopic fiber. The construction below puts every full normalized operator on one Hilbert space, specifies the action outside each core, and balances core growth against the normalized physical tail.

Common-space cores and physical tails

Definition 48 (Transported full normalized family). *Let K_X be compact. For each j , let $\hat{H}_j(X)$ be a bounded self-adjoint operator on $\mathcal{H}_j$, normalized as*

$$\hat{H}_j(X) = E_j^{-1}[H_j(X) - c_j(X)I], \quad E_j > 0. \quad (2588)$$

Let $U_j : \mathcal{H}_j \rightarrow \mathcal{H}$ be a declared unitary transport to one fixed Hilbert space, and put

$$T_j(X) = U_j \hat{H}_j(X) U_j^*, \quad T_*(X) = \hat{H}_*(X). \quad (2589)$$

Let P_L be nested physical-shell projections on $\mathcal{H}$, with $P_L \rightarrow I$ strongly, and put $Q_L = I - P_L$. For each L , declare one bounded self-adjoint complementary action $D_L(X)$ on $Q_L \mathcal{H}$. Define the full-space core extensions

$$\mathcal{C}_{j,L}(X) = P_L T_j(X) P_L + Q_L D_L(X) Q_L \quad (2590)$$

and

$$\mathcal{C}_{*,L}(X) = P_L T_*(X) P_L + Q_L D_L(X) Q_L. \quad (2591)$$

Thus the common complement is explicit and no finite matrix is extended by zero.

If only a core unitary is initially available, it must be completed as

$$U_{j,L} = V_{j,L} \oplus W_{j,L} \quad (2592)$$

with a declared complementary unitary $W_{j,L}$. The operators in Definition 48 are then formed using this complete transport.

Definition 49 (Growing-core and normalized-tail errors). *Define*

$$a_j(L) = \sup_{X \in K_X} \|\mathcal{C}_{j,L}(X) - \mathcal{C}_{*,L}(X)\| \quad (2593)$$

and

$$\tau_j(L) = \sup_{X \in K_X} \|T_j(X) - \mathcal{C}_{j,L}(X)\|, \quad (2594)$$

$$\tau_*(L) = \sup_{X \in K_X} \|T_*(X) - \mathcal{C}_{*,L}(X)\|. \quad (2595)$$

If $R_j^{\text{phys}}(L)$ is a raw physical tail of H_j , then the normalized estimate is

$$\tau_j(L) \leq E_j^{-1} R_j^{\text{phys}}(L). \quad (2596)$$

Proposition 87 (Sharp common-space split). *For every j, L ,*

$$\begin{aligned} & \sup_{X \in K_X} \|T_j(X) - T_*(X)\| \\ & \leq a_j(L) + \tau_j(L) + \tau_*(L). \end{aligned} \quad (2597)$$

Proof. Insert $\mathcal{C}_{j,L}(X)$ and $\mathcal{C}_{*,L}(X)$ between the two full operators and apply the triangle inequality. Every term is defined on $\mathcal{H}$, including its complementary action. Taking the supremum in X proves Eq. (2597). $\square$

This split extends the fixed-shell estimates in Proposition 10 and Theorem 19; it does not infer full-shell convergence from them.

The balanced diagonal

Theorem 139 (Full-shell normalized collapse). *Suppose there are $L_j \rightarrow \infty$ such that*

$$a_j(L_j) \rightarrow 0, \quad \tau_j(L_j) \rightarrow 0, \quad \tau_*(L_j) \rightarrow 0. \quad (2598)$$

Then

$$\epsilon_j := \sup_{X \in K_X} \|T_j(X) - T_*(X)\| \rightarrow 0. \quad (2599)$$

The assertion is operator-norm convergence on the fixed transported Hilbert space.

Proof. Use $L = L_j$ in Eq. (2597) and apply Eq. (2598). $\square$

Theorem 140 (Explicit exponential balance). *Assume, uniformly in j, L ,*

$$\tau_j(L) \leq C E_j^{-1} e^{-\delta L}, \quad \tau_*(L) \leq C_* e^{-\delta_* L}, \quad (2600)$$

with $\delta, \delta_ > 0$, and*

$$a_j(L) \leq A_j e^{\kappa L} \quad (2601)$$

with $A_j \rightarrow 0$ and $\kappa \geq 0$. Suppose $E_j \rightarrow 0$, and set

$$s_j = \log E_j^{-1}, \quad r_j = \log A_j^{-1}. \quad (2602)$$

For $\kappa > 0$, the two displayed envelopes admit a balanced diagonal if and only if

$$r_j - \frac{\kappa}{\delta} s_j \rightarrow \infty. \quad (2603)$$

One explicit real-valued choice before integer rounding is

$$L_j^{\text{bal}} = \frac{s_j}{\delta} + \frac{1}{2\kappa} \left(r_j - \frac{\kappa}{\delta} s_j \right). \quad (2604)$$

For $\kappa = 0$, it is enough that $A_j \rightarrow 0$ and $\delta L_j - s_j \rightarrow \infty$.

A convenient stronger sufficient condition is

$$\liminf_{j \rightarrow \infty} \frac{r_j}{s_j} > \frac{\kappa}{\delta}. \quad (2605)$$

Choose

$$\frac{1}{\delta} < c < \liminf_{j \rightarrow \infty} \frac{r_j}{\kappa s_j} \quad (2606)$$

when $\kappa > 0$, and set

$$L_j = \lceil c s_j \rceil. \quad (2607)$$

Then the hypotheses of Theorem 139 hold.

Proof. The normalized physical tail vanishes exactly under the envelope condition

$$\delta L_j - s_j \rightarrow \infty, \quad (2608)$$

while the core envelope vanishes exactly under

$$r_j - \kappa L_j \rightarrow \infty. \quad (2609)$$

If both hold, their positive linear combination gives Eq. (2603). Conversely, Eq. (2604) makes the two margins positive constant multiples of the left side of Eq. (2603). Integer rounding changes each margin by a bounded amount. The master tail vanishes because $L_j \rightarrow \infty$.

Under Eq. (2605), the interval in Eq. (2606) is nonempty. The cutoff in Eq. (2607) gives

$$E_j^{-1} e^{-\delta L_j} = O(E_j^{c\delta-1}) \quad (2610)$$

and

$$A_j e^{\kappa L_j} = O(A_j E_j^{-c\kappa}), \quad (2611)$$

both tending to zero. $\square$

The equivalence in Eq. (2603) concerns the stated upper-envelope certificate. Accidental cancellation in a particular model is not used.

Corollary 72 (Physical-angle cutoff law). *Let $E(\theta) \rightarrow 0$ as $\theta \rightarrow 0$. A normalized exponential tail requires*

$$\delta L_c(\theta) - \log E(\theta)^{-1} \rightarrow \infty. \quad (2612)$$

If

$$E(\theta) \asymp |\theta|^p \quad (2613)$$

and the core compatibility in Theorem 140 holds, then one may take

$$L_c(\theta) = \lceil c p \log |\theta|^{-1} \rceil, \quad c\delta > 1. \quad (2614)$$

A fixed cutoff is insufficient whenever its raw omitted tail stays nonzero, because Eq. (2596) then grows like $E(\theta)^{-1}$.

Buffered resonant island

Theorem 141 (Uniform resolvent, Riesz, and rank control). *Assume one rectifiable contour Γ_0 encloses a master island $I_*(X)$ for every $X \in K_X$, and*

$$d_\Gamma = \inf_{X \in K_X} \text{dist}(\Gamma_0, \text{spec } T_*(X)) > 0. \quad (2615)$$

Assume also the buffered-shell construction of Theorem 35: every complementary state that can approach Γ_0 is included in the declared active block, and the buffer boundary remains uniformly separated.

If $\epsilon_j < d_\Gamma$, then for $z \in \Gamma_0$,

$$\|(z - T_j(X))^{-1} - (z - T_*(X))^{-1}\| \leq \frac{\epsilon_j}{d_\Gamma(d_\Gamma - \epsilon_j)}. \quad (2616)$$

Define the Riesz projections by

$$P_j(X) = \frac{1}{2\pi i} \int_{\Gamma_0} (z - T_j(X))^{-1} dz. \quad (2617)$$

$$P_*(X) = \frac{1}{2\pi i} \int_{\Gamma_0} (z - T_*(X))^{-1} dz. \quad (2618)$$

They satisfy

$$\|P_j(X) - P_*(X)\| \leq \frac{\ell_{\Gamma_0} \epsilon_j}{2\pi d_\Gamma (d_\Gamma - \epsilon_j)}. \quad (2619)$$

When the right side is below one, $P_j(X)$ and $P_*(X)$ are unitarily equivalent. If either projection has finite rank, both ranks are equal. Without finite rank, the conclusion is projection equivalence rather than an unsupported finite-rank assertion.

Proof. The spectral theorem bounds the master resolvent by d_Γ^{-1} . Norm perturbation bounds the transported resolvent by $(d_\Gamma - \epsilon_j)^{-1}$. The resolvent identity proves Eq. (2616); contour integration proves Eq. (2619). Projections at norm distance below one are unitarily equivalent. The buffered hypothesis prevents a state omitted from the active block from crossing the contour. This is the rank-homotopy condition in Theorems 37 and 38. $\square$

If the buffer boundary loses separation, the contour and rank conclusions stop at that parameter. Enlarging the block to include every newly resonant state is mandatory.

Islands and continuous master functions

Theorem 142 (Inheritance under full-shell collapse). *For each $X \in K_X$, define the pointwise master gap*

$$\Delta_*(X) := \text{dist}(I_*(X), \text{spec } T_*(X) \setminus I_*(X)). \quad (2620)$$

Assume the uniform gap

$$\Delta_* := \inf_{X \in K_X} \Delta_*(X) > 0. \quad (2621)$$

If $\epsilon_j < \Delta_*/3$, the corresponding transported island $I_j(X)$ obeys, uniformly in X ,

$$d_H(I_j(X), I_*(X)) \leq \epsilon_j, \quad (2622)$$

$$|E_{j,\pm}(X) - E_{*,\pm}(X)| \leq \epsilon_j, \quad (2623)$$

$$|W_j(X) - W_*(X)| \leq 2\epsilon_j, \quad (2624)$$

and

$$|\Delta_j(X) - \Delta_*(X)| \leq 2\epsilon_j, \quad \Delta_j(X) \geq \Delta_*(X) - 2\epsilon_j. \quad (2625)$$

For every Hausdorff-Lipschitz spectral functional Φ ,

$$|\Phi(I_j(X)) - \Phi(I_*(X))| \leq \text{Lip}(\Phi) \epsilon_j. \quad (2626)$$

Proof. Operator-norm perturbation gives full spectral Hausdorff error at most ϵ_j . The uniform gap and the one-third condition select the same island partition. The edge, width, and factor-two gap estimates follow from Theorem 60; the Lipschitz estimate is its definition. These are also the consequences summarized in Corollary 8 for a fixed shell. $\square$

Derivative topology tiers

Definition 50 (Growing-shell C^r budget). *For $r = 1, 2$, take all transported momentum or parameter derivatives through order r in Definitions 48 and 49. Denote the resulting maxima by $a_j^{(r)}(L)$, $\tau_j^{(r)}(L)$, and $\tau_*^{(r)}(L)$, and put*

$$\epsilon_j^{(r)}(L) = a_j^{(r)}(L) + \tau_j^{(r)}(L) + \tau_*^{(r)}(L). \quad (2627)$$

Proposition 88 (Derivative inheritance requires derivative tails). *The C^0 conclusion in Theorem 139 controls spectra and continuous spectral functionals only. If a diagonal additionally satisfies*

$$\epsilon_j^{(1)}(L_j) \longrightarrow 0, \quad (2628)$$

then simple isolated branch velocities converge in the fixed metric and transport chart. If

$$\epsilon_j^{(2)}(L_j) \longrightarrow 0, \quad (2629)$$

then invariant Hessian spectra and full Hodge tensors converge under their fixed convention. Weighted Schur tails generally contain polynomial factors in L_j , so each derivative tier requires its own balanced diagonal.

Proof. The first assertion is the obstruction in Proposition 11. Under Eq. (2628), differentiated operators and the isolated Riesz projectors converge, so the Hellmann-Feynman formula gives velocity convergence. Under Eq. (2629), the differentiated resolvent chain and Theorem 18 give invariant Hessian and Hodge convergence. The polynomial-times-exponential bounds are precisely the weighted derivative Schur estimates in Theorem 56. $\square$

Neither derivative tier implies an unsmoothed or coherence-weighted DOS conclusion.

Natural full-distance exponential kernel

Corollary 73 (A nonempty the microscopic-root theorem balanced regime). *For the natural full-distance kernel, let*

$$\delta = \mu - h_\Gamma > 0 \quad (2630)$$

be the hopping-decay minus hyperbolic-growth margin from Theorem 120. Assume the normalized kinetic scale E_j and transported local profile obey

$$a_j(L) \leq A_j e^{\kappa L}. \quad (2631)$$

Then Theorem 140 applies with normalized tail

$$\tau_j(L) \leq C E_j^{-1} e^{-\delta L}. \quad (2632)$$

The balanced class is nonempty. For example, choose $s_j \rightarrow \infty$,

$$E_j = e^{-s_j}, \quad A_j = e^{-\beta s_j}, \quad \beta > \frac{\kappa}{\delta}, \quad (2633)$$

choose

$$\frac{1}{\delta} < c < \frac{\beta}{\kappa} \quad (2634)$$

when $\kappa > 0$, and set $L_j = \lceil cs_j \rceil$. For $E(\theta) \asymp |\theta|^p$ and $A(\theta) = O(|\theta|^{p\beta})$, the corresponding physical-angle choice is Eq. (2614).

The exact character-sector root is unchanged. The complete regular-operator island and gap in Theorem 123, and the Hodge locking controlled in Theorem 127, are transported only within their already declared representations.

Proof. The block Schur sum in Theorem 54, the exponential extension in Theorem 65, and the physical kernel estimate in Theorem 120 give Eq. (2632), including E_j^{-1} . Equation (2633) satisfies Eq. (2605). The explicit cutoff therefore makes the core, physical tail, and master tail vanish. $\square$

The local profile bound in Eq. (2631) is a quantitative hypothesis to be verified for the scan. Exponential summability alone does not prove it.

established finite covers

Corollary 74 (Finite-cover full-shell approximation). *Let N_j be finite covers, and let $H_{N_j}^b(X)$ denote the normalized retained finite operator on the declared sector. Assume*

$$2L_j + b < r_{\text{inj}, N_j} \quad (2635)$$

for the required observation buffer b . Assume the same retained sector passes the the finite-cover approximation theorem or the tempered-tower construction no-loss and no-pollution certificate, with uniform error $\eta_{N_j} \rightarrow 0$. Then

$$\sup_{X \in K_X} d_H(\text{spec } H_{N_j}^b(X), \text{spec } T_*(X)) \leq \eta_{N_j} + \epsilon_j. \quad (2636)$$

The edge and isolated-island conclusions of Theorem 142 hold with $\eta_{N_j} + \epsilon_j$.

Proof. The buffer gives exact local lifting of the cutoff operator by Theorem 58. It does not give upper spectral inclusion. That inclusion and no loss come from Theorems 129 and 132. Their finite-to-infinite error is η_{N_j} ; Theorem 139 supplies the infinite-to-master error ϵ_j . The Hausdorff triangle inequality proves Eq. (2636), and Theorem 130 proves the spectral consequences. $\square$

Projector comparison across different cover Hilbert spaces still requires a two-sided common-space or quasiunitary intertwiner. Hausdorff convergence alone does not compare Riesz projectors. This is consistent with Theorem 136.

Statement repairs and firewalls

Proposition 89 (Repair of fixed-shell collapse claims). *The earlier normalized-collapse statements are replaced by the following topology-qualified statements.*

1. *the fixed-star collapse theorem proves fixed transported-shell C^0 collapse under its stated tail hypothesis.*
2. *Full normalized microscopic collapse requires a common-space extension and the balanced growing diagonal of Theorem 139.*
3. *Global finite-cover collapse additionally requires retained-sector no loss and no pollution as in Corollary 74.*
4. *Velocity, Hessian, and Hodge conclusions require the separate C^1 or C^2 conditions in Proposition 88.*

This is the full-shell replacement for Corollary 9.

Proof. The four items are respectively the fixed-shell theorem, the common-space triangle estimate, the established finite-cover triangle estimate, and the derivative-topology proposition proved above. $\square$

Proposition 90 (Representation and observable separation theorem). *Character, automorphic, complete reduced-regular, and established retained finite-cover spectra retain the vocabulary of the reduced-completeness theorem and are not merged by operator collapse. Rank stability stops at a contour crossing or at loss of the the resonant-shell analysis resonant buffer. C^0 collapse gives no velocity, Hessian, Hodge, unsmoothed DOS, or coherence-weighted DOS conclusion. The theorem creates no root, arithmetic exponent class, exact crystallographic identity, or named-material conclusion. The exact square five-state target remains root-free. the density-limit theorem DOS claims and the sequence-class theorem arithmetic claims remain outside this subsection.*

Proof. The representation distinction is preserved by the common-space hypothesis, which compares only already identified operators. The rank boundary is the buffer assumption in Theorem 141. The topology boundaries follow from Proposition 88. None of the remaining structures appears in the hypotheses. $\square$

Vanishing-broadening and coherence-weighted density limits

Finite-cover trace convergence produces measures. It does not turn a finite atomic measure into an ordinary density. This subsection states the extra local-law and regularity hypotheses under which vanishing broadening or a true unsmoothed density is rigorous.

Definition 51 (Finite and limiting DOS measures). *Let H_N act on a retained space of dimension $n_N > 0$, with eigenvalues listed with multiplicity. Its normalized DOS measure is*

$$\nu_N = \frac{1}{n_N} \sum_{k=1}^{n_N} \delta_{E_{k,N}}. \quad (2637)$$

Let ν be a limiting trace measure, and assume all measures are supported in one compact interval I . Write

$$F_N(E) = \nu_N((-\infty, E]), \quad F(E) = \nu((-\infty, E]). \quad (2638)$$

Every nonzero ν_N is atomic and has no Lebesgue density.

Weak convergence of ν_N to ν is supplied in the finite-propagation setting by Theorem 62. Faithful support is the separate conclusion of Proposition 41.

Definition 52 (Admissible broadening kernel). *Let $K \geq 0$ satisfy*

$$K \in W^{1,1}(\mathbb{R}), \quad \int_{\mathbb{R}} K(u) du = 1, \quad (2639)$$

and, for $0 < \alpha \leq 1$,

$$m_\alpha(K) = \int_{\mathbb{R}} |u|^\alpha K(u) du < \infty. \quad (2640)$$

Put

$$K_\eta(x) = \eta^{-1} K(x/\eta), \quad \rho_{N,\eta} = K_\eta * \nu_N, \quad \rho_\eta = K_\eta * \nu. \quad (2641)$$

The finite function $\rho_{N,\eta}$ is a broadened observable, not an unsmoothed finite DOS density.

Quantitative vanishing broadening

Theorem 143 (CDF local law and joint limit). *Assume*

$$d\nu(E) = \rho(E) dE, \quad \rho \in C^{0,\alpha}(\mathbb{R}), \quad 0 < \alpha \leq 1, \quad (2642)$$

where ρ is supported in I . Define the quantitative CDF error

$$\kappa_N = \sup_{E \in \mathbb{R}} |F_N(E) - F(E)|. \quad (2643)$$

Then, for every $\eta > 0$,

$$\|\rho_{N,\eta} - \rho\|_\infty \leq \frac{\|K'\|_1 \kappa_N}{\eta} + [\rho]_{C^{0,\alpha}} m_\alpha(K) \eta^\alpha. \quad (2644)$$

Consequently, if

$$\eta_N \longrightarrow 0, \quad \frac{\kappa_N}{\eta_N} \longrightarrow 0, \quad (2645)$$

then

$$\|\rho_{N,\eta_N} - \rho\|_\infty \longrightarrow 0. \quad (2646)$$

Set

$$A = \|K'\|_1, \quad B = [\rho]_{C^{0,\alpha}} m_\alpha(K). \quad (2647)$$

When $A, B > 0$, the minimizing scale for the right side of Eq. (2644) is

$$\eta_N^{\text{opt}} = \left(\frac{A \kappa_N}{\alpha B} \right)^{1/(\alpha+1)}, \quad (2648)$$

and the optimized bound is

$$(\alpha+1) \alpha^{-\alpha/(\alpha+1)} (A \kappa_N)^{\alpha/(\alpha+1)} \times B^{1/(\alpha+1)}. \quad (2649)$$

Thus the sharp rate supplied by these two terms is $O(\kappa_N^{\alpha/(\alpha+1)})$.

Proof. Let $\sigma_N = \nu_N - \nu$, and let $G_N(E) = F_N(E) - F(E)$. For fixed x , Stieltjes integration by parts gives

$$(K_\eta * \sigma_N)(x) = \int_{\mathbb{R}} K_\eta(x - E) dG_N(E). \quad (2650)$$

$$(K_\eta * \sigma_N)(x) = \int_{\mathbb{R}} G_N(E) K'_\eta(x - E) dE. \quad (2651)$$

Consequently,

$$|(K_\eta * \sigma_N)(x)| \leq \kappa_N \|K'_\eta\|_1 = \frac{\|K'\|_1 \kappa_N}{\eta}. \quad (2652)$$

The $W^{1,1}$ kernel vanishes at infinity in its absolutely continuous representative, so there is no boundary contribution.

For the bias, positivity and unit mass give

$$|(K_\eta * \rho)(x) - \rho(x)| \leq [\rho]_{C^{0,\alpha}} m_\alpha(K) \eta^\alpha. \quad (2653)$$

Adding Eqs. (2652) and (2653) proves Eq. (2644). The joint condition proves convergence. Differentiating $A \kappa_N / \eta + B \eta^\alpha$ gives Eq. (2648); substitution gives Eq. (2649). $\square$

The first term in Eq. (2644) is a local-law term, not a consequence of weak convergence. No exchange of $N \rightarrow \infty$ and $\eta \rightarrow 0$ is supported without Eq. (2645).

A genuine unsmoothed-density theorem

Theorem 144 (Uniform convergence of existing densities). *Let $d\nu_N = \rho_N(E) dE$ on I , and suppose*

$$\sup_N \|\rho_N\|_\infty < \infty, \quad \sup_N [\rho_N]_{C^{0,\alpha}(I)} < \infty. \quad (2654)$$

If ν_N converges weakly to ν , then ν has a $C^{0,\alpha}$ density ρ , and

$$\|\rho_N - \rho\|_{L^\infty(I)} \rightarrow 0. \quad (2655)$$

This theorem does not apply to the atomic measures in Eq. (2637).

Proof. Equation (2654) makes the family uniformly bounded and equicontinuous. Arzelà–Ascoli gives a uniformly convergent subsequence from every subsequence. If its limit is f , then for every continuous test function g ,

$$\begin{aligned} \int_I g(E) f(E) dE &= \lim_N \int_I g(E) \rho_N(E) dE \\ &= \int_I g(E) d\nu(E). \end{aligned} \quad (2656)$$

Thus $d\nu = f dE$. Any two subsequential uniform limits agree almost everywhere and, by continuity, everywhere. Hence the entire sequence converges uniformly to the unique density $\rho = f$. The common Hölder bound passes to the limit. $\square$

Coherence-weighted local laws

Definition 53 (Preselected positive coherence measure). Let $0 \leq C_N \leq I$ and $0 \leq C \leq I$ be chosen at the reference point, using the specified target construction of Theorem 13. Define

$$\nu_N^C(B) = \tau_N(C_N^{1/2} \mathbf{1}_B(H_N) C_N^{1/2}) \quad (2657)$$

and

$$\nu^C(B) = \tau(C^{1/2} \mathbf{1}_B(H) C^{1/2}). \quad (2658)$$

By traciality these equal $\tau_N(C_N \mathbf{1}_B(H_N))$ and $\tau(C \mathbf{1}_B(H))$. They are positive measures of mass at most one.

Theorem 145 (Coherence transport and weighted joint law). Assume a common spectral interval $I = [-M, M]$. For a real polynomial

$$q(t) = \sum_{r=0}^d c_r t^r, \quad L_q(M) := \sum_{r=1}^d r |c_r| M^{r-1}. \quad (2659)$$

Suppose the declared Hamiltonian, trace, and coherence transports obey

$$\begin{aligned} |\tau_N(C_N q(H_N)) - \tau(C q(H))| \\ \leq r_N(q) + e_{C,N} \|q\|_{L^\infty(I)} + L_q(M) e_{H,N}, \end{aligned} \quad (2660)$$

where the right side tends to zero for each fixed q . Then

$$\nu_N^C \Rightarrow \nu^C. \quad (2661)$$

Let

$$\kappa_N^C = \sup_{E \in \mathbb{R}} |\nu_N^C((-\infty, E]) - \nu^C((-\infty, E])|. \quad (2662)$$

If

$$d\nu^C(E) = \rho_C(E) dE, \quad \rho_C \in C^{0,\alpha}(\mathbb{R}), \quad (2663)$$

then

$$\begin{aligned} \|K_\eta * \nu_N^C - \rho_C\|_\infty &\leq \frac{\|K'\|_1 \kappa_N^C}{\eta} \\ &\quad + [\rho_C]_{C^{0,\alpha}} m_\alpha(K) \eta^\alpha. \end{aligned} \quad (2664)$$

Thus the same joint condition and optimized $O((\kappa_N^C)^{\alpha/(\alpha+1)})$ rate hold with κ_N^C in place of κ_N .

Proof. Polynomial approximation is uniform on I . Positivity bounds the total masses, so Eq. (2660) extends from polynomials to all continuous tests, proving weak convergence. Apply the Stieltjes integration-by-parts and Hölder-bias proof of Theorem 143 to the two positive weighted measures. Their masses need not equal one; the kernel vanishes at infinity, so the same calculation has no boundary term. $\square$

When C_N is built from the preselected Riesz target, the norm estimates in Theorem 14, Theorem 34, and Theorem 141 control $e_{C,N}$. Exact finite-fiber participation and near-covariant coherence are respectively Theorems 32 and 33. Singular active rank must retain the margin in Theorem 52. None of these projector statements implies the separate weighted CDF rate $\kappa_N^C \rightarrow 0$.

A smooth-sector absolute-continuity certificate

Theorem 146 (Coarea density in a declared smooth sector). Let $\mathcal{M}$ be a compact q -dimensional Riemannian parameter manifold with C^1 nonnegative density $w(k)$. Let I_0 be a compact energy window whose branch preimages stay away from the boundary of every chart. Assume finitely many isolated branches

$$E_a \in C^2(\mathcal{M}), \quad 1 \leq a \leq r, \quad (2665)$$

and C^1 coherence weights $0 \leq c_a \leq 1$. Suppose

$$|\nabla E_a(k)| \geq v_0 > 0 \quad (2666)$$

whenever $E_a(k) \in I_0$. Define sector measures by

$$\nu(B) = \sum_{a=1}^r \int_{\mathcal{M}} \mathbf{1}_B(E_a(k)) w(k) d\text{vol}(k) \quad (2667)$$

and

$$\nu^C(B) = \sum_{a=1}^r \int_{\mathcal{M}} \mathbf{1}_B(E_a(k)) c_a(k) w(k) d\text{vol}(k). \quad (2668)$$

Then both measures are absolutely continuous on I_0 , with

$$\rho(E) = \sum_{a=1}^r \int_{E_a^{-1}(E)} \frac{w(k)}{|\nabla E_a(k)|} d\sigma_E(k) \quad (2669)$$

and

$$\rho_C(E) = \sum_{a=1}^r \int_{E_a^{-1}(E)} \frac{c_a(k)w(k)}{|\nabla E_a(k)|} d\sigma_E(k). \quad (2670)$$

Uniform bounds on the displayed C^2/C^1 data and regular level-set charts make these densities Lipschitz, hence $C^{0,\alpha}$ for every $0 < \alpha \leq 1$, on a strictly interior regular-value subwindow.

Proof. The coarea formula applied branch by branch gives Eqs. (2669) and (2670). The velocity margin makes the denominators uniformly finite. On a regular-value chart, the vector field

$$V_a = \frac{\nabla E_a}{|\nabla E_a|^2} \quad (2671)$$

transports the level set at energy E to that at nearby energy. Its flow, the hypersurface Jacobian, and the two integrands have uniformly bounded first variation under the stated compact C^2/C^1 bounds. The two fiber integrals are therefore Lipschitz in E . $\square$

A van Hove critical point violates Eq. (2666) and ends this certificate. Moreover, Theorem 134 does not turn a non-type-I reduced bulk into a smooth finite-dimensional Bloch manifold. The coarea theorem concerns only the declared smooth sector. For the full reduced trace measure, absolute continuity and Hölder regularity are independent spectral hypotheses or independent theorems.

Counterexamples and permanent statement policy

Proposition 91 (Weak and spectral convergence do not control peaks). *Each of the following obstruction mechanisms is explicit.*

1. *Empirical grid measures converge weakly to Lebesgue measure while every finite measure remains atomic and has no ordinary density.*
2. *On $[0, 2\pi]$, the probability densities*

$$\rho_N(x) = \frac{1 + \frac{1}{2} \sin(Nx)}{2\pi} \quad (2672)$$

converge weakly to $1/(2\pi)$, but their uniform distance from that limit is $1/(4\pi)$.

3. *If mass $\varepsilon_N \rightarrow 0$ is placed uniformly on an interval of width ε_N^2 , the measures can converge weakly while the added peak height is $\varepsilon_N^{-1} \rightarrow \infty$.*
4. *Diagonal matrices with spectral set $\{0, 1\}$ can have opposite normalized multiplicities. Their supports, edges, and Hausdorff spectral data agree, while their DOS measures approach different atoms.*

5. *Keeping the same Hamiltonian and changing a positive contraction that selects complementary eigenspaces leaves the unweighted measure unchanged but changes the coherence-weighted measure.*

Thus weak convergence, moments, faithful support, Hausdorff convergence, edge-gap convergence, and operator-norm location control do not imply a local law or uniform unsmoothed-density convergence.

Proof. The first item follows because a finite sum of Dirac masses is singular. The oscillatory term in the second item has zero weak limit by integration against continuous tests, while its amplitude is fixed. The third item has vanishing total added mass and diverging height. The fourth changes only multiplicity, which spectral-set metrics do not record. The fifth follows directly from the definition in Eq. (2657). $\square$

The moment and weak-DOS theorem, no-pollution theorem, and exponential extension in the spectral-loss analysis remain respectively Theorem 62, Theorem 64, and Theorem 65. Global island support from Theorem 130, operator-wise tower verification from Theorems 132 and 136, and full-shell collapse from Theorem 139 do not strengthen them into a local law.

Proposition 92 (Permanent DOS statement policy). *Without both a Hölder limiting density and a displayed quantitative CDF local law, every DOS-peak statement is labeled a fixed-broadening DOS observable. An unsmoothed-density limit is supported only by Theorem 144, Theorem 143, or another theorem proving equivalent regularity and rate hypotheses. A coherence-weighted limit additionally requires its own weighted CDF law. Riesz transport does not replace that law.*

Proof. The distinction between the finite atomic measure and its convolution is Definition 52. The need for a rate is Eq. (2644); the need for weighted information is Eq. (2664). Proposition 91 rules out all weaker implications. $\square$

Finite-cover matrix elements still require Theorem 58; intrinsic island transport and projector topology retain Theorems 60 and 131. The established finite-cover and island results in Corollary 74 and Theorem 142 remain spectral, not local-law, statements. No root, Hodge, representation-completeness, crystallographic, arithmetic, or material conclusion follows. The square five-state target remains root-free. the sequence-class theorem arithmetic and the exact-cell classification analysis remain outside this subsection.

Sequence-dependent arithmetic universality classes

Exact crystallographic growth is classified here as data of a chosen exact sequence. The geometric-effective exponent, the exact exponent, its arithmetic correction, and

any spectral root remain separate layers under the no-leakage theorem 90.

Exact classes and their invariances

Definition 54 (Exact exponent and envelope classes). *The Euclidean and hyperbolic exact-sequence domains are those in Definition 17. Let $s_j \rightarrow 0$, let $N_j^{\text{sc}} \geq 1$ be an exact crystallographic cell count, and put*

$$\ell_j = \log s_j^{-1}. \quad (2673)$$

Let $p > 0$ be the declared geometric-effective exponent, meaning

$$\frac{\log N_j^{\text{eff}}}{\ell_j} \rightarrow p. \quad (2674)$$

Define the running exact exponent and arithmetic correction by

$$\beta_j = \frac{\log N_j^{\text{sc}}}{\ell_j}, \quad \eta_j = \beta_j - p. \quad (2675)$$

The sequence belongs to the convergent class $\mathfrak{U}_p(\beta)$, $\beta \in [0, \infty]$, when

$$\beta_j \rightarrow \beta. \quad (2676)$$

It belongs to the envelope class $\mathfrak{U}_p[\beta_-, \beta_+]$ when

$$\liminf_{j \rightarrow \infty} \beta_j = \beta_-, \quad \limsup_{j \rightarrow \infty} \beta_j = \beta_+. \quad (2677)$$

The endpoints are extended nonnegative real numbers. These classes belong to the sequence, not to the lattice without a specified sequence.

If

$$N_j^{\text{sc}} = \mathcal{L}_j N_j^{\text{eff}}, \quad (2678)$$

then

$$\beta_j = p + \frac{\log \mathcal{L}_j}{\ell_j} + o(1). \quad (2679)$$

This is the paired-exponent identity of Theorem 93 in sequence-class notation.

Proposition 93 (Subpower invariance). *Suppose positive factors u_j, v_j define*

$$\tilde{N}_j = u_j N_j^{\text{sc}}, \quad \tilde{s}_j = v_j s_j, \quad (2680)$$

with

$$\log u_j = o(\ell_j), \quad \log v_j = o(\ell_j). \quad (2681)$$

Then every convergent class and every extended-real envelope class is unchanged. This includes bounded positive multiplicative changes, finite-index group replacement, constant layer or cell multiplicity, and the use of θ_j instead of $2s_j = 2 \sin(\theta_j/2)$.

Every subsequence of $\mathfrak{U}_p(\beta)$ retains β . A subsequence of an envelope class retains the same class when it retains both endpoint limits.

Proof. The modified denominator satisfies

$$\log \tilde{s}_j^{-1} = \ell_j - \log v_j = \ell_j[1 + o(1)], \quad (2682)$$

and the modified numerator is

$$\log \tilde{N}_j = \beta_j \ell_j + o(\ell_j). \quad (2683)$$

Thus the new exponent is $\beta_j[1 + o(1)] + o(1)$. This preserves every finite limit point and sends an unbounded subsequence to infinity, proving the extended envelope statement. A finite index and a fixed multiplicity give bounded factors. Since $2 \sin(\theta/2)/\theta \rightarrow 1$, the angle convention also satisfies Eq. (2681). The subsequence statements follow from the definitions. $\square$

Bounded arithmetic locking is sufficient but not necessary for $\mathfrak{U}_p(p)$. Indeed, bounded $\mathcal{L}_j$ makes the correction in Eq. (2679) vanish, but so does the unbounded subpower choice

$$\mathcal{L}_j = \exp(\sqrt{\ell_j}). \quad (2684)$$

This sharpens the locking alternatives in Proposition 1 and Corollary 56.

Height rays and envelopes

Definition 55 (Height ray). *Let $H_j \rightarrow \infty$, and suppose*

$$N_j^{\text{sc}} = q_j R_j, \quad \log R_j = o(\log H_j). \quad (2685)$$

Define

$$a_j = \frac{\log q_j}{\log H_j}, \quad b_j = \frac{\ell_j}{\log H_j}. \quad (2686)$$

Theorem 147 (Height-ray quotient and envelope). *If*

$$a_j \rightarrow a, \quad b_j \rightarrow b > 0, \quad (2687)$$

then

$$\beta = \frac{a}{b}, \quad \eta = \frac{a}{b} - p. \quad (2688)$$

More generally, suppose $0 \leq a_- \leq a_+$ and $0 < b_- \leq b_+ < \infty$, with

$$a_- + o(1) \leq a_j \leq a_+ + o(1) \quad (2689)$$

and

$$b_- + o(1) \leq b_j \leq b_+ + o(1). \quad (2690)$$

Then every limiting exact exponent belongs to

$$\left[\frac{a_-}{b_+}, \frac{a_+}{b_-} \right]. \quad (2691)$$

This is an inclusion and does not assert realization of every interior value.

Proof. Equations (2685) and (2686) give

$$\beta_j = \frac{a_j + o(1)}{b_j}. \quad (2692)$$

Taking limits proves Eq. (2688). Under the interval hypotheses, positivity of the denominators gives the lower quotient a_-/b_+ and upper quotient a_+/b_- for every convergent subsequence, which proves Eq. (2691). $\square$

Complete Euclidean realizability

For a primitive positive pair (m_j, n_j) , the exact Euclidean identities are

$$s_j = \frac{n_j}{\sqrt{m_j^2 + n_j^2}} \quad (2693)$$

and

$$N_j^{\text{sc}} = \frac{2(m_j^2 + n_j^2)}{d_j}, \quad d_j \in \{1, 2\}. \quad (2694)$$

Theorem 148 (Euclidean realizable exponent set). *Among unrestricted primitive exact Euclidean sequences with $s_j \rightarrow 0$, the complete realizable convergent exponent set is*

$$[2, \infty]. \quad (2695)$$

For each finite $\beta \in [2, \infty)$, it is realized by increasing primes m_j and

$$n_j = 1 \quad \text{when } \beta = 2, \quad (2696)$$

or, when $\beta > 2$, by

$$\gamma = 1 - \frac{2}{\beta}, \quad n_j = \lfloor m_j^\gamma \rfloor. \quad (2697)$$

The exponent $\beta = \infty$ is realized by

$$n_j = \left\lfloor \frac{m_j}{\log m_j} \right\rfloor. \quad (2698)$$

Proof. First, Eqs. (2693) and (2694) imply the exact identity

$$N_j^{\text{sc}} = \frac{2}{d_j} n_j^2 s_j^{-2}. \quad (2699)$$

Since $n_j \geq 1$ and $d_j \leq 2$, every running exponent has

$$\beta_j \geq 2. \quad (2700)$$

Thus no convergent exponent below two is realizable.

For finite $\beta > 2$, one has $0 < \gamma < 1$. Eventually $1 \leq n_j < m_j$; primality of m_j makes the pair primitive. The floor changes $\log n_j$ by $o(\log m_j)$, and $n_j/m_j \rightarrow 0$. Hence

$$\log s_j^{-1} = (1 - \gamma) \log m_j + o(\log m_j) \quad (2701)$$

and

$$\log N_j^{\text{sc}} = 2 \log m_j + O(1). \quad (2702)$$

Their quotient is $2/(1 - \gamma) = \beta$. The case $n_j = 1$ gives $\beta = 2$ by the same formulas.

For Eq. (2698), the pair is again primitive for large j , and

$$\log s_j^{-1} = \log \log m_j + o(\log \log m_j), \quad (2703)$$

while Eq. (2702) remains valid. The quotient tends to infinity. In every case $d_j \in \{1, 2\}$ changes the logarithm only by a bounded amount. This proves both realization and completeness of Eq. (2695). $\square$

Corollary 75 (Every Euclidean exponent envelope). *For every*

$$2 \leq \beta_- \leq \beta_+ \leq \infty, \quad (2704)$$

there is a primitive exact Euclidean sequence in $\mathfrak{U}_2[\beta_-, \beta_+]$.

Proof. Choose increasing primes and apply the construction in Theorem 148 with exponent β_- on the even subsequence and exponent β_+ on the odd subsequence. The primes can be selected successively so the full sequence is increasing. Every term lies in one of the two constructions, so no exponent lies below β_- or above β_+ , while both endpoints are attained. This proves the exact lower and upper envelope. If the endpoints agree, use the single convergent construction. $\square$

Theorem 148 strengthens the obstruction in Theorem 94: exponent two is the lower endpoint, not an unconditional exponent for unrestricted exact Euclidean sequences.

The established hyperbolic class

Theorem 149 (The explicit centered height class). *For the centered hyperbolic sequence of the arithmetic-height result, put*

$$H_j = H_{\mathcal{M}}(x_j) = \varepsilon_j n_j^2. \quad (2705)$$

The exact arithmetic data obey

$$q_j = H_j \mathcal{F}_j, \quad \log \mathcal{F}_j = o(\log H_j), \quad (2706)$$

and

$$s_j = H_j^{-1/4+o(1)}. \quad (2707)$$

Consequently the height class is

$$(a, b) = \left(1, \frac{1}{4}\right), \quad (2708)$$

the exact sequence class is

$$\mathfrak{U}_1(4), \quad (2709)$$

and the arithmetic correction relative to the hyperbolic geometric exponent ($p=1$) is

$$\eta = 4 - 1 = 3. \quad (2710)$$

Proof. The local-to-global degree formula in Lemma 2 and its subpower Euler factor give Eq. (2706); the exact sequence height is Eq. 1100. Thus $\log q_j / \log H_j \rightarrow 1$. The fixed layer multiplicity gives $N_j^{\text{sc}} = 2n_b q_j$, so the height remainder in Definition 55 is a positive constant. Equation (2707) gives $\ell_j / \log H_j \rightarrow 1/4$. Theorem 147 gives exponent four, agreeing with Eqs. (1109) and the leading result in Theorem 3. The geometric hyperbolic exponent is one, so the correction is three. Subpower and finite-index factors preserve the class by Proposition 93. $\square$

Proposition 94 (Open hyperbolic classification). *The established hyperbolic realizable set contains exponent four in the centered (2,3,8) class. The present results do not prove that four is unique, that all centered commensurator rays have one exponent, that exact exponent one is universal, or that the hyperbolic realizable set has a finite upper bound. Every additional ray must supply its own height-index and angle-height estimates. Commensurator density alone supplies neither.*

Proof. Theorem 149 proves realization of four. The locking failure in Eq. 1112 and the obstruction in Theorem 95 rule out promotion to exponent-one universality. No routed theorem supplies height estimates for every other ray, so no broader classification follows. $\square$

Conditional magic-complexity transfer

Theorem 150 (Sequence-class transfer to magic complexity). *Suppose an independently established spectral branch samples an exact sequence with*

$$\ell_j = c\omega_j^{-1/2} + o(\omega_j^{-1/2}), \quad c > 0. \quad (2711)$$

If the sequence belongs to $\mathfrak{U}_p(\beta)$, then

$$\sqrt{\omega_j} \log N_j^{\text{sc}} \longrightarrow \beta c. \quad (2712)$$

For an envelope class, the lower and upper limits are respectively $c\beta_-$ and $c\beta_+$, with the extended-real convention.

On the explicit the magic-surface sampling result sequence, $\beta = 4$, so the coefficient is $4c$. This recovers the coefficient in Eq. 1131 and does not alter the off-sequence detuning in Proposition 2.

Proof. By definition,

$$\sqrt{\omega_j} \log N_j^{\text{sc}} = \beta_j \sqrt{\omega_j} \ell_j. \quad (2713)$$

Equation (2711) makes the second factor converge to c . The convergent and envelope conclusions follow. Exact sampling and its complexity are conditional on Theorems 4 and 5; the arithmetic calculation does not prove their spectral premise. $\square$

Class ledger and subsequent rule

Proposition 95 (Four-layer class ledger). *The following data have distinct owners.*

1. *The geometric-effective exponent p belongs to N_j^{eff} .*
2. *The exact crystallographic class β , or its envelope, belongs to the chosen exact sequence N_j^{sc} .*
3. *The arithmetic correction is $\eta = \beta - p$, or the corresponding shifted envelope.*
4. *A spectral root, branch, and weak-coupling law belong to an independently established Hamiltonian theorem.*

Arithmetic may sample an existing root but cannot create one.

Proof. The first three statements are Definition 54 and Eq. (2679). The fourth follows from the no-leakage and exact guards in Theorem 90 and Corollary 54. The normalization separation theorem is Theorem 96. The dimension-independent version of this bookkeeping is Corollary 58. $\square$

Proposition 96 (Permanent exact-cell statement rule). *Every exact-cell exponent statement carried into the exact-cell classification must name a sequence class $\mathfrak{U}_p(\beta)$ or display an envelope $\mathfrak{U}_p[\beta_-, \beta_+]$. Exact hyperbolic exponent one remains only a conditional class. The explicit centered hyperbolic sequence is $\mathfrak{U}_1(4)$, with arithmetic correction three, and cannot be relabeled as exponent one.*

No class statement creates representation completeness, finite-cover convergence, a spectral root, Hodge cancellation, arithmetic-to-spectral leakage, crystallographic universality, or material identification. The exact square five-state target remains root-free.

Proof. The Euclidean obstruction and hyperbolic obstruction are Theorems 94 and 95. Exact magic cells remain conditional as stated in Proposition 55. The symmetric separation theorem is Corollary 57, and the higher-dimensional layer separation theorem is Corollary 59. These results condition the class language above and no stronger statement. $\square$

Sequence classes for exact and magic cells

This subsection supersedes every subsequent statement that passes from a geometric effective count to an exact

crystallographic count without an arithmetic hypothesis. It also supersedes every magic-cell statement that uses an exact-cell exponent without naming both the spectral branch and the arithmetic sequence. The exact locking criteria are those of Proposition 1; the centered exponent-four calculation is Theorem 3; and the sequence-class formulation is Definition 54.

Geometric, crystallographic, and spectral classes

Definition 56 (Three classes). *For $X \in \{E, H\}$, let $p_E = 2$ and $p_H = 1$. Every count statement is assigned exactly one primary owner.*

1. A geometric-effective statement concerns N_X^{eff} and is supported by local geometry.
2. An exact-crystallographic statement concerns $N_{X,j}^{\text{sc}}$ along a declared exact sequence and names either $\mathfrak{U}_p(\beta)$ or $\mathfrak{U}_p[\beta_-, \beta_+]$.
3. A spectral-conditional statement assumes an independently established branch or root. If it counts exact cells, it additionally names the exact-crystallographic class.

The adjective exact does not by itself identify an owner, and the adjective magic does not merge the spectral and crystallographic owners.

Definition 57 (Admissible asymptotic statement). *Let $s_j \rightarrow 0$ and put*

$$\ell_j = \log s_j^{-1}, \quad \beta_{X,j}^{\text{sc}} = \frac{\log N_{X,j}^{\text{sc}}}{\ell_j}. \quad (2714)$$

A statement is supported if it has one of the following forms:

1. it explicitly says effective and states a law for N_X^{eff} ;
2. it explicitly names an exact class or envelope;
3. it explicitly says spectral conditional, identifies the established branch, and names the arithmetic class whenever exact cells are counted.

Theorem 151 (Three-tier endpoint vocabulary). *The unconditional geometric-effective endpoint laws are*

$$N_E^{\text{eff}} = |\theta|^{-2+o(1)}, \quad N_H^{\text{eff}} = |\theta|^{-1+o(1)}. \quad (2715)$$

Along an exact sequence the crystallographic law is instead

$$\beta_X^{\text{sc}} = p_X + \eta_X^{\text{ar}}, \quad (2716)$$

or the corresponding lower and upper envelopes. A magic-cell law further requires an independently established spectral branch or root. None of these three owners implies either of the other two without the hypotheses of an explicit bridge theorem.

Proof. Equation (2715) is the local geometric content retained by Theorem 93. On an exact sequence,

$$N_{X,j}^{\text{sc}} = L_{X,j} N_{X,j}^{\text{eff}}. \quad (2717)$$

Taking logarithms and dividing by ℓ_j gives

$$\beta_{X,j}^{\text{sc}} = \frac{\log N_{X,j}^{\text{eff}}}{\ell_j} + \frac{\log L_{X,j}}{\ell_j}. \quad (2718)$$

Thus the second term is independent arithmetic data. The no-leakage theorem 90 forbids promoting it from geometry. Likewise, neither factorization (2717) nor a count asymptotic asserts that a Hamiltonian has a root. This proves all three separations. $\square$

Exact substitution theorem

Theorem 152 (Exact statement substitution rule). *Assume*

$$N_{X,j}^{\text{eff}} = s_j^{-p_X+o(1)}, \quad N_{X,j}^{\text{sc}} = L_{X,j} N_{X,j}^{\text{eff}}. \quad (2719)$$

Then the following statements hold.

1. The exact sequence is in $\mathfrak{U}_{p_X}(\beta)$ if and only if

$$\frac{\log L_{X,j}}{\ell_j} \rightarrow \beta - p_X. \quad (2720)$$

2. Exact exponent inheritance, $\beta = p_X$, is equivalent to subpower locking $\log L_{X,j} = o(\ell_j)$.
3. If $N_{X,j}^{\text{eff}} = \Theta(s_j^{-p_X})$, then $N_{X,j}^{\text{sc}} = \Theta(s_j^{-p_X})$ if and only if $L_{X,j}$ is bounded above and bounded away from zero.
4. If $L_{X,j} \rightarrow L_\infty \in (0, \infty)$, then the leading amplitude is multiplied by L_∞ .

Consequently, a statement asserting $N_{H,j}^{\text{sc}} = |\theta_j|^{-1+o(1)}$ is supported only for the conditional class $\mathfrak{U}_1(1)$, and a two-sided $\Theta(|\theta_j|^{-1})$ statement additionally requires bounded locking.

Proof. The first assertion follows immediately from Eq. (2718). Setting $\beta = p_X$ proves the second. Under the two-sided effective estimate, multiplication by $L_{X,j}$ preserves both sides exactly when $L_{X,j}$ has positive uniform lower and finite uniform upper bounds. Passing to a positive finite limit proves the amplitude statement. Replacing s_j by $|\theta_j|$ changes logarithms only by a sub-leading term in the endpoint chart, as already encoded in the subpower invariance of the sequence-class theorem. The hyperbolic conclusion follows from $p_H = 1$. $\square$

Corollary 76 (Centered sequence classification). *For the explicit centered hyperbolic sequence,*

$$\begin{aligned} q_j &= |\theta_j|^{-4+o(1)}, \\ N_{H,j}^{\text{sc}} &= |\theta_j|^{-4+o(1)}, \\ L_{H,j} &= |\theta_j|^{-3+o(1)}. \end{aligned} \quad (2721)$$

Hence this sequence belongs to $\mathfrak{U}_1(4)$ and has $\eta_H^{\text{ar}} = 3$. It violates exponent-one locking. This conclusion does not assert that four is the only realizable hyperbolic exact exponent.

Proof. The first two asymptotics are Eq. 1109. Dividing the second by the geometric effective law in Eq. (2715) gives the third. Equation (2720) then gives $\beta = 4$ and $\eta_H^{\text{ar}} = 4 - 1 = 3$. The open-classification statement is precisely Proposition 94. $\square$

Proposition 97 (Centered-sequence locking classification). *The geometric and crystallographic scaling laws are classified as follows.*

1. *Geometric-effective exponent one is already unconditional.*
2. *Exact exponent one is the conditional class $\mathfrak{U}_1(1)$.*
3. *The centered exact sequence is the proved class $\mathfrak{U}_1(4)$, so exponent-one locking fails on that sequence.*
4. *Other exact hyperbolic height rays may realize different sequence classes.*

The centered sequence therefore remains in class $\mathfrak{U}_1(4)$ under any broader classification.

Proof. The first statement is Eq. (2715). The second is Theorem 152. The third is Corollary 76. The fourth is the sharp scope of Proposition 94. Together they give the stated classification. $\square$

Magic complexity with an arithmetic class

Theorem 153 (Class-dependent magic-complexity transfer). *Suppose a richer microscopic model independently certifies a branch sampled at (θ_j, ω_j) and*

$$\log |\theta_j|^{-1} = c\omega_j^{-1/2} + o(\omega_j^{-1/2}), \quad c > 0. \quad (2722)$$

If the exact sampling lies in $\mathfrak{U}_p(\beta)$, then

$$\sqrt{\omega_j} \log N_j^{\text{sc}} \longrightarrow \beta c. \quad (2723)$$

If it lies in the envelope $\mathfrak{U}_p[\beta_-, \beta_+]$, then

$$\begin{aligned} \liminf_j \sqrt{\omega_j} \log N_j^{\text{sc}} &= \beta_- c, \\ \limsup_j \sqrt{\omega_j} \log N_j^{\text{sc}} &= \beta_+ c. \end{aligned} \quad (2724)$$

In particular, the explicit centered the magic-surface sampling result sampling has coefficient $4c$, whereas the conditional locked class $\mathfrak{U}_1(1)$ has coefficient c .

Proof. By the class hypothesis, $\log N_j^{\text{sc}} = (\beta + o(1)) \log |\theta_j|^{-1}$. Substitution of Eq. (2722) proves Eq. (2723). Applying the same positive multiplication to lower and upper limits proves Eq. (2724). The two specializations use $\beta = 4$ and $\beta = 1$, respectively. This is the sequence-class form of Theorem 150 and preserves the sharp coefficient in Eq. 1131. $\square$

Corollary 77 (Repair of the original complexity-transmutation statement). *The exact-cell conclusion of original Theorem 130 is retained only in the following form: first certify the spectral branch, then declare its exact sampling class, and finally use coefficient βc . The coefficient is $4c$ on the centered sequence and c only in $\mathfrak{U}_1(1)$. Arithmetic sampling cannot establish the branch.*

Proof. This is Theorem 153 with the logical order of its hypotheses made explicit. The exact square five-state model remains root-free, so no application to that model is supported. $\square$

Literal theorem-family dispositions

Proposition 98 (Disposition of original Theorem 178). *The retained endpoint comparison is geometric-effective and is exactly Eq. (2715). Its exponents are two and one. No exact-cell conclusion is part of this unconditional statement.*

Proof. This is the geometric component of Theorem 93, classified by Definition 56. $\square$

Proposition 99 (Disposition of original Theorem 179). *The exact endpoint exponents are*

$$\beta_E^{\text{sc}} = 2 + \eta_E^{\text{ar}}, \quad \beta_H^{\text{sc}} = 1 + \eta_H^{\text{ar}}, \quad (2725)$$

or the corresponding lower and upper envelopes. No default value is assigned to either arithmetic correction.

Proof. Apply Eq. (2716) with $p_E = 2$ and $p_H = 1$. The envelope statement follows by taking lower and upper limits in Eq. (2718). $\square$

Proposition 100 (Disposition of original Theorem 180). *The three inheritance claims are separated: subpower locking transfers an exponent, bounded locking transfers a two-sided Θ law, and a positive finite locking limit transfers the leading amplitude.*

Proof. These are items two through four of Theorem 152, consistent with Corollary 56. $\square$

Proposition 101 (Disposition of original Theorem 181). *Endpoint universality at the geometric-effective layer does not imply crystallographic universality. Exact endpoint comparisons must name a pair of classes or envelopes before their exponents can be compared.*

Proof. The independent arithmetic term in Eq. (2718) may differ between endpoints and sequences. The no-leakage theorem 90 forbids its deletion. $\square$

Proposition 102 (Disposition of original Theorem 182). *Every magic sampling statement is conditional on an independently established spectral branch or root. An exact magic-cell statement is additionally conditional on a declared arithmetic class or envelope.*

Proof. The hypotheses of Theorem 153 separate the spectral branch from its arithmetic sampling. Neither hypothesis is a consequence of the other. $\square$

Proposition 103 (Disposition of original Theorem 183). *The branch $N_M^{\text{sc}} = \Theta(|\theta|^{-1})$, including the family around eq:VIIB.Nsc.theta.H, is retained only under bounded locking in $\mathfrak{U}_1(1)$. On the centered exact magic sequence it is replaced by*

$$N_{M,j}^{\text{sc}} = |\theta_j|^{-4+o(1)}. \quad (2726)$$

Proof. The conditional inverse-linear statement is item three of Theorem 152. The centered replacement is Corollary 76, restricted to the independently established sampling considered in the magic-surface sampling result. $\square$

Proposition 104 (Disposition of original Theorem 184). *The effective magic-cell exponents are two and one. The exact exponents are $2+\eta_E^{\text{ar}}$ and $1+\eta_H^{\text{ar}}$, or their envelopes. On the explicit centered hyperbolic sampling the exact exponent is four. Every spectral adjective in this synthesis remains conditional on a established model branch.*

Proof. Combine Propositions 98, 99, and 102 with Corollary 76. $\square$

Proposition 105 (Disposition of original Theorem 197). *For a declared hyperbolic exact sequence with $p_H = 1$,*

$$\mathfrak{U}_1(1) \iff \log L_{H,j} = o(\log |\theta_j|^{-1}). \quad (2727)$$

This is an if-and-only-if criterion for the conditional exponent-one class, not a generic hyperbolic conclusion. The centered sequence has

$$\frac{\log L_{H,j}}{\log |\theta_j|^{-1}} \longrightarrow 3 \quad (2728)$$

and therefore belongs to $\mathfrak{U}_1(4)$.

Proof. Equation (2727) is Eq. (2720) with $p_X = \beta = 1$. The centered limit is the locking failure in Eq. 1112; insertion into Eq. (2718) gives exponent four. $\square$

Proposition 106 (Disposition of original Theorem 199). *The final endpoint synthesis is supported only in the following layered form.*

1. *The hyperbolic geometric-effective exponent is one.*
2. *The hyperbolic exact-crystallographic endpoint is a declared class $\mathfrak{U}_1(\beta)$ or envelope $\mathfrak{U}_1[\beta_-, \beta_+]$.*
3. *The explicit centered family is $\mathfrak{U}_1(4)$.*
4. *A magic-complexity statement is spectral conditional and, after a root has been established, has arithmetic coefficient β .*

Proof. Items one and two follow from Theorems 151 and 152. Item three is Corollary 76. Item four is Theorem 153. $\square$

Supersession map and permanent separation theorem

Z. Incommensurate Twists, Finite Covers, and the Euclidean Limit

The preceding subsections established the local geometric theory, the exact hyperbolic commensurability criterion, and the distinction between effective moiré domains and exact crystallographic supercells. Three fundamental consistency questions remain. First, the negatively curved theory must recover the Euclidean moiré problem continuously when the curvature tends to zero. This recovery must be demonstrated not only at the level of the metric, but also for the twist displacement, the local moiré length, the effective area, and the running scaling exponent. Second, a generic relative twist does not satisfy the exact commensurability condition. Such configurations nevertheless possess a perfectly well-defined local moiré geometry. A theory of hyperbolic moiré matter must therefore remain meaningful even when no finite exact supercell exists. Third, practical spectral calculations necessarily employ finite systems. In hyperbolic geometry, naive open-cluster truncation is particularly dangerous because the boundary remains extensive. It is therefore necessary to construct finite approximants that recover the infinite hyperbolic bilayer locally without introducing spurious boundary-dominated physics. The present subsection resolves these three issues. We first prove the exact Euclidean limit and its leading curvature corrections. We then show that the Euclidean and small-angle limits are nonuniform and derive the corresponding double-scaling criterion. We next characterize generic incommensurate twists and establish quantitative convergence bounds for commensurate approximants. Finally, we construct finite-cover sequences with arbitrarily large injectivity radius and formulate the correct local-thermodynamic approximation scheme for incommensurate hyperbolic bilayers. The hyperbolic curvature is $K = -\frac{1}{R^2}$. The Euclidean metric limit corresponds to $K \longrightarrow 0^-$, or equivalently, $R \longrightarrow \infty$. The limit must be taken while keeping the physical microscopic registry length and the relative twist fixed. Thus, $a = \text{const.}$, and

$$0 < \left| \sin\left(\frac{\theta}{2}\right) \right| \leq 1 \quad (2729)$$

throughout the limiting procedure. For compact notation define

$$s = \left| \sin\left(\frac{\theta}{2}\right) \right|. \quad (2730)$$

The exact hyperbolic twist displacement is

$$d_\theta^{(H)}(r) = 2R \operatorname{arsinh} \left[s \sinh\left(\frac{r}{R}\right) \right]. \quad (2731)$$

At fixed geodesic radius, $r = \text{const.}$, one has $\frac{r}{R} \rightarrow 0$. The hyperbolic sine therefore admits the expansion

$$\sinh\left(\frac{r}{R}\right) = \frac{r}{R} + \frac{r^3}{6R^3} + \frac{r^5}{120R^5} + O(R^{-7}). \quad (2732)$$

Consequently,

$$s \sinh\left(\frac{r}{R}\right) = \frac{sr}{R} + \frac{sr^3}{6R^3} + O(R^{-5}). \quad (2733)$$

The inverse hyperbolic sine satisfies

$$\text{arsinh } x = x - \frac{x^3}{6} + \frac{3x^5}{40} + O(x^7). \quad (2734)$$

Substitution gives

$$d_\theta^{(H)}(r) = 2sr + \frac{s(1-s^2)}{3} \frac{r^3}{R^2} + O(R^{-4}). \quad (2735)$$

Using

$$1 - s^2 = \cos^2\left(\frac{\theta}{2}\right), \quad (2736)$$

the leading curvature correction becomes

$$d_\theta^{(H)}(r) = 2r \left| \sin\left(\frac{\theta}{2}\right) \right| + \frac{r^3}{3R^2} \left| \sin\left(\frac{\theta}{2}\right) \right| \cos^2\left(\frac{\theta}{2}\right) + O(R^{-4}). \quad (2737)$$

Therefore,

$$\lim_{R \rightarrow \infty} d_\theta^{(H)}(r) = 2r \left| \sin\left(\frac{\theta}{2}\right) \right|. \quad (2738)$$

The right-hand side is exactly the Euclidean twist-displacement law,

$$d_\theta^{(E)}(r) = 2r \left| \sin\left(\frac{\theta}{2}\right) \right|. \quad (2739)$$

Thus the finite-angle displacement field converges continuously to its Euclidean counterpart. The hyperbolic local moiré length is

$$\xi_M^{(H)} = R \text{arsinh} \left[\frac{\sinh(a/2R)}{s} \right]. \quad (2740)$$

We now prove directly that

$$\lim_{R \rightarrow \infty} \xi_M^{(H)} = \frac{a}{2s}. \quad (2741)$$

Introduce the dimensionless variable $\alpha = \frac{a}{2R}$. Then $R = \frac{a}{2\alpha}$, and the Euclidean limit is $\alpha \rightarrow 0^+$. The local moiré length becomes

$$\xi_M^{(H)} = \frac{a}{2} \frac{\text{arsinh} [\sinh \alpha / s]}{\alpha}. \quad (2742)$$

Both numerator and denominator vanish in the limit,

$$\lim_{\alpha \rightarrow 0} \text{arsinh} \left[\frac{\sinh \alpha}{s} \right] = 0, \quad (2743)$$

and $\lim_{\alpha \rightarrow 0} \alpha = 0$. L'Hôpital's rule therefore applies. Differentiating the numerator gives

$$\frac{d}{d\alpha} \text{arsinh} \left[\frac{\sinh \alpha}{s} \right] = \frac{\cosh \alpha}{\sqrt{s^2 + \sinh^2 \alpha}}. \quad (2744)$$

At vanishing argument, $\cosh 0 = 1$, and $\sinh 0 = 0$. Hence,

$$\lim_{\alpha \rightarrow 0} \frac{\cosh \alpha}{\sqrt{s^2 + \sinh^2 \alpha}} = \frac{1}{s}. \quad (2745)$$

Therefore,

$$\lim_{\alpha \rightarrow 0} \frac{\text{arsinh} [\sinh \alpha / s]}{\alpha} = \frac{1}{s}. \quad (2746)$$

Substitution into Eq. (2742) gives

$$\lim_{R \rightarrow \infty} \xi_M^{(H)}(\theta, R) = \frac{a}{2 |\sin(\theta/2)|}. \quad (2747)$$

This is precisely the Euclidean local moiré length,

$$\xi_M^{(E)} = \frac{a}{2 |\sin(\theta/2)|}. \quad (2748)$$

The Euclidean limit is therefore exact rather than merely asymptotic at the level of scaling exponents. The convergence rate can also be obtained analytically. Using the expansion derived previously, one has

$$\xi_M^{(H)} = \frac{a}{2s} - \frac{a^3 (1-s^2)}{48R^2 s^3} + O(R^{-4}). \quad (2749)$$

Equivalently,

$$\xi_M^{(H)} = \xi_M^{(E)} - \frac{a^3 \cos^2(\theta/2)}{48R^2 |\sin(\theta/2)|^3} + O(R^{-4}). \quad (2750)$$

Thus the leading negative-curvature correction decreases the moiré length. This is consistent with the exact inequality $\xi_M^{(H)} \leq \xi_M^{(E)}$. The exact hyperbolic effective area is

$$A_M^{(H)} = 2\pi R^2 \left[\sqrt{1 + \frac{\sinh^2(a/2R)}{s^2}} - 1 \right]. \quad (2751)$$

Using the dimensionless microscopic curvature parameter, this becomes

$$A_M^{(H)} = \frac{\pi a^2}{2\alpha^2} \left[\sqrt{1 + \frac{\sinh^2 \alpha}{s^2}} - 1 \right]. \quad (2752)$$

Rationalizing the square-root expression gives

$$A_M^{(H)} = \frac{\pi a^2}{2\alpha^2} \frac{\sinh^2 \alpha / s^2}{1 + \sqrt{1 + \sinh^2 \alpha / s^2}}. \quad (2753)$$

Therefore,

$$A_M^{(H)} = \frac{\pi a^2}{2s^2} \left[\frac{\sinh \alpha}{\alpha} \right]^2 \frac{1}{1 + \sqrt{1 + \sinh^2 \alpha / s^2}}. \quad (2754)$$

As the curvature vanishes, $\frac{\sinh \alpha}{\alpha} \rightarrow 1$, while

$$\sqrt{1 + \frac{\sinh^2 \alpha}{s^2}} \rightarrow 1. \quad (2755)$$

Hence,

$$\lim_{R \rightarrow \infty} A_M^{(H)} = \frac{\pi a^2}{4s^2}. \quad (2756)$$

The Euclidean effective area is therefore

$$A_M^{(E)} = \frac{\pi a^2}{4 \sin^2(\theta/2)}. \quad (2757)$$

The area ratio admits the expansion

$$\begin{aligned} \frac{A_M^{(H)}}{A_M^{(E)}} &= 1 + \alpha^2 \left[\frac{1}{3} - \frac{1}{4s^2} \right] \\ &+ \alpha^4 \left[\frac{2}{45} - \frac{1}{6s^2} + \frac{1}{8s^4} \right] + O(\alpha^6). \end{aligned} \quad (2758)$$

The Euclidean convergence is therefore controlled by even powers of the microscopic curvature parameter. At fixed nonzero twist,

$$A_M^{(H)} = A_M^{(E)} \left[1 + O\left(\frac{a^2}{R^2}\right) \right]. \quad (2759)$$

The effective microscopic count is

$$N_M^{\text{eff}} = 2n_b \frac{A_M^{\text{eff}}}{A_0}. \quad (2760)$$

Provided that the microscopic family is chosen so that the quotient density has a well-defined Euclidean limit,

$$\rho_0^{(E)} = \lim_{R \rightarrow \infty} \frac{n_b}{A_0(R)}, \quad (2761)$$

one obtains

$$\lim_{R \rightarrow \infty} N_M^{\text{eff},H} = 2\rho_0^{(E)} A_M^{(E)}. \quad (2762)$$

Thus the effective microscopic counting law also reduces continuously to its Euclidean counterpart whenever the microscopic lattice family itself possesses a controlled flat-space limit. The preceding limits are metric statements. They do not imply that an arbitrary fixed hyperbolic lattice becomes a square Euclidean lattice when the curvature radius is sent to infinity. This distinction is essential. For a torsion-free compact quotient of fixed genus, the primitive area is

$$A_0 = 4\pi R^2 (g_0 - 1). \quad (2763)$$

If the genus is held fixed while $R \rightarrow \infty$, then $A_0 \rightarrow \infty$. The quotient therefore does not approach a finite Euclidean primitive cell. Likewise, for a fixed regular hyperbolic tessellation, the microscopic edge length and curvature radius satisfy a fixed dimensionless relation. For a regular tessellation one has

$$\cosh\left(\frac{a}{2R}\right) = \frac{\cos(\pi/p)}{\sin(\pi/q)}. \quad (2764)$$

For fixed integers specifying the tessellation, the right-hand side is constant. Therefore, $\frac{a}{R} = \text{const.}$ within that fixed regular lattice family. Consequently, the operation $R \rightarrow \infty$ simultaneously sends the microscopic edge length to infinity rather than holding it fixed. Hence the limit proved in Eq. (2747) must be interpreted as a local metric Euclidean limit. A crystallographic Euclidean limit requires a family of hyperbolic lattice models satisfying

$$\frac{a(R)}{R} \rightarrow 0, \quad (2765)$$

together with controlled convergence of the local adjacency structure and microscopic couplings. If the desired flat-space reference is the square lattice, one must additionally require that the local lattice graph and its embedding converge to the Euclidean square-lattice structure. This is an independent microscopic assumption.

Thus, metric flat-space limit $\neq$ automatic square-lattice crystallographic limit. The Euclidean limit is nonuniform in the twist angle. To expose this singularity, define the normalized local-length ratio

$$\mathcal{R}_\xi = \frac{\xi_M^{(H)}}{\xi_M^{(E)}}. \quad (2766)$$

At fixed nonzero twist, $\lim_{R \rightarrow \infty} \mathcal{R}_\xi = 1$. Taking the small-angle limit afterwards gives

$$\lim_{\theta \rightarrow 0} \lim_{R \rightarrow \infty} \mathcal{R}_\xi = 1. \quad (2767)$$

Now reverse the order. At every fixed finite curvature radius, the small-angle hyperbolic length obeys

$$\xi_M^{(H)} = R \ln |\theta|^{-1} + O(R), \quad (2768)$$

whereas the Euclidean length obeys $\xi_M^{(E)} \sim \frac{a}{|\theta|}$. Therefore,

$$\mathcal{R}_\xi \sim \frac{R|\theta|}{a} \ln |\theta|^{-1}. \quad (2769)$$

Since

$$\lim_{\theta \rightarrow 0} |\theta| \ln |\theta|^{-1} = 0, \quad (2770)$$

one obtains $\lim_{\theta \rightarrow 0} \mathcal{R}_\xi = 0$ for every fixed finite curvature radius. Hence,

$$\lim_{R \rightarrow \infty} \lim_{\theta \rightarrow 0} \mathcal{R}_\xi = 0. \quad (2771)$$

Thus,

$$\lim_{\theta \rightarrow 0} \lim_{R \rightarrow \infty} \mathcal{R}_\xi \neq \lim_{R \rightarrow \infty} \lim_{\theta \rightarrow 0} \mathcal{R}_\xi. \quad (2772)$$

The flat-space and small-angle limits do not commute. The same singularity appears at the area level. Define

$$\mathcal{R}_A = \frac{A_M^{(H)}}{A_M^{(E)}}. \quad (2773)$$

At fixed nonzero twist, $\lim_{R \rightarrow \infty} \mathcal{R}_A = 1$. Hence,

$$\lim_{\theta \rightarrow 0} \lim_{R \rightarrow \infty} \mathcal{R}_A = 1. \quad (2774)$$

At fixed negative curvature, however, $A_M^{(H)} \sim C_H |\theta|^{-1}$, while $A_M^{(E)} \sim C_E |\theta|^{-2}$. Therefore, $\mathcal{R}_A \sim C |\theta|$, and hence $\lim_{\theta \rightarrow 0} \mathcal{R}_A = 0$. Consequently,

$$\lim_{R \rightarrow \infty} \lim_{\theta \rightarrow 0} \mathcal{R}_A = 0. \quad (2775)$$

The effective-area crossover is therefore a genuine singular perturbation rather than an ordinary analytic curvature correction uniformly valid down to zero twist. The noncommutativity can be resolved by retaining the appropriate dimensionless combination of twist and curvature. For weak microscopic curvature and small twist, define $x = \frac{a}{R|\theta|}$. Up to an inessential factor of order unity, this quantity is the ratio between the Euclidean moiré length and the curvature radius. The joint limit

$$R \rightarrow \infty, \quad |\theta| \rightarrow 0 \quad (2776)$$

therefore separates into three asymptotic paths. If $x \rightarrow 0$, then Euclidean regime. If

$$x \rightarrow x_*, \quad 0 < x_* < \infty, \quad (2777)$$

then nontrivial crossover regime. If $x \rightarrow \infty$, then curvature-dominated regime. The Euclidean limit is therefore uniform only when

$$\frac{a}{R|\theta|} \rightarrow 0. \quad (2778)$$

In curvature notation, this condition is

$$\frac{a\sqrt{|K|}}{|\theta|} \rightarrow 0. \quad (2779)$$

Thus arbitrarily weak negative curvature becomes non-perturbative at sufficiently small twist. We now turn from the metric limit to crystallographic periodicity. For a fixed twist center, define the set of exact centered commensurate angles by

$$\Theta_{\text{exact}}(o) = \{\theta : g_\theta \in \text{Comm}_G(\Gamma)\}. \quad (2780)$$

A twist is exact commensurate when $\theta \in \Theta_{\text{exact}}(o)$, and incommensurate when $\theta \notin \Theta_{\text{exact}}(o)$. For an incommensurate twist,

$$[\Gamma : \Gamma \cap g_\theta \Gamma g_\theta^{-1}] = \infty. \quad (2781)$$

Thus no finite exact crystallographic supercell exists. Nevertheless, the metric quantities $d_\theta(r)$, $\xi_M(\theta, R)$, and $A_M^{\text{eff}}(\theta, R)$ remain finite and exactly defined for every nonzero twist. Hence, absence of an exact supercell $\not\Rightarrow$ absence of local moiré physics. The exact commensurability set is nongeneric even when it is dense. The crystal group is finitely generated, $\Gamma = \langle \gamma_1, \dots, \gamma_m \rangle$. Every commensurating element determines a conjugation

isomorphism between finite-index subgroups of the original crystal group. A finitely generated group has only countably many finite-index subgroups. For any two such finitely generated subgroups, a homomorphism is determined by the images of finitely many generators inside a countable target group. Hence there exist only countably many such subgroup isomorphisms. For a non-elementary Fuchsian lattice, an ambient orientation-preserving isometry implementing a prescribed conjugation is unique because the centralizer of a finite-index non-elementary subgroup is trivial. Consequently,

$$|\text{Comm}_G(\Gamma)| \leq \aleph_0. \quad (2782)$$

Thus,

$$|\Theta_{\text{exact}}(o)| \leq \aleph_0. \quad (2783)$$

Since the continuum set of twist angles is uncountable, exact commensurability occupies a measure-zero subset of angle space. Therefore, a generic twist is incommensurate. This remains true even when the exact angle set is dense. For a commensurator-active center, the exact centered-twist group is dense in the rotation stabilizer. Therefore, $\Theta_{\text{exact}}(o) = \mathbb{S}^1$ modulo the angular periodicity convention. At the same time, the exact set is countable. Thus, $\Theta_{\text{exact}}(o)$ can be dense and measure zero simultaneously. This structure is directly analogous to the distinction between rational and irrational parameters in Euclidean commensurability problems. In such a case every incommensurate target twist can be approximated arbitrarily closely by exact commensurate twists, although the target itself possesses no finite exact supercell. We now quantify how accurately a nearby twist approximates a target twist inside a finite hyperbolic region. Consider two relative angles, θ and $\theta' = \theta + \delta\theta$. For an arbitrary point at radial distance $r = d(o, x)$, the two rotated images are $g_\theta x$ and $g_{\theta'} x$. Because $g_\theta^{-1} g_{\theta'} = g_{\delta\theta}$, their geodesic separation is exactly the twist-displacement law evaluated at the angular difference. Therefore,

$$d(g_\theta x, g_{\theta'} x) = 2R \operatorname{arsinh} \left[\sinh\left(\frac{r}{R}\right) \left| \sin\left(\frac{\delta\theta}{2}\right) \right| \right]. \quad (2784)$$

For every point inside a disk satisfying $r \leq L$, monotonicity gives the uniform bound

$$d(g_\theta x, g_{\theta'} x) \leq \varepsilon_L(\delta\theta), \quad (2785)$$

where

$$\varepsilon_L(\delta\theta) = 2R \operatorname{arsinh} \left[\sinh\left(\frac{L}{R}\right) \left| \sin\left(\frac{\delta\theta}{2}\right) \right| \right]. \quad (2786)$$

Thus angular convergence implies uniform geometric convergence on every fixed compact hyperbolic disk. For sufficiently small angular mismatch, $|\delta\theta| \ll 1$, one has

$$\left| \sin\left(\frac{\delta\theta}{2}\right) \right| = \frac{|\delta\theta|}{2} + O(|\delta\theta|^3). \quad (2787)$$

For fixed finite observation radius, the error therefore becomes

$$\varepsilon_L(\delta\theta) = R \sinh\left(\frac{L}{R}\right) |\delta\theta| + O(|\delta\theta|^3). \quad (2788)$$

The angular sensitivity grows according to the hyperbolic Jacobi factor. For $L \gg R$, one obtains

$$\varepsilon_L \sim \frac{R}{2} e^{L/R} |\delta\theta| \quad (2789)$$

as long as the resulting displacement remains in the linear-response regime. Thus exponentially increasing angular precision is required to approximate increasingly large hyperbolic regions. For a prescribed positional tolerance, $\varepsilon > 0$, define the approximation horizon as the largest radius over which two nearby twists remain within this tolerance. It is determined by $\varepsilon_{L_{\text{app}}}(\delta\theta) = \varepsilon$. Using the exact error relation gives

$$\sinh\left(\frac{\varepsilon}{2R}\right) = \sinh\left(\frac{L_{\text{app}}}{R}\right) \left| \sin\left(\frac{\delta\theta}{2}\right) \right|. \quad (2790)$$

Therefore,

$$L_{\text{app}} = R \operatorname{arsinh} \left[\frac{\sinh(\varepsilon/2R)}{|\sin(\delta\theta/2)|} \right]. \quad (2791)$$

For asymptotically accurate angular approximation, $|\delta\theta| \rightarrow 0$, one obtains

$$L_{\text{app}} = R \ln \left[\frac{4 \sinh(\varepsilon/2R)}{|\delta\theta|} \right] + O(R\delta\theta^2). \quad (2792)$$

Thus improving the twist-angle approximation by a multiplicative factor increases the reliable real-space radius only additively. The inverse relation is equally important for numerical work. To guarantee $\varepsilon_L \leq \varepsilon$, one must satisfy

$$\left| \sin\left(\frac{\delta\theta}{2}\right) \right| \leq \frac{\sinh(\varepsilon/2R)}{\sinh(L/R)}. \quad (2793)$$

Hence,

$$|\delta\theta| \leq 2 \arcsin \left[\frac{\sinh(\varepsilon/2R)}{\sinh(L/R)} \right]. \quad (2794)$$

For a large target radius, $L \gg R$, this becomes

$$|\delta\theta| \lesssim 4 \sinh\left(\frac{\varepsilon}{2R}\right) e^{-L/R}. \quad (2795)$$

For a small positional tolerance, $\varepsilon \ll R$, one further obtains

$$|\delta\theta| \lesssim \frac{2\varepsilon}{R} e^{-L/R}. \quad (2796)$$

This exponential angular-resolution requirement is a direct consequence of negative curvature. Consider a sequence of angular approximants satisfying $\delta\theta_n \rightarrow 0$. Suppose simultaneously that the observation radius grows according to $L_n \rightarrow \infty$. The exact uniform error is

$$\varepsilon_n = 2R \operatorname{arsinh} \left[\sinh\left(\frac{L_n}{R}\right) \left| \sin\left(\frac{\delta\theta_n}{2}\right) \right| \right]. \quad (2797)$$

A necessary and sufficient condition for $\varepsilon_n \rightarrow 0$ is

$$\sinh\left(\frac{L_n}{R}\right) \left| \sin\left(\frac{\delta\theta_n}{2}\right) \right| \rightarrow 0. \quad (2798)$$

When both the angular error is small and the observation radius is large, this reduces to

$$|\delta\theta_n| e^{L_n/R} \rightarrow 0. \quad (2799)$$

Thus ordinary angular convergence is sufficient on every fixed compact region but insufficient on an arbitrarily rapidly expanding region. For every fixed observation radius, $L < \infty$, one has

$$\theta_n \rightarrow \theta \implies \varepsilon_L(\theta_n - \theta) \rightarrow 0. \quad (2800)$$

However, no nontrivial angular approximant converges uniformly over the entire hyperbolic plane. For every fixed nonzero angular error, $\delta\theta \neq 0$, one has

$$\lim_{L \rightarrow \infty} \varepsilon_L(\delta\theta) = \infty. \quad (2801)$$

Therefore, angular approximation is locally uniform but not globally uniform. This is another manifestation of exponential geodesic separation. Suppose the fixed-center exact twist group is dense. Let an arbitrary target angle satisfy $\theta \notin \Theta_{\text{exact}}(o)$. Density guarantees the existence of a sequence satisfying $\theta_n \in \Theta_{\text{exact}}(o)$, and $\theta_n \rightarrow \theta$. Each approximant possesses an exact finite-index moiré group,

$$\Gamma_{M,n} = \Gamma \cap g_{\theta_n} \Gamma g_{\theta_n}^{-1}, \quad (2802)$$

and an exact supercell index,

$$q_{M,n} = [\Gamma : \Gamma_{M,n}]. \quad (2803)$$

For every fixed local radius, the geometry of the exact approximant converges to the incommensurate target configuration. Thus, incommensurate local geometry = compact-limit of exact commensurate geometry. whenever the centered exact twist set is dense. If the fixed-center exact-twist subgroup is finite, then

$$|\Theta_{\text{exact}}(o)| < \infty. \quad (2804)$$

Let the target angle be distinct from every exact angle. Because the set is finite, the angular distance to the exact set has a strictly positive minimum,

$$\Delta\theta_{\min} = \min_{\theta' \in \Theta_{\text{exact}}(o)} |\theta - \theta'| > 0. \quad (2805)$$

No sequence of exact centered commensurate twists can then converge to the target angle. Thus commensurate-supercell approximation is not a universally available numerical strategy. A different finite-volume construction is required for generic incommensurate twists in this case. We now construct such an approximation. Let $\Gamma_n < \Gamma$ be a finite-index subgroup. Define its degree by $d_n = [\Gamma : \Gamma_n]$. The corresponding compact finite cover is

$$\Sigma_n = \Gamma_n \backslash \mathbb{H}_R^2. \quad (2806)$$

Its area satisfies $A_n = d_n A_0$. For a torsion-free primitive group, its genus is

$$g_n = 1 + d_n (g_0 - 1). \quad (2807)$$

Thus increasing the finite-cover degree increases both the available crystallographic area and the topology of the compact quotient. Choose a lift of the twist center in the universal cover. The injectivity radius of the finite cover at the corresponding projected point is

$$r_{\text{inj}}^{(n)}(o) = \frac{1}{2} \min_{\gamma \in \Gamma_n \setminus \{e\}} d(o, \gamma o). \quad (2808)$$

If $r_{\text{inj}}^{(n)}(o) > L$, then the quotient projection is injective on the geodesic disk, $B_{\mathbb{H}}(o, L)$. Consequently that disk embeds isometrically into the finite cover. Thus every microscopic lattice configuration contained inside the disk is reproduced without periodic self-identification. The required cover sequence exists because a cocompact surface group is residually finite. We give the construction explicitly. For an arbitrary target radius, define the finite set

$$S_L = \{\gamma \in \Gamma \setminus \{e\} : d(o, \gamma o) \leq 2L\}. \quad (2809)$$

The orbit of the base point is discrete. The closed ball of radius $2L$ is compact. Therefore, $|S_L| < \infty$. Residual finiteness means that for every nonidentity element there exists a finite quotient in which that element remains nontrivial. Thus, for every element of the finite set, there exists a homomorphism $\varphi_\gamma : \Gamma \rightarrow F_\gamma$, where the target group is finite and $\varphi_\gamma(\gamma) \neq e$. Define

$$\Gamma_L = \bigcap_{\gamma \in S_L} \ker \varphi_\gamma. \quad (2810)$$

Because the intersection is finite and each kernel has finite index, $[\Gamma : \Gamma_L] < \infty$. Moreover, $\Gamma_L \cap S_L = \emptyset$. Therefore every nonidentity element of the subgroup moves the base point farther than twice the prescribed radius, $d(o, \gamma o) > 2L$ for every $\gamma \in \Gamma_L \setminus \{e\}$. Consequently, $r_{\text{inj}}^{(L)}(o) > L$. Thus finite covers exist with arbitrarily large based injectivity radius. Choose an increasing sequence satisfying $L_n \rightarrow \infty$. For each target radius, construct a finite-index subgroup satisfying $r_{\text{inj}}^{(n)}(o) > L_n$. Then every fixed compact region eventually embeds isometrically into every sufficiently large cover. For every fixed finite observation radius, $L < \infty$, there exists an integer satisfying $n_0(L)$ such that $B_{\mathbb{H}}(o, L)$ is embedded isometrically in Σ_n for every $n \geq n_0(L)$. The finite covers therefore converge locally to the universal hyperbolic crystal. For the second twisted layer, define the conjugate subgroup

$$\Gamma_n^{(2)} = g_\theta \Gamma_n g_\theta^{-1}. \quad (2811)$$

Its finite index equals that of the first cover,

$$\left[g_\theta \Gamma g_\theta^{-1} : \Gamma_n^{(2)} \right] = d_n. \quad (2812)$$

Because conjugation is an isometry, the local injectivity radii of the two covers are identical under the corresponding rotated base points. Hence both microscopic layers admit simultaneously controlled local finite approximations. The two finite single-layer covers must not be mistaken for a common periodic bilayer supercell. A common quotient would require a finite-index subgroup satisfying $H \subset \Gamma$ and $H \subset g_\theta \Gamma g_\theta^{-1}$. Therefore,

$$H \subset \Gamma \cap g_\theta \Gamma g_\theta^{-1}. \quad (2813)$$

If the target twist is incommensurate, this intersection has infinite index. No common subgroup of finite index can then exist. Therefore, generic incommensurate bilayer $\neq$ single periodic finite quotient. Finite-cover approximation of an incommensurate bilayer must consequently be understood locally rather than as an artificial periodic supercell. Define the local bilayer pattern of radius as

$$\mathcal{P}_\theta(L) = [\Lambda_1 \cap B_{\mathbb{H}}(o, L)] \sqcup [\Lambda_2 \cap B_{\mathbb{H}}(o, L)]. \quad (2814)$$

If the finite-cover injectivity radius satisfies $r_{\text{inj}}^{(n)}(o) > L$, then the first component of this pattern is represented exactly in the first finite cover. The second component is represented exactly in the corresponding conjugate cover. Both local representatives may then be lifted to the common universal cover, where all interlayer geodesic distances are evaluated intrinsically. This construction defines a paired finite-cover approximation. It is not an exact periodic bilayer. Its purpose is instead to reproduce the finite-radius local geometry of the infinite incommensurate bilayer without open-boundary distortion. Consider an observable whose geometric support is contained inside $B_{\mathbb{H}}(o, L)$. Suppose the microscopic model possesses a finite interaction or hopping range $r_{\text{int}} < \infty$. To evaluate the observable without contamination from quotient identifications, it is sufficient that the finite-cover injectivity radius satisfy

$$r_{\text{inj}}^{(n)}(o) > L + r_{\text{int}}. \quad (2815)$$

Then every microscopic degree of freedom capable of influencing the observable lies within an isometrically embedded neighborhood. Consequently, any strictly finite-range local geometric observable is reproduced exactly by a sufficiently large paired finite-cover approximant. If the microscopic coupling is not strictly finite range but decays exponentially,

$$|t(d)| \leq t_0 e^{-d/\lambda}, \quad (2816)$$

one may introduce a finite buffer radius $B > 0$. The contribution from distances larger than the buffer is exponentially suppressed. Schematically, the omitted coupling strength satisfies

$$\epsilon_{\text{tail}} \lesssim C e^{-B/\lambda}, \quad (2817)$$

provided the physical decay rate dominates the relevant geometric growth of the number of coupled degrees of freedom. The finite-cover condition becomes

$$r_{\text{inj}}^{(n)}(o) > L + B. \quad (2818)$$

The limits must then be taken in the controlled order $n \rightarrow \infty$, followed by $B \rightarrow \infty$. Suppose an incommensurate target twist is approximated by an exact commensurate angle. Let the angular error be $\delta\theta_n = \theta_n - \theta$. Inside a disk of radius, the maximum displacement error is

$$\varepsilon_{L,n} = 2R \operatorname{arsinh} \left[\sinh \left(\frac{L}{R} \right) \left| \sin \left(\frac{\delta\theta_n}{2} \right) \right| \right]. \quad (2819)$$

Suppose an interlayer hopping function is Lipschitz continuous on the relevant distance interval with constant $\Lambda_t < \infty$. Thus,

$$|t(d_1) - t(d_2)| \leq \Lambda_t |d_1 - d_2|. \quad (2820)$$

The triangle inequality gives

$$|d(x, g_{\theta_n} y) - d(x, g_{\theta} y)| \leq d(g_{\theta_n} y, g_{\theta} y). \quad (2821)$$

Hence, inside the radius under consideration, $|\Delta t| \leq \Lambda_t \varepsilon_{L,n}$. Thus the geometric convergence bound transfers directly to local tight-binding matrix elements. Suppose each local degree of freedom couples to at most $z_{\max}$ matrix elements affected by the angular approximation. Then a standard row-sum estimate gives

$$\|H_{\theta_n}^{(L)} - H_{\theta}^{(L)}\| \leq z_{\max} \Lambda_t \varepsilon_{L,n}. \quad (2822)$$

Therefore, $\theta_n \rightarrow \theta$ implies $H_{\theta_n}^{(L)} \rightarrow H_{\theta}^{(L)}$ in operator norm on every fixed local region. This establishes a direct bridge from geometric angular approximation to the spectral calculations developed later. If the system size grows simultaneously with the angular approximation, the sufficient condition becomes

$$z_{\max} \Lambda_t \varepsilon_{L,n} \rightarrow 0. \quad (2823)$$

For large observation radius and small angular error, this requires

$$|\delta \theta_n| e^{L_n/R} \rightarrow 0. \quad (2824)$$

Thus the same exponential geometric sensitivity controls convergence of the local Hamiltonian. A naive finite truncation uses a geodesic disk, $B_{\mathbb{H}}(o, L)$. The perimeter-to-area ratio is

$$\frac{P_{\mathbb{H}}(L)}{A_{\mathbb{H}}(L)} = \frac{1}{R} \coth\left(\frac{L}{2R}\right). \quad (2825)$$

Hence,

$$\lim_{L \rightarrow \infty} \frac{P_{\mathbb{H}}(L)}{A_{\mathbb{H}}(L)} = \frac{1}{R}. \quad (2826)$$

The boundary therefore remains extensive. Consequently, a global observable averaged over all sites of a large open disk can retain an order-one boundary contribution. Thus, large open disk $\not\rightarrow$ bulk thermodynamic limit. This is a fundamental difference from ordinary Euclidean lattice calculations. Finite covers avoid this problem because the compact quotient has no physical boundary. For a commensurate bilayer, the exact common quotient $\Gamma_M \backslash \mathbb{H}_R^2$ is the natural boundaryless finite system. For an incommensurate bilayer, no common quotient exists. The correct construction is therefore

to use boundaryless finite covers of the individual layers while evaluating the coupled local geometry through their common lifts to the universal cover. Global quantities derived from an artificial identification of the two incommensurate covers should not be interpreted as exact bulk observables. The appropriate infinite-volume concept for generic incommensurate hyperbolic systems is consequently local. Let an observable have compact support radius $L_{\mathcal{O}} < \infty$. Choose finite covers satisfying

$$r_{\text{inj}}^{(n)}(o) > L_{\mathcal{O}} + r_{\text{int}}. \quad (2827)$$

Then the observable is independent of the chosen cover for all sufficiently large cover degree. Thus the local infinite-volume value may be defined by the stabilized limit

$$\langle \mathcal{O} \rangle_{\infty} = \lim_{n \rightarrow \infty} \langle \mathcal{O} \rangle_{\Sigma_n}. \quad (2828)$$

The limit is local rather than boundary-to-volume based. For incommensurate systems approximated by commensurate twists, three limits may be present: $n \rightarrow \infty$, $\theta_m \rightarrow \theta$, and $L \rightarrow \infty$. The safe local procedure is to fix the observation radius first. Then choose sufficiently large finite covers,

$$r_{\text{inj}}^{(n)} > L + r_{\text{int}}, \quad (2829)$$

and sufficiently accurate angular approximants,

$$\varepsilon_L(\theta_m - \theta) \ll 1. \quad (2830)$$

Only after these two convergence operations should the observation radius be increased. Symbolically, the controlled procedure is

$$L \text{ fixed} \rightarrow n \rightarrow \infty \rightarrow \theta_m \rightarrow \theta \rightarrow L \rightarrow \infty. \quad (2831)$$

When a coordinated diagonal sequence is used instead, it must satisfy

$$r_{\text{inj}}^{(n)} - L_n \rightarrow \infty \quad (2832)$$

together with

$$|\theta_n - \theta| e^{L_n/R} \rightarrow 0. \quad (2833)$$

These conditions prevent finite-cover wraparound and angular mismatch from surviving in the growing-domain limit. The preceding construction may be summarized formally.

a. Theorem 13. Local finite-cover approximation of a hyperbolic bilayer. Let the microscopic hyperbolic crystal group be a torsion-free cocompact Fuchsian lattice. For every finite observation radius, there exists a finite-index normal subgroup satisfying $r_{\text{inj}}(o) > L$. Consequently, the radius- L microscopic crystal pattern embeds isometrically in the corresponding compact finite cover. For an arbitrary relative twist, including a generic incommensurate twist, both layers admit such finite-cover representations through conjugate finite-index subgroups. Their joint local bilayer geometry is recovered exactly by lifting the two embedded patches to the common universal cover. For a finite-range microscopic model, every observable supported inside a fixed radius is reproduced exactly once

$$r_{\text{inj}} > L + r_{\text{int}}. \quad (2834)$$

For an angular commensurate approximant, the additional geometric error is bounded by

$$\varepsilon_L = 2R \operatorname{arsinh} [\sinh(L/R) |\sin(\delta\theta/2)|]. \quad (2835)$$

Thus convergence on a growing radius requires

$$|\delta\theta|e^{L/R} \rightarrow 0. \quad (2836)$$

The infinite incommensurate bilayer is therefore locally approximable by finite systems even when no exact common supercell exists. The metric part of the present subsection may likewise be summarized as a separate theorem.

b. Theorem 14. Euclidean limit and singular small-angle crossover. For the hyperbolic local moiré length

$$\xi_M^{(H)} = R \operatorname{arsinh} \left[\frac{\sinh(a/2R)}{|\sin(\theta/2)|} \right], \quad (2837)$$

the fixed-angle vanishing-curvature limit is

$$\lim_{R \rightarrow \infty} \xi_M^{(H)} = \frac{a}{2|\sin(\theta/2)|} = \xi_M^{(E)}. \quad (2838)$$

The corresponding effective-area limit is

$$\lim_{R \rightarrow \infty} A_M^{(H)} = \frac{\pi a^2}{4 \sin^2(\theta/2)} = A_M^{(E)}. \quad (2839)$$

However, the normalized hyperbolic-to-Euclidean length ratio satisfies

$$\lim_{\theta \rightarrow 0} \lim_{R \rightarrow \infty} \frac{\xi_M^{(H)}}{\xi_M^{(E)}} = 1, \quad (2840)$$

whereas

$$\lim_{R \rightarrow \infty} \lim_{\theta \rightarrow 0} \frac{\xi_M^{(H)}}{\xi_M^{(E)}} = 0. \quad (2841)$$

Thus the flat-space and small-angle limits do not commute. The crossover is controlled by $\frac{a\sqrt{|K|}}{|\theta|}$. The Euclidean regime is obtained only if this ratio tends to zero. The curvature-dominated regime is obtained when it diverges. A finite nonzero limiting value defines the universal crossover regime. The preceding results establish several logically distinct facts. First, exact commensurability is not required for local moiré geometry. Second, exact commensurate twists form at most a countable subset of the continuous twist parameter space. Third, when the fixed-center commensurate set is dense, generic incommensurate configurations can be approximated locally by exact commensurate supercells. Fourth, when the exact centered-twist set is finite, such an approximation is impossible, but paired finite covers still provide controlled local approximants. Fifth, hyperbolic angular sensitivity makes approximation nonuniform with system size. The central convergence requirement is therefore

$$|\delta\theta|e^{L/R} \rightarrow 0. \quad (2842)$$

Finally, naive open-cluster thermodynamic limits are not reliable because

$$\lim_{L \rightarrow \infty} \frac{P_{\mathbb{H}}(L)}{A_{\mathbb{H}}(L)} = \frac{1}{R}. \quad (2843)$$

Finite-cover and local-observable formulations are therefore not merely numerical conveniences but are required by the geometry. The flat-space consistency chain is

$$K \rightarrow 0^- \rightarrow d_{\theta}^{(H)} \rightarrow d_{\theta}^{(E)} \rightarrow \xi_M^{(H)} \rightarrow \xi_M^{(E)} \rightarrow A_M^{(H)} \rightarrow A_M^{(E)}. \quad (2844)$$

The generic incommensurate chain is $\theta \notin \Theta_{\text{exact}} \rightarrow$ no finite common subgroup $\rightarrow$ no exact supercell $\rightarrow$ well-defined local moiré geometry. When exact commensurate approximants exist, the approximation chain is

$$\theta_n \rightarrow \theta \rightarrow \varepsilon_{L,n} \rightarrow 0 \rightarrow \mathcal{P}_{\theta_n}(L) \rightarrow \mathcal{P}_{\theta}(L) \quad (2845)$$

for every fixed observation radius. For growing observation domains, the additional condition is

$$|\theta_n - \theta|e^{L_n/R} \rightarrow 0. \quad (2846)$$

When exact commensurate approximants do not exist, the finite-cover chain is $\Gamma \rightarrow \Gamma_n \rightarrow r_{\text{inj}}^{(n)} \rightarrow \infty \rightarrow$ exact local embedding $\rightarrow$ infinite-volume local observable.

The complete consistency structure of the geometric theory can therefore be summarized as

$$\begin{aligned}
\text{Euclidean limit : } & \frac{a\sqrt{|K|}}{|\theta|} \rightarrow 0, \\
\text{curvature-dominated limit : } & \frac{a\sqrt{|K|}}{|\theta|} \rightarrow \infty, \\
\text{incommensurate local limit : } & r_{\text{inj}} \rightarrow \infty, \\
\text{commensurate-approximation limit : } & |\delta\theta|e^{L/R} \rightarrow 0.
\end{aligned} \tag{2847}$$

These conditions close the geometric and finite-size foundations required before introducing the bilayer Hamiltonian. In particular, the Euclidean expression

$$\xi_M^{(E)} = \frac{a}{2|\sin(\theta/2)|} \tag{2848}$$

is recovered rigorously from the negatively curved theory, while generic incommensurate configurations remain mathematically well defined through their local geometry even in the absence of any exact periodic supercell. The finite-cover construction then provides a controlled route from this infinite incommensurate geometry to finite numerical systems without invoking an artificial Euclidean-style periodic approximation.

II. CURVATURE-DEPENDENT TIGHT-BINDING THEORY OF TWISTED BILAYERS

A. Microscopic Bilayer Tight-Binding Hamiltonian

The preceding geometric theory established that negative curvature changes the spatial organization of a twisted bilayer before any quantum dynamics is introduced. In particular, the relative twist generates a curvature-enhanced registry mismatch, a logarithmic local moiré length in the curvature-dominated regime, and a corresponding crossover of the effective moiré-area scaling law. The next question is whether this geometric reconstruction produces a corresponding microscopic reconstruction of the quantum Hamiltonian. A complete answer requires more than replacing an ordinary Euclidean distance by a hyperbolic distance inside a hopping amplitude. One must specify the Hilbert space of the bilayer, distinguish intralayer and interlayer geometry, ensure Hermiticity and thermodynamic well-posedness, identify the exact role played by the rigid twist, recover the Euclidean square-lattice Hamiltonian as a controlled limit, and determine explicitly how the curvature-enhanced registry field enters the tunneling matrix elements. We construct such a theory below. The minimal model contains one spinless orbital per microscopic lattice site. Additional spin, valley, sublattice, or orbital degrees of freedom can subsequently be included by promoting all scalar hopping amplitudes to finite-dimensional

matrices without modifying the geometric structure derived here. Let the underlying two-dimensional geometric manifold be $\mathcal{M}_K$, $K \leq 0$. The two cases of primary interest are $\mathcal{M}_0 = \mathbb{R}^2$, and

$$\mathcal{M}_K = \mathbb{H}_R^2, \quad K = -\frac{1}{R^2}. \tag{2849}$$

To distinguish the two physical layers, we embed them into the product space

$$\mathfrak{M}_K = \mathcal{M}_K \times \mathbb{R}_z. \tag{2850}$$

The product metric is chosen as

$$ds_{\mathfrak{M}}^2 = ds_K^2 + dz^2. \tag{2851}$$

The two layers are located at fixed transverse coordinates $z_1 = -\frac{h}{2}$, and $z_2 = +\frac{h}{2}$, where the positive parameter $h > 0$ is the interlayer separation. The product construction has an important advantage: it separates intrinsic in-plane geometry from the synthetic layer direction without requiring a particular embedding of the hyperbolic plane into ordinary three-dimensional Euclidean space. Let the reference microscopic lattice be $\Lambda_K = \{x_i\}_{i \in I}$, where the site labels form a countable index set. The first layer is $\Lambda_1 = \Lambda_K$. The second layer is obtained through the rigid relative isometry introduced in the geometric theory, $\Lambda_2 = g_\theta \Lambda_K$. Thus a site with reference label j in the second layer occupies the in-plane position $x_j^{(2)} = g_\theta x_j$. The corresponding physical bilayer coordinates are

$$X_i^{(1)} = \left(x_i, -\frac{h}{2}\right), \tag{2852}$$

and

$$X_j^{(2)} = \left(g_\theta x_j, +\frac{h}{2}\right). \tag{2853}$$

The intrinsic distance between two sites belonging to the same layer is

$$d_{ij}^{(\ell)} = d_K \left(x_i^{(\ell)}, x_j^{(\ell)}\right). \tag{2854}$$

Because the second layer is related to the first by an isometry,

$$d_K(g_\theta x_i, g_\theta x_j) = d_K(x_i, x_j). \tag{2855}$$

Therefore,

$$d_{ij}^{(1)} = d_{ij}^{(2)} = d_K(x_i, x_j). \tag{2856}$$

Rigid twisting consequently leaves every intralayer metric distance unchanged. For two sites belonging to different layers, the product metric gives the exact three-dimensional bilayer distance

$$D_{ij}^\perp(\theta, K, h) = \sqrt{h^2 + d_K^2(x_i, g_\theta x_j)}. \quad (2857)$$

Equation (2857) is the central geometric input into the microscopic interlayer tunneling problem. If the transverse separation is absorbed phenomenologically into the definition of the hopping function, one may equivalently use the purely in-plane separation

$$d_{ij}^{\perp, \parallel} = d_K(x_i, g_\theta x_j). \quad (2858)$$

The product-metric formulation will be retained below because it makes the layer geometry explicit. The single-particle Hilbert space of the first layer is $\mathcal{H}_1 = \ell^2(\Lambda_1)$. Likewise, $\mathcal{H}_2 = \ell^2(\Lambda_2)$. The full bilayer single-particle space is the direct sum

$$\mathcal{H}_{\text{BL}} = \mathcal{H}_1 \oplus \mathcal{H}_2. \quad (2859)$$

A general single-particle state can therefore be written as

$$|\Psi\rangle = \sum_i \psi_{1i} |1, i\rangle + \sum_j \psi_{2j} |2, j\rangle. \quad (2860)$$

Normalization requires

$$\sum_i |\psi_{1i}|^2 + \sum_j |\psi_{2j}|^2 = 1. \quad (2861)$$

Introduce the annihilation operators $c_{\ell i}$, and creation operators $c_{\ell i}^\dagger$. They obey the canonical anticommutation relations $\{c_{\ell i}, c_{\ell' j}^\dagger\} = \delta_{\ell\ell'} \delta_{ij}$, $\{c_{\ell i}, c_{\ell' j}\} = 0$, and $\{c_{\ell i}^\dagger, c_{\ell' j}^\dagger\} = 0$. The total particle-number operator is

$$\hat{N} = \sum_\ell \sum_i c_{\ell i}^\dagger c_{\ell i}. \quad (2862)$$

The microscopic number-conserving quadratic Hamiltonian is decomposed as

$$\hat{H} = \hat{H}_1 + \hat{H}_2 + \hat{H}_\perp. \quad (2863)$$

The intralayer Hamiltonian of layer ℓ is

$$\hat{H}_\ell = - \sum_{i,j} t_{ij}^{(\ell)} c_{\ell i}^\dagger c_{\ell j} + \text{h.c.} \quad (2864)$$

The interlayer contribution is

$$\hat{H}_\perp = \sum_{i,j} T_{ij} c_{1i}^\dagger c_{2j} + \text{h.c.} \quad (2865)$$

Any overall sign convention for the interlayer amplitude may be absorbed into the definition of the matrix elements. A uniform onsite term may be included through $\hat{H}_\mu = -\mu \hat{N}$, but it will be omitted unless explicitly required. For an isotropic single-orbital model, the intralayer hopping matrix element can be taken to depend only on intrinsic geodesic distance,

$$t_{ij}^{(\ell)} = t_K \left[d_K \left(x_i^{(\ell)}, x_j^{(\ell)} \right) \right]. \quad (2866)$$

Because the relative twist is an isometry,

$$t_{ij}^{(2)} = t_K \left[d_K (g_\theta x_i, g_\theta x_j) \right]. \quad (2867)$$

Using distance invariance gives $t_{ij}^{(2)} = t_{ij}^{(1)}$. Thus a rigid relative twist does not modify the isolated monolayer Hamiltonians. All nontrivial twist dependence enters through the interlayer tunneling matrix. The simplest specialization is the nearest-neighbor model. Let the microscopic nearest-neighbor spacing be $a > 0$. Define

$$t_K(d) = \begin{cases} t, & d = a, \\ 0, & d \neq a. \end{cases} \quad (2868)$$

The intralayer Hamiltonian becomes

$$\hat{H}_\ell = -t \sum_{\langle i,j \rangle_\ell} \left(c_{\ell i}^\dagger c_{\ell j} + c_{\ell j}^\dagger c_{\ell i} \right). \quad (2869)$$

This form is equally valid for Euclidean and hyperbolic lattices. Only the adjacency graph differs. For a regular hyperbolic tessellation whose lattice sites are chosen at the vertices, let the number of faces meeting at each vertex be q . Every vertex then has coordination number $z = q$. Define the adjacency matrix by

$$A_{ij} = \begin{cases} 1, & i \text{ and } j \text{ are nearest neighbors,} \\ 0, & \text{otherwise.} \end{cases} \quad (2870)$$

The one-particle intralayer Hamiltonian is therefore $h_0 = -tA$. The same adjacency operator describes both rigidly twisted layers in their respective intrinsic coordinate systems. The minimal isotropic interlayer model assumes that tunneling depends only on the physical bilayer separation. Thus,

$$T_{ij}(\theta, K, h) = \mathcal{T} \left[D_{ij}^\perp(\theta, K, h) \right]. \quad (2871)$$

Using the product-metric distance gives

$$T_{ij}(\theta, K, h) = \mathcal{T} \left[\sqrt{h^2 + d_K^2(x_i, g_\theta x_j)} \right]. \quad (2872)$$

The function $\mathcal{T}(D)$ contains the microscopic tunneling physics, whereas all dependence on twist and intrinsic curvature enters through the geometric distance. A useful minimal choice is

$$\mathcal{T}(D) = w \exp \left[-\frac{D - h}{\lambda_\perp} \right], \quad (2873)$$

where the vertical tunneling amplitude is $\mathcal{T}(h) = w$, and the positive tunneling decay length is $\lambda_\perp > 0$. The subtraction of the vertical separation in the exponential ensures that the normalization parameter is exactly the tunneling amplitude of vertically aligned orbitals. A finite-range alternative is $\mathcal{T}(D) = 0$ whenever $D > D_c$. The finite-range model is particularly useful for rigorous finite-cover calculations. Introduce the intralayer one-particle matrices

$$(h_\ell)_{ij} = -t_{ij}^{(\ell)} - \left[t_{ji}^{(\ell)} \right]^*, \quad (2874)$$

with the precise factor convention chosen so that each undirected bond is counted once. Define the interlayer matrix by

$$(\tau_\theta)_{ij} = T_{ij}(\theta, K, h). \quad (2875)$$

The bilayer single-particle Hamiltonian then has the exact block form

$$h_{\text{BL}} = \begin{pmatrix} h_1 & \tau_\theta \\ \tau_\theta^\dagger & h_2 \end{pmatrix}. \quad (2876)$$

The corresponding second-quantized Hamiltonian is the fermionic second quantization of this block operator. Rigid twisting does not change the spectrum of an isolated layer. To prove this statement, define the pullback unitary

$$U_\theta : \ell^2(\Lambda_K) \longrightarrow \ell^2(g_\theta \Lambda_K) \quad (2877)$$

through $U_\theta |i\rangle = |g_\theta i\rangle$. Because the relative transformation is bijective, $U_\theta^\dagger U_\theta = I$, and $U_\theta U_\theta^\dagger = I$. Thus, U_θ is unitary. The second-layer hopping matrix element satisfies

$$\langle g_\theta i | h_2 | g_\theta j \rangle = \langle i | h_1 | j \rangle. \quad (2878)$$

Therefore,

$$h_2 = U_\theta h_1 U_\theta^\dagger. \quad (2879)$$

Consequently,

$$\text{Spec}(h_2) = \text{Spec}(h_1). \quad (2880)$$

The rigid twist cannot by itself modify an isolated monolayer spectrum. Any twist-induced spectral reconstruction of the bilayer must therefore be generated by interlayer hybridization. The preceding unitary equivalence permits both layers to be represented on the same reference lattice. Define the bilayer unitary transformation

$$\mathcal{U}_\theta = \begin{pmatrix} I & 0 \\ 0 & U_\theta^\dagger \end{pmatrix}. \quad (2881)$$

The transformed Hamiltonian is

$$\tilde{h}_{\text{BL}} = \mathcal{U}_\theta h_{\text{BL}} \mathcal{U}_\theta^\dagger. \quad (2882)$$

Using Eq. (2879), one obtains

$$\tilde{h}_{\text{BL}}(\theta, K) = \begin{pmatrix} h_0(K) & \tilde{\tau}_\theta(K) \\ \tilde{\tau}_\theta^\dagger(K) & h_0(K) \end{pmatrix}, \quad (2883)$$

where $h_0 = h_1$, and $\tilde{\tau}_\theta = \tau_\theta U_\theta$. Both intralayer blocks are now identical. This representation isolates the twist dependence completely inside the interlayer block. Thus, rigid twist $\longrightarrow$ interlayer matrix reconstruction only. In the common reference labeling, the interlayer matrix element is

$$(\tilde{\tau}_\theta)_{ij} = \mathcal{T} \left[\sqrt{h^2 + d_K^2(x_i, g_\theta x_j)} \right]. \quad (2884)$$

Thus the entire rigid-twist problem is reduced to a family of geometry dependent matrices indexed by the relative angle and curvature. The microscopic Hamiltonian therefore takes the unified form

$$\tilde{h}_{\text{BL}} = \begin{pmatrix} h_0 & \tau(\theta, K, h) \\ \tau^\dagger(\theta, K, h) & h_0 \end{pmatrix}. \quad (2885)$$

This equation applies to both flat and negatively curved bilayers. The geometric consequences of curvature become particularly transparent in the coupling between sites carrying the same reference label in the two layers. For $i = j$, the in-plane mismatch is

$$d_\theta(r_i) = d_K(x_i, g_\theta x_i), \quad (2886)$$

where $r_i = d_K(o, x_i)$. For negative curvature, the exact twist geometry gives

$$d_\theta(r) = 2R \operatorname{arsinh} \left[\sinh\left(\frac{r}{R}\right) \left| \sin\left(\frac{\theta}{2}\right) \right| \right]. \quad (2887)$$

The corresponding physical interlayer distance is

$$D_{\text{loc}}(r) = \sqrt{h^2 + d_\theta^2(r)}. \quad (2888)$$

Hence the same-label tunneling amplitude is

$$T_{\text{loc}}(r) = \mathcal{T}[D_{\text{loc}}(r)]. \quad (2889)$$

Substitution gives the completely explicit radial tunneling profile

$$T_{\text{loc}}(r) = \mathcal{T} \left[\sqrt{h^2 + 4R^2 \operatorname{arsinh}^2(\sinh(r/R) |\sin(\theta/2)|)} \right]. \quad (2890)$$

This relation transfers the curvature-controlled registry geometry directly into the microscopic tunneling matrix. For fixed radius and small relative twist, the exact hyperbolic mismatch obeys

$$d_\theta(r) = R \sinh\left(\frac{r}{R}\right) |\theta| + O(|\theta|^3). \quad (2891)$$

Squaring gives

$$d_\theta^2(r) = R^2 \sinh^2\left(\frac{r}{R}\right) \theta^2 + O(\theta^4). \quad (2892)$$

The local physical interlayer distance therefore becomes

$$D_{\text{loc}}(r) = \sqrt{h^2 + R^2 \sinh^2\left(\frac{r}{R}\right) \theta^2 + O(\theta^4)}. \quad (2893)$$

For nonzero layer separation, expansion of the square root gives

$$D_{\text{loc}}(r) = h + \frac{R^2 \sinh^2(r/R)}{2h} \theta^2 + O(\theta^4). \quad (2894)$$

Thus the leading distance modulation is quadratic in the twist angle for a distance-only scalar tunneling model. Assume that the microscopic tunneling function is differentiable at the vertical separation. Taylor expansion gives

$$\mathcal{T}(h + \delta D) = \mathcal{T}(h) + \mathcal{T}'(h) \delta D + O(\delta D^2). \quad (2895)$$

Using Eq. (2894), one obtains

$$T_{\text{loc}}(r) = \mathcal{T}(h) + \frac{\mathcal{T}'(h) R^2 \sinh^2(r/R)}{2h} \theta^2 + O(\theta^4). \quad (2896)$$

For a monotonically decreasing tunneling function, $\mathcal{T}'(h) < 0$, the twist suppresses same-label tunneling away from the twist center. The magnitude of this suppression is enhanced by negative curvature through

$$R^2 \sinh^2\left(\frac{r}{R}\right). \quad (2897)$$

For the exponential tunneling model,

$$\mathcal{T}'(h) = -\frac{w}{\lambda_\perp}. \quad (2898)$$

Hence,

$$T_{\text{loc}}(r) = w \left[1 - \frac{R^2 \sinh^2(r/R)}{2h\lambda_\perp} \theta^2 + O(\theta^4) \right]. \quad (2899)$$

The relative tunneling modulation is therefore

$$\frac{w - T_{\text{loc}}(r)}{w} = \frac{R^2 \sinh^2(r/R)}{2h\lambda_\perp} \theta^2 + O(\theta^4). \quad (2900)$$

This is the first direct microscopic manifestation of the curvature-induced moire geometry. In Euclidean geometry, the corresponding same-label displacement is

$$d_\theta^{(E)}(r) = 2r \left| \sin\left(\frac{\theta}{2}\right) \right|. \quad (2901)$$

For small twist,

$$d_\theta^{(E)}(r) = r|\theta| + O(|\theta|^3). \quad (2902)$$

Therefore,

$$D_{\text{loc}}^{(E)} = h + \frac{r^2}{2h} \theta^2 + O(\theta^4). \quad (2903)$$

The Euclidean exponential-tunneling profile becomes

$$T_{\text{loc}}^{(E)}(r) = w \left[1 - \frac{r^2}{2h\lambda_\perp} \theta^2 + O(\theta^4) \right]. \quad (2904)$$

The ratio of the leading hyperbolic and Euclidean modulation coefficients is

$$\mathcal{C}_T(r) = \frac{R^2 \sinh^2(r/R)}{r^2}. \quad (2905)$$

Introducing $x = \frac{r}{R}$, this becomes

$$\mathcal{C}_T(x) = \left[\frac{\sinh x}{x} \right]^2. \quad (2906)$$

The small-radius expansion is

$$\mathcal{C}_T(x) = 1 + \frac{x^2}{3} + \frac{2x^4}{45} + O(x^6). \quad (2907)$$

At large radius,

$$\mathcal{C}_T(x) \sim \frac{e^{2x}}{4x^2}. \quad (2908)$$

Thus negative curvature exponentially amplifies the spatial sensitivity of the microscopic interlayer tunneling to a small relative twist. The preceding result establishes the microscopic chain

$$K < 0 \longrightarrow R \sinh(r/R) \longrightarrow d_\theta(r) \longrightarrow D_{\text{loc}}(r) \longrightarrow T_{\text{loc}}(r) \longrightarrow \tau_\theta. \quad \text{Spec} \left[\tilde{h}_{\text{BL}}(\theta) \right] = \text{Spec} \left[\tilde{h}_{\text{BL}}(-\theta) \right]. \quad (2909)$$

Thus the geometry modifies the Hamiltonian before any diagonalization is performed. This is fundamentally different from a phenomenological curvature-dependent band parameter inserted after the spectrum has been calculated. The intralayer hopping amplitudes satisfy

$$t_{ji}^{(\ell)} = \left[t_{ij}^{(\ell)} \right]^*. \quad (2910)$$

The interlayer Hamiltonian contains both the tunneling operator and its Hermitian conjugate. Therefore, $\hat{H}^\dagger = \hat{H}$. At the one-particle level, $h_{\text{BL}}^\dagger = h_{\text{BL}}$. Every term in the Hamiltonian contains one creation and one annihilation operator. Using the canonical anticommutation relations gives $[\hat{H}, \hat{N}] = 0$. The model therefore possesses a global charge symmetry. For the minimal scalar model, assume all hopping amplitudes are real, $t_{ij}^{(\ell)} \in \mathbb{R}$, and $T_{ij} \in \mathbb{R}$. The spinless time-reversal operator acts by complex conjugation. Hence,

$$\mathcal{T} h_{\text{BL}} \mathcal{T}^{-1} = h_{\text{BL}}. \quad (2911)$$

No topologically nontrivial Chern structure can arise from the minimal real-hopping model without additional time-reversal-breaking ingredients. Such ingredients can be introduced later if required, but they are not needed for the curvature-controlled band-flattening mechanism itself. Distance invariance gives

$$d_K(x_i, g_{-\theta} x_j) = d_K(g_\theta x_i, x_j). \quad (2912)$$

Symmetry of geodesic distance then implies

$$d_K(x_i, g_{-\theta} x_j) = d_K(x_j, g_\theta x_i). \quad (2913)$$

For a real scalar tunneling function,

$$\tau_{-\theta} = \tau_\theta^T = \tau_\theta^\dagger. \quad (2914)$$

Define the layer-exchange operator

$$P = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}. \quad (2915)$$

Then

$$P \tilde{h}_{\text{BL}}(\theta) P^{-1} = \tilde{h}_{\text{BL}}(-\theta). \quad (2916)$$

Therefore the spectra satisfy

The energy spectrum is consequently an even function of the signed twist. At an exact commensurate twist, the common hyperbolic translation group is

$$\Gamma_M = \Gamma \cap g_\theta \Gamma g_\theta^{-1}. \quad (2918)$$

For an element belonging to this group, define the induced unitary translation operator through

$$\hat{U}_\gamma c_{\ell i} \hat{U}_\gamma^{-1} = c_{\ell, \gamma i}. \quad (2919)$$

Intrinsic distances are preserved,

$$d_K(\gamma x, \gamma y) = d_K(x, y). \quad (2920)$$

Thus all intralayer hopping amplitudes are invariant. Because the same element preserves both layer lattices, interlayer separations are likewise invariant. Consequently,

$$\hat{U}_\gamma \hat{H} \hat{U}_\gamma^{-1} = \hat{H} \quad (2921)$$

for every element of the exact moiré group. Hence,

$$[\hat{H}, \hat{U}_\gamma] = 0. \quad (2922)$$

The crystallographic group developed in the preceding section is therefore not merely a geometric construction. It is the exact symmetry group of the microscopic bilayer Hamiltonian. If the coincidence index is finite, $q_M < \infty$, and the microscopic primitive quotient contains a finite number of basis orbitals, n_b , then one layer contains $n_b q_M$ orbitals per exact supercell. The total bilayer single-particle dimension per exact supercell is therefore

$$\dim \mathcal{H}_M = 2n_b q_M. \quad (2923)$$

Thus the exact crystallographic count obtained geometrically is precisely the dimension of the microscopic tight-binding problem before symmetry reduction. If geometric–arithmetic locking holds in the curvature-dominated regime, the matrix dimension consequently scales according to

$$\dim \mathcal{H}_M = \Theta(|\theta|^{-1}). \quad (2924)$$

This differs parametrically from the Euclidean square-lattice inverse-square supercell growth. At perfect alignment, $\theta = 0$. The two intrinsic lattices coincide. The co-rotating Hamiltonian becomes

$$\tilde{h}_{\text{BL}}(0) = \begin{pmatrix} h_0 & \tau_0 \\ \tau_0^\dagger & h_0 \end{pmatrix}. \quad (2925)$$

For real distance-only tunneling, $\tau_0 = \tau_0^\dagger$. Introduce the bonding–antibonding transformation

$$W = \frac{1}{\sqrt{2}} \begin{pmatrix} I & I \\ I & -I \end{pmatrix}. \quad (2926)$$

This transformation is unitary, $W^\dagger W = I$. Direct multiplication gives

$$W^\dagger \tilde{h}_{\text{BL}}(0) W = \begin{pmatrix} h_0 + \tau_0 & 0 \\ 0 & h_0 - \tau_0 \end{pmatrix}. \quad (2927)$$

The aligned bilayer therefore separates exactly into bonding and antibonding sectors. For a purely vertical interlayer tunneling model, $\tau_0 = wI$, one obtains

$$h_\pm = h_0 \pm wI. \quad (2928)$$

This provides a simple exact benchmark for all numerical implementations. The unified Hamiltonian must reproduce the conventional twisted square bilayer when the curvature vanishes. Set $K = 0$. The reference lattice is $\Lambda_0 = a\mathbb{Z}^2$. A lattice site may be labeled by two integers,

$$x_{mn} = a \begin{pmatrix} m \\ n \end{pmatrix}. \quad (2929)$$

The rigid twist is the ordinary planar rotation, $g_\theta = R_\theta$. The interlayer distance becomes

$$D_{ij}^\perp = \sqrt{h^2 + |x_i - R_\theta x_j|^2}. \quad (2930)$$

The nearest-neighbor intralayer Hamiltonian is

$$\hat{H}_\ell^{(E)} = -t \sum_{\langle i,j \rangle} \left(c_{\ell i}^\dagger c_{\ell j} + \text{h.c.} \right). \quad (2931)$$

The interlayer Hamiltonian is

$$\hat{H}_\perp^{(E)} = \sum_{i,j} \mathcal{T} \left[\sqrt{h^2 + |x_i - R_\theta x_j|^2} \right] c_{1i}^\dagger c_{2j} + \text{h.c.} \quad (2932)$$

Thus,

$$\hat{H}^{(E)} = \hat{H}_1^{(E)} + \hat{H}_2^{(E)} + \hat{H}_\perp^{(E)}. \quad (2933)$$

This is the required microscopic Euclidean square-bilayer Hamiltonian. For the isolated nearest-neighbor square lattice, introduce the Fourier transformation

$$c_{\ell,\mathbf{r}} = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}} c_{\ell,\mathbf{k}}. \quad (2934)$$

The intralayer dispersion is

$$\varepsilon_0(\mathbf{k}) = -2t [\cos(k_x a) + \cos(k_y a)]. \quad (2935)$$

For purely vertical tunneling at zero twist, the bonding and antibonding bands are

$$\varepsilon_\pm(\mathbf{k}) = -2t [\cos(k_x a) + \cos(k_y a)] \pm w. \quad (2936)$$

The more general twisted band structure will be developed in the subsequent spectral sections. For negative curvature, let the microscopic lattice be generated by a cocompact Fuchsian crystal group. The nearest-neighbor Hamiltonian remains

$$\hat{H}_\ell^{(H)} = -t \sum_{\langle i,j \rangle} \left(c_{\ell i}^\dagger c_{\ell j} + \text{h.c.} \right). \quad (2937)$$

The interlayer contribution is

$$\hat{H}_\perp^{(H)} = \sum_{i,j} \mathcal{T} \left[\sqrt{h^2 + d_{\mathbb{H}}^2(x_i, g_\theta x_j)} \right] c_{1i}^\dagger c_{2j} + \text{h.c.} \quad (2938)$$

Thus the minimal hyperbolic bilayer Hamiltonian is

$$\hat{H}^{(H)} = -t \sum_{\ell=1}^2 \sum_{\langle i,j \rangle_\ell} \left(c_{\ell i}^\dagger c_{\ell j} + \text{h.c.} \right) + \sum_{i,j} T_{ij}^{(H)} c_{1i}^\dagger c_{2j} + \text{h.c.}, \quad (2939)$$

with

$$T_{ij}^{(H)} = \mathcal{T} \left[\sqrt{h^2 + d_{\mathbb{H}}^2(x_i, g_\theta x_j)} \right]. \quad (2940)$$

Negative curvature introduces an important issue absent from ordinary finite-range formulations. The number of

lattice sites at large geodesic distance grows exponentially. Therefore an exponentially decaying hopping amplitude is not automatically summable merely because it decays exponentially. A quantitative condition is required. Let the microscopic lattice be uniformly discrete. Thus there exists a positive separation satisfying $\rho > 0$ such that distinct sites obey

$$d_K(x_i, x_j) \geq \rho. \quad (2941)$$

Consider the set of sites contained inside a geodesic disk of radius L ,

$$N_\Lambda(L) = \#[\Lambda \cap B(o, L)]. \quad (2942)$$

The balls of radius $\frac{\rho}{2}$ centered on these sites are pairwise disjoint. All such balls lie inside

$$B\left(o, L + \frac{\rho}{2}\right). \quad (2943)$$

Therefore,

$$N_\Lambda(L) A_{\mathbb{H}}\left(\frac{\rho}{2}\right) \leq A_{\mathbb{H}}\left(L + \frac{\rho}{2}\right). \quad (2944)$$

Using the asymptotic hyperbolic area law gives

$$N_\Lambda(L) \leq C_\rho e^{L/R} \quad (2945)$$

for sufficiently large radius and a finite positive constant. Thus the metric lattice-point growth exponent is bounded by $h_\Lambda \leq \frac{1}{R}$. For a cocompact lattice with nonzero asymptotic density, this exponential rate is attained. A sufficient condition for boundedness of the one-particle hopping operator is

$$\sup_i \sum_j |t_{ij}| < \infty. \quad (2946)$$

Likewise, the interlayer operator is bounded if

$$\sup_i \sum_j |T_{ij}| < \infty. \quad (2947)$$

Under these conditions, the Schur test gives finite operator norms. For example,

$$\|\tau_\theta\| \leq \sup_i \sum_j |T_{ij}|. \quad (2948)$$

Finite-range hopping satisfies these conditions automatically on a finite-coordination lattice. Consider a general exponentially decaying hopping law satisfying

$$|t(d)| \leq t_0 e^{-\mu d}. \quad (2949)$$

Partition the lattice into shells of radial width equal to the uniform separation scale. The number of sites in the n -th shell is bounded by

$$N_n \leq C e^{n\rho/R}. \quad (2950)$$

The total row sum therefore obeys

$$\sum_j |t_{ij}| \leq C t_0 \sum_{n=0}^{\infty} e^{n\rho/R} e^{-\mu n\rho}. \quad (2951)$$

This geometric series converges if and only if the sufficient inequality $\mu > \frac{1}{R}$ is satisfied. For a decay length defined by $\mu = \frac{1}{\lambda}$, the condition becomes $\lambda < R$. The same sufficient condition controls interlayer hopping because the full product distance obeys

$$D_{ij}^\perp \geq d_K(x_i, g_\theta x_j). \quad (2952)$$

Thus a microscopic tunneling decay that is perfectly acceptable in Euclidean space can become non-summable in a sufficiently strongly curved infinite lattice. This constraint is a direct consequence of exponential hyperbolic volume growth. For the strongest mathematical control, one may impose a finite geodesic hopping range, $r_c < \infty$. Then $t(d) = 0$ for $d > r_c$, and $\mathcal{T}(D) = 0$ for $D > D_c$. Every site then couples to only finitely many microscopic degrees of freedom. The resulting one-particle Hamiltonian is a bounded self-adjoint operator on the bilayer single-particle Hilbert space. This finite-range model will therefore serve as the rigorous baseline, while exponentially decaying hopping can be introduced subject to the summability condition derived above. The microscopic Hamiltonian contains several characteristic energy and length scales. Using the intralayer hopping as the reference energy scale, define $\omega = \frac{w}{t}$. Define the dimensionless layer separation by $\eta_h = \frac{h}{a}$. Define the dimensionless interlayer decay length by

$$\eta_\lambda = \frac{\lambda_\perp}{a}. \quad (2953)$$

The microscopic curvature is $\alpha = \frac{a}{2R}$. The complete minimal model can therefore be represented schematically as

$$\frac{\hat{H}}{t} = \mathcal{H}(\theta, \alpha, \omega, \eta_h, \eta_\lambda). \quad (2954)$$

The geometric crossover identified previously is controlled by the combination

$$\chi = \frac{\sinh \alpha}{|\sin(\theta/2)|}. \quad (2955)$$

Thus the same dimensionless parameter that controls the geometric moiré crossover also controls the spatial modulation entering the microscopic interlayer tunneling matrix. A valid unified Hamiltonian must reduce continuously to the Euclidean square-bilayer model. Consider a fixed finite set of lattice sites contained inside a bounded geodesic region. In the vanishing-curvature limit, the hyperbolic geodesic distance obeys

$$d_K(x, y) = d_E(x, y) + O(R^{-2}) \quad (2956)$$

for fixed physical separations. Consequently,

$$D_{ij}^{\perp, H} = D_{ij}^{\perp, E} + O(R^{-2}). \quad (2957)$$

Assume the hopping functions are locally Lipschitz. Thus,

$$|\mathcal{T}(D_1) - \mathcal{T}(D_2)| \leq L_{\mathcal{T}} |D_1 - D_2|. \quad (2958)$$

Then

$$|T_{ij}^{(H)} - T_{ij}^{(E)}| = O(R^{-2}) \quad (2959)$$

on every fixed local region. The same statement holds for a smooth distance-dependent intralayer hopping. For a finite local matrix, one therefore obtains

$$\|h_{\text{BL}}^{(H)} - h_{\text{BL}}^{(E)}\| = O(R^{-2}). \quad (2960)$$

Thus the microscopic quantum Hamiltonian inherits the controlled Euclidean limit already established for the geometry. For an incommensurate or infinite hyperbolic system, numerical calculations may use the finite-cover framework developed previously. Let a finite-index subgroup define a compact cover of degree d_n . The single-layer Hilbert-space dimension is $n_b d_n$. The bilayer matrix dimension is therefore

$$\dim \mathcal{H}_n^{\text{BL}} = 2n_b d_n. \quad (2961)$$

For a finite-range Hamiltonian, a local observable supported inside a radius smaller than the injectivity radius minus the hopping range is represented exactly. Thus the finite-cover approximation acts directly at the Hamiltonian level, not merely at the geometric level. The corotating representation yields an important exact separation. The derivative of the intralayer Hamiltonian with respect to the twist vanishes,

$$\frac{\partial h_0}{\partial \theta} = 0. \quad (2962)$$

By contrast,

$$\frac{\partial \tau_\theta}{\partial \theta} \neq 0 \quad (2963)$$

for a generic twisted configuration. Thus,

$$\frac{\partial \tilde{h}_{\text{BL}}}{\partial \theta} = \begin{pmatrix} 0 & \partial_\theta \tau_\theta \\ \partial_\theta \tau_\theta^\dagger & 0 \end{pmatrix}. \quad (2964)$$

Every rigid-twist spectral response is therefore generated by the off-diagonal layer-hybridization channel. This observation will be crucial when deriving the effective magic-band condition. Curvature influences the Hamiltonian through two conceptually distinct channels. The first is the intrinsic monolayer geometry, $h_0 = h_0(K)$, which determines the connectivity, generalized translation structure, and isolated-layer spectrum. The second is the registry-induced interlayer coupling,

$$\tau_\theta = \tau(\theta, K). \quad (2965)$$

Therefore,

$$\frac{\partial \tilde{h}_{\text{BL}}}{\partial K} = \begin{pmatrix} \partial_K h_0 & \partial_K \tau_\theta \\ \partial_K \tau_\theta^\dagger & \partial_K h_0 \end{pmatrix}. \quad (2966)$$

This distinction will later allow curvature-induced spectral changes to be separated into monolayer and moiré-hybridization contributions. Define the difference from the aligned bilayer Hamiltonian by

$$\delta h_\theta = \tilde{h}_{\text{BL}}(\theta) - \tilde{h}_{\text{BL}}(0). \quad (2967)$$

Since the intralayer blocks cancel exactly,

$$\delta h_\theta = \begin{pmatrix} 0 & \tau_\theta - \tau_0 \\ \tau_\theta^\dagger - \tau_0^\dagger & 0 \end{pmatrix}. \quad (2968)$$

Thus,

$$\|\delta h_\theta\| = \left\| \begin{pmatrix} 0 & \delta \tau_\theta \\ \delta \tau_\theta^\dagger & 0 \end{pmatrix} \right\|. \quad (2969)$$

Squaring the block matrix gives

$$(\delta h_\theta)^2 = \begin{pmatrix} \delta \tau_\theta \delta \tau_\theta^\dagger & 0 \\ 0 & \delta \tau_\theta^\dagger \delta \tau_\theta \end{pmatrix}. \quad (2970)$$

Therefore,

$$\|\delta h_\theta\| = \|\delta \tau_\theta\|. \quad (2971)$$

The operator norm of the complete twist-induced Hamiltonian reconstruction is exactly the operator norm of the change in the interlayer tunneling matrix. This gives a precise microscopic formulation of the statement that rigid twistronics is driven by interlayer hybridization. The leading weak-twist local tunneling change is

$$\delta T_{\text{loc}}^{(H)}(r) = \frac{\mathcal{T}'(h)R^2 \sinh^2(r/R)}{2h} \theta^2 + O(\theta^4). \quad (2972)$$

The corresponding Euclidean response is

$$\delta T_{\text{loc}}^{(E)}(r) = \frac{\mathcal{T}'(h)r^2}{2h} \theta^2 + O(\theta^4). \quad (2973)$$

Their leading ratio is therefore

$$\lim_{\theta \rightarrow 0} \frac{\delta T_{\text{loc}}^{(H)}(r)}{\delta T_{\text{loc}}^{(E)}(r)} = \left[\frac{R \sinh(r/R)}{r} \right]^2. \quad (2974)$$

This quantity is strictly larger than unity for every positive radius in negative curvature,

$$\left[\frac{R \sinh(r/R)}{r} \right]^2 > 1. \quad (2975)$$

Hence negative curvature does not merely modify the geometric moiré length. It amplifies the microscopic tunneling response itself. The principal results can be summarized formally.

a. Theorem 15. Curvature-dependent microscopic bilayer Hamiltonian. Consider two identical rigid lattices on a two-dimensional manifold of constant nonpositive curvature, separated by a fixed transverse distance and related by an orientation-preserving isometry. For a scalar distance-dependent intralayer hopping and a scalar distance-dependent interlayer tunneling function, the bilayer Hamiltonian is unitarily equivalent to

$$\tilde{h}_{\text{BL}}(\theta, K) = \begin{pmatrix} h_0(K) & \tau_\theta(K) \\ \tau_\theta^\dagger(K) & h_0(K) \end{pmatrix}. \quad (2976)$$

The interlayer matrix is

$$(\tau_\theta)_{ij} = \mathcal{T} \left[\sqrt{h^2 + d_K^2(x_i, g_\theta x_j)} \right]. \quad (2977)$$

Rigid twisting leaves the intralayer block invariant, $\partial_\theta h_0 = 0$. The complete twist-induced Hamiltonian reconstruction is therefore

$$\delta h_\theta = \begin{pmatrix} 0 & \delta \tau_\theta \\ \delta \tau_\theta^\dagger & 0 \end{pmatrix}, \quad (2978)$$

with exact operator-norm identity

$$\|\delta h_\theta\| = \|\delta \tau_\theta\|. \quad (2979)$$

For negative curvature, the same-label local tunneling profile is

$$T_{\text{loc}}(r) = \mathcal{T} \left[\sqrt{h^2 + 4R^2 \text{arsinh}^2(\sinh(r/R) |\sin(\theta/2)|)} \right]. \quad (2980)$$

At weak twist,

$$T_{\text{loc}}(r) = \mathcal{T}(h) + \frac{\mathcal{T}'(h)R^2 \sinh^2(r/R)}{2h} \theta^2 + O(\theta^4). \quad (2981)$$

The ratio between the hyperbolic and Euclidean leading tunneling responses is

$$\mathcal{C}_T(r) = \left[\frac{R \sinh(r/R)}{r} \right]^2. \quad (2982)$$

Thus negative curvature exponentially amplifies the spatial modulation of the microscopic interlayer tunneling. For finite-range hopping, the one-particle Hamiltonian is bounded. For exponentially decaying long-range hopping, a sufficient hyperbolic summability condition is $\mu > \frac{1}{R}$. Finally, in the vanishing-curvature limit the Hamiltonian converges locally to the ordinary twisted Euclidean bilayer Hamiltonian. The complete microscopic theory established in this subsection can be organized into two nested causal chains. The purely geometric chain is

$$K \longrightarrow d_K \longrightarrow d_\theta(r) \longrightarrow \xi_M. \quad (2983)$$

The quantum-mechanical continuation is

$$d_K \longrightarrow D_{ij}^\perp \longrightarrow T_{ij} \longrightarrow \tau_\theta \longrightarrow h_{\text{BL}}. \quad (2984)$$

Combining them gives curvature $\longrightarrow$ geodesic registry amplification $\longrightarrow$ interlayer tunneling reconstruction $\longrightarrow$ bilayer spectral reconstruction. The first three stages have now been established microscopically. The remaining problem is spectral. In the following sections, the block Hamiltonian derived here will be used to determine how curvature modifies hybridization, bandwidth, density of states, generalized dispersion, and ultimately the conditions under which magic flat bands emerge.

B. Geometry-Dependent Intralayer and Interlayer Couplings

The microscopic Hamiltonian derived in the preceding subsection separates the twisted-bilayer problem into identical intralayer blocks and a geometry-dependent interlayer tunneling matrix. A quantitative spectral theory nevertheless requires a more precise specification of the coupling kernels. This step is essential for causal interpretation. If the hopping amplitudes, decay lengths, interlayer separation, or energy normalization are themselves allowed to vary implicitly with curvature, then a curvature-dependent spectrum cannot be attributed uniquely to the geometry of the underlying space. The central objective of the present subsection is therefore to distinguish two logically different sources of curvature dependence. The first is geometric curvature dependence. The microscopic coupling law is held fixed, while the geodesic distances entering that law change with the geometry. The second is explicit microscopic curvature dependence. In this case the bare hopping law or one of its microscopic parameters is itself changed as the curvature changes. Only the first contribution belongs to the minimal geometry-driven twistrionic mechanism developed in this work. We construct below a hierarchy of coupling models ranging from the strict nearest-neighbor limit to exponentially decaying and general admissible short-range kernels. We establish the conditions under which the corresponding infinite hyperbolic Hamiltonians are bounded, derive controlled finite-range truncation estimates, separate overall energy-scale normalization from genuine geometric effects, and identify a universal weak-twist response that is independent of the detailed analytical form of the tunneling profile. Let the intralayer distance between two reference sites be denoted by

$$d_{ij}^{\parallel}(K) = d_K(x_i, x_j). \quad (2985)$$

The physical interlayer distance is

$$D_{ij}^{\perp}(K, \theta) = \sqrt{h^2 + d_K^2(x_i, g_{\theta}x_j)}. \quad (2986)$$

We define a bare intralayer radial coupling kernel by

$$t_{\parallel} : [0, \infty) \longrightarrow \mathbb{C}. \quad (2987)$$

The corresponding hopping matrix element is

$$t_{ij}(K) = t_{\parallel} [d_{ij}^{\parallel}(K)]. \quad (2988)$$

Similarly, define the bare interlayer radial tunneling kernel by

$$t_{\perp} : [h, \infty) \longrightarrow \mathbb{C}. \quad (2989)$$

The interlayer matrix element is

$$T_{ij}(K, \theta) = t_{\perp} [D_{ij}^{\perp}(K, \theta)]. \quad (2990)$$

Equations (2988) and (2990) provide the fundamental separation between microscopic coupling physics and geometry. The functions $t_{\parallel}$ and $t_{\perp}$ encode the microscopic orbital-overlap law. The quantities $d_{ij}^{\parallel}(K)$ and $D_{ij}^{\perp}(K, \theta)$ encode the geometry. This distinction will be maintained throughout the paper. We now define the baseline protocol used to identify curvature-induced physics. Let the microscopic coupling laws depend on a finite set of bare parameters, $\lambda_{\parallel} = (t_*, \lambda_{\parallel}, \dots)$, and

$$\lambda_{\perp} = (w_*, \lambda_{\perp}, h, \dots). \quad (2991)$$

The most general curvature derivative of an intralayer matrix element is

$$\frac{dt_{ij}}{dK} = \left. \frac{\partial t_{\parallel}}{\partial K} \right|_d + \frac{dt_{\parallel}}{dd} \frac{dd_{ij}^{\parallel}}{dK}. \quad (2992)$$

Similarly,

$$\frac{dT_{ij}}{dK} = \left. \frac{\partial t_{\perp}}{\partial K} \right|_D + \frac{dt_{\perp}}{dD} \frac{dD_{ij}^{\perp}}{dK}. \quad (2993)$$

The first term in each expression represents explicit microscopic curvature dependence. The second term represents purely geometric curvature dependence. The geometry-only baseline is defined by imposing

$$\left. \frac{\partial t_{\parallel}}{\partial K} \right|_d = 0, \quad \left. \frac{\partial t_{\perp}}{\partial K} \right|_D = 0. \quad (2994)$$

Under this protocol,

$$\frac{dt_{ij}}{dK} = t'_{\parallel}(d_{ij}^{\parallel}) \frac{dd_{ij}^{\parallel}}{dK}, \quad (2995)$$

and

$$\frac{dT_{ij}}{dK} = t'_{\perp}(D_{ij}^{\perp}) \frac{dD_{ij}^{\perp}}{dK}. \quad (2996)$$

Because the transverse separation is held fixed,

$$\frac{dh}{dK} = 0, \quad (2997)$$

one obtains

$$\frac{dD_{ij}^{\perp}}{dK} = \frac{d_{ij}^{\perp, \parallel}}{D_{ij}^{\perp}} \frac{dd_{ij}^{\perp, \parallel}}{dK}, \quad (2998)$$

where

$$d_{ij}^{\perp,\parallel} = d_K(x_i, g_\theta x_j). \quad (2999)$$

Thus every curvature dependence of the minimal tunneling matrix is generated by the curvature dependence of geodesic distances. For comparison, suppose the microscopic tunneling kernel is parameterized according to

$$t_\perp = t_\perp(D; w(K), \lambda_\perp(K), h(K)). \quad (3000)$$

Then the total curvature derivative contains additional contributions,

$$\begin{aligned} \frac{dT_{ij}}{dK} &= \frac{\partial t_\perp}{\partial D} \frac{dD_{ij}^\perp}{dK} + \frac{\partial t_\perp}{\partial w} \frac{dw}{dK} \\ &+ \frac{\partial t_\perp}{\partial \lambda_\perp} \frac{d\lambda_\perp}{dK} + \frac{\partial t_\perp}{\partial h} \frac{dh}{dK}. \end{aligned} \quad (3001)$$

A calculation allowing the last three terms to vary cannot attribute the resulting spectral response uniquely to the underlying spatial curvature. Accordingly, the principal results of this work will be obtained with

$$\frac{dw}{dK} = 0, \quad \frac{d\lambda_\perp}{dK} = 0, \quad \frac{d\lambda_\parallel}{dK} = 0, \quad \frac{dh}{dK} = 0, \quad (3002)$$

whenever a fixed-physical-length curvature family is used. Alternative normalizations will be introduced only as explicit control calculations. The conclusions should not rely on a single special hopping profile. We therefore define an admissible intralayer coupling class. The bare radial hopping envelope satisfies

$$|t_\parallel(d)| \leq F_\parallel(d), \quad (3003)$$

where the nonnegative envelope obeys the hyperbolic summability condition

$$\int_0^\infty e^{r/R} F_\parallel(r) dr < \infty. \quad (3004)$$

Likewise, an admissible interlayer envelope satisfies

$$|t_\perp[\sqrt{h^2 + r^2}]| \leq F_\perp(r), \quad (3005)$$

with

$$\int_0^\infty e^{r/R} F_\perp(r) dr < \infty. \quad (3006)$$

The exponential factor in these conditions is not arbitrary. It compensates for the exponential growth of the number of sites with geodesic radius in a two-dimensional

hyperbolic crystal. These integrability conditions define the natural short-range universality class for bounded radial hopping operators on the hyperbolic lattice. The strict nearest-neighbor model is obtained by choosing

$$t_\parallel(d) = \begin{cases} t, & d = a, \\ 0, & d \neq a. \end{cases} \quad (3007)$$

The corresponding Hamiltonian is

$$\hat{H}_\parallel = -t \sum_{\ell=1}^2 \sum_{\langle i,j \rangle_\ell} (c_{\ell i}^\dagger c_{\ell j} + \text{h.c.}). \quad (3008)$$

For a regular vertex lattice of coordination number z , the adjacency operator satisfies the row-sum bound $\|A\| \leq z$. Hence, $\|h_0\| \leq z|t|$. For a regular tessellation whose vertices are used as microscopic sites, $z = q$. The local second moment is particularly simple. For a nearest-neighbor regular graph, $(A^2)_{ii} = z$. Therefore,

$$\langle i | h_0^2 | i \rangle = zt^2. \quad (3009)$$

The natural root-mean-square local kinetic scale is consequently $E_{\text{kin}}^{\text{rms}} = |t|\sqrt{z}$. This observation becomes important when comparing lattices with different coordination numbers. If the microscopic edge length is held fixed, $a = \text{const.}$, and the bare nearest-neighbor hopping is fixed, $t = \text{const.}$, then the hopping amplitude on every existing nearest-neighbor bond has no explicit curvature dependence,

$$\frac{dt}{dK} = 0. \quad (3010)$$

Curvature can nevertheless modify the isolated monolayer Hamiltonian if the underlying lattice graph itself changes. Thus two sources of monolayer curvature response must be distinguished: bond-amplitude change and graph-geometry change. The baseline nearest-neighbor protocol removes the first while retaining the second. For a regular hyperbolic tessellation, the geodesic edge length and curvature radius are not independent. The exact relation is

$$\cosh\left(\frac{a}{2R}\right) = \frac{\cos(\pi/p)}{\sin(\pi/q)}. \quad (3011)$$

Therefore,

$$\frac{a}{R} = 2 \operatorname{arccosh} \left[\frac{\cos(\pi/p)}{\sin(\pi/q)} \right]. \quad (3012)$$

For a fixed regular tessellation, $p = \text{const.}$, $q = \text{const.}$, the dimensionless microscopic curvature is therefore fixed, $\frac{a}{R} = \text{const.}$ Changing the curvature radius

alone simply rescales every geometric length and does not generate a new dimensionless lattice geometry. Consequently, physically meaningful curvature variation in a strictly regular-tessellation family requires changing the tessellation data. Changing the tessellation data can introduce a separate coordination-number effect. To isolate curvature as cleanly as possible, one may therefore keep the vertex coordination fixed, $q = q_0$, while varying the polygon parameter, $p \rightarrow p'$. Then the coordination remains $z = q_0$, while the dimensionless curvature changes according to

$$\frac{a}{R} = 2 \operatorname{arcosh} \left[\frac{\cos(\pi/p)}{\sin(\pi/q_0)} \right]. \quad (3013)$$

This protocol suppresses the trivial kinetic-energy change associated with varying coordination number. It is therefore particularly useful for numerical tests of geometry-driven spectral reconstruction. To go beyond the nearest-neighbor limit, consider the radial intralayer kernel

$$t_{\parallel}(d) = t \exp \left[-\frac{d-a}{\lambda_{\parallel}} \right] \quad (3014)$$

for distinct lattice sites. The normalization is chosen so that $t_{\parallel}(a) = t$. The positive decay length satisfies $\lambda_{\parallel} > 0$. The hopping ratio between two bond lengths is

$$\frac{t_{\parallel}(d_2)}{t_{\parallel}(d_1)} = \exp \left[-\frac{d_2 - d_1}{\lambda_{\parallel}} \right]. \quad (3015)$$

Thus all geometric dependence enters through geodesic-distance differences. The number of sites contained in a geodesic shell grows exponentially. There exists a finite geometric constant such that

$$N_{\text{shell}}(r, r + \Delta r) \leq C_{\Lambda} e^{r/R} \quad (3016)$$

for fixed shell width and sufficiently large radius. The absolute row sum of the exponential hopping operator is therefore bounded by a series of the form

$$\sum_j |t_{ij}| \leq C \sum_{n=0}^{\infty} \exp \left[\frac{n\Delta r}{R} \right] \exp \left[-\frac{n\Delta r}{\lambda_{\parallel}} \right]. \quad (3017)$$

The ratio of the geometric series is

$$r_{\parallel} = \exp \left[-\Delta r \left(\frac{1}{\lambda_{\parallel}} - \frac{1}{R} \right) \right]. \quad (3018)$$

Convergence requires

$$\frac{1}{\lambda_{\parallel}} > \frac{1}{R}. \quad (3019)$$

Equivalently, $\lambda_{\parallel} < R$. Thus exponential hopping must decay faster than the exponential hyperbolic growth of the number of available sites. The preceding result can be expressed more generally. Define the lattice growth entropy by

$$h_{\Lambda} = \limsup_{r \rightarrow \infty} \frac{1}{r} \ln N_{\Lambda}(r). \quad (3020)$$

For a cocompact two-dimensional hyperbolic lattice, $h_{\Lambda} = \frac{1}{R}$ at the exponential-growth level. If the hopping envelope satisfies $F(r) \sim e^{-\mu r}$, then a sufficient and essentially sharp summability requirement is $\mu > h_{\Lambda}$. For the hyperbolic plane this reproduces $\mu > \frac{1}{R}$. This formulation will be useful when extending the model to non-constant coordination or more general hyperbolic crystal families. The physically preferred scalar interlayer kernel is defined using the full product-space distance. We impose the normalization $t_{\perp}(h) = w$, where the positive or complex parameter is the vertically aligned tunneling amplitude. A convenient exponentially decaying model is

$$t_{\perp}(D) = w \exp \left[-\frac{D-h}{\lambda_{\perp}} \right]. \quad (3021)$$

This satisfies $t_{\perp}(h) = w$. The corresponding matrix element is

$$T_{ij} = w \exp \left[-\frac{\sqrt{h^2 + d_K^2(x_i, g_{\theta} x_j)} - h}{\lambda_{\perp}} \right]. \quad (3022)$$

Let the in-plane separation be denoted by

$$d = d_K(x_i, g_{\theta} x_j). \quad (3023)$$

For $d \ll h$, the full physical distance satisfies

$$\begin{aligned} D &= h \sqrt{1 + \frac{d^2}{h^2}} \\ &= h + \frac{d^2}{2h} - \frac{d^4}{8h^3} + \frac{d^6}{16h^5} + O(d^8). \end{aligned} \quad (3024)$$

Hence,

$$D - h = \frac{d^2}{2h} - \frac{d^4}{8h^3} + O(d^6). \quad (3025)$$

The exponential tunneling becomes

$$T(d) = w \exp \left[-\frac{d^2}{2h\lambda_{\perp}} + \frac{d^4}{8h^3\lambda_{\perp}} + O(d^6) \right]. \quad (3026)$$

Thus, near perfect local registry, a physically isotropic distance-dependent tunneling law is approximately Gaussian in the in-plane registry mismatch rather than exponential in that mismatch. The leading expansion is

$$T(d) = w \left[1 - \frac{d^2}{2h\lambda_\perp} + O(d^4) \right]. \quad (3027)$$

This even dependence is a consequence of rotational invariance in the full product geometry. For $d \gg h$, the full distance satisfies

$$D = d\sqrt{1 + \frac{h^2}{d^2}} = d + \frac{h^2}{2d} - \frac{h^4}{8d^3} + O(d^{-5}). \quad (3028)$$

Therefore,

$$D - h = d - h + \frac{h^2}{2d} + O(d^{-3}). \quad (3029)$$

The tunneling consequently behaves as

$$T(d) = we^{h/\lambda_\perp} e^{-d/\lambda_\perp} \left[1 - \frac{h^2}{2d\lambda_\perp} + O(d^{-2}) \right]. \quad (3030)$$

Thus the full-distance kernel asymptotically inherits an ordinary exponential decay with the in-plane geodesic separation. The same hyperbolic summability threshold therefore applies. At large in-plane separation, $D \geq d$, and the exponential tunneling satisfies an envelope of the form

$$|T(d)| \leq C_\perp e^{-d/\lambda_\perp}. \quad (3031)$$

The interlayer row sum converges provided $\lambda_\perp < R$. More generally, if

$$|T(d)| \lesssim e^{-\mu_\perp d}, \quad (3032)$$

the sufficient condition is $\mu_\perp > \frac{1}{R}$. One may alternatively define a phenomenological kernel directly as a $\tilde{t}_\perp(d)$. function of in-plane mismatch. However, if $\tilde{t}'_\perp(0) \neq 0$, then the small-twist response contains a term proportional to the absolute value of the twist angle, $\delta T \propto |\theta|$. This produces a cusp at perfect alignment. By contrast, the product-distance construction gives

$$D - h = O(\theta^2), \quad (3033)$$

and therefore generates an analytic even response, $\delta T = O(\theta^2)$. For isotropic orbitals separated by a finite transverse distance, the product-distance formulation therefore provides the physically cleaner baseline. We now show that the leading curvature amplification does not

depend on the exponential ansatz. Assume only that the physical tunneling kernel is differentiable at the vertical separation. The same-label in-plane mismatch has the small-angle form

$$d_\theta(r) = J_K(r)|\theta| + O(|\theta|^3), \quad (3034)$$

where the transverse Jacobi amplitude is

$$J_K(r) = \begin{cases} r, & K = 0, \\ R \sinh(r/R), & K = -R^{-2}. \end{cases} \quad (3035)$$

The physical interlayer distance therefore satisfies

$$D_{\text{loc}}(r) = h + \frac{J_K^2(r)}{2h} \theta^2 + O(\theta^4). \quad (3036)$$

Taylor expansion of a general smooth tunneling law gives

$$T_{\text{loc}}(r) = t_\perp(h) + t'_\perp(h) [D_{\text{loc}}(r) - h] + O([D_{\text{loc}}(r) - h]^2). \quad (3037)$$

Consequently,

$$T_{\text{loc}}(r) = t_\perp(h) + \frac{t'_\perp(h)}{2h} J_K^2(r) \theta^2 + O(\theta^4). \quad (3038)$$

This result is independent of the detailed functional form of the tunneling profile. If $t'_\perp(h) \neq 0$, the ratio of the leading hyperbolic and Euclidean weak-twist responses is

$$C_{\text{hop}}(r) = \frac{J_H^2(r)}{J_E^2(r)} = \left[\frac{R \sinh(r/R)}{r} \right]^2. \quad (3039)$$

Thus the microscopic curvature amplification factor is universal within the entire class of smooth physical tunneling kernels having a nonzero first derivative at the aligned physical distance. The detailed kernel determines the prefactor, $\frac{t'_\perp(h)}{2h}$, while the curvature amplification itself is purely geometric. A special case occurs when $t'_\perp(h) = 0$. Then the quadratic response vanishes. Suppose the second derivative is nonzero, $t''_\perp(h) \neq 0$. Using

$$D_{\text{loc}} - h = \frac{J_K^2(r)}{2h} \theta^2 + O(\theta^4), \quad (3040)$$

the first nonzero tunneling correction becomes

$$T_{\text{loc}}(r) - T_{\text{loc}}(r)|_{\theta=0} = \frac{t''_\perp(h) J_K^4(r)}{8h^2} \theta^4 + O(\theta^6). \quad (3041)$$

Thus the microscopic tunneling response naturally separates into distinct analytic classes. The generic class

has leading response $\delta T \propto J_K^2 \theta^2$, whereas the first exceptional class has $\delta T \propto J_K^4 \theta^4$. The geometry therefore determines the spatial amplification within each response class, while the local analytical structure of the microscopic kernel determines which response class is realized. For numerical calculations, an exponentially decaying kernel must be truncated at a finite distance. Define the truncated intralayer kernel by

$$\mathbf{t}_{\parallel}^{(L)}(d) = \mathbf{t}_{\parallel}(d) \Theta(L - d), \quad (3042)$$

where the step function imposes the finite geodesic range. Likewise,

$$\mathbf{t}_{\perp}^{(L)}(D) = \mathbf{t}_{\perp}(D) \Theta(L - D). \quad (3043)$$

The omitted intralayer operator is

$$\Delta h_{\parallel}^{(L)} = h_{\parallel} - h_{\parallel}^{(L)}. \quad (3044)$$

The omitted interlayer operator is

$$\Delta \tau^{(L)} = \tau - \tau^{(L)}. \quad (3045)$$

Assume

$$|\mathbf{t}_{\parallel}(d)| \leq C_{\parallel} e^{-\mu_{\parallel} d}, \quad (3046)$$

with $\mu_{\parallel} > \frac{1}{R}$. The number of lattice sites in a shell is bounded by

$$N_{\text{shell}} \leq C_{\Lambda} e^{r/R}. \quad (3047)$$

The omitted row sum beyond the truncation radius therefore satisfies

$$\begin{aligned} \sup_i \sum_{j: d_{ij} > L} |t_{ij}| &\leq C_{\text{tail}} \int_L^{\infty} e^{r/R} e^{-\mu_{\parallel} r} dr \\ &= C_{\text{tail}} \int_L^{\infty} e^{-(\mu_{\parallel} - R^{-1})r} dr. \end{aligned} \quad (3048)$$

The integral is

$$\int_L^{\infty} e^{-(\mu_{\parallel} - R^{-1})r} dr = \frac{e^{-(\mu_{\parallel} - R^{-1})L}}{\mu_{\parallel} - R^{-1}}. \quad (3049)$$

Hence,

$$\sup_i \sum_{j: d_{ij} > L} |t_{ij}| \leq \frac{C_{\text{tail}}}{\mu_{\parallel} - R^{-1}} e^{-(\mu_{\parallel} - R^{-1})L}. \quad (3050)$$

The Schur test therefore gives

$$\|\Delta h_{\parallel}^{(L)}\| \leq C_{\parallel}^{\text{tr}} e^{-(\mu_{\parallel} - R^{-1})L}. \quad (3051)$$

The effective truncation decay rate is thus not the bare hopping-decay rate but the difference

$$\mu_{\text{eff}} = \mu_{\parallel} - \frac{1}{R}. \quad (3052)$$

Negative curvature therefore slows convergence of a long-range finite-range truncation even when the microscopic kernel itself is held fixed. Suppose the interlayer tunneling obeys

$$|T_{ij}| \leq C_{\perp} e^{-\mu_{\perp} d_{ij}^{\perp}}, \quad (3053)$$

with

$$\mu_{\perp} > \frac{1}{R}. \quad (3054)$$

The same shell-counting argument yields

$$\|\Delta \tau^{(L)}\| \leq C_{\perp}^{\text{tr}} e^{-(\mu_{\perp} - R^{-1})L}. \quad (3055)$$

The complete bilayer truncation error therefore satisfies

$$\|H - H^{(L)}\| \leq 2C_{\parallel}^{\text{tr}} e^{-(\mu_{\parallel} - R^{-1})L} + C_{\perp}^{\text{tr}} e^{-(\mu_{\perp} - R^{-1})L}. \quad (3056)$$

The numerical model can therefore be systematically converged by increasing the geodesic cutoff. Consider a finite exact supercell or finite-cover Hamiltonian. Let its eigenvalues be ordered according to $E_1 \leq E_2 \leq \dots \leq E_N$. Let the truncated Hamiltonian have eigenvalues $E_1^{(L)} \leq E_2^{(L)} \leq \dots \leq E_N^{(L)}$. For Hermitian matrices, eigenvalue perturbation theory gives

$$|E_n - E_n^{(L)}| \leq \|H - H^{(L)}\|. \quad (3057)$$

Therefore,

$$\max_n |E_n - E_n^{(L)}| \leq \epsilon_L, \quad (3058)$$

where

$$\epsilon_L = \|H - H^{(L)}\|. \quad (3059)$$

If a target band has bandwidth $W = E_{\text{max}} - E_{\text{min}}$, the truncated bandwidth satisfies

$$|W - W^{(L)}| \leq 2\epsilon_L. \quad (3060)$$

Thus apparent band flattening cannot be regarded as converged until the finite-range truncation error is parametrically smaller than the measured bandwidth. Suppose the dominant truncation error has the form $\epsilon_L \leq Ce^{-\gamma L}$, with

$$\gamma = \mu - \frac{1}{R} > 0. \quad (3061)$$

To achieve a target operator-norm accuracy satisfying $\epsilon_L \leq \varepsilon$, it is sufficient to choose

$$L \geq \frac{1}{\gamma} \ln \left(\frac{C}{\varepsilon} \right). \quad (3062)$$

As the decay rate approaches the hyperbolic volume-growth rate, $\mu \downarrow \frac{1}{R}$, the required cutoff diverges, $L \rightarrow \infty$. Thus the summability threshold is also a numerical-complexity threshold. A change of all hopping amplitudes by a common multiplicative constant does not constitute new spectral geometry. Let the Hamiltonian be written as $H = t_* \mathcal{H}$, where the positive parameter sets the global energy scale and the dimensionless operator contains all spectral shape information. Under the rescaling $t_* \rightarrow ct_*$, one has $E_n \rightarrow cE_n$. The eigenvectors remain unchanged, $|\psi_n\rangle \rightarrow |\psi_n\rangle$. The microscopic time unit is $\tau_* = \frac{\hbar}{t_*}$. Thus an overall coupling rescaling changes the physical clock but does not change dimensionless band geometry. For this reason all comparisons of band flattening should use either normalized bandwidths or scale-invariant ratios. Define the interlayer-to-intralayer coupling ratio by $\omega = \frac{w}{t_*}$. Define the dimensionless decay lengths by

$$\eta_{\parallel} = \frac{\lambda_{\parallel}}{a}, \quad \eta_{\perp} = \frac{\lambda_{\perp}}{a}. \quad (3063)$$

Define the dimensionless layer separation by $\eta_h = \frac{h}{a}$. Finally define the microscopic curvature by $\kappa_a = \frac{a}{R}$. The geometry-only Hamiltonian may then be written schematically as

$$\frac{H}{t_*} = \mathcal{H}(\theta, \kappa_a, \omega, \eta_{\parallel}, \eta_{\perp}, \eta_h). \quad (3064)$$

A clean curvature comparison holds $\omega, \eta_{\parallel}, \eta_{\perp}, \eta_h$ fixed while varying only the dimensionless geometric curvature and the twist. The principal physical protocol used in this work is the bare-fixed protocol. It imposes $t_* =$

const., $w = \text{const.}$, $\eta_{\parallel} = \text{const.}$, $\eta_{\perp} = \text{const.}$, and $\eta_h = \text{const.}$ across the curvature family. No curvature-dependent renormalization of the microscopic overlap law is introduced. Under this protocol every change in a hopping matrix element originates from either a change of geodesic distance or a change of the lattice connectivity required by the chosen hyperbolic crystal family. If the coordination number varies across a set of hyperbolic lattices, part of the spectral change can arise trivially from the increased number of nearest-neighbor bonds. A useful control calculation holds the local second moment fixed. From $\langle i | h_0^2 | i \rangle = zt^2$, choose the normalized nearest-neighbor hopping

$$t(z) = \frac{t_{\text{rms}}}{\sqrt{z}}. \quad (3065)$$

Then

$$\langle i | h_0^2 | i \rangle = t_{\text{rms}}^2 \quad (3066)$$

for every coordination number. An alternative row-sum normalization is

$$t(z) = \frac{t_{\text{row}}}{z}. \quad (3067)$$

Then the nearest-neighbor row sum is fixed,

$$\sum_j |t_{ij}| = t_{\text{row}}. \quad (3068)$$

These normalizations should be regarded as diagnostic controls rather than the primary physical model. If the same qualitative curvature-induced spectral reconstruction survives both the bare-fixed and coordination-normalized protocols, it cannot be attributed solely to a trivial change in the number of neighbors. A still stronger normalization could divide the monolayer Hamiltonian by its spectral norm,

$$\tilde{h}_0 = \frac{E_*}{\|h_0\|} h_0. \quad (3069)$$

This enforces $\|\tilde{h}_0\| = E_*$. However, such normalization removes part of the genuine curvature-induced monolayer spectral response. It should therefore not be used as the primary Hamiltonian. Its purpose is only to test whether a bilayer effect survives after the overall monolayer bandwidth has been artificially fixed. The total interlayer coupling strength can be quantified by the positive operator $\tau_{\theta} \tau_{\theta}^{\dagger}$. For a finite exact supercell or finite cover with single-layer dimension N , define the mean squared hybridization amplitude by

$$W_{\perp}^2 = \frac{1}{N} \text{Tr} \left(\tau_{\theta} \tau_{\theta}^{\dagger} \right). \quad (3070)$$

Equivalently,

$$W_{\perp}^2 = \frac{1}{N} \sum_{i,j} |T_{ij}|^2. \quad (3071)$$

Under the bare-fixed protocol, this quantity is allowed to vary with curvature and twist. Such variation is a genuine consequence of the geometry because the bare coupling kernel itself has not changed. To distinguish redistribution of tunneling from a change in its total integrated strength, define a control tunneling matrix

$$\hat{\tau}_{\theta} = \frac{W_{\perp}^{\text{ref}}}{W_{\perp}} \tau_{\theta}. \quad (3072)$$

It satisfies

$$\frac{1}{N} \text{Tr} \left(\hat{\tau}_{\theta} \hat{\tau}_{\theta}^{\dagger} \right) = (W_{\perp}^{\text{ref}})^2. \quad (3073)$$

This control removes the change in the average squared hybridization while retaining its spatial pattern. If a curvature-induced flat-band phenomenon survives this normalization, its origin cannot be reduced to a simple increase or decrease of total interlayer coupling strength. Again, this normalized model is a diagnostic control rather than the baseline physical Hamiltonian. At fixed curvature, the cleanest isolation of twist physics is obtained by subtracting the aligned Hamiltonian. Define

$$\delta_{\theta} H(K) = H(K, \theta) - H(K, 0). \quad (3074)$$

In the co-rotating representation, the intralayer terms cancel exactly. Therefore,

$$\delta_{\theta} H(K) = \begin{pmatrix} 0 & \tau(K, \theta) - \tau(K, 0) \\ \tau^{\dagger}(K, \theta) - \tau^{\dagger}(K, 0) & 0 \end{pmatrix}. \quad (3075)$$

Thus monolayer curvature physics cannot contaminate this differential twist response. At fixed twist, define

$$\delta_K H(\theta) = H(K, \theta) - H(0, \theta). \quad (3076)$$

This quantity contains both the monolayer curvature response and the curvature modification of the interlayer tunneling. It therefore does not isolate the curvature–twist interaction by itself. To isolate the nonadditive coupling between curvature and twist, define the mixed finite difference

$$\delta_{K\theta} H = H(K, \theta) - H(K, 0) - H(0, \theta) + H(0, 0). \quad (3077)$$

The same quantity can be written as

$$\delta_{K\theta} H = \delta_{\theta} H(K) - \delta_{\theta} H(0). \quad (3078)$$

This construction removes the pure-curvature response and the pure-twist response simultaneously. In the co-rotating representation,

$$\delta_{K\theta} H = \begin{pmatrix} 0 & \Delta_{K\theta} \tau \\ \Delta_{K\theta} \tau^{\dagger} & 0 \end{pmatrix}, \quad (3079)$$

where

$$\Delta_{K\theta} \tau = \tau(K, \theta) - \tau(K, 0) - \tau(0, \theta) + \tau(0, 0). \quad (3080)$$

Consequently,

$$\|\delta_{K\theta} H\| = \|\Delta_{K\theta} \tau\|. \quad (3081)$$

This quantity is a direct microscopic diagnostic of genuinely curvature-assisted twistrionic reconstruction. For the same-label tunneling channel, the Euclidean Jacobi amplitude is $J_E(r) = r$. The hyperbolic Jacobi amplitude is

$$J_H(r) = R \sinh\left(\frac{r}{R}\right). \quad (3082)$$

For weak curvature,

$$J_H(r) = r + \frac{r^3}{6R^2} + \frac{r^5}{120R^4} + O(R^{-6}). \quad (3083)$$

Squaring gives

$$J_H^2(r) = r^2 + \frac{r^4}{3R^2} + \frac{2r^6}{45R^4} + O(R^{-6}). \quad (3084)$$

The generic weak-twist tunneling difference between hyperbolic and Euclidean geometry is therefore

$$\begin{aligned} \delta T_{K\theta}(r) &= \frac{t'_{\perp}(h)}{2h} [J_H^2(r) - J_E^2(r)] \theta^2 + O(\theta^4) \\ &= \frac{t'_{\perp}(h)}{6h} \frac{r^4}{R^2} \theta^2 + O\left(\frac{r^6 \theta^2}{R^4}\right) + O(\theta^4). \end{aligned} \quad (3085)$$

Thus the leading genuine curvature–twist interaction appears at order

$$\delta T_{K\theta} \sim K r^4 \theta^2 \quad (3086)$$

up to the sign convention relating negative curvature to the inverse curvature-radius squared. This provides an analytic perturbative benchmark for numerical calculations. The geometry-only construction must recover the

Euclidean coupling matrices continuously. For a fixed local region,

$$d_K(x, y) = d_E(x, y) + O(R^{-2}). \quad (3087)$$

Assume the intralayer kernel is Lipschitz continuous on the relevant distance interval,

$$|\mathbf{t}_{\parallel}(d_1) - \mathbf{t}_{\parallel}(d_2)| \leq L_{\parallel} |d_1 - d_2|. \quad (3088)$$

Then

$$|t_{ij}^{(H)} - t_{ij}^{(E)}| = O(R^{-2}). \quad (3089)$$

Likewise, if the interlayer kernel is Lipschitz continuous,

$$|\mathbf{t}_{\perp}(D_1) - \mathbf{t}_{\perp}(D_2)| \leq L_{\perp} |D_1 - D_2|, \quad (3090)$$

then

$$|T_{ij}^{(H)} - T_{ij}^{(E)}| = O(R^{-2}). \quad (3091)$$

Therefore, on every fixed finite local Hilbert space,

$$\|H^{(H)} - H^{(E)}\| = O(R^{-2}). \quad (3092)$$

This verifies that no artificial coupling discontinuity is introduced by the curvature-dependent construction. For later perturbative and spectral estimates it is useful to define distance-weighted hopping moments. The intralayer moment of order m is

$$M_m^{\parallel} = \sup_i \sum_j \left[d_{ij}^{\parallel} \right]^m |t_{ij}|. \quad (3093)$$

The interlayer moment is

$$M_m^{\perp} = \sup_i \sum_j \left[d_{ij}^{\perp, \parallel} \right]^m |T_{ij}|. \quad (3094)$$

For exponentially decaying kernels satisfying $\mu > \frac{1}{R}$, every finite polynomial moment converges, $M_m < \infty$ for every finite nonnegative integer m . This follows because the exponential factor dominates any polynomial prefactor. These finite moments control gradient expansions, perturbative derivatives, and finite-range approximation errors. The distinction between geometry and bare parameters can now be stated at the Hamiltonian level. Let the Hamiltonian depend on curvature through both geometry and bare microscopic parameters,

$$H = H[G(K), \boldsymbol{\lambda}(K)], \quad (3095)$$

where the geometric data are represented schematically by $G(K)$, and the microscopic parameters by $\boldsymbol{\lambda}(K)$. The total curvature derivative is

$$\frac{dH}{dK} = \frac{\delta H}{\delta G} \Big|_{\boldsymbol{\lambda}} \frac{dG}{dK} + \frac{\partial H}{\partial \boldsymbol{\lambda}} \Big|_G \cdot \frac{d\boldsymbol{\lambda}}{dK}. \quad (3096)$$

The first term is the geometry-induced contribution. The second is the parameter-induced contribution. The baseline theory imposes

$$\frac{d\boldsymbol{\lambda}}{dK} = 0. \quad (3097)$$

Therefore,

$$\frac{dH}{dK} = \frac{\delta H}{\delta G} \Big|_{\boldsymbol{\lambda}} \frac{dG}{dK}. \quad (3098)$$

Every curvature effect reported under this protocol is therefore generated by geometry. A convincing curvature-induced spectral phenomenon should survive a hierarchy of controls. The first comparison uses fixed bare microscopic parameters, $\mathcal{C}_1$: bare-fixed curvature family. The second removes trivial coordination-number effects, $\mathcal{C}_2$: fixed coordination or local-moment normalization. The third removes changes in total interlayer hybridization strength, $\mathcal{C}_3$: fixed interlayer mean-square coupling. The fourth isolates the mixed curvature-twist response, $\mathcal{C}_4$: $\delta_{K\theta} H$. A phenomenon that remains visible under all four controls cannot be explained by an arbitrary change of the microscopic energy scale, coordination number, or total tunneling amplitude. Its origin is then genuinely associated with the geometry-dependent organization of the tunneling network. The exponential kernel is a convenient numerical choice but should not be treated as part of the universal mechanism. Consider a family of smooth monotonically decreasing physical kernels $\mathbf{t}_{\perp}^{(\nu)}(D)$, indexed by a profile label. Normalize every profile according to $\mathbf{t}_{\perp}^{(\nu)}(h) = w$. To compare their local response fairly, one may additionally fix the dimensionless logarithmic slope at vertical alignment,

$$\zeta_{\perp} = -h \frac{\mathbf{t}'_{\perp}(h)}{\mathbf{t}_{\perp}(h)}. \quad (3099)$$

Then the weak-twist response becomes

$$\frac{T_{\text{loc}}(r) - w}{w} = -\frac{\zeta_{\perp}}{2h^2} J_K^2(r) \theta^2 + O(\theta^4). \quad (3100)$$

All normalized profiles with the same logarithmic slope therefore share the same leading geometry-controlled tunneling reconstruction. This defines a microscopic coupling universality class. Differences among coupling profiles enter at higher order. Write

$$D_{\text{loc}} = h + A_2\theta^2 + A_4\theta^4 + O(\theta^6). \quad (3101)$$

The leading coefficient is

$$A_2 = \frac{J_K^2(r)}{2h}. \quad (3102)$$

The tunneling expansion is

$$T_{\text{loc}} = \mathbf{t}_{\perp}(h) + \mathbf{t}'_{\perp}(h)A_2\theta^2 + \left[\mathbf{t}_{\perp}(h)A_4 + \frac{1}{2}\mathbf{t}''_{\perp}(h)A_2^2 \right] \theta^4 + O(\theta^6). \quad (3103)$$

Thus the leading quadratic response depends only on the first local slope of the kernel, whereas profile curvature enters at quartic order. This separation will later be used to test whether the location of a magic band is controlled primarily by geometry or by nonuniversal short-distance details of the tunneling profile. The three coupling descriptions used throughout the paper can therefore be organized as follows. The strict graph model is $\mathbf{t}(d) \propto \delta_{d,a}$. The smooth microscopic model is $\mathbf{t}(d) \propto e^{-d/\lambda}$. The controlled numerical approximation is

$$\mathbf{t}^{(L)}(d) = \mathbf{t}(d)\Theta(L-d). \quad (3104)$$

The strict nearest-neighbor model isolates graph geometry. The smooth exponential model includes metric-shell structure beyond nearest neighbors. The finite-range approximation provides a systematically convergent numerical implementation of the smooth model. Agreement of the principal physical conclusions across all three levels constitutes a strong robustness test. The central results of the present subsection can be summarized formally.

a. Theorem 16. Geometry-dependent coupling universality. Consider a twisted bilayer on a two-dimensional constant-curvature manifold with microscopic couplings depending on intrinsic or product-space geodesic distance. Assume the bare kernels themselves possess no explicit curvature dependence. Then every curvature derivative of a microscopic coupling matrix element is generated entirely by the corresponding geodesic-distance derivative. For intralayer hopping,

$$\frac{dt_{ij}}{dK} = \mathbf{t}'_{\parallel}(d_{ij}^{\parallel}) \frac{dd_{ij}^{\parallel}}{dK}. \quad (3105)$$

For interlayer hopping,

$$\frac{dT_{ij}}{dK} = \mathbf{t}'_{\perp}(D_{ij}^{\perp}) \frac{dD_{ij}^{\perp}}{dK} \frac{dd_{ij}^{\perp}}{dK}. \quad (3106)$$

For a smooth physical interlayer tunneling kernel satisfying $\mathbf{t}'_{\perp}(h) \neq 0$, the weak-twist local response is

$$T_{\text{loc}}(r) = \mathbf{t}_{\perp}(h) + \frac{\mathbf{t}'_{\perp}(h)}{2h} J_K^2(r)\theta^2 + O(\theta^4), \quad (3107)$$

where the geometric Jacobi amplitude is $J_K(r) = r$ in flat space and

$$J_K(r) = R \sinh(r/R) \quad (3108)$$

in negative curvature. Consequently, the ratio between the hyperbolic and Euclidean leading responses is the profile-independent factor

$$\mathcal{C}_{\text{hop}}(r) = \left[\frac{R \sinh(r/R)}{r} \right]^2. \quad (3109)$$

For exponentially decaying radial kernels, the infinite hyperbolic hopping operator is bounded under the sufficient decay condition $\mu > \frac{1}{R}$. The finite-range truncation error then obeys

$$\|H - H^{(L)}\| \leq C e^{-(\mu - R^{-1})L} \quad (3110)$$

up to the sum of the corresponding intra- and interlayer contributions. Thus the dependence of the microscopic Hamiltonian on curvature can be cleanly separated into geometry-induced distance reconstruction, explicit microscopic parameter renormalization, and optional normalization controls. A second result formalizes the causal decomposition required for later spectral claims.

b. Theorem 17. Mixed curvature-twist isolation. Let the bilayer Hamiltonian be represented on a matched family of local or finite-cover Hilbert spaces. Define the mixed response by

$$\delta_{K\theta}H = H(K, \theta) - H(K, 0) - H(0, \theta) + H(0, 0). \quad (3111)$$

For a rigid bilayer in the co-rotating representation, every pure intralayer term cancels, and

$$\delta_{K\theta}H = \begin{pmatrix} 0 & \Delta_{K\theta}\tau \\ \Delta_{K\theta}\tau^\dagger & 0 \end{pmatrix}. \quad (3112)$$

Hence,

$$\|\delta_{K\theta}H\| = \|\Delta_{K\theta}\tau\|. \quad (3113)$$

Therefore any nonzero mixed response originates entirely from the curvature-dependent reconstruction of the interlayer tunneling network. In the weak-curvature and weak-twist regime, the same-label contribution satisfies

$$\delta T_{K\theta}(r) = \frac{t'_\perp(h)}{6h} \frac{r^4}{R^2} \theta^2 + \text{higher-order terms.} \quad (3114)$$

Thus the leading nonadditive curvature–twist coupling is analytically identifiable at the microscopic Hamiltonian level. The subsequent spectral analysis will therefore use three nested model levels. The primary physical model is the bare-fixed geometry-driven Hamiltonian,

$$H_{\text{phys}} = H(K, \theta; t_*, w, \lambda_\parallel, \lambda_\perp, h), \quad (3115)$$

with all bare microscopic parameters fixed. The first robustness control is the coordination-normalized model, H_{coord} in which trivial changes of local kinetic strength are removed. The second robustness control is the hybridization-normalized model, H_{hyb} in which the mean squared interlayer coupling is held fixed. A genuine geometry-induced moiré effect should remain qualitatively present in all three models, although its quantitative location may shift. The complete coupling hierarchy established in this subsection is bare orbital-overlap law $\rightarrow$ geodesic distance matrix $\rightarrow$ hopping matrix $\rightarrow$ bilayer Hamiltonian. Within the geometry-only protocol, curvature enters according to

$$K \rightarrow d_K \rightarrow t_{ij}(K), T_{ij}(K, \theta), \quad (3116)$$

while the bare parameters remain fixed. For the twist-sensitive interlayer sector, the microscopic chain is

$$K < 0 \rightarrow J_K(r) = R \sinh(r/R) \rightarrow d_\theta(r) \rightarrow D_{\text{loc}}(r) \rightarrow T_{\text{loc}}(r) \rightarrow T_{\text{BL}}(r, \theta). \quad (3117)$$

The generic leading response is therefore

$$\delta T_{\text{loc}} \propto J_K^2(r) \theta^2. \quad (3118)$$

Negative curvature changes this response through the purely geometric factor

$$\frac{J_H^2(r)}{J_E^2(r)} = \left[\frac{R \sinh(r/R)}{r} \right]^2. \quad (3119)$$

The finite-range approximation is controlled by

$$\|H - H^{(L)}\| = O[e^{-(\mu - R^{-1})L}], \quad (3120)$$

provided the microscopic decay outruns hyperbolic volume growth. Finally, the genuinely nonadditive curvature-assisted twist response is isolated by

$$\delta_{K\theta} H = H(K, \theta) - H(K, 0) - H(0, \theta) + H(0, 0). \quad (3121)$$

This hierarchy ensures that every curvature-dependent spectral effect identified in the subsequent sections can be traced either to a well-defined geometric reconstruction of the hopping network or to an explicitly declared normalization protocol. No curvature dependence of the bare microscopic hopping parameters is required to generate the effect.

C. Symmetries of the Twisted Bilayer Hamiltonian

The microscopic Hamiltonian constructed in the preceding subsections contains several distinct classes of symmetry. Their roles must be separated carefully. Internal symmetries constrain the microscopic degrees of freedom without acting on the underlying lattice geometry. Time-reversal symmetry is antiunitary and constrains complex-conjugate sectors. Layer exchange acts on the bilayer degree of freedom and, by itself, generally reverses the relative twist. Point-group operations act on the microscopic lattice geometry. Hyperbolic translations form a non-Abelian Fuchsian group, and at exact commensurate twists their finite-index intersection constitutes the exact moiré translation group. Finally, combinations of spatial operations with layer exchange can survive even when the corresponding spatial operation is broken as a layer-preserving symmetry. These distinctions are essential for determining which spectral degeneracies are genuinely symmetry enforced. The central objectives of the present subsection are therefore to derive the complete symmetry structure of the twisted bilayer Hamiltonian, determine the residual symmetry group after twisting, construct its action on Abelian and non-Abelian hyperbolic Bloch sectors, and establish precise criteria for distinguishing symmetry-enforced degeneracies, symmetry-protected crossings, time-reversal-related spectral pairing, and accidental degeneracies. We begin from the co-rotating representation derived previously. The microscopic single-particle Hamiltonian is

$$h_{\text{BL}}(\theta, K) = \begin{pmatrix} h_0(K) & \tau_\theta(K) \\ \tau_\theta^\dagger(K) & h_0(K) \end{pmatrix}. \quad (3122)$$

The identical diagonal blocks describe the isolated monolayers. The interlayer matrix contains all rigid-twist dependence,

$$\frac{\partial h_0}{\partial \theta} = 0. \quad (3123)$$

The twist derivative is therefore

$$\frac{\partial h_{\text{BL}}}{\partial \theta} = \begin{pmatrix} 0 & \partial_\theta \tau_\theta \\ \partial_\theta \tau_\theta^\dagger & 0 \end{pmatrix}. \quad (3124)$$

The subsequent symmetry analysis applies both to the Euclidean and hyperbolic realizations of this block structure. The second-quantized Hamiltonian conserves total particle number. The number operator is

$$\hat{N} = \sum_{\ell=1}^2 \sum_i c_{\ell i}^\dagger c_{\ell i}. \quad (3125)$$

Every term in the microscopic Hamiltonian contains one creation and one annihilation operator. Consequently, $[\hat{H}, \hat{N}] = 0$. The Hamiltonian therefore possesses a global charge symmetry, $U(1)_{\text{charge}}$. The corresponding unitary transformation is

$$\hat{U}_\varphi = \exp(i\varphi \hat{N}). \quad (3126)$$

It acts according to

$$\hat{U}_\varphi c_{\ell i} \hat{U}_\varphi^{-1} = e^{-i\varphi} c_{\ell i}. \quad (3127)$$

Hence,

$$\hat{U}_\varphi \hat{H} \hat{U}_\varphi^{-1} = \hat{H}. \quad (3128)$$

This symmetry fixes particle-number sectors but does not by itself enforce single-particle band degeneracies. We first consider the minimal spinless scalar-orbital model. Assume the intralayer hopping amplitudes are real, $t_{ij} \in \mathbb{R}$, and the interlayer tunneling amplitudes are real, $T_{ij} \in \mathbb{R}$. The spinless time-reversal operator is represented by complex conjugation, $\Theta = \mathcal{K}$, where the antiunitary operator on the right denotes complex conjugation in the real-space orbital basis. Its square is $\Theta^2 = +1$. Time reversal acts on the one-particle Hamiltonian according to

$$\Theta h_{\text{BL}} \Theta^{-1} = h_{\text{BL}}^*. \quad (3129)$$

For the real-hopping model, $h_{\text{BL}}^* = h_{\text{BL}}$, and therefore

$$\Theta h_{\text{BL}} \Theta^{-1} = h_{\text{BL}}. \quad (3130)$$

Thus the baseline model belongs to the spinless time-reversal-invariant class. Spinless time reversal does not produce conventional Kramers degeneracy. Suppose an eigenstate satisfies

$$h_{\text{BL}} |\psi\rangle = E |\psi\rangle. \quad (3131)$$

Time reversal gives

$$h_{\text{BL}} \Theta |\psi\rangle = E \Theta |\psi\rangle. \quad (3132)$$

However, $\Theta^2 = +1$ does not imply orthogonality between the state and its time-reversed partner. Therefore,

$$\Theta^2 = +1 \text{ not } \Rightarrow \text{twofold Kramers degeneracy}. \quad (3133)$$

This distinction will be important when identifying symmetry-protected degeneracies later. If physical spin is introduced and spin-orbit coupling preserves ordinary fermionic time reversal, the antiunitary symmetry instead satisfies $\Theta_{\text{spin}}^2 = -1$. At a time-reversal-invariant generalized momentum sector, Kramers theorem then enforces $\dim \mathcal{E}_E \geq 2$. The present work does not require this spinful extension, and all curvature-controlled flat-band mechanisms below can already be studied in the spinless model. Introduce Pauli matrices acting on the layer degree of freedom. The layer-exchange operator is

$$\mathcal{P}_L = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}. \quad (3134)$$

It satisfies $\mathcal{P}_L^2 = I$. The layer-polarization operator is

$$\mathcal{Z}_L = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}. \quad (3135)$$

The two operators anticommute, $\{\mathcal{P}_L, \mathcal{Z}_L\} = 0$. For a scalar distance-dependent tunneling kernel, the interlayer matrix obeys $\tau_{-\theta} = \tau_\theta^\dagger$. Acting with the layer-exchange matrix gives

$$\mathcal{P}_L h_{\text{BL}}(\theta) \mathcal{P}_L^{-1} = \begin{pmatrix} h_0 & \tau_\theta^\dagger \\ \tau_\theta & h_0 \end{pmatrix} = h_{\text{BL}}(-\theta). \quad (3136)$$

Therefore,

$$\mathcal{P}_L h_{\text{BL}}(\theta) \mathcal{P}_L^{-1} = h_{\text{BL}}(-\theta). \quad (3137)$$

Layer exchange is consequently not, in general, a symmetry of a fixed signed-twist Hamiltonian. Instead it establishes the spectral equivalence

$$\text{Spec } h_{\text{BL}}(\theta) = \text{Spec } h_{\text{BL}}(-\theta). \quad (3138)$$

At perfect alignment, $\theta = 0$, one has $\tau_0 = \tau_0^\dagger$. Therefore, $[h_{\text{BL}}(0), \mathcal{P}_L] = 0$. The Hilbert space decomposes into layer-even and layer-odd sectors, $\mathcal{H}_{\text{BL}} = \mathcal{H}_+ \oplus \mathcal{H}_-$. Their layer-exchange eigenvalues are

$$p_L = +1, \quad p_L = -1. \quad (3139)$$

The corresponding Hamiltonians are $h_\pm = h_0 \pm \tau_0$. Layer exchange therefore separates bonding and antibonding sectors. It does not force those sectors to be degenerate. The residual spatial symmetry is most transparent when

the relative twist is distributed symmetrically between the two layers. Starting from the asymmetric representation, apply a common global isometry corresponding to half the negative relative angle. The two layer lattices then become $\Lambda_- = g_{-\theta/2}\Lambda$, and $\Lambda_+ = g_{+\theta/2}\Lambda$. The relative twist remains θ . Because the transformation applied to both layers is a global isometry, the symmetric and asymmetric gauges are unitarily equivalent. Thus,

$$\text{Spec } h_{\text{sym}} = \text{Spec } h_{\text{asym}}. \quad (3140)$$

The symmetric gauge will be used to expose the point-group structure. Let the full monolayer crystallographic symmetry group be denoted by $\Delta < \text{Isom}(\mathbb{H}_R^2)$. Unlike the orientation-preserving Fuchsian translation group, the full group may contain orientation-reversing elements. The translation group is a normal subgroup, $\Gamma \triangleleft \Delta$. The point group of the monolayer is the finite quotient $P = \Delta/\Gamma$. The crystallographic structure is therefore described by the exact sequence

$$1 \longrightarrow \Gamma \longrightarrow \Delta \longrightarrow P \longrightarrow 1. \quad (3141)$$

If the extension splits, the space group is symmorphic and may be written $\Delta \simeq \Gamma \rtimes P$. A split extension is not assumed in the general derivations below. In the symmetric twist gauge, define $g_- = g_{-\theta/2}$, and $g_+ = g_{+\theta/2}$. The two layer space groups are

$$\Delta_- = g_- \Delta g_-^{-1}, \quad \Delta_+ = g_+ \Delta g_+^{-1}. \quad (3142)$$

Their translation groups are

$$\Gamma_- = g_- \Gamma g_-^{-1}, \quad \Gamma_+ = g_+ \Gamma g_+^{-1}. \quad (3143)$$

An isometry that preserves both layers individually must belong to both layer space groups. Therefore the maximal layer-preserving residual space group is

$$\Delta_M^+ = \Delta_- \cap \Delta_+. \quad (3144)$$

Its translation subgroup is $\Gamma_M = \Gamma_- \cap \Gamma_+$. This is the same exact moiré translation group obtained previously, expressed in the symmetric twist gauge. The residual translation group is normal inside the layer-preserving residual space group. Let $h \in \Delta_M^+$, and $\gamma \in \Gamma_M$. Because $h \in \Delta_-$, and $\Gamma_- \triangleleft \Delta_-$, one has $h\gamma h^{-1} \in \Gamma_-$. Likewise, $h \in \Delta_+$, and $\Gamma_+ \triangleleft \Delta_+$, so $h\gamma h^{-1} \in \Gamma_+$. Therefore,

$$h\gamma h^{-1} \in \Gamma_- \cap \Gamma_+. \quad (3145)$$

Hence, $\Gamma_M \triangleleft \Delta_M^+$. The layer-preserving residual point group is therefore defined by the quotient

$$P_M^+ = \Delta_M^+ / \Gamma_M. \quad (3146)$$

At an exact commensurate twist, the translation group has finite index. Consequently the quotient in Eq. (3146) is finite. The residual exact crystallographic structure therefore satisfies

$$1 \longrightarrow \Gamma_M \longrightarrow \Delta_M^+ \longrightarrow P_M^+ \longrightarrow 1. \quad (3147)$$

For every residual layer-preserving symmetry, define its action on microscopic orbitals by

$$U_h |\ell, i\rangle = |\ell, hi\rangle. \quad (3148)$$

Because the transformation is an isometry,

$$d_K(hx, hy) = d_K(x, y). \quad (3149)$$

The distance-dependent intralayer amplitudes therefore satisfy $t_{hi,hj} = t_{ij}$. The interlayer physical distances satisfy $D_{hi,hj}^\perp = D_{ij}^\perp$. Hence, $T_{hi,hj} = T_{ij}$. Therefore,

$$U_h h_{\text{BL}} U_h^{-1} = h_{\text{BL}} \quad (3150)$$

for every element of the residual layer-preserving space group. Equivalently, $[h_{\text{BL}}, U_h] = 0$. Let the common twist center be denoted by o . The stabilizer of this point inside the monolayer crystallographic group is $P_o = \text{Stab}_\Delta(o)$. For a generic high-symmetry center of a regular lattice, the orientation-preserving part of this stabilizer is cyclic. We denote it by $C_m = \langle c \rangle$. The generator satisfies $c^m = e$. Geometrically, the generator is a centered rotation through

$$\varphi_m = \frac{2\pi}{m}. \quad (3151)$$

All orientation-preserving rotations about the same point belong to the compact stabilizer subgroup and commute. Thus, $cg_\phi = g_\phi c$. Since $c\Lambda = \Lambda$, one obtains

$$c\Lambda_+ = cg_+\Lambda = g_+c\Lambda = g_+\Lambda = \Lambda_+. \quad (3152)$$

Similarly, $c\Lambda_- = \Lambda_-$. Therefore, $C_m \subset P_M^+$. A centered rotational point symmetry common to the original monolayer therefore survives an arbitrary centered relative twist. This is a purely geometric consequence of the Abelian structure of the rotation stabilizer about a fixed point. For a square lattice twisted about a common fourfold center, $m = 4$. Therefore, $C_4 \subset P_M^+$ for every centered twist. Thus the generic centered twisted square bilayer retains a common fourfold rotational symmetry even though its mirror symmetries need not remain layer preserving. Suppose the monolayer point group also contains an orientation-reversing symmetry fixing the twist center. Denote such an operation by σ . For a reflection-like operation, $\sigma^2 = e$. Its action on centered rotations reverses their orientation,

$$\sigma g_\phi \sigma^{-1} = g_{-\phi}. \quad (3153)$$

In particular, $\sigma g_+ = g_- \sigma$. Because the reference monolayer is reflection symmetric, $\sigma \Lambda = \Lambda$. Therefore,

$$\sigma \Lambda_+ = \sigma g_+ \Lambda = g_- \sigma \Lambda = g_- \Lambda = \Lambda_-. \quad (3154)$$

Likewise, $\sigma \Lambda_- = \Lambda_+$. Thus a monolayer reflection is generically not a layer-preserving symmetry of the twisted bilayer. Instead, it exchanges the two twisted layers. Because the two layers are identical, the layer exchange can compensate for the spatial interchange. Define $\tilde{\sigma} = \mathcal{P}_L U_\sigma$. The layer operator commutes with the spatial operator, $[\mathcal{P}_L, U_\sigma] = 0$. Therefore, $\tilde{\sigma}^2 = \mathcal{P}_L^2 U_\sigma^2$. Using the two involution properties gives $\tilde{\sigma}^2 = I$. For identical layers with identical microscopic hopping laws,

$$\tilde{\sigma} h_{\text{BL}}(\theta) \tilde{\sigma}^{-1} = h_{\text{BL}}(\theta). \quad (3155)$$

Thus, $[h_{\text{BL}}(\theta), \tilde{\sigma}] = 0$. Twisting therefore converts an ordinary monolayer mirror into a layer-exchanging mirror symmetry rather than necessarily destroying it. The centered rotation generator satisfies $c^m = e$. The layer-exchanging reflection satisfies $\tilde{\sigma}^2 = e$. Its conjugation action on the rotation generator is

$$\begin{aligned} \tilde{\sigma} c \tilde{\sigma}^{-1} &= \mathcal{P}_L U_\sigma U_c U_\sigma^{-1} \mathcal{P}_L^{-1} \\ &= U_{\sigma c \sigma^{-1}} = U_{c^{-1}}. \end{aligned} \quad (3156)$$

Therefore, $\tilde{\sigma} c \tilde{\sigma}^{-1} = c^{-1}$. The group generated by the surviving rotation and the combined layer-exchanging reflection is therefore

$$D_m^{\text{BL}} = \langle c, \tilde{\sigma} \mid c^m = \tilde{\sigma}^2 = e, \quad \tilde{\sigma} c \tilde{\sigma}^{-1} = c^{-1} \rangle. \quad (3157)$$

Abstractly, $D_m^{\text{BL}} \simeq D_m$. Thus a relative twist can preserve the abstract dihedral symmetry while changing the physical realization of its reflection elements from layer-preserving to layer-exchanging operations. The layer-preserving point group generically contains C_m . The full bilayer symmetry group, when layer exchange is admitted as a physical symmetry, contains D_m^{BL} . Thus one must distinguish P_M^+ from the enlarged layer-aware point group $\tilde{P}_M$. Generically,

$$C_m \subseteq P_M^+ \subseteq \tilde{P}_M, \quad (3158)$$

and, when the only additional operations are the combined reflections, $\tilde{P}_M \simeq D_m$. The preceding construction can be formulated without reference to a specific reflection. Let $\varepsilon \in \mathbb{Z}_2$ specify whether a symmetry preserves or exchanges the two layer labels. Define the full bilayer symmetry group as the set

$$\tilde{\Delta}_M = \{(h, \varepsilon) : h \Lambda_\ell = \Lambda_{\varepsilon(\ell)}\}. \quad (3159)$$

The layer-preserving subgroup is the kernel of the layer-action map. Hence,

$$1 \longrightarrow \Delta_M^+ \longrightarrow \tilde{\Delta}_M \longrightarrow E_\theta \longrightarrow 1, \quad (3160)$$

where $E_\theta \leq \mathbb{Z}_2$. If no layer-exchanging spatial symmetry exists, $E_\theta = \{e\}$. If at least one such symmetry exists, $E_\theta = \mathbb{Z}_2$. Consider a layer-polarizing potential,

$$h_V = \frac{V}{2} \mathcal{Z}_L. \quad (3161)$$

Under layer exchange,

$$\mathcal{P}_L \mathcal{Z}_L \mathcal{P}_L^{-1} = -\mathcal{Z}_L. \quad (3162)$$

Therefore,

$$\mathcal{P}_L h_V \mathcal{P}_L^{-1} = -h_V. \quad (3163)$$

The combined layer-exchanging reflection likewise satisfies

$$\tilde{\sigma} h_V \tilde{\sigma}^{-1} = -h_V. \quad (3164)$$

Consequently, $V \neq 0$ breaks every symmetry requiring layer exchange. The common centered rotation can remain intact. Thus the layer bias produces the controlled symmetry reduction $D_m^{\text{BL}} \longrightarrow C_m$. This provides a useful numerical diagnostic for determining whether a spectral degeneracy is protected by the layer-exchanging part of the point group. The preceding point-group derivations assume the microscopic coupling kernels depend only on geodesic distance. Suppose instead that the interlayer tunneling contains an orientation dependent factor,

$$T_{ij} = \mathbf{t}_\perp(D_{ij}, \Omega_{ij}). \quad (3165)$$

A spatial symmetry survives only if

$$\mathbf{t}_\perp(D, \Omega) = \mathbf{t}_\perp(D, h\Omega) \quad (3166)$$

for every relevant point-group operation. Thus isotropic distance-only hopping provides the maximal geometric symmetry baseline. Anisotropic orbital structure may reduce the residual group. We now derive the general degeneracy theorem. Consider a finite exact supercell or finite periodic cover with symmetry group $\mathcal{G}$. Let the Hilbert space carry a unitary representation $U : \mathcal{G} \longrightarrow \mathcal{U}(\mathcal{H})$. Suppose $[H, U_g] = 0$ for every group element. The Hilbert space decomposes into isotypic components,

$$\mathcal{H} = \bigoplus_{\lambda} (V_{\lambda} \otimes M_{\lambda}), \quad (3167)$$

where the representation space of the irreducible representation has $d_{\lambda} = \dim V_{\lambda}$, dimension and the multiplicity space is denoted by M_{λ} . The group representation takes the form

$$U_g = \bigoplus_{\lambda} [D_{\lambda}(g) \otimes I_{M_{\lambda}}]. \quad (3168)$$

Because the Hamiltonian belongs to the commutant of the group representation, Schur's lemma gives

$$H = \bigoplus_{\lambda} [I_{d_{\lambda}} \otimes h_{\lambda}]. \quad (3169)$$

Thus every eigenvalue of a multiplicity-space Hamiltonian is repeated at least d_{λ} times in the full Hilbert space. Therefore,

$$d_{\lambda} > 1 \implies \text{symmetry-enforced degeneracy}. \quad (3170)$$

This degeneracy cannot be lifted by any perturbation preserving the full symmetry group. Let a perturbation satisfy $[\delta H, U_g] = 0$ for every symmetry element. The same Schur decomposition gives

$$\delta H = \bigoplus_{\lambda} [I_{d_{\lambda}} \otimes \delta h_{\lambda}]. \quad (3171)$$

Hence,

$$H + \delta H = \bigoplus_{\lambda} I_{d_{\lambda}} \otimes (h_{\lambda} + \delta h_{\lambda}). \quad (3172)$$

The irreducible-representation factor remains proportional to the identity. Consequently, a multidimensional-irrep degeneracy cannot split without symmetry breaking. The exact hyperbolic moiré translation group is generally non-Abelian. To expose its degeneracy structure rigorously in a finite periodic system, choose a normal finite-index subgroup, $\Gamma_{\text{PBC}} \triangleleft \Gamma_M$. The finite translation group is

$$Q_M = \Gamma_M / \Gamma_{\text{PBC}}. \quad (3173)$$

The periodic-cover Hamiltonian commutes with the regular action of this finite group, $[H, U_q] = 0$ for every quotient-group element. Let an irreducible translation representation satisfy

$$\rho : Q_M \longrightarrow U(d_{\rho}). \quad (3174)$$

If $d_{\rho} > 1$, then the corresponding energy levels occur with at least translation-representation multiplicity d_{ρ} . This type of degeneracy has no analogue in an ordinary Euclidean Bravais translation group, whose irreducible unitary representations are one-dimensional. In the infinite hyperbolic system, a non-Abelian Bloch sector is specified by a unitary irreducible representation

$$\rho : \Gamma_M \longrightarrow U(d_{\rho}). \quad (3175)$$

The generalized Bloch transformation law is matrix valued. It may be written schematically as

$$\Psi(\gamma x) = \rho(\gamma) \Psi(x). \quad (3176)$$

The one-dimensional Abelian Bloch theory corresponds to $d_{\rho} = 1$. Higher-dimensional representations correspond to genuinely non-Abelian translation sectors. For a torsion-free compact hyperbolic quotient of genus $g_M \geq 2$, the translation group admits the standard surface-group presentation

$$\Gamma_M = \left\langle a_1, b_1, \dots, a_{g_M}, b_{g_M} \left| \prod_{j=1}^{g_M} [a_j, b_j] = e \right. \right\rangle. \quad (3177)$$

The commutator is defined by

$$[a, b] = aba^{-1}b^{-1}. \quad (3178)$$

A one-dimensional unitary representation is a homomorphism $\chi : \Gamma_M \longrightarrow U(1)$. Because the target group is Abelian, every group commutator is mapped to unity. Therefore the surface-group relation imposes no additional constraint on the individual generator phases. Define

$$\chi(a_j) = e^{ik_{a_j}}, \quad \chi(b_j) = e^{ik_{b_j}}. \quad (3179)$$

The Abelian hyperbolic crystal momentum is therefore the vector

$$\mathbf{k} = (k_{a_1}, k_{b_1}, \dots, k_{a_{g_M}}, k_{b_{g_M}}). \quad (3180)$$

Each component is periodic, $k_{\mu} \sim k_{\mu} + 2\pi$. Thus the Abelian generalized Brillouin zone is $\mathbb{T}^{2g_M}$. Time reversal complex conjugates a one-dimensional character, $\chi \mapsto \chi^*$. For the phase variables, $e^{ik_{\mu}} \mapsto e^{-ik_{\mu}}$. Therefore, $\Theta : \mathbf{k} \mapsto -\mathbf{k}$. The band energies consequently satisfy

$$E_n(\mathbf{k}) = E_n(-\mathbf{k}). \quad (3181)$$

This equality is a time-reversal pairing relation. It does not imply a twofold degeneracy at the same generalized

momentum unless the two momentum points are identical. A generalized momentum is time-reversal invariant when

$$\mathbf{k} \equiv -\mathbf{k} \pmod{2\pi}. \quad (3182)$$

For every component this requires $2k_\mu = 0 \pmod{2\pi}$. Hence, $k_\mu \in \{0, \pi\}$. There are $N_{\text{TRIM}} = 2^{2g_M}$ Abelian time-reversal-invariant generalized momentum points. For the spinless model, ordinary time reversal does not enforce Kramers degeneracy at these points because $\Theta^2 = +1$. For a non-Abelian unitary representation, time reversal acts by complex conjugation, $\rho \mapsto \rho^*$. Thus the energy spectra satisfy

$$\text{Spec } h_\rho = \text{Spec } h_{\rho^*}. \quad (3183)$$

If $\rho \not\simeq \rho^*$, the two representations label distinct translation sectors. Time reversal then produces symmetry-related spectral pairing between those sectors. If $\rho \simeq \rho^*$, the representation sector is time-reversal invariant. For the baseline spinless model, self-conjugacy alone does not imply Kramers doubling. Because the moiré translation group is normal in the residual space group, every residual point-group operation induces an automorphism of the translation group. Let a point-group element be represented by a space-group lift $s_p \in \Delta_M^+$. Its action on translations is

$$\alpha_p(\gamma) = s_p^{-1} \gamma s_p. \quad (3184)$$

Given a translation irrep, define its transformed representation by

$$\rho^p(\gamma) = \rho[\alpha_p(\gamma)]. \quad (3185)$$

Thus the point group acts on the representation space of generalized hyperbolic momenta. The spatial symmetry operator maps one translation sector to another. Consequently, $U_p h_\rho U_p^{-1} = h_{\rho^p}$. Therefore, $\text{Spec } h_\rho = \text{Spec } h_{\rho^p}$. The orbit

$$\text{Star}(\rho) = \{\rho^p : p \in P_M^+\} \quad (3186)$$

is the non-Abelian hyperbolic analogue of a momentum star in ordinary band theory. Distinct sectors in the same star possess identical symmetry-related spectra. The Abelianization of the surface group is $\Gamma_M^{\text{ab}} \simeq \mathbb{Z}^{2g_M}$. Let the homology coordinate vector of a group element be denoted by $\mathbf{n}(\gamma) \in \mathbb{Z}^{2g_M}$. A Bloch character can then be written as

$$\chi_{\mathbf{k}}(\gamma) = \exp[i\mathbf{k}^T \mathbf{n}(\gamma)]. \quad (3187)$$

The point-group automorphism induces an integer matrix satisfying

$$\mathbf{n}[\alpha_p(\gamma)] = M_p \mathbf{n}(\gamma). \quad (3188)$$

Therefore,

$$\chi_{\mathbf{k}}^p(\gamma) = \chi_{\mathbf{k}}[\alpha_p(\gamma)] = \exp[i\mathbf{k}^T M_p \mathbf{n}(\gamma)]. \quad (3189)$$

Thus the transformed generalized momentum is

$$\mathbf{k} \mapsto M_p^T \mathbf{k} \pmod{2\pi}. \quad (3190)$$

For an orientation-preserving automorphism of the compact quotient, the induced homology matrix preserves the intersection form. Denoting that form by

$$J = \begin{pmatrix} 0 & I_{g_M} \\ -I_{g_M} & 0 \end{pmatrix}, \quad (3191)$$

one has $M_p^T J M_p = J$. Thus the orientation-preserving point group acts through an integral symplectic representation on the Abelian hyperbolic Brillouin zone. A generalized Abelian momentum is invariant under a point-group operation when

$$M_p^T \mathbf{k} \equiv \mathbf{k} \pmod{2\pi}. \quad (3192)$$

The little point group of the momentum is

$$P_{\mathbf{k}} = \{p \in P_M^+ : M_p^T \mathbf{k} \equiv \mathbf{k}\}. \quad (3193)$$

At such a generalized momentum, the little-group operators act within the same Bloch sector and commute with the Bloch Hamiltonian. Therefore,

$$[h(\mathbf{k}), U_p(\mathbf{k})] = 0 \quad (3194)$$

for every little-group element. For a general translation irrep, the little point group is defined by representation equivalence. Thus,

$$P_\rho = \{p \in P_M^+ : \rho^p \simeq \rho\}. \quad (3195)$$

For every little-group element, there exists an intertwining unitary satisfying

$$W_p \rho(\gamma) W_p^{-1} = \rho(s_p^{-1} \gamma s_p). \quad (3196)$$

The little-group representation carried by the Bloch states determines additional symmetry-enforced degeneracies beyond those generated by the translation irrep itself. The extension

$$1 \longrightarrow \Gamma_M \longrightarrow \Delta_M^+ \longrightarrow P_M^+ \longrightarrow 1 \quad (3197)$$

need not split. Choose a set-theoretic section $s : P_M^+ \rightarrow \Delta_M^+$. Products of lifts may differ from the lift of the product by a translation,

$$s(p)s(q) = \gamma(p, q)s(pq), \quad (3198)$$

where $\gamma(p, q) \in \Gamma_M$. For an Abelian Bloch character, the translation contributes a phase. Thus the little-group operators obey

$$U_p(\mathbf{k})U_q(\mathbf{k}) = \omega_{\mathbf{k}}(p, q)U_{pq}(\mathbf{k}), \quad (3199)$$

with factor system

$$\omega_{\mathbf{k}}(p, q) = \chi_{\mathbf{k}}[\gamma(p, q)]. \quad (3200)$$

Associativity of the space-group product imposes

$$[U_p U_q] U_r = U_p [U_q U_r]. \quad (3201)$$

Using the projective multiplication law gives

$$\omega_{\mathbf{k}}(p, q)\omega_{\mathbf{k}}(pq, r)U_{pqr} = \omega_{\mathbf{k}}(q, r)\omega_{\mathbf{k}}(p, qr)U_{pqr}. \quad (3202)$$

Therefore,

$$\omega_{\mathbf{k}}(p, q)\omega_{\mathbf{k}}(pq, r) = \omega_{\mathbf{k}}(q, r)\omega_{\mathbf{k}}(p, qr). \quad (3203)$$

The factor system is thus a two-cocycle. A nontrivial factor system can force multidimensional projective irreducible representations even when the ordinary point group has only one-dimensional irreducible representations. Suppose the projective little group at a generalized momentum possesses no one-dimensional representation compatible with its factor system. Then every allowed projective representation has dimension satisfying $d_{\text{proj}} > 1$. Schur's lemma then implies a symmetry-enforced degeneracy of at least d_{proj} . Thus nonsplit hyperbolic space-group structure can generate a non-Euclidean analogue of nonsymmorphic band sticking. The ordinary cyclic group has only one-dimensional complex irreducible representations. For $C_m = \langle c \rangle$, the irreducible representations are labeled by an integer satisfying $j = 0, 1, \dots, m-1$. The rotation eigenvalue is

$$\lambda_j = \exp\left(\frac{2\pi i j}{m}\right). \quad (3204)$$

Since $\dim D_j = 1$, the centered rotation group alone does not force a local spectral degeneracy. It can, however, protect crossings between bands carrying different rotation eigenvalues. The generalized bilayer dihedral group can possess two-dimensional irreducible representations. For an appropriate integer label, define

$$\varphi_j = \frac{2\pi j}{m}. \quad (3205)$$

A two-dimensional representation may be written as

$$D_j(c) = \begin{pmatrix} e^{i\varphi_j} & 0 \\ 0 & e^{-i\varphi_j} \end{pmatrix}, \quad (3206)$$

and

$$D_j(\tilde{\sigma}) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (3207)$$

The dihedral conjugation relation is verified explicitly,

$$D_j(\tilde{\sigma}) D_j(c) D_j^{-1}(\tilde{\sigma}) = D_j(c^{-1}). \quad (3208)$$

For representations that are genuinely two dimensional, $\dim D_j = 2$. Therefore every state transforming in such a representation belongs to a symmetry-enforced doublet. For the centered square bilayer, $m = 4$. The bilayer point group contains D_4^{BL} . Its two-dimensional representation with $j = 1$ has rotation matrix

$$D(c_4) = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}. \quad (3209)$$

Therefore high-symmetry sectors transforming according to this two-dimensional representation possess an exact double degeneracy. A layer bias breaks the layer-exchanging reflection and reduces the group to C_4 . The two-dimensional dihedral representation then decomposes into two one-dimensional cyclic representations,

$$D_E \downarrow C_4 = D_{+i} \oplus D_{-i}. \quad (3210)$$

The degeneracy can therefore split once the layer-exchanging symmetry is broken. This provides a direct symmetry diagnostic. Time reversal complex conjugates a rotation eigenvalue. Thus,

$$\lambda_j \mapsto \lambda_j^* = \lambda_{-j}. \quad (3211)$$

At a time-reversal-invariant momentum sector, states with complex-conjugate rotation eigenvalues can therefore be related by time reversal. For spinless time reversal, such pairing does not automatically imply Kramers degeneracy unless the two states belong to distinct symmetry sectors that cannot hybridize. The distinction is time-reversal spectral pairing $\neq$ Kramers degeneracy. More general antiunitary operations can be constructed by combining time reversal with a spatial symmetry. Define $\mathcal{A}_s = U_s \Theta$. Its square is

$$\mathcal{A}_s^2 = U_s \Theta U_s \Theta^{-1} \Theta^2. \quad (3212)$$

For a spinless real-space symmetry represented by a real permutation matrix,

$$\Theta U_s \Theta^{-1} = U_s. \quad (3213)$$

Thus, $\mathcal{A}_s^2 = U_s^2$. Suppose the spatial symmetry squares to a translation, $s^2 = \gamma_s$, with $\gamma_s \in \Gamma_M$. Inside an Abelian Bloch sector, $U_{\gamma_s} = \chi_{\mathbf{k}}(\gamma_s)$. Therefore, $\mathcal{A}_s^2 = \chi_{\mathbf{k}}(\gamma_s)$. If $\chi_{\mathbf{k}}(\gamma_s) = -1$, then $\mathcal{A}_s^2 = -1$. Kramers' orthogonality argument then applies even though the microscopic time-reversal operator itself squares to positive unity. Let $|\psi\rangle$ be an eigenstate. Then

$$\langle \psi | \mathcal{A}_s \psi \rangle = - \langle \psi | \mathcal{A}_s \psi \rangle. \quad (3214)$$

Therefore, $\langle \psi | \mathcal{A}_s \psi \rangle = 0$. The corresponding energy level is at least doubly degenerate. Thus,

$$\mathcal{A}_s^2 = -1 \implies \text{symmetry-enforced doublet}. \quad (3215)$$

For the simple combined mirror constructed above, $\tilde{\sigma}^2 = +1$. Therefore this symmetry alone does not enforce Kramers-like doubling. Its importance instead lies in enlarging the point group from a cyclic group to a dihedral group and in protecting crossings between different mirror sectors. Let a unitary symmetry satisfy $[H, Q] = 0$. Suppose two states carry different symmetry eigenvalues,

$$Q |u_a\rangle = q_a |u_a\rangle, \quad Q |u_b\rangle = q_b |u_b\rangle. \quad (3216)$$

Let a perturbation preserve the symmetry, $[\delta H, Q] = 0$. The off-diagonal matrix element satisfies

$$\begin{aligned} \langle u_a | \delta H | u_b \rangle &= \langle u_a | Q^{-1} Q \delta H Q^{-1} Q | u_b \rangle \\ &= q_a^* q_b \langle u_a | \delta H | u_b \rangle. \end{aligned} \quad (3217)$$

If $q_a^* q_b \neq 1$, then

$$\langle u_a | \delta H | u_b \rangle = 0. \quad (3218)$$

Thus states belonging to inequivalent one-dimensional symmetry sectors cannot hybridize. If their energies cross as an external parameter is varied, the crossing cannot be gapped without breaking the protecting symmetry. A generic two-band Hermitian Hamiltonian can be written as

$$h_{\text{eff}} = \varepsilon_0 I + d_x \sigma_x + d_y \sigma_y + d_z \sigma_z. \quad (3219)$$

Its eigenvalues are

$$E_{\pm} = \varepsilon_0 \pm \sqrt{d_x^2 + d_y^2 + d_z^2}. \quad (3220)$$

A degeneracy requires $d_x = d_y = d_z = 0$. Thus a generic complex Hermitian two-band degeneracy has codimension 3. Suppose the two states have opposite eigenvalues of a unitary symmetry that can be represented by $Q = \sigma_z$. Commutation requires $[h_{\text{eff}}, \sigma_z] = 0$. Using the Pauli algebra gives $d_x = d_y = 0$. The Hamiltonian reduces to

$$h_{\text{eff}} = \varepsilon_0 I + d_z \sigma_z. \quad (3221)$$

A crossing now requires only $d_z = 0$. Thus the codimension is reduced from three to one. A single tuning parameter can therefore generate a stable symmetry-protected crossing. Suppose an antiunitary symmetry satisfies $\mathcal{A}^2 = +1$. In an appropriate basis it can be represented as complex conjugation. Then $h_{\text{eff}} = h_{\text{eff}}^*$. Because the Pauli matrix multiplying the imaginary coefficient is excluded, one obtains $d_y = 0$. A degeneracy therefore requires only $d_x = d_z = 0$. The codimension is reduced to 2. Such a real structure can stabilize isolated crossings in a two-dimensional parameter space. The Abelian hyperbolic Brillouin zone has dimension $d_{\text{BZ}} = 2g_M$. For a generic two-band Hermitian crossing of codimension three, the expected dimension of a transverse degeneracy manifold is $d_{\text{node}} = 2g_M - 3$. Thus high-dimensional hyperbolic reciprocal space can support degeneracy manifolds even in the absence of point-group protection. Such degeneracies must not be confused with symmetry-enforced degeneracies. Their existence follows from parameter-space codimension rather than from a multidimensional irreducible representation. Suppose a generalized momentum or non-Abelian translation sector has little group P_ρ . If the Bloch state transforms under an irreducible little-group representation of dimension $d_\lambda > 1$, then Schur's lemma implies degeneracy $\geq d_\lambda$. If the translation representation itself is also multidimensional, the minimal representation-theoretic multiplicity can be enhanced. In the simplest direct-product situation, $d_{\text{min}} \geq d_\rho d_\lambda$. For a nonsplit extension, the appropriate quantity is instead the dimension of the relevant irreducible representation or corepresentation of the full little space group. Two different types of equal-energy states must be distinguished. If two translation sectors are related by a point-group operation but are distinct, $\rho^p \neq \rho$, then

$$\text{Spec } h_\rho = \text{Spec } h_{\rho^p}. \quad (3222)$$

This is a symmetry-related star degeneracy in the complete finite system. It is analogous to the equality of energies at symmetry-related momenta. By contrast, if $\rho^p \simeq \rho$, the point-group operation acts within the same generalized momentum sector. A multidimensional irreducible representation of that little group then produces

a local symmetry-enforced degeneracy. Thus, star pairing $\neq$ little-group enforced degeneracy. A further distinction arises only in hyperbolic crystals. A high-dimensional non-Abelian translation irrep can itself enforce $d_\rho > 1$. This degeneracy exists before point-group symmetry is considered. Point-group operations can subsequently enlarge or reorganize the multiplet. Therefore, non-Abelian translation degeneracy and point-group degeneracy are mathematically independent mechanisms. Suppose layer exchange is an exact symmetry. Its irreducible representations are one dimensional, $p_L = \pm 1$. Therefore, $\mathcal{P}_L$ alone does not enforce a degeneracy. It can protect a crossing between opposite layer-parity sectors because the two sectors cannot hybridize, but the energies of the two sectors need not be equal. This distinction is particularly important at zero twist. The combined mirror has eigenvalues $\tilde{\sigma} = +1$ or $\tilde{\sigma} = -1$. Along a generalized momentum subspace invariant under the combined mirror, states carrying opposite mirror eigenvalues cannot hybridize. Therefore, if their dispersions intersect, $E_+ = E_-$, the crossing is protected as long as $[H, \tilde{\sigma}] = 0$. A layer bias breaks this condition and permits an avoided crossing. Similarly, if two bands carry different rotation eigenvalues, $\lambda_j \neq \lambda_{j'}$, then a rotation-preserving perturbation satisfies $\langle j | \delta H | j' \rangle = 0$. A crossing between those sectors cannot acquire an off-diagonal mass without breaking the rotational symmetry. We therefore distinguish two precise notions. A degeneracy is symmetry enforced when representation theory requires it for every Hamiltonian respecting the symmetry. This occurs when $d_\lambda > 1$, or when an antiunitary symmetry satisfies $\mathcal{A}^2 = -1$. A crossing is symmetry protected when the equality of two energies is not required by symmetry but hybridization between the corresponding sectors is forbidden. The latter condition has the form

$$\langle u_a | \delta H | u_b \rangle = 0 \quad (3223)$$

for every perturbation preserving the relevant symmetry. Thus, symmetry enforced $\neq$ symmetry protected. Suppose two degenerate states transform according to the same one-dimensional representation and no antiunitary symmetry squares to negative unity. Then symmetry permits an off-diagonal matrix element. A generic symmetry-preserving perturbation can therefore split the degeneracy. Such a degeneracy is accidental. In the two-band description, it corresponds to a solution of

$$d_x = d_y = d_z = 0 \quad (3224)$$

without an additional group-theoretic reason forcing those conditions. The classification above suggests a direct numerical test. Suppose a degeneracy occurs at energy E_* . Apply a small perturbation preserving all candidate protecting symmetries, δH_{sym} . If the degeneracy is representation enforced, $\Delta E_{\text{sym}} = 0$ to all orders consistent with the exact symmetry. Now introduce a perturbation that breaks one candidate symmetry, δH_{br} . If the

splitting behaves as $\Delta E_{\text{br}} \neq 0$, the broken symmetry is implicated in protecting the degeneracy. This procedure will be used later to distinguish magic-band flattening from symmetry-enforced band sticking. Let a degenerate doublet span the subspace

$$\mathcal{D} = \text{span} \{ |u_1\rangle, |u_2\rangle \}. \quad (3225)$$

Project a symmetry-breaking perturbation into this subspace,

$$V_{\text{eff}} = P_{\mathcal{D}} \delta H P_{\mathcal{D}}. \quad (3226)$$

Write

$$V_{\text{eff}} = v_0 I + v_x \sigma_x + v_y \sigma_y + v_z \sigma_z. \quad (3227)$$

The first-order splitting is

$$\Delta E = 2\sqrt{v_x^2 + v_y^2 + v_z^2}. \quad (3228)$$

If the perturbation preserves the multidimensional irrep symmetry, Schur's lemma imposes $v_x = v_y = v_z = 0$. Only the common shift remains, $V_{\text{eff}} = v_0 I$. This gives an explicit perturbative form of symmetry protection. At an exact commensurate twist, the Hamiltonian possesses the non-Abelian translation symmetry Γ_M . It also possesses the layer-preserving residual space group Δ_M^+ , and, when layer-exchanging operations exist, the enlarged group $\tilde{\Delta}_M$. The exact symmetry hierarchy is therefore

$$\Gamma_M \triangleleft \Delta_M^+ \subseteq \tilde{\Delta}_M. \quad (3229)$$

Time reversal extends the unitary symmetry group to a magnetic corepresentation structure,

$$\mathcal{G}_{\text{mag}} = \tilde{\Delta}_M + \Theta \tilde{\Delta}_M. \quad (3230)$$

For a generic incommensurate twist,

$$[\Gamma : \Gamma \cap g_\theta \Gamma g_\theta^{-1}] = \infty. \quad (3231)$$

There is therefore no finite-index exact moiré translation group. No finite exact moiré Brillouin zone exists. However, finite centered point symmetries may survive. In particular, when the twist is centered on an order- m monolayer rotation center, C_m remains layer preserving. For identical layers with reflection symmetry, the combined layer-exchanging reflection also remains. Therefore the finite bilayer point group can remain D_m^{BL} even when no exact translational supercell exists. Thus, point-group symmetry $\nRightarrow$ exact translational commensurability. Conversely, exact commensurability

does not imply that every monolayer point-group symmetry survives. Exact translation commensurability requires $g_\theta \in \text{Comm}(\Gamma)$. Residual point-group symmetry requires the additional intersection $\Delta_- \cap \Delta_+$. These are distinct group-theoretic conditions. Therefore, commensurability $\neq$ maximal point-group symmetry. The possible degeneracy mechanisms can now be classified. A non-Abelian translation irrep produces d_ρ -fold translation degeneracy. A multidimensional little-point-group irrep produces d_λ -fold point-group degeneracy. A nontrivial projective little-group representation produces d_{proj} -fold nonsymmorphic degeneracy. An antiunitary symmetry satisfying $\mathcal{A}^2 = -1$ produces Kramers-like double degeneracy. Ordinary spinless time reversal produces only $\rho \leftrightarrow \rho^*$ or $\mathbf{k} \leftrightarrow -\mathbf{k}$ spectral pairing. Layer exchange by itself produces sector decomposition, $p_L = \pm 1$, but not an enforced equality of the two sector energies. Different unitary symmetry eigenvalues can protect a crossing but do not force it to occur. The preceding results can be summarized formally.

a. Theorem 18. Symmetry-enforced multiplicity.

Let an exact finite-cover bilayer Hamiltonian commute with a unitary symmetry group. Suppose an energy eigenspace transforms according to an irreducible representation of dimension d_λ . Then the corresponding energy level has multiplicity satisfying $\text{mult}(E) \geq d_\lambda$. The degeneracy remains under every perturbation preserving the symmetry. If the symmetry representation arises from a non-Abelian translation sector, the same statement applies with $d_\lambda = d_\rho$. If an additional little-group representation acts independently, the multiplicity may be further enhanced.

b. Theorem 19. Antiunitary doublet criterion. Let an antiunitary symmetry commute with the Hamiltonian, $\mathcal{A}H\mathcal{A}^{-1} = H$. Suppose that within a symmetry-invariant sector it satisfies $\mathcal{A}^2 = -1$. Then every eigenstate in that sector has an orthogonal partner at the same energy. Hence, $\text{mult}(E) \geq 2$. Ordinary spinless time reversal does not satisfy the hypothesis, but a combined antiunitary spatial symmetry can satisfy it when its square produces a translation represented by negative unity.

c. Theorem 20. Protection of crossings between inequivalent symmetry sectors. Let two bands belong to inequivalent one-dimensional irreducible representations of a unitary symmetry. Then every perturbation preserving that symmetry has vanishing inter-sector matrix element,

$$\langle u_a | \delta H | u_b \rangle = 0. \quad (3232)$$

Consequently, if the two dispersions become equal at a point or manifold, $E_a = E_b$, the crossing cannot be gapped without either breaking the symmetry or moving the crossing away from the considered parameter point. Thus the crossing is symmetry protected but not necessarily symmetry enforced.

d. Theorem 21. Twist-induced realization of bilayer dihedral symmetry. Suppose the monolayer possesses a

centered point symmetry generated by an order- m rotation and a reflection. For a symmetric relative twist of two identical layers, the centered rotation preserves each layer individually. The reflection exchanges the two layers. Therefore the combined layer-exchanging reflection and the centered rotation generate $D_m^{\text{BL}} \simeq D_m$. The defining relations are $c^m = e$, $\tilde{\sigma}^2 = e$, and $\tilde{\sigma}c\tilde{\sigma}^{-1} = c^{-1}$. Thus twisting does not necessarily destroy the abstract dihedral symmetry. Instead, it converts orientation-reversing monolayer operations into layer-exchanging bilayer operations.

e. Theorem 22. Symmetry structure of the curvature-dependent twisted bilayer. Consider two identical rigid layers with isotropic distance-dependent hopping on a constant-curvature lattice. The minimal spinless bilayer Hamiltonian possesses charge conservation, $U(1)_{\text{charge}}$, and spinless time reversal, $\Theta^2 = +1$. At an exact commensurate twist, the maximal common translation symmetry is $\Gamma_M = \Gamma_- \cap \Gamma_+$, while the layer-preserving residual space group is

$$\Delta_M^+ = \Delta_- \cap \Delta_+. \quad (3233)$$

These groups satisfy $\Gamma_M \triangleleft \Delta_M^+$. The residual point group is

$$P_M^+ = \Delta_M^+ / \Gamma_M. \quad (3234)$$

If the twist center possesses an order- m rotational symmetry, the rotation subgroup survives every centered twist, $C_m \subseteq P_M^+$. If the monolayer also possesses an orientation-reversing symmetry fixing the twist center and the two layers are physically identical, the full layer-aware point group contains D_m^{BL} . A layer bias reduces this symmetry according to $D_m^{\text{BL}} \rightarrow C_m$. The exact Hamiltonian decomposes into irreducible symmetry sectors. Multidimensional unitary irreducible representations enforce degeneracies. Combined antiunitary symmetries whose square equals negative unity enforce doublets. Different one-dimensional symmetry sectors can cross without hybridizing, producing symmetry-protected crossings. Ordinary spinless time reversal and ordinary layer exchange alone do not force local twofold degeneracy. The complete degeneracy hierarchy may be written as follows. Non-Abelian translation symmetry gives

$$\Gamma_M \rightarrow d_\rho > 1 \rightarrow \text{translation-enforced multiplet.} \quad (3235)$$

Residual point-group symmetry gives

$$P_\rho \rightarrow d_\lambda > 1 \rightarrow \text{little-group-enforced multiplet.} \quad (3236)$$

Nonsplit space-group structure gives

$$\omega_\rho \neq 1 \rightarrow d_{\text{proj}} > 1 \rightarrow \text{projective band sticking.} \quad (3237)$$

A Kramers-type antiunitary condition gives

$$\mathcal{A}^2 = -1 \longrightarrow \text{symmetry-enforced doublet.} \quad (3238)$$

Different one-dimensional symmetry sectors give

$$q_a \neq q_b \longrightarrow \langle u_a | \delta H | u_b \rangle = 0 \longrightarrow \text{protected crossing.} \quad (3239)$$

By contrast, $\Theta^2 = +1$ produces spectral pairing but not ordinary Kramers doubling, and $\mathcal{P}_L^2 = +1$ produces layer-parity sectors but not an enforced degeneracy. The symmetry structure derived above imposes an important constraint on the interpretation of flat bands and band crossings. A narrow band may result from genuine suppression of kinetic dispersion. It may also appear narrow because a symmetry-enforced multiplet has been folded into an exact supercell. Similarly, a zero group velocity near a degeneracy can be generated by a symmetry-protected crossing rather than by a magic-angle flattening mechanism. Therefore every candidate magic state must be tested against the symmetry classification established here. The subsequent spectral analysis must distinguish symmetry-enforced degeneracy, symmetry-protected crossing, band-folding multiplicity and genuine curvature-controlled bandwidth suppression. A robust magic-band criterion must remain meaningful after the symmetry-enforced multiplicities have been resolved into irreducible sectors. Accordingly, the bandwidth used in the later magic-band analysis should be defined within a fixed irreducible representation sector whenever such a decomposition is available. The central causal structure for the remainder of the paper therefore contains two conceptually distinct but complementary pathways. The first pathway originates from the underlying geometry, which determines the residual symmetry group of the twisted bilayer. This residual symmetry structure fixes the relevant irreducible-representation content of the Hilbert space and thereby imposes symmetry-enforced constraints on the spectrum, including protected multiplets, symmetry-related spectral pairing, and symmetry-protected crossings. These spectral structures arise from representation theory and must therefore be identified independently of any genuine suppression of kinetic dispersion. The second pathway also originates from the geometry but proceeds through the microscopic reconstruction of the interlayer tunneling matrix. The curvature-dependent registry field modifies the spatial distribution and strength of interlayer hybridization, which in turn renormalizes the effective dispersion and can ultimately generate magic-band flattening. This mechanism is physically distinct from symmetry-enforced degeneracy because it operates through the modification of the kinetic and hybridization structure of the Hamiltonian rather than through representation-theoretic multiplicity. Only after these two causal pathways have been separated can a flat, nearly flat, or degenerate spectral feature be attributed unambiguously to a curvature-controlled magic-

band mechanism rather than to symmetry alone. In particular, all subsequent bandwidth, velocity, and flatness diagnostics must be evaluated after resolving the Hamiltonian into the appropriate irreducible symmetry sectors, so that symmetry-enforced multiplicities are not misidentified as genuine dispersion suppression.

D. Euclidean Bloch Theory and the Moiré Brillouin Zone

The hyperbolic bilayer theory developed in the preceding sections must be anchored to a complete Euclidean benchmark. The purpose of the present subsection is therefore not merely to reproduce the familiar square-lattice dispersion. Instead, we construct the full reciprocal-space hierarchy of an exactly commensurate twisted square bilayer and identify separately the roles of primitive Bloch periodicity, exact moiré periodicity, band folding, and interlayer hybridization. This distinction is essential for the later analysis of magic flat bands. A reduction of the Brillouin-zone area automatically produces a larger number of folded bands, but folding alone is a relabeling of the single-layer spectrum and cannot reduce the intrinsic dispersion of an isolated band. Genuine spectral reconstruction arises only when the interlayer coupling mixes the folded states. The Euclidean analysis below therefore establishes the reference structure primitive lattice $\longrightarrow$ primitive reciprocal lattice $\longrightarrow$ exact coincidence lattice $\longrightarrow$ moiré reciprocal lattice $\longrightarrow$ band folding $\longrightarrow$ interlayer hybridization. The reference square lattice is $\Lambda = a\mathbb{Z}^2$. A primitive direct-lattice basis is

$$\mathbf{a}_1 = a \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{a}_2 = a \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (3240)$$

Introduce the direct-basis matrix

$$A = (\mathbf{a}_1 \ \mathbf{a}_2) = aI_2. \quad (3241)$$

Its determinant is $\det A = a^2$. Thus the primitive-cell area is $\mathcal{A}_0 = a^2$. The reciprocal basis is defined by

$$\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi\delta_{ij}. \quad (3242)$$

In matrix form, $B = 2\pi A^{-T}$. For the square lattice,

$$A^{-T} = \frac{1}{a} I_2. \quad (3243)$$

Hence,

$$B = \frac{2\pi}{a} I_2. \quad (3244)$$

The primitive reciprocal vectors are therefore

$$\mathbf{b}_1 = \frac{2\pi}{a} \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{b}_2 = \frac{2\pi}{a} \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (3245)$$

The primitive reciprocal lattice is

$$\Lambda^* = \{n_1 \mathbf{b}_1 + n_2 \mathbf{b}_2 : n_1, n_2 \in \mathbb{Z}\}. \quad (3246)$$

Its primitive-cell area is

$$\mathcal{A}_{\text{BZ}} = |\det B| = \left(\frac{2\pi}{a}\right)^2. \quad (3247)$$

This is consistent with the general duality relation $\mathcal{A}_0 \mathcal{A}_{\text{BZ}} = (2\pi)^2$. A convenient primitive Brillouin zone is

$$\text{BZ}_0 = \left\{ \mathbf{k} : -\frac{\pi}{a} \leq k_x < \frac{\pi}{a}, \quad -\frac{\pi}{a} \leq k_y < \frac{\pi}{a} \right\}. \quad (3248)$$

Equivalent momenta satisfy

$$\mathbf{k} \sim \mathbf{k} + \mathbf{G}, \quad \mathbf{G} \in \Lambda^*. \quad (3249)$$

For one isolated square layer, the nearest-neighbor tight-binding Hamiltonian is

$$\hat{H}_0 = -t \sum_{\mathbf{R}} \left[c_{\mathbf{R}}^\dagger c_{\mathbf{R}+\mathbf{a}_1} + c_{\mathbf{R}}^\dagger c_{\mathbf{R}+\mathbf{a}_2} + \text{h.c.} \right]. \quad (3250)$$

The Fourier transform is defined by

$$c_{\mathbf{R}} = \frac{1}{\sqrt{N}} \sum_{\mathbf{k} \in \text{BZ}_0} e^{i\mathbf{k} \cdot \mathbf{R}} c_{\mathbf{k}}. \quad (3251)$$

The inverse transformation is

$$c_{\mathbf{k}} = \frac{1}{\sqrt{N}} \sum_{\mathbf{R}} e^{-i\mathbf{k} \cdot \mathbf{R}} c_{\mathbf{R}}. \quad (3252)$$

Substituting into the hopping term gives

$$\sum_{\mathbf{R}} c_{\mathbf{R}}^\dagger c_{\mathbf{R}+\mathbf{a}_1} = \frac{1}{N} \sum_{\mathbf{R}} \sum_{\mathbf{k}, \mathbf{k}'} e^{-i\mathbf{k} \cdot \mathbf{R}} e^{i\mathbf{k}' \cdot (\mathbf{R}+\mathbf{a}_1)} c_{\mathbf{k}}^\dagger c_{\mathbf{k}'}. \quad (3253)$$

The lattice sum produces

$$\sum_{\mathbf{R}} e^{i(\mathbf{k}' - \mathbf{k}) \cdot \mathbf{R}} = N \delta_{\mathbf{k}, \mathbf{k}'} \quad (3254)$$

up to reciprocal-lattice equivalence. Therefore,

$$\sum_{\mathbf{R}} c_{\mathbf{R}}^\dagger c_{\mathbf{R}+\mathbf{a}_1} = \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{a}_1} c_{\mathbf{k}}^\dagger c_{\mathbf{k}}. \quad (3255)$$

Similarly,

$$\sum_{\mathbf{R}} c_{\mathbf{R}}^\dagger c_{\mathbf{R}+\mathbf{a}_2} = \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{a}_2} c_{\mathbf{k}}^\dagger c_{\mathbf{k}}. \quad (3256)$$

Including the Hermitian conjugates gives

$$\hat{H}_0 = \sum_{\mathbf{k}} \varepsilon_0(\mathbf{k}) c_{\mathbf{k}}^\dagger c_{\mathbf{k}}. \quad (3257)$$

The exact nearest-neighbor dispersion is

$$\varepsilon_0(\mathbf{k}) = -2t [\cos(k_x a) + \cos(k_y a)]. \quad (3258)$$

At the zone center,

$$\mathbf{k}_\Gamma = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad (3259)$$

the energy is

$$\varepsilon_0(\mathbf{k}_\Gamma) = -4t. \quad (3260)$$

At the primitive zone corner,

$$\mathbf{k}_M = \begin{pmatrix} \pi/a \\ \pi/a \end{pmatrix}, \quad (3261)$$

one obtains

$$\varepsilon_0(\mathbf{k}_M) = +4t. \quad (3262)$$

Thus the nearest-neighbor monolayer bandwidth is $W_0 = 8|t|$. This quantity provides the natural Euclidean kinetic-energy benchmark for the later flat-band analysis. The primitive group velocity is

$$\mathbf{v}_0(\mathbf{k}) = \frac{1}{\hbar} \nabla_{\mathbf{k}} \varepsilon_0(\mathbf{k}). \quad (3263)$$

Differentiation gives

$$\mathbf{v}_0(\mathbf{k}) = \frac{2ta}{\hbar} \begin{pmatrix} \sin(k_x a) \\ \sin(k_y a) \end{pmatrix}. \quad (3264)$$

This primitive velocity will later be compared with the hybridization-renormalized moiré velocity. The first layer is chosen as the reference lattice. Its primitive reciprocal vectors are $\mathbf{b}_i^{(1)} = \mathbf{b}_i$. The second layer is rotated

through the relative angle. Its direct primitive vectors are $\mathbf{a}_i^{(2)} = R_\theta \mathbf{a}_i$. Because rotations preserve the reciprocal duality relation, its reciprocal vectors are $\mathbf{b}_i^{(2)} = R_\theta \mathbf{b}_i$. Thus the second reciprocal lattice is $\Lambda_2^* = R_\theta \Lambda^*$. Let the laboratory-frame momentum of the second layer be denoted by $\mathbf{q}$. The phase accumulated along a rotated primitive vector is

$$\mathbf{q} \cdot R_\theta \mathbf{a}_i = (R_{-\theta} \mathbf{q}) \cdot \mathbf{a}_i. \quad (3265)$$

Hence the second-layer dispersion is

$$\varepsilon_2(\mathbf{q}) = \varepsilon_0(R_{-\theta} \mathbf{q}). \quad (3266)$$

Explicitly,

$$\varepsilon_2(\mathbf{q}) = -2t \left[\cos(a[R_{-\theta} \mathbf{q}]_x) + \cos(a[R_{-\theta} \mathbf{q}]_y) \right]. \quad (3267)$$

The isolated spectra of the two layers are identical. They differ only by their orientation in laboratory momentum space. For an exactly commensurate twist, the common real-space lattice is

$$\Lambda_M = \Lambda_1 \cap \Lambda_2. \quad (3268)$$

The coincidence index is

$$\Sigma = [\Lambda_1 : \Lambda_M] = [\Lambda_2 : \Lambda_M]. \quad (3269)$$

Using the integer parameterization derived previously,

$$\tan\left(\frac{\theta}{2}\right) = \frac{n}{m}, \quad \gcd(m, n) = 1, \quad (3270)$$

the exact coincidence index is

$$\Sigma = \frac{m^2 + n^2}{d}, \quad (3271)$$

with

$$d = \begin{cases} 1, & m, n \text{ of opposite parity,} \\ 2, & m, n \text{ both odd.} \end{cases} \quad (3272)$$

The exact supercell area is $\mathcal{A}_M = \Sigma a^2$. For opposite parity, a primitive coincidence basis is

$$\mathbf{T}_1 = a \begin{pmatrix} m \\ n \end{pmatrix}, \quad \mathbf{T}_2 = a \begin{pmatrix} -n \\ m \end{pmatrix}. \quad (3273)$$

The direct supercell basis matrix is

$$C_M = a \begin{pmatrix} m & -n \\ n & m \end{pmatrix}. \quad (3274)$$

Its determinant is

$$\det C_M = a^2 (m^2 + n^2) = a^2 \Sigma. \quad (3275)$$

When both integers are odd, the primitive cell is reduced by the Gaussian factor described previously. A primitive basis is

$$\mathbf{T}_1 = \frac{a}{2} \begin{pmatrix} m+n \\ n-m \end{pmatrix}, \quad (3276)$$

and

$$\mathbf{T}_2 = \frac{a}{2} \begin{pmatrix} m-n \\ m+n \end{pmatrix}. \quad (3277)$$

Thus,

$$C_M = \frac{a}{2} \begin{pmatrix} m+n & m-n \\ n-m & m+n \end{pmatrix}. \quad (3278)$$

Its determinant is

$$\det C_M = \frac{a^2}{2} (m^2 + n^2) = a^2 \Sigma. \quad (3279)$$

In both parity sectors, $\mathbf{T}_1 \cdot \mathbf{T}_2 = 0$. The primitive vectors have equal length,

$$|\mathbf{T}_1| = |\mathbf{T}_2| = a\sqrt{\Sigma}. \quad (3280)$$

Thus the exact coincidence lattice is itself square. Its real-space lattice constant is $L_M^{\text{sc}} = a\sqrt{\Sigma}$. The exact reciprocal lattice is defined as the dual of the exact coincidence lattice, $\Lambda_M^* = (\Lambda_M)^*$. Its reciprocal basis matrix is $B_M = 2\pi C_M^{-T}$. For $d = 1$, one has

$$C_M^{-T} = \frac{1}{a(m^2 + n^2)} \begin{pmatrix} m & -n \\ n & m \end{pmatrix}. \quad (3281)$$

Therefore,

$$\mathbf{G}_1 = \frac{2\pi}{a(m^2 + n^2)} \begin{pmatrix} m \\ n \end{pmatrix}, \quad (3282)$$

and

$$\mathbf{G}_2 = \frac{2\pi}{a(m^2 + n^2)} \begin{pmatrix} -n \\ m \end{pmatrix}. \quad (3283)$$

Since the coincidence index is $\Sigma = m^2 + n^2$, these vectors have magnitude

$$|\mathbf{G}_1| = |\mathbf{G}_2| = \frac{2\pi}{a\sqrt{\Sigma}}. \quad (3284)$$

For the both-odd sector, direct inversion gives

$$\mathbf{G}_1 = \frac{2\pi}{a(m^2 + n^2)} \begin{pmatrix} m+n \\ n-m \end{pmatrix}, \quad (3285)$$

and

$$\mathbf{G}_2 = \frac{2\pi}{a(m^2 + n^2)} \begin{pmatrix} m-n \\ m+n \end{pmatrix}. \quad (3286)$$

Because

$$\Sigma = \frac{m^2 + n^2}{2}, \quad (3287)$$

their magnitudes again satisfy

$$|\mathbf{G}_1| = |\mathbf{G}_2| = \frac{2\pi}{a\sqrt{\Sigma}}. \quad (3288)$$

Thus the reciprocal geometry is unified by the coincidence index. The exact moiré reciprocal-cell area is

$$\mathcal{A}_{\text{mBZ}} = |\det B_M|. \quad (3289)$$

Using reciprocal-area duality gives

$$\mathcal{A}_{\text{mBZ}} = \frac{(2\pi)^2}{\mathcal{A}_M}. \quad (3290)$$

Since $\mathcal{A}_M = \Sigma a^2$, one obtains

$$\mathcal{A}_{\text{mBZ}} = \frac{1}{\Sigma} \left(\frac{2\pi}{a} \right)^2. \quad (3291)$$

Hence,

$$\frac{\mathcal{A}_{\text{BZ}}}{\mathcal{A}_{\text{mBZ}}} = \Sigma. \quad (3292)$$

This exact area ratio is the reciprocal-space origin of the Σ -fold band folding. Because the exact coincidence lattice remains square, the exact mBZ is also square. It may be parameterized by $\mathbf{k} = u\mathbf{G}_1 + v\mathbf{G}_2$, with

$$-\frac{1}{2} \leq u < \frac{1}{2}, \quad -\frac{1}{2} \leq v < \frac{1}{2}. \quad (3293)$$

The side length of the square mBZ is

$$G_M = \frac{2\pi}{a\sqrt{\Sigma}}. \quad (3294)$$

Define the mBZ center by $\mathbf{k}_{\Gamma_M} = \mathbf{0}$. A pair of edge centers may be chosen as $\mathbf{k}_{X_M} = \frac{\mathbf{G}_1}{2}$, and $\mathbf{k}_{Y_M} = \frac{\mathbf{G}_2}{2}$. The corner is

$$\mathbf{k}_{M_M} = \frac{\mathbf{G}_1 + \mathbf{G}_2}{2}. \quad (3295)$$

The conventional square-zone spectral path is therefore

$$\Gamma_M \longrightarrow X_M \longrightarrow M_M \longrightarrow \Gamma_M. \quad (3296)$$

Because the coincidence lattice is a sublattice of the first microscopic lattice, $\Lambda_M \subset \Lambda_1$, the reciprocal inclusion is reversed, $\Lambda_1^* \subset \Lambda_M^*$. Similarly, $\Lambda_2^* \subset \Lambda_M^*$. Thus the exact moiré reciprocal lattice is denser than either primitive reciprocal lattice. The exact moiré reciprocal lattice can be written directly in terms of the two primitive reciprocal lattices. The result is

$$\Lambda_M^* = \Lambda_1^* + \Lambda_2^*. \quad (3297)$$

We now prove this identity. Define $L^* = \Lambda_1^* + \Lambda_2^*$. The dual of this sum is the set of vectors satisfying

$$\mathbf{x} \cdot (\mathbf{G}_1 + \mathbf{G}_2) \in 2\pi\mathbb{Z} \quad (3298)$$

for every $\mathbf{G}_1 \in \Lambda_1^*$, and $\mathbf{G}_2 \in \Lambda_2^*$. Choosing the second reciprocal vector equal to zero gives $\mathbf{x} \in \Lambda_1$. Choosing the first reciprocal vector equal to zero gives $\mathbf{x} \in \Lambda_2$. Therefore,

$$(\Lambda_1^* + \Lambda_2^*)^* = \Lambda_1 \cap \Lambda_2. \quad (3299)$$

Taking the dual once more gives

$$\Lambda_1^* + \Lambda_2^* = (\Lambda_1 \cap \Lambda_2)^*. \quad (3300)$$

Since the intersection is the exact coincidence lattice, $\Lambda_1 \cap \Lambda_2 = \Lambda_M$, one obtains the desired result,

$$\Lambda_M^* = \Lambda_1^* + \Lambda_2^*. \quad (3301)$$

This relation is the exact reciprocal-space counterpart of real-space commensurability. A crucial distinction must now be made. A commonly used geometric construction defines reciprocal mismatch vectors by subtracting corresponding reciprocal vectors from the two layers. In a fixed reciprocal-basis pairing,

$$\Delta \mathbf{b}_i = R_\theta \mathbf{b}_i - \mathbf{b}_i. \quad (3302)$$

Because the two primitive reciprocal vectors have equal magnitude, $|\mathbf{b}_i| = \frac{2\pi}{a}$, one obtains

$$|\Delta \mathbf{b}_i| = \frac{4\pi}{a} \left| \sin \left(\frac{\theta}{2} \right) \right|. \quad (3303)$$

The determinant satisfies

$$|\det(R_\theta - I)| = 4 \sin^2 \left(\frac{\theta}{2} \right). \quad (3304)$$

Therefore the area of the corresponding reciprocal beat cell is

$$\mathcal{A}_{\text{beat}}^* = 4 \sin^2 \left(\frac{\theta}{2} \right) \left(\frac{2\pi}{a} \right)^2. \quad (3305)$$

The associated real-space beat-cell area is

$$\mathcal{A}_{\text{beat}} = \frac{a^2}{4 \sin^2(\theta/2)}. \quad (3306)$$

This geometric beat cell is not, in general, the exact atomic coincidence supercell. For the square lattice, relative rotations differing by a fourfold crystallographic symmetry are physically equivalent. Define the point-group-reduced twist angle by

$$\hat{\theta} = \min_{p \in \mathbb{Z}} \left| \theta - p \frac{\pi}{2} \right|, \quad (3307)$$

with the convention $0 \leq \hat{\theta} \leq \frac{\pi}{4}$. Let the closest square-lattice point-group rotation be $S_* = R_{p_*\pi/2}$. The symmetry-reduced reciprocal mismatch matrix is

$$\Delta B_{\text{geom}} = (R_\theta - S_*) B. \quad (3308)$$

Because the point-group matrix is orthogonal, the mismatch magnitude is controlled only by the reduced twist,

$$|\Delta \mathbf{b}_{\text{geom}}| = \frac{4\pi}{a} \sin \left(\frac{\hat{\theta}}{2} \right). \quad (3309)$$

Thus the physically relevant long-wavelength beat scale near any square-symmetry-equivalent alignment is determined by the reduced angle. Let the reduced commensurate angle be parameterized by

$$\tan \left(\frac{\hat{\theta}}{2} \right) = \frac{n}{m}. \quad (3310)$$

The exact coincidence index satisfies

$$\Sigma = \frac{n^2}{d} \csc^2 \left(\frac{\hat{\theta}}{2} \right). \quad (3311)$$

The exact supercell area is therefore

$$\mathcal{A}_M = a^2 \frac{n^2}{d} \csc^2 \left(\frac{\hat{\theta}}{2} \right). \quad (3312)$$

The geometric beat-cell area is

$$\mathcal{A}_{\text{beat}} = \frac{a^2}{4 \sin^2(\hat{\theta}/2)}. \quad (3313)$$

Their ratio is

$$\frac{\mathcal{A}_M}{\mathcal{A}_{\text{beat}}} = \frac{4n^2}{d}. \quad (3314)$$

Equivalently, the reciprocal areas satisfy

$$\frac{\mathcal{A}_{\text{beat}}^*}{\mathcal{A}_{\text{mBZ}}} = \frac{4n^2}{d}. \quad (3315)$$

Thus the geometric beat reciprocal pattern and the exact crystallographic mBZ are related by a discrete arithmetic factor. They must not be identified. Because $\Lambda_1^* \subset \Lambda_M^*$, one can form the finite quotient $\mathcal{Q}_1 = \Lambda_M^* / \Lambda_1^*$. The order of this quotient is $|\mathcal{Q}_1| = \Sigma$. To prove this result, note that the reciprocal primitive-cell areas satisfy

$$\mathcal{A}_{\text{BZ}} = \left(\frac{2\pi}{a} \right)^2, \quad (3316)$$

and

$$\mathcal{A}_{\text{mBZ}} = \frac{1}{\Sigma} \mathcal{A}_{\text{BZ}}. \quad (3317)$$

Therefore the reciprocal-lattice index is

$$[\Lambda_M^* : \Lambda_1^*] = \frac{\mathcal{A}_{\text{BZ}}}{\mathcal{A}_{\text{mBZ}}} = \Sigma. \quad (3318)$$

Similarly, $\mathcal{Q}_2 = \Lambda_M^* / \Lambda_2^*$ has order $|\mathcal{Q}_2| = \Sigma$. Choose a complete set of representatives $\left\{ \mathbf{Q}_\alpha^{(1)} \right\}_{\alpha=1}^\Sigma$ for the quotient $\mathcal{Q}_1 = \Lambda_M^* / \Lambda_1^*$. Likewise choose $\left\{ \mathbf{Q}_\beta^{(2)} \right\}_{\beta=1}^\Sigma$ for the quotient $\mathcal{Q}_2 = \Lambda_M^* / \Lambda_2^*$. Every primitive first-layer momentum can then be represented uniquely, up to boundary identifications, as

$$\mathbf{q} = \mathbf{k} + \mathbf{Q}_\alpha^{(1)} + \mathbf{G}_1, \quad (3319)$$

where $\mathbf{k} \in \text{mBZ}$, and $\mathbf{G}_1 \in \Lambda_1^*$. Likewise, every second-layer momentum is

$$\mathbf{q} = \mathbf{k} + \mathbf{Q}_\beta^{(2)} + \mathbf{G}_2, \quad (3320)$$

with $\mathbf{G}_2 \in \Lambda_2^*$. The primitive Brillouin zone decomposes into Σ translated copies of the exact mBZ. For every periodic integrable function,

$$f(\mathbf{q} + \mathbf{G}_1) = f(\mathbf{q}), \quad (3321)$$

one has

$$\int_{\text{BZ}_0} d^2q f(\mathbf{q}) = \sum_{\alpha=1}^{\Sigma} \int_{\text{mBZ}} d^2k f(\mathbf{k} + \mathbf{Q}_\alpha^{(1)}). \quad (3322)$$

This identity proves that band folding does not remove or create states. It merely reorganizes the same primitive Bloch states into a smaller Brillouin zone with a larger number of bands. At fixed moiré momentum, define the folded first-layer basis

$$|1, \alpha; \mathbf{k}\rangle = |1, \mathbf{k} + \mathbf{Q}_\alpha^{(1)}\rangle. \quad (3323)$$

The isolated first-layer folded Hamiltonian is diagonal,

$$[h_1^{\text{fold}}(\mathbf{k})]_{\alpha\alpha'} = \delta_{\alpha\alpha'} \varepsilon_0(\mathbf{k} + \mathbf{Q}_\alpha^{(1)}). \quad (3324)$$

Thus the single primitive band generates exactly Σ folded branches in the exact mBZ. Define

$$|2, \beta; \mathbf{k}\rangle = |2, \mathbf{k} + \mathbf{Q}_\beta^{(2)}\rangle. \quad (3325)$$

The second-layer folded Hamiltonian is

$$[h_2^{\text{fold}}(\mathbf{k})]_{\beta\beta'} = \delta_{\beta\beta'} \varepsilon_0 \left[R_{-\theta}(\mathbf{k} + \mathbf{Q}_\beta^{(2)}) \right]. \quad (3326)$$

The unhybridized bilayer therefore possesses 2Σ folded branches per exact moiré momentum. Before interlayer tunneling is introduced, the exact folded Hamiltonian is

$$h_{\text{fold}}^{(0)}(\mathbf{k}) = \begin{pmatrix} h_1^{\text{fold}}(\mathbf{k}) & 0 \\ 0 & h_2^{\text{fold}}(\mathbf{k}) \end{pmatrix}. \quad (3327)$$

Its dimension is

$$\dim h_{\text{fold}}^{(0)} = 2\Sigma. \quad (3328)$$

This equals the exact real-space microscopic orbital count per bilayer supercell. Thus reciprocal-space folding and real-space supercell counting agree exactly. The preceding statement can be proved constructively. Choose representatives of the finite direct-lattice quotient,

$$\{\mathbf{r}_a\}_{a=1}^{\Sigma} \subset \Lambda_1, \quad (3329)$$

such that

$$\Lambda_1 = \bigcup_{a=1}^{\Sigma} (\mathbf{r}_a + \Lambda_M). \quad (3330)$$

The finite quotient Λ_1/Λ_M has order Σ . Its dual character group is naturally isomorphic to Λ_M^*/Λ_1^* . Define the matrix

$$U_{a\alpha} = \frac{1}{\sqrt{\Sigma}} \exp \left[i \mathbf{Q}_\alpha^{(1)} \cdot \mathbf{r}_a \right]. \quad (3331)$$

The character orthogonality relation is

$$\sum_{a=1}^{\Sigma} \exp \left[i \left(\mathbf{Q}_\beta^{(1)} - \mathbf{Q}_\alpha^{(1)} \right) \cdot \mathbf{r}_a \right] = \Sigma \delta_{\alpha\beta}. \quad (3332)$$

Therefore,

$$[U^\dagger U]_{\alpha\beta} = \frac{1}{\Sigma} \sum_{a=1}^{\Sigma} e^{i(\mathbf{Q}_\beta^{(1)} - \mathbf{Q}_\alpha^{(1)}) \cdot \mathbf{r}_a} = \delta_{\alpha\beta}. \quad (3333)$$

Thus,

$$U^\dagger U = U U^\dagger = I_\Sigma. \quad (3334)$$

The real-space supercell orbital basis and the folded primitive-momentum basis are therefore related by an exact finite Fourier transform. Choose internal sites for layer one, $\{\mathbf{r}_{1a}\}_{a=1}^{\Sigma}$, and for layer two, $\{\mathbf{r}_{2b}\}_{b=1}^{\Sigma}$. Every microscopic site can then be written uniquely as

$$\mathbf{x}_\ell = \mathbf{R}_M + \mathbf{r}_{\ell a}, \quad (3335)$$

where $\mathbf{R}_M \in \Lambda_M$. In the periodic Bloch gauge, define

$$|\ell, a; \mathbf{k}\rangle = \frac{1}{\sqrt{N_M}} \sum_{\mathbf{R}_M} e^{i\mathbf{k} \cdot \mathbf{R}_M} |\ell, \mathbf{R}_M + \mathbf{r}_{\ell a}\rangle. \quad (3336)$$

For every exact moiré reciprocal vector, $\mathbf{G}_M \in \Lambda_M^*$, one has $e^{i\mathbf{G}_M \cdot \mathbf{R}_M} = 1$. Therefore,

$$|\ell, a; \mathbf{k} + \mathbf{G}_M\rangle = |\ell, a; \mathbf{k}\rangle. \quad (3337)$$

The corresponding Bloch Hamiltonian is exactly periodic over the mBZ. Define the real-space hopping matrix by

$$h_{ab}^{\ell\ell'}(\mathbf{R}_M) = \langle \ell, 0, a | H | \ell', \mathbf{R}_M, b \rangle. \quad (3338)$$

The Bloch matrix element is then

$$\left[h^{\ell\ell'}(\mathbf{k}) \right]_{ab} = \sum_{\mathbf{R}_M} h_{ab}^{\ell\ell'}(\mathbf{R}_M) e^{i\mathbf{k} \cdot \mathbf{R}_M}. \quad (3339)$$

The full exact bilayer Bloch Hamiltonian is

$$H_M(\mathbf{k}) = \begin{pmatrix} h_1^M(\mathbf{k}) & \tau_M(\mathbf{k}) \\ \tau_M^\dagger(\mathbf{k}) & h_2^M(\mathbf{k}) \end{pmatrix}. \quad (3340)$$

Each block has dimension $\Sigma \times \Sigma$. Therefore,

$$\dim H_M(\mathbf{k}) = 2\Sigma. \quad (3341)$$

The real-space hopping amplitudes satisfy

$$h_{ab}^{\ell\ell'}(\mathbf{R}) = \left[h_{ba}^{\ell'\ell}(-\mathbf{R}) \right]^*. \quad (3342)$$

Therefore,

$$H_M^\dagger(\mathbf{k}) = H_M(\mathbf{k}). \quad (3343)$$

Let U_1 denote the quotient Fourier transform of the first layer and U_2 the corresponding transform for the second layer. Define

$$\mathcal{U} = \begin{pmatrix} U_1 & 0 \\ 0 & U_2 \end{pmatrix}. \quad (3344)$$

The folded Hamiltonian is

$$H_{\text{fold}}(\mathbf{k}) = \mathcal{U}^\dagger H_M(\mathbf{k}) \mathcal{U}. \quad (3345)$$

Thus,

$$\text{Spec } H_{\text{fold}}(\mathbf{k}) = \text{Spec } H_M(\mathbf{k}). \quad (3346)$$

The two descriptions are exactly equivalent. If the interlayer tunneling is switched off, $\tau_M = 0$, the quotient Fourier transform diagonalizes each primitive-translation block. One obtains

$$U_1^\dagger h_1^M(\mathbf{k}) U_1 = \text{diag} \left[\varepsilon_0(\mathbf{k} + \mathbf{Q}_1^{(1)}), \dots, \varepsilon_0(\mathbf{k} + \mathbf{Q}_\Sigma^{(1)}) \right]. \quad (3347)$$

Similarly,

$$U_2^\dagger h_2^M(\mathbf{k}) U_2 = \text{diag} \left[\varepsilon_2(\mathbf{k} + \mathbf{Q}_1^{(2)}), \dots, \varepsilon_2(\mathbf{k} + \mathbf{Q}_\Sigma^{(2)}) \right]. \quad (3348)$$

Thus band folding is exactly a unitary rearrangement of primitive Bloch states. Consider one first-layer folded branch,

$$E_\alpha^{\text{fold}}(\mathbf{k}) = \varepsilon_0(\mathbf{k} + \mathbf{Q}_\alpha^{(1)}). \quad (3349)$$

Its gradient is

$$\nabla_{\mathbf{k}} E_\alpha^{\text{fold}}(\mathbf{k}) = \nabla_{\mathbf{q}} \varepsilon_0(\mathbf{q}) \Big|_{\mathbf{q}=\mathbf{k}+\mathbf{Q}_\alpha^{(1)}}. \quad (3350)$$

Therefore,

$$\mathbf{v}_\alpha^{\text{fold}}(\mathbf{k}) = \mathbf{v}_0(\mathbf{k} + \mathbf{Q}_\alpha^{(1)}). \quad (3351)$$

The local dispersion of the folded branch is exactly the primitive dispersion evaluated at a shifted momentum. Thus, band folding alone $\nrightarrow$ genuine dispersion suppression. This result provides an essential control for the later identification of magic flat bands. We now reintroduce the interlayer coupling. Let a first-layer Bloch state have momentum $\mathbf{q}_1$, and a second-layer Bloch state have momentum $\mathbf{q}_2$. The interlayer matrix element is

$$\mathcal{M}(\mathbf{q}_1, \mathbf{q}_2) = \langle 1, \mathbf{q}_1 | H_\perp | 2, \mathbf{q}_2 \rangle. \quad (3352)$$

For every common translation, $\mathbf{R}_M \in \Lambda_M$, the interlayer Hamiltonian is invariant,

$$U_{\mathbf{R}_M} H_\perp U_{\mathbf{R}_M}^{-1} = H_\perp. \quad (3353)$$

The two Bloch states transform according to

$$U_{\mathbf{R}_M} |1, \mathbf{q}_1\rangle = e^{-i\mathbf{q}_1 \cdot \mathbf{R}_M} |1, \mathbf{q}_1\rangle, \quad (3354)$$

and

$$U_{\mathbf{R}_M} |2, \mathbf{q}_2\rangle = e^{-i\mathbf{q}_2 \cdot \mathbf{R}_M} |2, \mathbf{q}_2\rangle. \quad (3355)$$

Therefore,

$$\mathcal{M}(\mathbf{q}_1, \mathbf{q}_2) = e^{i(\mathbf{q}_1 - \mathbf{q}_2) \cdot \mathbf{R}_M} \mathcal{M}(\mathbf{q}_1, \mathbf{q}_2). \quad (3356)$$

A nonzero matrix element therefore requires

$$e^{i(\mathbf{q}_1 - \mathbf{q}_2) \cdot \mathbf{R}_M} = 1 \quad (3357)$$

for every common translation. By definition of the reciprocal lattice, this is equivalent to $\mathbf{q}_1 - \mathbf{q}_2 \in \Lambda_M^*$. Thus interlayer tunneling conserves crystal momentum modulo the exact moiré reciprocal lattice. Write the two primitive momenta as

$$\mathbf{q}_1 = \mathbf{k} + \mathbf{Q}_\alpha^{(1)}, \quad \mathbf{q}_2 = \mathbf{k}' + \mathbf{Q}_\beta^{(2)}. \quad (3358)$$

Since both quotient representatives belong to the exact reciprocal lattice, $\mathbf{Q}_\alpha^{(1)} \in \Lambda_M^*$, and $\mathbf{Q}_\beta^{(2)} \in \Lambda_M^*$, the selection rule implies

$$\mathbf{k} \equiv \mathbf{k}' \pmod{\Lambda_M^*}. \quad (3359)$$

Restricting both variables to the same mBZ gives $\mathbf{k} = \mathbf{k}'$. The exact bilayer Hamiltonian therefore block diagonalizes in the conserved moiré momentum. Define

$$\mathcal{T}_{\alpha\beta}(\mathbf{k}) = \langle 1, \alpha; \mathbf{k} | H_\perp | 2, \beta; \mathbf{k} \rangle. \quad (3360)$$

The full folded Hamiltonian is

$$H_{\text{fold}}(\mathbf{k}) = \begin{pmatrix} E_1^{\text{fold}}(\mathbf{k}) & \mathcal{T}(\mathbf{k}) \\ \mathcal{T}^\dagger(\mathbf{k}) & E_2^{\text{fold}}(\mathbf{k}) \end{pmatrix}, \quad (3361)$$

where

$$E_1^{\text{fold}}(\mathbf{k}) = \text{diag} \left[\varepsilon_0 \left(\mathbf{k} + \mathbf{Q}_\alpha^{(1)} \right) \right], \quad (3362)$$

and

$$E_2^{\text{fold}}(\mathbf{k}) = \text{diag} \left[\varepsilon_0 \left(R_{-\theta} \left[\mathbf{k} + \mathbf{Q}_\beta^{(2)} \right] \right) \right]. \quad (3363)$$

This is the exact Euclidean moiré Bloch Hamiltonian for one orbital per microscopic square-lattice site. The interlayer block in the real-space supercell basis is

$$[\tau_M(\mathbf{k})]_{ab} = \sum_{\mathbf{R}_M} T_{ab}(\mathbf{R}_M) e^{i\mathbf{k} \cdot \mathbf{R}_M}, \quad (3364)$$

where

$$T_{ab}(\mathbf{R}_M) = \langle 1, 0, a | H_\perp | 2, \mathbf{R}_M, b \rangle. \quad (3365)$$

The folded-momentum hybridization matrix is obtained through the quotient Fourier transforms,

$$\mathcal{T}(\mathbf{k}) = U_1^\dagger \tau_M(\mathbf{k}) U_2. \quad (3366)$$

Thus the supercell and folded-momentum descriptions contain exactly the same interlayer physics. When the

interlayer tunneling can be represented by a smooth local registry field periodic under the exact supercell,

$$T(\mathbf{r} + \mathbf{R}_M) = T(\mathbf{r}), \quad (3367)$$

it admits the Fourier expansion

$$T(\mathbf{r}) = \sum_{\mathbf{G} \in \Lambda_M^*} T_{\mathbf{G}} e^{i\mathbf{G} \cdot \mathbf{r}}. \quad (3368)$$

The Fourier coefficients are

$$T_{\mathbf{G}} = \frac{1}{\mathcal{A}_M} \int_{\Omega_M} d^2r T(\mathbf{r}) e^{-i\mathbf{G} \cdot \mathbf{r}}. \quad (3369)$$

The momentum-space coupling generated by a Fourier harmonic satisfies

$$\mathbf{q}_1 = \mathbf{q}_2 + \mathbf{G}, \quad \mathbf{G} \in \Lambda_M^*. \quad (3370)$$

Thus the exact reciprocal lattice provides the complete allowed set of interlayer momentum transfers. If the tunneling profile is smooth on the microscopic lattice scale, its Fourier coefficients typically decrease with reciprocal-vector magnitude. The dominant coupling channels can then be approximated by a finite first star,

$$\mathcal{S}_1 = \{\mathbf{G} \in \Lambda_M^* : |\mathbf{G}| = G_M\}. \quad (3371)$$

For the square mBZ,

$$\mathcal{S}_1 = \{\pm \mathbf{G}_1, \pm \mathbf{G}_2\}. \quad (3372)$$

A minimal Fourier tunneling model is therefore

$$T(\mathbf{r}) \approx T_0 + T_{\mathbf{G}_1} e^{i\mathbf{G}_1 \cdot \mathbf{r}} + T_{-\mathbf{G}_1} e^{-i\mathbf{G}_1 \cdot \mathbf{r}} + T_{\mathbf{G}_2} e^{i\mathbf{G}_2 \cdot \mathbf{r}} + T_{-\mathbf{G}_2} e^{-i\mathbf{G}_2 \cdot \mathbf{r}}. \quad (3373)$$

For a real tunneling field, $T_{-\mathbf{G}} = T_{\mathbf{G}}^*$. The physical role of the interlayer coupling can be seen most clearly near a crossing between one folded state from each layer. Let the two unhybridized energies be $\varepsilon_1(\mathbf{k})$ and $\varepsilon_2(\mathbf{k})$. Let the corresponding interlayer matrix element be $V(\mathbf{k})$. The effective two-state Hamiltonian is

$$H_{2\text{lev}}(\mathbf{k}) = \begin{pmatrix} \varepsilon_1(\mathbf{k}) & V(\mathbf{k}) \\ V^*(\mathbf{k}) & \varepsilon_2(\mathbf{k}) \end{pmatrix}. \quad (3374)$$

Define the detuning

$$\Delta(\mathbf{k}) = \varepsilon_1(\mathbf{k}) - \varepsilon_2(\mathbf{k}). \quad (3375)$$

The characteristic equation is

$$\det [E - H_{2\text{lev}}] = 0. \quad (3376)$$

Explicitly,

$$[E - \varepsilon_1][E - \varepsilon_2] - |V|^2 = 0. \quad (3377)$$

Solving gives

$$E_{\pm} = \frac{\varepsilon_1 + \varepsilon_2}{2} \pm \sqrt{\frac{\Delta^2}{4} + |V|^2}. \quad (3378)$$

At exact resonance, $\Delta(\mathbf{k}_*) = 0$, the energies become

$$E_{\pm}(\mathbf{k}_*) = \varepsilon_* \pm |V(\mathbf{k}_*)|. \quad (3379)$$

Thus the avoided-crossing gap is

$$\Delta_{\text{hyb}} = 2|V(\mathbf{k}_*)|. \quad (3380)$$

Band folding creates the crossing. Interlayer tunneling determines whether that crossing remains exact or is reconstructed into an avoided crossing. Write the coupling phase as $V = |V|e^{i\phi}$. Define a mixing angle by

$$\tan(2\vartheta) = \frac{2|V|}{\Delta}. \quad (3381)$$

The normalized eigenstates may be chosen as

$$|+\rangle = \cos \vartheta |1\rangle + e^{-i\phi} \sin \vartheta |2\rangle, \quad (3382)$$

and

$$|-\rangle = -e^{i\phi} \sin \vartheta |1\rangle + \cos \vartheta |2\rangle. \quad (3383)$$

At resonance, $\vartheta = \frac{\pi}{4}$. The states are therefore equally distributed between the two folded branches. Differentiate the exact two-level energies. Define the unhybridized velocities

$$\mathbf{v}_1 = \frac{1}{\hbar} \nabla_{\mathbf{k}} \varepsilon_1, \quad \mathbf{v}_2 = \frac{1}{\hbar} \nabla_{\mathbf{k}} \varepsilon_2. \quad (3384)$$

The hybridized velocity is

$$\mathbf{v}_{\pm} = \frac{\mathbf{v}_1 + \mathbf{v}_2}{2} \pm \frac{\frac{\Delta}{4}(\mathbf{v}_1 - \mathbf{v}_2) + \frac{1}{\hbar} \text{Re}[V^* \nabla_{\mathbf{k}} V]}{\sqrt{\Delta^2/4 + |V|^2}}. \quad (3385)$$

At exact resonance, $\Delta = 0$, and therefore

$$\mathbf{v}_{\pm} = \frac{\mathbf{v}_1 + \mathbf{v}_2}{2} \pm \frac{1}{\hbar} \frac{\text{Re}[V^* \nabla_{\mathbf{k}} V]}{|V|}. \quad (3386)$$

If the tunneling matrix element is slowly varying in momentum, $\nabla_{\mathbf{k}} V \approx 0$, then

$$\mathbf{v}_{\pm} \approx \frac{\mathbf{v}_1 + \mathbf{v}_2}{2}. \quad (3387)$$

Thus strongly counterpropagating folded branches satisfying $\mathbf{v}_1 \approx -\mathbf{v}_2$ can generate strong velocity suppression after hybridization. This is the simplest microscopic precursor of a magic-band mechanism. Far from resonance, assume $|V| \ll |\Delta|$. The square root can be expanded. For the branch adiabatically connected to the first state, one obtains

$$E_1^{\text{eff}} = \varepsilon_1 + \frac{|V|^2}{\varepsilon_1 - \varepsilon_2} + O(|V|^4). \quad (3388)$$

For the second branch,

$$E_2^{\text{eff}} = \varepsilon_2 - \frac{|V|^2}{\varepsilon_1 - \varepsilon_2} + O(|V|^4). \quad (3389)$$

Thus even away from exact crossings, interlayer tunneling produces a momentum-dependent energy correction. Differentiate the second-order energy correction. One obtains

$$\nabla_{\mathbf{k}} \delta E_1 = \frac{2 \text{Re}[V^* \nabla_{\mathbf{k}} V]}{\Delta} - \frac{|V|^2}{\Delta^2} \nabla_{\mathbf{k}} \Delta. \quad (3390)$$

Using

$$\nabla_{\mathbf{k}} \Delta = \hbar(\mathbf{v}_1 - \mathbf{v}_2), \quad (3391)$$

the effective velocity becomes

$$\mathbf{v}_1^{\text{eff}} = \mathbf{v}_1 + \frac{2}{\hbar} \frac{\text{Re}[V^* \nabla_{\mathbf{k}} V]}{\Delta} - \frac{|V|^2}{\Delta^2} (\mathbf{v}_1 - \mathbf{v}_2) + O(|V|^4). \quad (3392)$$

The last term explicitly demonstrates that hybridization can suppress or enhance the original velocity depending on the relative orientation of the two folded dispersions. The hybridization mechanism can be generalized beyond two states. Write the folded Hamiltonian as

$$H_{\text{fold}} = \begin{pmatrix} h_1 & \mathcal{T} \\ \mathcal{T}^\dagger & h_2 \end{pmatrix}. \quad (3393)$$

The eigenvalue equation is

$$\begin{pmatrix} E - h_1 & -\mathcal{T} \\ -\mathcal{T}^\dagger & E - h_2 \end{pmatrix} \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} = 0. \quad (3394)$$

The second row gives

$$\psi_2 = (E - h_2)^{-1} \mathcal{T}^\dagger \psi_1 \quad (3395)$$

whenever the inverse exists. Substituting into the first row gives

$$[E - h_1 - \Sigma_1(E, \mathbf{k})] \psi_1 = 0, \quad (3396)$$

where the exact hybridization self-energy is

$$\Sigma_1(E, \mathbf{k}) = \mathcal{T}(\mathbf{k}) [E - h_2(\mathbf{k})]^{-1} \mathcal{T}^\dagger(\mathbf{k}). \quad (3397)$$

Thus the exact energy-dependent effective first-layer Hamiltonian is

$$h_1^{\text{eff}}(E, \mathbf{k}) = h_1(\mathbf{k}) + \Sigma_1(E, \mathbf{k}). \quad (3398)$$

An analogous expression holds for the second layer. The self-energy depends on momentum through both the folded second-layer dispersion and the interlayer matrix. Thus, $\nabla_{\mathbf{k}} \Sigma_1 \neq 0$ in a generic twisted bilayer. The effective dispersion therefore obeys schematically

$$\nabla_{\mathbf{k}} E = \nabla_{\mathbf{k}} \varepsilon_{\text{bare}} + \nabla_{\mathbf{k}} \Sigma. \quad (3399)$$

A magic-band condition can arise when the hybridization-induced velocity correction cancels the bare kinetic contribution. This mechanism is fundamentally different from ordinary band folding. Let an exact normalized moiré Bloch eigenstate satisfy

$$H_M(\mathbf{k}) |u_{n\mathbf{k}}\rangle = E_n(\mathbf{k}) |u_{n\mathbf{k}}\rangle. \quad (3400)$$

For a nondegenerate band, the Hellmann–Feynman theorem gives

$$\nabla_{\mathbf{k}} E_n = \langle u_{n\mathbf{k}} | \nabla_{\mathbf{k}} H_M(\mathbf{k}) | u_{n\mathbf{k}} \rangle. \quad (3401)$$

Therefore the group velocity is

$$\mathbf{v}_n(\mathbf{k}) = \frac{1}{\hbar} \langle u_{n\mathbf{k}} | \nabla_{\mathbf{k}} H_M(\mathbf{k}) | u_{n\mathbf{k}} \rangle. \quad (3402)$$

In the folded representation,

$$\nabla_{\mathbf{k}} H_{\text{fold}} = \begin{pmatrix} \nabla_{\mathbf{k}} E_1^{\text{fold}} & \nabla_{\mathbf{k}} \mathcal{T} \\ \nabla_{\mathbf{k}} \mathcal{T}^\dagger & \nabla_{\mathbf{k}} E_2^{\text{fold}} \end{pmatrix}. \quad (3403)$$

The exact velocity therefore contains both primitive-band velocity contributions and explicitly interlayer-hybridization contributions. To distinguish a genuinely hybridized state from a trivially folded primitive state, define the folded-basis components by

$$|u_{n\mathbf{k}}\rangle = \sum_{\alpha=1}^{\Sigma} u_{1\alpha}^{(n)} |1, \alpha; \mathbf{k}\rangle + \sum_{\beta=1}^{\Sigma} u_{2\beta}^{(n)} |2, \beta; \mathbf{k}\rangle. \quad (3404)$$

The corresponding unfolded weights are

$$W_{1\alpha}^{(n)}(\mathbf{k}) = \left| u_{1\alpha}^{(n)}(\mathbf{k}) \right|^2, \quad (3405)$$

and

$$W_{2\beta}^{(n)}(\mathbf{k}) = \left| u_{2\beta}^{(n)}(\mathbf{k}) \right|^2. \quad (3406)$$

Normalization gives

$$\sum_{\alpha} W_{1\alpha}^{(n)} + \sum_{\beta} W_{2\beta}^{(n)} = 1. \quad (3407)$$

A trivially folded state has its spectral weight concentrated on one primitive momentum. A strongly hybridized moiré state distributes weight over multiple folded momentum sectors and often over both layers. A useful measure of hybridization in folded momentum space is

$$\text{IPR}_k^{(n)} = \sum_{\alpha} \left[W_{1\alpha}^{(n)} \right]^2 + \sum_{\beta} \left[W_{2\beta}^{(n)} \right]^2. \quad (3408)$$

For a state concentrated entirely in one folded component, $\text{IPR}_k = 1$. For a state uniformly distributed over all folded components, $\text{IPR}_k = \frac{1}{2\Sigma}$. The inverse quantity,

$$N_{\text{part}}^{(k)} = \frac{1}{\text{IPR}_k}, \quad (3409)$$

estimates the number of significantly hybridized folded momentum sectors. For the ten-atom commensurate structure, the coincidence index is $\Sigma = 5$. Using the representation $(m, n) = (2, 1)$, the direct supercell vectors are

$$\mathbf{T}_1 = a \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad \mathbf{T}_2 = a \begin{pmatrix} -1 \\ 2 \end{pmatrix}. \quad (3410)$$

The exact reciprocal vectors are

$$\mathbf{G}_1 = \frac{2\pi}{5a} \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad \mathbf{G}_2 = \frac{2\pi}{5a} \begin{pmatrix} -1 \\ 2 \end{pmatrix}. \quad (3411)$$

Their magnitude is

$$G_M = \frac{2\pi}{a\sqrt{5}}. \quad (3412)$$

The exact mBZ area is

$$\mathcal{A}_{\text{mBZ}} = \frac{1}{5} \left(\frac{2\pi}{a} \right)^2. \quad (3413)$$

Each layer therefore contributes five folded branches. The exact Bloch Hamiltonian has dimension

$$\dim H_M^{(\Sigma=5)} = 10. \quad (3414)$$

This reciprocal-space dimension exactly reproduces the ten microscopic orbitals of the bilayer supercell. For the thirty-four-atom structure, $\Sigma = 17$. The reduced-angle representative is $(m, n) = (4, 1)$. The direct supercell basis is

$$\mathbf{T}_1 = a \begin{pmatrix} 4 \\ 1 \end{pmatrix}, \quad \mathbf{T}_2 = a \begin{pmatrix} -1 \\ 4 \end{pmatrix}. \quad (3415)$$

The reciprocal basis is

$$\mathbf{G}_1 = \frac{2\pi}{17a} \begin{pmatrix} 4 \\ 1 \end{pmatrix}, \quad \mathbf{G}_2 = \frac{2\pi}{17a} \begin{pmatrix} -1 \\ 4 \end{pmatrix}. \quad (3416)$$

Their magnitude is

$$G_M = \frac{2\pi}{a\sqrt{17}}. \quad (3417)$$

The exact mBZ area is

$$\mathcal{A}_{\text{mBZ}} = \frac{1}{17} \left(\frac{2\pi}{a} \right)^2. \quad (3418)$$

The exact Bloch Hamiltonian dimension is

$$\dim H_M^{(\Sigma=17)} = 34. \quad (3419)$$

Thus the original real-space counting problem and the reciprocal-space folding problem are exactly equivalent. The mBZ side length is

$$G_M = \frac{2\pi}{a\sqrt{\Sigma}}. \quad (3420)$$

Along a bounded-numerator Euclidean commensurate small-angle sequence, $\Sigma = \Theta(|\theta|^{-2})$. Therefore, $G_M = \Theta(|\theta|)$. The reciprocal moiré scale therefore collapses linearly with the small twist. This is the reciprocal-space

counterpart of the real-space inverse-angle growth of the exact Euclidean supercell. The product of the number of folded branches and the mBZ area is $\Sigma \mathcal{A}_{\text{mBZ}} = \mathcal{A}_{\text{BZ}}$. Thus Brillouin-zone compression and band multiplication are exactly compensating. For the bilayer,

$$2\Sigma \mathcal{A}_{\text{mBZ}} = 2\mathcal{A}_{\text{BZ}}. \quad (3421)$$

This identity expresses conservation of the total number of single-particle states under exact supercell folding. For one isolated monolayer, define the primitive density of states by

$$\rho_0(E) = \frac{1}{\mathcal{A}_{\text{BZ}}} \int_{\text{BZ}_0} d^2q \delta[E - \varepsilon_0(\mathbf{q})]. \quad (3422)$$

Using the exact folding identity gives

$$\rho_0(E) = \frac{1}{\Sigma \mathcal{A}_{\text{mBZ}}} \sum_{\alpha=1}^{\Sigma} \int_{\text{mBZ}} d^2k \delta[E - \varepsilon_0(\mathbf{k} + \mathbf{Q}_\alpha)]. \quad (3423)$$

Therefore, $\rho_{\text{fold}}(E) = \rho_0(E)$. Pure band folding does not enhance the physical density of states. Any new density-of-states peak that appears after interlayer coupling is therefore a genuine spectral-reconstruction effect. The exact Euclidean benchmark therefore separates two mechanisms. The first is kinematic folding,

$$\Sigma \uparrow \implies \mathcal{A}_{\text{mBZ}} \downarrow \implies \text{number of folded branches} \uparrow. \quad (3424)$$

The second is dynamical hybridization, $\mathcal{T} \neq 0 \implies$ avoided crossings $\longrightarrow$ velocity renormalization $\longrightarrow$ bandwidth reconstruction. Only the second mechanism can generate a genuinely new dispersion.

a. Theorem 23. Exact Euclidean moiré Bloch reduction. Consider two exactly commensurately twisted square lattices with coincidence index Σ . Then the exact common direct lattice has primitive-cell area $\mathcal{A}_M = \Sigma a^2$, while the exact moiré reciprocal lattice satisfies $\Lambda_M^* = \Lambda_1^* + \Lambda_2^*$. Its Brillouin-zone area is

$$\mathcal{A}_{\text{mBZ}} = \frac{1}{\Sigma} \mathcal{A}_{\text{BZ}}. \quad (3425)$$

The reciprocal quotients satisfy

$$|\Lambda_M^* / \Lambda_\ell^*| = \Sigma. \quad (3426)$$

Therefore each primitive monolayer band folds into exactly Σ branches. For a one-orbital bilayer, the exact moiré Bloch Hamiltonian has dimension 2Σ . Band folding is unitarily equivalent to a relabeling of primitive Bloch states and by itself preserves primitive velocities and the density of states. Genuine spectral reconstruction arises only through the interlayer hybridization matrix.

b. Theorem 24. Exact interlayer momentum-selection rule. For an exactly commensurate Euclidean twisted bilayer, every interlayer matrix element between primitive Bloch states vanishes unless

$$\mathbf{q}_1 - \mathbf{q}_2 \in \Lambda_M^*. \quad (3427)$$

Equivalently, interlayer tunneling preserves the reduced crystal momentum, $\mathbf{k} \in \text{mBZ}$. The exact folded Hamiltonian therefore has the block form

$$H_{\text{fold}}(\mathbf{k}) = \begin{pmatrix} E_1^{\text{fold}}(\mathbf{k}) & \mathcal{T}(\mathbf{k}) \\ \mathcal{T}^\dagger(\mathbf{k}) & E_2^{\text{fold}}(\mathbf{k}) \end{pmatrix}. \quad (3428)$$

At an isolated resonance between two folded branches, the hybridization gap is $\Delta_{\text{hyb}} = 2|V|$. Away from resonance, the leading spectral correction is

$$\delta E = \frac{|V|^2}{\Delta} + O(|V|^4). \quad (3429)$$

Thus interlayer coupling, rather than band folding itself, generates the new energy and velocity scales of the moiré spectrum.

c. Theorem 25. Folding-flattening separation. Let a primitive monolayer band be folded into an exact moiré Brillouin zone without interlayer coupling. Each folded branch is of the form

$$E_\alpha^{(0)}(\mathbf{k}) = \varepsilon_0(\mathbf{k} + \mathbf{Q}_\alpha). \quad (3430)$$

Its velocity satisfies

$$\nabla_{\mathbf{k}} E_\alpha^{(0)} = \nabla_{\mathbf{q}} \varepsilon_0 \quad (3431)$$

at the corresponding unfolded momentum. Therefore exact band folding alone cannot produce intrinsic velocity suppression. After interlayer tunneling is introduced, the exact eigenvalue derivative is

$$\nabla_{\mathbf{k}} E_n = \langle u_{n\mathbf{k}} | \nabla_{\mathbf{k}} H_{\text{fold}} | u_{n\mathbf{k}} \rangle. \quad (3432)$$

The difference between this quantity and the primitive velocity is generated by interlayer-state mixing and the momentum dependence of the tunneling matrix. Consequently, genuine moiré flattening $\implies$ hybridization-induced dispersion reconstruction. The exact Euclidean twisted-square-bilayer theory can now be summarized by

$$\Lambda \longrightarrow \Lambda^* \longrightarrow \Lambda_M = \Lambda_1 \cap \Lambda_2 \longrightarrow \Lambda_M^* = \Lambda_1^* + \Lambda_2^*. \quad (3433)$$

The real-space index relation is $[\Lambda : \Lambda_M] = \Sigma$, while the reciprocal relation is $[\Lambda_M^* : \Lambda^*] = \Sigma$. Therefore,

$$\mathcal{A}_M = \Sigma \mathcal{A}_0, \quad \mathcal{A}_{\text{mBZ}} = \frac{\mathcal{A}_{\text{BZ}}}{\Sigma}. \quad (3434)$$

The band-folding hierarchy is $\mathcal{A}_{\text{mBZ}} \downarrow \longrightarrow \Sigma$ folded states per layer $\longrightarrow 2\Sigma$ bilayer states per $\mathbf{k}$. The spectral-reconstruction hierarchy is folded-state crossings $\longrightarrow \mathcal{T}_{\alpha\beta} \longrightarrow$ hybridization gaps $\longrightarrow$ velocity renormalization $\longrightarrow$ bandwidth reconstruction. The crucial conceptual separation is therefore moiré Brillouin-zone reduction $\neq$ magic-band flattening. The former is a crystallographic consequence of the enlarged exact supercell. The latter is a dynamical consequence of interlayer hybridization. This Euclidean construction provides the benchmark against which the hyperbolic generalized Bloch theory must subsequently be compared.

E. Generalized Hyperbolic Bloch and Representation-Space Formulation

The Euclidean construction developed in the preceding subsection relies on a structural fact that fails fundamentally in hyperbolic crystallography: the Euclidean translation group is Abelian. Consequently all of its finite-dimensional unitary irreducible representations are one-dimensional and may be labeled by an ordinary crystal momentum. The translation group of a compact hyperbolic crystal is instead a non-Abelian Fuchsian group. Its irreducible unitary representations may have dimension larger than one. A complete generalized Bloch formulation must therefore distinguish one-dimensional Abelian characters from genuinely non-Abelian higher-rank representations. This distinction changes the meaning of reciprocal space itself. The Abelian sector is described by a higher-dimensional torus associated with the first homology of the compact quotient. Higher-dimensional representations are instead parameterized by moduli spaces of noncommuting unitary holonomies. The resulting spectral problem is therefore not described by a single two-dimensional Brillouin zone. The purpose of the present subsection is to construct this representation-space formulation directly from the microscopic bilayer Hamiltonian. We derive the generalized Bloch Hamiltonian for arbitrary finite-dimensional unitary representations, establish its Hermiticity and gauge covariance, prove the non-Abelian analogue of crystal-momentum conservation, construct the exact finite-group Fourier decomposition on periodic covers, derive the corresponding state-counting theorem, formulate hyperbolic zone folding through induced representations, and obtain a representation-space differential calculus that generalizes ordinary band velocity and effective-mass theory. A central result will be that the Euclidean relation $\mathbf{k} \longrightarrow H(\mathbf{k})$ must be replaced by $[\rho] \longrightarrow H[\rho]$, where

$$\rho : \Gamma_M \longrightarrow U(d_\rho) \quad (3435)$$

is a unitary irreducible representation of the exact hyperbolic moiré translation group and where representations related by unitary conjugation describe the same generalized crystal momentum. We begin with an exactly commensurate hyperbolic bilayer. The exact common translation group is

$$\Gamma_M = \Gamma \cap g_\theta \Gamma g_\theta^{-1}. \quad (3436)$$

We first restrict to the torsion-free cocompact case. The compact quotient is

$$\Sigma_M = \Gamma_M \backslash \mathbb{H}_R^2. \quad (3437)$$

Its genus is

$$g_M = 1 + q_M (g_0 - 1), \quad (3438)$$

where $q_M = [\Gamma : \Gamma_M]$. The primitive microscopic quotient contains n_b orbitals per layer. Therefore the exact moiré quotient contains $n_b q_M$ orbitals in each layer. The total number of physical orbitals inside one exact bilayer moiré fundamental domain is consequently $\nu_M = 2n_b q_M$. This quantity will play the role of the number of internal orbitals of the generalized Bloch Hamiltonian. The fundamental group of the compact genus- g_M quotient admits the standard presentation

$$\Gamma_M = \left\langle a_1, b_1, \dots, a_{g_M}, b_{g_M} \left| \prod_{j=1}^{g_M} [a_j, b_j] = e \right. \right\rangle. \quad (3439)$$

The commutator is

$$[a, b] = aba^{-1}b^{-1}. \quad (3440)$$

The single global relation

$$\prod_{j=1}^{g_M} [a_j, b_j] = e \quad (3441)$$

is the group-theoretic remnant of the polygon-edge identifications that compactify the hyperbolic fundamental domain. Unlike a Euclidean lattice translation group, the generators generally do not commute, $[a_i, b_j] \neq e$ for generic pairs of generators. This noncommutativity is the fundamental origin of the non-Abelian generalized Bloch structure. Choose the exact moiré fundamental domain as the reference cell. Label the internal physical orbitals by $\alpha = 1, \dots, \nu_M$. Every microscopic orbital in the infinite bilayer can be written as $|\gamma, \alpha\rangle$, $\gamma \in \Gamma_M$. The infinite single-particle Hilbert space is therefore

$$\mathcal{H}_\infty = \ell^2(\Gamma_M) \otimes \mathbb{C}^{\nu_M}. \quad (3442)$$

A general state is

$$|\Psi\rangle = \sum_{\gamma \in \Gamma_M} \sum_{\alpha=1}^{\nu_M} \psi_\alpha(\gamma) |\gamma, \alpha\rangle. \quad (3443)$$

Square summability requires

$$\sum_{\gamma \in \Gamma_M} \sum_{\alpha=1}^{\nu_M} |\psi_\alpha(\gamma)|^2 < \infty. \quad (3444)$$

The exact moiré translations act by the left regular representation. For $\eta \in \Gamma_M$, define

$$U_\eta |\gamma, \alpha\rangle = |\eta\gamma, \alpha\rangle. \quad (3445)$$

The operators are unitary, $U_\eta^\dagger = U_{\eta^{-1}}$, and obey the group multiplication law $U_{\eta_1} U_{\eta_2} = U_{\eta_1 \eta_2}$. Exact translational symmetry means $[H, U_\eta] = 0$ for every $\eta \in \Gamma_M$. Consider two microscopic orbitals $|\gamma, \alpha\rangle$ and $|\gamma', \beta\rangle$. Translation invariance gives

$$\begin{aligned} \langle \eta\gamma, \alpha | H | \eta\gamma', \beta \rangle &= \langle \gamma, \alpha | U_\eta^\dagger H U_\eta | \gamma', \beta \rangle \\ &= \langle \gamma, \alpha | H | \gamma', \beta \rangle. \end{aligned} \quad (3446)$$

Choose $\eta = \gamma^{-1}$. Then

$$\langle \gamma, \alpha | H | \gamma', \beta \rangle = \langle e, \alpha | H | \gamma^{-1} \gamma', \beta \rangle. \quad (3447)$$

Thus the Hamiltonian depends only on the relative group element $\delta = \gamma^{-1} \gamma'$. Define the matrix-valued hopping kernel

$$h_{\alpha\beta}(\delta) = \langle e, \alpha | H | \delta, \beta \rangle. \quad (3448)$$

For every group element, $h(\delta) \in \mathbb{C}^{\nu_M \times \nu_M}$. The action of the Hamiltonian on a wave function is

$$(H\psi)_\alpha(\gamma) = \sum_{\gamma'} \sum_{\beta} h_{\alpha\beta}(\gamma^{-1} \gamma') \psi_\beta(\gamma'). \quad (3449)$$

Set $\gamma' = \gamma\delta$. Then

$$(H\psi)_\alpha(\gamma) = \sum_{\delta \in \Gamma_M} \sum_{\beta} h_{\alpha\beta}(\delta) \psi_\beta(\gamma\delta). \quad (3450)$$

Thus the translation-invariant tight-binding Hamiltonian is a matrix-valued right-convolution operator on the non-Abelian group. This equation is the exact noncommutative analogue of the Euclidean real-space hopping representation $H(\mathbf{R} - \mathbf{R}')$. Hermiticity requires

$$\langle e, \alpha | H | \delta, \beta \rangle = [\langle \delta, \beta | H | e, \alpha \rangle]^*. \quad (3451)$$

Translation invariance gives

$$\langle \delta, \beta | H | e, \alpha \rangle = \langle e, \beta | H | \delta^{-1}, \alpha \rangle. \quad (3452)$$

Therefore, $h(\delta^{-1}) = h^\dagger(\delta)$. This condition will guarantee Hermiticity of every generalized Bloch Hamiltonian. The internal moiré orbital space decomposes into two layers,

$$\mathbb{C}^{\nu_M} = \mathbb{C}^{n_b q_M} \oplus \mathbb{C}^{n_b q_M}. \quad (3453)$$

The group hopping matrix therefore has the block structure

$$h(\delta) = \begin{pmatrix} h_0(\delta) & \tau_{12}(\delta) \\ \tau_{21}(\delta) & h_0(\delta) \end{pmatrix}. \quad (3454)$$

The Hermiticity condition implies $h_0(\delta^{-1}) = h_0^\dagger(\delta)$, and $\tau_{21}(\delta^{-1}) = \tau_{12}^\dagger(\delta)$. All curvature- and twist-dependent tunneling information is contained in the collection of matrices $\{\tau_{12}(\delta)\}_{\delta \in \Gamma_M}$. For comparison, let the translation group be $\Gamma_E = \mathbb{Z}^2$. Every irreducible unitary representation is one dimensional. For a crystal momentum $\mathbf{k} \in \mathbb{T}^2$, the representation is

$$\rho_{\mathbf{k}}(\mathbf{n}) = e^{i\mathbf{k} \cdot \mathbf{n}}. \quad (3455)$$

Substitution into the convolution kernel gives

$$H(\mathbf{k}) = \sum_{\mathbf{n} \in \mathbb{Z}^2} h(\mathbf{n}) e^{i\mathbf{k} \cdot \mathbf{n}}, \quad (3456)$$

which is precisely the ordinary Bloch Hamiltonian. The hyperbolic construction below must reduce exactly to this expression when the translation group becomes Abelian. Let $\rho : \Gamma_M \rightarrow U(d)$ be a finite-dimensional unitary representation. A ρ -automorphic wave function is a vector-valued function obeying

$$\Psi(\gamma x) = \rho(\gamma) \Psi(x). \quad (3457)$$

The representation dimension d is not an additional microscopic orbital degree of freedom. It is the dimension of the translation-representation fiber. The physical internal orbitals remain those contained in the exact moiré fundamental domain. The representation defines the quotient bundle

$$E_\rho = (\mathbb{H}_R^2 \times \mathbb{C}^d) / \Gamma_M, \quad (3458)$$

where the group action is

$$\gamma : (x, v) \mapsto (\gamma x, \rho(\gamma)v). \quad (3459)$$

A section of this rank- d flat unitary bundle is equivalent to a ρ -automorphic function on the universal cover. Thus the generalized Bloch problem may be interpreted as a tight-binding problem on the compact quotient with matrix-valued holonomy around noncontractible cycles. For $d = 1$, the bundle is a flat line bundle. For $d > 1$, it is a genuine higher-rank flat bundle. With the right-convolution convention in Eq. (3450), consider the vector-valued ansatz

$$\Psi_\alpha(\gamma) = \rho(\gamma) u_\alpha, \quad (3460)$$

where $u_\alpha \in \mathbb{C}^d$. Substitution gives

$$\begin{aligned} (H\Psi)_\alpha(\gamma) &= \sum_{\delta, \beta} h_{\alpha\beta}(\delta) \rho(\gamma\delta) u_\beta \\ &= \sum_{\delta, \beta} h_{\alpha\beta}(\delta) \rho(\gamma) \rho(\delta) u_\beta \\ &= \rho(\gamma) \sum_{\delta, \beta} h_{\alpha\beta}(\delta) \rho(\delta) u_\beta. \end{aligned} \quad (3461)$$

The group-coordinate dependence therefore factors out exactly. Define the finite-dimensional representation-space Hamiltonian by

$$H[\rho] = \sum_{\delta \in \Gamma_M} h(\delta) \otimes \rho(\delta). \quad (3462)$$

The generalized Bloch eigenvalue problem is $H[\rho]u = Eu$. The matrix dimension is

$$\dim H[\rho] = \nu_M d = 2n_b q_M d. \quad (3463)$$

For $d = 1$, this reduces to the Abelian hyperbolic Bloch Hamiltonian. For $d > 1$, the translation holonomies are matrix valued and the Bloch Hamiltonian is genuinely non-Abelian. Using the layer decomposition of the group kernel gives

$$H[\rho] = \begin{pmatrix} H_0[\rho] & T_{12}[\rho] \\ T_{21}[\rho] & H_0[\rho] \end{pmatrix}, \quad (3464)$$

where

$$H_0[\rho] = \sum_{\delta} h_0(\delta) \otimes \rho(\delta), \quad (3465)$$

and

$$T_{12}[\rho] = \sum_{\delta} \tau_{12}(\delta) \otimes \rho(\delta). \quad (3466)$$

Similarly,

$$T_{21}[\rho] = \sum_{\delta} \tau_{21}(\delta) \otimes \rho(\delta). \quad (3467)$$

Thus the geometry-to-tunneling mechanism derived in real space survives without modification in every generalized momentum sector. Because the representation is unitary, $\rho(\delta)^\dagger = \rho(\delta^{-1})$. Taking the adjoint gives

$$\begin{aligned} H[\rho]^\dagger &= \sum_{\delta} h^\dagger(\delta) \otimes \rho^\dagger(\delta) \\ &= \sum_{\delta} h(\delta^{-1}) \otimes \rho(\delta^{-1}). \end{aligned} \quad (3468)$$

Changing variables according to $\delta \mapsto \delta^{-1}$ gives $H[\rho]^\dagger = H[\rho]$. Thus every unitary representation sector defines a Hermitian generalized Bloch Hamiltonian. Suppose the group hopping kernel is absolutely summable,

$$\sum_{\delta \in \Gamma_M} \|h(\delta)\| < \infty. \quad (3469)$$

Because $\|\rho(\delta)\| = 1$, one obtains

$$\|H[\rho]\| \leq \sum_{\delta} \|h(\delta) \otimes \rho(\delta)\| = \sum_{\delta} \|h(\delta)\|. \quad (3470)$$

Hence,

$$\sup_{\rho} \|H[\rho]\| \leq \sum_{\delta} \|h(\delta)\| < \infty. \quad (3471)$$

The hyperbolic summability condition derived previously therefore controls all finite-dimensional unitary representation sectors simultaneously. If the hopping range is finite, there exists a finite subset $S \subset \Gamma_M$ such that $h(\delta) = 0$ for $\delta \notin S$. Then

$$H[\rho] = \sum_{\delta \in S} h(\delta) \otimes \rho(\delta). \quad (3472)$$

Thus the generalized Bloch Hamiltonian is a finite non-commutative polynomial in the holonomy matrices. We now derive the Abelian sector. A one-dimensional unitary representation is a character $\chi : \Gamma_M \rightarrow U(1)$. Because the target group is Abelian, $\chi([\gamma_1, \gamma_2]) = 1$ for every pair of group elements. Consequently every character factors through the Abelianization

$$\Gamma_M^{\text{ab}} = \Gamma_M / [\Gamma_M, \Gamma_M]. \quad (3473)$$

For the genus- g_M surface group, $\Gamma_M^{\text{ab}} \simeq \mathbb{Z}^{2g_M}$. Define $\chi(a_j) = e^{ik_{a_j}}$, and $\chi(b_j) = e^{ik_{b_j}}$. The Abelian generalized momentum vector is

$$\mathbf{k} = (k_{a_1}, k_{b_1}, \dots, k_{a_{g_M}}, k_{b_{g_M}}). \quad (3474)$$

Each phase is periodic, $k_\mu \sim k_\mu + 2\pi$. Therefore the Abelian generalized Brillouin zone is

$$\mathcal{B}_1 = \text{Hom}(\Gamma_M, U(1)) \simeq \mathbb{T}^{2g_M}. \quad (3475)$$

This torus is the Jacobian-type momentum space of the compact quotient. Let the Abelianization map be $\mathfrak{a} : \Gamma_M \rightarrow \mathbb{Z}^{2g_M}$. For every group element define $\mathbf{n}(\delta) = \mathfrak{a}(\delta)$. Then the character is

$$\chi_{\mathbf{k}}(\delta) = \exp[i\mathbf{k}^T \mathbf{n}(\delta)]. \quad (3476)$$

The Abelian hyperbolic Bloch Hamiltonian is therefore

$$H(\mathbf{k}) = \sum_{\delta \in \Gamma_M} h(\delta) \exp[i\mathbf{k}^T \mathbf{n}(\delta)]. \quad (3477)$$

This is the direct higher-genus analogue of the Euclidean Fourier series. The eigenvalue problem is

$$H(\mathbf{k})u_n(\mathbf{k}) = E_n(\mathbf{k})u_n(\mathbf{k}). \quad (3478)$$

At each Abelian generalized momentum there are $\nu_M = 2n_b q_M$ energy eigenvalues counting multiplicity. Thus the Abelian theory defines ν_M energy sheets over a $2g_M$ dimensional torus. Using the genus relation,

$$g_M = 1 + q_M(g_0 - 1), \quad (3479)$$

the Abelian momentum-space dimension is

$$D_A = 2g_M = 2 + 2q_M(g_0 - 1). \quad (3480)$$

Hence,

$$D_A = 2 + 2q_M(g_0 - 1). \quad (3481)$$

This is fundamentally different from Euclidean supercell theory. In Euclidean space the Brillouin-zone dimension remains equal to two independently of the supercell index. In the hyperbolic case, the topology of the compact quotient changes with the exact coincidence index, and the dimension of the Abelian holonomy torus increases with the supercell. The result above implies that the exact hyperbolic generalized momentum space must not be visualized as a two-dimensional reciprocal polygon whose area simply shrinks when the real-space supercell grows. The direct-space enlargement $q_M \uparrow$ produces $g_M \uparrow$, and therefore

$$\dim \mathcal{B}_1 = 2g_M \uparrow. \quad (3482)$$

Thus the hyperbolic reciprocal problem undergoes a topological enlargement of holonomy space rather than a simple fixed-dimensional area compression. The Abelian theory has an important structural limitation. Suppose two translations satisfy

$$\delta_1 \delta_2^{-1} \in [\Gamma_M, \Gamma_M]. \quad (3483)$$

Then $\mathbf{n}(\delta_1) = \mathbf{n}(\delta_2)$. Consequently,

$$\chi_{\mathbf{k}}(\delta_1) = \chi_{\mathbf{k}}(\delta_2) \quad (3484)$$

for every Abelian momentum. Therefore the one-dimensional Bloch theory cannot distinguish group elements that differ only by a commutator word.

a. Theorem 26. Abelian commutator blindness. Let $\delta_1, \delta_2 \in \Gamma_M$. The following statements are equivalent:

$$\chi(\delta_1) = \chi(\delta_2) \quad \text{for every } \chi \in \text{Hom}(\Gamma_M, U(1)), \quad (3485)$$

and

$$\delta_1 \delta_2^{-1} \in [\Gamma_M, \Gamma_M]. \quad (3486)$$

Thus one-dimensional hyperbolic Bloch theory detects only the Abelianized translation structure. It does not, in general, reconstruct the full non-Abelian group geometry. We now turn to a representation of rank $d > 1$. Define the generator holonomies

$$A_j = \rho(a_j) \in U(d), \quad (3487)$$

and

$$B_j = \rho(b_j) \in U(d). \quad (3488)$$

The surface-group relation requires

$$\prod_{j=1}^{g_M} A_j B_j A_j^{-1} B_j^{-1} = I_d. \quad (3489)$$

This is the fundamental non-Abelian holonomy constraint. Define the rank- d unitary representation variety by

$$\text{Rep}_d(\Gamma_M) = \text{Hom}(\Gamma_M, U(d)). \quad (3490)$$

Explicitly,

$$\text{Rep}_d(\Gamma_M) = \left\{ (A_1, B_1, \dots, A_{g_M}, B_{g_M}) \in U(d)^{2g_M} : \prod_{j=1}^{g_M} [A_j, B_j] = I_d \right\} \quad (3491)$$

Two representations related by a common unitary conjugation are physically equivalent generalized momentum data. Let $W \in U(d)$. Define

$$\rho^W(\gamma) = W \rho(\gamma) W^{-1}. \quad (3492)$$

The generator matrices transform as $A_j \mapsto W A_j W^{-1}$, and $B_j \mapsto W B_j W^{-1}$. Therefore the physical representation-space coordinate is the conjugacy class $[\rho]$. The rank- d generalized momentum moduli space is consequently

$$\mathcal{B}_d = \text{Hom}^{\text{irr}}(\Gamma_M, U(d)) / U(d), \quad (3493)$$

when attention is restricted to irreducible representations. Reducible representations form additional lower-dimensional or singular strata and can be treated separately. For $d = 1$, conjugation is trivial and $\mathcal{B}_1 = \mathbb{T}^{2g_M}$. For a conjugated representation,

$$\begin{aligned} H[\rho^W] &= \sum_{\delta} h(\delta) \otimes W \rho(\delta) W^{-1} \\ &= (I_{\nu_M} \otimes W) H[\rho] (I_{\nu_M} \otimes W^{-1}). \end{aligned} \quad (3494)$$

Thus,

$$H[\rho^W] = (I_{\nu_M} \otimes W) H[\rho] (I_{\nu_M} \otimes W^{-1}). \quad (3495)$$

Consequently,

$$\text{Spec } H[\rho^W] = \text{Spec } H[\rho]. \quad (3496)$$

The energy spectrum therefore descends from the raw representation variety to the quotient moduli space. We now derive the local real dimension of the smooth irreducible part of $\mathcal{B}_d$. Each unitary matrix contributes d^2 real parameters. There are $2g_M$ generator matrices. The unconstrained parameter count is therefore $2g_M d^2$. The product of commutators has determinant unity, $\det([A, B]) = 1$. Hence the surface relation takes values in $SU(d)$, whose real dimension is $d^2 - 1$. At a regular point, the surface relation therefore impose $d^2 - 1$ independent real constraints. The remaining dimension before quotienting is

$$2g_M d^2 - (d^2 - 1). \quad (3497)$$

For an irreducible unitary representation, Schur's lemma implies that its stabilizer under conjugation consists only of scalar unitary matrices,

$$\text{Stab}(\rho) = U(1)I_d. \quad (3498)$$

The conjugation group has dimension d^2 , while the stabilizer has dimension 1. Thus the conjugacy orbit has dimension $d^2 - 1$. Subtracting this orbit dimension yields

$$\begin{aligned} \dim_{\mathbb{R}} \mathcal{B}_d &= 2g_M d^2 - (d^2 - 1) - (d^2 - 1) \\ &= (2g_M - 2)d^2 + 2. \end{aligned} \quad (3499)$$

Therefore,

$$\dim_{\mathbb{R}} \mathcal{B}_d = (2g_M - 2)d^2 + 2 \quad (3500)$$

at smooth irreducible points. For $d = 1$, this reduces to

$$\dim_{\mathbb{R}} \mathcal{B}_1 = 2g_M, \quad (3501)$$

as required. Using

$$g_M = 1 + q_M(g_0 - 1), \quad (3502)$$

one obtains

$$\dim_{\mathbb{R}} \mathcal{B}_d = 2q_M(g_0 - 1)d^2 + 2. \quad (3503)$$

Thus,

$$\dim_{\mathbb{R}} \mathcal{B}_d = 2q_M(g_0 - 1)d^2 + 2. \quad (3504)$$

The complexity of generalized momentum space therefore grows linearly with the exact moiré index and quadratically with representation rank. If geometric-arithmetic locking establishes

$$q_M(\theta) = \Theta(|\theta|^{-1}), \quad (3505)$$

then for fixed representation rank,

$$\dim_{\mathbb{R}} \mathcal{B}_d = \Theta(d^2|\theta|^{-1}). \quad (3506)$$

This result is conditional on the exact crystallographic locking law. It does not follow from the effective geometric moiré area alone. Every higher-dimensional representation possesses an Abelian shadow. Define

$$\det \rho : \Gamma_M \longrightarrow U(1). \quad (3507)$$

Because the determinant is multiplicative,

$$\det [\rho(\gamma_1 \gamma_2)] = \det \rho(\gamma_1) \det \rho(\gamma_2), \quad (3508)$$

the determinant is a one-dimensional character. Thus there is a natural map $\pi_{\det} : \mathcal{B}_d \longrightarrow \mathcal{B}_1$ defined by $[\rho] \mapsto \det \rho$. The Abelian base has dimension $2g_M$. Subtracting this from the total representation-space dimension gives

$$\begin{aligned} D_{\text{NA}} &= (2g_M - 2)d^2 + 2 - 2g_M \\ &= (2g_M - 2)(d^2 - 1). \end{aligned} \quad (3509)$$

Therefore the genuinely non-Abelian holonomy degrees of freedom have local dimension

$$D_{\text{NA}} = (2g_M - 2)(d^2 - 1). \quad (3510)$$

Using the moiré index gives

$$D_{\text{NA}} = 2q_M(g_0 - 1)(d^2 - 1). \quad (3511)$$

Thus the generalized momentum space separates dimensionally into an Abelian holonomy backbone and a genuinely non-Abelian internal-holonomy sector. A genuinely irreducible higher-dimensional representation cannot be reduced to a set of scalar Bloch phases.

b. Theorem 27. No scalar momentum for irreducible higher-dimensional sectors. Suppose $\rho : \Gamma_M \longrightarrow U(d)$ is irreducible and $d > 1$. Then there exist group elements satisfying

$$[\rho(\gamma_1), \rho(\gamma_2)] \neq 0. \quad (3512)$$

Assume instead that $[\rho(\gamma_1), \rho(\gamma_2)] = 0$ for every pair of group elements. All matrices in the image of the representation are unitary and therefore normal. A commuting family of normal matrices can be simultaneously diagonalized. Hence there exists a basis in which

$$\rho(\gamma) = \text{diag} [\chi_1(\gamma), \dots, \chi_d(\gamma)] \quad (3513)$$

for every group element. Each coordinate axis is then invariant under the complete representation. Thus the representation decomposes into one-dimensional characters, $\rho = \chi_1 \oplus \dots \oplus \chi_d$. This contradicts irreducibility for $d > 1$. Therefore a higher-dimensional irreducible representation must contain noncommuting holonomies. The preceding theorem implies that a genuine non-Abelian generalized momentum cannot be encoded by replacing the Euclidean vector $\mathbf{k}$ with a longer vector of phases. The additional degrees of freedom are intrinsically matrix valued. Thus, $d > 1 \implies$ generalized momentum is holonomy-matrix data rather than phase data. For each smooth representation sector, define the eigenvalues by

$$H[\rho]|u_n[\rho]\rangle = E_n[\rho]|u_n[\rho]\rangle. \quad (3514)$$

Because the matrix dimension is $\nu_M d$, there are locally $\nu_M d$ energy eigenvalues counting multiplicity at

each rank- d representation. The natural generalized band is therefore a function $E_n^{(d)} : \mathcal{B}_d \rightarrow \mathbb{R}$. For $d = 1$, this reduces to an ordinary scalar energy function on the Abelian higher-dimensional torus. The finite-dimensional automorphic sectors form the disjoint hierarchy

$$\mathfrak{B}_{\text{fd}} = \bigsqcup_{d=1}^{\infty} \mathcal{B}_d. \quad (3515)$$

The first component, $\mathcal{B}_1$, is the Abelian hyperbolic Brillouin torus. The components with $d > 1$ contain non-Abelian translation sectors. The symbol in Eq. (3515) should not be interpreted as an assertion that the regular representation of the infinite Fuchsian group decomposes as an ordinary discrete direct sum over these finite-dimensional sectors. The exact complete decomposition of the infinite non-Abelian regular representation is a more subtle harmonic-analysis problem. The finite-dimensional spaces above instead parameterize exact automorphic boundary conditions and become completely controlled on finite quotient systems, as derived below. The exact Euclidean selection rule was ordinary momentum conservation modulo the moiré reciprocal lattice. Its non-Abelian replacement follows directly from symmetry. Let V_ρ and V_σ be two irreducible representation spaces of the common hyperbolic translation group. Let an interlayer operator satisfy $U_\gamma T U_\gamma^{-1} = T$ for every common translation. The inter-sector block

$$T_{\rho\sigma} : V_\sigma \rightarrow V_\rho \quad (3516)$$

must satisfy the intertwining relation

$$\rho(\gamma) T_{\rho\sigma} = T_{\rho\sigma} \sigma(\gamma) \quad (3517)$$

for every group element.

c. Theorem 28. Conservation of the translation irrep. If ρ and σ are inequivalent irreducible representations of Γ_M , then every translation-invariant operator satisfies $T_{\rho\sigma} = 0$. The inter-sector matrix is an intertwiner between two irreducible representations. By Schur's lemma,

$$\text{Hom}_{\Gamma_M}(V_\sigma, V_\rho) = \{0\} \quad (3518)$$

for inequivalent irreducible representations. Therefore the inter-sector matrix element must vanish. Thus the exact non-Abelian analogue of momentum conservation is $\rho_{\text{in}} \simeq \rho_{\text{out}}$. For one-dimensional representations, $\rho = \chi_{\mathbf{k}}$, and $\sigma = \chi_{\mathbf{k}'}$. Equivalence means equality, $\chi_{\mathbf{k}} = \chi_{\mathbf{k}'}$. Therefore, $\mathbf{k} \equiv \mathbf{k}' \pmod{2\pi}$. For $\Gamma = \mathbb{Z}^2$, this reduces further to ordinary Euclidean crystal-momentum conservation modulo a reciprocal-lattice vector. Thus the hierarchy is exact:

$$\mathbf{q}_1 - \mathbf{q}_2 \in \Lambda^* \iff \chi_{\mathbf{k}_1} = \chi_{\mathbf{k}_2} \implies \rho_{\text{in}} \simeq \rho_{\text{out}}. \quad (3519)$$

The strongest exact representation-theoretic decomposition is obtained on a finite periodic cover. Choose a normal finite-index subgroup $\Gamma_N \triangleleft \Gamma_M$. Define the finite quotient group $Q_N = \Gamma_M / \Gamma_N$. Its order is

$$|Q_N| = [\Gamma_M : \Gamma_N]. \quad (3520)$$

The finite-cover Hilbert space is

$$\mathcal{H}_N = \mathbb{C}[Q_N] \otimes \mathbb{C}^{\nu_M}. \quad (3521)$$

Its total dimension is

$$\dim \mathcal{H}_N = \nu_M |Q_N|. \quad (3522)$$

Let $\hat{Q}_N$ denote the set of inequivalent irreducible unitary representations of the finite group. For each representation, $\rho : Q_N \rightarrow U(d_\rho)$. The dimensions satisfy the fundamental identity

$$\sum_{\rho \in \hat{Q}_N} d_\rho^2 = |Q_N|. \quad (3523)$$

This identity is the non-Abelian generalization of counting discrete momenta in an ordinary finite Brillouin zone. For two finite-group irreducible representations, $\rho, \sigma \in \hat{Q}_N$, matrix-element orthogonality gives

$$\sum_{q \in Q_N} \rho(q)_{ij} \sigma(q)_{kl}^* = \frac{|Q_N|}{d_\rho} \delta_{\rho\sigma} \delta_{ik} \delta_{jl}. \quad (3524)$$

This relation replaces orthogonality of Euclidean plane waves. For a scalar function on the quotient group, $f : Q_N \rightarrow \mathbb{C}$, define the group Fourier transform by

$$\hat{f}(\rho) = \sqrt{\frac{d_\rho}{|Q_N|}} \sum_{q \in Q_N} f(q) \rho(q). \quad (3525)$$

The transformed object is a $d_\rho \times d_\rho$ matrix rather than a scalar Fourier coefficient. Schur orthogonality gives the Parseval identity

$$\sum_{q \in Q_N} |f(q)|^2 = \sum_{\rho \in \hat{Q}_N} \left\| \hat{f}(\rho) \right\|_F^2. \quad (3526)$$

The finite group algebra decomposes according to

$$\mathbb{C}[Q_N] \simeq \bigoplus_{\rho \in \hat{Q}_N} V_\rho \otimes V_\rho^*. \quad (3527)$$

Each irreducible representation appears with multiplicity d_ρ . The dimension count is

$$\dim \mathbb{C}[Q_N] = \sum_{\rho} d_{\rho} d_{\rho} = \sum_{\rho} d_{\rho}^2 = |Q_N|. \quad (3528)$$

Because the finite-cover Hamiltonian commutes with the left regular action, it acts trivially on one of the representation factors. Thus,

$$H_N \simeq \bigoplus_{\rho \in \hat{Q}_N} [I_{d_{\rho}} \otimes H[\rho]]. \quad (3529)$$

The generalized Bloch block has dimension

$$\dim H[\rho] = \nu_M d_{\rho}. \quad (3530)$$

Every eigenvalue of this block appears in the full finite-cover spectrum with multiplicity at least d_{ρ} . The number of states contributed by the sector is

$$d_{\rho} (\nu_M d_{\rho}) = \nu_M d_{\rho}^2. \quad (3531)$$

Summing over all sectors gives

$$N_{\text{states}} = \nu_M \sum_{\rho} d_{\rho}^2 = \nu_M |Q_N|. \quad (3532)$$

Therefore,

$$N_{\text{states}} = \nu_M |Q_N|. \quad (3533)$$

This exactly reproduces the real-space finite-cover Hilbert-space dimension. If the finite translation quotient is Abelian, every irreducible representation is one dimensional, $d_{\rho} = 1$. Then

$$\sum_{\rho} 1 = |Q_N|. \quad (3534)$$

Hence the decomposition contains $|Q_N|$ ordinary scalar momentum sectors. This is precisely conventional Euclidean finite-size Bloch theory. In a non-Abelian quotient, the number of distinct representation sectors need not equal the number of cells. The missing sector count is compensated exactly by the squared irrep dimensions. The isotypic component associated with a finite-group irrep can be selected explicitly. Let the irreducible character be

$$\chi_{\rho}(q) = \text{Tr } \rho(q). \quad (3535)$$

Define

$$P_{\rho} = \frac{d_{\rho}}{|Q_N|} \sum_{q \in Q_N} \chi_{\rho}(q^{-1}) U_q. \quad (3536)$$

Character orthogonality gives $P_{\rho}^2 = P_{\rho}$, $P_{\rho}^{\dagger} = P_{\rho}$, and $P_{\rho} P_{\sigma} = 0$ for $\rho \neq \sigma$. Completeness gives

$$\sum_{\rho \in \hat{Q}_N} P_{\rho} = I. \quad (3537)$$

Since the Hamiltonian commutes with all translations, $[H, P_{\rho}] = 0$. Thus the finite-cover Hamiltonian can be decomposed exactly without any approximate momentum assignment. Let the eigenvalues of the block be

$$E_{n,\rho}, \quad n = 1, \dots, \nu_M d_{\rho}. \quad (3538)$$

Each occurs with multiplicity d_{ρ} . The normalized finite-cover density of states is therefore

$$\rho_N(E) = \frac{1}{\nu_M |Q_N|} \sum_{\rho \in \hat{Q}_N} d_{\rho} \sum_{n=1}^{\nu_M d_{\rho}} \delta(E - E_{n,\rho}). \quad (3539)$$

The weight of an entire representation sector is proportional to d_{ρ}^2 . This fact is essential when comparing Abelian and non-Abelian numerical spectra. For any sufficiently regular function of the Hamiltonian, $F(H)$, the full finite-cover trace decomposes as

$$\text{Tr}_{\mathcal{H}_N} F(H_N) = \sum_{\rho \in \hat{Q}_N} d_{\rho} \text{Tr } F(H[\rho]). \quad (3540)$$

This identity is the exact non-Abelian analogue of summing a Euclidean band quantity over discrete momenta. For a general representation define the Wilson character of a group word $w \in \Gamma_M$ by $W_{\rho}(w) = \text{Tr } \rho(w)$. Under representation-gauge conjugation,

$$\rho(w) \mapsto W \rho(w) W^{-1}, \quad (3541)$$

and therefore $W_{\rho}(w)$ is gauge invariant. It is also invariant under conjugation of the group word,

$$W_{\rho}(\gamma w \gamma^{-1}) = W_{\rho}(w). \quad (3542)$$

Thus Wilson characters naturally encode the non-Abelian holonomy information accessible to spectral invariants. Using

$$H[\rho] = \sum_{\delta} h(\delta) \otimes \rho(\delta), \quad (3543)$$

the second power is

$$H[\rho]^2 = \sum_{\delta_1, \delta_2} h(\delta_1) h(\delta_2) \otimes \rho(\delta_1 \delta_2). \quad (3544)$$

Iterating gives

$$H[\rho]^n = \sum_{\delta_1, \dots, \delta_n} h(\delta_1) \cdots h(\delta_n) \otimes \rho(\delta_1 \cdots \delta_n). \quad (3545)$$

Taking the trace over orbital and representation spaces gives

$$\begin{aligned} \text{Tr } H[\rho]^n &= \sum_{\delta_1, \dots, \delta_n} \text{Tr}_{\text{orb}} [h(\delta_1) \cdots h(\delta_n)] \\ &\times W_\rho(\delta_1 \cdots \delta_n). \end{aligned} \quad (3546)$$

This identity is exact.

d. Theorem 29. Representation dependence of spectral moments. Every finite spectral moment of the generalized Bloch Hamiltonian can be written as a sum of real-space hopping-path amplitudes multiplied by Wilson characters of the corresponding translation words. Thus representation-space dispersion is generated by the interference between microscopic hopping paths carrying different non-Abelian holonomies. For $d = 1$, the Wilson factor reduces to the scalar phase

$$W_{\chi\mathbf{k}}(w) = \exp[i\mathbf{k}^T \mathbf{n}(w)]. \quad (3547)$$

For $d > 1$, the same spectral moment depends on traces of noncommuting matrix holonomies. This is the precise representation-theoretic origin of non-Abelian hyperbolic band dispersion. Suppose two hopping paths generate words w_1 and w_2 having the same Abelianization, $\mathbf{n}(w_1) = \mathbf{n}(w_2)$. Then every Abelian character assigns identical phases,

$$W_{\chi\mathbf{k}}(w_1) = W_{\chi\mathbf{k}}(w_2). \quad (3548)$$

A higher-dimensional representation can nevertheless satisfy

$$W_\rho(w_1) \neq W_\rho(w_2). \quad (3549)$$

Thus genuinely non-Abelian Bloch sectors can distinguish interference processes that are invisible to every Abelian generalized momentum. We now derive the representation-theoretic meaning of hyperbolic supercells. Let $\Gamma_{\text{sc}} < \Gamma_M$ be a finite-index subgroup. Define the supercell index $N_{\text{sc}} = [\Gamma_M : \Gamma_{\text{sc}}]$. A scalar Abelian supercell boundary condition is specified by a character $\chi : \Gamma_{\text{sc}} \rightarrow U(1)$. The corresponding representation of the original translation group is the induced representation $\text{Ind}_{\Gamma_{\text{sc}}}^{\Gamma_M} \chi$. Its dimension is

$$\dim \text{Ind}_{\Gamma_{\text{sc}}}^{\Gamma_M} \chi = N_{\text{sc}}. \quad (3550)$$

Thus an Abelian Bloch state of a hyperbolic supercell corresponds, from the viewpoint of the smaller primitive translation group, to a generally higher-dimensional representation.

e. Theorem 30. Non-Abelian zone folding by supercells. Let $\Gamma_{\text{sc}} < \Gamma_M$ be a finite-index hyperbolic supercell translation group and let χ be an Abelian character of the supercell group. Then the supercell Bloch problem is representation-theoretically equivalent to the primitive-cell Hamiltonian evaluated in the induced representation

$$\rho_{\text{ind}} = \text{Ind}_{\Gamma_{\text{sc}}}^{\Gamma_M} \chi. \quad (3551)$$

If the induced representation decomposes according to

$$\rho_{\text{ind}} \simeq \bigoplus_{\rho} m_{\rho}(\chi) \rho, \quad (3552)$$

then the corresponding supercell Bloch Hamiltonian decomposes into the same primitive-group non-Abelian Bloch sectors. The multiplicity is determined by Frobenius reciprocity,

$$m_{\rho}(\chi) = \dim \text{Hom}_{\Gamma_{\text{sc}}} \left(\text{Res}_{\Gamma_{\text{sc}}}^{\Gamma_M} \rho, \chi \right). \quad (3553)$$

Thus hyperbolic supercell calculations are not merely large real-space approximations. They provide a systematic representation-theoretic mechanism for accessing higher-dimensional non-Abelian Bloch sectors. For the trivial character, $\chi = 1$, and a normal supercell subgroup, $\Gamma_{\text{sc}} \triangleleft \Gamma_M$, the induced representation reduces to the regular representation of the finite quotient,

$$\text{Ind}_{\Gamma_{\text{sc}}}^{\Gamma_M} 1 \simeq \mathbb{C} [\Gamma_M / \Gamma_{\text{sc}}]. \quad (3554)$$

The regular-representation decomposition derived above is therefore the zero-supercell-momentum limit of the more general induced-representation construction. For an Abelian Euclidean translation group, every induced representation decomposes into one-dimensional characters. Thus,

$$\text{Ind}_{\Lambda_M}^{\Lambda} \chi_{\mathbf{k}} = \bigoplus_{\alpha=1}^{\Sigma} \chi_{\mathbf{k}+\mathbf{Q}_{\alpha}}. \quad (3555)$$

This is precisely the ordinary Euclidean relation

$$E_{\alpha}^{\text{fold}}(\mathbf{k}) = \varepsilon(\mathbf{k} + \mathbf{Q}_{\alpha}). \quad (3556)$$

The non-Abelian supercell construction is therefore an exact generalization of conventional band folding. To formulate dispersion in higher-rank representation space, consider a smooth path of representations,

$$\rho_s : \Gamma_M \rightarrow U(d), \quad (3557)$$

with $\rho_0 = \rho$. Define the right logarithmic derivative

$$X(\gamma) = \left. \frac{d\rho_s(\gamma)}{ds} \right|_{s=0} \rho(\gamma)^{-1}. \quad (3558)$$

Because the representation is unitary, $X(\gamma) \in \mathfrak{u}(d)$. The representation property is

$$\rho_s(\gamma\delta) = \rho_s(\gamma)\rho_s(\delta). \quad (3559)$$

Differentiate and multiply by

$$\rho(\gamma\delta)^{-1} = \rho(\delta)^{-1}\rho(\gamma)^{-1}. \quad (3560)$$

One obtains

$$X(\gamma\delta) = X(\gamma) + \rho(\gamma)X(\delta)\rho(\gamma)^{-1}. \quad (3561)$$

Thus,

$$X(\gamma\delta) = X(\gamma) + \text{Ad}_{\rho(\gamma)} X(\delta). \quad (3562)$$

This is the one-cocycle condition for the adjoint representation. Consider a pure gauge deformation

$$\rho_s(\gamma) = e^{sY} \rho(\gamma) e^{-sY}, \quad (3563)$$

where $Y \in \mathfrak{u}(d)$. Differentiation gives

$$\dot{\rho}(\gamma) = Y\rho(\gamma) - \rho(\gamma)Y. \quad (3564)$$

Therefore,

$$X_Y(\gamma) = Y - \text{Ad}_{\rho(\gamma)} Y. \quad (3565)$$

Such cocycles are coboundaries. Let $Z^1(\Gamma_M, \text{Ad } \rho)$ denote the space of cocycles and $B^1(\Gamma_M, \text{Ad } \rho)$ the space of gauge coboundaries. At a smooth irreducible point, the physical tangent space is

$$T_{[\rho]} \mathcal{B}_d \simeq H^1(\Gamma_M, \text{Ad } \rho) = \frac{Z^1}{B^1}. \quad (3566)$$

Thus generalized momentum variations are cohomology classes of matrix-valued infinitesimal holonomy deformations. Differentiate

$$H[\rho_s] = \sum_{\delta} h(\delta) \otimes \rho_s(\delta). \quad (3567)$$

Using

$$\dot{\rho}(\delta) = X(\delta)\rho(\delta), \quad (3568)$$

one obtains

$$\dot{H}_X = \sum_{\delta} h(\delta) \otimes X(\delta)\rho(\delta). \quad (3569)$$

This is the non-Abelian analogue of differentiating a Euclidean Bloch Hamiltonian with respect to crystal momentum. For a gauge coboundary,

$$X_Y(\delta) = Y - \rho(\delta)Y\rho(\delta)^{-1}. \quad (3570)$$

Then

$$\begin{aligned} \dot{H}_{X_Y} &= \sum_{\delta} h(\delta) \otimes [Y\rho(\delta) - \rho(\delta)Y] \\ &= [I_{\nu_M} \otimes Y, H[\rho]]. \end{aligned} \quad (3571)$$

Thus,

$$\dot{H}_{X_Y} = [I_{\nu_M} \otimes Y, H[\rho]]. \quad (3572)$$

Pure gauge motion therefore changes the matrix representation but not its spectrum. Let a nondegenerate generalized Bloch eigenstate satisfy

$$H[\rho]|u_n\rangle = E_n[\rho]|u_n\rangle. \quad (3573)$$

Differentiation gives the Hellmann–Feynman relation

$$\frac{dE_n}{ds} = \langle u_n | \dot{H}_X | u_n \rangle. \quad (3574)$$

Using the explicit derivative,

$$\frac{dE_n}{ds} = \sum_{\delta} \langle u_n | h(\delta) \otimes X(\delta)\rho(\delta) | u_n \rangle. \quad (3575)$$

This defines the first derivative of the energy along a physical representation-space tangent direction. For a pure gauge tangent, $\dot{H} = [I \otimes Y, H]$. Hence,

$$\begin{aligned} \frac{dE_n}{ds} &= \langle u_n | [I \otimes Y, H] | u_n \rangle \\ &= E_n \langle u_n | I \otimes Y | u_n \rangle - E_n \langle u_n | I \otimes Y | u_n \rangle = 0. \end{aligned} \quad (3576)$$

Therefore, dE_n annihilates all gauge directions and is a well-defined covector on $T_{[\rho]} \mathcal{B}_d$. Define the directional spectral response by

$$v_n[X] = \frac{1}{\hbar} dE_n[X]. \quad (3577)$$

Explicitly,

$$v_n[X] = \frac{1}{\hbar} \langle u_n | \dot{H}_X | u_n \rangle. \quad (3578)$$

This quantity should be interpreted as a generalized dispersion response in holonomy space. For $d > 1$, it is not, in general, an ordinary two-dimensional physical velocity vector. Its natural mathematical object is a covector on the representation moduli space. For an Abelian character,

$$\rho_{\mathbf{k}}(\delta) = e^{i\mathbf{k}^T \mathbf{n}(\delta)}. \quad (3579)$$

Consider a path $\mathbf{k}(s) = \mathbf{k} + s\mathbf{v}$. Then

$$\dot{\rho}(\delta) = i\mathbf{v}^T \mathbf{n}(\delta) \rho(\delta). \quad (3580)$$

Therefore,

$$X(\delta) = i\mathbf{v}^T \mathbf{n}(\delta). \quad (3581)$$

The Hamiltonian derivative becomes

$$\dot{H} = i \sum_{\delta} [\mathbf{v}^T \mathbf{n}(\delta)] h(\delta) e^{i\mathbf{k}^T \mathbf{n}(\delta)}. \quad (3582)$$

This is precisely

$$\dot{H} = \mathbf{v}^T \nabla_{\mathbf{k}} H(\mathbf{k}). \quad (3583)$$

Thus ordinary group velocity is recovered as the rank-one special case of the representation-space differential. Let the representation follow a smooth path and let the band remain nondegenerate. Standard second-order perturbation theory gives

$$\frac{d^2 E_n}{ds^2} = \langle u_n | \ddot{H} | u_n \rangle + 2 \sum_{m \neq n} \frac{|\langle u_m | \dot{H} | u_n \rangle|^2}{E_n - E_m}. \quad (3584)$$

The first term is the direct curvature of the representation-dependent Hamiltonian. The second term is the virtual mixing with all other generalized Bloch bands. This is the non-Abelian analogue of the inverse-effective-mass formula. For two independent tangent directions $X, Y \in T_{[\rho]} \mathcal{B}_d$, one may define a bilinear spectral Hessian

$$\mathcal{H}_n(X, Y) = \nabla^2 E_n(X, Y). \quad (3585)$$

At a stationary representation-space point, $dE_n = 0$, the Hessian controls the leading dispersion. A representation-space flat state therefore requires suppression not merely along one arbitrarily chosen path but

across all physically relevant tangent directions. Because the representation variety has no unique Euclidean coordinate system, the statement

$$\frac{\partial E}{\partial x_i} = 0 \quad (3586)$$

in a particular parameterization is insufficient by itself. The coordinate-independent first-order condition is

$$dE_n = 0 \quad \text{as a covector on } T_{[\rho]} \mathcal{B}_d. \quad (3587)$$

Equivalently, $v_n[X] = 0$ for every physical tangent class

$$[X] \in H^1(\Gamma_M, \text{Ad } \rho). \quad (3588)$$

This condition will later provide the correct higher-rank generalization of vanishing group velocity. A scalar quantity such as $|\nabla E|$ requires a metric on representation space. Thus a claim involving the magnitude of a non-Abelian generalized velocity must specify a metric G_ρ on $T_{[\rho]} \mathcal{B}_d$. By contrast, the condition $dE = 0$ is metric independent. Therefore the most robust theoretical definition of an exact stationary generalized band point is the vanishing of the full energy differential, not the vanishing of a coordinate-dependent norm. For an isolated nondegenerate generalized band, choose local coordinates λ^μ on a smooth patch of $\mathcal{B}_d$. Define the Berry connection by

$$\mathcal{A}_\mu^{(n)} = i \langle u_n | \partial_\mu u_n \rangle. \quad (3589)$$

The Berry curvature is

$$\mathcal{F}_{\mu\nu}^{(n)} = \partial_\mu \mathcal{A}_\nu^{(n)} - \partial_\nu \mathcal{A}_\mu^{(n)}. \quad (3590)$$

For a degenerate multiplet, the connection becomes matrix valued,

$$[\mathcal{A}_\mu]_{ab} = i \langle u_a | \partial_\mu u_b \rangle. \quad (3591)$$

Thus non-Abelian translation holonomy and non-Abelian Berry transport are conceptually distinct structures and must not be conflated. The former labels generalized crystal momentum. The latter describes adiabatic transport of eigenstates over the representation moduli space. For an isolated state, define the projector

$$P_n = |u_n\rangle \langle u_n|. \quad (3592)$$

The quantum metric is

$$g_{\mu\nu}^{(n)} = \text{Re} [\langle \partial_\mu u_n | (I - P_n) | \partial_\nu u_n \rangle]. \quad (3593)$$

This quantity provides a natural local measure of how rapidly the generalized Bloch eigenstate changes in representation space. It will later be useful for distinguishing energetic flatness from wave-function rigidity. The real-space mixed curvature–twist response derived previously is $\delta_{K\theta}h(\delta)$. Its representation-space transform is

$$\delta_{K\theta}H[\rho] = \sum_{\delta} \delta_{K\theta}h(\delta) \otimes \rho(\delta). \quad (3594)$$

The norm satisfies

$$\|\delta_{K\theta}H[\rho]\| \leq \sum_{\delta} \|\delta_{K\theta}h(\delta)\|. \quad (3595)$$

Therefore,

$$\sup_{\rho} \|\delta_{K\theta}H[\rho]\| \leq \sum_{\delta} \|\delta_{K\theta}h(\delta)\|. \quad (3596)$$

The microscopic curvature-assisted tunneling reconstruction is therefore uniformly inherited by all finite-dimensional unitary representation sectors. Because rigid twisting leaves the intralayer kernel unchanged, $\partial_{\theta}h_0(\delta) = 0$. Therefore,

$$\partial_{\theta}H[\rho] = \begin{pmatrix} 0 & \partial_{\theta}T_{12}[\rho] \\ \partial_{\theta}T_{21}[\rho] & 0 \end{pmatrix}. \quad (3597)$$

Thus the earlier microscopic conclusion survives exactly: the twist dependence of every generalized Bloch Hamiltonian is generated through its interlayer block. Let an unhybridized representation-sector Hamiltonian be

$$H_0[\rho] = \begin{pmatrix} H_1[\rho] & 0 \\ 0 & H_2[\rho] \end{pmatrix}. \quad (3598)$$

The full Hamiltonian is

$$H[\rho] = H_0[\rho] + \begin{pmatrix} 0 & T[\rho] \\ T^{\dagger}[\rho] & 0 \end{pmatrix}. \quad (3599)$$

Thus hybridization occurs separately inside every conserved translation irrep. There is no coupling between inequivalent generalized momenta. This is the precise hyperbolic replacement of the statement that an exact Euclidean moiré Hamiltonian decomposes into independent $\mathbf{k}$ -blocks. The exact eigenvalue equation is

$$\begin{pmatrix} E - H_1[\rho] & -T[\rho] \\ -T^{\dagger}[\rho] & E - H_2[\rho] \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = 0. \quad (3600)$$

If the second block is invertible,

$$u_2 = [E - H_2[\rho]]^{-1} T^{\dagger}[\rho] u_1. \quad (3601)$$

Substitution gives

$$[E - H_1[\rho] - \Sigma_1(E, [\rho])] u_1 = 0, \quad (3602)$$

where

$$\Sigma_1(E, [\rho]) = T[\rho] [E - H_2[\rho]]^{-1} T^{\dagger}[\rho]. \quad (3603)$$

The self-energy is now a function over representation moduli rather than ordinary reciprocal momentum. The energy differential becomes schematically

$$dE = dE_{\text{bare}} + dE_{\text{hyb}}. \quad (3604)$$

In representation-space language, magic-band formation requires hybridization-induced cancellation of the bare energy differential,

$$dE_{\text{hyb}} \simeq -dE_{\text{bare}} \quad (3605)$$

over the physically relevant tangent directions. Thus the Euclidean velocity-cancellation mechanism generalizes naturally from a vector equation to a covector equation on representation moduli. In the Abelian sector, first-order stationary dispersion requires

$$\frac{\partial E}{\partial k_{\mu}} = 0 \quad (3606)$$

for $\mu = 1, \dots, 2g_M$. In the rank- d sector, the coordinate-independent condition is

$$dE = 0 \quad \text{on} \quad H^1(\Gamma_M, \text{Ad } \rho). \quad (3607)$$

Thus higher-rank flatness is intrinsically a much stronger multidirectional constraint. At a smooth rank- d point, the number of independent first-order representation directions is

$$D_d = (2g_M - 2)d^2 + 2. \quad (3608)$$

A generic stationary condition formally imposes D_d first-derivative constraints. Therefore the occurrence of a fully stationary generalized band over a large representation space is highly nongeneric unless additional structure correlates the derivatives. This observation provides a strong reason to search for a geometric mechanism that simultaneously renormalizes many representation directions, rather than interpreting isolated low-velocity cuts as evidence of a global magic state. Suppose a numerical calculation follows a one-dimensional path $[\rho(s)] \subset \mathcal{B}_d$. Vanishing derivative along that path gives only

$$\frac{dE}{ds} = 0. \quad (3609)$$

This implies $dE[X] = 0$ for one tangent vector X . It does not imply $dE = 0$ on the full tangent space. Therefore

a visually flat dispersion along a selected representation-space path is not sufficient to establish a genuinely flat hyperbolic band. A sequence of normal finite covers $\Gamma_N \triangleleft \Gamma_M$ with injectivity radius satisfying $r_{\text{inj}}^{(N)} \rightarrow \infty$ provides increasingly accurate local realizations of the infinite hyperbolic crystal. Each quotient accesses exactly the irreducible representations that factor through

$$Q_N = \Gamma_M / \Gamma_N. \quad (3610)$$

The union of such sectors gives a systematic finite-quotient hierarchy. However, no assumption is made here that finite-image irreducible representations are automatically dense in every component of the full continuous unitary character variety. That is a separate representation-theoretic question. The finite-cover construction is used because it gives an exact finite decomposition and a controlled local thermodynamic limit, not because a stronger unproved representation-density statement is required. For a generic incommensurate twist, the common translation subgroup has infinite index,

$$[\Gamma : \Gamma \cap g_\theta \Gamma g_\theta^{-1}] = \infty. \quad (3611)$$

There is then no finite exact moiré quotient Σ_M and no finite exact moiré translation group from which a global compact representation-space Bloch formulation of the type above can be built. The local Hamiltonian remains perfectly well defined. Its spectral analysis must instead use either commensurate approximants or paired finite-cover constructions. Thus, generalized hyperbolic Bloch decomposition and local incommensurate spectral theory must be treated as distinct but compatible frameworks. If the common crystallographic group contains torsion, the quotient is an orbifold rather than a smooth surface. The group presentation then contains additional elliptic generators, $c_\alpha^{m_\alpha} = e$. A unitary representation must satisfy

$$\rho(c_\alpha)^{m_\alpha} = I_d. \quad (3612)$$

The generalized Bloch Hamiltonian retains the form

$$H[\rho] = \sum_{\delta} h(\delta) \otimes \rho(\delta). \quad (3613)$$

Only the allowed representation variety is modified by the additional orbifold constraints. Thus the non-Abelian formulation is not restricted conceptually to torsion-free crystal groups.

f. Theorem 31. Representation-space Bloch reduction. Let the exact hyperbolic moiré translation group be a torsion-free cocompact Fuchsian group and let the microscopic bilayer Hamiltonian be translation invariant with absolutely summable matrix-valued hopping kernel $h(\delta)$. For every finite-dimensional unitary representation $\rho : \Gamma_M \rightarrow U(d)$, the automorphic eigenvalue problem reduces exactly to the finite matrix

$$H[\rho] = \sum_{\delta \in \Gamma_M} h(\delta) \otimes \rho(\delta). \quad (3614)$$

The matrix is Hermitian and satisfies

$$\dim H[\rho] = 2n_b q_M d. \quad (3615)$$

Representations related by unitary conjugation yield unitarily equivalent Hamiltonians. Consequently the energy spectrum is a function of the moduli point $[\rho] \in \mathcal{B}_d$. For $d = 1$, the construction reduces to the Abelian higher-genus Bloch Hamiltonian on $\mathbb{T}^{2g_M}$. For $d > 1$, it defines genuinely non-Abelian generalized bands.

g. Theorem 32. Non-Abelian translation selection rule. Let a bilayer coupling preserve the exact common translation group. Then it cannot couple two inequivalent irreducible representations of that group. Therefore the exact selection rule is $\rho_{\text{initial}} \simeq \rho_{\text{final}}$. For one-dimensional representations this reduces to equality of Abelian hyperbolic crystal momenta. For an ordinary Euclidean translation group it reduces to crystal-momentum conservation modulo reciprocal-lattice vectors. Thus representation equivalence is the fundamental form of translational selection, and ordinary momentum conservation is its Abelian special case.

h. Theorem 33. Moiré growth of generalized momentum space. At a smooth irreducible rank- d representation of an exact genus- g_M hyperbolic moiré quotient,

$$\dim_{\mathbb{R}} \mathcal{B}_d = (2g_M - 2)d^2 + 2. \quad (3616)$$

Using the exact covering relation,

$$g_M = 1 + q_M(g_0 - 1), \quad (3617)$$

this becomes

$$\dim_{\mathbb{R}} \mathcal{B}_d = 2q_M(g_0 - 1)d^2 + 2. \quad (3618)$$

Therefore exact moiré enlargement increases not only the number of microscopic orbitals in the fundamental domain but also the dimensionality of generalized momentum space. If the exact coincidence index inherits the curvature-dominated law

$$q_M = \Theta(|\theta|^{-1}), \quad (3619)$$

then for fixed rank

$$\dim_{\mathbb{R}} \mathcal{B}_d = \Theta(d^2 |\theta|^{-1}). \quad (3620)$$

This behavior has no direct Euclidean supercell analogue.

i. Theorem 34. Holonomy-controlled generalized dispersion. Let the representation vary smoothly along a physical tangent cocycle

$$[X] \in H^1(\Gamma_M, \text{Ad } \rho). \quad (3621)$$

Then the generalized Bloch Hamiltonian varies according to

$$\dot{H}_X = \sum_{\delta} h(\delta) \otimes X(\delta) \rho(\delta). \quad (3622)$$

For a nondegenerate band,

$$dE_n[X] = \langle u_n | \dot{H}_X | u_n \rangle. \quad (3623)$$

Gauge coboundaries generate only unitary conjugation and satisfy $dE_n[X_{\text{gauge}}] = 0$. Hence the generalized velocity is a well-defined covector on the representation moduli space. The Abelian ordinary momentum derivative is recovered exactly for $d = 1$.

j. Theorem 35. Supercell access to non-Abelian Bloch sectors. Let a finite-index supercell group satisfy

$$\Gamma_{\text{sc}} < \Gamma_M. \quad (3624)$$

A one-dimensional supercell Bloch character

$$\chi : \Gamma_{\text{sc}} \longrightarrow U(1) \quad (3625)$$

defines an induced representation of the primitive translation group,

$$\rho_{\text{ind}} = \text{Ind}_{\Gamma_{\text{sc}}}^{\Gamma_M} \chi, \quad (3626)$$

with dimension equal to the supercell index. Its irreducible decomposition gives precisely the primitive-group generalized Bloch sectors contained in that supercell calculation. For a Euclidean Abelian translation group all components are one-dimensional and the theorem reduces to conventional zone folding. For a non-Abelian Fuchsian translation group the decomposition may contain higher-dimensional irreducible representations. Thus non-Abelian hyperbolic Bloch states arise naturally as the representation-theoretic content of hyperbolic supercell folding. The generalized reciprocal-space construction begins with the exact common translation group. The group determines the compact quotient surface and its fundamental group relations. Its one-dimensional representations determine the Abelian Jacobian momentum torus, while its higher-dimensional irreducible representations determine non-Abelian holonomy moduli. The microscopic hopping kernel is transformed into each representation sector by the exact matrix-valued group Fourier rule

$$H[\rho] = \sum_{\delta} h(\delta) \otimes \rho(\delta). \quad (3627)$$

The interlayer coupling respects the same representation label, so hybridization occurs inside a fixed generalized momentum sector. The spectrum therefore depends simultaneously on the microscopic orbital Hamiltonian and on the non-Abelian holonomy structure of the translation group. The Euclidean theory is characterized by an Abelian translation group, $\Gamma_E = \mathbb{Z}^2$, one-dimensional irreducible representations, $\rho_{\mathbf{k}} = e^{i\mathbf{k} \cdot \mathbf{R}}$, and a fixed two-dimensional reciprocal torus. The hyperbolic theory instead possesses a surface group,

$$\Gamma_M = \pi_1(\Sigma_{g_M}), \quad (3628)$$

an Abelian representation sector of dimension $2g_M$, and higher-rank representation spaces with dimension

$$(2g_M - 2)d^2 + 2. \quad (3629)$$

Therefore the replacement

$$\mathbb{T}^2 \longrightarrow \mathbb{T}^{2g_M} \longrightarrow \mathcal{B}_d \quad (3630)$$

is not merely a higher-dimensional version of Euclidean reciprocal space. The final step introduces noncommuting holonomy data that cannot be represented by any scalar momentum vector. The representation-space theory substantially strengthens the later definition of a hyperbolic magic band. It is not sufficient to identify a small slope along one arbitrary one-dimensional path through an Abelian momentum torus. It is also not sufficient to identify a narrow spectrum only in the one-dimensional representation sector. A genuine hyperbolic magic-band claim must specify which representation sectors are being tested and must distinguish three increasingly strong statements. The weakest statement is Abelian path flattening. The intermediate statement is flattening throughout the Abelian higher-genus Brillouin torus. The strongest finite-dimensional statement is suppression of dispersion across the physically relevant directions of higher-rank representation moduli. At first order, the strongest condition is $dE_n^{(d)} = 0$ on the relevant representation-space tangent directions. The curvature-controlled magic mechanism must therefore ultimately be tested against the generalized differential

$$dE = dE_{\text{bare}} + dE_{\text{hyb}}, \quad (3631)$$

and not merely against a visually small slope in a selected plot. The exact crystallographic route is determined by the common Fuchsian translation group. The topology of its quotient fixes the number of noncontractible cycles. These cycles support Abelian and non-Abelian unitary holonomies. The holonomies define the generalized momentum sector, and the microscopic hopping kernel becomes a finite representation-space Hamiltonian. Independently, curvature reconstructs the real-space interlayer tunneling matrix. That reconstructed tunneling kernel is then transformed into every allowed representation sector through the same group Fourier rule. The resulting theoretical structure can therefore be expressed in prose as follows. Negative curvature first determines the hyperbolic crystal group and the exact common moiré translation subgroup. The topology of the corresponding compact quotient determines the representation space in which generalized crystal momentum lives. The same geometry independently modifies the real-space registry field and therefore the interlayer tunneling amplitudes. These two geometric effects meet inside the representation-space Hamiltonian $H[\rho]$, where symmetry fixes the allowed sector and tunneling reconstruction determines the dispersion within that sector. Magic-band formation can then be identified only after

the representation-theoretic multiplicities have been resolved and the hybridization-induced energy differential has been shown to suppress the bare dispersion over the relevant generalized momentum directions. This separation establishes the mathematical foundation required for a curvature-controlled theory of hyperbolic magic bands.

F. Numerical Spectral Framework and Controlled Approximants

The geometric and representation-theoretic constructions developed above define the twisted hyperbolic bilayer as an infinite quantum system. A numerical calculation, however, necessarily replaces this infinite problem by a sequence of finite matrices. In hyperbolic geometry this replacement is substantially more delicate than in an ordinary Euclidean crystal. The difficulty is not merely computational. A naive open truncation of a hyperbolic lattice contains an extensive boundary, a finite commensurate supercell samples only a restricted set of generalized translation representations, an insufficient hopping cutoff can generate artificial band flattening, and an angular commensurate approximant can cease to approximate the target twist exponentially rapidly with increasing spatial radius. Consequently, numerical convergence cannot be established by increasing the number of sites alone. The purpose of the present subsection is to construct a controlled hierarchy of finite approximants and to identify quantitative criteria under which geometric, spectral, and representation-theoretic observables may be regarded as converged. The primary numerical object will be a boundaryless periodic finite cover generated by a finite quotient of the exact hyperbolic moiré translation group. Open clusters will be used only as auxiliary real-space diagnostics and will require an explicit boundary-state analysis. The numerical architecture is organized around the sequence

$$\Gamma_M \longrightarrow \Gamma_N \longrightarrow Q_N = \Gamma_M / \Gamma_N \longrightarrow \Sigma_N \longrightarrow H_N \longrightarrow \text{Spec } H_N \quad (3632)$$

where Γ_N is a finite-index normal subgroup of the exact moiré translation group, Q_N is the corresponding finite quotient, and Σ_N is the associated compact periodic cover. The numerical calculation is then supplemented by the independent control chain hopping cutoff $\longrightarrow$ cover injectivity radius $\longrightarrow$ solver residual $\longrightarrow$ symmetry verification $\longrightarrow$ cover convergence. Only after all terms in this hierarchy are controlled will spectral flattening or magic-band formation be interpreted as a property of the infinite hyperbolic bilayer. Consider first an exactly commensurate hyperbolic bilayer with exact translation group Γ_M . Choose a normal finite-index subgroup $\Gamma_N \triangleleft \Gamma_M$. Its index is $d_N = [\Gamma_M : \Gamma_N]$. Normality allows the quotient $Q_N = \Gamma_M / \Gamma_N$ to inherit a group structure. Its order is $|Q_N| = d_N$. The finite quotient is therefore the exact discrete translation group of the

periodic finite approximant. The corresponding compact hyperbolic surface is

$$\Sigma_N = \Gamma_N \backslash \mathbb{H}_R^2. \quad (3633)$$

Because Γ_N has finite index, the area is

$$A_N = d_N A_M^{\text{sc}}, \quad (3634)$$

where the exact moiré-cell area is $A_M^{\text{sc}} = q_M A_0$. Hence,

$$A_N = d_N q_M A_0. \quad (3635)$$

For a torsion-free quotient, the genus satisfies

$$g_N = 1 + d_N (g_M - 1). \quad (3636)$$

Using

$$g_M - 1 = q_M (g_0 - 1), \quad (3637)$$

one obtains

$$g_N = 1 + d_N q_M (g_0 - 1). \quad (3638)$$

Thus increasing the periodic-cover size increases the topology of the boundaryless finite system rather than creating a larger open disk. The exact moiré quotient contains $\nu_M = 2n_b q_M$ physical bilayer orbitals. The periodic cover contains d_N copies of the exact moiré cell. Therefore,

$$N_H^{(N)} = \nu_M d_N = 2n_b q_M d_N. \quad (3639)$$

This is the dimension of the full finite-cover one-particle Hamiltonian. The matrix dimension must therefore be distinguished from the number of generalized momentum sectors. For a non-Abelian finite quotient, these two numbers are not equal. A computationally useful description of the finite cover is obtained by a surjective homomorphism $\phi_N : \Gamma_M \longrightarrow Q_N$, whose kernel is $\ker \phi_N = \Gamma_N$. Let the exact moiré translation group be generated by

$$\Gamma_M = \langle s_1, \dots, s_r \mid \mathcal{R}_1, \dots, \mathcal{R}_p \rangle. \quad (3640)$$

The finite quotient is specified by finite-group elements $q_\mu = \phi_N(s_\mu)$ that satisfy every defining relation, $\mathcal{R}_a(q_1, \dots, q_r) = e_{Q_N}$. For a surface group, this includes the constraint

$$\prod_{j=1}^{g_M} [\phi_N(a_j), \phi_N(b_j)] = e_{Q_N}. \quad (3641)$$

Verification of the defining relations is an exact algebraic test of the finite quotient before any Hamiltonian is constructed. Let $|q\rangle$, $q \in Q_N$, be the quotient-cell basis. For every group element define the right-regular permutation matrix by

$$\mathcal{R}_N(\delta)|q\rangle = |q\phi_N(\delta)^{-1}\rangle. \quad (3642)$$

The matrices satisfy

$$\mathcal{R}_N(\delta_1)\mathcal{R}_N(\delta_2) = \mathcal{R}_N(\delta_1\delta_2). \quad (3643)$$

They are unitary permutation matrices, $\mathcal{R}_N(\delta)^\dagger = \mathcal{R}_N(\delta^{-1})$. Every row and every column of a right-regular matrix contains exactly one nonzero entry. Thus the group-theoretic translation part of the finite-cover Hamiltonian is maximally sparse. Let the infinite translation-invariant Hamiltonian have matrix-valued group kernel $h(\delta)$. The finite-cover Hamiltonian is obtained by replacing the infinite regular representation by the finite right-regular representation,

$$H_N = \sum_{\delta \in \Gamma_M} h(\delta) \otimes \mathcal{R}_N(\delta). \quad (3644)$$

For a finite-range model, only a finite set contributes. Define the hopping support

$$S_c = \{\delta \in \Gamma_M : h(\delta) \neq 0\}. \quad (3645)$$

Then

$$H_N = \sum_{\delta \in S_c} h(\delta) \otimes \mathcal{R}_N(\delta). \quad (3646)$$

This representation allows the Hamiltonian to be assembled without constructing an open real-space hyperbolic disk. The group kernel satisfies $h(\delta^{-1}) = h(\delta)^\dagger$. Therefore,

$$\begin{aligned} H_N^\dagger &= \sum_{\delta} h(\delta)^\dagger \otimes \mathcal{R}_N(\delta)^\dagger \\ &= \sum_{\delta} h(\delta^{-1}) \otimes \mathcal{R}_N(\delta^{-1}). \end{aligned} \quad (3647)$$

Changing the summation variable gives $H_N^\dagger = H_N$. Thus periodic identification introduces no loss of Hermiticity. Let the number of physical hopping channels emerging from one orbital be bounded by $z_{\text{eff}} < \infty$. For a finite-range Hamiltonian with fixed microscopic parameters, z_{eff} does not grow with the cover degree. The number of nonzero matrix elements therefore satisfies

$$\text{nnz}(H_N) = O\left(z_{\text{eff}} N_H^{(N)}\right). \quad (3648)$$

For fixed hopping range,

$$\text{nnz}(H_N) = O\left(N_H^{(N)}\right). \quad (3649)$$

Thus the finite-cover Hamiltonian remains sparse even though its topological genus increases with system size. For a truncated long-range hopping model with geodesic cutoff L_c , the number of sites within the hopping neighborhood grows approximately as

$$z_{\text{eff}}(L_c) = O\left(e^{L_c/R}\right). \quad (3650)$$

Consequently,

$$\text{nnz}\left(H_N^{(L_c)}\right) = O\left[N_H^{(N)} e^{L_c/R}\right]. \quad (3651)$$

The hopping cutoff therefore controls not only the physical truncation error but also the computational complexity. To formulate cover convergence precisely, define the maximal physical propagation length of the finite-range Hamiltonian by

$$L_H = \sup\{d(x_\alpha, \delta x_\beta) : h_{\alpha\beta}(\delta) \neq 0\}. \quad (3652)$$

For a strictly nearest-neighbor model, $L_H \sim L_c$. For a truncated long-range model, $L_H \sim L_c$. The quantity L_H , rather than matrix dimension alone, determines whether periodic wraparound contaminates local hopping processes. Define the systole of the periodic cover as the length of its shortest noncontractible closed geodesic,

$$\ell_{\text{sys}}^{(N)} = \inf_{\gamma \in \Gamma_N \setminus \{e\}} \ell(\gamma), \quad (3653)$$

where $\ell(\gamma)$ is the translation length of the deck transformation. The global injectivity radius is

$$r_{\text{inj}}^{(N)} = \frac{1}{2} \ell_{\text{sys}}^{(N)}. \quad (3654)$$

A geodesic ball of radius smaller than the injectivity radius is locally indistinguishable from the universal hyperbolic cover. Thus the relevant finite-size parameter is not merely d_N , but also $r_{\text{inj}}^{(N)}$. Two covers with the same number of sites may have very different short noncontractible cycles and therefore very different finite-size errors. Suppose $r_{\text{inj}}^{(N)} > L_H$. Then no two distinct hopping processes of physical range at most L_H can be identified through a nontrivial deck transformation of the cover. Therefore the local hopping neighborhood of every site is exactly the same as in the infinite system. The finite-cover Hamiltonian is then locally identical to the infinite Hamiltonian at the one-hop level. A product of m hopping processes can propagate through a physical distance no larger than mL_H . A nontrivial quotient loop

of m hops is excluded whenever $\ell_{\text{sys}}^{(N)} > mL_H$. Equivalently, $2r_{\text{inj}}^{(N)} > mL_H$. This condition will yield an exact convergence theorem for spectral moments. A controlled thermodynamic limit is obtained from a nested sequence of normal finite-index subgroups,

$$\Gamma_M = \Gamma_0 \triangleright \Gamma_1 \triangleright \Gamma_2 \triangleright \cdots, \quad (3655)$$

with

$$\bigcap_{N=1}^{\infty} \Gamma_N = \{e\}. \quad (3656)$$

For a residually finite surface group such towers can be constructed. Normality is particularly useful because exclusion of one group element automatically excludes all of its conjugates. Fix an arbitrary positive length L . A compact hyperbolic surface contains only finitely many conjugacy classes of closed geodesics with length not exceeding L . Choose representatives $\gamma_1, \dots, \gamma_{M(L)}$. Because the subgroup intersection is trivial, every nonidentity representative is excluded from all sufficiently deep terms of the tower. Thus there exists $N_0(L)$ such that $\gamma_j \notin \Gamma_N$ for every $N \geq N_0(L)$ and every short-geodesic representative. Because Γ_N is normal, none of their conjugates belongs to the subgroup. Hence, $\ell_{\text{sys}}^{(N)} > L$ for all sufficiently large N . Since L was arbitrary, $\ell_{\text{sys}}^{(N)} \rightarrow \infty$, and therefore $r_{\text{inj}}^{(N)} \rightarrow \infty$. The periodic covers thus become locally indistinguishable from the infinite hyperbolic crystal on arbitrarily large length scales. Consider an observable supported inside a physical region of radius L_O . Suppose the Hamiltonian has propagation range L_H . The observable depends only on the Hamiltonian within the enlarged region $L_O + L_H$. Therefore the finite cover reproduces the infinite-system observable exactly whenever

$$r_{\text{inj}}^{(N)} > L_O + L_H. \quad (3657)$$

For strictly local finite-range observables, convergence is therefore not merely asymptotic. It becomes exact once the injectivity radius exceeds the required causal neighborhood. Define the normalized local trace of the infinite translation-invariant operator by

$$\tau_{\infty}(A) = \frac{1}{\nu_M} \sum_{\alpha=1}^{\nu_M} \langle e, \alpha | A | e, \alpha \rangle. \quad (3658)$$

For a finite periodic cover, define

$$\tau_N(A_N) = \frac{1}{\nu_M d_N} \text{Tr } A_N. \quad (3659)$$

These are the natural bulk-normalized traces. Consider the m -th spectral moment,

$$\mu_m^{(N)} = \tau_N(H_N^m). \quad (3660)$$

The infinite-system moment is

$$\mu_m^{(\infty)} = \tau_{\infty}(H^m). \quad (3661)$$

Every term in H^m corresponds to a closed sequence of m hopping steps. A spurious quotient contribution would require a nontrivial deck transformation represented by a path of physical length at most mL_H . If $2r_{\text{inj}}^{(N)} > mL_H$, such a deck transformation does not exist. Therefore, $\mu_m^{(N)} = \mu_m^{(\infty)}$ for all sufficiently large covers.

a. Theorem 36. Exact stabilization of finite-range spectral moments. Let H be a bounded translation-invariant finite-range Hamiltonian with propagation radius L_H , and let the periodic covers satisfy $r_{\text{inj}}^{(N)} \rightarrow \infty$. Then for every fixed nonnegative integer m , there exists a finite N_m such that

$$\tau_N(H_N^m) = \tau_{\infty}(H^m) \quad (3662)$$

for every $N \geq N_m$. A sufficient condition is $2r_{\text{inj}}^{(N)} > mL_H$. Thus every fixed finite spectral moment eventually becomes exactly independent of the finite cover. Let the finite-cover eigenvalues be $E_1^{(N)}, \dots, E_{N_H^{(N)}}^{(N)}$. Define the normalized empirical spectral measure by

$$d\nu_N(E) = \frac{1}{N_H^{(N)}} \sum_{j=1}^{N_H^{(N)}} \delta(E - E_j^{(N)}) dE. \quad (3663)$$

Its moments are

$$\int E^m d\nu_N(E) = \mu_m^{(N)}. \quad (3664)$$

Because the Hamiltonians are uniformly bounded, all spectral measures have support in a common compact interval. Moment stabilization therefore implies weak convergence of the finite-cover spectral measures to the infinite bulk density-of-states measure. For every continuous bounded function on the common spectral interval, $f(E)$, one obtains

$$\lim_{N \rightarrow \infty} \frac{1}{N_H^{(N)}} \text{Tr } f(H_N) = \tau_{\infty}[f(H)]. \quad (3665)$$

Thus periodic finite covers provide a boundary-free controlled route to the bulk density of states. This result is fundamentally stronger than convergence of a global open-disk density of states, which is contaminated by

an extensive boundary. Weak convergence of the density of states does not by itself imply pointwise convergence of every individual finite-cover eigenvalue. In particular, it does not automatically guarantee convergence of $E_{\min}^{(N)}$, $E_{\max}^{(N)}$, or an isolated finite-size spectral gap. These quantities must be tested separately. Therefore, This distinction prevents an apparently converged density of states from being used as the sole evidence for a magic-band gap. For an exponentially decaying hopping model, define a truncated Hamiltonian $H^{(L_c)}$ with propagation radius L_c . Suppose the hopping envelope decays according to $|h(d)| \lesssim e^{-\mu d}$, with

$$\gamma = \mu - \frac{1}{R} > 0. \quad (3666)$$

The operator truncation error satisfies

$$\|H - H^{(L_c)}\| \leq C_{\text{tr}} e^{-\gamma L_c}. \quad (3667)$$

A controlled finite-cover sequence must send the cutoff to infinity while keeping it asymptotically smaller than the injectivity radius. Thus choose $L_c^{(N)} \rightarrow \infty$, and

$$\frac{L_c^{(N)}}{r_{\text{inj}}^{(N)}} \rightarrow 0. \quad (3668)$$

Then

$$\|H - H^{(L_c^{(N)})}\| \rightarrow 0, \quad (3669)$$

while every fixed-order spectral moment eventually satisfies the no-aliasing condition.

b. Theorem 37. Controlled long-range periodic-cover limit. Suppose the infinite hopping Hamiltonian obeys a hyperbolically summable exponential bound with $\mu > R^{-1}$. Let periodic covers satisfy $r_{\text{inj}}^{(N)} \rightarrow \infty$. Choose cutoffs satisfying $L_c^{(N)} \rightarrow \infty$ and

$$\frac{L_c^{(N)}}{r_{\text{inj}}^{(N)}} \rightarrow 0. \quad (3670)$$

Then the hopping truncation error vanishes,

$$\|H - H^{(L_c^{(N)})}\| \rightarrow 0, \quad (3671)$$

and for every fixed spectral moment m , quotient aliasing is absent for all sufficiently large N . The corresponding finite-cover density-of-states measures therefore converge to the infinite long-range bulk spectral measure. Because the periodic cover possesses finite translation group Q_N ,

the finite-cover Hamiltonian can be decomposed into irreducible translation sectors. Let $\hat{Q}_N$ denote the inequivalent irreducible unitary representations. For $\rho \in \hat{Q}_N$, let the representation dimension be d_ρ . The finite-group dimension identity is

$$\sum_{\rho \in \hat{Q}_N} d_\rho^2 = d_N. \quad (3672)$$

The finite-cover Hamiltonian decomposes as

$$H_N \simeq \bigoplus_{\rho \in \hat{Q}_N} [I_{d_\rho} \otimes H[\rho]]. \quad (3673)$$

The representation block has dimension

$$\dim H[\rho] = \nu_M d_\rho. \quad (3674)$$

This exact block decomposition may reduce the numerical problem substantially and simultaneously labels every finite-cover eigenstate by its generalized translation sector. The number of physical states contributed by one irrep sector is $N_\rho = \nu_M d_\rho^2$. Summing gives

$$\sum_{\rho} N_\rho = \nu_M d_N = N_H^{(N)}. \quad (3675)$$

Thus symmetry reduction conserves the complete finite-cover state count exactly. If the quotient characters are available, define

$$P_\rho = \frac{d_\rho}{d_N} \sum_{q \in Q_N} \chi_\rho(q^{-1}) U_q. \quad (3676)$$

The projectors satisfy $P_\rho^2 = P_\rho$, $P_\rho^\dagger = P_\rho$, and $P_\rho P_\sigma = 0$ for $\rho \neq \sigma$. Furthermore,

$$\sum_{\rho} P_\rho = I. \quad (3677)$$

Numerically computed eigenvectors can therefore be classified by the weights

$$w_\rho^{(n)} = \langle \psi_n | P_\rho | \psi_n \rangle. \quad (3678)$$

An ideally symmetry-pure state satisfies $w_\rho^{(n)} = 1$ for one representation and zero for all others. For sufficiently small finite covers, the complete Hamiltonian can be converted to a dense Hermitian matrix and diagonalized exactly up to floating-point precision. The eigenvalue problem is $H_N V = V D$, where $D = \text{diag}(E_1, \dots, E_{N_H})$. The eigenvectors satisfy $V^\dagger V = I$. Generic dense Hermitian diagonalization requires computational time scaling as

$O(N_H^3)$, and memory scaling as $O(N_H^2)$. Dense exact diagonalization is therefore used primarily for small covers, benchmark structures, and validation of sparse iterative methods. Despite its unfavorable asymptotic scaling, complete diagonalization provides several unique checks. It gives the full state count, N_H , the exact finite-cover density of states,

$$\rho_N(E) = \frac{1}{N_H} \sum_{j=1}^{N_H} \delta(E - E_j), \quad (3679)$$

and exact trace identities,

$$\sum_j E_j = \text{Tr } H_N, \quad (3680)$$

and

$$\sum_j E_j^2 = \text{Tr } H_N^2. \quad (3681)$$

These identities provide stringent implementation tests. For larger covers, only a subset of the spectrum should be computed. Given a normalized starting vector $|v\rangle$, the order- m Krylov space is

$$\mathcal{K}_m(H, v) = \text{span} \{ |v\rangle, H|v\rangle, \dots, H^{m-1}|v\rangle \}. \quad (3682)$$

Because the Hamiltonian is sparse, each matrix-vector multiplication costs $O(\text{nnz } H)$. For a fixed-range Hamiltonian this is asymptotically linear in matrix dimension. For a Hermitian Hamiltonian, the Krylov basis may be generated by the Lanczos recurrence. Initialize

$$|q_1\rangle = \frac{|v\rangle}{\|v\|}, \quad \beta_1 = 0, \quad |q_0\rangle = 0. \quad (3683)$$

At step j , compute

$$|w_j\rangle = H|q_j\rangle - \beta_j|q_{j-1}\rangle. \quad (3684)$$

The diagonal coefficient is $\alpha_j = \langle q_j | H | q_j \rangle$. Subtract the component along the current basis vector,

$$|w_j\rangle \leftarrow |w_j\rangle - \alpha_j|q_j\rangle. \quad (3685)$$

After optional reorthogonalization, $\beta_{j+1} = \|w_j\|$. If the coefficient is nonzero,

$$|q_{j+1}\rangle = \frac{|w_j\rangle}{\beta_{j+1}}. \quad (3686)$$

Define $Q_m = (q_1 \ q_2 \ \dots \ q_m)$. The projected Hamiltonian is the real symmetric tridiagonal matrix $T_m = Q_m^\dagger H Q_m$. Explicitly,

$$T_m = \begin{pmatrix} \alpha_1 & \beta_2 & 0 & \dots \\ \beta_2 & \alpha_2 & \beta_3 & \dots \\ 0 & \beta_3 & \alpha_3 & \ddots \\ \vdots & \vdots & \ddots & \ddots \end{pmatrix}. \quad (3687)$$

The exact Lanczos relation is

$$H Q_m = Q_m T_m + \beta_{m+1} |q_{m+1}\rangle e_m^T. \quad (3688)$$

Let $T_m y_j = \vartheta_j y_j$. The corresponding Ritz vector is $|\tilde{\psi}_j\rangle = Q_m y_j$. Its residual is

$$\begin{aligned} |r_j\rangle &= H|\tilde{\psi}_j\rangle - \vartheta_j|\tilde{\psi}_j\rangle \\ &= \beta_{m+1} |q_{m+1}\rangle e_m^T y_j. \end{aligned} \quad (3689)$$

Hence, $\|r_j\| = |\beta_{m+1} e_m^T y_j|$. This residual provides an a posteriori eigenpair-verification criterion. For a Hermitian matrix and normalized approximate eigenvector,

$$\text{dist}(\vartheta_j, \text{Spec } H) \leq \|r_j\|. \quad (3690)$$

Thus a candidate near-flat-band eigenvalue should not be accepted merely because the iterative solver has returned a number. Its residual must be parametrically smaller than all physically relevant spectral scales being extracted. Finite-precision Lanczos arithmetic can destroy exact Krylov orthogonality. The orthogonality error may be monitored using

$$\epsilon_{\text{orth}} = \|Q_m^\dagger Q_m - I\|. \quad (3691)$$

Loss of orthogonality can produce repeated or ghost Ritz values. Therefore full or selective reorthogonalization should be used whenever closely spaced or degenerate eigenvalues are important. This is particularly relevant in hyperbolic systems because representation-theoretic multiplicities are physically meaningful and must not be confused with numerical duplication. A single starting vector can have zero projection onto an entire symmetry sector. To robustly recover degenerate or nearly degenerate multiplets, introduce a block of starting vectors, $V_1 = (v_1 \ \dots \ v_p)$. The block Krylov space is

$$\mathcal{K}_m(H, V_1) = \text{span} \{ V_1, H V_1, \dots, H^{m-1} V_1 \}. \quad (3692)$$

Block Lanczos is therefore the preferred method when symmetry requires multiplets of dimension larger than one. An even stronger strategy is to initialize the Krylov vectors inside a single exact symmetry sector. Choose

$$|v_\rho\rangle = \frac{P_\rho |v\rangle}{\|P_\rho |v\rangle\|}. \quad (3693)$$

Because $[H, P_\rho] = 0$, all Krylov vectors remain in the same sector,

$$\mathcal{K}_m(H, v_\rho) \subset \text{Ran } P_\rho. \quad (3694)$$

This prevents unrelated symmetry sectors from contaminating a targeted spectral calculation. Standard Lanczos converges most rapidly to extremal eigenvalues. It is therefore well suited for calculating $E_{\min}$, $E_{\max}$, or states close to isolated spectral edges. Interior eigenvalues require a spectral transformation. Let the desired target energy be σ . Assume $\sigma \notin \text{Spec } H$. Define the shift-invert operator $A_\sigma = (H - \sigma I)^{-1}$. If

$$H|\psi_j\rangle = E_j|\psi_j\rangle, \quad (3695)$$

then

$$A_\sigma|\psi_j\rangle = \mu_j|\psi_j\rangle, \quad (3696)$$

with

$$\mu_j = \frac{1}{E_j - \sigma}. \quad (3697)$$

Thus eigenvalues closest to the shift become the largest-magnitude eigenvalues of the transformed operator. From a converged shift-invert eigenvalue, $\tilde{\mu}_j$, the physical energy is reconstructed as

$$\tilde{E}_j = \sigma + \frac{1}{\tilde{\mu}_j}. \quad (3698)$$

The eigenvector is unchanged. A shift-invert matrix-vector multiplication requires solving

$$(H - \sigma I)x = b. \quad (3699)$$

A sparse direct factorization may be performed once and reused for every Krylov step. However, fill-in can substantially exceed the number of nonzero entries in the original Hamiltonian. Therefore there is no universal complexity law depending only on $\text{nnz}(H)$. Memory usage must be monitored explicitly for every cover family. For very large systems, the linear system may itself be solved iteratively. Let the approximate solution satisfy

$$\|(H - \sigma I)x - b\| \leq \epsilon_{\text{inner}} \|b\|. \quad (3700)$$

The inner tolerance must be made sufficiently smaller than the requested outer eigenpair accuracy. Otherwise the apparent convergence of the outer Krylov iteration can be limited by the linear-solver error. Regardless of whether exact or inexact shift-invert is used, final

eigenpairs should always be established using the original Hamiltonian,

$$r_j^{(H)} = H\tilde{\psi}_j - \tilde{E}_j\tilde{\psi}_j. \quad (3701)$$

The relevant numerical admissibility criterion is

$$\frac{\|r_j^{(H)}\|}{E_{\text{scale}}} \ll 1, \quad (3702)$$

where the reference energy scale must be reported explicitly. To resolve a target band centered near energy E_* , one may choose $\sigma \simeq E_*$. The shift should not coincide numerically with an exact eigenvalue because the linear operator would become singular. For a dense cluster of nearly flat states, multiple nearby shifts should be used to confirm that the same spectral manifold is recovered. When factorization of the shift-invert operator becomes prohibitively expensive, an alternative is a polynomial filter. Let the Hamiltonian spectrum lie in $[E_{\min}, E_{\max}]$. Rescale the Hamiltonian to

$$\tilde{H} = \frac{H - c}{a_E}, \quad (3703)$$

where

$$c = \frac{E_{\max} + E_{\min}}{2}, \quad (3704)$$

and

$$a_E = \frac{E_{\max} - E_{\min}}{2}. \quad (3705)$$

Then $\text{Spec } \tilde{H} \subset [-1, 1]$. Chebyshev polynomial filters can then preferentially amplify a desired energy window while retaining sparse matrix-vector multiplication. For large finite covers, the complete density of states can be calculated without computing all eigenvalues. Define the Chebyshev moments

$$\mu_m = \frac{1}{N_H} \text{Tr } T_m(\tilde{H}), \quad (3706)$$

where T_m is the Chebyshev polynomial. The rescaled density of states is approximated by

$$\rho_M(x) = \frac{1}{\pi\sqrt{1-x^2}} \left[g_0\mu_0 + 2 \sum_{m=1}^M g_m\mu_m T_m(x) \right], \quad (3707)$$

where g_m denotes a damping kernel such as a Jackson-type kernel. The method requires only repeated sparse matrix-vector multiplication. For sufficiently large matrices, the trace moments may be estimated using random

normalized vectors. Let $|r_s\rangle$, $s = 1, \dots, N_r$, satisfy an unbiased stochastic trace condition. Then

$$\mu_m \simeq \frac{1}{N_r} \sum_{s=1}^{N_r} \langle r_s | T_m(\tilde{H}) | r_s \rangle. \quad (3708)$$

Convergence must be checked independently with respect to both M and N_r . A DOS peak should not be interpreted as a physical flat-band singularity unless it survives both convergence tests. Suppose the numerical calculation uses an approximate Hamiltonian $\tilde{H} = H + \Delta H$. Let

$$\tilde{H}|\tilde{\psi}\rangle = \tilde{E}|\tilde{\psi}\rangle + |r\rangle. \quad (3709)$$

The residual with respect to the exact target Hamiltonian is

$$H|\tilde{\psi}\rangle - \tilde{E}|\tilde{\psi}\rangle = |r\rangle - \Delta H|\tilde{\psi}\rangle. \quad (3710)$$

Hence,

$$\|H|\tilde{\psi}\rangle - \tilde{E}|\tilde{\psi}\rangle\| \leq \|r\| + \|\Delta H\|. \quad (3711)$$

Therefore,

$$\text{dist}(\tilde{E}, \text{Spec } H) \leq \|r\| + \|\Delta H\|. \quad (3712)$$

This is the basic numerical verification inequality used below. For an exponentially decaying hopping profile,

$$\epsilon_{\text{tr}} = \|H - H^{(L_c)}\| \leq C_{\text{tr}} e^{-\gamma L_c}. \quad (3713)$$

The truncation cutoff should therefore be increased until $\epsilon_{\text{tr}} \ll W_{\text{target}}$, where W_{target} is the smallest physically relevant bandwidth being claimed. Otherwise truncation itself can create an apparent flat band. Suppose an incommensurate target twist is replaced by a commensurate approximant, θ_N . Define $\delta\theta_N = \theta_N - \theta$. Inside a radius L , the geometric displacement error is

$$\epsilon_{\text{geo}}(L, \delta\theta_N) = 2R \text{arsinh}[\sinh(L/R) |\sin(\delta\theta_N/2)|]. \quad (3714)$$

For small angular error,

$$\epsilon_{\text{geo}} \simeq R \sinh(L/R) |\delta\theta_N|. \quad (3715)$$

For $L \gg R$, this behaves as

$$\epsilon_{\text{geo}} \simeq \frac{R}{2} e^{L/R} |\delta\theta_N|. \quad (3716)$$

If the hopping kernel is Lipschitz continuous with constant Λ_t , then a local hopping element satisfies $|\Delta t| \leq \Lambda_t \epsilon_{\text{geo}}$. For bounded local coordination, z_{eff} , the local Hamiltonian error is bounded schematically by $\epsilon_\theta \lesssim z_{\text{eff}} \Lambda_t \epsilon_{\text{geo}}$. Thus a growing-domain commensurate approximation must satisfy $|\delta\theta_N| e^{L_N/R} \rightarrow 0$. This condition is substantially stronger than ordinary angular convergence. The total controlled finite-matrix error may be decomposed schematically as

$$\epsilon_{\text{tot}} \leq \epsilon_{\text{tr}} + \epsilon_{\text{cover}} + \epsilon_\theta + \epsilon_{\text{solver}} + \epsilon_{\text{fp}}, \quad (3717)$$

where ϵ_{tr} is the hopping-truncation error, ϵ_{cover} is finite-cover error for the chosen observable, ϵ_θ is angular-approximation error, ϵ_{solver} is iterative eigensolver error, and ϵ_{fp} collects floating-point effects. For strictly local finite-range observables satisfying the injectivity condition, $\epsilon_{\text{cover}} = 0$. For global spectral edges, the finite-cover contribution must instead be estimated empirically from a sequence of increasing periodic covers. Let an observable extracted from cover N be $\mathcal{O}_N$. A naive finite-size analysis plots $\mathcal{O}_N$ only against the number of sites. For a hyperbolic periodic cover, this is insufficient. The geometrically relevant scaling variable is $r_{\text{inj}}^{(N)}$. Therefore convergence should be reported as $\mathcal{O} = \mathcal{O}(r_{\text{inj}})$ in addition to $\mathcal{O} = \mathcal{O}(N_H)$. This distinction detects finite covers containing anomalously short noncontractible cycles. For two consecutive covers define

$$\Delta_{\mathcal{O}}^{(N)} = \frac{|\mathcal{O}_{N+1} - \mathcal{O}_N|}{\max[E_{\text{ref}}, |\mathcal{O}_{N+1}|, |\mathcal{O}_N|]}. \quad (3718)$$

Here E_{ref} is a nonzero normalization preventing an artificial singularity when the observable approaches zero. A numerical claim should require convergence across multiple independent cover sizes rather than one apparently small successive difference. Because different finite quotients of similar degree can possess different systoles, a stronger test compares multiple inequivalent covers satisfying $d_N^{(1)} \approx d_N^{(2)}$ but $r_{\text{inj}}^{(1)} \neq r_{\text{inj}}^{(2)}$. If a spectral feature is controlled by local infinite-system physics, its finite-size error should correlate primarily with injectivity radius rather than with accidental quotient topology. This provides a direct numerical test of finite-cover artifacts. Energy ordering alone is not sufficient for tracking spectral branches as twist, curvature, or coupling strength is changed. Let $|\psi_n(\lambda)\rangle$ be an eigenstate at parameter value λ . Define the overlap matrix between neighboring parameter values by

$$\mathcal{M}_{nm} = |\langle \psi_n(\lambda) | \psi_m(\lambda + \delta\lambda) \rangle|^2. \quad (3719)$$

Branches should be matched by maximizing the total overlap rather than by matching eigenvalue indices alone. This prevents artificial branch swapping at avoided crossings. Inside a degenerate or nearly degenerate manifold,

individual eigenvectors are basis dependent. Suppose the target cluster is represented by the projector

$$P(\lambda) = \sum_{n \in \mathcal{C}} |\psi_n(\lambda)\rangle \langle \psi_n(\lambda)|. \quad (3720)$$

The appropriate continuity diagnostic is the projector distance

$$D_P = \|P(\lambda + \delta\lambda) - P(\lambda)\|. \quad (3721)$$

This quantity is invariant under arbitrary unitary rotations of the basis inside the degenerate multiplet. It is therefore preferable to direct eigenvector overlap whenever representation theory enforces degeneracy. For every exact symmetry operator U_g , define the numerical commutation error

$$\epsilon_{\text{sym}}(g) = \frac{\|H_N U_g - U_g H_N\|}{\|H_N\|}. \quad (3722)$$

The maximal symmetry error is

$$\epsilon_{\text{sym}}^{\max} = \max_{g \in \mathcal{G}_{\text{test}}} \epsilon_{\text{sym}}(g). \quad (3723)$$

The tested set should include generators of the exact translation group, the residual point group, and any layer-exchanging symmetry used in the degeneracy classification. A symmetry-protected degeneracy cannot be trusted if the implemented Hamiltonian violates the protecting symmetry at a comparable or larger scale than the observed splitting. Define the normalized Hermiticity defect

$$\epsilon_{\text{Herm}} = \frac{\|H_N - H_N^\dagger\|}{\|H_N\|}. \quad (3724)$$

For a correctly assembled real or complex Hermitian model, ϵ_{Herm} must be at the level expected from numerical precision. This test should be performed before diagonalization. The first two matrix moments can also be calculated directly from the sparse matrix. Define $M_1 = \text{Tr } H_N$, and $M_2 = \text{Tr } H_N^2$. After exact diagonalization, verify

$$M_1 = \sum_j E_j, \quad M_2 = \sum_j E_j^2. \quad (3725)$$

The second identity is particularly effective at detecting missing or double-counted hopping terms. Let a target band cluster occupy the interval $[E_-, E_+]$. Define its bandwidth by

$$W_N = E_+^{(N)} - E_-^{(N)}. \quad (3726)$$

Let the minimum separation from neighboring states be

$$\Delta_N = \min \left(E_-^{(N)} - E_{\text{below}}^{(N)}, E_{\text{above}}^{(N)} - E_+^{(N)} \right). \quad (3727)$$

A natural dimensionless flatness ratio is

$$\mathcal{F}_N = \frac{\Delta_N}{W_N}. \quad (3728)$$

A large value is meaningful only if both numerator and denominator have independently converged with respect to cover size, hopping cutoff, and solver tolerance. Suppose the total eigenvalue uncertainty is ϵ_E . A measured bandwidth obeying $W_N \lesssim 2\epsilon_E$ cannot be distinguished numerically from an exactly flat level. Therefore a quantitative flat-band claim requires $\epsilon_E \ll W_N$ for a finite but nonzero bandwidth claim. If the proposed result is exact flatness, one must instead demonstrate analytic symmetry or interference constraints forcing the bandwidth to vanish. Although periodic covers are the primary bulk method, open clusters remain useful for real-space visualization. Let Ω_L be an open geodesic cluster of radius L . Its boundary is physical. The boundary-to-area ratio obeys

$$\lim_{L \rightarrow \infty} \frac{P_{\mathbb{H}}(L)}{A_{\mathbb{H}}(L)} = \frac{1}{R}. \quad (3729)$$

Therefore the number of boundary-region sites remains an extensive fraction of the complete cluster. Define a boundary shell of physical thickness b by

$$\partial_b \Omega_L = \{x \in \Omega_L : d(x, \partial \Omega_L) \leq b\}. \quad (3730)$$

The corresponding core is

$$\Omega_L^{\text{core}}(b) = \Omega_L \setminus \partial_b \Omega_L. \quad (3731)$$

For a geodesic disk,

$$\Omega_L^{\text{core}}(b) = B_{\mathbb{H}}(o, L - b). \quad (3732)$$

The geometric boundary-shell fraction is

$$f_b(L) = 1 - \frac{A_{\mathbb{H}}(L - b)}{A_{\mathbb{H}}(L)}. \quad (3733)$$

Using exponential hyperbolic area growth gives

$$\lim_{L \rightarrow \infty} f_b(L) = 1 - e^{-b/R}. \quad (3734)$$

Thus even a fixed-thickness shell occupies a nonzero fraction of a very large hyperbolic disk. This is why a raw

boundary-weight threshold cannot be imported from Euclidean finite clusters. Let $P_\partial^{(b)}$ project onto all orbitals lying within the chosen boundary shell. For a normalized eigenstate, define the boundary weight

$$W_\partial^{(n)}(b) = \langle \psi_n | P_\partial^{(b)} | \psi_n \rangle. \quad (3735)$$

The core weight is

$$W_{\text{core}}^{(n)}(b) = 1 - W_\partial^{(n)}(b). \quad (3736)$$

A spatially extended state approximately sampling the sites uniformly has $W_\partial^{\text{ext}} \simeq f_b$. Because $f_b \not\rightarrow 0$, a perfectly ordinary bulk-extended state can have a substantial raw boundary weight. Therefore, W_∂ large $\neq$ boundary localized. Boundary-state filtering must compare the state weight against the geometric baseline. Define

$$\mathcal{B}_n(b) = \frac{W_\partial^{(n)}(b)}{f_b}. \quad (3737)$$

An approximately uniform state has $\mathcal{B}_n \simeq 1$. A state statistically depleted from the boundary has $\mathcal{B}_n < 1$, while a boundary-enriched state has $\mathcal{B}_n > 1$. A stronger diagnostic compares the odds of finding the wave function in the boundary shell with the geometric site odds. Define

$$\mathcal{S}_\partial^{(n)} = \ln \left[\frac{W_\partial^{(n)}}{1 - W_\partial^{(n)}} \right] - \ln \left[\frac{f_b}{1 - f_b} \right]. \quad (3738)$$

For an approximately uniform extended state, $\mathcal{S}_\partial \simeq 0$. Boundary enrichment gives $\mathcal{S}_\partial > 0$. Strong localization to the outer shell drives $\mathcal{S}_\partial \rightarrow +\infty$. This diagnostic automatically compensates for the extensive geometric boundary fraction. A bulk spectral analysis based on an open cluster should not discard states using one universal numerical threshold. Instead, the classification should be checked across several shell thicknesses satisfying $b \gtrsim L_H$. A state is identified as robustly boundary enriched only when $\mathcal{S}_\partial^{(n)}(b)$ remains systematically positive and increases relative to bulk states over a controlled interval of shell thicknesses. The numerical threshold must be reported and varied as part of the robustness analysis. A cleaner bulk observable is obtained directly from the core region. Define

$$\rho_{\text{core}}(E) = \frac{1}{N_{\text{core}}} \sum_{i \in \text{core}} \langle i | \delta(E - H) | i \rangle. \quad (3739)$$

Using the eigenbasis gives

$$\rho_{\text{core}}(E) = \frac{1}{N_{\text{core}}} \sum_n W_{\text{core}}^{(n)} \delta(E - E_n). \quad (3740)$$

Boundary-localized states are automatically suppressed by their small core weight. A compact periodic finite cover has $\partial\Sigma_N = \emptyset$. Therefore, boundary-state filtering is unnecessary for periodic-cover spectra. This is one of the principal reasons periodic finite covers are preferred for bulk band, density-of-states, and flatness calculations. Open-boundary filtering is retained only for calculations intentionally probing real-space edge phenomena or for visualization of finite patches. For a normalized real-space eigenstate, define

$$\text{IPR}_n = \sum_{i=1}^{N_H} |\psi_n(i)|^4. \quad (3741)$$

The participation number is

$$P_n = \frac{1}{\text{IPR}_n}. \quad (3742)$$

A perfectly uniform state over N_H orbitals has

$$\text{IPR} = \frac{1}{N_H}, \quad (3743)$$

whereas a state localized on an order-one number of orbitals has $\text{IPR} = O(1)$. The IPR must be interpreted together with boundary enrichment because a boundary-localized state can have either small or large IPR depending on whether it occupies a narrow or extended boundary shell. Define

$$p_n = \frac{P_n}{N_H} = \frac{1}{N_H \text{IPR}_n}. \quad (3744)$$

An extended state has $p_n = O(1)$, whereas a strongly localized state satisfies $p_n \rightarrow 0$ along a growing-cover sequence. This provides a cover-size scaling criterion for localization. For bilayer eigenstates, define the layer projectors P_1, P_2 . The layer polarization is

$$\mathcal{P}_L^{(n)} = \langle \psi_n | P_1 - P_2 | \psi_n \rangle. \quad (3745)$$

It satisfies $-1 \leq \mathcal{P}_L^{(n)} \leq 1$. Strongly hybridized bilayer states typically satisfy $\mathcal{P}_L \simeq 0$, while layer-polarized states approach the endpoints. This observable helps distinguish genuine interlayer-hybridization flat bands from almost decoupled monolayer states. Define the interlayer component of the Hamiltonian by $H_\perp$. The hybridization expectation of an eigenstate is

$$\mathcal{W}_\perp^{(n)} = \langle \psi_n | H_\perp | \psi_n \rangle. \quad (3746)$$

A curvature-controlled magic-band candidate should be analyzed jointly through $W_n, \mathcal{P}_L^{(n)}, \mathcal{W}_\perp^{(n)}$, and the appropriate representation-sector label. This prevents

an accidentally isolated or weakly coupled level from being misclassified as a hybridization-induced magic state. Suppose the target twist is incommensurate but exact commensurate angles are dense at the chosen center. Choose a sequence $\theta_N \rightarrow \theta$. Every approximant defines its own exact moiré group, $\Gamma_M^{(N)}$, coincidence index, $q_M^{(N)}$, and periodic finite-cover hierarchy. However, convergence of the angle alone is insufficient. For a spatial scale L_N , the joint condition is

$$|\theta_N - \theta| e^{L_N/R} \rightarrow 0. \quad (3747)$$

This condition must be checked whenever a growing spatial region is used to compare commensurate and incommensurate systems. When no dense exact centered commensurate sequence exists, the target incommensurate bilayer must not be forced into an artificial common supercell. Instead, choose a finite cover of the first-layer group, $\Gamma_N^{(1)} < \Gamma$, and its twisted conjugate,

$$\Gamma_N^{(2)} = g_\theta \Gamma_N^{(1)} g_\theta^{-1}. \quad (3748)$$

The two finite layers are lifted to the common universal hyperbolic plane when interlayer matrix elements are evaluated. This paired-cover construction reproduces local incommensurate geometry without claiming the existence of a nonexistent common periodic translation group. Let the target observable have support inside radius L , and let the hopping range be L_H . If both layer covers satisfy

$$r_{\text{inj}}^{(N)} > L + L_H, \quad (3749)$$

then the coupled local Hamiltonian patch is represented exactly. Thus paired covers provide a controlled local thermodynamic limit even when $\Gamma \cap g_\theta \Gamma g_\theta^{-1}$ has infinite index. The complete numerical solver hierarchy is therefore organized by spectral objective. Complete exact spectra on small covers are obtained from dense Hermitian diagonalization. Extremal states on large sparse covers are obtained from Lanczos or related Hermitian Krylov methods. Interior states in a narrow target energy window are obtained from shift-invert Lanczos or another spectral transformation. Large-system densities of states are obtained from polynomial or stochastic trace methods. Degenerate symmetry multiplets are treated with block Krylov spaces or explicit representation-sector decomposition. This hierarchy avoids computing unnecessary portions of an exponentially growing finite-cover Hilbert space. A reported eigenpair should satisfy simultaneously

$$\frac{\|H - H^\dagger\|}{\|H\|} \ll 1, \quad \frac{\|H\psi - E\psi\|}{E_{\text{scale}}} \ll 1, \quad (3750)$$

and, when a symmetry sector is claimed, $1 - w_p \ll 1$. If the state belongs to a protected multiplet, all members of

the multiplet must converge at comparable residual tolerance. A candidate flat band must satisfy a stronger hierarchy. The hopping cutoff must be converged, $W \gg \epsilon_{\text{tr}}$, the eigensolver error must be smaller than the reported bandwidth, $W \gg \epsilon_{\text{solver}}$, and the finite-cover dependence must satisfy $W_N \rightarrow W_\infty$ over a sequence with $r_{\text{inj}}^{(N)} \rightarrow \infty$. The candidate must also remain distinct from a symmetry-enforced degeneracy. Thus the bandwidth should be evaluated inside a fixed irreducible sector whenever possible. A claimed gap must satisfy $\Delta \gg \epsilon_{\text{tot}}$, and must remain open under increasing cover size,

$$\liminf_{N \rightarrow \infty} \Delta_N > 0. \quad (3751)$$

A finite-size level spacing that vanishes with increasing cover size must not be interpreted as a thermodynamic spectral gap. For a cover small enough to permit complete diagonalization, the same target eigenvalues should also be calculated using the iterative method intended for larger systems. Define

$$\epsilon_{\text{cross}} = \max_{j \in \mathcal{W}} |E_j^{\text{ED}} - E_j^{\text{iter}}|, \quad (3752)$$

where $\mathcal{W}$ is the tested spectral window. The iterative algorithm should not be used for production data until ϵ_{cross} is demonstrably smaller than the physical energy scales under discussion. When multiple inequivalent finite quotients are available, the same local or density-of-states observable should be calculated on more than one quotient family. Let $\mathcal{O}_N^{(A)}$ and $\mathcal{O}_N^{(B)}$ denote two independent cover sequences. A bulk result should satisfy

$$\lim_{N \rightarrow \infty} \mathcal{O}_N^{(A)} = \lim_{N \rightarrow \infty} \mathcal{O}_N^{(B)}. \quad (3753)$$

This test guards against a result controlled by a special sequence of finite quotient groups.

c. Theorem 38. Boundary-free periodic-cover convergence. Let H be a bounded finite-range hyperbolic bilayer Hamiltonian, and let Σ_N be a normal finite-cover sequence satisfying $r_{\text{inj}}^{(N)} \rightarrow \infty$. Then every fixed-radius local observable is reproduced exactly for all sufficiently large covers. Every fixed spectral moment of the normalized finite-cover density of states likewise stabilizes exactly. Consequently the empirical finite-cover spectral measures converge weakly to the infinite bulk spectral measure. No boundary-state subtraction or boundary-to-volume extrapolation is required.

d. Theorem 39. established finite-matrix spectral approximation. Let the exact target finite Hamiltonian be H and the implemented Hamiltonian be $\tilde{H} = H + \Delta H$. Let a normalized numerical eigenpair satisfy

$$\tilde{H}|\tilde{\psi}\rangle = \tilde{E}|\tilde{\psi}\rangle + |r\rangle. \quad (3754)$$

Then

$$\text{dist}(\tilde{E}, \text{Spec } H) \leq \|\Delta H\| + \|r\|. \quad (3755)$$

Therefore every reported spectral scale must exceed the combined Hamiltonian-model error and eigensolver residual. For a truncated exponentially decaying model,

$$\|\Delta H\| \leq C_{\text{tr}} e^{-(\mu - R^{-1})L_c} \quad (3756)$$

up to additional explicitly controlled angular or model-approximation terms.

e. Theorem 40. Angular accuracy required on growing hyperbolic domains. Let a sequence of twist approximants satisfy $\theta_N \rightarrow \theta$. Let the spatial region whose Hamiltonian is to be approximated have radius L_N . Then uniform geometric convergence over the growing region requires

$$\sinh(L_N/R) |\sin[(\theta_N - \theta)/2]| \rightarrow 0. \quad (3757)$$

In the small-angle-error and large-radius regime this becomes

$$|\theta_N - \theta| e^{L_N/R} \rightarrow 0. \quad (3758)$$

Thus angular convergence by itself is insufficient to guarantee convergence of a growing hyperbolic numerical system.

f. Theorem 41. Boundary enrichment rather than raw boundary weight. For an open hyperbolic disk and fixed boundary-shell thickness b , the geometric fraction of sites in the shell approaches a nonzero constant,

$$f_b \rightarrow 1 - e^{-b/R}. \quad (3759)$$

Therefore an extended state generically carries an order-one fraction of its norm in the boundary shell. A raw criterion based only on W_∂ cannot distinguish boundary localization from a spatially extended state. A boundary-state diagnostic must instead compare wavefunction weight to the geometric baseline, for example through

$$\mathcal{B} = \frac{W_\partial}{f_b} \quad (3760)$$

or through the log-odds enrichment

$$\mathcal{S}_\partial = \ln \left[\frac{W_\partial/(1 - W_\partial)}{f_b/(1 - f_b)} \right]. \quad (3761)$$

Periodic finite covers avoid this ambiguity entirely. The numerical calculation should begin by fixing the physical microscopic model and the hopping cutoff. The exact or approximate hyperbolic translation group is then

represented through a finite quotient or paired finite-cover construction. Before diagonalization, the finite matrix is checked for Hermiticity, translation relations, point-group covariance, and state counting. Small covers are solved by complete diagonalization. The resulting spectra are used to verify symmetry multiplicities, trace identities, DOS normalization, and the accuracy of the iterative solver. Larger covers are solved by sparse Lanczos methods. Interior target bands are obtained through shift-invert or another spectral transformation. Large-scale density-of-states calculations use polynomial or stochastic trace methods. Every target spectral quantity is then repeated over increasing covers, increasing hopping cutoff, and tighter solver tolerance. For open-cluster visualizations, boundary enrichment and core local density of states are calculated independently. No candidate magic band is accepted until the same feature survives this complete convergence hierarchy.

For an exact commensurate configuration, the numerical calculation begins with the twist angle θ , from which the exact moiré translation group Γ_M is determined. A finite-index normal subgroup $\Gamma_N \triangleleft \Gamma_M$ is then chosen to construct the finite quotient $Q_N = \Gamma_M/\Gamma_N$, which defines a boundaryless periodic finite approximant of the exact hyperbolic moiré system. The corresponding sparse finite-cover Hamiltonian H_N is assembled from the microscopic hopping kernel and the finite-group translation representation. Because H_N commutes with the finite translation group, it can subsequently be decomposed into generalized translation sectors $H[\rho]$, where ρ runs over the irreducible unitary representations of the quotient group. Diagonalization of these symmetry-resolved blocks finally produces the physical eigenvalues $E_{n,\rho}$. Thus the exact commensurate calculation proceeds conceptually from crystallographic determination of Γ_M , through controlled finite-cover construction and sparse Hamiltonian assembly, to representation-resolved spectral computation.

For an incommensurate target twist, the numerical hierarchy is fundamentally different because no finite exact common moiré translation group exists. The starting object is therefore the local infinite-bilayer geometry at the target angle θ . This geometry is approximated using paired finite covers of the two individual layer groups, with interlayer matrix elements evaluated through their common lifts to the universal hyperbolic plane. The resulting local finite Hamiltonian H_N^{loc} is used to calculate local spectral observables without imposing an artificial common periodic supercell. If exact commensurate twists are additionally employed as numerical approximants, convergence of the angles alone is insufficient. For an approximant θ_N intended to reproduce the target system over a spatial scale L_N , the joint condition $|\theta_N - \theta| e^{L_N/R} \rightarrow 0$ must hold. This condition expresses the exponential amplification of angular mismatch in negative curvature. Consequently, an incommensurate spectrum can never be justified merely by diagonalizing a large exact supercell at a numerically nearby twist angle;

the angular approximation and the spatial scale being resolved must converge together.

The numerical framework also separates several mechanisms that can otherwise produce superficially similar narrow spectral features. Genuine physical flattening must remain stable as the hopping cutoff is increased and the microscopic Hamiltonian approaches its infinite-range or intended finite-range limit. A finite-cover artifact, by contrast, changes systematically as the injectivity radius of the periodic cover is increased. A symmetry-enforced degeneracy can remain exactly degenerate because of the irreducible-representation structure even though no genuine kinetic-energy suppression has occurred. Likewise, a feature generated by an open boundary can disappear from the core local spectral measure or exhibit a strong enhancement relative to the geometric boundary fraction. A small bandwidth observed in a finite matrix is therefore not, by itself, evidence for a bulk magic band. The physical interpretation must first exclude truncation effects, cover artifacts, symmetry-enforced multiplicities, and boundary-generated states.

The complete numerical verification procedure begins by defining the physical Hamiltonian on the infinite hyperbolic bilayer. Finite-range hopping, or exponentially decaying hopping satisfying the previously derived hyperbolic summability condition, ensures that the microscopic Hamiltonian is represented by a bounded sparse operator. Exact commensurate systems are then approximated by boundaryless periodic covers generated by finite quotients of the common Fuchsian translation group. The relevant measure of local finite-size accuracy is the injectivity radius of the cover rather than the number of sites alone. In particular, for finite-range Hamiltonians, fixed-order spectral moments become exactly equal to their infinite-system values once the systole exceeds the maximum physical length of the hopping paths contributing to those moments. This provides a controlled and boundary-free route to the thermodynamic density-of-states limit.

The sparse finite-cover Hamiltonian is solved using a hierarchy of complementary spectral methods. Complete Hermitian diagonalization is used on sufficiently small systems to provide full-spectrum benchmarks, verify state counting, test trace identities, and certify symmetry multiplicities. Lanczos and block-Lanczos methods are then used to obtain extremal and symmetry-resolved states on larger sparse covers. When the physically relevant states lie in the interior of the spectrum, shift-invert or another spectral transformation converts the target eigenvalues into extremal eigenvalues of a transformed operator, while every reported eigenpair is ultimately recertified through the residual of the original Hamiltonian. For systems too large for explicit eigenvalue calculation over the complete spectrum, polynomial trace methods such as kernel-polynomial expansions provide controlled estimates of the density of states.

The numerical error budget must keep conceptually distinct approximations separate. Finite-range trunca-

tion, finite-cover effects, angular approximation, eigen-solver residuals, and floating-point errors represent different sources of uncertainty and must be converged independently. In particular, convergence of the hopping cutoff does not establish convergence with respect to cover size, and convergence of the cover does not compensate for an inaccurate commensurate approximation to an incommensurate target angle. For incommensurate configurations, paired finite covers therefore provide the primary controlled local approximation. Exact commensurate approximants may be used as an additional diagnostic only when their angular error decreases rapidly enough relative to the increasing spatial scale to satisfy the hyperbolic joint-convergence condition.

Open hyperbolic clusters are not used as the primary approximation to the bulk thermodynamic system because their boundary remains extensive even in the large-radius limit. When such open patches are employed for real-space visualization, local-density analysis, or explicit studies of edge phenomena, boundary states must be identified relative to the nonvanishing geometric fraction of sites contained in the boundary region. A raw boundary-weight threshold is insufficient because an ordinary extended state can itself carry an order-one fraction of its total probability near the boundary. Boundary enrichment, core local density of states, participation measures, and their scaling with system size must therefore be examined together.

The resulting computational architecture can consequently be understood as a sequence in which the infinite microscopic Hamiltonian is first replaced by a geometrically controlled finite representation, that representation is converted into a sparse symmetry-resolved matrix, and established eigensolvers or spectral-measure methods are then applied. The resulting spectral quantities are subsequently tested against increasing cover injectivity radius, increasing hopping cutoff, decreasing solver residual, and, where relevant, improving angular approximation. Only after stability under this complete hierarchy has been demonstrated is a finite-system spectral feature promoted to a bulk conclusion.

Accordingly, the subsequent analysis of curvature-controlled spectral reconstruction and magic flat-band formation uses only spectral features that survive the complete numerical verification procedure. This requirement ensures that an observed suppression of bandwidth, enhancement of the density of states, emergence of a gap, or formation of a nearly dispersionless manifold can be attributed to the microscopic curvature-twist mechanism itself rather than to finite-size geometry, numerical truncation, representation-theoretic degeneracy, boundary localization, or solver error.

III. MOIRÉ DIFFRACTION AND GENERALIZED RECIPROCAL-SPACE STRUCTURE

A. Euclidean Lattice Structure Factor

Diffraction provides the most direct reciprocal-space probe of the geometric structures derived in the preceding sections. Nevertheless, several conceptually different reciprocal objects must be distinguished before a diffraction pattern can be interpreted correctly. The reciprocal lattices of the two individual layers determine the primary atomic Bragg peaks. The difference between corresponding reciprocal vectors determines a geometric beat wavevector and therefore the local moire length. At an exact commensurate twist, the common real-space coincidence lattice defines an exact crystallographic moire reciprocal lattice. Finally, genuine moire satellite intensity at reciprocal vectors outside the two bare monolayer reciprocal lattices requires a modulation that reduces the separate translation symmetries of the two layers to the common moire translation group. These four structures are related, but they are not identical. The purpose of the present subsection is therefore to derive the complete Euclidean diffraction hierarchy and to determine precisely which reciprocal-space features arise from rigid lattice superposition, which encode geometric interference, and which require genuine moire reconstruction. Consider a bilayer with microscopic basis positions τ_α , $\alpha = 1, \dots, n_b$. The reference Bravais lattice is

$$\Lambda = \{\mathbf{R} = n_1 \mathbf{a}_1 + n_2 \mathbf{a}_2 : n_1, n_2 \in \mathbb{Z}\}. \quad (3762)$$

For the square lattice,

$$\mathbf{a}_1 = a \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{a}_2 = a \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (3763)$$

The first layer has in-plane positions

$$\mathbf{r}_{1,\mathbf{R}\alpha} = \mathbf{R} + \tau_\alpha. \quad (3764)$$

Allowing for an arbitrary rigid in-plane shift $\mathbf{d}$, the second layer has positions

$$\mathbf{r}_{2,\mathbf{R}\alpha} = R_\theta (\mathbf{R} + \tau_\alpha) + \mathbf{d}. \quad (3765)$$

The two layers lie at transverse coordinates

$$z_1 = -\frac{h}{2}, \quad z_2 = +\frac{h}{2}. \quad (3766)$$

Thus the full scattering coordinates are $\mathbf{X}_{\ell,\mathbf{R}\alpha} = (\mathbf{r}_{\ell,\mathbf{R}\alpha}, z_\ell)$. Let the scattering vector be $\mathbf{Q} = (\mathbf{q}, q_z)$, where $\mathbf{q}$ is the in-plane momentum transfer. The microscopic scattering density may be written as

$$\rho(\mathbf{X}) = \sum_{\ell=1}^2 \sum_{\mathbf{R}} \sum_{\alpha=1}^{n_b} f_{\ell\alpha}(\mathbf{Q}) \delta^{(3)}[\mathbf{X} - \mathbf{X}_{\ell,\mathbf{R}\alpha}]. \quad (3767)$$

where $f_{\ell\alpha}(\mathbf{Q})$ is the microscopic scattering form factor. The kinematic scattering amplitude is the Fourier transform of the microscopic density,

$$\mathcal{A}(\mathbf{Q}) = \int d^3X \rho(\mathbf{X}) e^{-i\mathbf{Q}\cdot\mathbf{X}}. \quad (3768)$$

Substitution gives

$$\mathcal{A}(\mathbf{Q}) = \sum_{\ell,\mathbf{R},\alpha} f_{\ell\alpha}(\mathbf{Q}) e^{-i\mathbf{q}\cdot\mathbf{r}_{\ell,\mathbf{R}\alpha}} e^{-iq_z z_\ell}. \quad (3769)$$

The amplitude decomposes into the two layer contributions, $\mathcal{A} = \mathcal{A}_1 + \mathcal{A}_2$. The unnormalized diffraction intensity is

$$I(\mathbf{Q}) = |\mathcal{A}(\mathbf{Q})|^2. \quad (3770)$$

For a finite sample containing N_{tot} microscopic scatterers, a convenient normalized structure factor is

$$S(\mathbf{Q}) = \frac{1}{N_{\text{tot}}} \left| \sum_{j=1}^{N_{\text{tot}}} f_j(\mathbf{Q}) e^{-i\mathbf{Q}\cdot\mathbf{X}_j} \right|^2. \quad (3771)$$

For identical unit scattering amplitudes and purely in-plane scattering, $f_j = 1$, $q_z = 0$, this reduces to

$$S(\mathbf{q}) = \frac{1}{N_{\text{tot}}} \left| \sum_{j=1}^{N_{\text{tot}}} e^{-i\mathbf{q}\cdot\mathbf{r}_j} \right|^2. \quad (3772)$$

Up to normalization, this is the structure factor used throughout the subsequent diffraction analysis. Using $\mathcal{A} = \mathcal{A}_1 + \mathcal{A}_2$, the intensity is

$$I = |\mathcal{A}_1|^2 + |\mathcal{A}_2|^2 + \mathcal{A}_1 \mathcal{A}_2^* + \mathcal{A}_1^* \mathcal{A}_2. \quad (3773)$$

Therefore,

$$I = I_1 + I_2 + 2 \text{Re} [\mathcal{A}_1 \mathcal{A}_2^*]. \quad (3774)$$

The first two terms are the individual-layer diffraction intensities. The third term is the coherent bilayer interference contribution. These terms must be separated when identifying moire-related intensity. For the first layer,

$$\mathcal{A}_1 = e^{+iq_z h/2} \sum_{\mathbf{R}} \sum_{\alpha} f_{1\alpha}(\mathbf{Q}) e^{-i\mathbf{q}\cdot(\mathbf{R}+\tau_\alpha)}. \quad (3775)$$

Define the microscopic basis form factor

$$F_1(\mathbf{q}, \mathbf{Q}) = \sum_{\alpha} f_{1\alpha}(\mathbf{Q}) e^{-i\mathbf{q} \cdot \boldsymbol{\tau}_{\alpha}}. \quad (3776)$$

Then

$$\mathcal{A}_1 = e^{+iq_z h/2} F_1(\mathbf{q}, \mathbf{Q}) \sum_{\mathbf{R} \in \Lambda} e^{-i\mathbf{q} \cdot \mathbf{R}}. \quad (3777)$$

The reciprocal lattice is

$$\Lambda^* = \{\mathbf{G} = m_1 \mathbf{b}_1 + m_2 \mathbf{b}_2 : m_1, m_2 \in \mathbb{Z}\}. \quad (3778)$$

with

$$\mathbf{b}_1 = \frac{2\pi}{a} \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{b}_2 = \frac{2\pi}{a} \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (3779)$$

Poisson summation gives

$$\sum_{\mathbf{R} \in \Lambda} e^{-i\mathbf{q} \cdot \mathbf{R}} = \frac{(2\pi)^2}{A_0} \sum_{\mathbf{G} \in \Lambda^*} \delta^{(2)}(\mathbf{q} - \mathbf{G}). \quad (3780)$$

where $A_0 = a^2$. Thus

$$\mathcal{A}_1(\mathbf{Q}) = e^{+iq_z h/2} \frac{(2\pi)^2}{A_0} \sum_{\mathbf{G} \in \Lambda^*} F_1(\mathbf{G}, \mathbf{Q}) \delta^{(2)}(\mathbf{q} - \mathbf{G}). \quad (3781)$$

The ideal infinite first layer therefore produces Bragg peaks at $\mathbf{q} = \mathbf{G} \in \Lambda^*$. The second-layer amplitude is

$$\begin{aligned} \mathcal{A}_2 &= e^{-iq_z h/2} e^{-i\mathbf{q} \cdot \mathbf{d}} \\ &\times \sum_{\mathbf{R}, \alpha} f_{2\alpha}(\mathbf{Q}) e^{-i\mathbf{q} \cdot R_{\theta}(\mathbf{R} + \boldsymbol{\tau}_{\alpha})}. \end{aligned} \quad (3782)$$

Using

$$\mathbf{q} \cdot R_{\theta} \mathbf{x} = (R_{-\theta} \mathbf{q}) \cdot \mathbf{x}, \quad (3783)$$

one obtains

$$\begin{aligned} \mathcal{A}_2 &= e^{-iq_z h/2} e^{-i\mathbf{q} \cdot \mathbf{d}} F_2(R_{-\theta} \mathbf{q}, \mathbf{Q}) \\ &\times \sum_{\mathbf{R}} e^{-i(R_{-\theta} \mathbf{q}) \cdot \mathbf{R}}. \end{aligned} \quad (3784)$$

Poisson summation gives

$$\begin{aligned} \mathcal{A}_2 &= e^{-iq_z h/2} e^{-i\mathbf{q} \cdot \mathbf{d}} \frac{(2\pi)^2}{A_0} \\ &\times \sum_{\mathbf{G} \in \Lambda^*} F_2(\mathbf{G}, \mathbf{Q}) \delta^{(2)}(\mathbf{q} - R_{\theta} \mathbf{G}). \end{aligned} \quad (3785)$$

The second-layer reciprocal lattice is therefore $\Lambda_2^* = R_{\theta} \Lambda^*$. For the rigid bilayer in the single-scattering density model, the diffraction amplitude therefore has support on $\Lambda_1^* \cup \Lambda_2^*$. Thus the primary reciprocal-space signature of twisting is the superposition of two rotated reciprocal lattices. This statement must be distinguished from the presence of true additional moire satellite peaks. The visual existence of a long-wavelength moire envelope in real space does not, by itself, imply nonzero first-order scattering amplitude at every moire reciprocal vector. An experiment or numerical Fourier transform always contains a finite observation window. Let the real-space window be $W_L(\mathbf{r})$, with characteristic linear size L . The finite density is

$$\rho_L(\mathbf{r}) = W_L(\mathbf{r}) \rho_{\infty}(\mathbf{r}). \quad (3786)$$

Fourier multiplication becomes convolution,

$$\tilde{\rho}_L(\mathbf{q}) = \frac{1}{(2\pi)^2} \widetilde{W}_L * \tilde{\rho}_{\infty}. \quad (3787)$$

Therefore an ideal reciprocal delta peak at $\mathbf{G}$ becomes

$$\delta^{(2)}(\mathbf{q} - \mathbf{G}) \longrightarrow \widetilde{W}_L(\mathbf{q} - \mathbf{G}). \quad (3788)$$

If the windows form a scale family satisfying

$$W_L(\mathbf{r}) = W_1\left(\frac{\mathbf{r}}{L}\right), \quad (3789)$$

then

$$\widetilde{W}_L(\mathbf{q}) = L^2 \widetilde{W}_1(L\mathbf{q}). \quad (3790)$$

Hence the reciprocal linewidth scales as $\delta q_{\text{win}} = O(L^{-1})$. This finite-width scale will later determine the minimum resolvable twist. Consider the same reciprocal vector in the two layers. The first-layer peak is located at $\mathbf{G}$, whereas the second-layer peak is $R_{\theta} \mathbf{G}$. Define the reciprocal splitting vector

$$\Delta \mathbf{G}_{\theta} = R_{\theta} \mathbf{G} - \mathbf{G}. \quad (3791)$$

Its squared magnitude is

$$\begin{aligned} |\Delta \mathbf{G}_{\theta}|^2 &= |R_{\theta} \mathbf{G}|^2 + |\mathbf{G}|^2 - 2\mathbf{G} \cdot R_{\theta} \mathbf{G} \\ &= 2G^2 (1 - \cos \theta). \end{aligned} \quad (3792)$$

Using

$$1 - \cos \theta = 2 \sin^2 \left(\frac{\theta}{2} \right), \quad (3793)$$

one obtains

$$|\Delta \mathbf{G}_\theta| = 2G \left| \sin \left(\frac{\theta}{2} \right) \right|. \quad (3794)$$

This expression is exact at arbitrary twist. In two dimensions,

$$e^{i\theta} - 1 = 2ie^{i\theta/2} \sin \left(\frac{\theta}{2} \right). \quad (3795)$$

Therefore the matrix identity is

$$R_\theta - I = 2 \sin \left(\frac{\theta}{2} \right) R_{\theta/2+\pi/2} \quad (3796)$$

for positive twist in the chosen orientation convention. Hence

$$\Delta \mathbf{G}_\theta = 2 \sin \left(\frac{\theta}{2} \right) R_{\theta/2+\pi/2} \mathbf{G}. \quad (3797)$$

Thus the reciprocal beat vector is obtained by scaling the microscopic reciprocal vector by $2 \sin \left(\frac{\theta}{2} \right)$ and rotating it through $\frac{\pi}{2} + \frac{\theta}{2}$. For the first square reciprocal shell, $G_0 = \frac{2\pi}{a}$. The peak splitting is therefore

$$\Delta q_1 = \frac{4\pi}{a} \left| \sin \left(\frac{\theta}{2} \right) \right|. \quad (3798)$$

Define the geometric moiré wavevector by $q_M^{\text{geom}} = \Delta q_1$. The associated real-space beat length is

$$L_M^{\text{beat}} = \frac{2\pi}{q_M^{\text{geom}}}. \quad (3799)$$

Using Eq. (3798) gives

$$L_M^{\text{beat}} = \frac{a}{2 |\sin(\theta/2)|}. \quad (3800)$$

This is exactly the Euclidean local moiré length obtained previously, $L_M^{\text{beat}} = \xi_M^{(E)}$. Thus the real-space local registry scale and the reciprocal peak splitting are exact Fourier-dual quantities. Define

$$\mathbf{q}_1^{\text{beat}} = (R_\theta - I) \mathbf{b}_1, \quad (3801)$$

and

$$\mathbf{q}_2^{\text{beat}} = (R_\theta - I) \mathbf{b}_2. \quad (3802)$$

The beat reciprocal-basis matrix is

$$B_{\text{beat}} = (R_\theta - I) B. \quad (3803)$$

Its determinant is

$$|\det B_{\text{beat}}| = |\det (R_\theta - I)| |\det B|. \quad (3804)$$

Since

$$|\det (R_\theta - I)| = 4 \sin^2 \left(\frac{\theta}{2} \right), \quad (3805)$$

one obtains

$$A_{\text{beat}}^* = 4 \sin^2 \left(\frac{\theta}{2} \right) \left(\frac{2\pi}{a} \right)^2. \quad (3806)$$

The reciprocal dual therefore has area

$$A_{\text{beat}} = \frac{a^2}{4 \sin^2 (\theta/2)}. \quad (3807)$$

This is the geometric beat area derived previously. For an exactly commensurate twist, $\Lambda_M = \Lambda_1 \cap \Lambda_2$. The exact crystallographic reciprocal lattice is $\Lambda_M^* = (\Lambda_M)^*$. As established previously, $\Lambda_M^* = \Lambda_1^* + \Lambda_2^*$. Thus

$$\Lambda_1^* \subset \Lambda_M^*, \quad \Lambda_2^* \subset \Lambda_M^*. \quad (3808)$$

The combined bilayer density is therefore periodic under the exact coincidence lattice and is allowed, by common translation symmetry, to possess Fourier components anywhere on Λ_M^* . However, the rigid uncoupled atomic densities occupy only the two reciprocal sublattices. Define the set

$$\mathcal{E}_M = \Lambda_M^* \setminus (\Lambda_1^* \cup \Lambda_2^*). \quad (3809)$$

For the ideal rigid bilayer and the linear density-scattering model, $\mathcal{A}_0(\mathbf{Q}) = 0$ for $\mathbf{q} \in \mathcal{E}_M$, apart from finite-window leakage or higher-order scattering effects. Thus many reciprocal vectors allowed by the exact moiré periodicity are systematically extinct in the perfectly rigid linear-scattering limit. This distinction is central: allowed exact moiré reciprocal vector $\nrightarrow$ nonzero rigid-bilayer diffraction intensity. For a commensurate square twist, write

$$\tan \left(\frac{\hat{\theta}}{2} \right) = \frac{n}{m}, \quad (3810)$$

where $\hat{\theta}$ is the point-group-reduced angle. The exact coincidence index is

$$\Sigma = \frac{m^2 + n^2}{d}, \quad (3811)$$

with $d = 1$ for opposite parity and $d = 2$ when both integers are odd. Since

$$\sin\left(\frac{\hat{\theta}}{2}\right) = \frac{n}{\sqrt{m^2 + n^2}}, \quad (3812)$$

one obtains

$$\Sigma = \frac{n^2}{d} \csc^2\left(\frac{\hat{\theta}}{2}\right). \quad (3813)$$

The exact supercell linear dimension is $L_M^{\text{sc}} = a\sqrt{\Sigma}$. The beat length is

$$L_M^{\text{beat}} = \frac{a}{2 \sin\left(\frac{\hat{\theta}}{2}\right)}. \quad (3814)$$

Therefore,

$$\frac{L_M^{\text{sc}}}{L_M^{\text{beat}}} = \frac{2n}{\sqrt{d}}. \quad (3815)$$

Likewise,

$$\frac{A_M^{\text{sc}}}{A_M^{\text{beat}}} = \frac{4n^2}{d}. \quad (3816)$$

Thus the local diffraction beat scale and the exact crystallographic supercell scale differ by an arithmetic multiplier. The exact square moiré reciprocal basis has magnitude

$$G_M = \frac{2\pi}{a\sqrt{\Sigma}}. \quad (3817)$$

The primary reciprocal splitting is

$$\Delta q_1 = \frac{4\pi}{a} \sin\left(\frac{\hat{\theta}}{2}\right). \quad (3818)$$

Using the exact coincidence-index formula yields

$$\frac{\Delta q_1}{G_M} = \frac{2n}{\sqrt{d}}. \quad (3819)$$

Thus the geometric beat splitting is generally not the primitive exact moiré reciprocal vector. It is an arithmetic higher harmonic of the exact reciprocal lattice. For $d = 1$, the exact reciprocal basis is

$$\mathbf{G}_1 = \frac{2\pi}{a(m^2 + n^2)} \begin{pmatrix} m \\ n \end{pmatrix}, \quad (3820)$$

and

$$\mathbf{G}_2 = \frac{2\pi}{a(m^2 + n^2)} \begin{pmatrix} -n \\ m \end{pmatrix}. \quad (3821)$$

Direct substitution into the reciprocal splitting gives

$$R_\theta \mathbf{b}_1 - \mathbf{b}_1 = 2n \mathbf{G}_2. \quad (3822)$$

and

$$R_\theta \mathbf{b}_2 - \mathbf{b}_2 = -2n \mathbf{G}_1. \quad (3823)$$

Thus the familiar reciprocal splitting corresponds exactly to the $2n$ -th harmonic of the primitive crystallographic reciprocal basis. When both m and n are odd, the primitive reciprocal basis is

$$\mathbf{G}_1 = \frac{2\pi}{a(m^2 + n^2)} \begin{pmatrix} m+n \\ n-m \end{pmatrix}, \quad (3824)$$

and

$$\mathbf{G}_2 = \frac{2\pi}{a(m^2 + n^2)} \begin{pmatrix} m-n \\ m+n \end{pmatrix}. \quad (3825)$$

One obtains

$$R_\theta \mathbf{b}_1 - \mathbf{b}_1 = n(\mathbf{G}_2 - \mathbf{G}_1). \quad (3826)$$

and

$$R_\theta \mathbf{b}_2 - \mathbf{b}_2 = -n(\mathbf{G}_1 + \mathbf{G}_2). \quad (3827)$$

The beat reciprocal vectors are therefore exact integer combinations of the crystallographic reciprocal vectors, but they need not themselves be a primitive reciprocal basis. True satellite intensity outside the two bare reciprocal sublattices appears when the microscopic scattering density acquires a modulation with the common moiré periodicity. A natural example is a displacement field. Let the relaxed position of an atom be

$$\mathbf{r} = \mathbf{r}^{(0)} + \mathbf{u}_\ell(\mathbf{r}^{(0)}). \quad (3828)$$

where the displacement field satisfies

$$\mathbf{u}_\ell(\mathbf{r} + \mathbf{R}_M) = \mathbf{u}_\ell(\mathbf{r}). \quad (3829)$$

It therefore has the Fourier expansion

$$\mathbf{u}_\ell(\mathbf{r}) = \sum_{\mathbf{Q} \in \Lambda_M^*} \mathbf{u}_{\ell, \mathbf{Q}} e^{i\mathbf{Q} \cdot \mathbf{r}}. \quad (3830)$$

The scattering phase is $e^{-i\mathbf{q} \cdot [\mathbf{r}^{(0)} + \mathbf{u}(\mathbf{r}^{(0)})]}$. For weak displacement, $|\mathbf{q} \cdot \mathbf{u}| \ll 1$, the exponential becomes

$$e^{-i\mathbf{q} \cdot \mathbf{r}} = e^{-i\mathbf{q} \cdot \mathbf{r}^{(0)}} \left[1 - i\mathbf{q} \cdot \mathbf{u}(\mathbf{r}^{(0)}) - \frac{1}{2} (\mathbf{q} \cdot \mathbf{u})^2 + \dots \right]. \quad (3831)$$

The first-order correction to the scattering amplitude is

$$\delta \mathcal{A}_\ell^{(1)}(\mathbf{q}) = -i \sum_{\mathbf{r}^{(0)}} f e^{-i\mathbf{q} \cdot \mathbf{r}^{(0)}} \mathbf{q} \cdot \mathbf{u}_\ell(\mathbf{r}^{(0)}). \quad (3832)$$

Substituting the displacement Fourier expansion gives

$$\delta \mathcal{A}_\ell^{(1)}(\mathbf{q}) = -i \sum_{\mathbf{Q}} \mathbf{q} \cdot \mathbf{u}_{\ell, \mathbf{Q}} \times \sum_{\mathbf{r}^{(0)}} f e^{-i(\mathbf{q} - \mathbf{Q}) \cdot \mathbf{r}^{(0)}}. \quad (3833)$$

The atomic lattice sum becomes nonzero when $\mathbf{q} - \mathbf{Q} = \mathbf{G}_\ell$, where $\mathbf{G}_\ell \in \Lambda_\ell^*$. Therefore the first-order satellite condition is

$$\mathbf{q}_{\text{sat}} = \mathbf{G}_\ell + \mathbf{Q}, \quad \mathbf{Q} \in \Lambda_M^*. \quad (3834)$$

This is the microscopic origin of moiré diffraction satellites generated by periodic structural reconstruction. At $\mathbf{q} = \mathbf{G}_\ell + \mathbf{Q}$, the leading satellite amplitude is proportional to

$$\delta \mathcal{A}_\ell^{(1)} \propto -i (\mathbf{G}_\ell + \mathbf{Q}) \cdot \mathbf{u}_{\ell, \mathbf{Q}} F_\ell(\mathbf{G}_\ell). \quad (3835)$$

Thus longitudinal components of the displacement relative to the scattering vector contribute directly to the satellite amplitude. If the primary Bragg amplitude at $\mathbf{G}_\ell$ is $\mathcal{A}_G$, then to leading order

$$\frac{I_{\mathbf{G}+\mathbf{Q}}}{I_G} \sim |(\mathbf{G} + \mathbf{Q}) \cdot \mathbf{u}_\mathbf{Q}|^2. \quad (3836)$$

Thus weak structural modulation produces satellite amplitude linear in the displacement and satellite intensity quadratic in the displacement. A second mechanism is a periodically modulated scattering weight. Let

$$f(\mathbf{r}) = f_0 [1 + \eta(\mathbf{r})]. \quad (3837)$$

with

$$\eta(\mathbf{r}) = \sum_{\mathbf{Q} \in \Lambda_M^*} \eta_{\mathbf{Q}} e^{i\mathbf{Q} \cdot \mathbf{r}}. \quad (3838)$$

The scattering amplitude contains

$$\delta \mathcal{A} \propto \sum_{\mathbf{Q}} \eta_{\mathbf{Q}} \sum_{\mathbf{r}^{(0)}} e^{-i(\mathbf{q} - \mathbf{Q}) \cdot \mathbf{r}^{(0)}}. \quad (3839)$$

Hence the same satellite selection rule follows, $\mathbf{q} = \mathbf{G} + \mathbf{Q}$. The leading intensity ratio is

$$\frac{I_{\mathbf{G}+\mathbf{Q}}}{I_G} \sim |\eta_{\mathbf{Q}}|^2. \quad (3840)$$

Thus structural displacement and local scattering-amplitude modulation generate the same satellite positions but different polarization and intensity selection rules. The quadratic term in the displacement expansion contains products $\mathbf{u}_{\mathbf{Q}_1} \mathbf{u}_{\mathbf{Q}_2}$. It therefore produces reciprocal transfers $\mathbf{Q} = \mathbf{Q}_1 + \mathbf{Q}_2$. More generally, an order- p modulation process can produce

$$\mathbf{q} = \mathbf{G} + \sum_{j=1}^p \mathbf{Q}_j. \quad (3841)$$

For a weak modulation amplitude u , the corresponding amplitude scales schematically as $\mathcal{A}^{(p)} = O(u^p)$, and the isolated intensity contribution scales as $I^{(p)} = O(u^{2p})$. High-order satellites therefore provide a sensitive probe of strong interlayer reconstruction. The geometric moiré wavevector also appears naturally in nonlinear observables. Let the leading harmonic of the two layer densities be represented schematically as

$$\rho_1(\mathbf{r}) \sim \cos(\mathbf{G} \cdot \mathbf{r}). \quad (3842)$$

and

$$\rho_2(\mathbf{r}) \sim \cos(R_\theta \mathbf{G} \cdot \mathbf{r}). \quad (3843)$$

Their sum contains only the two microscopic Fourier harmonics. However, their product is

$$\begin{aligned} \rho_1 \rho_2 &\sim \frac{1}{2} \cos[(R_\theta \mathbf{G} - \mathbf{G}) \cdot \mathbf{r}] \\ &+ \frac{1}{2} \cos[(R_\theta \mathbf{G} + \mathbf{G}) \cdot \mathbf{r}]. \end{aligned} \quad (3844)$$

Thus the slowly varying registry component appears at

$$\mathbf{q}_M = R_\theta \mathbf{G} - \mathbf{G}. \quad (3845)$$

This explains why the geometric moiré pattern is directly visible in stacking-sensitive or nonlinear real-space observables even when the linear atomic density contains only the two microscopic reciprocal peaks. For a real scattering density, $\rho(\mathbf{r}) \in \mathbb{R}$. Its Fourier transform satisfies $\mathcal{A}(-\mathbf{q}) = \mathcal{A}^*(\mathbf{q})$. Therefore, $S(-\mathbf{q}) = S(\mathbf{q})$. The in-plane diffraction intensity is inversion symmetric in the absence of a mechanism violating the assumptions underlying this relation. Let $C_4 = R_{\pi/2}$. The square reciprocal lattice satisfies $C_4 \Lambda^* = \Lambda^*$. Since all two-dimensional rotations about the same center commute,

$$C_4 R_\theta = R_\theta C_4. \quad (3846)$$

Therefore, $C_4\Lambda_2^* = \Lambda_2^*$. The bare reciprocal union is consequently fourfold symmetric,

$$C_4 [\Lambda_1^* \cup \Lambda_2^*] = \Lambda_1^* \cup \Lambda_2^*. \quad (3847)$$

For isotropic identical scattering form factors, $S(C_4\mathbf{q}) = S(\mathbf{q})$. In the symmetric twist gauge, a monolayer reflection exchanges the two layers. For identical layers and purely in-plane scattering, layer exchange does not change the measured total density. Hence the total intensity satisfies the corresponding reflection symmetry, $S(\sigma\mathbf{q}) = S(\mathbf{q})$. Thus the in-plane diffraction intensity can realize the full bilayer dihedral symmetry even when the reflection is not layer preserving. A special symmetry enhancement occurs when the reduced twist is $\hat{\theta} = \frac{\pi}{4}$. Then $R_{\pi/4}\Lambda_1^* = \Lambda_2^*$. Applying the same rotation again gives

$$R_{\pi/4}\Lambda_2^* = R_{\pi/2}\Lambda_1^* = \Lambda_1^*. \quad (3848)$$

Therefore,

$$R_{\pi/4} [\Lambda_1^* \cup \Lambda_2^*] = \Lambda_1^* \cup \Lambda_2^*. \quad (3849)$$

For identical in-plane layer weights,

$$S(R_{\pi/4}\mathbf{q}) = S(\mathbf{q}). \quad (3850)$$

The bare diffraction pattern therefore exhibits an eightfold symmetry enhancement even though a periodic two-dimensional Bravais lattice cannot possess an ordinary eightfold crystallographic point symmetry. This observation becomes particularly relevant for incommensurate square-bilayer quasicrystalline limits. The layer amplitudes contain transverse phases $e^{+iq_z h/2}$ and $e^{-iq_z h/2}$. At an in-plane momentum where both layer amplitudes contribute coherently, the interference term contains $e^{iq_z h}$. Thus the coherent bilayer intensity can oscillate as a function of out-of-plane momentum transfer. For purely in-plane diffraction, $q_z = 0$, the transverse phase disappears. The exact first-shell splitting is

$$\Delta q_1(\theta) = \frac{4\pi}{a} \sin\left(\frac{|\theta|}{2}\right). \quad (3851)$$

Differentiation gives

$$\frac{d\Delta q_1}{d|\theta|} = \frac{2\pi}{a} \cos\left(\frac{|\theta|}{2}\right). \quad (3852)$$

For small twist,

$$\Delta q_1 = \frac{2\pi}{a}|\theta| - \frac{\pi}{12a}|\theta|^3 + \frac{\pi}{960a}|\theta|^5 + O(|\theta|^7). \quad (3853)$$

The reciprocal splitting is therefore asymptotically linear in the twist. Using the cosecant expansion gives

$$L_M^{\text{beat}} = \frac{a}{|\theta|} \left[1 + \frac{\theta^2}{24} + \frac{7\theta^4}{5760} + O(\theta^6) \right]. \quad (3854)$$

Thus reciprocal peak splitting and real-space moiré divergence are exact inverse manifestations of the same small-angle geometry.

a. Theorem 42. Rigid-bilayer diffraction decomposition. For two identical rigid Euclidean lattices related by a relative rotation, the kinematic single-scattering amplitude decomposes into two monolayer reciprocal-lattice combs. The reciprocal support is

$$\text{supp } \mathcal{A}_0 = \Lambda_1^* \cup \Lambda_2^*. \quad (3855)$$

For corresponding reciprocal vectors of magnitude G , the exact peak splitting is

$$\Delta q = 2G |\sin(\theta/2)|. \quad (3856)$$

For the first square reciprocal shell, the associated real-space beat scale is

$$L_M^{\text{beat}} = \frac{a}{2 |\sin(\theta/2)|}. \quad (3857)$$

At a commensurate twist, the exact crystallographic reciprocal lattice is larger than the bare diffraction support, $\Lambda_M^* \supseteq \Lambda_1^* \cup \Lambda_2^*$, and additional reciprocal vectors are systematically extinct in the perfectly rigid linear-scattering limit.

b. Theorem 43. Moiré satellite selection. Suppose a microscopic scattering density is modulated by a field periodic under the exact moiré lattice. Let its Fourier harmonics belong to $\mathbf{Q} \in \Lambda_M^*$. Then a microscopic Bragg peak at $\mathbf{G}_\ell$ generates satellite amplitude at $\mathbf{q}_{\text{sat}} = \mathbf{G}_\ell + \mathbf{Q}$. For a weak displacement field, the first-order satellite amplitude is linear in the displacement Fourier coefficient,

$$\mathcal{A}_{\text{sat}} = O(\mathbf{q} \cdot \mathbf{u}_{\mathbf{Q}}). \quad (3858)$$

whereas the leading isolated satellite intensity is quadratic,

$$I_{\text{sat}} = O(|\mathbf{q} \cdot \mathbf{u}_{\mathbf{Q}}|^2). \quad (3859)$$

Thus nonzero intensity on the additional moiré reciprocal vectors is a probe of genuine modulation or reconstruction rather than a necessary consequence of rigid superposition alone.

B. Diffraction Signatures of Commensurate and Incommensurate Moiré States

The preceding subsection established that reciprocal peak splitting and exact crystallographic reciprocal periodicity are different objects. This distinction becomes particularly important when comparing commensurate and incommensurate twists. An exactly commensurate bilayer possesses a finite real-space coincidence cell and therefore a discrete rank-two exact moiré reciprocal lattice. A generic incommensurate bilayer has no finite exact supercell. Its two atomic reciprocal lattices nevertheless remain perfectly well defined, and the geometric splitting between corresponding Bragg peaks continues to

define a local moiré scale. Furthermore, an ideal incommensurate structure need not produce diffuse diffraction. A deterministic quasiperiodic structure can possess sharp pure-point reciprocal features belonging to a higher-rank reciprocal module. Diffuse broadening instead probes finite coherence, disorder, phason fluctuations, strain inhomogeneity, or finite observation size. The present subsection therefore develops quantitative diffraction criteria capable of distinguishing periodic commensurability, quasiperiodic incommensurability, finite coherence, and purely geometric beat scales. For the square lattice, an exact finite coincidence lattice exists if and only if $R_\theta \in SO(2, \mathbb{Q})$. Equivalently, $\cos \theta \in \mathbb{Q}$, $\sin \theta \in \mathbb{Q}$. Equivalently,

$$\tan\left(\frac{\theta}{2}\right) \in \mathbb{Q}. \quad (3860)$$

The commensurate twists are therefore parameterized by coprime integers

$$\tan\left(\frac{\theta}{2}\right) = \frac{n}{m}. \quad (3861)$$

A twist is incommensurate when $R_\theta \notin SO(2, \mathbb{Q})$. Then

$$[\Lambda : \Lambda \cap R_\theta \Lambda] = \infty. \quad (3862)$$

There is no finite exact real-space supercell and therefore no finite-area exact moiré Brillouin zone. Nevertheless,

$$\Delta \mathbf{G}_\theta = R_\theta \mathbf{G} - \mathbf{G} \quad (3863)$$

remains well defined for every reciprocal vector. Thus local moiré diffraction geometry does not require exact commensurability. Define the additive reciprocal module generated by the two layers as $\mathcal{M}_\theta^* = \Lambda_1^* + \Lambda_2^*$. Since every reciprocal lattice is inversion symmetric, $-\Lambda_\ell^* = \Lambda_\ell^*$, the same module may be written as $\mathcal{M}_\theta^* = \Lambda_1^* - \Lambda_2^*$. Explicitly,

$$\mathcal{M}_\theta^* = \{m_1 \mathbf{b}_1 + m_2 \mathbf{b}_2 + n_1 R_\theta \mathbf{b}_1 + n_2 R_\theta \mathbf{b}_2 : m_i, n_i \in \mathbb{Z}\}. \quad (3864)$$

This object provides the natural reciprocal indexing space for nonlinear or reconstructed bilayer observables. At an exact commensurate twist, $\mathcal{M}_\theta^* = \Lambda_M^*$. It is a discrete rank-two lattice. Thus every reciprocal vector generated by arbitrary integer combinations of the two microscopic reciprocal lattices can be indexed using two exact moiré reciprocal basis vectors,

$$\mathbf{Q} = h \mathbf{G}_1 + k \mathbf{G}_2, \quad h, k \in \mathbb{Z}. \quad (3865)$$

This two-integer indexing property is the reciprocal-space expression of exact translational commensurability. For a generic irrational rotation, the four vectors $\mathbf{b}_1$, $\mathbf{b}_2$, $R_\theta \mathbf{b}_1$, $R_\theta \mathbf{b}_2$ are rationally independent up to the two-dimensional embedding. The reciprocal module then has integer rank $\text{rank}_{\mathbb{Z}} \mathcal{M}_\theta^* = 4$. It is no longer a discrete two-dimensional reciprocal lattice. For generic

irrational rotations, its closure is dense in the reciprocal plane, $\overline{\mathcal{M}_\theta^*} = \mathbb{R}^2$. Exceptional irrational angles can exhibit additional integer relations and a lower module rank, but exact periodic commensurability still requires a discrete rank-two module. Suppose, for contradiction, that the reciprocal module of an incommensurate bilayer were a discrete full-rank lattice, $\mathcal{M}_\theta^* \subset \mathbb{R}^2$. Because $\Lambda_1^* \subset \mathcal{M}_\theta^*$, a discrete full-rank superlattice containing Λ_1^* has finite index, $[\mathcal{M}_\theta^* : \Lambda_1^*] < \infty$. Likewise, $[\mathcal{M}_\theta^* : \Lambda_2^*] < \infty$. The two microscopic reciprocal lattices would therefore be commensurate. Duality would imply commensurability of the two direct lattices. This contradicts the assumed incommensurate twist. Therefore an incommensurate reciprocal module cannot be a discrete rank-two lattice. The preceding higher-rank module must not be confused with the bare single-scattering diffraction support. For a perfectly rigid bilayer,

$$\text{supp } \mathcal{A}_0 = \Lambda_1^* \cup \Lambda_2^* \quad (3866)$$

even at an incommensurate twist. The full reciprocal module becomes physically populated when nonlinear stacking-sensitive observables, structural relaxation, scattering-amplitude modulation, or higher-order scattering processes generate sums and differences of microscopic reciprocal vectors. Thus, reciprocal module $\neq$ bare first-order diffraction support. An incommensurate deterministic modulation can generate a countable set of sharp reciprocal peaks of the form

$$\mathbf{Q} = m_1 \mathbf{b}_1 + m_2 \mathbf{b}_2 + n_1 R_\theta \mathbf{b}_1 + n_2 R_\theta \mathbf{b}_2. \quad (3867)$$

Although this set can be dense mathematically, experimentally observable intensity need not be dense at finite resolution because high-order coefficients become progressively weaker. Therefore incommensurate $\nrightarrow$ diffuse. Ideal quasiperiodicity and finite coherence are logically distinct. For a reciprocal-module vector represented by integers

$$\mathbf{Q} = m_1 \mathbf{b}_1 + m_2 \mathbf{b}_2 + n_1 R_\theta \mathbf{b}_1 + n_2 R_\theta \mathbf{b}_2, \quad (3868)$$

define an algebraic order

$$\mathcal{O}(\mathbf{Q}) = |m_1| + |m_2| + |n_1| + |n_2|. \quad (3869)$$

For a weak interlayer modulation, Fourier amplitudes are generally suppressed as the order increases. Thus any finite intensity threshold selects only a finite effective subset of the mathematically infinite reciprocal module. This explains why a quasiperiodic diffraction pattern can remain experimentally sparse despite having a dense reciprocal module. For any microscopic reciprocal vector of magnitude G ,

$$\Delta q_G = 2G \left| \sin\left(\frac{\theta}{2}\right) \right|. \quad (3870)$$

This equation contains no commensurability assumption. Therefore the reciprocal splitting is a local geometric observable that is defined equally for commensurate and

incommensurate states. The associated beat length is

$$L_G^{\text{beat}} = \frac{2\pi}{\Delta q_G}. \quad (3871)$$

For the first square reciprocal shell,

$$L_1^{\text{beat}} = \frac{a}{2|\sin(\theta/2)|}. \quad (3872)$$

At a commensurate twist,

$$L_M^{\text{sc}} = \frac{2n}{\sqrt{d}} L_M^{\text{beat}}. \quad (3873)$$

Thus diffraction peak splitting alone determines the local beat length but does not uniquely determine the exact supercell size without arithmetic information. This is the reciprocal-space counterpart of the distinction between effective moiré domains and exact coincidence cells. Define

$$\eta_{\text{arith}} = \frac{\Delta q_1}{G_M}. \quad (3874)$$

For a commensurate square bilayer,

$$\eta_{\text{arith}} = \frac{2n}{\sqrt{d}}. \quad (3875)$$

Its square is

$$\eta_{\text{arith}}^2 = \frac{4n^2}{d}. \quad (3876)$$

The same quantity is exactly the area ratio

$$\eta_{\text{arith}}^2 = \frac{A_M^{\text{sc}}}{A_M^{\text{beat}}}. \quad (3877)$$

Thus reciprocal diffraction provides a direct route to the geometric–arithmetic separation of the moiré scale. Equation (3870) can be inverted. For a reciprocal shell of known magnitude G ,

$$|\theta| = 2 \arcsin \left(\frac{\Delta q_G}{2G} \right). \quad (3878)$$

For the first square shell,

$$|\theta| = 2 \arcsin \left(\frac{a\Delta q_1}{4\pi} \right). \quad (3879)$$

Thus the twist can be extracted without first determining the exact commensurate supercell. Differentiate the inversion formula. One obtains

$$\frac{d\theta}{d(\Delta q)} = \frac{1}{G\sqrt{1 - (\Delta q/2G)^2}}. \quad (3880)$$

Using

$$\frac{\Delta q}{2G} = \sin \left(\frac{\theta}{2} \right), \quad (3881)$$

this becomes

$$\frac{d\theta}{d(\Delta q)} = \frac{1}{G \cos(\theta/2)}. \quad (3882)$$

Therefore an experimental splitting uncertainty $\sigma_{\Delta q}$ gives

$$\sigma_\theta \simeq \frac{\sigma_{\Delta q}}{G \cos(\theta/2)}. \quad (3883)$$

At small twist,

$$\sigma_\theta \simeq \frac{\sigma_{\Delta q}}{G}. \quad (3884)$$

Since

$$L_M^{\text{beat}} = \frac{2\pi}{\Delta q}, \quad (3885)$$

one has

$$\frac{dL_M^{\text{beat}}}{L_M^{\text{beat}}} = -\frac{d(\Delta q)}{\Delta q}. \quad (3886)$$

Thus the relative uncertainty of the moiré length equals the relative uncertainty of the measured reciprocal splitting. For a finite observation region, the reciprocal peak width is $\delta q_{\text{win}} \sim \frac{c_W}{L}$, where c_W depends on the window geometry and the chosen linewidth definition. Two reciprocal peaks can be resolved only when $\Delta q_G \gtrsim \delta q_{\text{win}}$. Using the exact splitting gives

$$2G \sin \left(\frac{|\theta|}{2} \right) \gtrsim \delta q_{\text{win}}. \quad (3887)$$

The minimum resolvable twist is therefore approximately

$$\theta_{\text{res}} = 2 \arcsin \left(\frac{\delta q_{\text{win}}}{2G} \right). \quad (3888)$$

For small twist,

$$\theta_{\text{res}} \simeq \frac{\delta q_{\text{win}}}{G}. \quad (3889)$$

For the first square reciprocal shell,

$$\theta_{\text{res}} \simeq \frac{a}{2\pi} \delta q_{\text{win}} \simeq \frac{c_W a}{2\pi L}. \quad (3890)$$

Thus the smallest directly resolvable twist decreases inversely with the linear observation size. Using

$$L_M^{\text{beat}} = \frac{2\pi}{\Delta q}, \quad (3891)$$

the peak-resolution condition may be rewritten as

$$L \gtrsim \frac{c_W}{2\pi} L_M^{\text{beat}}. \quad (3892)$$

Thus an observation window must span an order-one fraction of a moiré period before the reciprocal splitting

can be resolved. For an exact commensurate bilayer with genuine moiré-periodic reconstruction, every reciprocal peak generated by the full periodic structure belongs to Λ_M^* . Therefore every peak can be represented using two integers,

$$\mathbf{q}_j = h_j \mathbf{G}_1 + k_j \mathbf{G}_2. \quad (3893)$$

A complete commensurate diffraction pattern therefore admits a globally consistent rank-two indexing. Given measured reciprocal peaks $\{\mathbf{q}_j\}$, and a candidate reciprocal basis $B_M = (\mathbf{G}_1 \ \mathbf{G}_2)$, define the indexing residual of peak j by

$$r_j = \min_{\mathbf{n} \in \mathbb{Z}^2} \|\mathbf{q}_j - B_M \mathbf{n}\|. \quad (3894)$$

Normalize by the experimental reciprocal linewidth,

$$\mathcal{R}_j = \frac{r_j}{\delta q_j}. \quad (3895)$$

A candidate commensurate indexing requires $\mathcal{R}_j \lesssim 1$ for every resolved peak attributed to the common periodic structure. A finite subset of an incommensurate higher-rank diffraction module can always lie close to a rank-two approximant. Therefore a single successful finite-resolution indexing does not prove exact commensurability. A stronger test must demonstrate stability as $\delta q \rightarrow 0$, and as the reciprocal observation radius increases, $q_{\max} \rightarrow \infty$. An exact commensurate basis remains globally valid. A generic incommensurate pattern produces additional reciprocal vectors that eventually violate every fixed rank-two indexing. Define the minimal number of integer generators needed to index the ideal reciprocal module by

$$r_{\text{rec}} = \text{rank}_{\mathbb{Z}} \mathcal{M}_{\theta}^*. \quad (3896)$$

For an exact commensurate two-dimensional bilayer, $r_{\text{rec}} = 2$. For a generic incommensurate square bilayer, $r_{\text{rec}} = 4$. Thus the periodic-quasiperiodic distinction can be formulated as a change in reciprocal integer rank rather than merely as a change in the visual appearance of a Fourier image. Let an otherwise ideal modulation possess a finite real-space coherence length ξ_{coh} . A finite correlation envelope broadens reciprocal satellites. Schematically, $\kappa_{\text{sat}} = O(\xi_{\text{coh}}^{-1})$, where κ_{sat} is the intrinsic satellite linewidth after instrumental broadening has been removed. This length must not be confused with the geometric moiré length,

$$L_M^{\text{beat}} = \frac{2\pi}{\Delta q}. \quad (3897)$$

The two quantities measure different physics: L_M^{beat} = spatial period of the registry beat, whereas ξ_{coh} = distance over which that reciprocal phase remains coherent. An ideal commensurate crystal has infinite translational coherence in the absence of defects. However, a real commensurate sample can possess $\xi_{\text{coh}} < \infty$ because

of disorder, strain gradients, dislocations, finite domains, or instrumental limitations. Conversely, an ideal deterministic quasiperiodic incommensurate structure can retain sharp pure-point diffraction over arbitrarily large distances. Therefore, commensurate/incommensurate $\neq$ coherent/incoherent. These are independent classifications. Two reciprocal length scales must also be distinguished experimentally. The first is the directly observed splitting of corresponding atomic peaks,

$$\Delta q_{\text{atomic}} = 2G \sin\left(\frac{\theta}{2}\right). \quad (3898)$$

The second is the primitive exact satellite spacing

$$G_M = \frac{2\pi}{a\sqrt{\Sigma}} \quad (3899)$$

at a commensurate square twist. Their ratio is arithmetic,

$$\Delta q_{\text{atomic}} = \frac{2n}{\sqrt{d}} G_M. \quad (3900)$$

Thus the smallest observed satellite spacing can reveal information that cannot be obtained from the primary peak splitting alone. If the primitive exact reciprocal spacing can be identified, the coincidence index follows directly from reciprocal area scaling. For the square lattice,

$$\Sigma = \left(\frac{G_0}{G_M}\right)^2. \quad (3901)$$

where $G_0 = \frac{2\pi}{a}$. For one microscopic orbital per site in each layer, the bilayer supercell contains $N_M^{\text{sc}} = 2\Sigma$. Thus exact reciprocal-lattice indexing gives the crystallographic atom count directly. For $\Sigma = 5$, the exact reciprocal primitive magnitude is

$$G_M = \frac{2\pi}{a\sqrt{5}}. \quad (3902)$$

For the $(m, n) = (2, 1)$ representation, $d = 1$, $n = 1$. Therefore, $\Delta q_1 = 2G_M$. Thus the primary atomic reciprocal splitting corresponds to the second harmonic of the exact primitive moiré reciprocal scale. For $\Sigma = 17$, and the canonical parameterization $(m, n) = (4, 1)$, one again has $d = 1$, $n = 1$. Hence,

$$G_M = \frac{2\pi}{a\sqrt{17}}, \quad (3903)$$

and $\Delta q_1 = 2G_M$. Although the two structures have different exact supercell sizes and twist angles, their first-shell beat vector occupies the same harmonic order relative to their respective exact reciprocal primitive vectors. For a commensurate sequence with $n > 1$, the primary splitting satisfies

$$\frac{\Delta q_1}{G_M} > 2 \quad (3904)$$

for the opposite-parity class. Thus increasingly high arithmetic branches can possess the same local geometric beat scale while requiring substantially larger exact crystallographic supercells. This is directly visible in diffraction if the primitive exact satellite lattice can be resolved. The exact peak splitting has the expansion

$$\Delta q_G = G|\theta| - \frac{G}{24}|\theta|^3 + \frac{G}{1920}|\theta|^5 + O(|\theta|^7). \quad (3905)$$

For the first square shell,

$$\Delta q_1 = \frac{2\pi}{a}|\theta| - \frac{\pi}{12a}|\theta|^3 + \frac{\pi}{960a}|\theta|^5 + O(|\theta|^7). \quad (3906)$$

Therefore small-angle twist metrology from diffraction is asymptotically linear but admits controlled nonlinear corrections. Suppose interlayer relaxation generates dominant moiré displacement harmonics $\mathbf{u}_\mathbf{Q}$. The satellite intensity is approximately

$$I_{\mathbf{G}+\mathbf{Q}} \propto |(\mathbf{G} + \mathbf{Q}) \cdot \mathbf{u}_\mathbf{Q}|^2. \quad (3907)$$

Hence the ratio

$$\mathcal{I}_\mathbf{Q} = \frac{I_{\mathbf{G}+\mathbf{Q}}}{I_\mathbf{G}} \quad (3908)$$

provides a direct measure of the Fourier amplitude of structural reconstruction in the weak-distortion limit. The peak position determines the modulation wavelength. The peak intensity determines the modulation amplitude. The peak linewidth determines the modulation coherence length. These three diffraction observables therefore encode distinct real-space properties. For a resolved satellite peak, define its displacement from the parent Bragg peak,

$$\mathbf{Q}_{\text{sat}} = \mathbf{q}_{\text{sat}} - \mathbf{G}. \quad (3909)$$

Its magnitude determines the real-space modulation scale,

$$L_{\text{sat}} = \frac{2\pi}{|\mathbf{Q}_{\text{sat}}|}. \quad (3910)$$

Its integrated intensity determines the modulation strength, $I_{\text{sat}} \propto |u_\mathbf{Q}|^2$ up to the polarization and atomic form factors. Its linewidth determines coherence, $\xi_{\text{coh}} \sim \kappa_{\text{sat}}^{-1}$. Thus position, intensity, linewidth, must not be conflated when interpreting a diffraction image. For a generic centered square-bilayer twist, each reciprocal lattice is fourfold symmetric. Therefore the union retains at least C_4 in reciprocal space. At the special reduced twist $\hat{\theta} = \frac{\pi}{4}$, the two reciprocal lattices are exchanged by a forty-five-degree rotation. The ideal equal-weight diffraction pattern therefore has C_8 rotational symmetry. If the corresponding structure is incommensurate, this provides a quasicrystalline diffraction signature: a noncrystallographic rotational symmetry can coexist with sharp reciprocal peaks because periodic rank-two translational

order is absent. The diffraction classification can now be stated in a form that separates periodicity, local moiré scale, and coherence. An exact commensurate state satisfies

$$\mathcal{M}_\theta^* = \Lambda_M^*, \quad r_{\text{rec}} = 2. \quad (3911)$$

Its complete reconstructed diffraction can be indexed by two reciprocal integers. A generic ideal incommensurate state satisfies $r_{\text{rec}} = 4$, with no finite exact reciprocal primitive cell. Nevertheless, its local peak splitting satisfies the same exact geometric law,

$$\Delta q = 2G \sin\left(\frac{|\theta|}{2}\right). \quad (3912)$$

Thus reciprocal peak splitting alone distinguishes twist magnitude but does not distinguish exact commensurability. The latter requires arithmetic indexing information.

a. Theorem 44. Reciprocal-rank criterion for commensurability. For two twisted square lattices, define the additive reciprocal module by

$$\mathcal{M}_\theta^* = \Lambda_1^* + \Lambda_2^*. \quad (3913)$$

If the twist is exactly commensurate, then $\mathcal{M}_\theta^* = \Lambda_M^*$ is a discrete rank-two reciprocal lattice. Conversely, if this module is a discrete full-rank lattice, then the two microscopic reciprocal lattices are commensurate and the direct lattices possess a finite common coincidence lattice. Thus exact commensurability $\iff$ discrete rank-two reciprocal module. For generic irrational twists, the module has rank four and no finite moiré reciprocal primitive cell exists.

b. Theorem 45. Reciprocal splitting-moire-length duality. For any pair of corresponding reciprocal vectors of magnitude G , the twist-induced reciprocal splitting is

$$\Delta q_G = 2G |\sin(\theta/2)|. \quad (3914)$$

The associated beat scale is

$$L_G^{\text{beat}} = \frac{2\pi}{\Delta q_G}. \quad (3915)$$

For the first square reciprocal shell,

$$L_M^{\text{beat}} = \frac{a}{2|\sin(\theta/2)|}. \quad (3916)$$

This relation is independent of whether the bilayer is commensurate or incommensurate. Therefore reciprocal peak splitting measures the local geometric moiré scale rather than the exact crystallographic supercell size.

c. Theorem 46. Exact supercell scale from diffraction harmonics. For an exactly commensurate square bilayer parameterized by coprime integers m, n and parity factor d , the exact crystallographic reciprocal scale and the local beat splitting satisfy

$$\frac{\Delta q_1}{G_M} = \frac{2n}{\sqrt{d}}. \quad (3917)$$

Equivalently,

$$\frac{L_M^{\text{sc}}}{L_M^{\text{beat}}} = \frac{2n}{\sqrt{d}}, \quad \frac{A_M^{\text{sc}}}{A_M^{\text{beat}}} = \frac{4n^2}{d}. \quad (3918)$$

Thus diffraction separates continuously determined geometric information from discrete arithmetic commensurability information.

d. Theorem 47. Finite-window diffraction resolution.

Let a reciprocal peak have finite-window linewidth δq_{win} . The twist-induced splitting of a reciprocal shell of magnitude G is resolvable only if

$$2G \sin\left(\frac{|\theta|}{2}\right) \gtrsim \delta q_{\text{win}}. \quad (3919)$$

The corresponding minimum resolvable twist is

$$\theta_{\text{res}} = 2 \arcsin\left(\frac{\delta q_{\text{win}}}{2G}\right). \quad (3920)$$

For small twist and observation size L ,

$$\theta_{\text{res}} = O\left(\frac{a}{L}\right). \quad (3921)$$

Therefore apparent peak merging at small angle can arise from finite reciprocal resolution rather than from the disappearance of the moiré scale. The reciprocal-space structure of a twisted Euclidean bilayer therefore contains several nested levels of information. The primary atomic Bragg peaks determine the reciprocal lattices of the two layers and their relative orientation. The splitting between corresponding Bragg peaks determines the geometric beat scale and therefore the local moiré length. Exact commensurability introduces a discrete reciprocal lattice that contains both microscopic reciprocal lattices as sublattices. The primitive spacing of this exact reciprocal lattice contains arithmetic information not accessible from the geometric beat scale alone. Genuine satellite intensity at additional moiré reciprocal vectors probes a further level of physics. Such intensity arises when structural relaxation, stacking-dependent scattering, amplitude modulation, or higher-order processes populate Fourier components that are extinct in the perfectly rigid linear-scattering model. Satellite positions then encode modulation wavevectors, their intensities encode modulation amplitudes, and their linewidths encode coherence lengths. For an incommensurate bilayer, the local geometric splitting remains well-defined even though no finite exact reciprocal cell exists. The full quasiperiodic reciprocal structure is described by a higher-rank module rather than a rank-two reciprocal lattice. Consequently, the distinction between commensurate and incommensurate states is not determined by the existence of a visible moiré wavelength, nor by peak sharpness alone. It is determined by whether the complete reciprocal structure admits a globally consistent discrete rank-two indexing. The complete diffraction logic may therefore be summarized in prose as follows. Twist first rotates the

two microscopic reciprocal lattices and creates a measurable splitting between corresponding Bragg peaks. This splitting defines the local geometric moiré length. If the twist is arithmetically commensurate, the two reciprocal lattices embed into a larger exact moiré reciprocal lattice, whose primitive scale can differ from the direct Bragg-peak splitting by an integer-arithmetic factor. Interlayer reconstruction can then transfer intensity into additional moiré reciprocal vectors, producing true satellites. If the twist is incommensurate, no finite exact reciprocal primitive cell exists, although a higher-rank quasiperiodic reciprocal module and a well-defined local beat scale remain. Diffraction therefore separates local geometry, exact crystallography, structural reconstruction, and coherence into distinct experimentally and numerically accessible quantities.

C. Hyperbolic Structure Factor and Representation-Space Diffraction

The Euclidean diffraction theory developed in the preceding subsections relies on the existence of a globally defined additive reciprocal vector $\mathbf{q} \in \mathbb{R}^2$. A plane wave transforms under an ordinary translation according to a one-dimensional character, and the usual structure factor is therefore obtained by an Abelian Fourier transform. The corresponding construction cannot be transferred unchanged to a hyperbolic crystal. The exact translation group of a compact hyperbolic lattice is a non-Abelian Fuchsian group, and the natural harmonic variables are therefore unitary representations rather than scalar reciprocal vectors. In addition, two logically distinct representation structures appear in a twisted bilayer. Electronic generalized Bloch states are classified by representations of the exact common moiré translation group Γ_M , whereas structural modulations and diffraction satellites are naturally classified relative to the larger parent translation groups of the individual layers. The purpose of the present subsection is to construct a generalized hyperbolic diffraction theory that preserves this distinction. We derive an exact representation-space analogue of the static structure factor, establish a non-Abelian Wiener-Khinchin theorem, formulate the hyperbolic analogue of reciprocal-lattice selection rules through induced representations, derive the representation-theoretic counterpart of a moiré beat wavevector, and obtain a rigorous distinction between exact commensurate and incommensurate diffraction. The central replacement is not merely $\mathbf{q} \rightarrow$ a more complicated momentum vector. Instead, the Euclidean reciprocal coordinate must be replaced by a representation class, $\mathbf{q} \rightarrow [\varpi]$, where $\varpi : \Gamma \rightarrow U(d_\varpi)$ is a unitary representation of the relevant hyperbolic translation group. For $d_\varpi = 1$, the representation is an Abelian character. For $d_\varpi > 1$, the reciprocal degree of freedom is intrinsically matrix valued. Before defining the generalized structure factor, it is essential to distinguish two different representation-theoretic roles. Let the

two microscopic layer translation groups be $\Gamma_1 = \Gamma$, and $\Gamma_2 = g_\theta \Gamma g_\theta^{-1}$. Their exact common translation group is $\Gamma_M = \Gamma_1 \cap \Gamma_2$. Electronic generalized Bloch states are classified by unitary representations

$$\rho_M : \Gamma_M \longrightarrow U(d_{\rho_M}). \quad (3922)$$

By contrast, a structural modulation of layer ℓ is measured relative to the parent translation symmetry Γ_ℓ . Its generalized reciprocal harmonics are therefore classified by representations

$$\varpi_\ell : \Gamma_\ell \longrightarrow U(d_{\varpi_\ell}). \quad (3923)$$

Thus, ρ_M = electronic generalized momentum, whereas ϖ_ℓ = structural or diffraction harmonic. These labels must not be identified. A structural moiré modulation can be nontrivial under the parent group Γ_ℓ while remaining exactly invariant under Γ_M . Consequently, it can generate nontrivial diffraction satellites without violating conservation of the electronic generalized momentum ρ_M . Choose a microscopic fundamental domain of one parent hyperbolic crystal. Let its internal scattering sites be x_α , $\alpha = 1, \dots, n_b$. Every site in the infinite crystal can be represented as $x_{\gamma\alpha} = \gamma x_\alpha$, $\gamma \in \Gamma$. Let the microscopic scattering contrast associated with that site be $f_\alpha(\gamma)$. The complete translation-indexed scattering field is therefore $f : \Gamma \longrightarrow \mathbb{C}^{n_b}$. For a perfectly rigid homogeneous monolayer, $f_\alpha(\gamma) = f_\alpha^{(0)}$ for every group element. A moiré modulation generates a nonuniform field satisfying only the smaller translation symmetry. Suppose the bilayer is exactly commensurate. For layer one, the common moiré translation group is a finite-index subgroup, $\Gamma_M < \Gamma_1$. The index is $q_M = [\Gamma_1 : \Gamma_M]$. A structural field possessing exact moiré periodicity satisfies $f(h\gamma) = f(\gamma)$ for every $h \in \Gamma_M$. Equivalently, the field is a function on the finite coset space $X_M = \Gamma_M \backslash \Gamma_1$. Its cardinality is $|X_M| = q_M$. Thus the exact hyperbolic moiré modulation contains a finite number of parent-cell registry classes. The parent group acts on the coset space through right translation. For $g \in \Gamma_1$, define

$$\mathcal{U}(g)|\Gamma_M \gamma\rangle = |\Gamma_M \gamma g^{-1}\rangle. \quad (3924)$$

This is a q_M -dimensional unitary permutation representation. Representation-theoretically it is the induced representation

$$\mathcal{P}_M = \text{Ind}_{\Gamma_M}^{\Gamma_1} \mathbf{1}, \quad (3925)$$

where $\mathbf{1}$ is the trivial representation of the moiré subgroup. Its dimension is exactly $\dim \mathcal{P}_M = q_M$. This finite-dimensional induced representation is the exact representation-theoretic analogue of the Euclidean moiré reciprocal superlattice. Decompose the induced representation into irreducible parent-group representations,

$$\text{Ind}_{\Gamma_M}^{\Gamma_1} \mathbf{1} \simeq \bigoplus_{\varpi} m_{\varpi} V_{\varpi}. \quad (3926)$$

Here V_{ϖ} is the carrier space of the irrep ϖ , and m_{ϖ} is its multiplicity. Dimension conservation gives

$$q_M = \sum_{\varpi} m_{\varpi} d_{\varpi}. \quad (3927)$$

The set of generalized reciprocal harmonics is therefore

$$\mathfrak{R}_M^{(1)} = \{[\varpi] : m_{\varpi} > 0\}. \quad (3928)$$

This object replaces the ordinary finite set of reciprocal superlattice harmonics inside one parent Brillouin zone. The multiplicity of an irrep in the induced representation is determined by Frobenius reciprocity,

$$m_{\varpi} = \dim \text{Hom}_{\Gamma_1} (V_{\varpi}, \text{Ind}_{\Gamma_M}^{\Gamma_1} \mathbf{1}). \quad (3929)$$

Frobenius reciprocity gives

$$m_{\varpi} = \dim \text{Hom}_{\Gamma_M} (\text{Res}_{\Gamma_M}^{\Gamma_1} V_{\varpi}, \mathbf{1}). \quad (3930)$$

The right-hand side is exactly the dimension of the Γ_M -invariant subspace,

$$m_{\varpi} = \dim V_{\varpi}^{\Gamma_M}, \quad (3931)$$

where

$$V_{\varpi}^{\Gamma_M} = \{v \in V_{\varpi} : \varpi(h)v = v \quad \forall h \in \Gamma_M\}. \quad (3932)$$

Therefore a parent-group irrep is an allowed exact moiré diffraction harmonic if and only if $V_{\varpi}^{\Gamma_M} \neq \{0\}$. This is the hyperbolic representation-theoretic analogue of a reciprocal superlattice Bragg condition.

a. Theorem 48. Representation-theoretic moiré reciprocal structure. Let Γ_M be a finite-index exact moiré translation subgroup of the parent hyperbolic translation group Γ . Then the complete finite-dimensional reciprocal content associated with the moiré enlargement is the induced representation

$$\mathcal{P}_M = \text{Ind}_{\Gamma_M}^{\Gamma} \mathbf{1}. \quad (3933)$$

An irreducible representation ϖ occurs in this reciprocal content if and only if $V_{\varpi}^{\Gamma_M} \neq 0$. Its multiplicity is

$$m_{\varpi} = \dim V_{\varpi}^{\Gamma_M}, \quad (3934)$$

and the exact reciprocal state-counting identity is

$$q_M = \sum_{\varpi} m_{\varpi} d_{\varpi}. \quad (3935)$$

Thus the ordinary list of moiré reciprocal vectors is replaced by a finite collection of irreducible Fuchsian-group representations with representation-theoretic multiplicities. A particularly transparent case occurs when $\Gamma_M \triangleleft \Gamma$. Then the quotient $Q_M = \Gamma/\Gamma_M$ is a finite group of order $|Q_M| = q_M$. Every representation contributing to the induced representation is trivial on Γ_M and therefore factors through the quotient, $\varpi = \tilde{\varpi} \circ \pi$, where $\pi : \Gamma \rightarrow Q_M$ is the quotient map. The induced representation is the regular representation of Q_M ,

$$\text{Ind}_{\Gamma_M}^{\Gamma} \mathbf{1} \simeq \mathbb{C}[Q_M]. \quad (3936)$$

The decomposition becomes

$$\mathbb{C}[Q_M] \simeq \bigoplus_{\varpi \in \hat{Q}_M} V_{\varpi} \otimes V_{\varpi}^*. \quad (3937)$$

Therefore $m_{\varpi} = d_{\varpi}$, and

$$q_M = \sum_{\varpi \in \hat{Q}_M} d_{\varpi}^2. \quad (3938)$$

This is the exact non-Abelian replacement of counting q_M discrete moiré reciprocal momenta in an Abelian supercell. The common subgroup need not be normal. Define its normal core by

$$\text{Core}_{\Gamma}(\Gamma_M) = \bigcap_{g \in \Gamma} g\Gamma_M g^{-1}. \quad (3939)$$

Because Γ_M has finite index, its normal core also has finite index,

$$[\Gamma : \text{Core}_{\Gamma}(\Gamma_M)] < \infty. \quad (3940)$$

The permutation representation on $\Gamma_M \backslash \Gamma$ therefore factors through the finite group

$$G_M = \Gamma / \text{Core}_{\Gamma}(\Gamma_M). \quad (3941)$$

Consequently the complete exact commensurate diffraction decomposition can always be reduced to finite-group representation theory, even when the moiré subgroup itself is not normal. The representation-theoretic Bragg condition must reproduce the ordinary Euclidean result. Let $\Gamma = \Lambda \simeq \mathbb{Z}^2$, and let $\Gamma_M = \Lambda_M$. Every irrep is a one-dimensional character,

$$\chi_{\mathbf{k}}(\mathbf{R}) = e^{i\mathbf{k} \cdot \mathbf{R}}. \quad (3942)$$

The invariant-subspace condition becomes

$$\chi_{\mathbf{k}}(\mathbf{R}_M) = 1 \quad \forall \mathbf{R}_M \in \Lambda_M. \quad (3943)$$

Hence

$$\mathbf{k} \cdot \mathbf{R}_M \in 2\pi\mathbb{Z} \quad (3944)$$

for every exact moiré translation. By definition this is equivalent to $\mathbf{k} \in \Lambda_M^*$. Modulo the parent reciprocal lattice, $\mathbf{k} \in \Lambda_M^*/\Lambda^*$. Thus the usual exact moiré reciprocal vectors are precisely the one-dimensional Abelian special case of the generalized Bragg condition. For the normal finite-quotient case, let the scattering modulation be a complex scalar field $f : Q_M \rightarrow \mathbb{C}$. The field values can encode stacking-dependent scattering contrast, displacement-induced amplitude, local orbital response, or any other quantity periodic only under the moiré subgroup. For each irrep

$$\varpi : Q_M \rightarrow U(d_{\varpi}), \quad (3945)$$

define the normalized non-Abelian Fourier amplitude

$$\hat{f}(\varpi) = \sqrt{\frac{d_{\varpi}}{|Q_M|}} \sum_{q \in Q_M} f(q) \varpi(q). \quad (3946)$$

Unlike an ordinary Fourier coefficient,

$$\hat{f}(\varpi) \in \mathbb{C}^{d_{\varpi} \times d_{\varpi}}. \quad (3947)$$

Thus the diffraction amplitude of a non-Abelian reciprocal sector is matrix valued. Finite-group irreducible representations satisfy

$$\sum_{q \in Q_M} \varpi(q)_{ab} \zeta(q)_{cd}^* = \frac{|Q_M|}{d_{\varpi}} \delta_{\varpi\zeta} \delta_{ac} \delta_{bd}. \quad (3948)$$

This identity replaces orthogonality of Euclidean plane waves. It implies the Plancherel relation

$$\sum_{q \in Q_M} |f(q)|^2 = \sum_{\varpi \in \hat{Q}_M} \left\| \hat{f}(\varpi) \right\|_F^2, \quad (3949)$$

where

$$\|A\|_F^2 = \text{Tr}(A^{\dagger} A) \quad (3950)$$

is the Frobenius norm. Thus the complete scattering power is exactly conserved by the non-Abelian Fourier transform. Define the generalized diffraction intensity of the irrep sector by

$$I_{\varpi} = \left\| \hat{f}(\varpi) \right\|_F^2. \quad (3951)$$

The total intensity satisfies

$$\sum_{\varpi} I_{\varpi} = \sum_{q \in Q_M} |f(q)|^2. \quad (3952)$$

The normalized representation-space structure-factor weight is therefore

$$\mathcal{S}_{\varpi} = \frac{I_{\varpi}}{\sum_{\varsigma} I_{\varsigma}}. \quad (3953)$$

It satisfies $0 \leq \mathcal{S}_{\varpi} \leq 1$, and

$$\sum_{\varpi} \mathcal{S}_{\varpi} = 1. \quad (3954)$$

This probability-like distribution is the finite-quotient generalized diffraction pattern in representation space. Consider a perfectly homogeneous parent-layer scattering field, $f(q) = f_0$ for all quotient cells. Then

$$\hat{f}(\varpi) = f_0 \sqrt{\frac{d_{\varpi}}{|Q_M|}} \sum_{q \in Q_M} \varpi(q). \quad (3955)$$

For a nontrivial irrep,

$$\sum_{q \in Q_M} \varpi(q) = 0. \quad (3956)$$

Therefore $\hat{f}(\varpi) = 0$ for every nontrivial representation. Only the trivial irrep carries weight. Thus the hyperbolic representation-space theory reproduces the same physical conclusion obtained in Euclidean diffraction: rigid homogeneous layer $\nrightarrow$ nontrivial moiré satellites. Non-trivial generalized diffraction sectors require a genuine modulation of the parent-layer scattering field. Let

$$f(q) = f_0 + \epsilon \eta(q), \quad (3957)$$

where $|\epsilon| \ll 1$. For every nontrivial irrep,

$$\hat{f}(\varpi) = \epsilon \hat{\eta}(\varpi), \quad (3958)$$

because the uniform part vanishes identically. Therefore,

$$I_{\varpi} = |\epsilon|^2 \|\hat{\eta}(\varpi)\|_F^2 \quad (3959)$$

for every nontrivial generalized reciprocal sector. Thus generalized hyperbolic satellite amplitudes are linear in a weak modulation, whereas their isolated intensities are quadratic. This is the exact representation-space analogue of the Euclidean weak satellite law. The translation symmetry may allow an irrep because $m_{\varpi} > 0$, while

the actual scattering modulation may have no projection onto that sector, $\hat{\eta}(\varpi) = 0$. Such a sector is a generalized systematic extinction. Therefore,

$$m_{\varpi} > 0 \nrightarrow I_{\varpi} > 0. \quad (3960)$$

The representation-theoretic selection rule determines which reciprocal sectors are allowed. The microscopic basis and modulation determine which of those allowed sectors actually carry intensity. The ordinary Euclidean structure factor can equivalently be defined as the Fourier transform of a two-point correlation function. The same construction has an exact non-Abelian generalization. For a finite quotient define the translation autocorrelation

$$C(g) = \frac{1}{|Q_M|} \sum_{q \in Q_M} f(q)^* f(qg). \quad (3961)$$

The group element $g \in Q_M$ replaces the Euclidean displacement vector. The generalized structure tensor is

$$\mathbb{S}(\varpi) = \sum_{g \in Q_M} C(g) \varpi(g). \quad (3962)$$

It is a $d_{\varpi} \times d_{\varpi}$ matrix. Substitute the definition of the autocorrelation,

$$\mathbb{S}(\varpi) = \frac{1}{|Q_M|} \sum_g \sum_q f(q)^* f(qg) \varpi(g). \quad (3963)$$

Set $r = qg$. Then $g = q^{-1}r$. Using the representation property,

$$\varpi(q^{-1}r) = \varpi(q)^{\dagger} \varpi(r), \quad (3964)$$

gives

$$\begin{aligned} \mathbb{S}(\varpi) &= \frac{1}{|Q_M|} \sum_{q,r} f(q)^* f(r) \varpi(q)^{\dagger} \varpi(r) \\ &= \frac{1}{|Q_M|} \left[\sum_q f(q) \varpi(q) \right]^{\dagger} \left[\sum_r f(r) \varpi(r) \right]. \end{aligned} \quad (3965)$$

Using Eq. (3946),

$$\mathbb{S}(\varpi) = \frac{1}{d_{\varpi}} \hat{f}(\varpi)^{\dagger} \hat{f}(\varpi). \quad (3966)$$

Therefore, $\mathbb{S}(\varpi) \succeq 0$. The scalar sector intensity is recovered from

$$I_{\varpi} = d_{\varpi} \text{Tr} \mathbb{S}(\varpi). \quad (3967)$$

b. Theorem 49. Non-Abelian Wiener–Khinchin theorem. For a finite hyperbolic translation quotient, the group Fourier transform of the translation autocorrelation is a positive-semidefinite matrix equal to the squared non-Abelian scattering amplitude,

$$\mathbb{S}(\varpi) = \frac{1}{d_\varpi} \hat{f}(\varpi)^\dagger \hat{f}(\varpi). \quad (3968)$$

Thus the Euclidean scalar static structure factor is replaced by a positive-semidefinite representation-space structure tensor. For a crystal containing n_b microscopic basis orbitals, let $f_\alpha(q)$ denote the scattering contrast of orbital α . Define the cross-correlation

$$C_{\alpha\beta}(g) = \frac{1}{|Q_M|} \sum_q f_\alpha(q)^* f_\beta(qg). \quad (3969)$$

The generalized structure tensor becomes

$$\mathbb{S}_{\alpha\beta}(\varpi) = \sum_g C_{\alpha\beta}(g) \varpi(g). \quad (3970)$$

Thus

$$\mathbb{S}(\varpi) \in \text{End}(\mathbb{C}^{n_b} \otimes V_\varpi). \quad (3971)$$

A probe with orbital form-factor vector $\mathbf{F} = (F_1, \dots, F_{n_b})^T$ measures the contracted intensity

$$I_\varpi[\mathbf{F}] = d_\varpi \text{Tr}_{V_\varpi} [\mathbf{F}^\dagger \mathbb{S}(\varpi) \mathbf{F}]. \quad (3972)$$

This is the non-Abelian generalization of the ordinary crystallographic basis structure factor. For a bilayer, define layer-resolved scattering fields $f_1(q)$, $f_2(q)$. The layer cross-correlations are

$$C_{\ell\ell'}(g) = \frac{1}{|Q_M|} \sum_q f_\ell(q)^* f_{\ell'}(qg). \quad (3973)$$

Their representation transforms are

$$\mathbb{S}_{\ell\ell'}(\varpi) = \sum_g C_{\ell\ell'}(g) \varpi(g). \quad (3974)$$

The complete bilayer structure tensor is

$$\mathbb{S}_{\text{BL}}(\varpi) = \begin{pmatrix} \mathbb{S}_{11}(\varpi) & \mathbb{S}_{12}(\varpi) \\ \mathbb{S}_{21}(\varpi) & \mathbb{S}_{22}(\varpi) \end{pmatrix}. \quad (3975)$$

Hermiticity gives

$$\mathbb{S}_{21}(\varpi) = \mathbb{S}_{12}(\varpi)^\dagger. \quad (3976)$$

For a probe carrying transverse momentum transfer q_z , the two layers acquire amplitudes

$$c_1 = e^{+iq_z h/2}, \quad c_2 = e^{-iq_z h/2}. \quad (3977)$$

Define

$$\mathbf{c}(q_z) = \begin{pmatrix} e^{+iq_z h/2} \\ e^{-iq_z h/2} \end{pmatrix}. \quad (3978)$$

The representation-resolved bilayer intensity is

$$I_\varpi(q_z) = d_\varpi \text{Tr} [\mathbf{c}^\dagger(q_z) \mathbb{S}_{\text{BL}}(\varpi) \mathbf{c}(q_z)]. \quad (3979)$$

Expanding gives

$$I_\varpi(q_z) = d_\varpi \text{Tr} \mathbb{S}_{11} + d_\varpi \text{Tr} \mathbb{S}_{22} + 2d_\varpi \text{Re} [e^{-iq_z h} \text{Tr} \mathbb{S}_{12}]. \quad (3980)$$

This is the representation-space analogue of the coherent Euclidean bilayer interference term. Define the normalized interlayer coherence in sector ϖ by

$$\mathcal{C}_\varpi = \frac{\text{Tr} \mathbb{S}_{12}(\varpi)}{\sqrt{\text{Tr} \mathbb{S}_{11}(\varpi) \text{Tr} \mathbb{S}_{22}(\varpi)}}. \quad (3981)$$

The Cauchy–Schwarz inequality gives $|\mathcal{C}_\varpi| \leq 1$. Thus each generalized reciprocal sector carries not only an intensity but also a complex measure of interlayer coherence. The Euclidean moiré beat vector arises from the difference between two one-dimensional translation characters. The correct non-Abelian replacement is a tensor-product representation. Let $\varpi_1 : \Gamma_1 \rightarrow U(d_1)$ and $\varpi_2 : \Gamma_2 \rightarrow U(d_2)$ describe reciprocal harmonics of the two layers. Restrict both representations to the common subgroup, $\text{Res}_{\Gamma_M}^{\Gamma_1} \varpi_1$, and $\text{Res}_{\Gamma_M}^{\Gamma_2} \varpi_2$. The bilinear interference channel transforms under

$$\mathcal{B}_{12} = \varpi_2|_{\Gamma_M} \otimes (\varpi_1|_{\Gamma_M})^*. \quad (3982)$$

This tensor product is the representation-theoretic analogue of a reciprocal-vector difference. A slowly varying moiré interference component exists when the tensor product contains a common-translation singlet. Thus the generalized beat condition is

$$[V_{\varpi_2} \otimes V_{\varpi_1}^*]^{\Gamma_M} \neq 0. \quad (3983)$$

Its multiplicity is

$$b_{12} = \dim [V_{\varpi_2} \otimes V_{\varpi_1}^*]^{\Gamma_M}. \quad (3984)$$

Using the canonical tensor–Hom isomorphism,

$$V_{\varpi_2} \otimes V_{\varpi_1}^* \simeq \text{Hom}(V_{\varpi_1}, V_{\varpi_2}), \quad (3985)$$

one obtains

$$b_{12} = \dim \text{Hom}_{\Gamma_M} (V_{\varpi_1}|_{\Gamma_M}, V_{\varpi_2}|_{\Gamma_M}). \quad (3986)$$

Thus generalized moiré interference is controlled by the number of intertwiners between the two layer representations after restriction to the common translation group.

c. Theorem 50. Non-Abelian moiré beat condition.

Let ϖ_1 and ϖ_2 be reciprocal harmonics of the two parent hyperbolic layers. A common-translation-invariant bilinear interference channel exists if and only if

$$\text{Hom}_{\Gamma_M} (V_{\varpi_1}|_{\Gamma_M}, V_{\varpi_2}|_{\Gamma_M}) \neq 0. \quad (3987)$$

The number of linearly independent generalized beat channels is

$$b_{12} = \dim \text{Hom}_{\Gamma_M} (V_{\varpi_1}|_{\Gamma_M}, V_{\varpi_2}|_{\Gamma_M}). \quad (3988)$$

Therefore the Euclidean difference wavevector is replaced by an intertwining problem between restricted non-Abelian translation representations. For an ordinary Euclidean lattice,

$$\varpi_1 = \chi_{\mathbf{k}_1}, \quad \varpi_2 = \chi_{\mathbf{k}_2}. \quad (3989)$$

The tensor-product beat representation is

$$\chi_{\mathbf{k}_2} \otimes \chi_{\mathbf{k}_1}^* = \chi_{\mathbf{k}_2 - \mathbf{k}_1}. \quad (3990)$$

The common-subgroup invariant condition becomes $e^{i(\mathbf{k}_2 - \mathbf{k}_1) \cdot \mathbf{R}_M} = 1$ for every moiré translation. Therefore, $\mathbf{k}_2 - \mathbf{k}_1 \in \Lambda_M^*$. Thus the familiar Euclidean reciprocal-difference condition is recovered exactly as the rank-one limit of Theorem 50. The generalized beat construction also clarifies an apparent tension between structural satellites and electronic momentum conservation. A structural moiré modulation may transform nontrivially under the parent group, $\varpi \neq \mathbf{1}$, while remaining invariant under the common subgroup, Γ_M acts trivially on the physical modulation pattern. The corresponding electronic perturbation therefore commutes with the common translation group,

$$[V_{\text{moire}}, U_\gamma] = 0 \quad \forall \gamma \in \Gamma_M. \quad (3991)$$

Consequently it cannot mix inequivalent electronic generalized Bloch representations,

$$\rho_M \not\sim \sigma_M \implies \langle \rho_M | V_{\text{moire}} | \sigma_M \rangle = 0. \quad (3992)$$

Thus nontrivial structural diffraction and conservation of electronic generalized momentum are fully compatible. Let a residual point-group operation induce an automorphism $\alpha_p : \Gamma \longrightarrow \Gamma$. It acts on a representation according to

$$\varpi^p(\gamma) = \varpi[\alpha_p(\gamma)]. \quad (3993)$$

If the scattering field is invariant under the point-group operation, the representation-resolved intensities satisfy $I_{\varpi^p} = I_\varpi$. Thus a point-group orbit

$$\text{Star}(\varpi) = \{\varpi^p : p \in P\} \quad (3994)$$

is the hyperbolic analogue of a star of symmetry-related reciprocal vectors. For a real scattering field, complex conjugation maps $\varpi \longrightarrow \varpi^*$. The Fourier amplitudes satisfy the representation-space analogue of Friedel pairing, $I_{\varpi^*} = I_\varpi$. For self-conjugate representations, $\varpi \simeq \varpi^*$, the corresponding generalized reciprocal sector is individually invariant under the real-density constraint. Suppose a modulation transforms according to a one-dimensional point-group character η_p . If a diffraction sector is invariant under a point-group element but the microscopic form factor acquires a nontrivial phase, $\eta_p \neq 1$, the projected amplitude may be forced to vanish. Thus point-group selection rules can produce representation-space systematic extinctions in addition to translation-induced selection rules. The complete generalized intensity therefore depends on three logically separate ingredients: translation admissibility, point-group selection, and microscopic basis interference. The representation framework yields a particularly sharp distinction between exact commensurate and incommensurate hyperbolic twists. If the twist is exactly commensurate,

$$q_M = [\Gamma_1 : \Gamma_M] < \infty. \quad (3995)$$

Therefore,

$$\dim \text{Ind}_{\Gamma_M}^{\Gamma_1} \mathbf{1} = q_M < \infty. \quad (3996)$$

The complete structural moiré reciprocal content is therefore encoded by a finite-dimensional permutation representation. For a generic incommensurate twist,

$$[\Gamma_1 : \Gamma_M] = \infty. \quad (3997)$$

The induced representation acts on $\ell^2(\Gamma_M \setminus \Gamma_1)$, and has infinite dimension,

$$\dim \text{Ind}_{\Gamma_M}^{\Gamma_1} \mathbf{1} = \infty. \quad (3998)$$

Thus no finite list of exact generalized reciprocal harmonics can encode the complete incommensurate structure.

d. Theorem 51. Finite-reciprocal-representation criterion for hyperbolic commensurability. Let the two hyperbolic layers have common translation subgroup Γ_M . Then exact commensurability is equivalent to

$$\dim \text{Ind}_{\Gamma_M}^{\Gamma_\ell} \mathbf{1} < \infty. \quad (3999)$$

Indeed,

$$\dim \text{Ind}_{\Gamma_M}^{\Gamma_\ell} \mathbf{1} = [\Gamma_\ell : \Gamma_M]. \quad (4000)$$

Thus the Euclidean distinction between a finite reciprocal supercell and a higher-rank incommensurate reciprocal module becomes, in hyperbolic crystallography, a distinction between finite-dimensional and infinite-dimensional induced translation representations. Along a sequence of exact commensurate twists, the dimension of the structural reciprocal representation is

$$D_{\text{diff}}(\theta) = q_M(\theta). \quad (4001)$$

If geometric–arithmetic locking holds,

$$q_M(\theta) = \Theta(|\theta|^{-1}), \quad (4002)$$

then

$$D_{\text{diff}}(\theta) = \Theta(|\theta|^{-1}). \quad (4003)$$

Thus the curvature-dominated exact real-space supercell scaling is inherited directly by the dimension of the exact generalized diffraction representation. This statement is conditional on exact crystallographic locking and is not implied by the effective geometric moiré area alone. The finite-quotient construction provides the strongest exact numerical definition of the generalized structure factor. For an infinite hyperbolic lattice, a translation-invariant correlation function may instead be written as $C : \Gamma \rightarrow \mathbb{C}$. Whenever the correlation is absolutely summable,

$$\sum_{\gamma \in \Gamma} |C(\gamma)| < \infty, \quad (4004)$$

its transform in any finite-dimensional unitary representation is well-defined by

$$\mathbb{S}[\varpi] = \sum_{\gamma \in \Gamma} C(\gamma) \varpi(\gamma). \quad (4005)$$

Unitarity gives $\|\varpi(\gamma)\| = 1$, and therefore

$$\|\mathbb{S}[\varpi]\| \leq \sum_{\gamma} |C(\gamma)|. \quad (4006)$$

Thus absolute correlation summability gives a representation-uniform bound. Hyperbolic volume growth imposes a nontrivial condition on the existence of the ordinary absolutely convergent structure-factor transform. Suppose the number of translation images in a shell of radius r obeys

$$N_{\text{shell}}(r) \lesssim e^{r/R}. \quad (4007)$$

Suppose the connected correlation decays exponentially,

$$|C(r)| \lesssim C_0 e^{-r/\xi_{\text{corr}}}. \quad (4008)$$

Then the absolute shell sum is bounded schematically by

$$\begin{aligned} \sum_{\gamma} |C(\gamma)| &\lesssim \int_0^\infty dr e^{r/R} e^{-r/\xi_{\text{corr}}} \\ &= \int_0^\infty dr e^{-(\xi_{\text{corr}}^{-1} - R^{-1})r}. \end{aligned} \quad (4009)$$

This integral converges when $\frac{1}{\xi_{\text{corr}}} > \frac{1}{R}$. Equivalently, $\xi_{\text{corr}} < R$. Thus an exponentially decaying correlation function need not be absolutely summable on a hyperbolic lattice unless its decay outruns hyperbolic volume growth. The preceding result defines two analytically distinct regimes. When $\xi_{\text{corr}} < R$, the ordinary representation sum is absolutely convergent and produces a bounded continuous structure tensor on every finite-rank representation sector. When $\xi_{\text{corr}} \geq R$, the simple ℓ^1 criterion fails. The failure does not imply that diffraction ceases to exist. Instead the spectral object must be defined through finite-cover limits, distributional transforms, or a singular spectral measure rather than by an absolutely convergent group sum. This is the diffraction analogue of the hyperbolic long-range-hopping threshold derived previously.

e. Theorem 52. Volume-growth criterion for a regular generalized structure factor. Let a translation correlation satisfy

$$|C(\gamma)| \leq C_0 e^{-\mu d(o, \gamma o)}. \quad (4010)$$

For a cocompact two-dimensional hyperbolic translation group, a sufficient condition for absolute summability is $\mu > \frac{1}{R}$. Under this condition,

$$\mathbb{S}[\varpi] = \sum_{\gamma} C(\gamma) \varpi(\gamma) \quad (4011)$$

converges absolutely and uniformly in every finite-dimensional unitary representation. Thus hyperbolic diffraction contains an intrinsic competition between correlation decay and exponential geometric volume growth.

A finite-cover definition remains meaningful independently of absolute summability. Let a finite quotient contain $N = |Q_N|$ translation cells. Define the unnormalized representation amplitude

$$F_N(\varpi) = \sum_{q \in Q_N} f(q) \varpi(q). \quad (4012)$$

A normalized generalized Bragg weight is

$$\mathcal{B}_N(\varpi) = \frac{\|F_N(\varpi)\|_F^2}{d\varpi N^2}. \quad (4013)$$

For a perfectly coherent ordered component in a fixed representation channel,

$$\mathcal{B}_N(\varpi) \longrightarrow \mathcal{B}_\infty(\varpi) > 0 \quad (4014)$$

along an appropriate compatible cover sequence. For an incoherent zero-mean component with only short-range correlations, one generically expects the unnormalized amplitude power to grow only linearly with N , giving $\mathcal{B}_N(\varpi) \longrightarrow 0$. Thus coherent generalized Bragg order can be separated from diffuse representation-space spectral weight by finite-cover scaling. The translation representation label does not by itself resolve all spatial structure inside a hyperbolic fundamental domain. This is directly analogous to the Euclidean distinction between Bloch momentum and reciprocal harmonics within the microscopic unit cell. For an exact common quotient,

$$\Sigma_M = \Gamma_M \backslash \mathbb{H}_R^2, \quad (4015)$$

consider a unitary representation $\rho : \Gamma_M \longrightarrow U(d_\rho)$. It defines a rank- d_ρ flat bundle

$$E_\rho = (\mathbb{H}_R^2 \times \mathbb{C}^{d_\rho}) / \Gamma_M. \quad (4016)$$

The associated covariant Laplace–Beltrami operator is denoted by $-\Delta_\rho$. Because the quotient is compact and the operator is elliptic and self-adjoint, its spectrum is discrete,

$$-\Delta_\rho \Phi_{n,\rho} = \lambda_{n,\rho} \Phi_{n,\rho}. \quad (4017)$$

The eigenvalues satisfy

$$0 \leq \lambda_{0,\rho} \leq \lambda_{1,\rho} \leq \lambda_{2,\rho} \leq \dots \quad (4018)$$

The eigenmodes satisfy the automorphic condition

$$\Phi_{n,\rho}(\gamma x) = \rho(\gamma) \Phi_{n,\rho}(x). \quad (4019)$$

A complete compact-quotient harmonic label therefore consists of the pair $\mathbf{q} = ([\rho], n)$. The representation class describes translation holonomy. The discrete mode index describes spatial harmonic structure inside the quotient. Thus the Euclidean reciprocal vector is replaced structurally by $\mathbf{q} \longrightarrow ([\rho], n)$. This distinction is essential because $[\rho]$ alone is a generalized Bloch or translation label rather than a complete spatial-frequency label. Let a projected scattering source in the ρ sector be the section $s_\rho(x) \in E_\rho$. Its spectral coefficient is

$$\mathcal{A}_{n,\rho} = \int_{\Sigma_M} dA(x) \Phi_{n,\rho}^\dagger(x) s_\rho(x). \quad (4020)$$

The generalized harmonic intensity is

$$I_{n,\rho} = |\mathcal{A}_{n,\rho}|^2. \quad (4021)$$

For a multiplet of internal scattering channels, the amplitude becomes a vector or matrix and the physical intensity is obtained by the appropriate Hermitian norm. The twisted-Laplacian eigenmodes form a complete orthonormal basis in the corresponding square-integrable bundle-section space. Therefore,

$$\int_{\Sigma_M} dA \|s_\rho(x)\|^2 = \sum_n |\mathcal{A}_{n,\rho}|^2. \quad (4022)$$

Thus generalized diffraction conserves total spectral weight both in the translation-irrep decomposition and in the within-sector Laplacian harmonic decomposition. A scalar diagnostic analogous to the magnitude of an ordinary reciprocal wavevector may be defined from the Laplacian eigenvalue,

$$\kappa_{n,\rho} = \sqrt{\lambda_{n,\rho}}. \quad (4023)$$

Its dimension is inverse length. A corresponding spectral length may be introduced as

$$L_{n,\rho}^{\text{spec}} = \frac{2\pi}{\kappa_{n,\rho}}, \quad (4024)$$

when $\kappa_{n,\rho} \neq 0$. This length is a spectral diagnostic and must not automatically be identified with a global translation period. Hyperbolic space generally possesses no globally additive reciprocal wavevector whose magnitude alone characterizes a mode. The complete representation- and harmonic-resolved diffraction measure may be written schematically as

$$d\mathcal{D}(\kappa) = \sum_\rho \sum_n W_\rho I_{n,\rho} \delta(\kappa - \kappa_{n,\rho}) d\kappa, \quad (4025)$$

where W_ρ is the representation multiplicity or Plancherel weight appropriate to the finite-cover or automorphic decomposition being used. For a finite normal quotient,

$W_\rho = d_\rho$ in the full regular-representation trace decomposition. Thus generalized diffraction contains both matrix-valued translation information and scalar geometric spectral information. Although a global Euclidean reciprocal vector does not exist, the local geodesic moiré length derived previously defines an inverse-length spectral proxy. Recall

$$\xi_M^{(H)} = R \operatorname{arsinh} \left[\frac{\sinh(a/2R)}{|\sin(\theta/2)|} \right]. \quad (4026)$$

Define the local geometric reciprocal scale by

$$q_M^{\text{loc}} = \frac{2\pi}{\xi_M^{(H)}}. \quad (4027)$$

Therefore,

$$q_M^{\text{loc}} = \frac{2\pi}{R \operatorname{arsinh} \left[\frac{\sinh(a/2R)}{|\sin(\theta/2)|} \right]}. \quad (4028)$$

This quantity is not asserted to be a global Fuchsian reciprocal vector. It is the inverse local registry length and therefore provides the natural local diffraction scale against which representation-space spectral features can be compared. Take $R \rightarrow \infty$ at fixed lattice spacing and nonzero twist. Using

$$\lim_{R \rightarrow \infty} \xi_M^{(H)} = \frac{a}{2|\sin(\theta/2)|}, \quad (4029)$$

one obtains

$$\lim_{R \rightarrow \infty} q_M^{\text{loc}} = \frac{2\pi}{a/[2|\sin(\theta/2)|]} = \frac{4\pi}{a} |\sin(\theta/2)|. \quad (4030)$$

Hence,

$$\lim_{R \rightarrow \infty} q_M^{\text{loc}} = \Delta q_1^{(E)}. \quad (4031)$$

Thus the local hyperbolic inverse-length scale reduces exactly to the Euclidean first-shell reciprocal peak splitting. Let

$$s = \left| \sin \left(\frac{\theta}{2} \right) \right|. \quad (4032)$$

The hyperbolic moiré length has the expansion

$$\xi_M^{(H)} = \frac{a}{2s} - \frac{a^3(1-s^2)}{48R^2s^3} + O(R^{-4}). \quad (4033)$$

Factor out the Euclidean length,

$$\xi_M^{(H)} = \frac{a}{2s} \left[1 - \frac{a^2(1-s^2)}{24R^2s^2} + O(R^{-4}) \right]. \quad (4034)$$

Taking the reciprocal gives

$$q_M^{\text{loc}} = \frac{4\pi s}{a} \left[1 + \frac{a^2(1-s^2)}{24R^2s^2} + O(R^{-4}) \right]. \quad (4035)$$

Thus weak negative curvature increases the local inverse moiré length at fixed twist. Define

$$b = \sinh \left(\frac{a}{2R} \right). \quad (4036)$$

At fixed negative curvature and small twist, $s \simeq \frac{|\theta|}{2}$.

When $\frac{b}{s} \gg 1$, one has

$$\operatorname{arsinh} \left(\frac{b}{s} \right) = \ln \left(\frac{2b}{s} \right) + O \left(\frac{s^2}{b^2} \right). \quad (4037)$$

Therefore,

$$q_M^{\text{loc}} \sim \frac{2\pi}{R \ln(4b/|\theta|)}. \quad (4038)$$

Thus the local reciprocal scale vanishes only logarithmically as the twist approaches zero at fixed negative curvature. This differs qualitatively from the Euclidean law

$$\Delta q_1^{(E)} \sim \frac{2\pi}{a} |\theta|. \quad (4039)$$

Introduce the crossover variable $\chi = \frac{b}{s}$. Because

$$q_M^{\text{loc}} \propto \frac{1}{\operatorname{arsinh} \chi}, \quad (4040)$$

define the running twist exponent by

$$\nu_q = \frac{d \ln q_M^{\text{loc}}}{d \ln s}. \quad (4041)$$

Since

$$\frac{d\chi}{d \ln s} = -\chi, \quad (4042)$$

one obtains

$$\begin{aligned} \nu_q &= -\frac{1}{\operatorname{arsinh} \chi} \frac{1}{\sqrt{1+\chi^2}} \frac{d\chi}{d \ln s} \\ &= \frac{\chi}{\sqrt{1+\chi^2} \operatorname{arsinh} \chi}. \end{aligned} \quad (4043)$$

Therefore,

$$\nu_q(\chi) = \frac{\chi}{\sqrt{1 + \chi^2} \operatorname{arsinh} \chi}. \quad (4044)$$

In the Euclidean regime, $\chi \rightarrow 0$, one obtains $\nu_q \rightarrow 1$. In the curvature-dominated regime, $\chi \rightarrow \infty$, one finds

$$\nu_q \sim \frac{1}{\ln(2\chi)} \rightarrow 0. \quad (4045)$$

Thus the local reciprocal response flows continuously from a linear Euclidean twist law to an asymptotically logarithmic hyperbolic law.

f. Theorem 53. Curvature-induced crossover of the local reciprocal moiré scale. The inverse local registry length of a negatively curved twisted bilayer is

$$q_M^{\text{loc}} = \frac{2\pi}{R \operatorname{arsinh} \left[\frac{\sinh(a/2R)}{|\sin(\theta/2)|} \right]}. \quad (4046)$$

Its exact Euclidean limit is

$$q_M^{\text{loc}} \rightarrow \frac{4\pi}{a} |\sin(\theta/2)|. \quad (4047)$$

At fixed negative curvature and asymptotically small twist,

$$q_M^{\text{loc}} \sim \frac{2\pi}{R \ln [4 \sinh(a/2R)/|\theta|]}. \quad (4048)$$

Therefore negative curvature changes the local reciprocal response from an asymptotically linear twist law to an asymptotically inverse-logarithmic law. This result concerns the local inverse registry length and does not assume the existence of a globally additive hyperbolic reciprocal vector. In the continuum weak-curvature regime define

$$u = \frac{R|\theta|}{a}. \quad (4049)$$

At fixed u with $\frac{a}{R} \rightarrow 0$, the geometric moiré length satisfies

$$\frac{\xi_M}{R} \rightarrow \operatorname{arsinh} \left(\frac{1}{u} \right). \quad (4050)$$

Therefore,

$$Rq_M^{\text{loc}} \rightarrow \frac{2\pi}{\operatorname{arsinh}(1/u)}. \quad (4051)$$

This defines a universal local reciprocal crossover function. The leading Euclidean reciprocal scale is

$$q_E = \frac{2\pi}{\xi_E} \simeq \frac{2\pi}{R} u. \quad (4052)$$

Hence the ratio is

$$\mathcal{R}_q(u) = \frac{q_M^{\text{loc}}}{q_E} = \frac{1}{u \operatorname{arsinh}(1/u)}. \quad (4053)$$

For $u \rightarrow \infty$, one obtains $\mathcal{R}_q(u) \rightarrow 1$. For $u \rightarrow 0$, one obtains

$$\mathcal{R}_q(u) \sim \frac{1}{u \ln(2/u)} \rightarrow \infty. \quad (4054)$$

Thus, at fixed twist relative to the Euclidean prediction, negative curvature shifts spectral weight toward a parametrically larger local inverse-length scale because the hyperbolic moiré domain is shorter than its Euclidean counterpart. For numerical calculations, choose a normal periodic-cover subgroup $\Gamma_N \triangleleft \Gamma$. The finite quotient is

$$Q_N = \Gamma/\Gamma_N. \quad (4055)$$

For every quotient irrep compute

$$\hat{f}_N(\varpi) = \sqrt{\frac{d_\varpi}{|Q_N|}} \sum_{q \in Q_N} f_N(q) \varpi(q). \quad (4056)$$

The generalized finite-cover structure factor is

$$I_N(\varpi) = \left\| \hat{f}_N(\varpi) \right\|_F^2. \quad (4057)$$

The complete state-counting identity is

$$\sum_{\varpi \in \hat{Q}_N} d_\varpi^2 = |Q_N|. \quad (4058)$$

Thus finite-cover diffraction may be computed using precisely the same finite-group representation data that appear in the hyperbolic spectral Hamiltonian. Let $C_N(\gamma)$ be a correlation function measured on a periodic finite cover. Suppose the corresponding real-space separation satisfies $d(o, \gamma o) < r_{\text{inj}}^{(N)}$. Then the group displacement is represented without quotient aliasing. For every fixed finite collection of such displacements, $C_N(\gamma) \rightarrow C_\infty(\gamma)$ and, for strictly local finite-range observables, equality is eventually exact. Therefore generalized diffraction constructed from finite-range correlations has a controlled periodic-cover thermodynamic limit. For an incommensurate twist, no finite exact common quotient exists. One may nevertheless calculate local bilayer correlations on paired finite covers. Let $\Gamma_N^{(1)} < \Gamma_1$, and

$$\Gamma_N^{(2)} = g_\theta \Gamma_N^{(1)} g_\theta^{-1}. \quad (4059)$$

Each layer admits its own finite-group Fourier decomposition. Interlayer correlation functions are evaluated after lifting both covers to the common universal hyperbolic plane. Thus the incommensurate generalized diffraction problem remains locally well defined even though no finite common representation $\text{Ind}_{\Gamma_M}^{\Gamma_\ell} \mathbf{1}$ exists. If a sequence

of exact commensurate twists approaches an incommensurate target, $\theta_j \rightarrow \theta$, each approximant possesses a finite induced reciprocal representation,

$$\mathcal{P}_j = \text{Ind}_{\Gamma_{M,j}}^{\Gamma} \mathbf{1}. \quad (4060)$$

Its dimension is

$$\dim \mathcal{P}_j = q_{M,j}. \quad (4061)$$

For a genuinely incommensurate limiting target, the exact reciprocal representation becomes infinite dimensional. Therefore an approximant sequence typically satisfies $q_{M,j} \rightarrow \infty$ whenever the exact common subgroup approaches infinite index. This provides a representation-space analogue of the proliferation of reciprocal harmonics in Euclidean quasiperiodic approximants. The real-space geometric mismatch between an approximant twist and the target twist over radius L is

$$\epsilon_L = 2R \operatorname{arsinh} \left[\sinh(L/R) \left| \sin \left(\frac{\theta_j - \theta}{2} \right) \right| \right]. \quad (4062)$$

Thus convergence of local correlation functions over a growing region requires

$$|\theta_j - \theta| e^{L_j/R} \rightarrow 0. \quad (4063)$$

The same condition is therefore required for convergence of generalized diffraction computed from commensurate approximants. The framework above provides an exact analogue of the Euclidean distinction between bare Bragg peaks and true moiré satellites. A rigid homogeneous parent layer occupies only the trivial parent-group representation. A structural modulation that lowers the translation symmetry from Γ_ℓ to Γ_M populates nontrivial irreducible components of $\text{Ind}_{\Gamma_M}^{\Gamma_\ell} \mathbf{1}$. Therefore, nontrivial $I_\varpi \iff$ nontrivial parent-symmetry reconstruction provided finite-window leakage and probe-specific background contributions have been removed. This gives a direct group-theoretic definition of a hyperbolic moiré satellite. Define the total nontrivial diffraction fraction by

$$\mathcal{W}_{\text{sat}} = 1 - \mathcal{S}_1, \quad (4064)$$

where $\mathcal{S}_1$ is the normalized weight of the trivial parent-group representation. Equivalently,

$$\mathcal{W}_{\text{sat}} = \sum_{\varpi \neq 1} \mathcal{S}_\varpi. \quad (4065)$$

For a perfectly homogeneous parent layer, $\mathcal{W}_{\text{sat}} = 0$. For a weak modulation, $\mathcal{W}_{\text{sat}} = O(|\epsilon|^2)$. Thus $\mathcal{W}_{\text{sat}}$ provides a basis-independent scalar measure of the total structural reconstruction in representation space. The distribution of satellite weight among generalized reciprocal sectors contains additional information. Define

$$\text{IPR}_{\text{rep}} = \sum_{\varpi} \mathcal{S}_\varpi^2. \quad (4066)$$

The corresponding participation number is

$$N_{\text{rep}} = \frac{1}{\text{IPR}_{\text{rep}}}. \quad (4067)$$

A diffraction pattern concentrated in one representation sector has $N_{\text{rep}} = 1$. A pattern spread over many sectors has $N_{\text{rep}} \gg 1$. This quantity provides a natural reciprocal-complexity measure for hyperbolic moiré reconstruction. A complementary basis-independent measure is the Shannon entropy

$$\mathcal{S}_{\text{rep}}^{\text{ent}} = - \sum_{\varpi} \mathcal{S}_\varpi \ln \mathcal{S}_\varpi. \quad (4068)$$

For a single reciprocal representation, $\mathcal{S}_{\text{rep}}^{\text{ent}} = 0$. An increasingly broad distribution of generalized reciprocal weight increases the entropy. This observable can be used to quantify the proliferation of reciprocal channels as commensurate approximants approach an incommensurate twist. Let the representation-resolved scattering field depend on curvature and twist, $f = f(K, \theta)$. Define the mixed curvature-twist diffraction response

$$\delta_{K\theta} \hat{f}(\varpi) = \hat{f}(K, \theta; \varpi) - \hat{f}(K, 0; \varpi) - \hat{f}(0, \theta; \varpi) + \hat{f}(0, 0; \varpi). \quad (4069)$$

The corresponding mixed intensity diagnostic is

$$\mathcal{D}_{K\theta}(\varpi) = \left\| \delta_{K\theta} \hat{f}(\varpi) \right\|_F^2. \quad (4070)$$

This removes the pure-curvature and pure-twist contributions at the amplitude level. A nonzero value therefore detects a genuinely nonadditive curvature-assisted reconstruction of the generalized diffraction pattern. The complete structural pathway can now be written as

$$K \rightarrow \text{hyperbolic translation group} \rightarrow \Gamma_M \rightarrow \text{Ind}_{\Gamma_M}^{\Gamma_\ell} \mathbf{1} \rightarrow \mathfrak{R}_M^{(\ell)}. \quad (4071)$$

Independently, curvature modifies the real-space registry field,

$$K \rightarrow d_\theta(r) \rightarrow \xi_M \rightarrow f(K, \theta). \quad (4072)$$

The two pathways meet in the generalized structure factor,

$$f(K, \theta) \rightarrow \hat{f}(\varpi) \rightarrow \mathbb{S}(\varpi) \rightarrow I_\varpi. \quad (4073)$$

Thus curvature affects diffraction both by changing the admissible representation structure of an exact commensurate supercell and by changing the microscopic weight carried by those representation sectors.

g. Theorem 54. Hyperbolic representation-space structure factor. Let a hyperbolic moiré modulation be represented on a finite quotient Q of the relevant translation group by a scattering field $f(q)$. For every unitary irrep $\varpi : Q \rightarrow U(d_\varpi)$, define

$$\hat{f}(\varpi) = \sqrt{\frac{d_\varpi}{|Q|}} \sum_{q \in Q} f(q) \varpi(q). \quad (4074)$$

Then the generalized static structure tensor is

$$\mathbb{S}(\varpi) = \frac{1}{d_\varpi} \hat{f}(\varpi)^\dagger \hat{f}(\varpi), \quad (4075)$$

and the scalar sector intensity is

$$I_\varpi = d_\varpi \text{Tr } \mathbb{S}(\varpi) = \left\| \hat{f}(\varpi) \right\|_F^2. \quad (4076)$$

The total spectral power satisfies

$$\sum_{\varpi} I_\varpi = \sum_{q \in Q} |f(q)|^2. \quad (4077)$$

Thus the Euclidean scalar structure factor is replaced by a family of positive-semidefinite matrices indexed by non-Abelian translation representations.

h. Theorem 55. Exact generalized satellite criterion. Let the parent translation group be Γ , and let the exact commensurate moiré translation group be the finite-index subgroup Γ_M . A parent-group irrep can occur as an exact moiré diffraction harmonic if and only if $V_\varpi^{\Gamma_M} \neq 0$. Equivalently, $\varpi \subset \text{Ind}_{\Gamma_M}^{\Gamma} \mathbf{1}$. However, nonzero diffraction intensity additionally requires $P_\varpi f \neq 0$. Thus the generalized satellite condition separates exactly into translation-allowed sector and nonvanishing microscopic structure factor. This is the full non-Abelian counterpart of the distinction between an allowed reciprocal vector and a systematically extinct Euclidean diffraction peak.

i. Theorem 56. Representation-space commensurability classification. For a twisted hyperbolic bilayer,

$$q_M = [\Gamma_\ell : \Gamma_M]. \quad (4078)$$

If $q_M < \infty$, the exact structural reciprocal content is finite dimensional,

$$\dim \text{Ind}_{\Gamma_M}^{\Gamma_\ell} \mathbf{1} = q_M. \quad (4079)$$

If $q_M = \infty$, the exact induced reciprocal representation is infinite dimensional,

$$\dim \text{Ind}_{\Gamma_M}^{\Gamma_\ell} \mathbf{1} = \infty. \quad (4080)$$

Therefore the exact commensurate-incommensurate distinction can be expressed directly as a finite- versus infinite-dimensional generalized reciprocal representation.

j. Theorem 57. Recovery of ordinary reciprocal diffraction. For an Abelian Euclidean translation group, $\Gamma = \mathbb{Z}^2$, all irreducible unitary representations are one dimensional. The generalized reciprocal condition $V_\chi^{\Gamma_M} \neq 0$ reduces exactly to $\mathbf{k} \in \Lambda_M^*$. The induced representation decomposes into the ordinary set of q_M folded reciprocal characters. The group Fourier transform becomes the standard discrete Fourier transform. The generalized beat condition becomes

$$\mathbf{k}_2 - \mathbf{k}_1 \in \Lambda_M^*. \quad (4081)$$

The representation-space structure tensor becomes the ordinary scalar static structure factor. Thus every construction introduced in the present subsection reduces continuously and algebraically to conventional Euclidean reciprocal-space diffraction when the translation group becomes Abelian. The generalized hyperbolic diffraction theory therefore contains four different levels of information that must remain conceptually separate. The first level is the local geometric registry scale, ξ_M or equivalently its inverse-length proxy

$$q_M^{\text{loc}} = 2\pi/\xi_M. \quad (4082)$$

This quantity exists independently of exact commensurability. The second level is exact crystallography, encoded by the subgroup index

$$q_M = [\Gamma : \Gamma_M] \quad (4083)$$

and by the finite induced representation $\text{Ind}_{\Gamma_M}^{\Gamma} \mathbf{1}$. The third level is microscopic structural reconstruction, encoded by the actual representation amplitudes $\hat{f}(\varpi)$ and generalized structure tensors $\mathbb{S}(\varpi)$. The fourth level is coherence and spectral regularity, encoded by the finite-cover scaling of the representation weights and by the decay of translation correlations. Therefore, local moiré geometry $\neq$ exact commensurability $\neq$ satellite intensity $\neq$ coherence. The Euclidean Fourier-transform picture is therefore generalized in two steps. First, ordinary translation phases are replaced by unitary representations of the Fuchsian translation group. Exact moiré periodicity is encoded by the finite-dimensional induced representation from the common translation subgroup to the parent layer group. Its irreducible decomposition supplies the generalized reciprocal harmonics, and Frobenius reciprocity gives the exact analogue of the reciprocal Bragg condition. Second, microscopic scattering contrasts are transformed into these representation sectors by a non-Abelian group Fourier transform. The resulting diffraction amplitude is matrix valued, and the static structure factor becomes a positive-semidefinite structure tensor. Its trace gives the observable sector weight, while its matrix structure retains information that is absent from any scalar reciprocal vector. The hyperbolic analogue of a Euclidean beat wavevector is not a subtraction of two vectors. It is the tensor product of one layer representation with the dual of the other. A moiré interference channel exists when this tensor product contains a singlet of the common translation subgroup. The ordinary Euclidean relation $\mathbf{k}_2 - \mathbf{k}_1 \in \Lambda_M^*$ is recovered exactly when all representations become one dimensional. Exact commensurability is correspondingly characterized by a finite-dimensional induced reciprocal representation. Incommensurability produces an infinite-dimensional induced representation and therefore no finite exact generalized reciprocal cell. This representation-theoretic criterion is the direct hyperbolic counterpart of the distinction between a discrete rank-two Euclidean reciprocal lattice and a higher-rank

quasiperiodic reciprocal module. Finally, the local inverse registry length provides a complementary geometric spectral scale. Its Euclidean limit is precisely the ordinary Bragg-peak splitting, whereas at fixed negative curvature its small-angle dependence crosses over from a linear twist law to an inverse-logarithmic law. The generalized diffraction problem therefore carries two distinct curvature signatures: the representation content changes through the exact common Fuchsian subgroup, while the local spectral scale changes through curvature-amplified registry geometry. This construction replaces the ordinary notion of a reciprocal-space diffraction image by a representation-resolved spectral measure while retaining the Euclidean structure factor as its Abelian limit.

D. Curvature Signatures in Moiré Diffraction

The generalized diffraction framework developed above provides the mathematical language required to distinguish a negatively curved moiré system from its Euclidean counterpart. The central problem is now to identify observables that respond directly to curvature rather than to microscopic form factors, relaxation amplitudes, finite-size effects, or arbitrary normalization conventions. A useful curvature diagnostic should satisfy several requirements. It should vanish, or reduce to a simple null relation, in the Euclidean limit. It should become nonzero for negative curvature without requiring an explicit curvature dependence of the microscopic scattering amplitude. It should preferably depend on diffraction-peak positions or spectral labels rather than absolute intensities, since peak intensities are more sensitive to probe-specific form factors and structural reconstruction. Finally, it should admit a controlled finite-cover and finite-resolution implementation. We therefore distinguish three levels of curvature-sensitive diffraction information. The primary level consists of the trajectory of the fundamental moiré-sensitive reciprocal scale as a function of twist. The secondary level consists of logarithmic slopes and multi-angle dimensionless ratios that eliminate unknown overall calibration factors. The third level consists of exact crystallographic representation counting along commensurate sequences. Intensity redistribution among generalized reciprocal sectors provides an additional diagnostic, but is not by itself a universal curvature signature. The strongest geometric prediction is that negative curvature changes the small-angle dependence of the fundamental moiré reciprocal scale from a linear law to an inverse-logarithmic law. To isolate curvature, all microscopic parameters must first be held fixed. We compare a Euclidean reference system and a negatively curved system with the same microscopic lattice spacing a , the same interlayer separation, h , the same microscopic hopping and scattering laws, and the same relative twist angle θ . The negative curvature is parameterized by $K = -\frac{1}{R^2}$. The comparison there-

fore belongs to the geometry-only protocol developed previously, $K \rightarrow$ geometry, bare microscopic parameters fixed. This restriction is essential. If the microscopic hopping amplitudes, scattering amplitudes, lattice spacing, or interlayer separation are simultaneously changed with curvature, an observed diffraction shift cannot be assigned uniquely to geometry. For a family of exact regular tessellations, a/R may be constrained by the tessellation data. In that setting a controlled comparison requires either a family of tessellations with matched microscopic scale or an explicit accounting of the change in a/R . A fixed- $\{p, q\}$ tessellation should not be interpreted as an independent continuous curvature sweep at fixed a . Define

$$s = \left| \sin \left(\frac{\theta}{2} \right) \right|. \quad (4084)$$

In the Euclidean bilayer, the local moiré length is

$$\xi_M^{(E)} = \frac{a}{2s}. \quad (4085)$$

The corresponding fundamental geometric reciprocal scale is

$$q_M^{(E)} = \frac{2\pi}{\xi_M^{(E)}} = \frac{4\pi s}{a}. \quad (4086)$$

For small twist,

$$q_M^{(E)} = \frac{2\pi}{a} |\theta| - \frac{\pi}{12a} |\theta|^3 + O(|\theta|^5). \quad (4087)$$

Thus the leading Euclidean reciprocal response is linear in the twist angle. For negative curvature, the exact local registry length is

$$\xi_M^{(H)} = R \operatorname{arsinh} \left[\frac{\sinh(a/2R)}{s} \right]. \quad (4088)$$

The corresponding inverse-length diffraction proxy is

$$q_M^{(H)} = \frac{2\pi}{R \operatorname{arsinh} \left[\frac{\sinh(a/2R)}{s} \right]}. \quad (4089)$$

This quantity should be interpreted carefully. It is not a globally additive reciprocal vector of the Fuchsian crystal. It is the reciprocal scale associated with the local registry length and therefore predicts the characteristic spectral scale of a moiré-sensitive diffraction mode. For a physical probe whose dominant low-frequency satellite follows the fundamental registry mode, the observed spectral peak is expected to track $q_M^{(H)}$, possibly up to a probe-dependent calibration factor. The curvature dependence derived below concerns the geometry of the registry scale itself and is therefore independent of such an overall intensity normalization. Introduce $\alpha = \frac{a}{2R}$,

and $b = \sinh \alpha$. The curvature–twist crossover variable is $\chi = \frac{b}{s}$. Then

$$q_M^{(H)} = \frac{2\pi}{R \operatorname{arsinh} \chi}. \quad (4090)$$

For weak curvature and small twist,

$$b \simeq \frac{a}{2R}, \quad s \simeq \frac{|\theta|}{2}, \quad (4091)$$

and therefore

$$\chi \simeq \frac{a}{R|\theta|}. \quad (4092)$$

Thus the single ratio $a/(R|\theta|)$ determines whether diffraction is locally Euclidean or curvature dominated. Define the ratio between the negatively curved and Euclidean local reciprocal scales,

$$\mathcal{C}_q(K, \theta) = \frac{q_M^{(H)}}{q_M^{(E)}}. \quad (4093)$$

Using Eqs. (4086) and (4089), one obtains

$$\mathcal{C}_q = \frac{\alpha}{s \operatorname{arsinh} \left(\frac{\sinh \alpha}{s} \right)}. \quad (4094)$$

We now determine its sign relative to unity. For $0 < s < 1$, define $t = \frac{1}{s} > 1$. Because $\sinh x$ is strictly convex for positive x , and $\sinh 0 = 0$, $\sinh(t\alpha) > t \sinh \alpha$. Therefore,

$$\sinh \left(\frac{\alpha}{s} \right) > \frac{\sinh \alpha}{s}. \quad (4095)$$

Since arsinh is strictly increasing,

$$\frac{\alpha}{s} > \operatorname{arsinh} \left(\frac{\sinh \alpha}{s} \right). \quad (4096)$$

Multiplication by $s > 0$ gives

$$\alpha > s \operatorname{arsinh} \left(\frac{\sinh \alpha}{s} \right). \quad (4097)$$

Hence,

$$\mathcal{C}_q > 1 \quad \text{for } K < 0, \quad 0 < s < 1. \quad (4098)$$

Thus negative curvature shifts the fundamental local moiré diffraction scale outward relative to the Euclidean prediction. Equivalently, $q_M^{(H)} > q_M^{(E)}$. The corresponding real-space statement is $\xi_M^{(H)} < \xi_M^{(E)}$.

a. Theorem 58. Curvature-induced reciprocal-scale enhancement. For fixed microscopic lattice spacing a and fixed nonzero twist with $0 < s < 1$, the geometry-defined fundamental reciprocal scale satisfies

$$q_M(K < 0, \theta) > q_M(K = 0, \theta). \quad (4099)$$

Equality is recovered only in the flat limit $R \rightarrow \infty$. Therefore a systematic outward deviation of the moiré-sensitive diffraction scale from

$$\frac{4\pi}{a} \left| \sin \left(\frac{\theta}{2} \right) \right| \quad (4100)$$

is a direct geometric signature of negative curvature within the geometry-only comparison protocol. The exact hyperbolic registry length has the expansion

$$\begin{aligned} \xi_M^{(H)} &= \frac{a}{2s} - \frac{a^3(1-s^2)}{48R^2s^3} \\ &+ \frac{a^5(1-s^2)(9-s^2)}{3840R^4s^5} + O(R^{-6}). \end{aligned} \quad (4101)$$

Factorizing the Euclidean length gives

$$\begin{aligned} \xi_M^{(H)} &= \frac{a}{2s} \left[1 - \frac{a^2(1-s^2)}{24R^2s^2} \right. \\ &\left. + \frac{a^4(1-s^2)(9-s^2)}{1920R^4s^4} + O(R^{-6}) \right]. \end{aligned} \quad (4102)$$

Taking the reciprocal yields

$$\begin{aligned} q_M^{(H)} &= q_M^{(E)} \left[1 + \frac{a^2(1-s^2)}{24R^2s^2} \right. \\ &\left. - \frac{a^4(1-s^2)(17+7s^2)}{5760R^4s^4} + O(R^{-6}) \right]. \end{aligned} \quad (4103)$$

Since

$$1 - s^2 = \cos^2 \left(\frac{\theta}{2} \right), \quad (4104)$$

the leading correction becomes

$$\frac{q_M^{(H)} - q_M^{(E)}}{q_M^{(E)}} = \frac{a^2}{24R^2} \cot^2 \left(\frac{\theta}{2} \right) + O(R^{-4}). \quad (4105)$$

Using $K = -\frac{1}{R^2}$, this can be written as

$$\frac{q_M(K, \theta)}{q_M(0, \theta)} = 1 - \frac{Ka^2}{24} \cot^2 \left(\frac{\theta}{2} \right) + O(K^2a^4). \quad (4106)$$

For $K < 0$, the leading correction is positive. Define the zero-curvature diffraction susceptibility by

$$\Xi_K^{\text{diff}}(\theta) = - \left. \frac{\partial \ln q_M(K, \theta)}{\partial K} \right|_{K=0}. \quad (4107)$$

Equation (4106) gives

$$\Xi_K^{\text{diff}}(\theta) = \frac{a^2}{24} \cot^2 \left(\frac{\theta}{2} \right). \quad (4108)$$

At small twist,

$$\Xi_K^{\text{diff}} \sim \frac{a^2}{6\theta^2}. \quad (4109)$$

Thus the weak-curvature response becomes increasingly sensitive at small twist. However, this divergence does not imply that the perturbative expansion remains valid arbitrarily close to zero twist. The expansion parameter is

$$\varepsilon_{\text{curv}} = \frac{a^2}{R^2} \cot^2 \left(\frac{\theta}{2} \right). \quad (4110)$$

The weak-curvature expansion requires $\varepsilon_{\text{curv}} \ll 1$. At fixed negative curvature, this condition eventually fails as $\theta \rightarrow 0$, and the exact inverse-logarithmic regime must be used instead. In the weak-curvature regime, define the measured relative peak shift

$$\delta_q = \frac{q_M^{(H)}}{q_M^{(E)}} - 1. \quad (4111)$$

Equation (4106) can be inverted to give

$$K_{\text{est}} = -\frac{24}{a^2} \tan^2 \left(\frac{\theta}{2} \right) \delta_q + O(\delta_q^2). \quad (4112)$$

A positive outward relative shift, $\delta_q > 0$, therefore corresponds to $K_{\text{est}} < 0$. This provides a direct weak-curvature inversion formula using only a peak position and a known microscopic lattice scale. The natural crossover between Euclidean and curvature-dominated diffraction occurs when $\chi = 1$. Thus

$$s_c = \sinh \left(\frac{a}{2R} \right). \quad (4113)$$

If

$$\sinh \left(\frac{a}{2R} \right) < 1, \quad (4114)$$

the exact crossover twist is

$$\theta_c = 2 \arcsin \left[\sinh \left(\frac{a}{2R} \right) \right]. \quad (4115)$$

For weak curvature,

$$\theta_c = \frac{a}{R} + \frac{a^3}{12R^3} + O(R^{-5}). \quad (4116)$$

Thus the leading crossover criterion is $|\theta| \sim \frac{a}{R}$. For $|\theta| \gg \frac{a}{R}$, the diffraction response is approximately Euclidean. For $|\theta| \ll \frac{a}{R}$, the response is curvature dominated. If

$$\sinh(a/2R) \geq 1, \quad (4117)$$

then $\chi \geq 1$ for every allowed value $s \leq 1$, and no parametrically Euclidean local registry regime exists within the full angular interval. At fixed negative curvature and small twist, $s \simeq \frac{|\theta|}{2}$. For $\chi \gg 1$, the asymptotic expansion

$$\text{arsinh } \chi = \ln(2\chi) + O(\chi^{-2}) \quad (4118)$$

gives

$$q_M^{(H)} \sim \frac{2\pi}{R \ln \left[\frac{4 \sinh(a/2R)}{|\theta|} \right]}. \quad (4119)$$

Therefore,

$$q_M^{(H)} \propto \frac{1}{\ln(1/|\theta|)} \quad (4120)$$

up to curvature- and lattice-dependent constants. This differs qualitatively from $q_M^{(E)} \propto |\theta|$. The distinction is therefore not merely a perturbative peak shift. Negative curvature changes the functional universality class of the small-angle diffraction trajectory. Peak positions alone require knowledge of the absolute reciprocal calibration. A stronger test is obtained from a logarithmic derivative. Define

$$\nu_s = \frac{d \ln q_M}{d \ln s}. \quad (4121)$$

For the Euclidean result,

$$q_M^{(E)} = \frac{4\pi}{a} s, \quad (4122)$$

and therefore $\nu_s^{(E)} = 1$. For the negatively curved system,

$$q_M^{(H)} = \frac{2\pi}{R \text{arsinh } \chi}, \quad \chi = \frac{b}{s}. \quad (4123)$$

Since

$$\frac{d\chi}{d\ln s} = -\chi, \quad (4124)$$

one obtains

$$\nu_s^{(H)} = \frac{\chi}{\sqrt{1+\chi^2} \operatorname{arsinh} \chi}. \quad (4125)$$

For every finite positive χ , $0 < \nu_s^{(H)} < 1$. The limits are $\nu_s^{(H)} \rightarrow 1$ ($\chi \rightarrow 0$), and

$$\nu_s^{(H)} \sim \frac{1}{\ln(2\chi)} \rightarrow 0 \quad (\chi \rightarrow \infty). \quad (4126)$$

Thus the logarithmic slope continuously flows from the Euclidean value 1 toward zero. Experiments or numerical calculations naturally vary θ rather than s . Define

$$\nu_\theta = \frac{d\ln q_M}{d\ln |\theta|}. \quad (4127)$$

Since

$$s = \sin\left(\frac{|\theta|}{2}\right), \quad (4128)$$

one has

$$\frac{d\ln s}{d\ln |\theta|} = \frac{|\theta|}{2} \cot\left(\frac{|\theta|}{2}\right). \quad (4129)$$

Therefore,

$$\nu_\theta^{(H)} = \frac{|\theta|}{2} \cot\left(\frac{|\theta|}{2}\right) \frac{\chi}{\sqrt{1+\chi^2} \operatorname{arsinh} \chi}. \quad (4130)$$

The corresponding Euclidean slope is

$$\nu_\theta^{(E)} = \frac{|\theta|}{2} \cot\left(\frac{|\theta|}{2}\right). \quad (4131)$$

Hence their ratio is particularly simple,

$$\frac{\nu_\theta^{(H)}}{\nu_\theta^{(E)}} = \frac{\chi}{\sqrt{1+\chi^2} \operatorname{arsinh} \chi}. \quad (4132)$$

The trivial nonlinear dependence of $s = \sin(|\theta|/2)$ therefore cancels exactly. Define the dimensionless curvature index

$$\mathcal{I}_{\text{curv}} = 1 - \frac{\nu_\theta}{\frac{|\theta|}{2} \cot\left(\frac{|\theta|}{2}\right)}. \quad (4133)$$

For flat space, $K = 0 \Rightarrow \mathcal{I}_{\text{curv}} = 0$. For negative curvature, $K < 0 \Rightarrow 0 < \mathcal{I}_{\text{curv}} < 1$. In the curvature-dominated small-angle regime, $\mathcal{I}_{\text{curv}} \rightarrow 1$. This provides an amplitude-independent curvature null test.

b. Theorem 59. Universal diffraction-slope criterion. Let the fundamental moiré-sensitive diffraction scale follow the local registry mode. Then

$$K = 0 \Rightarrow \frac{d\ln q_M}{d\ln s} = 1, \quad (4134)$$

whereas

$$K < 0 \Rightarrow 0 < \frac{d\ln q_M}{d\ln s} < 1. \quad (4135)$$

At fixed negative curvature,

$$\lim_{\theta \rightarrow 0} \frac{d\ln q_M}{d\ln s} = 0. \quad (4136)$$

Thus the logarithmic slope distinguishes negative curvature without requiring knowledge of the absolute diffraction intensity. The running exponent itself obeys a closed differential equation. From

$$\nu_s = \frac{\chi}{\sqrt{1+\chi^2} \operatorname{arsinh} \chi}, \quad (4137)$$

one obtains

$$\frac{d\ln \nu_s}{d\ln \chi} = \frac{1}{1+\chi^2} - \nu_s. \quad (4138)$$

Since $d\ln \chi = -d\ln s$, the flow with twist is

$$\frac{d\nu_s}{d\ln s} = \nu_s \left[\nu_s - \frac{1}{1+\chi^2} \right]. \quad (4139)$$

Thus the curvature crossover is not only characterized by two limiting exponents but by a universal continuous flow between them. The hyperbolic peak trajectory admits an exact algebraic inversion. Starting from

$$q_M^{(H)} = \frac{2\pi}{R \operatorname{arsinh}(b/s)}, \quad (4140)$$

one obtains

$$\operatorname{arsinh}\left(\frac{b}{s}\right) = \frac{2\pi}{R q_M^{(H)}}. \quad (4141)$$

Applying sinh gives

$$\frac{b}{s} = \sinh\left(\frac{2\pi}{R q_M^{(H)}}\right). \quad (4142)$$

Therefore,

$$s \sinh \left(\frac{2\pi}{Rq_M^{(H)}} \right) = \sinh \left(\frac{a}{2R} \right). \quad (4143)$$

The right-hand side is independent of twist. This gives a particularly strong multi-angle diffraction test. Consider two twist angles θ_1, θ_2 , with corresponding reciprocal scales q_1, q_2 . Define

$$s_i = \left| \sin \left(\frac{\theta_i}{2} \right) \right|. \quad (4144)$$

The exact hyperbolic theory predicts

$$s_1 \sinh \left(\frac{2\pi}{Rq_1} \right) = s_2 \sinh \left(\frac{2\pi}{Rq_2} \right). \quad (4145)$$

Importantly, the microscopic lattice spacing a has canceled. Therefore R can, in principle, be extracted from the trajectory of two or more moiré-sensitive diffraction peaks without first knowing the absolute microscopic length scale. After R has been determined, the lattice scale is recovered from

$$a = 2R \operatorname{arsinh} \left[s_i \sinh \left(\frac{2\pi}{Rq_i} \right) \right]. \quad (4146)$$

The value obtained must be independent of i .

c. Theorem 60. Multi-angle curvature inversion. Suppose a family of measured or numerically extracted fundamental moiré-sensitive reciprocal scales q_i follows the geometry-defined hyperbolic registry scale. Then the curvature radius is constrained by

$$s_i \sinh \left(\frac{2\pi}{Rq_i} \right) = s_j \sinh \left(\frac{2\pi}{Rq_j} \right) \quad (4147)$$

for every pair i, j . Once R is obtained, all angles must give the same reconstructed microscopic scale,

$$a_i = 2R \operatorname{arsinh} \left[s_i \sinh \left(\frac{2\pi}{Rq_i} \right) \right]. \quad (4148)$$

Thus the curvature hypothesis is overconstrained by multi-angle diffraction data. Define

$$\mathcal{U}_i(R, a) = \frac{s_i \sinh \left(\frac{2\pi}{Rq_i} \right)}{\sinh(a/2R)}. \quad (4149)$$

The exact geometry predicts $\mathcal{U}_i = 1$ for every twist angle. Thus an entire diffraction data set should collapse onto a single constant when represented through the hyperbolic scaling variables. The Euclidean limit follows by expanding the hyperbolic sine for $R \rightarrow \infty$,

$$\sinh \left(\frac{2\pi}{Rq} \right) \simeq \frac{2\pi}{Rq}, \quad (4150)$$

and

$$\sinh \left(\frac{a}{2R} \right) \simeq \frac{a}{2R}. \quad (4151)$$

Equation (4149) then reduces to $q = \frac{4\pi s}{a}$, as required. Suppose a physical probe measures a spectral peak $\kappa_M = \zeta q_M$, where ζ is an unknown but angle-independent calibration factor over the range of interest. The Euclidean prediction is then

$$\frac{\kappa_M}{s} = \frac{4\pi\zeta}{a}, \quad (4152)$$

which is independent of twist. For two angles define

$$\mathcal{P}_{ij} = \ln \left[\frac{\kappa_i/s_i}{\kappa_j/s_j} \right]. \quad (4153)$$

Then the flat-space null hypothesis predicts

$$K = 0 \implies \mathcal{P}_{ij} = 0. \quad (4154)$$

For the hyperbolic geometric trajectory,

$$\mathcal{P}_{ij} = \ln \left[\frac{s_j \operatorname{arsinh}(b/s_j)}{s_i \operatorname{arsinh}(b/s_i)} \right]. \quad (4155)$$

The unknown multiplicative calibration ζ cancels exactly. This makes $\mathcal{P}_{ij}$ particularly useful when the physical detector coordinate is related to the theoretical local spectral scale by an unknown constant conversion factor. The ratio

$$\mathcal{C}_q = \frac{q_M^{(H)}}{q_M^{(E)}} \quad (4156)$$

also reveals a nonuniform flat-space limit. At fixed nonzero twist, $\lim_{R \rightarrow \infty} \mathcal{C}_q = 1$. Therefore,

$$\lim_{\theta \rightarrow 0} \lim_{R \rightarrow \infty} \mathcal{C}_q = 1. \quad (4157)$$

By contrast, for every fixed finite R ,

$$q_M^{(H)} \sim \frac{2\pi}{R \ln(C/|\theta|)}, \quad (4158)$$

whereas

$$q_M^{(E)} \sim \frac{2\pi}{a} |\theta|. \quad (4159)$$

Hence,

$$\mathcal{C}_q \sim \frac{a}{R|\theta| \ln(C/|\theta|)}, \quad (4160)$$

and therefore

$$\lim_{\theta \rightarrow 0} \mathcal{C}_q = \infty \quad \text{for every finite } R. \quad (4161)$$

Consequently,

$$\lim_{R \rightarrow \infty} \lim_{\theta \rightarrow 0} \mathcal{C}_q = \infty. \quad (4162)$$

Thus,

$$\lim_{\theta \rightarrow 0} \lim_{R \rightarrow \infty} \mathcal{C}_q \neq \lim_{R \rightarrow \infty} \lim_{\theta \rightarrow 0} \mathcal{C}_q. \quad (4163)$$

This noncommutativity is a direct reciprocal-space manifestation of the singular small-angle effect of negative curvature. Let the effective reciprocal resolution of the probe be δq . A fundamental moiré feature is resolvable only when $q_M \gtrsim \delta q$. For the Euclidean system, $\frac{4\pi s}{a} \gtrsim \delta q$. Thus $s \gtrsim \frac{a\delta q}{4\pi}$. The exact minimum resolvable Euclidean twist is therefore

$$\theta_{\text{res}}^{(E)} = 2 \arcsin \left(\frac{a\delta q}{4\pi} \right). \quad (4164)$$

At high reciprocal resolution,

$$\theta_{\text{res}}^{(E)} \simeq \frac{a\delta q}{2\pi}. \quad (4165)$$

For the negatively curved system, the threshold condition is

$$\frac{2\pi}{R \text{arsinh}(b/s)} \gtrsim \delta q. \quad (4166)$$

At equality,

$$\text{arsinh} \left(\frac{b}{s_{\text{res}}} \right) = \frac{2\pi}{R\delta q}. \quad (4167)$$

Therefore,

$$s_{\text{res}} = \frac{b}{\sinh \left(\frac{2\pi}{R\delta q} \right)}. \quad (4168)$$

Provided the right-hand side does not exceed unity, the exact threshold is

$$\theta_{\text{res}}^{(H)} = 2 \arcsin \left[\frac{\sinh(a/2R)}{\sinh \left(\frac{2\pi}{R\delta q} \right)} \right]. \quad (4169)$$

For $\frac{2\pi}{R\delta q} \gg 1$, one has

$$\sinh \left(\frac{2\pi}{R\delta q} \right) \simeq \frac{1}{2} \exp \left(\frac{2\pi}{R\delta q} \right). \quad (4170)$$

Consequently,

$$\theta_{\text{res}}^{(H)} \sim 4 \sinh \left(\frac{a}{2R} \right) \exp \left(-\frac{2\pi}{R\delta q} \right). \quad (4171)$$

Thus the resolution threshold exhibits a qualitatively different scaling from the Euclidean law.

d. Theorem 61. Curvature-dependent twist-resolution law. For a reciprocal probe that resolves the fundamental local registry mode, the minimum twist detectable with reciprocal resolution δq obeys

$$\theta_{\text{res}}^{(E)} \sim \frac{a\delta q}{2\pi} \quad (4172)$$

in Euclidean space, whereas at fixed negative curvature and sufficiently high reciprocal resolution,

$$\theta_{\text{res}}^{(H)} \sim 4 \sinh \left(\frac{a}{2R} \right) \exp \left(-\frac{2\pi}{R\delta q} \right). \quad (4173)$$

The distinction is therefore algebraic in flat space but exponential in the hyperbolic resolution variable. This statement applies to a moiré-sensitive registry mode and should not be confused with the separation of the two bare microscopic monolayer Bragg patterns. The physical generalized diffraction spectrum may contain several representation sectors and several Laplace–Beltrami harmonic indices. Let $I_{n,\rho}$ denote the corresponding intensity and let

$$\kappa_{n,\rho} = \sqrt{\lambda_{n,\rho}} \quad (4174)$$

be the spectral reciprocal magnitude. Choose a low-frequency moiré-sensitive satellite window $\mathcal{W}_M$. Define the total weight of a mode by $w_{n,\rho} = W_\rho I_{n,\rho}$, where W_ρ is the appropriate representation multiplicity. The spectral centroid is

$$\bar{\kappa}_M = \frac{\sum_{(n,\rho) \in \mathcal{W}_M} w_{n,\rho} \kappa_{n,\rho}}{\sum_{(n,\rho) \in \mathcal{W}_M} w_{n,\rho}}. \quad (4175)$$

The corresponding variance is

$$\sigma_\kappa^2 = \frac{\sum_{(n,\rho) \in \mathcal{W}_M} w_{n,\rho} (\kappa_{n,\rho} - \bar{\kappa}_M)^2}{\sum_{(n,\rho) \in \mathcal{W}_M} w_{n,\rho}}. \quad (4176)$$

A sharply defined fundamental diffraction packet satisfies

$$\frac{\sigma_\kappa}{\bar{\kappa}_M} \ll 1. \quad (4177)$$

Only in this regime should a single experimentally extracted spectral position be directly compared with q_M^{loc} . Define

$$\zeta(K, \theta) = \frac{\bar{\kappa}_M}{q_M^{\text{loc}}(K, \theta)}. \quad (4178)$$

For a diffraction observable constructed directly from the fundamental registry field, the natural convention gives $\zeta = 1$. For a physical probe, ζ can depend on the coupling between the registry modulation and the measured scattering channel. A geometry-dominated diffraction signature requires at minimum that ζ remain smooth and non-singular across the curvature-twist crossover. A particularly robust situation occurs if

$$\frac{\partial \ln \zeta}{\partial \ln |\theta|} \ll \left| \frac{\partial \ln q_M^{\text{loc}}}{\partial \ln |\theta|} \right|. \quad (4179)$$

Then the measured peak trajectory inherits the geometric scaling exponent. For a measured centroid define

$$\mathcal{T}_E(\theta) = \frac{a\bar{\kappa}_M}{4\pi s}. \quad (4180)$$

If the calibration is chosen such that $\zeta = 1$, the flat-space null hypothesis predicts

$$K = 0 \implies \mathcal{T}_E(\theta) = 1. \quad (4181)$$

Negative curvature predicts

$$\mathcal{T}_E(\theta) = \mathcal{C}_q(K, \theta) > 1. \quad (4182)$$

If an unknown constant calibration factor is present, the absolute value of $\mathcal{T}_E$ is not universal, but its twist dependence remains a curvature diagnostic. Negative curvature cannot be detected from an arbitrarily small local patch. If $L_{\text{obs}} \ll R$, the metric is locally Euclidean up to curvature corrections of relative order

$$O\left(\frac{L_{\text{obs}}^2}{R^2}\right). \quad (4183)$$

A practical curvature-sensitive diffraction measurement therefore requires the moiré registry pattern to probe distances comparable with the curvature radius. A useful condition is $\chi \gtrsim 1$. Equivalently, $|\theta| \lesssim \frac{a}{R}$ in the weak-curvature regime. The observation region must simultaneously satisfy $L_{\text{obs}} \gtrsim \xi_M$, so that at least an order-one fraction of the fundamental moiré modulation is sampled. Thus a curvature-sensitive window requires

$$L_{\text{obs}} \gtrsim \xi_M, \quad \chi \gtrsim 1, \quad \delta q \lesssim q_M^{\text{loc}}. \quad (4184)$$

For a periodic hyperbolic finite cover, the observation radius must also remain below the injectivity radius. Therefore a controlled numerical calculation requires

$$r_{\text{inj}}^{(N)} > L_{\text{obs}} + L_{\text{int}}, \quad (4185)$$

where L_{int} denotes the effective interaction or correlation range entering the diffraction observable. A particularly safe hierarchy is

$$r_{\text{inj}}^{(N)} \gg \xi_M \gtrsim R. \quad (4186)$$

in the curvature-sensitive crossover regime. The finite-cover sequence must therefore grow sufficiently rapidly to resolve the curvature-induced moiré length before quotient topology contaminates the observable. Peak-position scaling does not require exact commensurability. A second and logically independent curvature signature arises along exact commensurate sequences. For a Euclidean square bilayer, $D_{\text{diff}}^{(E)} = \Sigma$, where Σ is the exact coincidence index. For a commensurate sequence with bounded arithmetic numerator n , $\Sigma = \Theta(|\theta|^{-2})$. For the exact hyperbolic system,

$$D_{\text{diff}}^{(H)} = q_M = [\Gamma : \Gamma_M]. \quad (4187)$$

If geometric-arithmetic locking holds along an exact small-angle sequence, $q_M = \Theta(A_M^{\text{eff}})$, then the curvature-dominated geometric area law gives

$$D_{\text{diff}}^{(H)} = \Theta(|\theta|^{-1}). \quad (4188)$$

Thus exact reciprocal representation space grows with a different small-angle exponent. Define

$$\beta_{\text{diff}} = -\frac{d \ln D_{\text{diff}}}{d \ln |\theta|}. \quad (4189)$$

Along the Euclidean bounded-arithmetic sequences, $\beta_{\text{diff}}^{(E)} = 2$. Along hyperbolic exact sequences satisfying crystallographic locking, $\beta_{\text{diff}}^{(H)} = 1$. This provides a second curvature signature that is independent of the local peak-position law. However, this result is conditional. It requires the existence of arbitrarily small exact commensurate twists at the chosen center and the geometric-arithmetic locking relation. Neither condition follows from local geometry alone.

e. Theorem 62. Curvature change of exact diffraction-space growth. For bounded-arithmetic commensurate square-lattice sequences,

$$D_{\text{diff}}^{(E)} = \Theta(|\theta|^{-2}). \quad (4190)$$

For an exact hyperbolic commensurate sequence satisfying geometric-arithmetic locking,

$$D_{\text{diff}}^{(H)} = \Theta(|\theta|^{-1}). \quad (4191)$$

Hence negative curvature changes the asymptotic growth exponent of the exact structural reciprocal representation from two to one. This theorem concerns the dimension of the exact induced diffraction representation and should not be identified with the number of peaks above a finite experimental intensity threshold. Peak positions provide the most geometry-protected signature. Curvature can also redistribute diffraction weight among generalized representation sectors. Recall the normalized sector weights

$$\mathcal{S}_\varpi, \quad \sum_\varpi \mathcal{S}_\varpi = 1. \quad (4192)$$

The total nontrivial satellite fraction is $\mathcal{W}_{\text{sat}} = 1 - \mathcal{S}_1$. The representation participation number is

$$N_{\text{rep}} = \frac{1}{\sum_\varpi \mathcal{S}_\varpi^2}. \quad (4193)$$

The representation entropy is

$$\mathcal{S}_{\text{rep}}^{\text{ent}} = - \sum_\varpi \mathcal{S}_\varpi \ln \mathcal{S}_\varpi. \quad (4194)$$

These quantities measure structural reconstruction and reciprocal complexity. However, $\mathcal{W}_{\text{sat}}$, N_{rep} , $\mathcal{S}_{\text{rep}}^{\text{ent}}$ are not universal functions of curvature alone. They also depend on the microscopic scattering channel, structural relaxation, basis form factors, and the coupling of the probe to the moiré registry field. They should therefore be used as secondary confirmation rather than as the primary curvature diagnostic. A useful way to remove additive pure-curvature and pure-twist effects is to define the amplitude-level mixed response

$$\begin{aligned} \delta_{K\theta} \hat{f}(\varpi) &= \hat{f}(K, \theta; \varpi) - \hat{f}(K, 0; \varpi) \\ &\quad - \hat{f}(0, \theta; \varpi) + \hat{f}(0, 0; \varpi). \end{aligned} \quad (4195)$$

The corresponding norm is

$$\mathcal{D}_{K\theta}(\varpi) = \left\| \delta_{K\theta} \hat{f}(\varpi) \right\|_F^2. \quad (4196)$$

A nonzero value identifies a nonadditive curvature-twist reconstruction of the diffraction amplitude. Nevertheless, this quantity remains more model dependent than the geometric peak-position trajectory. The diffraction signatures can now be ordered by theoretical robustness. The most direct geometry-only observable is the fundamental reciprocal trajectory, $q_M(\theta)$. The strongest calibration-independent observable is its logarithmic slope,

$$\frac{d \ln q_M}{d \ln s}. \quad (4197)$$

A complementary multi-angle observable is the exact hyperbolic collapse

$$s \sinh \left(\frac{2\pi}{Rq_M} \right) = \text{constant}. \quad (4198)$$

Along suitable exact commensurate sequences, the induced-representation dimension provides the crystallographic observable $D_{\text{diff}}(\theta)$. Finally, representation-space intensity redistribution provides $\mathcal{W}_{\text{sat}}$, N_{rep} , $\mathcal{S}_{\text{rep}}^{\text{ent}}$. The first three classes depend primarily on geometry. The final class additionally depends on microscopic scattering physics. The complete diffraction discrimination problem can therefore be expressed as two competing hypotheses. The Euclidean null hypothesis predicts

$$q_M = \frac{4\pi s}{a}, \quad \frac{d \ln q_M}{d \ln s} = 1, \quad \frac{q_M(\theta_i)/s_i}{q_M(\theta_j)/s_j} = 1. \quad (4199)$$

The negative-curvature hypothesis predicts

$$\begin{aligned} q_M &= \frac{2\pi}{R \operatorname{arsinh} \left[\frac{\sinh(a/2R)}{s} \right]}, \\ 0 &< \frac{d \ln q_M}{d \ln s} < 1, \end{aligned} \quad (4200)$$

and the exact multi-angle relation

$$s_i \sinh \left(\frac{2\pi}{Rq_i} \right) = s_j \sinh \left(\frac{2\pi}{Rq_j} \right). \quad (4201)$$

The distinction therefore survives even if all absolute diffraction intensities are discarded. A numerical test of the curvature signature should use a sequence of periodic hyperbolic covers with increasing injectivity radius. For each curvature and twist, the generalized diffraction spectrum is computed and the fundamental moiré-sensitive spectral packet is identified. The centroid $\bar{\kappa}_M$ and width σ_κ are then extracted. The peak should satisfy

$$\frac{\sigma_\kappa}{\bar{\kappa}_M} \ll 1 \quad (4202)$$

and converge as $r_{\text{inj}}^{(N)} \rightarrow \infty$. The logarithmic trajectory

$$\frac{d \ln \bar{\kappa}_M}{d \ln s} \quad (4203)$$

is then compared against the Euclidean value 1 and the hyperbolic crossover function. The hopping cutoff, representation truncation, spectral broadening, and finite-cover size must be converged independently. A curvature signature is accepted only when the deviation from the Euclidean trajectory exceeds the complete numerical uncertainty. Let the uncertainty in the extracted reciprocal centroid be σ_q . The curvature-induced shift is

$$\Delta q_K = q_M^{(H)} - q_M^{(E)}. \quad (4204)$$

A statistically resolvable geometry-induced deviation requires $\Delta q_K \gg \sigma_q$. In the weak-curvature regime,

$$\Delta q_K \simeq q_M^{(E)} \frac{a^2}{24R^2} \cot^2 \left(\frac{\theta}{2} \right). \quad (4205)$$

Thus decreasing the twist initially amplifies the relative curvature signal. However, once

$$\frac{a}{R|\theta|} = O(1), \quad (4206)$$

the perturbative formula must be replaced by the exact crossover law. A crucial interpretational distinction must be maintained. The ordinary Euclidean result

$$\Delta q_{\text{atomic}} = 2G \left| \sin \left(\frac{\theta}{2} \right) \right| \quad (4207)$$

describes the separation between corresponding reciprocal vectors of two rigid Euclidean monolayers. In a hyperbolic crystal there is no globally additive two-dimensional reciprocal lattice on which this formula can simply be modified by replacing G . The curvature signature derived in the present subsection instead concerns the moiré-sensitive local or representation-space spectral scale. Therefore, $q_M^{\text{loc}} \neq$ a naive hyperbolic replacement of a bare Bragg vector. A physical experiment must couple to registry-sensitive structural, electronic, optical, or synthetic-lattice spectral response in order to observe the predicted crossover directly.

f. Theorem 63. Geometry-protected diffraction hierarchy. Within the geometry-only protocol, negative curvature produces the following hierarchy of reciprocal-space signatures. First, the fundamental local moiré reciprocal scale satisfies $q_M^{(H)} > q_M^{(E)}$. Second, its normalized logarithmic slope satisfies

$$0 < \frac{d \ln q_M^{(H)}}{d \ln s} < 1, \quad (4208)$$

whereas the Euclidean value is exactly unity. Third, the complete hyperbolic peak trajectory satisfies the multi-angle collapse

$$s \sinh \left(\frac{2\pi}{Rq_M^{(H)}} \right) = \sinh \left(\frac{a}{2R} \right). \quad (4209)$$

Fourth, along exact commensurate sequences satisfying geometric–arithmetic locking, the exact diffraction–representation dimension grows as

$$D_{\text{diff}}^{(H)} = \Theta(|\theta|^{-1}) \quad (4210)$$

instead of the Euclidean bounded-arithmetic law

$$D_{\text{diff}}^{(E)} = \Theta(|\theta|^{-2}). \quad (4211)$$

The first three signatures require only local geometric information. The fourth additionally requires exact crystallographic commensurability. The diffraction results can now be expressed through a universal double-scaling variable. For weak microscopic curvature, define

$$u = \frac{R|\theta|}{a}. \quad (4212)$$

Then

$$\frac{\xi_M^{(H)}}{R} \longrightarrow \text{arsinh} \left(\frac{1}{u} \right), \quad (4213)$$

and therefore

$$Rq_M^{(H)} \longrightarrow \frac{2\pi}{\text{arsinh}(1/u)}. \quad (4214)$$

Define the universal reciprocal crossover function

$$\mathcal{Q}(u) = \frac{1}{\text{arsinh}(1/u)}. \quad (4215)$$

Then

$$Rq_M^{(H)} = 2\pi \mathcal{Q}(u). \quad (4216)$$

For $u \gg 1$, one has

$$\text{arsinh} \left(\frac{1}{u} \right) = \frac{1}{u} - \frac{1}{6u^3} + O(u^{-5}), \quad (4217)$$

and therefore

$$\mathcal{Q}(u) = u + \frac{1}{6u} + O(u^{-3}). \quad (4218)$$

This recovers the Euclidean linear regime. For $u \ll 1$, one has

$$\operatorname{arsinh}\left(\frac{1}{u}\right) = \ln\left(\frac{2}{u}\right) + O(u^2), \quad (4219)$$

and hence

$$\mathcal{Q}(u) \sim \frac{1}{\ln(2/u)}. \quad (4220)$$

Thus the entire Euclidean-to-hyperbolic diffraction crossover belongs to a single universal scaling function. The principal reciprocal-space effect of negative curvature is therefore not a simple rigid displacement of an otherwise Euclidean diffraction pattern. Curvature changes the relation between twist and the spatial scale over which the two layers lose registry. Because the hyperbolic Jacobi field grows exponentially with distance, a fixed relative twist produces registry mismatch more rapidly than in flat space. The real-space moiré domain is consequently shortened, and its inverse spectral scale is shifted outward. In the weak-curvature regime this effect appears as a perturbative correction proportional to

$$\frac{a^2}{R^2} \cot^2\left(\frac{\theta}{2}\right). \quad (4221)$$

As the twist is reduced, this correction grows until perturbation theory fails near $|\theta| \sim \frac{a}{R}$. Below this scale the diffraction response enters a genuinely curvature-dominated regime in which the fundamental inverse registry length vanishes only as

$$q_M^{(H)} \sim \frac{1}{R \ln(1/|\theta|)}. \quad (4222)$$

The corresponding logarithmic diffraction exponent flows continuously from the Euclidean value one to zero. This provides a direct reciprocal-space signature that is substantially stronger than a curvature-dependent intensity change. A change of intensity can arise from many microscopic mechanisms. By contrast, the nonlinear trajectory of the fundamental moiré spectral scale, its logarithmic slope, and the exact multi-angle inversion identity are geometric consequences of the hyperbolic registry law. The strongest experimentally or numerically accessible null test is therefore not an absolute peak intensity but the functional dependence of the moiré-sensitive reciprocal scale on twist. Flat space predicts $q_M/s = \text{const.}$, whereas negative curvature predicts a systematic violation of this relation and, ultimately, an inverse-logarithmic small-angle trajectory. This distinction supplies an observable bridge between the geometric moiré theory and the spectral reconstruction developed in the following sections.

IV. CURVATURE-CONTROLLED SPECTRAL RECONSTRUCTION

A. Benchmark Band Structures of Euclidean Twisted Square Bilayers

The geometric, microscopic, symmetry-resolved, and numerical constructions developed above now allow us to address the first genuinely spectral question: how does a twisted bilayer spectrum reorganize when a Euclidean moiré system is continuously deformed into a negatively curved one? We deliberately separate this problem from the subsequent magic-band analysis. Spectral reconstruction is more general than spectral flattening. A spectrum may undergo avoided crossings, gap opening, level repulsion, density-of-states redistribution, representation-sector reorganization, or strong wave-function hybridization without yet becoming flat. We therefore first construct a quantitative theory of spectral reconstruction itself. The parameter family is written as

$$H = H\left(K, \theta, \frac{w}{t}\right), \quad (4223)$$

where $K \leq 0$ is the Gaussian curvature, θ is the relative twist angle, t fixes the intralayer kinetic-energy scale, and w sets the characteristic interlayer tunneling scale. It is convenient to introduce $\omega = \frac{w}{t}$, and, for negative curvature,

$$\kappa = -K = \frac{1}{R^2} \geq 0. \quad (4224)$$

Thus

$$K = 0 \iff \kappa = 0, \quad (4225)$$

whereas

$$K < 0 \iff \kappa > 0. \quad (4226)$$

All bare microscopic parameters other than the explicitly varied (κ, θ, ω) are held fixed unless a normalization control is stated explicitly. In particular, the geometry-only comparison keeps t , w , h , and the hopping decay lengths fixed. The spectral problem will be analyzed at three complementary levels. First, exactly commensurate Euclidean twisted square bilayers provide a complete benchmark in which the ordinary moiré Brillouin zone exists and all bands can be plotted directly. Second, exactly commensurate hyperbolic bilayers are resolved through finite-dimensional representation-space Hamiltonians $H[\rho]$. Third, periodic finite covers provide a common numerical framework in which the complete density of states and representation-resolved spectra can be

compared without introducing an extensive hyperbolic boundary. The Euclidean calculations serve two distinct purposes. They first verify the crystallographic constructions through explicit ten-site, thirty-four-site, and sub-hundred-site spectra. More importantly, they provide a null benchmark against which curvature-induced reconstruction can later be measured. We therefore keep this subsection deliberately focused on exact band counting, folding, hybridization, and benchmark spectral diagnostics. For one orbital per microscopic square-lattice site and nearest-neighbor intralayer hopping t , the monolayer Hamiltonian is

$$H_0 = -t \sum_{\langle ij \rangle} (c_i^\dagger c_j + c_j^\dagger c_i). \quad (4227)$$

The primitive Bloch dispersion is

$$\varepsilon_0(\mathbf{q}) = -2t [\cos(q_x a) + \cos(q_y a)]. \quad (4228)$$

Its extrema are

$$\varepsilon_{\min} = -4t, \quad \varepsilon_{\max} = 4t, \quad (4229)$$

and hence the primitive bandwidth is $W_0 = 8|t|$. The group velocity is

$$\mathbf{v}_0(\mathbf{q}) = \frac{1}{\hbar} \nabla_{\mathbf{q}} \varepsilon_0(\mathbf{q}), \quad (4230)$$

or explicitly,

$$\mathbf{v}_0(\mathbf{q}) = \frac{2ta}{\hbar} \begin{pmatrix} \sin(q_x a) \\ \sin(q_y a) \end{pmatrix}. \quad (4231)$$

These quantities provide the reference kinetic scales against which all subsequent spectral reconstruction is measured. For a commensurate square-lattice twist parameterized by coprime integers (m, n) ,

$$\tan\left(\frac{\theta}{2}\right) = \frac{n}{m}, \quad (4232)$$

the exact coincidence index is

$$\Sigma = \frac{m^2 + n^2}{d}, \quad (4233)$$

where

$$d = \begin{cases} 1, & m, n \text{ have opposite parity,} \\ 2, & m, n \text{ are both odd.} \end{cases} \quad (4234)$$

The exact bilayer supercell contains $N_{\text{sc}} = 2\Sigma$ microscopic orbitals. The exact moire Brillouin-zone area is

$$A_{\text{mBZ}} = \frac{1}{\Sigma} \left(\frac{2\pi}{a} \right)^2. \quad (4235)$$

The primitive reciprocal scale of the square coincidence lattice is

$$G_M = \frac{2\pi}{a\sqrt{\Sigma}}. \quad (4236)$$

Thus decreasing the Brillouin-zone area by a factor Σ is exactly compensated by the appearance of Σ folded branches from each layer. Let the Σ microscopic sites of layer ℓ inside one exact moire cell be $\mathbf{r}_{\ell s}$, $s = 1, \dots, \Sigma$. Let $\mathbf{R}_M \in \Lambda_M$ run over exact moire translations. The intralayer Bloch block is

$$[h_\ell(\mathbf{k})]_{ss'} = \sum_{\mathbf{R}_M} t_{\parallel} (|\mathbf{r}_{\ell s} - \mathbf{r}_{\ell s'} - \mathbf{R}_M|) e^{i\mathbf{k} \cdot \mathbf{R}_M}. \quad (4237)$$

The interlayer block is

$$[T(\mathbf{k})]_{ss'} = \sum_{\mathbf{R}_M} t_{\perp} \left(\sqrt{h^2 + |\mathbf{r}_{1s} - \mathbf{r}_{2s'} - \mathbf{R}_M|^2} \right) e^{i\mathbf{k} \cdot \mathbf{R}_M}. \quad (4238)$$

The exact Euclidean bilayer Bloch Hamiltonian is therefore

$$H_M(\mathbf{k}) = \begin{pmatrix} h_1(\mathbf{k}) & T(\mathbf{k}) \\ T^\dagger(\mathbf{k}) & h_2(\mathbf{k}) \end{pmatrix}. \quad (4239)$$

Its matrix dimension is exactly

$$\dim H_M = 2\Sigma = N_{\text{sc}}. \quad (4240)$$

Thus real-space atom counting and reciprocal-space band counting coincide exactly. Let $\{\mathbf{Q}_\alpha^{(1)}\}_{\alpha=1}^{\Sigma}$ be representatives of Λ_M^*/Λ_1^* , and let $\{\mathbf{Q}_\beta^{(2)}\}_{\beta=1}^{\Sigma}$ be representatives of Λ_M^*/Λ_2^* . The unhybridized folded energies are

$$E_{1\alpha}^{(0)}(\mathbf{k}) = \varepsilon_0(\mathbf{k} + \mathbf{Q}_\alpha^{(1)}), \quad (4241)$$

and

$$E_{2\beta}^{(0)}(\mathbf{k}) = \varepsilon_0 \left[R_{-\theta}(\mathbf{k} + \mathbf{Q}_\beta^{(2)}) \right]. \quad (4242)$$

The exact folded Hamiltonian is

$$H_{\text{fold}}(\mathbf{k}) = \begin{pmatrix} E_1^{\text{fold}}(\mathbf{k}) & \mathcal{T}(\mathbf{k}) \\ \mathcal{T}^\dagger(\mathbf{k}) & E_2^{\text{fold}}(\mathbf{k}) \end{pmatrix}, \quad (4243)$$

where the diagonal matrices contain the folded energies and $\mathcal{T}$ is the interlayer matrix in the folded basis. The real-space and folded Hamiltonians are related by an exact unitary transformation,

$$H_{\text{fold}}(\mathbf{k}) = \mathcal{U}^\dagger(\mathbf{k}) H_M(\mathbf{k}) \mathcal{U}(\mathbf{k}). \quad (4244)$$

Consequently,

$$\text{Spec } H_{\text{fold}}(\mathbf{k}) = \text{Spec } H_M(\mathbf{k}). \quad (4245)$$

For the square coincidence lattice, the primitive reciprocal vectors are orthogonal and have equal magnitude.

We define $\Gamma = \mathbf{0}$, $\text{diag } X = \frac{\mathbf{G}_1}{2}$, $Y = \frac{\mathbf{G}_2}{2}$, and

$$M = \frac{\mathbf{G}_1 + \mathbf{G}_2}{2}. \quad (4246)$$

The standard benchmark band path is therefore

$$\Gamma \longrightarrow X \longrightarrow M \longrightarrow \Gamma. \quad (4247)$$

Every Euclidean benchmark spectrum below is evaluated on this same symmetry-normalized path. For the ten-site structure, $\Sigma = 5$. Using the $(m, n) = (2, 1)$ representation,

$$\mathbf{T}_1 = a \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad \mathbf{T}_2 = a \begin{pmatrix} -1 \\ 2 \end{pmatrix}. \quad (4248)$$

The reciprocal basis is

$$\mathbf{G}_1 = \frac{2\pi}{5a} \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad \mathbf{G}_2 = \frac{2\pi}{5a} \begin{pmatrix} -1 \\ 2 \end{pmatrix}. \quad (4249)$$

Hence,

$$G_M^{(10)} = \frac{2\pi}{a\sqrt{5}}, \quad (4250)$$

and $\dim H_M^{(10)} = 10$. Each layer contributes five folded branches. The displayed twist $\theta = 53.130102354\dots^\circ$ is crystallographically equivalent to the reduced angle $\hat{\theta} = 36.869897646\dots^\circ$. Both descriptions therefore produce the same physical benchmark spectrum after the appropriate square-lattice point-group transformation. For the thirty-four-site structure, $\Sigma = 17$, with reduced representative $(m, n) = (4, 1)$. The direct basis is

$$\mathbf{T}_1 = a \begin{pmatrix} 4 \\ 1 \end{pmatrix}, \quad \mathbf{T}_2 = a \begin{pmatrix} -1 \\ 4 \end{pmatrix}. \quad (4251)$$

The reciprocal basis is

Σ	N_{sc}	(m, n)	$\hat{\theta}$	$G_M a / (2\pi)$
41	82	(9, 1)	12.680383492°	41 ^{-1/2}
25	50	(7, 1)	16.260204708°	25 ^{-1/2}
37	74	(6, 1)	18.924644416°	37 ^{-1/2}
13	26	(5, 1)	22.619864948°	13 ^{-1/2}
17	34	(4, 1)	28.072486936°	17 ^{-1/2}
5	10	(3, 1)	36.869897646°	5 ^{-1/2}
29	58	(5, 2)	43.602818973°	29 ^{-1/2}

TABLE I. Exact Euclidean square-bilayer benchmark structures with fewer than one hundred bilayer orbitals. The listed angles lie in the irreducible interval $0 \leq \hat{\theta} \leq \pi/4$.

$$\mathbf{G}_1 = \frac{2\pi}{17a} \begin{pmatrix} 4 \\ 1 \end{pmatrix}, \quad \mathbf{G}_2 = \frac{2\pi}{17a} \begin{pmatrix} -1 \\ 4 \end{pmatrix}. \quad (4252)$$

Therefore,

$$G_M^{(34)} = \frac{2\pi}{a\sqrt{17}}, \quad (4253)$$

and $\dim H_M^{(34)} = 34$. The corresponding twist is $\hat{\theta} = 28.072486936\dots^\circ$. The nontrivial crystallographically inequivalent square-bilayer structures with fewer than one hundred bilayer orbitals are summarized by

This finite family provides a direct numerical analysis of the exact commensurability theory before negative curvature is introduced. Set $w = 0$. Then $\mathcal{T}(\mathbf{k}) = 0$, and the spectrum is exactly the union of folded monolayer energies. For every folded branch,

$$E_\alpha^{(0)}(\mathbf{k}) = \varepsilon_0(\mathbf{k} + \mathbf{Q}_\alpha). \quad (4254)$$

Its derivative is

$$\nabla_{\mathbf{k}} E_\alpha^{(0)} = \nabla_{\mathbf{q}} \varepsilon_0(\mathbf{q}) \Big|_{\mathbf{q}=\mathbf{k}+\mathbf{Q}_\alpha}. \quad (4255)$$

Thus folding alone does not suppress the primitive velocity. Likewise, the density of states is exactly preserved, $\rho_{\text{fold}}(E) = \rho_0(E)$. The $w = 0$ spectra therefore provide a strict numerical null test: any new gap, velocity suppression, DOS enhancement, or bandwidth reorganization appearing at $w \neq 0$ must arise from hybridization rather than from the mere increase in the number of plotted bands. Let one folded layer block have dimension Σ , and write the co-rotating Euclidean Hamiltonian as

$$H = \begin{pmatrix} A & T \\ T^\dagger & A \end{pmatrix}, \quad (4256)$$

where $A = E^{\text{fold}}(\mathbf{k})$. The trace is

$$\text{Tr } H = 2 \text{Tr } A. \quad (4257)$$

Hence interlayer tunneling does not change the spectral centroid at fixed $\mathbf{k}$. Squaring the Hamiltonian gives

$$H^2 = \begin{pmatrix} A^2 + TT^\dagger & AT + TA \\ T^\dagger A + AT^\dagger & A^2 + T^\dagger T \end{pmatrix}. \quad (4258)$$

Therefore,

$$\text{Tr } H^2 = 2 \text{Tr } A^2 + 2 \text{Tr } TT^\dagger. \quad (4259)$$

Define the normalized second spectral moment by

$$\mu_2 = \frac{1}{2\Sigma} \text{Tr } H^2. \quad (4260)$$

Then

$$\mu_2 = \frac{1}{\Sigma} \text{Tr } A^2 + \frac{1}{\Sigma} \text{Tr } TT^\dagger. \quad (4261)$$

Thus the exact increase of the raw second moment generated by hybridization is

$$\Delta\mu_2 = \frac{1}{\Sigma} \text{Tr } TT^\dagger \geq 0. \quad (4262)$$

If $T = w\mathcal{T}$, then

$$\Delta\mu_2 = w^2 \frac{1}{\Sigma} \text{Tr } \mathcal{T}\mathcal{T}^\dagger. \quad (4263)$$

This identity provides an exact spectral analysis of the numerical hybridization strength.

a. Theorem 64. Exact Euclidean benchmark decomposition. For an exactly commensurate square bilayer with coincidence index Σ , the full Bloch Hamiltonian has dimension 2Σ . At $w = 0$, its spectrum is an exact folding of the primitive monolayer spectrum and therefore preserves both the unfolded velocity field and the monolayer density of states. At $w \neq 0$, the first spectral moment is unchanged by the interlayer block, whereas the second raw moment increases exactly by

$$\Delta\mu_2 = \frac{1}{\Sigma} \text{Tr } TT^\dagger. \quad (4264)$$

Consequently, the Euclidean benchmark separates crystallographic band multiplication from genuine dynamical spectral reconstruction without any ambiguity. Consider an isolated unhybridized level $|a\rangle$ with energy ε_a . Away from degeneracy, the leading hybridization correction is

$$\delta E_a^{(2)} = \sum_{b \neq a} \frac{|V_{ab}|^2}{\varepsilon_a - \varepsilon_b} + O(w^4), \quad (4265)$$

where

$$V = \begin{pmatrix} 0 & T \\ T^\dagger & 0 \end{pmatrix}. \quad (4266)$$

At an isolated two-state resonance,

$$H_{2\text{lev}} = \begin{pmatrix} \varepsilon_1 & V \\ V^* & \varepsilon_2 \end{pmatrix}. \quad (4267)$$

Defining $\Delta = \varepsilon_1 - \varepsilon_2$, the exact eigenvalues are

$$E_{\pm} = \frac{\varepsilon_1 + \varepsilon_2}{2} \pm \sqrt{\frac{\Delta^2}{4} + |V|^2}. \quad (4268)$$

At resonance, $\Delta = 0$, the gap is $\Delta_{\text{hyb}} = 2|V|$. The ten-site, thirty-four-site, and remaining sub-hundred-site structures will therefore be benchmarked by comparing the $w = 0$ folded spectra with the $w \neq 0$ spectra using the same mBZ path and the same energy normalization. No spectral narrowing observed in the $w = 0$ plots will be interpreted as a physical reconstruction. We now turn to the central problem of this section. Negative curvature changes both the real-space distance matrix and the representation space in which the exact spectral problem is formulated. We therefore need a spectral comparison that remains meaningful even when ordinary crystal momentum no longer exists. The key distinction is between microscopic spectral continuity and crystallographic reorganization. On a matched finite Hilbert space, continuously varying curvature produces a continuous finite Hermitian matrix and hence a continuous spectrum. By contrast, the exact commensurate subgroup Γ_M can change arithmetically as (K, θ) is varied. Its index, quotient genus, and representation content can therefore change discretely. Such a change is a discontinuity of the exact crystallographic description, not a discontinuity of the local microscopic Hamiltonian. We now make this distinction quantitative. Write the physical Hamiltonian in the co-rotating representation as

$$H(\kappa, \theta, \omega) = t \begin{pmatrix} \mathcal{H}_0(\kappa) & \omega\mathcal{T}(\kappa, \theta) \\ \omega\mathcal{T}^\dagger(\kappa, \theta) & \mathcal{H}_0(\kappa) \end{pmatrix}. \quad (4269)$$

The complete dimensionless spectral problem is therefore

$$\mathcal{H}(\kappa, \theta, \omega) |u_n\rangle = e_n(\kappa, \theta, \omega) |u_n\rangle, \quad (4270)$$

with $E_n = te_n$. The remaining dimensionless microscopic control parameters, such as

$$\frac{h}{a}, \quad \frac{\lambda_{\parallel}}{a}, \quad \frac{\lambda_{\perp}}{a}, \quad (4271)$$

are held fixed during the primary (κ, θ, ω) sweep. For an exactly commensurate hyperbolic bilayer, the exact common translation group is

$$\Gamma_M = \Gamma \cap g_\theta \Gamma g_\theta^{-1}. \quad (4272)$$

For a unitary representation

$$\rho : \Gamma_M \longrightarrow U(d_\rho), \quad (4273)$$

the exact generalized Bloch Hamiltonian is

$$H[\rho; \kappa, \theta, \omega] = \sum_{\delta \in \Gamma_M} h(\delta; \kappa, \theta, \omega) \otimes \rho(\delta). \quad (4274)$$

Its dimension is

$$\dim H[\rho] = 2n_b q_M d_\rho. \quad (4275)$$

The sector-resolved spectrum is

$$H[\rho] |u_{n,\rho}\rangle = E_{n,\rho} |u_{n,\rho}\rangle. \quad (4276)$$

For $d_\rho = 1$, this gives the Abelian higher-genus bands. For $d_\rho > 1$, the spectral label is intrinsically non-Abelian. Choose a normal finite-index subgroup $\Gamma_N \triangleleft \Gamma_M$, and define $Q_N = \Gamma_M / \Gamma_N$. The finite-cover Hamiltonian is

$$H_N = \sum_{\delta \in \Gamma_M} h(\delta) \otimes \mathcal{R}_N(\delta), \quad (4277)$$

and decomposes exactly as

$$H_N \simeq \bigoplus_{\rho \in \hat{Q}_N} [I_{d_\rho} \otimes H[\rho]]. \quad (4278)$$

If $M_\rho = n_b q_M d_\rho$, then

$$\dim H[\rho] = 2M_\rho. \quad (4279)$$

The normalized finite-cover spectral measure is

$$d\nu_N(E) = \frac{1}{2n_b q_M |Q_N|} \sum_{\rho \in \hat{Q}_N} d_\rho \sum_{n=1}^{2M_\rho} \delta(E - E_{n,\rho}) dE. \quad (4280)$$

This is the principal bulk spectral object used below. We next derive the curvature vertex entering the Hamiltonian explicitly. Consider two points with geodesic polar coordinates (r_i, ϕ_i) , (r_j, ϕ_j) , and define their angular separation $\varphi_{ij} = \phi_i - \phi_j$. For curvature $K = -\kappa$, the hyperbolic law of cosines gives

$$\cosh(\sqrt{\kappa} d_{\kappa,ij}) = \cosh(\sqrt{\kappa} r_i) \cosh(\sqrt{\kappa} r_j) - \sinh(\sqrt{\kappa} r_i) \sinh(\sqrt{\kappa} r_j) \cos \varphi_{ij}. \quad (4281)$$

At $\kappa = 0$, the Euclidean chord distance is

$$d_{0,ij}^2 = r_i^2 + r_j^2 - 2r_i r_j \cos \varphi_{ij}. \quad (4282)$$

Write the curvature expansion as

$$d_{\kappa,ij}^2 = d_{0,ij}^2 + \kappa A_{ij} + O(\kappa^2). \quad (4283)$$

The left-hand side of Eq. (4281) becomes

$$\begin{aligned} \cosh(\sqrt{\kappa} d_{\kappa,ij}) &= 1 + \frac{\kappa d_{0,ij}^2}{2} \\ &+ \kappa^2 \left[\frac{A_{ij}}{2} + \frac{d_{0,ij}^4}{24} \right] + O(\kappa^3). \end{aligned} \quad (4284)$$

The right-hand side expands to

$$\begin{aligned} &1 + \frac{\kappa}{2} (r_i^2 + r_j^2 - 2r_i r_j \cos \varphi_{ij}) \\ &+ \kappa^2 \left[\frac{r_i^4 + r_j^4}{24} + \frac{r_i^2 r_j^2}{4} - \frac{r_i^3 r_j + r_i r_j^3}{6} \cos \varphi_{ij} \right] + O(\kappa^3). \end{aligned} \quad (4285)$$

Equating the coefficients of κ^2 gives

$$A_{ij} = \frac{r_i^2 r_j^2 \sin^2 \varphi_{ij}}{3}. \quad (4286)$$

Therefore the universal weak-curvature distance law is

$$d_{\kappa,ij}^2 = d_{0,ij}^2 + \frac{\kappa r_i^2 r_j^2 \sin^2 \varphi_{ij}}{3} + O(\kappa^2). \quad (4287)$$

Equivalently, for $d_{0,ij} \neq 0$,

$$d_{\kappa,ij} = d_{0,ij} + \frac{\kappa r_i^2 r_j^2 \sin^2 \varphi_{ij}}{6d_{0,ij}} + O(\kappa^2). \quad (4288)$$

This formula gives an explicit pairwise curvature vertex for every distance-dependent hopping matrix element. For a twisted bilayer, define

$$\varphi_{ij}(\theta) = \phi_i - \phi_j - \theta. \quad (4289)$$

The Euclidean in-plane separation is

$$d_{ij}^{(E)}(\theta) = \sqrt{r_i^2 + r_j^2 - 2r_i r_j \cos \varphi_{ij}(\theta)}. \quad (4290)$$

The physical Euclidean interlayer distance is

$$D_{ij}^{(E)}(\theta) = \sqrt{h^2 + [d_{ij}^{(E)}(\theta)]^2}. \quad (4291)$$

Using Eq. (4287), the corresponding hyperbolic interlayer distance satisfies

$$D_{ij}^{(H)}(\kappa, \theta) = D_{ij}^{(E)}(\theta) + \kappa \frac{r_i^2 r_j^2 \sin^2 \varphi_{ij}(\theta)}{6D_{ij}^{(E)}(\theta)} + O(\kappa^2). \quad (4292)$$

For a differentiable radial tunneling kernel $T(D)$,

$$T_{ij}^{(H)} = T \left[D_{ij}^{(H)} \right]. \quad (4293)$$

Taylor expansion gives

$$T_{ij}^{(H)}(\kappa, \theta) = T \left[D_{ij}^{(E)}(\theta) \right] + \kappa T' \left[D_{ij}^{(E)}(\theta) \right] \frac{r_i^2 r_j^2 \sin^2 \varphi_{ij}(\theta)}{6D_{ij}^{(E)}(\theta)} + O(\kappa^2). \quad (4294)$$

Thus the exact first curvature derivative at flat space is

$$\left. \frac{\partial T_{ij}}{\partial \kappa} \right|_{\kappa=0} = T' \left[D_{ij}^{(E)}(\theta) \right] \frac{r_i^2 r_j^2 \sin^2 \varphi_{ij}(\theta)}{6D_{ij}^{(E)}(\theta)}. \quad (4295)$$

For the exponential tunneling law

$$T(D) = w \exp \left[-\frac{D-h}{\lambda_{\perp}} \right], \quad (4296)$$

one has

$$T'(D) = -\frac{1}{\lambda_{\perp}} T(D). \quad (4297)$$

Therefore,

$$T_{ij}^{(H)} = T_{ij}^{(E)} \left[1 - \kappa \frac{r_i^2 r_j^2 \sin^2 \varphi_{ij}(\theta)}{6\lambda_{\perp} D_{ij}^{(E)}(\theta)} + O(\kappa^2) \right]. \quad (4298)$$

Thus negative curvature does not act as a uniform renormalization of w . It changes different tunneling matrix elements by different amounts according to their real-space geometry. For the same reference label in the two layers, $r_i = r_j = r$, and $\varphi_{ii}(\theta) = -\theta$. The Euclidean lateral mismatch is

$$d_{\text{same}}^{(E)} = 2r \left| \sin \left(\frac{\theta}{2} \right) \right|. \quad (4299)$$

Hence

$$D_{\text{same}}^{(E)} = \sqrt{h^2 + 4r^2 \sin^2 \left(\frac{\theta}{2} \right)}. \quad (4300)$$

The first curvature correction is

$$T_{\text{same}}^{(H)} = T \left(D_{\text{same}}^{(E)} \right) + \kappa T' \left(D_{\text{same}}^{(E)} \right) \frac{r^4 \sin^2 \theta}{6D_{\text{same}}^{(E)}} + O(\kappa^2). \quad (4301)$$

At simultaneously weak curvature and weak twist,

$$D_{\text{same}}^{(E)} = h + O(\theta^2), \quad (4302)$$

so that

$$T_{\text{same}}^{(H)} - T_{\text{same}}^{(E)} = \kappa \frac{T'(h)}{6h} r^4 \theta^2 + O(\kappa \theta^4) + O(\kappa^2 \theta^2). \quad (4303)$$

For exponential tunneling,

$$T_{\text{same}}^{(H)} - T_{\text{same}}^{(E)} = -\frac{w}{6h\lambda_{\perp}} \kappa r^4 \theta^2 + O(\kappa \theta^4) + O(\kappa^2 \theta^2). \quad (4304)$$

This explicitly recovers the curvature-twist microscopic response derived earlier and now identifies the matrix element that enters the spectral problem.

b. Theorem 65. Universal weak-curvature hopping vertex. For a distance-dependent bilayer coupling on a matched local geometry, the first negative-curvature correction to an interlayer matrix element is

$$\left. \frac{\partial T_{ij}}{\partial \kappa} \right|_{\kappa=0} = T' \left[D_{ij}^{(E)} \right] \frac{r_i^2 r_j^2 \sin^2 \varphi_{ij}}{6D_{ij}^{(E)}}. \quad (4305)$$

The correction depends on the geometric area factor $r_i r_j |\sin \varphi_{ij}|$, rather than only on the original Euclidean distance. Consequently, curvature reconstructs the spatial organization of the tunneling matrix even when all bare tunneling parameters are held fixed. Write the group kernel as

$$h(\delta; \kappa, \theta, \omega) = h^{(0)}(\delta; \theta, \omega) + \kappa v_{\kappa}(\delta; \theta, \omega) + O(\kappa^2). \quad (4306)$$

The representation-space Hamiltonian becomes

$$H[\rho; \kappa] = H^{(0)}[\rho] + \kappa V_{\kappa}[\rho] + O(\kappa^2), \quad (4307)$$

where

$$V_{\kappa}[\rho] = \sum_{\delta \in \Gamma_M} v_{\kappa}(\delta) \otimes \rho(\delta). \quad (4308)$$

Thus the same real-space curvature vertex is transformed into every generalized momentum sector by the group Fourier rule. For a nondegenerate Euclidean-limit eigenstate,

$$H^{(0)}[\rho] |n, \rho\rangle = E_{n,\rho}^{(0)} |n, \rho\rangle, \quad (4309)$$

first-order perturbation theory gives

$$E_{n,\rho}(\kappa) = E_{n,\rho}^{(0)} + \kappa \langle n, \rho | V_\kappa[\rho] | n, \rho \rangle + O(\kappa^2). \quad (4310)$$

If

$$H[\rho; \kappa] = H_0 + \kappa V + \kappa^2 W + O(\kappa^3), \quad (4311)$$

then the second-order eigenvalue is

$$E_n(\kappa) = E_n^{(0)} + \kappa V_{nn} + \kappa^2 \left[W_{nn} + \sum_{m \neq n} \frac{|V_{mn}|^2}{E_n^{(0)} - E_m^{(0)}} \right] + O(\kappa^3). \quad (4312)$$

At a degeneracy, the first curvature splitting is obtained by diagonalizing $PV_\kappa[\rho]P$ inside the degenerate eigenspace P . This provides a direct quantitative prediction for the initial spectral flow from $K = 0$ to $K < 0$. Let $\mathbf{p} = (\kappa, \theta, \omega)$ denote the control parameters on a fixed finite cover or fixed representation block. Suppose $H(\mathbf{p})$ is continuous in operator norm. For two parameter points, $\mathbf{p}, \mathbf{p}'$, Weyl's inequality gives, for the ordered eigenvalues,

$$|E_j(\mathbf{p}) - E_j(\mathbf{p}')| \leq \|H(\mathbf{p}) - H(\mathbf{p}')\|. \quad (4313)$$

Thus every ordered finite-dimensional eigenvalue is continuous. If the Hamiltonian is locally Lipschitz,

$$\|H(\mathbf{p}) - H(\mathbf{p}')\| \leq L_H \|\mathbf{p} - \mathbf{p}'\|, \quad (4314)$$

then

$$|E_j(\mathbf{p}) - E_j(\mathbf{p}')| \leq L_H \|\mathbf{p} - \mathbf{p}'\|. \quad (4315)$$

Consequently, a genuine finite jump in an eigenvalue cannot be generated by a smooth curvature deformation on a matched finite Hilbert space.

c. Theorem 66. Microscopic spectral continuity. For any fixed finite cover or fixed finite-dimensional representation sector with a continuous distance-dependent hopping law, the ordered spectrum is continuous in (κ, θ, ω) . Therefore an apparent discontinuous jump in a finite numerical spectral plot must arise from at least one of the following mechanisms: change of Hilbert-space identification, change of exact commensurate quotient, discontinuous hopping truncation, branch reassignment or a numerical failure of convergence. A thermodynamic spectral nonanalyticity can occur only after an infinite-system limit is taken. To compare complete spectra without

matching individual bands, define the normalized empirical measures

$$\nu_{\mathbf{p}} = \frac{1}{N} \sum_{j=1}^N \delta(E - E_j(\mathbf{p})). \quad (4316)$$

For two finite Hermitian matrices of equal dimension, define

$$\mathfrak{D}_{\text{spec}}(\mathbf{p}, \mathbf{p}') = W_1(\nu_{\mathbf{p}}, \nu_{\mathbf{p}'}), \quad (4317)$$

where W_1 is the one-dimensional Wasserstein distance. Because the optimal matching of one-dimensional empirical spectra is the ordered matching,

$$W_1 = \frac{1}{N} \sum_{j=1}^N |E_j(\mathbf{p}) - E_j(\mathbf{p}')|. \quad (4318)$$

Using Weyl's inequality,

$$\mathfrak{D}_{\text{spec}}(\mathbf{p}, \mathbf{p}') \leq \|H(\mathbf{p}) - H(\mathbf{p}')\|. \quad (4319)$$

We therefore define the curvature reconstruction distance by

$$\mathfrak{D}_K(\kappa; \theta, \omega) = W_1[\nu_{\kappa, \theta, \omega}, \nu_{0, \theta, \omega}]. \quad (4320)$$

This scalar quantifies the complete spectral displacement induced by negative curvature without requiring a band-by-band correspondence. Define the Lorentzian-broadened density of states by

$$\rho_\eta(E; \mathbf{p}) = -\frac{1}{\pi N} \text{Im Tr} [E + i\eta - H(\mathbf{p})]^{-1}. \quad (4321)$$

The resolvent identity gives

$$R_{\mathbf{p}} - R_{\mathbf{p}'} = R_{\mathbf{p}} [H(\mathbf{p}) - H(\mathbf{p}')] R_{\mathbf{p}'}, \quad (4322)$$

where

$$R_{\mathbf{p}} = [E + i\eta - H(\mathbf{p})]^{-1}. \quad (4323)$$

Since

$$\|R_{\mathbf{p}}\| \leq \frac{1}{\eta}, \quad (4324)$$

we obtain

$$|\rho_\eta(E; \mathbf{p}) - \rho_\eta(E; \mathbf{p}')| \leq \frac{\|H(\mathbf{p}) - H(\mathbf{p}')\|}{\pi\eta^2}. \quad (4325)$$

Thus a finite-broadening DOS cannot jump under a smooth finite-matrix curvature deformation. The preceding continuity theorem applies only when the Hilbert spaces are matched. Exact commensurability introduces an additional arithmetic structure. The exact group $\Gamma_M(\kappa, \theta)$ can change as the geometric parameters change. Consequently,

$$q_M = [\Gamma : \Gamma_M] \quad (4326)$$

can jump between different integer values. The quotient genus,

$$g_M = 1 + q_M(g_0 - 1), \quad (4327)$$

then changes simultaneously. The representation-space dimension

$$\dim_{\mathbb{R}} \mathcal{B}_d = 2q_M(g_0 - 1)d^2 + 2 \quad (4328)$$

therefore changes discretely. This is a crystallographic reorganization of the exact global representation space. It does not imply a discontinuity of any fixed-radius local observable. The correct comparison across such parameter points is therefore not $E_j^{(q_M)}$ versus $E_j^{(q_M)}$, because the two spectra contain different numbers of states. Instead we compare normalized quantities such as $d\nu(E)$, $\tau(H^m)$, local spectral functions, or converged finite-cover DOS measures.

d. Theorem 67. Crystallographic discontinuity versus local spectral continuity. A discrete change of the exact coincidence index q_M changes the dimension and topology of the exact commensurate representation space. Such a change can reorganize the global band-label structure discontinuously. Nevertheless, if the microscopic Hamiltonian converges locally in operator norm, every fixed-radius local spectral observable remains continuous. Thus arithmetic discontinuity of q_M and physical discontinuity of the local spectrum are mathematically distinct phenomena. The local structure of a spectral reconstruction can be made explicit near a pair of interacting levels. Let

$$H_{\text{eff}}(\kappa) = \begin{pmatrix} \varepsilon_1(\kappa) & V(\kappa) \\ V^*(\kappa) & \varepsilon_2(\kappa) \end{pmatrix}. \quad (4329)$$

Expand

$$\begin{aligned} \varepsilon_1(\kappa) &= \varepsilon_1^{(0)} + \kappa a_1 + O(\kappa^2), \\ \varepsilon_2(\kappa) &= \varepsilon_2^{(0)} + \kappa a_2 + O(\kappa^2), \end{aligned} \quad (4330)$$

and

$$V(\kappa) = V_0 + \kappa v_\kappa + O(\kappa^2). \quad (4331)$$

Define

$$\Delta_0 = \varepsilon_1^{(0)} - \varepsilon_2^{(0)}, \quad \delta a = a_1 - a_2. \quad (4332)$$

Then

$$\Delta(\kappa) = \Delta_0 + \kappa \delta a + O(\kappa^2). \quad (4333)$$

The exact two-level eigenvalues are

$$E_{\pm}(\kappa) = \frac{\varepsilon_1(\kappa) + \varepsilon_2(\kappa)}{2} \pm \sqrt{\frac{\Delta^2(\kappa)}{4} + |V(\kappa)|^2}. \quad (4334)$$

If $\delta a \neq 0$, the curvature at which the uncoupled levels become resonant is, to leading order,

$$\kappa_{\text{res}} = -\frac{\Delta_0}{\delta a}. \quad (4335)$$

At this point, the actual avoided-crossing gap is

$$\Delta_{\text{min}} = 2|V(\kappa_{\text{res}})|. \quad (4336)$$

Therefore curvature reconstructs the spectrum through two independent channels: $\kappa \rightarrow \Delta(\kappa)$, which moves the bare levels relative to one another, and $\kappa \rightarrow V(\kappa)$, which changes the hybridization gap. A true crossing requires $\Delta(\kappa_*) = 0$ and simultaneously $V(\kappa_*) = 0$. If symmetry permits $V \neq 0$, curvature produces an avoided crossing rather than a discontinuity. If symmetry forces $V \equiv 0$, the two branches cross at

$$\kappa_* = -\frac{\Delta_0}{\delta a}. \quad (4337)$$

The ordered eigenvalues remain continuous, although their derivatives may be nonanalytic at the crossing. A particularly useful global diagnostic follows from exact trace identities. Inside one representation sector, write

$$H_\rho = \begin{pmatrix} A_\rho & T_\rho \\ T_\rho^\dagger & A_\rho \end{pmatrix}, \quad (4338)$$

where each layer block has dimension M_ρ . Define

$$\mu_p^{(\rho)} = \frac{1}{2M_\rho} \text{Tr } H_\rho^p. \quad (4339)$$

The first moment is

$$\mu_1^{(\rho)} = \frac{1}{M_\rho} \text{Tr } A_\rho. \quad (4340)$$

The second moment is

$$\mu_2^{(\rho)} = \frac{1}{M_\rho} \text{Tr } A_\rho^2 + \frac{1}{M_\rho} \text{Tr } T_\rho T_\rho^\dagger. \quad (4341)$$

The third power has trace

$$\text{Tr } H_\rho^3 = 2 \text{Tr } A_\rho^3 + 3 \text{Tr } [A_\rho (T_\rho T_\rho^\dagger + T_\rho^\dagger T_\rho)]. \quad (4342)$$

Therefore,

$$\mu_3^{(\rho)} = \frac{1}{M_\rho} \text{Tr } A_\rho^3 + \frac{3}{2M_\rho} \text{Tr } [A_\rho (T_\rho T_\rho^\dagger + T_\rho^\dagger T_\rho)]. \quad (4343)$$

The second moment therefore measures the total mean-square hybridization, whereas the third moment already depends on how the hybridization matrix is oriented relative to the intralayer kinetic operator. This distinction becomes central when curvature reorganizes the tunneling pattern without simply increasing its total norm. Define

$$W_\perp^2(\kappa, \theta) = \frac{1}{M_\rho} \text{Tr } T_\rho T_\rho^\dagger. \quad (4344)$$

Then

$$\mu_2^{(\rho)} = \mu_{2,0}^{(\rho)} + W_\perp^2, \quad (4345)$$

where

$$\mu_{2,0}^{(\rho)} = \frac{1}{M_\rho} \text{Tr } A_\rho^2. \quad (4346)$$

Thus a bare-fixed curvature sweep contains two effects: change of total hybridization strength, and change of hybridization geometry. We separate them explicitly. Choose a reference hybridization strength $W_\perp^{\text{ref}}$. Define the normalized interlayer matrix

$$\hat{T}_\rho(\kappa, \theta) = \frac{W_\perp^{\text{ref}}}{W_\perp(\kappa, \theta)} T_\rho(\kappa, \theta). \quad (4347)$$

Then

$$\frac{1}{M_\rho} \text{Tr } \hat{T}_\rho \hat{T}_\rho^\dagger = (W_\perp^{\text{ref}})^2 \quad (4348)$$

for every curvature. If the intralayer block is also held fixed or independently normalized, then both $\mu_1^{(\rho)}$ and $\mu_2^{(\rho)}$ remain invariant under the normalized curvature sweep. Nevertheless, $\mu_3^{(\rho)}$ and all higher spectral moments may still change because the matrix orientation of $\hat{T}_\rho$ relative to A_ρ changes. This gives a strict spectral criterion for geometry-driven reconstruction: a curvature-dependent spectrum can reorganize even when its centroid and mean-square energy are held fixed.

e. Theorem 68. Spectral-shape reconstruction at fixed total hybridization. Suppose A_ρ is fixed and the interlayer matrix is normalized so that

$$\frac{1}{M_\rho} \text{Tr } T_\rho T_\rho^\dagger = \text{constant}. \quad (4349)$$

Then the first and second spectral moments are curvature independent, but the third moment is

$$\mu_3^{(\rho)} = \frac{1}{M_\rho} \text{Tr } A_\rho^3 + \frac{3}{2M_\rho} \text{Tr } [A_\rho (T_\rho T_\rho^\dagger + T_\rho^\dagger T_\rho)], \quad (4350)$$

and may vary continuously with curvature. Therefore a curvature-dependent redistribution of spectral weight that survives fixed- $W_\perp$ normalization is a genuine reconstruction of the tunneling geometry rather than a trivial increase or decrease of the overall coupling scale. Define the normalized matrix

$$\tilde{T}_\rho = \frac{T_\rho}{W_\perp}. \quad (4351)$$

Its Frobenius norm satisfies

$$\frac{1}{M_\rho} \left\| \tilde{T}_\rho \right\|_F^2 = 1. \quad (4352)$$

We define the pure shape-reconstruction distance by

$$\mathfrak{D}_T^{\text{shape}}(\kappa; \theta) = \frac{1}{\sqrt{M_\rho}} \left\| \tilde{T}_\rho(\kappa, \theta) - \tilde{T}_\rho(0, \theta) \right\|_F. \quad (4353)$$

This quantity vanishes if curvature changes only the overall magnitude of the interlayer matrix,

$$T_\rho(\kappa, \theta) = c(\kappa) T_\rho(0, \theta), \quad (4354)$$

because the scalar factor cancels after normalization. Thus $\mathfrak{D}_T^{\text{shape}} > 0$ detects a change in the spatial organization of interlayer hybridization. The pure curvature response contains monolayer curvature effects as well as curvature-induced tunneling reconstruction. To isolate the nonadditive curvature-twist contribution, define

$$\begin{aligned} \delta_{\kappa\theta} H &= H(\kappa, \theta, \omega) - H(\kappa, 0, \omega) \\ &\quad - H(0, \theta, \omega) + H(0, 0, \omega). \end{aligned} \quad (4355)$$

Equivalently, define the additive reference Hamiltonian

$$H_{\text{add}}(\kappa, \theta, \omega) = H(\kappa, 0, \omega) + H(0, \theta, \omega) - H(0, 0, \omega). \quad (4356)$$

Then

$$\delta_{\kappa\theta} H = H(\kappa, \theta, \omega) - H_{\text{add}}(\kappa, \theta, \omega). \quad (4357)$$

In the co-rotating representation, the pure intralayer terms cancel and

$$\delta_{\kappa\theta}H = \begin{pmatrix} 0 & \Delta_{\kappa\theta}T \\ \Delta_{\kappa\theta}T^\dagger & 0 \end{pmatrix}, \quad (4358)$$

where

$$\Delta_{\kappa\theta}T = T(\kappa, \theta) - T(\kappa, 0) - T(0, \theta) + T(0, 0). \quad (4359)$$

Therefore,

$$\|\delta_{\kappa\theta}H\| = \|\Delta_{\kappa\theta}T\|. \quad (4360)$$

This is an exact operator-level isolation of curvature-assisted twisting. Let ν_{full} be the normalized spectral measure of $H(\kappa, \theta, \omega)$, and let ν_{add} be the normalized spectral measure of $H_{\text{add}}(\kappa, \theta, \omega)$. Define

$$\mathfrak{D}_{\kappa\theta}^{\text{spec}} = W_1(\nu_{\text{full}}, \nu_{\text{add}}). \quad (4361)$$

Using Eq. (4319),

$$\mathfrak{D}_{\kappa\theta}^{\text{spec}} \leq \|\delta_{\kappa\theta}H\| = \|\Delta_{\kappa\theta}T\|. \quad (4362)$$

Thus the entire nonadditive spectral reconstruction is bounded by the curvature-dependent reconstruction of the interlayer tunneling matrix. For the same-label channel, the leading microscopic contribution is

$$\Delta_{\kappa\theta}T_{\text{same}} = \kappa T' \left(D_{\text{same}}^{(E)} \right) \frac{r^4 \sin^2 \theta}{6D_{\text{same}}^{(E)}} + O(\kappa^2). \quad (4363)$$

At weak twist,

$$\Delta_{\kappa\theta}T_{\text{same}} = \kappa \frac{T'(h)}{6h} r^4 \theta^2 + O(\kappa \theta^4) + O(\kappa^2 \theta^2). \quad (4364)$$

Hence the leading symmetry-even curvature-twist spectral reconstruction is quadratic in the twist and linear in the curvature magnitude.

f. Theorem 69. Exact nonadditive spectral-reconstruction bound. For a matched finite Hamiltonian family,

$$\mathfrak{D}_{\kappa\theta}^{\text{spec}} \leq \|\Delta_{\kappa\theta}T\|. \quad (4365)$$

In a layer-exchange-symmetric centered bilayer, spectral observables are even functions of θ . Their analytic weak-twist mixed response therefore begins at order $O(\kappa\theta^2)$, even though individual matrix elements in a fixed nonsymmetric gauge may contain odd-in- θ contributions. This provides a quantitative distinction between pure curvature effects and genuinely curvature-assisted

twistronic reconstruction. The reconstruction can also be expressed exactly through an energy-dependent self-energy. Write a representation-sector Hamiltonian as

$$H_\rho = \begin{pmatrix} A_{1,\rho} & T_\rho \\ T_\rho^\dagger & A_{2,\rho} \end{pmatrix}. \quad (4366)$$

The eigenvalue equation is

$$\begin{pmatrix} E - A_{1,\rho} & -T_\rho \\ -T_\rho^\dagger & E - A_{2,\rho} \end{pmatrix} \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} = 0. \quad (4367)$$

If $E - A_{2,\rho}$ is invertible,

$$\psi_2 = (E - A_{2,\rho})^{-1} T_\rho^\dagger \psi_1. \quad (4368)$$

Substitution gives

$$[E - A_{1,\rho} - \Sigma_{1,\rho}(E)] \psi_1 = 0, \quad (4369)$$

with exact self-energy

$$\Sigma_{1,\rho}(E) = T_\rho (E - A_{2,\rho})^{-1} T_\rho^\dagger. \quad (4370)$$

Define

$$R_2 = (E - A_{2,\rho})^{-1}. \quad (4371)$$

At fixed E , differentiation with respect to curvature gives

$$\begin{aligned} \partial_\kappa \Sigma_{1,\rho} &= (\partial_\kappa T_\rho) R_2 T_\rho^\dagger \\ &+ T_\rho R_2 (\partial_\kappa A_{2,\rho}) R_2 T_\rho^\dagger \\ &+ T_\rho R_2 (\partial_\kappa T_\rho^\dagger). \end{aligned} \quad (4372)$$

Under an intralayer-fixed control, $\partial_\kappa A_{2,\rho} = 0$, so that

$$\partial_\kappa \Sigma_{1,\rho} = (\partial_\kappa T_\rho) R_2 T_\rho^\dagger + T_\rho R_2 (\partial_\kappa T_\rho^\dagger). \quad (4373)$$

Thus curvature reconstructs the effective dispersion through the energy-dependent hybridization self-energy even if the isolated monolayer Hamiltonian is held fixed. Using $T_\rho = w\mathcal{T}_\rho$, the self-energy becomes

$$\Sigma_{1,\rho} = w^2 \mathcal{T}_\rho R_2 \mathcal{T}_\rho^\dagger. \quad (4374)$$

Therefore off-resonant spectral reconstruction begins quadratically in w/t ,

$$\delta E_{\text{off}} = O\left(\frac{w^2}{t}\right). \quad (4375)$$

Near an exact degeneracy, however, degenerate perturbation theory produces a linear splitting, $\Delta E_{\text{res}} = O(w)$. Thus the same $\omega = w/t$ sweep naturally separates two regimes: off-resonant reconstruction $\sim \omega^2$, and resonant hybridization $\sim \omega$. This distinction must be retained when comparing spectra at different twists and curvatures. Since

$$H = t[\mathcal{H}_0 + \omega \mathcal{V}_\perp], \quad (4376)$$

where

$$\mathcal{V}_\perp = \begin{pmatrix} 0 & \mathcal{T} \\ \mathcal{T}^\dagger & 0 \end{pmatrix}, \quad (4377)$$

the Hellmann–Feynman theorem gives

$$\frac{\partial E_n}{\partial \omega} = t \langle u_n | \mathcal{V}_\perp | u_n \rangle. \quad (4378)$$

Therefore the slope of a spectral branch with respect to w/t directly measures its interlayer hybridization expectation value. This provides a useful color scale for the (κ, θ, ω) spectral-flow plots. The most important bridge from spectral reconstruction to the later flat-band analysis is the curvature derivative of dispersion itself. Let $\rho(s)$ be a smooth path in generalized representation space. Its tangent direction will be denoted by X . Define

$$H_s = \frac{\partial H}{\partial s}, \quad H_\kappa = \frac{\partial H}{\partial \kappa}, \quad H_{\kappa s} = \frac{\partial^2 H}{\partial \kappa \partial s}. \quad (4379)$$

For a nondegenerate eigenvalue,

$$\frac{\partial E_n}{\partial s} = \langle n | H_s | n \rangle. \quad (4380)$$

Differentiating once more with respect to curvature gives

$$\begin{aligned} \frac{\partial^2 E_n}{\partial \kappa \partial s} &= \langle n | H_{\kappa s} | n \rangle \\ &+ 2 \operatorname{Re} \sum_{m \neq n} \frac{\langle n | H_\kappa | m \rangle \langle m | H_s | n \rangle}{E_n - E_m}. \end{aligned} \quad (4381)$$

The generalized velocity along X is

$$v_n[X] = \frac{1}{\hbar} \frac{\partial E_n}{\partial s}. \quad (4382)$$

Hence

$$\begin{aligned} \frac{\partial v_n[X]}{\partial \kappa} &= \frac{1}{\hbar} \langle n | H_{\kappa s} | n \rangle \\ &+ \frac{2}{\hbar} \operatorname{Re} \sum_{m \neq n} \frac{\langle n | H_\kappa | m \rangle \langle m | H_s | n \rangle}{E_n - E_m}. \end{aligned} \quad (4383)$$

The first term is the direct curvature modification of the generalized dispersion operator. The second term is an interband mixing term and becomes strongly enhanced near small spectral denominators. Thus curvature can reconstruct a band velocity even when the direct curvature derivative of its diagonal energy is small. For the Euclidean Abelian sector, $s = k_\mu$, and Eq. (4383) reduces to the ordinary curvature derivative of the band group velocity. For a hyperbolic non-Abelian sector, the same formula acts on a tangent direction of representation moduli space.

g. Theorem 70. Curvature reconstruction of generalized dispersion. For a nondegenerate generalized Bloch eigenstate, the curvature response of the dispersion differential is

$$\begin{aligned} \partial_\kappa (dE_n[X]) &= \langle n | \partial_\kappa \dot{H}_X | n \rangle \\ &+ 2 \operatorname{Re} \sum_{m \neq n} \frac{\langle n | \partial_\kappa H | m \rangle \langle m | \dot{H}_X | n \rangle}{E_n - E_m}. \end{aligned} \quad (4384)$$

Negative curvature therefore modifies generalized dispersion through both a direct geometric tunneling vertex and a resonantly enhanced interband mixing term. This formula provides the quantitative bridge between curvature-controlled spectral reconstruction and the later analysis of dispersion suppression. A spectral crossing observed in a full finite-cover plot must be classified before it is interpreted. Let two states carry irreducible translation labels ρ_1, ρ_2 . If $\rho_1 \not\sim \rho_2$, translation symmetry forbids hybridization between the two sectors. Their energies may therefore cross as curvature varies. If $\rho_1 \simeq \rho_2$, a generic symmetry-allowed coupling produces an avoided crossing. Thus every curvature-driven crossing is classified according to different representation sectors $\rightarrow$ symmetry-allowed crossing, whereas same representation sector $\rightarrow$ generic avoided crossing, unless an additional symmetry forces the matrix element to vanish. This classification prevents representation-theoretic degeneracies from being mistaken for dynamical flattening. For each parameter triple (θ, κ, ω) , we extract the following quantities. The complete normalized spectral measure is $\nu(E; \theta, \kappa, \omega)$. The curvature reconstruction distance is $\mathfrak{D}_K(\kappa; \theta, \omega)$. The nonadditive curvature-twist reconstruction distance is $\mathfrak{D}_{\kappa\theta}^{\text{spec}}$. The mean-square hybridization is $W_\perp^2(\kappa, \theta)$. The pure tunneling-shape reconstruction is $\mathfrak{D}_T^{\text{shape}}(\kappa; \theta)$. The spectral moments are $\mu_1, \mu_2, \mu_3, \dots$. For continuous representation-space bands, we additionally evaluate $dE_n[X]$, and $\partial_\kappa dE_n[X]$. Together these observables distinguish a uniform energy rescaling from a true reorganization of spectral shape. The primary spectral calculation uses the bare-fixed family $H_{\text{bare}}(\kappa, \theta, \omega)$, with all microscopic energy and length parameters fixed. The first robustness family removes trivial changes of the intralayer kinetic scale, $H_{\text{coord}}(\kappa, \theta, \omega)$, using the coordination or local second-moment normalization defined above. The second robustness family fixes the total mean-square interlayer hy-

bridization, $H_{\text{hyb}}(\kappa, \theta, \omega)$, through Eq. (4347). A spectral feature that survives all three families cannot be explained by a trivial change of the overall monolayer bandwidth, coordination number, or total interlayer tunneling norm. Its origin is then the curvature-dependent spatial organization of the Hamiltonian. The hybridization-normalized family yields a particularly strong diagnostic. Suppose A_ρ and $W_\perp$ are held fixed. Then $\partial_\kappa \mu_1^{(\rho)} = 0$, and $\partial_\kappa \mu_2^{(\rho)} = 0$. If nevertheless $\partial_\kappa \mu_3^{(\rho)} \neq 0$, or $\mathfrak{D}_{\text{spec}}(\kappa, 0) > 0$, the spectral reconstruction cannot be attributed to an overall energy scale. It is generated by changes in the orientation and spatial texture of the hybridization matrix. This constitutes a direct operational test of geometry-driven spectral reconstruction. Every hyperbolic spectrum used for a bulk conclusion must be converged along $r_{\text{inj}}^{(N)} \rightarrow \infty$. For a finite-range Hamiltonian with propagation radius L_H , the normalized m -th spectral moment is exactly equal to its infinite-system value once $2r_{\text{inj}}^{(N)} > mL_H$. Thus $\mu_m^{(N)} = \mu_m^{(\infty)}$ for every sufficiently large cover at fixed m . For exponentially decaying hopping, the geodesic cutoff L_c is increased simultaneously with the cover while maintaining $L_c^{(N)} \rightarrow \infty$, and

$$\frac{L_c^{(N)}}{r_{\text{inj}}^{(N)}} \rightarrow 0. \quad (4385)$$

The corresponding truncation error obeys

$$\|H - H^{(L_c)}\| \leq C \exp[-(\mu - R^{-1})L_c], \quad (4386)$$

provided $\mu > R^{-1}$. Therefore every curvature-dependent spectral shift must exceed both the finite-cover and hopping-truncation error before it is regarded as physical. For fixed (θ, ω) , we consider the spectral flow $\kappa : 0 \rightarrow \kappa_{\text{max}}$. At each curvature, the exact or finite-cover spectrum is computed. A continuously evolving isolated branch satisfies

$$E_n(\kappa) = E_n(0) + \int_0^\kappa d\kappa' \langle u_n(\kappa') | \partial_{\kappa'} H | u_n(\kappa') \rangle. \quad (4387)$$

This equation is exact as long as the branch remains differentiable. At a degeneracy, the spectral projector rather than an individual eigenvector is continued. The projector flow therefore provides the correct branch-tracking object through degenerate regions. For fixed (κ, ω) , the twist flow is $\theta : \theta_1 \rightarrow \theta_2$. Because identical layers satisfy the layer-exchange equivalence

$$\text{Spec } H(\kappa, \theta, \omega) = \text{Spec } H(\kappa, -\theta, \omega), \quad (4388)$$

every analytic scalar spectral observable is even in θ . Therefore its small-twist expansion has the form

$$\mathcal{O}(\kappa, \theta, \omega) = \mathcal{O}_0(\kappa, \omega) + \theta^2 \mathcal{O}_2(\kappa, \omega) + O(\theta^4). \quad (4389)$$

This symmetry provides an additional numerical consistency check. A linear-in- θ shift of a symmetry-invariant scalar spectrum near $\theta = 0$ signals either branch misidentification, explicit layer-asymmetry, or numerical error. For fixed (κ, θ) , the interlayer-coupling flow is $\omega : 0 \rightarrow \omega_{\text{max}}$. The exact derivative

$$\frac{\partial E_n}{\partial \omega} = t \langle u_n | \mathcal{V}_\perp | u_n \rangle \quad (4390)$$

provides the local hybridization sensitivity. The second derivative is

$$\frac{\partial^2 E_n}{\partial \omega^2} = 2t^2 \sum_{m \neq n} \frac{|\langle m | \mathcal{V}_\perp | n \rangle|^2}{E_n - E_m}, \quad (4391)$$

whenever $\mathcal{V}_\perp$ itself is independent of ω . Thus strong curvature dependence of the denominators can strongly enhance the nonlinear response to w/t even when the bare tunneling amplitude is unchanged. The complete spectral reconstruction is therefore a surface over $\mathcal{P} = \{(\theta, \kappa, \omega)\}$. For every spectral branch or representation sector we obtain

$$E_{n,\rho} = E_{n,\rho}(\theta, \kappa, \omega). \quad (4392)$$

Its local response vector is

$$\nabla_{\mathcal{P}} E_{n,\rho} = \begin{pmatrix} \partial_\theta E_{n,\rho} \\ \partial_\kappa E_{n,\rho} \\ \partial_\omega E_{n,\rho} \end{pmatrix}. \quad (4393)$$

The Hessian is

$$\mathcal{H}_{ab}^{(n,\rho)} = \frac{\partial^2 E_{n,\rho}}{\partial p_a \partial p_b}, \quad p_a \in \{\theta, \kappa, \omega\}. \quad (4394)$$

The off-diagonal entries $\partial_\kappa \partial_\theta E$, $\partial_\kappa \partial_\omega E$, $\partial_\theta \partial_\omega E$ measure genuinely coupled responses that cannot be inferred from one-parameter sweeps alone. In particular, $\partial_\kappa \partial_\omega E_n$ quantifies how curvature changes the susceptibility of a spectral state to interlayer coupling. Because $H_\omega = t\mathcal{V}_\perp$, the exact mixed derivative for a nondegenerate level is

$$\begin{aligned} \frac{\partial^2 E_n}{\partial \kappa \partial \omega} &= \langle n | H_{\kappa\omega} | n \rangle \\ &+ 2 \text{Re} \sum_{m \neq n} \frac{\langle n | H_\kappa | m \rangle \langle m | H_\omega | n \rangle}{E_n - E_m}. \end{aligned} \quad (4395)$$

For a geometry-dependent tunneling matrix, $H_{\kappa\omega} \neq 0$. Thus curvature changes both the direct interlayer matrix and the wave-function-mediated susceptibility to w/t . This produces a quantitative measure of curvature-enhanced or curvature-suppressed hybridization that is

distinct from the bare value of w . The full spectral logic established in V.A and V.B can now be stated without invoking flat-band criteria. For Euclidean commensurate systems, exact supercell $\rightarrow$ band folding $\rightarrow$ interlayer hybridization $\rightarrow$ spectral reconstruction. For the negatively curved system,

$$\begin{aligned}
K < 0 &\rightarrow \text{geodesic-distance reconstruction} \\
&\rightarrow T_{ij}(K, \theta) \\
&\rightarrow T[\rho; K, \theta] \\
&\rightarrow \Sigma(E, [\rho]; K, \theta) \\
&\rightarrow E_{n,\rho}(K, \theta, \omega).
\end{aligned} \tag{4396}$$

The exact crystallographic route proceeds in parallel,

$$K, \theta \rightarrow \Gamma_M \rightarrow q_M \rightarrow g_M \rightarrow \mathcal{B}_d \rightarrow \text{allowed spectral sectors.} \tag{4397}$$

These two routes must be kept distinct. The first changes the Hamiltonian matrix elements. The second changes the exact global representation structure in which the Hamiltonian is resolved. They meet only at the final representation-space spectral problem. We therefore classify the observed $K = 0 \rightarrow K < 0$ reconstruction into four mathematically distinct cases. A smooth deformation satisfies $E_{n,\rho}(\kappa) \in C^1$ over the curvature interval of interest. An avoided crossing satisfies $\Delta_{\min} > 0$ and remains continuous and differentiable. A symmetry-protected crossing satisfies $\Delta_{\min} = 0$, while the two states remain in distinct symmetry sectors. The ordered spectrum is continuous but may fail to be differentiable at the crossing. An arithmetic commensurability change satisfies $q_M^- \neq q_M^+$. In this case the exact global band-label space itself changes and raw ordered eigenvalues on the two sides cannot be compared directly. The correct objects are normalized spectral measures and local observables. A genuinely thermodynamic nonanalytic reconstruction would require a sequence of finite covers for which a singularity survives $r_{\text{inj}}^{(N)} \rightarrow \infty$. No finite-cover level jump alone is sufficient evidence for such a nonanalyticity.

h. Theorem 71. Complete curvature-controlled spectral classification. For a geometry-driven twisted bilayer with continuous microscopic hopping kernels, negative curvature can reconstruct the spectrum through three independent mechanisms: local Hamiltonian deformation, hybridization-induced level repulsion and self-energy reconstruction, and arithmetic reorganization of the exact representation space. The first two are continuous on every matched finite Hilbert space. The third can change discretely because the exact commensurability index is integer valued. A true bulk spectral nonanalyticity can be established only after finite-cover convergence. This separates microscopic spectral physics from global crystallographic arithmetic. The Euclidean benchmark calculation therefore consists of the exact spectra of the ten-site and thirty-four-site

structures, supplemented by the remaining $N_{\text{sc}} < 100$ commensurate cells. For each structure we calculate both $w = 0$ and $w \neq 0$ spectra along the same normalized $\Gamma - X - M - \Gamma$ path. The hyperbolic calculation then evaluates the converged representation- or finite-cover-resolved spectra over a controlled grid of (θ, κ, ω) . For every parameter point we record $\nu(E)$, μ_1 , μ_2 , μ_3 , $W_\perp$, $\mathfrak{D}_T^{\text{shape}}$, $\mathfrak{D}_K$, $\mathfrak{D}_{\kappa\theta}^{\text{spec}}$, and, whenever a continuous generalized momentum sector is available,

$$dE_n[X], \quad \partial_\kappa dE_n[X]. \tag{4398}$$

This collection is sufficient to determine whether the change from $K = 0$ to $K < 0$ is a trivial rescaling, a hybridization-strength effect, a genuine tunneling-shape reconstruction, a symmetry-protected crossing, or an arithmetic reorganization of the exact crystallographic spectrum. No magic-band criterion is imposed at this stage. The purpose of the present section is to establish the full curvature-dependent spectral reconstruction on which that later criterion must act.

B. Bandwidth, Density of States, and Spectral Gaps

The spectral reconstruction derived above must now be converted into quantitative observables that can be compared across Euclidean Bloch bands, hyperbolic representation-space bands, and periodic finite-cover spectra. We use three primary diagnostics: the bandwidth, the density of states, and the spectral isolation gap. We then derive exact relations among these objects and the generalized energy differential. The main objective is to distinguish four physically different phenomena: global energetic flattening, local stationary dispersion, density-of-states enhancement, and spectral isolation. None of these properties is mathematically equivalent to the others. In particular, a large density-of-states peak may arise from an isolated saddle point without a globally narrow band, a narrow band may fail to be spectrally isolated, and a vanishing derivative along one selected path does not imply suppression of the complete generalized dispersion. We therefore construct a hierarchy in which every flatness statement can be tested independently at the energetic, differential, and spectral-gap levels. For an ordinary Euclidean commensurate bilayer, the spectral parameter is

$$\lambda = \mathbf{k} \in \text{mBZ}. \tag{4399}$$

For an Abelian hyperbolic sector,

$$\lambda = \mathbf{k} \in \mathbb{T}^{2g_M}. \tag{4400}$$

For a rank- d non-Abelian generalized Bloch sector, the spectral parameter is a representation class

$$\lambda = [\rho] \in \mathcal{B}_d, \quad (4401)$$

where

$$\mathcal{B}_d = \text{Hom}^{\text{irr}}(\Gamma_M, U(d)) / U(d). \quad (4402)$$

Whenever a smooth isolated band is studied, we restrict the analysis to a connected smooth domain $\mathfrak{D}_d \subset \mathcal{B}_d$ over which the corresponding spectral projector can be continued adiabatically. The energy is therefore written uniformly as

$$E_n = E_n(\lambda; \kappa, \theta, \omega), \quad (4403)$$

where

$$\kappa = -K \geq 0, \quad \omega = \frac{w}{t}. \quad (4404)$$

The Euclidean system is recovered at $\kappa = 0$. An individual eigenvalue label is insufficient near symmetry-enforced or accidental degeneracies. We therefore formulate the diagnostics first for an isolated spectral cluster. Let the target cluster contain r continuously tracked states and let

$$P_{\mathcal{C}}(\lambda) = \sum_{a=1}^r |u_a(\lambda)\rangle \langle u_a(\lambda)| \quad (4405)$$

be its spectral projector. If the cluster is isolated by a contour $\mathcal{C}$ in the complex energy plane, the same projector can be written in the basis-independent Riesz form

$$P_{\mathcal{C}}(\lambda) = \frac{1}{2\pi i} \oint_{\mathcal{C}} dz [z - H(\lambda)]^{-1}. \quad (4406)$$

This expression remains well defined even when states inside the cluster cross or become degenerate. For a single isolated nondegenerate band, $r = 1$, and we write $P_{\mathcal{C}} = P_n$. All flatness diagnostics below are therefore invariant under arbitrary unitary rotations inside a protected degenerate multiplet. For an isolated Euclidean moiré band, we define

$$E_n^{\max} = \max_{\mathbf{k} \in \text{mBZ}} E_n(\mathbf{k}), \quad (4407)$$

and

$$E_n^{\min} = \min_{\mathbf{k} \in \text{mBZ}} E_n(\mathbf{k}). \quad (4408)$$

The exact Euclidean bandwidth is

$$W_n^{(E)} = E_n^{\max} - E_n^{\min}. \quad (4409)$$

For the target cluster $\mathcal{C}$, define the upper and lower cluster edges by

$$E_{\mathcal{C}}^{\max} = \max_{\mathbf{k} \in \text{mBZ}} \lambda_{\max}[P_{\mathcal{C}}(\mathbf{k})H(\mathbf{k})P_{\mathcal{C}}(\mathbf{k})], \quad (4410)$$

and

$$E_{\mathcal{C}}^{\min} = \min_{\mathbf{k} \in \text{mBZ}} \lambda_{\min}[P_{\mathcal{C}}(\mathbf{k})H(\mathbf{k})P_{\mathcal{C}}(\mathbf{k})]. \quad (4411)$$

The cluster bandwidth is

$$W_{\mathcal{C}}^{(E)} = E_{\mathcal{C}}^{\max} - E_{\mathcal{C}}^{\min}. \quad (4412)$$

Thus a symmetry-protected r -dimensional multiplet is assigned one basis-independent global spectral width. For a smooth rank- d generalized band over $\mathfrak{D}_d$, we define

$$E_{n,d}^{\max} = \sup_{[\rho] \in \mathfrak{D}_d} E_n[\rho], \quad (4413)$$

and

$$E_{n,d}^{\min} = \inf_{[\rho] \in \mathfrak{D}_d} E_n[\rho]. \quad (4414)$$

The generalized bandwidth is

$$W_n^{(d)} = E_{n,d}^{\max} - E_{n,d}^{\min}. \quad (4415)$$

For an isolated rank- r spectral cluster,

$$\begin{aligned} W_{\mathcal{C}}^{(d)} &= \sup_{[\rho] \in \mathfrak{D}_d} \lambda_{\max}[P_{\mathcal{C}}[\rho]H[\rho]P_{\mathcal{C}}[\rho]] \\ &\quad - \inf_{[\rho] \in \mathfrak{D}_d} \lambda_{\min}[P_{\mathcal{C}}[\rho]H[\rho]P_{\mathcal{C}}[\rho]]. \end{aligned} \quad (4416)$$

This definition measures the complete energy variation over the generalized momentum domain rather than along one arbitrarily selected representation-space path. In the Abelian limit, $d = 1$, and $\mathfrak{D}_1 = \mathbb{T}^{2g_M}$, so that Eq. (4415) becomes a direct higher-genus generalization of the ordinary Bloch bandwidth. A finite quotient contains only a discrete collection of generalized translation sectors. Let

$$\mathcal{C}_N = \{(n, \rho)\}_{\text{target}} \quad (4417)$$

denote the set of finite-cover eigenstates belonging to the target band or cluster, identified through representation labels and projector overlap. Define

$$E_{\mathcal{C},N}^{\max} = \max_{(n,\rho) \in \mathcal{C}_N} E_{n,\rho}^{(N)}, \quad (4418)$$

and

$$E_{\mathcal{C},N}^{\min} = \min_{(n,\rho) \in \mathcal{C}_N} E_{n,\rho}^{(N)}. \quad (4419)$$

The finite-cover bandwidth is

$$W_{\mathcal{C},N} = E_{\mathcal{C},N}^{\max} - E_{\mathcal{C},N}^{\min}. \quad (4420)$$

Representation multiplicities do not change this width because repeated copies of the same representation-sector eigenvalue do not change either spectral edge. The bulk bandwidth is accepted only after $W_{\mathcal{C},N} \rightarrow W_{\mathcal{C},\infty}$ along increasing injectivity radius. The curvature dependence of the bandwidth can be calculated without finite-difference subtraction when the extrema remain nondegenerate. Suppose the upper band edge occurs at $\lambda_+(\kappa)$, and the lower band edge occurs at $\lambda_-(\kappa)$. Then

$$W_n(\kappa) = E_n[\lambda_+(\kappa), \kappa] - E_n[\lambda_-(\kappa), \kappa]. \quad (4421)$$

At a differentiable extremum,

$$d_\lambda E_n[\lambda_\pm(\kappa), \kappa] = 0. \quad (4422)$$

Therefore the envelope theorem gives

$$\frac{dW_n}{d\kappa} = \left. \frac{\partial E_n}{\partial \kappa} \right|_{\lambda_+} - \left. \frac{\partial E_n}{\partial \kappa} \right|_{\lambda_-}. \quad (4423)$$

Using the Hellmann–Feynman theorem,

$$\frac{dW_n}{d\kappa} = \langle u_{n,+} | H_\kappa | u_{n,+} \rangle - \langle u_{n,-} | H_\kappa | u_{n,-} \rangle, \quad (4424)$$

where

$$H_\kappa = \frac{\partial H}{\partial \kappa}. \quad (4425)$$

Thus curvature narrows the band at first order precisely when

$$\langle u_{n,+} | H_\kappa | u_{n,+} \rangle < \langle u_{n,-} | H_\kappa | u_{n,-} \rangle. \quad (4426)$$

This is an explicit spectral-edge criterion rather than a qualitative statement that curvature reduces the bandwidth. We define the logarithmic bandwidth susceptibility by

$$\chi_{W,n} = -\frac{\partial \ln W_n}{\partial \kappa}. \quad (4427)$$

Using Eq. (4424),

$$\chi_{W,n} = \frac{\langle u_{n,-} | H_\kappa | u_{n,-} \rangle - \langle u_{n,+} | H_\kappa | u_{n,+} \rangle}{W_n}. \quad (4428)$$

Hence $\chi_{W,n} > 0$ is the local criterion for curvature-induced bandwidth suppression. The quantity is dimensionful in κ . A dimensionless response can be defined as $\tilde{\chi}_{W,n} = \kappa \chi_{W,n}$. The first-order bandwidth derivative can vanish even when curvature produces a strong second-order reconstruction. Let x^μ denote local coordinates on the relevant spectral parameter domain, and let $x_*(\kappa)$ be a nondegenerate band extremum satisfying

$$\partial_\mu E_n(x_*, \kappa) = 0. \quad (4429)$$

Define the spectral Hessian

$$M_{\mu\nu} = \partial_\mu \partial_\nu E_n. \quad (4430)$$

Differentiating the stationarity condition gives

$$M_{\mu\nu} \frac{dx_*^\nu}{d\kappa} + \partial_\mu \partial_\kappa E_n = 0. \quad (4431)$$

If M is invertible,

$$\frac{dx_*^\mu}{d\kappa} = - (M^{-1})^{\mu\nu} \partial_\nu \partial_\kappa E_n. \quad (4432)$$

The total second curvature derivative of the edge energy is therefore

$$\frac{d^2 E_{n,*}}{d\kappa^2} = \partial_\kappa^2 E_n - (\partial_\kappa \partial_\mu E_n) (M^{-1})^{\mu\nu} (\partial_\nu \partial_\kappa E_n). \quad (4433)$$

The first term contains the direct second-order curvature response and virtual interband mixing. The second term is the correction generated by motion of the extremum in generalized momentum space. For a nondegenerate eigenstate,

$$\partial_\kappa^2 E_n = \langle n | H_{\kappa\kappa} | n \rangle + 2 \sum_{m \neq n} \frac{|\langle m | H_\kappa | n \rangle|^2}{E_n - E_m}. \quad (4434)$$

Thus a small first-order bandwidth response does not imply weak curvature reconstruction.

a. Theorem 72. Exact curvature response of a nondegenerate bandwidth. For a continuously tracked isolated band with unique nondegenerate upper and lower extrema,

$$\frac{dW_n}{d\kappa} = \langle u_{n,+} | H_\kappa | u_{n,+} \rangle - \langle u_{n,-} | H_\kappa | u_{n,-} \rangle. \quad (4435)$$

The second derivative is obtained by subtracting Eq. (4433) evaluated at the lower edge from the corresponding result at the upper edge. Consequently, curvature-induced flattening is controlled by the difference between the curvature susceptibilities of the two spectral edges rather than by the average energy shift of the band. For an exact Euclidean commensurate bilayer with 2Σ bands, we define the normalized density of states by

$$\rho^{(E)}(E) = \frac{1}{2\Sigma A_{\text{mBZ}}} \sum_{n=1}^{2\Sigma} \int_{\text{mBZ}} d^2k \delta[E - E_n(\mathbf{k})]. \quad (4436)$$

The normalization is

$$\int_{-\infty}^{\infty} \rho^{(E)}(E) dE = 1. \quad (4437)$$

The density of states per exact moiré cell is

$$D^{(E)}(E) = 2\Sigma \rho^{(E)}(E), \quad (4438)$$

and therefore

$$\int D^{(E)}(E) dE = 2\Sigma. \quad (4439)$$

The two normalizations will be kept distinct: $\rho(E)$ is normalized to unity and is convenient for comparing systems of different supercell size, whereas $D(E)$ counts states per moiré cell. For one isolated band,

$$\rho_n^{(E)}(E) = \frac{1}{A_{\text{mBZ}}} \int_{\text{mBZ}} d^2k \delta[E - E_n(\mathbf{k})]. \quad (4440)$$

It satisfies

$$\int \rho_n^{(E)}(E) dE = 1. \quad (4441)$$

The complete normalized DOS may therefore be written as

$$\rho^{(E)}(E) = \frac{1}{2\Sigma} \sum_{n=1}^{2\Sigma} \rho_n^{(E)}(E). \quad (4442)$$

For a cluster containing r bands, define

$$\rho_C^{(E)}(E) = \frac{1}{r} \sum_{n \in \mathcal{C}} \rho_n^{(E)}(E), \quad (4443)$$

so that

$$\int \rho_C^{(E)}(E) dE = 1. \quad (4444)$$

For a normal finite cover with quotient Q_N , the exact decomposition is

$$H_N \simeq \bigoplus_{\rho \in \hat{Q}_N} [I_{d_\rho} \otimes H[\rho]]. \quad (4445)$$

The representation block $H[\rho]$ has dimension $\nu_M d_\rho$, where $\nu_M = 2n_b q_M$. We define the conditional DOS inside one irrep sector by

$$\hat{\rho}_{N,\rho}(E) = \frac{1}{\nu_M d_\rho} \sum_{n=1}^{\nu_M d_\rho} \delta(E - E_{n,\rho}). \quad (4446)$$

It is normalized according to

$$\int \hat{\rho}_{N,\rho}(E) dE = 1. \quad (4447)$$

The fraction of all finite-cover states carried by the irrep is

$$p_\rho = \frac{d_\rho^2}{|Q_N|}. \quad (4448)$$

The complete normalized finite-cover DOS is therefore

$$\rho_N(E) = \sum_{\rho \in \hat{Q}_N} p_\rho \hat{\rho}_{N,\rho}(E). \quad (4449)$$

Explicitly,

$$\rho_N(E) = \frac{1}{\nu_M |Q_N|} \sum_{\rho \in \hat{Q}_N} d_\rho \sum_{n=1}^{\nu_M d_\rho} \delta(E - E_{n,\rho}). \quad (4450)$$

The normalization is

$$\int \rho_N(E) dE = 1. \quad (4451)$$

This representation weighting is mandatory: counting every irrep block with equal weight would give an incorrect physical density of states. Along a convergent normal-cover sequence, $r_{\text{inj}}^{(N)} \rightarrow \infty$, the normalized measures converge weakly,

$$\rho_N(E) dE \Rightarrow d\nu_\infty(E). \quad (4452)$$

When the limiting measure possesses a density, we write

$$d\nu_\infty(E) = \rho_\infty(E) dE. \quad (4453)$$

All hyperbolic DOS statements below refer either to ρ_N after explicit finite-cover convergence or to this limiting bulk spectral measure. For a smooth continuous generalized band over a domain $\mathfrak{D}_d$, choose an explicit Riemannian metric G and its measure $d\mu_G(\lambda)$. Let

$$V_G = \int_{\mathfrak{D}_d} d\mu_G. \quad (4454)$$

We define the normalized representation-space DOS of one band by

$$\rho_{n,d}^{\text{rep}}(E) = \frac{1}{V_G} \int_{\mathfrak{D}_d} d\mu_G(\lambda) \delta[E - E_n(\lambda)]. \quad (4455)$$

This quantity is a dispersion diagnostic over the chosen generalized momentum domain. For $d = 1$, the natural measure is the normalized Haar measure on $\mathbb{T}^{2g_M}$. For $d > 1$, the measure G must be stated explicitly; we do not identify an arbitrarily sampled representation-space histogram with the complete physical bulk DOS. At a regular energy value satisfying $dE_n \neq 0$, the coarea formula gives

$$\rho_{n,d}^{\text{rep}}(E) = \frac{1}{V_G} \int_{E_n^{-1}(E)} \frac{dA_{G,E}}{\|dE_n\|_{G^{-1}}}. \quad (4456)$$

Here $dA_{G,E}$ is the induced measure on the constant-energy hypersurface. Equation (4456) gives a direct quantitative relation between generalized dispersion and density of states. If $\|dE_n\|_{G^{-1}}$ becomes small over an extended constant-energy manifold, the DOS increases. Thus DOS enhancement is generated not merely by a large number of folded states but by suppression of the generalized energy differential. For an ordinary two-dimensional Euclidean band, Eq. (4456) reduces to

$$\rho_n^{(E)}(E) = \frac{1}{A_{\text{mBZ}}} \oint_{E_n(\mathbf{k})=E} \frac{d\ell}{|\nabla_{\mathbf{k}} E_n|}. \quad (4457)$$

Using

$$\mathbf{v}_n = \frac{1}{\hbar} \nabla_{\mathbf{k}} E_n, \quad (4458)$$

this becomes

$$\rho_n^{(E)}(E) = \frac{1}{\hbar A_{\text{mBZ}}} \oint_{E_n(\mathbf{k})=E} \frac{d\ell}{|\mathbf{v}_n|}. \quad (4459)$$

The ordinary inverse-velocity enhancement of the DOS is therefore the rank-one Euclidean limit of the generalized representation-space formula. Let the generalized

momentum-space dimension be $D = \dim_{\mathbb{R}} \mathfrak{D}_d$. Near a nondegenerate stationary minimum at λ_* , choose local orthonormal coordinates x^μ and write

$$E_n(x) = E_* + \frac{1}{2} x^T M x + O(|x|^3), \quad (4460)$$

where M is positive definite. To leading order,

$$\rho_n^{\text{rep}}(E) \simeq \frac{1}{V_G \sqrt{\det M}} \int_{\mathbb{R}^D} d^D y \delta \left[E - E_* - \frac{|y|^2}{2} \right]. \quad (4461)$$

Using spherical coordinates,

$$d^D y = S_{D-1} r^{D-1} dr, \quad (4462)$$

where

$$S_{D-1} = \frac{2\pi^{D/2}}{\Gamma(D/2)}, \quad (4463)$$

we obtain

$$\begin{aligned} \rho_n^{\text{rep}}(E) &\simeq \frac{S_{D-1}}{V_G \sqrt{\det M}} \\ &\times [2(E - E_*)]^{D/2-1} \Theta(E - E_*). \end{aligned} \quad (4464)$$

Thus the DOS prefactor diverges as $(\det M)^{-1/2}$ when the generalized spectral Hessian becomes soft. This gives a quantitative higher-dimensional generalization of van-Hove enhancement. For the Abelian hyperbolic sector, $D = 2g_M$, whereas at a smooth rank- d irreducible point,

$$D = (2g_M - 2)d^2 + 2. \quad (4465)$$

Negative curvature can therefore change not only the Hessian but also the dimensionality of the generalized dispersion space sampled by an exact commensurate structure. The connection between global bandwidth and DOS enhancement admits a model-independent inequality. Let the normalized DOS of an isolated band satisfy

$$\int \rho_n(E) dE = 1, \quad (4466)$$

and suppose its support lies inside $[E_n^{\min}, E_n^{\max}]$ of width $W_n = E_n^{\max} - E_n^{\min}$. Then

$$1 = \int_{E_n^{\min}}^{E_n^{\max}} \rho_n(E) dE \leq W_n \|\rho_n\|_\infty. \quad (4467)$$

Therefore,

$$\|\rho_n\|_\infty \geq \frac{1}{W_n}. \quad (4468)$$

For a cluster containing r states per generalized momentum and using the unnormalized physical cluster DOS D_C ,

$$\int D_C(E) dE = r, \quad (4469)$$

so that

$$\|D_C\|_\infty \geq \frac{r}{W_C}. \quad (4470)$$

b. Theorem 73. Bandwidth-enforced DOS enhancement. Every normalized isolated band of bandwidth $W_n > 0$ satisfies

$$\|\rho_n\|_\infty W_n \geq 1. \quad (4471)$$

Consequently, $W_n \rightarrow 0$ forces the unbroadened band DOS to become singular. The converse is false: a divergent DOS can be generated by an isolated critical point even when the global bandwidth remains large. Therefore DOS enhancement is a necessary consequence of an isolated flat-band limit but is not, by itself, sufficient evidence for global flatness. To distinguish global narrowing from a sharply localized van-Hove feature, we define a second DOS diagnostic. For a normalized band DOS with finite square integral, define

$$W_{\rho,n}^{\text{eff}} = \left[\int \rho_n^2(E) dE \right]^{-1}. \quad (4472)$$

Using Cauchy-Schwarz,

$$1 = \left[\int \rho_n(E) dE \right]^2 \leq W_n \int \rho_n^2(E) dE. \quad (4473)$$

Therefore, $W_{\rho,n}^{\text{eff}} \leq W_n$. We define the DOS concentration ratio

$$\eta_{\rho,n} = \frac{W_{\rho,n}^{\text{eff}}}{W_n}. \quad (4474)$$

It satisfies $0 < \eta_{\rho,n} \leq 1$. A nearly uniform DOS distributed across the complete narrow band gives an order-one value of $\eta_{\rho,n}$. A sharp DOS singularity embedded inside a much wider dispersive band gives $\eta_{\rho,n} \ll 1$. Thus W_n and $W_{\rho,n}^{\text{eff}}$ distinguish global energetic flattening from local spectral concentration. When the ideal DOS contains singular distributions, the same construction is applied to a stated finite broadening η , and convergence

with respect to $\eta \rightarrow 0^+$ is analyzed separately. The curvature dependence of the DOS admits an exact conservation law. For a finite matched Hilbert space of dimension N_H , define

$$\rho(E, \kappa) = \frac{1}{N_H} \sum_{j=1}^{N_H} \delta[E - E_j(\kappa)]. \quad (4475)$$

Differentiating gives

$$\frac{\partial \rho}{\partial \kappa} = -\frac{1}{N_H} \sum_j \frac{dE_j}{d\kappa} \delta'[E - E_j(\kappa)]. \quad (4476)$$

Define the curvature spectral current

$$J_\kappa(E) = \frac{1}{N_H} \sum_j \frac{dE_j}{d\kappa} \delta[E - E_j(\kappa)]. \quad (4477)$$

Then

$$\frac{\partial \rho(E, \kappa)}{\partial \kappa} + \frac{\partial J_\kappa(E, \kappa)}{\partial E} = 0. \quad (4478)$$

Using Hellmann-Feynman,

$$\frac{dE_j}{d\kappa} = \langle j | H_\kappa | j \rangle, \quad (4479)$$

and therefore

$$J_\kappa(E) = \frac{1}{N_H} \text{Tr} [H_\kappa \delta(E - H)]. \quad (4480)$$

Thus curvature redistributes spectral weight through a conserved current in energy space. Define the integrated DOS by

$$\mathcal{N}(E, \kappa) = \int_{-\infty}^E \rho(E', \kappa) dE'. \quad (4481)$$

Integration of Eq. (4478) gives

$$\frac{\partial \mathcal{N}(E, \kappa)}{\partial \kappa} = -J_\kappa(E, \kappa), \quad (4482)$$

assuming the spectral current vanishes below the complete spectrum. Therefore $J_\kappa(E)$ measures the net rate at which spectral states cross the energy E as curvature changes. This provides a direct quantitative description of curvature-induced DOS redistribution. Define the normalized moments

$$\mu_m(\kappa) = \int E^m \rho(E, \kappa) dE. \quad (4483)$$

Multiplying the continuity equation by E^m and integrating by parts gives

$$\frac{d\mu_m}{d\kappa} = m \int E^{m-1} J_\kappa(E) dE. \quad (4484)$$

Using the trace representation,

$$\frac{d\mu_m}{d\kappa} = \frac{m}{N_H} \text{Tr} [H^{m-1} H_\kappa]. \quad (4485)$$

For $m = 1$, we obtain

$$\frac{d\mu_1}{d\kappa} = \frac{1}{N_H} \text{Tr} H_\kappa. \quad (4486)$$

For $m = 2$, we obtain

$$\frac{d\mu_2}{d\kappa} = \frac{2}{N_H} \text{Tr} (H H_\kappa). \quad (4487)$$

These relations connect the DOS reconstruction directly to the spectral-moment hierarchy derived in V.B.

c. Theorem 74. Exact curvature continuity equation for the density of states. For every differentiable finite-dimensional Hamiltonian family $H(\kappa)$, the normalized spectral density satisfies

$$\partial_\kappa \rho + \partial_E J_\kappa = 0, \quad (4488)$$

with

$$J_\kappa(E) = \frac{1}{N_H} \text{Tr} [H_\kappa \delta(E - H)]. \quad (4489)$$

Thus curvature does not create or destroy spectral states. It transports them through energy space. The same relation survives the finite-cover thermodynamic limit whenever the corresponding spectral measures and current measures converge. For numerical calculations, define

$$\rho_\eta(E) = -\frac{1}{\pi N_H} \text{Im Tr} [E + i\eta - H]^{-1}. \quad (4490)$$

Let

$$R_\eta(E) = [E + i\eta - H]^{-1}. \quad (4491)$$

Differentiation gives

$$\partial_\kappa R_\eta = R_\eta H_\kappa R_\eta. \quad (4492)$$

Therefore,

$$\partial_\kappa \rho_\eta(E) = -\frac{1}{\pi N_H} \text{Im Tr} [R_\eta H_\kappa R_\eta]. \quad (4493)$$

Using

$$\|R_\eta\| \leq \frac{1}{\eta}, \quad (4494)$$

we obtain the uniform bound

$$|\partial_\kappa \rho_\eta(E)| \leq \frac{\|H_\kappa\|}{\pi \eta^2}. \quad (4495)$$

This provides a numerical scale against which a curvature-induced DOS change can be established. Let the ordered bands at a fixed generalized momentum be $E_1(\lambda) \leq E_2(\lambda) \leq \dots$. For an isolated candidate band n , define the lower local separation

$$\delta_n^-(\lambda) = E_n(\lambda) - E_{n-1}(\lambda), \quad (4496)$$

and the upper local separation

$$\delta_n^+(\lambda) = E_{n+1}(\lambda) - E_n(\lambda). \quad (4497)$$

The local direct gap is

$$\delta_n^{\text{dir}}(\lambda) = \min [\delta_n^-(\lambda), \delta_n^+(\lambda)]. \quad (4498)$$

The global direct gap is

$$\Delta_n^{\text{dir}} = \inf_\lambda \delta_n^{\text{dir}}(\lambda). \quad (4499)$$

A positive value, $\Delta_n^{\text{dir}} > 0$, means that the band is separated from its neighbors at every individual generalized momentum. Global isolation additionally requires a positive spectral interval. Define

$$E_n^{\min} = \inf_\lambda E_n(\lambda), \quad E_n^{\max} = \sup_\lambda E_n(\lambda). \quad (4500)$$

The lower indirect gap is

$$\Delta_n^- = E_n^{\min} - E_{n-1}^{\max}, \quad (4501)$$

and the upper indirect gap is

$$\Delta_n^+ = E_{n+1}^{\min} - E_n^{\max}. \quad (4502)$$

We define the signed global isolation gap by

$$\Delta_n = \min [\Delta_n^-, \Delta_n^+]. \quad (4503)$$

The three regimes are therefore

$$\begin{aligned}\Delta_n > 0 &\implies \text{globally isolated band,} \\ \Delta_n = 0 &\implies \text{spectral touching,} \\ \Delta_n < 0 &\implies \text{indirect spectral overlap.}\end{aligned}\quad (4504)$$

A positive direct gap with a negative indirect gap therefore describes bands that never touch at equal generalized momentum but overlap in their global energy ranges. This distinction is essential for flat-band isolation. For a cluster $\mathcal{C} = \{a, a+1, \dots, b\}$, define

$$E_{\mathcal{C}}^{\min} = \inf_{\lambda} E_a(\lambda), \quad (4505)$$

and

$$E_{\mathcal{C}}^{\max} = \sup_{\lambda} E_b(\lambda). \quad (4506)$$

The lower isolation gap is

$$\Delta_{\mathcal{C}}^{-} = E_{\mathcal{C}}^{\min} - \sup_{\lambda} E_{a-1}(\lambda), \quad (4507)$$

and the upper isolation gap is

$$\Delta_{\mathcal{C}}^{+} = \inf_{\lambda} E_{b+1}(\lambda) - E_{\mathcal{C}}^{\max}. \quad (4508)$$

The signed cluster isolation gap is

$$\Delta_{\mathcal{C}} = \min [\Delta_{\mathcal{C}}^{-}, \Delta_{\mathcal{C}}^{+}]. \quad (4509)$$

This definition remains valid when states inside the target multiplet cross or remain exactly degenerate. Suppose the minimum upper direct gap occurs at a unique generalized momentum $\lambda_g^{+}(\kappa)$. Then

$$\Delta_{n,+}^{\text{dir}} = E_{n+1}(\lambda_g^{+}) - E_n(\lambda_g^{+}). \quad (4510)$$

At the minimizing point, the envelope theorem gives

$$\begin{aligned}\frac{d\Delta_{n,+}^{\text{dir}}}{d\kappa} &= \langle u_{n+1,g} | H_{\kappa} | u_{n+1,g} \rangle \\ &- \langle u_{n,g} | H_{\kappa} | u_{n,g} \rangle.\end{aligned}\quad (4511)$$

Thus curvature opens the direct gap at first order when the upper state has the larger curvature energy susceptibility. The upper indirect isolation gap is

$$\Delta_n^{+} = E_{n+1}^{\min} - E_n^{\max}. \quad (4512)$$

Assuming differentiable unique extrema,

$$\begin{aligned}\frac{d\Delta_n^{+}}{d\kappa} &= \langle u_{n+1,\min} | H_{\kappa} | u_{n+1,\min} \rangle \\ &- \langle u_{n,\max} | H_{\kappa} | u_{n,\max} \rangle.\end{aligned}\quad (4513)$$

Similarly,

$$\begin{aligned}\frac{d\Delta_n^{-}}{d\kappa} &= \langle u_{n,\min} | H_{\kappa} | u_{n,\min} \rangle \\ &- \langle u_{n-1,\max} | H_{\kappa} | u_{n-1,\max} \rangle.\end{aligned}\quad (4514)$$

If one of the two gaps is strictly smaller than the other, it controls Δ_n locally and its derivative equals the derivative of the active branch. At a switching point, $\Delta_n^{+} = \Delta_n^{-}$, the minimum is continuous but can develop a cusp.

d. Theorem 75. Spectral-edge criterion for curvature-induced gap opening. For a globally isolated band whose active upper isolation gap is Δ_n^{+} , curvature increases the isolation gap at first order precisely when

$$\langle u_{n+1,\min} | H_{\kappa} | u_{n+1,\min} \rangle > \langle u_{n,\max} | H_{\kappa} | u_{n,\max} \rangle. \quad (4515)$$

For a lower active gap, the corresponding condition is

$$\langle u_{n,\min} | H_{\kappa} | u_{n,\min} \rangle > \langle u_{n-1,\max} | H_{\kappa} | u_{n-1,\max} \rangle. \quad (4516)$$

Gap opening is therefore controlled by differences of curvature susceptibilities at different spectral edges rather than by a uniform energy shift. Let an approximate Hamiltonian satisfy

$$\tilde{H} = H + \delta H, \quad (4517)$$

with $\|\delta H\| \leq \epsilon$. Every eigenvalue obeys

$$|\tilde{E}_j - E_j| \leq \epsilon. \quad (4518)$$

Therefore a bandwidth satisfies

$$|\widetilde{W}_n - W_n| \leq 2\epsilon. \quad (4519)$$

Likewise, an indirect isolation gap satisfies

$$|\widetilde{\Delta}_n - \Delta_n| \leq 2\epsilon, \quad (4520)$$

provided the same target cluster is followed continuously. Thus a numerically observed positive gap is established only if $\widetilde{\Delta}_n > 2\epsilon$. A measured finite bandwidth is resolved only if $\widetilde{W}_n > 2\epsilon$. If $\widetilde{W}_n \leq 2\epsilon$, the data cannot distinguish a finite narrow bandwidth from an exactly flat level. We identify $\epsilon = \epsilon_{\text{tot}}$, with the complete numerical error budget,

$$\epsilon_{\text{tot}} \leq \epsilon_{\text{tr}} + \epsilon_{\text{cover}} + \epsilon_{\theta} + \epsilon_{\text{solver}} + \epsilon_{\text{fp}}. \quad (4521)$$

Therefore the established physical intervals are

$$W_n \in \left[\max \left(0, \widetilde{W}_n - 2\epsilon_{\text{tot}} \right), \widetilde{W}_n + 2\epsilon_{\text{tot}} \right], \quad (4522)$$

and

$$\Delta_n \in \left[\widetilde{\Delta}_n - 2\epsilon_{\text{tot}}, \widetilde{\Delta}_n + 2\epsilon_{\text{tot}} \right]. \quad (4523)$$

These bounds are used directly in every subsequent flatness claim. For an isolated band satisfying $W_n > 0$ and $\Delta_n > 0$, we define the conventional flatness ratio by

$$\mathcal{F}_n = \frac{\Delta_n}{W_n}. \quad (4524)$$

For a cluster,

$$\mathcal{F}_C = \frac{\Delta_C}{W_C}. \quad (4525)$$

A large value requires two simultaneous conditions: $W_C \ll 1$ in the chosen energy units, and $\Delta_C > 0$. A small bandwidth with $\Delta_C \leq 0$ is therefore not an isolated flat band. Because $\mathcal{F}_n$ diverges when $W_n \rightarrow 0$, we also define the bounded isolation-flatness index

$$\Phi_n = \frac{\Delta_n}{\Delta_n + W_n}, \quad (4526)$$

whenever $\Delta_n > 0$. The relation to the conventional ratio is

$$\Phi_n = \frac{\mathcal{F}_n}{1 + \mathcal{F}_n}. \quad (4527)$$

Hence $0 < \Phi_n < 1$ for a finite isolated band. The limit $\Phi_n \rightarrow 1$ corresponds to an isolated flat-band limit $W_n \rightarrow 0$ with a finite positive gap. The limit $\Phi_n \rightarrow 0$ corresponds to loss of spectral isolation. Using the error bounds above, a rigorous lower bound on the physical flatness ratio is

$$\mathcal{F}_n^{\text{cert}} = \frac{\widetilde{\Delta}_n - 2\epsilon_{\text{tot}}}{\widetilde{W}_n + 2\epsilon_{\text{tot}}}, \quad (4528)$$

provided $\widetilde{\Delta}_n > 2\epsilon_{\text{tot}}$. Then $\mathcal{F}_n \geq \mathcal{F}_n^{\text{cert}}$. Thus no flatness ratio is quoted without a corresponding error-established lower bound.

e. Theorem 76. Error-established isolation and flatness. If the total operator-level uncertainty is bounded by ϵ_{tot} , then

$$\left| \widetilde{W}_n - W_n \right| \leq 2\epsilon_{\text{tot}}, \quad (4529)$$

and

$$\left| \widetilde{\Delta}_n - \Delta_n \right| \leq 2\epsilon_{\text{tot}}. \quad (4530)$$

Therefore $\widetilde{\Delta}_n > 2\epsilon_{\text{tot}}$ is sufficient to certify a positive spectral gap, and Eq. (4528) gives a rigorous lower bound on the physical flatness ratio. A small global bandwidth does not by itself guarantee small local dispersion everywhere. To quantify the generalized energy differential, choose a metric G on the spectral parameter domain and define

$$\mathcal{V}_n^\infty = \frac{1}{\hbar} \sup_{\lambda \in \mathfrak{D}} \|dE_n(\lambda)\|_{G^{-1}}. \quad (4531)$$

We also define the root-mean-square generalized dispersion,

$$\mathcal{V}_n^{\text{rms}} = \frac{1}{\hbar} \left[\frac{1}{V_G} \int_{\mathfrak{D}} d\mu_G \|dE_n\|_{G^{-1}}^2 \right]^{1/2}. \quad (4532)$$

For an ordinary Euclidean band,

$$\mathcal{V}_n^\infty = \sup_{\mathbf{k}} |\mathbf{v}_n(\mathbf{k})|, \quad (4533)$$

and

$$\mathcal{V}_n^{\text{rms}} = \left[\frac{1}{A_{\text{mBZ}}} \int_{\text{mBZ}} d^2k |\mathbf{v}_n(\mathbf{k})|^2 \right]^{1/2}. \quad (4534)$$

For a rank- d hyperbolic band, these quantities measure the complete dispersion over all physical representation-space directions selected by G . Let

$$D_G = \text{diam}_G(\mathfrak{D}) \quad (4535)$$

be the geodesic diameter of the generalized momentum domain. Let $\lambda_{\text{max}}, \lambda_{\text{min}}$ be points at which the band attains its maximum and minimum. Choose a minimizing geodesic $\gamma : [0, L] \rightarrow \mathfrak{D}$ joining these points, with $L \leq D_G$. Then

$$\begin{aligned} W_n &= E_n(\lambda_{\text{max}}) - E_n(\lambda_{\text{min}}) \\ &= \int_0^L dE_n[\dot{\gamma}(s)] ds. \end{aligned} \quad (4536)$$

Therefore,

$$W_n \leq D_G \sup_{\lambda} \|dE_n\|_{G^{-1}}. \quad (4537)$$

Equivalently,

$$W_n \leq \hbar D_G \mathcal{V}_n^\infty. \quad (4538)$$

Hence $\mathcal{V}_n^\infty \rightarrow 0$ forces $W_n \rightarrow 0$. The converse does not hold without additional regularity assumptions: a function can have small total range while containing rapidly oscillating local derivatives.

f. Theorem 77. Global energetic flatness from generalized velocity suppression. For every differentiable isolated generalized band on a connected compact spectral domain,

$$W_n \leq \hbar D_G \mathcal{V}_n^\infty. \quad (4539)$$

Thus suppression of the full representation-space energy differential provides a rigorous sufficient condition for global energetic flattening. A flat appearance along one selected one-dimensional path gives no such bound on the full generalized bandwidth. Combining Eqs. (4468) and (4538) gives

$$\|\rho_n\|_\infty \geq \frac{1}{W_n} \geq \frac{1}{\hbar D_G \mathcal{V}_n^\infty}. \quad (4540)$$

Therefore extended suppression of generalized velocity forces a corresponding DOS enhancement. This chain is one directional:

$$\mathcal{V}_n^\infty \rightarrow 0 \implies W_n \rightarrow 0 \implies \|\rho_n\|_\infty \rightarrow \infty. \quad (4541)$$

The reverse implications are not generally valid. For

$$\mathcal{F}_n = \frac{\Delta_n}{W_n}, \quad (4542)$$

the logarithmic curvature derivative is

$$\frac{\partial \ln \mathcal{F}_n}{\partial \kappa} = \frac{1}{\Delta_n} \frac{\partial \Delta_n}{\partial \kappa} - \frac{1}{W_n} \frac{\partial W_n}{\partial \kappa}. \quad (4543)$$

We define the curvature flatness susceptibility

$$\chi_{\mathcal{F},n} = \frac{\partial \ln \mathcal{F}_n}{\partial \kappa}. \quad (4544)$$

Thus $\chi_{\mathcal{F},n} > 0$ means that negative curvature increases the isolation gap relative to the bandwidth. Using the active upper indirect gap, one obtains explicitly

$$\chi_{\mathcal{F},n} = \frac{\langle u_{n+1,\min} | H_\kappa | u_{n+1,\min} \rangle - \langle u_{n,\max} | H_\kappa | u_{n,\max} \rangle}{\Delta_n} - \frac{\langle u_{n,\max} | H_\kappa | u_{n,\max} \rangle - \langle u_{n,\min} | H_\kappa | u_{n,\min} \rangle}{W_n}. \quad (4545)$$

This formula separates gap opening and bandwidth suppression into two independent curvature mechanisms. A positive $\chi_{\mathcal{F},n}$ can therefore arise from bandwidth narrowing, gap opening, or both. For

$$\Phi_n = \frac{\Delta_n}{\Delta_n + W_n}, \quad (4546)$$

differentiation gives

$$\frac{\partial \Phi_n}{\partial \kappa} = \frac{W_n \partial_\kappa \Delta_n - \Delta_n \partial_\kappa W_n}{(\Delta_n + W_n)^2}. \quad (4547)$$

Therefore $\partial_\kappa \Phi_n > 0$ if and only if

$$\frac{\partial_\kappa \Delta_n}{\Delta_n} > \frac{\partial_\kappa W_n}{W_n}. \quad (4548)$$

This is equivalent to $\chi_{\mathcal{F},n} > 0$. Thus the bounded and unbounded flatness diagnostics contain the same local curvature information. To separate overall energy rescaling from genuine spectral improvement, we define the dimensionless logarithmic response vector

$$\mathbf{R}_n^{\text{spec}} = \begin{pmatrix} \partial_\kappa \ln W_n \\ \partial_\kappa \ln \Delta_n \end{pmatrix}. \quad (4549)$$

A uniform energy rescaling $H \rightarrow c(\kappa)H$ gives

$$\partial_\kappa \ln W_n = \partial_\kappa \ln \Delta_n = \partial_\kappa \ln c. \quad (4550)$$

Hence $\chi_{\mathcal{F},n} = 0$. A nonzero curvature flatness susceptibility therefore cannot be generated by a purely uniform rescaling of the complete Hamiltonian. This provides a strict test for nontrivial spectral-shape reconstruction.

g. Theorem 78. Flatness susceptibility removes uniform energy rescaling. If curvature changes the Hamiltonian only through a positive scalar factor,

$$H(\kappa) = c(\kappa)H(0), \quad (4551)$$

then

$$W_n(\kappa) = c(\kappa)W_n(0), \quad (4552)$$

and

$$\Delta_n(\kappa) = c(\kappa)\Delta_n(0). \quad (4553)$$

Consequently,

$$\mathcal{F}_n(\kappa) = \mathcal{F}_n(0), \quad \chi_{\mathcal{F},n} = 0. \quad (4554)$$

Therefore curvature-dependent variation of Δ_n/W_n is a direct signature of spectral-shape reconstruction rather than overall energy-scale renormalization. Suppose a representation-theoretic multiplet has exact degeneracy

$$E_1(\lambda) = E_2(\lambda) = \cdots = E_r(\lambda) \quad (4555)$$

at each λ . The internal splitting is then $W_{\text{internal}} = 0$. This does not imply that the multiplet is flat. The physically relevant bandwidth is instead

$$W_C = \sup_{\lambda} E_C(\lambda) - \inf_{\lambda} E_C(\lambda). \quad (4556)$$

Thus exact representation multiplicity removes internal level splitting but does not remove generalized momentum dispersion. Every flatness calculation is therefore performed after symmetry multiplicity has been factored into the cluster projector. For a purely folded Euclidean band,

$$E_{\alpha}^{(0)}(\mathbf{k}) = \varepsilon_0(\mathbf{k} + \mathbf{Q}_{\alpha}). \quad (4557)$$

The velocity is unchanged at the corresponding unfolded momentum,

$$\nabla_{\mathbf{k}} E_{\alpha}^{(0)} = \nabla_{\mathbf{q}} \varepsilon_0. \quad (4558)$$

Although an individual plotted segment inside a reduced mBZ may appear short in energy over one selected path, the unfolded spectral family retains the original kinetic dispersion. We therefore regard a reduction of W_n as physical only when it survives the unfolded-weight analysis and cannot be reproduced at $w = 0$. At a nondegenerate saddle point, $dE_n = 0$, but the Hessian has mixed signature. Such a point can produce a large or singular DOS even when $W_n = O(t)$. Therefore the combination

$$\|\rho_n\|_{\infty} \gg t^{-1} \quad (4559)$$

is not sufficient for a flat-band claim. We require a simultaneous reduction of the global bandwidth, W_n , and the complete generalized differential, $\mathcal{V}_n^{\infty}$. The DOS then serves as a complementary spectral consequence rather than the sole diagnostic. For a finite-cover sequence define $W_{C,N}$, $\Delta_{C,N}$. We define the upper thermodynamic bandwidth by

$$W_C^{\text{sup}} = \limsup_{N \rightarrow \infty} W_{C,N}, \quad (4560)$$

and the lower thermodynamic isolation gap by

$$\Delta_C^{\text{inf}} = \liminf_{N \rightarrow \infty} \Delta_{C,N}. \quad (4561)$$

A persistent bulk spectral gap is established by $\Delta_C^{\text{inf}} > 0$. A thermodynamic flat-band limit requires $W_C^{\text{sup}} = 0$. When both quantities are finite and positive, define the conservative bulk flatness ratio

$$\mathcal{F}_C^{(\infty)} = \frac{\Delta_C^{\text{inf}}}{W_C^{\text{sup}}}. \quad (4562)$$

This definition remains meaningful even before separate convergence of every finite-cover sequence has been numerically established. If $W_C^{\text{sup}} = 0$ and $\Delta_C^{\text{inf}} > 0$, the target cluster is an isolated exactly flat thermodynamic spectral manifold.

h. Theorem 79. Conservative thermodynamic flatness criterion. Let a sequence of symmetry-resolved finite covers represent the same adiabatically tracked target cluster. If

$$\limsup_{N \rightarrow \infty} W_{C,N} = W_C^{\text{sup}}, \quad (4563)$$

and

$$\liminf_{N \rightarrow \infty} \Delta_{C,N} = \Delta_C^{\text{inf}} > 0, \quad (4564)$$

then

$$\mathcal{F}_C^{(\infty)} = \frac{\Delta_C^{\text{inf}}}{W_C^{\text{sup}}} \quad (4565)$$

is a lower-conservative measure of bulk spectral isolation relative to dispersion. A finite-size level spacing that vanishes with increasing injectivity radius cannot contribute a positive Δ_C^{inf} . To quantify spectral reconstruction relative to the Euclidean system, we define the bandwidth compression factor

$$\mathcal{R}_{W,n}(\kappa, \theta, \omega) = \frac{W_n(\kappa, \theta, \omega)}{W_n(0, \theta, \omega)}. \quad (4566)$$

Curvature-induced narrowing corresponds to $\mathcal{R}_{W,n} < 1$. The isolation-gap ratio is

$$\mathcal{R}_{\Delta,n}(\kappa, \theta, \omega) = \frac{\Delta_n(\kappa, \theta, \omega)}{\Delta_n(0, \theta, \omega)}, \quad (4567)$$

whenever the Euclidean reference gap is nonzero. The generalized velocity compression factor is

$$\mathcal{R}_{V,n} = \frac{\mathcal{V}_n^{\infty}(\kappa, \theta, \omega)}{\mathcal{V}_n^{\infty}(0, \theta, \omega)}. \quad (4568)$$

Finally, the flatness enhancement factor is

$$\mathcal{R}_{\mathcal{F},n} = \frac{\mathcal{F}_n(\kappa, \theta, \omega)}{\mathcal{F}_n(0, \theta, \omega)}. \quad (4569)$$

A pure energy rescaling gives $\mathcal{R}_{\mathcal{F},n} = 1$. Thus $\mathcal{R}_{\mathcal{F},n} \neq 1$ directly identifies nonuniform spectral reconstruction. For a fixed broadening or for a regular continuous DOS, define

$$\mathcal{R}_{\rho,n} = \frac{\|\rho_n(E; \kappa, \theta, \omega)\|_\infty}{\|\rho_n(E; 0, \theta, \omega)\|_\infty}. \quad (4570)$$

A curvature-induced DOS enhancement satisfies $\mathcal{R}_{\rho,n} > 1$. However, because DOS enhancement can arise from a local saddle-point reconstruction, we do not identify $\mathcal{R}_{\rho,n} > 1$ with global flatness unless it is accompanied by $\mathcal{R}_{W,n} < 1$ and suppression of the full generalized velocity diagnostic. We therefore represent each candidate band or cluster by the triplet

$$\mathfrak{S}_n = (W_n, \Delta_n, \mathcal{V}_n^\infty). \quad (4571)$$

The DOS provides an additional derived function $\rho_n(E)$. The triplet separates three logically independent requirements. The first component measures global energetic dispersion. The second measures global spectral isolation. The third measures the largest local generalized dispersion response. An isolated globally flat band therefore approaches

$$W_n \longrightarrow 0, \quad \Delta_n > 0, \quad \mathcal{V}_n^\infty \longrightarrow 0. \quad (4572)$$

The DOS enhancement then follows as a spectral consequence. The unified diagnostics allow us to distinguish several reconstruction regimes quantitatively. A predominantly rigid energy rescaling satisfies

$$\frac{\partial_\kappa W_n}{W_n} = \frac{\partial_\kappa \Delta_n}{\Delta_n}, \quad (4573)$$

and therefore $\chi_{\mathcal{F},n} = 0$. A bandwidth-dominated reconstruction satisfies $\partial_\kappa W_n < 0$, with comparatively weak gap response. A gap-dominated reconstruction satisfies $\partial_\kappa \Delta_n > 0$, without strong bandwidth suppression. A simultaneous isolation-and-compression regime satisfies

$$\partial_\kappa W_n < 0, \quad \partial_\kappa \Delta_n > 0. \quad (4574)$$

This regime gives the strongest positive curvature flatness susceptibility, $\chi_{\mathcal{F},n} > 0$. A DOS-only reconstruction satisfies $\mathcal{R}_{\rho,n} > 1$ without a substantial reduction of W_n ; such a feature is interpreted as spectral concentration rather than global flattening. The diagnostics also distinguish resonant and off-resonant interlayer reconstruction. Away from resonances,

$$\delta E = O\left(\frac{w^2}{t}\right), \quad (4575)$$

so that the leading bandwidth change is generically

$$\delta W_n = O\left(\frac{w^2}{t}\right). \quad (4576)$$

Near an isolated resonance,

$$\Delta_{\text{hyb}} = 2|V| = O(w), \quad (4577)$$

and gap opening is linear in w . Thus a numerical sweep in $\omega = \frac{w}{t}$ can distinguish the two mechanisms through the scaling laws $\Delta_{\text{res}} \propto \omega$, and $\delta W_{\text{off}} \propto \omega^2$. Curvature modifies both coefficients through the geometry-dependent tunneling matrix and the reconstructed energy denominators. The genuinely nonadditive effect of curvature and twist can be isolated at the level of the bandwidth. Define

$$\begin{aligned} \delta_{\kappa\theta} W_n &= W_n(\kappa, \theta) - W_n(\kappa, 0) \\ &\quad - W_n(0, \theta) + W_n(0, 0). \end{aligned} \quad (4578)$$

Similarly,

$$\begin{aligned} \delta_{\kappa\theta} \Delta_n &= \Delta_n(\kappa, \theta) - \Delta_n(\kappa, 0) \\ &\quad - \Delta_n(0, \theta) + \Delta_n(0, 0). \end{aligned} \quad (4579)$$

For identical centered layers, scalar spectral observables are even in θ . Therefore, whenever the functions are analytic at the origin,

$$\delta_{\kappa\theta} W_n = O(\kappa\theta^2), \quad (4580)$$

and

$$\delta_{\kappa\theta} \Delta_n = O(\kappa\theta^2). \quad (4581)$$

The corresponding mixed flatness response is

$$\begin{aligned} \delta_{\kappa\theta} \ln \mathcal{F}_n &= \ln \mathcal{F}_n(\kappa, \theta) - \ln \mathcal{F}_n(\kappa, 0) \\ &\quad - \ln \mathcal{F}_n(0, \theta) + \ln \mathcal{F}_n(0, 0). \end{aligned} \quad (4582)$$

A nonzero value isolates curvature-assisted twistrionic flattening from the sum of pure-curvature and pure-twist effects. For each periodic cover we compute $W_{\mathcal{C},N}$, $\Delta_{\mathcal{C},N}$, $\rho_N(E)$. We additionally compute

$$\mathcal{F}_{\mathcal{C},N} = \frac{\Delta_{\mathcal{C},N}}{W_{\mathcal{C},N}}. \quad (4583)$$

The convergence test requires $r_{\text{inj}}^{(N)} \rightarrow \infty$, $\epsilon_{\text{tr}}^{(N)} \rightarrow 0$, and $\epsilon_{\text{solver}}^{(N)} \rightarrow 0$. For an incommensurate approximant sequence we additionally require

$$|\theta_N - \theta|_{e^{L_N/R}} \rightarrow 0. \quad (4584)$$

A curvature-dependent suppression of $W_{\mathcal{C},N}$ is regarded as a bulk effect only if the same trend persists under increasing $r_{\text{inj}}^{(N)}$ and decreasing total numerical error. Let $\{Q_N^{(A)}\}$ and $\{Q_N^{(B)}\}$ be two inequivalent normal-cover families. We require

$$\lim_{N \rightarrow \infty} W_{\mathcal{C},N}^{(A)} = \lim_{N \rightarrow \infty} W_{\mathcal{C},N}^{(B)}, \quad (4585)$$

and

$$\liminf_{N \rightarrow \infty} \Delta_{\mathcal{C},N}^{(A)} = \liminf_{N \rightarrow \infty} \Delta_{\mathcal{C},N}^{(B)}. \quad (4586)$$

Likewise, the normalized spectral measures must converge to the same bulk limit, $d\nu_{\infty}^{(A)} = d\nu_{\infty}^{(B)}$. This removes finite-quotient topology as an alternative explanation for an apparently narrow band or spectral gap. For every selected twist and curvature, we therefore calculate the symmetry-resolved spectrum together with the three primary diagnostics W_n , $\rho(E)$, Δ_n . For Euclidean benchmark structures, we evaluate W_n and Δ_n over the complete mBZ rather than only along the displayed high-symmetry path. The path spectrum is used for visualization. The full two-dimensional mBZ is used for quantitative extrema. For Abelian hyperbolic bands, the corresponding extrema are evaluated over $\mathbb{T}^{2g_M}$, or over a converged sampling thereof. For higher-rank sectors, we use a stated representation-space domain and metric, together with finite-cover sampling whenever a complete continuous parameterization is unavailable. The complete finite-cover DOS is always weighted by the exact state factors $\frac{d^2 \rho}{|Q_N|}$. The complete quantitative hierarchy can now be stated precisely. Global energetic flattening is measured by W_n . Spectral isolation is measured by Δ_n . Generalized local dispersion is measured by

$$\mathcal{V}_n^{\infty} = \frac{1}{\hbar} \sup_{\lambda} \|dE_n\|_{G^{-1}}. \quad (4587)$$

Spectral concentration is measured by $\rho_n(E)$ and by $W_{\rho,n}^{\text{eff}}$. The conventional isolation-to-bandwidth ratio is

$$\mathcal{F}_n = \frac{\Delta_n}{W_n}. \quad (4588)$$

These quantities obey the rigorous implication chain

$$\mathcal{V}_n^{\infty} \rightarrow 0 \implies W_n \rightarrow 0 \implies \|\rho_n\|_{\infty} \rightarrow \infty, \quad (4589)$$

whereas spectral isolation requires the independent condition $\Delta_n > 0$. Thus a fully isolated flat-band limit requires simultaneously $W_n \rightarrow 0$, $\mathcal{V}_n^{\infty} \rightarrow 0$, and $\liminf \Delta_n > 0$. No single DOS peak, symmetry degeneracy, folded-band multiplicity, or one-dimensional low-slope cut satisfies these conditions by itself.

i. Theorem 80. Unified bandwidth-DOS-gap flatness criterion. Let an adiabatically tracked band or spectral cluster remain well defined along a converged Euclidean or hyperbolic spectral family. Then its global energetic dispersion is quantified by $W_{\mathcal{C}}$, its global spectral isolation by $\Delta_{\mathcal{C}}$, and its complete first-order generalized dispersion by $\mathcal{V}_{\mathcal{C}}^{\infty}$. The inequalities

$$W_{\mathcal{C}} \leq \hbar D_G \mathcal{V}_{\mathcal{C}}^{\infty}, \quad \|\rho_{\mathcal{C}}\|_{\infty} \geq \frac{1}{W_{\mathcal{C}}} \quad (4590)$$

hold for a normalized isolated cluster. Consequently, complete generalized-velocity suppression forces global bandwidth suppression, which in turn forces DOS enhancement. A spectrally isolated flat-band limit additionally requires

$$\liminf \Delta_{\mathcal{C}} > 0. \quad (4591)$$

This establishes a strict hierarchy generalized dispersion suppression $\implies$ global energetic flattening $\implies$ DOS concentration, with spectral isolation supplied by the independent positive-gap condition.

C. Group Velocity and Effective Dispersion

The preceding spectral analysis establishes the global bandwidth, density-of-states distribution, and spectral isolation of the reconstructed bands. We now resolve the local differential structure of those spectra. This step is indispensable because a globally narrow band, a locally stationary band, and a band with a small effective dispersion tensor are not equivalent statements. For an ordinary Euclidean crystal, the relevant local object is the group velocity

$$\mathbf{v}_n(\mathbf{k}) = \frac{1}{\hbar} \nabla_{\mathbf{k}} E_n(\mathbf{k}). \quad (4592)$$

For a hyperbolic crystal, an ordinary two-dimensional crystal momentum does not exist globally. The energy is instead a function on the unitary representation space of the exact common Fuchsian translation group. Consequently, the correct local dispersion object is the complete differential

$$dE_n \in T_{[\rho]}^* \mathcal{B}_d, \quad (4593)$$

rather than a two-component momentum gradient. We derive below an explicit hierarchy connecting these two

descriptions. The Euclidean group velocity emerges exactly as the rank-one Abelian special case, whereas the non-Abelian theory produces a coordinate-independent dispersion covector, a canonical geometry-induced norm, and a generalized effective-dispersion tensor. Let the exact Euclidean moiré Bloch Hamiltonian satisfy

$$H_M(\mathbf{k}) |u_{n\mathbf{k}}\rangle = E_n(\mathbf{k}) |u_{n\mathbf{k}}\rangle, \quad (4594)$$

with normalization

$$\langle u_{n\mathbf{k}} | u_{n\mathbf{k}} \rangle = 1. \quad (4595)$$

Differentiation with respect to k_μ gives

$$\frac{\partial H_M}{\partial k_\mu} |u_{n\mathbf{k}}\rangle + H_M |\partial_\mu u_{n\mathbf{k}}\rangle = \frac{\partial E_n}{\partial k_\mu} |u_{n\mathbf{k}}\rangle + E_n |\partial_\mu u_{n\mathbf{k}}\rangle. \quad (4596)$$

Multiplication from the left by $\langle u_{n\mathbf{k}} |$ gives

$$\frac{\partial E_n}{\partial k_\mu} = \langle u_{n\mathbf{k}} | \frac{\partial H_M}{\partial k_\mu} | u_{n\mathbf{k}} \rangle. \quad (4597)$$

Therefore,

$$\mathbf{v}_n(\mathbf{k}) = \frac{1}{\hbar} \langle u_{n\mathbf{k}} | \nabla_{\mathbf{k}} H_M(\mathbf{k}) | u_{n\mathbf{k}} \rangle. \quad (4598)$$

This expression is exact for every isolated nondegenerate Euclidean moiré band. In the exact folded basis,

$$H_{\text{fold}}(\mathbf{k}) = \begin{pmatrix} E_1^{\text{fold}}(\mathbf{k}) & \mathcal{T}(\mathbf{k}) \\ \mathcal{T}^\dagger(\mathbf{k}) & E_2^{\text{fold}}(\mathbf{k}) \end{pmatrix}. \quad (4599)$$

Write the normalized eigenstate as

$$|u_{n\mathbf{k}}\rangle = \sum_{\alpha=1}^{\Sigma} u_{1\alpha}^{(n)} |1, \alpha; \mathbf{k}\rangle + \sum_{\beta=1}^{\Sigma} u_{2\beta}^{(n)} |2, \beta; \mathbf{k}\rangle. \quad (4600)$$

The normalization condition is

$$\sum_{\alpha=1}^{\Sigma} |u_{1\alpha}^{(n)}|^2 + \sum_{\beta=1}^{\Sigma} |u_{2\beta}^{(n)}|^2 = 1. \quad (4601)$$

Differentiating the folded Hamiltonian gives

$$\nabla_{\mathbf{k}} H_{\text{fold}} = \begin{pmatrix} \nabla_{\mathbf{k}} E_1^{\text{fold}} & \nabla_{\mathbf{k}} \mathcal{T} \\ \nabla_{\mathbf{k}} \mathcal{T}^\dagger & \nabla_{\mathbf{k}} E_2^{\text{fold}} \end{pmatrix}. \quad (4602)$$

Substitution into Eq. (4598) yields

$$\begin{aligned} \hbar \mathbf{v}_n &= \sum_{\alpha} |u_{1\alpha}^{(n)}|^2 \nabla_{\mathbf{k}} E_{1\alpha}^{\text{fold}} \\ &+ \sum_{\beta} |u_{2\beta}^{(n)}|^2 \nabla_{\mathbf{k}} E_{2\beta}^{\text{fold}} \\ &+ 2 \text{Re} \sum_{\alpha, \beta} \left[u_{1\alpha}^{(n)} \right]^* u_{2\beta}^{(n)} \nabla_{\mathbf{k}} \mathcal{T}_{\alpha\beta}. \end{aligned} \quad (4603)$$

We therefore define the folded kinetic contribution

$$\hbar \mathbf{v}_n^{\text{kin}} = \sum_{\alpha} |u_{1\alpha}^{(n)}|^2 \nabla_{\mathbf{k}} E_{1\alpha}^{\text{fold}} + \sum_{\beta} |u_{2\beta}^{(n)}|^2 \nabla_{\mathbf{k}} E_{2\beta}^{\text{fold}}, \quad (4604)$$

and the explicit tunneling-gradient contribution

$$\hbar \mathbf{v}_n^{\text{tun}} = 2 \text{Re} \sum_{\alpha, \beta} \left[u_{1\alpha}^{(n)} \right]^* u_{2\beta}^{(n)} \nabla_{\mathbf{k}} \mathcal{T}_{\alpha\beta}. \quad (4605)$$

Hence,

$$\mathbf{v}_n = \mathbf{v}_n^{\text{kin}} + \mathbf{v}_n^{\text{tun}}. \quad (4606)$$

Equation (4606) provides an exact decomposition of the moiré group velocity into a wave-function redistribution term and a direct momentum dependence of the interlayer tunneling matrix. At zero interlayer coupling, $\mathcal{T} = 0$. An isolated folded state has

$$u_{1\alpha}^{(n)} = \delta_{\alpha\alpha_0}, \quad u_{2\beta}^{(n)} = 0, \quad (4607)$$

or the corresponding layer-two form. Therefore,

$$\mathbf{v}_n = \frac{1}{\hbar} \nabla_{\mathbf{k}} E_{1\alpha_0}^{\text{fold}}. \quad (4608)$$

Since

$$E_{1\alpha_0}^{\text{fold}}(\mathbf{k}) = \varepsilon_0(\mathbf{k} + \mathbf{Q}_{\alpha_0}), \quad (4609)$$

we obtain

$$\mathbf{v}_n = \frac{1}{\hbar} \nabla_{\mathbf{q}} \varepsilon_0(\mathbf{q}) \Big|_{\mathbf{q}=\mathbf{k}+\mathbf{Q}_{\alpha_0}}. \quad (4610)$$

Thus the reduction of the Brillouin-zone area does not reduce the underlying primitive velocity. Any genuine suppression of $\mathbf{v}_n$ requires hybridization-induced redistribution of the eigenvector, an explicit tunneling-gradient contribution, or both. Near an isolated crossing, the folded spectrum can be reduced to

$$H_2(\mathbf{k}) = \begin{pmatrix} \varepsilon_1(\mathbf{k}) & V(\mathbf{k}) \\ V^*(\mathbf{k}) & \varepsilon_2(\mathbf{k}) \end{pmatrix}. \quad (4611)$$

Define

$$\bar{\varepsilon} = \frac{\varepsilon_1 + \varepsilon_2}{2}, \quad (4612)$$

and $\Delta = \varepsilon_1 - \varepsilon_2$. The eigenvalues are

$$E_{\pm} = \bar{\varepsilon} \pm \Omega, \quad (4613)$$

where

$$\Omega = \sqrt{\frac{\Delta^2}{4} + |V|^2}. \quad (4614)$$

Differentiation gives

$$\nabla_{\mathbf{k}} \Omega = \frac{\frac{\Delta}{4} \nabla_{\mathbf{k}} \Delta + \text{Re}[V^* \nabla_{\mathbf{k}} V]}{\Omega}. \quad (4615)$$

Therefore,

$$\begin{aligned} \nabla_{\mathbf{k}} E_{\pm} &= \frac{\nabla_{\mathbf{k}} \varepsilon_1 + \nabla_{\mathbf{k}} \varepsilon_2}{2} \\ &\pm \frac{\frac{\Delta}{4} (\nabla_{\mathbf{k}} \varepsilon_1 - \nabla_{\mathbf{k}} \varepsilon_2) + \text{Re}[V^* \nabla_{\mathbf{k}} V]}{\sqrt{\Delta^2/4 + |V|^2}}. \end{aligned} \quad (4616)$$

The exact group velocities are

$$\mathbf{v}_{\pm} = \frac{1}{\hbar} \nabla_{\mathbf{k}} E_{\pm}. \quad (4617)$$

At exact resonance, $\Delta = 0$, we obtain

$$\nabla_{\mathbf{k}} E_{\pm} = \frac{\nabla_{\mathbf{k}} \varepsilon_1 + \nabla_{\mathbf{k}} \varepsilon_2}{2} \pm \frac{\text{Re}[V^* \nabla_{\mathbf{k}} V]}{|V|}. \quad (4618)$$

If the tunneling amplitude is locally momentum independent, $\nabla_{\mathbf{k}} V = 0$, then

$$\mathbf{v}_{\pm} = \frac{\mathbf{v}_1 + \mathbf{v}_2}{2}. \quad (4619)$$

Thus counterpropagating folded states satisfying $\mathbf{v}_1 \simeq -\mathbf{v}_2$ produce a strongly suppressed hybridized velocity. The exact cancellation condition at resonance is

$$\frac{\nabla_{\mathbf{k}} \varepsilon_1 + \nabla_{\mathbf{k}} \varepsilon_2}{2} = \mp \frac{\text{Re}[V^* \nabla_{\mathbf{k}} V]}{|V|}. \quad (4620)$$

This equation explicitly separates cancellation produced by opposite bare velocities from cancellation produced by the momentum texture of the interlayer tunneling.

a. Theorem 81. Exact two-level velocity reconstruction. For an isolated pair of hybridizing folded states, the reconstructed velocity is given exactly by Eq. (4616). At resonance, velocity suppression is governed by two independent terms: the average bare velocity and the tunneling-gradient contribution. Therefore a small hybridized velocity cannot in general be inferred solely from a large hybridization gap. We now derive the effective dispersion beyond a two-level truncation. Write the exact folded Hamiltonian as

$$H = \begin{pmatrix} H_1 & T \\ T^\dagger & H_2 \end{pmatrix}. \quad (4621)$$

The eigenvalue equation is

$$\begin{pmatrix} E - H_1 & -T \\ -T^\dagger & E - H_2 \end{pmatrix} \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} = 0. \quad (4622)$$

If $E - H_2$ is invertible, the second row gives

$$\psi_2 = R_2(E) T^\dagger \psi_1, \quad (4623)$$

where

$$R_2(E) = (E - H_2)^{-1}. \quad (4624)$$

Substitution into the first row gives

$$[E - H_1 - \Sigma_1(E)] \psi_1 = 0, \quad (4625)$$

with exact self-energy

$$\Sigma_1(E) = T R_2(E) T^\dagger. \quad (4626)$$

We therefore define the energy-dependent effective Hamiltonian

$$H_1^{\text{eff}}(E) = H_1 + \Sigma_1(E). \quad (4627)$$

The effective spectral problem is nonlinear in E . This energy dependence produces an additional velocity-renormalization factor that is absent from a static effective Hamiltonian. Since

$$R_2 = (E - H_2)^{-1}, \quad (4628)$$

we have

$$\frac{\partial R_2}{\partial E} = -R_2^2. \quad (4629)$$

Therefore,

$$\frac{\partial \Sigma_1}{\partial E} = -TR_2^2 T^\dagger. \quad (4630)$$

For a real eigenvalue outside the spectrum of H_2 , R_2 is Hermitian and hence

$$-\frac{\partial \Sigma_1}{\partial E} = TR_2^2 T^\dagger \succeq 0. \quad (4631)$$

Thus

$$I - \frac{\partial \Sigma_1}{\partial E} = I + TR_2^2 T^\dagger \succ 0. \quad (4632)$$

This positive operator is not an arbitrary normalization device. It is exactly the metric induced on the reduced layer-one eigenvector by eliminating layer two. Using

$$\psi_2 = R_2 T^\dagger \psi_1, \quad (4633)$$

the full-state normalization becomes

$$\begin{aligned} 1 &= \langle \psi_1 | \psi_1 \rangle + \langle \psi_2 | \psi_2 \rangle \\ &= \langle \psi_1 | [I + TR_2^2 T^\dagger] | \psi_1 \rangle. \end{aligned} \quad (4634)$$

Using Eq. (4630),

$$1 = \langle \psi_1 | [I - \partial_E \Sigma_1] | \psi_1 \rangle. \quad (4635)$$

Hence the same operator that renormalizes the effective dispersion also restores the exact probability carried by the eliminated layer. Let λ denote any continuous spectral parameter. In the Euclidean case, $\lambda = k_\mu$. In a generalized hyperbolic sector, λ may parameterize a physical representation-space direction. Differentiate

$$[E - H_1 - \Sigma_1(E, \lambda)] \psi_1 = 0. \quad (4636)$$

Taking the expectation value in the reduced eigenstate gives

$$\begin{aligned} 0 &= \frac{dE}{d\lambda} \langle \psi_1 | [I - \partial_E \Sigma_1] | \psi_1 \rangle \\ &\quad - \langle \psi_1 | [\partial_\lambda H_1 + (\partial_\lambda \Sigma_1)_E] | \psi_1 \rangle. \end{aligned} \quad (4637)$$

Therefore,

$$\frac{dE}{d\lambda} = \frac{\langle \psi_1 | [\partial_\lambda H_1 + (\partial_\lambda \Sigma_1)_E] | \psi_1 \rangle}{\langle \psi_1 | [I - \partial_E \Sigma_1] | \psi_1 \rangle}. \quad (4638)$$

Define the exact residue factor

$$Z_{\text{eff}} = [\langle \psi_1 | (I - \partial_E \Sigma_1) | \psi_1 \rangle]^{-1}. \quad (4639)$$

Then

$$\frac{dE}{d\lambda} = Z_{\text{eff}} \langle \psi_1 | [\partial_\lambda H_1 + (\partial_\lambda \Sigma_1)_E] | \psi_1 \rangle. \quad (4640)$$

Because

$$-\partial_E \Sigma_1 \succeq 0, \quad (4641)$$

we have $0 < Z_{\text{eff}} \leq 1$ for the Hermitian off-resonant problem. Thus interlayer hybridization produces two distinct dispersion renormalizations: $\partial_\lambda \Sigma_1$, which reconstructs the numerator, and Z_{eff} , which suppresses the complete effective dispersion through the energy dependence of the self-energy. At fixed energy,

$$\Sigma_1 = TR_2 T^\dagger. \quad (4642)$$

Therefore,

$$\begin{aligned} \partial_\lambda \Sigma_1 &= (\partial_\lambda T) R_2 T^\dagger + TR_2 (\partial_\lambda H_2) R_2 T^\dagger \\ &\quad + TR_2 (\partial_\lambda T^\dagger). \end{aligned} \quad (4643)$$

The first and third terms originate from the momentum or holonomy texture of the tunneling matrix. The middle term propagates the dispersion of the second layer through the virtual hybridization process. Consequently,

$$\begin{aligned} \frac{dE}{d\lambda} &= Z_{\text{eff}} \langle \psi_1 | \partial_\lambda H_1 | \psi_1 \rangle \\ &\quad + Z_{\text{eff}} \langle \psi_1 | (\partial_\lambda T) R_2 T^\dagger | \psi_1 \rangle \\ &\quad + Z_{\text{eff}} \langle \psi_1 | TR_2 (\partial_\lambda H_2) R_2 T^\dagger | \psi_1 \rangle \\ &\quad + Z_{\text{eff}} \langle \psi_1 | TR_2 (\partial_\lambda T^\dagger) | \psi_1 \rangle. \end{aligned} \quad (4644)$$

This decomposition provides a completely explicit microscopic separation between bare-layer dispersion, virtual propagation through the opposite layer, and tunneling-gradient reconstruction.

b. Theorem 82. Exact energy-dependent effective-dispersion identity. After one layer is eliminated exactly, the derivative of a nondegenerate eigenvalue along any continuous physical spectral parameter satisfies

$$\frac{dE}{d\lambda} = Z_{\text{eff}} \langle \psi_1 | [\partial_\lambda H_1 + (\partial_\lambda \Sigma_1)_E] | \psi_1 \rangle, \quad (4645)$$

with

$$Z_{\text{eff}} = [\langle \psi_1 | (I - \partial_E \Sigma_1) | \psi_1 \rangle]^{-1}. \quad (4646)$$

The factor Z_{eff} is fixed exactly by the norm carried by the eliminated layer. Thus hybridization-induced velocity suppression contains an intrinsically dynamical self-energy contribution even before any bandwidth minimum is imposed. Suppose the relevant energy lies far from

the second-layer states. Let the typical detuning satisfy $|\Delta| \gg \|T\|$. Then

$$R_2 = O(\Delta^{-1}), \quad \Sigma_1 = O\left(\frac{w^2}{\Delta}\right). \quad (4647)$$

Furthermore,

$$-\partial_E \Sigma_1 = O\left(\frac{w^2}{\Delta^2}\right). \quad (4648)$$

Therefore,

$$Z_{\text{eff}} = 1 - O\left(\frac{w^2}{\Delta^2}\right). \quad (4649)$$

The effective dispersion becomes

$$\frac{dE}{d\lambda} = \frac{d\varepsilon_{\text{bare}}}{d\lambda} + O\left(\frac{w^2}{\Delta}\right)_\lambda + O\left(\frac{w^2}{\Delta^2}\right) \frac{d\varepsilon_{\text{bare}}}{d\lambda}. \quad (4650)$$

Hence off-resonant velocity reconstruction is generically quadratic in w/t . Near resonance, this expansion fails and the full avoided-crossing theory must be used. For an isolated nondegenerate Euclidean band, define the inverse effective-mass tensor by

$$[m_n^{*-1}]_{\mu\nu} = \frac{1}{\hbar^2} \frac{\partial^2 E_n}{\partial k_\mu \partial k_\nu}. \quad (4651)$$

Differentiating the Hellmann–Feynman relation gives

$$\begin{aligned} \frac{\partial^2 E_n}{\partial k_\mu \partial k_\nu} &= \langle n | H_{\mu\nu} | n \rangle \\ &+ 2 \operatorname{Re} \sum_{m \neq n} \frac{\langle n | H_\mu | m \rangle \langle m | H_\nu | n \rangle}{E_n - E_m}, \end{aligned} \quad (4652)$$

where

$$H_\mu = \frac{\partial H}{\partial k_\mu}, \quad H_{\mu\nu} = \frac{\partial^2 H}{\partial k_\mu \partial k_\nu}. \quad (4653)$$

Thus

$$\begin{aligned} [m_n^{*-1}]_{\mu\nu} &= \frac{1}{\hbar^2} \langle n | H_{\mu\nu} | n \rangle \\ &+ \frac{2}{\hbar^2} \operatorname{Re} \sum_{m \neq n} \frac{\langle n | H_\mu | m \rangle \langle m | H_\nu | n \rangle}{E_n - E_m}. \end{aligned} \quad (4654)$$

The second term becomes strongly enhanced near small interband energy denominators. Therefore a reconstructed effective mass is not determined solely by the curvature of the bare folded dispersion. Interband hybridization provides an equally fundamental contribution. At a stationary point,

$$\nabla_{\mathbf{k}} E_n = 0, \quad (4655)$$

diagonalize the Hessian,

$$M_n = \nabla_{\mathbf{k}} \nabla_{\mathbf{k}} E_n. \quad (4656)$$

Let its eigenvalues be $\lambda_{n,1}^{(E)}, \lambda_{n,2}^{(E)}$. The principal inverse masses are

$$\frac{1}{m_{n,a}^*} = \frac{\lambda_{n,a}^{(E)}}{\hbar^2}, \quad a = 1, 2. \quad (4657)$$

A minimum satisfies

$$\lambda_{n,1}^{(E)} > 0, \quad \lambda_{n,2}^{(E)} > 0. \quad (4658)$$

A maximum satisfies

$$\lambda_{n,1}^{(E)} < 0, \quad \lambda_{n,2}^{(E)} < 0. \quad (4659)$$

A saddle satisfies $\lambda_{n,1}^{(E)} \lambda_{n,2}^{(E)} < 0$. A soft dispersion direction appears when $\lambda_{n,a}^{(E)} \rightarrow 0$. Thus velocity suppression and Hessian softening represent independent orders of dispersion suppression. We now construct the corresponding hyperbolic quantity. Let

$$\rho : \Gamma_M \rightarrow U(d) \quad (4660)$$

be a unitary representation. Let ρ_s be a smooth representation path satisfying $\rho_0 = \rho$. Define the right logarithmic derivative

$$X(\gamma) = \left. \frac{d\rho_s(\gamma)}{ds} \right|_{s=0} \rho(\gamma)^{-1}. \quad (4661)$$

The representation property implies the cocycle relation

$$X(\gamma\delta) = X(\gamma) + \operatorname{Ad}_{\rho(\gamma)} X(\delta). \quad (4662)$$

Thus

$$X \in Z^1(\Gamma_M, \operatorname{Ad} \rho). \quad (4663)$$

The generalized Bloch Hamiltonian is

$$H[\rho] = \sum_{\delta \in \Gamma_M} h(\delta) \otimes \rho(\delta). \quad (4664)$$

Its derivative along the cocycle is

$$\dot{H}_X = \sum_{\delta \in \Gamma_M} h(\delta) \otimes X(\delta) \rho(\delta). \quad (4665)$$

For a nondegenerate eigenstate,

$$H[\rho] |u_n\rangle = E_n[\rho] |u_n\rangle, \quad (4666)$$

we obtain

$$dE_n[X] = \langle u_n | \dot{H}_X | u_n \rangle. \quad (4667)$$

This is the exact non-Abelian analogue of

$$\mathbf{v}_n = \frac{1}{\hbar} \nabla_{\mathbf{k}} E_n. \quad (4668)$$

A gauge variation has the form

$$\rho_s = e^{sY} \rho e^{-sY}. \quad (4669)$$

Its cocycle is

$$X_Y(\gamma) = Y - \text{Ad}_{\rho(\gamma)} Y. \quad (4670)$$

The corresponding Hamiltonian derivative is

$$\dot{H}_{X_Y} = [I \otimes Y, H[\rho]]. \quad (4671)$$

Therefore, $dE_n[X_Y] = 0$. Hence dE_n descends to the quotient space

$$H^1(\Gamma_M, \text{Ad } \rho) = \frac{Z^1(\Gamma_M, \text{Ad } \rho)}{B^1(\Gamma_M, \text{Ad } \rho)}. \quad (4672)$$

The exact coordinate-independent stationary condition is therefore

$$dE_n = 0 \quad \text{on} \quad H^1(\Gamma_M, \text{Ad } \rho). \quad (4673)$$

A vanishing covector is metric independent. Quantifying the magnitude of a nonzero generalized dispersion, however, requires a metric. We remove this ambiguity by using the metric induced directly by the hyperbolic geometry. Let

$$\Sigma_M = \Gamma_M \backslash \mathbb{H}_R^2 \quad (4674)$$

be the compact hyperbolic moiré quotient. A tangent class

$$[X] \in H^1(\Gamma_M, \text{Ad } \rho) \quad (4675)$$

has a unique harmonic representative

$$\alpha_X \in \Omega^1(\Sigma_M, \text{Ad } E_\rho) \quad (4676)$$

at a smooth irreducible unitary point. We define the normalized Hodge inner product by

$$G_\rho(X, Y) = \frac{1}{A_M} \int_{\Sigma_M} \text{Re Tr} [\alpha_X^\dagger \wedge * \alpha_Y], \quad (4677)$$

where

$$A_M = \text{Area}(\Sigma_M). \quad (4678)$$

Because harmonic representatives are used, the metric depends only on the cohomology classes. Thus G_ρ provides a canonical geometry-induced metric on the physical generalized momentum tangent space. No arbitrary coordinate choice is required. Let $\{X_a\}_{a=1}^{D_d}$ be a local tangent basis, where

$$D_d = (2g_M - 2)d^2 + 2. \quad (4679)$$

Define

$$G_{ab} = G_\rho(X_a, X_b). \quad (4680)$$

Let $G^{ab} = (G^{-1})^{ab}$. The components of the energy covector are

$$\mathcal{E}_a^{(n)} = dE_n[X_a]. \quad (4681)$$

We define the Hodge-normalized generalized dispersion rate by

$$\Omega_n^{\text{gen}} = \frac{1}{\hbar} \left[G^{ab} \mathcal{E}_a^{(n)} \mathcal{E}_b^{(n)} \right]^{1/2}. \quad (4682)$$

Equivalently,

$$\Omega_n^{\text{gen}} = \frac{1}{\hbar} \|dE_n\|_{G_\rho^{-1}}. \quad (4683)$$

The same quantity can be characterized variationally,

$$\Omega_n^{\text{gen}} = \frac{1}{\hbar} \sup_{[X] \neq 0} \frac{|dE_n[X]|}{\sqrt{G_\rho(X, X)}}. \quad (4684)$$

Thus Ω_n^{gen} is the largest first-order spectral response among all unit Hodge-norm generalized momentum directions. Unlike the Euclidean group velocity, its natural physical dimension is a frequency because generalized holonomy coordinates are dimensionless. We therefore

call it a generalized dispersion rate rather than an ordinary real-space propagation velocity. For an Abelian Euclidean translation group,

$$\rho_{\mathbf{k}}(\mathbf{R}) = e^{i\mathbf{k} \cdot \mathbf{R}}. \quad (4685)$$

A momentum displacement $\delta\mathbf{k}$ generates

$$X_{\delta\mathbf{k}}(\mathbf{R}) = i\delta\mathbf{k} \cdot \mathbf{R}. \quad (4686)$$

The representation-space derivative becomes

$$dE_n[X_{\delta\mathbf{k}}] = \delta\mathbf{k} \cdot \nabla_{\mathbf{k}} E_n. \quad (4687)$$

Thus the generalized energy covector reduces exactly to the ordinary momentum gradient. After the Hodge metric is expressed in the conventional reciprocal coordinates, the physical Euclidean velocity is recovered as

$$\mathbf{v}_n = \frac{1}{\hbar} \nabla_{\mathbf{k}} E_n. \quad (4688)$$

Therefore the Hodge-normalized construction does not replace ordinary group velocity where ordinary crystal momentum exists. It extends the same differential concept to the non-Abelian hyperbolic setting.

c. Theorem 83. Canonical hyperbolic generalized-dispersion diagnostic. At a smooth irreducible representation point, the hyperbolic metric induces a Hodge metric on $H^1(\Gamma_M, \text{Ad } \rho)$. The coordinate-independent magnitude of first-order generalized dispersion is

$$\Omega_n^{\text{gen}} = \frac{1}{\hbar} \|dE_n\|_{G_\rho^{-1}}. \quad (4689)$$

It vanishes if and only if $dE_n = 0$ on the complete physical tangent space. Thus a one-dimensional path derivative equal to zero is not sufficient for a hyperbolic stationary band unless the complete Hodge-normalized covector also vanishes. At a stationary representation point, $dE_n = 0$, the leading energy variation is second order. Let X and Y be physical tangent directions. Define

$$\mathcal{M}_n(X, Y) = \nabla^2 E_n(X, Y). \quad (4690)$$

For a smooth two-parameter representation family, let $\dot{H}_X$ and $\dot{H}_Y$ be the first derivatives, and let $\ddot{H}_{XY}$ denote the mixed second derivative. Standard perturbation theory gives

$$\begin{aligned} \mathcal{M}_n(X, Y) &= \langle n | \ddot{H}_{XY} | n \rangle \\ &+ \sum_{m \neq n} \frac{\langle n | \dot{H}_X | m \rangle \langle m | \dot{H}_Y | n \rangle}{E_n - E_m} \\ &+ \sum_{m \neq n} \frac{\langle n | \dot{H}_Y | m \rangle \langle m | \dot{H}_X | n \rangle}{E_n - E_m}. \end{aligned} \quad (4691)$$

Equivalently,

$$\begin{aligned} \mathcal{M}_n(X, Y) &= \langle n | \ddot{H}_{XY} | n \rangle \\ &+ 2 \text{Re} \sum_{m \neq n} \frac{\langle n | \dot{H}_X | m \rangle \langle m | \dot{H}_Y | n \rangle}{E_n - E_m}. \end{aligned} \quad (4692)$$

This is the exact non-Abelian analogue of the Euclidean inverse-effective-mass tensor. Choose a tangent basis and define

$$M_{ab}^{(n)} = \mathcal{M}_n(X_a, X_b). \quad (4693)$$

The physically invariant principal dispersion curvatures are obtained from the generalized eigenvalue problem

$$M^{(n)} \mathbf{c}_a = \Lambda_{n,a} G \mathbf{c}_a. \quad (4694)$$

The eigenvalues $\Lambda_{n,a}$, $a = 1, \dots, D_d$, are invariant under changes of local tangent coordinates. We define the generalized inverse-dispersion tensor eigenvalues by

$$\mathbf{m}_{n,a}^{-1} = \frac{\Lambda_{n,a}}{\hbar^2}. \quad (4695)$$

These quantities are not ordinary mechanical masses for $d > 1$. They measure the second-order energetic response to unit geometry-normalized holonomy deformations. At a full stationary point, the signs of $\{\Lambda_{n,a}\}$ classify the local generalized dispersion. A strict generalized minimum satisfies $\Lambda_{n,a} > 0 \ \forall a$. A strict generalized maximum satisfies $\Lambda_{n,a} < 0 \ \forall a$. A generalized saddle contains both signs. A second-order soft direction satisfies $\Lambda_{n,a} = 0$ for at least one physical tangent direction. A fully second-order-flat stationary point satisfies $\Lambda_{n,a} = 0 \ \forall a$. This is substantially stronger than the first-order condition $dE_n = 0$. We define the maximal principal curvature

$$\Lambda_n^{\text{max}} = \max_a |\Lambda_{n,a}|, \quad (4696)$$

the root-mean-square curvature

$$\Lambda_n^{\text{rms}} = \left[\frac{1}{D_d} \sum_{a=1}^{D_d} \Lambda_{n,a}^2 \right]^{1/2}, \quad (4697)$$

and the geometric curvature scale

$$\Lambda_n^{\text{geom}} = \left| \prod_{a=1}^{D_d} \Lambda_{n,a} \right|^{1/D_d}. \quad (4698)$$

The determinant invariant is

$$\mathcal{D}_n^{(2)} = \frac{\det M^{(n)}}{\det G} = \prod_{a=1}^{D_d} \Lambda_{n,a}. \quad (4699)$$

The ratio is coordinate invariant because both determinants acquire the same Jacobian factor under a tangent-basis transformation. A vanishing value, $\mathcal{D}_n^{(2)} = 0$, signals at least one second-order soft generalized dispersion direction. Near a nondegenerate stationary point, V.C established that the representation-space DOS contains the factor

$$\left| \det M^{(n)} \right|^{-1/2}. \quad (4700)$$

Using the Hodge metric, we may write this invariantly as

$$\frac{\sqrt{\det G}}{\sqrt{|\det M^{(n)}|}} = \frac{1}{\sqrt{|\mathcal{D}_n^{(2)}|}}. \quad (4701)$$

Thus $\mathcal{D}_n^{(2)} \rightarrow 0$ produces a parametrically enhanced stationary-point DOS prefactor. This connects the local effective-dispersion tensor directly to the spectral concentration derived in V.C.

d. Theorem 84. Geometry-normalized generalized effective dispersion. At a full stationary representation point, the local second-order dispersion is characterized by the generalized eigenvalues

$$M^{(n)} \mathbf{c}_a = \Lambda_{n,a} G \mathbf{c}_a. \quad (4702)$$

The set $\{\Lambda_{n,a}\}$ is coordinate invariant. Its vanishing directions identify genuine second-order soft modes, and the invariant

$$\mathcal{D}_n^{(2)} = \frac{\det M^{(n)}}{\det G} \quad (4703)$$

controls the leading stationary-point DOS enhancement. Thus effective dispersion and DOS reconstruction are two manifestations of the same geometry-normalized Hessian. We now connect these diagnostics to the curvature-controlled Hamiltonian derived in V.B. For a Euclidean or continuously matched Abelian sector,

$$\mathbf{v}_n = \frac{1}{\hbar} \nabla_{\mathbf{k}} E_n. \quad (4704)$$

Differentiation with respect to $\kappa = -K$ gives

$$\partial_\kappa v_{n,\mu} = \frac{1}{\hbar} \partial_\kappa \partial_{k_\mu} E_n. \quad (4705)$$

For a nondegenerate eigenstate,

$$\begin{aligned} \partial_\kappa \partial_{k_\mu} E_n &= \langle n | H_{\kappa\mu} | n \rangle \\ &+ 2 \operatorname{Re} \sum_{m \neq n} \frac{\langle n | H_\kappa | m \rangle \langle m | H_\mu | n \rangle}{E_n - E_m}. \end{aligned} \quad (4706)$$

Therefore,

$$\begin{aligned} \partial_\kappa v_{n,\mu} &= \frac{1}{\hbar} \langle n | H_{\kappa\mu} | n \rangle \\ &+ \frac{2}{\hbar} \operatorname{Re} \sum_{m \neq n} \frac{\langle n | H_\kappa | m \rangle \langle m | H_\mu | n \rangle}{E_n - E_m}. \end{aligned} \quad (4707)$$

The first term is the direct curvature modification of the velocity operator. The second term arises from curvature-induced wave-function rotation through all other bands. Near a small interband gap, the second contribution can become parametrically large. For a fixed physical tangent direction X ,

$$dE_n[X] = \langle n | \dot{H}_X | n \rangle. \quad (4708)$$

Differentiation with respect to curvature gives

$$\begin{aligned} \partial_\kappa dE_n[X] &= \langle n | \partial_\kappa \dot{H}_X | n \rangle \\ &+ 2 \operatorname{Re} \sum_{m \neq n} \frac{\langle n | H_\kappa | m \rangle \langle m | \dot{H}_X | n \rangle}{E_n - E_m}. \end{aligned} \quad (4709)$$

Because

$$\dot{H}_X = \sum_{\delta} h(\delta) \otimes X(\delta) \rho(\delta), \quad (4710)$$

we have

$$\partial_\kappa \dot{H}_X = \sum_{\delta} \partial_\kappa h(\delta) \otimes X(\delta) \rho(\delta) \quad (4711)$$

when the representation coordinate is held fixed. Thus the microscopic curvature vertex derived in V.B feeds directly into every generalized dispersion direction.

e. Theorem 85. Curvature reconstruction of the full generalized dispersion covector. For every nondegenerate generalized Bloch state and every physical tangent class X ,

$$\begin{aligned} \partial_\kappa dE_n[X] &= \langle n | \partial_\kappa \dot{H}_X | n \rangle \\ &+ 2 \operatorname{Re} \sum_{m \neq n} \frac{\langle n | H_\kappa | m \rangle \langle m | \dot{H}_X | n \rangle}{E_n - E_m}. \end{aligned} \quad (4712)$$

Negative curvature therefore suppresses or enhances generalized dispersion through both a direct reconstruction

of the hopping kernel and an interband-mixing contribution amplified by small spectral denominators. The scalar diagnostic

$$[\Omega_n^{\text{gen}}]^2 = \frac{1}{\hbar^2} G^{ab} \mathcal{E}_a \mathcal{E}_b \quad (4713)$$

contains two curvature-dependent structures:

$$\mathcal{E}_a = dE_n[X_a], \quad (4714)$$

and the geometric metric G^{ab} . Differentiation gives

$$\partial_\kappa [\Omega_n^{\text{gen}}]^2 = \frac{2}{\hbar^2} G^{ab} \mathcal{E}_a \partial_\kappa \mathcal{E}_b + \frac{1}{\hbar^2} (\partial_\kappa G^{ab}) \mathcal{E}_a \mathcal{E}_b. \quad (4715)$$

The first term is the curvature reconstruction of the energy differential. The second term is the curvature reconstruction of the geometry used to measure generalized momentum directions. At an exactly stationary point, $\mathcal{E}_a = 0 \quad \forall a$, both terms vanish and first-order dispersion remains zero instantaneously. Away from exact stationarity, the two effects must be separated. For $\Omega_n^{\text{gen}} > 0$, we define

$$\chi_{\Omega,n} = -\partial_\kappa \ln \Omega_n^{\text{gen}}. \quad (4716)$$

Using Eq. (4715),

$$\chi_{\Omega,n} = -\frac{G^{ab} \mathcal{E}_a \partial_\kappa \mathcal{E}_b}{G^{cd} \mathcal{E}_c \mathcal{E}_d} - \frac{1}{2} \frac{(\partial_\kappa G^{ab}) \mathcal{E}_a \mathcal{E}_b}{G^{cd} \mathcal{E}_c \mathcal{E}_d}. \quad (4717)$$

We identify

$$\chi_{\Omega,n}^{\text{spec}} = -\frac{G^{ab} \mathcal{E}_a \partial_\kappa \mathcal{E}_b}{G^{cd} \mathcal{E}_c \mathcal{E}_d}, \quad (4718)$$

and

$$\chi_{\Omega,n}^{\text{geom}} = -\frac{1}{2} \frac{(\partial_\kappa G^{ab}) \mathcal{E}_a \mathcal{E}_b}{G^{cd} \mathcal{E}_c \mathcal{E}_d}. \quad (4719)$$

Therefore,

$$\chi_{\Omega,n} = \chi_{\Omega,n}^{\text{spec}} + \chi_{\Omega,n}^{\text{geom}}. \quad (4720)$$

This decomposition prevents a change of the generalized momentum metric from being misidentified as a change of the underlying energy covector. To isolate pure spectral reconstruction, choose a reference curvature κ_0 . Freeze the Hodge metric at $G_\rho^{(0)} = G_\rho(\kappa_0)$. Define the fixed-metric generalized dispersion rate

$$\Omega_n^{\text{fix}}(\kappa) = \frac{1}{\hbar} \|dE_n(\kappa)\|_{[G_\rho^{(0)}]^{-1}}. \quad (4721)$$

The ratio

$$\mathcal{R}_{\Omega,n}^{\text{spec}}(\kappa) = \frac{\Omega_n^{\text{fix}}(\kappa)}{\Omega_n^{\text{fix}}(0)} \quad (4722)$$

measures pure spectral-differential reconstruction. By contrast,

$$\mathcal{R}_{\Omega,n}^{\text{full}}(\kappa) = \frac{\Omega_n^{\text{gen}}(\kappa)}{\Omega_n^{\text{gen}}(0)} \quad (4723)$$

contains both spectral and geometric metric effects. The difference between the two ratios quantifies the contribution of curvature-induced deformation of generalized momentum geometry. For the Euclidean band, define

$$V_n^{(E)} = \sup_{\mathbf{k}} |\mathbf{v}_n(\mathbf{k})|. \quad (4724)$$

For the hyperbolic generalized band, define

$$\Omega_n^{(H)} = \sup_{[\rho] \in \mathfrak{D}_d} \Omega_n^{\text{gen}}[\rho]. \quad (4725)$$

Within a fixed spectral family, we define the dimensionless generalized dispersion compression factor

$$\mathcal{C}_n^{\text{disp}}(\kappa) = \frac{\Omega_n^{(H)}(\kappa)}{\Omega_n^{(H)}(0)}. \quad (4726)$$

Curvature-induced first-order dispersion suppression corresponds to $\mathcal{C}_n^{\text{disp}} < 1$. This quantity is evaluated only after the same spectral cluster has been matched continuously across curvature. Let

$$D_G = \text{diam}_G(\mathfrak{D}_d) \quad (4727)$$

be the Hodge-metric diameter of the generalized momentum domain. For any two points ρ_1 and ρ_2 , choose a minimizing geodesic γ . Then

$$|E_n(\rho_2) - E_n(\rho_1)| = \left| \int_\gamma dE_n \right| \leq \hbar \Omega_n^{(H)} \text{Length}_G(\gamma). \quad (4728)$$

Therefore, $W_n \leq \hbar D_G \Omega_n^{(H)}$. This gives the precise relation between the local generalized dispersion diagnostic introduced here and the global bandwidth analyzed in V.C. Although the numerical value of $\Omega_n^{(H)}$ depends on the Hodge normalization, the condition $dE_n = 0$ does not. If

$$dE_n = 0 \quad \text{at every point of a connected domain,} \quad (4729)$$

then for every differentiable path γ ,

$$\frac{d}{ds} E_n [\gamma(s)] = 0. \quad (4730)$$

Therefore, $E_n [\gamma(s)] = \text{constant}$. Since the domain is connected, $W_n = 0$. Thus exact vanishing of the complete generalized energy covector over a connected representation domain is sufficient for exact global flatness.

f. Theorem 86. Exact differential criterion for a globally flat generalized band. Let E_n be continuously differentiable over a connected generalized momentum domain. Then $dE_n = 0$ everywhere implies $E_n = \text{constant}$ over the complete domain and therefore $W_n = 0$. This implication is coordinate and metric independent. We now distinguish three levels of local dispersion suppression. First-order suppression at one point requires $dE_n = 0$. Second-order suppression additionally requires $\mathcal{M}_n = 0$ as a bilinear form on the physical tangent space. If both conditions hold, the local expansion begins at third order,

$$E_n(\lambda + \delta\lambda) - E_n(\lambda) = O(\|\delta\lambda\|^3). \quad (4731)$$

More generally, if the first nonvanishing derivative has order p ,

$$E_n - E_* = O(\|\delta\lambda\|^p), \quad (4732)$$

we identify p as the local dispersion order. The ordinary quadratic effective-mass description corresponds to $p = 2$. A higher-order stationary point satisfies $p > 2$. Such a point produces a stronger local spectral concentration than a generic quadratic stationary point. To make the previous statement quantitative, suppose a D -dimensional generalized momentum neighborhood has leading isotropic dispersion

$$E - E_* = Ar^p, \quad A > 0. \quad (4733)$$

The local spectral measure is proportional to

$$\rho(E) \propto \int_0^\infty r^{D-1} dr \delta(E - E_* - Ar^p). \quad (4734)$$

The root satisfying the delta-function constraint is

$$r_* = \left(\frac{E - E_*}{A} \right)^{1/p}. \quad (4735)$$

Since

$$\left| \frac{d}{dr} Ar^p \right|_{r=r_*} = pAr_*^{p-1}, \quad (4736)$$

we obtain

$$\rho(E) \propto \frac{1}{pA^{D/p}} (E - E_*)^{D/p-1} \Theta(E - E_*). \quad (4737)$$

Thus increasing the dispersion order p increases the low-energy DOS singularity. A divergent edge DOS occurs when

$$\frac{D}{p} - 1 < 0, \quad (4738)$$

or equivalently $p > D$. A finite nonzero edge DOS occurs when $p = D$. A vanishing edge DOS occurs when $p < D$. This shows explicitly that the dimensionality of generalized momentum space and the order of dispersion suppression compete in determining the observable DOS.

g. Theorem 87. Generalized dispersion-order DOS law. For a local isotropic generalized dispersion $E - E_* = Ar^p$ in spectral dimension D , the local DOS obeys

$$\rho(E) \propto (E - E_*)^{D/p-1}. \quad (4739)$$

Therefore higher-order flattening does not automatically generate a divergent DOS when the generalized momentum dimension is large. The condition for a power-law divergence is $p > D$. This distinction is particularly important for hyperbolic representation spaces whose dimension grows with the exact moiré index. At a stationary point, curvature reconstructs the second-order dispersion tensor. Let

$$M_{ab}^{(n)} = \nabla_a \nabla_b E_n. \quad (4740)$$

Differentiating with respect to curvature gives

$$\partial_\kappa M_{ab}^{(n)} = \partial_\kappa \nabla_a \nabla_b E_n. \quad (4741)$$

In fixed local tangent coordinates at the evaluation point,

$$\begin{aligned} \partial_\kappa M_{ab}^{(n)} &= \langle n | H_{\kappa ab} | n \rangle \\ &+ 2 \operatorname{Re} \sum_{m \neq n} \frac{\langle n | H_{\kappa a} | m \rangle \langle m | H_b | n \rangle}{E_n - E_m} \\ &+ 2 \operatorname{Re} \sum_{m \neq n} \frac{\langle n | H_a | m \rangle \langle m | H_{\kappa b} | n \rangle}{E_n - E_m} \\ &+ 2 \operatorname{Re} \sum_{m \neq n} \frac{\langle n | H_\kappa | m \rangle \langle m | H_{ab} | n \rangle}{E_n - E_m} \\ &- 2 \operatorname{Re} \sum_{m \neq n} \frac{(\partial_\kappa E_n - \partial_\kappa E_m) \langle n | H_a | m \rangle \langle m | H_b | n \rangle}{(E_n - E_m)^2} \\ &+ \mathcal{R}_{ab}^{(3)}. \end{aligned} \quad (4742)$$

Here $\mathcal{R}_{ab}^{(3)}$ collects the remaining terms generated by curvature-induced derivatives of the eigenvectors inside the matrix elements; equivalently, the complete quantity can be obtained by differentiating the exact spectral Hessian covariantly. The displayed terms already show the key enhancement mechanism: small interband denominators enter at first and second powers. Thus curvature can produce strong effective-mass renormalization before a macroscopic bandwidth collapse becomes visible. Let the generalized Hessian eigenproblem be

$$M(\kappa)\mathbf{c}_a = \Lambda_a(\kappa)G(\kappa)\mathbf{c}_a, \quad (4743)$$

with normalization $\mathbf{c}_a^\dagger G \mathbf{c}_a = 1$. For a nondegenerate principal curvature,

$$\frac{d\Lambda_a}{d\kappa} = \mathbf{c}_a^\dagger [\partial_\kappa M - \Lambda_a \partial_\kappa G] \mathbf{c}_a. \quad (4744)$$

Therefore a principal direction softens with increasing negative curvature when

$$\mathbf{c}_a^\dagger [\partial_\kappa M - \Lambda_a \partial_\kappa G] \mathbf{c}_a < 0 \quad (4745)$$

for a positive initial curvature eigenvalue. A second-order soft point occurs at $\Lambda_a(\kappa_*) = 0$. This condition is more specific than a minimum of the global bandwidth. Suppose a Euclidean band is not stationary at a generalized momentum point, $dE_n^{(0)} \neq 0$. For weak curvature,

$$dE_n(\kappa) = dE_n^{(0)} + \kappa \Xi_n^{(1)} + O(\kappa^2), \quad (4746)$$

where

$$\Xi_n^{(1)}[X] = \partial_\kappa dE_n[X]|_{\kappa=0}. \quad (4747)$$

A curvature-induced full stationary point requires

$$dE_n^{(0)} + \kappa_* \Xi_n^{(1)} + O(\kappa_*^2) = 0 \quad (4748)$$

as an equality of covectors. At leading order this equation has a scalar solution κ_* only if the two covectors are collinear, $\Xi_n^{(1)} = -cdE_n^{(0)}$ for some $c > 0$. Then

$$\kappa_* = \frac{1}{c} + O(\kappa_*^2). \quad (4749)$$

In a multidimensional representation space, generic curvature variation cannot cancel unrelated components independently with a single scalar parameter. Thus a complete generalized stationary state requires correlated renormalization of many tangent directions. This is precisely the structure that a geometric reconstruction mechanism must generate if a globally suppressed hyperbolic dispersion is to occur.

h. Theorem 88. Multidirectional curvature-locking condition. Let the generalized energy differential have the weak-curvature expansion

$$dE_n(\kappa) = dE_n^{(0)} + \kappa \Xi_n^{(1)} + O(\kappa^2). \quad (4750)$$

A first-order curvature-induced full stationary point exists only if $\Xi_n^{(1)} \parallel dE_n^{(0)}$ with opposite orientation. For a tangent space of dimension greater than one, this constitutes a nontrivial directional-locking condition. Suppression observed along one selected path does not establish this multidirectional cancellation. To quantify the preceding condition, choose an orthonormal Hodge tangent basis, $G_{ab} = \delta_{ab}$. Define the normalized initial dispersion vector

$$\hat{\mathcal{E}}_a = \frac{\mathcal{E}_a^{(0)}}{\left[\sum_b \left(\mathcal{E}_b^{(0)} \right)^2 \right]^{1/2}}. \quad (4751)$$

Define the normalized curvature-response vector

$$\hat{\Xi}_a = \frac{\Xi_a^{(1)}}{\left[\sum_b \left(\Xi_b^{(1)} \right)^2 \right]^{1/2}}. \quad (4752)$$

We define the curvature-locking coefficient

$$\mathcal{L}_n^{\text{dir}} = - \sum_{a=1}^{D_d} \hat{\mathcal{E}}_a \hat{\Xi}_a. \quad (4753)$$

It satisfies $-1 \leq \mathcal{L}_n^{\text{dir}} \leq 1$. Perfect first-order antiparallel reconstruction gives $\mathcal{L}_n^{\text{dir}} = 1$. Orthogonal curvature response gives $\mathcal{L}_n^{\text{dir}} = 0$. A response that increases the existing dispersion direction gives $\mathcal{L}_n^{\text{dir}} < 0$. Thus $\mathcal{L}_n^{\text{dir}}$ quantifies whether curvature acts coherently against the complete generalized dispersion rather than merely reducing one component accidentally. Directional alignment alone does not determine whether the response is large enough to cancel the original dispersion. Define

$$\mathcal{A}_n^{\text{dir}} = \frac{\left\| \Xi_n^{(1)} \right\|_{G^{-1}}}{\left\| dE_n^{(0)} \right\|_{G^{-1}}}. \quad (4754)$$

Its dimension is inverse curvature. The leading curvature required for cancellation is

$$\kappa_n^{\text{lin}} = \frac{1}{\mathcal{A}_n^{\text{dir}}} \quad (4755)$$

when $\mathcal{L}_n^{\text{dir}} = 1$. For imperfect locking, the residual first-order dispersion norm is

$$\begin{aligned} \left\| dE_n^{(0)} + \kappa \Xi_n^{(1)} \right\|_{G^{-1}}^2 &= \left\| dE_n^{(0)} \right\|_{G^{-1}}^2 + \kappa^2 \left\| \Xi_n^{(1)} \right\|_{G^{-1}}^2 \\ &- 2\kappa \mathcal{L}_n^{\text{dir}} \left\| dE_n^{(0)} \right\|_{G^{-1}} \left\| \Xi_n^{(1)} \right\|_{G^{-1}}. \end{aligned} \quad (4756)$$

Minimization with respect to curvature gives

$$\kappa_{\text{opt}}^{(1)} = \mathcal{L}_n^{\text{dir}} \frac{\left\| dE_n^{(0)} \right\|_{G^{-1}}}{\left\| \Xi_n^{(1)} \right\|_{G^{-1}}} \quad (4757)$$

when $\mathcal{L}_n^{\text{dir}} > 0$. The minimum residual norm is

$$\|dE_n\|_{\min} = \left\| dE_n^{(0)} \right\|_{G^{-1}} \sqrt{1 - (\mathcal{L}_n^{\text{dir}})^2}. \quad (4758)$$

This is a strong quantitative result. Unless $\mathcal{L}_n^{\text{dir}} = 1$, a single curvature parameter cannot eliminate the complete first-order generalized dispersion within linear response.

i. Theorem 89. Curvature-locking bound on achievable first-order flattening. Let

$$dE_n(\kappa) = dE_n^{(0)} + \kappa \Xi_n^{(1)} + O(\kappa^2). \quad (4759)$$

Then the optimal linear-response curvature is

$$\kappa_{\text{opt}}^{(1)} = \mathcal{L}_n^{\text{dir}} \frac{\left\| dE_n^{(0)} \right\|_{G^{-1}}}{\left\| \Xi_n^{(1)} \right\|_{G^{-1}}}, \quad (4760)$$

and the smallest attainable first-order dispersion is

$$\|dE_n\|_{\min} = \left\| dE_n^{(0)} \right\|_{G^{-1}} \sqrt{1 - (\mathcal{L}_n^{\text{dir}})^2}. \quad (4761)$$

Exact first-order cancellation is possible in linear response only for perfect antiparallel locking, $\mathcal{L}_n^{\text{dir}} = 1$. Thus the emergence of a globally stationary hyperbolic band requires a highly correlated curvature response across the complete representation-space tangent structure. The curvature-response covector is

$$\Xi_n^{(1)}[X] = \partial_\kappa dE_n[X]|_{\kappa=0}. \quad (4762)$$

Using Eq. (4709),

$$\begin{aligned} \Xi_n^{(1)}[X] &= \langle n | \partial_\kappa \dot{H}_X \Big|_{\kappa=0} | n \rangle \\ &+ 2 \text{Re} \sum_{m \neq n} \frac{\langle n | H_\kappa^{(0)} | m \rangle \langle m | \dot{H}_X^{(0)} | n \rangle}{E_n^{(0)} - E_m^{(0)}}. \end{aligned} \quad (4763)$$

The first term reflects direct curvature reconstruction of the microscopic hopping matrix. The second term reflects curvature-induced rotation of the eigenstate. Because the same real-space curvature field modifies many hopping channels simultaneously, these tangent responses are not independent random variables. This provides a concrete microscopic route by which a single geometric parameter can generate correlated multidirectional dispersion renormalization. A finite quotient does not provide a continuous representation manifold. We therefore require a discrete estimator that converges to the representation-space differential when nearby sectors become available. Let ρ_a, ρ_b be two symmetry-resolved finite-cover sectors belonging to the same adiabatically tracked band family. Let their representation-space distance be $d_G(\rho_a, \rho_b)$. Define the pairwise finite-difference dispersion rate

$$\Omega_n^{(ab)} = \frac{|E_{n,\rho_a} - E_{n,\rho_b}|}{\hbar d_G(\rho_a, \rho_b)}. \quad (4764)$$

For a set of locally neighboring sectors $\mathcal{N}_N(\rho)$, define

$$\Omega_{n,N}^{\text{disc}}(\rho) = \max_{\rho' \in \mathcal{N}_N(\rho)} \frac{|E_{n,\rho'} - E_{n,\rho}|}{\hbar d_G(\rho', \rho)}. \quad (4765)$$

If the finite quotient sequence samples a smooth representation-space component with

$$\max_{\rho' \in \mathcal{N}_N(\rho)} d_G(\rho', \rho) \longrightarrow 0, \quad (4766)$$

then $\Omega_{n,N}^{\text{disc}}(\rho) \longrightarrow \Omega_n^{\text{gen}}(\rho)$ provided the corresponding directional sampling becomes asymptotically complete. We do not assume this density property automatically for arbitrary finite-image representation sequences. When it cannot be established, the finite-difference quantity is reported only as a finite-cover diagnostic and not as the complete continuous representation-space norm. Suppose three sectors lie approximately along the same local representation-space geodesic, ρ_-, ρ_0, ρ_+ , with approximately equal Hodge separation

$$d_G(\rho_0, \rho_+) \simeq d_G(\rho_-, \rho_0) = \delta s. \quad (4767)$$

The directional second derivative is estimated by

$$\mathcal{M}_{n,N}^{\text{disc}} = \frac{E_{n,\rho_+} - 2E_{n,\rho_0} + E_{n,\rho_-}}{(\delta s)^2}. \quad (4768)$$

A converged second-order softness claim requires $\mathcal{M}_{n,N}^{\text{disc}} \longrightarrow 0$ under increasing finite-cover resolution and decreasing sector spacing. Suppose every finite-cover energy has uncertainty

$$|E_{n,\rho}^{\text{num}} - E_{n,\rho}| \leq \epsilon_E. \quad (4769)$$

Then the pairwise energy difference has uncertainty bounded by $\epsilon_{\Delta E} \leq 2\epsilon_E$. Therefore,

$$\left| \Omega_{n,\text{num}}^{(ab)} - \Omega_n^{(ab)} \right| \leq \frac{2\epsilon_E}{\hbar d_G(\rho_a, \rho_b)}. \quad (4770)$$

Thus excessively small representation-space separation amplifies numerical energy uncertainty. A reliable finite-difference derivative requires

$$|E_{n,\rho_a} - E_{n,\rho_b}| \gg 2\epsilon_E. \quad (4771)$$

The derivative limit must therefore be taken jointly with increasing spectral accuracy. Define

$$\mathcal{R}_{\Omega,N}(\kappa) = \frac{\Omega_{n,N}^{\text{disc}}(\kappa)}{\Omega_{n,N}^{\text{disc}}(0)}. \quad (4772)$$

A curvature-dependent generalized dispersion suppression is accepted only if

$$\limsup_{N \rightarrow \infty} \mathcal{R}_{\Omega,N}(\kappa) < 1 \quad (4773)$$

and the result is stable under independent quotient families. This criterion prevents a sparse or anisotropic sampling of representation space from producing an artificial low-velocity signal. V.C established the global bandwidth W_n . We now combine it with the complete first-order dispersion scale. Define

$$\mathcal{J}_n^{(1)} = \frac{W_n}{\hbar D_G \Omega_n^{(H)}}. \quad (4774)$$

From

$$W_n \leq \hbar D_G \Omega_n^{(H)}, \quad (4775)$$

we obtain $0 \leq \mathcal{J}_n^{(1)} \leq 1$. A value $\mathcal{J}_n^{(1)} \simeq 1$ means that the global bandwidth nearly saturates the maximum-gradient bound. A value $\mathcal{J}_n^{(1)} \ll 1$ means that large local derivatives occupy only a restricted portion of the generalized momentum domain and do not control the total band range. Thus a bandwidth alone cannot determine how uniformly dispersion is suppressed. We define

$$\mathcal{U}_n^{\text{disp}} = \frac{\Omega_n^{\text{rms}}}{\Omega_n^{\text{max}}}, \quad (4776)$$

where

$$\Omega_n^{\text{max}} = \sup_{\lambda} \Omega_n^{\text{gen}}(\lambda), \quad \Omega_n^{\text{rms}} = \left[\frac{1}{V_G} \int_{\mathfrak{D}_d} d\mu_G (\Omega_n^{\text{gen}})^2 \right]^{1/2}. \quad (4777)$$

The ratio satisfies $0 \leq \mathcal{U}_n^{\text{disp}} \leq 1$. A very small value indicates a band that is nearly stationary over most of the representation domain but retains isolated steep regions. A value closer to unity indicates that the dispersion magnitude is distributed more uniformly. Therefore the pair $(\Omega_n^{\text{max}}, \mathcal{U}_n^{\text{disp}})$ distinguishes uniformly suppressed dispersion from a band containing isolated residual high-velocity sectors. For a curvature sweep, define

$$\mathcal{R}_n^{(2)}(\kappa) = \frac{\Lambda_n^{\text{rms}}(\kappa)}{\Lambda_n^{\text{rms}}(0)}. \quad (4778)$$

First-order dispersion suppression is measured by

$$\mathcal{R}_n^{(1)}(\kappa) = \frac{\Omega_n^{\text{max}}(\kappa)}{\Omega_n^{\text{max}}(0)}. \quad (4779)$$

Thus $\mathcal{R}_n^{(1)} < 1$ indicates first-order flattening, whereas $\mathcal{R}_n^{(2)} < 1$ indicates second-order softening. A particularly strong reconstruction satisfies both $\mathcal{R}_n^{(1)} \ll 1$ and $\mathcal{R}_n^{(2)} \ll 1$. The global spectral flatness ratio is

$$\mathcal{F}_n = \frac{\Delta_n}{W_n}. \quad (4780)$$

The differential diagnostics introduced here add information not contained in $\mathcal{F}_n$. We therefore represent a candidate reconstructed band by

$$\mathfrak{D}_n = (W_n, \Delta_n, \Omega_n^{\text{max}}, \Lambda_n^{\text{rms}}). \quad (4781)$$

The four entries measure, respectively, global energy range, spectral isolation, largest first-order generalized dispersion, typical second-order generalized dispersion. A narrow but nonstationary band has small W_n but finite Ω_n^{max} . A stationary but strongly curved band can have $\Omega_n^{\text{max}} \simeq 0$ near a selected point but large Λ_n^{rms} . An isolated deeply flattened band requires simultaneous suppression of the first- and second-order dispersion diagnostics together with a positive gap. The self-energy formula gives an explicit microscopic route to such suppression. Write

$$dE_n = Z_{\text{eff}} [dE_n^{\text{bare}} + dE_n^{\Sigma}]. \quad (4782)$$

The bare contribution is

$$dE_n^{\text{bare}} = \langle \psi_1 | dH_1 | \psi_1 \rangle, \quad (4783)$$

and the hybridization contribution is

$$dE_n^{\Sigma} = \langle \psi_1 | d\lambda \Sigma_1 | \psi_1 \rangle. \quad (4784)$$

Complete first-order cancellation requires

$$dE_n^\Sigma = -dE_n^{\text{bare}} \quad (4785)$$

as a covector equality over all physical tangent directions. The residue factor Z_{eff} then multiplies an already suppressed numerator. Thus strong dispersion suppression can occur through two cooperating mechanisms:

$$dE_{\text{bare}} + dE_\Sigma \longrightarrow 0, \quad (4786)$$

and $Z_{\text{eff}} < 1$. The first is a directional cancellation mechanism. The second is an energy-dependent hybridization renormalization. From

$$Z_{\text{eff}} = [1 - \langle \partial_E \Sigma_1 \rangle]^{-1}, \quad (4787)$$

we obtain

$$\partial_\kappa \ln Z_{\text{eff}} = Z_{\text{eff}} \partial_\kappa \langle \partial_E \Sigma_1 \rangle, \quad (4788)$$

where the derivative includes both explicit curvature dependence and the curvature-induced change of the reduced eigenstate. Thus curvature can suppress the generalized velocity even if the momentum-gradient numerator is unchanged, provided $\partial_\kappa Z_{\text{eff}} < 0$. For the off-resonant estimate,

$$1 - Z_{\text{eff}} = O\left(\frac{w^2}{\Delta^2}\right). \quad (4789)$$

Near a curvature-driven resonance, this correction can become large and the perturbative estimate fails. V.B introduced the hybridization-normalized control in which $\text{Tr } TT^\dagger = \text{constant}$. Even under this constraint, $\nabla_{\mathbf{k}} T$ and

$$T(E - H_2)^{-1} T^\dagger \quad (4790)$$

can change strongly with curvature. Therefore $\mathbf{v}_n^{\text{tun}}$, dE_n^Σ , and Z_{eff} remain curvature dependent even when the total mean-square hybridization is fixed. Consequently, velocity suppression that survives the fixed- $W_\perp$ control is a direct signature of tunneling-shape reconstruction rather than a trivial increase in total coupling strength.

j. Theorem 90. Dispersion reconstruction at fixed hybridization norm. Holding $\text{Tr } TT^\dagger$ fixed does not fix the generalized dispersion because the exact derivative depends independently on

$$dT, \quad T(E - H_2)^{-1} T^\dagger, \quad \partial_E \Sigma_1. \quad (4791)$$

Therefore curvature-induced velocity suppression that remains under hybridization normalization cannot be reduced to a scalar renormalization of w/t . For

every Euclidean benchmark structure, we calculate $E_n(\mathbf{k})$, $\mathbf{v}_n(\mathbf{k})$, and $[m_n^{*-1}]_{\mu\nu}$. We additionally decompose the velocity according to

$$\mathbf{v}_n = \mathbf{v}_n^{\text{kin}} + \mathbf{v}_n^{\text{tun}}. \quad (4792)$$

For every selected avoided crossing, we calculate $\Delta(\mathbf{k})$, $V(\mathbf{k})$, $\nabla_{\mathbf{k}} V(\mathbf{k})$, and compare the full numerical velocity to Eq. (4616). This provides an exact analysis of the velocity reconstruction mechanism. For every sampled representation sector, we calculate the complete energy covector

$$\mathcal{E}_a^{(n)} = dE_n[X_a], \quad (4793)$$

the Hodge metric G_{ab} , and the generalized dispersion rate

$$\Omega_n^{\text{gen}} = \frac{1}{\hbar} \sqrt{G^{ab} \mathcal{E}_a^{(n)} \mathcal{E}_b^{(n)}}. \quad (4794)$$

At stationary points, we calculate

$$M_{ab}^{(n)} = \nabla_a \nabla_b E_n, \quad (4795)$$

solve

$$M^{(n)} \mathbf{c}_a = \Lambda_{n,a} G \mathbf{c}_a, \quad (4796)$$

and report the full set $\{\Lambda_{n,a}\}$. We additionally calculate $\mathcal{L}_n^{\text{dir}}$, and $\mathcal{A}_n^{\text{dir}}$, to determine whether curvature suppresses the full generalized dispersion coherently or merely along selected directions. A generalized velocity claim is accepted only after the energy differences or analytic derivatives are converged below the scale being reported. For finite-cover derivatives, we require

$$2\epsilon_E \ll |E_{n,\rho_a} - E_{n,\rho_b}|. \quad (4797)$$

For Hessian estimates, the numerical energy uncertainty must satisfy

$$\epsilon_E \ll |\mathcal{M}_n^{\text{disc}}| (\delta s)^2. \quad (4798)$$

The hopping cutoff, cover injectivity radius, symmetry error, and representation-sector assignment are converged independently. Any apparent velocity suppression that disappears under increasing representation-space resolution is treated as a sampling artifact. The Euclidean and hyperbolic theories can now be placed in one exact hierarchy. For an ordinary Euclidean crystal,

$$\begin{aligned} H(\mathbf{k}) &\longrightarrow E_n(\mathbf{k}) \longrightarrow \nabla_{\mathbf{k}} E_n \\ &\longrightarrow \mathbf{v}_n = \frac{1}{\hbar} \nabla_{\mathbf{k}} E_n \longrightarrow \nabla_{\mathbf{k}}^2 E_n \longrightarrow m_n^*. \end{aligned} \quad (4799)$$

For an exact hyperbolic moiré crystal,

$$\begin{aligned} H[\rho] &\longrightarrow E_n[\rho] \longrightarrow dE_n \in H^1(\Gamma_M, \text{Ad } \rho)^* \\ &\longrightarrow \Omega_n^{\text{gen}} = \frac{1}{\hbar} \|dE_n\|_{G_\rho^{-1}} \longrightarrow \mathcal{M}_n = \nabla^2 E_n \\ &\longrightarrow \{\Lambda_{n,a}\}. \end{aligned} \quad (4800)$$

The correspondence is therefore $\nabla_{\mathbf{k}} E \longleftrightarrow dE$, $|\mathbf{v}| \longleftrightarrow \Omega_n^{\text{gen}}$, and

$$\hbar^2 m^{*-1} \longleftrightarrow G^{-1} \mathcal{M}. \quad (4801)$$

The complete curvature-controlled reconstruction mechanism can finally be written as

$$\begin{aligned} K < 0 &\longrightarrow \text{geodesic tunneling reconstruction} \\ &\longrightarrow \partial_\kappa H \longrightarrow \partial_\kappa dE_n \longrightarrow \{\mathcal{L}_n^{\text{dir}}, \mathcal{A}_n^{\text{dir}}\} \\ &\longrightarrow \Omega_n^{\text{gen}} \longrightarrow \{\Lambda_{n,a}\} \\ &\longrightarrow \text{effective-dispersion reconstruction.} \end{aligned} \quad (4802)$$

We therefore do not diagnose hyperbolic dispersion from a visually flat one-dimensional spectral cut. We require suppression of the complete energy covector over all physically relevant representation-space tangent directions, and we resolve the residual second-order dispersion through the geometry-normalized Hessian eigenvalues. The most restrictive first-order signature is the simultaneous behavior $\Omega_n^{\text{gen}} \longrightarrow 0$, together with $\mathcal{L}_n^{\text{dir}} \longrightarrow 1$. The first condition states that the reconstructed generalized dispersion is small. The second establishes that the suppression originates from a coherent curvature response opposing the original multidirectional dispersion rather than from an accidental cancellation along a selected path. At a stationary point, the next diagnostic is $\Lambda_n^{\text{rms}} \longrightarrow 0$, which identifies second-order softening. These quantities provide the complete differential bridge from the curvature-controlled spectral reconstruction established in V.A–V.C to the subsequent analysis of the parameter conditions under which the reconstructed bands become anomalously flat.

D. Wave-Function Localization and Participation Geometry

The spectral reconstruction developed above must finally be separated from real-space localization. This distinction is essential because a narrow finite-size energy cluster can arise for two physically inequivalent reasons. The first mechanism is genuine suppression of the dispersion of a coherent band. In this case the spectral width becomes small while the corresponding spectral subspace remains extended over the bulk translation structure. The second mechanism is spectral crowding of spatially localized states. A large finite system can contain many

localized states with similar energies, producing arbitrarily small level spacings even though no coherent band has become flat. We therefore do not use a small bandwidth, a high density of states, or an individual inverse participation ratio as a sufficient flat-band criterion. Instead we construct a hierarchy of real-space participation, finite-size scaling, boundary enrichment, spectral-projector participation, and translation-sector coverage. The resulting analysis distinguishes global band flattening, bulk wave-function extension, intra-moiré-cell concentration, true real-space localization, boundary accumulation and finite-size spectral crowding. and These phenomena are mathematically distinct and are diagnosed separately. Consider a periodic finite cover containing N_c translation cells. Let each cell contain ν internal orbitals, including layer, sublattice, and exact moiré-cell degrees of freedom. The one-particle Hilbert-space dimension is therefore $N_H = \nu N_c$. We label a real-space orbital by

$$(x, \alpha), \quad x = 1, \dots, N_c, \quad \alpha = 1, \dots, \nu. \quad (4803)$$

A normalized eigenstate is written as

$$|\psi_n\rangle = \sum_{x=1}^{N_c} \sum_{\alpha=1}^{\nu} \psi_{n;x\alpha} |x, \alpha\rangle. \quad (4804)$$

Normalization gives

$$\sum_{x=1}^{N_c} \sum_{\alpha=1}^{\nu} |\psi_{n;x\alpha}|^2 = 1. \quad (4805)$$

Define the orbital probability

$$p_{n;x\alpha} = |\psi_{n;x\alpha}|^2. \quad (4806)$$

Then $p_{n;x\alpha} \geq 0$ and

$$\sum_{x,\alpha} p_{n;x\alpha} = 1. \quad (4807)$$

The orbital inverse participation ratio is

$$\text{IPR}_n^{\text{orb}} = \sum_{x=1}^{N_c} \sum_{\alpha=1}^{\nu} p_{n;x\alpha}^2. \quad (4808)$$

Equivalently,

$$\text{IPR}_n^{\text{orb}} = \sum_{x,\alpha} |\psi_{n;x\alpha}|^4. \quad (4809)$$

The participation number is

$$P_n^{\text{orb}} = \frac{1}{\text{IPR}_n^{\text{orb}}}. \quad (4810)$$

The normalized orbital participation ratio is

$$p_n^{\text{orb}} = \frac{P_n^{\text{orb}}}{N_H} = \frac{1}{N_H \text{IPR}_n^{\text{orb}}}. \quad (4811)$$

By the Cauchy–Schwarz inequality,

$$\left(\sum_{i=1}^{N_H} p_i \right)^2 \leq N_H \sum_{i=1}^{N_H} p_i^2. \quad (4812)$$

Because

$$\sum_{i=1}^{N_H} p_i = 1, \quad (4813)$$

we obtain

$$\text{IPR}_n^{\text{orb}} \geq \frac{1}{N_H}. \quad (4814)$$

The upper bound follows from $p_i^2 \leq p_i$, which gives $\text{IPR}_n^{\text{orb}} \leq 1$. Therefore,

$$\frac{1}{N_H} \leq \text{IPR}_n^{\text{orb}} \leq 1. \quad (4815)$$

Consequently, $1 \leq P_n^{\text{orb}} \leq N_H$, and

$$\frac{1}{N_H} \leq p_n^{\text{orb}} \leq 1. \quad (4816)$$

A perfectly uniform state satisfies

$$p_i = \frac{1}{N_H}, \quad (4817)$$

and therefore

$$\text{IPR}_n^{\text{orb}} = \frac{1}{N_H}, \quad P_n^{\text{orb}} = N_H, \quad p_n^{\text{orb}} = 1. \quad (4818)$$

A state localized on an order-one number N_{loc} of orbitals has

$$\text{IPR}_n^{\text{orb}} = O(N_{\text{loc}}^{-1}), \quad (4819)$$

and therefore

$$p_n^{\text{orb}} = O\left(\frac{N_{\text{loc}}}{N_H}\right) \rightarrow 0 \quad (4820)$$

along an increasing-cover sequence. Orbital participation alone does not distinguish localization between different moiré cells from concentration inside one moiré cell. Define the total probability in translation cell x by

$$P_{n;x} = \sum_{\alpha=1}^{\nu} p_{n;x\alpha}. \quad (4821)$$

Then

$$\sum_{x=1}^{N_c} P_{n;x} = 1. \quad (4822)$$

The cell IPR is

$$\text{IPR}_n^{\text{cell}} = \sum_{x=1}^{N_c} P_{n;x}^2. \quad (4823)$$

The cell participation number is

$$P_n^{\text{cell}} = \frac{1}{\text{IPR}_n^{\text{cell}}}, \quad (4824)$$

and the normalized cell participation is

$$p_n^{\text{cell}} = \frac{1}{N_c \text{IPR}_n^{\text{cell}}}. \quad (4825)$$

A state extended uniformly over all translation cells satisfies

$$\text{IPR}_n^{\text{cell}} = \frac{1}{N_c}, \quad (4826)$$

irrespective of how strongly its probability is concentrated among the internal orbitals of each cell. This distinction is crucial for moiré systems. A state can be strongly concentrated near one preferred stacking region inside every moiré cell while remaining fully extended across the entire crystal. Such a state is not real-space localized in the thermodynamic sense. For every cell with $P_{n;x} > 0$, define the conditional internal probability

$$q_{n;\alpha|x} = \frac{p_{n;x\alpha}}{P_{n;x}}. \quad (4827)$$

It satisfies

$$\sum_{\alpha=1}^{\nu} q_{n;\alpha|x} = 1. \quad (4828)$$

Define the internal IPR of cell x by

$$\mathcal{I}_n^{\text{int}}(x) = \sum_{\alpha=1}^{\nu} q_{n;\alpha|x}^2. \quad (4829)$$

Because

$$p_{n;x\alpha} = P_{n;x} q_{n;\alpha|x}, \quad (4830)$$

the orbital IPR becomes

$$\text{IPR}_n^{\text{orb}} = \sum_{x,\alpha} P_{n;x}^2 q_{n;\alpha|x}^2 = \sum_x P_{n;x}^2 \mathcal{I}_n^{\text{int}}(x). \quad (4831)$$

The internal IPR obeys

$$\frac{1}{\nu} \leq \mathcal{I}_n^{\text{int}}(x) \leq 1. \quad (4832)$$

Therefore,

$$\frac{1}{\nu} \text{IPR}_n^{\text{cell}} \leq \text{IPR}_n^{\text{orb}} \leq \text{IPR}_n^{\text{cell}}. \quad (4833)$$

We define the effective internal participation number by

$$P_n^{\text{int}} = \frac{\text{IPR}_n^{\text{cell}}}{\text{IPR}_n^{\text{orb}}}. \quad (4834)$$

Equation (4833) gives $1 \leq P_n^{\text{int}} \leq \nu$. Thus the pair $(p_n^{\text{cell}}, P_n^{\text{int}})$ separates thermodynamic spatial extension from internal moiré-cell concentration.

a. Theorem 91. Exact participation decomposition. For every normalized finite-cover eigenstate,

$$\text{IPR}_n^{\text{orb}} = \sum_x P_{n;x}^2 \mathcal{I}_n^{\text{int}}(x), \quad (4835)$$

and

$$\frac{\text{IPR}_n^{\text{cell}}}{\nu} \leq \text{IPR}_n^{\text{orb}} \leq \text{IPR}_n^{\text{cell}}. \quad (4836)$$

Consequently, strong localization inside each moiré cell does not imply localization across moiré cells. A thermodynamically extended state can simultaneously satisfy

$$\text{IPR}_n^{\text{cell}} = O(N_c^{-1}) \quad (4837)$$

and $P_n^{\text{int}} = O(1)$. This is the characteristic participation structure of a Bloch-extended state with strong intra-cell moiré concentration. Let P_1, P_2 be the projectors onto the two layers. Define the layer weights

$$w_1^{(n)} = \langle \psi_n | P_1 | \psi_n \rangle, \quad (4838)$$

and

$$w_2^{(n)} = \langle \psi_n | P_2 | \psi_n \rangle. \quad (4839)$$

They satisfy $w_1^{(n)} + w_2^{(n)} = 1$. The layer polarization introduced previously is

$$\mathcal{P}_L^{(n)} = w_1^{(n)} - w_2^{(n)}. \quad (4840)$$

Therefore,

$$w_1^{(n)} = \frac{1 + \mathcal{P}_L^{(n)}}{2}, \quad w_2^{(n)} = \frac{1 - \mathcal{P}_L^{(n)}}{2}. \quad (4841)$$

Define the layer IPR by

$$\text{IPR}_L^{(n)} = \left[w_1^{(n)} \right]^2 + \left[w_2^{(n)} \right]^2. \quad (4842)$$

Substitution gives the exact identity

$$\text{IPR}_L^{(n)} = \frac{1 + \left[\mathcal{P}_L^{(n)} \right]^2}{2}. \quad (4843)$$

The effective number of occupied layers is therefore

$$P_L^{(n)} = \frac{1}{\text{IPR}_L^{(n)}} = \frac{2}{1 + \left[\mathcal{P}_L^{(n)} \right]^2}. \quad (4844)$$

Hence $1 \leq P_L^{(n)} \leq 2$. A fully layer-polarized state has

$$\left| \mathcal{P}_L^{(n)} \right| = 1, \quad P_L^{(n)} = 1. \quad (4845)$$

A layer-balanced state has $\mathcal{P}_L^{(n)} = 0$ and $P_L^{(n)} = 2$. Thus layer polarization and layer participation contain exactly the same two-layer information but in complementary forms. The ordinary IPR is the second member of a complete hierarchy. For $q > 0, q \neq 1$, define the generalized participation moment

$$I_q^{(n)} = \sum_{i=1}^{N_H} p_{n;i}^q. \quad (4846)$$

The associated Rényi participation number is

$$P_q^{(n)} = \left[I_q^{(n)} \right]^{-\frac{1}{q-1}}. \quad (4847)$$

For $q = 2$, we recover $I_2^{(n)} = \text{IPR}_n^{\text{orb}}$, and $P_2^{(n)} = P_n^{\text{orb}}$. For $q \rightarrow 1$, the participation entropy is

$$S_1^{(n)} = - \sum_{i=1}^{N_H} p_{n;i} \ln p_{n;i}, \quad (4848)$$

and the corresponding effective support is $P_1^{(n)} = e^{S_1^{(n)}}$. Euclidean fractal dimensions should not be imported directly into a hyperbolic lattice because the physical volume grows exponentially with radius. We therefore define a volume-normalized generalized participation dimension

$$D_q^{\text{vol}} = \lim_{N_H \rightarrow \infty} \frac{\ln P_q}{\ln N_H}. \quad (4849)$$

Equivalently,

$$D_q^{\text{vol}} = -\frac{1}{q-1} \lim_{N_H \rightarrow \infty} \frac{\ln I_q}{\ln N_H}. \quad (4850)$$

For $q = 2$,

$$D_2^{\text{vol}} = - \lim_{N_H \rightarrow \infty} \frac{\ln \text{IPR}}{\ln N_H}. \quad (4851)$$

A fully ergodic extended state satisfies $D_q^{\text{vol}} = 1$. A strongly localized state satisfies $D_q^{\text{vol}} = 0$. A nonergodic extended or multifractal state satisfies $0 < D_q^{\text{vol}} < 1$. Thus the finite-size law

$$\text{IPR}_n \sim N_H^{-D_2^{\text{vol}}} \quad (4852)$$

provides a unified classification independent of whether the underlying real-space geometry is Euclidean or hyperbolic. For a cocompact hyperbolic lattice, the number of sites within geodesic radius L grows asymptotically as

$$N(L) \asymp C e^{h_\Lambda L}, \quad (4853)$$

where the lattice growth entropy satisfies $h_\Lambda = \frac{1}{R}$ for the asymptotic two-dimensional hyperbolic growth exponent. Using Eq. (4852),

$$\text{IPR}_n(L) \sim \exp[-D_2^{\text{vol}} h_\Lambda L]. \quad (4854)$$

For a fully extended state,

$$\text{IPR}_n(L) \sim e^{-L/R}. \quad (4855)$$

For a localized state,

$$\text{IPR}_n(L) \rightarrow \text{IPR}_n^{(\infty)} > 0. \quad (4856)$$

Thus the correct hyperbolic finite-size scaling is exponential in the linear radius but algebraic in the total Hilbert-space dimension.

b. Theorem 92. Hyperbolic participation scaling. If $N(L) \asymp e^{L/R}$, and a state has volume participation dimension D_2^{vol} , then

$$\text{IPR}(L) \sim e^{-D_2^{\text{vol}} L/R}. \quad (4857)$$

Consequently, plotting the IPR against the linear radius without accounting for exponential volume growth can substantially misidentify an extended hyperbolic state. Negative curvature imposes an additional geometric constraint on normalizable localized states. Assume that an asymptotically radial state centered at x_0 obeys

$$|\psi(r)|^2 \asymp A e^{-2r/\xi_\psi}, \quad (4858)$$

where $r = d(x, x_0)$. The number of sites in a shell of radius r behaves as

$$n_{\text{shell}}(r) \asymp C_\Lambda e^{h_\Lambda r}. \quad (4859)$$

The normalization integral therefore scales as

$$\begin{aligned} 1 &\asymp A \int_0^\infty dr e^{h_\Lambda r} e^{-2r/\xi_\psi} \\ &= A \int_0^\infty dr e^{-(2/\xi_\psi - h_\Lambda)r}. \end{aligned} \quad (4860)$$

Normalizability requires $\frac{2}{\xi_\psi} > h_\Lambda$. Therefore, $\xi_\psi < \frac{2}{h_\Lambda}$.

For a two-dimensional hyperbolic lattice, $h_\Lambda = \frac{1}{R}$, and hence $\xi_\psi < 2R$. This condition is absent in Euclidean two-dimensional space, where polynomial volume growth is always dominated by any finite exponential decay length. Under the same radial approximation,

$$\text{IPR} \asymp A^2 C_\Lambda \int_0^\infty dr e^{h_\Lambda r} e^{-4r/\xi_\psi}. \quad (4861)$$

Hence

$$\text{IPR} \asymp \frac{A^2 C_\Lambda}{4/\xi_\psi - h_\Lambda}. \quad (4862)$$

From normalization,

$$A \asymp \frac{2/\xi_\psi - h_\Lambda}{C_\Lambda}. \quad (4863)$$

Therefore,

$$\text{IPR} \asymp \frac{(2/\xi_\psi - h_\Lambda)^2}{C_\Lambda (4/\xi_\psi - h_\Lambda)}. \quad (4864)$$

As $\xi_\psi \rightarrow \frac{2}{h_\Lambda}$ from below,

$$\text{IPR} \propto \left(\frac{2}{\xi_\psi} - h_\Lambda \right)^2. \quad (4865)$$

Thus the participation number diverges continuously as the exponential localization length approaches the hyperbolic entropy threshold.

c. Theorem 93. Hyperbolic entropy barrier to exponential localization. For a radial state satisfying

$$|\psi(r)|^2 \asymp e^{-2r/\xi_\psi} \quad (4866)$$

on a lattice with shell-growth entropy h_Λ , normalizability requires $\xi_\psi < \frac{2}{h_\Lambda}$. For the hyperbolic plane, $\xi_\psi < 2R$. Therefore localization competes directly with hyperbolic volume entropy. A decay profile that would appear exponentially localized from its radial amplitude alone can fail to define a localized thermodynamic state if its decay does not outrun the exponential growth of available sites. The IPR measures support volume but does not contain direct information about the geometric diameter of that support. We therefore define a basis-independent probability radius. For a normalized state and a candidate center x , define the cumulative weight

$$F_n(x, r) = \sum_{i: d(x_i, x) \leq r} p_{n;i}. \quad (4867)$$

For a fixed probability level $0 < \zeta < 1$, define

$$r_\zeta^{(n)}(x) = \inf \{r : F_n(x, r) \geq \zeta\}. \quad (4868)$$

The optimal localization radius is

$$r_\zeta^{(n)} = \inf_x r_\zeta^{(n)}(x). \quad (4869)$$

A thermodynamically localized state satisfies

$$\sup_N r_\zeta^{(n)} < \infty \quad (4870)$$

for a fixed ζ close to unity. An extended state has $r_\zeta^{(n)} \rightarrow \infty$. For a uniform state in a geodesic disk of radius L ,

$$F(r) = \frac{A_{\mathbb{H}}(r)}{A_{\mathbb{H}}(L)}. \quad (4871)$$

At large radii,

$$A_{\mathbb{H}}(r) \propto e^{r/R}. \quad (4872)$$

Therefore,

$$F(r) \simeq e^{(r-L)/R}. \quad (4873)$$

Setting $F(r_\zeta) = \zeta$ gives

$$r_\zeta = L + R \ln \zeta + o(1). \quad (4874)$$

Thus $r_\zeta = L - O(R)$ for a bulk-uniform hyperbolic state. Since $L \sim R \ln N$, we obtain

$$r_\zeta = R \ln N + O(R). \quad (4875)$$

By contrast, a localized state satisfies $r_\zeta = O(\xi_\psi)$. The joint scaling of IPR and r_ζ therefore distinguishes participation volume from geometric localization length. A low IPR does not guarantee bulk extension on an open hyperbolic cluster. Let the outer boundary shell of thickness b contain a fraction

$$f_b(L) = \frac{N_{\partial b}(L)}{N(L)}. \quad (4876)$$

For a hyperbolic disk,

$$f_b(L) \rightarrow 1 - e^{-b/R}. \quad (4877)$$

Therefore, $N_{\partial b} = \Theta(N)$. Consider a state uniformly extended over this boundary shell. Then

$$p_i \simeq \frac{1}{N_{\partial b}} \quad (4878)$$

inside the shell, and zero elsewhere. Its IPR is

$$\text{IPR}_\partial \simeq \frac{1}{N_{\partial b}}. \quad (4879)$$

Because $N_{\partial b} = \Theta(N)$, we obtain

$$\text{IPR}_\partial = \Theta(N^{-1}). \quad (4880)$$

Thus a boundary state can exhibit exactly the same volume scaling of the IPR as a bulk-extended state. This is a specifically severe problem in negative curvature. Let

$$W_\partial^{(n)}(b) = \sum_{i \in \partial_b \Omega} p_{n;i} \quad (4881)$$

be the wave-function weight in the boundary shell. The geometric baseline is

$$f_b = \frac{N_{\partial b}}{N}. \quad (4882)$$

Define the boundary enrichment ratio

$$\mathcal{B}_n(b) = \frac{W_{\partial}^{(n)}(b)}{f_b}. \quad (4883)$$

A uniform extended state satisfies $\mathcal{B}_n \simeq 1$. A boundary-enriched state satisfies $\mathcal{B}_n > 1$. We also use the boundary log-odds score

$$\mathcal{S}_{\partial}^{(n)} = \ln \left[\frac{W_{\partial}^{(n)}}{1 - W_{\partial}^{(n)}} \right] - \ln \left[\frac{f_b}{1 - f_b} \right]. \quad (4884)$$

A bulk-uniform state satisfies $\mathcal{S}_{\partial}^{(n)} \simeq 0$, whereas a boundary-localized state satisfies $\mathcal{S}_{\partial}^{(n)} > 0$.

d. Theorem 94. IPR cannot distinguish bulk and boundary extension on an open hyperbolic disk. Because a fixed-thickness hyperbolic boundary shell contains $\Theta(N)$ sites, a state uniformly extended along that shell satisfies

$$\text{IPR} = \Theta(N^{-1}), \quad (4885)$$

which is the same volume scaling as a bulk-uniform state. Therefore the IPR alone is insufficient for bulk localization analysis on an open hyperbolic cluster. Boundary enrichment or boundaryless periodic covers are required. For an open cluster, define the core weight

$$W_{\text{core}}^{(n)} = 1 - W_{\partial}^{(n)}. \quad (4886)$$

For $W_{\text{core}}^{(n)} > 0$, define the normalized core probability

$$p_{n;i}^{\text{core}} = \frac{p_{n;i}}{W_{\text{core}}^{(n)}}, \quad i \in \Omega_{\text{core}}. \quad (4887)$$

The core IPR is

$$\text{IPR}_n^{\text{core}} = \sum_{i \in \Omega_{\text{core}}} [p_{n;i}^{\text{core}}]^2. \quad (4888)$$

If the core contains N_{core} orbitals, define

$$p_n^{\text{core}} = \frac{1}{N_{\text{core}} \text{IPR}_n^{\text{core}}}. \quad (4889)$$

A genuine bulk-localized state has $p_n^{\text{core}} \rightarrow 0$ if its localization center lies inside the core. A boundary state instead has $W_{\text{core}}^{(n)} \rightarrow 0$. Thus boundary localization and bulk localization are separated by the pair $(W_{\text{core}}^{(n)}, p_n^{\text{core}})$. An additional complication arises when an eigenvalue is degenerate. Let

$$\mathcal{E} = \text{span} \{|u_1\rangle, \dots, |u_r\rangle\} \quad (4890)$$

be an r -dimensional degenerate eigenspace. A new orthonormal basis may be generated by

$$|\tilde{u}_a\rangle = \sum_{b=1}^r U_{ba} |u_b\rangle, \quad U \in U(r). \quad (4891)$$

Individual quantities $\text{IPR}(u_a)$ are not invariant under this transformation. Therefore a large IPR of one chosen vector inside an exactly degenerate flat-band subspace does not prove physical localization. This is particularly important for exact flat bands, which can admit both Bloch-like extended bases and compact-localized bases spanning the same spectral subspace. Let the basis-independent projector onto an isolated cluster $\mathcal{C}$ be

$$P_{\mathcal{C}} = \sum_{a=1}^r |u_a\rangle \langle u_a|. \quad (4892)$$

Its rank is $r_{\mathcal{C}} = \text{Tr } P_{\mathcal{C}}$. Define the normalized diagonal projector density

$$q_i^{(\mathcal{C})} = \frac{(P_{\mathcal{C}})_{ii}}{r_{\mathcal{C}}}. \quad (4893)$$

Because

$$\sum_i (P_{\mathcal{C}})_{ii} = r_{\mathcal{C}}, \quad (4894)$$

we have

$$\sum_i q_i^{(\mathcal{C})} = 1. \quad (4895)$$

We define the projector IPR by

$$\text{IPR}[P_{\mathcal{C}}] = \sum_i [q_i^{(\mathcal{C})}]^2. \quad (4896)$$

The corresponding normalized projector participation is

$$p[P_{\mathcal{C}}] = \frac{1}{N_H \text{IPR}[P_{\mathcal{C}}]}. \quad (4897)$$

These quantities depend only on the spectral subspace and are invariant under $U(r_{\mathcal{C}})$ rotations inside the cluster. For a single nondegenerate state, $r_{\mathcal{C}} = 1$, and

$$\text{IPR}[P_{\mathcal{C}}] = \text{IPR}_n^{\text{orb}}. \quad (4898)$$

Define the projector weight in cell x by

$$Q_x^{(\mathcal{C})} = \frac{1}{r_{\mathcal{C}}} \sum_{\alpha=1}^{\nu} \langle x, \alpha | P_{\mathcal{C}} | x, \alpha \rangle. \quad (4899)$$

Then

$$\sum_{x=1}^{N_c} Q_x^{(C)} = 1. \quad (4900)$$

The cell-resolved projector IPR is

$$\text{IPR}_{\text{cell}}[P_C] = \sum_{x=1}^{N_c} \left[Q_x^{(C)} \right]^2. \quad (4901)$$

Define

$$p_{\text{cell}}[P_C] = \frac{1}{N_c \text{IPR}_{\text{cell}}[P_C]}. \quad (4902)$$

This quantity is the preferred participation diagnostic for a degenerate band subspace. Suppose the finite-cover translation group acts transitively on the translation cells and the target spectral projector satisfies

$$[P_C, U_g] = 0 \quad \forall g. \quad (4903)$$

Translation covariance implies $Q_{gx}^{(C)} = Q_x^{(C)}$. Transitivity therefore gives $Q_x^{(C)} = Q \quad \forall x$. Normalization then gives $N_c Q = 1$, so that $Q = \frac{1}{N_c}$. Consequently,

$$\text{IPR}_{\text{cell}}[P_C] = \frac{1}{N_c}, \quad (4904)$$

and $p_{\text{cell}}[P_C] = 1$. This result is exact and does not depend on the basis chosen inside the spectral subspace.

e. Theorem 95. Translation-covariant flat-band projector participation. Let an isolated spectral subspace be invariant under a translation group acting transitively on N_c cells. Then its normalized cell projector density is exactly uniform, $Q_x = \frac{1}{N_c}$, and therefore

$$\text{IPR}_{\text{cell}}[P] = \frac{1}{N_c}, \quad p_{\text{cell}}[P] = 1. \quad (4905)$$

Thus an exactly flat periodic band may admit compact-localized basis states with large individual IPR while its spectral projector remains completely translation extended. Individual-state localization inside an exactly degenerate band therefore cannot be used to reject genuine flat-band physics. For a numerically identified cluster, define the normalized translation-covariance defect

$$\epsilon_{\text{cov}}[P_C] = \max_{g \in \mathcal{G}_{\text{test}}} \frac{\| [P_C, U_g] \|_F}{2 \| P_C \|_F}. \quad (4906)$$

An exact translation-resolved band satisfies $\epsilon_{\text{cov}} = 0$. A cluster produced by an open boundary, an explicitly

translation-breaking perturbation, or an incomplete selection of localized states can satisfy $\epsilon_{\text{cov}} > 0$. This provides a basis-independent complement to the IPR. For a normal finite quotient Q_N , let P_ρ be the exact projector onto the translation irrep ρ . Because $[P_C, P_\rho] = 0$ for an exact symmetry-resolved spectral cluster, define

$$r_C(\rho) = \frac{1}{d_\rho} \text{rank}(P_C P_\rho). \quad (4907)$$

The factor d_ρ removes the regular-representation multiplicity. Thus $r_C(\rho)$ counts the number of reduced representation-block eigenvalues belonging to the target cluster. The physical state weight of irrep ρ in the finite quotient is

$$p_\rho = \frac{d_\rho^2}{|Q_N|}. \quad (4908)$$

For a target one-band family, define the representation coverage

$$\mathcal{C}_{\text{rep}}^{(N)} = \sum_{\rho \in \widehat{Q}_N} p_\rho \mathbf{1}[r_C(\rho) \geq 1]. \quad (4909)$$

It satisfies $0 \leq \mathcal{C}_{\text{rep}}^{(N)} \leq 1$. A complete band sampled across all finite-quotient sectors satisfies $\mathcal{C}_{\text{rep}}^{(N)} = 1$. A sparse collection of accidental finite-size states occupies only a subset of sectors and gives $\mathcal{C}_{\text{rep}}^{(N)} < 1$. For a cluster expected to contain r_* bands, the indicator in Eq. (4909) is replaced by $\mathbf{1}[r_C(\rho) \geq r_*]$. We now derive why small level spacings alone cannot establish a flat band. Consider an energy interval

$$\mathcal{I} = \left[E_0 - \frac{\Delta E}{2}, E_0 + \frac{\Delta E}{2} \right]. \quad (4910)$$

Let $M_N(\mathcal{I})$ be the number of finite-system eigenvalues in this interval. The mean level spacing is approximately

$$\bar{\delta}_N = \frac{\Delta E}{M_N(\mathcal{I}) - 1}. \quad (4911)$$

If the thermodynamic density of states per orbital satisfies $\rho(E_0) > 0$, then

$$M_N(\mathcal{I}) \simeq N_H \rho(E_0) \Delta E \quad (4912)$$

for sufficiently small but fixed ΔE . Therefore,

$$\bar{\delta}_N \simeq \frac{1}{N_H \rho(E_0)}. \quad (4913)$$

Hence $\bar{\delta}_N \rightarrow 0$ in every thermodynamic system with a finite nonzero DOS, whether the corresponding states are localized or extended.

f. Theorem 96. Vanishing finite-size level spacing is not a flat-band criterion. Whenever the thermodynamic DOS is finite and nonzero at E_0 ,

$$\bar{\delta}_N = \Theta(N_H^{-1}). \quad (4914)$$

Therefore arbitrarily small finite-size level spacing occurs generically in the thermodynamic limit and does not imply suppression of a coherent band dispersion. A flat-band claim must be based on the width of an adiabatically connected translation-resolved spectral family rather than the spacing between adjacent finite-system eigenvalues. Define the scaled level spacing

$$\delta_N^{\text{sc}} = N_H \bar{\delta}_N. \quad (4915)$$

For an ordinary spectral region with finite DOS,

$$\delta_N^{\text{sc}} \longrightarrow \frac{1}{\rho(E_0)}. \quad (4916)$$

Now consider one coherent band containing $M_N = \beta N_H$ states and having bandwidth W_N . Its typical internal spacing is

$$\bar{\delta}_N^{\text{band}} \simeq \frac{W_N}{\beta N_H}. \quad (4917)$$

Therefore,

$$N_H \bar{\delta}_N^{\text{band}} \simeq \frac{W_N}{\beta}. \quad (4918)$$

If $W_N \longrightarrow 0$, then $N_H \bar{\delta}_N^{\text{band}} \longrightarrow 0$. This differs from ordinary thermodynamic level crowding with finite bandwidth, for which the scaled spacing remains finite. Define

$$\mathcal{L}_n^{(N)} = N_H \text{IPR}_n^{\text{orb}}. \quad (4919)$$

For a fully extended ergodic state, $\mathcal{L}_n^{(N)} = O(1)$. For a localized state, $\mathcal{L}_n^{(N)} = \Theta(N_H)$. For a state with volume fractal dimension D_2^{vol} , $\mathcal{L}_n^{(N)} \sim N_H^{1-D_2^{\text{vol}}}$. Thus $\mathcal{L}_n^{(N)}$ measures the departure from ergodic volume participation. For a degenerate cluster we instead use

$$\mathcal{L}_C^{(N)} = N_C \text{IPR}_{\text{cell}}[P_C]. \quad (4920)$$

An exactly translation-covariant band projector satisfies $\mathcal{L}_C^{(N)} = 1$. The pair $(\mathcal{L}_N, \delta_N^{\text{sc}})$ provides a useful finite-size classification. An ordinary extended dispersive band has $\mathcal{L}_N = O(1)$ and $\delta_N^{\text{sc}} = O(1)$. A coherent flattening band has $\mathcal{L}_N = O(1)$ or, for a degenerate cluster, $\mathcal{L}_C^{(N)} = O(1)$, while $\delta_N^{\text{sc}} \longrightarrow 0$ if the band width vanishes. A localization-induced crowded spectrum instead

has $\mathcal{L}_N \longrightarrow \infty$ for typical nondegenerate states, even though $\bar{\delta}_N \longrightarrow 0$. Thus small raw level spacing and small coherent band width occupy different finite-size scaling sectors. To connect localization directly to the DOS, choose a normalized spectral broadening kernel $K_\eta(E)$ satisfying

$$\int K_\eta(E) dE = 1. \quad (4921)$$

Define the broadened DOS

$$\rho_\eta(E) = \frac{1}{N_H} \sum_n K_\eta(E - E_n). \quad (4922)$$

Define the energy-resolved average IPR

$$\overline{\text{IPR}}_\eta(E) = \frac{\sum_n \text{IPR}_n K_\eta(E - E_n)}{\sum_n K_\eta(E - E_n)}. \quad (4923)$$

The corresponding average normalized participation is

$$\bar{p}_\eta(E) = \frac{\sum_n p_n^{\text{orb}} K_\eta(E - E_n)}{\sum_n K_\eta(E - E_n)}. \quad (4924)$$

A DOS peak generated by extended states satisfies

$$\liminf_{N \rightarrow \infty} \bar{p}_\eta(E) > 0 \quad (4925)$$

for fixed physically resolved broadening. A DOS peak generated by strongly localized states satisfies $\bar{p}_\eta(E) \longrightarrow 0$. Thus the pair $(\rho_\eta(E), \bar{p}_\eta(E))$ distinguishes DOS enhancement by coherent spectral flattening from DOS enhancement by localized-state crowding. Define the local density of states at orbital i by

$$\rho_{i,\eta}(E) = \sum_n p_{n,i} K_\eta(E - E_n). \quad (4926)$$

The total broadened spectral weight is

$$\sum_i \rho_{i,\eta}(E) = \sum_n K_\eta(E - E_n). \quad (4927)$$

Define the LDOS participation ratio

$$\text{IPR}_{\text{LDOS}}(E, \eta) = \frac{\sum_i \rho_{i,\eta}^2(E)}{\left[\sum_i \rho_{i,\eta}(E) \right]^2}. \quad (4928)$$

If a single isolated level dominates the window,

$$K_\eta(E - E_n) \gg K_\eta(E - E_m) \quad (4929)$$

for all $m \neq n$, then $\rho_{i,\eta}(E) \propto p_{n,i}$. Consequently, $\text{IPR}_{\text{LDOS}} \rightarrow \text{IPR}_n$. Thus localization can be extracted directly from spatially resolved spectral weight without reconstructing an arbitrary eigenvector basis. The curvature dependence of localization can be obtained analytically. For a nondegenerate normalized eigenstate,

$$\text{IPR}_n = \sum_i |\psi_{n,i}|^4. \quad (4930)$$

Differentiation gives

$$\partial_\kappa \text{IPR}_n = 4 \text{Re} \sum_i |\psi_{n,i}|^2 \psi_{n,i}^* \partial_\kappa \psi_{n,i}. \quad (4931)$$

Choose the parallel-transport gauge $\langle n | \partial_\kappa n \rangle = 0$. First-order perturbation theory gives

$$|\partial_\kappa n\rangle = \sum_{m \neq n} |m\rangle \frac{\langle m | H_\kappa | n \rangle}{E_n - E_m}. \quad (4932)$$

Therefore,

$$\begin{aligned} \partial_\kappa \text{IPR}_n &= 4 \text{Re} \sum_{m \neq n} \frac{\langle m | H_\kappa | n \rangle}{E_n - E_m} \\ &\times \sum_i |\psi_{n,i}|^2 \psi_{n,i}^* \psi_{m,i}. \end{aligned} \quad (4933)$$

Thus curvature changes the IPR through interband wave-function mixing. Small spectral denominators can strongly enhance this response. Define the curvature participation susceptibility by

$$\chi_{P,n} = \partial_\kappa \ln P_n. \quad (4934)$$

Since $P_n = \text{IPR}_n^{-1}$, we obtain

$$\chi_{P,n} = -\partial_\kappa \ln \text{IPR}_n. \quad (4935)$$

Therefore, $\chi_{P,n} > 0$ means that negative curvature increases real-space participation, whereas $\chi_{P,n} < 0$ means that curvature localizes the state more strongly. This quantity is evaluated independently from the curvature bandwidth susceptibility derived in V.C. Define the third participation moment

$$I_3^{(n)} = \sum_i p_{n,i}^3. \quad (4936)$$

Using the Cauchy-Schwarz inequality on Eq. (4931),

$$|\partial_\kappa \text{IPR}_n| \leq 4 \left[\sum_i |\psi_{n,i}|^6 \right]^{1/2} \|\partial_\kappa n\|. \quad (4937)$$

Hence,

$$|\partial_\kappa \text{IPR}_n| \leq 4 \sqrt{I_3^{(n)}} \|\partial_\kappa n\|. \quad (4938)$$

The quantum metric in the curvature direction is

$$g_{\kappa\kappa}^{(n)} = \langle \partial_\kappa n | (I - P_n) | \partial_\kappa n \rangle. \quad (4939)$$

In the parallel-transport gauge,

$$g_{\kappa\kappa}^{(n)} = \|\partial_\kappa n\|^2. \quad (4940)$$

Therefore,

$$|\partial_\kappa \text{IPR}_n| \leq 4 \sqrt{I_3^{(n)} g_{\kappa\kappa}^{(n)}}. \quad (4941)$$

Using perturbation theory,

$$g_{\kappa\kappa}^{(n)} = \sum_{m \neq n} \frac{|\langle m | H_\kappa | n \rangle|^2}{(E_n - E_m)^2}. \quad (4942)$$

Thus curvature-induced localization changes are bounded by the quantum geometric susceptibility of the eigenstate.

g. Theorem 97. Quantum-geometric bound on participation reconstruction. For a nondegenerate eigenstate,

$$|\partial_\kappa \text{IPR}_n| \leq 4 \sqrt{I_3^{(n)} g_{\kappa\kappa}^{(n)}}. \quad (4943)$$

Because

$$g_{\kappa\kappa}^{(n)} = \sum_{m \neq n} \frac{|\langle m | H_\kappa | n \rangle|^2}{(E_n - E_m)^2}, \quad (4944)$$

strong IPR reconstruction is naturally enhanced near curvature-driven spectral resonances. This provides a direct bridge between the spectral reconstruction derived in V.B-V.D and the real-space participation geometry. For a degenerate or nearly degenerate target cluster, individual eigenvectors are not the correct objects. Using the Riesz projector,

$$P_C = \frac{1}{2\pi i} \oint_{\mathcal{C}} dz (z - H)^{-1}, \quad (4945)$$

differentiate with respect to curvature. Since

$$\partial_\kappa (z - H)^{-1} = (z - H)^{-1} H_\kappa (z - H)^{-1}, \quad (4946)$$

we obtain

$$\partial_\kappa P_C = \frac{1}{2\pi i} \oint_{\mathcal{C}} dz R(z) H_\kappa R(z), \quad (4947)$$

where

$$R(z) = (z - H)^{-1}. \quad (4948)$$

The curvature derivative of the diagonal projector density is

$$\partial_\kappa q_i^{(c)} = \frac{1}{r_C} \langle i | \partial_\kappa P_C | i \rangle, \quad (4949)$$

provided the cluster rank remains fixed. Therefore,

$$\partial_\kappa \text{IPR}[P_C] = 2 \sum_i q_i^{(c)} \partial_\kappa q_i^{(c)}. \quad (4950)$$

This expression is basis independent and remains valid through internal level crossings inside the isolated cluster. We now define the precise false-positive mechanism that the participation analysis is designed to exclude. Suppose a finite system contains $M_N = \Theta(N_H)$ spatially localized states within a fixed energy interval of width $W_{\text{loc}} = O(1)$. The mean level spacing then scales as

$$\bar{\delta}_N = O(N_H^{-1}). \quad (4951)$$

Nevertheless, for each localized state, $\text{IPR}_n = O(1)$, and therefore

$$p_n = O(N_H^{-1}) \rightarrow 0. \quad (4952)$$

Thus spectral crowding and spatial localization can occur simultaneously. No coherent dispersion suppression is required. For a nondegenerate or weakly split coherent band on a periodic cover, a bulk-extended eigenstate satisfies

$$\text{IPR}_n^{\text{cell}} = O(N_c^{-1}), \quad (4953)$$

and hence

$$\liminf_{N \rightarrow \infty} p_n^{\text{cell}} > 0. \quad (4954)$$

If the band simultaneously flattens, $W_{n,N} \rightarrow 0$, while its representation coverage remains complete, $\mathcal{C}_{\text{rep}}^{(N)} \rightarrow 1$, the shrinking spectral width is associated with a coherent

bulk band rather than a collection of isolated real-space resonances. An exactly flat periodic band requires special treatment. Suppose $W_C = 0$ and the corresponding eigenspace has macroscopic rank. A compact-localized orthonormal basis may have

$$\text{IPR}(u_a) = O(1). \quad (4955)$$

A translation-adapted basis spanning the same eigenspace may instead have

$$\text{IPR}(\tilde{u}_a) = O(N^{-1}). \quad (4956)$$

The two descriptions are related by a unitary rotation inside the exactly degenerate subspace. Therefore individual-state IPR scaling is not a physical invariant of an exact flat band. The correct invariants are P_C , $\epsilon_{\text{cov}}[P_C]$, $p_{\text{cell}}[P_C]$, and $\mathcal{C}_{\text{rep}}$. A genuine translation-covariant exact flat band satisfies $\epsilon_{\text{cov}} = 0$, $p_{\text{cell}}[P_C] = 1$, and complete representation coverage. We can now formulate the central distinction of this subsection. A localization-induced crowded cluster is characterized by at least one of the following thermodynamic signatures: $p_n^{\text{cell}} \rightarrow 0$ for typical nondegenerate states, $D_2^{\text{vol}} < 1$, $r_\zeta = O(1)$, $\mathcal{S}_\partial > 0$ for boundary-generated states, or incomplete translation-sector organization. A coherent flattened band instead requires a spectral family with $W_C \ll t$, together with bulk spectral convergence and coherent representation structure. For a nondegenerate near-flat band we additionally require

$$\liminf_{N \rightarrow \infty} p_n^{\text{cell}} > 0. \quad (4957)$$

For an exactly degenerate flat band, the individual-state condition is replaced by

$$\epsilon_{\text{cov}}[P_C] \rightarrow 0, \quad p_{\text{cell}}[P_C] = O(1), \quad (4958)$$

and $\mathcal{C}_{\text{rep}} \rightarrow 1$.

h. Theorem 98. Localization-crowding exclusion criterion. Let $\mathcal{C}_N$ be a sequence of finite-size spectral clusters with small or decreasing energy width. The observation $\bar{\delta}_N \rightarrow 0$ does not establish band flattening. For a nondegenerate coherent band, localization-induced spectral crowding is excluded if the same adiabatically tracked family satisfies

$$\liminf_{N \rightarrow \infty} p_n^{\text{cell}} > 0, \quad (4959)$$

together with $\mathcal{C}_{\text{rep}}^{(N)} \rightarrow 1$, and boundary-independent finite-cover convergence. For an exactly degenerate flat band, the individual-state condition is replaced by the basis-independent projector conditions $\epsilon_{\text{cov}}[P_C] \rightarrow 0$, and

$$p_{\text{cell}}[P_C] = O(1). \quad (4960)$$

Thus compact localization of a chosen flat-band basis is distinguished from localization-induced spectral crowding of physically distinct finite-size states. To visualize this distinction quantitatively, define the spectral compression variable

$$\mathcal{W}_N = \frac{W_{C,N}}{t}, \quad (4961)$$

and the participation variable $\mathcal{P}_N = p_{C,N}^{\text{cell}}$. For a nondegenerate cluster, $\mathcal{P}_N = p_{n,N}^{\text{cell}}$. For a degenerate cluster,

$$\mathcal{P}_N = p_{\text{cell}}[P_{C,N}]. \quad (4962)$$

The thermodynamic regimes are then separated as follows. A coherent dispersive band satisfies $\mathcal{W}_\infty = O(1)$, and $\mathcal{P}_\infty > 0$. A coherent near-flat band satisfies $0 < \mathcal{W}_\infty \ll 1$, and $\mathcal{P}_\infty > 0$. A coherent exactly flat band satisfies $\mathcal{W}_\infty = 0$ with a translation-covariant projector. A localized crowded regime satisfies $\mathcal{P}_N \rightarrow 0$ for nondegenerate states, even if $\bar{\delta}_N \rightarrow 0$. This two-axis analysis prevents spectral density from being confused with band coherence. We now connect participation directly to the curvature sweep. Define the cell-participation compression ratio

$$\mathcal{R}_{P,n}(\kappa) = \frac{p_n^{\text{cell}}(\kappa)}{p_n^{\text{cell}}(0)}. \quad (4963)$$

Then $\mathcal{R}_{P,n} > 1$ means that negative curvature increases inter-cell extension, whereas $\mathcal{R}_{P,n} < 1$ means that negative curvature promotes spatial concentration. Define the internal participation ratio

$$\mathcal{R}_{\text{int},n}(\kappa) = \frac{P_n^{\text{int}}(\kappa)}{P_n^{\text{int}}(0)}. \quad (4964)$$

The pair $(\mathcal{R}_{P,n}, \mathcal{R}_{\text{int},n})$ therefore distinguishes curvature-induced global localization from curvature-induced redistribution inside each moiré cell. For example, $\mathcal{R}_{P,n} \simeq 1$ with $\mathcal{R}_{\text{int},n} < 1$ indicates stronger concentration onto fewer internal orbitals while the state remains globally extended. This is precisely the type of wave-function reconstruction that can accompany coherent moiré-band flattening without producing true thermodynamic localization. Define the curvature bandwidth ratio from V.C,

$$\mathcal{R}_{W,n}(\kappa) = \frac{W_n(\kappa)}{W_n(0)}. \quad (4965)$$

The pair $(\mathcal{R}_{W,n}, \mathcal{R}_{P,n})$ separates two physically different routes to a narrow finite spectrum. Coherent spectral flattening with preserved bulk extension satisfies

$\mathcal{R}_{W,n} < 1$ while $\mathcal{R}_{P,n} = O(1)$. Localization-assisted spectral crowding satisfies $\mathcal{R}_{P,n} \ll 1$ or, in the thermodynamic limit, $p_n^{\text{cell}} \rightarrow 0$. Thus a decreasing bandwidth accompanied by vanishing participation is not interpreted as evidence for a coherent magic-band mechanism. V.D introduced the maximal generalized dispersion rate Ω_n^{max} . We now combine it with participation. A coherent dispersion-suppressed band satisfies $\Omega_n^{\text{max}} \rightarrow 0$ while retaining bulk participation. A localized state can also exhibit a small holonomy response because a wave function confined to a small region is insensitive to distant boundary conditions. Therefore $\Omega_n^{\text{max}} \ll 1$ is not independently sufficient for coherent flattening. The complete first-order diagnostic is the pair $(\Omega_n^{\text{max}}, p_n^{\text{cell}})$. For a nondegenerate coherent flat-band limit we require $\Omega_n^{\text{max}} \rightarrow 0$ and

$$\liminf_{N \rightarrow \infty} p_n^{\text{cell}} > 0. \quad (4966)$$

For an exactly degenerate flat band, the participation condition is again replaced by the translation-covariant cluster projector criterion.

i. Theorem 99. Small generalized velocity does not distinguish flatness from localization. A spatially localized state can become exponentially insensitive to global holonomy or boundary-phase variations, producing $dE_n \approx 0$ without belonging to a coherent flat band. Therefore a small generalized dispersion covector must be interpreted jointly with real-space participation. For a nondegenerate spectral family, coherent flattening requires both $\|dE_n\| \rightarrow 0$ and

$$\liminf p_n^{\text{cell}} > 0. \quad (4967)$$

For an exactly degenerate flat band, the second condition is replaced by translation-covariant projector participation and complete representation coverage. For every candidate spectral cluster we therefore calculate the following real-space quantities. The orbital IPR is IPR^{orb} . The cell IPR is IPR^{cell} . The normalized cell participation is p^{cell} . The internal participation number is P^{int} . The layer participation is

$$P_L = \frac{2}{1 + \mathcal{P}_L^2}. \quad (4968)$$

The volume participation exponent is D_2^{vol} . The geometric probability radius is r_ζ . For open clusters we additionally calculate $\mathcal{B}_n(b)$, and $\mathcal{S}_\partial^{(n)}(b)$. For degenerate clusters we calculate $p_{\text{cell}}[P_C]$, $\epsilon_{\text{cov}}[P_C]$, and $\mathcal{C}_{\text{rep}}$. No single one of these diagnostics is used in isolation. For each selected parameter point (θ, κ, ω) , we calculate the participation diagnostics on a sequence of normal periodic covers satisfying $r_{\text{inj}}^{(N)} \rightarrow \infty$. The orbital and cell IPRs are fitted to

$$\text{IPR}_N = AN_H^{-D_2^{\text{vol}}} [1 + BN_H^{-\omega_{\text{fs}}} + o(N_H^{-\omega_{\text{fs}}})], \quad (4969)$$

where $\omega_{\text{fs}} > 0$ is the leading finite-size correction exponent. The logarithmic two-size estimator is

$$D_{2,\text{eff}}^{(N_1, N_2)} = -\frac{\ln[\text{IPR}(N_2)/\text{IPR}(N_1)]}{\ln(N_2/N_1)}. \quad (4970)$$

A converged volume exponent satisfies $D_{2,\text{eff}}^{(N_1, N_2)} \rightarrow D_2^{\text{vol}}$ as both cover sizes increase. Because two covers with comparable state count can have different injectivity radii and quotient topology, we compare independent cover families. Let $p_{n,A}^{\text{cell}}(N)$ and $p_{n,B}^{\text{cell}}(N)$ denote the same adiabatically tracked spectral family on two independent cover towers. A bulk participation result requires

$$\lim_{N \rightarrow \infty} p_{n,A}^{\text{cell}} = \lim_{N \rightarrow \infty} p_{n,B}^{\text{cell}}. \quad (4971)$$

Likewise, $D_{2,A}^{\text{vol}} = D_{2,B}^{\text{vol}}$. This removes accidental quotient topology as an explanation for the observed localization behavior. For an incommensurate target twist, no exact global moiré translation group exists. We therefore calculate participation directly in the paired finite-cover real-space basis. The total number of physical orbitals is $N_H^{(N)}$, and the IPR is still

$$\text{IPR}_n^{(N)} = \sum_{i=1}^{N_H^{(N)}} |\psi_{n,i}^{(N)}|^4. \quad (4972)$$

The volume exponent is

$$D_2^{\text{vol}} = -\lim_{N \rightarrow \infty} \frac{\ln \text{IPR}_n^{(N)}}{\ln N_H^{(N)}}. \quad (4973)$$

For commensurate angular approximants, the simultaneous geometric condition

$$|\theta_N - \theta| e^{L_N/R} \rightarrow 0 \quad (4974)$$

must be satisfied before participation exponents from different approximants are compared. Otherwise an apparent change in localization can originate from an uncontrolled change of local registry geometry. We now combine the diagnostics of V.C, V.D, and V.E. For an adiabatically tracked target cluster define

$$\mathfrak{Q}_C = (W_C, \Delta_C, \Omega_C^{\text{max}}, \mathcal{P}_C), \quad (4975)$$

where $\mathcal{P}_C = p_n^{\text{cell}}$ for a nondegenerate band, and

$$\mathcal{P}_C = p_{\text{cell}}[P_C] \quad (4976)$$

for an exactly degenerate cluster. The four entries measure global spectral width, spectral isolation, complete first-order generalized dispersion and bulk participation geometry. A nondegenerate coherent isolated flat-band limit requires

$$W_C \rightarrow 0, \quad \liminf \Delta_C > 0, \quad \Omega_C^{\text{max}} \rightarrow 0, \quad (4977)$$

and $\liminf \mathcal{P}_C > 0$. For an exactly degenerate band, the final condition is supplemented by

$$\epsilon_{\text{cov}}[P_C] \rightarrow 0, \quad \mathcal{C}_{\text{rep}} \rightarrow 1. \quad (4978)$$

j. Theorem 100. Complete spectral-participation separation of coherent flattening and localization crowding. Let an isolated sequence of finite-cover spectral clusters be tracked continuously in (θ, κ, ω) . For a nondegenerate band, a shrinking spectral width represents coherent bulk flattening rather than localization-induced crowding only if the thermodynamic sequence simultaneously satisfies

$$W_C \rightarrow 0, \quad \liminf \Delta_C > 0, \quad \Omega_C^{\text{max}} \rightarrow 0, \quad (4979)$$

and $\liminf p_C^{\text{cell}} > 0$. For an exactly degenerate flat band, individual-state participation is basis dependent and is replaced by

$$p_{\text{cell}}[P_C] = O(1), \quad \epsilon_{\text{cov}}[P_C] \rightarrow 0, \quad \mathcal{C}_{\text{rep}} \rightarrow 1. \quad (4980)$$

By contrast, localization-induced spectral crowding is identified by vanishing real-space participation, bounded geometric localization radius, boundary enrichment, or incomplete coherent representation structure even when the raw finite-size level spacing tends to zero. The complete real-space diagnostic chain is therefore

$$\begin{aligned} |\psi_n\rangle &\rightarrow p_{n;x\alpha} \rightarrow \{\text{IPR}^{\text{orb}}, \text{IPR}^{\text{cell}}\} \\ &\rightarrow \{p^{\text{cell}}, P^{\text{int}}, P_L\} \rightarrow \{D_2^{\text{vol}}, r_\zeta\}. \end{aligned} \quad (4981)$$

For a degenerate spectral manifold, the basis-independent chain is

$$P_C \rightarrow Q_x^{(C)} \rightarrow p_{\text{cell}}[P_C] \rightarrow \epsilon_{\text{cov}}[P_C] \rightarrow \mathcal{C}_{\text{rep}}. \quad (4982)$$

The complete reviewer-level discrimination then becomes small finite-size level spacing $\not\Rightarrow$ flat band, large DOS $\not\Rightarrow$ flat band, large individual IPR $\not\Rightarrow$ absence of an exact flat band, small holonomy sensitivity $\not\Rightarrow$ coherent flattening. Instead, coherent curvature-controlled flattening requires simultaneous agreement between spectral, differential, and participation diagnostics. For a nondegenerate band, the defining thermodynamic pattern is

$$W \longrightarrow 0, \quad \Delta > 0, \quad \Omega^{\max} \longrightarrow 0, \quad p^{\text{cell}} = O(1). \quad (4983)$$

For an exactly degenerate flat-band manifold, the final line is replaced by the translation-covariant projector conditions

$$p_{\text{cell}}[P_{\mathcal{C}}] = O(1), \quad \epsilon_{\text{cov}}[P_{\mathcal{C}}] \longrightarrow 0, \quad \mathcal{C}_{\text{rep}} \longrightarrow 1. \quad (4984)$$

Localization-induced spectral crowding instead exhibits

$$\bar{\delta}_N \longrightarrow 0, \quad p_n^{\text{cell}} \longrightarrow 0, \quad (4985)$$

for typical nondegenerate localized states, or a positive boundary enrichment score when the crowding originates from open hyperbolic edges. This separation ensures that any narrow spectral manifold carried forward into the subsequent analysis has survived the strongest real-space localization, boundary, degeneracy, and finite-size crowding tests.

V. CURVATURE-CONTROLLED MAGIC FLAT BANDS

A. Operational Definition of a Magic State

The preceding analysis provides all ingredients required to formulate a nonvisual and quantitatively falsifiable definition of a magic state. We do not identify a magic configuration from the appearance of a narrow band along a selected spectral cut. Instead, we require simultaneous suppression of the complete energetic dispersion, suppression of the generalized first-order dispersion, enhancement of coherent spectral density, positive spectral isolation, and survival of the localization, symmetry, and finite-size tests established above. This distinction is essential because several unrelated mechanisms can generate superficially flat spectral features. Exact band folding can increase the number of plotted branches without reducing the primitive dispersion. Symmetry can enforce exact degeneracy without suppressing kinetic dispersion. Localized states can become insensitive to global holonomy and therefore exhibit a small generalized velocity. A dense collection of localized finite-size levels can generate a large density of states and arbitrarily small level spacings. Finally, numerical truncation or an insufficiently large hyperbolic cover can artificially reduce the measured bandwidth. We therefore construct the magic criterion as an intersection of independent spectral conditions rather than as a single visually inferred property. We write the control parameters as

$$\mathbf{p} = (\theta, K, \omega), \quad \omega = \frac{w}{t}, \quad K \leq 0. \quad (4986)$$

Equivalently, for negative curvature we may use

$$\kappa = -K = \frac{1}{R^2} \geq 0. \quad (4987)$$

The Hamiltonian is therefore $H = H(\theta, K, \omega)$. All microscopic parameters not explicitly included in $\mathbf{p}$ are held fixed in the primary geometry-only comparison. For the square-lattice benchmark, the fixed kinetic reference scale is

$$E_{\text{kin}} = W_0 = 8|t|. \quad (4988)$$

For a more general microscopic model we define E_{kin} once from the corresponding untwisted flat-space monolayer bandwidth and keep that reference fixed throughout the curvature sweep. A magic state is associated with an adiabatically tracked spectral band or isolated spectral cluster. Let $\mathcal{C} = \{n_1, \dots, n_r\}$ denote the target cluster. Its basis-independent spectral projector is

$$P_{\mathcal{C}}(\lambda; \mathbf{p}) = \sum_{a=1}^r |u_a(\lambda; \mathbf{p})\rangle \langle u_a(\lambda; \mathbf{p})|. \quad (4989)$$

When the cluster is separated from the rest of the spectrum, the same projector can be written as

$$P_{\mathcal{C}} = \frac{1}{2\pi i} \oint_{\mathcal{C}} dz (z - H)^{-1}. \quad (4990)$$

This definition remains valid through crossings and exact degeneracies inside the target manifold. Define the global lower cluster edge by

$$E_{\mathcal{C}}^- = \inf_{\lambda} \lambda_{\min} [P_{\mathcal{C}}(\lambda) H(\lambda) P_{\mathcal{C}}(\lambda)]. \quad (4991)$$

Define the global upper cluster edge by

$$E_{\mathcal{C}}^+ = \sup_{\lambda} \lambda_{\max} [P_{\mathcal{C}}(\lambda) H(\lambda) P_{\mathcal{C}}(\lambda)]. \quad (4992)$$

The exact global cluster bandwidth is

$$W_{\mathcal{C}} = E_{\mathcal{C}}^+ - E_{\mathcal{C}}^-. \quad (4993)$$

For a single isolated band, $r = 1$, and

$$W_{\mathcal{C}} = \sup_{\lambda} E_n(\lambda) - \inf_{\lambda} E_n(\lambda). \quad (4994)$$

The parameter λ denotes the complete relevant spectral domain: ordinary moiré momentum in the Euclidean problem, Abelian higher-genus momentum in the rank-one hyperbolic sector, or the complete physical representation-space domain in a higher-rank sector.

We therefore never define W_C from a single plotted path unless that path is proved to contain the complete spectral extrema. Let the complete energy interval occupied by the target cluster be

$$\mathcal{I}_C = [E_C^-, E_C^+]. \quad (4995)$$

Let the highest spectral energy below the cluster be

$$E_{\text{below}} = \sup_{\lambda, m \notin C, E_m < E_C^-} E_m(\lambda), \quad (4996)$$

and let the lowest spectral energy above the cluster be

$$E_{\text{above}} = \inf_{\lambda, m \notin C, E_m > E_C^+} E_m(\lambda). \quad (4997)$$

Define the lower isolation gap by

$$\Delta_C^- = E_C^- - E_{\text{below}}, \quad (4998)$$

and the upper isolation gap by

$$\Delta_C^+ = E_{\text{above}} - E_C^+. \quad (4999)$$

The signed global isolation gap is

$$\Delta_C = \min(\Delta_C^-, \Delta_C^+). \quad (5000)$$

The cluster is globally isolated only when $\Delta_C > 0$. A local direct gap is insufficient for the operational definition because two bands can remain separated at every common generalized momentum while overlapping globally in energy. For a Euclidean band,

$$\mathbf{v}_n(\mathbf{k}) = \frac{1}{\hbar} \nabla_{\mathbf{k}} E_n(\mathbf{k}). \quad (5001)$$

For a hyperbolic generalized band, the local spectral derivative is the covector

$$dE_n \in T_{[\rho]}^* \mathcal{B}_d. \quad (5002)$$

Using the Hodge metric G_ρ introduced above, the local generalized dispersion rate is

$$\Omega_n^{\text{gen}} = \frac{1}{\hbar} \|dE_n\|_{G_\rho^{-1}}. \quad (5003)$$

For a cluster we define the maximal first-order dispersion by

$$\Omega_C^{\text{max}} = \sup_{\lambda, a \in C} \Omega_a^{\text{gen}}(\lambda). \quad (5004)$$

In the Euclidean problem the corresponding quantity is

$$\Omega_C^{\text{max}} = \sup_{\mathbf{k}, a \in C} \frac{|\nabla_{\mathbf{k}} E_a|}{\hbar L_{\text{ref}}}, \quad (5005)$$

if the same dimensionless spectral-coordinate convention is desired. More directly, we may retain the physical Euclidean velocity

$$v_C^{\text{max}} = \sup_{\mathbf{k}, a \in C} |\mathbf{v}_a(\mathbf{k})|. \quad (5006)$$

For the unified magic functional we convert the generalized derivative into an energy variation by multiplying by the metric diameter

$$D_G = \text{diam}_G(\mathfrak{D}). \quad (5007)$$

Define

$$E_{\text{disp}, C} = \hbar D_G \Omega_C^{\text{max}}. \quad (5008)$$

The bandwidth inequality derived above gives $W_C \leq E_{\text{disp}, C}$ for a connected differentiable band family. Thus $E_{\text{disp}, C}$ measures the largest first-order energy variation permitted by the complete generalized dispersion. First-order stationarity alone does not guarantee a weakly dispersive band. At a stationary point, let $M_C = \nabla^2 E_C$ denote the generalized spectral Hessian. Let

$$M\mathbf{c}_a = \Lambda_a G\mathbf{c}_a \quad (5009)$$

be the corresponding geometry-normalized generalized eigenproblem. Define

$$\Lambda_C^{\text{max}} = \sup_{\lambda, a \in C, j} |\Lambda_{a,j}(\lambda)|. \quad (5010)$$

The maximal quadratic energy variation across the spectral domain is bounded to leading order by

$$E_{\text{disp}, C}^{(2)} = \frac{1}{2} D_G^2 \Lambda_C^{\text{max}}. \quad (5011)$$

This quantity will be used as a stronger second-order magic certificate, but it is not required for the minimal operational score. A magic state must not be confused with a localized level whose spectral response is small because it does not explore the translation structure. For a nondegenerate band on a finite cover we define the normalized cell participation

$$p_C^{\text{cell}} = \frac{1}{N_c \text{IPR}_C^{\text{cell}}}. \quad (5012)$$

For an exactly degenerate cluster we instead use the basis-independent spectral-projector participation

$$p_{\text{cell}}[P_C] = \frac{1}{N_c \text{IPR}_{\text{cell}}[P_C]}. \quad (5013)$$

Let $\mathcal{C}_{\text{rep}}$ denote the representation-sector coverage of the tracked band family. Let $\epsilon_{\text{cov}}[P_C]$ denote the translation-covariance defect of the cluster projector. For a nondegenerate band we define

$$\mathcal{C}_C^{\text{coh}} = p_C^{\text{cell}} \mathcal{C}_{\text{rep}}. \quad (5014)$$

For a degenerate cluster we define

$$\mathcal{C}_C^{\text{coh}} = \frac{p_{\text{cell}}[P_C] \mathcal{C}_{\text{rep}}}{1 + \epsilon_{\text{cov}}[P_C]}. \quad (5015)$$

In both cases, $0 \leq \mathcal{C}_C^{\text{coh}} \leq 1$. A perfectly translation-coherent band family has $\mathcal{C}_C^{\text{coh}} = 1$. A strongly localized nondegenerate state satisfies $p_C^{\text{cell}} \rightarrow 0$ and therefore $\mathcal{C}_C^{\text{coh}} \rightarrow 0$. Thus localization cannot produce a high operational magic score merely by making the eigenvalue insensitive to global boundary conditions. A large DOS is also not sufficient by itself because localized states can generate strong spectral crowding. We therefore define a coherence-weighted DOS. Let $\rho_{C,\eta}(E)$ be the normalized broadened DOS of the target cluster. Define

$$\rho_{C,\eta}^{\text{coh}}(E) = \mathcal{C}_C^{\text{coh}} \rho_{C,\eta}(E). \quad (5016)$$

Its maximal coherent spectral density is

$$\rho_{C,\eta}^{\text{coh,max}} = \max_E \rho_{C,\eta}^{\text{coh}}(E). \quad (5017)$$

Thus an arbitrarily large DOS generated by a sequence of strongly localized states is suppressed by the vanishing participation factor. For an exactly degenerate translation-covariant flat-band subspace, the projector participation remains finite even if one chooses a compact-localized eigenbasis. Therefore the coherent DOS does not incorrectly reject compact-localized representations of a genuine exact flat band. For finite numerical spectra we use a Lorentzian kernel

$$K_\eta(E) = \frac{1}{\pi} \frac{\eta}{E^2 + \eta^2}. \quad (5018)$$

Then

$$\rho_{C,\eta}(E) = \frac{1}{N_C} \sum_{n \in \mathcal{C}} K_\eta(E - E_n), \quad (5019)$$

with

$$\int_{-\infty}^{\infty} \rho_{C,\eta}(E) dE = 1. \quad (5020)$$

For a resolved finite-width band we require $\epsilon_E \ll \eta \ll W_C$, whenever this hierarchy can be satisfied. For an exactly flat candidate with $W_C = 0$, the correct procedure is different. We first establish exact or thermodynamic flatness independently and then take $\eta \rightarrow 0^+$ after the finite-cover limit. For a normalized exactly flat spectral manifold at energy E_* ,

$$\rho_{C,\eta}(E) = K_\eta(E - E_*). \quad (5021)$$

Therefore,

$$\rho_{C,\eta}^{\text{max}} = \frac{1}{\pi\eta}. \quad (5022)$$

Hence $\rho_{C,\eta}^{\text{max}} \rightarrow \infty$ as $\eta \rightarrow 0^+$. The operational definition contains a hard isolation condition. We define

$$\mathcal{G}_\Delta = \begin{cases} 1, & \Delta_C > 0, \\ 0, & \Delta_C \leq 0. \end{cases} \quad (5023)$$

Thus an unisolated band receives zero magic score, irrespective of how small its bandwidth or generalized velocity becomes. For the thermodynamic system we require the stronger condition

$$\liminf_{N \rightarrow \infty} \frac{\Delta_{C,N}}{E_{\text{kin}}} > 0. \quad (5024)$$

This prevents a sequence with $W_{C,N} \rightarrow 0$ and simultaneously $\Delta_{C,N} \rightarrow 0$ from being classified as an isolated magic band merely because Δ/W becomes large. We now define four dimensionless residuals whose simultaneous suppression constitutes the minimal operational magic condition. The energetic-flatness residual is

$$r_W = \frac{W_C}{\Delta_C}. \quad (5025)$$

The first-order dispersion residual is

$$r_v = \frac{E_{\text{disp},C}}{\Delta_C} = \frac{\hbar D_G \Omega_C^{\text{max}}}{\Delta_C}. \quad (5026)$$

The coherent-DOS residual is

$$r_\rho = \frac{1}{E_{\text{kin}} \rho_{C,\eta}^{\text{coh,max}}}. \quad (5027)$$

The coherence residual is

$$r_{\text{coh}} = \frac{1 - \mathcal{C}_c^{\text{coh}}}{\mathcal{C}_c^{\text{coh}}}. \quad (5028)$$

For $\mathcal{C}_c^{\text{coh}} = 0$, we set $r_{\text{coh}} = +\infty$. The complete residual vector is

$$\mathbf{r}_{\text{magic}} = (r_W, r_v, r_\rho, r_{\text{coh}}). \quad (5029)$$

A strict operational magic limit requires $\mathbf{r}_{\text{magic}} \rightarrow \mathbf{0}$ together with the positive thermodynamic isolation condition. The condition $r_W \rightarrow 0$ means $W_c \ll \Delta_c$. Thus the spectral manifold is globally narrow relative to its isolation gap. The condition $r_v \rightarrow 0$ means

$$\hbar D_G \Omega_c^{\text{max}} \ll \Delta_c. \quad (5030)$$

Thus the complete first-order generalized dispersion is suppressed rather than merely one plotted component. The condition $r_\rho \rightarrow 0$ means

$$E_{\text{kin}} \rho_{\mathcal{C},\eta}^{\text{coh,max}} \rightarrow \infty. \quad (5031)$$

Thus the high spectral density is large compared with the microscopic kinetic-energy scale and is carried by the coherent band subspace. Finally, $r_{\text{coh}} \rightarrow 0$ means $\mathcal{C}_c^{\text{coh}} \rightarrow 1$. Therefore the spectral concentration remains translation coherent rather than becoming a collection of localized finite-size resonances. We now define a bounded scalar functional that converts the residual vector into a single ranking variable. For $\Delta_c > 0$ and $\mathcal{C}_c^{\text{coh}} > 0$, define

$$\mathcal{M}_c(\theta, K, \omega) = \left[\prod_{a \in \{W, v, \rho, \text{coh}\}} \frac{1}{1 + r_a} \right]^{1/4}. \quad (5032)$$

If $\Delta_c \leq 0$ or $\mathcal{C}_c^{\text{coh}} = 0$, we define $\mathcal{M}_c = 0$. Substituting the explicit residuals gives

$$\mathcal{M}_c = \mathcal{G}_\Delta \left[\frac{\Delta_c}{\Delta_c + W_c} \times \frac{\Delta_c}{\Delta_c + \hbar D_G \Omega_c^{\text{max}}} \times \frac{E_{\text{kin}} \rho_{\mathcal{C},\eta}^{\text{coh,max}}}{1 + E_{\text{kin}} \rho_{\mathcal{C},\eta}^{\text{coh,max}}} \times \mathcal{C}_c^{\text{coh}} \right]^{1/4}. \quad (5033)$$

This functional is completely explicit. Its first factor measures isolation relative to global bandwidth. Its second factor measures isolation relative to complete first-order dispersion. Its third factor measures coherent DOS enhancement relative to the fixed microscopic kinetic scale. Its fourth factor removes localization and incomplete translation-sector coverage. Each factor in Eq. (5033) lies between zero and unity. Therefore, $0 \leq \mathcal{M}_c \leq 1$. If

$$W_c \rightarrow 0, \quad \Omega_c^{\text{max}} \rightarrow 0, \quad \rho_{\mathcal{C},\eta}^{\text{coh,max}} \rightarrow \infty, \quad \mathcal{C}_c^{\text{coh}} \rightarrow 1, \quad (5034)$$

and

$$\liminf \Delta_c > 0, \quad (5035)$$

then $\mathcal{M}_c \rightarrow 1$. Conversely, if $\mathcal{M}_c \rightarrow 1$, every factor in Eq. (5033) must tend to unity. Therefore,

$$\frac{W_c}{\Delta_c} \rightarrow 0, \quad \frac{\hbar D_G \Omega_c^{\text{max}}}{\Delta_c} \rightarrow 0, \quad E_{\text{kin}} \rho_{\mathcal{C},\eta}^{\text{coh,max}} \rightarrow \infty, \quad (5036)$$

and $\mathcal{C}_c^{\text{coh}} \rightarrow 1$.

a. Theorem 101. Exact operational magic-limit criterion. For an isolated and continuously tracked spectral cluster, $\mathcal{M}_c \rightarrow 1$ if and only if all four dimensionless residuals satisfy

$$r_W, r_v, r_\rho, r_{\text{coh}} \rightarrow 0 \quad (5037)$$

within a thermodynamic sequence whose isolation gap obeys

$$\liminf \frac{\Delta_c}{E_{\text{kin}}} > 0. \quad (5038)$$

Thus the unit magic limit is equivalent to simultaneous global energetic flattening, multidirectional first-order dispersion suppression, coherent DOS enhancement, and translation-coherent spectral support. The conventional flatness ratio is

$$\mathcal{F}_c = \frac{\Delta_c}{W_c}. \quad (5039)$$

Therefore,

$$r_W = \frac{1}{\mathcal{F}_c}. \quad (5040)$$

The energetic factor in the magic functional is

$$\frac{1}{1 + r_W} = \frac{\mathcal{F}_c}{1 + \mathcal{F}_c} = \frac{\Delta_c}{\Delta_c + W_c}. \quad (5041)$$

Thus the first factor is exactly the bounded version of the conventional flatness ratio derived above. However, $\mathcal{F}_c \gg 1$ does not by itself imply $\mathcal{M}_c \simeq 1$, because the velocity, DOS-coherence, and participation factors can remain small. A first-order stationary band can retain strong quadratic dispersion. We therefore define the second-order residual

$$r_2 = \frac{E_{\text{disp},c}^{(2)}}{\Delta_c} = \frac{D_G^2 \Lambda_c^{\text{max}}}{2\Delta_c}. \quad (5042)$$

The strong second-order magic functional is

$$\mathcal{M}_C^{(2)} = \mathcal{G}_\Delta \left[\frac{1}{(1+r_W)(1+r_v)(1+r_2)(1+r_\rho)(1+r_{\text{coh}})} \right]^{1/5}. \quad (5043)$$

Using

$$\frac{1}{1+r_{\text{coh}}} = \mathcal{C}_C^{\text{coh}}, \quad (5044)$$

this becomes

$$\begin{aligned} \mathcal{M}_C^{(2)} = \mathcal{G}_\Delta & \left[\frac{\Delta_C}{\Delta_C + W_C} \times \frac{\Delta_C}{\Delta_C + \hbar D_G \Omega_C^{\text{max}}} \right. \\ & \times \frac{\Delta_C}{\Delta_C + \frac{1}{2} D_G^2 \Lambda_C^{\text{max}}} \times \left. \frac{E_{\text{kin}} \rho_{C,\eta}^{\text{coh,max}}}{1 + E_{\text{kin}} \rho_{C,\eta}^{\text{coh,max}}} \times \mathcal{C}_C^{\text{coh}} \right]^{1/5}. \end{aligned} \quad (5045)$$

The minimal functional $\mathcal{M}_C$ is used for operational detection. The stronger functional $\mathcal{M}_C^{(2)}$ is used when second-order dispersion is available with sufficient numerical accuracy. An operational flatness functional should not change under a trivial overall rescaling of the Hamiltonian. Consider

$$H \longrightarrow cH, \quad c > 0. \quad (5046)$$

Then $W_C \longrightarrow cW_C$, and $\Delta_C \longrightarrow c\Delta_C$. The generalized energy differential scales as $dE \longrightarrow cdE$. Therefore, $E_{\text{disp},C} \longrightarrow cE_{\text{disp},C}$. Likewise,

$$\Lambda_C^{\text{max}} \longrightarrow c\Lambda_C^{\text{max}}, \quad E_{\text{disp},C}^{(2)} \longrightarrow cE_{\text{disp},C}^{(2)}. \quad (5047)$$

The DOS transforms according to

$$\rho_{cH}(E) = \frac{1}{c} \rho_H \left(\frac{E}{c} \right). \quad (5048)$$

Thus

$$\rho_{cH}^{\text{max}} = \frac{1}{c} \rho_H^{\text{max}}. \quad (5049)$$

The kinetic reference energy transforms as $E_{\text{kin}} \longrightarrow cE_{\text{kin}}$. Therefore, $E_{\text{kin}} \rho_C^{\text{coh,max}}$ is invariant. The coherence factor is a property of normalized projectors and probabilities and therefore remains unchanged. Consequently, r_W , r_v , r_2 , r_ρ , r_{coh} are all invariant under uniform positive rescaling of the complete energy spectrum.

b. Theorem 102. Energy-scale invariance of the magic functional. Under

$$H \longrightarrow cH, \quad c > 0, \quad (5050)$$

with the DOS broadening and reference kinetic energy rescaled in the same physical energy units, $\mathcal{M}_C \longrightarrow \mathcal{M}_C$. Likewise, $\mathcal{M}_C^{(2)} \longrightarrow \mathcal{M}_C^{(2)}$. Therefore the operational magic score cannot be increased by a trivial overall reduction of the energy scale. For every normalized isolated band, $\rho_C^{\text{max}} W_C \geq 1$. Thus $W_C \longrightarrow 0$ forces the ideal unbroadened DOS to increase. Nevertheless the converse does not hold. A saddle point can produce $\rho_C^{\text{max}} \longrightarrow \infty$ while $W_C = O(E_{\text{kin}})$. Furthermore, a dense collection of localized states can produce a large finite-cover DOS without defining a coherent band. The DOS factor is therefore retained as an independent diagnostic, but it is multiplied by the coherence factor through ρ^{coh} . Suppose a sequence of localized states satisfies $p_{C,N}^{\text{cell}} \longrightarrow 0$. Then $\mathcal{C}_{C,N}^{\text{coh}} \longrightarrow 0$. Hence

$$\rho_{C,\eta}^{\text{coh,max}} = \mathcal{C}_C^{\text{coh}} \rho_{C,\eta}^{\text{max}} \quad (5051)$$

cannot remain macroscopically large merely because the raw spectral density is large, unless the DOS divergence overcomes the vanishing coherent fraction. More importantly, the magic functional contains the independent factor $[\mathcal{C}_C^{\text{coh}}]^{1/4}$. Therefore,

$$\mathcal{C}_C^{\text{coh}} \longrightarrow 0 \implies \mathcal{M}_C \longrightarrow 0. \quad (5052)$$

This remains true even if

$$W_C \longrightarrow 0, \quad \Omega_C^{\text{max}} \longrightarrow 0, \quad \rho_{C,\eta}^{\text{max}} \longrightarrow \infty. \quad (5053)$$

c. Theorem 103. Localization veto. A sequence of nondegenerate localized spectral states satisfying $p_{C,N}^{\text{cell}} \longrightarrow 0$ cannot approach the unit operational magic limit. Specifically,

$$\lim_{N \rightarrow \infty} \mathcal{M}_{C,N} = 0 \quad (5054)$$

whenever the coherent participation factor vanishes. Thus localization-induced level crowding is excluded independently of the raw bandwidth, raw DOS, or holonomy sensitivity. An exact flat band can admit compact-localized basis states. Individual-state IPR is therefore not a valid veto inside an exactly degenerate band. For the target projector,

$$P_C \longrightarrow U_C P_C U_C^\dagger = P_C \quad (5055)$$

under an arbitrary unitary basis rotation inside the degenerate cluster. Therefore, $p_{\text{cell}}[P_C]$ and $\epsilon_{\text{cov}}[P_C]$ are

invariant. For an exactly translation-covariant flat-band projector,

$$p_{\text{cell}}[P_C] = 1, \quad \epsilon_{\text{cov}}[P_C] = 0, \quad (5056)$$

and $C_{\text{rep}} = 1$. Hence $C_C^{\text{coh}} = 1$ independently of whether the chosen eigenbasis is Bloch-like or compact localized.

d. Theorem 104. Basis-invariant exact-flat-band admissibility. The operational magic functional is invariant under arbitrary unitary rotations inside an isolated degenerate flat-band subspace when the coherence factor is evaluated from the spectral projector. Therefore compact-localized basis representations of a translation-covariant exact flat band are not misclassified as localization-induced artifacts. Let a symmetry enforce an r -dimensional multiplet $E_1(\lambda) = \dots = E_r(\lambda)$. The internal level splitting is zero, $\Delta E_{\text{internal}} = 0$. However, the cluster bandwidth remains

$$W_C = \sup_{\lambda} E_C(\lambda) - \inf_{\lambda} E_C(\lambda). \quad (5057)$$

Likewise, the complete generalized dispersion remains dE_C and can be nonzero. Thus symmetry multiplicity does not automatically improve either r_W or r_v . The use of the cluster projector eliminates arbitrary internal basis splittings without eliminating the physical dispersion of the multiplet. At zero interlayer coupling, exact supercell folding is a unitary relabeling of the primitive spectrum. Let

$$H_{\text{fold}} = U^\dagger H_{\text{primitive}} U. \quad (5058)$$

Then

$$\text{Spec } H_{\text{fold}} = \text{Spec } H_{\text{primitive}}. \quad (5059)$$

The physical DOS is unchanged, $\rho_{\text{fold}} = \rho_{\text{primitive}}$. The unfolded velocity is unchanged,

$$\nabla_{\mathbf{k}} E_{\alpha}^{\text{fold}} = \nabla_{\mathbf{q}} \varepsilon_0. \quad (5060)$$

Therefore a folded branch segment is not treated as an independent candidate band unless it is part of an adiabatically tracked physical spectral manifold. This prevents Brillouin-zone compression from artificially increasing $\mathcal{M}$. A magic state in the twisted bilayer must arise from the interlayer reconstruction mechanism rather than from an already flat uncoupled monolayer. Define the zero-hybridization reference bandwidth

$$W_C^{(0)} = W_C(\theta, K, \omega = 0). \quad (5061)$$

Define the hybridization compression ratio

$$\mathcal{R}_{\text{hyb}}^W = \frac{W_C(\theta, K, \omega)}{W_C^{(0)}}. \quad (5062)$$

A hybridization-induced narrowing satisfies $\mathcal{R}_{\text{hyb}}^W < 1$. The corresponding generalized-dispersion ratio is

$$\mathcal{R}_{\text{hyb}}^v = \frac{\Omega_C^{\text{max}}(\theta, K, \omega)}{\Omega_C^{\text{max}}(\theta, K, 0)}. \quad (5063)$$

A hybridization-driven dispersion suppression satisfies $\mathcal{R}_{\text{hyb}}^v < 1$. These quantities do not enter the minimal numerical value of $\mathcal{M}$; instead they establish the causal origin of a high magic score. Likewise, a high magic score at $K < 0$ is not by itself sufficient to claim that curvature caused the flattening. Define the flat-space reference score

$$\mathcal{M}_C^{(0)}(\theta, \omega) = \mathcal{M}_C(\theta, K = 0, \omega). \quad (5064)$$

Define the curvature enhancement ratio

$$\mathcal{R}_K^{\mathcal{M}} = \frac{\mathcal{M}_C(\theta, K, \omega)}{\mathcal{M}_C^{(0)}(\theta, \omega)}. \quad (5065)$$

A curvature-enhanced magic state satisfies $\mathcal{R}_K^{\mathcal{M}} > 1$. The causal statement is strengthened by requiring the same enhancement to survive the fixed-hybridization and coordination-normalized controls introduced above. Thus we separate two statements: magic $\iff \mathcal{M}$ is large and curvature controlled $\iff \mathcal{M}$ is enhanced by $K < 0$ under the causal controls. We now state the asymptotic operational definition independently of any finite numerical threshold. A target cluster is an exact thermodynamic magic flat-band manifold if

$$\begin{aligned} \lim_{N \rightarrow \infty} \frac{W_{C,N}}{E_{\text{kin}}} &= 0, \quad \lim_{N \rightarrow \infty} \frac{\hbar D_{G,N} \Omega_{C,N}^{\text{max}}}{E_{\text{kin}}} = 0, \\ \liminf_{N \rightarrow \infty} \frac{\Delta_{C,N}}{E_{\text{kin}}} &> 0, \end{aligned} \quad (5066)$$

and the coherent participation structure remains nonvanishing. For a nondegenerate band,

$$\liminf_{N \rightarrow \infty} p_{C,N}^{\text{cell}} > 0. \quad (5067)$$

For a degenerate flat-band manifold,

$$\liminf_{N \rightarrow \infty} p_{\text{cell}}[P_{C,N}] > 0, \quad \epsilon_{\text{cov}}[P_{C,N}] \longrightarrow 0, \quad (5068)$$

and $C_{\text{rep}}^{(N)} \longrightarrow 1$. The coherent DOS then develops the corresponding spectral concentration when the energy resolution is removed after the thermodynamic limit.

e. Theorem 105. Exact thermodynamic operational definition. A spectral cluster is accepted as an exact coherent magic flat-band manifold only when global bandwidth suppression, complete first-order dispersion suppression, finite positive thermodynamic isolation, and coherent bulk participation occur simultaneously. Neither vanishing finite-size level spacing, diverging raw DOS, symmetry degeneracy, nor vanishing holonomy sensitivity alone is sufficient. A numerical magic score must be a lower bound on the physical score rather than a value obtained by inserting uncertain eigenvalues directly into the ideal formula. Let the measured bandwidth be $\widetilde{W}_C$, and let the total eigenvalue-scale uncertainty be ϵ_E . The exact bandwidth obeys

$$W_C \leq \widetilde{W}_C + 2\epsilon_E. \quad (5069)$$

Define the conservative upper bandwidth

$$W_C^U = \widetilde{W}_C + 2\epsilon_E. \quad (5070)$$

For the measured gap $\widetilde{\Delta}_C$, the exact gap obeys

$$\Delta_C \geq \widetilde{\Delta}_C - 2\epsilon_E. \quad (5071)$$

Define

$$\Delta_C^L = \widetilde{\Delta}_C - 2\epsilon_E. \quad (5072)$$

If $\Delta_C^L \leq 0$, spectral isolation is not established. Suppose the generalized dispersion is measured as $\widetilde{\Omega}_C^{\max}$ with uncertainty ϵ_Ω . Define

$$\Omega_C^U = \widetilde{\Omega}_C^{\max} + \epsilon_\Omega. \quad (5073)$$

Then $\Omega_C^{\max} \leq \Omega_C^U$. For a finite-difference derivative between representation sectors separated by δs , the energy-error contribution satisfies

$$\epsilon_\Omega^{(E)} \leq \frac{2\epsilon_E}{\hbar\delta s}. \quad (5074)$$

The discretization contribution is estimated independently by convergence with decreasing δs . Let the measured broadened coherent DOS maximum be $\widehat{\rho}_{C,\eta}^{\text{coh,max}}$. Let its numerical uncertainty be ϵ_ρ . Define

$$\rho_{C,\eta}^L = \max \left[0, \widehat{\rho}_{C,\eta}^{\text{coh,max}} - \epsilon_\rho \right]. \quad (5075)$$

For a resolvent-based Lorentzian DOS, an operator error ϵ_H contributes at most

$$\epsilon_\rho^{(H)} \leq \frac{\epsilon_H}{\pi\eta^2} \quad (5076)$$

up to the normalization convention. Stochastic, polynomial-truncation, and sampling uncertainties are added explicitly to obtain the complete ϵ_ρ . Let $\widetilde{C}_C^{\text{coh}}$ be the measured coherence factor and let ϵ_{coh} bound its uncertainty. Define

$$C_C^L = \max \left[0, \widetilde{C}_C^{\text{coh}} - \epsilon_{\text{coh}} \right]. \quad (5077)$$

If $\Delta_C^L > 0$ and $C_C^L > 0$, we define the established magic score by

$$\mathcal{M}_C^{\text{cert}} = \left[\frac{\Delta_C^L}{\Delta_C^L + W_C^U} \times \frac{\Delta_C^L}{\Delta_C^L + \hbar D_G \Omega_C^U} \times \frac{E_{\text{kin}} \rho_{C,\eta}^L}{1 + E_{\text{kin}} \rho_{C,\eta}^L} \times C_C^L \right]^{1/4}. \quad (5078)$$

If either conservative lower bound vanishes, we set $\mathcal{M}_C^{\text{cert}} = 0$. The functional is monotone increasing in the gap, coherent DOS, and coherence factor, and monotone decreasing in the bandwidth and generalized dispersion. Therefore the conservative substitutions above give a lower bound.

f. Theorem 106. established lower bound on the physical magic score. Under the stated uncertainty bounds, $\mathcal{M}_C \geq \mathcal{M}_C^{\text{cert}}$. Consequently, a reported finite numerical magic score can be interpreted as a established minimum degree of simultaneous energetic, differential, spectral-density, and coherence flattening. At a given parameter point, several isolated spectral clusters may exist. Let $\mathfrak{C}_{\text{adm}}(\mathbf{p})$ denote the set of clusters satisfying the hard spectral-isolation and numerical-verification requirements. We define the parameter-level magic functional by

$$\mathcal{M} \left(\theta, K, \frac{w}{t} \right) = \max_{C \in \mathfrak{C}_{\text{adm}}} \mathcal{M}_C \left(\theta, K, \frac{w}{t} \right). \quad (5079)$$

The maximizing cluster is

$$C_* \in \arg \max_{C \in \mathfrak{C}_{\text{adm}}} \mathcal{M}_C. \quad (5080)$$

For numerical data we analogously define

$$\mathcal{M}^{\text{cert}} \left(\theta, K, \frac{w}{t} \right) = \max_{C \in \mathfrak{C}_{\text{adm}}} \mathcal{M}_C^{\text{cert}}. \quad (5081)$$

This is the quantity used to locate operational magic configurations. For fixed K and $\omega = \frac{w}{t}$, we define the operational magic-angle set by

$$\Theta_{\text{magic}}(K, \omega) = \arg \max_{\theta} \mathcal{M}(\theta, K, \omega). \quad (5082)$$

If the maximizer is unique,

$$\Theta_{\text{magic}}(K, \omega) = \{\theta_{\text{m}}(K, \omega)\}. \quad (5083)$$

If several angles give indistinguishable maxima within the established uncertainty, we report a magic interval or discrete magic-angle set rather than forcing a unique value. Thus the operational definition does not assume a priori that a unique magic angle exists. The complete magic set is

$$\mathfrak{P}_{\text{magic}} = \arg \max_{(\theta, K, \omega)} \mathcal{M}(\theta, K, \omega). \quad (5084)$$

Depending on the model, this set can be an isolated point, a discrete collection of points, a one-dimensional ridge or a higher-dimensional manifold. The existence and geometry of this set are determined rather than assumed. Inside a smooth region with one fixed maximizing cluster and positive isolation gap, let $p_a \in \{\theta, K, \omega\}$. For an interior local maximum, $\partial_{p_a} \mathcal{M}_C = 0$ for every active control direction. Equivalently,

$$\partial_{p_a} \ln \mathcal{M}_C = 0. \quad (5085)$$

From Eq. (5033),

$$\begin{aligned} 4\partial_{p_a} \ln \mathcal{M}_C &= \partial_{p_a} \ln \left[\frac{\Delta}{\Delta + W} \right] + \partial_{p_a} \ln \left[\frac{\Delta}{\Delta + E_{\text{disp}}} \right] \\ &+ \partial_{p_a} \ln \left[\frac{E_{\text{kin}} \rho_{\text{coh}}^{\text{max}}}{1 + E_{\text{kin}} \rho_{\text{coh}}^{\text{max}}} \right] + \partial_{p_a} \ln \mathcal{C}_{\text{coh}}. \end{aligned} \quad (5086)$$

For compact notation we temporarily suppress the cluster subscript. The first term is

$$\partial_{p_a} \ln \left[\frac{\Delta}{\Delta + W} \right] = - \frac{\Delta \partial_{p_a} W - W \partial_{p_a} \Delta}{\Delta(\Delta + W)}. \quad (5087)$$

The second term is

$$\partial_{p_a} \ln \left[\frac{\Delta}{\Delta + E_{\text{disp}}} \right] = - \frac{\Delta \partial_{p_a} E_{\text{disp}} - E_{\text{disp}} \partial_{p_a} \Delta}{\Delta(\Delta + E_{\text{disp}})}. \quad (5088)$$

Define $X_\rho = E_{\text{kin}} \rho_{\text{coh}}^{\text{max}}$. Since E_{kin} is fixed in the geometry-only protocol,

$$\partial_{p_a} \ln \left[\frac{X_\rho}{1 + X_\rho} \right] = \frac{1}{1 + X_\rho} \partial_{p_a} \ln \rho_{\text{coh}}^{\text{max}}. \quad (5089)$$

Therefore the exact stationarity equation is

$$\begin{aligned} 0 &= - \frac{\Delta \partial_{p_a} W - W \partial_{p_a} \Delta}{\Delta(\Delta + W)} \\ &- \frac{\Delta \partial_{p_a} E_{\text{disp}} - E_{\text{disp}} \partial_{p_a} \Delta}{\Delta(\Delta + E_{\text{disp}})} \\ &+ \frac{1}{1 + E_{\text{kin}} \rho_{\text{coh}}^{\text{max}}} \partial_{p_a} \ln \rho_{\text{coh}}^{\text{max}} \\ &+ \partial_{p_a} \ln \mathcal{C}_{\text{coh}}. \end{aligned} \quad (5090)$$

This is the exact local balance condition defining an interior operational magic optimum. Equation (5090) shows that an operational magic point is not generally the minimum of the bandwidth alone. The first contribution favors decreasing W . The second favors decreasing the complete generalized dispersion $E_{\text{disp}} = \hbar D_G \Omega^{\text{max}}$. The third favors increasing coherent spectral density. The fourth penalizes loss of translation coherence or the onset of localization. The derivatives of the isolation gap enter both spectral-compression terms. Thus a point with the absolutely smallest bandwidth need not maximize the magic functional if it is simultaneously approaching a gap closure, localization transition, or representation-sector loss.

g. Theorem 107. Operational magic is a constrained spectral optimum rather than a bandwidth minimum. At a smooth interior optimum, the control derivatives satisfy Eq. (5090). Therefore $\partial_p W = 0$ is neither necessary nor sufficient for an operational magic point. The optimum is determined by the simultaneous competition between bandwidth, isolation gap, complete generalized dispersion, coherent DOS, and bulk participation. A stronger but nongeneric situation occurs when all primary diagnostics reach favorable extrema at the same parameter point. Let $\mathbf{p}_* = (\theta_*, K_*, \omega_*)$. A coincident-extremum point satisfies

$$\nabla_{\mathbf{p}} W_C \Big|_{\mathbf{p}_*} = 0, \quad \nabla_{\mathbf{p}} \Omega_C^{\text{max}} \Big|_{\mathbf{p}_*} = 0, \quad (5091)$$

and

$$\nabla_{\mathbf{p}} \rho_C^{\text{coh, max}} \Big|_{\mathbf{p}_*} = 0. \quad (5092)$$

The corresponding second-derivative conditions are

$$\nabla_{\mathbf{p}}^2 W_C \succeq 0, \quad \nabla_{\mathbf{p}}^2 \Omega_C^{\text{max}} \succeq 0, \quad \nabla_{\mathbf{p}}^2 \rho_C^{\text{coh, max}} \preceq 0. \quad (5093)$$

The point must additionally satisfy $\Delta_C > 0$ and a finite coherent participation factor. This stronger coincidence criterion is useful when the numerical data support it, but it is not imposed as a prerequisite for the more general operational definition. Because bandwidth, velocity, gap, and DOS are independent observables, no single parameter point must optimize every quantity separately. Define the objective vector

$$\mathbf{O}_C = \left(W_C, E_{\text{disp},C}, -\rho_C^{\text{coh,max}}, -\Delta_C, -\mathcal{C}_C^{\text{coh}} \right). \quad (5094)$$

A parameter point $\mathbf{p}_1$ dominates $\mathbf{p}_2$ if every component satisfies $O_j(\mathbf{p}_1) \leq O_j(\mathbf{p}_2)$ and at least one inequality is strict. An operational magic candidate must lie on the corresponding Pareto front. The scalar functional $\mathcal{M}$ provides a symmetric dimensionless ranking of this physically admissible Pareto set. We do not interpret the numerical value of $\mathcal{M}$ independently of its constituent observables. The multiplicative form treats the four residual axes symmetrically. For a robustness test, define positive weights

$$\alpha_W, \alpha_v, \alpha_\rho, \alpha_{\text{coh}} > 0 \quad (5095)$$

satisfying

$$\alpha_W + \alpha_v + \alpha_\rho + \alpha_{\text{coh}} = 1. \quad (5096)$$

Define

$$\begin{aligned} \mathcal{M}_C^{(\alpha)} &= \mathcal{G}_\Delta (1 + r_W)^{-\alpha_W} (1 + r_v)^{-\alpha_v} \\ &\times (1 + r_\rho)^{-\alpha_\rho} (1 + r_{\text{coh}})^{-\alpha_{\text{coh}}}. \end{aligned} \quad (5097)$$

The primary definition corresponds to

$$\alpha_W = \alpha_v = \alpha_\rho = \alpha_{\text{coh}} = \frac{1}{4}. \quad (5098)$$

A physically robust magic configuration should remain near the optimum under moderate variation of the positive weights. This prevents a single arbitrary scalarization convention from generating the result. Suppose $r_a \rightarrow 0 \quad \forall a$. Then for every positive normalized weight vector, $\mathcal{M}_C^{(\alpha)} \rightarrow 1$. Conversely, if all weights are strictly positive and $\mathcal{M}_C^{(\alpha)} \rightarrow 1$, then each residual must vanish. Therefore the asymptotic definition of an exact magic state is independent of the chosen positive scalarization weights.

h. Theorem 108. Scalarization-independent exact magic limit. The exact limit $\mathbf{r}_{\text{magic}} \rightarrow 0$ is equivalent to $\mathcal{M}_C^{(\alpha)} \rightarrow 1$ for every strictly positive normalized weight vector. Thus the residual-vector definition is fundamental, while the equal-weight functional is an operational ranking device rather than an arbitrary replacement of the underlying physical conditions. For cover N , define $\mathcal{M}_{C,N}$. The bulk lower magic score is

$$\mathcal{M}_C^{\text{bulk}} = \liminf_{N \rightarrow \infty} \mathcal{M}_{C,N}^{\text{cert}}. \quad (5099)$$

A finite-cover peak in $\mathcal{M}_N$ is accepted only if the location and height converge with increasing injectivity radius.

Thus $r_{\text{inj}}^{(N)} \rightarrow \infty$ is part of the operational definition. For exponentially decaying hopping we simultaneously require $L_C^{(N)} \rightarrow \infty$, and

$$\frac{L_C^{(N)}}{r_{\text{inj}}^{(N)}} \rightarrow 0. \quad (5100)$$

For an incommensurate angular approximant sequence we additionally require

$$|\theta_N - \theta| e^{L_N/R} \rightarrow 0. \quad (5101)$$

Thus the magic score inherits the complete numerical convergence hierarchy rather than bypassing it. Let

$$\left\{ Q_N^{(A)} \right\}, \quad \left\{ Q_N^{(B)} \right\} \quad (5102)$$

be inequivalent finite-cover towers. A bulk operational result requires

$$\liminf_{N \rightarrow \infty} \mathcal{M}_N^{(A)} = \liminf_{N \rightarrow \infty} \mathcal{M}_N^{(B)}. \quad (5103)$$

Likewise, the maximizing parameter location must converge consistently. This prevents special quotient topology from generating a false magic configuration. We now combine the complete set of requirements. For every candidate cluster and parameter point we record

$$\mathfrak{C}_{\text{magic}} = \left(W_C, \Delta_C, \Omega_C^{\text{max}}, \rho_C^{\text{coh,max}}, \mathcal{C}_C^{\text{coh}}, \mathcal{M}_C \right). \quad (5104)$$

When second-order data are available, we enlarge this certificate to

$$\mathfrak{C}_{\text{magic}}^{(2)} = \left(W_C, \Delta_C, \Omega_C^{\text{max}}, \Lambda_C^{\text{max}}, \rho_C^{\text{coh,max}}, \mathcal{C}_C^{\text{coh}}, \mathcal{M}_C^{(2)} \right). \quad (5105)$$

A single scalar value is never reported without this underlying certificate vector. The operational definition rejects the principal false-positive mechanisms explicitly. Pure band folding is rejected because it does not suppress the complete unfolded dispersion. Symmetry-enforced degeneracy is rejected because the cluster bandwidth and projector dispersion remain finite unless the complete multiplet is actually flat. Localization-induced crowding is rejected because the coherent participation factor vanishes. Boundary states are rejected through periodic covers or boundary enrichment analysis. A van-Hove DOS peak is rejected if the global bandwidth and generalized dispersion remain large. A closing-gap narrow band is rejected by $\mathcal{G}_\Delta = 0$. A numerical truncation artifact is rejected because $\mathcal{M}^{\text{cert}}$ uses conservative upper and lower error bounds. An exact compact-localized flat-band basis is retained because the participation criterion is applied to the spectral projector rather than to an arbitrary eigenvector basis.

i. Theorem 109. False-positive-resistant operational magic criterion. A high value of $\mathcal{M}_C^{\text{cert}}$ cannot be generated solely by any one of the following mechanisms: band folding, symmetry multiplicity, localized level crowding, boundary accumulation, a van-Hove singularity or a vanishing isolation gap. The operational score becomes large only when these alternatives have been removed simultaneously and the same coherent spectral manifold is narrow, isolated, weakly dispersive, and spectrally concentrated. We distinguish two levels of verification. The minimal operational magic criterion uses

$$(W_C, \Delta_C, \Omega_C^{\text{max}}, \rho_C^{\text{coh, max}}, \mathcal{C}_C^{\text{coh}}). \quad (5106)$$

The strong operational criterion additionally requires small second-order dispersion,

$$\frac{D_G^2 \Lambda_C^{\text{max}}}{2\Delta_C} \ll 1. \quad (5107)$$

Thus the strong criterion distinguishes a truly weakly curved flat manifold from a band that is first-order stationary but retains a large quadratic dispersion. We distinguish the spectral object from the external control values. A magic state is the spectral cluster $\mathcal{C}_*$ that maximizes the operational functional while satisfying the hard verification conditions. A magic parameter point is $\mathbf{p}_* = (\theta_*, K_*, \omega_*)$ at which that maximum is realized. A magic angle is the θ -coordinate of the optimum only after K and ω have been specified. Thus the term magic angle is not treated as a fundamental one-parameter concept once curvature and interlayer coupling are independent control variables. We can now state the complete definition. For fixed microscopic model parameters, an isolated spectral cluster $\mathcal{C}$ at $\mathbf{p} = (\theta, K, w/t)$ is called an operational magic state when it belongs to the admissible thermodynamic spectral family and locally or globally maximizes the established functional $\mathcal{M}_C^{\text{cert}}$ while satisfying $\Delta_C^L > 0$, nonvanishing coherent participation, $\mathcal{C}_C^L > 0$, and the complete finite-cover, truncation, angular, symmetry, and eigensolver convergence conditions. The strict magic-flat limit is

$$\begin{aligned} \frac{W_C}{\Delta_C} &\rightarrow 0, \quad \frac{\hbar D_G \Omega_C^{\text{max}}}{\Delta_C} \rightarrow 0, \\ E_{\text{kin}} \rho_C^{\text{coh, max}} &\rightarrow \infty, \quad \mathcal{C}_C^{\text{coh}} \rightarrow 1, \\ \liminf \frac{\Delta_C}{E_{\text{kin}}} &> 0. \end{aligned} \quad (5108)$$

For the strong second-order limit we additionally require

$$\frac{D_G^2 \Lambda_C^{\text{max}}}{2\Delta_C} \rightarrow 0. \quad (5109)$$

j. Theorem 110. Complete operational characterization of a magic flat-band state. The operational magic criterion is simultaneously sensitive to global bandwidth, complete generalized dispersion, coherent spectral-density enhancement, spectral isolation, and bulk participation. In the strict limit, $\mathcal{M}_C \rightarrow 1$ is equivalent to suppression of the complete residual vector $\mathbf{r}_{\text{magic}} \rightarrow 0$ within a spectrally isolated thermodynamic sequence. The operational definition is invariant under overall energy rescaling, basis rotations inside degenerate spectral manifolds, and ordinary supercell relabeling. It assigns zero asymptotic weight to localization-induced spectral crowding and does not identify a gapless narrow band as an isolated magic state. The complete magic-state diagnosis proceeds from the microscopic Hamiltonian to an isolated spectral manifold,

$$H(\theta, K, w/t) \rightarrow P_C. \quad (5110)$$

The spectral projector determines $W_C, \Delta_C, dE_C, \rho_C, \mathcal{C}_C^{\text{coh}}$. These quantities produce the dimensionless residual vector

$$\mathbf{r}_{\text{magic}} = \left(\frac{W_C}{\Delta_C}, \frac{\hbar D_G \Omega_C^{\text{max}}}{\Delta_C}, \frac{1}{E_{\text{kin}} \rho_C^{\text{coh, max}}}, \frac{1 - \mathcal{C}_C^{\text{coh}}}{\mathcal{C}_C^{\text{coh}}} \right). \quad (5111)$$

The residuals are then combined into

$$\mathcal{M}_C = \mathcal{G}_\Delta \left[\prod_a (1 + r_a)^{-1} \right]^{1/4}. \quad (5112)$$

Finally,

$$\mathcal{M}(\theta, K, w/t) = \max_C \mathcal{M}_C \quad (5113)$$

provides the operational magic-flatness landscape over the complete control parameter space. Only a spectral manifold satisfying this complete hierarchy is carried forward as a curvature-controlled magic-state candidate.

B. Euclidean Magic-Angle Benchmark

We now specialize the operational magic-state construction to the Euclidean limit, $K = 0$. This limit provides the indispensable reference against which every curvature-dependent magic-angle shift will subsequently be measured. The Euclidean benchmark has two logically distinct purposes. First, it must reproduce the complete twisted-square-lattice spectral reconstruction already derived from the exact coincidence lattice, the exact moiré Brillouin zone, and interlayer hybridization. Second, it must determine the magic condition without using curvature as an adjustable mechanism. We therefore define

the Euclidean magic-angle branches by $\theta_\nu^* \left(0, \frac{w}{t}\right)$ where ν labels distinct operational magic branches. We do not assume that a magic branch exists for every value of w/t , nor do we assume that it is unique. Existence, multiplicity, and the location of the branches are determined from the exact Euclidean spectral problem. For one spinless orbital per microscopic lattice site, the reference square-lattice dispersion is

$$\varepsilon_0(\mathbf{k}) = -2t [\cos(k_x a) + \cos(k_y a)]. \quad (5114)$$

An exactly commensurate twist is parameterized by coprime integers (m, n_c) ,

$$\tan\left(\frac{\theta}{2}\right) = \frac{n_c}{m}, \quad (5115)$$

where we use n_c rather than n to avoid confusion with the band index. The coincidence index is

$$\Sigma = \frac{m^2 + n_c^2}{d}, \quad (5116)$$

with

$$d = \begin{cases} 1, & m, n_c \text{ have opposite parity,} \\ 2, & m, n_c \text{ are both odd.} \end{cases} \quad (5117)$$

Using

$$\sin\left(\frac{\theta}{2}\right) = \frac{n_c}{\sqrt{m^2 + n_c^2}}, \quad (5118)$$

we obtain the exact identity

$$\Sigma = \frac{n_c^2}{d} \csc^2\left(\frac{\theta}{2}\right). \quad (5119)$$

The bilayer supercell contains $N_{\text{sc}} = 2\Sigma$ microscopic orbitals. The exact moiré reciprocal basis satisfies

$$|\mathbf{G}_1| = |\mathbf{G}_2| = G_M, \quad (5120)$$

where

$$G_M = \frac{2\pi}{a\sqrt{\Sigma}}. \quad (5121)$$

The two primitive reciprocal vectors are orthogonal, $\mathbf{G}_1 \cdot \mathbf{G}_2 = 0$. Thus the exact moiré Brillouin zone is a square. For every exact commensurate angle, the Euclidean Bloch Hamiltonian is

$$H_\Sigma(\mathbf{k}; \theta, \omega) = \begin{pmatrix} E_1^{\text{fold}}(\mathbf{k}) & \mathcal{T}_\theta(\mathbf{k}) \\ \mathcal{T}_\theta^\dagger(\mathbf{k}) & E_2^{\text{fold}}(\mathbf{k}) \end{pmatrix}, \quad (5122)$$

where $\omega = \frac{w}{t}$. Its exact dimension is

$$\dim H_\Sigma = 2\Sigma. \quad (5123)$$

The eigenvalue equation is

$$H_\Sigma(\mathbf{k}; \theta, \omega) |u_{\ell\mathbf{k}}\rangle = E_\ell(\mathbf{k}; \theta, \omega) |u_{\ell\mathbf{k}}\rangle. \quad (5124)$$

At $\omega = 0$, the spectrum reduces exactly to folded monolayer branches,

$$E_\ell^{(0)}(\mathbf{k}) = \varepsilon_0(\mathbf{k} + \mathbf{Q}_\ell). \quad (5125)$$

Their velocities satisfy

$$\nabla_{\mathbf{k}} E_\ell^{(0)} = \nabla_{\mathbf{q}} \varepsilon_0 \Big|_{\mathbf{q}=\mathbf{k}+\mathbf{Q}_\ell}. \quad (5126)$$

Therefore no magic condition can be generated by the exact moiré relabeling alone. A nontrivial Euclidean magic state must satisfy $\omega > 0$. For every continuously tracked spectral cluster $\mathcal{C}$, we evaluate the operational functional derived above at $K = 0$,

$$\mathcal{M}_{\mathcal{C}}^{(E)}(\theta, \omega) = \mathcal{M}_{\mathcal{C}}(\theta, 0, \omega). \quad (5127)$$

The complete Euclidean magic landscape is

$$\mathcal{M}^{(E)}(\theta, \omega) = \max_{\mathcal{C}} \mathcal{M}_{\mathcal{C}}^{(E)}(\theta, \omega). \quad (5128)$$

We define the Euclidean magic-angle set by

$$\Theta_{\text{magic}}^{(E)}(\omega) = \arg \max_{\theta} \mathcal{M}^{(E)}(\theta, \omega). \quad (5129)$$

If the ν -th branch is isolated and unique, we write

$$\theta_\nu^*(0, \omega) \in \Theta_{\text{magic}}^{(E)}(\omega). \quad (5130)$$

Thus the Euclidean magic angle is not defined by a visually flat cut. It is the twist at which the same coherent spectral manifold simultaneously optimizes bandwidth suppression, full group-velocity suppression, coherent DOS enhancement, and positive isolation. The exact crystallographic spectrum exists on the discrete set

$$\Theta_{\text{exact}}^\square = \left\{ 2 \arctan\left(\frac{n_c}{m}\right) : \gcd(m, n_c) = 1 \right\} \quad (5131)$$

modulo the square-lattice point group. For a finite supercell cutoff $N_{\text{sc}} \leq N_{\text{max}}$, define

$$\Theta_{\text{exact}}(N_{\text{max}}) = \left\{ \theta \in \Theta_{\text{exact}}^{\square} : 2\Sigma(\theta) \leq N_{\text{max}} \right\}. \quad (5132)$$

The finite-supercell operational magic angle is

$$\theta_{\nu, N_{\text{max}}}^* = \arg \max_{\theta \in \Theta_{\text{exact}}(N_{\text{max}})} \mathcal{M}_{\nu}^{(E)}(\theta, \omega). \quad (5133)$$

The converged exact-commensurate benchmark requires $\theta_{\nu, N_{\text{max}}}^* \rightarrow \theta_{\nu}^*$ as increasingly large commensurate structures are admitted. The ten-site and thirty-four-site structures provide exact implementation benchmarks, but they are not assumed a priori to coincide with an asymptotic magic angle. The small-angle Euclidean scaling is controlled by the low-energy dispersion near a square-lattice band extremum. We first expand around $\mathbf{k} = \mathbf{0}$. Using

$$\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} + O(x^8), \quad (5134)$$

the monolayer dispersion becomes

$$\begin{aligned} \varepsilon_0(\mathbf{k}) &= -4t + ta^2(k_x^2 + k_y^2) - \frac{ta^4}{12}(k_x^4 + k_y^4) \\ &+ \frac{ta^6}{360}(k_x^6 + k_y^6) + O(k^8 a^8). \end{aligned} \quad (5135)$$

Define $A = ta^2$. Then the leading dispersion is

$$\varepsilon_0 + 4t = Ak^2 + O(k^4). \quad (5136)$$

The bare effective mass is therefore

$$m_0^* = \frac{\hbar^2}{2ta^2}. \quad (5137)$$

This quadratic band edge is the decisive distinction between the square lattice and a linearly dispersing Dirac reference system. The natural momentum scale of the exact square moiré Brillouin zone is

$$G_M = \frac{2\pi}{a\sqrt{\Sigma}}. \quad (5138)$$

The corresponding quadratic kinetic scale is $E_M = AG_M^2$. Substituting $A = ta^2$ gives

$$E_M = \frac{4\pi^2 t}{\Sigma}. \quad (5139)$$

Using Eq. (5119),

$$E_M = \frac{4\pi^2 dt}{n_c^2} \sin^2 \left(\frac{\theta}{2} \right). \quad (5140)$$

At small twist,

$$E_M = \frac{\pi^2 dt}{n_c^2} \theta^2 + O(\theta^4). \quad (5141)$$

Thus the Euclidean square-lattice moiré kinetic scale collapses quadratically with twist, $E_M \propto t\theta^2$. This quadratic law is the fundamental scale governing the Euclidean square-lattice magic-angle benchmark. Write the moiré-periodic interlayer tunneling field as

$$T(\mathbf{r}) = \sum_{\mathbf{G} \in \Lambda_M^*} T_{\mathbf{G}} e^{i\mathbf{G} \cdot \mathbf{r}}. \quad (5142)$$

The first reciprocal star is

$$\mathcal{S}_1 = \{+\mathbf{G}_1, -\mathbf{G}_1, +\mathbf{G}_2, -\mathbf{G}_2\}. \quad (5143)$$

For the scalar one-orbital benchmark, C_4 symmetry gives equal first-star amplitudes,

$$|T_{\mathbf{G}_1}| = |T_{-\mathbf{G}_1}| = |T_{\mathbf{G}_2}| = |T_{-\mathbf{G}_2}| = w_1. \quad (5144)$$

We parameterize

$$w_1(\theta) = \gamma_1(\theta)w. \quad (5145)$$

For a smooth microscopic tunneling kernel,

$$\gamma_1(\theta) = \gamma_0 + O(\theta^2) \quad (5146)$$

along a small-angle branch, provided the first-star Fourier component has a regular $G_M \rightarrow 0$ limit. The ratio between the first-star hybridization scale and the moiré kinetic-energy scale is

$$\alpha_{\square} = \frac{w_1}{E_M}. \quad (5147)$$

Using Eq. (5139),

$$\alpha_{\square} = \frac{\gamma_1(\theta)}{4\pi^2} \frac{w}{t} \Sigma. \quad (5148)$$

Equivalently,

$$\alpha_{\square} = \frac{\gamma_1(\theta)n_c^2}{4\pi^2 d} \frac{w}{t} \csc^2 \left(\frac{\theta}{2} \right). \quad (5149)$$

For small twist,

$$\alpha_{\square} = \frac{\gamma_0 n_c^2}{\pi^2 d} \frac{w/t}{\theta^2} [1 + O(\theta^2)]. \quad (5150)$$

Thus the effective hybridization grows as

$$\alpha_{\square} \propto \frac{w/t}{\theta^2}. \quad (5151)$$

a. Theorem 111. Quadratic Euclidean moiré coupling law. For the square-lattice band-edge continuum,

$$E_M = ta^2 G_M^2 = \frac{4\pi^2 t}{\Sigma}, \quad (5152)$$

and the dimensionless interlayer coupling is

$$\alpha_{\square} = \frac{w_1}{E_M} = \frac{\gamma_1 \omega \Sigma}{4\pi^2}. \quad (5153)$$

Along a bounded-numerator small-angle commensurate sequence,

$$\alpha_{\square} = \Theta\left(\frac{\omega}{\theta^2}\right). \quad (5154)$$

Therefore reducing the Euclidean twist at fixed w/t drives the system toward strong moiré hybridization through a quadratic rather than linear kinetic-scale collapse. Define the rescaled momentum

$$\mathbf{p} = \frac{\mathbf{k}}{G_M}. \quad (5155)$$

Then $\mathbf{k} = G_M \mathbf{p}$. The quadratic kinetic term becomes

$$Ak^2 = E_M p^2. \quad (5156)$$

The quartic term becomes

$$-\frac{ta^4}{12} (k_x^4 + k_y^4) = -E_M \frac{a^2 G_M^2}{12} (p_x^4 + p_y^4). \quad (5157)$$

Since

$$a^2 G_M^2 = \frac{4\pi^2}{\Sigma}, \quad (5158)$$

the quartic correction is suppressed by $O(\Sigma^{-1})$. Therefore the low-energy moiré Hamiltonian takes the scaling form

$$\frac{H_{\text{low}} + 4t}{E_M} = \mathcal{H}_0(\mathbf{p}) + \alpha_{\square} \mathcal{V}_1 + O(\Sigma^{-1}). \quad (5159)$$

Here $\mathcal{H}_0$ is the dimensionless quadratic kinetic operator and $\mathcal{V}_1$ is the normalized moiré tunneling operator. Thus the leading small-angle spectral problem depends on w/t and θ only through the combination $\alpha_{\square}$. Let a target continuum band have bandwidth $W_{\nu}(\theta, \omega)$. The scaling form implies

$$\frac{W_{\nu}}{E_M} = \mathcal{W}_{\nu}(\alpha_{\square}) + O(\Sigma^{-1}). \quad (5160)$$

Similarly, for a spectral isolation gap,

$$\frac{\Delta_{\nu}}{E_M} = \mathcal{D}_{\nu}(\alpha_{\square}) + O(\Sigma^{-1}). \quad (5161)$$

The normalized DOS obeys

$$E_M \rho_{\nu}(E) = \mathcal{R}_{\nu}\left(\frac{E - E_{\nu,0}}{E_M}, \alpha_{\square}\right) + O(\Sigma^{-1}). \quad (5162)$$

The group-velocity scale satisfies

$$\frac{\hbar v_{\nu}}{E_M/G_M} = \mathcal{V}_{\nu}(\mathbf{p}, \alpha_{\square}) + O(\Sigma^{-1}). \quad (5163)$$

Consequently, the operational magic functional satisfies

$$\mathcal{M}_{\nu}^{(E)}(\theta, \omega) = \mathfrak{M}_{\nu}(\alpha_{\square}) + O(\Sigma^{-1}). \quad (5164)$$

This provides a stringent numerical benchmark. Spectra calculated at different values of w/t and θ must collapse when expressed in the variables

$$\frac{E + 4t}{E_M}, \quad \alpha_{\square}. \quad (5165)$$

Failure of this collapse at sufficiently small twist signals either non-negligible lattice corrections, substantial higher-star tunneling structure, or an incorrect implementation of the Euclidean continuum limit.

b. Theorem 112. Small-angle Euclidean spectral-collapse theorem. If the square-lattice band edge is quadratic and the normalized tunneling profile possesses a regular small-angle limit, then every dimensionless low-energy spectral observable has the asymptotic form

$$\mathcal{O}(\theta, \omega) = \mathcal{O}_0(\alpha_{\square}) + O(\Sigma^{-1}). \quad (5166)$$

Hence the leading Euclidean magic condition is a condition on one dimensionless coupling rather than on θ and w/t separately. Suppose the continuum operational functional possesses an isolated local maximum at $\alpha_{\square} = \alpha_{\nu}^*$, with

$$\left. \frac{d\mathfrak{M}_{\nu}}{d\alpha_{\square}} \right|_{\alpha_{\nu}^*} = 0. \quad (5167)$$

For a strict local maximum,

$$\left. \frac{d^2 \mathfrak{M}_\nu}{d\alpha_\square^2} \right|_{\alpha_\nu^*} < 0. \quad (5168)$$

The values $\{\alpha_\nu^*\}$ are the dimensionless Euclidean magic couplings. They are determined by the microscopic dimensionless tunneling structure and the operational magic criterion. Once they are known, the physical magic angles follow analytically. At a magic point, $\alpha_\square = \alpha_\nu^*$. Using Eq. (5148),

$$\alpha_\nu^* = \frac{\gamma_1(\theta_\nu^*) \omega \Sigma_\nu^*}{4\pi^2}. \quad (5169)$$

Therefore,

$$\Sigma_\nu^* = \frac{4\pi^2 \alpha_\nu^*}{\gamma_1(\theta_\nu^*)} \frac{t}{w}. \quad (5170)$$

The corresponding bilayer supercell size is

$$N_{sc,\nu}^* = \frac{8\pi^2 \alpha_\nu^*}{\gamma_1(\theta_\nu^*)} \frac{t}{w}. \quad (5171)$$

Thus the Euclidean square-lattice magic supercell grows inversely with the interlayer coupling,

$$N_{sc,\nu}^* = \Theta\left(\frac{t}{w}\right). \quad (5172)$$

Using Eq. (5149), the magic condition becomes

$$\alpha_\nu^* = \frac{\gamma_1(\theta_\nu^*) n_c^2}{4\pi^2 d} \omega \csc^2\left(\frac{\theta_\nu^*}{2}\right). \quad (5173)$$

Therefore,

$$\sin^2\left(\frac{\theta_\nu^*}{2}\right) = \frac{\gamma_1(\theta_\nu^*) n_c^2}{4\pi^2 d \alpha_\nu^*} \frac{w}{t}. \quad (5174)$$

If the first-star form factor is approximated by its small-angle limit,

$$\gamma_1(\theta) = \gamma_0 + O(\theta^2), \quad (5175)$$

then the leading explicit result is

$$\theta_\nu^*\left(0, \frac{w}{t}\right) = 2 \arcsin \left[\frac{n_c}{2\pi \sqrt{d \alpha_\nu^*}} \sqrt{\gamma_0 \frac{w}{t}} \right] + O\left[\left(\frac{w}{t}\right)^{3/2}\right]. \quad (5176)$$

For weak interlayer coupling,

$$\theta_\nu^*\left(0, \frac{w}{t}\right) = \frac{n_c}{\pi \sqrt{d \alpha_\nu^*}} \sqrt{\gamma_0 \frac{w}{t}} + O\left[\left(\frac{w}{t}\right)^{3/2}\right]. \quad (5177)$$

This is the leading Euclidean magic-angle scaling law of the quadratic square-lattice band-edge problem.

c. Theorem 113. Euclidean square-lattice magic-angle scaling. If the small-angle dimensionless continuum problem possesses a finite isolated operational magic coupling α_ν^* , then

$$\theta_\nu^* \propto \left(\frac{w}{t}\right)^{1/2}. \quad (5178)$$

Simultaneously,

$$\Sigma_\nu^* \propto \frac{t}{w}, \quad N_{sc,\nu}^* \propto \frac{t}{w}. \quad (5179)$$

The square-root magic-angle law follows directly from the quadratic band-edge dispersion and is therefore distinct from a linear-dispersion moiré system. The preceding square-root law is a special case of a more general kinematic result. Suppose the monolayer dispersion near the relevant reference point obeys

$$\varepsilon(\mathbf{k}_0 + \mathbf{q}) - \varepsilon(\mathbf{k}_0) = c_p |\mathbf{q}|^p + o(|\mathbf{q}|^p), \quad (5180)$$

where $p > 0$. Since $G_M = \Theta(\theta)$, the moiré kinetic scale is

$$E_M^{(p)} = \Theta(t\theta^p). \quad (5181)$$

The dimensionless coupling is therefore

$$\alpha_p = \Theta\left(\frac{w/t}{\theta^p}\right). \quad (5182)$$

A finite magic root $\alpha_p = \alpha_{p,\nu}^*$ implies

$$\theta_\nu^* = \Theta\left[\left(\frac{w}{t}\right)^{1/p}\right]. \quad (5183)$$

Thus $p = 1$ gives $\theta^* \propto \frac{w}{t}$, whereas the square-lattice band edge has $p = 2$, and therefore

$$\theta^* \propto \sqrt{\frac{w}{t}}. \quad (5184)$$

d. Theorem 114. Dispersion-order universality of Euclidean magic angles. For a moiré system whose relevant untwisted dispersion begins at order p in momentum, every finite dimensionless magic-coupling root generates the weak-coupling scaling

$$\theta_\nu^* = \Theta\left[(w/t)^{1/p}\right]. \quad (5185)$$

The magic-angle exponent is therefore fixed by the order of the underlying kinetic dispersion before details of

the microscopic tunneling determine the numerical prefactor. We now derive an analytic weak-hybridization benchmark that can be checked directly against the exact numerical bands. Consider one central quadratic state with dispersion

$$\varepsilon_c(\mathbf{k}) = Ak^2, \quad (5186)$$

and the four first-star states

$$\varepsilon_{\mathbf{G}}(\mathbf{k}) = A|\mathbf{k} + \mathbf{G}|^2, \quad \mathbf{G} \in \mathcal{S}_1. \quad (5187)$$

The second-order energy correction is

$$\delta E^{(2)}(\mathbf{k}) = \sum_{\mathbf{G} \in \mathcal{S}_1} \frac{w_1^2}{Ak^2 - A|\mathbf{k} + \mathbf{G}|^2}. \quad (5188)$$

Since

$$Ak^2 - A|\mathbf{k} + \mathbf{G}|^2 = -A(G_M^2 + 2\mathbf{k} \cdot \mathbf{G}), \quad (5189)$$

we obtain

$$\begin{aligned} \frac{1}{Ak^2 - A|\mathbf{k} + \mathbf{G}|^2} &= -\frac{1}{AG_M^2} \left[1 - \frac{2\mathbf{k} \cdot \mathbf{G}}{G_M^2} \right. \\ &\quad \left. + \frac{4(\mathbf{k} \cdot \mathbf{G})^2}{G_M^4} + O\left(\frac{k^3}{G_M^3}\right) \right]. \end{aligned} \quad (5190)$$

The first star satisfies

$$\sum_{\mathbf{G} \in \mathcal{S}_1} \mathbf{G} = 0. \quad (5191)$$

It also satisfies

$$\sum_{\mathbf{G} \in \mathcal{S}_1} G_\mu G_\nu = 2G_M^2 \delta_{\mu\nu}. \quad (5192)$$

Therefore,

$$\sum_{\mathbf{G} \in \mathcal{S}_1} (\mathbf{k} \cdot \mathbf{G})^2 = 2G_M^2 k^2. \quad (5193)$$

Substitution yields

$$\delta E^{(2)}(\mathbf{k}) = -\frac{4w_1^2}{AG_M^2} - \frac{8w_1^2}{AG_M^4} k^2 + O(k^4). \quad (5194)$$

Using $E_M = AG_M^2$ and

$$\alpha_{\square} = \frac{w_1}{AG_M^2}, \quad (5195)$$

the reconstructed low-energy dispersion becomes

$$\begin{aligned} E_{\text{eff}}(\mathbf{k}) &= E_{\Gamma} - 4\alpha_{\square}^2 E_M \\ &\quad + A[1 - 8\alpha_{\square}^2] k^2 + O(k^4) + O(w_1^4). \end{aligned} \quad (5196)$$

Thus the quadratic coefficient is reduced according to

$$\frac{A_{\text{eff}}}{A} = 1 - 8\alpha_{\square}^2 + O(\alpha_{\square}^4). \quad (5197)$$

Since

$$A = \frac{\hbar^2}{2m_0^*}, \quad A_{\text{eff}} = \frac{\hbar^2}{2m_{\text{eff}}^*}, \quad (5198)$$

we obtain

$$\frac{m_0^*}{m_{\text{eff}}^*} = 1 - 8\alpha_{\square}^2 + O(\alpha_{\square}^4). \quad (5199)$$

Thus interlayer hybridization increases the magnitude of the effective mass at weak coupling. The perturbative quadratic coefficient formally vanishes at $\alpha_{\text{pc}} = \frac{1}{\sqrt{8}}$. We

interpret α_{pc} only as the first-star perturbative mass-cancellation scale. Because $\alpha_{\text{pc}} = O(1)$, it lies outside the parametrically weak regime required for the truncated series itself to establish an exact magic point. The physical operational magic coupling α_{ν}^* must therefore be obtained from the full Hamiltonian. The value $1/\sqrt{8}$ provides a benchmark for the onset and sign of mass renormalization, not a substituted numerical magic angle.

e. Theorem 115. First-star square-lattice mass-renormalization benchmark. For an isotropic scalar first-star truncation of a quadratic square-lattice band edge,

$$\frac{m_0^*}{m_{\text{eff}}^*} = 1 - 8\alpha_{\square}^2 + O(\alpha_{\square}^4). \quad (5200)$$

Therefore the exact numerical Euclidean spectrum must initially exhibit quadratic suppression of the inverse effective mass as $\alpha_{\square}$ is increased from zero. Failure to reproduce this weak-coupling trend would signal an inconsistency between the microscopic tunneling implementation and the square-lattice continuum reduction. The preceding factor 8 can be generalized without assuming that only the first star contributes. Let

$$\mathbf{g} = \frac{\mathbf{G}}{G_M}, \quad (5201)$$

and define

$$\tau_{\mathbf{g}} = \frac{T_{\mathbf{G}}}{w_1}. \quad (5202)$$

To second order in tunneling, the quadratic correction tensor is

$$\frac{\delta A_{\mu\nu}}{A} = -4\alpha_{\square}^2 \sum_{\mathbf{g}} |\tau_{\mathbf{g}}|^2 \frac{g_{\mu}g_{\nu}}{|\mathbf{g}|^6}. \quad (5203)$$

For a C_4 -symmetric tunneling spectrum,

$$\delta A_{\mu\nu} = -C_2 A \alpha_{\square}^2 \delta_{\mu\nu}, \quad (5204)$$

where

$$C_2 = 2 \sum_{\mathbf{g}} \frac{|\tau_{\mathbf{g}}|^2}{|\mathbf{g}|^4}. \quad (5205)$$

Thus

$$\frac{A_{\text{eff}}}{A} = 1 - C_2 \alpha_{\square}^2 + O(\alpha_{\square}^4). \quad (5206)$$

For the four equal first-star vectors, $C_2 = 8$. This coefficient provides a direct bridge between the microscopic Fourier content of the interlayer tunneling and the leading Euclidean mass renormalization. At the moiré zone center, the little group contains the fourfold rotation. The Hessian of a nondegenerate C_4 -symmetric band must satisfy

$$R_{\pi/2}^T M R_{\pi/2} = M. \quad (5207)$$

For a real symmetric 2×2 Hessian, this implies $M = \beta I_2$. Therefore,

$$\frac{\partial^2 E}{\partial k_x^2} = \frac{\partial^2 E}{\partial k_y^2} = \beta, \quad (5208)$$

and

$$\frac{\partial^2 E}{\partial k_x \partial k_y} = 0. \quad (5209)$$

The complete quadratic flattening condition at the zone center therefore reduces to the scalar condition $\beta(\theta, \omega) = 0$. In the first-star weak-coupling approximation,

$$\beta = 2A [1 - 8\alpha_{\square}^2 + O(\alpha_{\square}^4)]. \quad (5210)$$

Thus the fourfold symmetry converts a two-component effective-mass condition into one scalar equation. The condition $\beta = 0$ suppresses the quadratic dispersion at one high-symmetry point. It does not by itself imply $W_C = 0$. The global operational magic condition additionally requires suppression of the dispersion throughout

the complete moiré Brillouin zone. We therefore distinguish $\theta_{\nu}^{\text{mass}}$ defined by $\beta(\theta_{\nu}^{\text{mass}}, \omega) = 0$, from the operational angle θ_{ν}^* defined by the maximum of the complete magic functional. Agreement, $\theta_{\nu}^{\text{mass}} \simeq \theta_{\nu}^*$, would demonstrate that the global magic state is controlled by a zone-center effective-mass cancellation. A systematic separation between the two angles would instead identify a more global or multi-resonance flattening mechanism. Within the isotropic quadratic approximation,

$$E(\mathbf{k}) = E_{\Gamma} + A_{\text{eff}} k^2. \quad (5211)$$

The local DOS per physical area is

$$\mathcal{D}(E) = \frac{1}{(2\pi)^2} \int d^2k \delta[E - E_{\Gamma} - A_{\text{eff}} k^2]. \quad (5212)$$

Using polar coordinates,

$$d^2k = 2\pi k dk, \quad (5213)$$

we obtain

$$\mathcal{D}(E) = \frac{1}{4\pi |A_{\text{eff}}|} \quad (5214)$$

on the appropriate side of the quadratic extremum. The bare value is

$$\mathcal{D}_0 = \frac{1}{4\pi A}. \quad (5215)$$

Therefore,

$$\frac{\mathcal{D}}{\mathcal{D}_0} = \frac{1}{|1 - C_2 \alpha_{\square}^2|} \quad (5216)$$

to leading perturbative order. Thus effective-mass enhancement, velocity suppression, and DOS enhancement are analytically locked in the weak-coupling Euclidean benchmark. The exact square mBZ may be written as

$$\mathbf{k} = u\mathbf{G}_1 + v\mathbf{G}_2, \quad -\frac{1}{2} \leq u, v < \frac{1}{2}. \quad (5217)$$

At the corner

$$M = \frac{\mathbf{G}_1 + \mathbf{G}_2}{2}, \quad (5218)$$

we have

$$|M|^2 = \frac{G_M^2}{2}. \quad (5219)$$

Therefore the quadratic bandwidth is approximately

$$W_{\Gamma}^{(2)} = \frac{|A_{\text{eff}}|G_M^2}{2}. \quad (5220)$$

Using $E_M = AG_M^2$, we obtain

$$\frac{W_{\Gamma}^{(2)}}{E_M} = \frac{1}{2} |1 - C_2 \alpha_{\square}^2|. \quad (5221)$$

For the minimal first-star model,

$$\frac{W_{\Gamma}^{(2)}}{E_M} = \frac{1}{2} |1 - 8\alpha_{\square}^2| + O(\alpha_{\square}^4). \quad (5222)$$

This is the quantitative weak-coupling bandwidth benchmark. For

$$E(\mathbf{k}) = E_{\Gamma} + A_{\text{eff}}k^2, \quad (5223)$$

the group velocity is

$$\mathbf{v} = \frac{2A_{\text{eff}}}{\hbar} \mathbf{k}. \quad (5224)$$

At the mBZ corner,

$$v_M = \frac{\sqrt{2}|A_{\text{eff}}|G_M}{\hbar}. \quad (5225)$$

Define the bare moiré velocity scale

$$v_M^{(0)} = \frac{\sqrt{2}AG_M}{\hbar}. \quad (5226)$$

Then

$$\frac{v_M}{v_M^{(0)}} = |1 - C_2 \alpha_{\square}^2|. \quad (5227)$$

The same coefficient therefore controls the quadratic-bandwidth suppression and the local velocity suppression.

f. Theorem 116. Weak-coupling Euclidean flattening lock. Within the quadratic C_4 -symmetric regime,

$$\begin{aligned} \frac{m_0^*}{m_{\text{eff}}^*} &= 1 - C_2 \alpha_{\square}^2, \\ \frac{v}{v_0} &= |1 - C_2 \alpha_{\square}^2|, \\ \frac{W}{W_0^{\text{moiré}}} &= |1 - C_2 \alpha_{\square}^2|, \end{aligned} \quad (5228)$$

and

$$\frac{\mathcal{D}}{\mathcal{D}_0} = \frac{1}{|1 - C_2 \alpha_{\square}^2|}. \quad (5229)$$

Thus the initial Euclidean magic-band tendency is not four unrelated numerical observations. It is a single hybridization-driven renormalization visible simultaneously in effective mass, velocity, bandwidth, and DOS. When $A_{\text{eff}} = 0$, the leading C_4 -symmetric dispersion has the general quartic form

$$E(\mathbf{k}) = E_{\star} + B_1 (k_x^4 + k_y^4) + B_2 k_x^2 k_y^2 + O(k^6). \quad (5230)$$

The bare square lattice contributes

$$B_1^{(0)} = -\frac{ta^4}{12}, \quad (5231)$$

whereas interlayer hybridization renormalizes both B_1 and B_2 . Throughout the square mBZ,

$$|k_x| \leq \frac{G_M}{2}, \quad |k_y| \leq \frac{G_M}{2}. \quad (5232)$$

Therefore,

$$k_x^4 + k_y^4 \leq \frac{G_M^4}{8}, \quad k_x^2 k_y^2 \leq \frac{G_M^4}{16}. \quad (5233)$$

Consequently the quartic residual bandwidth satisfies

$$W_{\text{quartic}} \leq G_M^4 \left[\frac{|B_1|}{8} + \frac{|B_2|}{16} \right] + O(G_M^6). \quad (5234)$$

Since $G_M = \Theta(\theta)$, we obtain $W_{\text{quartic}} = O(t\theta^4)$. By contrast, an ordinary untuned quadratic moiré band has

$$W_{\text{quadratic}} = O(t\theta^2). \quad (5235)$$

Therefore quadratic mass cancellation changes the small-angle bandwidth power from two to four. Using

$$\theta_{\nu}^{\star} = \Theta\left(\sqrt{w/t}\right), \quad (5236)$$

the quartic residual scaling becomes

$$W_{\text{quartic}} = O\left[t\left(\frac{w}{t}\right)^2\right]. \quad (5237)$$

Equivalently,

$$\frac{W_{\text{quartic}}}{t} = O\left[\left(\frac{w}{t}\right)^2\right]. \quad (5238)$$

Thus the square-lattice magic scaling predicts two nested weak-coupling laws, $\theta^{\star} = O(\omega^{1/2})$, and, after quadratic cancellation, $W^{\star} = O(t\omega^2)$.

g. Theorem 117. Quadratic-to-quartic Euclidean flattening crossover. If the C_4 -symmetric quadratic coefficient of a Euclidean moiré band is tuned to zero while the quartic coefficients remain finite, then

$$W : O(\theta^2) \longrightarrow O(\theta^4). \quad (5239)$$

Along a square-lattice magic-angle branch satisfying $\theta^* = O(\omega^{1/2})$, the residual quartic bandwidth scales as $W^* = O(t\omega^2)$. This provides an independent power-law test of the Euclidean magic mechanism beyond the location of the magic angle itself. The perturbative mass benchmark is not sufficient in the strong hybridization regime. We therefore return to the exact Euclidean self-energy. For one layer,

$$\Sigma_1(E, \mathbf{k}) = \mathcal{T}(\mathbf{k}) [E - h_2(\mathbf{k})]^{-1} \mathcal{T}^\dagger(\mathbf{k}). \quad (5240)$$

The exact effective eigenproblem is

$$[E - h_1 - \Sigma_1(E, \mathbf{k})] \psi_1 = 0. \quad (5241)$$

The exact energy derivative is

$$\frac{dE}{dk_\mu} = Z_{\text{eff}} \langle \psi_1 | [\partial_\mu h_1 + (\partial_\mu \Sigma_1)_E] | \psi_1 \rangle, \quad (5242)$$

where

$$Z_{\text{eff}} = [\langle \psi_1 | (I - \partial_E \Sigma_1) | \psi_1 \rangle]^{-1}. \quad (5243)$$

Thus the exact local Euclidean magic condition is

$$\langle \psi_1 | [\nabla_{\mathbf{k}} h_1 + \nabla_{\mathbf{k}} \Sigma_1] | \psi_1 \rangle = 0. \quad (5244)$$

Since $Z_{\text{eff}} > 0$, the residue factor changes the magnitude of the reconstructed velocity but does not alter the exact zero of the numerator. Therefore the exact stationary condition is a cancellation between the bare kinetic covector and the hybridization self-energy covector. Define

$$\mathbf{v}_{\text{bare}} = \frac{Z_{\text{eff}}}{\hbar} \langle \psi_1 | \nabla_{\mathbf{k}} h_1 | \psi_1 \rangle, \quad (5245)$$

and

$$\mathbf{v}_{\text{hyb}} = \frac{Z_{\text{eff}}}{\hbar} \langle \psi_1 | \nabla_{\mathbf{k}} \Sigma_1 | \psi_1 \rangle. \quad (5246)$$

Then

$$\mathbf{v} = \mathbf{v}_{\text{bare}} + \mathbf{v}_{\text{hyb}}. \quad (5247)$$

Define the local velocity-cancellation residual

$$\mathcal{R}_v(\mathbf{k}) = \frac{|\mathbf{v}_{\text{bare}} + \mathbf{v}_{\text{hyb}}|}{|\mathbf{v}_{\text{bare}}| + |\mathbf{v}_{\text{hyb}}|}. \quad (5248)$$

It satisfies $0 \leq \mathcal{R}_v \leq 1$. Perfect local cancellation gives $\mathcal{R}_v = 0$. Define the global Euclidean cancellation residual by

$$\mathcal{R}_{v,\mathcal{C}}^{\text{max}} = \sup_{\mathbf{k} \in \text{mBZ}} \mathcal{R}_{v,\mathcal{C}}(\mathbf{k}). \quad (5249)$$

A globally weakly dispersive state requires suppression of this quantity over the complete relevant momentum domain rather than at one isolated crossing. The exact folded eigenstate is

$$|u_{\ell\mathbf{k}}\rangle = \sum_{\alpha=1}^{\Sigma} u_{1\alpha}^{(\ell)} |1, \alpha; \mathbf{k}\rangle + \sum_{\beta=1}^{\Sigma} u_{2\beta}^{(\ell)} |2, \beta; \mathbf{k}\rangle. \quad (5250)$$

Define the unfolded weights

$$W_{1\alpha}^{(\ell)} = |u_{1\alpha}^{(\ell)}|^2, \quad W_{2\beta}^{(\ell)} = |u_{2\beta}^{(\ell)}|^2. \quad (5251)$$

The folded-momentum IPR is

$$\text{IPR}_k^{(\ell)} = \sum_{\alpha} [W_{1\alpha}^{(\ell)}]^2 + \sum_{\beta} [W_{2\beta}^{(\ell)}]^2. \quad (5252)$$

The folded-momentum participation number is

$$N_{\text{part},k}^{(\ell)} = \frac{1}{\text{IPR}_k^{(\ell)}}. \quad (5253)$$

A trivially folded state satisfies $N_{\text{part},k} \simeq 1$. A strongly reconstructed state satisfies $N_{\text{part},k} \gg 1$. Thus the Euclidean magic benchmark requires not only a narrow band but a corresponding increase in folded-momentum hybridization. Define the mBZ-averaged folded participation by

$$\overline{N_{\text{part},k}} = \frac{1}{A_{\text{mBZ}}} \int_{\text{mBZ}} d^2k N_{\text{part},k}(\mathbf{k}). \quad (5254)$$

For an operational magic branch, we evaluate

$$\overline{N_{\text{part},k}}^* = \overline{N_{\text{part},k}}(\theta_\nu^*, \omega). \quad (5255)$$

A magic candidate generated merely by selecting one almost-decoupled folded branch would satisfy $\overline{N_{\text{part},k}}^* \simeq 1$ and is rejected by the hybridization-causality analysis. A genuine Euclidean moiré flattening state must exhibit nontrivial folded-momentum reconstruction over the relevant portion of the mBZ. Let P_1, P_2 be the two layer projectors. The layer polarization is

$$\mathcal{P}_L = \langle u | P_1 - P_2 | u \rangle. \quad (5256)$$

The layer participation number is

$$P_L = \frac{2}{1 + \mathcal{P}_L^2}. \quad (5257)$$

A strongly layer-hybridized state satisfies $\mathcal{P}_L \simeq 0$, and therefore $P_L \simeq 2$. The interlayer-energy expectation is

$$E_{\perp}^{(\ell)} = \langle u_{\ell} | H_{\perp} | u_{\ell} \rangle. \quad (5258)$$

Using

$$H = H_0 + \omega t \mathcal{V}_{\perp}, \quad (5259)$$

Hellmann–Feynman gives

$$\frac{\partial E_{\ell}}{\partial \omega} = t \langle u_{\ell} | \mathcal{V}_{\perp} | u_{\ell} \rangle. \quad (5260)$$

Therefore a strongly hybridized magic state is also identifiable through its nontrivial spectral response to w/t . Equation (5170) implies

$$\omega \Sigma_{\nu}^* = \frac{4\pi^2 \alpha_{\nu}^*}{\gamma_1 (\theta_{\nu}^*)}. \quad (5261)$$

Therefore, in the smooth small-angle limit,

$$\lim_{\omega \rightarrow 0} \omega \Sigma_{\nu}^* = \frac{4\pi^2 \alpha_{\nu}^*}{\gamma_0}. \quad (5262)$$

Similarly,

$$\lim_{\omega \rightarrow 0} \omega N_{\text{sc},\nu}^* = \frac{8\pi^2 \alpha_{\nu}^*}{\gamma_0}. \quad (5263)$$

These limits provide parameter-free numerical collapse tests once α_{ν}^*/γ_0 has been extracted from one coupling regime. From Eq. (5177),

$$\frac{\theta_{\nu}^*}{\sqrt{\omega}} = \frac{n_c}{\pi \sqrt{d \alpha_{\nu}^*}} \sqrt{\gamma_0} + O(\omega). \quad (5264)$$

Hence

$$\lim_{\omega \rightarrow 0} \frac{\theta_{\nu}^*}{\sqrt{w/t}} = \frac{n_c}{\pi \sqrt{d \alpha_{\nu}^*}} \sqrt{\gamma_0}. \quad (5265)$$

The Euclidean magic benchmark therefore produces two independent collapse laws, $\theta_{\nu}^* \propto \sqrt{\omega}$, and $N_{\text{sc},\nu}^* \propto \omega^{-1}$.

Their simultaneous verification is substantially stronger than fitting a single magic-angle curve. If the dimensionless continuum problem contains several magic roots, $\alpha_1^*, \alpha_2^*, \dots$, then each generates a separate physical branch. For two branches μ and ν , the weak-coupling angle ratio is

$$\frac{\theta_{\nu}^*}{\theta_{\mu}^*} \longrightarrow \sqrt{\frac{\alpha_{\mu}^*}{\alpha_{\nu}^*}}. \quad (5266)$$

The corresponding supercell-index ratio is

$$\frac{\Sigma_{\nu}^*}{\Sigma_{\mu}^*} \longrightarrow \frac{\alpha_{\nu}^*}{\alpha_{\mu}^*}. \quad (5267)$$

Thus the sequence of dimensionless roots completely fixes the asymptotic hierarchy of Euclidean magic angles and supercell sizes.

h. Theorem 118. Euclidean magic-branch ratio law. For any two finite continuum magic couplings α_{μ}^* and α_{ν}^* ,

$$\lim_{\omega \rightarrow 0} \frac{\theta_{\nu}^*}{\theta_{\mu}^*} = \sqrt{\frac{\alpha_{\mu}^*}{\alpha_{\nu}^*}}, \quad (5268)$$

whereas

$$\lim_{\omega \rightarrow 0} \frac{N_{\text{sc},\nu}^*}{N_{\text{sc},\mu}^*} = \frac{\alpha_{\nu}^*}{\alpha_{\mu}^*}. \quad (5269)$$

Thus the magic-angle hierarchy and the supercell-size hierarchy are inverse square-root and linear images of the same dimensionless root structure. The exact square-lattice calculation samples rational half-angle twists. Consider a fixed-numerator sequence $n_c = \text{constant}$, with large m . Then

$$\theta_m = 2 \arctan \left(\frac{n_c}{m} \right). \quad (5270)$$

For large m ,

$$\theta_m = \frac{2n_c}{m} - \frac{2n_c^3}{3m^3} + O(m^{-5}). \quad (5271)$$

Within a fixed parity subsequence, let $m \rightarrow m + \Delta m$, where $\Delta m = 2$. Then

$$\Delta \theta_m = \theta_{m+2} - \theta_m = -\frac{4n_c}{m^2} + O(m^{-3}). \quad (5272)$$

Since $\theta \sim \frac{2n_c}{m}$, we obtain $|\Delta \theta| = O(\theta^2)$. At a magic branch with $\theta^* = O(\sqrt{\omega})$, the commensurate angular

spacing therefore satisfies $|\Delta\theta^*| = O(\omega)$. Thus exact commensurate sequences resolve the asymptotic $\sqrt{\omega}$ magic-angle trajectory with an absolute angle error smaller by one additional power of $\sqrt{\omega}$. Because

$$\alpha_{\square} = \frac{\gamma_1 \omega \Sigma}{4\pi^2}, \quad (5273)$$

the relative commensurate sampling error is

$$\frac{\Delta\alpha_{\square}}{\alpha_{\square}} = \frac{\Delta\Sigma}{\Sigma} + \frac{\Delta\gamma_1}{\gamma_1}. \quad (5274)$$

For a fixed-numerator large- m parity subsequence, $\Sigma = \Theta(m^2)$, whereas $\Delta\Sigma = \Theta(m)$. Therefore,

$$\frac{\Delta\Sigma}{\Sigma} = O(m^{-1}) = O(\theta). \quad (5275)$$

At the magic scale,

$$\frac{\Delta\alpha_{\square}}{\alpha_{\square}} = O(\sqrt{\omega}) \quad (5276)$$

up to the smooth tunneling-form-factor correction. Thus exact commensurate discretization does not obstruct convergence of the continuum magic root. The previously constructed exact structures with $N_{\text{sc}} < 100$ provide a mandatory implementation benchmark. However, the asymptotic magic-supercell relation gives

$$N_{\text{sc},\nu}^* \simeq \frac{8\pi^2 \alpha_{\nu}^*}{\gamma_0 \omega}. \quad (5277)$$

Therefore the condition $N_{\text{sc},\nu}^* < 100$ would require

$$\omega > \frac{2\pi^2 \alpha_{\nu}^*}{25\gamma_0}. \quad (5278)$$

Consequently, for weak interlayer coupling a genuine asymptotic magic configuration may require substantially more than one hundred microscopic bilayer orbitals. The ten-site, thirty-four-site, and complete sub-hundred-site families are therefore used to verify crystallography, folding, hybridization, eigenvalue counting, and the initial flattening trend. They are not assumed to exhaust the weak-coupling magic-angle regime. The numerical benchmark is divided into four levels. The first level uses the ten-site structure, $\Sigma = 5$, $N_{\text{sc}} = 10$. The second level uses the thirty-four-site structure, $\Sigma = 17$, $N_{\text{sc}} = 34$. The third level contains every exact square bilayer with $N_{\text{sc}} < 100$. The fourth level consists of increasing small-angle exact commensurate supercells selected specifically to resolve $\alpha_{\square} \simeq \alpha_{\nu}^*$. Only the fourth level is used for the asymptotic weak-coupling magic-angle scaling. For every exact angle and every selected value of ω , we compute the complete spectrum

$E_{\ell}(\mathbf{k})$, over the complete two-dimensional mBZ. We determine the target-cluster bandwidth $W_{\mathcal{C}}$, the signed global isolation gap $\Delta_{\mathcal{C}}$, the maximal group velocity $v_{\mathcal{C}}^{\text{max}}$, the effective-mass tensor $[m_{\mathcal{C}}^{*-1}]_{\mu\nu}$, the DOS $\rho_{\mathcal{C}}(E)$, the folded-momentum participation $N_{\text{part},k}$, the layer polarization $\mathcal{P}_L$, and the operational magic score $\mathcal{M}_{\mathcal{C}}^{(E)}$. No benchmark angle is classified from one observable alone. The Euclidean implementation must first reproduce

$$\frac{m_0^*}{m_{\text{eff}}^*} = 1 - C_2 \alpha_{\square}^2 + O(\alpha_{\square}^4) \quad (5279)$$

at small $\alpha_{\square}$. It must simultaneously reproduce

$$\frac{W}{E_M} = \frac{1}{2} |1 - C_2 \alpha_{\square}^2| + \dots, \quad (5280)$$

and

$$\frac{\mathcal{D}}{\mathcal{D}_0} = \frac{1}{|1 - C_2 \alpha_{\square}^2|} + \dots. \quad (5281)$$

The same calculations must show $N_{\text{part},k} > 1$ as the flattening develops. These relations provide independent spectral, differential, and wave-function verification of the same Euclidean hybridization mechanism. For several values of $\omega = \frac{w}{t}$, we choose exact commensurate angles with matched $\alpha_{\square}$. The rescaled band structures must satisfy

$$\frac{E_{\ell} + 4t}{E_M} \longrightarrow \mathcal{E}_{\ell}(\mathbf{p}, \alpha_{\square}) \quad (5282)$$

as $\Sigma \longrightarrow \infty$. Likewise,

$$\frac{W_{\mathcal{C}}}{E_M}, \quad \frac{\Delta_{\mathcal{C}}}{E_M} \quad (5283)$$

must collapse onto functions of $\alpha_{\square}$. The operational magic maxima extracted from different ω values must converge to the same dimensionless root,

$$\alpha_{\square}(\theta_{\nu}^*, \omega) \longrightarrow \alpha_{\nu}^*. \quad (5284)$$

i. Theorem 119. Euclidean magic-collapse benchmark. A consistent small-angle square-lattice implementation must satisfy the simultaneous limits

$$\begin{aligned} \alpha_{\square}(\theta_{\nu}^*, \omega) &\longrightarrow \alpha_{\nu}^*, \\ \frac{\theta_{\nu}^*}{\sqrt{\omega}} &\longrightarrow \text{constant}, \\ \omega N_{\text{sc},\nu}^* &\longrightarrow \text{constant}. \end{aligned} \quad (5285)$$

These three limits test the same Euclidean magic coupling in dimensionless, angular, and crystallographic

variables. At fixed $K = 0$ and fixed ω , an interior operational magic angle satisfies

$$\frac{\partial \mathcal{M}_c^{(E)}}{\partial \theta} = 0. \quad (5286)$$

Equivalently,

$$\frac{\partial \ln \mathcal{M}_c^{(E)}}{\partial \theta} = 0. \quad (5287)$$

Using the operational functional derived above, this gives

$$\begin{aligned} 0 = & -\frac{\Delta \partial_\theta W - W \partial_\theta \Delta}{\Delta(\Delta + W)} \\ & - \frac{\Delta \partial_\theta E_{\text{disp}} - E_{\text{disp}} \partial_\theta \Delta}{\Delta(\Delta + E_{\text{disp}})} \\ & + \frac{1}{1 + E_{\text{kin}} \rho_{\text{coh}}^{\text{max}}} \partial_\theta \ln \rho_{\text{coh}}^{\text{max}} \\ & + \partial_\theta \ln \mathcal{C}_{\text{coh}}. \end{aligned} \quad (5288)$$

This is the exact Euclidean operational magic-angle equation. A bandwidth minimum alone would satisfy only $\partial_\theta W = 0$, which is not equivalent to Eq. (5288). In the asymptotic continuum regime,

$$\mathcal{M}_c^{(E)} = \mathfrak{M}_c(\alpha_\square) + O(\Sigma^{-1}). \quad (5289)$$

Therefore,

$$\frac{\partial \mathcal{M}_c^{(E)}}{\partial \theta} = \frac{d\mathfrak{M}_c}{d\alpha_\square} \frac{\partial \alpha_\square}{\partial \theta} + O(\Sigma^{-1}). \quad (5290)$$

For nonzero twist,

$$\frac{\partial \alpha_\square}{\partial \theta} \neq 0. \quad (5291)$$

Hence the leading magic condition reduces to

$$\frac{d\mathfrak{M}_c}{d\alpha_\square} = 0. \quad (5292)$$

Thus the physical angle is a geometric reparameterization of a dimensionless spectral optimum. A Euclidean benchmark is accepted only if its flattening disappears when interlayer tunneling is removed. Define

$$\mathcal{R}_W^{\text{hyb}} = \frac{W_c(\theta, 0, \omega)}{W_c(\theta, 0, 0)}. \quad (5293)$$

A hybridization-driven flattening satisfies $\mathcal{R}_W^{\text{hyb}} < 1$. Define

$$\mathcal{R}_v^{\text{hyb}} = \frac{v_c^{\text{max}}(\theta, 0, \omega)}{v_c^{\text{max}}(\theta, 0, 0)}. \quad (5294)$$

A hybridization-driven velocity suppression satisfies $\mathcal{R}_v^{\text{hyb}} < 1$. Finally, the folded-momentum participation must satisfy $N_{\text{part},k} > 1$ over the reconstructed part of the band. These three conditions establish the microscopic causality of the Euclidean magic benchmark.

j. Theorem 120. Euclidean hybridization-causality criterion. An operational Euclidean magic state is classified as twist-hybridization-induced only if the same tracked spectral family satisfies $W(\omega) < W(0)$, $v^{\text{max}}(\omega) < v^{\text{max}}(0)$, and exhibits nontrivial folded-momentum participation. Thus neither exact commensurability nor band folding alone can generate the accepted Euclidean magic state. The minimal Euclidean magic benchmark consists of $W_c \ll \Delta_c$, $\hbar D_G \Omega_c^{\text{max}} \ll \Delta_c$, $E_{\text{kin}} \rho_c^{\text{coh,max}} \gg 1$, and $\mathcal{C}_c^{\text{coh}} = O(1)$. The stronger second-order benchmark additionally requires

$$\frac{D_G^2 \Lambda_c^{\text{max}}}{2\Delta_c} \ll 1. \quad (5295)$$

For a zone-center mass-controlled branch, this corresponds locally to $A_{\text{eff}} \simeq 0$ and residual quartic dispersion. For each chosen ω , we first calculate the exact band structures of the ten-site and thirty-four-site benchmark cells. We then calculate the complete previously enumerated family with $N_{\text{sc}} < 100$. These calculations verify atom counting, reciprocal folding, symmetry, hybridization gaps, unfolded weights, and the initial effective-mass renormalization. We subsequently extend the exact commensurate sequence until $\alpha_\square$ passes continuously through the region containing the operational dimensionless maximum. For every target cluster we calculate $\mathcal{M}_c^{(E)}(\theta, \omega)$. The discrete maximizing angle is recorded as $\theta_{\nu,N}^*$. The corresponding dimensionless coupling is

$$\alpha_{\nu,N}^* = \alpha_\square(\theta_{\nu,N}^*, \omega). \quad (5296)$$

We require convergence $\alpha_{\nu,N}^* \rightarrow \alpha_\nu^*$. The physical magic angle is then inferred from the converged dimensionless root rather than from interpolation of one finite-size spectrum. For every accepted Euclidean magic branch we record

$$\mathfrak{B}_\nu^{(E)} = (\alpha_\nu^*, \theta_\nu^*, \Sigma_\nu^*, N_{\text{sc},\nu}^*, W_\nu^*, \Delta_\nu^*, v_\nu^*, \rho_\nu^*, N_{\text{part},k}^*). \quad (5297)$$

We additionally record the operational scores $\mathcal{M}_\nu^*$ and $\mathcal{M}_\nu^{(2)*}$ when the second-order dispersion tensor is numerically resolved. This certificate allows the Euclidean reference to be reproduced without relying on visual inspection of band plots.

k. Theorem 121. Complete Euclidean magic-angle benchmark. For the twisted square-lattice bilayer at $K = 0$, the small-angle band-edge problem is governed by the dimensionless coupling

$$\alpha_{\square} = \frac{\gamma_1 \omega \Sigma}{4\pi^2}. \quad (5298)$$

If the dimensionless continuum spectrum contains an isolated operational magic root α_{ν}^* , then

$$\theta_{\nu}^* = \frac{n_c}{\pi \sqrt{d\alpha_{\nu}^*}} \sqrt{\frac{w}{t}} + O\left[(w/t)^{3/2}\right], \quad (5299)$$

while

$$N_{sc,\nu}^* = \frac{8\pi^2 \alpha_{\nu}^*}{\gamma_0} \frac{t}{w} + O(1). \quad (5300)$$

At weak hybridization the effective quadratic dispersion obeys

$$\frac{A_{\text{eff}}}{A} = 1 - C_2 \alpha_{\square}^2 + O(\alpha_{\square}^4), \quad (5301)$$

with $C_2 = 8$ in the minimal equal-amplitude first-star model. If the quadratic coefficient is tuned to zero while the quartic coefficients remain finite, the residual bandwidth crosses from $W = O(t\theta^2)$ to $W = O(t\theta^4)$. The Euclidean magic benchmark is accepted only if these scaling relations are accompanied by nontrivial folded-momentum hybridization, suppression of the complete group velocity, positive spectral isolation, and convergence of the operational magic functional. The complete Euclidean mechanism can therefore be organized quantitatively as

$$\begin{aligned} \theta &\longrightarrow \Sigma(\theta) \longrightarrow G_M = \frac{2\pi}{a\sqrt{\Sigma}} \\ &\longrightarrow E_M = ta^2 G_M^2 \longrightarrow \alpha_{\square} = \frac{w_1}{E_M} \\ &\longrightarrow \text{hybridization self-energy} \\ &\longrightarrow \{m_{\text{eff}}^*, v, W, \rho\} \longrightarrow \mathcal{M}^{(E)}(\theta, \omega). \end{aligned} \quad (5302)$$

The corresponding weak-coupling magic scaling is

$$\theta_{\nu}^* \sim \sqrt{\frac{w}{t}}, \quad \Sigma_{\nu}^* \sim \frac{t}{w}, \quad N_{sc,\nu}^* \sim \frac{t}{w}. \quad (5303)$$

These relations do not determine the numerical value of α_{ν}^* without solving the dimensionless microscopic spectral problem. Instead, they separate the universal square-lattice kinematic structure from the model-dependent dimensionless magic root. The Euclidean

benchmark is therefore complete only after the numerical calculation demonstrates the simultaneous collapse

$$\begin{aligned} \alpha_{\square}(\theta_{\nu}^*, \omega) &\longrightarrow \alpha_{\nu}^*, \\ \frac{\theta_{\nu}^*}{\sqrt{w}} &\longrightarrow \text{constant}, \\ \omega N_{sc,\nu}^* &\longrightarrow \text{constant}, \end{aligned} \quad (5304)$$

together with the spectral and wave-function conditions encoded in the operational magic functional. Only after this curvature-free reference has been established can a shift of the magic condition at $K < 0$ be attributed to curvature rather than to an incorrect Euclidean baseline.

C. From Magic Angles to Curvature-Dependent Magic Curves

The Euclidean analysis identifies magic configurations through a sequence of dimensionless roots $\{\alpha_{\nu}^*\}$, which generate the corresponding Euclidean magic angles

$$\theta_{\nu}^* \left(0, \frac{w}{t}\right). \quad (5305)$$

Negative curvature changes this one-parameter structure fundamentally. The twist angle is no longer the only geometric quantity controlling the moire reconstruction. The curvature radius introduces an independent length scale,

$$R = \frac{1}{\sqrt{|K|}}, \quad K < 0, \quad (5306)$$

and the magic condition must therefore be promoted to

$$\theta_{\nu}^* = \theta_{\nu}^* \left(K, \frac{w}{t}\right). \quad (5307)$$

The complete collection of these branches defines a magic manifold in $\left(\theta, K, \frac{w}{t}\right)$ space. We first establish the existence and differential structure of this manifold directly from the operational magic functional. We then derive an explicit geometric scaling prediction using the exact hyperbolic moire-length law. Let the operational magic functional of the ν -th continuously tracked spectral cluster be

$$\mathcal{M}_{\nu} = \mathcal{M}_{\nu}(\theta, K, \omega), \quad \omega = \frac{w}{t}. \quad (5308)$$

For fixed K and ω , an interior operational magic angle satisfies

$$\frac{\partial \mathcal{M}_{\nu}}{\partial \theta} = 0. \quad (5309)$$

Because $\mathcal{M}_\nu > 0$ inside the admissible isolated region, it is convenient to introduce

$$\mathcal{L}_\nu = \ln \mathcal{M}_\nu. \quad (5310)$$

The stationary condition becomes

$$F_\nu(\theta, K, \omega) \equiv \frac{\partial \mathcal{L}_\nu}{\partial \theta} = 0. \quad (5311)$$

A strict local magic maximum additionally satisfies

$$\frac{\partial^2 \mathcal{L}_\nu}{\partial \theta^2} < 0. \quad (5312)$$

Suppose

$$F_\nu(\theta_0, K_0, \omega_0) = 0, \quad \left. \frac{\partial F_\nu}{\partial \theta} \right|_{(\theta_0, K_0, \omega_0)} \neq 0. \quad (5313)$$

The implicit-function theorem then gives a unique smooth local surface $\theta = \theta_\nu^*(K, \omega)$ passing through the reference magic point. Thus a nondegenerate Euclidean magic angle does not disappear instantaneously when weak curvature is introduced. It extends locally into a curvature-dependent magic curve.

a. Theorem 122. Local continuation of a Euclidean magic angle. Let $\theta_\nu^*(0, \omega_0)$ be a strict isolated Euclidean operational magic maximum. If

$$\partial_\theta^2 \ln \mathcal{M}_\nu \big|_{(\theta_\nu^*, 0, \omega_0)} \neq 0, \quad (5314)$$

then there exists a neighborhood of $(K, \omega) = (0, \omega_0)$ in which the magic configuration is represented by a unique smooth branch $\theta_\nu^* = \theta_\nu^*(K, \omega)$. The result is independent of any local geometric approximation and follows directly from the fully operational spectral definition. Differentiate

$$F_\nu[\theta_\nu^*(K, \omega), K, \omega] = 0 \quad (5315)$$

with respect to curvature. We obtain

$$\frac{\partial F_\nu}{\partial \theta} \frac{\partial \theta_\nu^*}{\partial K} + \frac{\partial F_\nu}{\partial K} = 0. \quad (5316)$$

Therefore,

$$\frac{\partial \theta_\nu^*}{\partial K} = -\frac{\partial_K \partial_\theta \ln \mathcal{M}_\nu}{\partial_\theta^2 \ln \mathcal{M}_\nu}. \quad (5317)$$

Similarly,

$$\frac{\partial \theta_\nu^*}{\partial \omega} = -\frac{\partial_\omega \partial_\theta \ln \mathcal{M}_\nu}{\partial_\theta^2 \ln \mathcal{M}_\nu}. \quad (5318)$$

These equations are exact. They show that the displacement of a magic angle is controlled by a mixed spectral response rather than by the curvature dependence of the bandwidth alone. Differentiating a second time with respect to K gives

$$\begin{aligned} 0 = & \partial_\theta^2 \mathcal{L}_\nu \frac{\partial^2 \theta_\nu^*}{\partial K^2} + \partial_\theta^3 \mathcal{L}_\nu \left(\frac{\partial \theta_\nu^*}{\partial K} \right)^2 \\ & + 2\partial_\theta^2 \partial_K \mathcal{L}_\nu \frac{\partial \theta_\nu^*}{\partial K} + \partial_\theta \partial_K^2 \mathcal{L}_\nu. \end{aligned} \quad (5319)$$

Hence,

$$\begin{aligned} \frac{\partial^2 \theta_\nu^*}{\partial K^2} = & -\frac{1}{\partial_\theta^2 \mathcal{L}_\nu} \left[\partial_\theta \partial_K^2 \mathcal{L}_\nu + 2\partial_\theta^2 \partial_K \mathcal{L}_\nu \frac{\partial \theta_\nu^*}{\partial K} \right. \\ & \left. + \partial_\theta^3 \mathcal{L}_\nu \left(\frac{\partial \theta_\nu^*}{\partial K} \right)^2 \right]. \end{aligned} \quad (5320)$$

The mixed K - ω response is

$$\begin{aligned} \frac{\partial^2 \theta_\nu^*}{\partial K \partial \omega} = & -\frac{1}{\partial_\theta^2 \mathcal{L}_\nu} \left[\partial_\theta \partial_K \partial_\omega \mathcal{L}_\nu + \partial_\theta^2 \partial_K \mathcal{L}_\nu \frac{\partial \theta_\nu^*}{\partial \omega} \right. \\ & \left. + \partial_\theta^2 \partial_\omega \mathcal{L}_\nu \frac{\partial \theta_\nu^*}{\partial K} + \partial_\theta^3 \mathcal{L}_\nu \frac{\partial \theta_\nu^*}{\partial K} \frac{\partial \theta_\nu^*}{\partial \omega} \right]. \end{aligned} \quad (5321)$$

Thus the complete local geometry of the three-dimensional magic manifold is determined by derivatives of the same operational functional. Define the optimized score by

$$\mathcal{M}_\nu^*(K, \omega) = \mathcal{M}_\nu[\theta_\nu^*(K, \omega), K, \omega]. \quad (5322)$$

Using $\partial_\theta \mathcal{M}_\nu = 0$ on the magic branch, the envelope theorem gives

$$\frac{\partial \ln \mathcal{M}_\nu^*}{\partial K} = \left. \frac{\partial \ln \mathcal{M}_\nu}{\partial K} \right|_{\theta=\theta_\nu^*}. \quad (5323)$$

Likewise,

$$\frac{\partial \ln \mathcal{M}_\nu^*}{\partial \omega} = \left. \frac{\partial \ln \mathcal{M}_\nu}{\partial \omega} \right|_{\theta=\theta_\nu^*}. \quad (5324)$$

The second optimized curvature derivative is

$$\frac{\partial^2 \mathcal{L}_\nu^*}{\partial K^2} = \partial_K^2 \mathcal{L}_\nu - \frac{(\partial_K \partial_\theta \mathcal{L}_\nu)^2}{\partial_\theta^2 \mathcal{L}_\nu}. \quad (5325)$$

Thus a curvature sweep can affect the magic state in two independent ways: it can displace the optimal angle, and it can increase or decrease the maximum achievable magic quality after the angle has been reoptimized.

b. Theorem 123. Shift and enhancement are independent magic responses. The curvature derivative $\partial_K \theta_\nu^*$ measures displacement of the magic branch, whereas $\partial_K \mathcal{M}_\nu^*$ measures curvature enhancement or suppression of the optimized magic quality. A nonzero magic-angle shift therefore does not by itself imply that negative curvature produces a better flat band. We now derive an analytic prediction for the magic manifold from the exact hyperbolic registry geometry. Let

$$s = \left| \sin \left(\frac{\theta}{2} \right) \right|, \quad \zeta = \frac{a}{2R}. \quad (5326)$$

Define $b = \sinh \zeta$. The exact local hyperbolic moiré length is

$$\xi_M^{(H)} = R \operatorname{arsinh} \left(\frac{b}{s} \right). \quad (5327)$$

The corresponding local inverse-length scale is

$$q_M^{\text{loc}} = \frac{2\pi}{R \operatorname{arsinh} (b/s)}. \quad (5328)$$

In Euclidean space,

$$q_M^{(E)} = \frac{4\pi s}{a}. \quad (5329)$$

Their ratio is

$$\mathcal{C}_q = \frac{q_M^{\text{loc}}}{q_M^{(E)}}. \quad (5330)$$

Substitution gives

$$\mathcal{C}_q = \frac{\zeta}{s \operatorname{arsinh} (b/s)}. \quad (5331)$$

This quantity approaches unity in the flat-space limit. The local inverse registry scale must not be identified directly with the primitive exact Euclidean moiré reciprocal vector. For the Euclidean commensurate sequence,

$$G_M = \frac{2\pi}{a\sqrt{\Sigma}}, \quad (5332)$$

whereas

$$q_M^{(E)} = \frac{4\pi s}{a}. \quad (5333)$$

Using

$$\Sigma = \frac{n_c^2}{d} s^{-2}, \quad (5334)$$

we obtain

$$\frac{q_M^{(E)}}{G_M} = \frac{2n_c}{\sqrt{d}}. \quad (5335)$$

We therefore retain this Euclidean arithmetic normalization and define the geometry-renormalized kinetic scale by

$$E_M^{\text{geo}}(K, \theta) = E_M^{(E)}(\theta) \mathcal{C}_q^2(K, \theta). \quad (5336)$$

This construction has the exact Euclidean limit

$$\lim_{K \rightarrow 0^-} E_M^{\text{geo}} = E_M^{(E)}. \quad (5337)$$

It uses the local hyperbolic registry scale only to renormalize the already correctly normalized Euclidean kinetic scale. It does not assert the existence of a global hyperbolic reciprocal vector. Let the dominant moiré tunneling harmonic be

$$w_1(K, \theta) = \gamma_1(K, \theta) w. \quad (5338)$$

The function $\gamma_1(K, \theta)$ contains the curvature-dependent redistribution of the microscopic interlayer tunneling profile into the dominant moiré harmonic. The exact real-space tunneling reconstruction derived previously implies that γ_1 need not remain constant with curvature. We therefore retain it explicitly. The geometry-renormalized dimensionless coupling is

$$\alpha_{\square}^{\text{geo}}(K, \theta, \omega) = \frac{w_1(K, \theta)}{E_M^{\text{geo}}(K, \theta)}. \quad (5339)$$

Using the Euclidean result,

$$\alpha_{\square}^{(E)} = \frac{\gamma_1 n_c^2}{4\pi^2 d} \omega s^{-2}, \quad (5340)$$

we obtain

$$\alpha_{\square}^{\text{geo}} = \frac{\gamma_1 n_c^2}{4\pi^2 d} \omega s^{-2} \mathcal{C}_q^{-2}. \quad (5341)$$

Using Eq. (5331), the factors of s cancel exactly. Therefore,

$$\alpha_{\square}^{\text{geo}} = \frac{\gamma_1(K, \theta) n_c^2}{4\pi^2 d} \omega \frac{\operatorname{arsinh}^2(b/s)}{\zeta^2}. \quad (5342)$$

This is the central curvature-dependent dimensionless coupling. Assume first that the same dimensionless spectral branch is controlled by a magic root α_ν^* that remains

finite under the geometric crossover. The leading magic-curve equation is

$$\alpha_{\square}^{\text{geo}}(K, \theta_{\nu}^*, \omega) = \alpha_{\nu}^*. \quad (5343)$$

Substitution gives

$$\text{arsinh} \left[\frac{\sinh \zeta}{\sin(\theta_{\nu}^*/2)} \right] = \frac{2\pi\zeta}{n_c} \sqrt{\frac{d\alpha_{\nu}^*}{\gamma_1(K, \theta_{\nu}^*)\omega}}. \quad (5344)$$

This equation defines a curvature-dependent magic surface. When the curvature dependence of the dominant tunneling harmonic is either negligible or removed by a corresponding normalization, $\gamma_1 = \gamma_0$, define

$$\Lambda_{\nu} = \frac{2\pi}{n_c} \sqrt{\frac{d\alpha_{\nu}^*}{\gamma_0\omega}}. \quad (5345)$$

Equation (5344) becomes

$$\text{arsinh} \left[\frac{\sinh \zeta}{\sin(\theta_{\nu}^*/2)} \right] = \Lambda_{\nu} \zeta. \quad (5346)$$

Applying the hyperbolic sine gives

$$\frac{\sinh \zeta}{\sin(\theta_{\nu}^*/2)} = \sinh(\Lambda_{\nu} \zeta). \quad (5347)$$

Therefore,

$$\sin \left(\frac{\theta_{\nu}^*}{2} \right) = \frac{\sinh \zeta}{\sinh(\Lambda_{\nu} \zeta)}. \quad (5348)$$

Hence the explicit magic surface is

$$\theta_{\nu}^*(K, \omega) = 2 \arcsin \left[\frac{\sinh(a\sqrt{|K|}/2)}{\sinh \left(\frac{\pi a \sqrt{|K|}}{n_c} \sqrt{\frac{d\alpha_{\nu}^*}{\gamma_0\omega}} \right)} \right]. \quad (5349)$$

Equation (5349) is the explicit curvature-dependent continuation of the Euclidean magic-angle law within the geometry-controlled scaling theory.

c. Theorem 124. Explicit curvature-dependent magic curve. Under a curvature-independent normalized dominant tunneling harmonic and a finite dimensionless magic root α_{ν}^* , the local geometry-controlled magic branch is

$$\theta_{\nu}^*(K, \omega) = 2 \arcsin \left[\frac{\sinh(a\sqrt{|K|}/2)}{\sinh \left(\frac{\pi a \sqrt{|K|}}{n_c} \sqrt{\frac{d\alpha_{\nu}^*}{\gamma_0\omega}} \right)} \right]. \quad (5350)$$

Its exact $K \rightarrow 0$ limit reproduces the Euclidean square-root magic-angle scaling. Take $\zeta \rightarrow 0$. Then

$$\sinh \zeta = \zeta + O(\zeta^3), \quad (5351)$$

and

$$\sinh(\Lambda_{\nu} \zeta) = \Lambda_{\nu} \zeta + O(\zeta^3). \quad (5352)$$

Therefore,

$$\lim_{K \rightarrow 0^-} \sin \left(\frac{\theta_{\nu}^*}{2} \right) = \frac{1}{\Lambda_{\nu}}. \quad (5353)$$

Using Eq. (5345),

$$\sin \left(\frac{\theta_{\nu}^*(0, \omega)}{2} \right) = \frac{n_c}{2\pi} \sqrt{\frac{\gamma_0\omega}{d\alpha_{\nu}^*}}. \quad (5354)$$

This is exactly the Euclidean result obtained in VI.B. The explicit magic equation has a particularly simple geometric interpretation. The exact hyperbolic registry length is

$$\xi_M^{(H)} = R \text{arsinh} \left(\frac{\sinh \zeta}{s} \right). \quad (5355)$$

Along the magic curve,

$$\text{arsinh} \left(\frac{\sinh \zeta}{s_{\nu}^*} \right) = \Lambda_{\nu} \zeta. \quad (5356)$$

Therefore,

$$\xi_{M,\nu}^* = R \Lambda_{\nu} \zeta. \quad (5357)$$

Using $R\zeta = \frac{a}{2}$, we obtain

$$\xi_{M,\nu}^* = \frac{\pi a}{n_c} \sqrt{\frac{d\alpha_{\nu}^*}{\gamma_0\omega}}. \quad (5358)$$

Remarkably, the curvature radius has disappeared. Thus, within the fixed-harmonic geometric scaling theory, curvature does not change the local registry length selected by the magic condition. It changes the twist required to realize that length. The corresponding magic reciprocal scale is

$$q_{M,\nu}^{\text{loc},*} = \frac{2\pi}{\xi_{M,\nu}^*} = \frac{2n_c}{a} \sqrt{\frac{\gamma_0\omega}{d\alpha_{\nu}^*}}. \quad (5359)$$

Thus the geometry-controlled magic curve is an iso-registry-length and iso-local-reciprocal-scale contour.

d. Theorem 125. Iso-registry characterization of the magic curve. For a curvature-independent dominant tunneling harmonic, the entire curvature-dependent magic branch satisfies

$$\xi_M [K, \theta_\nu^*(K, \omega)] = \xi_{M,\nu}^*(\omega), \quad (5360)$$

where

$$\xi_{M,\nu}^*(\omega) = \frac{\pi a}{n_c} \sqrt{\frac{d\alpha_\nu^*}{\gamma_0 \omega}}. \quad (5361)$$

Negative curvature therefore shifts the magic angle by changing the twist-to-registry-length map rather than by changing the target registry length itself. Define

$$\lambda_\nu = \frac{\xi_{M,\nu}^*}{R}. \quad (5362)$$

Using Eq. (5358),

$$\lambda_\nu = \frac{\pi a}{n_c R} \sqrt{\frac{d\alpha_\nu^*}{\gamma_0 \omega}}. \quad (5363)$$

Equivalently,

$$\lambda_\nu^2 = \frac{\pi^2 a^2 d\alpha_\nu^*}{n_c^2 \gamma_0} \frac{|K|}{\omega}. \quad (5364)$$

Thus the complete Euclidean-to-hyperbolic magic crossover is governed by the combined variable

$$\lambda_\nu^2 \propto \frac{|K|}{w/t}. \quad (5365)$$

The Euclidean magic regime is $\lambda_\nu \ll 1$, whereas the curvature-dominated magic regime is $\lambda_\nu \gg 1$. The crossover occurs when $R \sim \xi_{M,\nu}^*$. This gives the direct physical interpretation of the magic-curve crossover: curvature becomes nonperturbative when the Euclidean magic moire length reaches the curvature radius. Set $\lambda_\nu = 1$. Then

$$|K_\nu^{\text{cross}}| = \frac{1}{\left(\xi_{M,\nu}^*\right)^2}. \quad (5366)$$

Using Eq. (5358),

$$|K_\nu^{\text{cross}}| = \frac{n_c^2 \gamma_0}{\pi^2 a^2 d\alpha_\nu^*} \omega. \quad (5367)$$

Therefore,

$$|K_\nu^{\text{cross}}| \propto \frac{w}{t}. \quad (5368)$$

The crossover line in the $(|K|, w/t)$ plane is asymptotically linear.

e. Theorem 126. Curvature-coupling crossover law. The geometry-controlled magic branch changes from its Euclidean regime to its curvature-dominated regime when

$$|K| \left(\xi_{M,\nu}^*\right)^2 = O(1). \quad (5369)$$

For a quadratic square-lattice magic branch,

$$|K_\nu^{\text{cross}}| = \Theta(w/t). \quad (5370)$$

Thus curvature and interlayer coupling enter the magic crossover through a single combined scaling variable. Let the Euclidean magic half-angle be

$$s_{\nu,E}^* = \sin\left(\frac{\theta_\nu^*(0, \omega)}{2}\right). \quad (5371)$$

From the Euclidean limit,

$$s_{\nu,E}^* = \frac{1}{\Lambda_\nu}. \quad (5372)$$

Because $\lambda_\nu = \Lambda_\nu \zeta$, we have

$$s_{\nu,E}^* = \frac{\zeta}{\lambda_\nu}. \quad (5373)$$

The exact ratio is therefore

$$\frac{s_\nu^*(K, \omega)}{s_{\nu,E}^*(\omega)} = \frac{\sinh \zeta}{\zeta} \frac{\lambda_\nu}{\sinh \lambda_\nu}. \quad (5374)$$

In the weak microscopic-curvature limit,

$$\zeta = \frac{a}{2R} \ll 1, \quad (5375)$$

we have

$$\frac{\sinh \zeta}{\zeta} = 1 + O(\zeta^2). \quad (5376)$$

Therefore,

$$\frac{s_\nu^*(K, \omega)}{s_{\nu,E}^*(\omega)} \longrightarrow \mathcal{H}(\lambda_\nu), \quad (5377)$$

where

$$\mathcal{H}(\lambda) = \frac{\lambda}{\sinh \lambda}. \quad (5378)$$

For asymptotically small magic angles, $\theta \simeq 2s$, and therefore

$$\frac{\theta_\nu^*(K, \omega)}{\theta_\nu^*(0, \omega)} \simeq \frac{\lambda_\nu}{\sinh \lambda_\nu}. \quad (5379)$$

This is the universal curvature-dependent magic-curve crossover function. For $\lambda \ll 1$, we have

$$\frac{\lambda}{\sinh \lambda} = 1 - \frac{\lambda^2}{6} + \frac{7\lambda^4}{360} + O(\lambda^6). \quad (5380)$$

Thus

$$\frac{\theta_\nu^*(K, \omega)}{\theta_\nu^*(0, \omega)} = 1 - \frac{|K| (\xi_{M,\nu}^*)^2}{6} + O[K^2 (\xi_{M,\nu}^*)^4] \quad (5381)$$

in the small-angle continuum regime. Thus negative curvature lowers the geometric magic angle at fixed normalized tunneling strength. The same result can be derived without taking the small-angle limit. Let

$$\theta_{\nu,E}^* = \theta_\nu^*(0, \omega), \quad (5382)$$

and write

$$s_E = \sin\left(\frac{\theta_{\nu,E}^*}{2}\right), \quad c_E = \cos\left(\frac{\theta_{\nu,E}^*}{2}\right). \quad (5383)$$

The weak-curvature reciprocal enhancement is

$$C_q = 1 + \frac{a^2 |K|}{24} \cot^2\left(\frac{\theta}{2}\right) + O(K^2 a^4). \quad (5384)$$

For a curvature-independent tunneling harmonic, the magic condition requires

$$\delta \ln \alpha_{\square}^{\text{geo}} = 0. \quad (5385)$$

Therefore,

$$-2\delta \ln s - 2\delta \ln C_q = 0. \quad (5386)$$

Hence,

$$\delta \ln s = -\frac{a^2 |K|}{24} \cot^2\left(\frac{\theta_{\nu,E}^*}{2}\right). \quad (5387)$$

Using

$$\delta \ln s = \frac{1}{2} \cot\left(\frac{\theta_{\nu,E}^*}{2}\right) \delta \theta, \quad (5388)$$

we obtain

$$\theta_\nu^*(K, \omega) = \theta_{\nu,E}^* - \frac{a^2 |K|}{12} \cot\left(\frac{\theta_{\nu,E}^*}{2}\right) + O(K^2 a^4). \quad (5389)$$

Equivalently, because $K < 0$,

$$\theta_\nu^*(K, \omega) = \theta_{\nu,E}^* + \frac{a^2 K}{12} \cot\left(\frac{\theta_{\nu,E}^*}{2}\right) + O(K^2 a^4). \quad (5390)$$

f. Theorem 127. Leading geometric magic-angle shift. Within the fixed-harmonic geometric scaling theory,

$$\left. \frac{\partial \theta_\nu^*}{\partial |K|} \right|_{K=0} = -\frac{a^2}{12} \cot\left(\frac{\theta_{\nu,E}^*}{2}\right). \quad (5391)$$

Thus every nontrivial reduced-angle Euclidean magic branch initially shifts toward smaller twist as the magnitude of negative curvature is increased. Using

$$s_\nu^* = \frac{\sinh \zeta}{\sinh(\Lambda_\nu \zeta)}, \quad (5392)$$

we obtain

$$\frac{d \ln s_\nu^*}{d \zeta} = \coth \zeta - \Lambda_\nu \coth(\Lambda_\nu \zeta). \quad (5393)$$

Define

$$f(x) = x \coth x. \quad (5394)$$

Its derivative is

$$f'(x) = \coth x - x \operatorname{csch}^2 x. \quad (5395)$$

Multiplying by $\sinh^2 x$ gives

$$\sinh^2 x f'(x) = \sinh x \cosh x - x. \quad (5396)$$

Since

$$\sinh x \cosh x = \frac{1}{2} \sinh(2x) > x \quad (5397)$$

for $x > 0$, we have $f'(x) > 0$. A physical Euclidean magic solution requires

$$s_{\nu,E}^* = \Lambda_\nu^{-1} \leq 1, \quad (5398)$$

and hence $\Lambda_\nu \geq 1$. Therefore,

$$\Lambda_\nu \coth(\Lambda_\nu \zeta) \geq \coth \zeta, \quad (5399)$$

with strict inequality for $\Lambda_\nu > 1$. Consequently,

$$\frac{d \theta_\nu^*}{d |K|} < 0 \quad (5400)$$

for the nontrivial fixed-harmonic geometric branch. Because

$$\zeta = \frac{a}{2} \sqrt{|K|}, \quad (5401)$$

we have

$$\frac{d\zeta}{d|K|} = \frac{\zeta}{2|K|}. \quad (5402)$$

Furthermore,

$$\frac{d\theta}{d \ln s} = 2 \tan \left(\frac{\theta}{2} \right). \quad (5403)$$

Therefore,

$$\frac{\partial \theta_\nu^*}{\partial |K|} = \frac{\zeta}{|K|} \tan \left(\frac{\theta_\nu^*}{2} \right) [\coth \zeta - \Lambda_\nu \coth (\Lambda_\nu \zeta)]. \quad (5404)$$

This expression is negative for every nontrivial physical branch. Because

$$\Lambda_\nu \propto \omega^{-1/2}, \quad (5405)$$

we have

$$\frac{d\Lambda_\nu}{d \ln \omega} = -\frac{\Lambda_\nu}{2}. \quad (5406)$$

At fixed curvature,

$$\frac{d \ln s_\nu^*}{d \ln \omega} = \frac{\Lambda_\nu \zeta}{2} \coth (\Lambda_\nu \zeta). \quad (5407)$$

Hence,

$$\frac{\partial \theta_\nu^*}{\partial \ln \omega} = \tan \left(\frac{\theta_\nu^*}{2} \right) \Lambda_\nu \zeta \coth (\Lambda_\nu \zeta). \quad (5408)$$

In the Euclidean limit, $\Lambda_\nu \zeta \rightarrow 0$, and $x \coth x \rightarrow 1$. Therefore,

$$\frac{\partial \ln \theta_\nu^*}{\partial \ln \omega} \rightarrow \frac{1}{2} \quad (5409)$$

at small angle, recovering the square-root law. We now consider $\lambda_\nu = \Lambda_\nu \zeta \gg 1$. Then

$$\sinh (\Lambda_\nu \zeta) = \frac{1}{2} e^{\Lambda_\nu \zeta} [1 + O(e^{-2\Lambda_\nu \zeta})]. \quad (5410)$$

Thus

$$s_\nu^* = 2 \sinh \zeta e^{-\Lambda_\nu \zeta} [1 + O(e^{-2\Lambda_\nu \zeta})]. \quad (5411)$$

For asymptotically small magic angle, $\theta_\nu^* \simeq 2s_\nu^*$. Therefore,

$$\theta_\nu^* \sim 4 \sinh \left(\frac{a}{2R} \right) \exp \left[-\frac{\pi a}{n_c R} \sqrt{\frac{d\alpha_\nu^*}{\gamma_0 \omega}} \right]. \quad (5412)$$

This is qualitatively different from the Euclidean law $\theta_{\nu,E}^* \propto \omega^{1/2}$. At any fixed finite negative curvature radius,

$$\theta_\nu^* \propto \exp \left[-\frac{\text{const}}{\sqrt{\omega}} \right] \quad (5413)$$

in the asymptotically weak-coupling curvature-dominated regime.

g. Theorem 128. Curvature transmutation of magic-angle universality. Within the geometry-controlled dimensionless-root theory, the weak-coupling Euclidean square-lattice magic law is algebraic,

$$\theta_\nu^* \sim \omega^{1/2}, \quad K = 0. \quad (5414)$$

At every fixed finite negative curvature satisfying the curvature-dominated condition,

$$\theta_\nu^* \sim \exp \left[-\frac{C_\nu(K)}{\sqrt{\omega}} \right], \quad K < 0, \quad (5415)$$

with

$$C_\nu(K) = \frac{\pi a \sqrt{|K|}}{n_c} \sqrt{\frac{d\alpha_\nu^*}{\gamma_0}}. \quad (5416)$$

Negative curvature therefore converts the Euclidean algebraic magic-angle law into an essential singularity. Define

$$\beta_\nu^{\text{magic}} = \frac{d \ln \theta_\nu^*}{d \ln \omega}. \quad (5417)$$

At asymptotically small angle,

$$\ln \theta_\nu^* \simeq \ln s_\nu^* + \text{constant}. \quad (5418)$$

Therefore,

$$\beta_\nu^{\text{magic}} \simeq \frac{\lambda_\nu}{2} \coth \lambda_\nu. \quad (5419)$$

For $\lambda_\nu \ll 1$, we obtain

$$\beta_\nu^{\text{magic}} = \frac{1}{2} + \frac{\lambda_\nu^2}{6} + O(\lambda_\nu^4). \quad (5420)$$

For $\lambda_\nu \gg 1$, we obtain

$$\beta_\nu^{\text{magic}} \sim \frac{\lambda_\nu}{2}. \quad (5421)$$

Thus no finite power-law exponent describes the asymptotic curvature-dominated magic curve. The divergence of the running exponent is the differential signature of the essential singularity. The fixed- γ_0 theory isolates the purely geometric effect. The physical tunneling harmonic generally satisfies $\gamma_1 = \gamma_1(K, \theta)$. Define the flat-space curvature susceptibility of the dominant tunneling harmonic by

$$\Gamma_{T,\nu} = \left. \frac{\partial \ln \gamma_1}{\partial |K|} \right|_{K=0, \theta=\theta_{\nu,E}^*}. \quad (5422)$$

Allow also for a possible intrinsic curvature response of the dimensionless spectral root,

$$\Gamma_{\alpha,\nu} = \left. \frac{\partial \ln \alpha_\nu^*}{\partial |K|} \right|_{K=0}. \quad (5423)$$

The generalized magic condition is

$$\alpha_\square^{\text{geo}} = \alpha_\nu^*(K). \quad (5424)$$

At weak curvature and neglecting the mild θ -dependence of γ_1 , differentiation gives

$$\left. \frac{\partial \theta_\nu^*}{\partial |K|} \right|_0 = (\Gamma_{T,\nu} - \Gamma_{\alpha,\nu}) \tan\left(\frac{\theta_{\nu,E}^*}{2}\right) - \frac{a^2}{12} \cot\left(\frac{\theta_{\nu,E}^*}{2}\right). \quad (5425)$$

This result separates the curvature shift into three terms. The first comes from curvature reconstruction of the tunneling harmonic. The second comes from an intrinsic curvature shift of the dimensionless spectral root. The third is the universal geometric registry contribution.

h. Theorem 129. Three-component magic-angle curvature response. At weak negative curvature,

$$\partial_{|K|} \theta_\nu^* = \partial_{|K|} \theta_\nu^*|_{\text{tun}} + \partial_{|K|} \theta_\nu^*|_{\text{root}} + \partial_{|K|} \theta_\nu^*|_{\text{geom}}, \quad (5426)$$

with

$$\begin{aligned} \partial_{|K|} \theta_\nu^*|_{\text{tun}} &= \Gamma_{T,\nu} \tan\left(\frac{\theta_{\nu,E}^*}{2}\right), \\ \partial_{|K|} \theta_\nu^*|_{\text{root}} &= -\Gamma_{\alpha,\nu} \tan\left(\frac{\theta_{\nu,E}^*}{2}\right), \end{aligned} \quad (5427)$$

and

$$\partial_{|K|} \theta_\nu^*|_{\text{geom}} = -\frac{a^2}{12} \cot\left(\frac{\theta_{\nu,E}^*}{2}\right). \quad (5428)$$

Thus the sign of the physical magic-angle shift is not fixed by geometry alone when curvature substantially reconstructs the relevant tunneling harmonic or the dimensionless spectral root. If $\Gamma_{\alpha,\nu} = 0$, the first-order magic shift changes sign when

$$\Gamma_{T,\nu} \tan\left(\frac{\theta_{\nu,E}^*}{2}\right) = \frac{a^2}{12} \cot\left(\frac{\theta_{\nu,E}^*}{2}\right). \quad (5429)$$

Therefore the critical tunneling susceptibility is

$$\Gamma_{T,\nu}^{\text{crit}} = \frac{a^2}{12} \cot^2\left(\frac{\theta_{\nu,E}^*}{2}\right). \quad (5430)$$

If $\Gamma_{T,\nu} < \Gamma_{T,\nu}^{\text{crit}}$, the magic angle decreases with increasing negative curvature. If $\Gamma_{T,\nu} > \Gamma_{T,\nu}^{\text{crit}}$, the tunneling reconstruction is strong enough to reverse the geometric shift. This provides a direct numerical criterion for distinguishing geometry-dominated and tunneling-dominated magic-curve motion. The exact effective hyperbolic moiré area is

$$A_M^{\text{eff}} = 2\pi R^2 \left[\sqrt{1 + \frac{\sinh^2 \zeta}{s^2}} - 1 \right]. \quad (5431)$$

Along the fixed-harmonic magic curve,

$$\frac{\sinh \zeta}{s_\nu^*} = \sinh \lambda_\nu. \quad (5432)$$

Therefore,

$$\sqrt{1 + \frac{\sinh^2 \zeta}{(s_\nu^*)^2}} = \cosh \lambda_\nu. \quad (5433)$$

Hence,

$$A_{M,\nu}^{\text{eff},*} = 2\pi R^2 [\cosh \lambda_\nu - 1]. \quad (5434)$$

Using

$$\lambda_\nu = \frac{\xi_{M,\nu}^*}{R}, \quad (5435)$$

we can write

$$A_{M,\nu}^{\text{eff},*} = 2\pi R^2 \left[\cosh\left(\frac{\xi_{M,\nu}^*}{R}\right) - 1 \right]. \quad (5436)$$

Normalize by the Euclidean disk area associated with the same magic registry length,

$$A_{\nu,E}^* = \pi (\xi_{M,\nu}^*)^2. \quad (5437)$$

Then

$$\frac{A_{M,\nu}^{\text{eff},\star}}{A_{\nu,E}^{\star}} = \mathcal{A}(\lambda_{\nu}), \quad (5438)$$

where

$$\mathcal{A}(\lambda) = \frac{2(\cosh \lambda - 1)}{\lambda^2}. \quad (5439)$$

For $\lambda \ll 1$, we obtain

$$\mathcal{A}(\lambda) = 1 + \frac{\lambda^2}{12} + \frac{\lambda^4}{360} + O(\lambda^6). \quad (5440)$$

For $\lambda \gg 1$, we obtain

$$\mathcal{A}(\lambda) \sim \frac{e^{\lambda}}{\lambda^2}. \quad (5441)$$

Thus the twist angle required for magic becomes exponentially smaller while the geometric area enclosed by the corresponding fixed registry length becomes exponentially larger relative to its Euclidean value. The geometric area must not be identified automatically with the exact hyperbolic crystallographic index. Suppose, however, that an exact commensurate magic sequence satisfies the geometric–arithmetic locking condition

$$c_- \frac{A_M^{\text{eff}}}{A_0} \leq q_M \leq c_+ \frac{A_M^{\text{eff}}}{A_0}, \quad (5442)$$

with positive constants independent of the asymptotic sequence. Then

$$q_{M,\nu}^{\star} = \Theta \left[\frac{R^2}{A_0} (\cosh \lambda_{\nu} - 1) \right]. \quad (5443)$$

In the Euclidean regime,

$$q_{M,\nu}^{\star} = \Theta \left[\frac{(\xi_{M,\nu}^{\star})^2}{A_0} \right]. \quad (5444)$$

Since

$$(\xi_{M,\nu}^{\star})^2 = \Theta(\omega^{-1}), \quad (5445)$$

this reproduces

$$q_{M,\nu}^{\star} = \Theta(\omega^{-1}). \quad (5446)$$

In the curvature-dominated regime,

$$q_{M,\nu}^{\star} = \Theta \left[\frac{R^2}{A_0} e^{\xi_{M,\nu}^{\star}/R} \right]. \quad (5447)$$

Because

$$\xi_{M,\nu}^{\star} = \Theta(\omega^{-1/2}), \quad (5448)$$

we obtain the conditional asymptotic law

$$q_{M,\nu}^{\star} = \Theta \left[\frac{R^2}{A_0} \exp \left(\frac{C_{\nu}}{R\sqrt{\omega}} \right) \right]. \quad (5449)$$

Thus the computational and crystallographic complexity of an exact hyperbolic magic sequence can become exponentially larger than its Euclidean counterpart.

i. Theorem 130. Conditional magic-supercell complexity transmutation. Under geometric–arithmetic locking, the weak-coupling Euclidean magic supercell satisfies

$$q_M^{\star} = \Theta(\omega^{-1}), \quad (5450)$$

whereas at fixed negative curvature in the curvature-dominated regime,

$$q_M^{\star} = \Theta[\exp(C/\sqrt{\omega})] \quad (5451)$$

up to the explicitly displayed curvature- and primitive-cell prefactors. The exact crystallographic statement is conditional on locking and on the existence of the corresponding exact commensurate sequence. The displayed inverse-linear law is the class $\mathfrak{U}_1(1)$ specialization. More generally, an exact class $\mathfrak{U}_1(\beta)$ transfers a established magic coefficient c to $\sqrt{\omega} \log N_M^{\text{sc},\star} \rightarrow \beta c$; the centered class has $\beta = 4$. Arithmetic never supplies the spectral root itself. Consider two magic roots satisfying $\alpha_{\mu}^{\star} < \alpha_{\nu}^{\star}$. Then $\Lambda_{\mu} < \Lambda_{\nu}$. Because $\sinh x$ is strictly increasing for positive x ,

$$\sinh(\Lambda_{\mu}\zeta) < \sinh(\Lambda_{\nu}\zeta). \quad (5452)$$

Therefore, $s_{\mu}^{\star} > s_{\nu}^{\star}$. Hence, $\theta_{\mu}^{\star} > \theta_{\nu}^{\star}$. Thus the ordering of magic branches cannot reverse in the fixed-harmonic geometric theory. Their exact half-angle ratio is

$$\frac{s_{\nu}^{\star}}{s_{\mu}^{\star}} = \frac{\sinh(\Lambda_{\mu}\zeta)}{\sinh(\Lambda_{\nu}\zeta)}. \quad (5453)$$

In the Euclidean limit,

$$\frac{\theta_{\nu}^{\star}}{\theta_{\mu}^{\star}} \longrightarrow \sqrt{\frac{\alpha_{\mu}^{\star}}{\alpha_{\nu}^{\star}}}. \quad (5454)$$

In the curvature-dominated regime,

$$\frac{\theta_{\nu}^{\star}}{\theta_{\mu}^{\star}} \sim \exp[-(\Lambda_{\nu} - \Lambda_{\mu})\zeta]. \quad (5455)$$

Negative curvature therefore exponentially separates successive weak-coupling magic branches. A physical angle requires $0 < s_\nu^* \leq 1$. Using

$$s_\nu^* = \frac{\sinh \zeta}{\sinh(\Lambda_\nu \zeta)}, \quad (5456)$$

this requires

$$\sinh(\Lambda_\nu \zeta) \geq \sinh \zeta. \quad (5457)$$

Hence $\Lambda_\nu \geq 1$. Using Eq. (5345),

$$\omega \leq \frac{4\pi^2 d\alpha_\nu^*}{n_c^2 \gamma_0}. \quad (5458)$$

If the twist is restricted to the square-lattice irreducible interval $0 < \theta \leq \frac{\pi}{4}$, then $s \leq s_{\max}$, where

$$s_{\max} = \sin\left(\frac{\pi}{8}\right). \quad (5459)$$

The stronger branch-existence condition is

$$\sinh(\Lambda_\nu \zeta) \geq \frac{\sinh \zeta}{s_{\max}}. \quad (5460)$$

Therefore,

$$\Lambda_\nu \geq \frac{1}{\zeta} \operatorname{arsinh}\left(\frac{\sinh \zeta}{s_{\max}}\right). \quad (5461)$$

This condition determines where a particular analytic branch leaves the chosen irreducible angular domain. The exact operational curve is $\theta_{\nu, \text{op}}^*(K, \omega)$ defined by $\partial_\theta \mathcal{M}_\nu = 0$. The geometric scaling prediction is $\theta_{\nu, \text{geo}}^*(K, \omega)$ defined by Eq. (5344) or, under fixed harmonic normalization, Eq. (5349). We define the nongeometric magic displacement by

$$\delta\theta_{\nu}^{\text{ng}} = \theta_{\nu, \text{op}}^* - \theta_{\nu, \text{geo}}^*. \quad (5462)$$

The normalized residual is

$$\mathcal{R}_\nu^{\text{ng}} = \frac{\delta\theta_\nu^{\text{ng}}}{\theta_{\nu, \text{op}}^*}. \quad (5463)$$

A small value, $|\mathcal{R}_\nu^{\text{ng}}| \ll 1$, demonstrates that the curvature-dependent magic shift is controlled primarily by the registry geometry. A systematic nonzero residual identifies additional spectral, representation-theoretic, or tunneling-harmonic reconstruction beyond the local geometric scaling law.

j. Theorem 131. Geometry-to-spectrum magic residual. The difference

$$\delta\theta_\nu^{\text{ng}} = \theta_{\nu, \text{op}}^* - \theta_{\nu, \text{geo}}^* \quad (5464)$$

provides a direct quantitative separation between the curvature shift predicted solely from the exact registry geometry and the shift produced by the complete microscopic spectrum. Thus the geometric magic curve is a falsifiable prediction rather than a definition imposed on the numerical data. The local geometric and operational magic surfaces exist independently of exact finite-index commensurability. For a hyperbolic parent group Γ , an exact commensurate twist requires $g_\theta \in \text{Comm}(\Gamma)$. At fixed curvature and coupling, define the continuous operational magic angle by $\theta_\nu^*(K, \omega)$. An exact crystallographic magic point exists only if

$$g_{\theta_\nu^*(K, \omega)} \in \text{Comm}(\Gamma). \quad (5465)$$

Thus the exact crystallographic magic set is

$$\mathfrak{M}_{\text{magic}}^{\text{exact}} = \mathfrak{M}_{\text{magic}}^{\text{op}} \cap \mathfrak{C}_{\text{exact}}, \quad (5466)$$

where

$$\mathfrak{C}_{\text{exact}} = \{(\theta, K, \omega) : g_\theta \in \text{Comm}(\Gamma)\}. \quad (5467)$$

A continuous local magic curve therefore need not contain an arbitrarily dense sequence of exact crystallographic points. If the fixed-center exact twist set is finite, then no sequence of nontrivial exact magic twists can approach zero. Nevertheless, the local magic curve remains well defined through the microscopic Hamiltonian and paired-cover construction. If the center is twist-active and exact commensurate twists are dense, the operational curve can be sampled by an exact sequence $\theta_j \rightarrow \theta_\nu^*(K, \omega)$. For a growing hyperbolic computational domain of radius L_j , the required angular accuracy remains

$$|\theta_j - \theta_\nu^*| e^{L_j/R} \rightarrow 0. \quad (5468)$$

Thus exact-angle density alone is insufficient to guarantee convergence of a magic-band calculation on growing hyperbolic domains. The branch equation $\theta = \theta_\nu^*(K, \omega)$ defines a two-dimensional surface in (θ, K, ω) space. We denote this surface by

$$\mathcal{S}_\nu^{\text{magic}} = \{(\theta, K, \omega) : \theta = \theta_\nu^*(K, \omega)\}. \quad (5469)$$

At fixed ω , its intersection with a constant-coupling plane is a magic curve,

$$\mathcal{C}_\nu^{\text{magic}}(\omega) = \{(\theta, K) : \theta = \theta_\nu^*(K, \omega)\}. \quad (5470)$$

At fixed curvature, the corresponding section is

$$\mathcal{C}_\nu^{\text{magic}}(K) = \{(\theta, \omega) : \theta = \theta_\nu^*(K, \omega)\}. \quad (5471)$$

The complete magic manifold is the union over all branches,

$$\mathcal{S}^{\text{magic}} = \bigcup_\nu \mathcal{S}_\nu^{\text{magic}}. \quad (5472)$$

A magic branch is physically meaningful only if it remains a local maximum transverse to the surface. Define the transverse curvature

$$\kappa_\nu^\perp = -\partial_\theta^2 \ln \mathcal{M}_\nu|_{\theta=\theta_\nu^*}. \quad (5473)$$

A stable magic ridge requires $\kappa_\nu^\perp > 0$. The angular width of the magic region around the optimum can be defined for a small logarithmic score drop $\delta\mathcal{L} > 0$. Using

$$\mathcal{L}_\nu(\theta) = \mathcal{L}_\nu^* - \frac{1}{2}\kappa_\nu^\perp (\theta - \theta_\nu^*)^2 + O[(\theta - \theta_\nu^*)^3], \quad (5474)$$

the half-width is

$$\delta\theta_\nu^{\text{magic}} \simeq \sqrt{\frac{2\delta\mathcal{L}}{\kappa_\nu^\perp}}. \quad (5475)$$

Thus the magic manifold carries not only a location but a quantitatively defined transverse stability scale. Differentiating Eq. (5475) gives

$$\frac{\partial \ln \delta\theta_\nu^{\text{magic}}}{\partial K} = -\frac{1}{2} \frac{\partial \ln \kappa_\nu^\perp}{\partial K} \quad (5476)$$

for fixed score threshold. Thus curvature can make a magic condition more or less fine tuned even if the maximal magic score itself remains unchanged. A large $\kappa_\nu^\perp$ corresponds to a sharp magic angle. A small $\kappa_\nu^\perp$ corresponds to a broad magic plateau. This quantity accompanies the magic-curve location. For each fixed value of $\omega = \frac{w}{t}$, we select a sequence of curvatures

$$K_0 = 0, \quad K_1, \quad \dots, \quad K_N < 0. \quad (5477)$$

At each curvature we evaluate a two-dimensional parameter grid in θ and the relevant spectral-sector variables. For each continuously tracked spectral cluster we calculate W_ν , Δ_ν , Ω_ν^{max} , $\rho_\nu^{\text{coh,max}}$, $\mathcal{C}_\nu^{\text{coh}}$, and $\mathcal{M}_\nu^{\text{cert}}$. The magic angle at each curvature is obtained from

$$\theta_\nu^*(K, \omega) = \arg \max_\theta \mathcal{M}_\nu^{\text{cert}}(\theta, K, \omega). \quad (5478)$$

The same cluster is followed between neighboring parameter points by projector overlap rather than eigenvalue ordering. Let the numerical score near the optimum be approximated by

$$\mathcal{L}(\theta) = \mathcal{L}^* - \frac{\kappa_\perp}{2} (\theta - \theta^*)^2. \quad (5479)$$

Suppose the uncertainty in the logarithmic magic score is $\epsilon_{\mathcal{L}}$. The corresponding angular uncertainty is

$$\epsilon_\theta \gtrsim \sqrt{\frac{2\epsilon_{\mathcal{L}}}{\kappa_\perp}}. \quad (5480)$$

Thus a very broad magic ridge intrinsically limits the precision with which a unique magic angle can be quoted. For every numerically determined operational magic point, compute the Euclidean magic registry length

$$\xi_{\nu,E}^*(\omega) = \frac{\pi a}{n_c} \sqrt{\frac{d\alpha_\nu^*}{\gamma_0 \omega}}. \quad (5481)$$

Define

$$\lambda_\nu = \frac{\xi_{\nu,E}^*}{R}. \quad (5482)$$

The geometry-controlled prediction is

$$\frac{\theta_{\nu,\text{op}}^*(K, \omega)}{\theta_\nu^*(0, \omega)} \simeq \frac{\lambda_\nu}{\sinh \lambda_\nu} \quad (5483)$$

in the weak microscopic-curvature and small-angle regime after the tunneling harmonic has been controlled. The same data must also satisfy

$$\xi_M [K, \theta_{\nu,\text{op}}^*(K, \omega)] \simeq \xi_{\nu,E}^*(\omega) \quad (5484)$$

if the curvature shift is predominantly geometric. These two tests are independent representations of the same predicted magic manifold. At fixed negative curvature, define

$$X = \frac{1}{\sqrt{\omega}}. \quad (5485)$$

In the curvature-dominated regime,

$$\ln \theta_\nu^* = -\frac{\pi a}{n_c R} \sqrt{\frac{d\alpha_\nu^*}{\gamma_0}} X + \ln \left[4 \sinh \left(\frac{a}{2R} \right) \right] + o(1). \quad (5486)$$

Therefore a plot of $\ln \theta_\nu^*$ against $1/\sqrt{\omega}$ must become asymptotically linear. The predicted slope is

$$S_\nu(K) = -\frac{\pi a}{n_c R} \sqrt{\frac{d\alpha_\nu^*}{\gamma_0}}. \quad (5487)$$

This provides a direct numerical test of the essential-singularity prediction without fitting a multi-parameter nonlinear function. A reported curvature-dependent magic branch must satisfy four independent levels of validation. The first is spectral verification: the bandwidth, isolation gap, generalized dispersion, coherent DOS, and participation conditions defining $\mathcal{M}$ must converge. The second is branch verification: the same spectral projector must be tracked continuously along the curve. The third is geometric verification: the operational branch must be compared with Eq. (5349) or the more general implicit equation Eq. (5344). The fourth is crystallographic verification: when exact commensurate points are claimed, the corresponding twists must satisfy the exact commensurator condition and all finite-cover observables must converge. Only after all four tests are satisfied do we identify a measured curve as a curvature-controlled magic manifold.

k. Theorem 132. Complete curvature-dependent magic-manifold structure. A nondegenerate Euclidean magic angle extends locally into an exact operational surface $\theta_\nu^* = \theta_\nu^*(K, \omega)$ whenever the transverse magic curvature is nonzero. Under the geometry-controlled dimensionless-root reduction, this surface obeys the explicit scaling law

$$\sin\left(\frac{\theta_\nu^*}{2}\right) = \frac{\sinh(a/2R)}{\sinh\left[\frac{\pi a}{n_c R} \sqrt{\frac{d\alpha_\nu^*}{\gamma_0 \omega}}\right]}. \quad (5488)$$

The corresponding crossover variable is

$$\lambda_\nu = \frac{\xi_{M,\nu}^*}{R} = \Theta \left[\sqrt{\frac{|K|}{\omega}} \right]. \quad (5489)$$

For $\lambda_\nu \ll 1$, the Euclidean law $\theta_\nu^* \propto \omega^{1/2}$ is recovered. For $\lambda_\nu \gg 1$, the magic angle obeys the curvature-dominated essential singularity

$$\theta_\nu^* \propto \exp\left[-\frac{C_\nu(K)}{\sqrt{\omega}}\right]. \quad (5490)$$

Thus negative curvature does not merely shift a pre-existing Euclidean magic angle. It changes the functional universality of the complete magic manifold. The complete construction can now be expressed in two complementary forms. The exact operational route is

$$\begin{aligned} H(\theta, K, \omega) &\longrightarrow \{W, \Delta, dE, \rho, \mathcal{C}_{\text{coh}}\} \longrightarrow \mathcal{M}(\theta, K, \omega) \\ &\longrightarrow \partial_\theta \mathcal{M} = 0 \longrightarrow \theta_\nu^*(K, \omega). \end{aligned} \quad (5491)$$

The geometric scaling route is

$$\begin{aligned} K < 0 &\longrightarrow \xi_M(\theta, K) \longrightarrow q_M^{\text{loc}}(\theta, K) \longrightarrow E_M^{\text{geo}}(\theta, K) \\ &\longrightarrow \alpha_{\square}^{\text{geo}}(\theta, K, \omega) \longrightarrow \alpha_{\square}^{\text{geo}} = \alpha_\nu^* \longrightarrow \theta_{\nu, \text{geo}}^*(K, \omega). \end{aligned} \quad (5492)$$

Their difference,

$$\delta\theta_\nu^{\text{ng}} = \theta_{\nu, \text{op}}^* - \theta_{\nu, \text{geo}}^*, \quad (5493)$$

quantifies the part of the magic shift not explained by the exact local registry geometry. The strongest geometry-controlled prediction is the universal collapse

$$\frac{\theta_\nu^*(K, \omega)}{\theta_\nu^*(0, \omega)} \simeq \frac{\lambda_\nu}{\sinh \lambda_\nu}, \quad \lambda_\nu = \frac{\xi_{M,\nu}^*}{R}. \quad (5494)$$

The corresponding magic registry length is

$$\xi_{M,\nu}^* = \frac{\pi a}{n_c} \sqrt{\frac{d\alpha_\nu^*}{\gamma_0(w/t)}}, \quad (5495)$$

and the curvature crossover occurs when $R \sim \xi_{M,\nu}^*$. Consequently, the Euclidean concept of isolated magic angles is promoted to a family of curvature-dependent surfaces whose weak-coupling behavior crosses from an algebraic square-root law to an essential singularity. The central test is therefore no longer whether one value of θ produces a narrow Euclidean band. The decisive test is whether the complete numerical magic manifold collapses onto the predicted curvature-controlled scaling functions while simultaneously retaining the spectral, representation-space, localization, and thermodynamic verification established above.

D. Curvature-Induced Birth, Shift, and Termination of Magic Branches

The curvature-dependent magic surfaces derived above describe the smooth continuation of an already existing magic branch. A substantially stronger possibility is that negative curvature changes the number and organization of the magic branches themselves. We therefore ask whether variation of

$$K, \quad \theta, \quad \omega = \frac{w}{t} \quad (5496)$$

can produce branch displacement, branch creation, branch annihilation, branch termination and reordering of the complete magic hierarchy. These phenomena cannot be identified from a sequence of visually selected bandwidth minima. They require a bifurcation theory of the operational magic functional. For an admissible continuously tracked spectral cluster $\mathcal{C}_\nu$, define

$$\mathcal{M}_\nu = \mathcal{M}_\nu(\theta, K, \omega), \quad (5497)$$

and introduce

$$\mathcal{L}_\nu = \ln \mathcal{M}_\nu. \quad (5498)$$

Inside the region where $\Delta_\nu > 0$ and $\mathcal{C}_\nu^{\text{coh}} > 0$, the function $\mathcal{L}_\nu$ is smooth. Define

$$F_\nu(\theta, K, \omega) = \frac{\partial \mathcal{L}_\nu}{\partial \theta}. \quad (5499)$$

Every interior stationary magic branch satisfies $F_\nu = 0$. A local maximum additionally satisfies

$$\partial_\theta F_\nu = \partial_\theta^2 \mathcal{L}_\nu < 0. \quad (5500)$$

The transverse magic curvature is

$$\kappa_\nu^\perp = -\partial_\theta^2 \mathcal{L}_\nu. \quad (5501)$$

Thus $\kappa_\nu^\perp > 0$ characterizes a stable interior magic maximum. Let u denote any smooth one-dimensional direction in control space, $u = u(K, \omega)$. Along a regular branch,

$$F_\nu[\theta_\nu^*(u), u] = 0. \quad (5502)$$

If $\partial_\theta F_\nu \neq 0$, differentiation gives

$$\frac{d\theta_\nu^*}{du} = -\frac{\partial_u F_\nu}{\partial_\theta F_\nu}. \quad (5503)$$

Therefore an ordinary branch shift is mathematically regular as long as $\kappa_\nu^\perp \neq 0$. The failure of this condition identifies the first location at which ordinary branch continuation can break down. The geometry-only magic curve derived above has

$$s_\nu^* = \sin\left(\frac{\theta_\nu^*}{2}\right) = \frac{\sinh \zeta}{\sinh(\Lambda_\nu \zeta)}, \quad (5504)$$

where

$$\zeta = \frac{a}{2R}, \quad \Lambda_\nu > 1 \quad (5505)$$

for a nontrivial physical branch. Differentiation gives

$$\frac{d \ln s_\nu^*}{d\zeta} = \coth \zeta - \Lambda_\nu \coth(\Lambda_\nu \zeta). \quad (5506)$$

Define

$$f(x) = x \coth x. \quad (5507)$$

Its derivative is

$$f'(x) = \coth x - x \operatorname{csch}^2 x. \quad (5508)$$

Multiplication by $\sinh^2 x$ gives

$$\sinh^2 x f'(x) = \sinh x \cosh x - x. \quad (5509)$$

For $x > 0$, we have

$$\sinh x \cosh x = \frac{1}{2} \sinh(2x) > x. \quad (5510)$$

Therefore, $f'(x) > 0$. Since $\Lambda_\nu \zeta > \zeta$, we obtain

$$\Lambda_\nu \zeta \coth(\Lambda_\nu \zeta) > \zeta \coth \zeta. \quad (5511)$$

Dividing by $\zeta > 0$ gives

$$\Lambda_\nu \coth(\Lambda_\nu \zeta) > \coth \zeta. \quad (5512)$$

Hence

$$\frac{ds_\nu^*}{d\zeta} < 0. \quad (5513)$$

The geometry-only branch is therefore single valued and strictly monotone.

a. Theorem 133. No interior magic-branch bifurcation from pure registry remapping. If the dimensionless magic root α_ν^* and the normalized dominant tunneling harmonic are held fixed, the hyperbolic registry map produces exactly one smooth monotone continuation of every Euclidean magic root. Consequently, pure replacement of the Euclidean registry length by the hyperbolic registry length cannot by itself produce an interior saddle-node creation or annihilation of magic branches. Therefore any numerically observed interior branch birth or annihilation must originate from at least one additional mechanism, curvature-dependent tunneling reconstruction, curvature dependence of the dimensionless spectral root, spectral-gap admissibility, localization or coherence loss, or representation-space reorganization. This provides a direct falsification test for a purely geometric explanation of the magic manifold. To study branch bifurcations explicitly, write

$$V = E_{\text{disp}} = \hbar D_G \Omega^{\text{max}}, \quad X = E_{\text{kin}} \rho_{\text{coh}}^{\text{max}}, \quad (5514)$$

and $C = \mathcal{C}_{\text{coh}}$. Inside the admissible region,

$$\mathcal{M} = \left[\frac{\Delta}{\Delta + W} \frac{\Delta}{\Delta + V} \frac{X}{1 + X} C \right]^{1/4}. \quad (5515)$$

Therefore,

$$\begin{aligned} 4\mathcal{L} &= \ln \Delta - \ln(\Delta + W) \\ &+ \ln \Delta - \ln(\Delta + V) \\ &+ \ln X - \ln(1 + X) + \ln C. \end{aligned} \quad (5516)$$

Differentiating with respect to θ gives

$$\begin{aligned} 4F &= \frac{\Delta_\theta}{\Delta} - \frac{\Delta_\theta + W_\theta}{\Delta + W} + \frac{\Delta_\theta}{\Delta} - \frac{\Delta_\theta + V_\theta}{\Delta + V} \\ &+ \frac{X_\theta}{X(1 + X)} + \frac{C_\theta}{C}. \end{aligned} \quad (5517)$$

Thus every operational magic stationary point satisfies the fully explicit balance $F = 0$. The first two lines describe the competition between gap, bandwidth, and generalized dispersion. The third line describes coherent DOS reconstruction and real-space or representation-space coherence. Differentiating once more gives

$$\begin{aligned} 4F_\theta &= \frac{\Delta_{\theta\theta}}{\Delta} - \frac{\Delta_\theta^2}{\Delta^2} - \frac{\Delta_{\theta\theta} + W_{\theta\theta}}{\Delta + W} + \frac{(\Delta_\theta + W_\theta)^2}{(\Delta + W)^2} \\ &+ \frac{\Delta_{\theta\theta}}{\Delta} - \frac{\Delta_\theta^2}{\Delta^2} - \frac{\Delta_{\theta\theta} + V_{\theta\theta}}{\Delta + V} + \frac{(\Delta_\theta + V_\theta)^2}{(\Delta + V)^2} \\ &+ \frac{X_{\theta\theta}}{X(1 + X)} - \frac{(1 + 2X)X_\theta^2}{X^2(1 + X)^2} + \frac{C_{\theta\theta}}{C} - \frac{C_\theta^2}{C^2}. \end{aligned} \quad (5518)$$

Because $F_\theta = \partial_\theta^2 \mathcal{L}$, the local maximum condition is $F_\theta < 0$. A generic branch bifurcation becomes possible when this transverse curvature vanishes. Let u again denote a control direction. The mixed derivative is

$$\begin{aligned} 4F_u &= \frac{\Delta_{\theta u}}{\Delta} - \frac{\Delta_\theta \Delta_u}{\Delta^2} - \frac{\Delta_{\theta u} + W_{\theta u}}{\Delta + W} + \frac{(\Delta_\theta + W_\theta)(\Delta_u + W_u)}{(\Delta + W)^2} \\ &+ \frac{\Delta_{\theta u}}{\Delta} - \frac{\Delta_\theta \Delta_u}{\Delta^2} - \frac{\Delta_{\theta u} + V_{\theta u}}{\Delta + V} + \frac{(\Delta_\theta + V_\theta)(\Delta_u + V_u)}{(\Delta + V)^2} \\ &+ \frac{X_{\theta u}}{X(1 + X)} - \frac{(1 + 2X)X_\theta X_u}{X^2(1 + X)^2} + \frac{C_{\theta u}}{C} - \frac{C_\theta C_u}{C^2}. \end{aligned} \quad (5519)$$

Equations (5517), (5518), and (5519) provide a direct numerical criterion for detecting a curvature-driven magic bifurcation. Consider a critical point (θ_c, u_c) satisfying $F_c = 0$, and $(F_\theta)_c = 0$. Assume the generic nondegeneracy conditions $(F_u)_c \neq 0$, and $(F_{\theta\theta})_c \neq 0$. Introduce $x = \theta - \theta_c$, and $\delta u = u - u_c$. Taylor expansion gives

$$F = a\delta u + \frac{b}{2}x^2 + O(x^3, x\delta u, \delta u^2), \quad (5520)$$

where $a = (F_u)_c$, and $b = (F_{\theta\theta})_c$. The stationary equation $F = 0$ gives

$$x^2 = -\frac{2a}{b}\delta u. \quad (5521)$$

Therefore, on the side satisfying $-\frac{a\delta u}{b} > 0$, there are two stationary branches,

$$\theta_\pm^* = \theta_c \pm \sqrt{-\frac{2a}{b}\delta u} + O(\delta u). \quad (5522)$$

On the opposite side, no nearby stationary branch exists. Near the fold,

$$F_\theta = bx + O(x^2, \delta u). \quad (5523)$$

Since $F_\theta = \partial_\theta^2 \mathcal{L}$, the branch satisfying $bx < 0$ is a local maximum. The branch satisfying $bx > 0$ is a local minimum. Thus a generic fold creates or annihilates one operational magic maximum together with one unstable separating minimum. It does not generically create two stable magic maxima at once. The separation between the two stationary branches is

$$\Delta\theta_{\text{fold}} = 2\sqrt{-\frac{2a}{b}\delta u} + O(\delta u). \quad (5524)$$

Therefore,

$$\Delta\theta_{\text{fold}} \propto |\delta u|^{1/2}. \quad (5525)$$

The transverse curvature on the maximum branch is

$$\kappa_\perp = -F_\theta = \sqrt{2|ab\delta u|} + O(|\delta u|). \quad (5526)$$

Hence, $\kappa_\perp \propto |\delta u|^{1/2}$. For a fixed allowed logarithmic score decrease $\delta\mathcal{L} > 0$, the magic width satisfies

$$\delta\theta_{\text{magic}} \simeq \sqrt{\frac{2\delta\mathcal{L}}{\kappa_\perp}}. \quad (5527)$$

Therefore,

$$\delta\theta_{\text{magic}} \propto |\delta u|^{-1/4}. \quad (5528)$$

The newly born magic branch becomes anomalously broad as the fold is approached. Because $F = \partial_\theta \mathcal{L}$, integration of the local normal form gives

$$\mathcal{L} = \mathcal{L}_c + \mathcal{L}_u \delta u + a\delta u x + \frac{b}{6}x^3 + \dots \quad (5529)$$

At a stationary point,

$$a\delta u = -\frac{b}{2}x^2. \quad (5530)$$

Therefore the branch-dependent part is

$$\delta\mathcal{L}_{\text{branch}} = -\frac{b}{3}x^3. \quad (5531)$$

The absolute score difference between the two stationary branches is

$$|\Delta\mathcal{L}_{\text{fold}}| = \frac{2^{5/2}}{3} \frac{|a|^{3/2}}{|b|^{1/2}} |\delta u|^{3/2} + o(|\delta u|^{3/2}). \quad (5532)$$

Thus

$$|\Delta \ln \mathcal{M}| \propto |\delta u|^{3/2}. \quad (5533)$$

b. Theorem 134. Universal magic-branch fold scaling. At a generic saddle-node bifurcation of the operational magic landscape,

$$F = 0, \quad F_\theta = 0, \quad F_u \neq 0, \quad F_{\theta\theta} \neq 0, \quad (5534)$$

one stable magic maximum and one unstable minimum are created or annihilated. The universal local scaling laws are

$$\Delta\theta \propto |\delta u|^{1/2}, \quad \kappa_\perp \propto |\delta u|^{1/2}, \quad \delta\theta_{\text{magic}} \propto |\delta u|^{-1/4}, \quad (5535)$$

and

$$\Delta \ln \mathcal{M} \propto |\delta u|^{3/2}. \quad (5536)$$

These exponents are independent of the microscopic details of the Hamiltonian as long as the fold is generic. The full problem contains at least two independent controls, $|K|$ and ω . A higher-order organization of the fold lines can therefore occur at a cusp point. Let (θ_c, u_c, v_c) satisfy $F = 0$, $F_\theta = 0$, and $F_{\theta\theta} = 0$. Assume $F_{\theta\theta\theta} \neq 0$. The two control parameters must unfold the constant and linear terms independently. A sufficient nondegeneracy condition is

$$\det \begin{pmatrix} F_u & F_v \\ F_{\theta u} & F_{\theta v} \end{pmatrix}_c \neq 0. \quad (5537)$$

After smooth local rescaling of the variables, the stationary equation takes the canonical form

$$F = -x^3 + Ux + V + O(4), \quad (5538)$$

where $x = \theta - \theta_c$. The stationary equation is therefore

$$x^3 - Ux - V = 0. \quad (5539)$$

The cubic $x^3 - Ux - V = 0$ has discriminant

$$\mathcal{D}_{\text{cusp}} = 4U^3 - 27V^2. \quad (5540)$$

For $\mathcal{D}_{\text{cusp}} < 0$, there is one stationary point. For $\mathcal{D}_{\text{cusp}} = 0$, two stationary points merge. For $\mathcal{D}_{\text{cusp}} > 0$, there are three distinct stationary points. The two fold boundaries are

$$V_\pm = \pm \frac{2}{3\sqrt{3}} U^{3/2}, \quad U \geq 0. \quad (5541)$$

The local stability is

$$F_\theta = -3x^2 + U. \quad (5542)$$

Inside the three-root region, the two outer roots satisfy $F_\theta < 0$, and are therefore local maxima of the magic score. The middle root satisfies $F_\theta > 0$, and is a local minimum. Thus a cusp allows two distinct stable magic branches to coexist at the same values of curvature and interlayer coupling. Integrating the normal form gives

$$\mathcal{L} = \mathcal{L}_c - \frac{x^4}{4} + \frac{Ux^2}{2} + Vx + \dots. \quad (5543)$$

At $V = 0$, the stationary points are $x_0 = 0$, and $x_\pm = \pm\sqrt{U}$. The two outer maxima have

$$\mathcal{L}_\pm = \mathcal{L}_c + \frac{U^2}{4}, \quad (5544)$$

whereas the central minimum has $\mathcal{L}_0 = \mathcal{L}_c$. The two magic maxima are therefore exactly degenerate in score at the symmetric cusp section.

c. Theorem 135. Cusp-mediated coexistence of two magic branches. A generic codimension-two cusp of the operational magic landscape produces a parameter region containing two simultaneously stable magic maxima separated by one unstable minimum. The region is determined locally by $4U^3 > 27V^2$. Thus curvature and interlayer coupling can create a genuine bistable magic regime in which two distinct magic angles coexist for the same external control parameters. For small nonzero V , the two outer maxima remain close to $x_\pm = \pm\sqrt{U}$. The envelope theorem gives

$$\frac{d\mathcal{L}_\pm^*}{dV} = x_\pm. \quad (5545)$$

At $V = 0$,

$$\left. \frac{d\mathcal{L}_+^*}{dV} \right|_0 = \sqrt{U}, \quad (5546)$$

and

$$\left. \frac{d\mathcal{L}_-^*}{dV} \right|_0 = -\sqrt{U}. \quad (5547)$$

Therefore,

$$\mathcal{L}_+^* - \mathcal{L}_-^* = 2\sqrt{U}V + O(V^2). \quad (5548)$$

Thus $V > 0$ selects the $+$ branch as the global optimum, whereas $V < 0$ selects the $-$ branch. At $V = 0$, the identity of the globally optimal magic branch switches. The corresponding magic-angle jump is $\Delta\theta_{\text{hier}} = 2\sqrt{U}$ to leading order in the canonical coordinate. As $U \rightarrow 0^+$, the jump vanishes according to $\Delta\theta_{\text{hier}} \propto U^{1/2}$. This produces a critical endpoint of the magic-hierarchy transition. A magic branch can also appear or disappear without a stationary-point fold. The operational definition contains the hard isolation requirement $\Delta > 0$. Suppose an otherwise smooth candidate branch reaches $\Delta = 0$ at a control value $u = u_g$. Assume

$$W \rightarrow W_g > 0, \quad V \rightarrow V_g > 0, \quad X \rightarrow X_g > 0, \quad (5549)$$

and $C \rightarrow C_g > 0$. For small positive gap,

$$\frac{\Delta}{\Delta + W} = \frac{\Delta}{W_g} + O(\Delta^2), \quad (5550)$$

and

$$\frac{\Delta}{\Delta + V} = \frac{\Delta}{V_g} + O(\Delta^2). \quad (5551)$$

Therefore,

$$\mathcal{M} = \left[\frac{X_g}{1 + X_g} C_g \right]^{1/4} \frac{\Delta^{1/2}}{(W_g V_g)^{1/4}} + o(\Delta^{1/2}). \quad (5552)$$

Thus an operational magic branch born through spectral isolation begins with a square-root score. If

$$\Delta = A_g |u - u_g| + o(|u - u_g|), \quad (5553)$$

then

$$\mathcal{M} \propto |u - u_g|^{1/2}. \quad (5554)$$

The strong functional contains the additional factor

$$\frac{\Delta}{\Delta + E_{\text{disp}}^{(2)}}. \quad (5555)$$

Assume

$$E_{\text{disp}}^{(2)} \rightarrow E_{2,g} > 0. \quad (5556)$$

Then

$$\mathcal{M}^{(2)} = \left[\frac{X_g}{1 + X_g} C_g \right]^{1/5} \frac{\Delta^{3/5}}{(W_g V_g E_{2,g})^{1/5}} + o(\Delta^{3/5}). \quad (5557)$$

Hence $\mathcal{M}^{(2)} \propto |u - u_g|^{3/5}$ for a linearly opening gap.

d. Theorem 136. Universal operational termination exponents at a generic gap closure. If the isolation gap closes linearly while the bandwidth, generalized dispersion scale, coherent DOS, and coherence remain finite, then $\mathcal{M} \propto \Delta^{1/2}$, whereas $\mathcal{M}^{(2)} \propto \Delta^{3/5}$. Thus gap-mediated branch birth or termination has a scaling law distinct from saddle-node creation. A second admissibility mechanism is loss of coherent bulk participation. Let $C = C_{\text{coh}} \rightarrow 0$. Suppose the raw normalized DOS remains finite,

$$\rho_{\text{raw}}^{\text{max}} \rightarrow \rho_c < \infty. \quad (5558)$$

Since $\rho_{\text{coh}}^{\text{max}} = C \rho_{\text{raw}}^{\text{max}}$, we have

$$X = E_{\text{kin}} \rho_{\text{coh}}^{\text{max}} = E_{\text{kin}} \rho_c C + o(C). \quad (5559)$$

Therefore,

$$\frac{X}{1 + X} = E_{\text{kin}} \rho_c C + o(C). \quad (5560)$$

Let

$$A_W = \frac{\Delta}{\Delta + W}, \quad A_V = \frac{\Delta}{\Delta + V}, \quad (5561)$$

with finite nonzero limits. Then

$$\mathcal{M}^4 = A_W A_V E_{\text{kin}} \rho_c C^2 + o(C^2). \quad (5562)$$

Hence

$$\mathcal{M} = [A_W A_V E_{\text{kin}} \rho_c]^{1/4} C^{1/2} + o(C^{1/2}). \quad (5563)$$

For the strong second-order functional, $\mathcal{M}^{(2)} \propto C^{2/5}$. If $C \propto |u - u_\ell|^{\beta_\ell}$, then

$$\mathcal{M} \propto |u - u_\ell|^{\beta_\ell/2}, \quad (5564)$$

and

$$\mathcal{M}^{(2)} \propto |u - u_\ell|^{2\beta_\ell/5}. \quad (5565)$$

e. Theorem 137. Localization termination of a magic branch. If a coherent spectral branch loses translation-space participation while its raw DOS remains finite, the operational magic score vanishes as $\mathcal{M} \propto (\mathcal{C}_{\text{coh}})^{1/2}$. Thus a narrow localized manifold cannot remain operationally magic merely because both its bandwidth and holonomy sensitivity are small. The physical reduced twist domain may be written as $\theta_{\min} \leq \theta \leq \theta_{\max}$. An interior branch can reach either endpoint without undergoing a saddle-node bifurcation. At the lower boundary, a local constrained maximum satisfies $F \leq 0$ at $\theta = \theta_{\min}$. At the upper boundary, the corresponding condition is $F \geq 0$ at $\theta = \theta_{\max}$. Thus the disappearance of an interior stationary solution at an angular boundary need not represent physical annihilation. The global optimum may continue as a boundary optimum.

E. Crystallographic termination

The operational magic branch is defined from the local or thermodynamic Hamiltonian. Exact crystallographic periodicity imposes the additional condition $g_\theta \in \text{Comm}_G(\Gamma)$. Therefore an exact magic point satisfies

$$[\Gamma : \Gamma \cap g_\theta \Gamma g_\theta^{-1}] < \infty. \quad (5566)$$

The exact coincidence index is $q_M = [\Gamma : \Gamma_M]$. The exact quotient genus is

$$g_M = 1 + q_M(g_0 - 1). \quad (5567)$$

The internal orbital number is $\nu_M = 2n_b q_M$. Thus a discontinuous change in q_M changes both the exact microscopic cell and the topology of generalized momentum space. A local operational branch can remain perfectly smooth across such a change, while an exact commensurate branch may cease to exist. We therefore distinguish operational branch termination, crystallographic branch termination. In particular, failure to find an exact commensurate continuation cannot by itself be interpreted as annihilation of the physical local magic state. Every exact hyperbolic branch carries a representation signature. We record

$$\mathfrak{R}_\nu = (q_M, g_M, d_\rho, \lambda_{\text{sym}}, r_C), \quad (5568)$$

where d_ρ is the translation-irrep rank, λ_{sym} denotes the residual point-group representation, and r_C is the target spectral-cluster rank. A branch is regarded as continuously identical only when its projector can be followed continuously within the corresponding representation sector. If the exact representation signature changes discontinuously, we classify the event as a representation-space reconstruction rather than ordinary motion of one

fixed band. At a smooth irreducible rank- d_ρ point, the representation-space dimension is

$$D_\rho = (2g_M - 2)d_\rho^2 + 2. \quad (5569)$$

Using

$$g_M = 1 + q_M(g_0 - 1), \quad (5570)$$

we obtain

$$D_\rho = 2q_M(g_0 - 1)d_\rho^2 + 2. \quad (5571)$$

Therefore a crystallographic jump $q_M^- \neq q_M^+$ produces

$$\Delta D_\rho = 2(g_0 - 1)d_\rho^2 \Delta q_M. \quad (5572)$$

Thus curvature can reorganize not only the energies but the number of independent generalized-momentum directions over which a magic condition must be satisfied. For an exponentially decaying hyperbolic hopping envelope,

$$|h(r)| \lesssim e^{-\mu_h r}, \quad (5573)$$

a sufficient absolute-summability condition is $\mu_h > \frac{1}{R}$. Since $\frac{1}{R} = \sqrt{|K|}$, the controlled long-range theory requires $|K| < \mu_h^2$. If this sufficient condition fails, we do not classify the event as physical termination of a magic branch. Instead, it marks the boundary of the present bounded-operator verification for that long-range hopping model. Finite-range hopping is not subject to this particular restriction. Let the optimized score of branch ν be

$$\mathcal{M}_\nu^*(K, \omega) = \mathcal{M}_\nu[\theta_\nu^*(K, \omega), K, \omega]. \quad (5574)$$

Define $S_\nu = \ln \mathcal{M}_\nu^*$. The global magic branch is $\nu_g = \arg \max_\nu S_\nu$. The globally preferred magic angle is therefore $\theta_g^* = \theta_{\nu_g}^*$. For two stable branches μ and ν , define the pairwise score difference

$$\Phi_{\mu\nu}(K, \omega) = S_\mu(K, \omega) - S_\nu(K, \omega). \quad (5575)$$

A hierarchy-switching wall satisfies $\Phi_{\mu\nu} = 0$. If $\partial_\omega \Phi_{\mu\nu} \neq 0$, the wall locally satisfies

$$\frac{d\omega_{\mu\nu}^{\text{wall}}}{dK} = -\frac{\partial_K \Phi_{\mu\nu}}{\partial_\omega \Phi_{\mu\nu}}. \quad (5576)$$

Because each score is already optimized over θ , the envelope theorem gives

$$\partial_K S_\nu = \partial_K \ln \mathcal{M}_\nu|_{\theta=\theta_\nu^*}, \quad (5577)$$

and

$$\partial_\omega S_\nu = \partial_\omega \ln \mathcal{M}_\nu|_{\theta=\theta_\nu^*}. \quad (5578)$$

Thus the hierarchy-wall slope is directly computable from the same spectral observables defining the magic functional. Suppose two branches satisfy $S_\mu = S_\nu$ at a hierarchy wall, while $\theta_\mu^* \neq \theta_\nu^*$. If $\Phi_{\mu\nu}$ changes sign across the wall, the globally optimal branch switches. The global magic score

$$\mathcal{M}_{\text{global}}^* = \max_\nu \mathcal{M}_\nu^* \quad (5579)$$

remains continuous at the crossing. However, the globally selected magic angle jumps by

$$\Delta\theta_{\mu\nu}^{\text{jump}} = \theta_\nu^* - \theta_\mu^*. \quad (5580)$$

Furthermore, if

$$\partial_K S_\mu \neq \partial_K S_\nu, \quad (5581)$$

the derivative of the optimized global score is discontinuous.

a. Theorem 138. First-order magic-hierarchy transition. When two simultaneously stable magic branches exchange global score ordering while remaining separated in angle, the optimized magic score is continuous but the identity and angle of the globally preferred magic state change discontinuously. The transition is characterized by $S_\mu = S_\nu$, $\theta_\mu^* \neq \theta_\nu^*$, and a sign change of $\Phi_{\mu\nu}$. This defines a magic-hierarchy transition without requiring any nonanalyticity of the microscopic finite-dimensional Hamiltonian. The cusp normal form gives an explicit realization of the preceding transition. For $U > 0$, and $V = 0$, two stable maxima occur at $x_\pm = \pm\sqrt{U}$. Their angle separation is $\Delta\theta = 2\sqrt{U}$. Their score difference changes sign linearly with V ,

$$S_+ - S_- = 2\sqrt{U}V + O(V^2). \quad (5582)$$

Therefore $V = 0$, $U > 0$ is a hierarchy-switching line. At $U = 0$, $V = 0$, the two maxima and the intervening minimum coalesce. The angle jump vanishes as $\Delta\theta \propto U^{1/2}$.

b. Theorem 139. Cusp termination of a first-order magic-hierarchy line. A generic cusp provides a critical endpoint at which a discontinuous switch between two globally competing magic branches terminates continuously. The branch-angle discontinuity obeys $\Delta\theta_{\text{hier}} \propto U^{1/2}$ as the cusp is approached. This supplies a direct

bifurcation-theoretic mechanism for curvature-driven restructuring of the complete magic hierarchy. Define $N_{\text{mag}}(K, \omega)$ as the number of admissible interior local maxima satisfying $F_\nu = 0$, $F_{\nu,\theta} < 0$, $\Delta_\nu > 0$, and $\mathcal{C}_\nu^{\text{coh}} > 0$. Away from bifurcation and admissibility boundaries, N_{mag} is locally constant. A generic saddle-node changes N_{mag} by one because one stable maximum is created or annihilated together with one unstable minimum. A gap-opening boundary can also change N_{mag} by one without a stationary-point bifurcation. A localization transition can remove an otherwise stationary branch from the admissible set. Thus the integer-valued function $N_{\text{mag}}(K, \omega)$ provides the simplest global classification of branch creation and annihilation.

c. Theorem 140. Piecewise-constant magic-branch number. For a smooth finite-dimensional or converged thermodynamic magic landscape, the number of admissible stable magic branches is constant on every connected region that contains no fold, cusp, gap-closing boundary, coherence boundary, or physical parameter boundary. Changes in N_{mag} therefore identify sharply defined bifurcation or admissibility loci in the $(K, w/t)$ plane. The no-bifurcation theorem assumed a fixed normalized tunneling harmonic and fixed dimensionless root. We now determine exactly what is required to violate it. The reduced magic equation is $\mathcal{G}(\theta, K, \omega) = 0$, where

$$\mathcal{G} = \text{arsinh}\left(\frac{b}{s}\right) - \frac{2\pi\zeta}{n_c} \sqrt{\frac{d\alpha_\nu^*(K, \theta)}{\gamma_1(K, \theta)\omega}}. \quad (5583)$$

Here

$$b = \sinh \zeta, \quad s = \sin\left(\frac{\theta}{2}\right). \quad (5584)$$

Let $c = \cos\left(\frac{\theta}{2}\right)$. The geometric derivative is

$$\frac{\partial}{\partial\theta} \text{arsinh}\left(\frac{b}{s}\right) = -\frac{bc}{2s\sqrt{s^2 + b^2}}. \quad (5585)$$

On the magic solution itself, the second term of Eq. (5583) equals

$$L = \text{arsinh}\left(\frac{b}{s}\right). \quad (5586)$$

Therefore,

$$\mathcal{G}_\theta = -\frac{bc}{2s\sqrt{s^2 + b^2}} - \frac{L}{2} [\partial_\theta \ln \alpha_\nu^* - \partial_\theta \ln \gamma_1]. \quad (5587)$$

A reduced scaling branch can develop a fold only when $\mathcal{G}_\theta = 0$. Hence,

$$\partial_\theta \ln\left(\frac{\gamma_1}{\alpha_\nu^*}\right) = \frac{bc}{s\sqrt{s^2 + b^2} \text{arsinh}(b/s)}. \quad (5588)$$

The right-hand side is strictly positive for $0 < \theta < \pi$. Thus a reduced-model fold requires sufficiently strong angular reconstruction of the effective tunneling-to-root ratio.

d. Theorem 141. Microscopic criterion for curvature-induced branch birth beyond pure geometry. The local registry map alone cannot generate an interior fold. Within the reduced dimensionless-root theory, branch creation or annihilation becomes possible only if

$$\frac{\gamma_1(K, \theta)}{\alpha_\nu^*(K, \theta)} \quad (5589)$$

develops sufficient angle dependence to satisfy

$$\partial_\theta \ln \left(\frac{\gamma_1}{\alpha_\nu^*} \right) = \frac{bc}{s\sqrt{s^2 + b^2} \operatorname{arsinh}(b/s)}. \quad (5590)$$

Therefore an observed branch fold directly measures spectral or tunneling reconstruction beyond simple curvature-induced remapping of the moiré length. Suppose two competing magic branches belong to inequivalent irreducible translation or point-group sectors, $\rho_\mu \not\sim \rho_\nu$. Translation symmetry forbids a matrix element coupling the two sectors. Thus their underlying spectral manifolds may cross as control parameters are varied without developing an avoided crossing from a symmetry-preserving perturbation. Consequently, a hierarchy wall satisfying $S_\mu = S_\nu$ can be symmetry protected. This gives a particularly clean mechanism for a discontinuous global magic-angle switch. If the two branches belong to the same representation sector, by contrast, hybridization can reconstruct the spectral clusters and convert an apparent branch crossing into an avoided branch exchange. Let the stable operational branch scores be ordered as

$$S_{\pi_1} \geq S_{\pi_2} \geq \dots \geq S_{\pi_{N_{\text{mag}}}}. \quad (5591)$$

The permutation

$$\Pi_{\text{mag}} = (\pi_1, \pi_2, \dots, \pi_{N_{\text{mag}}}) \quad (5592)$$

defines the complete magic hierarchy. A hierarchy restructuring occurs when either N_{mag} changes or Π_{mag} changes. Thus branch birth and score reordering are two distinct forms of hierarchy reconstruction. It is possible for N_{mag} to remain constant while Π_{mag} changes. This occurs when two pre-existing stable branches cross in optimized score. Therefore

$$\Delta N_{\text{mag}} = 0 \quad \text{but} \quad \Delta \Pi_{\text{mag}} \neq 0 \quad (5593)$$

defines a pure hierarchy transition. By contrast, $\Delta N_{\text{mag}} \neq 0$ identifies branch creation or termination.

The fixed-root geometric theory derived above preserves the ordering of branch angles. If $\alpha_\mu^* < \alpha_\nu^*$, then $\theta_\mu^*(K, \omega) > \theta_\nu^*(K, \omega)$ for every curvature on which both branches exist. Therefore a reversal of the angle ordering, $\theta_\mu^* - \theta_\nu^*$ changing sign, cannot arise from fixed-root registry remapping alone. It is another direct signature of curvature-dependent spectral-root or tunneling reconstruction. Suppose an operational branch remains smooth in score and angle while its wave-function character changes. Let $P_\nu(u)$ be its spectral projector. For two nearby control values, define

$$\mathcal{O}_\nu(u, \delta u) = \frac{1}{r_\nu} \operatorname{Tr} [P_\nu(u) P_\nu(u + \delta u)]. \quad (5594)$$

For a smooth branch, $\mathcal{O}_\nu = 1 - O(\delta u^2)$. A rapid exchange of spectral character produces a pronounced reduction in this overlap even when the optimized magic score remains smooth. We classify such an event as branch transmutation rather than branch annihilation. This distinction is particularly important near same-symmetry avoided crossings. The complete magic phase diagram in the (K, ω) plane therefore contains several distinct loci. A fold line satisfies $F = 0$, $F_\theta = 0$. A cusp satisfies $F = F_\theta = F_{\theta\theta} = 0$. A gap boundary satisfies $\Delta = 0$. A coherence boundary satisfies $C = 0$. A hierarchy wall satisfies $S_\mu = S_\nu$. An exact crystallographic locus additionally satisfies $g_\theta \in \operatorname{Comm}_G(\Gamma)$. These loci are not equivalent and must be plotted separately.

e. Theorem 142. Complete taxonomy of curvature-controlled magic restructuring. The magic landscape admits five mathematically distinct reconstruction events: regular branch shift, saddle-node birth or annihilation, gap- or coherence-mediated admissibility termination, crystallographic or representation-space termination and magic-hierarchy switching between coexisting branches. Only the second changes the local number of stationary extrema through a generic fold. Only the final mechanism can change the globally preferred magic branch without changing the total number of stable branches. For a numerically sampled magic surface, a candidate fold is located by solving simultaneously $F = 0$ and $F_\theta = 0$. The nondegeneracy conditions $|F_u| > \epsilon_{Fu}$ and $|F_{\theta\theta}| > \epsilon_{F\theta\theta}$ must remain larger than their numerical uncertainties. On the two-branch side, the measured separation must satisfy

$$(\Delta\theta)^2 = A_{\text{fold}} |u - u_c| + o(|u - u_c|), \quad (5595)$$

where

$$A_{\text{fold}} = \left| \frac{8F_u}{F_{\theta\theta}} \right|_c. \quad (5596)$$

This gives a direct test of the square-root bifurcation law without fitting the exponent freely. A candidate cusp

requires $F \simeq 0$, $F_\theta \simeq 0$, and $F_{\theta\theta} \simeq 0$, while $F_{\theta\theta\theta}$ remains nonzero and the control-unfolding determinant in Eq. (5537) remains finite. The two fold lines emerging from the cusp must obey

$$|V| = \frac{2}{3\sqrt{3}}U^{3/2} + o(U^{3/2}). \quad (5597)$$

The branch-angle jump on the hierarchy-switching line must obey

$$\Delta\theta_{\text{hier}} = 2\sqrt{U} + o(U^{1/2}). \quad (5598)$$

The simultaneous observation of the $3/2$ fold-line exponent and the $1/2$ branch-jump exponent provides a much stronger cusp identification than visual inspection of a three-dimensional plot. For every disappearing branch, we evaluate F , F_θ , Δ , C , q_M , ϵ_{tot} . A saddle-node termination satisfies $F \rightarrow 0$, $F_\theta \rightarrow 0$, while $\Delta > 0$, $C > 0$. A gap termination satisfies $\Delta \rightarrow 0^+$ without requiring $F_\theta \rightarrow 0$. A localization termination satisfies $C \rightarrow 0$ while the spectral stationary condition may remain regular. A crystallographic termination is accompanied by loss of the exact finite-index common subgroup but need not produce any singularity in local spectral observables. A numerical failure is identified when ϵ_{tot} becomes comparable with or larger than the reported branch-separation or gap scale. No disappearing finite-size feature is classified as branch annihilation until these alternatives have been excluded. Let $u_{c,N}$ be the apparent fold location on cover N . A bulk fold requires $u_{c,N} \rightarrow u_{c,\infty}$ as $r_{\text{inj}}^{(N)} \rightarrow \infty$. The local normal-form coefficients must also converge, $F_u^{(N)} \rightarrow F_u^{(\infty)}$, and $F_{\theta\theta}^{(N)} \rightarrow F_{\theta\theta}^{(\infty)}$. Therefore the predicted fold coefficient

$$A_{\text{fold}}^{(N)} = \left| \frac{8F_u^{(N)}}{F_{\theta\theta}^{(N)}} \right| \quad (5599)$$

must itself converge. The same bifurcation must be reproduced on inequivalent finite-cover families before it is interpreted as a bulk phenomenon. To quantify the extent of branch reordering, define the rank of branch ν by

$$r_\nu^{\text{mag}}(K, \omega) = 1 + \sum_{\mu \neq \nu} \mathbf{1}[S_\mu > S_\nu]. \quad (5600)$$

The globally optimal branch has $r_\nu^{\text{mag}} = 1$. For two parameter points $\mathbf{p}_1$ and $\mathbf{p}_2$, define

$$\mathcal{R}_{\text{hier}}(\mathbf{p}_1, \mathbf{p}_2) = \frac{1}{2} \sum_\nu |r_\nu^{\text{mag}}(\mathbf{p}_2) - r_\nu^{\text{mag}}(\mathbf{p}_1)|. \quad (5601)$$

A value $\mathcal{R}_{\text{hier}} = 0$ means the branch ordering is unchanged. A positive value measures the total rank displacement of the magic hierarchy. This discrete diagnostic is complementary to the continuous magic-angle shift.

Let a fold generate a stable branch for $u > u_c$ after orienting the local control coordinate appropriately. Define

$$\mathcal{B}_\nu = |\theta_\nu^* - \theta_c|. \quad (5602)$$

The fold law gives

$$\mathcal{B}_\nu = B_\nu(u - u_c)^{1/2} + o\left((u - u_c)^{1/2}\right), \quad (5603)$$

where

$$B_\nu = \sqrt{\left| \frac{2F_u}{F_{\theta\theta}} \right|_c}. \quad (5604)$$

Thus the displacement of the newly created magic branch from the critical angle behaves as an order parameter with universal exponent $1/2$. Suppose no corresponding stable branch exists at $K = 0$ within the same projector family and parameter interval. Suppose a stable branch appears at finite negative curvature through a fold satisfying $F = F_\theta = 0$. If the finite-cover extrapolation gives $|K_c| > 0$, and the branch persists for $|K| > |K_c|$ with converged positive gap and coherent participation, then the branch is not a shifted Euclidean magic angle. It is a curvature-born magic branch.

f. Theorem 143. Operational criterion for curvature-induced birth of a genuinely new magic branch. A magic branch is curvature born, rather than continuously shifted from a Euclidean branch, if

$$N_{\text{mag}}(K = 0) < N_{\text{mag}}(K < K_c), \quad (5605)$$

and the increase is generated by a converged finite-curvature fold,

$$F = 0, \quad F_\theta = 0, \quad F_K \neq 0, \quad F_{\theta\theta} \neq 0. \quad (5606)$$

Near its birth point,

$$|\theta^* - \theta_c| \propto |K - K_c|^{1/2}. \quad (5607)$$

Such a branch cannot be produced by the fixed-root hyperbolic registry remapping derived in VI.C and therefore directly demonstrates additional curvature-induced spectral reconstruction. For every magic branch we therefore record

$$\mathfrak{B}_\nu^{\text{magic}} = \left(\theta_\nu^*, \mathcal{M}_\nu^*, \kappa_\nu^\perp, W_\nu, \Delta_\nu, \Omega_\nu^{\text{max}}, \mathcal{C}_\nu^{\text{coh}}, q_{M,\nu}, d_{\rho,\nu}, r_\nu^{\text{mag}} \right). \quad (5608)$$

A branch creation event additionally records K_c , θ_c , F_K , $F_{\theta\theta}$. A cusp records

K_c , ω_c , θ_c , $F_{\theta\theta\theta}$, together with the control-unfolding determinant. A termination event records whether the terminating quantity is F_θ , Δ , C , q_M , or a physical parameter boundary. This prevents distinct physical mechanisms from being grouped under one generic statement that a magic branch has disappeared. The complete curvature-dependent magic landscape therefore has a hierarchical structure. A regular branch obeys $F = 0$, $F_\theta \neq 0$, and moves smoothly with curvature. A saddle-node satisfies $F = F_\theta = 0$, and creates or annihilates one stable magic maximum. A cusp satisfies $F = F_\theta = F_{\theta\theta} = 0$, and organizes a region in which two stable magic branches coexist. A gap termination satisfies $\Delta \rightarrow 0^+$, whereas a localization termination satisfies $\mathcal{C}_{\text{coh}} \rightarrow 0$. A hierarchy transition satisfies $S_\mu = S_\nu$ while both branches remain stable. The complete causal chain is therefore

$$\begin{aligned} K &\longrightarrow \{h_0(K), \tau(\theta, K), \Gamma_M(K, \theta)\} \longrightarrow \{W, \Delta, dE, \rho, \mathcal{C}_{\text{coh}}\} \\ &\longrightarrow \mathcal{M}(\theta, K, \omega) \longrightarrow \{F, F_\theta, F_{\theta\theta}\} \\ &\longrightarrow \{\text{shift, fold, cusp, termination, hierarchy switch}\}. \end{aligned} \quad (5609)$$

The decisive distinction is therefore no longer merely $\theta^*(0) \longrightarrow \theta^*(K)$. The stronger possibility is curvature $\longrightarrow$ reconstruction of the topology of the magic landscape itself. The most restrictive evidence for such a reconstruction would be a finite-curvature branch birth satisfying the universal square-root fold law, a cusp organizing two coexisting magic maxima, or a hierarchy wall across which the globally preferred magic angle jumps while both branches remain individually stable. Because pure hyperbolic remapping of a fixed Euclidean magic root cannot produce any of these interior bifurcations, their observation would isolate a genuinely spectral consequence of curvature beyond the already established geometric moiré crossover.

F. Geometric Mechanism of Magic-Band Formation

The preceding analysis establishes where curvature-dependent magic states may occur and how their branches may shift, bifurcate, or terminate. We now derive the microscopic mechanism responsible for their formation. Our objective is to close the complete causal chain $K \longrightarrow$ moiré scattering geometry $\longrightarrow$ interlayer hybridization $\longrightarrow$ self-energy reconstruction $\longrightarrow$ velocity suppression $\longrightarrow$ bandwidth suppression. The central quantity will be a dimensionless effective coupling

$$\alpha_{\text{eff}} = \alpha_{\text{eff}}\left(\theta, K, \frac{w}{t}\right), \quad (5610)$$

constructed from the actual moiré scattering scale and the actual geometry-dependent active interlayer coupling. The resulting generalized magic condition will take the form

$$\alpha_{\text{eff}}\left(\theta, K, \frac{w}{t}\right) = \alpha_\nu^*, \quad (5611)$$

when the dimensionless scattering profile belongs to a fixed universality class. We derive every ingredient entering Eq. (5611) explicitly. Let

$$K = -\frac{1}{R^2} \leq 0, \quad (5612)$$

and define

$$s = \left| \sin\left(\frac{\theta}{2}\right) \right|. \quad (5613)$$

For negative curvature, introduce

$$\zeta = \frac{a}{2R}, \quad b = \sinh \zeta. \quad (5614)$$

The exact hyperbolic local moiré length is

$$\xi_M = R \operatorname{arsinh}\left(\frac{b}{s}\right). \quad (5615)$$

Define the dimensionless moiré radius

$$L = \frac{\xi_M}{R} = \operatorname{arsinh}\left(\frac{b}{s}\right). \quad (5616)$$

Equation (5616) implies the exact identity

$$\sinh L = \frac{b}{s}. \quad (5617)$$

Hence,

$$s = \frac{b}{\sinh L}. \quad (5618)$$

We now parameterize the complete effective moiré registry disk by

$$r = u\xi_M, \quad 0 \leq u \leq 1. \quad (5619)$$

Thus, $\frac{r}{R} = uL$. The exact hyperbolic twist mismatch is

$$d_\theta(r) = 2R \operatorname{arsinh}\left[s \sinh\left(\frac{r}{R}\right)\right]. \quad (5620)$$

Substituting Eqs. (5618) and (5619) gives

$$d_M(u; L, \zeta) = 2R \operatorname{arsinh}\left[b \frac{\sinh(uL)}{\sinh L}\right]. \quad (5621)$$

This is the exact registry-mismatch profile across one effective moiré domain. At the twist center, $d_M(0; L, \zeta) = 0$. At the edge,

$$\begin{aligned} d_M(1; L, \zeta) &= 2R \operatorname{arsinh}(b) \\ &= 2R\zeta = a. \end{aligned} \quad (5622)$$

Thus the normalized coordinate u always maps the domain center to exact registry and the domain boundary to the microscopic mismatch threshold a . The radial hyperbolic area element is

$$dA = 2\pi R \sinh\left(\frac{r}{R}\right) dr. \quad (5623)$$

Using $r = RL u$, we have $dr = RL du$. Therefore,

$$dA = 2\pi R^2 L \sinh(uL) du. \quad (5624)$$

The total effective moiré area is

$$A_M^{\text{eff}} = 2\pi R^2 (\cosh L - 1). \quad (5625)$$

Consequently the normalized radial probability measure is

$$p_L(u) = \frac{L \sinh(uL)}{\cosh L - 1}, \quad 0 \leq u \leq 1. \quad (5626)$$

Indeed,

$$\int_0^1 p_L(u) du = \frac{L}{\cosh L - 1} \int_0^1 \sinh(uL) du = 1. \quad (5627)$$

Thus every radial geometric or tunneling observable f possesses the exact moiré-domain average

$$\langle f \rangle_M = \int_0^1 p_L(u) f(u) du. \quad (5628)$$

In the Euclidean limit, $R \rightarrow \infty$, we have $\zeta \rightarrow 0$, and

$$L = \frac{\zeta}{s} + O(\zeta^3). \quad (5629)$$

For small L ,

$$\sinh(uL) = uL + O(L^3), \quad (5630)$$

and

$$\sinh L = L + O(L^3). \quad (5631)$$

Therefore,

$$\frac{\sinh(uL)}{\sinh L} = u + O(L^2). \quad (5632)$$

Since $b = \zeta + O(\zeta^3)$, Eq. (5621) gives $d_M(u) \rightarrow au$. Likewise,

$$\cosh L - 1 = \frac{L^2}{2} + O(L^4), \quad (5633)$$

so that $p_L(u) \rightarrow 2u$. Thus the complete Euclidean rescaled moiré geometry is recovered: $d_M^{(E)}(u) = au$, with normalized disk measure $p_E(u) = 2u$.

a. Theorem 144. Exact rescaled moiré-scattering geometry. For a centered twisted bilayer of constant negative curvature, the complete radial registry geometry inside the effective moiré domain is specified by

$$d_M(u) = 2R \operatorname{arsinh} \left[\sinh\left(\frac{a}{2R}\right) \frac{\sinh(uL)}{\sinh L} \right], \quad (5634)$$

and

$$p_L(u) = \frac{L \sinh(uL)}{\cosh L - 1}, \quad (5635)$$

where

$$L = \operatorname{arsinh} \left[\frac{\sinh(a/2R)}{|\sin(\theta/2)|} \right]. \quad (5636)$$

The Euclidean functions

$$d_M^{(E)}(u) = au, \quad p_E(u) = 2u \quad (5637)$$

are recovered exactly as $R \rightarrow \infty$. We next take the fixed-curvature small-angle limit. For $s \rightarrow 0$ at fixed R ,

$$L = \operatorname{arsinh} \left(\frac{b}{s} \right) \rightarrow \infty. \quad (5638)$$

For every fixed $u < 1$, we have

$$\frac{\sinh(uL)}{\sinh L} \sim e^{-(1-u)L}. \quad (5639)$$

Therefore,

$$d_M(u) \rightarrow 0 \quad \text{for every fixed } u < 1. \quad (5640)$$

At the same time, $d_M(1) = a$. Thus the mismatch does not approach a linear profile. Instead it develops a boundary layer near $u = 1$. Introduce the boundary coordinate $y = L(1 - u)$. Then $u = 1 - \frac{y}{L}$. For fixed y and $L \rightarrow \infty$,

$$\frac{\sinh(uL)}{\sinh L} = \frac{\sinh(L-y)}{\sinh L} \longrightarrow e^{-y}. \quad (5641)$$

Hence the mismatch converges to

$$d_\infty(y) = 2R \operatorname{arsinh}(be^{-y}), \quad 0 \leq y < \infty. \quad (5642)$$

The normalized area measure transforms according to $du = -\frac{dy}{L}$. Furthermore,

$$p_L(u)du = \frac{\sinh(L-y)}{\cosh L - 1} dy. \quad (5643)$$

Therefore,

$$p_L(u)du \longrightarrow e^{-y}dy. \quad (5644)$$

The limiting measure is normalized,

$$\int_0^\infty e^{-y}dy = 1. \quad (5645)$$

Thus both the physical hyperbolic area and the registry mismatch become concentrated in an $O(L^{-1})$ layer in the normalized radial coordinate.

b. Theorem 145. Curvature-dominated moiré boundary-layer theorem. At fixed negative curvature and asymptotically small twist, the rescaled moiré domain develops a universal boundary layer. With $y = L(1-u)$, the exact limiting geometry is

$$d_\infty(y) = 2R \operatorname{arsinh}[\sinh(a/2R)e^{-y}], \quad (5646)$$

and

$$d\mu_\infty(y) = e^{-y}dy. \quad (5647)$$

Consequently, negative curvature converts the Euclidean smooth radial registry profile into a boundary-layer scattering geometry rather than merely rescaling its spatial period. The physical interlayer distance is

$$D_M(u) = \sqrt{h^2 + d_M^2(u)}. \quad (5648)$$

For a general smooth distance-dependent interlayer kernel, $T = T(D)$, the exact radial tunneling field is

$$\mathcal{T}_L(u) = T \left[\sqrt{h^2 + 4R^2 \operatorname{arsinh}^2 \left(b \frac{\sinh(uL)}{\sinh L} \right)} \right]. \quad (5649)$$

The domain-averaged tunneling amplitude is

$$\bar{T}_L = \int_0^1 p_L(u) \mathcal{T}_L(u) du. \quad (5650)$$

Define the nonuniform moiré tunneling field by

$$\delta \mathcal{T}_L(u) = \mathcal{T}_L(u) - \bar{T}_L. \quad (5651)$$

By construction,

$$\int_0^1 p_L(u) \delta \mathcal{T}_L(u) du = 0. \quad (5652)$$

The total moiré-scattering power is

$$W_{\text{sc}}^2(K, \theta) = \int_0^1 p_L(u) |\delta \mathcal{T}_L(u)|^2 du. \quad (5653)$$

The quantity W_{sc} has dimensions of energy. It measures the nonuniform part of the interlayer network and excludes the spatially uniform bonding-antibonding component. Using Eqs. (5646) and (5647),

$$\mathcal{T}_E(u) = T \left(\sqrt{h^2 + a^2 u^2} \right). \quad (5654)$$

The Euclidean average is

$$\bar{T}_E = \int_0^1 2u \mathcal{T}_E(u) du. \quad (5655)$$

Hence,

$$W_{\text{sc},E}^2 = \int_0^1 2u |\mathcal{T}_E(u) - \bar{T}_E|^2 du. \quad (5656)$$

Thus the scattering power has a finite and explicitly calculable Euclidean limit. Using the boundary-layer theorem,

$$\mathcal{T}_\infty(y) = T \left[\sqrt{h^2 + 4R^2 \operatorname{arsinh}^2(b e^{-y})} \right]. \quad (5657)$$

Its average is

$$\bar{T}_\infty = \int_0^\infty e^{-y} \mathcal{T}_\infty(y) dy. \quad (5658)$$

Therefore,

$$W_{\text{sc},\infty}^2 = \int_0^\infty e^{-y} |\mathcal{T}_\infty(y) - \bar{T}_\infty|^2 dy. \quad (5659)$$

The right-hand side is independent of the twist angle. Thus the nontrivial scattering amplitude remains finite in the curvature-dominated small-angle limit even though the magic period becomes parametrically large. For an exact Euclidean moiré cell, expand

$$\mathcal{T}(\mathbf{r}) = \sum_{\mathbf{G} \in \Lambda_M^*} T_{\mathbf{G}} e^{i\mathbf{G} \cdot \mathbf{r}}. \quad (5660)$$

The uniform component is $T_0 = \bar{T}$. Parseval's identity gives

$$\frac{1}{A_M} \int_{A_M} |\mathcal{T} - \bar{T}|^2 dA = \sum_{\mathbf{G} \neq 0} |T_{\mathbf{G}}|^2. \quad (5661)$$

Hence,

$$W_{\text{sc}}^2 = \sum_{\mathbf{G} \neq 0} |T_{\mathbf{G}}|^2 \quad (5662)$$

under the corresponding normalization. Thus W_{sc} is precisely the total power carried by nontrivial moiré scattering harmonics. For a finite normal hyperbolic quotient Q_N , let

$$\delta T(q) = T(q) - \bar{T}. \quad (5663)$$

Define the group Fourier coefficient

$$\widehat{\delta T}(\rho) = \sqrt{\frac{d_\rho}{|Q_N|}} \sum_{q \in Q_N} \delta T(q) \rho(q). \quad (5664)$$

Finite-group Plancherel theory gives

$$\frac{1}{|Q_N|} \sum_{q \in Q_N} |\delta T(q)|^2 = \sum_{\rho \neq 1} \left\| \widehat{\delta T}(\rho) \right\|_F^2. \quad (5665)$$

The trivial representation is absent because the average has been removed.

c. Theorem 146. Geometry-to-scattering Parseval identity. The variance of the curvature-dependent interlayer field over one moiré domain equals the total power carried by its nontrivial scattering harmonics. In Euclidean space these are nonzero reciprocal-lattice harmonics. On a finite hyperbolic quotient they are nontrivial group-representation components. Therefore registry geometry $\rightarrow W_{\text{sc}} \rightarrow$ nontrivial hybridization channel is an exact Parseval statement rather than a qualitative analogy. Not every nontrivial scattering harmonic contributes equally to a given target band. Let P_ν project onto the target low-energy spectral manifold. Let Q_ν^{act} project onto the dominant moiré-scattered states that hybridize with P_ν within the same exact symmetry sector. Define the nonuniform interlayer operator

$$\delta\tau = \tau - \tau_{\text{triv}}, \quad (5666)$$

where τ_{triv} is the trivial scattering component. The active hybridization matrix is

$$V_\nu = P_\nu \delta\tau Q_\nu^{\text{act}}. \quad (5667)$$

Let $r_\nu = \text{rank } P_\nu$. Let z_ν denote the number of symmetry-equivalent active scattering channels per target state. We define the active RMS hybridization scale by

$$w_\nu^{\text{eff}} = \left[\frac{1}{r_\nu z_\nu} \text{Tr}(V_\nu V_\nu^\dagger) \right]^{1/2}. \quad (5668)$$

For the Euclidean equal-amplitude first-star model, $z_\nu = 4$, and $w_\nu^{\text{eff}} = |T_{\mathbf{G}_1}|$. Thus the normalization reduces exactly to the first-star amplitude used in the Euclidean benchmark. Define the dimensionless active tunneling factor

$$\gamma_\nu^{\text{eff}}(K, \theta) = \frac{w_\nu^{\text{eff}}}{w}. \quad (5669)$$

The total scattering power and active scattering scale are related through

$$(w_\nu^{\text{eff}})^2 = \eta_\nu^{\text{act}} W_{\text{sc}}^2, \quad (5670)$$

where $0 \leq \eta_\nu^{\text{act}} \leq 1$ is the fraction of nontrivial scattering power entering the active resonant shell after the channel normalization has been fixed. The local inverse registry length is

$$q_M^{\text{loc}} = \frac{2\pi}{\xi_M} = \frac{2\pi}{RL}. \quad (5671)$$

This is not a global hyperbolic reciprocal vector. To connect continuously to the exact Euclidean square-lattice benchmark, we retain the exact Euclidean arithmetic normalization. For a Euclidean commensurate branch labeled by (m, n_c) ,

$$\Sigma = \frac{n_c^2}{d} s^{-2}. \quad (5672)$$

Its exact primitive moiré reciprocal magnitude is

$$G_M^{(E)} = \frac{2\pi}{a\sqrt{\Sigma}}. \quad (5673)$$

Since

$$q_M^{(E)} = \frac{4\pi s}{a}, \quad (5674)$$

one finds

$$\frac{G_M^{(E)}}{q_M^{(E)}} = \frac{\sqrt{d}}{2n_c}. \quad (5675)$$

We therefore define the arithmetic-normalized local scattering scale

$$Q_M^{\text{eff}}(K, \theta) = \frac{\sqrt{d}}{2n_c} q_M^{\text{loc}}(K, \theta). \quad (5676)$$

Explicitly,

$$Q_M^{\text{eff}} = \frac{\pi\sqrt{d}}{n_c R L} = \frac{\pi\sqrt{d}}{n_c R \operatorname{arsinh} \left[\frac{\sinh(a/2R)}{|\sin(\theta/2)|} \right]}. \quad (5677)$$

Its flat-space limit is

$$\lim_{R \rightarrow \infty} Q_M^{\text{eff}} = \frac{2\pi\sqrt{d}}{n_c a} s = \frac{2\pi}{a\sqrt{\Sigma}} = G_M^{(E)}. \quad (5678)$$

Thus Q_M^{eff} reproduces the exact Euclidean moiré reciprocal scale while remaining explicitly a local spectral scale in the hyperbolic system. Near the square-lattice band edge,

$$\varepsilon_0 + 4t = ta^2 k^2 + O(k^4 a^4). \quad (5679)$$

The natural geometry-controlled kinetic-energy scale is therefore

$$E_M^{\text{eff}} = ta^2 (Q_M^{\text{eff}})^2. \quad (5680)$$

Substitution of Eq. (5677) gives

$$E_M^{\text{eff}} = \frac{\pi^2 d t a^2}{n_c^2 R^2 L^2}. \quad (5681)$$

Using $\zeta = \frac{a}{2R}$, this becomes

$$E_M^{\text{eff}} = \frac{4\pi^2 d t}{n_c^2} \frac{\zeta^2}{L^2}. \quad (5682)$$

In the Euclidean limit,

$$L = \frac{\zeta}{s} + O(\zeta^3), \quad (5683)$$

and therefore

$$E_M^{\text{eff}} \longrightarrow \frac{4\pi^2 d t}{n_c^2} s^2 = \frac{4\pi^2 t}{\Sigma}. \quad (5684)$$

This is exactly the Euclidean moiré kinetic scale derived previously. We now combine the active hybridization scale with the kinetic scale. Define

$$\alpha_\nu^{\text{eff}}(K, \theta, \omega) = \frac{w_\nu^{\text{eff}}(K, \theta)}{E_M^{\text{eff}}(K, \theta)}. \quad (5685)$$

Using

$$w_\nu^{\text{eff}} = \gamma_\nu^{\text{eff}} w, \quad \omega = \frac{w}{t}, \quad (5686)$$

we obtain

$$\alpha_\nu^{\text{eff}} = \frac{\gamma_\nu^{\text{eff}} n_c^2}{4\pi^2 d} \omega \left(\frac{L}{\zeta} \right)^2. \quad (5687)$$

Restoring the geometric variables,

$$\alpha_\nu^{\text{eff}} = \frac{\gamma_\nu^{\text{eff}}(K, \theta) n_c^2}{4\pi^2 d} \frac{w}{t} \left\{ \frac{\operatorname{arsinh} \left[\frac{\sinh(a/2R)}{|\sin(\theta/2)|} \right]}{a/(2R)} \right\}^2. \quad (5688)$$

Equation (5688) is the explicit effective coupling that closes the geometry-to-spectrum mechanism.

d. Theorem 147. Microscopic curvature-dependent effective coupling. For a quadratic square-lattice band edge and an active moiré scattering shell normalized to the Euclidean benchmark, the hybridization-to-kinetic ratio is

$$\alpha_\nu^{\text{eff}} = \frac{\gamma_\nu^{\text{eff}} n_c^2}{4\pi^2 d} \frac{w}{t} \left(\frac{L}{\zeta} \right)^2, \quad (5689)$$

with

$$L = \operatorname{arsinh} \left[\frac{\sinh \zeta}{|\sin(\theta/2)|} \right]. \quad (5690)$$

Every factor is directly computable from the microscopic geometry and the projected tunneling matrix. No global hyperbolic reciprocal vector is required. For $\zeta \rightarrow 0$, we have

$$\frac{L}{\zeta} = \frac{1}{s} + O(\zeta^2). \quad (5691)$$

Hence

$$\alpha_\nu^{\text{eff}} = \frac{\gamma_{\nu,E}^{\text{eff}} n_c^2}{4\pi^2 d} \frac{w}{t} \csc^2 \left(\frac{\theta}{2} \right) + O(|K|a^2). \quad (5692)$$

For small twist,

$$\alpha_\nu^{\text{eff}} \sim \frac{\gamma_{\nu,E}^{\text{eff}} n_c^2}{\pi^2 d} \frac{w/t}{\theta^2}. \quad (5693)$$

Thus the Euclidean square-lattice result is recovered exactly. At fixed finite R and asymptotically small twist,

$$L = \ln \left[\frac{4 \sinh(a/2R)}{|\theta|} \right] + o(1). \quad (5694)$$

If the active scattering factor approaches a finite nonzero limit,

$$\gamma_\nu^{\text{eff}}(K, \theta) \longrightarrow \gamma_{\nu,\infty}^{\text{eff}}(K), \quad (5695)$$

then

$$\alpha_\nu^{\text{eff}} \sim \frac{\gamma_{\nu,\infty}^{\text{eff}} n_c^2 R^2}{\pi^2 d a^2} \frac{w}{t} \ln^2 \left[\frac{4 \sinh(a/2R)}{|\theta|} \right]. \quad (5696)$$

Thus the Euclidean divergence $\alpha_{\text{eff}} \propto |\theta|^{-2}$ is replaced by the curvature-dominated law

$$\alpha_{\text{eff}} \propto \ln^2 |\theta|^{-1}. \quad (5697)$$

Negative curvature therefore changes the functional scaling of the dimensionless hybridization strength itself.

e. Theorem 148. Curvature transmutation of the effective-coupling law. For the quadratic square-lattice reference, $\alpha_{\text{eff}} \sim \omega |\theta|^{-2}$ in the Euclidean small-angle regime, whereas

$$\alpha_{\text{eff}} \sim \omega \ln^2 |\theta|^{-1} \quad (5698)$$

at fixed negative curvature, provided the normalized active scattering amplitude has a finite nonzero asymptotic limit. This transmutation is the coupling-space origin of the algebraic-to-essential-singularity magic-curve crossover derived above. The finite limit assumed above follows naturally from the rescaled scattering geometry. For every bounded continuous tunneling observable $F(d)$, the boundary-layer theorem gives

$$\lim_{L \rightarrow \infty} \int_0^1 p_L(u) F[d_M(u)] du = \int_0^\infty e^{-y} F[2R \operatorname{arsinh}(be^{-y})] dy. \quad (5699)$$

Therefore both $\bar{T}_L$ and $W_{\text{sc},L}^2$ approach finite limits. If the active fraction satisfies

$$\eta_\nu^{\text{act}} \longrightarrow \eta_{\nu,\infty}^{\text{act}} > 0, \quad (5700)$$

then

$$w_\nu^{\text{eff}} \longrightarrow w_{\nu,\infty}^{\text{eff}} > 0. \quad (5701)$$

Hence

$$\gamma_\nu^{\text{eff}} \longrightarrow \gamma_{\nu,\infty}^{\text{eff}} > 0. \quad (5702)$$

The nonzero active coupling is therefore consistent with the fact that the fixed- r weak-twist local response begins at order θ^2 . The two limits probe different spatial scalings. A fixed physical radius satisfies $r = O(1)$, whereas a moiré-scale point satisfies $r = O(\xi_M)$. The latter must be used in a magic-band theory. For an exponential interlayer kernel,

$$T(D) = w \exp \left[-\frac{D-h}{\lambda_\perp} \right], \quad (5703)$$

and small lateral mismatch $d \ll h$, we have

$$D = h + \frac{d^2}{2h} + O(d^4), \quad (5704)$$

so that

$$\frac{T(D)}{w} = 1 - \frac{d^2}{2h\lambda_\perp} + O(d^4). \quad (5705)$$

To leading order,

$$\frac{W_{\text{sc}}}{w} = \frac{1}{2h\lambda_\perp} \sqrt{\operatorname{Var}_M(d^2)} + O(d^4). \quad (5706)$$

For the Euclidean rescaled profile,

$$d = au, \quad p(u) = 2u. \quad (5707)$$

Therefore,

$$\langle d^2 \rangle_E = a^2 \int_0^1 2u^3 du = \frac{a^2}{2}, \quad (5708)$$

and

$$\langle d^4 \rangle_E = a^4 \int_0^1 2u^5 du = \frac{a^4}{3}. \quad (5709)$$

Thus

$$\operatorname{Var}_E(d^2) = \frac{a^4}{12}. \quad (5710)$$

Hence

$$\frac{W_{\text{sc},E}}{w} = \frac{a^2}{4\sqrt{3}h\lambda_\perp} + O\left(\frac{a^4}{h^2\lambda_\perp^2}\right). \quad (5711)$$

Now consider the curvature-dominated boundary layer together with weak microscopic curvature, $\zeta \ll 1$. Then

$$d_\infty(y) = ae^{-y} + O(\zeta^2 a). \quad (5712)$$

Using

$$d\mu = e^{-y} dy, \quad (5713)$$

we find

$$\langle d^2 \rangle_\infty = a^2 \int_0^\infty e^{-3y} dy = \frac{a^2}{3}, \quad (5714)$$

and

$$\langle d^4 \rangle_\infty = a^4 \int_0^\infty e^{-5y} dy = \frac{a^4}{5}. \quad (5715)$$

Therefore,

$$\text{Var}_\infty(d^2) = \frac{4a^4}{45}. \quad (5716)$$

Hence,

$$\frac{W_{\text{sc},\infty}}{w} = \frac{a^2}{3\sqrt{5}h\lambda_\perp} + O(\zeta^2) + O\left(\frac{a^4}{h^2\lambda_\perp^2}\right). \quad (5717)$$

The ratio is

$$\frac{W_{\text{sc},\infty}}{W_{\text{sc},E}} = \frac{4\sqrt{3}}{3\sqrt{5}} + O(\zeta^2) \approx 1.033 + O(\zeta^2). \quad (5718)$$

Thus even in a weak-microscopic-curvature smooth-kernel limit, curvature changes the normalized scattering profile while leaving the active scattering power finite. The much stronger small-angle effect enters through the kinetic scale E_M^{eff} , which changes from an algebraic to a logarithmic twist dependence. Equation (5687) can be written as

$$\alpha_\nu^{\text{eff}} = \mathcal{A}_\nu \omega \gamma_\nu^{\text{eff}} \mathcal{G}_K, \quad (5719)$$

where

$$\mathcal{A}_\nu = \frac{n_c^2}{4\pi^2 d}, \quad \mathcal{G}_K(\theta) = \left(\frac{L}{\zeta}\right)^2. \quad (5720)$$

Thus the effective coupling contains three independent factors: arithmetic normalization, active tunneling amplitude, and curvature-controlled scattering geometry. The factorization makes the causal content explicit. At fixed twist and fixed w/t ,

$$d \ln \alpha_\nu^{\text{eff}} = d \ln \gamma_\nu^{\text{eff}} - 2d \ln Q_M^{\text{eff}}. \quad (5721)$$

Therefore,

$$\partial_{|K|} \ln \alpha_\nu^{\text{eff}} = \partial_{|K|} \ln w_\nu^{\text{eff}} - 2\partial_{|K|} \ln Q_M^{\text{eff}}. \quad (5722)$$

The first contribution measures curvature-induced hybridization reconstruction. The second measures curvature-induced reconstruction of the moiré scattering momentum and hence of the kinetic energy denominator. These two contributions have no reason to possess the same sign. The local reciprocal scale obeys

$$\frac{Q_M^{\text{eff}}(K, \theta)}{Q_M^{\text{eff}}(0, \theta)} = 1 + \frac{a^2 |K|}{24} \cot^2\left(\frac{\theta}{2}\right) + O(K^2 a^4). \quad (5723)$$

Therefore,

$$\partial_{|K|} \ln Q_M^{\text{eff}}|_{K=0} = \frac{a^2}{24} \cot^2\left(\frac{\theta}{2}\right). \quad (5724)$$

Define the active tunneling susceptibility by

$$\Gamma_\nu^{\text{hyb}} = \partial_{|K|} \ln w_\nu^{\text{eff}}|_{K=0}. \quad (5725)$$

Then

$$\partial_{|K|} \ln \alpha_\nu^{\text{eff}}|_{K=0} = \Gamma_\nu^{\text{hyb}} - \frac{a^2}{12} \cot^2\left(\frac{\theta}{2}\right). \quad (5726)$$

Thus the curvature response of the effective coupling is explicitly the competition between hybridization enhancement and geometric kinetic-scale enhancement.

f. Theorem 149. Exact causal decomposition of the curvature response. For a quadratic reference band,

$$\partial_{|K|} \ln \alpha_{\text{eff}} = \partial_{|K|} \ln w_{\text{eff}} - 2\partial_{|K|} \ln Q_M^{\text{eff}}. \quad (5727)$$

At zero curvature,

$$\partial_{|K|} \ln \alpha_{\text{eff}}|_0 = \Gamma_{\text{hyb}} - \frac{a^2}{12} \cot^2\left(\frac{\theta}{2}\right). \quad (5728)$$

This equation separates the two microscopic routes by which negative curvature moves a magic condition. We now show how α_{eff} enters the spectral reconstruction. Inside one fixed symmetry and generalized translation sector, decompose the Hilbert space into

$$\mathcal{H} = \mathcal{H}_P \oplus \mathcal{H}_Q, \quad (5729)$$

where P projects onto the target low-energy manifold and $Q = I - P$ contains the active scattered states. The Hamiltonian is

$$H = \begin{pmatrix} H_P & V \\ V^\dagger & H_Q \end{pmatrix}. \quad (5730)$$

We scale energies by E_M^{eff} . Define

$$\hat{H}_P = \frac{H_P - E_0}{E_M^{\text{eff}}}, \quad \hat{H}_Q = \frac{H_Q - E_0}{E_M^{\text{eff}}}, \quad (5731)$$

and

$$\hat{V} = \frac{V}{w_\nu^{\text{eff}}}. \quad (5732)$$

The dimensionless Hamiltonian is therefore

$$\hat{H} = \begin{pmatrix} \hat{H}_P & \alpha_\nu^{\text{eff}} \hat{V} \\ \alpha_\nu^{\text{eff}} \hat{V}^\dagger & \hat{H}_Q \end{pmatrix}. \quad (5733)$$

Thus, after the scattering geometry and active tunneling profile have been normalized, the entire hybridization strength enters through one scalar parameter, α_ν^{eff} . Residual dependence on curvature is contained only in dimensionless shape parameters of $\hat{H}_P$, $\hat{H}_Q$, $\hat{V}$. Let

$$H \begin{pmatrix} \psi_P \\ \psi_Q \end{pmatrix} = E \begin{pmatrix} \psi_P \\ \psi_Q \end{pmatrix}. \quad (5734)$$

The Q -sector equation gives

$$\psi_Q = (E - H_Q)^{-1} V^\dagger \psi_P, \quad (5735)$$

whenever the inverse exists. Substitution gives

$$[E - H_P - \Sigma_P(E)] \psi_P = 0, \quad (5736)$$

with exact self-energy

$$\Sigma_P(E) = V (E - H_Q)^{-1} V^\dagger. \quad (5737)$$

In scaled units,

$$\frac{\Sigma_P}{E_M^{\text{eff}}} = (\alpha_\nu^{\text{eff}})^2 \hat{V} \left(\varepsilon - \hat{H}_Q \right)^{-1} \hat{V}^\dagger, \quad (5738)$$

where

$$\varepsilon = \frac{E - E_0}{E_M^{\text{eff}}}. \quad (5739)$$

This equation shows directly that the leading virtual-hybridization reconstruction is governed by $(\alpha_\nu^{\text{eff}})^2$. Let X denote a physical generalized-momentum tangent direction. The exact derivative of the reconstructed eigenvalue can be written as

$$dE[X] = dE_{\text{bare}}[X] + dE_{\text{hyb}}[X]. \quad (5740)$$

Using the energy-dependent effective Hamiltonian,

$$dE[X] = Z_{\text{eff}} \langle \psi_P | [dH_P[X] + d\Sigma_P[X]] | \psi_P \rangle, \quad (5741)$$

where

$$Z_{\text{eff}} = [\langle \psi_P | (I - \partial_E \Sigma_P) | \psi_P \rangle]^{-1}. \quad (5742)$$

Since $Z_{\text{eff}} > 0$ for the Hermitian off-resonant problem, exact first-order flattening along X requires

$$\langle \psi_P | dH_P[X] | \psi_P \rangle + \langle \psi_P | d\Sigma_P[X] | \psi_P \rangle = 0. \quad (5743)$$

Thus the magic condition is a direct cancellation of the bare kinetic dispersion by the geometry-induced hybridization self-energy. Suppose the target reference point is stationary and its bare quadratic dispersion is characterized by the Hessian M_0 . Let G be the Hodge or ordinary momentum-space metric. Define the geometry-normalized bare Hessian

$$\mathbb{M}_0 = G^{-1/2} M_0 G^{-1/2}. \quad (5744)$$

The second-order self-energy generates

$$M_{\text{hyb}} = (\alpha_\nu^{\text{eff}})^2 M_{\text{hyb}}^{(2)} + O[(\alpha_\nu^{\text{eff}})^4]. \quad (5745)$$

Define

$$\mathbb{C}_\nu = -\mathbb{M}_0^{-1/2} G^{-1/2} M_{\text{hyb}}^{(2)} G^{-1/2} \mathbb{M}_0^{-1/2}. \quad (5746)$$

Then

$$\mathbb{M}_{\text{eff}} = \mathbb{M}_0^{1/2} \left[I - (\alpha_\nu^{\text{eff}})^2 \mathbb{C}_\nu + O(\alpha_{\text{eff}}^4) \right] \mathbb{M}_0^{1/2}. \quad (5747)$$

Thus the dimensionless matrix $\mathbb{C}_\nu$ contains the complete active scattering geometry relevant to second-order dispersion suppression. For a quadratic Euclidean reference band,

$$\varepsilon_0 = Ak^2, \quad A = ta^2, \quad (5748)$$

let the active scattering vectors be $\mathbf{Q}_a = Q_M^{\text{eff}} \mathbf{g}_a$. Write their amplitudes as $T_a = w_{\text{eff}} \tau_a$. The second-order energy correction is

$$\delta E^{(2)} = \sum_a \frac{|T_a|^2}{Ak^2 - A|\mathbf{k} + \mathbf{Q}_a|^2}. \quad (5749)$$

Expanding to quadratic order in $\mathbf{k}$ gives

$$\frac{\delta A_{\mu\nu}}{A} = -4 (\alpha_{\text{eff}})^2 \sum_a |\tau_a|^2 \frac{g_{a\mu} g_{a\nu}}{|\mathbf{g}_a|^6}. \quad (5750)$$

Define

$$C_{\mu\nu} = 4 \sum_a |\tau_a|^2 \frac{g_{a\mu} g_{a\nu}}{|\mathbf{g}_a|^6}. \quad (5751)$$

Then

$$A_{\mu\nu}^{\text{eff}} = A \left[\delta_{\mu\nu} - (\alpha_{\text{eff}})^2 C_{\mu\nu} \right] + O(\alpha_{\text{eff}}^4). \quad (5752)$$

For a C_4 -symmetric active shell, $C_{\mu\nu} = C_\nu \delta_{\mu\nu}$. Hence,

$$\frac{A_{\text{eff}}}{A} = 1 - C_\nu (\alpha_{\text{eff}})^2 + O(\alpha_{\text{eff}}^4). \quad (5753)$$

For the equal-amplitude four-vector first star, $C_\nu = 8$.

g. Theorem 150. Scattering-geometry control of the inverse effective mass. To second order in the active hybridization, the inverse effective-mass tensor is reduced by the positive scattering tensor

$$C_{\mu\nu} = 4 \sum_a |\tau_a|^2 \frac{g_{a\mu} g_{a\nu}}{|\mathbf{g}_a|^6}. \quad (5754)$$

For a symmetry-isotropic shell,

$$m_{\text{eff}}^{-1} = m_0^{-1} [1 - C_\nu \alpha_{\text{eff}}^2 + O(\alpha_{\text{eff}}^4)]. \quad (5755)$$

Thus the same geometric scattering data that determine the hybridization network determine the leading suppression of the kinetic dispersion. Within the symmetry-isotropic perturbative regime, the quadratic dispersion vanishes when

$$1 - C_\nu (\alpha_{\text{eff}})^2 = 0. \quad (5756)$$

Therefore,

$$\alpha_{\nu, \text{pc}}^* = \frac{1}{\sqrt{C_\nu}}. \quad (5757)$$

For the minimal square first-star model,

$$\alpha_{\text{pc}}^* = \frac{1}{\sqrt{8}}. \quad (5758)$$

As established in the Euclidean benchmark, this is a perturbative mass-cancellation scale and is not automatically the exact global magic coupling. The exact operational root must be obtained from the complete spectrum. The preceding perturbative calculation suggests a more general structure. Let the scaled Hamiltonian depend on α_ν^{eff} and a finite collection of dimensionless shape variables $\mathbf{s}_\nu = (s_{\nu,1}, s_{\nu,2}, \dots)$. Then

$$\frac{H_\nu - E_{\nu,0}}{E_M^{\text{eff}}} = \hat{H}_\nu(\alpha_\nu^{\text{eff}}, \mathbf{s}_\nu). \quad (5759)$$

Examples of the shape variables include

$$\frac{T_2}{T_1}, \quad \frac{T_3}{T_1}, \quad \eta_\nu^{\text{act}}, \quad \frac{h}{a}, \quad \frac{\lambda_\perp}{a}. \quad (5760)$$

If the curvature-twist family approaches a universal shape, $\mathbf{s}_\nu \rightarrow \mathbf{s}_\nu^{(0)}$, then every dimensionless spectral observable becomes a function of α_ν^{eff} alone. In particular,

$$\frac{W_\nu}{E_M^{\text{eff}}} = \mathcal{W}_\nu(\alpha_\nu^{\text{eff}}), \quad \frac{\Delta_\nu}{E_M^{\text{eff}}} = \mathcal{D}_\nu(\alpha_\nu^{\text{eff}}), \quad (5761)$$

and

$$\frac{\hbar D_G \Omega_\nu^{\text{max}}}{E_M^{\text{eff}}} = \mathcal{V}_\nu(\alpha_\nu^{\text{eff}}). \quad (5762)$$

Likewise,

$$E_M^{\text{eff}} \rho_\nu^{\text{coh,max}} = \mathcal{R}_\nu(\alpha_\nu^{\text{eff}}). \quad (5763)$$

The operational magic functional therefore reduces to

$$\mathcal{M}_\nu = \mathfrak{M}_\nu(\alpha_\nu^{\text{eff}}). \quad (5764)$$

Let α_ν^* be an isolated maximum of the universal dimensionless functional,

$$\left. \frac{d\mathfrak{M}_\nu}{d\alpha} \right|_{\alpha=\alpha_\nu^*} = 0, \quad (5765)$$

with

$$\left. \frac{d^2\mathfrak{M}_\nu}{d\alpha^2} \right|_{\alpha=\alpha_\nu^*} < 0. \quad (5766)$$

Then the complete curvature-dependent magic manifold satisfies

$$\alpha_\nu^{\text{eff}}\left(\theta, K, \frac{w}{t}\right) = \alpha_\nu^* \quad (5767)$$

Substitution of Eq. (5688) gives

$$\frac{\gamma_\nu^{\text{eff}}(K, \theta) n_c^2}{4\pi^2 d} \frac{w}{t} \left\{ \frac{\text{arsinh}\left[\frac{\sinh(a/2R)}{|\sin(\theta/2)|}\right]}{a/(2R)} \right\}^2 = \alpha_\nu^* \quad (5768)$$

This is the analytic generalized magic condition sought above.

h. Theorem 151. Effective-coupling magic criterion. If the normalized active scattering profile belongs to a fixed dimensionless universality class, the complete operational magic condition reduces to the single scalar equation $\alpha_\nu^{\text{eff}} = \alpha_\nu^*$. The left-hand side is determined microscopically by curvature, twist, interlayer coupling, and the active scattering power. The right-hand side is the dimensionless spectral root fixed by the normalized Hamiltonian. This separates geometry-dependent kinematics from the universal dimensionless flattening condition. If the normalized scattering profile itself changes significantly, the root becomes shape dependent, $\alpha_\nu^* = \alpha_\nu^*(\mathbf{s}_\nu)$. The exact corrected condition is then

$$\alpha_\nu^{\text{eff}}(K, \theta, \omega) = \alpha_\nu^*[\mathbf{s}_\nu(K, \theta)] \quad (5769)$$

Expanding around a reference shape,

$$\mathbf{s}_\nu = \mathbf{s}_\nu^{(0)} + \delta\mathbf{s}_\nu, \quad (5770)$$

gives

$$\alpha_\nu^*(\mathbf{s}_\nu) = \alpha_\nu^*(\mathbf{s}_\nu^{(0)}) + \sum_a \left. \frac{\partial \alpha_\nu^*}{\partial s_{\nu,a}} \right|_0 \delta s_{\nu,a} + O(\|\delta\mathbf{s}_\nu\|^2). \quad (5771)$$

Thus deviations from single-parameter collapse are themselves quantitatively measurable. Along the generalized magic curve,

$$\ln \alpha_\nu^{\text{eff}} - \ln \alpha_\nu^* = 0. \quad (5772)$$

Differentiating gives

$$\begin{aligned} 0 &= [\partial_\theta \ln \alpha_\nu^{\text{eff}} - \partial_\theta \ln \alpha_\nu^*] d\theta \\ &+ [\partial_{|K|} \ln \alpha_\nu^{\text{eff}} - \partial_{|K|} \ln \alpha_\nu^*] d|K| \\ &+ [\partial_{\ln \omega} \ln \alpha_\nu^{\text{eff}} - \partial_{\ln \omega} \ln \alpha_\nu^*] d \ln \omega. \end{aligned} \quad (5773)$$

At fixed ω ,

$$\frac{d\theta_\nu^*}{d|K|} = -\frac{\partial_{|K|} \ln \alpha_\nu^{\text{eff}} - \partial_{|K|} \ln \alpha_\nu^*}{\partial_\theta \ln \alpha_\nu^{\text{eff}} - \partial_\theta \ln \alpha_\nu^*}. \quad (5774)$$

If the dimensionless root is curvature independent, $\partial_{|K|} \alpha_\nu^* = 0$, and the active tunneling factor is weakly angle dependent, the formula reduces directly to the geometric shift derived in VI.C. From

$$\ln \alpha_\nu^{\text{eff}} = \ln \gamma_\nu^{\text{eff}} + 2 \ln L - 2 \ln \zeta + \text{constant}, \quad (5775)$$

we obtain

$$\partial_\theta \ln \alpha_\nu^{\text{eff}} = \partial_\theta \ln \gamma_\nu^{\text{eff}} + \frac{2}{L} \partial_\theta L. \quad (5776)$$

Since

$$\partial_\theta L = -\frac{bc}{2s\sqrt{s^2 + b^2}}, \quad (5777)$$

where $c = \cos\left(\frac{\theta}{2}\right)$, we obtain

$$\partial_\theta \ln \alpha_\nu^{\text{eff}} = \partial_\theta \ln \gamma_\nu^{\text{eff}} - \frac{bc}{s\sqrt{s^2 + b^2}L}. \quad (5778)$$

In the Euclidean limit,

$$\partial_\theta \ln \alpha_\nu^{\text{eff}} \longrightarrow \partial_\theta \ln \gamma_{\nu,E}^{\text{eff}} - \cot\left(\frac{\theta}{2}\right). \quad (5779)$$

At fixed curvature and asymptotically small angle,

$$\frac{d \ln \alpha_\nu^{\text{eff}}}{d \ln s} = \frac{d \ln \gamma_\nu^{\text{eff}}}{d \ln s} - \frac{2}{L} + o(L^{-1}). \quad (5780)$$

If γ_ν^{eff} approaches a constant,

$$\frac{d \ln \alpha_\nu^{\text{eff}}}{d \ln s} \sim -\frac{2}{\ln(1/s)} \longrightarrow 0. \quad (5781)$$

Thus negative curvature makes the effective coupling progressively less sensitive to fractional changes in the twist angle. Within the one-parameter scaling regime,

$$\frac{dE_\nu}{E_M^{\text{eff}}} = d\varepsilon_\nu(\alpha_{\text{eff}}). \quad (5782)$$

Define the normalized generalized-dispersion magnitude

$$\mathcal{V}_\nu(\alpha) = \frac{\hbar D_G \Omega_\nu^{\text{max}}}{E_M^{\text{eff}}}. \quad (5783)$$

At weak hybridization,

$$\mathcal{V}_\nu(\alpha) = \mathcal{V}_\nu^{(0)} - c_\nu^{(v)} \alpha^2 + O(\alpha^4). \quad (5784)$$

For a symmetry-locked quadratic shell, $c_\nu^{(v)} > 0$. Therefore, $\frac{d\mathcal{V}_\nu}{d\alpha} < 0$ initially. At the perturbative cancellation point, $\alpha = \frac{1}{\sqrt{C_\nu}}$, the quadratic contribution vanishes.

The complete nonperturbative operational root α_ν^* is determined by minimizing the full multidirectional dispersion together with the remaining spectral conditions. For a connected generalized momentum domain,

$$W_\nu \leq \hbar D_G \Omega_\nu^{\max}. \quad (5785)$$

Using Eq. (5783),

$$\frac{W_\nu}{E_M^{\text{eff}}} \leq \mathcal{V}_\nu(\alpha_{\text{eff}}). \quad (5786)$$

Therefore, $\mathcal{V}_\nu \rightarrow 0$ implies $W_\nu \rightarrow 0$. This proves the implication self-energy cancellation $\rightarrow$ complete velocity suppression $\rightarrow$ global bandwidth suppression. No separate qualitative assumption about band flattening is required.

i. Theorem 152. Effective-coupling mechanism of bandwidth suppression. Suppose a curvature-twist family admits a one-parameter dimensionless collapse controlled by α_{eff} . If

$$dE_{\text{bare}} + dE_{\text{hyb}} \rightarrow 0 \quad (5787)$$

over the complete physical generalized-momentum tangent space as $\alpha_{\text{eff}} \rightarrow \alpha_\nu^*$, then $\Omega_\nu^{\max} \rightarrow 0$ and hence $W_\nu \rightarrow 0$. Thus the effective-coupling criterion has a direct dynamical meaning: $\alpha_{\text{eff}} = \alpha_\nu^*$ is the point at which the hybridization self-energy cancels the bare generalized dispersion. For a normalized isolated band,

$$\|\rho_\nu\|_\infty W_\nu \geq 1. \quad (5788)$$

Therefore, $W_\nu \rightarrow 0$ forces $\|\rho_\nu\|_\infty \rightarrow \infty$ in the ideal unbroadened isolated-band limit. Thus the complete mechanism predicts

$$\begin{aligned} \alpha_{\text{eff}} \rightarrow \alpha_\nu^* \rightarrow dE \rightarrow 0 \rightarrow W \rightarrow 0 \\ \rightarrow \rho_{\max} \rightarrow \infty. \end{aligned} \quad (5789)$$

The DOS enhancement is therefore a consequence of coherent dispersion suppression rather than the defining cause of the magic state. A hyperbolic generalized momentum space can have dimension greater than two. Let $\{X_a\}_{a=1}^{D_\rho}$ be an orthonormal Hodge tangent basis. Define the bare dispersion covector

$$v_a^{(0)} = dE_{\text{bare}}[X_a], \quad (5790)$$

and the hybridization response

$$v_a^{(\text{hyb})} = dE_{\text{hyb}}[X_a]. \quad (5791)$$

In the weak-hybridization regime,

$$v_a^{(\text{hyb})} = -(\alpha_{\text{eff}})^2 \sum_b \mathbb{C}_{ab} v_b^{(0)} + O(\alpha_{\text{eff}}^4). \quad (5792)$$

Therefore,

$$\mathbf{v}^{\text{eff}} = [I - \alpha_{\text{eff}}^2 \mathbb{C}] \mathbf{v}^{(0)} + O(\alpha_{\text{eff}}^4). \quad (5793)$$

Let the eigenvalues of $\mathbb{C}$ be $c_1, c_2, \dots, c_{D_\rho}$. A principal direction becomes first-order soft when

$$\alpha_{\text{eff}} = \frac{1}{\sqrt{c_a}}. \quad (5794)$$

Complete simultaneous first-order cancellation requires the relevant eigenvalues to be locked, $c_1 = c_2 = \dots = c_{\text{act}}$ or requires higher-order reconstruction to align the separate directional zeros. Thus the representation-space theory gives a precise mathematical meaning to multidirectional magic-band formation.

j. Theorem 153. Generalized directional magic condition. The perturbative generalized dispersion obeys

$$\mathbf{v}^{\text{eff}} = [I - \alpha_{\text{eff}}^2 \mathbb{C}] \mathbf{v}^{(0)} + O(\alpha_{\text{eff}}^4). \quad (5795)$$

A common scalar magic coupling can suppress all active directions simultaneously only when the curvature-induced scattering geometry produces the required directional locking of the eigenvalues of $\mathbb{C}$. This requirement is absent from a one-dimensional spectral cut and is specific to the full generalized hyperbolic momentum problem. The mean-square hybridization control introduced previously replaces

$$\tau \rightarrow \hat{\tau} = \frac{W_\perp^{\text{ref}}}{W_\perp} \tau. \quad (5796)$$

This keeps $\text{Tr} \hat{\tau} \hat{\tau}^\dagger$ fixed. However, it does not generally keep $P_\nu \hat{\delta\tau} Q_\nu^{\text{act}}$ fixed. Define

$$\hat{w}_\nu^{\text{eff}} = \left[\frac{1}{r_\nu z_\nu} \text{Tr} \left(P_\nu \hat{\delta\tau} Q_\nu^{\text{act}} \hat{\delta\tau}^\dagger P_\nu \right) \right]^{1/2}. \quad (5797)$$

Then $\hat{w}_\nu^{\text{eff}}$ can still vary with curvature because the scattering power is redistributed among different representation and folded-momentum channels. The normalized effective coupling is

$$\hat{\alpha}_\nu^{\text{eff}} = \frac{\hat{w}_\nu^{\text{eff}}}{E_M^{\text{eff}}}. \quad (5798)$$

A curvature-controlled magic shift surviving this normalization therefore cannot be explained by a scalar change in total hybridization strength. It must arise from scattering-pattern redistribution and/or geometric kinetic-scale reconstruction. Consider a family of smooth tunneling kernels $T^{(\mu)}(D)$ satisfying

$$T^{(\mu)}(h) = w. \quad (5799)$$

Fix their logarithmic slope,

$$\zeta_\perp = -h \frac{T^{(\mu)'}(h)}{T^{(\mu)}(h)}. \quad (5800)$$

The weak local twist expansion is

$$\frac{T_{\text{loc}}^{(\mu)}(r) - w}{w} = -\frac{\zeta_\perp}{2h^2} J_K^2(r) \theta^2 + O(\theta^4), \quad (5801)$$

where

$$J_K(r) = R \sinh\left(\frac{r}{R}\right) \quad (5802)$$

for negative curvature. Thus all such kernels share the same leading geometry dependence. Differences in $T''(h)$, $T'''(h)$, ... enter at higher order. Therefore the leading curvature dependence of γ_ν^{eff} and hence of α_ν^{eff} defines a coupling-profile universality class controlled by the geometry and the common logarithmic slope.

k. Theorem 154. Leading effective-coupling universality. For smooth distance-dependent tunneling kernels with the same vertical amplitude and logarithmic slope, the leading curvature-twist reconstruction of the active effective coupling is identical up to higher-order profile corrections. Consequently, a magic curve that remains stable under variation of the higher derivatives of the tunneling profile is controlled primarily by the geometric effective coupling rather than by nonuniversal short-distance details. The construction is not restricted to a quadratic reference band. Suppose

$$\varepsilon(\mathbf{k}_0 + \mathbf{q}) - \varepsilon(\mathbf{k}_0) = c_p |\mathbf{q}|^p + o(|\mathbf{q}|^p), \quad (5803)$$

where $p > 0$. The geometric kinetic scale is

$$E_{M,p}^{\text{eff}} = c_p (Q_M^{\text{eff}})^p. \quad (5804)$$

The generalized effective coupling is

$$\alpha_{\text{eff}}^{(p)} = \frac{w_{\text{eff}}}{c_p (Q_M^{\text{eff}})^p}. \quad (5805)$$

Therefore,

$$d \ln \alpha_{\text{eff}}^{(p)} = d \ln w_{\text{eff}} - p d \ln Q_M^{\text{eff}}. \quad (5806)$$

In the Euclidean small-angle regime, $Q_M^{\text{eff}} = \Theta(\theta)$, so that

$$\alpha_{\text{eff}}^{(p)} = \Theta(\omega \theta^{-p}). \quad (5807)$$

At fixed negative curvature,

$$Q_M^{\text{eff}} = \Theta \left[\frac{1}{R \ln(1/|\theta|)} \right], \quad (5808)$$

and therefore

$$\alpha_{\text{eff}}^{(p)} = \Theta[\omega R^p \ln^p(1/|\theta|)] \quad (5809)$$

up to fixed microscopic length normalization and the active tunneling factor. If $\alpha_{\text{eff}}^{(p)} = \alpha_\nu^*$, then the Euclidean weak-coupling law is

$$\theta_\nu^* = \Theta(\omega^{1/p}). \quad (5810)$$

At fixed negative curvature, the curvature-dominated solution satisfies

$$\ln \left(\frac{1}{|\theta_\nu^*|} \right) = \Theta(\omega^{-1/p}). \quad (5811)$$

Hence,

$$\theta_\nu^* = \exp \left[-\Theta(\omega^{-1/p}) \right]. \quad (5812)$$

l. Theorem 155. Dispersion-order transmutation of the magic universality class. For a reference dispersion of order p , the geometry-controlled effective-coupling condition produces $\theta^* \sim \omega^{1/p}$ in Euclidean space and

$$\theta^* \sim \exp \left[-C \omega^{-1/p} \right] \quad (5813)$$

at fixed negative curvature in the curvature-dominated regime. Thus the algebraic-to-essential-singularity crossover is not peculiar to the quadratic square lattice. The underlying dispersion order determines the exponent inside the essential singularity. The effective coupling can be extracted without fitting a band structure. For every finite cover or exact Euclidean supercell, we first determine the target projector P_ν . We then construct the non-trivial interlayer operator

$$\delta\tau = \tau - \tau_{\text{triv}}. \quad (5814)$$

The active scattered subspace is identified from the symmetry-resolved states satisfying the selected energy-window and scattering-channel criteria, Q_ν^{act} . The active coupling is then measured directly as

$$w_\nu^{\text{eff}} = \left[\frac{1}{r_\nu z_\nu} \text{Tr} (P_\nu \delta \tau Q_\nu^{\text{act}} \delta \tau^\dagger P_\nu) \right]^{1/2}. \quad (5815)$$

The geometric scale is obtained independently from

$$Q_M^{\text{eff}} = \frac{\pi \sqrt{d}}{n_c R L}. \quad (5816)$$

Therefore,

$$\alpha_\nu^{\text{eff,num}} = \frac{w_\nu^{\text{eff}}}{ta^2 (Q_M^{\text{eff}})^2}. \quad (5817)$$

No magic-band spectral fit enters this definition. Consequently, the prediction $\alpha_\nu^{\text{eff}} = \alpha_\nu^*$ can be tested independently against the actual spectral location of the magic state. For every operational magic point $(K_j, \theta_{\nu,j}^*, \omega_j)$, compute

$$\alpha_{\nu,j}^{\text{eff}} = \alpha_\nu^{\text{eff}} (K_j, \theta_{\nu,j}^*, \omega_j). \quad (5818)$$

The one-parameter geometric mechanism predicts $\alpha_{\nu,j}^{\text{eff}} \rightarrow \alpha_\nu^*$ for all points belonging to the same magic branch. Define the normalized collapse residual

$$\mathcal{R}_{\alpha,\nu} = \frac{\sqrt{\langle (\alpha_{\nu,j}^{\text{eff}} - \bar{\alpha}_\nu^{\text{eff}})^2 \rangle}}{\bar{\alpha}_\nu^{\text{eff}}}. \quad (5819)$$

A successful one-parameter collapse requires $\mathcal{R}_{\alpha,\nu} \ll 1$. A systematic nonzero residual is then analyzed against the shape variables $\mathbf{s}_\nu$. At every magic point, compute separately dE_{bare} , and dE_{hyb} . Define the cancellation residual

$$\mathcal{R}_{\text{cancel}} = \frac{\|dE_{\text{bare}} + dE_{\text{hyb}}\|_{G^{-1}}}{\|dE_{\text{bare}}\|_{G^{-1}} + \|dE_{\text{hyb}}\|_{G^{-1}}}. \quad (5820)$$

It satisfies $0 \leq \mathcal{R}_{\text{cancel}} \leq 1$. Exact first-order cancellation gives $\mathcal{R}_{\text{cancel}} = 0$. A geometry-controlled magic mechanism requires the simultaneous behavior $\alpha_{\text{eff}} \rightarrow \alpha_\nu^*$, and $\mathcal{R}_{\text{cancel}} \rightarrow 0$. This test is substantially stronger than observing only a minimum in the bandwidth. Define the dimensionless bandwidth

$$\mathcal{W}_\nu = \frac{W_\nu}{E_M^{\text{eff}}}. \quad (5821)$$

The geometric effective-coupling theory predicts $\mathcal{W}_\nu = \mathcal{W}_\nu(\alpha_{\text{eff}})$ within a fixed profile class. Thus data obtained

at different K , θ , w/t must collapse onto one dimensionless bandwidth curve when plotted against α_{eff} . The minimum must occur at the same α_ν^* identified from the operational magic functional. Likewise,

$$\mathcal{V}_\nu = \frac{\hbar D_G \Omega_\nu^{\text{max}}}{E_M^{\text{eff}}} \quad (5822)$$

must collapse as $\mathcal{V}_\nu = \mathcal{V}_\nu(\alpha_{\text{eff}})$. The coherent DOS must satisfy

$$\mathcal{R}_\nu = E_M^{\text{eff}} \rho_\nu^{\text{coh,max}} = \mathcal{R}_\nu(\alpha_{\text{eff}}). \quad (5823)$$

At the same effective coupling, $\alpha_{\text{eff}} = \alpha_\nu^*$, we therefore expect $\mathcal{V}_\nu \rightarrow \text{minimum}$, $\mathcal{W}_\nu \rightarrow \text{minimum}$, and $\mathcal{R}_\nu \rightarrow \text{maximum}$, subject to the independent positive-gap and coherence requirements.

m. Theorem 156. Multi-observable effective-coupling collapse. If α_{eff} is the correct control variable of a magic branch, then independently calculated spectral observables must collapse according to

$$\frac{W}{E_M^{\text{eff}}} = \mathcal{W}(\alpha_{\text{eff}}), \quad \frac{\hbar D_G \Omega^{\text{max}}}{E_M^{\text{eff}}} = \mathcal{V}(\alpha_{\text{eff}}), \quad (5824)$$

and

$$E_M^{\text{eff}} \rho^{\text{coh,max}} = \mathcal{R}(\alpha_{\text{eff}}). \quad (5825)$$

Their extrema must coincide at the same dimensionless magic root within the numerical and profile-shape corrections. This constitutes a direct falsifiable test of the effective-coupling mechanism. The complete sequence can now be written without any qualitative gap. Negative curvature first modifies geodesic deviation,

$$J_K(r) = R \sinh\left(\frac{r}{R}\right). \quad (5826)$$

This modifies the exact registry field,

$$d_\theta(r) = 2R \text{arsinh}[\sinh(r/R) |\sin(\theta/2)|]. \quad (5827)$$

The registry field determines ξ_M , $p_L(u)$, $d_M(u)$, which determine the microscopic tunneling field $\mathcal{T}_L(u)$. Its nontrivial scattering power determines W_{sc} , and projection onto the active spectral shell gives w_ν^{eff} . Independently, the same registry geometry determines the effective scattering momentum Q_M^{eff} and the kinetic scale

$$E_M^{\text{eff}} = ta^2 (Q_M^{\text{eff}})^2. \quad (5828)$$

Their ratio gives

$$\alpha_\nu^{\text{eff}} = \frac{w_\nu^{\text{eff}}}{E_M^{\text{eff}}}. \quad (5829)$$

The active hybridization then enters the Schur self-energy as

$$\frac{\Sigma}{E_M^{\text{eff}}} = O \left[(\alpha_\nu^{\text{eff}})^2 \right]. \quad (5830)$$

The resulting energy differential is

$$dE = dE_{\text{bare}} + dE_{\text{hyb}}. \quad (5831)$$

At $\alpha_\nu^{\text{eff}} = \alpha_\nu^*$, the active self-energy suppresses the bare generalized dispersion,

$$dE_{\text{bare}} + dE_{\text{hyb}} \longrightarrow 0. \quad (5832)$$

Therefore, $\Omega_\nu^{\text{max}} \longrightarrow 0$, which implies $W_\nu \longrightarrow 0$, and hence produces coherent DOS enhancement.

n. Theorem 157. Geometric mechanism of curvature-controlled magic-band formation. For a twisted bilayer with distance-dependent microscopic tunneling, the geometry-controlled route to a magic band is

$$\begin{aligned} K < 0 &\longrightarrow \{d_M(u), p_L(u)\} \longrightarrow \text{nontrivial moiré scattering power} \\ &\longrightarrow w_\nu^{\text{eff}} \longrightarrow \alpha_\nu^{\text{eff}} = \frac{w_\nu^{\text{eff}}}{E_M^{\text{eff}}} \longrightarrow \Sigma_{\text{hyb}} \longrightarrow dE_{\text{bare}} + dE_{\text{hyb}} \\ &\longrightarrow \Omega_\nu^{\text{max}} \longrightarrow W_\nu. \end{aligned} \quad (5833)$$

When the normalized active scattering profile belongs to a fixed universality class, the full magic condition reduces to

$$\alpha_\nu^{\text{eff}} \left(\theta, K, \frac{w}{t} \right) = \alpha_\nu^*. \quad (5834)$$

For the quadratic square-lattice reference,

$$\alpha_\nu^{\text{eff}} = \frac{\gamma_\nu^{\text{eff}}(K, \theta) n_c^2 w}{4\pi^2 d} \frac{w}{t} \left\{ \frac{\text{arsinh} \left[\frac{\sinh(a/2R)}{|\sin(\theta/2)|} \right]}{a/(2R)} \right\}^2. \quad (5835)$$

This expression recovers the Euclidean law

$$\alpha_{\text{eff}} \propto \frac{w/t}{\theta^2} \quad (5836)$$

and crosses at fixed negative curvature to

$$\alpha_{\text{eff}} \propto \frac{w}{t} \ln^2 \left(\frac{1}{|\theta|} \right). \quad (5837)$$

Consequently the same dimensionless magic root that generates the Euclidean algebraic magic-angle law generates an essential-singularity magic curve in the curvature-dominated regime. The geometric mechanism is therefore not a phenomenological correlation between curvature and a numerically narrow band. It is a closed microscopic chain from curvature-controlled geodesic registry, through an explicitly measurable active scattering coupling, to self-energy cancellation of the generalized kinetic dispersion.

G. Scaling Collapse and Magic-Band Universality

The geometric mechanism derived above suggests a substantially stronger statement than the existence of curvature-dependent magic angles. If the curvature, twist, and interlayer coupling enter the low-energy spectral problem only through a properly normalized effective coupling, then spectra obtained at different microscopic parameter values should not be independent functions of three variables. After the correct geometric energy scale has been removed, they should collapse onto a common one-parameter family. We therefore seek a scaling structure of the form

$$W_\nu(\theta, K, \omega) = W_0(\theta, K) \Phi_{W,\nu}[X_\nu(\theta, K, \omega)], \quad (5838)$$

where $\omega = \frac{w}{t}$, W_0 is a geometry-controlled kinetic scale, X_ν is a dimensionless scaling variable, and $\Phi_{W,\nu}$ is independent of the separate values of K , θ , and w/t within one scaling universality class. The central objective is not to postulate such a collapse phenomenologically. We derive the unique natural scaling variable from the microscopic effective coupling obtained above and determine the conditions under which a one-parameter collapse follows rigorously from the Hamiltonian. For the ν -th continuously tracked spectral family, the microscopic effective coupling is

$$\alpha_\nu^{\text{eff}} = \frac{w_\nu^{\text{eff}}}{E_M^{\text{eff}}}. \quad (5839)$$

For the quadratic square-lattice reference,

$$E_M^{\text{eff}} = ta^2 (Q_M^{\text{eff}})^2, \quad (5840)$$

with

$$Q_M^{\text{eff}} = \frac{\pi\sqrt{d}}{n_c R L}, \quad (5841)$$

where

$$L = \text{arsinh} \left[\frac{\sinh \zeta}{s} \right], \quad \zeta = \frac{a}{2R}, \quad (5842)$$

and

$$s = \left| \sin \left(\frac{\theta}{2} \right) \right|. \quad (5843)$$

Using $w_\nu^{\text{eff}} = \gamma_\nu^{\text{eff}} w$, we obtained

$$\alpha_\nu^{\text{eff}} = \frac{\gamma_\nu^{\text{eff}} n_c^2}{4\pi^2 d} \omega \left(\frac{L}{\zeta} \right)^2. \quad (5844)$$

Let α_ν^* denote the dimensionless operational magic root of the corresponding normalized spectral family. We define the master scaling variable by

$$X_\nu = \frac{\alpha_\nu^{\text{eff}}}{\alpha_\nu^*}. \quad (5845)$$

Therefore,

$$X_\nu = \frac{\gamma_\nu^{\text{eff}} n_c^2}{4\pi^2 d \alpha_\nu^*} \omega \left(\frac{L}{\zeta} \right)^2. \quad (5846)$$

By construction, the magic condition is $X_\nu = 1$. Thus the complete curvature-dependent magic manifold becomes a level set of one dimensionless coordinate. For $R \rightarrow \infty$, we have

$$L = \frac{\zeta}{s} + O(\zeta^3). \quad (5847)$$

Hence,

$$\left(\frac{L}{\zeta} \right)^2 = \frac{1}{s^2} + O(\zeta^2). \quad (5848)$$

Therefore,

$$X_\nu^{(E)} = \frac{\gamma_{\nu,E}^{\text{eff}} n_c^2}{4\pi^2 d \alpha_\nu^*} \omega \csc^2 \left(\frac{\theta}{2} \right). \quad (5849)$$

For small twist,

$$X_\nu^{(E)} = \frac{\gamma_{\nu,E}^{\text{eff}} n_c^2}{\pi^2 d \alpha_\nu^*} \frac{\omega}{\theta^2} [1 + O(\theta^2)]. \quad (5850)$$

Thus the Euclidean magic condition is $X_\nu^{(E)} = 1$, which immediately yields the previously derived law $\theta_\nu^* \propto \omega^{1/2}$. We now take the simultaneous weak-curvature and small-angle limit,

$$\frac{a}{R} \rightarrow 0, \quad |\theta| \rightarrow 0, \quad (5851)$$

while keeping

$$u = \frac{R|\theta|}{a} \quad (5852)$$

fixed. Because

$$\zeta = \frac{a}{2R} = \frac{|\theta|}{2u}, \quad (5853)$$

and

$$L \rightarrow \text{arsinh} \left(\frac{1}{u} \right), \quad (5854)$$

we obtain

$$\left(\frac{L}{\zeta} \right)^2 \rightarrow \frac{4u^2}{\theta^2} \text{arsinh}^2 \left(\frac{1}{u} \right). \quad (5855)$$

Define the geometric crossover function

$$G(u) = u \text{arsinh} \left(\frac{1}{u} \right). \quad (5856)$$

Then

$$X_\nu \rightarrow \frac{\gamma_\nu^{\text{eff}} n_c^2}{\pi^2 d \alpha_\nu^*} \frac{\omega}{\theta^2} G^2(u). \quad (5857)$$

This is the unified scaling variable sought above.

a. Theorem 158. Geometry-to-effective-coupling locking. In the weak-curvature/small-angle double-scaling limit,

$$\frac{\xi_M^{(H)}}{\xi_M^{(E)}} \rightarrow G(u), \quad (5858)$$

while the effective coupling satisfies

$$\frac{\alpha_\nu^{\text{eff},H}}{\alpha_\nu^{\text{eff},E}} = \frac{\gamma_{\nu,H}^{\text{eff}}}{\gamma_{\nu,E}^{\text{eff}}} G^2(u). \quad (5859)$$

Under a fixed-active-hybridization normalization, $\gamma_{\nu,H}^{\text{eff}} = \gamma_{\nu,E}^{\text{eff}}$, and therefore

$$\frac{\alpha_\nu^{\text{eff},H}}{\alpha_\nu^{\text{eff},E}} = G^2(u). \quad (5860)$$

Thus the same parameter-free geometric crossover function that controls the real-space moiré length controls the spectral effective coupling after being squared. Define the Euclidean moiré-to-curvature ratio

$$x = \frac{\xi_M^{(E)}}{R}. \quad (5861)$$

In the continuum small-angle limit, $x = \frac{1}{u}$. Hence

$$G(u) = \frac{\operatorname{arsinh} x}{x}. \quad (5862)$$

The effective-coupling crossover factor may therefore be written as

$$\mathcal{G}_\alpha(x) = \left[\frac{\operatorname{arsinh} x}{x} \right]^2. \quad (5863)$$

Thus

$$X_\nu = X_\nu^{(E)} \frac{\gamma_\nu^{\text{eff}}}{\gamma_{\nu,E}^{\text{eff}}} \mathcal{G}_\alpha(x). \quad (5864)$$

The Euclidean endpoint is $\mathcal{G}_\alpha(0) = 1$. For $x \gg 1$, we have

$$\mathcal{G}_\alpha(x) \sim \frac{\ln^2(2x)}{x^2}. \quad (5865)$$

The effective moiré energy is

$$E_M^{\text{eff}} = \frac{4\pi^2 d}{n_c^2} \frac{\zeta^2}{L^2}. \quad (5866)$$

In the double-scaling limit,

$$\frac{E_M^{\text{eff}}}{t} \longrightarrow \frac{\pi^2 d}{n_c^2} \frac{\theta^2}{G^2(u)}. \quad (5867)$$

The Euclidean reference is

$$\frac{E_M^{(E)}}{t} = \frac{\pi^2 d}{n_c^2} \theta^2 + O(\theta^4). \quad (5868)$$

Therefore,

$$\frac{E_M^{\text{eff}}}{E_M^{(E)}} \longrightarrow \frac{1}{G^2(u)}. \quad (5869)$$

Equations (5860) and (5869) are reciprocal statements. Curvature decreases the local moiré length, increases the corresponding kinetic energy scale, and therefore reduces the effective hybridization ratio at fixed active tunneling strength. The geometric crossover function satisfies $G(u) = uA(u)$, where

$$A(u) = \operatorname{arsinh} \left(\frac{1}{u} \right). \quad (5870)$$

Differentiation gives

$$\frac{dA}{du} = -\frac{1}{u\sqrt{1+u^2}}. \quad (5871)$$

Hence

$$\frac{d \ln G}{d \ln u} = 1 - \frac{1}{\sqrt{1+u^2} \operatorname{arsinh}(1/u)}. \quad (5872)$$

Define the running local-length exponent

$$\nu_\xi(u) = \frac{1}{\sqrt{1+u^2} \operatorname{arsinh}(1/u)}. \quad (5873)$$

Therefore,

$$\frac{d \ln G}{d \ln u} = 1 - \nu_\xi(u). \quad (5874)$$

At fixed R ,

$$\frac{d \ln u}{d \ln |\theta|} = 1. \quad (5875)$$

If the active tunneling factor is twist independent at leading order,

$$\frac{d \ln \gamma_\nu^{\text{eff}}}{d \ln |\theta|} = 0, \quad (5876)$$

then Eq. (5857) yields

$$\frac{d \ln X_\nu}{d \ln |\theta|} = -2\nu_\xi(u). \quad (5877)$$

Define the positive running effective-coupling exponent by

$$\beta_X = -\frac{d \ln X_\nu}{d \ln |\theta|}. \quad (5878)$$

Then $\beta_X(u) = 2\nu_\xi(u)$. For $u \gg 1$, we have $\nu_\xi \rightarrow 1$, and therefore $\beta_X \rightarrow 2$. For $u \ll 1$, we have

$$\nu_\xi \sim \frac{1}{\ln(2/u)}, \quad (5879)$$

so that

$$\beta_X \sim \frac{2}{\ln(2/u)} \rightarrow 0. \quad (5880)$$

b. Theorem 159. Universal flow of the effective-coupling exponent. Under fixed normalized active tunneling, the running twist exponent of the dimensionless effective coupling is exactly twice the running local-moiré length exponent, $\beta_X = 2\nu_\xi$. Thus the effective-coupling law flows continuously from $X \propto |\theta|^{-2}$ in Euclidean geometry to an asymptotically logarithmic dependence in negative curvature. This relation directly locks geometric scaling to the spectral control variable. At fixed θ ,

$$u = \frac{R|\theta|}{a} \propto |K|^{-1/2}. \quad (5881)$$

Therefore,

$$\frac{d \ln u}{d \ln |K|} = -\frac{1}{2}. \quad (5882)$$

Using Eq. (5874),

$$\frac{d \ln G^2}{d \ln |K|} = -(1 - \nu_\xi). \quad (5883)$$

Hence

$$\frac{d \ln X_\nu}{d \ln |K|} = \frac{d \ln \gamma_\nu^{\text{eff}}}{d \ln |K|} + \nu_\xi - 1. \quad (5884)$$

Under fixed active hybridization,

$$\frac{d \ln X_\nu}{d \ln |K|} = \nu_\xi - 1 \leq 0. \quad (5885)$$

Thus the geometry-only effect of increasing negative curvature is to reduce the effective coupling at fixed twist. Let $\mathbf{p} = (\theta, K, \omega)$ denote a microscopic parameter point. For one continuously tracked spectral family, subtract an arbitrary common energy origin $E_0(\mathbf{p})$ and divide by the geometry-controlled kinetic scale. Define

$$\hat{H}_\nu(\mathbf{p}) = \frac{H_\nu(\mathbf{p}) - E_0(\mathbf{p})I}{E_M^{\text{eff}}(\mathbf{p})}. \quad (5886)$$

A strict one-parameter scaling regime exists if there is a limiting $\hat{H}_\nu^*(X)$ dimensionless Hamiltonian such that

$$\hat{H}_\nu(\mathbf{p}) = \hat{H}_\nu^*[X_\nu(\mathbf{p})] + \mathcal{R}_\nu(\mathbf{p}), \quad (5887)$$

where

$$\|\mathcal{R}_\nu(\mathbf{p})\| \longrightarrow 0 \quad (5888)$$

in the scaling limit. The remainder contains lattice corrections, variation of the normalized tunneling shape,

finite-cover corrections, and any other dimensionless fields not absorbed into X_ν . If $\mathcal{R}_\nu = 0$, then every normalized eigenvalue satisfies

$$\frac{E_j(\mathbf{p}) - E_0(\mathbf{p})}{E_M^{\text{eff}}(\mathbf{p})} = \varepsilon_{j,\nu}[X_\nu(\mathbf{p})]. \quad (5889)$$

Consequently, every energy difference has the form

$$E_j - E_k = E_M^{\text{eff}}[\varepsilon_{j,\nu}(X) - \varepsilon_{k,\nu}(X)]. \quad (5890)$$

In particular, the bandwidth satisfies

$$W_\nu = E_M^{\text{eff}}\Phi_{W,\nu}(X_\nu), \quad (5891)$$

where

$$\Phi_{W,\nu}(X) = \varepsilon_\nu^{\text{max}}(X) - \varepsilon_\nu^{\text{min}}(X). \quad (5892)$$

Likewise,

$$\Delta_\nu = E_M^{\text{eff}}\Phi_{\Delta,\nu}(X_\nu), \quad (5893)$$

and

$$\hbar D_G \Omega_\nu^{\text{max}} = E_M^{\text{eff}}\Phi_{V,\nu}(X_\nu). \quad (5894)$$

c. Theorem 160. One-parameter spectral-collapse theorem. If the normalized Hamiltonian converges to a one-parameter family $\hat{H}_\nu^*(X)$, then every homogeneous spectral observable of energy degree one possesses the scaling form

$$\mathcal{O}_E(\theta, K, \omega) = E_M^{\text{eff}}\Phi_{\mathcal{O},\nu}(X_\nu). \quad (5895)$$

Thus the bandwidth, spectral gap, and generalized dispersion obtained from different values of K , θ , and w/t must collapse onto single-variable functions when expressed in the correct units. For a normalized cluster DOS,

$$\int \rho_\nu(E) dE = 1. \quad (5896)$$

Introduce the scaled energy coordinate

$$\epsilon = \frac{E - E_0}{E_M^{\text{eff}}}. \quad (5897)$$

Normalization requires

$$\rho_\nu(E) dE = \hat{\rho}_\nu(\epsilon) d\epsilon. \quad (5898)$$

Since

$$dE = E_M^{\text{eff}} d\epsilon, \quad (5899)$$

we obtain

$$\rho_\nu(E; \mathbf{p}) = \frac{1}{E_M^{\text{eff}}(\mathbf{p})} \Phi_{\rho, \nu}[\epsilon, X_\nu(\mathbf{p})]. \quad (5900)$$

Consequently,

$$E_M^{\text{eff}} \rho_\nu^{\text{max}} = \Phi_{\rho, \nu}^{\text{max}}(X_\nu). \quad (5901)$$

Thus DOS enhancement and bandwidth suppression are predicted to collapse under the same horizontal scaling coordinate X_ν . Substituting Eq. (5867) into Eq. (5891) gives

$$\frac{W_\nu}{t} = \frac{\pi^2 d}{n_c^2} \frac{\theta^2}{G^2(u)} \Phi_{W, \nu}(X_\nu). \quad (5902)$$

Using Eq. (5857),

$$\frac{W_\nu}{t} = \frac{\pi^2 d}{n_c^2} \frac{\theta^2}{G^2(u)} \times \Phi_{W, \nu} \left[\frac{\gamma_\nu^{\text{eff}} n_c^2 \omega}{\pi^2 d \alpha_\nu^* \theta^2} G^2(u) \right]. \quad (5903)$$

This is the explicit curvature-controlled bandwidth scaling law. No independent curvature-dependent fitting parameter appears once γ_ν^{eff} has been measured from the microscopic tunneling matrix. Define the Euclidean scaling coordinate

$$X_{\nu, E} = \frac{\gamma_{\nu, E}^{\text{eff}} n_c^2 \omega}{\pi^2 d \alpha_\nu^* \theta^2}. \quad (5904)$$

Define

$$\Gamma_\nu(K, \theta) = \frac{\gamma_\nu^{\text{eff}}(K, \theta)}{\gamma_{\nu, E}^{\text{eff}}}. \quad (5905)$$

Then

$$X_\nu = X_{\nu, E} \Gamma_\nu G^2(u). \quad (5906)$$

Likewise,

$$E_M^{\text{eff}} = \frac{E_M^{(E)}}{G^2(u)}. \quad (5907)$$

Therefore,

$$\frac{W_\nu}{E_M^{(E)}} = \frac{1}{G^2(u)} \Phi_{W, \nu}[X_{\nu, E} \Gamma_\nu G^2(u)]. \quad (5908)$$

Under the fixed-active-hybridization control, $\Gamma_\nu = 1$, and hence

$$\frac{W_\nu}{E_M^{(E)}} = \frac{1}{G^2(u)} \Phi_{W, \nu}[X_{\nu, E} G^2(u)]. \quad (5909)$$

d. Theorem 161. Horizontal-vertical geometric collapse. Within a fixed normalized active-scattering class, negative curvature acts on the Euclidean bandwidth scaling curve through two locked transformations:

$$X_E \longrightarrow X_E G^2(u), \quad (5910)$$

and

$$\frac{W}{E_M^{(E)}} \longrightarrow G^{-2}(u) \Phi_W[X_E G^2(u)]. \quad (5911)$$

Equivalently, the rescaled data

$$Y_W = G^2(u) \frac{W}{E_M^{(E)}} \quad (5912)$$

must satisfy

$$Y_W = \Phi_W(X), \quad X = X_E G^2(u). \quad (5913)$$

Thus the same geometric crossover function provides both the horizontal and vertical rescaling required for a universal spectral collapse. For $u \gg 1$, we have

$$G(u) = 1 - \frac{1}{6u^2} + O(u^{-4}). \quad (5914)$$

Therefore,

$$\frac{W_\nu}{t} \sim \frac{\pi^2 d}{n_c^2} \theta^2 \Phi_{W, \nu} \left[C_\nu \frac{\omega}{\theta^2} \right], \quad (5915)$$

where

$$C_\nu = \frac{\gamma_{\nu, E}^{\text{eff}} n_c^2}{\pi^2 d \alpha_\nu^*}. \quad (5916)$$

This is the Euclidean square-lattice scaling form. For $u \ll 1$, we have

$$G(u) = u \ln \left(\frac{2}{u} \right) + O \left[u^3 \ln \left(\frac{1}{u} \right) \right]. \quad (5917)$$

Because

$$u = \frac{R|\theta|}{a}, \quad (5918)$$

we obtain

$$\frac{\theta^2}{G^2(u)} \sim \frac{a^2}{R^2} \frac{1}{\ln^2 \left(\frac{2a}{R|\theta|} \right)}. \quad (5919)$$

Furthermore,

$$X_\nu \sim C_\nu \Gamma_\nu \omega \frac{R^2}{a^2} \ln^2 \left(\frac{2a}{R|\theta|} \right). \quad (5920)$$

Therefore,

$$\begin{aligned} \frac{W_\nu}{t} &\sim \frac{\pi^2 d}{n_c^2} \frac{a^2}{R^2} \frac{1}{\ln^2 \left(\frac{2a}{R|\theta|} \right)} \\ &\times \Phi_{W,\nu} \left[C_\nu \Gamma_\nu \omega \frac{R^2}{a^2} \ln^2 \left(\frac{2a}{R|\theta|} \right) \right]. \end{aligned} \quad (5921)$$

Thus the Euclidean and curvature-dominated bandwidth laws are two asymptotic branches of the same master function. The full universal function $\Phi_{W,\nu}(X)$ must be obtained from the complete spectral problem. Nevertheless, the perturbative equal-amplitude first-star model provides an analytic benchmark. For a C_4 -symmetric quadratic band edge,

$$\frac{A_{\text{eff}}}{A} = 1 - C_\nu \alpha_{\text{eff}}^2 + O(\alpha_{\text{eff}}^4). \quad (5922)$$

The perturbative cancellation coupling is

$$\alpha_{\text{pc}} = \frac{1}{\sqrt{C_\nu}}. \quad (5923)$$

Define

$$X_{\text{pc}} = \frac{\alpha_{\text{eff}}}{\alpha_{\text{pc}}}. \quad (5924)$$

Then

$$\frac{A_{\text{eff}}}{A} = 1 - X_{\text{pc}}^2 + O(X_{\text{pc}}^4). \quad (5925)$$

Within the quadratic mBZ approximation,

$$W = \frac{E_M^{\text{eff}}}{2} \left| \frac{A_{\text{eff}}}{A} \right|. \quad (5926)$$

Therefore,

$$\Phi_W^{(\text{pc})}(X_{\text{pc}}) = \frac{1}{2} |1 - X_{\text{pc}}^2|. \quad (5927)$$

The normalized velocity satisfies

$$\Phi_V^{(\text{pc})}(X_{\text{pc}}) = |1 - X_{\text{pc}}^2|. \quad (5928)$$

The local quadratic DOS enhancement obeys

$$\frac{\rho}{\rho_0} = \frac{1}{|1 - X_{\text{pc}}^2|}. \quad (5929)$$

Thus,

$$2\Phi_W^{(\text{pc})} = \Phi_V^{(\text{pc})} = \left(\frac{\rho}{\rho_0} \right)^{-1}. \quad (5930)$$

e. Theorem 162. Perturbative analytic magic-band master curve. For the equal-amplitude C_4 -symmetric active shell, the leading normalized bandwidth, velocity, and local DOS are controlled by one analytic function, $|1 - X_{\text{pc}}^2|$. Hence the weak-coupling theory predicts a parameter-free relation among three independently measurable observables. The complete nonperturbative scaling function need not retain this exact form, but its weak-hybridization branch must reproduce the corresponding perturbative expansion. Let

$$X_{\text{pc}} = 1 + \delta X, \quad |\delta X| \ll 1. \quad (5931)$$

Then

$$1 - X_{\text{pc}}^2 = -2\delta X - \delta X^2. \quad (5932)$$

Therefore,

$$\frac{W}{E_M^{\text{eff}}} = |\delta X| + O(\delta X^2). \quad (5933)$$

Likewise,

$$\frac{v}{v_0} = 2|\delta X| + O(\delta X^2), \quad (5934)$$

whereas

$$\frac{\rho}{\rho_0} = \frac{1}{2|\delta X|} + O(1). \quad (5935)$$

Thus the perturbative continuum theory predicts the critical exponents $W \propto |X - 1|$, and $\rho \propto |X - 1|^{-1}$. These singular laws are rounded by higher-order lattice dispersion, additional scattering shells, finite broadening, and any nonzero residual continuum bandwidth. The exact normalized bandwidth need not vanish at the operational magic root. We therefore write the general local expansion as

$$\Phi_{W,\nu}(X) = \Phi_{W,\nu}^* + A_\nu |X - 1|^{p_\nu} + o(|X - 1|^{p_\nu}), \quad (5936)$$

where

$$\Phi_{W,\nu}^* = \Phi_{W,\nu}(1) \geq 0. \quad (5937)$$

A smooth nondegenerate minimum has $p_\nu = 2$. A cusp generated by switching spectral extrema can give $p_\nu = 1$. The perturbative mass-cancellation benchmark belongs to the latter class. Therefore the near-magic exponent is itself an observable property of the master curve rather than an assumed universal number. Two qualitatively different situations must be separated. The first is $\Phi_{W,\nu}^* > 0$. Then the leading magic bandwidth remains proportional to the moiré kinetic scale,

$$W_\nu^* \sim E_M^{\text{eff}} \Phi_{W,\nu}^*. \quad (5938)$$

The second is $\Phi_{W,\nu}^* = 0$. Then the leading continuum scaling function itself becomes exactly flat at the magic root, and the observed residual bandwidth is controlled by irrelevant corrections. This distinction can be established numerically by studying the scaling of $\frac{W_\nu^*}{E_M^{\text{eff}}}$. A nonzero limit identifies a finite scaled-bandwidth magic class. A vanishing limit identifies a continuum-exact-flat class. The microscopic square-lattice dispersion is

$$\varepsilon_0 + 4t = ta^2k^2 - \frac{ta^4}{12}(k_x^4 + k_y^4) + O(k^6a^6). \quad (5939)$$

Write $\mathbf{k} = Q_M^{\text{eff}} \mathbf{p}$. Since

$$E_M^{\text{eff}} = ta^2 (Q_M^{\text{eff}})^2, \quad (5940)$$

division by E_M^{eff} gives

$$\frac{\varepsilon_0 + 4t}{E_M^{\text{eff}}} = p^2 - \frac{y_{\text{lat}}}{12}(p_x^4 + p_y^4) + O(y_{\text{lat}}^2), \quad (5941)$$

where

$$y_{\text{lat}} = a^2 (Q_M^{\text{eff}})^2 = \frac{E_M^{\text{eff}}}{t}. \quad (5942)$$

Thus y_{lat} is an explicit irrelevant scaling field measuring deviation from the quadratic continuum theory. Let $\mathbf{y} = (y_{\text{lat}}, y_1, y_2, \dots)$ collect the lattice and normalized-profile corrections. The dimensionless Hamiltonian admits the expansion

$$\hat{H}_\nu = \hat{H}_\nu^*(X) + y_{\text{lat}} \hat{\mathcal{O}}_{\text{lat}}(X) + \sum_a y_a \hat{\mathcal{O}}_a(X) + O(\|\mathbf{y}\|^2). \quad (5943)$$

Away from a spectral switching singularity, the normalized bandwidth therefore has the expansion

$$\frac{W_\nu}{E_M^{\text{eff}}} = \Phi_{W,\nu}(X) + y_{\text{lat}} \Psi_{W,\text{lat},\nu}(X) + \sum_a y_a \Psi_{W,a,\nu}(X) + O(\|\mathbf{y}\|^2). \quad (5944)$$

The same structure holds for the gap, generalized dispersion, and broadened DOS.

f. Theorem 163. Controlled correction-to-scaling structure. A one-parameter magic-band collapse is the leading term of a systematic multi-field expansion. The leading horizontal coordinate is $X = \alpha_{\text{eff}}/\alpha^*$. The microscopic lattice correction is explicitly $y_{\text{lat}} = E_M^{\text{eff}}/t$, while deviations of the normalized scattering profile enter as additional dimensionless fields. Thus failure of perfect collapse need not immediately falsify the effective-coupling mechanism; it must first be tested against the predicted correction fields. Suppose $\Phi_{W,\nu}(1) = 0$. Assume the leading correction at the magic point is the quartic lattice field,

$$\frac{W_\nu^*}{E_M^{\text{eff}}} = c_{4,\nu} y_{\text{lat}} + o(y_{\text{lat}}), \quad (5945)$$

with $c_{4,\nu} > 0$. Using Eq. (5942),

$$\frac{W_\nu^*}{t} = c_{4,\nu} \left(\frac{E_M^{\text{eff}}}{t} \right)^2 + o \left[\left(\frac{E_M^{\text{eff}}}{t} \right)^2 \right]. \quad (5946)$$

At the magic root, $\alpha_\nu^{\text{eff}} = \alpha_\nu^*$, so that

$$E_M^{\text{eff}} = \frac{w_\nu^{\text{eff}}}{\alpha_\nu^*}. \quad (5947)$$

Therefore,

$$\frac{E_M^{\text{eff},*}}{t} = \frac{\gamma_\nu^{\text{eff},*}}{\alpha_\nu^*} \omega. \quad (5948)$$

Hence,

$$\frac{W_\nu^*}{t} = c_{4,\nu} \left(\frac{\gamma_\nu^{\text{eff},*}}{\alpha_\nu^*} \right)^2 \omega^2 + o(\omega^2). \quad (5949)$$

g. Theorem 164. Curvature-independent residual-bandwidth exponent. If the leading continuum master function is exactly flat at $X = 1$, the dominant microscopic correction is quartic, and the normalized active hybridization approaches a finite limit, then

$$W_\nu^* \propto t \left(\frac{w}{t} \right)^2 \quad (5950)$$

along both Euclidean and hyperbolic magic branches. Curvature changes the magic angle required to satisfy $X = 1$, but not the leading weak-coupling exponent of the residual bandwidth under these assumptions. This provides an independent universality test beyond the magic-angle trajectory itself. The operational magic functional contains

$$\frac{\Delta}{\Delta + W}, \quad \frac{\Delta}{\Delta + \hbar D_G \Omega^{\text{max}}}, \quad (5951)$$

the coherent DOS factor, and the coherence factor. Under one-parameter spectral collapse,

$$\begin{aligned} W &= E_M \Phi_W(X), \\ \Delta &= E_M \Phi_\Delta(X), \end{aligned} \quad (5952)$$

and

$$\hbar D_G \Omega^{\max} = E_M \Phi_V(X). \quad (5953)$$

The first two factors therefore become exactly

$$\frac{\Phi_\Delta(X)}{\Phi_\Delta(X) + \Phi_W(X)}, \quad (5954)$$

and

$$\frac{\Phi_\Delta(X)}{\Phi_\Delta(X) + \Phi_V(X)}. \quad (5955)$$

Define

$$y_E = \frac{E_M^{\text{eff}}}{E_{\text{kin}}}. \quad (5956)$$

Since

$$E_M \rho_{\text{coh}}^{\max} = \Phi_\rho^{\text{coh}}(X), \quad (5957)$$

we obtain

$$E_{\text{kin}} \rho_{\text{coh}}^{\max} = \frac{\Phi_\rho^{\text{coh}}(X)}{y_E}. \quad (5958)$$

Hence the DOS factor becomes

$$\frac{E_{\text{kin}} \rho_{\text{coh}}^{\max}}{1 + E_{\text{kin}} \rho_{\text{coh}}^{\max}} = \frac{\Phi_\rho^{\text{coh}}(X)}{y_E + \Phi_\rho^{\text{coh}}(X)}. \quad (5959)$$

Therefore the complete operational score has the scaling form

$$\mathcal{M}_\nu = \mathfrak{M}_\nu(X_\nu, y_E, \mathbf{y}). \quad (5960)$$

In the continuum scaling limit, $y_E \rightarrow 0$, and $\mathbf{y} \rightarrow 0$. If $\Phi_\rho^{\text{coh}}(X) > 0$, then

$$\frac{\Phi_\rho^{\text{coh}}}{y_E + \Phi_\rho^{\text{coh}}} \rightarrow 1. \quad (5961)$$

Thus $\mathcal{M}_\nu \rightarrow \mathfrak{M}_\nu^{(\infty)}(X_\nu)$. This establishes the precise sense in which the complete operational magic functional

becomes a one-parameter universal function. The master bandwidth law is

$$W = E_M \Phi_W(X). \quad (5962)$$

At fixed curvature and coupling,

$$\frac{d \ln E_M}{d \ln |\theta|} = 2\nu_\xi. \quad (5963)$$

For fixed active tunneling,

$$\frac{d \ln X}{d \ln |\theta|} = -2\nu_\xi. \quad (5964)$$

Define the master-function logarithmic slope

$$\beta_\Phi(X) = \frac{d \ln \Phi_W}{d \ln X}. \quad (5965)$$

Then

$$\frac{d \ln W}{d \ln |\theta|} = \frac{d \ln E_M}{d \ln |\theta|} + \beta_\Phi \frac{d \ln X}{d \ln |\theta|}. \quad (5966)$$

Therefore,

$$\frac{d \ln W}{d \ln |\theta|} = 2\nu_\xi(u) [1 - \beta_\Phi(X)]. \quad (5967)$$

h. Theorem 165. Geometry–spectrum exponent factorization. Within a one-parameter scaling regime, the running twist exponent of the bandwidth factorizes into geometric running exponent $\times$ spectral master-function response. Explicitly,

$$\frac{d \ln W}{d \ln |\theta|} = 2\nu_\xi(1 - \beta_\Phi). \quad (5968)$$

Thus geometry controls how rapidly the scaling coordinate moves with twist, whereas Φ_W controls how the spectrum responds to that motion. This provides a direct quantitative separation of geometric and spectral universality. Along a magic branch, $X_\nu = 1$. Assume the active tunneling factor is independent of ω and θ at leading order. Differentiating $\ln X_\nu = 0$ along the magic branch gives

$$0 = d \ln \omega - 2\nu_\xi d \ln |\theta_\nu^*|. \quad (5969)$$

Therefore,

$$\frac{d \ln |\theta_\nu^*|}{d \ln \omega} = \frac{1}{2\nu_\xi}. \quad (5970)$$

On the magic branch,

$$L_\nu^* = \frac{\xi_{M,\nu}^*}{R}. \quad (5971)$$

Using

$$u_\nu^* = \frac{1}{\sinh L_\nu^*}, \quad (5972)$$

we obtain

$$\nu_\xi = \frac{\tanh L_\nu^*}{L_\nu^*}. \quad (5973)$$

Hence

$$\frac{d \ln |\theta_\nu^*|}{d \ln \omega} = \frac{L_\nu^*}{2} \coth L_\nu^*. \quad (5974)$$

This reproduces independently the running magic exponent derived from the explicit magic curve.

i. Theorem 166. Reciprocal locking of geometric and magic exponents. Under fixed normalized active tunneling,

$$\beta_{\text{magic}} = \frac{1}{\beta_X} = \frac{1}{2\nu_\xi}. \quad (5975)$$

Thus the running exponent governing the magic-angle response to w/t is the reciprocal of the running exponent governing the twist dependence of the effective coupling.

The Euclidean endpoint gives $\beta_{\text{magic}} \rightarrow \frac{1}{2}$, whereas the curvature-dominated regime gives a diverging effective power exponent, signaling the essential singularity. A visual data collapse is insufficient for a strong universality claim. We therefore derive an operator-level error bound. Let

$$\hat{H}_i = \hat{H}_\nu^*(X_i) + R_i, \quad (5976)$$

with $\|R_i\| \leq \epsilon_i^{\text{sh}}$. Let the reference master Hamiltonian have eigenvalues $\varepsilon_1^* \leq \varepsilon_2^* \leq \dots$. Weyl's inequality gives

$$|\hat{\varepsilon}_{j,i} - \varepsilon_j^*(X_i)| \leq \epsilon_i^{\text{sh}}. \quad (5977)$$

Therefore the normalized bandwidth satisfies

$$\left| \frac{W_i}{E_{M,i}^{\text{eff}}} - \Phi_W(X_i) \right| \leq 2\epsilon_i^{\text{sh}}. \quad (5978)$$

Likewise,

$$\left| \frac{\Delta_i}{E_{M,i}^{\text{eff}}} - \Phi_\Delta(X_i) \right| \leq 2\epsilon_i^{\text{sh}}. \quad (5979)$$

Let the computed Hamiltonian satisfy

$$\|H_i^{\text{num}} - H_i\| \leq \epsilon_i^{\text{num}}. \quad (5980)$$

Define the dimensionless numerical error

$$\hat{\epsilon}_i^{\text{num}} = \frac{\epsilon_i^{\text{num}}}{E_{M,i}^{\text{eff}}}. \quad (5981)$$

The total collapse uncertainty is $\epsilon_i^{\text{coll}} = \epsilon_i^{\text{sh}} + \hat{\epsilon}_i^{\text{num}}$. Therefore,

$$\left| \frac{W_i^{\text{num}}}{E_{M,i}^{\text{eff}}} - \Phi_W(X_i) \right| \leq 2\epsilon_i^{\text{coll}}. \quad (5982)$$

The same bound applies to the normalized isolation gap.

j. Theorem 167. Error-established bandwidth collapse. A one-parameter bandwidth universality claim is established only when every observed deviation from the master curve is bounded by the independently estimated dimensionless shape and numerical errors. Systematic residuals exceeding $2\epsilon^{\text{coll}}$ reject the proposed one-parameter master Hamiltonian. Thus data collapse becomes a falsifiable operator-level statement rather than a graphical impression. Let the dimensionless Lorentzian broadening be

$$\hat{\eta} = \frac{\eta}{E_M^{\text{eff}}}. \quad (5983)$$

We keep $\hat{\eta}$ fixed when comparing different parameter points. Define the scaled DOS

$$\hat{\rho}_{\hat{\eta}} = E_M^{\text{eff}} \rho_\eta. \quad (5984)$$

The resolvent bound gives

$$|\hat{\rho}_{i,\hat{\eta}} - \Phi_{\rho,\hat{\eta}}(X_i)| \leq \frac{\epsilon_i^{\text{coll}}}{\pi \hat{\eta}^2}. \quad (5985)$$

Thus DOS curves should be compared at equal dimensionless rather than equal absolute broadening. Let the master Hamiltonian possess an isolated target state separated by a dimensionless spectral gap $g_* > 0$. Suppose

$$\|R\| < \frac{g_*}{2}. \quad (5986)$$

Then spectral-projector perturbation theory gives

$$\|P - P_\star\| = O\left(\frac{\|R\|}{g_\star}\right). \quad (5987)$$

Let λ denote a normalized generalized momentum coordinate. Assume $\|\partial_\lambda R\| \leq \epsilon_\lambda^{\text{sh}}$. The Hellmann–Feynman derivative then satisfies

$$\left|\partial_\lambda \widehat{E} - \partial_\lambda \widehat{E}_\star\right| \leq \epsilon_\lambda^{\text{sh}} + O\left[\frac{\left\|\partial_\lambda \widehat{H}_\star\right\|}{g_\star} \epsilon_\lambda^{\text{sh}}\right]. \quad (5988)$$

Thus a velocity or generalized-dispersion collapse is more demanding than an energy collapse because it requires convergence of both the Hamiltonian and its tangent derivative. Consider two microscopic interlayer kernels, $T^{(A)}(D)$, $T^{(B)}(D)$, normalized according to

$$T^{(A)}(h) = T^{(B)}(h) = w. \quad (5989)$$

Suppose they possess the same vertical logarithmic slope,

$$-h \frac{T^{(A)'}(h)}{T^{(A)}(h)} = -h \frac{T^{(B)'}(h)}{T^{(B)}(h)} = \zeta_\perp. \quad (5990)$$

Their local weak-twist response agrees to quadratic order. However, equality of this local slope alone does not guarantee complete global spectral universality. For strong one-parameter collapse, we require the normalized active scattering profiles to converge,

$$\left\|\widehat{V}_\nu^{(A)} - U_\nu \widehat{V}_\nu^{(B)} U_\nu^\dagger\right\| \longrightarrow 0 \quad (5991)$$

for a symmetry-compatible unitary identification U_ν . Under this condition, the normalized Hamiltonians obey

$$\left\|\widehat{H}_\nu^{(A)}(X) - U_\nu \widehat{H}_\nu^{(B)}(X) U_\nu^\dagger\right\| \longrightarrow 0. \quad (5992)$$

Therefore their spectral master functions coincide.

k. Theorem 168. Microscopic magic-band universality class. Two microscopic twisted-bilayer models belong to the same strong magic-band universality class if, after scaling energies by their E_M^{eff} and horizontal coordinates by their $X = \alpha_{\text{eff}}/\alpha^\star$, their normalized active Hamiltonians become unitarily equivalent in the scaling limit. Then

$$\Phi_W^{(A)}(X) = \Phi_W^{(B)}(X), \quad \Phi_\Delta^{(A)}(X) = \Phi_\Delta^{(B)}(X), \quad (5993)$$

and

$$\Phi_V^{(A)}(X) = \Phi_V^{(B)}(X). \quad (5994)$$

Equality of the local tunneling slope is sufficient for leading local coupling universality but not, by itself, for this stronger global spectral statement. The weakest universality statement concerns only the geometric scaling coordinate. It requires $X = X_E G^2(u)$ under fixed normalized active coupling. The next level requires bandwidth collapse,

$$\frac{W}{E_M^{\text{eff}}} = \Phi_W(X). \quad (5995)$$

A stronger spectral statement requires simultaneous collapse of

$$\frac{W}{E_M^{\text{eff}}}, \quad \frac{\Delta}{E_M^{\text{eff}}}, \quad \frac{\hbar D_G \Omega^{\text{max}}}{E_M^{\text{eff}}}, \quad E_M^{\text{eff}} \rho^{\text{coh,max}}. \quad (5996)$$

The strongest microscopic universality statement additionally requires cross-model collapse after changing the microscopic tunneling profile within the same normalized coupling class. These levels are tested separately. Define the spectral scaling vector

$$\Phi_\nu(X) = \begin{pmatrix} \Phi_{W,\nu}(X) \\ \Phi_{\Delta,\nu}(X) \\ \Phi_{V,\nu}(X) \\ \Phi_{\rho,\nu}^{\text{coh}}(X) \\ \Phi_{C,\nu}(X) \end{pmatrix}, \quad (5997)$$

where

$$\Phi_{C,\nu}(X) = \mathcal{C}_\nu^{\text{coh}} \quad (5998)$$

in the scaling limit. Then the complete magic-band evolution is represented by a one-dimensional trajectory in the five-dimensional observable space. Different microscopic parameter sweeps, θ -sweep, K -sweep, and w/t -sweep, must generate the same trajectory when reparameterized by X . Thus the strongest collapse does not merely superpose several bandwidth curves. It makes physically different parameter paths indistinguishable after projection onto the scaling coordinate. Consider two control-space paths, $\mathcal{P}_1 : \mathbf{p}_1(s)$, and $\mathcal{P}_2 : \mathbf{p}_2(s)$. Suppose

$$X[\mathbf{p}_1(s_1)] = X[\mathbf{p}_2(s_2)]. \quad (5999)$$

One-parameter universality predicts

$$\frac{W[\mathbf{p}_1(s_1)]}{E_M^{\text{eff}}[\mathbf{p}_1(s_1)]} = \frac{W[\mathbf{p}_2(s_2)]}{E_M^{\text{eff}}[\mathbf{p}_2(s_2)]} \quad (6000)$$

up to controlled correction fields. Likewise,

$$\frac{\Delta_1}{E_{M,1}^{\text{eff}}} = \frac{\Delta_2}{E_{M,2}^{\text{eff}}}, \quad (6001)$$

and

$$E_{M,1}^{\text{eff}} \rho_1^{\text{coh,max}} = E_{M,2}^{\text{eff}} \rho_2^{\text{coh,max}}. \quad (6002)$$

This equal- X path-independence test is stronger than fitting all data to one flexible curve because it permits direct pairwise comparisons. For two numerical data points i and j satisfying

$$|X_i - X_j| \leq \delta X, \quad (6003)$$

define

$$Y_{W,i} = \frac{W_i}{E_{M,i}^{\text{eff}}}. \quad (6004)$$

The pairwise bandwidth-collapse residual is

$$\mathcal{R}_{W,ij} = \frac{|Y_{W,i} - Y_{W,j}|}{\frac{1}{2}(|Y_{W,i}| + |Y_{W,j}|) + \epsilon_{\text{reg}}}, \quad (6005)$$

where $\epsilon_{\text{reg}} > 0$ is used only to avoid an artificial divergence near an exactly vanishing bandwidth. The maximum pairwise residual over matched- X data is

$$\mathcal{R}_W^{\text{coll}} = \sup_{|X_i - X_j| \leq \delta X} \mathcal{R}_{W,ij}. \quad (6006)$$

A successful collapse requires $\mathcal{R}_W^{\text{coll}}$ to decrease systematically as $\delta X \rightarrow 0$, the finite-cover size increases, and the correction fields decrease. For data point i , let $\sigma_{W,i}$ denote the complete uncertainty of $Y_{W,i}$. For a pair i, j , define

$$\sigma_{ij} = \sqrt{\sigma_{W,i}^2 + \sigma_{W,j}^2}. \quad (6007)$$

The standardized collapse discrepancy is

$$Z_{ij}^W = \frac{|Y_{W,i} - Y_{W,j}|}{\sigma_{ij}}. \quad (6008)$$

Persistent values $Z_{ij}^W \gg 1$ for matched X reject one-parameter bandwidth universality. The same test is applied independently to the gap, generalized dispersion, and coherent DOS. We evaluate the scaling theory under the three Hamiltonian families defined previously. For the bare-fixed physical family, H_{phys} , both E_M^{eff} and w_ν^{eff} are allowed to change with curvature. For the coordination-normalized family, H_{coord} , trivial intralayer kinetic-scale variation is removed. For the hybridization-normalized family, H_{hyb} , the total mean-square interlayer coupling is fixed. For every family a , define its independently measured

$$X_\nu^{(a)} = \frac{w_\nu^{\text{eff},(a)}}{\alpha_\nu^* E_M^{\text{eff},(a)}}. \quad (6009)$$

A strong geometry-controlled universality claim requires

$$\frac{W_\nu^{(a)}}{E_M^{\text{eff},(a)}} \rightarrow \Phi_{W,\nu}(X_\nu^{(a)}) \quad (6010)$$

for all three control families in the scaling regime. This test distinguishes universality of the geometric mechanism from accidental collapse caused by one particular energy normalization. The scaling coordinate has been normalized by the branch-specific root α_ν^* . Therefore every branch has magic point $X_\nu = 1$. However, this does not imply automatically that

$$\Phi_{W,\mu} = \Phi_{W,\nu} \quad (6011)$$

for distinct branches. The minimal universality claim is fixed branch + different $K, \theta, \omega \rightarrow$ same $\Phi_{W,\nu}$. A stronger cross-branch universality would require

$$\hat{H}_\mu^*(X) \simeq U \hat{H}_\nu^*(X) U^\dagger \quad (6012)$$

after any required spectral reindexing. We therefore test cross-branch collapse rather than assuming it. For an exact commensurate hyperbolic branch, the scaled observables are obtained from representation-resolved finite covers. For an incommensurate target, the same local scaling variable X_ν remains well defined from the local registry geometry and active tunneling matrix. If commensurate approximants are used, they must satisfy

$$|\theta_N - \theta| e^{L_N/R} \rightarrow 0. \quad (6013)$$

The scaling collapse is accepted only if

$$\frac{W_N}{E_{M,N}^{\text{eff}}} \rightarrow \Phi_W(X) \quad (6014)$$

independently of the chosen admissible approximant sequence. Thus universality is formulated at the level of local bulk spectral physics rather than requiring exact crystallographic periodicity. Let $Q_N^{(A)}$ and $Q_N^{(B)}$ be two inequivalent normal-cover towers. At matched scaling coordinate X , define

$$Y_{W,N}^{(A)} = \frac{W_N^{(A)}}{E_{M,N}^{\text{eff},(A)}}, \quad (6015)$$

and

$$Y_{W,N}^{(B)} = \frac{W_N^{(B)}}{E_{M,N}^{\text{eff},(B)}}. \quad (6016)$$

A bulk universal function requires

$$\lim_{N \rightarrow \infty} Y_{W,N}^{(A)} = \lim_{N \rightarrow \infty} Y_{W,N}^{(B)} = \Phi_W(X). \quad (6017)$$

The corresponding cross-cover residual is

$$\mathcal{R}_{\text{cover}}(X) = \left| Y_{W,\infty}^{(A)} - Y_{W,\infty}^{(B)} \right|. \quad (6018)$$

Universality requires $\mathcal{R}_{\text{cover}}(X) = 0$ within established numerical uncertainty. The magic locus is $X_\nu = 1$. Using the double-scaling expression,

$$\frac{\gamma_\nu^{\text{eff}} n_c^2}{\pi^2 d \alpha_\nu^*} \frac{\omega}{\theta_\nu^{*2}} G^2 \left(\frac{R|\theta_\nu^*|}{a} \right) = 1. \quad (6019)$$

Thus the entire magic surface is the $X = 1$ contour of the same scaling coordinate that controls the bandwidth collapse. The magic-angle law and the bandwidth collapse are therefore not independent predictions. They are two projections of one universal scaling structure. For every operational magic point, define

$$X_\nu^* = X_\nu(K, \theta_\nu^*, \omega). \quad (6020)$$

One-parameter universality requires $X_\nu^* \rightarrow 1$. The root-collapse residual is

$$\mathcal{R}_{X,\nu} = \sqrt{\left\langle (X_\nu^* - 1)^2 \right\rangle}. \quad (6021)$$

A successful theory requires $\mathcal{R}_{X,\nu} \rightarrow 0$ under increasing finite-cover accuracy and decreasing correction fields. The geometric theory established independently that

$$\frac{\xi_M^{(H)}}{\xi_M^{(E)}} = G(u). \quad (6022)$$

The spectral theory now gives

$$\frac{E_M^{(H)}}{E_M^{(E)}} = G^{-2}(u), \quad (6023)$$

and, under fixed active tunneling,

$$\frac{\alpha_{\text{eff}}^{(H)}}{\alpha_{\text{eff}}^{(E)}} = G^2(u). \quad (6024)$$

Finally, the bandwidth satisfies

$$\frac{W^{(H)}}{E_M^{(H)}} = \Phi_W [X_E G^2(u)]. \quad (6025)$$

Thus the complete geometry-to-spectrum crossover is

$$G(u) \rightarrow G^{-2}(u) \rightarrow G^2(u) \rightarrow \Phi_W [X_E G^2(u)]. \quad (6026)$$

The first factor controls length. The second controls the kinetic energy. The third controls the effective hybridization ratio. The final function controls the spectral flattening.

l. Theorem 169. Curvature-controlled universality of magic-band formation. Consider a continuously tracked twisted-bilayer spectral family with a quadratic microscopic reference dispersion. Suppose that

$$\hat{H}_\nu(\theta, K, \omega) \rightarrow \hat{H}_\nu^*(X_\nu) \quad (6027)$$

in operator norm after division by the geometry-controlled kinetic scale, with

$$X_\nu = \frac{\alpha_\nu^{\text{eff}}}{\alpha_\nu^*}. \quad (6028)$$

Then the complete bandwidth obeys

$$W_\nu = E_M^{\text{eff}} \Phi_{W,\nu}(X_\nu), \quad (6029)$$

while the gap, generalized dispersion, and DOS obey

$$\Delta_\nu = E_M^{\text{eff}} \Phi_{\Delta,\nu}(X_\nu), \quad \hbar D_G \Omega_\nu^{\text{max}} = E_M^{\text{eff}} \Phi_{V,\nu}(X_\nu), \quad (6030)$$

and

$$E_M^{\text{eff}} \rho_\nu^{\text{coh,max}} = \Phi_{\rho,\nu}(X_\nu). \quad (6031)$$

In the weak-curvature/small-angle double-scaling limit,

$$X_\nu = \frac{\gamma_\nu^{\text{eff}} n_c^2}{\pi^2 d \alpha_\nu^*} \frac{w/t}{\theta^2} G^2 \left(\frac{R|\theta|}{a} \right), \quad (6032)$$

where

$$G(u) = u \operatorname{arsinh} \left(\frac{1}{u} \right) \quad (6033)$$

is the same parameter-free crossover function that controls the curvature-dependent moiré length. The magic manifold is the universal level set $X_\nu = 1$. Consequently, Euclidean and negatively curved spectra do not constitute independent families once expressed in the correct

scaled variables. They become different parameterizations of the same master spectral functions. The complete theory can now be summarized through the hierarchy

$$\begin{aligned} K, \theta &\longrightarrow G\left(\frac{R|\theta|}{a}\right) \longrightarrow E_M^{\text{eff}} \\ &\longrightarrow \alpha_{\text{eff}} \longrightarrow X = \frac{\alpha_{\text{eff}}}{\alpha^*} \\ &\longrightarrow \{\Phi_W(X), \Phi_\Delta(X), \Phi_V(X), \Phi_\rho(X)\}. \end{aligned} \quad (6034)$$

The corresponding bandwidth law is

$$W = E_M^{\text{eff}} \Phi_W(X). \quad (6035)$$

Equivalently, using the Euclidean kinetic scale,

$$G^2(u) \frac{W}{E_M^{(E)}} = \Phi_W[X_E \Gamma G^2(u)]. \quad (6036)$$

Under fixed normalized active hybridization, $\Gamma = 1$, and the collapse becomes parameter free once the Euclidean magic root has been determined. The Euclidean endpoint is $G(u) \longrightarrow 1$, so that

$$X \propto \frac{w/t}{\theta^2}. \quad (6037)$$

The curvature-dominated endpoint is

$$G(u) \sim u \ln\left(\frac{2}{u}\right), \quad (6038)$$

so that

$$X \propto \frac{w}{t} \frac{R^2}{a^2} \ln^2\left(\frac{2a}{R|\theta|}\right). \quad (6039)$$

The two apparently different magic-angle laws,

$$\theta^* \propto \left(\frac{w}{t}\right)^{1/2}, \quad \theta^* \propto \exp\left[-\frac{C(K)}{\sqrt{w/t}}\right], \quad (6040)$$

are therefore not separate mechanisms. They are the two asymptotic solutions of the same universal condition $X = 1$. The strongest test of curvature-controlled magic-band universality is thus not the observation of a shifted flat-band minimum. It is the simultaneous collapse

$$\left\{ \frac{W}{E_M^{\text{eff}}}, \frac{\Delta}{E_M^{\text{eff}}}, \frac{\hbar D_G \Omega^{\text{max}}}{E_M^{\text{eff}}}, E_M^{\text{eff}} \rho^{\text{coh, max}} \right\} = \{\Phi_W(X), \Phi_\Delta(X), \Phi_V(X), \Phi_\rho(X)\} \quad (6041)$$

for parameter sets with different curvature, twist, and interlayer coupling. When this collapse survives finite-cover convergence, hopping-cutoff convergence, symmetry resolution, localization tests, and microscopic coupling-profile controls, the curvature dependence is no longer described merely as a displacement of a magic angle. It is described by a single geometry-renormalized scaling coordinate and a common set of spectral master functions.

VI. GEOMETRY-TO-SPECTRUM CORRESPONDENCE

The geometric and spectral theories developed above can now be combined into a single quantitative correspondence. The purpose of this section is not to place geometric and electronic results next to one another. We instead derive explicit maps showing how the curvature-controlled moiré length and moiré-cell scaling enter the reconstructed spectrum and the magic-branch hierarchy. The first correspondence relates the local moiré length to the geometry-controlled kinetic scale, the effective hybridization ratio, and the reconstructed bandwidth. The second correspondence eliminates the microscopic coupling parameter between the geometric moiré-cell count and the spectral magic branch. This produces a direct relation between $N_M(\theta, K)$ and $\theta_\nu^*\left(K, \frac{w}{t}\right)$, and shows that the Euclidean-to-hyperbolic crossover of the geometric moiré exponent is inherited directly by the magic-branch scaling.

A. Moiré Length versus Spectral Reconstruction

We begin from the exact local moiré length

$$\xi_M = R \operatorname{arsinh}\left[\frac{\sinh(a/2R)}{|\sin(\theta/2)|}\right]. \quad (6042)$$

Introduce $L = \frac{\xi_M}{R}$. Then

$$L = \operatorname{arsinh}\left[\frac{\sinh(a/2R)}{|\sin(\theta/2)|}\right]. \quad (6043)$$

The first objective is to eliminate the twist angle from the spectral energy scale and express the reconstructed spectrum directly in terms of the geometric length ξ_M . The arithmetic-normalized effective scattering momentum introduced above is

$$Q_M^{\text{eff}} = \frac{\sqrt{d}}{2n_c} q_M^{\text{loc}}, \quad (6044)$$

where

$$q_M^{\text{loc}} = \frac{2\pi}{\xi_M}. \quad (6045)$$

Therefore,

$$Q_M^{\text{eff}} = \frac{\pi\sqrt{d}}{n_c\xi_M}. \quad (6046)$$

This relation is independent of any small-angle approximation. It is also independent of whether an exact finite hyperbolic reciprocal lattice exists, because Q_M^{eff} is a local geometry-normalized spectral scale rather than a global additive hyperbolic reciprocal vector. For a quadratic microscopic reference dispersion,

$$\varepsilon_0 + 4t = ta^2k^2 + O(k^4a^4), \quad (6047)$$

the corresponding geometry-controlled moiré kinetic scale is

$$E_M^{\text{eff}} = ta^2 (Q_M^{\text{eff}})^2. \quad (6048)$$

Substitution of Eq. (6046) gives the exact length representation

$$E_M^{\text{eff}} = \frac{\pi^2 d}{n_c^2} t \left(\frac{a}{\xi_M} \right)^2. \quad (6049)$$

Thus the geometry-to-kinetic map is simply $E_M^{\text{eff}} \propto \xi_M^{-2}$. The curvature dependence is contained entirely in the way θ determines ξ_M .

a. Theorem 170. Exact length-kinetic-energy correspondence. For the quadratic square-lattice reference, the local curvature-controlled moiré length determines the normalized low-energy kinetic scale through

$$E_M^{\text{eff}} \xi_M^2 = \frac{\pi^2 d}{n_c^2} ta^2. \quad (6050)$$

Therefore the product on the left is independent of curvature and twist. Negative curvature changes the kinetic scale only by changing the geometric moiré length required at a given twist. The active interlayer scattering scale is

$$w_\nu^{\text{eff}} = \gamma_\nu^{\text{eff}}(K, \theta) w. \quad (6051)$$

The effective coupling is

$$\alpha_\nu^{\text{eff}} = \frac{w_\nu^{\text{eff}}}{E_M^{\text{eff}}}. \quad (6052)$$

Using Eq. (6049) gives

$$\alpha_\nu^{\text{eff}} = \frac{\gamma_\nu^{\text{eff}} n_c^2}{\pi^2 d} \frac{w}{t} \left(\frac{\xi_M}{a} \right)^2. \quad (6053)$$

This equation is considerably more transparent than the equivalent curvature-twist expression. At fixed active hybridization, $\gamma_\nu^{\text{eff}} = \text{constant}$, the complete dimensionless competition between kinetic energy and interlayer coupling is controlled by $\alpha_\nu^{\text{eff}} \propto \xi_M^2$. Thus increasing the moiré length suppresses the kinetic energy as ξ_M^{-2} and simultaneously increases the hybridization-to-kinetic ratio as ξ_M^2 .

b. Theorem 171. Length amplification of the effective hybridization. At fixed w/t and fixed normalized active scattering amplitude,

$$\frac{d \ln E_M^{\text{eff}}}{d \ln \xi_M} = -2, \quad (6054)$$

whereas

$$\frac{d \ln \alpha_\nu^{\text{eff}}}{d \ln \xi_M} = 2. \quad (6055)$$

The same geometric increase of ξ_M therefore acts twice: it weakens the bare kinetic scale and strengthens the relative hybridization by the inverse amount. Define

$$X_\nu = \frac{\alpha_\nu^{\text{eff}}}{\alpha_\nu^*}. \quad (6056)$$

Using Eq. (6053),

$$X_\nu = \frac{\gamma_\nu^{\text{eff}} n_c^2}{\pi^2 d \alpha_\nu^*} \frac{w}{t} \left(\frac{\xi_M}{a} \right)^2. \quad (6057)$$

Thus the magic condition $X_\nu = 1$ can be written as a direct condition on the geometric moiré length,

$$\gamma_\nu^{\text{eff}}(K, \theta_\nu^*) \left(\frac{\xi_{M,\nu}^*}{a} \right)^2 = \frac{\pi^2 d \alpha_\nu^*}{n_c^2} \frac{t}{w}. \quad (6058)$$

If the active tunneling factor is constant within the scaling class, $\gamma_\nu^{\text{eff}} = \gamma_\nu$, then

$$\xi_{M,\nu}^* = \frac{\pi a}{n_c} \sqrt{\frac{d \alpha_\nu^*}{\gamma_\nu} \frac{t}{w}}. \quad (6059)$$

The magic condition therefore selects one preferred geometric registry length at each w/t . Curvature changes the twist angle required to reach this length but does not change the length itself in the fixed- γ_ν scaling theory. The spectral-collapse theory gives

$$W_\nu = E_M^{\text{eff}} \Phi_{W,\nu}(X_\nu). \quad (6060)$$

Substitution of Eqs. (6049) and (6057) gives the central geometry-to-spectrum equation

$$\begin{aligned} \frac{W_\nu}{t} &= \frac{\pi^2 d}{n_c^2} \left(\frac{a}{\xi_M} \right)^2 \\ &\times \Phi_{W,\nu} \left[\frac{\gamma_\nu^{\text{eff}} n_c^2 w}{\pi^2 d \alpha_\nu^*} \left(\frac{\xi_M}{a} \right)^2 \right]. \end{aligned} \quad (6061)$$

This is the explicit form of $\xi_M \rightarrow W$. The bandwidth is not an arbitrary function of curvature and twist once ξ_M has been specified. Within one scaling universality class, curvature and twist enter the bandwidth only through the moiré length and the active tunneling factor.

c. Theorem 172. Moiré-length spectral reconstruction law. If the normalized spectral family is controlled by the effective coupling X_ν , then

$$W_\nu = \frac{\pi^2 d t a^2}{n_c^2 \xi_M^2} \Phi_{W,\nu} \left[C_\nu \gamma_\nu^{\text{eff}} \frac{w}{t} \frac{\xi_M^2}{a^2} \right], \quad (6062)$$

where

$$C_\nu = \frac{n_c^2}{\pi^2 d \alpha_\nu^*}. \quad (6063)$$

Thus the geometry-to-spectrum correspondence consists of a ξ_M^{-2} vertical energy rescaling and a ξ_M^2 horizontal effective-coupling rescaling. The same derivation gives

$$\Delta_\nu = \frac{\pi^2 d t a^2}{n_c^2 \xi_M^2} \Phi_{\Delta,\nu}(X_\nu). \quad (6064)$$

Likewise,

$$\hbar D_G \Omega_\nu^{\text{max}} = \frac{\pi^2 d t a^2}{n_c^2 \xi_M^2} \Phi_{V,\nu}(X_\nu). \quad (6065)$$

The normalized coherent DOS satisfies

$$\rho_\nu^{\text{coh}}(E) = \frac{n_c^2 \xi_M^2}{\pi^2 d t a^2} \Phi_{\rho,\nu}(\epsilon, X_\nu), \quad (6066)$$

where

$$\epsilon = \frac{E - E_0}{E_M^{\text{eff}}}. \quad (6067)$$

Therefore the same increase of ξ_M that reduces the characteristic energy scale enhances the spectral density by the inverse factor. Equation (6061) implies

$$\frac{n_c^2}{\pi^2 d} \frac{W_\nu}{t} \left(\frac{\xi_M}{a} \right)^2 = \Phi_{W,\nu}(X_\nu). \quad (6068)$$

Similarly,

$$\frac{n_c^2}{\pi^2 d} \frac{\Delta_\nu}{t} \left(\frac{\xi_M}{a} \right)^2 = \Phi_{\Delta,\nu}(X_\nu), \quad (6069)$$

and

$$\frac{n_c^2}{\pi^2 d} \frac{\hbar D_G \Omega_\nu^{\text{max}}}{t} \left(\frac{\xi_M}{a} \right)^2 = \Phi_{V,\nu}(X_\nu). \quad (6070)$$

These equations provide direct data-collapse tests in which the horizontal coordinate is constructed from the measured geometric moiré length rather than from the bare twist angle. At a magic point, $X_\nu = 1$. Hence

$$\frac{W_\nu^* (\xi_{M,\nu}^*)^2}{t a^2} = \frac{\pi^2 d}{n_c^2} \Phi_{W,\nu}(1). \quad (6071)$$

If $\Phi_{W,\nu}(1) > 0$, then the magic bandwidth obeys $W_\nu^* \propto (\xi_{M,\nu}^*)^{-2}$. Thus a finite-scaled-bandwidth magic state possesses a universal length-bandwidth product. If the leading continuum master function satisfies $\Phi_{W,\nu}(1) = 0$, the quadratic energy scale cancels at leading order. Suppose the first nonvanishing lattice correction is quartic. Then

$$\frac{W_\nu^*}{t} = c_{4,\nu} \left(\frac{E_M^{\text{eff}}}{t} \right)^2 + o \left[\left(\frac{E_M^{\text{eff}}}{t} \right)^2 \right]. \quad (6072)$$

Using Eq. (6049),

$$\frac{W_\nu^*}{t} = c_{4,\nu} \frac{\pi^4 d^2}{n_c^4} \left(\frac{a}{\xi_{M,\nu}^*} \right)^4 + o \left[\left(\frac{a}{\xi_{M,\nu}^*} \right)^4 \right]. \quad (6073)$$

Therefore,

$$\frac{W_\nu^* (\xi_{M,\nu}^*)^4}{t a^4} \rightarrow c_{4,\nu} \frac{\pi^4 d^2}{n_c^4}. \quad (6074)$$

d. Theorem 173. Spectral distinction between two magic universality classes. The geometry-to-spectrum correspondence distinguishes two asymptotic magic-band classes. If $\Phi_{W,\nu}(1) > 0$, then $W_\nu^* \propto \xi_{M,\nu}^{*-2}$. If $\Phi_{W,\nu}(1) = 0$ and the leading irrelevant dispersion is quartic, then $W_\nu^* \propto \xi_{M,\nu}^{*-4}$. Thus the power of the magic bandwidth with respect to the directly measurable moiré length identifies whether the continuum master band is only strongly compressed or exactly flat at leading order. Define

$$\beta_{W,\nu}(X) = \frac{d \ln \Phi_{W,\nu}}{d \ln X}. \quad (6075)$$

Define the active-coupling length susceptibility by

$$\eta_{\gamma,\nu} = \frac{d \ln \gamma_{\nu}^{\text{eff}}}{d \ln \xi_M}. \quad (6076)$$

From Eq. (6057),

$$\frac{d \ln X_{\nu}}{d \ln \xi_M} = 2 + \eta_{\gamma,\nu}. \quad (6077)$$

Taking a logarithmic derivative of Eq. (6060) gives

$$\frac{d \ln W_{\nu}}{d \ln \xi_M} = -2 + (2 + \eta_{\gamma,\nu}) \beta_{W,\nu}(X_{\nu}). \quad (6078)$$

Under fixed normalized active tunneling, $\eta_{\gamma,\nu} = 0$, and therefore

$$-\frac{d \ln W_{\nu}}{d \ln \xi_M} = 2 [1 - \beta_{W,\nu}(X_{\nu})]. \quad (6079)$$

This equation separates the purely geometric inverse-square kinetic scaling from the spectral response encoded in the master function. The scaled Schur self-energy is

$$\frac{\Sigma_{\nu}}{E_M^{\text{eff}}} = (\alpha_{\nu}^{\text{eff}})^2 \hat{V}_{\nu} \left(\epsilon - \hat{H}_{Q,\nu} \right)^{-1} \hat{V}_{\nu}^{\dagger}. \quad (6080)$$

Suppose the relevant dimensionless detuning obeys

$$\text{dist} \left[\epsilon, \text{Spec} \hat{H}_{Q,\nu} \right] \geq g_{\nu} > 0. \quad (6081)$$

Then

$$\left\| \left(\epsilon - \hat{H}_{Q,\nu} \right)^{-1} \right\| \leq \frac{1}{g_{\nu}}. \quad (6082)$$

Therefore,

$$\frac{\|\Sigma_{\nu}\|}{E_M^{\text{eff}}} \leq \frac{(\alpha_{\nu}^{\text{eff}})^2}{g_{\nu}} \left\| \hat{V}_{\nu} \right\|^2. \quad (6083)$$

At fixed normalized active scattering, $(\alpha_{\nu}^{\text{eff}})^2 \propto \xi_M^4$. Hence the self-energy measured relative to the bare moiré kinetic scale can grow quartically with the geometric moiré length. This provides the microscopic amplification mechanism by which a geometric length variation produces a much stronger spectral reconstruction.

e. Theorem 174. Quartic relative self-energy amplification. Within an off-resonant active scattering shell of fixed dimensionless shape,

$$\frac{\|\Sigma_{\nu}\|}{E_M^{\text{eff}}} = O(\xi_M^4) \quad (6084)$$

at fixed w/t , up to the microscopic normalization by a . Therefore the geometry-to-spectrum map is intrinsically nonlinear: the same moiré-length increase that reduces the bare kinetic scale as ξ_M^{-2} increases the relative second-order hybridization self-energy as ξ_M^4 . Define the geometric running length exponent by

$$\nu_{\xi} = -\frac{d \ln \xi_M}{d \ln s}. \quad (6085)$$

For the exact hyperbolic geometry,

$$\nu_{\xi} = \frac{\tanh L}{L}. \quad (6086)$$

Using Eq. (6079),

$$\frac{d \ln W_{\nu}}{d \ln s} = \frac{d \ln W_{\nu}}{d \ln \xi_M} \frac{d \ln \xi_M}{d \ln s}. \quad (6087)$$

Therefore,

$$\frac{d \ln W_{\nu}}{d \ln s} = 2\nu_{\xi} [1 - \beta_{W,\nu}(X_{\nu})]. \quad (6088)$$

This recovers the geometry-spectrum exponent factorization obtained from the scaling collapse, now directly from the length correspondence.

f. Theorem 175. Geometry-to-spectrum exponent composition. The running twist exponent of the reconstructed bandwidth is the product of two factors: ν_{ξ} , which is fixed entirely by the curved-space moiré geometry, and $1 - \beta_{W,\nu}$, which is fixed entirely by the spectral master function. Therefore geometric and electronic scaling are not additive independent effects. They compose multiplicatively at the level of running exponents. In the Euclidean small-angle regime, $\xi_M \sim \frac{a}{|\theta|}$. Substitution into Eq. (6061) gives

$$\frac{W_{\nu}}{t} \sim \frac{\pi^2 d}{n_c^2} \theta^2 \Phi_{W,\nu} \left[C_{\nu} \gamma_{\nu}^{\text{eff}} \frac{w/t}{\theta^2} \right]. \quad (6089)$$

At fixed negative curvature,

$$\xi_M \sim R \ln \left[\frac{4 \sinh(a/2R)}{|\theta|} \right]. \quad (6090)$$

Hence,

$$\frac{W_\nu}{t} \sim \frac{\pi^2 da^2}{n_c^2 R^2} \frac{1}{\ln^2 \left[\frac{4 \sinh(a/2R)}{|\theta|} \right]} \quad (6091)$$

$$\times \Phi_{W,\nu} \left[C_\nu \gamma_\nu^{\text{eff}} \frac{w}{t} \frac{R^2}{a^2} \ln^2 \left[\frac{4 \sinh(a/2R)}{|\theta|} \right] \right].$$

Thus the Euclidean algebraic bandwidth scaling and the hyperbolic logarithmic bandwidth scaling are generated by one common length-dependent spectral law. For every numerical parameter point we calculate independently ξ_M , W_ν , γ_ν^{eff} , α_ν^{eff} . Define

$$Y_\nu^{(W)} = \frac{n_c^2}{\pi^2 d} \frac{W_\nu}{t} \left(\frac{\xi_M}{a} \right)^2. \quad (6092)$$

The geometry-to-spectrum correspondence predicts

$$Y_\nu^{(W)} = \Phi_{W,\nu}(X_\nu). \quad (6093)$$

Thus systems with distinct K , θ , w/t but equal X_ν must produce the same length-normalized bandwidth within the controlled correction fields. This provides a direct test of $\xi_M \rightarrow W$ without fitting the curvature dependence of the bandwidth separately. We now establish the second correspondence. The geometric theory determines how the effective moiré-cell size grows with twist and curvature. The spectral theory determines which geometric length is selected by the magic condition. Combining the two allows us to eliminate the microscopic coupling parameter and obtain a direct quantitative relation between the number of moiré-scale degrees of freedom and the location of the magic branch. The distinction between the effective geometric count and the exact crystallographic count remains essential throughout. The exact hyperbolic area of the effective registry domain is

$$A_M^{\text{eff}} = 2\pi R^2 \left[\cosh \left(\frac{\xi_M}{R} \right) - 1 \right]. \quad (6094)$$

If the primitive cell has covolume $A_0(K)$ and contains n_b microscopic sites per layer, the bilayer area-normalized effective microscopic count is

$$N_M^{\text{eff}} = 2n_b \frac{A_M^{\text{eff}}}{A_0(K)}. \quad (6095)$$

Therefore,

$$N_M^{\text{eff}} = \frac{4\pi n_b R^2}{A_0(K)} \left[\cosh \left(\frac{\xi_M}{R} \right) - 1 \right]. \quad (6096)$$

Thus the cell count is an explicit function of exactly the same geometric length that controls the spectral energy scale in VII.A.

g. Theorem 176. Common-length origin of geometric count and spectral reconstruction. Both the effective microscopic count and the reconstructed spectral scale are functions of one geometric variable, $L = \frac{\xi_M}{R}$. Explicitly,

$$N_M^{\text{eff}} = \frac{4\pi n_b R^2}{A_0} (\cosh L - 1), \quad (6097)$$

whereas

$$E_M^{\text{eff}} = \frac{\pi^2 d t a^2}{n_c^2 R^2 L^2}. \quad (6098)$$

The geometric cell size grows through the hyperbolic volume law $\cosh L - 1$, while the electronic kinetic scale decreases through L^{-2} . Define

$$\mathbf{n}_M = \frac{A_0(K)}{4\pi n_b R^2} N_M^{\text{eff}}. \quad (6099)$$

Equation (6096) gives

$$\mathbf{n}_M = \cosh L - 1. \quad (6100)$$

Therefore,

$$L = \text{arcosh}(1 + \mathbf{n}_M). \quad (6101)$$

Equivalently,

$$\xi_M = R \text{arcosh} \left[1 + \frac{A_0(K) N_M^{\text{eff}}}{4\pi n_b R^2} \right]. \quad (6102)$$

Thus the effective cell count determines the local moiré length uniquely. The geometric twist relation is

$$\sinh L = \frac{\sinh(a/2R)}{s}, \quad (6103)$$

where

$$s = \left| \sin \left(\frac{\theta}{2} \right) \right|. \quad (6104)$$

Using

$$\cosh L = 1 + \mathbf{n}_M, \quad (6105)$$

we obtain

$$\sinh^2 L = (1 + \mathbf{n}_M)^2 - 1. \quad (6106)$$

Therefore,

$$\sinh L = \sqrt{\mathbf{n}_M (\mathbf{n}_M + 2)}. \quad (6107)$$

Hence,

$$\left| \sin \left(\frac{\theta}{2} \right) \right| = \frac{\sinh(a/2R)}{\sqrt{\mathbf{n}_M (\mathbf{n}_M + 2)}}. \quad (6108)$$

Finally,

$$|\theta| = 2 \arcsin \left[\frac{\sinh(a/2R)}{\sqrt{\mathbf{n}_M (\mathbf{n}_M + 2)}} \right]. \quad (6109)$$

This is an exact geometric conversion between effective microscopic count and twist.

h. Theorem 177. Exact cell-count-twist correspondence. For the effective hyperbolic moiré domain, N_M^{eff} and $|\theta|$ are related bijectively through Eq. (6109) at fixed curvature and microscopic density. Thus any spectral condition selecting a particular twist automatically selects a unique effective moiré-cell count, and conversely. The magic condition is $X_\nu = 1$. Using Eq. (6057),

$$\gamma_\nu^{\text{eff},*} \left(\frac{\xi_{M,\nu}^*}{a} \right)^2 = \frac{\pi^2 d\alpha_\nu^*}{n_c^2} \frac{t}{w}. \quad (6110)$$

For a fixed active-scattering universality class, $\gamma_\nu^{\text{eff},*} = \gamma_\nu$, and therefore

$$\xi_{M,\nu}^* = \frac{\pi a}{n_c} \sqrt{\frac{d\alpha_\nu^*}{\gamma_\nu} \frac{t}{w}}. \quad (6111)$$

Introduce

$$\lambda_\nu^* = \frac{\xi_{M,\nu}^*}{R}. \quad (6112)$$

Then

$$\lambda_\nu^* = \frac{\pi a}{n_c R} \sqrt{\frac{d\alpha_\nu^*}{\gamma_\nu} \frac{t}{w}}. \quad (6113)$$

Substitution of Eq. (6112) into Eq. (6096) gives

$$N_{M,\nu}^{\text{eff},*} = \frac{4\pi n_b R^2}{A_0(K)} [\cosh(\lambda_\nu^*) - 1]. \quad (6114)$$

Thus

$$N_{M,\nu}^{\text{eff},*} = \frac{4\pi n_b R^2}{A_0(K)} \times \left\{ \cosh \left[\frac{\pi a}{n_c R} \sqrt{\frac{d\alpha_\nu^*}{\gamma_\nu} \frac{t}{w}} \right] - 1 \right\}. \quad (6115)$$

This is the direct magic-cell scaling law. The magic angle satisfies

$$\sinh \lambda_\nu^* = \frac{\sinh(a/2R)}{|\sin(\theta_\nu^*/2)|}. \quad (6116)$$

Therefore,

$$\left| \sin \left(\frac{\theta_\nu^*}{2} \right) \right| = \frac{\sinh(a/2R)}{\sinh \lambda_\nu^*}. \quad (6117)$$

Equations (6114) and (6117) provide a parametric representation of the complete geometry-to-magic correspondence. The same variable λ_ν^* determines both the spectral magic angle and the geometric magic-cell count.

i. Theorem 178. One-parameter magic geometry. Within a fixed active-scattering universality class, the pair $(\theta_\nu^*, N_{M,\nu}^{\text{eff},*})$ is parameterized exactly by

$$\lambda_\nu^* = \xi_{M,\nu}^*/R. \quad (6118)$$

Explicitly,

$$\left| \sin \left(\frac{\theta_\nu^*}{2} \right) \right| = \frac{\sinh(a/2R)}{\sinh \lambda_\nu^*}, \quad (6119)$$

and

$$N_{M,\nu}^{\text{eff},*} = \frac{4\pi n_b R^2}{A_0} (\cosh \lambda_\nu^* - 1). \quad (6120)$$

The geometric moiré-cell law and the spectral magic law are therefore two different projections of the same variable. For $\lambda_\nu^* \ll 1$, we have

$$\cosh \lambda_\nu^* - 1 = \frac{(\lambda_\nu^*)^2}{2} + \frac{(\lambda_\nu^*)^4}{24} + O[(\lambda_\nu^*)^6]. \quad (6121)$$

Hence,

$$N_{M,\nu}^{\text{eff},*} = \frac{2\pi n_b}{A_0} (\xi_{M,\nu}^*)^2 \left[1 + \frac{(\xi_{M,\nu}^*)^2}{12R^2} + O\left(\frac{\xi_{M,\nu}^{*4}}{R^4}\right) \right]. \quad (6122)$$

Using Eq. (6111),

$$N_{M,\nu}^{\text{eff},*} = \frac{2\pi^3 n_b a^2 d\alpha_\nu^*}{A_0 n_c^2 \gamma_\nu} \frac{t}{w} + O\left[|K| \left(\frac{t}{w} \right)^2 \right]. \quad (6123)$$

Thus

$$N_{M,\nu}^{\text{eff},*} \propto \left(\frac{w}{t} \right)^{-1} \quad (6124)$$

in the Euclidean regime. Since

$$\theta_\nu^* \propto \left(\frac{w}{t}\right)^{1/2}, \quad (6125)$$

we obtain

$$N_{M,\nu}^{\text{eff},*} \propto |\theta_\nu^*|^{-2}. \quad (6126)$$

For $\lambda_\nu^* \gg 1$, we have

$$\cosh \lambda_\nu^* - 1 = \frac{1}{2} e^{\lambda_\nu^*} \left[1 + O\left(e^{-\lambda_\nu^*}\right) \right]. \quad (6127)$$

Therefore,

$$N_{M,\nu}^{\text{eff},*} \sim \frac{2\pi n_b R^2}{A_0(K)} e^{\lambda_\nu^*}. \quad (6128)$$

Using Eq. (6113),

$$N_{M,\nu}^{\text{eff},*} \sim \frac{2\pi n_b R^2}{A_0(K)} \exp \left[\frac{\pi a}{n_c R} \sqrt{\frac{d\alpha_\nu^*}{\gamma_\nu} \frac{t}{w}} \right]. \quad (6129)$$

Thus the weak-coupling magic-cell count grows by an essential singularity at fixed negative curvature. At the same time,

$$|\theta_\nu^*| \sim 4 \sinh \left(\frac{a}{2R} \right) e^{-\lambda_\nu^*}. \quad (6130)$$

Eliminating λ_ν^* gives

$$N_{M,\nu}^{\text{eff},*} \sim \frac{8\pi n_b R^2 \sinh(a/2R)}{A_0(K)} \frac{1}{|\theta_\nu^*|}. \quad (6131)$$

Thus

$$N_{M,\nu}^{\text{eff},*} \propto |\theta_\nu^*|^{-1} \quad (6132)$$

in the curvature-dominated regime.

j. Theorem 179. Transmutation of magic-cell scaling. The effective microscopic count evaluated on the magic branch obeys

$$N_{M,\nu}^{\text{eff},*} \propto |\theta_\nu^*|^{-2} \quad (6133)$$

in the Euclidean regime and

$$N_{M,\nu}^{\text{eff},*} \propto |\theta_\nu^*|^{-1} \quad (6134)$$

in the curvature-dominated regime. Therefore the geometric exponent change $2 \rightarrow 1$ is inherited directly by

the spectral magic branch. We now sharpen the preceding endpoint laws into an exact running relation. At fixed curvature, define

$$\beta_M^* = -\frac{d \ln N_{M,\nu}^{\text{eff},*}}{d \ln s_\nu^*}. \quad (6135)$$

Because N_M^{eff} is the same geometric count defined earlier, its exact running exponent is

$$\beta_M = 1 + \frac{1}{\sqrt{1 + \chi^2}}. \quad (6136)$$

At the magic point,

$$\chi_\nu^* = \frac{\sinh(a/2R)}{s_\nu^*}. \quad (6137)$$

Using

$$\sinh \lambda_\nu^* = \chi_\nu^*, \quad (6138)$$

we obtain

$$\sqrt{1 + (\chi_\nu^*)^2} = \cosh \lambda_\nu^*. \quad (6139)$$

Therefore,

$$\beta_M^* = 1 + \text{sech } \lambda_\nu^*. \quad (6140)$$

This runs continuously from 2 to 1. The half-angle variable obeys

$$s = \sin \left(\frac{|\theta|}{2} \right). \quad (6141)$$

Hence,

$$\frac{d \ln s}{d \ln |\theta|} = \frac{|\theta|}{2} \cot \left(\frac{|\theta|}{2} \right). \quad (6142)$$

Therefore,

$$-\frac{d \ln N_{M,\nu}^{\text{eff},*}}{d \ln |\theta_\nu^*|} = \frac{|\theta_\nu^*|}{2} \cot \left(\frac{|\theta_\nu^*|}{2} \right) [1 + \text{sech } \lambda_\nu^*]. \quad (6143)$$

Equivalently,

$$-\frac{d \ln N_{M,\nu}^{\text{eff},*}}{d \ln |\theta_\nu^*|} = \frac{|\theta_\nu^*|}{2} \cot \left(\frac{|\theta_\nu^*|}{2} \right) \beta_M(\chi_\nu^*). \quad (6144)$$

For asymptotically small magic angles,

$$\frac{|\theta_\nu^*|}{2} \cot\left(\frac{|\theta_\nu^*|}{2}\right) \rightarrow 1. \quad (6145)$$

Hence,

$$-\frac{d \ln N_{M,\nu}^{\text{eff},*}}{d \ln |\theta_\nu^*|} \rightarrow \beta_M(\chi_\nu^*). \quad (6146)$$

k. Theorem 180. Exact geometric-magic exponent locking. The running exponent relating the effective moiré-cell count to the spectral magic angle is the geometric running moiré exponent itself, up to the exact finite-angle half-angle factor,

$$\beta_{\text{cell}}^{\text{magic}} = \frac{|\theta_\nu^*|}{2} \cot\left(\frac{|\theta_\nu^*|}{2}\right) \beta_M(\chi_\nu^*). \quad (6147)$$

In the small-angle limit, $\beta_{\text{cell}}^{\text{magic}} = \beta_M$. Thus the geometric crossover exponent is transferred directly into the spectral magic-branch scaling without introducing a new fitted exponent. The same result can be obtained independently by differentiating with respect to the microscopic coupling. For a general microscopic reference dispersion of order p , the magic condition gives

$$\xi_{M,\nu}^* \propto \omega^{-1/p}. \quad (6148)$$

Therefore,

$$\lambda_\nu^* \propto \omega^{-1/p}. \quad (6149)$$

Hence,

$$\frac{d\lambda_\nu^*}{d \ln \omega} = -\frac{\lambda_\nu^*}{p}. \quad (6150)$$

From

$$N_{M,\nu}^{\text{eff},*} \propto \cosh \lambda_\nu^* - 1, \quad (6151)$$

we obtain

$$-\frac{d \ln N_{M,\nu}^{\text{eff},*}}{d \ln \omega} = \frac{\lambda_\nu^*}{p} \frac{\sinh \lambda_\nu^*}{\cosh \lambda_\nu^* - 1}. \quad (6152)$$

Using

$$\frac{\sinh \lambda}{\cosh \lambda - 1} = \coth\left(\frac{\lambda}{2}\right), \quad (6153)$$

we obtain

Meanwhile,

$$-\frac{d \ln N_{M,\nu}^{\text{eff},*}}{d \ln \omega} = \frac{\lambda_\nu^*}{p} \coth\left(\frac{\lambda_\nu^*}{2}\right). \quad (6154)$$

$$s_\nu^* = \frac{\sinh(a/2R)}{\sinh \lambda_\nu^*}. \quad (6155)$$

Therefore,

$$\frac{d \ln s_\nu^*}{d \ln \omega} = \frac{\lambda_\nu^*}{p} \coth \lambda_\nu^*. \quad (6156)$$

Taking the ratio gives

$$-\frac{d \ln N_{M,\nu}^{\text{eff},*}}{d \ln s_\nu^*} = \frac{\coth(\lambda_\nu^*/2)}{\coth \lambda_\nu^*}. \quad (6157)$$

Using

$$\coth\left(\frac{\lambda}{2}\right) = \frac{\cosh \lambda + 1}{\sinh \lambda}, \quad (6158)$$

and

$$\coth \lambda = \frac{\cosh \lambda}{\sinh \lambda}, \quad (6159)$$

we obtain

$$-\frac{d \ln N_{M,\nu}^{\text{eff},*}}{d \ln s_\nu^*} = 1 + \text{sech } \lambda_\nu^*. \quad (6160)$$

The dispersion order p has cancelled exactly.

l. Theorem 181. Dispersion-order independence of the geometry-to-magic exponent. Although the microscopic dispersion order p controls the separate coupling laws

$$\xi_M^* \propto \omega^{-1/p}, \quad \theta_\nu^* = \theta_\nu^*(\omega), \quad (6161)$$

it cancels completely after the coupling parameter is eliminated between the geometric cell count and the magic angle. Therefore,

$$-\frac{d \ln N_{M,\nu}^{\text{eff},*}}{d \ln s_\nu^*} = 1 + \text{sech } \lambda_\nu^* \quad (6162)$$

for every dispersion order p satisfying the effective-coupling scaling assumptions. The relation between magic-cell scaling and magic-angle scaling is therefore

purely geometric. Define the Euclidean reference count associated with the same magic length,

$$N_{M,\nu}^{E,*} = \frac{2\pi n_b}{A_0} (\xi_{M,\nu}^*)^2. \quad (6163)$$

Under a density-matched comparison, define

$$\mathcal{N}(\lambda) = \frac{N_M^H}{N_M^E}. \quad (6164)$$

Using Eq. (6114),

$$\mathcal{N}(\lambda) = \frac{2(\cosh \lambda - 1)}{\lambda^2}. \quad (6165)$$

Define the Euclidean half-angle corresponding to the same magic length,

$$s_E^* = \frac{a}{2\xi_M^*} = \frac{\zeta}{\lambda}. \quad (6166)$$

The hyperbolic half-angle is

$$s_H^* = \frac{\sinh \zeta}{\sinh \lambda}. \quad (6167)$$

Therefore,

$$\mathcal{T}(\lambda, \zeta) = \frac{s_H^*}{s_E^*} = \frac{\sinh \zeta}{\zeta} \frac{\lambda}{\sinh \lambda}. \quad (6168)$$

The product of the cell and angle crossover amplitudes is

$$\mathcal{N} \mathcal{T}^2 = \frac{2(\cosh \lambda - 1)}{\lambda^2} \left(\frac{\sinh \zeta}{\zeta} \right)^2 \frac{\lambda^2}{\sinh^2 \lambda}. \quad (6169)$$

Using

$$\sinh^2 \lambda = (\cosh \lambda - 1)(\cosh \lambda + 1), \quad (6170)$$

we obtain

$$\mathcal{N}(\lambda) \mathcal{T}^2(\lambda, \zeta) = \left(\frac{\sinh \zeta}{\zeta} \right)^2 \text{sech}^2 \left(\frac{\lambda}{2} \right). \quad (6171)$$

In the weak microscopic-curvature limit,

$$\frac{\sinh \zeta}{\zeta} \rightarrow 1, \quad (6172)$$

and hence

$$\mathcal{N}(\lambda) \mathcal{T}^2(\lambda) = \text{sech}^2 \left(\frac{\lambda}{2} \right). \quad (6173)$$

m. Theorem 182. Parameter-free magic angle-cell amplitude locking. After the Euclidean reference density and magic length have been matched, the curvature-induced enhancement of the magic-cell count and the curvature-induced suppression of the magic half-angle are not independent. They satisfy the exact parametric identity

$$\mathcal{N} \mathcal{T}^2 = \left(\frac{\sinh \zeta}{\zeta} \right)^2 \text{sech}^2 \left(\frac{\lambda}{2} \right). \quad (6174)$$

Thus the cell-size crossover and the magic-angle crossover provide two independent observables of one underlying geometric variable. In the Euclidean regime, $\lambda_\nu^* \ll 1$. Using

$$\xi_{M,\nu}^* \simeq \frac{a}{|\theta_\nu^*|}, \quad (6175)$$

and

$$N_{M,\nu}^{\text{eff},*} \simeq \frac{2\pi n_b}{A_0} (\xi_{M,\nu}^*)^2, \quad (6176)$$

we obtain

$$N_{M,\nu}^{\text{eff},*} |\theta_\nu^*|^2 \rightarrow \frac{2\pi n_b a^2}{A_0}. \quad (6177)$$

In the curvature-dominated regime, $\lambda_\nu^* \gg 1$. Using Eqs. (6128) and (6130),

$$N_{M,\nu}^{\text{eff},*} |\theta_\nu^*| \rightarrow \frac{8\pi n_b R^2 \sinh(a/2R)}{A_0(K)}. \quad (6178)$$

Thus the endpoint scaling laws possess not only universal exponents but explicit geometric amplitudes. For an exact commensurate hyperbolic magic configuration, define the geometric-arithmetic locking factor

$$\mathcal{L}_M = \frac{q_M A_0}{A_M^{\text{eff}}}. \quad (6179)$$

The exact bilayer microscopic count satisfies

$$N_M^{\text{sc}} = \mathcal{L}_M N_M^{\text{eff}}. \quad (6180)$$

Therefore, on the magic branch,

$$N_{M,\nu}^{\text{sc},*} = \mathcal{L}_{M,\nu}^* N_{M,\nu}^{\text{eff},*}. \quad (6181)$$

Define

$$\delta_{\text{arith}} = -\frac{d \ln \mathcal{L}_M}{d \ln s}. \quad (6182)$$

Then

$$-\frac{d \ln N_M^{\text{sc}}}{d \ln s} = \beta_M + \delta_{\text{arith}}. \quad (6183)$$

Therefore the physical-angle magic exponent is

$$-\frac{d \ln N_{M,\nu}^{\text{sc},*}}{d \ln |\theta_\nu^*|} = \frac{|\theta_\nu^*|}{2} \cot \left(\frac{|\theta_\nu^*|}{2} \right) [\beta_M + \delta_{\text{arith}}]. \quad (6184)$$

If strong geometric–arithmetic locking holds,

$$\mathcal{L}_M \longrightarrow \mathcal{L}_*, \quad 0 < \mathcal{L}_* < \infty, \quad (6185)$$

then $\delta_{\text{arith}} \longrightarrow 0$. Consequently,

$$-\frac{d \ln N_{M,\nu}^{\text{sc},*}}{d \ln |\theta_\nu^*|} \longrightarrow \beta_M \quad (6186)$$

in the small-angle limit.

n. Theorem 183. Conditional transfer from geometric to exact magic-cell scaling. The exact crystallographic magic-cell exponent decomposes as

$$\beta_{\text{sc}}^{\text{magic}} = \beta_{\text{geom}}^{\text{magic}} + \delta_{\text{arith}} \quad (6187)$$

after the exact half-angle conversion has been included. Therefore the effective geometric magic-cell law becomes an exact crystallographic law only when the arithmetic locking contribution is asymptotically negligible. This condition is essential and cannot be removed from the exact hyperbolic claim. The exact common subgroup has index q_M . The quotient genus is

$$g_M = 1 + q_M(g_0 - 1). \quad (6188)$$

At a smooth irreducible rank- d_ρ representation point, the dimension of generalized momentum space is

$$D_\rho = 2q_M(g_0 - 1)d_\rho^2 + 2. \quad (6189)$$

Since

$$N_M^{\text{sc}} = 2n_b q_M, \quad (6190)$$

we may write

$$D_\rho = \frac{g_0 - 1}{n_b} d_\rho^2 N_M^{\text{sc}} + 2. \quad (6191)$$

Thus the exact magic-cell scaling simultaneously controls the dimensionality of the generalized momentum space over which flattening must be verified. Under geometric–arithmetic locking in the curvature-dominated regime,

$$N_{M,\nu}^{\text{sc},*} = \Theta(|\theta_\nu^*|^{-1}), \quad (6192)$$

The inverse-linear exact count is restricted to the bounded-locking class $\mathfrak{U}_1(1)$. For the explicit centered magic sequence one instead has $N_M^{\text{sc},*} = |\theta^*|^{-4+o(1)}$, and the corresponding subsequent density scale uses exponent four. It follows in the bounded-locking tier that

$$D_{\rho,\nu}^* = \Theta(d_\rho^2 |\theta_\nu^*|^{-1}). \quad (6193)$$

At fixed negative curvature and weak coupling,

$$N_{M,\nu}^{\text{sc},*} = \Theta \left[\exp \left(\frac{C_\nu}{\sqrt{\omega}} \right) \right], \quad (6194)$$

and hence

$$D_{\rho,\nu}^* = \Theta \left[d_\rho^2 \exp \left(\frac{C_\nu}{\sqrt{\omega}} \right) \right]. \quad (6195)$$

Thus the same curvature mechanism that produces an essential-singularity magic-angle trajectory can produce an exponentially growing exact representation-space complexity. We can now formulate the complete correspondence. Assume a reference dispersion of order p , $E_M \propto Q_M^p$, and a finite active tunneling scale. The effective coupling is $\alpha_{\text{eff}}^{(p)} \propto \omega \xi_M^p$. The magic condition $\alpha_{\text{eff}}^{(p)} = \alpha_\nu^*$ therefore selects

$$\xi_{M,\nu}^* \propto \omega^{-1/p}. \quad (6196)$$

The geometry then determines

$$\theta_\nu^* = \Theta_K(\xi_{M,\nu}^*), \quad (6197)$$

and

$$N_{M,\nu}^{\text{eff},*} = \mathcal{N}_K(\xi_{M,\nu}^*). \quad (6198)$$

For constant negative curvature,

$$\Theta_K(\xi) = 2 \arcsin \left[\frac{\sinh(a/2R)}{\sinh(\xi/R)} \right], \quad (6199)$$

and

$$\mathcal{N}_K(\xi) = \frac{4\pi n_b R^2}{A_0(K)} [\cosh(\xi/R) - 1]. \quad (6200)$$

Thus the spectral magic law is obtained by composing the dynamical effective-coupling condition with two purely geometric maps.

o. Theorem 184. Geometry-to-spectrum correspondence theorem. Suppose the magic condition selects a geometric length through $\xi_{M,\nu}^* = \xi_\nu^*(\omega)$. Then the magic angle and effective magic-cell count are not independent electronic observables. They are fixed by

$$\theta_\nu^* = \Theta_K [\xi_\nu^*(\omega)], \quad (6201)$$

and

$$N_{M,\nu}^{\text{eff},*} = \mathcal{N}_K [\xi_\nu^*(\omega)]. \quad (6202)$$

After eliminating ω ,

$$-\frac{d \ln N_{M,\nu}^{\text{eff},*}}{d \ln |\theta_\nu^*|} = \frac{|\theta_\nu^*|}{2} \cot \left(\frac{|\theta_\nu^*|}{2} \right) \beta_M (\chi_\nu^*). \quad (6203)$$

Therefore the Euclidean-to-hyperbolic geometric exponent crossover $2 \rightarrow 1$ is inherited directly by the scaling relation between the magic-cell count and the magic angle. The result is independent of the microscopic dispersion order p ; that order controls the parameterization of the magic trajectory by w/t , but it cancels when the geometric cell count is related directly to the magic angle. The correspondence can be tested without fitting independent exponents. For every operational magic point, calculate $\xi_{M,\nu}^*$, $N_{M,\nu}^{\text{eff},*}$, θ_ν^* and X_ν^* . The first consistency condition is $X_\nu^* \simeq 1$. The second is the magic-length relation

$$\gamma_\nu^{\text{eff},*} \left(\frac{\xi_{M,\nu}^*}{a} \right)^2 \simeq \frac{\pi^2 d \alpha_\nu^* t}{n_c^2 w}. \quad (6204)$$

The third is the exact cell relation

$$N_{M,\nu}^{\text{eff},*} = \frac{4\pi n_b R^2}{A_0} \left[\cosh \left(\frac{\xi_{M,\nu}^*}{R} \right) - 1 \right]. \quad (6205)$$

The fourth is the angle relation

$$\left| \sin \left(\frac{\theta_\nu^*}{2} \right) \right| = \frac{\sinh(a/2R)}{\sinh(\xi_{M,\nu}^*/R)}. \quad (6206)$$

These four equations contain no independently fitted magic-cell exponent. For adjacent magic points along a fixed-curvature branch, define the finite-difference estimator

$$\beta_{\text{cell},ij}^{\text{num}} = -\frac{\ln [N_{M,j}^{\text{eff},*}/N_{M,i}^{\text{eff},*}]}{\ln [|\theta_j^*|/|\theta_i^*|]}. \quad (6207)$$

The predicted value at the midpoint is

$$\beta_{\text{cell}}^{\text{pred}} = \frac{|\theta^*|}{2} \cot \left(\frac{|\theta^*|}{2} \right) \left[1 + \text{sech} \left(\frac{\xi_M^*}{R} \right) \right]. \quad (6208)$$

Define the exponent residual

$$\mathcal{R}_\beta = \beta_{\text{cell}}^{\text{num}} - \beta_{\text{cell}}^{\text{pred}}. \quad (6209)$$

The geometry-to-spectrum correspondence requires $\mathcal{R}_\beta \rightarrow 0$ under increasing spectral and finite-cover accuracy. For exact crystallographic counts, the corresponding residual must include the independently measured arithmetic correction δ_{arith} . The complete result can now be stated in a compact mathematical sequence. The local geometry determines

$$\xi_M = R \text{arsinh} \left[\frac{\sinh(a/2R)}{|\sin(\theta/2)|} \right]. \quad (6210)$$

The same length determines the kinetic scale,

$$E_M^{\text{eff}} = \frac{\pi^2 d t a^2}{n_c^2 \xi_M^2}, \quad (6211)$$

and the effective coupling,

$$\alpha_\nu^{\text{eff}} = \frac{\gamma_\nu^{\text{eff}} n_c^2 w}{\pi^2 d} \left(\frac{\xi_M}{a} \right)^2. \quad (6212)$$

The spectral magic condition selects $\alpha_\nu^{\text{eff}} = \alpha_\nu^*$. This fixes $\xi_{M,\nu}^*$. The same geometric length then determines both

$$\theta_\nu^* = 2 \arcsin \left[\frac{\sinh(a/2R)}{\sinh(\xi_{M,\nu}^*/R)} \right] \quad (6213)$$

and

$$N_{M,\nu}^{\text{eff},*} = \frac{4\pi n_b R^2}{A_0} \left[\cosh \left(\frac{\xi_{M,\nu}^*}{R} \right) - 1 \right]. \quad (6214)$$

Consequently, the geometric running exponent and the magic-branch running exponent satisfy

$$-\frac{d \ln N_{M,\nu}^{\text{eff},*}}{d \ln |\theta_\nu^*|} = \frac{|\theta_\nu^*|}{2} \cot \left(\frac{|\theta_\nu^*|}{2} \right) \left[1 + \text{sech} \left(\frac{\xi_{M,\nu}^*}{R} \right) \right]. \quad (6215)$$

At asymptotically small angle this becomes

$$-\frac{d \ln N_{M,\nu}^{\text{eff},*}}{d \ln |\theta_\nu^*|} = 1 + \text{sech} \left(\frac{\xi_{M,\nu}^*}{R} \right), \quad (6216)$$

which flows continuously from 2 to 1. The geometry-to-spectrum correspondence is therefore stronger than the statement that curvature modifies both geometry and electronic bands. The same geometric variable controls the moiré length, the effective cell count, the kinetic-energy scale, the hybridization ratio, the bandwidth scaling, and the magic-branch trajectory. Within the effective-coupling universality class, the spectral magic law is the image of the geometric moiré law under the single condition $\alpha_\nu^{\text{eff}} = \alpha_\nu^*$. For exact hyperbolic crystallographic supercells, this correspondence is promoted from the effective count to the literal coincidence-cell count only after the independently defined geometric–arithmetic locking factor has been shown to remain asymptotically controlled.

B. Curvature-Controlled Universality Classes

The geometry-to-spectrum correspondence derived above allows us to formulate the Euclidean and negatively curved moiré problems as two distinct asymptotic scaling classes. We use the term universality class in a precise asymptotic sense. We do not infer a thermodynamic phase transition at a finite value of curvature or twist. Instead, we identify the asymptotic fixed points of the exact curvature-controlled scaling flow and determine which geometric and spectral exponents remain invariant under finite changes of microscopic registry conventions, microscopic density, and smooth tunneling parameters. The central distinction is generated by the competition between the Euclidean moiré scale and the curvature radius. The exact geometric crossover variable is

$$\chi = \frac{\sinh(a/2R)}{|\sin(\theta/2)|}, \quad K = -\frac{1}{R^2}. \quad (6217)$$

The exact local moiré length is

$$\xi_M = R \operatorname{arsinh} \chi. \quad (6218)$$

The exact effective moiré area is

$$A_M^{\text{eff}} = 2\pi R^2 \left(\sqrt{1 + \chi^2} - 1 \right). \quad (6219)$$

The area-normalized effective microscopic count is

$$N_M^{\text{eff}} = 2n_b \frac{A_M^{\text{eff}}}{A_0(K)}. \quad (6220)$$

These equations contain both asymptotic classes and the complete crossover between them. Rationalizing Eq. (6219) gives

$$A_M^{\text{eff}} = 2\pi R^2 \frac{\chi^2}{1 + \sqrt{1 + \chi^2}}. \quad (6221)$$

We therefore define

$$F(\chi) = \frac{2}{1 + \sqrt{1 + \chi^2}}. \quad (6222)$$

Then

$$A_M^{\text{eff}} = \pi R^2 \chi^2 F(\chi). \quad (6223)$$

The function satisfies $F(0) = 1$, and

$$\lim_{\chi \rightarrow \infty} F(\chi) = 0. \quad (6224)$$

Differentiation gives

$$\frac{dF}{d\chi} = -\frac{2\chi}{\sqrt{1 + \chi^2} \left[1 + \sqrt{1 + \chi^2} \right]^2}. \quad (6225)$$

Hence

$$\frac{dF}{d\chi} < 0 \quad (6226)$$

for every $\chi > 0$. Thus the geometric crossover is strictly monotone. Introduce the logarithmic crossover coordinate $\ell = \ln \chi$. Then

$$\frac{dF}{d\ell} = \chi \frac{dF}{d\chi}. \quad (6227)$$

The explicit function can be inverted. From

$$\frac{2}{F} = 1 + \sqrt{1 + \chi^2}, \quad (6228)$$

we obtain

$$\sqrt{1 + \chi^2} = \frac{2}{F} - 1. \quad (6229)$$

Squaring gives

$$1 + \chi^2 = \left(\frac{2}{F} - 1 \right)^2. \quad (6230)$$

Therefore,

$$\chi^2 = \frac{4(1 - F)}{F^2}. \quad (6231)$$

Substitution into Eq. (6227) gives the closed autonomous equation

$$\frac{dF}{d\ell} = -\frac{2F(1 - F)}{2 - F}. \quad (6232)$$

Its fixed points satisfy $F(1 - F) = 0$. Therefore there are exactly two endpoint fixed points, $F_E^* = 1$, and $F_H^* = 0$. There is no additional fixed point satisfying $0 < F^* < 1$. Thus the complete one-parameter geometric flow connects exactly two asymptotic classes.

a. Theorem 185. Two-fixed-point structure of curvature-controlled moiré scaling. The exact normalized moiré-area amplitude obeys the autonomous flow

$$\frac{dF}{d \ln \chi} = -\frac{2F(1-F)}{2-F}, \quad (6233)$$

with precisely two endpoint fixed points, $F_E^* = 1$, $F_H^* = 0$. The first defines the Euclidean moiré scaling class. The second defines the curvature-dominated hyperbolic moiré scaling class. Every finite nonzero value of χ belongs to the universal crossover trajectory connecting these two fixed points rather than defining a third asymptotic fixed-point class. Set

$$F = 1 - \epsilon, \quad 0 < \epsilon \ll 1. \quad (6234)$$

Substitution into Eq. (6232) gives

$$-\frac{d\epsilon}{d\ell} = -\frac{2(1-\epsilon)\epsilon}{1+\epsilon}. \quad (6235)$$

Therefore,

$$\frac{d\epsilon}{d\ell} = 2\epsilon - 4\epsilon^2 + O(\epsilon^3). \quad (6236)$$

The leading flow is

$$\frac{d\epsilon}{d\ell} = 2\epsilon + O(\epsilon^2). \quad (6237)$$

Hence

$$\epsilon \propto e^{2\ell} = \chi^2. \quad (6238)$$

Indeed, the exact expansion gives

$$F(\chi) = 1 - \frac{\chi^2}{4} + \frac{\chi^4}{8} + O(\chi^6). \quad (6239)$$

Thus negative curvature enters the Euclidean fixed point through a quadratically relevant scaling field. For $F \ll 1$, Eq. (6232) gives

$$\frac{dF}{d\ell} = -F + \frac{1}{2}F^2 + O(F^3). \quad (6240)$$

Therefore,

$$\frac{dF}{d\ell} = -F + O(F^2). \quad (6241)$$

Hence

$$F \propto e^{-\ell} = \chi^{-1}. \quad (6242)$$

The explicit asymptotic expansion is

$$F(\chi) = \frac{2}{\chi} - \frac{2}{\chi^2} + O(\chi^{-3}). \quad (6243)$$

Thus the curvature-dominated fixed point is asymptotically stable as the twist is reduced at fixed negative curvature.

b. Theorem 186. Fixed-point stability and relevance of curvature. Along increasing $\ell = \ln \chi$, the Euclidean fixed point possesses linear eigenvalue $y_E = 2$, whereas the hyperbolic fixed point possesses linear eigenvalue $y_H = -1$. Therefore negative curvature is a relevant perturbation of the Euclidean small-angle scaling fixed point, while the curvature-dominated fixed point is stable under further reduction of the twist. Define the running effective-area exponent by

$$\beta_M = -\frac{d \ln A_M^{\text{eff}}}{d \ln s}, \quad s = \left| \sin \left(\frac{\theta}{2} \right) \right|. \quad (6244)$$

Because

$$\chi = \frac{\sinh(a/2R)}{s}, \quad (6245)$$

we have

$$\frac{d \ln \chi}{d \ln s} = -1. \quad (6246)$$

Direct differentiation of Eq. (6219) gives

$$\beta_M(\chi) = 1 + \frac{1}{\sqrt{1+\chi^2}}. \quad (6247)$$

Using Eq. (6229),

$$\frac{1}{\sqrt{1+\chi^2}} = \frac{F}{2-F}. \quad (6248)$$

Therefore,

$$\beta_M = \frac{2}{2-F}. \quad (6249)$$

At the two fixed points, $\beta_{M,E}^* = 2$, and $\beta_{M,H}^* = 1$. Near the Euclidean fixed point,

$$\beta_M = 2 - \frac{\chi^2}{2} + O(\chi^4). \quad (6250)$$

Near the hyperbolic fixed point,

$$\beta_M = 1 + \frac{1}{\chi} + O(\chi^{-3}). \quad (6251)$$

Define

$$\nu_\xi = -\frac{d \ln \xi_M}{d \ln s}. \quad (6252)$$

Using

$$\xi_M = R \operatorname{arsinh} \chi, \quad (6253)$$

we obtain

$$\frac{d \ln \xi_M}{d \ln s} = -\frac{\chi}{\sqrt{1 + \chi^2} \operatorname{arsinh} \chi}. \quad (6254)$$

Hence

$$\nu_\xi(\chi) = \frac{\chi}{\sqrt{1 + \chi^2} \operatorname{arsinh} \chi}. \quad (6255)$$

At the Euclidean endpoint, $\nu_{\xi,E}^* = 1$. At the hyperbolic endpoint, $\nu_{\xi,H}^* = 0$. Thus the two classes possess different length-scaling exponents as well as different area/count exponents. We now define the Euclidean class by the asymptotic basin $\chi \rightarrow 0$. In this regime,

$$\operatorname{arsinh} \chi = \chi - \frac{\chi^3}{6} + O(\chi^5). \quad (6256)$$

For weak microscopic curvature,

$$R\chi = \frac{a}{2s} + O(R^{-2}). \quad (6257)$$

Therefore,

$$\xi_M^{(E)} \sim \frac{a}{2s} \sim \frac{a}{|\theta|}. \quad (6258)$$

The effective area satisfies

$$A_M^{\text{eff},E} \sim \pi \left(\xi_M^{(E)} \right)^2, \quad (6259)$$

and hence

$$A_M^{\text{eff},E} \sim \frac{\pi a^2}{|\theta|^2}. \quad (6260)$$

For a fixed microscopic density,

$$\rho_0 = \frac{n_b}{A_0}, \quad (6261)$$

the effective bilayer count therefore satisfies

$$N_M^{\text{eff},E} \propto |\theta|^{-2}. \quad (6262)$$

The class fingerprint is consequently

$$(\nu_\xi^*, \beta_M^*)_E = (1, 2). \quad (6263)$$

c. Definition. Euclidean moiré universality class. We call a small-angle scaling sequence Euclidean when $\chi \rightarrow 0$, so that

$$\xi_M = \Theta(|\theta|^{-1}), \quad N_M^{\text{eff}} = \Theta(|\theta|^{-2}). \quad (6264)$$

Finite multiplicative changes in the microscopic registry threshold or microscopic density change only amplitudes and crossover positions and do not change this class. We define the curvature-dominated basin by $\chi \rightarrow \infty$. Then

$$\operatorname{arsinh} \chi = \ln(2\chi) + O(\chi^{-2}). \quad (6265)$$

Therefore,

$$\xi_M = R \ln \left[\frac{2 \sinh(a/2R)}{s} \right] + o(R). \quad (6266)$$

At small twist,

$$s = \frac{|\theta|}{2} + O(|\theta|^3), \quad (6267)$$

so that

$$\xi_M^{(H)} = R \ln \left[\frac{4 \sinh(a/2R)}{|\theta|} \right] + o(R). \quad (6268)$$

The large-radius hyperbolic area law is

$$A_{\mathbb{H}}(r) \sim \pi R^2 e^{r/R}. \quad (6269)$$

Therefore,

$$A_M^{\text{eff},H} \sim \pi R^2 \exp \left(\frac{\xi_M}{R} \right). \quad (6270)$$

Using Eq. (6268),

$$A_M^{\text{eff},H} \sim \frac{4\pi R^2 \sinh(a/2R)}{|\theta|}. \quad (6271)$$

Thus

$$N_M^{\text{eff},H} \propto |\theta|^{-1} \quad (6272)$$

at fixed negative curvature and fixed microscopic quotient density. The class fingerprint is

$$(\nu_\xi^*, \beta_M^*)_H = (0, 1). \quad (6273)$$

d. Definition. Hyperbolic moiré universality class. We call a small-angle scaling sequence hyperbolic when $\chi \rightarrow \infty$, so that

$$\xi_M = R \ln |\theta|^{-1} + O(R), \quad (6274)$$

and

$$N_M^{\text{eff}} = \Theta(|\theta|^{-1}). \quad (6275)$$

The logarithmic length divergence and inverse-linear effective-cell growth are the defining endpoint scaling laws of this class.

e. Theorem 187. Curvature changes the geometric moiré universality class. The Euclidean and hyperbolic endpoint classes possess inequivalent scaling fingerprints,

$$(\nu_\xi, \beta_M)_E = (1, 2), \quad (6276)$$

whereas

$$(\nu_\xi, \beta_M)_H = (0, 1). \quad (6277)$$

Therefore negative curvature changes both the functional divergence of the local moiré length and the power-law growth of the effective moiré cell. The two endpoint theories cannot be related by a finite multiplicative renormalization of length or area. At fixed negative curvature,

$$\xi_M^{(H)} \sim R \ln |\theta|^{-1}, \quad (6278)$$

whereas $\xi_M^{(E)} \sim a|\theta|^{-1}$. Therefore,

$$\frac{\xi_M^{(H)}}{\xi_M^{(E)}} \sim \frac{R}{a} |\theta| \ln |\theta|^{-1}. \quad (6279)$$

Since

$$\lim_{x \rightarrow 0^+} x \ln \left(\frac{1}{x} \right) = 0, \quad (6280)$$

we obtain

$$\lim_{|\theta| \rightarrow 0} \frac{\xi_M^{(H)}}{\xi_M^{(E)}} = 0. \quad (6281)$$

Similarly,

$$\frac{N_M^{\text{eff},H}}{N_M^{\text{eff},E}} = O(|\theta|), \quad (6282)$$

and hence

$$\lim_{|\theta| \rightarrow 0} \frac{N_M^{\text{eff},H}}{N_M^{\text{eff},E}} = 0. \quad (6283)$$

No finite nonzero constants C_ξ and C_N can therefore satisfy

$$\xi_M^{(H)} \sim C_\xi \xi_M^{(E)}, \quad N_M^{\text{eff},H} \sim C_N N_M^{\text{eff},E} \quad (6284)$$

in the small-angle limit. Thus the distinction is an exponent and functional-form change rather than a change of prefactor. In the weak-microscopic-curvature and small-angle regime,

$$\sinh \left(\frac{a}{2R} \right) = \frac{a}{2R} + O \left(\frac{a^3}{R^3} \right), \quad (6285)$$

and

$$s = \frac{|\theta|}{2} + O(|\theta|^3). \quad (6286)$$

Hence

$$\chi = \frac{a}{R|\theta|} \left[1 + O \left(\frac{a^2}{R^2} \right) + O(\theta^2) \right]. \quad (6287)$$

Because $\frac{1}{R} = \sqrt{|K|}$, we obtain

$$\chi \sim \frac{a\sqrt{|K|}}{|\theta|}. \quad (6288)$$

It is therefore natural to define the Euclidean curvature scaling field

$$g_K = \chi^2 \sim \frac{a^2|K|}{\theta^2}. \quad (6289)$$

Near the Euclidean fixed point,

$$1 - F = \frac{g_K}{4} + O(g_K^2). \quad (6290)$$

Thus g_K is the relevant perturbation that drives the system away from Euclidean scaling. Consider any path satisfying $|K| \rightarrow 0$, $|\theta| \rightarrow 0$. The asymptotic class is determined by

$$g_K = \frac{a^2|K|}{\theta^2}. \quad (6291)$$

If $g_K \rightarrow 0$, then $\chi \rightarrow 0$ and the path belongs to the Euclidean universality class. If $g_K \rightarrow \infty$, then $\chi \rightarrow \infty$ and the path belongs to the hyperbolic universality class. If

$$g_K \rightarrow g_0, \quad 0 < g_0 < \infty, \quad (6292)$$

then the path remains on the universal crossover trajectory and reaches neither endpoint fixed point.

f. Theorem 188. Basin criterion for the moiré universality class. The asymptotic geometry is classified completely by the dimensionless ratio

$$g_K = a^2|K|/\theta^2. \quad (6293)$$

The three asymptotic possibilities are

$$g_K \rightarrow 0 \Rightarrow \mathcal{U}_E, \quad g_K \rightarrow \infty \Rightarrow \mathcal{U}_H, \quad (6294)$$

and

$$0 < g_K \rightarrow g_0 < \infty \Rightarrow \text{universal crossover}. \quad (6295)$$

Thus $K = 0$ and $K < 0$ are not the only meaningful statements. The class is selected by the relative asymptotic rate at which curvature and twist approach zero. Suppose

$$a^2|K| \sim C|\theta|^\mu, \quad C > 0. \quad (6296)$$

Then

$$g_K \sim C|\theta|^{\mu-2}. \quad (6297)$$

Therefore, $\mu > 2 \Rightarrow \mathcal{U}_E$, whereas $\mu < 2 \Rightarrow \mathcal{U}_H$. The marginal scaling $\mu = 2$ retains a finite crossover coordinate. Thus the Euclidean fixed point has crossover scaling dimension two with respect to the twist. For every fixed nonzero twist,

$$\lim_{K \rightarrow 0^-} \chi = 0. \quad (6298)$$

Therefore,

$$\lim_{K \rightarrow 0^-} F = 1, \quad \lim_{K \rightarrow 0^-} \beta_M = 2. \quad (6299)$$

Taking the twist to zero afterwards gives

$$\lim_{|\theta| \rightarrow 0} \lim_{K \rightarrow 0^-} \beta_M = 2. \quad (6300)$$

By contrast, at every fixed negative curvature,

$$\lim_{|\theta| \rightarrow 0} \chi = \infty. \quad (6301)$$

Hence

$$\lim_{|\theta| \rightarrow 0} \beta_M = 1. \quad (6302)$$

Taking $K \rightarrow 0^-$ afterwards gives

$$\lim_{K \rightarrow 0^-} \lim_{|\theta| \rightarrow 0} \beta_M = 1. \quad (6303)$$

Therefore,

$$\lim_{|\theta| \rightarrow 0} \lim_{K \rightarrow 0^-} \beta_M \neq \lim_{K \rightarrow 0^-} \lim_{|\theta| \rightarrow 0} \beta_M. \quad (6304)$$

The same occurs for the local-length exponent,

$$\lim_{|\theta| \rightarrow 0} \lim_{K \rightarrow 0^-} \nu_\xi = 1, \quad (6305)$$

while

$$\lim_{K \rightarrow 0^-} \lim_{|\theta| \rightarrow 0} \nu_\xi = 0. \quad (6306)$$

Likewise,

$$\lim_{|\theta| \rightarrow 0} \lim_{K \rightarrow 0^-} F = 1, \quad (6307)$$

whereas

$$\lim_{K \rightarrow 0^-} \lim_{|\theta| \rightarrow 0} F = 0. \quad (6308)$$

g. Theorem 189. Singular asymptotic role of zero curvature. The metric and all finite-twist observables recover the Euclidean limit continuously as $K \rightarrow 0$. Nevertheless, the small-angle scaling fixed point is singular in the order of limits:

$$(K \rightarrow 0) \circ (\theta \rightarrow 0) \neq (\theta \rightarrow 0) \circ (K \rightarrow 0). \quad (6309)$$

Thus the curvature-controlled universality-class distinction is an asymptotic singularity rather than a finite-parameter discontinuity. For every finite $0 < \chi < \infty$

∞ , the functions $F(\chi)$, $\xi_M(\chi)$, $A_M^{\text{eff}}(\chi)$, $\beta_M(\chi)$ are smooth. Furthermore,

$$\frac{dF}{d\chi} < 0 \quad (6310)$$

and no finite- χ fixed point exists. Therefore no singular critical angle follows from the purely geometric sector. We consequently use the phrase universality-class change to mean that the two asymptotic endpoints possess inequivalent fixed-point exponents and functional scaling laws. We do not identify the finite crossover region itself as a thermodynamic phase boundary. Let the microscopic registry threshold be an arbitrary fixed positive length $\ell_* > 0$. The exact crossover variable becomes

$$\chi_* = \frac{\sinh(\ell_*/2R)}{s}. \quad (6311)$$

The local length becomes

$$\xi_{M,*} = R \operatorname{arsinh} \chi_*. \quad (6312)$$

The effective area is

$$A_{M,*}^{\text{eff}} = 2\pi R^2 \left[\sqrt{1 + \chi_*^2} - 1 \right]. \quad (6313)$$

For $\chi_* \ll 1$, we obtain

$$A_{M,*}^{\text{eff}} \sim \frac{\pi \ell_*^2}{4s^2}, \quad (6314)$$

and hence $\beta_M \rightarrow 2$. For $\chi_* \gg 1$, we obtain

$$A_{M,*}^{\text{eff}} \sim \frac{2\pi R^2 \sinh(\ell_*/2R)}{s}, \quad (6315)$$

and therefore $\beta_M \rightarrow 1$. Thus replacing a by a different finite positive registry convention changes amplitudes but leaves both fixed-point exponents invariant.

h. Theorem 190. Registry-threshold universality. Any fixed, finite, positive, twist-independent registry threshold produces the same endpoint classes, $(\nu_\xi, \beta_M)_E = (1, 2)$, and $(\nu_\xi, \beta_M)_H = (0, 1)$. The registry threshold is therefore a nonuniversal metric parameter rather than a class-defining variable. We now include the electronic problem. Suppose the relevant microscopic reference dispersion begins at order p ,

$$\varepsilon(\mathbf{k}_0 + \mathbf{q}) - \varepsilon(\mathbf{k}_0) = c_p |\mathbf{q}|^p + o(|\mathbf{q}|^p). \quad (6316)$$

The geometry-controlled kinetic scale is

$$E_{M,p}^{\text{eff}} = c_p (Q_M^{\text{eff}})^p. \quad (6317)$$

Because $Q_M^{\text{eff}} \propto \xi_M^{-1}$, we have $E_{M,p}^{\text{eff}} \propto \xi_M^{-p}$. Let $w_\nu^{\text{eff}} = \gamma_\nu^{\text{eff}} w$. Then

$$\alpha_\nu^{\text{eff}} = \frac{w_\nu^{\text{eff}}}{E_{M,p}^{\text{eff}}} \propto \gamma_\nu^{\text{eff}} \omega \xi_M^p. \quad (6318)$$

Define the running tunneling-profile exponent by

$$\eta_{\gamma,\nu} = -\frac{d \ln \gamma_\nu^{\text{eff}}}{d \ln s}. \quad (6319)$$

Define

$$\beta_{X,\nu} = -\frac{d \ln \alpha_\nu^{\text{eff}}}{d \ln s}. \quad (6320)$$

Using

$$\nu_\xi = -\frac{d \ln \xi_M}{d \ln s}, \quad (6321)$$

we obtain the exact composition law

$$\beta_{X,\nu} = p\nu_\xi + \eta_{\gamma,\nu}. \quad (6322)$$

If the normalized active tunneling approaches a finite nonzero limit, $0 < \gamma_{\nu,*} < \infty$, with $\eta_{\gamma,\nu} \rightarrow 0$, then $\beta_{X,E}^* = p$, whereas $\beta_{X,H}^* = 0$.

i. Theorem 191. Spectral inheritance of the geometric fixed points. For a regular active tunneling profile, $\eta_\gamma \rightarrow 0$, the effective-coupling exponent is determined entirely by the geometric length exponent, $\beta_X = p\nu_\xi$. Therefore the Euclidean class has an algebraically divergent effective coupling,

$$\alpha_{\text{eff}} \propto \omega |\theta|^{-p}, \quad (6323)$$

whereas the hyperbolic class has only logarithmic growth,

$$\alpha_{\text{eff}} \propto \omega \ln^p |\theta|^{-1}. \quad (6324)$$

Thus the geometric universality-class change is inherited by the spectral control variable before any explicit diagonalization is performed. Equation (6322) gives an explicit criterion for whether a tunneling profile changes the asymptotic spectral class. If $\eta_{\gamma,\nu} \rightarrow 0$, the tunneling profile changes amplitudes and the numerical magic root but does not change the endpoint effective-coupling exponent. If instead

$$\eta_{\gamma,\nu} \rightarrow \eta_{\gamma,\nu}^* \neq 0, \quad (6325)$$

then

$$\beta_{X,\nu}^* = p\nu_\xi^* + \eta_{\gamma,\nu}^*. \quad (6326)$$

Such a tunneling deformation is asymptotically relevant and defines a different spectral scaling class. Thus profile universality requires a finite normalized active scattering limit rather than merely smoothness of the microscopic kernel at one distance. Suppose $\alpha_\nu^{\text{eff}} = \alpha_\nu^*$ selects the ν -th magic branch. In the Euclidean class,

$$\alpha_\nu^{\text{eff}} \propto \omega |\theta|^{-p}. \quad (6327)$$

Therefore,

$$|\theta_\nu^*| \propto \omega^{1/p}. \quad (6328)$$

For the quadratic square-lattice reference, $p = 2$, and hence

$$|\theta_\nu^*| \propto \sqrt{\frac{w}{t}}. \quad (6329)$$

The selected magic length satisfies

$$\xi_{M,\nu}^* \propto \omega^{-1/p}. \quad (6330)$$

In two dimensions,

$$N_{M,\nu}^{\text{eff},*} \propto (\xi_{M,\nu}^*)^2, \quad (6331)$$

so that

$$N_{M,\nu}^{\text{eff},*} \propto \omega^{-2/p}. \quad (6332)$$

For $p = 2$,

$$N_{M,\nu}^{\text{eff},*} \propto \left(\frac{w}{t}\right)^{-1}. \quad (6333)$$

At fixed negative curvature,

$$\alpha_\nu^{\text{eff}} \propto \omega R^p \ln^p |\theta|^{-1}, \quad (6334)$$

up to fixed microscopic length factors. The magic condition therefore requires

$$\ln |\theta_\nu^*|^{-1} = \Theta \left(\omega^{-1/p} \right). \quad (6335)$$

Consequently,

$$|\theta_\nu^*| = \exp \left[-\Theta \left(\omega^{-1/p} \right) \right]. \quad (6336)$$

For $p = 2$,

$$|\theta_\nu^*| \sim 4 \sinh \left(\frac{a}{2R} \right) \exp \left[-\frac{\pi a}{n_c R} \sqrt{\frac{d\alpha_\nu^*}{\gamma_\nu} \frac{t}{w}} \right]. \quad (6337)$$

At the same time,

$$N_{M,\nu}^{\text{eff},*} \sim \frac{2\pi n_b R^2}{A_0} \exp \left(\frac{\xi_{M,\nu}^*}{R} \right). \quad (6338)$$

Thus

$$N_{M,\nu}^{\text{eff},*} = \exp \left[\Theta \left(\omega^{-1/p} \right) \right]. \quad (6339)$$

For $p = 2$,

$$N_{M,\nu}^{\text{eff},*} = \exp \left[\Theta \left(\sqrt{\frac{t}{w}} \right) \right]. \quad (6340)$$

j. Theorem 192. Curvature changes the magic-band scaling class. For a regular active tunneling profile and a finite dimensionless magic root, the Euclidean magic branch obeys the algebraic law $|\theta_\nu^*| \propto \omega^{1/p}$, whereas the fixed-negative-curvature magic branch obeys

$$|\theta_\nu^*| = \exp \left[-\Theta \left(\omega^{-1/p} \right) \right]. \quad (6341)$$

The two spectral laws cannot be mapped into one another by a finite renormalization of the magic-angle prefactor. Negative curvature therefore changes the asymptotic functional class of the magic trajectory from algebraic to essential. For $p = 2$, the Euclidean angle behaves as

$$\theta_{\nu,E}^* = C_{E,\nu} \omega^{1/2} [1 + o(1)], \quad (6342)$$

while the hyperbolic branch behaves as

$$\theta_{\nu,H}^* = C_H \exp \left[-C_\nu \omega^{-1/2} \right] [1 + o(1)]. \quad (6343)$$

Therefore,

$$\frac{\theta_{\nu,H}^*}{\theta_{\nu,E}^*} = \frac{C_H}{C_{E,\nu}} \omega^{-1/2} \exp \left[-C_\nu \omega^{-1/2} \right] [1 + o(1)]. \quad (6344)$$

For every $C_\nu > 0$, we have

$$\lim_{\omega \rightarrow 0^+} \frac{\theta_{\nu,H}^*}{\theta_{\nu,E}^*} = 0. \quad (6345)$$

Thus no finite multiplicative magic-angle renormalization connects the two endpoint theories. The natural magic crossover coordinate is the ratio

$$\lambda_\nu^* = \frac{\xi_{M,\nu}^*}{R}. \quad (6346)$$

For a general dispersion order p ,

$$\xi_{M,\nu}^* = C_{p,\nu} a \omega^{-1/p}, \quad (6347)$$

where $C_{p,\nu}$ is dimensionless. Hence

$$\lambda_\nu^* = C_{p,\nu} a \sqrt{|K|} \omega^{-1/p}. \quad (6348)$$

Define $g_\nu^{\text{magic}} = (\lambda_\nu^*)^2$. Then

$$g_\nu^{\text{magic}} = C_{p,\nu}^2 a^2 |K| \omega^{-2/p}. \quad (6349)$$

For the quadratic square-lattice branch,

$$C_{2,\nu} = \frac{\pi}{n_c} \sqrt{\frac{d\alpha_\nu^*}{\gamma_\nu}}, \quad (6350)$$

and hence

$$g_\nu^{\text{magic}} = \frac{\pi^2 d\alpha_\nu^*}{n_c^2 \gamma_\nu} \frac{a^2 |K|}{\omega}. \quad (6351)$$

If $g_\nu^{\text{magic}} \rightarrow 0$, then $\lambda_\nu^* \rightarrow 0$ and the magic branch belongs to the Euclidean spectral class. If $g_\nu^{\text{magic}} \rightarrow \infty$, then $\lambda_\nu^* \rightarrow \infty$ and the branch belongs to the hyperbolic spectral class. If

$$0 < g_\nu^{\text{magic}} \rightarrow g_{\nu,0} < \infty, \quad (6352)$$

the branch remains in the universal crossover regime.

k. Theorem 193. Magic-band universality is controlled by $a^2 |K| \omega^{-2/p}$. For a reference dispersion of order p , the Euclidean-to-hyperbolic magic crossover is controlled by $a^2 |K| \omega^{-2/p}$. The Euclidean magic class corresponds to the vanishing of this variable. The hyperbolic magic class corresponds to its divergence. For the square-lattice quadratic reference, $p = 2$, and the class-changing scaling field reduces to $\frac{a^2 |K|}{w/t}$. Suppose

$$a^2 |K| \sim C \omega^\mu, \quad C > 0. \quad (6353)$$

Then

$$g_\nu^{\text{magic}} \propto \omega^{\mu-2/p}. \quad (6354)$$

Therefore,

$$\mu > \frac{2}{p} \implies \text{Euclidean magic class}, \quad (6355)$$

whereas

$$\mu < \frac{2}{p} \implies \text{hyperbolic magic class}. \quad (6356)$$

The marginal trajectory $\mu = \frac{2}{p}$ remains on the universal magic crossover. For $p = 2$, $\mu_c = 1$. Thus the square-lattice magic problem is Euclidean when curvature vanishes faster than w/t , hyperbolic when it vanishes more slowly than w/t , and remains at finite crossover when

$$a^2 |K| \propto \frac{w}{t}. \quad (6357)$$

The Euclidean magic half-angle corresponding to the same selected magic length is

$$s_{\nu,E}^* = \frac{a}{2\xi_{M,\nu}^*}. \quad (6358)$$

The hyperbolic magic half-angle is

$$s_{\nu,H}^* = \frac{\sinh(a/2R)}{\sinh \lambda_\nu^*}. \quad (6359)$$

Since

$$\frac{a}{2R} = \zeta, \quad \xi_{M,\nu}^* = R \lambda_\nu^*, \quad (6360)$$

we have

$$s_{\nu,E}^* = \frac{\zeta}{\lambda_\nu^*}. \quad (6361)$$

Therefore,

$$\frac{s_{\nu,H}^*}{s_{\nu,E}^*} = \frac{\sinh \zeta}{\zeta} \frac{\lambda_\nu^*}{\sinh \lambda_\nu^*}. \quad (6362)$$

In the weak microscopic-curvature limit,

$$\frac{\sinh \zeta}{\zeta} \rightarrow 1, \quad (6363)$$

and hence

$$\frac{\theta_{\nu,H}^*}{\theta_{\nu,E}^*} \rightarrow \frac{\lambda_\nu^*}{\sinh \lambda_\nu^*} \quad (6364)$$

at asymptotically small angle. The corresponding effective magic-cell ratio is

$$\frac{N_{M,\nu}^{\text{eff},H,*}}{N_{M,\nu}^{\text{eff},E,*}} = \frac{2 [\cosh \lambda_\nu^* - 1]}{(\lambda_\nu^*)^2} \quad (6365)$$

for a density-matched comparison. Thus the two fixed points are connected by a parameter-free one-dimensional trajectory in the space of magic-angle and magic-cell observables. Define

$$\mathcal{T}_\nu = \frac{\theta_{\nu,H}^*}{\theta_{\nu,E}^*}. \quad (6366)$$

For fixed $\omega > 0$, $K \rightarrow 0$ implies $\lambda_\nu^* \rightarrow 0$. Therefore,

$$\lim_{\omega \rightarrow 0} \lim_{K \rightarrow 0^-} \mathcal{T}_\nu = 1. \quad (6367)$$

By contrast, at every fixed $K < 0$, $\omega \rightarrow 0$ gives $\lambda_\nu^* \rightarrow \infty$. Because

$$\frac{\lambda}{\sinh \lambda} \rightarrow 0, \quad (6368)$$

we obtain

$$\lim_{K \rightarrow 0^-} \lim_{\omega \rightarrow 0} \mathcal{T}_\nu = 0. \quad (6369)$$

Therefore,

$$\lim_{\omega \rightarrow 0} \lim_{K \rightarrow 0^-} \mathcal{T}_\nu \neq \lim_{K \rightarrow 0^-} \lim_{\omega \rightarrow 0} \mathcal{T}_\nu. \quad (6370)$$

l. Theorem 194. Singular zero-curvature boundary of the magic universality class. Although the magic curve approaches its Euclidean value continuously at fixed nonzero w/t , the zero-curvature and weak-coupling limits do not commute. Thus arbitrarily weak negative curvature becomes asymptotically dominant when the interlayer coupling is reduced sufficiently far. This is the spectral counterpart of the noncommuting $K \rightarrow 0$ and $\theta \rightarrow 0$ geometric limits. Define

$$\beta_{\text{cell}}^{\text{magic}} = -\frac{d \ln N_{M,\nu}^{\text{eff},*}}{d \ln |\theta_\nu^*|}. \quad (6371)$$

The exact relation derived above is

$$\beta_{\text{cell}}^{\text{magic}} = \frac{|\theta_\nu^*|}{2} \cot \left(\frac{|\theta_\nu^*|}{2} \right) \beta_M(\chi_\nu^*). \quad (6372)$$

At asymptotically small magic angle,

$$\frac{|\theta_\nu^*|}{2} \cot \left(\frac{|\theta_\nu^*|}{2} \right) \rightarrow 1. \quad (6373)$$

Hence, $\beta_{\text{cell}}^{\text{magic}} \rightarrow \beta_M$. Therefore, $\beta_{\text{cell},E}^{\text{magic}} = 2$, and $\beta_{\text{cell},H}^{\text{magic}} = 1$. This equality does not depend on the microscopic dispersion order p .

m. Theorem 195. Geometric exponent as a spectral universality invariant. After eliminating w/t between the magic condition and the geometric cell law, the microscopic dispersion order cancels. The asymptotic exponent relating the magic-cell count to the magic angle is therefore purely geometric, $N_{M,\nu}^{\text{eff},*} \propto |\theta_\nu^*|^{-2}$ in the Euclidean class, and

$$N_{M,\nu}^{\text{eff},*} \propto |\theta_\nu^*|^{-1} \quad (6374)$$

in the hyperbolic class. Thus the geometry-to-spectrum correspondence transfers the universality class directly into the magic-band hierarchy. The two-dimensional exponent change is the first nontrivial member of a more general geometric law. Let the spatial dimension be D . For Euclidean geometry, a moiré domain of radius ξ_M has volume $V_M^{(E)} \propto \xi_M^D$. Since $\xi_M^{(E)} \propto |\theta|^{-1}$, we obtain $V_M^{(E)} \propto |\theta|^{-D}$. For hyperbolic geometry,

$$V_{\mathbb{H}^D}(r) \propto \exp \left[\frac{(D-1)r}{R} \right] \quad (6375)$$

at large radius. Since

$$\xi_M^{(H)} = R \ln |\theta|^{-1} + O(R), \quad (6376)$$

we obtain

$$V_M^{(H)} \propto |\theta|^{-(D-1)}. \quad (6377)$$

Therefore,

$$\beta_{D,E}^* = D, \quad \beta_{D,H}^* = D-1. \quad (6378)$$

On a magic branch, $\xi_M^* \propto \omega^{-1/p}$. Hence

$$N_{D,E}^{\text{eff},*} \propto \omega^{-D/p}, \quad (6379)$$

whereas

$$N_{D,H}^{\text{eff},*} = \exp \left[\Theta \left(\omega^{-1/p} \right) \right] \quad (6380)$$

with the exponent in the exponential multiplied by $D-1$. Eliminating ω yields

$$N_{D,E}^{\text{eff},*} \propto |\theta^*|^{-D}, \quad N_{D,H}^{\text{eff},*} \propto |\theta^*|^{-(D-1)}. \quad (6381)$$

n. Theorem 196. Codimension-one universality shift. In spatial dimension D , constant negative curvature changes the effective moiré-volume exponent according to $D \rightarrow D-1$. The same exponent reduction appears directly in the relation between the magic-cell count

and the magic angle. Thus the two-dimensional transformation $2 \rightarrow 1$ is not an isolated accident of the square-lattice construction. The Euclidean geometric class should also be distinguished from an arbitrary exact coincidence sequence. For the square lattice,

$$\Sigma = \frac{n_c^2}{d} \csc^2 \left(\frac{\theta}{2} \right), \quad (6382)$$

and $N_M^{\text{sc},E} = 2\Sigma$. Therefore,

$$N_M^{\text{sc},E} = \frac{2n_c^2}{d} \csc^2 \left(\frac{\theta}{2} \right). \quad (6383)$$

For a bounded-numerator primitive sequence, $n_c = O(1)$, we obtain

$$N_M^{\text{sc},E} = \Theta(|\theta|^{-2}). \quad (6384)$$

If n_c itself grows along the chosen exact sequence, the exact crystallographic count can grow faster. Thus the Euclidean universality class is most cleanly defined through the geometric effective count, while an exact coincidence sequence carries an additional arithmetic factor. For an exact hyperbolic commensurate twist,

$$q_M = [\Gamma : \Gamma \cap g_\theta \Gamma g_\theta^{-1}]. \quad (6385)$$

The exact bilayer count is $N_M^{\text{sc}} = 2n_b q_M$. Define the locking factor

$$\mathcal{L}_M = \frac{N_M^{\text{sc}}}{N_M^{\text{eff}}} = \frac{q_M A_0}{A_M^{\text{eff}}}. \quad (6386)$$

Define the arithmetic running exponent by

$$\delta_{\text{arith}} = -\frac{d \ln \mathcal{L}_M}{d \ln s}. \quad (6387)$$

Then $\beta_{\text{sc}} = \beta_M + \delta_{\text{arith}}$. In the hyperbolic geometric class, $\beta_M \rightarrow 1$. Therefore, $\beta_{\text{sc}} \rightarrow 1 + \delta_{\text{arith}}^*$ if the arithmetic exponent has a limit. The exact crystallographic system belongs to the same hyperbolic universality class only when $\delta_{\text{arith}} \rightarrow 0$. Let r_M^{sc} be the area-equivalent exact-supercell radius. Define the radial defect

$$\Delta_M = r_M^{\text{sc}} - \xi_M. \quad (6388)$$

The exact crystallographic exponent satisfies

$$\beta_{\text{exact}} = 1 + \lim \frac{\Delta_M}{\xi_M} \quad (6389)$$

whenever the limit exists in the curvature-dominated regime. Therefore, $\Delta_M = o(\xi_M)$ is the sharp radial

condition for exact inheritance of the hyperbolic geometric exponent. A stronger bounded radial locking, $\Delta_M = O(1)$, gives two-sided multiplicative comparison,

$$c_- N_M^{\text{eff}} \leq N_M^{\text{sc}} \leq c_+ N_M^{\text{eff}}. \quad (6390)$$

A convergent radial defect, $\Delta_M \rightarrow \Delta_*$, gives the stronger amplitude relation

$$N_M^{\text{sc}} \sim e^{\Delta_*/R} N_M^{\text{eff}}. \quad (6391)$$

o. Theorem 197. Exact crystallographic hyperbolic universality criterion. An exact hyperbolic small-angle sequence belongs to the geometric hyperbolic universality class precisely when the arithmetic correction is asymptotically irrelevant, $\delta_{\text{arith}} \rightarrow 0$. Equivalently, when the radial limit exists, this requires

$$\frac{\Delta_M}{\xi_M} \rightarrow 0. \quad (6392)$$

Then

$$N_M^{\text{sc}} = |\theta|^{-1+o(1)}. \quad (6393)$$

Under bounded multiplicative locking the stronger law is

$$N_M^{\text{sc}} = \Theta(|\theta|^{-1}). \quad (6394)$$

These inverse-linear conclusions are the conditional $\mathfrak{U}_1(1)$ tier. The explicit centered sequence belongs to $\mathfrak{U}_1(4)$, has arithmetic correction three, and therefore does not satisfy these exponent-one exact laws. If instead

$$\frac{\Delta_M}{\xi_M} \rightarrow \delta \neq 0, \quad (6395)$$

then $\beta_{\text{exact}} = 1 + \delta$. For $\delta > 0$, the exact crystallographic count exhibits arithmetic super-growth,

$$N_M^{\text{sc}} \sim |\theta|^{-(1+\delta)}. \quad (6396)$$

For $-1 < \delta < 0$, the exact count exhibits arithmetic under-growth,

$$N_M^{\text{sc}} \sim |\theta|^{-(1+\delta)}. \quad (6397)$$

If $\frac{\Delta_M}{\xi_M}$ fails to converge, distinct upper and lower exact exponents can occur. Thus failure of exact locking does not invalidate the hyperbolic geometric universality class. It means only that exact Fuchsian arithmetic adds an independent scaling field at the crystallographic level. A local and effective hyperbolic moiré geometry exists for every twist. An exact crystallographic class requires an

exact sequence $\theta_j \rightarrow 0$ satisfying $g_{\theta_j} \in \text{Comm}_G(\Gamma)$. If no such sequence exists at the chosen twist center, then no asymptotic exact supercell exponent can be defined. The effective hyperbolic class remains valid because its derivation does not require global commensurability. We therefore distinguish effective geometric universality from exact crystallographic universality. The first is unconditional. The second requires both existence of an exact sequence and asymptotic control of the arithmetic locking factor. For an exact hyperbolic commensurate supercell,

$$g_M = 1 + q_M(g_0 - 1). \quad (6398)$$

At a smooth irreducible representation of rank d_ρ , the real dimension of generalized momentum space is

$$D_\rho = (2g_M - 2)d_\rho^2 + 2. \quad (6399)$$

Substitution gives

$$D_\rho = 2q_M(g_0 - 1)d_\rho^2 + 2. \quad (6400)$$

If exact geometric–arithmetic locking holds,

$$q_M = \Theta(|\theta|^{-1}) \quad (6401)$$

in the hyperbolic class. Therefore,

$$D_\rho = \Theta(d_\rho^2 |\theta|^{-1}). \quad (6402)$$

By contrast, ordinary Euclidean Bloch momentum always lives in a two-dimensional Brillouin zone, $\dim \text{mBZ} = 2$, even though the number of folded bands grows with the supercell. Thus exact negative-curvature commensurability can change not merely the supercell exponent but the dimensionality of the generalized translation parameter space over which complete flattening must be verified. For the two-dimensional quadratic square-lattice problem with regular active tunneling, the Euclidean endpoint has $F^* = 1$, $\nu_\xi^* = 1$, $\beta_M^* = 2$, $\beta_X^* = 2$, and $\theta_\nu^* \propto \omega^{1/2}$. The curvature-dominated endpoint has $F^* = 0$, $\nu_\xi^* = 0$, $\beta_M^* = 1$, $\beta_X^* = 0$, and

$$\theta_\nu^* \sim \exp[-C_\nu \omega^{-1/2}]. \quad (6403)$$

Thus the class change appears simultaneously in geometry, effective coupling, and the spectral magic trajectory. The geometric class does not by itself determine whether the continuum magic band is exactly flat. The normalized bandwidth obeys

$$W_\nu = E_M^{\text{eff}} \Phi_{W,\nu}(X_\nu). \quad (6404)$$

At the magic root, $X_\nu = 1$. If $\Phi_{W,\nu}(1) > 0$, the magic state belongs to the finite-scaled-bandwidth spectral subclass. If $\Phi_{W,\nu}(1) = 0$, the leading continuum master band is exactly flat and the observed bandwidth is generated by irrelevant corrections. These two possibilities are logically independent of whether the geometry lies in the Euclidean or hyperbolic class. Thus curvature controls the geometric universality class, while the value of $\Phi_{W,\nu}(1)$ classifies a separate spectral-flatness property of the master Hamiltonian. The theory now provides explicit criteria for perturbations. A finite change in the registry threshold is class preserving. A finite change in the microscopic density $\frac{n_b}{A_0}$ is class preserving because it changes only the count amplitude. A smooth change of the tunneling kernel is class preserving if $\eta_\gamma \rightarrow 0$. Arithmetic corrections are class preserving at the exact crystallographic level if $\delta_{\text{arith}} \rightarrow 0$. By contrast, a nonzero limiting η_γ^* changes the spectral effective-coupling exponent, and a nonzero limiting δ_{arith}^* changes the exact crystallographic exponent. The universality structure therefore separates metric amplitudes, spectral scaling fields, and arithmetic scaling fields explicitly. For numerical data it is useful to define asymptotic class indicators that do not require fitting unknown amplitudes. Define

$$\mathcal{I}_\xi = \frac{\ln(\xi_M/a)}{\ln|\theta|^{-1}}. \quad (6405)$$

In the Euclidean class, $\mathcal{I}_\xi \rightarrow 1$. In the hyperbolic class,

$$\ln \xi_M = \ln R + \ln \ln |\theta|^{-1} + O(1), \quad (6406)$$

and therefore $\mathcal{I}_\xi \rightarrow 0$. Similarly define

$$\mathcal{I}_N = \frac{\ln N_M^{\text{eff}}}{\ln|\theta|^{-1}}. \quad (6407)$$

Then $\mathcal{I}_N \rightarrow 2$ in the Euclidean class and $\mathcal{I}_N \rightarrow 1$ in the hyperbolic class. These ratios provide amplitude-free numerical classifiers. Define

$$\mathcal{I}_{\text{mag}} = \frac{\ln |\theta_\nu^*|^{-1}}{\ln \omega^{-1}}. \quad (6408)$$

For the Euclidean algebraic class, $|\theta_\nu^*| \propto \omega^{1/p}$, and hence $\mathcal{I}_{\text{mag}} \rightarrow \frac{1}{p}$. For the hyperbolic class

$$\ln |\theta_\nu^*|^{-1} = \Theta(\omega^{-1/p}). \quad (6409)$$

Since

$$\frac{\omega^{-1/p}}{\ln \omega^{-1}} \rightarrow \infty, \quad (6410)$$

we obtain $\mathcal{I}_{\text{mag}} \rightarrow \infty$. Thus the algebraic and essential magic classes can be distinguished without fitting the exponential coefficient. For two nearby small-angle data points, define

$$\nu_{\xi}^{(ij)} = -\frac{\ln(\xi_{M,j}/\xi_{M,i})}{\ln(s_j/s_i)}. \quad (6411)$$

Define

$$\beta_M^{(ij)} = -\frac{\ln(N_{M,j}^{\text{eff}}/N_{M,i}^{\text{eff}})}{\ln(s_j/s_i)}. \quad (6412)$$

The theory predicts $\nu_{\xi}^{(ij)} \rightarrow 1$, $\beta_M^{(ij)} \rightarrow 2$ in the Euclidean basin. It predicts $\nu_{\xi}^{(ij)} \rightarrow 0$, $\beta_M^{(ij)} \rightarrow 1$ in the hyperbolic basin. For magic points, define

$$\beta_{\text{cell}}^{(ij)} = -\frac{\ln[N_{M,j}^{\text{eff},*}/N_{M,i}^{\text{eff},*}]}{\ln[|\theta_j^*|/|\theta_i^*|]}. \quad (6413)$$

The same endpoint values must be recovered. This provides a direct numerical test that the geometric and spectral classes have locked to one another. Define the geometric prediction at a magic point by

$$\beta_{\text{pred}} = \frac{|\theta_{\nu}^*|}{2} \cot\left(\frac{|\theta_{\nu}^*|}{2}\right) \left[1 + \frac{1}{\sqrt{1 + (\chi_{\nu}^*)^2}}\right]. \quad (6414)$$

Define the measured magic-cell exponent by

$$\beta_{\text{meas}} = -\frac{d \ln N_{M,\nu}^{\text{eff},*}}{d \ln |\theta_{\nu}^*|}. \quad (6415)$$

We define the class-locking residual by

$$\mathcal{R}_{\mathcal{U}} = \beta_{\text{meas}} - \beta_{\text{pred}}. \quad (6416)$$

The geometry-to-spectrum universality hypothesis requires $\mathcal{R}_{\mathcal{U}} \rightarrow 0$ under increasing numerical accuracy and improved scaling resolution. For exact crystallographic counts, the predicted exponent is modified to

$$\beta_{\text{pred}}^{\text{sc}} = \frac{|\theta_{\nu}^*|}{2} \cot\left(\frac{|\theta_{\nu}^*|}{2}\right) [\beta_M + \delta_{\text{arith}}]. \quad (6417)$$

Thus the arithmetic correction can be measured independently rather than absorbed into a fitted effective exponent. Define the pair $\mathbf{U} = (F, \beta_M)$. Because

$$\beta_M = \frac{2}{2-F}, \quad (6418)$$

the two observables are algebraically locked. Thus the complete geometric crossover occupies the one-dimensional curve

$$\mathbf{U}(F) = \left(F, \frac{2}{2-F}\right), \quad 0 \leq F \leq 1. \quad (6419)$$

The Euclidean endpoint is $\mathbf{U}_E = (1, 2)$, whereas the hyperbolic endpoint is $\mathbf{U}_H = (0, 1)$. No fit parameter is required to locate a measured point on this universality trajectory. For regular active tunneling and dispersion order p , $\beta_X = p\nu_{\xi}$. Therefore define the extended class vector

$$\mathbf{U}_{\text{ext}} = (F, \nu_{\xi}, \beta_M, \beta_X). \quad (6420)$$

For the quadratic square-lattice reference, $\mathbf{U}_E^{(p=2)} = (1, 1, 2, 2)$, whereas $\mathbf{U}_H^{(p=2)} = (0, 0, 1, 0)$. These endpoint values distinguish the two classes using independent geometric and spectral observables. In the Euclidean class, $\beta_X^* = p$. Holding $X = 1$ on the magic manifold gives

$$d \ln \omega = p d \ln |\theta^*|, \quad (6421)$$

and therefore

$$\frac{d \ln |\theta^*|}{d \ln \omega} = \frac{1}{p}. \quad (6422)$$

At the hyperbolic fixed point, $\beta_X^* = 0$. Thus no finite power $|\theta^*| \propto \omega^{\beta}$ can solve the magic condition asymptotically. The disappearance of a finite scaling exponent is the fixed-point origin of the essential singularity.

p. Theorem 198. Fixed-point origin of the algebraic-to-essential magic crossover. The Euclidean fixed point possesses a finite effective-coupling exponent $\beta_{X,E}^* = p$, and therefore produces an algebraic magic law $\theta^* \propto \omega^{1/p}$. The hyperbolic fixed point satisfies $\beta_{X,H}^* = 0$, so no finite power-law magic exponent exists. The resulting solution is instead

$$\theta^* = \exp\left[-\Theta\left(\omega^{-1/p}\right)\right]. \quad (6423)$$

Thus the essential singularity follows directly from the change of the effective-coupling fixed-point exponent. We can now combine the geometric, spectral, and crystallographic levels.

q. Theorem 199. Curvature-controlled moiré universality-class theorem. Consider a centered twisted bilayer with a finite positive microscopic registry threshold, a regular microscopic density, and a smooth active interlayer scattering profile. The exact geometric crossover is controlled by

$$\chi = \frac{\sinh(a/2R)}{|\sin(\theta/2)|}. \quad (6424)$$

Its autonomous scaling flow possesses two and only two endpoint fixed points. The Euclidean class is characterized by

$$\chi \rightarrow 0, \quad \xi_M \sim |\theta|^{-1}, \quad N_M^{\text{eff}} \sim |\theta|^{-2}. \quad (6425)$$

For a reference dispersion of order p ,

$$\alpha_{\text{eff}} \sim \omega |\theta|^{-p}, \quad \theta^* \sim \omega^{1/p}. \quad (6426)$$

The hyperbolic class is characterized by

$$\chi \rightarrow \infty, \quad \xi_M \sim R \ln |\theta|^{-1}, \quad N_M^{\text{eff}} \sim |\theta|^{-1}. \quad (6427)$$

Its spectral scaling is

$$\alpha_{\text{eff}} \sim \omega \ln^p |\theta|^{-1}, \quad \theta^* = \exp \left[-\Theta \left(\omega^{-1/p} \right) \right]. \quad (6428)$$

The crossover between these two classes is controlled geometrically by

$$g_K = \frac{a^2 |K|}{\theta^2} \quad (6429)$$

and on a magic branch by

$$g_{\text{magic}} = C_{p,\nu}^2 a^2 |K| \omega^{-2/p}. \quad (6430)$$

The effective magic-cell exponent inherits the geometric class, $N_M^{\text{eff},*} \propto |\theta^*|^{-2}$ at the Euclidean fixed point and $N_M^{\text{eff},*} \propto |\theta^*|^{-1}$ at the hyperbolic fixed point. Exact crystallographic counting is instead sequence dependent: an exact class $\mathfrak{U}_1(\beta)$ has exponent $\beta_{\text{sc}} = \beta = 1 + \delta_{\text{arith}}$, with lower and upper envelopes when the limit fails to exist. It agrees with the effective hyperbolic exponent only in the conditional bounded-locking tier $\mathfrak{U}_1(1)$. The explicit centered sequence belongs to $\mathfrak{U}_1(4)$ and has arithmetic correction three. Thus curvature changes the asymptotic moiré universality class at the levels of geometric length, microscopic counting, effective electronic coupling, and the spectral magic trajectory. The complete fixed-point structure can be stated directly. The Euclidean moiré class satisfies

$$\begin{aligned} \xi_M &\sim |\theta|^{-1}, \quad N_M^{\text{eff}} \sim |\theta|^{-2}, \\ \alpha_{\text{eff}} &\sim \omega |\theta|^{-p}, \quad \theta^* \sim \omega^{1/p}. \end{aligned} \quad (6431)$$

The negatively curved moiré class satisfies

$$\begin{aligned} \xi_M &\sim R \ln |\theta|^{-1}, \quad N_M^{\text{eff}} \sim |\theta|^{-1}, \\ \alpha_{\text{eff}} &\sim \omega \ln^p |\theta|^{-1}, \quad \theta^* = \exp \left[-\Theta \left(\omega^{-1/p} \right) \right]. \end{aligned} \quad (6432)$$

For the physical square-lattice reference, $p = 2$, and therefore

$$\mathcal{U}_E: \quad \xi_M \sim |\theta|^{-1}, \quad N_M^{\text{eff}} \sim |\theta|^{-2}, \quad \theta^* \sim \sqrt{w/t}, \quad (6433)$$

whereas

$$\begin{aligned} \mathcal{U}_H: \quad \xi_M &\sim R \ln |\theta|^{-1}, \quad N_M^{\text{eff}} \sim |\theta|^{-1}, \\ \theta^* &\sim \exp \left[-C(K) \sqrt{t/w} \right]. \end{aligned} \quad (6434)$$

VII. CONCLUSION

Negative curvature changes the moiré scaling law from the Euclidean inverse-angle form to a logarithmic length law. The associated effective cell count scales as $|\theta|^{-1}$, while exact crystallographic counts retain an additional arithmetic dependence on the chosen commensurate sequence. For the explicit centered sequence this dependence gives the class $\mathfrak{U}_1(4)$ and arithmetic correction three.

The microscopic models display a sharp magic-no-magic dichotomy. The exact square five-state first-star Hamiltonian has no quadratic magic root. The centered surface-group bilayer with geodesic intralayer hopping and full-distance exponential interlayer tunneling has a positive simple dispersion-curvature root. Weighted Schur bounds continue this root to the full exponential kernel and control its displacement and derivative margin. The regular group-algebra operator also contains an isolated point island with a positive gap, active overlap, coherence, and a stable Riesz projector.

In the regular-octagon symmetry class, the hopping moment is proportional to the canonical Hodge metric. The same coupling therefore cancels the complete tangent Hessian, with all principal eigenvalues crossing zero at a common nonzero slope. The perturbation estimates quantify the residual anisotropy, root displacement, eigenvalue splitting, gap stability, and projector stability away from the symmetric point.

Finite covers recover the retained bulk sectors when no-loss and no-pollution estimates hold on the same sector. Operator-tempered and operator-Ramanujan towers give a testable tower-independent class, while balanced cutoff growth and normalized tails control the full-shell limit. These results connect curvature-controlled registry geometry, arithmetic commensurability, and microscopic spectral flattening within one explicit hyperbolic bilayer construction.

VIII. COMPUTER-ASSISTED VALIDATION IN AN ARITHMETIC REGULAR-OCTAGON HYPERBOLIC BILAYER

The analytical results derived above distinguish three logically independent layers of hyperbolic moiré physics: the local registry geometry, the exact arithmetic crystallography, and the microscopic spectral response. We now subject these three layers to a common computer-assisted validation programme. The purpose of this section is not to replace the preceding proofs by numerical evidence. Instead, we use exact algebraic computation, certified finite approximants, independent spectral algorithms, controlled perturbations, and explicit falsification tests to establish that the analytical mechanisms remain visible in a nontrivial microscopic realization and to identify precisely where the available computations do not yet license a stronger thermodynamic statement.

The validation architecture is deliberately hierarchical. Identities that can be checked in exact arithmetic are treated algebraically. Root existence, strict positivity, and perturbative exclusion statements are recertified by interval or residual bounds. Infinite-volume observables are approximated through certified non-Abelian covers rather than open hyperbolic disks. Large spectral measures are evaluated independently by the kernel polynomial method and stochastic Lanczos quadrature. Finally, every positive test is accompanied by negative controls in which one of the theorem hypotheses is deliberately violated. This separation is essential: numerical agreement at a finite scale is not identified with a theorem, whereas a certified failure of an incorrect reduction is treated as evidence for the necessity of the corresponding hypothesis.

A. Microscopic model, geometric assumptions, and observable sector

We work on the hyperbolic plane of curvature

$$K = -\frac{1}{R^2} < 0, \quad (6435)$$

and choose a centered regular-octagon realization of a genus-two surface group,

$$\Gamma = \langle a_1, b_1, a_2, b_2 \mid [a_1, b_1][a_2, b_2] = e \rangle. \quad (6436)$$

If $o \in \mathbb{H}_R^2$ denotes the polygon center, the single-layer crystal is the uniformly discrete orbit $\Lambda = \Gamma o$. The two layers are embedded in $\mathbb{H}_R^2 \times \mathbb{R}$, separated by a normal distance $h > 0$, and the second layer is transformed by the centered isometry g_θ ,

$$\Lambda_1 = \Lambda, \quad \Lambda_2 = g_\theta \Lambda. \quad (6437)$$

The continuous parameter θ is used for local registry and spectral scans. Exact arithmetic tests use the centered

commensurator sequence

$$\theta_j = 2 \arctan \left(\frac{\sqrt{3}}{j} \right), \quad (6438)$$

for which the finite-index common subgroup and its coincidence index can be evaluated independently of the local moiré construction. The scalar microscopic Hilbert space is

$$\mathcal{H}_{\text{sc}} = \ell^2(\Gamma) \otimes \mathbb{C}_{\text{layer}}^2. \quad (6439)$$

Let λ_Γ be the left regular representation and let $S = \{s_a, s_a^{-1}\}_{a=1}^4$ denote the equal-length first shell of the centered octagon orbit. The intralayer operator is

$$h_0 = -tA_S, \quad A_S = \sum_{a=1}^4 [\lambda_\Gamma(s_a) + \lambda_\Gamma(s_a)^*]. \quad (6440)$$

For $\gamma \in \Gamma$, the full physical interlayer separation is

$$D_\gamma = \sqrt{h^2 + d_{\mathbb{H}}(x_1, \gamma x_2)^2}, \quad r_\gamma = D_\gamma - h, \quad (6441)$$

and the exponentially decaying tunnelling coefficient is

$$q_\gamma = \exp(-\mu r_\gamma), \quad \mu = \lambda_\perp^{-1} > 0. \quad (6442)$$

The first-shell and complete tunnelling operators are

$$Q_1 = I + q_1 A_S, \quad Q_\infty = \sum_{\gamma \in \Gamma} q_\gamma \lambda_\Gamma(\gamma), \quad (6443)$$

and the bilayer Hamiltonian is

$$H_Q(w) = \begin{pmatrix} h_0 & wQ \\ wQ & h_0 \end{pmatrix}, \quad Q \in \{Q_1, Q_\infty\}. \quad (6444)$$

The decay rate is restricted to the regime in which the kernel and the derivative-weighted kernels required by the Hodge response are summable. The target is pre-registered as the layer-even sector of the exact layer-exchange symmetry. No target is changed after a root, gap, or flat branch has been located. To test whether the scalar mechanism survives a more realistic internal structure, we use an arithmetic regular-octagon three-orbital bilayer,

$$\mathcal{H}_{\text{ARO-3B}} = \ell^2(\Gamma) \otimes \mathbb{C}_{\text{layer}}^2 \otimes \text{span} \{|s\rangle, |p_1\rangle, |p_2\rangle\}. \quad (6445)$$

The p_1 and p_2 orbitals are defined in local orthonormal tangent frames. Directional orbitals at distinct sites are compared only after Levi-Civita parallel transport. The intralayer matrix element takes the hyperbolic Slater-Koster form

$$H_{\parallel, ij}^{\alpha\beta} = V_{\alpha\beta}(d_{\mathbb{H}}(x_i, x_j)) \mathcal{A}_{\alpha\beta}(\hat{d}_{ij}, \mathcal{P}_{j \rightarrow i}), \quad (6446)$$

where the independent channels include $V_{ss\sigma}$, $V_{sp\sigma}$, $V_{pp\sigma}$, and $V_{pp\pi}$. The interlayer block is constructed from the same full product-space distance as the scalar model,

$$T_{\perp, ij}^{\alpha\beta} = W_{\alpha\beta}(D_{ij}^{(\theta)}, \hat{D}_{ij}) \exp \left[-\frac{D_{ij}^{(\theta)} - h}{\lambda_\perp} \right]. \quad (6447)$$

The orbital-diagonal, symmetry-preserving subspace reduces continuously to the scalar group-algebra model. Bond anisotropy, orbital splitting, s - p mixing, and unequal $pp\sigma$ and $pp\pi$ amplitudes are then introduced as controlled perturbations. Open hyperbolic disks are not used as bulk approximants because their boundary fraction remains finite. Instead, we use three inequivalent non-Abelian congruence towers, denoted

$$\mathcal{T}_{23/11}, \quad \mathcal{T}_{31/3}, \quad \mathcal{T}_{7/2}, \quad (6448)$$

with certified lower bounds on the injectivity radius. At each level $\Gamma_N \triangleleft \Gamma$, the quotient $Q_N = \Gamma/\Gamma_N$ is treated through its exact finite-group decomposition. The representation-theoretic consistency relation is

$$\sum_{\rho \in \widehat{Q}_N} d_\rho^2 = |Q_N|, \quad (6449)$$

and the regular spectrum is reconstructed as

$$\text{Spec}(H_N^{\text{reg}}) = \bigsqcup_{\rho \in \widehat{Q}_N} d_\rho \text{Spec}(H_N[\rho]). \quad (6450)$$

Exact character data and Newton identities are used to reconstruct the characteristic factors without requiring explicit matrices for every group element. Sparse extremal spectra are computed by Lanczos or block-Lanczos methods. Spectral projectors are obtained from Riesz-contour calculations when a separated island is available. Large spectral measures are computed independently by Jackson-damped kernel polynomial expansions and fully reorthogonalized stochastic Lanczos quadrature. The corresponding cumulative distribution functions are denoted by

$$F_N^{\text{KPM}}(E), \quad F_N^{\text{SLQ}}(E). \quad (6451)$$

The weak-measure discrepancy is measured by

$$\kappa_N = \sup_E |F_N(E) - F_{\text{ref}}(E)|, \quad (6452)$$

and the frozen finite-broadening schedule is

$$\eta_N = \sqrt{\kappa_N}. \quad (6453)$$

This schedule is used only as a finite-broadening diagnostic. A local unsmoothed density is not declared unless a tower-uniform regularity theorem is independently available.

B. Intrinsic geometric validation and dimensional crossover

The exact centered-twist displacement law is

$$\sinh\left(\frac{d_\theta(r)}{2R}\right) = \sinh\left(\frac{r}{R}\right) \left| \sin\left(\frac{\theta}{2}\right) \right|. \quad (6454)$$

For a fixed registry threshold a , the corresponding moiré length is

$$\xi_M(\theta, R) = R \operatorname{arsinh} \left[\frac{\sinh(a/2R)}{|\sin(\theta/2)|} \right]. \quad (6455)$$

We evaluate the left-hand side by independent geodesic calculations in hyperboloid and disk coordinates and determine ξ_M by numerical inversion of the registry condition. Figure 1 shows that the numerical and analytical constructions agree over the entire frozen parameter set, with residuals at the level of floating-point or high-precision arithmetic. This rules out a coordinate or inversion artifact as the source of the logarithmic hyperbolic moiré scale.

Introducing

$$\chi = \frac{\sinh(a/2R)}{|\sin(\theta/2)|}, \quad (6456)$$

the normalized effective area is controlled by a parameter-free crossover function, while the running exponent interpolates between the locally Euclidean and curvature-dominated regimes. The numerical collapse in Fig. 2 is performed for several values of R without fitting a separate curve for each radius. The same calculation is repeated in higher effective dimensions, with the number of active exponentially expanding directions kept explicit.

C. Exact arithmetic growth and radial mismatch

The local geometric law does not determine the exact finite-index crystallographic count. For the centered arithmetic sequence,

$$q_j = \left[\Gamma : \Gamma \cap g_{\theta_j} \Gamma g_{\theta_j}^{-1} \right], \quad (6457)$$

the theorem predicts

$$q_j = |\theta_j|^{-4+o(1)}. \quad (6458)$$

Let $r_{M,j}^{\text{sc}}$ be the area-equivalent radius of the exact supercell and define

$$\Delta_j = r_{M,j}^{\text{sc}} - \xi_{M,j}. \quad (6459)$$

The arithmetic class then implies

$$\frac{\Delta_j}{\xi_{M,j}} \longrightarrow 3, \quad \frac{r_{M,j}^{\text{sc}}}{\xi_{M,j}} \longrightarrow 4. \quad (6460)$$

Figure 3 shows both aspects of the same mechanism. The pointwise logarithmic exponent approaches the theorem value from above, while the radius ratios converge to the arithmetic corrections 3 and 4. The slow, non-monotone finite-scale approach is expected because the exact index retains local arithmetic fluctuations; the certified statement is logarithmic asymptotic growth rather than smooth regular variation under every fixed index dilation.

Hyperbolic Moiré-Length Consistency

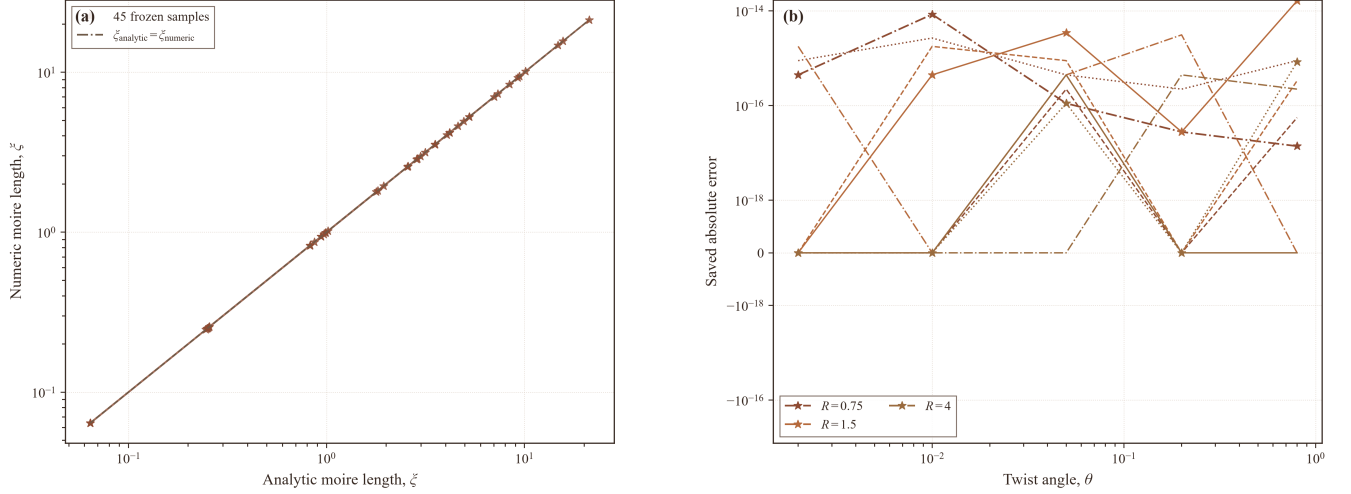


FIG. 1. **Intrinsic consistency of the hyperbolic moiré length.** **a**, Numerical solutions of the registry-threshold equation are compared with the closed expression in Eq. (6455) for 45 frozen samples. The points remain on the identity line over more than two decades in ξ_M . **b**, Absolute discrepancies at machine or high-precision round-off scale throughout the sampled twist interval. The agreement verifies the intrinsic distance implementation and the numerical inversion independently of the later spectral model.

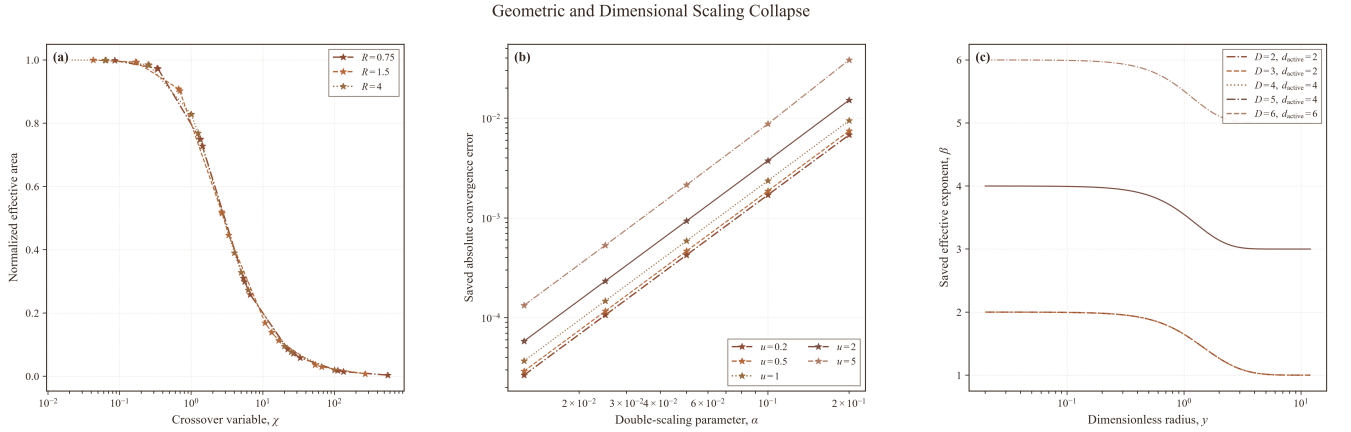


FIG. 2. **Curvature-controlled geometric and dimensional scaling collapse.** **a**, Normalized effective moiré areas for distinct curvature radii collapse onto a common function of χ , demonstrating that the crossover is controlled by one intrinsic geometric variable rather than by radius-dependent fitting parameters. **b**, Absolute errors in the double-scaling expansion decrease with the preregistered scaling parameter for several fixed values of the combined variable u . **c**, The running exponent crosses from d_{active} to $d_{\text{active}} - 1$, showing that exponential geodesic growth removes one power from the effective moiré-volume law. Coincident curves correspond to ambient dimensions with the same number of active expanding directions.

D. Microscopic magic–no-magic dichotomy

The minimal square five-state first-star model provides the negative side of the microscopic dichotomy. Its exact quadratic response is

$$c_{\square}(\alpha) = \frac{r^2 - r + 2}{r(r + 1)}, \quad r = \sqrt{1 + 16\alpha^2}, \quad (6461)$$

and satisfies

$$c_{\square}(\alpha) \geq 4\sqrt{2} - 5 > 0. \quad (6462)$$

Consequently, the perturbative suppression visible at weak coupling cannot be promoted to an exact quadratic magic root. The natural hyperbolic surface-group model behaves differently. In the layer-even first-shell character sector, the response is affine in w and has the explicit positive simple root

$$w^* = \frac{t}{q_1}. \quad (6463)$$

Figure 4 places the two calculations on the same logical level. The square response remains above its certified

Arithmetic Exponent and Radial Locking

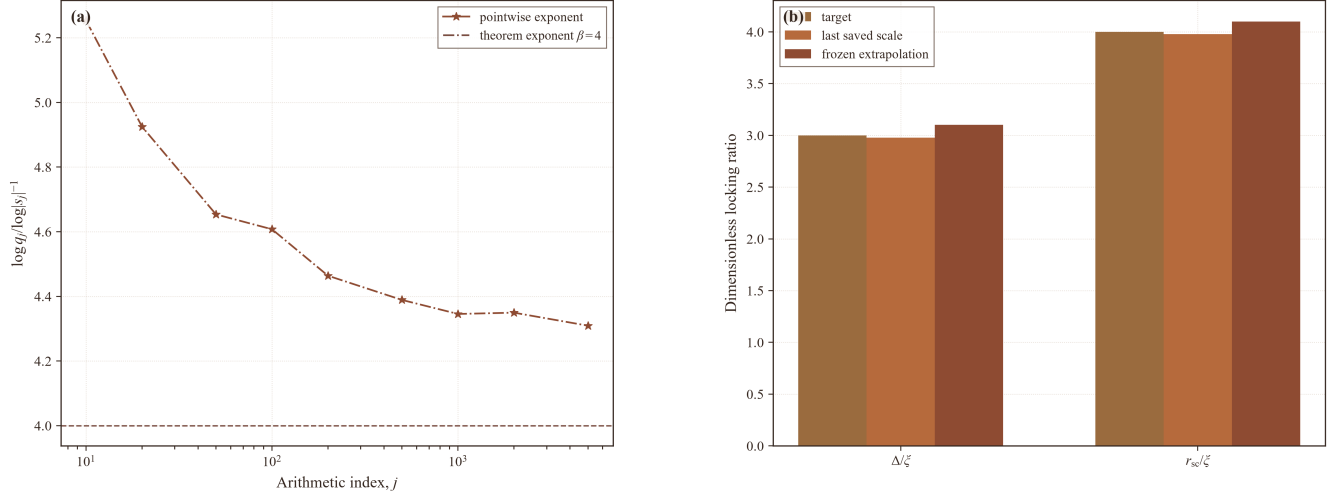


FIG. 3. **Arithmetic exponent four and radial locking defect.** **a**, The finite-scale logarithmic exponent $\log q_j / \log |s_j|^{-1}$ approaches the frozen theorem value $\beta = 4$. Arithmetic oscillations remain visible at accessible indices but do not change the asymptotic class. **b**, The last computed scale and the frozen extrapolation are compared with the theoretical targets $\Delta_j/\xi_{M,j} = 3$ and $r_{M,j}^{\text{sc}}/\xi_{M,j} = 4$. The joint agreement verifies that the exact exponent is generated by a radial crystallographic defect rather than by the local geometric area law.

positive floor, whereas the hyperbolic response crosses zero transversely at a positive coupling. The numerical calculation therefore verifies the model-specific dichotomy rather than suggesting that curvature creates a root in every Hamiltonian.

The full exponential kernel is restored by decomposing the group sum into word-length shells,

$$Q_\infty = \sum_{\ell=0}^{\infty} Q^{(\ell)}. \quad (6464)$$

Weighted Schur estimates control both the scalar kernel and the derivative-weighted Hodge kernel. If $F_1(w)$ is the first-shell response and $F_\infty(w)$ the full response, the displacement of the simple root is bounded by

$$|w_\infty^* - w_1^*| \leq \frac{\sup_w |F_\infty(w) - F_1(w)|}{\inf_w |F_1'(w)|}. \quad (6465)$$

The full root is not required to remain perturbatively close to the first-shell root; the certified statement is that it remains positive, unique in the declared interval, and bounded by the complete tail budget.

E. Complete Hodge cancellation and multiorbital robustness

The hyperbolic analogue of a momentum-space effective mass is a bilinear form on $V = H^1(\Gamma; \mathbb{R})$, equipped with the canonical Hodge metric G . Let

$$C_S = \sum_{a=1}^4 dk_a \otimes dk_a, \quad B_\infty = \sum_{\gamma \in \Gamma} q_\gamma n(\gamma) \otimes n(\gamma). \quad (6466)$$

The full tangent response is

$$\mathcal{K}(w) = 2tC_S - wB_\infty. \quad (6467)$$

In the certified regular-octagon symmetry class, the self-adjoint commutant is scalar, forcing $B_\infty = b_\infty C_S$. The unique common root is therefore

$$w_H = \frac{2t}{b_\infty}, \quad \mathcal{K}(w_H) = 0. \quad (6468)$$

All generalized principal Hessian eigenvalues vanish at the same coupling and cross zero with a common nonzero slope. In Fig. 6, the coupling coordinate is the root-centred coordinate stored by the validation calculation, so the origin represents the physical coupling w_H . The four diagonal Hodge components coincide throughout the symmetry-reduced scan. The residual audit then separates a coordinate-basis zero, a finite-precision symmetry residual, and exact algebraic cancellation.

Outside the exact symmetry class, a trace root need not cancel the traceless part of $\mathcal{K}$. We therefore perturb both the scalar and three-orbital models. The perturbation analysis monitors the root, isolation gap, Hodge-eigenvalue spread, and the separation between symmetry-enforced degeneracy and genuine kinetic flattening.

F. Finite-cover bulk hierarchy and full-shell limits

For an observable supported inside a radius smaller than the injectivity radius, local finite-cover agreement follows from exact lifting of the rooted labelled ball.

Square-Lattice No-Go and Hyperbolic Positive Root

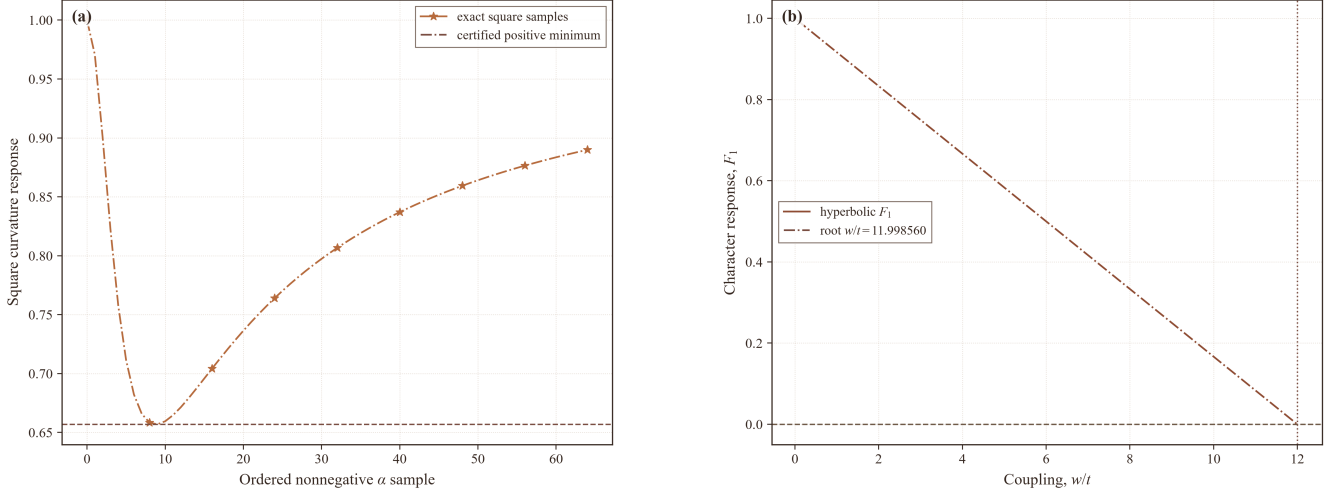


FIG. 4. **Exact square-lattice no-go and positive hyperbolic root.** **a**, Exact samples of the square first-star curvature response remain strictly above the certified minimum $4\sqrt{2} - 5$. The finite minimum is not caused by a hidden gap closing and therefore excludes an exact quadratic magic root in this model. **b**, The hyperbolic layer-even character response crosses zero at $w/t = 11.998560$, with a nonzero slope. The horizontal zero line and the vertical root marker make the contrast with panel **a** explicit. The two panels validate the microscopic magic–no-magic dichotomy between two distinct, fully declared Hamiltonians.

Full-Kernel Shell Closure and Root-Gap Certification

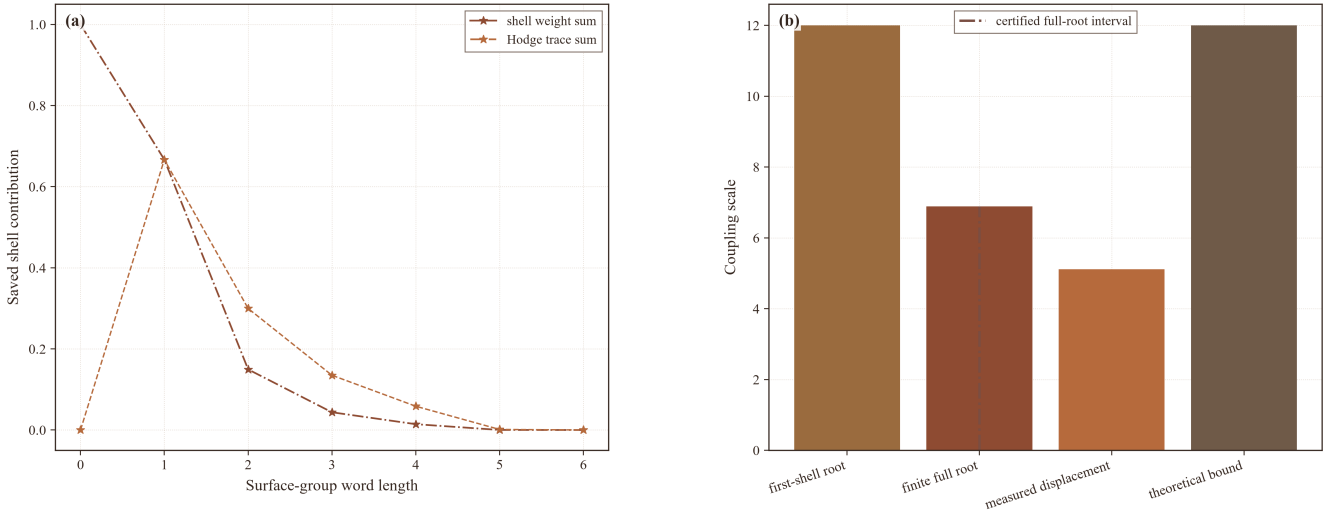


FIG. 5. **Full-kernel shell closure and certified root continuation.** **a**, Shell contributions to the scalar weight and Hodge trace decay rapidly with surface-group word length. The derivative-weighted tail is larger at intermediate shells but is nevertheless summable. **b**, The first-shell root, the finite full-kernel root, their measured displacement, and the analytical displacement bound are displayed on the same coupling scale. The full root undergoes a substantial but controlled renormalization and remains inside the certified interval. The figure therefore establishes root persistence under the complete physical kernel, not an assumption of a numerically negligible tail.

Strong global spectral convergence requires both a lower inclusion and a no-pollution upper inclusion. We write the total licensed finite-cover budget schematically as

$$\eta_N^{\text{spec}} = u_N + b_N + \tau_N + \epsilon_{\text{solver},N}, \quad (6469)$$

where u_N is the no-loss error, b_N is the no-pollution error, τ_N is the physical truncation error, and $\epsilon_{\text{solver},N}$ is the eigensolver residual. The extended calculation uses five certified levels in each of three inequivalent non-Abelian towers. Figure 8 shows that the injectivity-radius lower bound grows with congruence level and that matched

Hodge-Response Structure and Exact Cancellation

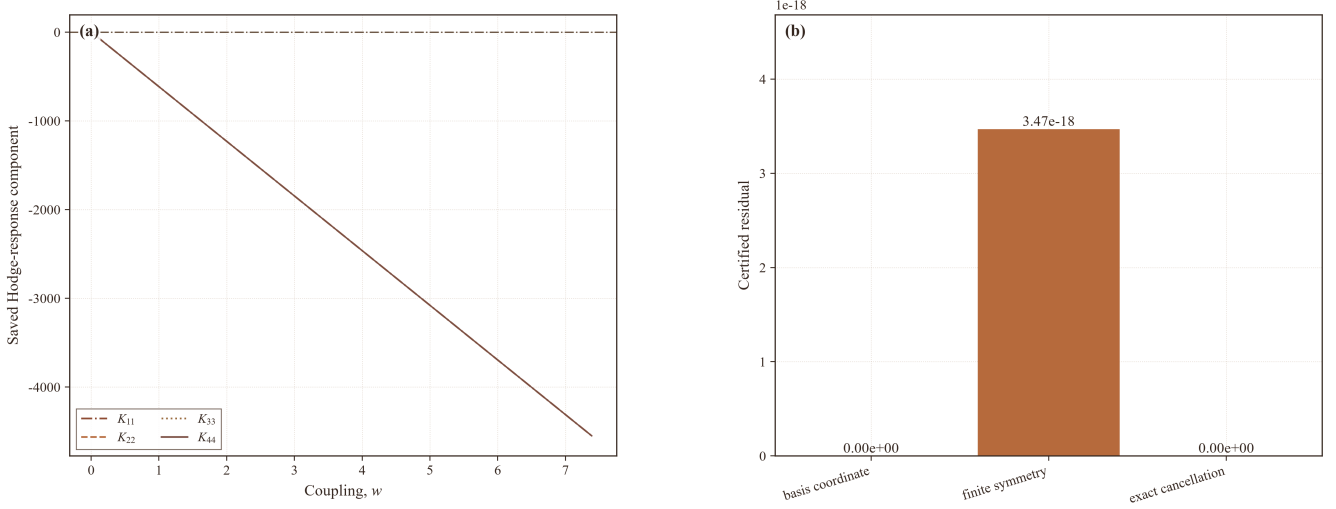


FIG. 6. **Symmetry-locked Hodge response and exact common cancellation.** **a**, The four saved canonical Hodge components K_{11} , K_{22} , K_{33} , and K_{44} remain locked to one common linear response in the root-centred coupling coordinate. Their simultaneous zero occurs at the certified physical coupling w_H . **b**, The basis-coordinate and exact-cancellation residuals vanish, while the finite numerical symmetry residual is 3.47×10^{-18} . The result verifies cancellation of the full Hodge tensor rather than a trace zero or a one-dimensional momentum-direction condition.

Symmetry Breaking and Multiorbital Robustness

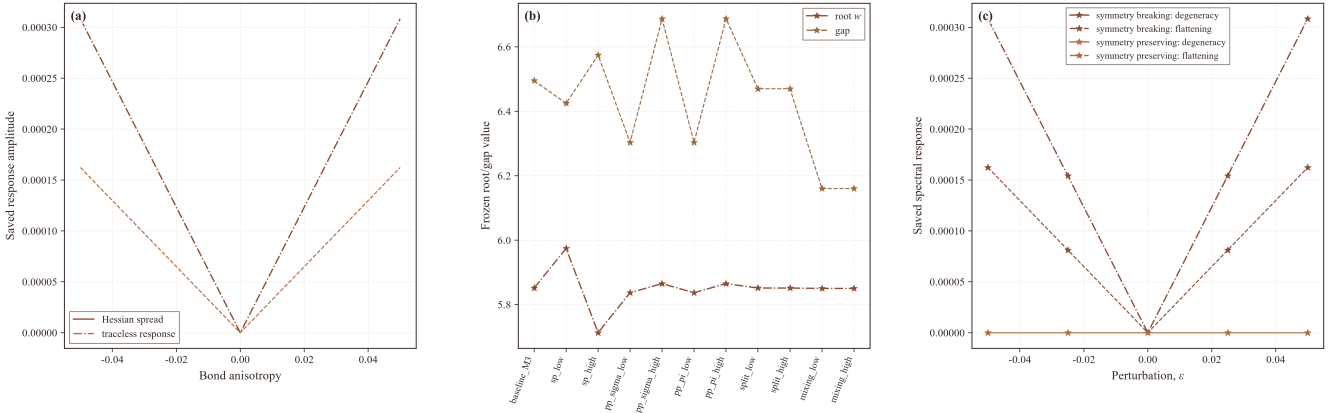


FIG. 7. **Symmetry breaking, orbital enrichment, and stability of the positive root.** **a**, The Hodge-eigenvalue spread and the traceless response vanish at the octagon-symmetric point and grow continuously under bond anisotropy. This identifies the exact obstruction to extending complete cancellation outside the proportionality class. **b**, The frozen root and positive gap are evaluated for the scalar baseline and for controlled s - p , $pp\sigma$, $pp\pi$, orbital-splitting, and interorbital-mixing perturbations. The root remains finite and the isolation gap remains positive throughout the tested multiorbital domain. **c**, symmetry-preserving perturbations leave the degeneracy response zero, whereas symmetry-breaking perturbations generate both degeneracy splitting and a controlled flattening response. The comparison separates symmetry degeneracy from the magic-band mechanism.

retained-sector spectral measures become mutually consistent. The validated conclusion is a weak bulk limit in the declared projective sector. The stronger full-regular-spectrum edge and gap statement remains inconclusive because the global no-pollution gate is not closed for every sector.

A true full-shell limit must simultaneously increase the physical shell and the cover scale. We preregister the

balanced diagonal

$$L_j = \left\lfloor \sqrt{r_{\text{inj},j}^{\text{lower}}} \right\rfloor, \quad (6470)$$

which realizes $L_j = 1, 2, 3, 4, 5$ and satisfies

$$L_j \rightarrow \infty, \quad \frac{L_j}{r_{\text{inj},j}} \rightarrow 0. \quad (6471)$$

Weak Bulk Spectral Consistency Across Certified Non-Abelian Towers

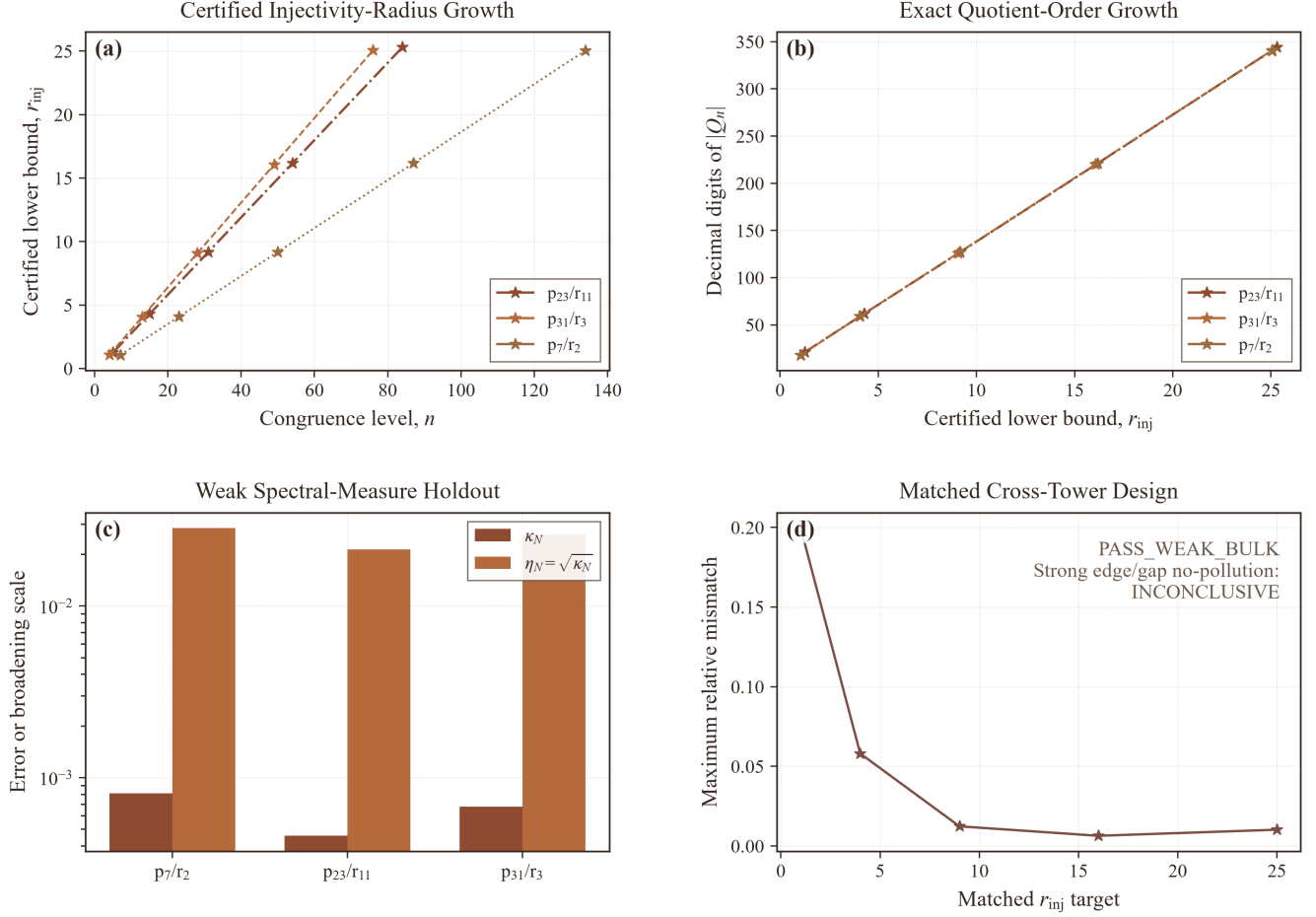


FIG. 8. **Weak bulk spectral consistency across certified non-Abelian towers.** **a**, Certified injectivity-radius lower bounds grow along the three congruence towers. **b**, The exact quotient orders increase rapidly with the certified geometric scale, providing progressively larger boundaryless approximants. **c**, Holdout cumulative-measure errors κ_N and the frozen broadening scales $\eta_N = \sqrt{\kappa_N}$ remain small in all three retained projective sectors. **d**, The maximum relative mismatch between matched towers decreases strongly with the matched injectivity-radius target. These results establish PASS_WEAK_BULK for the retained spectral measures. They do not assert strong convergence of every full-spectrum edge or gap, because complete no-pollution remains unresolved.

The full error hierarchy separates C^0 , C^1 , and C^2 quantities. Figure 9 demonstrates that the balanced sequence itself is now correctly realized and that all displayed bounds decrease. They nevertheless approach nonzero floors because the available physical packing and derivative-weighted Hodge tail bounds do not close. The figure therefore identifies a precise tail-theorem gap rather than providing evidence against the conditional convergence theorem.

G. Weak spectral measures and the local-density boundary

Weak convergence of spectral measures is strictly weaker than convergence of an unsmoothed density. The

KPM and SLQ calculations are performed on matrix-free projective actions with dimensions approaching 10^6 . The two methods use independent polynomial or Krylov approximations and therefore provide a nontrivial cross-check. Figure 10 demonstrates three distinct levels. First, the cumulative measures agree between KPM and SLQ. Second, the frozen broadening schedule is compatible with the measured CDF error. Third, the finite-broadening coherence-weighted densities are mutually consistent. The last observation is not promoted to an unsmoothed density theorem because no tower-uniform Hölder or local-limit regularity certificate has been established.

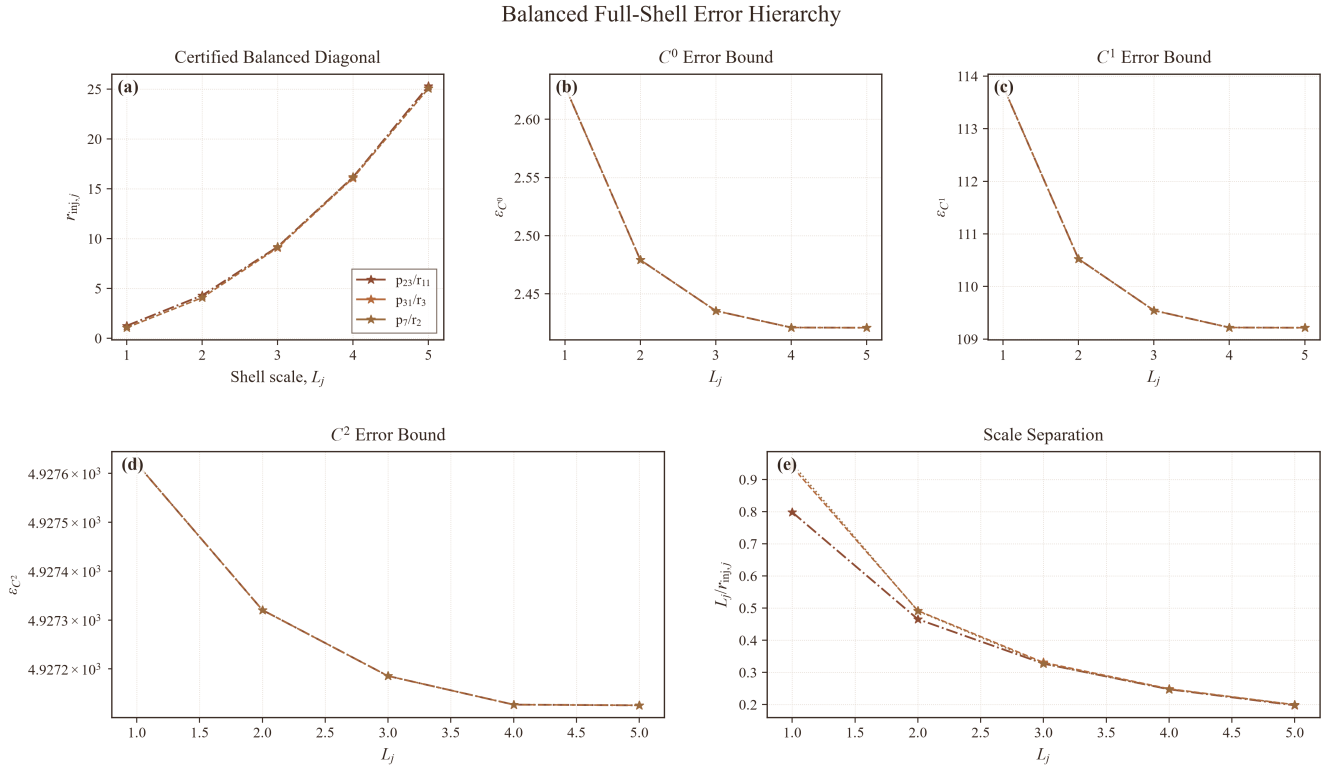


FIG. 9. **Balanced full-shell error hierarchy and unresolved tail closure.** **a**, The preregistered shell law realizes $L_j = 1, \dots, 5$ while the certified injectivity radius grows in all three towers. **b–d**, The C^0 , C^1 , and C^2 error bounds decrease along the balanced diagonal but retain nonzero floors under the currently available physical and Hodge-tail estimates. **e**, The ratios $L_j/r_{\text{inj},j}$ decrease as required by scale separation. Thus the numerical sequence now satisfies the correct geometric design, but the full-shell conclusion remains INCONCLUSIVE because the model-specific kernel-tail theorem is not yet closed.

H. External reproduction and negative-control certification

The internal spectral pipeline is checked against public hyperbolic-lattice data. Public tessellation and supercell records are parsed by their record grammar rather than by hard-coded line offsets, reconstructed as Hermitian graph Hamiltonians, and compared with the published broadened DOS. The same operator is independently mapped to its circuit-Laplacian representation.

A validation framework is useful only if it rejects incorrect reductions. We therefore repeat selected calculations with a wrong metric, a wrong target transport, an unregistered shape variable, an incorrect normalization, an incomplete active shell, and a deliberately invalid incommensurate joint limit. We also verify directly that C^0 convergence of operators does not imply C^1 convergence of their derivatives.

I. Arithmetic magic complexity and incommensurate joint limits

Once a positive microscopic branch has been independently certified, it can be sampled on an exact arithmetic

sequence. If

$$\log |\theta_j|^{-1} = c_\nu \omega_j^{-1/2} + o\left(\omega_j^{-1/2}\right), \quad (6472)$$

then the $U_1(4)$ arithmetic class predicts

$$\sqrt{\omega_j} \log q_j \longrightarrow 4c_\nu. \quad (6473)$$

The finite sequence retains strong arithmetic oscillations at its shallowest levels, but its upper-scale trend approaches the frozen coefficient.

For an incommensurate target, an angular approximation $\theta_N = \theta + \delta\theta_N$ is controlled over an observation radius L_N only if

$$|\delta\theta_N| \exp\left(\frac{L_N}{R}\right) \longrightarrow 0. \quad (6474)$$

The exponential factor is a direct consequence of hyperbolic geodesic separation. Figure 14 compares preregistered passing and deliberately failing sequences on all three towers.

Vanishing-Broadening Spectral-Measure Hierarchy

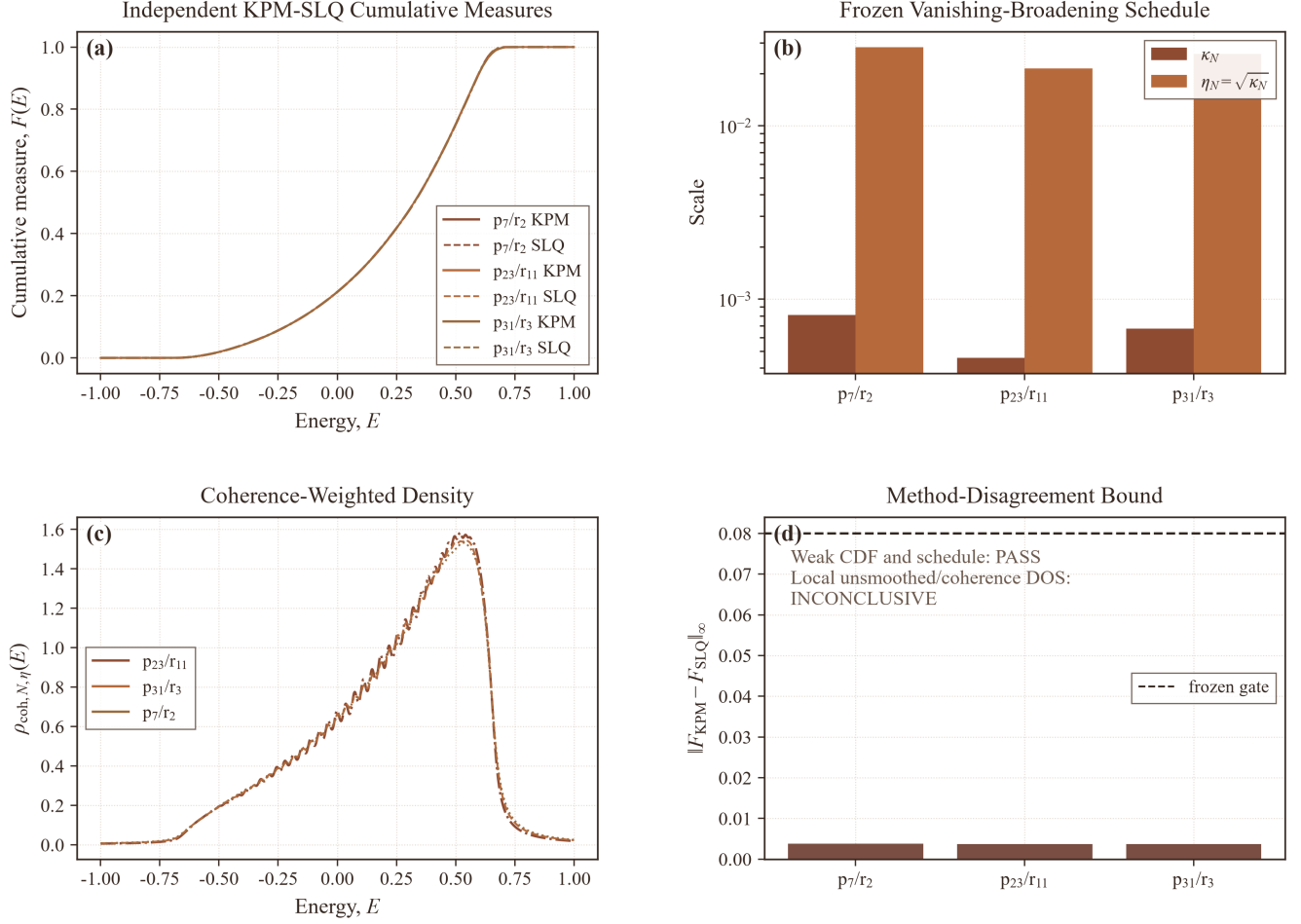


FIG. 10. **Weak spectral-measure convergence and the vanishing-broadening hierarchy.** **a**, Independent KPM and SLQ cumulative measures agree for the three certified projective sectors. **b**, The measured CDF errors κ_N and the preregistered scales $\eta_N = \sqrt{\kappa_N}$ satisfy the finite-broadening schedule. **c**, Coherence-weighted broadened densities obtained from the three towers have the same global profile. **d**, The KPM–SLQ disagreement lies well below the frozen method gate. The weak CDF and finite-broadening statements pass, whereas local unsmoothed and coherence-weighted density convergence remains inconclusive in the absence of a tower-uniform regularity theorem.

J. Operational flat target and curvature-dependent magic landscape

The operational target is accepted only when a common preregistered projector simultaneously satisfies a narrow-band condition, a positive isolation gap, a bounded dispersion diagnostic, and positive coherent participation. We denote the corresponding data by

$$\mathcal{B}_\nu = (W_\nu, \Delta_\nu, \Omega_{\max, \nu}, \rho_{\text{coh}, \nu}, C_{\text{coh}, \nu}). \quad (6475)$$

No single small bandwidth is interpreted as a magic band if the target loses its gap, projector rank, or coherence.

Figure 15 shows the complete transported spectrum at the selected point. The target branch remains nearly dispersionless around $E/t \simeq 3$, while the remaining complete spectral branches lie near $E/t \simeq 9.5$. The large

separation makes the target a well-defined isolated spectral object rather than an accidental cluster of finite-size levels.

The operational magic score M is evaluated on a frozen $(\theta, w/t)$ grid at several curvature values. The score is interpreted only inside the preregistered admissibility domain. The resulting landscape tests whether negative curvature merely shifts an existing Euclidean optimum or reorganizes the full hierarchy of favourable parameter regions.

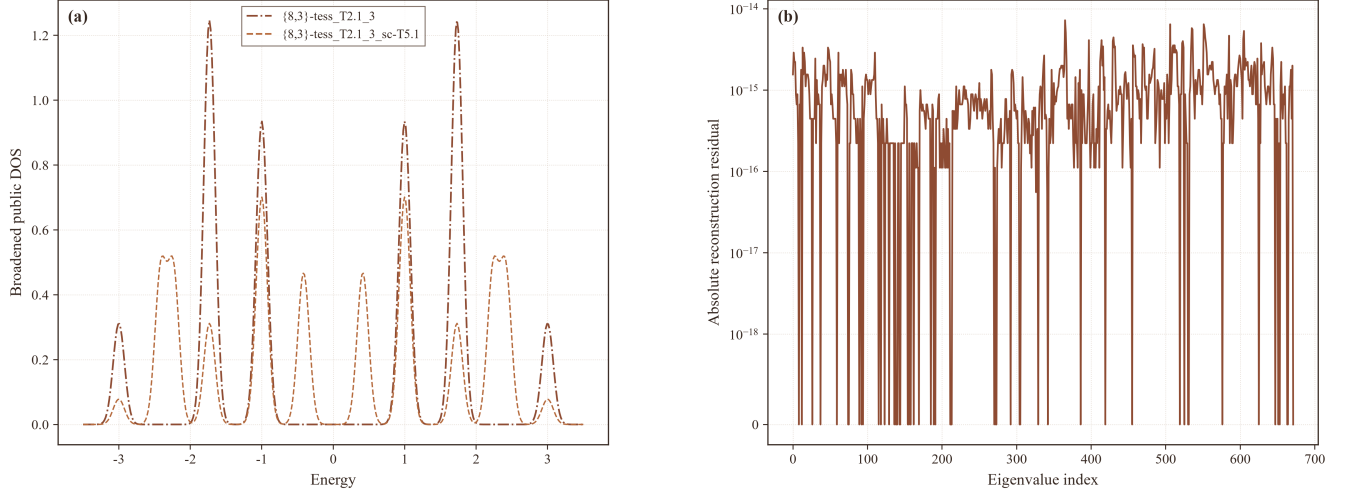


FIG. 11. **Independent reproduction of HyperBloch and circuit spectra.** **a**, Broadened densities of states reconstructed from public $\{8,3\}$ tessellation and supercell files reproduce the expected spectral structures. **b**, Absolute eigenvalue-reconstruction residuals remain at approximately 10^{-15} , consistent with floating-point round-off. The external agreement tests the graph parser, Hermitian assembly, spectral normalization, and circuit mapping independently of the hyperbolic-bilayer results.

K. Restricted universality and microscopic profile coordinates

The operator-collapse theorem is a fixed-comparison-class statement. It does not assert that all changes in curvature radius, twist angle, tunnelling profile, and cut-off can be absorbed into one scalar coordinate. We test the hierarchy

$$\begin{aligned} H_0 : \quad & \hat{H} = \mathcal{H}_*(X), \\ H_2 : \quad & \hat{H} = \mathcal{H}_*(X) + Y_\theta \Psi_\theta, \\ H_3 : \quad & \hat{H} = \mathcal{H}_*(X, K) + Y_\theta \Psi_\theta(X, K), \end{aligned} \quad (6476)$$

and

$$H_4 : \quad \hat{H} = \mathcal{H}_*(X; \mathcal{C}) + Y_\theta \Psi_\theta + Y_{\text{prof}} \Psi_{\text{prof}}, \quad Y_{\text{prof}} \sim \frac{\lambda_\perp}{a}. \quad (6477)$$

Here $\mathcal{C}$ denotes the fixed geometric and microscopic comparison class. Training data are used only to determine the finite correction operators; the conclusions are evaluated on independent holdout families. The principal numerical result is that the previously large radius-dependent residual is almost completely removed by the dimensionless tunnelling-profile coordinate $\lambda_\perp/a$, with an independent median reduction of approximately 99.9936%. In contrast, the unrestricted joint holdout does not pass a common global gate. The validated conclusion is therefore restricted-class universality with a controlled microscopic profile field, not unrestricted one-parameter collapse.

To determine whether curvature nevertheless defines an independent local operator direction, we form the tan-

gent Gram matrix

$$\mathcal{G}_{ab} = \left\langle \frac{\partial \hat{H}}{\partial z_a}, \frac{\partial \hat{H}}{\partial z_b} \right\rangle, \quad (z_1, z_2) = (X, K), \quad (6478)$$

using the fixed operator metric. A stable nonzero second singular value certifies local linear independence of the two operator tangents. Global physical relevance, however, additionally requires that the selected scalar observables retain a stable rank-two Jacobian and that the associated holdouts pass.

L. Integrated validation status and scope of the conclusions

The combined analytical and computational results support the following hierarchy. First, the curvature-controlled local geometry is verified independently of the microscopic Hamiltonian. The exact displacement law, moiré-length inversion, effective-area collapse, dimensional crossover, and incommensurate joint-limit condition all survive independent numerical implementations. Second, exact arithmetic crystallography remains distinct from local geometry. The centered sequence realizes the logarithmic exponent four and the radial ratios 3 and 4, while retaining finite-scale arithmetic oscillations. Sampling an independently certified spectral branch on this sequence transfers the same arithmetic exponent into the magic-complexity coefficient. Third, the microscopic spectral theory exhibits a genuine model-dependent magic-no-magic dichotomy. The exact square five-state model remains root-free, whereas the natural

Negative Controls and Expected Falsification

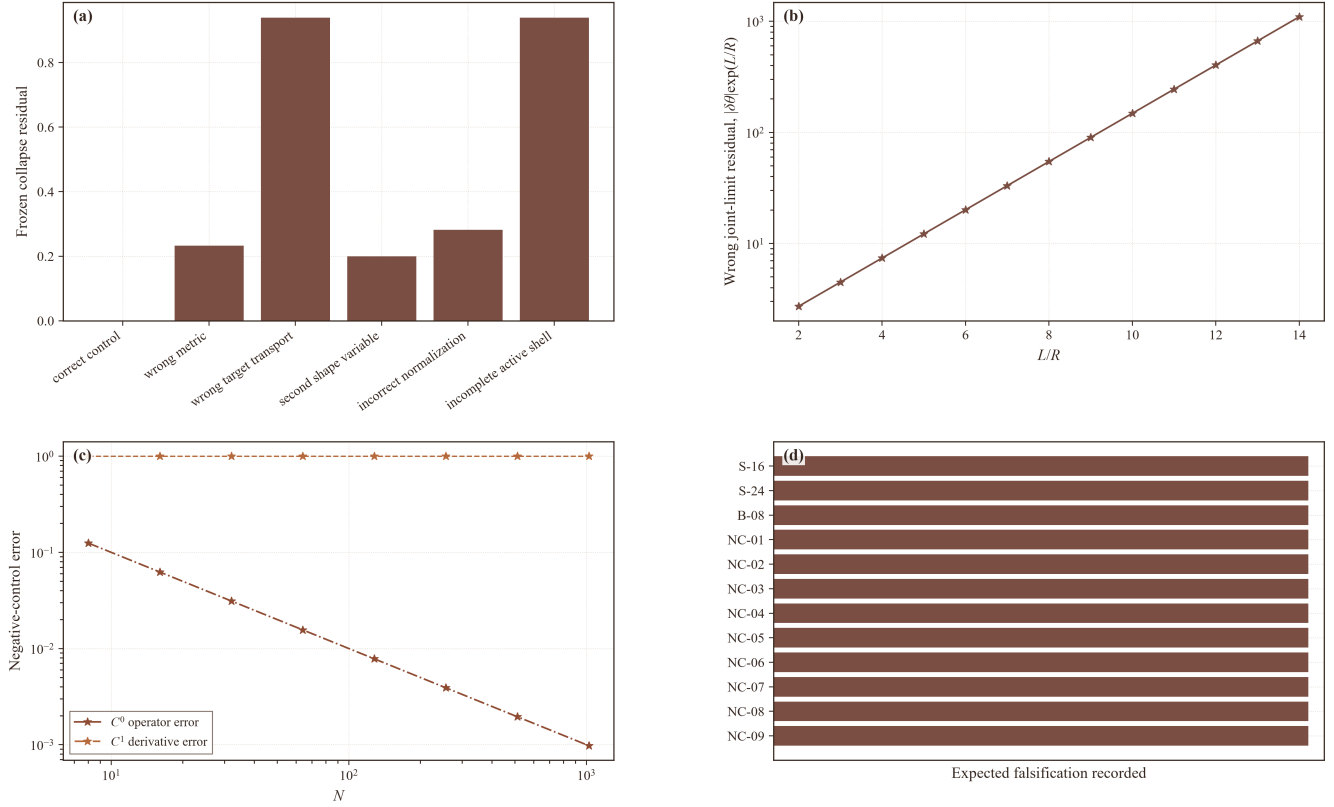


FIG. 12. **Negative controls and expected falsification of invalid reductions.** **a**, Frozen collapse residuals remain small only for the correct control and become finite when the metric, target transport, shape content, normalization, or active shell is altered. **b**, A deliberately incorrect joint-limit sequence grows exponentially with L/R , as predicted when the angular error is not reduced rapidly enough. **c**, The C^0 operator error decreases with system size while the C^1 derivative error remains finite, explicitly falsifying the implication $C^0 \Rightarrow C^1$. **d**, All preregistered expected-failure tasks are recorded. The negative controls demonstrate that the validation pipeline does not return a positive result for arbitrary conventions or incomplete hypotheses.

hyperbolic surface-group Hamiltonian has a positive simple root that persists under the complete exponential kernel. In the regular-octagon symmetry class, the same microscopic family cancels the full canonical Hodge tensor. Controlled orbital and symmetry-breaking perturbations preserve a positive root and isolation gap while correctly generating a traceless Hodge residual away from the exact symmetry class. Fourth, the thermodynamic conclusions retain explicit scope conditions. Three non-Abelian towers support a retained-sector weak bulk limit and independent KPM–SLQ cumulative measures. Strong full-spectrum no-pollution, full-shell $C^0/C^1/C^2$ closure, and a tower-uniform unsmoothed coherence-density theorem remain unresolved. These are recorded as explicit mathematical boundaries rather than interpreted as failures of the conditional theorems. Finally, spectral universality is class restricted. The data reject an unconditional

collapse across arbitrary geometric and microscopic deformations, but strongly support one-parameter collapse within the fixed comparison class. The dominant former radius residual is resolved by the dimensionless profile coordinate $\lambda_{\perp}/a$. A locally independent curvature tangent exists at the operator level, but a globally identifiable second curvature coordinate is not established by the present observable holdouts.

Every arrow in Eq. (??) is supported by an independent calculation or certification layer, while every stronger unresolved implication is kept outside the final claim. This combination of positive construction, negative controls, and explicit inconclusive boundaries supplies a computer-assisted validation of curvature-controlled hyperbolic twistrionics without replacing the analytical scope conditions on which the theory depends.

[1] K. Takeuchi, “Arithmetic triangle groups,” J. Math. Soc. Japan **29**, 91–106 (1977), doi:10.2969/jmsj/02910091,

Theorem 3.

Exact Sampled Magic Complexity

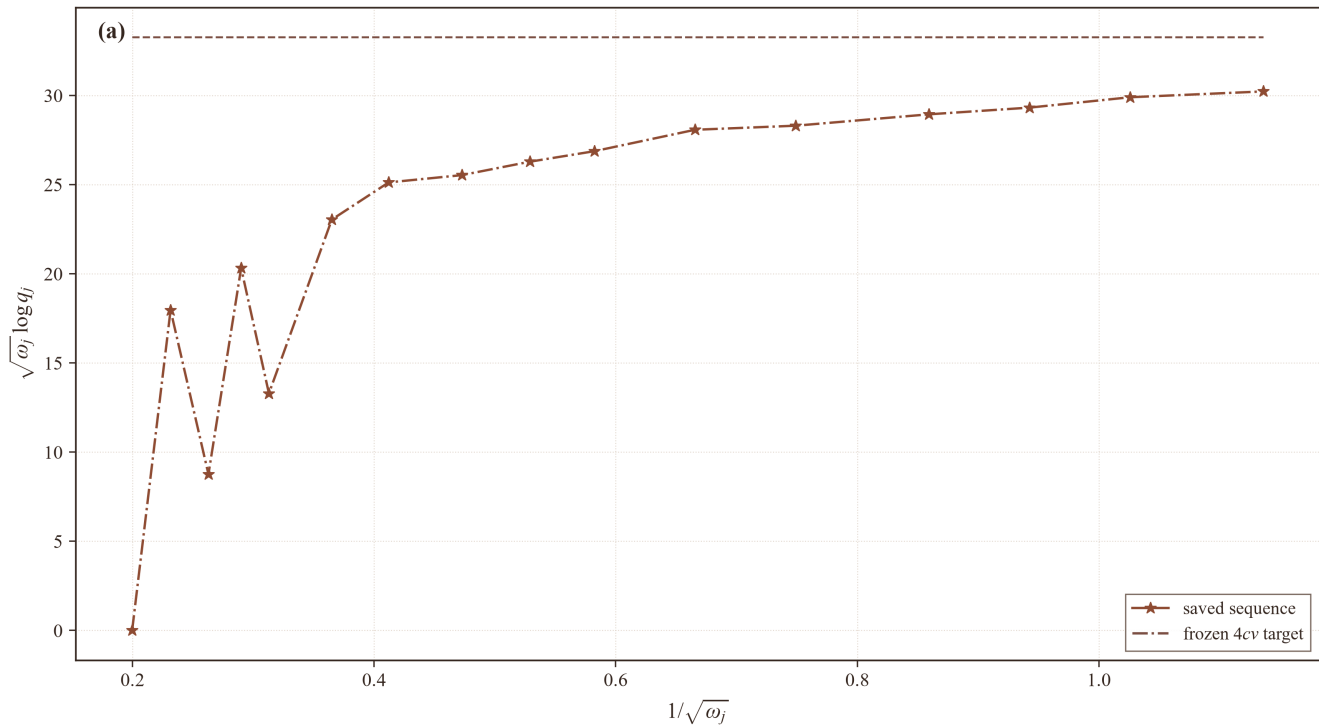


FIG. 13. **Exact arithmetic complexity along a sampled magic sequence.** The quantity $\sqrt{\omega_j} \log q_j$ is evaluated on the exact centered sequence after the positive microscopic branch has been fixed independently. Small-index arithmetic oscillations are visible, followed by systematic approach to the frozen target $4c_\nu$. The figure validates the transfer of the crystallographic exponent four into the logarithmic magic-complexity coefficient; the arithmetic sampling does not itself create the spectral root.

- [2] K. Takeuchi, “Commensurability classes of arithmetic triangle groups,” J. Fac. Sci. Univ. Tokyo Sect. IA Math. **24**, 201–212 (1977), Proposition 2 and Table (1), doi:10.15083/00039716.
- [3] For the tree, elementary-divisor distance, projective-line count, and strong approximation used here, see J. Voight,

Quaternion Algebras, Graduate Texts in Mathematics Vol. 288 (Springer, Cham, 2021), Secs. 23.5 and 26.4 and Main Theorem 28.5.3, doi:10.1007/978-3-030-56694-4. See also M.-F. Vignéras, *Arithmétique des Algèbres de Quaternions*, Lecture Notes in Mathematics Vol. 800 (Springer, Berlin, 1980), doi:10.1007/BFb0091027.

Joint Incommensurate-Limit Falsification

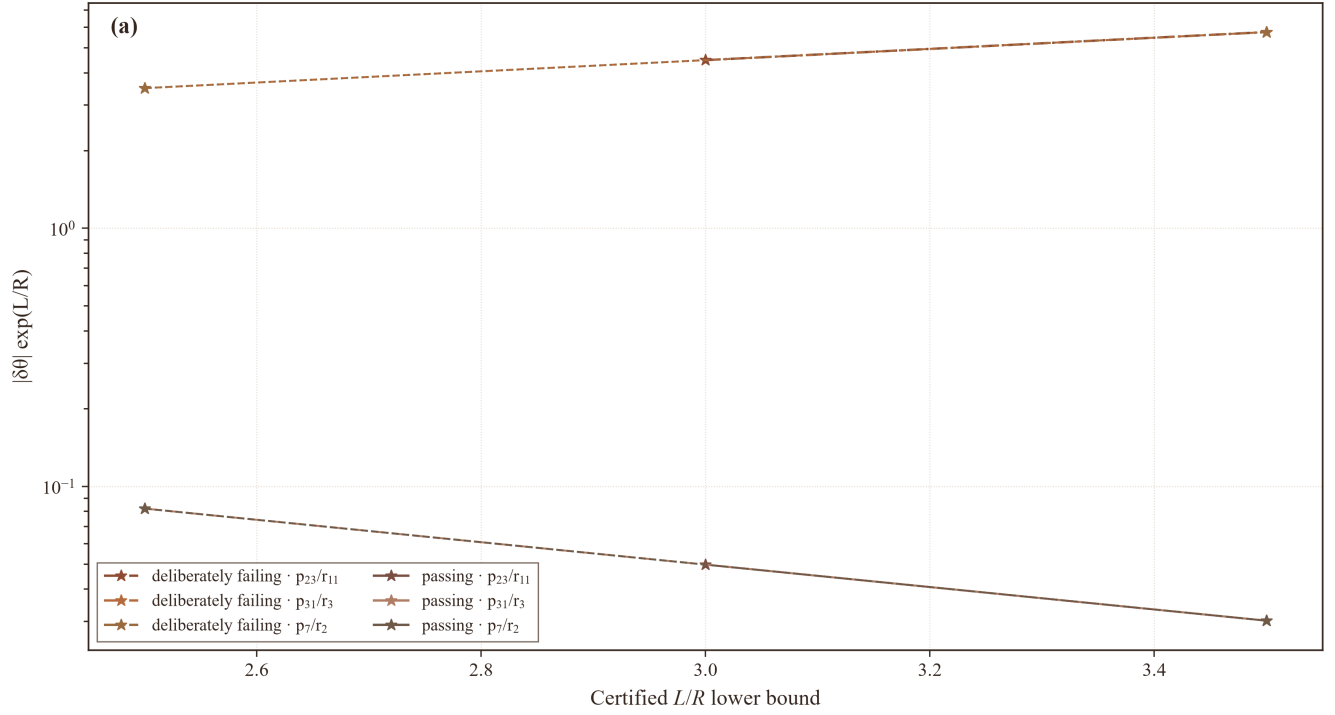


FIG. 14. **Falsification test for the incommensurate angular-spatial joint limit.** Passing sequences decrease in $|\delta\theta| \exp(L/R)$ as the certified lower bound on L/R increases, whereas deliberately failing sequences grow. The three non-Abelian towers reproduce the same separation between the two classes. The result demonstrates that a large nearby commensurate cover is not by itself a controlled approximation: its angular error must decrease exponentially relative to the probed hyperbolic radius.

Operational Magic-Target Spectrum

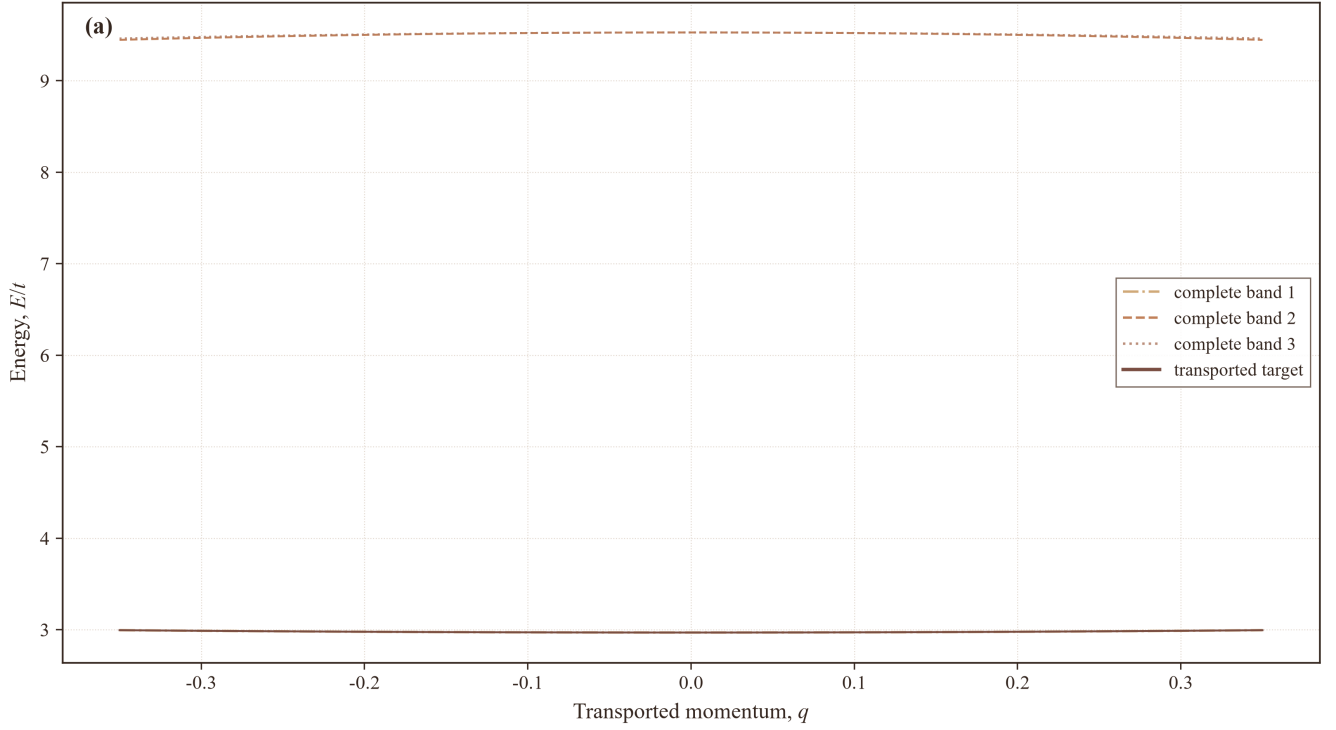


FIG. 15. **Isolated operational magic target in the complete transported spectrum.** The preregistered target forms a narrow branch near $E/t = 3$, while the remaining complete bands are grouped near $E/t = 9.5$. The target bandwidth is small compared with the external separation, and the branch is tracked by the same transported projector over the displayed momentum interval. In combination with the independently stored gap, overlap, and coherence certificates, this spectrum provides the direct microscopic manifestation of the positive hyperbolic root.

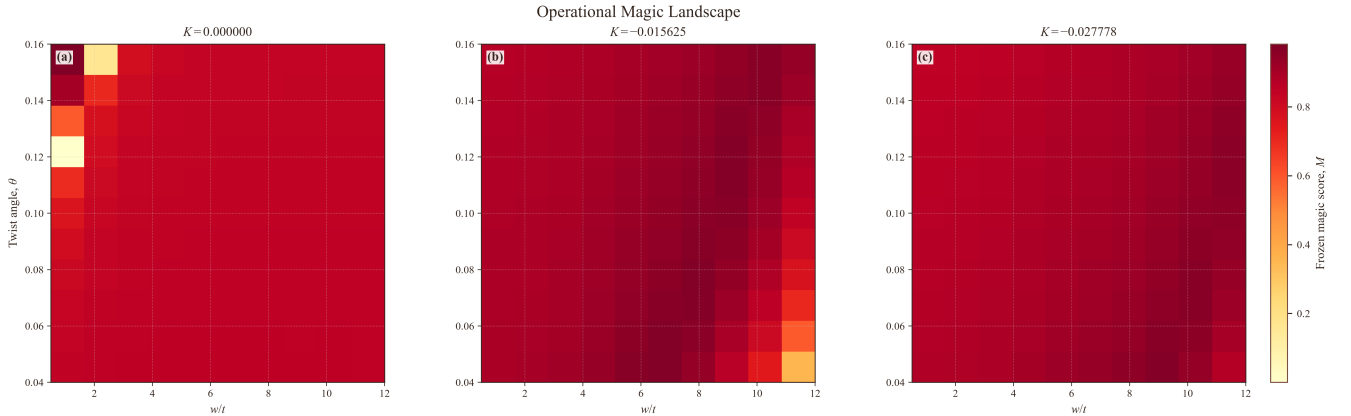


FIG. 16. **Curvature-dependent reconstruction of the operational magic landscape.** The frozen score M is shown as a function of twist angle and interlayer coupling for $K = 0$, $K = -0.015625$, and $K = -0.027778$. The location and extent of high-score regions change strongly with curvature: the Euclidean low-coupling structure is replaced by a broad intermediate curvature region, while stronger negative curvature suppresses or displaces parts of that region. The figure establishes curvature-dependent landscape reconstruction. It does not, by itself, certify a fold, cusp, or curvature-born branch, which require independent derivative and thermodynamic-limit conditions.

Restricted-Class Operator Collapse and Profile-Coordinate Correction

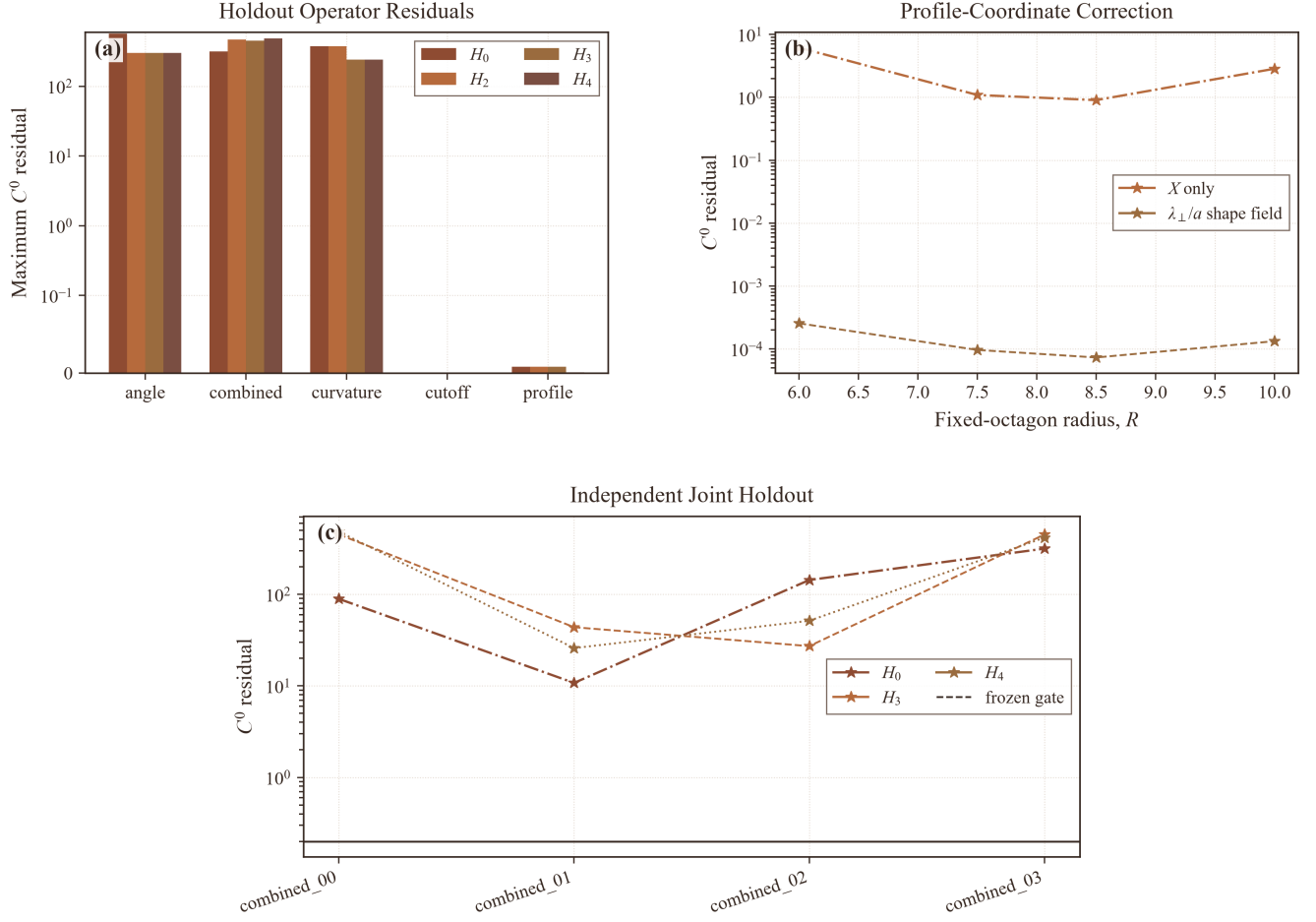


FIG. 17. **Restricted-class operator collapse and microscopic profile-coordinate correction.** **a**, Maximum C^0 holdout residuals for the preregistered hypotheses H_0 , H_2 , H_3 , and H_4 are resolved by deformation channel. Unrestricted angular, combined, and curvature comparisons retain large residuals, whereas fixed-class cutoff and profile comparisons are small. **b**, Replacing an X -only description by the derived shape coordinate $\lambda_{\perp}/a$ reduces the radius-family residual by several orders of magnitude. **c**, Independent joint holdouts remain above the frozen global gate for the unrestricted hypotheses. The data therefore establish PASS_RESTRICTED_CLASS: one-parameter collapse is valid inside the declared comparison class, and the dominant former radius residual is a microscopic tunnelling-profile effect rather than a universally identifiable curvature coordinate.

Operator-Tangent Rank Structure of the Curvature-Related Sector

Local operator-rank evidence; global curvature relevance remains INCONCLUSIVE

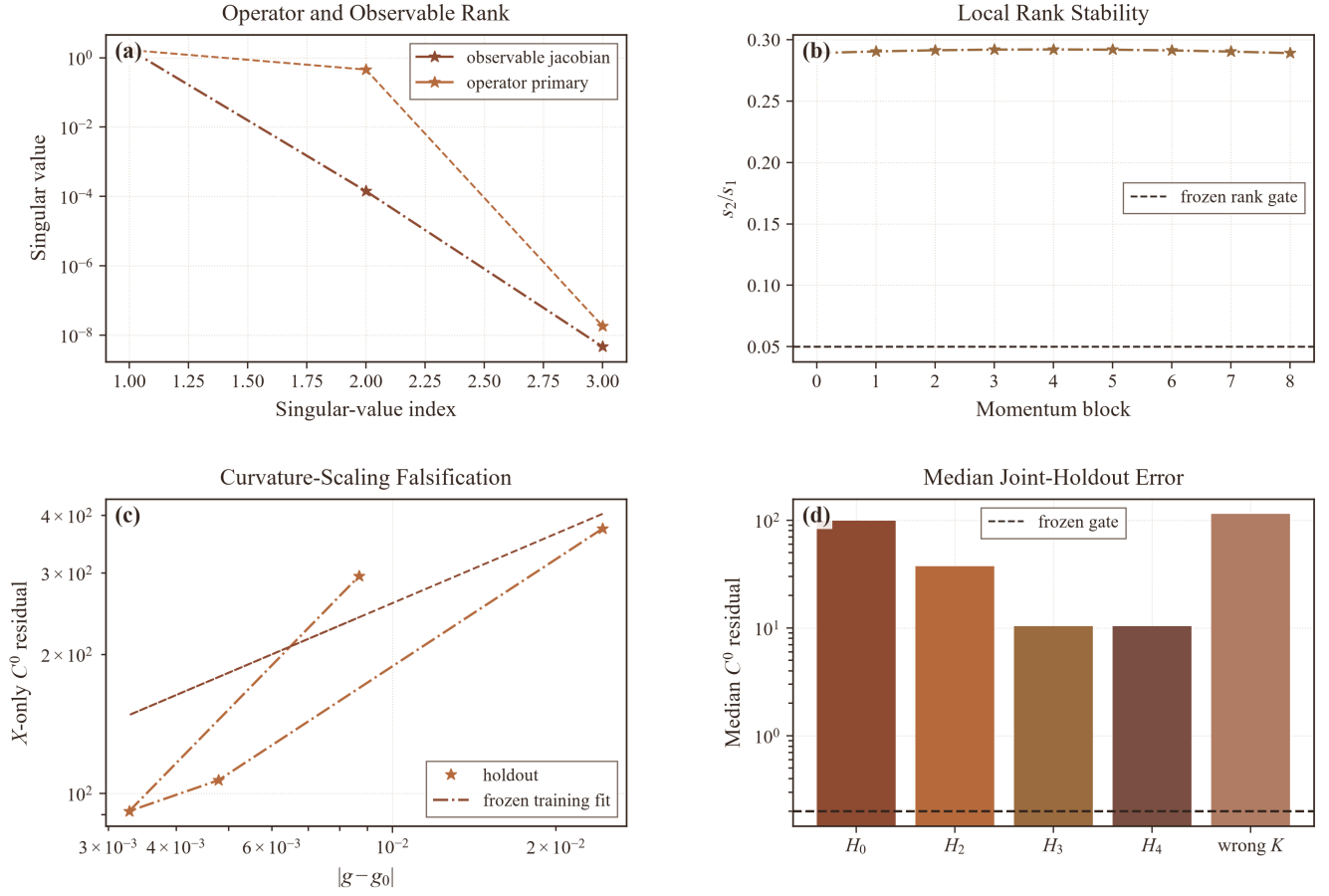


FIG. 18. **Local operator-tangent rank and the boundary of global curvature identifiability.** **a**, The primary operator tangent has a stable second singular value, whereas the scalar-observable Jacobian loses its second direction by several orders of magnitude. **b**, The operator ratio s_2/s_1 remains well above the frozen local rank gate over all momentum blocks, certifying a local rank-two tangent structure. **c**, The X -only curvature holdout does not follow a universally predictive frozen scaling law. **d**, Median joint-holdout residuals for H_0 , H_2 , H_3 , H_4 , and a deliberately wrong curvature convention remain above the global gate. Curvature is therefore locally independent at the operator level, but its globally observable status as a universal second relevant coordinate remains inconclusive.